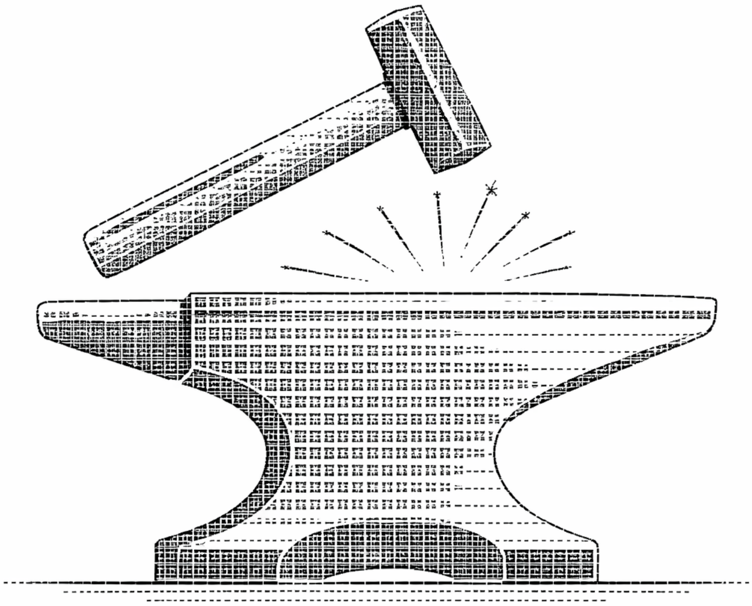
CRYSTALS, GREEN-FUNCTIONS, THERMAL QUANTUM FIELD THEORY AND APPLICATIONS IN QUASI-EQUILIBRIUM MATERIALS SCIENCE



by

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Contents

James Gleick about Isaac Newton:

> "*Newton formed his argument in classic greek geometrical style. Axioms, Lemmas, Corollaries, QED. As the best model available for perfection in knowledge, it gave his physical program the stamp of certainty. He proved facts about triangles and tangents, chords and parallelograms and from there, by a long chain of arguments, proved facts about the moon and the tides.*"

David Hilbert in a lecture given at the International Congress of Mathematicians in Paris 1900:

> "*Zudem ist es ein Irrtum zu glauben, dass die Strenge in der Beweisführung die Feindin der Einfachheit wäre.*"

Galileo Galilei:

> "*The book of nature is written in the language of mathematics.*"

Lucius Annaeus Seneca in his Hercules Furens:

> "*Non est ad astra mollis e terris via.*"

If you find any mistakes, please contact[1] me so that I can correct them.

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This still needs to be done or digitalized

Legend:

- |————————| = Important for the doctoral thesis

Chapter 5:

- section 5.2: the derivation of the electron-phonon-coupling, as in Felicianos article.

Chapter 6: The derivation of the Bloch-Theorem and the reference to Slater-determinants solving the non-interacting problem.

Chapter 7:

- Proof of Corollary 7.32.
- Proof of Corollary 7.36.
- Looking up what a derivation is in mathematics and adding a remark on that, as in Remark 7.36.1.
- Proof of Corollary 7.40.
- Proof of Lemma 7.41 (only needs to be digitalized).
- Looking up what a derivation is in mathematics and adding a remark on that in Corollary 7.44.
- Elaborating on Definition 7.46.
- Proof of Corollary 7.48.

Chapter 8:

- Updating Definition 8.14 (is important for Eliashberg-theory from the Operator-formalism)
- Proof of Lemma 8.31 (only needs to be digitalized).

Chapter 9:

- Proof of Lemma 9.6.
- Proof of Theorem 9.7.
- Proof of Theorem 9.9.
- Elaborating on Remark 9.10.1.
- Proof of Corollary 9.16.
- Definition 9.18 (Thermal equilibrium of two systems).
- Proof of Theorem 9.19.
- Elaborating on Hypothesis 9.24.
- Proof of Theorem 9.26.
- Proof of Theorem 9.28.

- Introduction to the grand canonical ensemble in section 9.5.
- Proof of the equivalence of ensembles in section 9.6.
- Elaborating on Legendre-transformations in thermodynamics in section 9.7.
- Elaborating on the transition from microcanonical to canonical ensemble in section 9.8.2.
- Proof of Lemma 9.35.
- Elaborating on Theorem 9.41.
- Elaborating on the transition from canonical to grand canonical ensemble in section 9.8.3.
- Proof of Corollary 9.42 (only needs to be digitalized).

Chapter 10:

- Thermodynamic relations for mixed-systems as in Remark 10.7.3.

Chapter 12:

- Adding proper reference in Remark 12.4.1.
- Putting the references into the right place under Equation 12.7.
- Giving nice final Equations in Theorem 12.9.
- Adding the differential equations for phononic Green-functions as in Theorem 12.12.
- Double-Checking the results in Theorem 12.13.
- Finishing Theorem 12.14 and adding the identities from the Rammer-book.

Chapter 13:

- Adding the shape in Theorem 13.8 and making a comment about the limiting case that the poles approach each other and everything is well defined due to the continuity in the Residue-theorem.
- Proof of Theorem 13.10.
- Proof of Theorem 13.11.

Chapter 14:

- Finding a reference to Fact 14.2.
- Adding a comment on the positivity of the spectral function for positive frequencies in the case of hermitian adjoint operators (even (!) for the bosonic case).
- Adding the identities in Theorem 14.9.
- Reworking section 14.3 and adding the identities for phononic Green-functions.

Chapter 15:

- Adding the expression for the free time evolution in Theorem 15.3.

- Proof of Theorem 15.4 and checking the prefactors. [2]
- Proof of Theorem 15.5. [3]
- Straightening the references in Remark 15.5.1 .
- Straightening the references in Remark 15.6.1.
- Straightening the references in Remark 15.6.2 (also in the footnote).
- Straightening the reference in Remark 15.8.2 (also in the footnote).
- Finding a proper reference for the Residue-Theorem i.e. Theorem 15.13.

Chapter 16 :

- Proof of Lemma 16.6.
- Proof of Theorem 16.9.
- Proof of Lemma 16.10.
- Proof of Lemma 16.11.
- Proof of Lemma 16.17 (only needs to be digitalized).
- Proof of the trace-log-identity in Corollary 16.18.
- Proof of the Wick-Theorem i.e. Theorem 16.21.
- Adding references from later chapters to Fact 16.25 and making the reference appropriate.
- Proof of Theorem 16.27.
- Fixing the references in Definition 16.29.
- Fixing the references in Remark 16.29.1.
- Proof of Lemma 16.31 (only needs to be referenced to the symmetries chapter).
- Proof of Lemma 16.32 (only needs to be digitalized).
- Elaborating on Remark 16.32.2.
- Proof of Lemma 16.36 (only needs to be digitalized).
- Adding references to Remark 16.39.1.
- Fixing the references in Remark 16.40.2.
- Proof of Corollary 16.41.
- Elaborating on Remark 16.44.2 and Remark 16.45.1.
- Highlighting that Theorem 16.45 only holds in the case of no SSB.

[2] Only the formulation must be implemented. The proof can be referenced.
[3] Again: Only the formulation SHOULD be implemented. The proof is impossible to find in the literature.

- Making a nice figure for Remark 16.47.2.
- Proof of Lemma 16.49 (only needs to be digitalized).
- Updating the proofs in subsections 16.1.7 and 16.1.8.
- Updating section 16.2.

Chapter 17:

- Just fixing the thiings from my master-thesis and then referencing the Thomale-paper for the rest of the details.
- Implementing the consequences of the symmetries in section 17.1.2 (only needs to be copied from the Thomale-paper).

Chapter **??**: Getting a grip on the Feynman-diagrams (cf. Salmhofer book and "Work in Progress"-chapter).

Chapter 19:

- Digitalizing the proofs in the chapter.
- Adding the references in Remark 19.4.1.
- Adding the reference in Lemma 19.9.
- Fixing the reference in Remark 19.13.2.
- Adding the value in Remark 19.14.1.
- Figuring out which identity I was referring to in Remark 19.14.3.
- Digitalizing the Equation in Corollary 19.15.
- Straightening the references in Lemma 19.16.
- Elaborating on subsection 19.1.3.
- Straightening the references in subsection 19.1.5.
- Fixing the reference in Definition 19.21.
- Fixing the proof of Theorem 19.22.
- Fixing the proof of Lemma 19.24 (only a reference is needed).
- Elaborating on Theorem 19.26.
- Elaborating on subsection 19.2.3.

Chapter 20 :

- Fixing the reference in the introduction.
- Fixing references to the models.
- Fixing the reference in Remark 20.1.1.

- Checking the reference in Theorem 20.2.
- Calculating the polarization in Theorem 20.4.

Chapter 21:

- Checking if the details are really compatible with the definition of the Green-functions introduced earlier.
- Adding references to earlier identities (?).

Chapter 23 :

- Adding a reference / figure for the DOS in the interacting case.
- Checking the details of the Proof of Lemma 23.8.
- Adding an equation for the electronic Entropy and phononic thermodynamic variables and also in the interacting

Chapter 24 :

- Adding internal energy expression in Fact 24.19.
- Digitalizing the expression in Lemma 24.22.
- Adding the expression of the interaction energy temperature-derivative in subsubsection 24.2.3.4.

Chapter 25: Implementing (i.e. digitalizing) the corrections from my notes.

Part I.

Fundamentals on Quantum Mechanics and Crystals

Motivation and Overview

After this part of the notes you will understand ...

- what donuts in Figure 1 have to do with materials theory,
- how and why one determines the band-structure of a crystal, and how and why it is usually represented by pictures like in Figure 2,

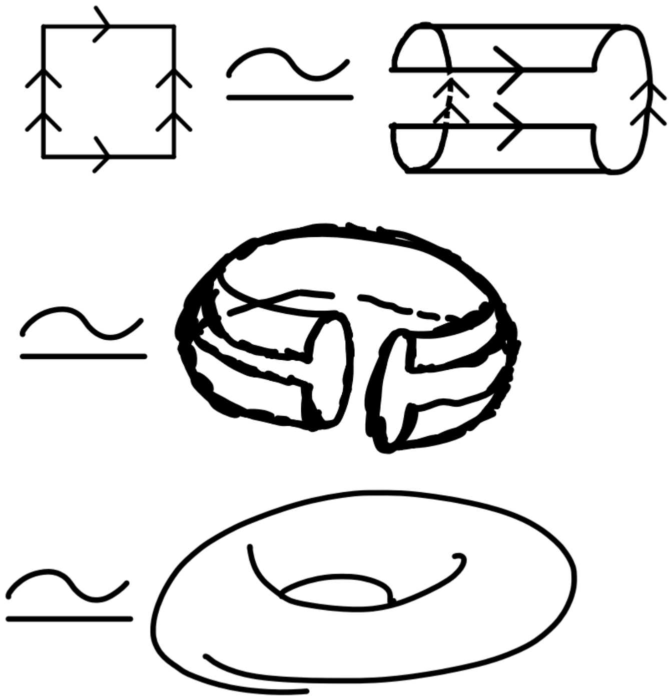


Figure 1.: Identifying the boundaries of a square yields a torus („donut"). One also says that the two objects are „homeomorphic" to each other.

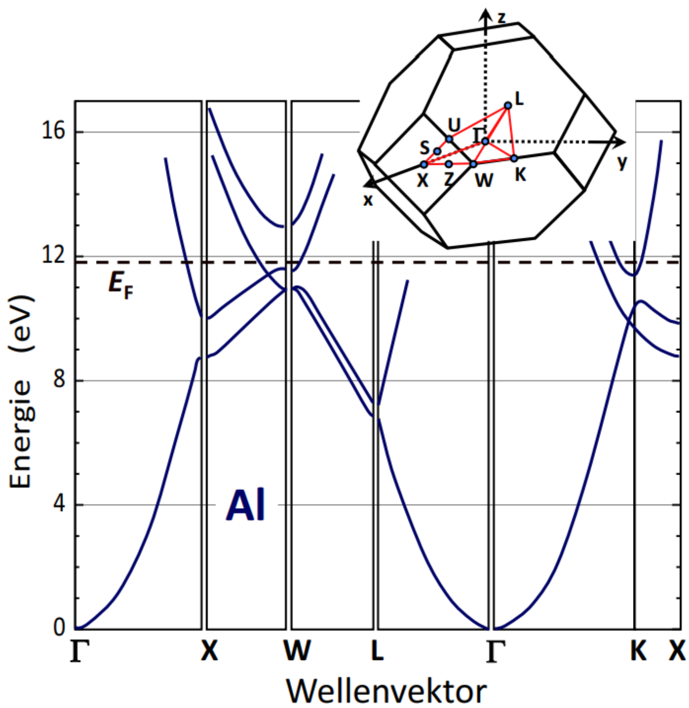


Figure 2.: The Band-Structure of Aluminium along the high-symmetry **k**-points. Taken from [GM].

1. Fourier-Expansions

Fact 1.1 The Fourier-transformation exists in all kinds of shapes and colours. One can expand

- finite sequences of numbers in the complex plane
- periodic functions on a finite interval $(0, L)$,
- antiperiodic functions on a finite interval $(0, \hbar\beta)$ leading to the so-called Matsubara-frequencies.
- sufficiently fast decaying functions[1] on $\mathbb{R}$,

Here we will get to know the different versions that are important for materials theory.

1.1. Fourier for a Discrete Function on a Finite Set of Points

Definition 1.2 For a sequence of N (complex or real-valued) numbers $\left\{\xi_1, \dots, \xi_N\right\} \subset \mathbb{C}^{d_x}$, one introduces their *discrete Fourier-transformated sequence* as another sequence of numbers $\left\{\hat{\xi}_1, \dots, \hat{\xi}_N\right\} \subset \mathbb{C}^{d_x}$ as

$$\hat{\xi}_j = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} \xi_n \, \mathrm{e}^{+\mathrm{i}2\pi \frac{n}{N} j} \tag{1.1}$$

Remark 1.2.1 There are many prefactor-conventions in this definition, which can be quite annoying when one looks at literature, and they don't state which one they used. The sign in the exponent could have been chosen the other way around in Equations (1.1) and (1.2). The sign-convention chosen above refers usually to j representing a „space-like“-coordinate, whereas the other sign-convention refers to „time-like“-coordinates. The prefactor $\frac{1}{\sqrt{N}}$ in Equation (1.1) can also be exchanged for 1, or $\frac{1}{N}$, or sometimes $\frac{1}{\sqrt{Na}} \equiv 1/\sqrt{L}$ or $\frac{1}{Na} \equiv 1/L$ with corresponding changes in the Equation (1.2) accordingly, whereas a represents a so-called "lattice constant" (will be introduced later). If one chooses the last prefactor convention, one also sometimes encounters integral notations, although one is strictly speaking working with discrete variables. This trend is particularly present in thermal quantum field theory (cf. Equation TBReferenced in Chapter TBReferenced), when one introduces Graßmann-variables in integral notation for their temporal argument, although they actually refer to discrete variables. Notice further that one of the prefactor-conventions here corresponds to the usual Fourier-transformation for periodic functions (with slightly changed notation).

Remark 1.2.2 As we will see below, one introduces the same notation for position-*operators* of ions that are centered around a crystal-site. Consequently, one finds Fourier-transforms of operators and the inverse transformations.

[1] Notice that these functions coincide at $x = \pm\infty$, such that they are "periodic to zero at infinity", which is the main idea behind the derivation of the underlying formulae.

Lemma 1.3 The inverse transformation reads

$$\xi_j = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} \hat{\xi}_n \, \mathrm{e}^{-\mathrm{i}2\pi \frac{n}{N} j} \tag{1.2}$$

Proof 1.3 Inserting the one equation into the other gives

$$\begin{aligned}
\xi_j &= \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} \hat{\xi}_n \, \mathrm{e}^{-\mathrm{i}2\pi \frac{n}{N} j} \\
&= \frac{1}{N} \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} \xi_m \, \mathrm{e}^{-\mathrm{i}2\pi \frac{n}{N} j} \, \mathrm{e}^{+\mathrm{i}2\pi \frac{m}{N} n} \\
&= \frac{1}{N} \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} \xi_m \, \exp\left[2\pi\mathrm{i} \frac{1}{N} (mn - jn)\right] \\
&= \frac{1}{N} \sum_{m=0}^{N-1} \xi_m \sum_{n=0}^{N-1} \left(\exp\left[2\pi\mathrm{i} \frac{1}{N} (m - j)\right]\right)^n = \xi_j
\end{aligned} \tag{1.3}$$

whereas we used in the last step the finite geometric series

$$\sum_{n=0}^{N-1} \left(\exp\left[2\pi\mathrm{i} \frac{1}{N} (m - j)\right]\right)^n = \begin{cases} N, & m = j \\ \frac{1-\mathrm{e}^{2\pi\mathrm{i}(m-j)}}{1-\mathrm{e}^{2\pi\mathrm{i}\frac{1}{N}(m-j)}} = 0, & m \neq j \end{cases} . \tag{1.4}$$

1.2. Fourier for a Continuous, Periodic Function on a Finite Interval

This section briefly lists basic terms and concepts that are mostly assumed to be known to clarify the notation.

Notation 1.4 We consider the set of periodic, square-integrable functions on (a, b) as

$$L^2_{\mathrm{per}}(a, b) = \left\{ f : (a, b) \longrightarrow \mathbb{C} \quad \middle| \quad f(a) = f(b) \right\} \tag{1.5}$$

Lemma 1.5 The set $L^2_{\mathrm{per}}(a, b)$ is, equipped with point-wise addition and scalar-multiplication a complex vector space.

Proof 1.5 One easily verifies the different properties.

Lemma 1.6 The mapping

$$\begin{aligned} \langle\ ,\ \rangle : L^2_{\text{per}}(a,b) \times L^2_{\text{per}}(a,b) &\longrightarrow \mathbb{C} \\ (f,g) &\longmapsto \langle f,g\rangle \equiv \langle f|g\rangle \end{aligned} \tag{1.6}$$

with

$$\langle f,g\rangle := \int_a^b \overline{f(x)}g(x)\,dx.$$

defines a hermitian scalar-product on $L^2_{\text{per}}(a,b)$.

Proof 1.6 See Kaul, Fischer, Band 2.

Theorem 1.7 For $f \in L^2_{\text{per}}(a,b)$ one finds the *Fourier-expansion* in terms of complex exponentials as

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{\pm \mathrm{i}\frac{2\pi n x}{b-a}}, \tag{1.7}$$

where the Fourier coefficients c_n are given by

$$c_n = \frac{1}{b-a}\int_a^b f(x) e^{\mp \mathrm{i}\frac{2\pi n x}{b-a}}\,dx. \tag{1.8}$$

which converges in the L^2-sense[2].

Proof 1.7 A proof, that periodic functions can be expanded in this way, is given in the literature[3].

Remark 1.7.1 The sign is convention; In analogy to electromagnetic waves one usually agrees to use the one sign-convention "$\pm \equiv +$" if ξ represents a space-like-argument, and the other sign-convention "$\pm \equiv -$" if it is a time-like-argument. Notice that just as in the discrete case, one oftentimes shifts around prefactors into / out of the Fourier-coefficients.

Remark 1.7.2 One immediately finds for the functions defined on a three-torus (i.e. periodic in three arguments[4]) similar expansions.

[2]This means that

$$\lim_{N\to\infty}\int_a^b \left| f(x) - \sum_{n=-N}^{N} c_n e^{\pm \mathrm{i}\frac{2\pi n x}{b-a}} \right|^2 dx = 0. \tag{1.9}$$

[3]In the book [5], one finds the expansion and the identities for the coefficients for periodic functions that are defined on the interval $[0, 2\pi]$. So one will not circumvent a little bit of effort with the substitution rule, in order to reproduce the results above.

[4]Notice that wave-functions in a crystal (after applying of Born-von-Karman-boundary-conditions fall into this category.

Notation 1.8 For $f \in L^2_{\text{per}}([a, b])$ we denote its periodic standard-continuation on $\mathbb{R}$ with the same symbol for notational simplicity.

Fact 1.9 The plane-waves provide a basis for $L^2_{\text{per}}([a, b])$. Together with the scalar-product, this turns the space into a Hilbert-space.

1.3. Fourier for a Continuous, Statistical Function on a Finite Interval

Notation 1.10 For reasons, that are difficult to motivate several chapters in advance, we consider functions

$$\mathscr{G}_\zeta \in L^2_{\text{per}}(-\hbar\beta, +\hbar\beta) \tag{1.10}$$

with time-like argument

$$\begin{aligned} \mathscr{G}_\zeta :\ (-\hbar\beta, +\hbar\beta) &\longrightarrow \mathbb{C} \\ \tau \quad &\longmapsto \mathscr{G}_\zeta(\tau) \end{aligned}$$

and which carry a *statistical index* $\zeta \in \{-1, +1\}$ such that the *statistics-condition* is fulfilled

$$\forall \tau \in (-\hbar\beta, 0) : \quad \mathscr{G}_\zeta(\tau + \hbar\beta) = \zeta \mathscr{G}_\zeta(\tau). \tag{1.11}$$

Remark 1.10.1 We call a function that obeys this condition to be *ζ-periodic.*

Remark 1.10.2 The statistical index ζ represents *bosonic* ($\zeta = +1$) and *fermionic* ($\zeta = -1$) functions.

Remark 1.10.3 Denoting it's analytic standard-continuation on $\mathbb{R}$ with the same symbol, we obtain the property

$$\forall \tau \in \mathbb{R} : \quad \mathscr{G}_\zeta(\tau + \hbar\beta) = \zeta \mathscr{G}_\zeta(\tau). \tag{1.12}$$

Lemma 1.11 By a Fourier-expansion one finds the expansion for the periodic standard-continuation as

$$\mathscr{G}_\zeta(\tau) = \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} \mathscr{G}_\zeta(\mathrm{i}\omega_n)\, \mathrm{e}^{-\mathrm{i}\omega_n \tau} \tag{1.13}$$

with coefficients

$$\mathcal{G}_\zeta(\mathrm{i}\omega_n) = \int_0^{\hbar\beta} \mathrm{d}\tau\, \mathcal{G}_\zeta(\tau)\, \mathrm{e}^{\mathrm{i}\omega_n \tau}. \tag{1.14}$$

and *Matsubara-frequencies* that depend on the statistics

$$\Omega_\zeta = \frac{2\mathbb{Z} + 1 - \theta(\zeta)}{\hbar\beta}\pi. \tag{1.15}$$

Proof 1.11 See Appendix, Proofs, page 306.

Corollary 1.12 For future reference we note the identity

$$\forall \omega_m \in \Omega_{+1} : \int_0^{\hbar\beta} \mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}\tau\omega_m} = \delta_{\omega_m,0}\hbar\beta \tag{1.16}$$

Proof 1.12 See Appendix, Proofs, page 307.

Remark 1.12.1 Notice that the difference of two fermionic Matsubara-frequencies is again a bosonic Matsubara-frequency.

Definition 1.13 For each $\zeta \in \{-1, +1\}$ we denote the corresponding subspace of $L^2_{\mathrm{per}}(-\hbar\beta, +\hbar\beta)$ for which Equation (1.11) is fulfilled with $L_\mathrm{B} \equiv L_1$ and $L_\mathrm{F} \equiv L_{-1}$ respectively.

Lemma 1.14 The space of periodic functions can be decomposed into the direct sum of bosonic and fermionic functions as

$$L^2_{\mathrm{per}}(-\hbar\beta, +\hbar\beta) = L_\mathrm{B} \oplus L_\mathrm{F} \equiv \bigoplus_{\zeta\in\{-1,+1\}} L_\zeta \tag{1.17}$$

Proof 1.14 The plane-waves can be divided into bosonic and fermionic parts by considering the corresponding Matsubara-frequencies. The bosonic and fermionic plane-waves are linearly independent, and they span the whole space of periodic functions.

Lemma 1.15 On L_ζ we obtain the resolution of identity

$$\mathbb{1}_{L_\zeta} \equiv \sum_{\omega_n \in \Omega_\zeta} |\, \mathrm{i}\omega_n \rangle\langle\, \mathrm{i}\omega_n \,| \tag{1.18}$$

with plane waves

$$\langle \tau \mid \mathrm{i}\omega_n \rangle \equiv \mathrm{e}^{-\mathrm{i}\omega_n \tau}, \qquad \langle \mathrm{i}\omega_n \mid \tau \rangle \equiv \mathrm{e}^{+\mathrm{i}\omega_n \tau} \tag{1.19}$$

and the delta-function

$$\delta(\tau) = \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} \mathrm{e}^{-\mathrm{i}\omega_n \tau} \tag{1.20}$$

Proof 1.15 See Appendix, Proofs, page 307.

1.4. Fourier for a Continuous, Decaying Function on an Infinite Interval

Notation 1.16 For „sufficiently smooth“ complex-valued functions and distributions that decay „sufficiently fast“ at space-time-infinity

$$\begin{aligned} f :\ & \mathbb{R}^3 \times \mathbb{R} \longrightarrow \mathbb{C} \\ & (\vec{x}, t) \quad \longmapsto f(\vec{x}, t) \equiv \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \int \frac{\mathrm{d}\omega}{2\pi}\, \mathrm{e}^{\mathrm{i}\left(+\vec{k}\cdot\vec{x} - \omega t\right)}\, f(\vec{k}, \omega) \end{aligned}$$

we write for the ("physical") *Fourier-transformation*

$$\begin{aligned} (\mathscr{F} f) :\ & \mathbb{R}^3 \times \mathbb{R} \longrightarrow \mathbb{C} \\ & (\vec{k}, \omega) \ \longmapsto (\mathscr{F} f)(\vec{k}, \omega) \equiv f(\vec{k}, \omega) \equiv \int \mathrm{d}^3 x \int \mathrm{d}t\, \mathrm{e}^{-\mathrm{i}\left(+\vec{k}\cdot\vec{x} - \omega t\right)}\, f(\vec{x}, t) \end{aligned}$$

Proof 1.16 A proof for this expansion, can be found in [5].

2. Quantum Mechanics

There is at least three formalisms with which one can describe quantum-mechanical systems:

1. The „wavefunction-formalism“,
2. The „density-matrix-formalism“,
3. The „path-integral-formalism“.

In this chapter we will focus on the wavefunction-formalism, the other formalisms build upon it.

2.1. Fundamentals about Hilbert-Spaces

Definition 2.1 A Hilbert-space $\mathcal{H}$ is a complex vector space, with a hermitian[1] scalar-product

$$\begin{aligned} \langle\ ,\ \rangle : \mathcal{H} \times \mathcal{H} &\to \mathbb{C} \\ (\psi, \chi) &\mapsto \langle \psi, \chi \rangle \equiv \langle \psi | \chi \rangle \end{aligned}$$

which is complete and has a countable orthonormal basis[2].

Notation 2.2 The countable orthonormal basis from the definition above is usually denoted as $\{|1\rangle, |2\rangle, \dots, |d_{\mathcal{H}}\rangle\}$ such that

$$\mathcal{H} = \operatorname{span}_{\mathbb{C}}\{|1\rangle, |2\rangle, \dots, |d_{\mathcal{H}}\rangle\} \tag{2.1}$$

and most oftentimes the basis is specified further, depending on the context of the problem.

Fact 2.3 Some common single-particle Hilbert-spaces[3] to work with are in $d_{\mathbf{x}} \in \{1, 2, 3\}$ spatial-dimensions ...

- $\mathcal{H} = L^2(\mathbb{R}^{d_{\mathbf{x}}})$ with $\langle \psi | \chi \rangle = \displaystyle\int \mathrm{d}^{d_{\mathbf{x}}}\mathbf{x}\, \overline{\psi}(\mathbf{x}) \chi(\mathbf{x})$ for a Schrödinger-particle in space,
- $\mathcal{H} = L^2(\mathbb{T}^{d_{\mathbf{x}}})$ with $\langle \psi | \chi \rangle = \displaystyle\int_{\mathrm{vol}} \mathrm{d}^{d_{\mathbf{x}}}\mathbf{x}\, \overline{\psi}(\mathbf{x}) \chi(\mathbf{x})$ for a Schrödinger-particle in a $d_{\mathbf{x}}$-dimensionsional „crystal“[4],

[1] Recall that a hermitian scalar-product is linear w.r.t. the second argument, obeys $\langle \psi_1, \psi_2 \rangle = \langle \psi_2, \psi_1 \rangle^*$ and is positive semidefinite.
[2] Note that the latter condition is equivalent to the Hilbert-space being „separable“. The precise meaning of this can be looked up. Usually (in almost all situations) one is however doing quite well with the abovementioned definition.
[3] Proving that these spaces fall into the definition above is far from trivial. For example the separability of $L^2(\mathbb{R})$ takes (according to "folklore") about 6 pages in J. von Neumanns book.
[4] The precise meaning of the word „crystal“ will be introduced later. At this point we note, that one can identify coordinates $\mathbf{x}$ in a „crystal“ (with periodic boundary conditions) with angles on a torus.

- $\mathcal{H} = \mathbb{C}^2$ with complex standard-scalarproduct for a two-level-system,
- $\mathcal{H} = L^2(\mathbb{R}^{d_x}) \otimes \mathbb{C}^2$ with scalar-product $\langle \psi | \chi \rangle = \sum_{\sigma \in \{\uparrow,\downarrow\}} \int \mathrm{d}^{d_x}\mathbf{x}\, \overline{\psi}_\sigma(\mathbf{x}) \chi_\sigma(\mathbf{x}) \equiv \sumint_{\sigma,\mathbf{x}} \overline{\psi}_\sigma(\mathbf{x}) \chi_\sigma(\mathbf{x})$ for a Pauli-particle in space,
- $\mathcal{H} = L^2(\mathbb{T}^{d_x}) \otimes \mathbb{C}^2$ with scalar-product $\langle \psi | \chi \rangle = \sum_{\sigma \in \{\uparrow,\downarrow\}} \int_{\mathrm{vol}} \mathrm{d}^{d_x}\mathbf{x}\, \overline{\psi}_\sigma(\mathbf{x}) \chi_\sigma(\mathbf{x}) \equiv \sumint_{\sigma,\mathbf{x}} \overline{\psi}_\sigma(\mathbf{x}) \chi_\sigma(\mathbf{x})$ for a Pauli-particle in a „crystal“,

Remark 2.3.1 For the Hilbert space in physics, it is common practice to consider those elements as equivalent / equal, that differ only on sets of measure zero, thereby working with equivalence classes. However, since these distinctions have no effect on the observables considered in this work, we shall not concern ourselves with them here.

Remark 2.3.2 When there is more than one particle present, new structures and phenomena related to permutation symmetry appear. It turns out that the generalization to n-particle quantum mechanics depends on whether the particles under consideration are distinguishable or indistinguishable. In the former case the Hilbert space $\mathcal{H}$ of the many-particle system is simply the tensor product

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \cdots \otimes \mathcal{H}_n. \tag{2.2}$$

Remark 2.3.3 The Hilbert-spaces for indistiguishable many-particle-systems are a little bit tedious to introduce (approximately 20 pages in the book from Negele and Orland). We will consider them in a later chapter. Apart from the single-particle-examples however, everything in this chapter also applies for many-particle-systems.

Notation 2.4 For vectors $\psi \in \mathcal{H}$ one agrees on the notation $|\psi\rangle \equiv \psi$. One calls these kind of vectors „ket“-vectors. Through the canonical isomorphism[5] to the dual-space

$$\begin{aligned} \mathrm{J}_{\mathcal{H}} : & \mathcal{H} \longrightarrow \mathcal{H}^* \\ & \psi \mapsto \mathrm{J}_{\mathcal{H}}(\psi) \equiv \langle \psi, \bullet \rangle \equiv \langle \psi | \bullet \rangle \end{aligned}$$

one introduces so-called „bra“-vectors. Taking a bra- and a ket-vector $\langle \chi |$ and $|\psi\rangle$ and „calculating their bra-c-ket“ $\langle \chi | \psi \rangle$ is therefore equivalent to calculating the scalar-product between two vectors.

Notation 2.5 An *operator* A is a linear mapping defined on the Hilbert-space

$$\begin{aligned} A : & \mathcal{H} \mapsto \mathcal{H} \\ & |\psi\rangle \mapsto A|\psi\rangle \end{aligned}$$

[5]Isomorph, as a *real* vector space, not as a complex vector-space, since the mapping $\mathrm{J}_{\mathcal{H}}$ is anti-linear. (Cf. Zirnbauer-Lecture-Notes.)

Remark 2.5.1 In the case that $\mathcal{H}$ represents a single-particle Hilbert-space, one also speaks of a *single-particle-operator*. In the case that $\mathcal{H}$ represents a many-particle Hilbert-space, and the action of A is not trivial on the different sectors, one also speaks of a *many-particle-operator*.

Remark 2.5.2 Sometimes one also encounters notations as $|A\psi\rangle \equiv A|\psi\rangle$. For the corresponding bra-vector one writes $\mathrm{J}_{\mathcal{H}}(A\psi) \equiv \langle A\psi|\bullet\rangle$.

Notation 2.6 Let $\mathcal{H} = \mathrm{span}_{\mathbb{C}}\{|1\rangle, |2\rangle, \dots\}$ be spanned by an orthonormal basis[6]. Then we have the matrix of mat A w.r.t. this basis defined as

$$(\mathrm{mat}\ A)_{i,j} \equiv \langle i|A|j\rangle. \tag{2.3}$$

Remark 2.6.1 As a special case of an operator, let us introduce the *projector* $\mathrm{P}_\chi \equiv |\chi\rangle\langle\chi|$ of a state $\chi \in \mathcal{H}$ as a linear mapping

$$\begin{aligned} \mathrm{P}_\chi : &\mathcal{H} \longrightarrow \mathcal{H} \\ &\psi \longmapsto \mathrm{P}_\chi(\psi) \equiv |\chi\rangle\langle\chi|\psi\rangle. \end{aligned}$$

Remark 2.6.2 In Section 9.1 we will understand how the matrix-representation of projectors are related to dyadic products in linear algebra.

Remark 2.6.3 Note that

$$\mathbb{1} \equiv \mathrm{Id}_{\mathcal{H}} \equiv \sum_n |n\rangle\langle n|, \tag{2.4}$$

for an orthonormal system of the Hilbert-space, which can serve as a useful mnemonic in the following sense:

- A vector $\psi \equiv |\psi\rangle$ can be expanded w.r.t. the basis as

$$|\psi\rangle = \mathbb{1}|\psi\rangle = \sum_n |n\rangle\langle n|\psi\rangle \tag{2.5}$$

  which is the same expression that one encounters in linear algebra for finite-dimensional vector spaces.

- The quantity $A|\psi\rangle$ represents again a vector that can be expanded w.r.t. the basis as

$$A|\psi\rangle = A\mathbb{1}|\psi\rangle = \sum_n A|n\rangle\langle n|\psi\rangle \tag{2.6}$$

  this yields for the m-th component of the vector $A|\psi\rangle$

[6] Case-dependent: Could be a finite basis, but also infinite.

$$\langle m|A|\psi\rangle = \sum_n \langle m|A|n\rangle\langle n|\psi\rangle \tag{2.7}$$

with matrix-elements $\langle m|A|n\rangle \in \mathbb{C}$.

Notation 2.7 In the case of a single, spinless particle, i.e. for $V \in \left\{\mathbb{T}^{d_x}, \mathbb{R}^{d_x}\right\}$, with $d_{\mathbf{x}} \in \{1,2,3\}$, as a mnemonic, one usually introduces for $\mathcal{H} = L^2(V)$ so-called *position-eigenbasis* $|\mathbf{x}\rangle$ with $\mathbf{x} \in V$, giving a special version of the unity-operator

$$\mathrm{Id}_{\mathcal{H}} \equiv \int_V \mathrm{d}^{d_x}\mathbf{x}\, |\mathbf{x}\rangle\langle\mathbf{x}|. \tag{2.8}$$

By direct comparison of the expressions

$$\int \mathrm{d}^{d_x}\mathbf{x}\, \overline{\chi}(\mathbf{x})\psi(\mathbf{x}) = \langle\chi|\psi\rangle = \langle\chi|\, \mathrm{Id}_{\mathcal{H}}\, |\psi\rangle = \int \mathrm{d}^{d_x}\mathbf{x}\, \langle\chi|\mathbf{x}\rangle\, \langle\mathbf{x}|\psi\rangle, \tag{2.9}$$

one sees that it must hold

$$\langle\chi|\mathbf{x}\rangle = \overline{\chi}(\mathbf{x}), \qquad \langle\mathbf{x}|\psi\rangle = \psi(\mathbf{x}). \tag{2.10}$$

Remark 2.7.1 Although one finds in almost every physics-textbook that one must handle these "functions" with a lot of care, note that a regularization of a delta-function is square-integrable. Within the framework of solid-state-physics, one finds the well-defined, discrete „twins“ of position-eigenstates in so-called „Wannier-states“. Adopting a continuum-notation one finds well defined expressions and justifications for the „state-vectors“ $|\mathbf{x}\rangle$ above.

Notation 2.8 In the case of a single, spin-$\frac{1}{2}$ particle, i.e. for $\mathcal{H} = L^2(V) \otimes \mathbb{C}^2$, with $V \in \left\{\mathbb{T}^{d_x}, \mathbb{R}^{d_x}\right\}$, and $d_{\mathbf{x}} \in \{1,2,3\}$ and some basis for $\mathbb{C}^2 \equiv \mathrm{span}_{\mathbb{C}}\{|\uparrow\rangle, |\downarrow\rangle\}$ (let it be the eigenstates of σ_z here for concreteness), one also finds a resolution of identity w.r.t. to the product-basis

$$\mathrm{Id}_{\mathcal{H}} \equiv \sum\!\!\!\!\!\!\int_{\substack{\mathbf{x}\in V\\ \sigma\in\{\uparrow,\downarrow\}}} |\mathbf{x},\sigma\rangle\langle\mathbf{x},\sigma| \equiv \sum\!\!\!\!\!\!\int |x\rangle\langle x| \tag{2.11}$$

with coordinate-representation of states $|\psi\rangle \in \mathcal{H}$ w.r.t. our chosen basis given by

$$\langle x|\psi\rangle \equiv \psi(x) \equiv \psi_\sigma(\mathbf{x}). \tag{2.12}$$

Remark 2.8.1 In the case of more than one particle (with, or without spin) present, one can also introduce position-eigenstates as a product-basis of single-particle-position-eigenstates. This is elaborated in section 33.1.1.

Notation 2.9 Let A be a linear operator. The hermitian adjoint of A is the unique linear mapping $A^\dagger : \mathcal{H} \to \mathcal{H}$ that fulfills the following condition:

$$\forall \psi, \chi \in \mathcal{H} \times \mathcal{H} : \quad \langle \chi | A^\dagger | \psi \rangle = \langle \psi | A | \chi \rangle^* \equiv \langle \psi | A \chi \rangle^* = \langle A \chi | \psi \rangle \tag{2.13}$$

where we introduce the notations

$$\langle \psi | A^\dagger \equiv \langle A \psi | \equiv \mathrm{J}_{\mathcal{H}} \left(A | \psi \rangle \right). \tag{2.14}$$

Remark 2.9.1 If $A = A^\dagger$, then[7] we call the operator A *self-adjoint* or *hermitian*[8]. For reasons that will become clear in the next chapter, we will call self-adjoing operators *observables*[9]

Remark 2.9.2 Some properties of hermitian operators are:

- Their eigenvalues are real,
- eigenvectors associated to different eigenvalues are orthogonal,
- they can be diagonalized in an orthonormal basis of eigenvectors (spectral-theorem).

2.2. Canonical Quantization (from Classical Physics to Quantum Physics)

Hypothesis 2.10 Let

$$\begin{aligned} H &\equiv H(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N, \mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_N, t) \\ &\equiv H(x_{1,1}, \dots, x_{1,d_\mathbf{x}}, x_{2,1}, \dots, x_{2,d_\mathbf{x}}, \dots, p_{1,1}, \dots, p_{1,d_\mathbf{x}}, p_{2,1}, \dots, p_{2,d_\mathbf{x}}, \dots, t) \\ &\equiv H(x_1, x_2, \dots, x_{d_\mathbf{x}N}, p_1, p_2, \dots, p_{d_\mathbf{x}N}, t) \\ &\equiv H(x, p, t) \end{aligned} \tag{2.15}$$

be the Hamilton-function of a classical, non-relativistic, N-(Newton-)particle system in $d_\mathbf{x}$ dimensions. Let us further assume, that H is a linear-combination of expressions of the type

1. $p_i p_j$,
2. $p_i f_j(x_1, x_2, \dots, t)$
3. $F(x_1, x_2, \dots, t)$

with functions f_i and F. The *canonically-quantized Hamilton-operator*, which we also denote by H[10], is obtained by replacing the expressions in the following way:

1. $p_i p_j \mapsto \frac{\hbar}{\mathrm{i}} \frac{\partial}{\partial x_i} \frac{\hbar}{\mathrm{i}} \frac{\partial}{\partial x_j} = -\hbar^2 \frac{\partial}{\partial x_i} \frac{\partial}{\partial x_j}$,
2. $p_i f_j(x_1, x_2, \dots, t) = \frac{1}{2} \left(p_i f_j(x_1, x_2, \dots, t) + f_j(x_1, x_2, \dots, t) p_i \right) \mapsto \frac{1}{2} \left(\frac{\hbar}{\mathrm{i}} \frac{\partial}{\partial x_i} f_i(x_1, x_2, \dots, t) + f_i(x_1, x_2, \dots, t) \frac{\hbar}{\mathrm{i}} \frac{\partial}{\partial x_i} \right)$

[7] Recall that two operators are considered to be identical if their action on an arbitrary vector of the vector space is the same.
[8] As we are physicist, we throw both of these kinds of operators into the same pot.
[9] Mnemonic: „[observ]able-vari[able]“.
[10] One has to conclude from the context, whether one is speaking about a Hamilton-function or a Hamilton-operator.

3. $F(x_1, x_2, \dots, t) \mapsto F(x_1, x_2, \dots, t)$

Remark 2.10.1 There is no mathematical way of proving that this is the correct technique to quantize a classical system. The only possibility is to compare the results of the quantization scheme with the results of experiments.

Remark 2.10.2 Note that the canonical quantization is a mapping from functions to operators. Sometimes one encounters the notation $x \equiv \hat{x}$ or $p_x \equiv \hat{p_x} \equiv \hbar/\mathrm{i}\partial_x$ to highlight this fact.

Remark 2.10.3 As an example, consider the Hamilton-function of a single, non-relativistic particle in a time-independent potential $V \equiv V(\mathbf{x})$

$$H(\mathbf{x}, \mathbf{p}) = \frac{\mathbf{p}^2}{2m} + V(\mathbf{x}). \tag{2.16}$$

Then, the canonically-quantized Hamilton-operator reads

$$H = -\hbar^2 \frac{\Delta}{2m} + V(\mathbf{x}). \tag{2.17}$$

Lemma 2.11 Notice further, that one has on the operator-level the so-called canonical commutation relations (CCR)

$$\left[x_i,\, p_j\right] = \mathrm{i}\hbar\delta_{i,j}. \tag{2.18}$$

Proof 2.11 See Appendix, Proofs, page 286.

Definition 2.12 We consider an N-particle system in $d_{\mathbf{x}} \in \{1, 2, 3\}$ dimensions. The Schrödinger-equation associated to this system reads

$$\mathrm{i}\hbar \left(\partial_t \psi\right)(\mathbf{x}, t) = (H\psi)(\mathbf{x}, t). \tag{2.19}$$

with canonically-quantized (and potentially time-dependent) Hamilton-operator $H(t)$.

Remark 2.12.1 Particles[11] (with mass m etc.), that obey the Schrödinger-equation, are called *Schrödinger-particles*.

Theorem 2.13 The formal solution to equation (2.19) is given by application of *time-evolution operator*

$$\psi(\mathbf{x}, t_\mathrm{f}) = U(t_\mathrm{f}, t_\mathrm{i})\psi(\mathbf{x}, t_\mathrm{i}) \tag{2.20}$$

[11] In the broadest sense of the word.

with

$$U(t_\mathrm{f},t_\mathrm{i}) = \mathrm{T}_t \exp\left(-\frac{\mathrm{i}}{\hbar}\int_{t_\mathrm{i}}^{t_\mathrm{f}} \mathrm{d}t'\, H(t')\right) = \sum_{n=0}^{\infty} \frac{1}{n!}\left(\frac{\mathrm{i}}{\hbar}\right)^n \int_{[t_\mathrm{i},t_\mathrm{f}]^n} \mathrm{d}t_1 \cdots \mathrm{d}t_n\ \mathrm{T}_t \prod_{j=1}^{n} H(t_j). \tag{2.21}$$

Remark 2.13.1 Notice, that the formal solution to this equation is for time-independent potentials given by

$$\psi(\mathbf{x},t) = \mathrm{e}^{-\frac{\mathrm{i}}{\hbar}H(t-t_0)}\psi(\mathbf{x},t_0) \equiv U(t,t_0)\psi(\mathbf{x},t_0) \tag{2.22}$$

assuming we have $\psi(\mathbf{x},t_0)$ given (which is most often not the case). The operator U is called *time-evolution-operator* and can be generalized for time-dependent potentials.

Definition 2.14 A time-dependent pure state $t \mapsto \psi(t) \equiv \psi_t \in \mathcal{H}$ is called *stationary*, if it satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle\, \psi_t \mid A \mid \psi_t\,\rangle \equiv \frac{\mathrm{d}}{\mathrm{d}t}\langle\, A\,\rangle_{\psi_t} = 0 \quad \text{for all hermitian operators } A. \tag{2.23}$$

Remark 2.14.1 As examples:

- An electron moving through free space is not in stationary state (the expectation-value of the position operator changes with time).
- The ground state of the electronic system of a solid is (in the wavefunction-formalism (and as we shall see later, in the "density-matrix" formalism as well), and in absence of interactions) a stationary state (in the non-interacting case: a product of a so-called „Slater-determinant“ (to be introduced later) and plane wave for the spatial and temporal part respectively; the expectation-value of all observables are then constant in time, if there is no external perturbation).

Theorem 2.15 For a time-independent potential-function $V \equiv V(\mathbf{x},t) \equiv V(\mathbf{x})$ a product ansatz

$$\psi(\mathbf{x},t) \equiv \psi(\mathbf{x})\,\phi(t) \tag{2.24}$$

leads to[12] the time-independent-Schrödinger-equation

$$E\psi(\mathbf{x}) = (H\psi)\,(\mathbf{x}), \tag{2.25}$$

with

$$\phi(t) = \mathrm{e}^{-\mathrm{i}Et/\hbar}. \tag{2.26}$$

Proof 2.15 See Appendix, Proofs, page 286.

[12]With "leads to" we mean: We can solve the time-dependent Schrödinger-equation iff we can solve the time-independent Schrödinger-equation.

Corollary 2.16 The product ansatz above yields stationary-states.

Proof 2.16 See Appendix, Proofs, page 287.

Hypothesis 2.17 The product-ansatz in Equation (2.24) gives the „correct“ solution of the Schrödinger-equation, iff the state of interest is a stationary-state that can be described in the wavefunction-formalism.

Remark 2.17.1 Notice that the canonically quantized Hamiltonian of a single particle in $d_{\mathbf{x}} \in \{1, 2, 3\}$ dimensions with Hamilton-function given as $H = \frac{p^2}{2m} + V(\mathbf{x})$ is thus given by

$$H = -\frac{\hbar^2}{2m}\Delta_{\mathbf{x}} + V(\mathbf{x}). \tag{2.27}$$

Definition 2.18 The time-dependent Pauli equation for an N-particle system reads

$$i\frac{\partial}{\partial t}\psi(\mathbf{x}_1, s_1; \dots; \mathbf{x}_N, s_N; t) = \hat{H}\,\psi(\mathbf{x}_1, s_1; \dots; \mathbf{x}_N, s_N; t), \tag{2.28}$$

where the Hamiltonian is

$$\hat{H} = \sum_{j=1}^{N}\left[\frac{1}{2m_j}\left(-i\nabla_j - q_j\mathbf{A}(\mathbf{x}_j, t)\right)^2 + q_j\phi(\mathbf{x}_j, t) - \frac{q_j}{2m_j}\sigma_j \cdot \mathbf{B}(\mathbf{x}_j, t)\right] + \hat{V}_{\text{int}}. \tag{2.29}$$

Here, σ_j are the Pauli matrices acting on the spin degree of freedom of particle j, $\mathbf{A}$ and ϕ denote the vector and scalar potentials, and $\mathbf{B} = \nabla \times \mathbf{A}$ is the magnetic field.

Remark 2.18.1 For a system of N electrons (all of mass m and charge $-e$), this simplifies to

$$\hat{H} = \sum_{j=1}^{N}\left[\frac{1}{2m}\left(-i\nabla_j + e\mathbf{A}(\mathbf{x}_j, t)\right)^2 - e\phi(\mathbf{x}_j, t) + \frac{e}{2m}\sigma_j \cdot \mathbf{B}(\mathbf{x}_j, t)\right] + \sum_{i<j}\frac{e^2}{|\mathbf{x}_i - \mathbf{x}_j|}. \tag{2.30}$$

Remark 2.18.2 The single-particle Pauli-equation reads

$$i\hbar\partial_t\psi = \left(\frac{1}{2m}\left[(\mathbf{p} - q\mathbf{A})^2 - q\hbar\boldsymbol{\sigma}\cdot\mathbf{B}\right] + q\phi\right)\psi \tag{2.31}$$

with $\psi(\mathbf{x}, t) \equiv \left(\psi_\uparrow, \psi_\downarrow\right)^{\mathrm{T}}(\mathbf{x}, t)$ etc.

Remark 2.18.3 Analogously one speaks about *Pauli*-particles.

Fact 2.19 A time-independent vector potential $\mathbf{A}(\mathbf{x})$ leads to a time-independent Hamiltonian $\hat{H}$ in the Pauli equation, and thus allows for a product ansatz of the form $\psi(\mathbf{x}_1, s_1; \dots; \mathbf{x}_N, s_N; t) = \psi(\mathbf{x}_1, s_1; \dots; \mathbf{x}_N, s_N)\phi(t)$, which yields the time-independent Pauli equation for a spatial wavefunction ψ.

2.3. The Postulates (for Wavefunction-States)

I A *wavefunction-state* of a physical system is a normalized solution $\psi(\bullet, t)$ of the underlying Schrödinger- / Pauli-equation, s.t. for each $t \in \mathbb{R}$, the coordinates[13] of the vector are complex-valued

$$\begin{aligned} \mathbf{x} &\mapsto \psi(\mathbf{x}, t) \equiv \langle \mathbf{x} | \psi_t \rangle \in \mathbb{C} \quad \text{for Schrödinger-equations} \\ x &\mapsto \psi(x, t) \equiv \langle x | \psi_t \rangle \in \mathbb{C} \quad \text{for Pauli-equations} \end{aligned} \tag{2.32}$$

and the vector is normalized, i.e. $\langle \psi_t | \psi_t \rangle = 1$.

II a) For any measurable quantity[14] there is an observable A.

b) The outcome of a measurement of A, i.e. the measured quantity, is one of the eigenvalues α_n of A.

c) Let $\{|\phi_{n,i}\rangle\}_{i=1,\dots,m}$ denote an orthonormal basis of the eigenspace of the eigenvalue (of A) α_n. A measurement of the observable A with a system in the state $|\psi_t\rangle$ gives the measured value α_n with probability

$$p_n = \sum_{i=1}^{m} |\langle \phi_{n,i} | \psi_t \rangle|^2. \tag{2.33}$$

Remark 2.19.1 For states that are not representated by wavefunctions, one obtains generalizations of the postulates. We will elaborate on this at a later point (when we acutally need it).

Remark 2.19.2 For wavefunctions obeying the Pauli-equation, the *Born-probability-interpretation* can be summarized as follows:

$$|\psi(x_1, x_2, \dots, x_N)|^2 \prod_{j=1}^{N} \mathrm{d}\mathbf{x}_j = \left\{ \begin{array}{l} \text{The probability for finding the } N \text{ particles in the } 3N\text{-dimensional volume } \prod_{j=1}^{N} \mathrm{d}\mathbf{x}_j, \\ \text{surrounding the point } (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N) \\ \text{in the } 3N\text{-dimensional configuration space.} \end{array} \right\}.$$

For wavefunctions obeying the Schrödinger-equation, the same interpretation holds, but with $x_i \mapsto \mathbf{x}_i$ on the left hand side of the equation.

Remark 2.19.3 One finds these postulates in different books in all kinds of colours. It is, in my opinion, very difficult, if not impossible, to give a pedagogical and simultaneously complete introduction / description of quantum mechanics.

Notation 2.20 Further one finds the *expectation-value* of the observable A in the state $|\psi_t\rangle$ as

$$\langle A \rangle = \langle \psi_t | A | \psi_t \rangle. \tag{2.34}$$

[13] Notice that, depending on the system (with or without spin) and the number of particles, these coordinates are vectors with entries as many as $\mathbf{x}$ has.

[14] For example position, momentum, spin, angular-momentum, etc. There are also more complicated observables, such as Fermi-surfaces (through ARPES-measurements). Proving that these observables are indeed observables in the mathematical sense is a non-trivial, but straightforward task.

2.4. Time-Dependencies in Many-Body Quantum Mechanics

Notation 2.21 Let $\mathcal{H}$ denote the Hilbert-space (of a single- or many-particle) quantum system with a (possibly spin-dependent) time-independent Hamiltonian H (in the Schrödinger-picture) and consider the time-evolution in the Schrödinger- / Pauli-equation

$$\mathrm{i}\hbar\partial_t|\psi(t)\rangle = H|\psi(t)\rangle.$$

Definition 2.22 For arbitrary times $t_\mathrm{i}, t_\mathrm{f} \in \mathbb{R}$ we define the time-evolution operator $U(t_\mathrm{f}, t_\mathrm{i})$ of the system as

$$U(t_\mathrm{f}, t_\mathrm{i}) = \exp -\frac{\mathrm{i}}{\hbar}\left(t_\mathrm{f} - t_\mathrm{i}\right) H.$$

Theorem 2.23 The time-evolution operator defined above gives a formal solution to the Schrödinger-equation for a given initial-condition.

Proof 2.23 By taking the derivative, one sees that one indeeds obtains a solution for a given initial condition.

Definition 2.24 Let $X \equiv X_\mathrm{S} \equiv X_\mathrm{S}(t)$ be a possibly time-dependent operator, then one defines the corresponding Heisenberg-operator as

$$X_\mathrm{H}(t) := U(t,0)^\dagger\, X\, U(t,0).$$

Theorem 2.25 One immediately concludes the Heisenberg equation of motions

$$\frac{\mathrm{d}}{\mathrm{d}t}X_\mathrm{H}(t) = -\frac{\mathrm{i}}{\hbar}[X, H]_\mathrm{H}(t) + \left(\frac{\partial X_\mathrm{S}}{\partial t}\right)_\mathrm{H}(t) \tag{2.35}$$

Proof 2.25 See Appendix, Proofs, page 288.

Definition 2.26 In the so-called *Schrödinger-picture* one evaluates expectation-values of an observable X as

$$\langle X\rangle(t) \equiv \langle\psi_t|X|\psi_t\rangle, \tag{2.36}$$

In the *Heisenberg-picture* one evaluates expectation-values of an observable as

$$\langle X\rangle(t) \equiv \langle\psi_{t=0}|X_\mathrm{H}(t)|\psi_{t=0}\rangle \tag{2.37}$$

whereas $|\psi_{t=0}\rangle$ refers to some (known) state at time equal zero.

Remark 2.26.1 For pure states, both pictures are equivalent.

Remark 2.26.2 Sometimes it turns out to be useful to switch to yet another picture, the so-called „interaction-picture“. As the number of occasions where this is useful is rare (e.g. „Fermi’s golden rule“, or „one-body-Green-functions for interacting systems“) we will not introduce it at this point.

3. Crystals

We focus on „crystalline solids“ (elaborated below) in this lecture.

3.1. Crystals, Lattices and Bases

Definition 3.1 For $d_{\mathbf{x}} \in \{1, 2, 3\}$ and linearly independent vectors $\bigcup_{j=1}^{d_{\mathrm{x}}} \{\mathbf{a}_j\}$ we introduce the corresponding *infinite point lattice*

$$\mathrm{L}_\infty \equiv \left\{ \mathbf{R_n} \equiv \sum_{i=1}^{d_{\mathrm{x}}} n_i \mathbf{a}_i \quad | \quad \mathbf{n} \in \mathbb{Z}^{d_{\mathrm{x}}} \right\} \subset \mathbb{R}^{d_{\mathrm{x}}}. \tag{3.1}$$

Remark 3.1.1 The vectors $\bigcup_{j=1}^{d_{\mathrm{x}}} \{\mathbf{a}_j\}$ are usually called *lattice vectors* or *basis (vectors)*.

Definition 3.2 We define the *unit-cell* UC of a lattice w.r.t. the basis $\bigcup_{j=1}^{d_{\mathrm{x}}} \{\mathbf{a}_j\}$ as the set

$$\mathrm{UC} \equiv \mathrm{span}_{[0,1]} \bigcup_{j=1}^{d_{\mathrm{x}}} \{\mathbf{a}_j\} \subset \mathbb{R}^{d_{\mathrm{x}}} \tag{3.2}$$

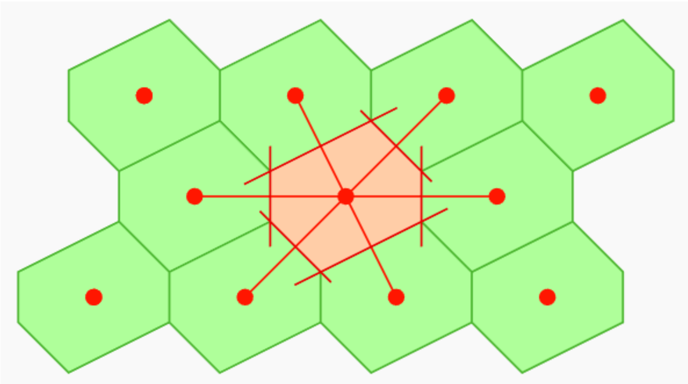
and we define its volume as

$$\mathrm{vol(UC)} = \int \mathrm{d}^{d_{\mathrm{x}}}\mathbf{r}\ 1_{\mathrm{UC}}(\mathbf{r}). \tag{3.3}$$

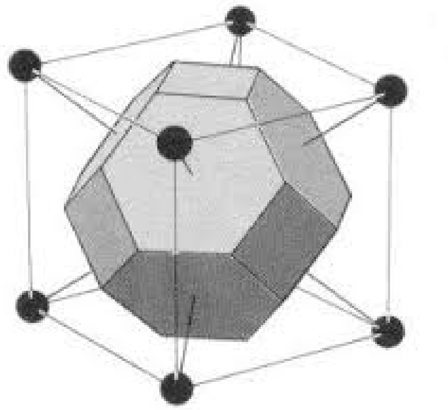
Definition 3.3 The *Wigner-Seitz-cell* around the lattice point $\mathbf{R_m}$ is defined as the set

$$A_{\mathbf{R_m}} \equiv A(\mathbf{R_m}) \equiv A_{\mathbf{m}} = \left\{ \mathbf{r} \in \mathbb{R}^d \mid \forall \mathbf{n} \neq \mathbf{m} :\ |\mathbf{r} - \mathbf{R_m}| < |\mathbf{r} - \mathbf{R_n}| \right\}. \tag{3.4}$$

Remark 3.3.1 Therefore, all points in $A_{\mathbf{R_m}}$ are closer to the point $\mathbf{R_m}$ than to the neighboring grid-points.



(a) 2D construction for an „oblique-lattice“.



(b) 3D construction for a „bcc-lattice“.

Figure 3.1.: (a) 2D and (b) 3D constructions of a Wigner-Seitz-cell.

Definition 3.4 In one dimension, $d_x = 1$, there exists only one kind of (*Bravais-*)lattice: Fixing the lattice constant a determines the lattice completely.

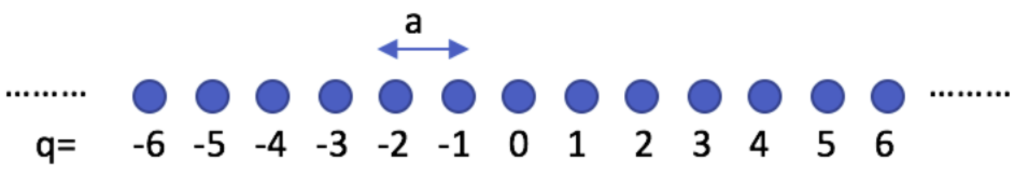


Figure 3.2.: One dimensional lattice (schematically).

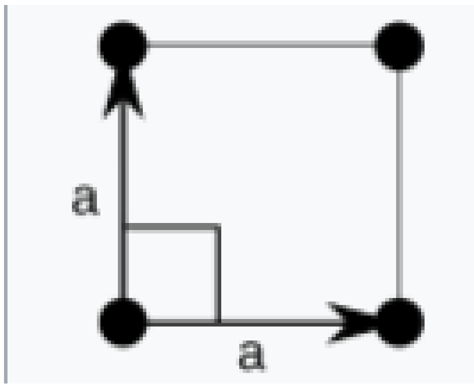
Definition 3.5 In two dimension, $d_x = 2$, one introduces 5 so-called *(infinite) Bravais-lattices*[1], by considering the following cases of unit-vector combinations: Let $\varphi = \angle(\mathbf{a}_1, \mathbf{a}_2)$. Then one introduces …

1. *Square lattice*

   One choice of basis vectors is:

   $$\mathbf{a}_1 = (a, 0), \qquad \mathbf{a}_2 = (0, a),$$

   where $a = b$, and the angle $\varphi = 90°$.



The square lattice.

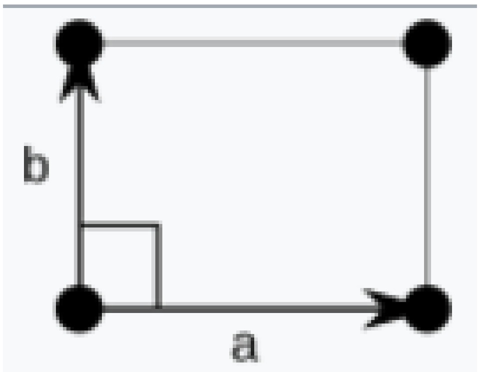
2. *Rectangular lattice*

[1] International Tables A page 720: There are 5 Bravais-Lattices in two dimensions.

One choice of basis vectors is:

$$\mathbf{a}_1 = (a, 0), \qquad \mathbf{a}_2 = (0, b),$$

where $a \neq b$, and the angle $\varphi = 90°$.



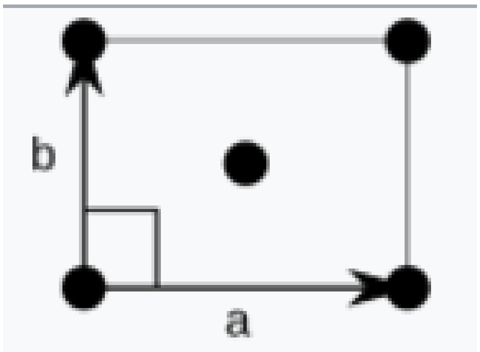
The rectangular lattice.

3. *Centered rectangular lattice*

One choice of basis vectors is:

$$\mathbf{a}_1 = \left(\frac{a}{2}, \frac{b}{2}\right), \qquad \mathbf{a}_2 = (a, 0),$$

where $a \neq b$, and the angle $\varphi \neq 90°$.



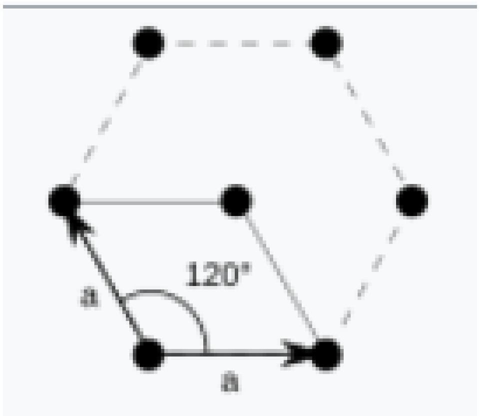
The centered rectangular lattice.

4. *Triangular (hexagonal) lattice*

One choice of basis vectors is:

$$\mathbf{a}_1 = (a, 0), \qquad \mathbf{a}_2 = \left(-\frac{a}{2}, \frac{\sqrt{3}}{2}a\right),$$

and the angle $\varphi = 120°$.



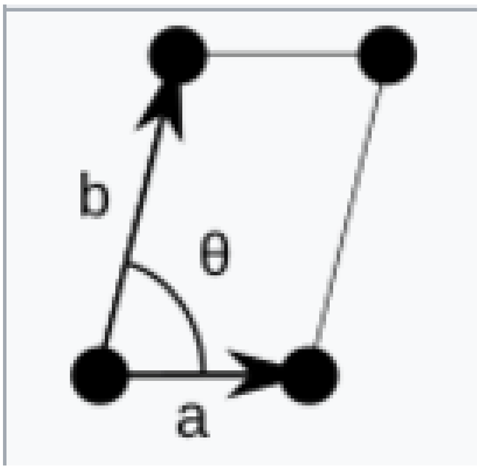
The triangular lattice.

5. *Oblique lattice*

One choice of basis vectors is:

$$\mathbf{a}_1 = (a, 0), \qquad \mathbf{a}_2 = (b\cos\varphi,\ b\sin\varphi),$$

where $a \neq b$, and $0° < \varphi < 180°$.



The oblique lattice.

Definition 3.6 In three dimensions, $d_{\mathbf{x}} = 3$, one introduces 14 so-called *Bravais-lattices*, by considering the following cases of unit-vector combinations: We denote the angles as

$$\alpha = \angle(\mathbf{a}_2, \mathbf{a}_3), \quad \beta = \angle(\mathbf{a}_1, \mathbf{a}_3), \quad \gamma = \angle(\mathbf{a}_1, \mathbf{a}_2) \tag{3.5}$$

and introduce the notation

$$\mathbf{a}_1 = \mathbf{a}, \quad \mathbf{a}_2 = \mathbf{b}, \quad \mathbf{a}_3 = \mathbf{c} \tag{3.6}$$

then one introduces the different lattices[2] …

1: the *triclinic lattice*: $a \neq b \neq c, \quad \alpha \neq \beta \neq \gamma,$

2-3: The *monoclinic* lattices

- a) the *monoclinic primitive lattice*: $a \neq b \neq c, \quad \alpha = \gamma = 90° \neq \beta$
- b) the *monoclinic base-centered lattice*: $a \neq b \neq c, \quad \alpha = \gamma = 90° \neq \beta$

4-7: The *orthorhombic* lattices

- a) the *orthorhombic primitive lattice*: $a \neq b \neq c, \quad \alpha = \beta = \gamma = 90°$
- b) the *orthorhombic base-centered lattice*: $a \neq b \neq c, \quad \alpha = \beta = \gamma = 90°$
- c) the *orthorhombic body-centered lattice*: $a \neq b \neq c, \quad \alpha = \beta = \gamma = 90°$
- d) the *orthorhombic face-centered lattice*: $a \neq b \neq c, \quad \alpha = \beta = \gamma = 90°$

8: the *hexagonal lattice*: $a = b \neq c, \quad \alpha = \beta = 90°, \quad \gamma = 120°$

9: the *trigonal lattice*: $a = b = c, \quad \alpha = \beta = \gamma \neq 90°$

10-11: The *tetragonal* lattices

- a) the *tetragonal primitive lattice*: $a = b \neq c, \quad \alpha = \beta = \gamma = 90°$
- b) the *tetragonal body-centered lattice*: $a = b \neq c, \quad \alpha = \beta = \gamma = 90°$

12-14: The *cubic* lattices

[2]Notice that e.g. the equations $a \neq b \neq c$ allow (strictly speaking) for the case $a = c \neq b$. But this is just poorly written in the books.

a) the *cubic primitive lattice*: $a = b = c, \quad \alpha = \beta = \gamma = 90°$

b) the *cubic body-centered lattice*: $a = b = c, \quad \alpha = \beta = \gamma = 90°$

c) the *cubic face-centered lattice*: $a = b = c, \quad \alpha = \beta = \gamma = 90°$

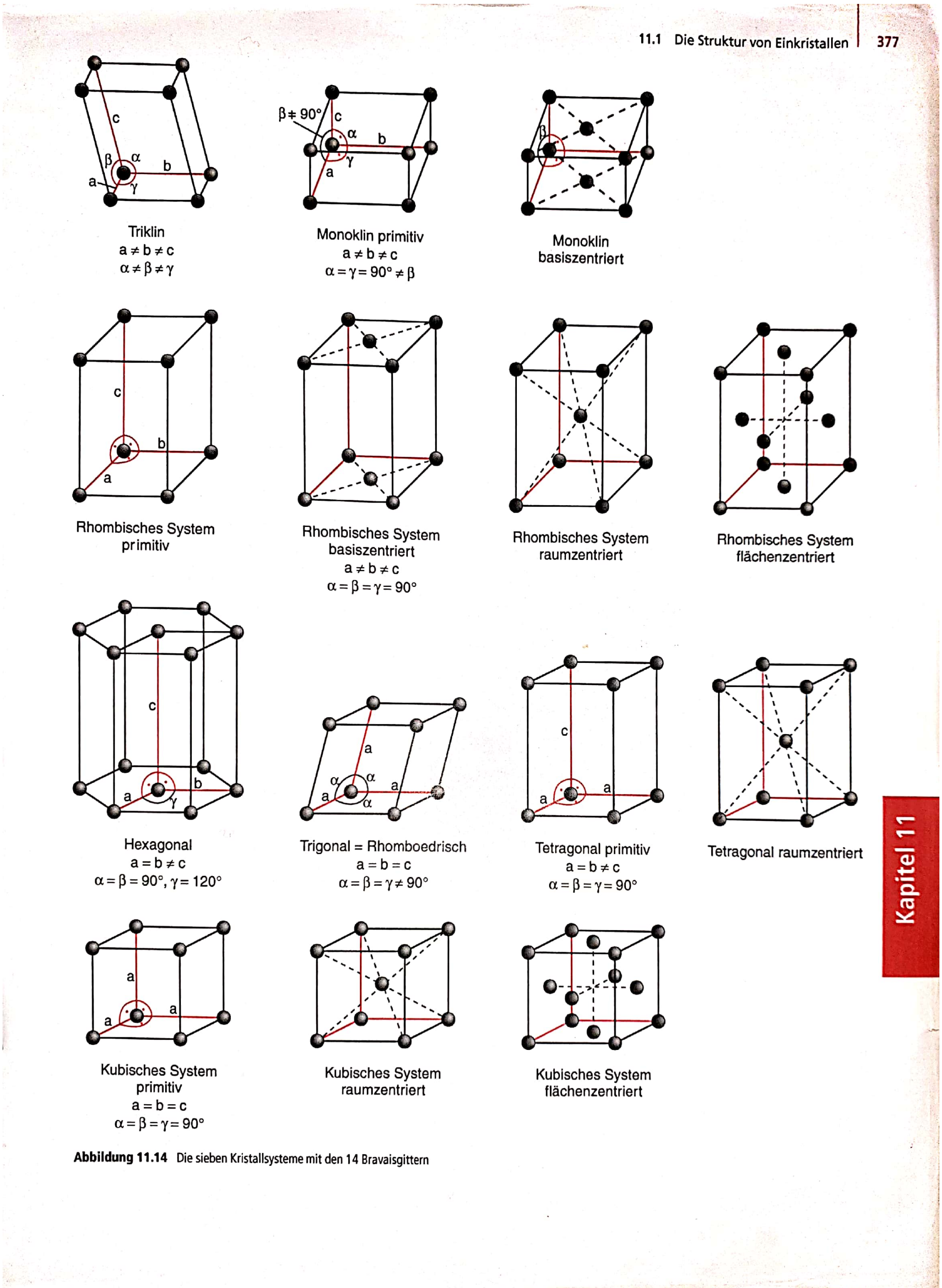


Figure 3.3.: Excerpt from Wolfang Demtröders Experimentalphysik 3.

Quoting from [Ashcroft]:

The seven crystal systems and fourteen Bravais lattices described above exhaust the possibilities. This *is far from obvious [...]. However, it is of no practical importance to understand why these are the only distinct cases. It is enough to know why the categories exist, and what they are.*

Definition 3.7 We define an *infinite crystal* C_∞ by taking an infinite Bravais-lattice L_∞, and enlarge the number of points in the following way

$$\mathrm{C}_\infty \equiv \left\{ \boldsymbol{R}_{\mathbf{n}\kappa} \equiv \mathbf{R}_\mathbf{n} + \boldsymbol{R}_\kappa \equiv R_{\mathbf{n},\kappa,\alpha} \quad \middle| \quad \mathbf{R}_\mathbf{n} \in \mathrm{L}_\infty \quad \wedge \quad \kappa \in \{1, 2, \dots r\} \quad \wedge \quad \alpha \in \{1, \dots, d_\mathbf{x}\} \right\} \subset \mathbb{R}^{d_\mathrm{x}} \tag{3.7}$$

with *atomic basis-vectors* $\{\boldsymbol{R}_1, \dots, \boldsymbol{R}_r\}$ satisfying the equation

$$\boldsymbol{R}_\mu \in \{\boldsymbol{R}_1, \dots, \boldsymbol{R}_r\} \subset \mathrm{UC}\,. \tag{3.8}$$

Remark 3.7.1 The *atomic basis-vectors* $\{\boldsymbol{R}_1, \dots, \boldsymbol{R}_r\}$ represent the individual positions of the different atoms within the unit cell, and must not be confused with the primitive-vectors of the underlying lattice.

Remark 3.7.2 Notice that therefore, every infinite Bravais-lattice can be interpreted as a crystal with a trivial mono-atomic basis, i.e. $r = 1$ and $\boldsymbol{R}_1 = 0$.

Remark 3.7.3 A simple mnemonic for the definitions reads

$$\text{Crystal} = \text{Lattice} + \text{Basis}. \tag{3.9}$$

Remark 3.7.4 Note further, that this decomposition is not unique. For example potassium in its crystalline-form at room-temperature can be considered as a fcc-lattice with a mono-atomic basis *or* as a sc-lattice with a diatomic basis.

Remark 3.7.5 There are little pitfalls in the phraseology of the community... Graphene's crystal-structure is an example. It's realized as a triangular lattice with two basis-atoms. Because of the point-structure that one ends up with, one usually calls this crystal the *honeycomb-lattice*, although it is neither a Bravais-lattice, nor an infinite point-lattice in the sense above.

Definition 3.8 We call the lattice vectors $\bigcup_{j=1}^{d_\mathrm{x}} \left\{\mathbf{a}_j\right\}$ *primitive*, if r in Equation (3.7) is minimal. The corresponding unit-cell is called a *primitive unit-cell*.

Remark 3.8.1 A *primitive unit-cell* is the smallest unit cell that one can obtain for a crystal.

Definition 3.9 For a Bravais-lattice L_∞ with basis $\bigcup_{j=1}^{d_x}\{\mathbf{a}_j\}$ we introduce the corresponding *reciprocal basis* in different dimensions

- $d = 1$: $\quad \mathbf{b}_1 = 2\pi \frac{\mathbf{a}_1}{|\mathbf{a}_1|}$.
- $d = 2$: $\quad \mathbf{b}_1 = 2\pi \frac{(-a_{2y}, a_{2x})}{|\mathbf{a}_1 \wedge \mathbf{a}_2|}, \quad \mathbf{b}_2 = 2\pi \frac{(a_{1y}, -a_{1x})}{|\mathbf{a}_1 \wedge \mathbf{a}_2|}$.
- $d = 3$: $\quad \mathbf{b}_1 = 2\pi \frac{\mathbf{a}_2 \times \mathbf{a}_3}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}, \qquad \mathbf{b}_2 = 2\pi \frac{\mathbf{a}_3 \times \mathbf{a}_1}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}, \qquad \mathbf{b}_3 = 2\pi \frac{\mathbf{a}_1 \times \mathbf{a}_2}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}$.

Subsequently we define the corresponding *infinite reciprocal lattice* RL_∞ as

$$\mathrm{RL}_\infty \equiv \left\{ \mathbf{G} \in \mathbb{R}^{d_x} \;\middle|\; \mathbf{G} = \sum_{j=1}^{d_x} n_j \mathbf{b}_j \quad \text{with} \quad \mathbf{n} \in \bigtimes_{j=1}^{d_x} \mathbb{Z} \right\} = \mathrm{span}_{\mathbb{Z}^{d_x}} \bigcup_{j=1}^{d_x} \{\mathbf{b}_j\} \,. \tag{3.10}$$

Remark 3.9.1 Recall that the wedge-product and the triple-product measure the (signed) volume of the parallelogram and the parallelepiped spanned by the vectors they are being applied on.

Corollary 3.10 It holds that

$$\mathrm{RL}_\infty \equiv \left\{ \mathbf{G} \in \mathbb{R}^{d_x} \;\middle|\; \forall \mathbf{R} \in L_\infty \,:\, \mathbf{G} \cdot \mathbf{R} \in 2\pi\mathbb{Z} \right\}. \tag{3.11}$$

Proof 3.10 With the anticommutativity-rules for the triple-product, one immediately sees that these vectors fulfill the defining condition.

Definition 3.11 The *first Brillouin-zone* FBZ is defined as

$$\mathrm{FBZ} = \left\{ \mathbf{r} \in \mathbb{R}^{d_x} \;\middle|\; \forall \mathbf{G}_\mathbf{n} \in \mathrm{RL}_\infty \,:\, |\mathbf{r} - 0| < |\mathbf{r} - \mathbf{G}_\mathbf{n}| \right\}. \tag{3.12}$$

Remark 3.11.1 Therefore, the first Brillouin-zone is the Wigner-Seitz-cell of the origin of the reciprocal lattice.

Definition 3.12 A *unit-cell-periodic-function* or also *lattice-periodic function* is a smooth function

$$\begin{aligned} f \,:\, & \mathbb{R}^{d_x} \longrightarrow \mathbb{C} \\ & \mathbf{x} \quad \longmapsto f(\mathbf{x}) \end{aligned}$$

satisfying

$$f(\mathbf{x} + \mathbf{R}_\mathbf{n}) = f(\mathbf{x}) \quad \text{for } \mathbf{R}_\mathbf{n} \in L_\infty \,. \tag{3.13}$$

Remark 3.12.1 Examples include ....

- The potential $\mathbf{x} \mapsto V(\mathbf{x}) \in \mathbb{R}$ that a "Bloch-electron"[3] experiences is a lattice-periodic function.
- A "Bloch-wavefunction"[4] $\mathbf{x} \mapsto \psi_{n,\mathbf{k}}(\mathbf{x}) \in \mathbb{C}$ can be factorized into a plane-wave $\mathrm{e}^{\mathrm{i}\mathbf{x}\mathbf{k}}$ and a lattice-periodic function $u_{nm\mathbf{k}}(\mathbf{x})$.
- The "charge-density"[5] $\mathbf{x} \mapsto \rho(\mathbf{x}) \in \mathbb{R}$ is a lattice-periodic function.

Corollary 3.13 For a lattice-periodic function f it follows

$$f(\mathbf{x}) = \sum_{\mathbf{G} \in \mathrm{RL}_\infty} f_{\mathbf{G}}\, \mathrm{e}^{\mathrm{i}\mathbf{G}\cdot\mathbf{x}} \tag{3.14}$$

with expansion coefficients defined by

$$f_{\mathbf{G}} = \frac{1}{\mathrm{Vol(UC)}} \int_{\mathrm{UC}} \mathrm{d}^{d_\mathbf{x}}\mathbf{x}\, f(\mathbf{x})\, \mathrm{e}^{-\mathrm{i}\mathbf{G}\cdot\mathbf{x}} \tag{3.15}$$

Proof 3.13 See Appendix, Proofs, page 289.

Symmetries in Crystals

Symmetry-considerations are not trivial matters. There are endless tables on which kind of lattices imply what kind of behaviour for different kinds of experiments. (Laue-backscattering, Raman-scattering etc.) These behaviours can be concluded from symmetry-considerations and group-theoretical methods corresponding to the underlying lattice. Those who are interested can search for "International Tables for Crystallography".

3.2. Boundaries, Equivalences and Fourier

Hypothesis 3.14 We consider a real material embedded into $\mathbb{R}^{d_\mathbf{x}}$ with some finite volume $0 < V_{d_\mathbf{x}} < \infty$ and some underlying crystal(-structure), and introduce the numbers $N_1, \ldots, N_{d_\mathbf{x}} \in \mathbb{N}$ such that the scalar-product ($d_\mathbf{x} = 1$), wedge-product ($d_\mathbf{x} = 2$) or triple-product ($d_\mathbf{x} = 3$) coincides the volume

- $d_\mathbf{x} = 1: \quad V_{d_\mathbf{x}=1} \equiv L = N_1 \sqrt{\mathbf{a}_1^2}.$
- $d_\mathbf{x} = 2: \quad V_{d_\mathbf{x}=2} \equiv A = N_1 N_2\, \mathbf{a}_1 \wedge \mathbf{a}_2 = N_1 N_2 \,\mathrm{Det} \begin{pmatrix} a_{1,x} & a_{1,y} \\ a_{2,x} & a_{2,y} \end{pmatrix}.$
- $d_\mathbf{x} = 3: \quad V_{d_\mathbf{x}=3} \equiv V = N_1 N_2 N_3\, \mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3) = N_1 N_2 N_3 \,\mathrm{Det} \begin{pmatrix} a_{1,x} & a_{1,y} & a_{1,z} \\ a_{2,x} & a_{2,y} & a_{2,z} \\ a_{3,x} & a_{3,y} & a_{3,z} \end{pmatrix}.$

[3]This term will be defined later.
[4]This term will be defined later.
[5]This term will be defined later.

and we denote the total number of cells as

$$\prod_{i=1}^{d_x} N_i = N_p \tag{3.16}$$

Remark 3.14.1 Recall that the wedge product captures the (signed) area spanned by the two vectors in the plane, whereas the triple-product is the (signed) volume of the parallelepiped defined by the three vectors given.

Remark 3.14.2 When people speak about infinite crystals, they are usually considering the so-called *thermodynamic limit* in which $V_{d_x} \to \infty$ and $\prod_{i=1}^{d_x} N_i \to \infty$ with the additional condition that the size of the unit-cell is a constant $\frac{1}{V_{d_x}} \prod_{j=1}^{d_x} N_j = \text{const.}$ In almost every (DFT) computer simulation however, crystals have a finite size with "periodic boundaries" (see below.)[6] One obtains „good approximations“ to densities of states and other observables by setting at least $N_i \approx 40$, unless there are some symmetries to consider.

We envision to catch the following two birds with one stone:

1. If we consider a piece of solid, and if we „look“ far into the bulk of the underlying material, we observe that in the Schrödinger-equation the potential arising from the ionic cores on the lattice site that a single-electron experiences „looks“ (at least within a finite region inside the bulk) like a periodic[7] function.
2. If we consider a piece of solid, it has a finite size.

Fact 3.15 One can model both these attributes in harmony to each other, by introducing periodic boundary conditions for crystal as a whole and restricting attention to solutions of the Schrödinger-eq. (4.2) which are periodic with the finite Bravais-lattice. We therefore identify wavefunctions in the bulk of the crystal as wavefunctions on a three-torus.

Definition 3.16 On $\mathbb{R}^{d_x}$ we define the following relation

$$\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^{d_x} : \mathbf{x} \sim_x \mathbf{y} \quad :\Longleftrightarrow \quad \exists \mathbf{n} \in \mathbb{Z}^{d_x} : \mathbf{x} = \mathbf{y} + \sum_{i=1}^{d_x} n_i N_i \mathbf{a}_i . \tag{3.17}$$

Introducing these periodic boundary conditions is known as *Born- von Karman-boundary-conditions*.

Lemma 3.17 This defines an equivalence-relation.

Proof 3.17 Reflexivity is clear. For symmetry, we note that

$$\mathbf{x} = \mathbf{y} + \sum_{i=1}^{d_x} n_i N_i \mathbf{a}_i \Longleftrightarrow \mathbf{y} = \mathbf{x} + \sum_{i=1}^{d_x} (-n_i) N_i \mathbf{a}_i . \tag{3.18}$$

[6]Exceptions: quasicrystals, crystals with impurities, open boundary conditions.
[7]Notice that there are many useful mathematical formulae for periodic functions.

For transitivity one adds the coefficients.

Remark 3.17.1 Recall that a set on which an equivalence-relation is defined, can be partitioned into its equivalence-classes. This is an integral insight, in the construction of the Hilbert-space and the finite point lattice, and the Fourier-transform.

Lemma 3.18 It holds

$$\mathbb{R}^{d_\mathrm{x}} / \sim_x \equiv \left\{ [\mathbf{x}] = \sum_{i=1}^{d_\mathrm{x}} [c_i \mathbf{a}_i] \;\middle|\; c_i \in [0, N_i) \right\} \subset \mathbb{R}^{d_\mathrm{x}}. \tag{3.19}$$

Proof 3.18 We have to show, that for $\tilde{\mathbf{x}} \in \mathbb{R}^{d_\mathrm{x}}$ there is a unique representative $[\mathbf{x}] \in \mathbb{R}^{d_\mathrm{x}} / \sim_x$, such that $\tilde{\mathbf{x}} \in [\mathbf{x}]$. For this, we note that the lattice-vectors, being linearly independent, form a basis of $\mathbb{R}^{d_\mathrm{x}}$. Therefore, we can decompose $\tilde{\mathbf{x}}$ as

$$\tilde{\mathbf{x}} = \sum_{i=1}^{d_\mathrm{x}} \tilde{c}_i \mathbf{a}_i \quad \text{with } \tilde{c}_i \in \mathbb{R}. \tag{3.20}$$

Now we define

$$c_i = \tilde{c}_i \bmod N_i \in [0, N_i) \tag{3.21}$$

and

$$\mathbf{x} = \sum_{i=1}^{d_\mathrm{x}} c_i \mathbf{a}_i \quad \text{with} [\mathbf{x}] \in \mathbb{R}^{d_\mathrm{x}} / \sim_x . \tag{3.22}$$

It follows further that $\tilde{\mathbf{x}} \sim_x \mathbf{x}$ and therefore $[\tilde{\mathbf{x}}] = [\mathbf{x}] \in \mathbb{R}^{d_\mathrm{x}} / \sim_x$.

Corollary 3.19 Since $L_\infty \subset \mathbb{R}^{d_\mathrm{x}}$, this induces also a partitioning of the lattice into equivalence classes as

$$\mathrm{L_f} \equiv \mathrm{L_\infty} / \sim_x \equiv \left\{ [\mathbf{R_n}] = \sum_{i=1}^{d_\mathrm{x}} [n_i \mathbf{a}_i] \;\middle|\; 0 \leq n_i < N_i \wedge n_i \in \mathbb{N} \right\} \tag{3.23}$$

where we introduced the *finite point lattice* $\mathrm{L_f}$.

Proof 3.19 Analogously to the proof of Lemma 3.18.

Remark 3.19.1 Analogously we introduce the *finite crystal* $\mathrm{C_f}$ as

$$\mathrm{C_f} \equiv \left\{ \boldsymbol{R}_{\mathbf{n}\kappa} \equiv \mathbf{R_n} + \boldsymbol{R}_\kappa \equiv R_{\mathbf{n},\kappa,\alpha} \;\middle|\; \mathbf{R_n} \in \mathrm{L_f} \quad \wedge \quad \kappa \in \{1, 2, \dots r\} \quad \wedge \quad \alpha \in \{1, \dots, d_\mathbf{x}\} \right\} \subset \mathbb{R}^{d_\mathrm{x}}. \tag{3.24}$$

In order to write down neat and simple expressions for the „discrete Fourier-transform" (will be introduced later), it is also convenient to introduce yet another lattice.

Definition 3.20 For a Bravais-lattice L_∞ with infinite reciprocal lattice and corresponding vectors $\mathbf{b}_i$ as in TBD, we introduce the *fine infinite reciprocal lattice* FRL_∞ as

$$\mathrm{FRL}_\infty \equiv \left\{ \mathbf{q} \equiv \sum_{i=1}^{d_\mathrm{x}} \frac{n_i}{N_i} \mathbf{b}_i \;\middle|\; n_i \in \mathbb{Z} \right\} \subset \mathbb{R}^{d_\mathrm{x}} \tag{3.25}$$

Definition 3.21 On FRL_∞ we define the following relation

$$\forall \mathbf{q}, \mathbf{q}' \in \mathrm{FRL}_\infty \;:\; \mathbf{q} \sim_q \mathbf{q}' \quad :\Longleftrightarrow \quad \exists \mathbf{n} \in \mathbb{Z}^{d_\mathrm{x}} \;:\; \mathbf{q} = \mathbf{q}' + \sum_{i=1}^{d_\mathrm{x}} n_i N_i \mathbf{b}_i \tag{3.26}$$

Lemma 3.22 This is an equivalence relation on FRL_∞.

Proof 3.22 Analogously to the proof of Lemma 3.18.

Corollary 3.23 It holds

$$\mathrm{Q} \equiv \mathrm{FRL}_\infty \,/ \sim_q \equiv \left\{ [\mathbf{q}] = \sum_{i=1}^{d_\mathrm{x}} \left[\frac{n_i}{N_i} \mathbf{b}_i \right] \;\middle|\; 0 \leq n_i < N_i \right\} \tag{3.27}$$

Proof 3.23 Analogously to the proof of Lemma 3.18.

Notation 3.24 This set Q above, is what within the physics-community called the *allowed wave-vectors within the first Brillouin-zone*. Oftentimes one re-centers the representatives of the equivalence-classes around the origin, mostly for aesthetic reasons. There are however exceptions where it has computational reasons, for example when one introduces approximations. Then, considering for example acoustic phonons, for which one has $\hbar\omega_\mathbf{q} \approx \mathrm{const.}\, q$ for $q \ll 1$ ("Goldstone-theorem"), it is favourable to consider this „symmetric" choice of equivalence-classes.

Notation 3.25 Summing over these equivalence-classes is usually denoted with an unconspicious

$$\sum_{\mathbf{q}} f(\mathbf{q}) \tag{3.28}$$

instead of

$$\sum_{[\mathbf{q}]\in \mathrm{Q}} f(\mathbf{q}) \tag{3.29}$$

with functions, that are independent on the choice of representative.

In order to proceed we now switch to the physics-notation.

Notation 3.26 In physics, one introduces the notation for the actual position(-operators) of the ionic coordinates (their average oscillates around the corresponding points in the crystal)

$$\left\{ \boldsymbol{\tau}_{\mathbf{n}\kappa} \equiv \mathbf{R}_{\mathbf{n}\kappa} + \boldsymbol{\Delta\tau}_{\mathbf{n}\kappa} \;\middle|\; [\mathbf{R}_\mathbf{n}] \in \mathrm{L}_\mathrm{f} \quad \wedge \quad \kappa \in \{1, 2, \ldots r\} \right\}$$

one introduces their Fourier-transforms as another sequence of numbers

$$\left\{ \tau_{\mathbf{q}\kappa} \;\middle|\; [\mathbf{q}] \in \mathrm{Q} \quad \wedge \quad \kappa \in \{1, 2, \ldots r\} \right\}$$

as finite sums

$$\tau_{\mathbf{q}\kappa} = \frac{1}{\sqrt{N}} \sum_{[\mathbf{R}_\mathbf{n}]\in \mathrm{L}_\mathrm{f}} \tau_{\mathbf{n}\kappa}\, \mathrm{e}^{+\mathrm{i}\mathbf{q}\cdot\mathbf{R}_\mathbf{n}} \tag{3.30}$$

whereas $N \equiv N_p \equiv \prod_{i=1}^{d_\mathrm{x}} N_i$.

Remark 3.26.1 Notice that this definition is well-defined, as both sides are independent of the choice of the representative that one chooses.

Lemma 3.27 The inverse transformation reads

$$\tau_{\mathbf{n}\kappa} = \frac{1}{\sqrt{N}} \sum_{[\mathbf{q}]\in \mathrm{Q}} \tau_{\mathbf{q}\kappa}\, \mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R}_\mathbf{n}} \tag{3.31}$$

Proof 3.27 With the geometric series (again).

High Symmetry Points

Fact 3.28 For each lattice, one introduces within the first Brillouin-zone so-called „*high-symmetry-points*" in order to plot band-structures.

Remark 3.28.1 These high-symmetry-points for each individual crystal are not unique. In different literatures one finds different high-symmetry-points.

Remark 3.28.2 It will turn out later, that fundamental quantities as so-called "phononic dispersion-relations" and "electronic band-structures", are (for each respective "branch" or "band") actually functions which take arguments from the first Brillouin-zone of the underlying crystal. Let us use ξ as a generic symbol for such an object of interest with domain located within the first Brillouin-zone:

$$\begin{aligned} \xi :\ & \mathrm{Q} \longrightarrow \mathbb{R} \\ & \mathbf{q} \longmapsto \xi(\mathbf{q}) \end{aligned}$$

with $\mathrm{Q} \subset \mathrm{FBZ}$. As one can hardly draw the graph of a function that takes three-dimensional arguments, one usually depics the values of these functions on **k**-path. What one then usually sees is a picture with the **k**-path on the abscissa with the corresponding values of ξ on the ordinate. Hence the picture advertised in the beginning.

Remark 3.28.3 One can also rewrite this using the usual tools (and notations) that one knows from math-lectures by introducing a concrete curve $[0, 1] \ni s \longmapsto \gamma(s) \in \mathrm{FBZ}$ that connects the different high-symmetry **k**-points and then interpreting the pictures as image of a composite mapping $[0, 1] \ni s \longmapsto \xi(\gamma(s)) \in \mathbb{R}$. However...

Remark 3.28.4 These high-symmetry-points are actually a topic on their own... The mentality is, to get a glimpse of for example the band-structure of a certain material. (Keep in mind that the band-structure is a four-dimensional object, taking three arguments and giving one number for each argument.) By reducing the dimension to a two-dimensional picture in a neat-fashion, one can partly try to draw conclusions about this material. (For example, if there is a band-gap along the high-symmetry-points, it indicates that there is a band-gap in the density of states, leading to the classification of the material being metallic or insulating. There are however exceptions when this does not work...

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High-throughput electronic band structure calculations: Challenges and tools

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Figure 3.4.: A paper that lists high symmetry points for the different Bravais-lattices.

302 W. Setyawan, S. Curtarolo / Computational Materials Science 49 (2010) 299–312

The following protocol should be followed. Unit cells must first be reduced to standard primitives, then they should be appropriately relaxed (if needed). Before the static run, the cells should be reduced again to standard primitive (symmetry and orientation might have changed during the relaxation). The user should then perform the static run and then project the eigenvalues along the directions which are specified in the "kpath" option. If the user is running AFLOW and VASP, the web interface can also prepare a template input file "aflow.in" which performs all the mentioned tasks.

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Appendix A

The choice of lattice vectors implemented in AFLOW is given here. When the primitive lattice is the same as the conventional one, it is simply called "lattice". Variables a, b, c, α, β, γ denote the axial lengths and interaxial angles of the conventional lattice vectors, while k_a, k_b, k_c, k_α, k_β, k_γ are those of the primitive reciprocal lattice vectors $\mathbf{b}_1$, $\mathbf{b}_2$, $\mathbf{b}_3$. The coordinates of symmetry **k**-points are given in fractions of $\mathbf{b}_1$, $\mathbf{b}_2$, $\mathbf{b}_3$.

A.1. Cubic (CUB, cP)

Lattice (see Fig. 1 and Table 2)

$\mathbf{a}_1 = (a, 0, 0)$
$\mathbf{a}_2 = (0, a, 0)$
$\mathbf{a}_3 = (0, 0, a)$

A.2. Face-centered cubic (FCC, cF)

(See Fig. 2 and Table 3.)

Conventional lattice	Primitive lattice
$\mathbf{a}_1 = (a, 0, 0)$	$\mathbf{a}_1 = (0, a/2, a/2)$
$\mathbf{a}_2 = (0, a, 0)$	$\mathbf{a}_2 = (a/2, 0, a/2)$
$\mathbf{a}_3 = (0, 0, a)$	$\mathbf{a}_3 = (a/2, a/2, 0)$

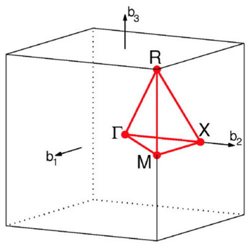


Fig. 1. Brillouin zone of CUB lattice. Path: Γ–X–M–Γ–R–$X|M$–R. An example of band structure using this path is given in Fig. 26.

Table 2
Symmetry **k**-points of CUB lattice.

$\times\mathbf{b}_1$	$\times\mathbf{b}_2$	$\times\mathbf{b}_3$		$\times\mathbf{b}_1$	$\times\mathbf{b}_2$	$\times\mathbf{b}_3$	
0	0	0	Γ	1/2	1/2	1/2	R
1/2	1/2	0	M	0	1/2	0	X

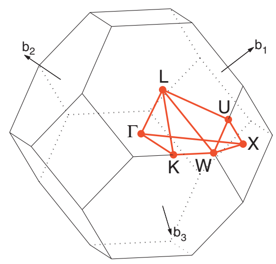


Fig. 2. Brillouin zone of FCC lattice. Path: Γ–X–W–K–Γ–L–U–W–L–$K|U$–X. An example of band structure using this path is given in Fig. 27.

Table 3
Symmetry **k**-points of FCC lattice.

$\times\mathbf{b}_1$	$\times\mathbf{b}_2$	$\times\mathbf{b}_3$		$\times\mathbf{b}_1$	$\times\mathbf{b}_2$	$\times\mathbf{b}_3$	
0	0	0	Γ	5/8	1/4	5/8	U
3/8	3/8	3/4	K	1/2	1/4	3/4	W
1/2	1/2	1/2	L	1/2	0	1/2	X

A.3. Body-centered cubic (BCC, cI)

(See Fig. 3 and Table 4.)

Conventional lattice	Primitive lattice
$\mathbf{a}_1 = (a, 0, 0)$	$\mathbf{a}_1 = (-a/2, a/2, a/2)$
$\mathbf{a}_2 = (0, a, 0)$	$\mathbf{a}_2 = (a/2, -a/2, a/2)$
$\mathbf{a}_3 = (0, 0, a)$	$\mathbf{a}_3 = (a/2, a/2, -a/2)$

A.4. Tetragonal (TET, tP)

Lattice (see Fig. 4 and Table 5)

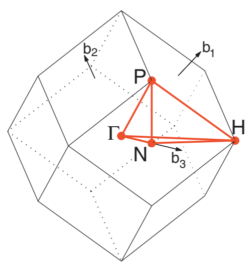


Fig. 3. Brillouin zone of BCC lattice. Path: Γ–H–N–Γ–P–$H|P$–N. An example of band structure using this path is given in Fig. 28.

Figure 3.5.: Excerpt of the paper above.

4. Quantum Mechanics of Crystals - Electrons and Cores

4.1. The General Problem

Notation 4.1 We consider a charge neutral *piece of solid material* and model it as a finite crystal as in Remark 3.19.1. Notice that this crystal has ...

- No defects, no impurities, no disorder, no surfaces, no interfaces, no grain-boundaries, no dislocations, no vacancies, ...
- a dimension $d_{\mathbf{x}} \in \{1, 2, 3\}$,
- an underlying crystal-structure, with lattice-vectors $\bigcup_{j=1}^{d_{\mathbf{x}}} \{\mathbf{a}_j\}$, and atomic basis-vectors $\{\boldsymbol{R}_1, \ldots, \boldsymbol{R}_r\}$ according to Definition 3.7,
- a finite size, which will determine the numbers of unit-cells in each direction $N_1, \ldots, N_{d_{\mathbf{x}}}$ and the total number of unit cells $\prod_{i=1}^{d_{\mathbf{x}}} N_i = N_p$ in Hypothesis 3.14,
- atoms or (inclusive) ionic cores with (individual) masses $\{M_1, \ldots, M_r\}$ charges $\{Z_1\mathrm{e}, \ldots, Z_r\mathrm{e}\}$ centered located on the crystal-points.
- $N_e = (Z_1 + \ldots + Z_r) \prod_{j=1}^{d_{\mathbf{x}}} N_j$ (indistinguishable) electrons with (individual) mass m and charge $-\mathrm{e}$.

Remark 4.1.1 Notice that within these notations, this gives us a total of $rN_p = N_a$ ions to take care of, which results in $rd_{\mathbf{x}}N_p = d_{\mathbf{x}}N_a$ ionic coordinates. To distinguish the ionic coordinates from the lattice-points, we denote them (the coordinates) with[1] $\boldsymbol{\tau}$ (with proper indices that are clear in their meaning from the context).

Remark 4.1.2 Similarly, we have to take care of $d_{\mathbf{x}}N_e$ electronic coordinates[2]. Neglecting the spin-coordinate, we collect the spatial coordinates of the electrons and ionic cores in the collective vectors

$$\begin{aligned}
\mathbf{r} &\equiv (\mathbf{r}_1, \ldots, \mathbf{r}_{N_e}) = \text{„electronic coordinates“} \\
\boldsymbol{\tau} &\equiv (\boldsymbol{\tau}_1, \ldots, \boldsymbol{\tau}_{N_a}) = \text{„ionic coordinates“}
\end{aligned}$$

[1] Notice that we denote the coordinates of the ions with another symbol than the lattice-vectors. This notation ("τ") was (probably) introduced by Feliciano Giustino (one of the main developers of the EPW-software) in his 2017-article „Electron-Phonon interactions from first principles“.

[2] We neglect the spin-coordinates for the moment. The fact, that in usual textbooks, for example, [Czycholl], the spin is being neglected is actually not really trivial. It is usually later re-implemented in the formulation of Bloch-wavefunctions by just „attaching“ the spin-part of a wavefunction to its spatial component. There are though exceptions, for example when the lattice-periodic potential possesses a non-trivial spin-dependence. Usual codes, for example [ELK, QE] possess an exchange-correlation functional with this property. This is the so-called „LSDA“-approximation.

Remark 4.1.3 From X-ray-spectroscopy, one can really conclude, that we are (in general) working with partially but not fully ionized cores. The ionic cares can be very well described with the orbital-model. Innermost shells are very similar to the shells of the generalized hydrogen-problem. One can however also compute their band-structure and sees that it is very flat.

Remark 4.1.4 With powder-diffraction (also called „Debye-Scherrer-method“) or with the „Laue-Backscattering-Method“ one can also conclude that the ions are really centered around lattice points.

Notation 4.2 We write for the kinetic-energy-operators

$$\begin{aligned}
\mathrm{T}_e &= \sum_{i=1}^{N_e} -\frac{\hbar^2}{2m}\Delta_{\mathbf{r}_i}, \\
\mathrm{T}_a &= \sum_{i=1}^{N_a} -\frac{\hbar^2}{2M_i}\Delta_{\tau_i} = \sum_{\substack{\mathbf{R_n}\in \mathrm{L_f} \\ \alpha\in\{1,\dots,d_{\mathbf{x}}\} \\ \kappa\in\{1,\dots,r\}}} -\frac{\hbar^2}{2M_\kappa}\frac{\partial^2}{\partial\tau^2_{\mathbf{n}\kappa\alpha}}
\end{aligned}$$

and for the interaction-operators

$$\begin{aligned}
V_{e-e} &\equiv \sum_{1\le i<j\le N_e} v_{e-e}(\mathbf{r}_i - \mathbf{r}_j) \\
V_{a-a} &\equiv \sum_{1\le i<j\le N_a} v_{a_i-a_j}(\boldsymbol{\tau}_i - \boldsymbol{\tau}_j) \\
V_{e-a} &\equiv \sum_{1\le i\le N_e}\sum_{1\le j\le N_a} v_{e-a_j}(\mathbf{r}_i - \boldsymbol{\tau}_j)
\end{aligned}$$

Remark 4.2.1 In modern density functional theory codes, one oftentimes approximates the interaction-potentials above through so-called „pseudopotentials“, instead of the proper Coulomb-interactions[3], which is mainly for computational convenience. (Leitmotiv: „Just solve it!“)

Hypothesis 4.3 Assuming the solid to be in a stationary state, the time-independent Schrödinger-equation now reads

[3] That would read in the current notation

$$\begin{aligned}
V_{e-e} &\equiv \sum_{1\le i<j\le N_e} v_{e-e}(\mathbf{r}_i - \mathbf{r}_j) &&\approx \sum_{1\le i<j\le N_e} \frac{e^2}{4\pi\epsilon_0|\mathbf{r}_i - \mathbf{r}_j|} \\
V_{a-a} &\equiv \sum_{1\le i<j\le N_a} v_{a_i-a_j}(\boldsymbol{\tau}_i - \boldsymbol{\tau}_j) &&\approx \sum_{1\le i<j\le N_a} \frac{Z_iZ_je^2}{4\pi\epsilon_0|\boldsymbol{\tau}_i - \boldsymbol{\tau}_j|} \\
V_{e-a} &\equiv \sum_{1\le i\le N_e}\sum_{1\le j\le N_a} v_{e-a_j}(\mathbf{r}_i - \boldsymbol{\tau}_j) &&\approx \sum_{1\le i\le N_e}\sum_{1\le j\le N_a} \frac{-Z_je^2}{4\pi\epsilon_0|\mathbf{r}_i - \boldsymbol{\tau}_j|}
\end{aligned} \tag{4.1}$$

$$H\psi(\mathbf{r},\boldsymbol{\tau}) = E\psi(\mathbf{r},\boldsymbol{\tau}) \tag{4.2}$$

with

$$H \equiv \mathrm{T}_e + \mathrm{T}_a + \mathrm{V}_{e-e} + \mathrm{V}_{a-a} + \mathrm{V}_{e-a} \tag{4.3}$$

Remark 4.3.1 This Schrödinger-equation is impossible to solve exactly (neither analytically nor numerically) for crystals that are bigger than a couple of unit-cells in each direction. One needs to refer to approximation-schemes.

4.2. The Adiabatic / Born-Oppenheimer-Approximation

The general aim is to „decouple“ the dynamics (in the sense of a differential equation) of the electrons from the dynamics of the lattice.

Notation 4.4 We decompose the Hamiltonian above as

$$\begin{aligned} H &= \mathrm{T}_e + \mathrm{V}_{e-e} + \mathrm{V}_{a-a} + \mathrm{V}_{e-a} \quad &+ \quad &\mathrm{T}_a \\ &\equiv \qquad H_0 &+ \quad &V \end{aligned}$$

Definition 4.5 Applying now a perturbative treatment and neglecting the terms in equation (27.14) (see below) is called the *Born-Oppenheimer-Approximation* a.k.a *adiabatic approximation*

Notation 4.6 Let us assume, that for a fixed ion-configuration $\boldsymbol{\tau}$ we are able to determine all the orthonormal solutions

$$\left\{\mathbb{R}^{d_x N_e} \ni \mathbf{r} \mapsto \phi_{\alpha,\tau}(\mathbf{r}) \in \mathbb{C}\right\}_{\alpha\in\mathscr{A}} \tag{4.4}$$

of the equation

$$\left(H_0\phi_{\alpha,\tau}\right)(\mathbf{r}) = \left(\left[\mathrm{T}_e + \mathrm{V}_{e-e} + \mathrm{V}_{a-a} + \mathrm{V}_{e-a}\right]\phi_{\alpha,\tau}\right)(\mathbf{r}) = \epsilon_{\alpha,\tau}\phi_{\alpha,\tau}(\mathbf{r}) \tag{4.5}$$

Remark 4.6.1 This assumption, of being able to solve the electronic problem, was actually quite optimistic when it was proposed in 1927. In fact, even nowadays, only perturbative solutions to this problem exist.

Remark 4.6.2 In order to solve the problem

$$\left(H_0\phi_{\alpha,\tau}\right)(\mathbf{r}) \equiv \epsilon_{\alpha,\tau}\phi_{\alpha,\tau}(\mathbf{r}) \tag{4.6}$$

perturbatively, it is nowadays usual to do a mean-field-approximation: Instead of considering the full electron-electron-interaction in Equation (4.1), one determines (with density functional theory) an effective single-particle-interaction, such that one has

$$\left(\left[\mathrm{T}_e + \mathrm{V}_{e-e} + \mathrm{V}_{a-a} + \mathrm{V}_{e-a}\right]\phi_{\alpha,\tau}\right)(\mathbf{r}) = \epsilon_{\alpha,\tau}\phi_{\alpha,\tau}(\mathbf{r}) \quad \overset{\text{Approximation}}{\Longleftrightarrow} \quad \left(\left[\mathrm{T}_e + \mathrm{V}_{\text{eff}} + \mathrm{V}_{a-a} + \mathrm{V}_{e-a}\right]\phi_{\alpha,\tau}\right)(\mathbf{r}) = \epsilon_{\alpha,\tau}\phi_{\alpha,\tau}(\mathbf{r}) \tag{4.7}$$

whereas the effective one-particle-potentials

$$\mathrm{V}_{\text{eff}}(\mathbf{r}) \equiv \sum_{i=1}^{N_e} \mathrm{V}_{\text{eff}}(\mathbf{r}_i) \equiv \sum_{i=1}^{N_e} \mathrm{V}_{\mathrm{H}}(\mathbf{r}_i) + \mathrm{V}_{\mathrm{XC}}(\mathbf{r}_i) \tag{4.8}$$

is composed of the so-called „Hartree“ and „exchange-correlation“-contributions. These potentials are determined through density functional theory, on which we will not elaborate further in this course.

Remark 4.6.3 With the procedure elucidated above, it becomes clear, that $\alpha \in \mathscr{A}$ represents a multi-index and the wavefunction $\mathbf{r} \mapsto \phi_{\alpha,\tau}(\mathbf{r})$ represents a Slater-determinant. These Slater-determinants are being build with single-particle-wavefunctions $\psi_{n\mathbf{k}\sigma}(x)$ (*Bloch-functions* or *Kohn-Sham-wavefunctions*, or, even more precise, *Bloch-Kohn-Sham-wavefunctions*) which will turn out to obey the one-particle Schrödinger-equation

$$\left(-\frac{\hbar^2}{2m}\Delta + V_{\mathrm{H}}(\mathbf{x}) + V_{\mathrm{XC}}(\mathbf{x}) + V_{\text{e-a}}(\mathbf{x}) + \text{const.}\right)\psi_{n\mathbf{k}\sigma}(\mathbf{x}) = E_{n\mathbf{k}}\psi_{n\mathbf{k}\sigma}(\mathbf{x}) \tag{4.9}$$

with $n \in \{1, 2, \dots\}$, $\mathbf{k} \in \mathrm{Q}$ and $\sigma \in \{\downarrow, \uparrow\}$ in the absence of spin-orbit-coupling[4].

Remark 4.6.4 One notices, that in order to tackle the Schrödinger-equations of crystals, one divides the problem into further and further smaller (more or less managable) problems:

- We will be able to solve the time-independent Schrödinger-equation for the full (electronic and ionic system) in Equation (2.19) …

  … *if* we are able to solve the time-independent Schrödinger-equation for the effective electronic problem for some fixed ion-configuration in Equation (4.2).

- We will be able to solve the time-independent Schrödinger-equation for the effective electronic problem for some fixed ion-configuration in Equation (4.2) …

  … *if* we are able to find a (lattice-periodic) single-particle-potential that represents a good approximation to the electron-electron-interactions in Equation (4.7),

  … *and (inclusive) if* we are able solve the time-independent Schrödinger-equation for a single-electron moving in a lattice-periodic potential with an effective one-particle-interaction in Equation (4.9).

Remark 4.6.5 There are also problems with the procedure outlined above... For example band-gaps and energies might be calculated inaccurately by a poor approximation of the electron-electron-interactions. One cure / catchphrase is here the so-called „GW-approximation“, on which we will not elaborate further in this document.

[4] In the presence of spin-orbit coupling the energies acquire a spin-dependence $E_{n\mathbf{k}} \mapsto E_{n\mathbf{k}\sigma}$.

Remark 4.6.6 Although the laborious procedure outlined above results in „concrete“ (numerical) expressions for the wavefunctions, it will later on turn out to be appropriate to not work directly with the wavefunctions themselves, but with the matrix-elements (w.r.t. those wavefunctions above) of the Hamiltonian and a „density“ that represents „the state of the thermodynamic equilibrium in the grand canonical ensemble in absence of spontaneous symmetry breaking“. We will briefly touch this formalism, when we compute specific heats in the low-temperature-limit in the formalism of second-quantization.

Notation 4.7 Having determined the wave-functions $\left\{\mathbf{r} \mapsto \phi_{\alpha,\tau}(\mathbf{r})\right\}_{\alpha\in\mathscr{A}}$ with eigenenergies $\left\{\epsilon_{\alpha,\tau} \equiv \epsilon_{\alpha}(\boldsymbol{\tau})\right\}_{\alpha\in\mathscr{A}}$ for an (arbitrarily) fixed ion position τ, we now consider the full time-independent Schrödinger-equation

$$H\psi(\mathbf{r},\boldsymbol{\tau}) = E\psi(\mathbf{r},\boldsymbol{\tau}) \tag{4.10}$$

and expand the wavefunctions w.r.t. complete basis

$$\psi(\mathbf{r},\boldsymbol{\tau}) = \sum_{\alpha\in\mathscr{A}} \chi_{\alpha}(\boldsymbol{\tau})\phi_{\alpha,\tau}(\mathbf{r}) \tag{4.11}$$

Theorem 4.8 The expansion in Equation (4.11) is a solution to the Schrödinger-Equation (4.10) in frame of the adiabatic approximation, if the coefficients $\chi_{\alpha}(\boldsymbol{\tau})$ fulfill the equation

$$\left(T_{\mathrm{a}} + \epsilon_{\alpha,\tau}\right)\chi_{\alpha}(\boldsymbol{\tau}) = E\chi_{\alpha}(\boldsymbol{\tau}) \tag{4.12}$$

Proof 4.8 See Appendix, Proofs, page 292.

Remark 4.8.1 Since the electronic energies represent for each index α a mapping $\tau \mapsto \epsilon_{\alpha,\tau} \equiv \epsilon_{\alpha}(\boldsymbol{\tau})$ one usually interpretes Equation (27.16) as the *ionic Schrödinger-equation*.

5. The Ionic Problem

5.1. The Schrödinger-Equation of the Ionic-System in the Harmonic-Approximation

We start with the effective Schrödinger-equation for the wavefunctions of the ions

$$H\chi(\boldsymbol{\tau}) = E\chi(\boldsymbol{\tau}) \qquad \text{with} \qquad H = \sum_{\substack{\mathbf{R_n}\in \mathrm{L_f}\\ \alpha\in\{1,\dots,d_{\mathbf{x}}\}\\ \kappa\in\{1,\dots,r\}}} -\frac{\hbar^2}{2M_\kappa}\frac{\partial^2}{\partial\tau^2_{\mathbf{n}\kappa\alpha}} + V_{\text{eff}}(\boldsymbol{\tau}) \tag{5.1}$$

with an effective potential[1]

$$\boldsymbol{\tau} \mapsto V_{\text{eff}}(\boldsymbol{\tau}) \in \mathbb{R} \tag{5.2}$$

that is the result of the solution to the electronic problem for different ion configurations $\boldsymbol{\tau}$.

Remark 5.0.1 Recall from Lemma **??** that we have the canonical commutation relations for the ionic coordinates

$$\left[\tau_{\mathbf{n}\kappa\alpha},\, p_{\tau_{\mathbf{n}'\kappa'\alpha'}}\right] = \mathrm{i}\hbar\delta_{\mathbf{n},\mathbf{n}'}\delta_{\kappa,\kappa'}\delta_{\alpha,\alpha'} \tag{5.3}$$

<u>Intermezzo: Coordinate-Change</u>

Lemma 5.1 If one changes coordinates as

$$x_i = x_i(x'_1,\dots,x'_n) \tag{5.4}$$

then it follows for the derivatives of the corresponding coordinates that

$$\frac{\partial}{\partial x'_i} = \sum_{j=1}^{n}\frac{\partial x_j}{\partial x'_i}\frac{\partial}{\partial x_j} \tag{5.5}$$

Proof 5.1 See any textbook on calculus in more than one dimensions.

[1]Notice that we changed the notation of the effective potential, relative to the last section.

Hypothesis 5.2 We assume that there is an minimal equilibirum-position for the ions $\boldsymbol{\tau}$ s.t.

$$\min_{\boldsymbol{\tau} \in \mathbb{R}^{d_{\mathbf{x}} N_a}} V_{\mathrm{eff}}(\boldsymbol{\tau}) \equiv V_{\mathrm{eff}}(\mathbf{R}_{\mathrm{eq}}) = \text{const.} \overset{\text{w.l.o.g.}}{=} 0 \quad \text{with} \quad \left(\nabla_{\boldsymbol{\tau}} V_{\mathrm{eff}}\right)(\mathbf{R}_{\mathrm{eq}}) = 0. \tag{5.6}$$

Notation 5.3 We change coordinates to the elongation of the ions from the individual lattice-sites

$$\Delta\boldsymbol{\tau}_{\mathbf{n}\kappa} \equiv \boldsymbol{\tau}_{\mathbf{n}\kappa} - \mathbf{R}_{\mathbf{n}\kappa} \quad \Longleftrightarrow \quad \Delta\tau_{\mathbf{n}\kappa\alpha} \equiv \tau_{\mathbf{n}\kappa\alpha} - R_{\mathbf{n}\kappa\alpha}. \tag{5.7}$$

Lemma 5.4 The derivatives change according to the chain-rule

$$\frac{\partial}{\partial \tau_{\mathbf{n}\kappa\alpha}} \equiv \frac{\partial}{\partial \Delta\tau_{\mathbf{n}\kappa\alpha}}. \tag{5.8}$$

and that this transformation preserves the CCR; it holds

$$\left[\Delta\tau_{\mathbf{n}\kappa\alpha},\, p_{\Delta\tau_{\mathbf{n}'\kappa'\alpha'}}\right] = \mathrm{i}\hbar\delta_{\mathbf{n},\mathbf{n}'}\delta_{\kappa,\kappa'}\delta_{\alpha,\alpha'} \tag{5.9}$$

Proof 5.4 Follows immediately from the linearity of the commutator together with Lemma 5.1.

Corollary 5.5 Changing the coordinates and expanding the effective potential in a series up to the second-order simplifies the Hamiltonian to

$$T_a = \sum_{\substack{\mathbf{R}_{\mathbf{n}} \in \mathrm{L}_{\mathrm{f}} \\ \kappa \in \{1,\ldots,r\} \\ \alpha \in \{1,\ldots,d_{\mathbf{x}}\}}} -\frac{\hbar^2}{2M_\kappa}\frac{\partial^2}{\partial \Delta\tau^2_{\mathbf{n}\kappa\alpha}}$$

$$\begin{aligned} V_{\mathrm{eff}}(\Delta\boldsymbol{\tau}) &\equiv V_{\mathrm{eff}}(0) + \sum_{\substack{\mathbf{n} \in \mathrm{L}_{\mathrm{f}} \\ \kappa \in \{1,\ldots,r\} \\ \alpha \in \{1,\ldots,d_{\mathbf{x}}\}}} \left(\frac{\partial V_{\mathrm{eff}}}{\partial \Delta\tau_{\mathbf{n}\kappa\alpha}}\right)(0)\,\Delta\tau_{\mathbf{n}\kappa\alpha} + \frac{1}{2}\sum_{\substack{\mathbf{n},\mathbf{n}' \in \mathrm{L}_{\mathrm{f}} \\ \kappa,\kappa' \in \{1,\ldots,r\} \\ \alpha,\alpha' \in \{1,\ldots,d_{\mathbf{x}}\}}} \left(\frac{\partial^2 V_{\mathrm{eff}}}{\partial \Delta\tau_{\mathbf{n}\kappa\alpha}\partial \Delta\tau_{\mathbf{n}'\kappa'\alpha'}}\right)(0)\,\Delta\tau_{\mathbf{n}\kappa\alpha}\Delta\tau_{\mathbf{n}'\kappa'\alpha'} + \ldots \\ &\approx 0 \qquad + 0 \qquad + \frac{1}{2}\sum_{\substack{\mathbf{n},\mathbf{n}' \in \mathrm{L}_{\mathrm{f}} \\ \kappa,\kappa' \in \{1,\ldots,r\} \\ \alpha,\alpha' \in \{1,\ldots,d_{\mathbf{x}}\}}} \left(\frac{\partial^2 V_{\mathrm{eff}}}{\partial \Delta\tau_{\mathbf{n}\kappa\alpha}\partial \Delta\tau_{\mathbf{n}'\kappa'\alpha'}}\right)(0)\,\Delta\tau_{\mathbf{n}\kappa\alpha}\Delta\tau_{\mathbf{n}'\kappa'\alpha'} \end{aligned}$$

Proof 5.5 Follows immediately by the assumptions and the Taylor-expansion.

Notation 5.6 From the Hesse-matrix at the equilibrium-position, one defines the *real-space-dynamical-matrix* as

$$D_{\mathbf{n}\kappa\alpha,\mathbf{n}'\kappa'\alpha'} \equiv \frac{1}{\sqrt{M_\kappa M_{\kappa'}}} \left(\frac{\partial^2 V_{\text{eff}}}{\partial \Delta\tau_{\mathbf{n}\kappa\alpha} \partial \Delta\tau_{\mathbf{n}'\kappa'\alpha'}} \right)(0). \tag{5.10}$$

Remark 5.6.1 Notice that ...

- D is real-valued (the potential is real-valued),
- D is symmetric since the Schwarz-theorem applies

  $$\frac{\partial^2 V_{\text{eff}}}{\partial \Delta\tau_{\mathbf{n}\kappa\alpha} \partial \Delta\tau_{\mathbf{n}'\kappa'\alpha'}}(\bullet) = \frac{\partial^2 V_{\text{eff}}}{\partial \Delta\tau_{\mathbf{n}'\kappa'\alpha'} \partial \Delta\tau_{\mathbf{n}\kappa\alpha}}(\bullet),$$

  which implies for the dynamical matrix the relation

  $$D_{\mathbf{n}\kappa\alpha,\mathbf{n}'\kappa'\alpha'} = D_{\mathbf{n}'\kappa'\alpha',\mathbf{n}\kappa\alpha} \tag{5.11}$$

  meaning that the real-space-dynamical matrix is symmetric.
- D is positive definite, i.e. $\mathbf{v}^{\mathrm{T}} \cdot D \cdot \mathbf{v} > 0$ for $\mathbf{v} \neq 0$, since we assume an equilibrium-position that minimizes the potential energy.

At this point we are facing a crossroad. We can now continue with the solution of the Schrödinger-equation in two ways. The first is valuable for pedagogical reasons, and the other is valuable for computational (i.e. numerical) reasons. We will present both. In either way, we have to go through a series of coordinate-transformations, which is quite exhausting to keep track of.

5.1.1. Road 1 through Real-Space: A Poor Man's Phonons.

Notation 5.7 We introduce new coordinates as

$$\widetilde{u}_{\mathbf{n}\kappa\alpha} \equiv \sqrt{M_\kappa}\Delta\tau_{\mathbf{n}\kappa\alpha} \tag{5.12}$$

Lemma 5.8 We obtain for the new derivatives the relation

$$\frac{\partial}{\partial \widetilde{u}_{\mathbf{n}\kappa\alpha}} = \frac{1}{\sqrt{M_\kappa}} \frac{\partial}{\partial \Delta\tau_{\mathbf{n}\kappa\alpha}} \tag{5.13}$$

and this transformation preserves the CCR, i.e. it holds

$$\left[\widetilde{u}_{\mathbf{n}\kappa\alpha}, \, p_{\widetilde{u}_{\mathbf{n}'\kappa'\alpha'}}\right] = \mathrm{i}\hbar\delta_{\mathbf{n},\mathbf{n}'}\delta_{\kappa,\kappa'}\delta_{\alpha,\alpha'} \tag{5.14}$$

Proof 5.8 Follows immediately from the linearity of the commutator, and the chain-rule for derivatives in Lemma 5.1.

Corollary 5.9 Switching to the new coordinates simplifies the Hamiltonian in the harmonic approximation to

$$\begin{aligned}
H_{\text{harm}} &= T_{\text{ions}} + V_{\text{eff}}(\Delta\boldsymbol{\tau}) \\
&= \frac{1}{2} \sum_{\substack{\mathbf{R_n} \in \mathrm{L_f} \\ \kappa\in\{1,\dots,r\} \\ \alpha\in\{1,\dots,d\}}} -\hbar^2 \frac{\partial^2}{\partial \widetilde{u}_{\mathbf{n}\kappa\alpha}^2} + \frac{1}{2} \sum_{\substack{\mathbf{n},\mathbf{n}'\in \mathrm{L_f} \\ \kappa,\kappa'\in\{1,\dots,r\} \\ \alpha,\alpha'\in\{1,\dots,d\}}} D_{\mathbf{n}\kappa\alpha,\mathbf{n}'\kappa'\alpha'} \widetilde{u}_{\mathbf{n}\kappa\alpha} \widetilde{u}_{\mathbf{n}'\kappa'\alpha'} \\
&= \frac{1}{2} \mathbf{p}_{\widetilde{\mathbf{u}}}^{\mathrm{T}} \cdot \mathbb{1} \cdot \mathbf{p}_{\widetilde{\mathbf{u}}} + \frac{1}{2} \widetilde{\mathbf{u}}^{\mathrm{T}} \cdot D \cdot \widetilde{\mathbf{u}}
\end{aligned}$$

Proof 5.9 Follows immediately by insertion and applying the correspondence principle backwards.

Lemma 5.10 Since the dynamical matrix is symmetric, one can find an orthogonal[2] matrix $C \in \mathrm{Mat}(\mathbb{R}^{d_{\mathrm{x}}N_a} \times \mathbb{R}^{d_{\mathrm{x}}N_a})$ s.t.

$$C D C^{\mathrm{T}} = \Omega = \mathrm{diag}(\omega_1^2, \omega_2^2, \dots, \omega_{d_{\mathrm{x}}N_a}^2) \tag{5.16}$$

with frequencies that are chosen to be positive $\omega_i \in \mathbb{R}_{>0}$.

Proof 5.10 See any textbook on linear algebra.

Notation 5.11 We introduce new coordinates as

$$\mathbf{u} \equiv C\widetilde{\mathbf{u}} \tag{5.17}$$

Lemma 5.12 We obtain new momenta as

$$\mathbf{p_u} = C^{\mathrm{T}} \mathbf{p}_{\widetilde{\mathbf{u}}} \quad \Longleftrightarrow \quad \mathbf{p}_{\widetilde{\mathbf{u}}} = C \mathbf{p_u} \tag{5.18}$$

and this coordinate-transformation is compatible with the CCR; it holds

$$\left[u_{\mathbf{n}\kappa\alpha},\, p_{u_{\mathbf{n}'\kappa'\alpha'}}\right] = \mathrm{i}\hbar \delta_{\mathbf{n},\mathbf{n}'} \delta_{\kappa,\kappa'} \delta_{\alpha,\alpha'} \tag{5.19}$$

Proof 5.12 Follows immediately from the linearity of the commutator, and the chain-rule for derivatives in Lemma 5.1.

[2] Recall that this means that the transpose of the matrix is its inverse

$$C \cdot C^{\mathrm{T}} = C^{\mathrm{T}} \cdot C = \mathbb{1}. \tag{5.15}$$

Corollary 5.13 Changing to the new coordinates brings the Hamiltonian into the diagonal form

$$H_{\text{harm.}} = \frac{1}{2}\mathbf{p}_{\mathbf{u}}^{\mathrm{T}} \cdot \mathbb{1} \cdot \mathbf{p}_{\mathbf{u}} + \frac{1}{2}\mathbf{u}^{\mathrm{T}} \cdot \Omega \cdot \mathbf{u}$$

Proof 5.13 By direct calculation, we obtain

$$\begin{aligned} H_{\text{harm.}} &= \frac{1}{2}\tilde{\mathbf{p}}^{\mathrm{T}} \cdot C^{\mathrm{T}} \cdot C \cdot \tilde{\mathbf{p}} + \frac{1}{2}\tilde{\mathbf{u}}^{\mathrm{T}} C^{\mathrm{T}} \cdot C \cdot D \cdot C^{\mathrm{T}} \cdot C\tilde{\mathbf{u}} \\ &= \frac{1}{2}\tilde{\mathbf{p}}^{\mathrm{T}} \cdot C^{\mathrm{T}} \cdot C \cdot \tilde{\mathbf{p}} + \frac{1}{2}\tilde{\mathbf{u}}^{\mathrm{T}} C^{\mathrm{T}} \cdot \Omega \cdot C\tilde{\mathbf{u}} \end{aligned}$$

Remark 5.13.1 Notice therefore, that within the harmonic approximation, the ionic system is can be described as a sum of harmonic oscillators

$$H_{\text{harm.}} = \frac{1}{2}\mathbf{p}_{\mathbf{u}}^{\mathrm{T}} \cdot \mathbb{1} \cdot \mathbf{p}_{\mathbf{u}} + \frac{1}{2}\mathbf{u}^{\mathrm{T}} \cdot \Omega \cdot \mathbf{u} = \frac{1}{2}\sum_{i=1}^{d_{\mathbf{x}}N_a} p_i^2 + \omega_i^2 u_i^2 \equiv \sum_{i=1}^{d_{\mathbf{x}}N_a} H_i$$

Corollary 5.14 Using the standard-techniques from the single-particle-harmonic-oscillator, i.e. introducing ladder-operators

$$b_j = \sqrt{\frac{\omega_j}{2\hbar}}u_j + \mathrm{i}\sqrt{\frac{1}{2\hbar\omega_j}}p_j, \qquad b_j^\dagger = \sqrt{\frac{\omega_j}{2\hbar}}u_j - \mathrm{i}\sqrt{\frac{1}{2\hbar\omega_j}}p_j \tag{5.20}$$

with commutation relations

$$[b_j, b_k^\dagger] = \delta_{j,k}, \qquad [b_j, b_k] = [b_j^\dagger, b_k^\dagger] = 0, \tag{5.21}$$

one arrives at the semi-final form the Hamilton-operator in the harmonic approximation

$$H_{\text{harm.}} = \sum_{j=1}^{d_{\mathbf{x}}N_a} \hbar\omega_j \left(b_j^\dagger b_j + \frac{1}{2}\right). \tag{5.22}$$

Proof 5.14 By using standard-theorems that we proved earlier in the lecture and on the exercise sheets, we immediately obtain the result.

Remark 5.14.1 We therefore see, that we can solve the effective Schrödinger-equation of the ionic problem iff we can diagonalize the real-space-dynamical-matrix D (which is a task that we can't for realistic system sizes). One circumvents this problem by going to reciprocal space.

5.1.2. Road 2 through Reciprocal-Space: State of the Art Description and Identities.

Theorem 5.15 Through the Coulomb-potential and the periodic boundary-conditions the dynamical matrix inherits the translational invariance

$$D_{\mathbf{n}\kappa\alpha,\,\mathbf{n}'\kappa'\alpha'} \equiv D_{\kappa\alpha,\,\kappa'\alpha'}(\mathbf{R_n} - \mathbf{R_{n'}}) \tag{5.23}$$

Proof 5.15 Follows from the distance-dependence of the Coulomb-interaction, when one explicitly writes down the sums.

Remark 5.15.1 Notice that this implies that the dynamical matrix is a function of the difference of the lattice-vectors, which implies the relations

$$D_{\mathbf{n}\kappa\alpha,\,\mathbf{n}'\kappa'\alpha'} = D_{\mathbf{n}-\mathbf{n}'\kappa\alpha,\,0\kappa'\alpha'} = D_{0\kappa\alpha,\,-(\mathbf{n}-\mathbf{n}')\kappa'\alpha'} \tag{5.24}$$

Notation 5.16 We change the coordinates and consider discrete Fourier-transforms with prefactor-conventions, to introduce the *Fourier-transformed Dynamical Matrix*

$$\begin{aligned}
\Delta\tau_{\mathbf{n}\kappa\alpha} &\equiv \frac{1}{\sqrt{N_p}}\sum_{\mathbf{q}\in\mathrm{Q}} \Delta\tau_{\mathbf{q}\kappa\alpha}\mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot\mathbf{R_n}} & \iff \quad & \Delta\tau_{\mathbf{q}\kappa\alpha} \equiv \frac{1}{\sqrt{N_p}}\sum_{\mathbf{R_n}\in\mathrm{L_f}} \Delta\tau_{\mathbf{n}\kappa\alpha}\mathrm{e}^{+\mathrm{i}\mathbf{q}\cdot\mathbf{R_n}} \\
D_{\kappa\alpha,\,\kappa'\alpha'}(\mathbf{R_n}-\mathbf{R_{n'}}) &\equiv \frac{1}{N_p}\sum_{\mathbf{q}\in\mathrm{Q}} D_{\kappa\alpha,\,\kappa'\alpha'}(\mathbf{q})\mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot(\mathbf{R_n}-\mathbf{R_{n'}})} & \iff \quad & D_{\kappa\alpha,\,\kappa'\alpha'}(\mathbf{q}) \equiv \sum_{\mathbf{R_n}\in\mathrm{L_f}} D_{\kappa\alpha,\,\kappa'\alpha'}(\mathbf{R_n})\mathrm{e}^{+\mathrm{i}\mathbf{q}\cdot\mathbf{R_n}}
\end{aligned} \tag{5.25}$$

Remark 5.16.1 One might wonder how it comes that one uses in two subsequent lines two different prefactor-conventions for the definition of the Fourier-transform (of a quantity). It is this choice in particular that makes the resulting final expression for the Hamiltonian look „particularly simple". This charm has prevailed and these two prefactor-conventions are therefore in all the books and more importantly in all the codes for first principles and ab initio calculations and one has to get used to them.

Remark 5.16.2 Notice that since $\Delta\tau_{\mathbf{n}\kappa\alpha} = \Delta\tau^{\dagger}_{\mathbf{n}\kappa\alpha}$ (as an operator) it follows that

$$\Delta\tau^{\dagger}_{\mathbf{q}\kappa\alpha} = \frac{1}{\sqrt{N_p}}\sum_{\mathbf{R_n}\in\mathrm{L_f}} \Delta\tau^{\dagger}_{\mathbf{n}\kappa\alpha}\mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot\mathbf{R_n}} = \frac{1}{\sqrt{N_p}}\sum_{\mathbf{R_n}\in\mathrm{L_f}} \Delta\tau_{\mathbf{n}\kappa\alpha}\mathrm{e}^{+\mathrm{i}(-\mathbf{q})\cdot\mathbf{R_n}} = \Delta\tau_{-\mathbf{q}\kappa\alpha} \tag{5.26}$$

(as an operator again) and one obtains the further auxiliary-identity

$$\Delta\tau_{\mathbf{n}\kappa\alpha} = \Delta\tau^{\dagger}_{\mathbf{n}\kappa\alpha} = \frac{1}{\sqrt{N_p}}\sum_{\mathbf{q}\in\mathrm{Q}} \Delta\tau^{\dagger}_{\mathbf{q}\kappa\alpha}\mathrm{e}^{+\mathrm{i}\mathbf{q}\cdot\mathbf{R_n}} = \frac{1}{\sqrt{N_p}}\sum_{\mathbf{q}\in\mathrm{Q}} \Delta\tau_{-\mathbf{q}\kappa\alpha}\mathrm{e}^{+\mathrm{i}\mathbf{q}\cdot\mathbf{R_n}} \tag{5.27}$$

which will turn out to be good to use in the following.

Lemma 5.17 The derivatives change accordingly again:

$$\frac{\partial}{\partial \Delta\tau_{\mathbf{n}\kappa\alpha}} = \frac{1}{\sqrt{N_p}} \sum_{\mathbf{q}\in \mathrm{Q}} \mathrm{e}^{+\mathrm{i}\mathbf{q}\mathbf{R_n}} \frac{\partial}{\partial \Delta\tau_{\mathbf{q}\kappa\alpha}} \quad \Longleftrightarrow \quad \frac{\partial}{\partial \Delta\tau_{\mathbf{q}\kappa\alpha}} = \frac{1}{\sqrt{N_p}} \sum_{\mathbf{R_n}\in \mathrm{L_f}} \mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R_n}} \frac{\partial}{\partial \Delta\tau_{\mathbf{n}\kappa\alpha}} \tag{5.28}$$

and this transformation preserves the CCR, i.e. one has

$$\begin{aligned} \left[\Delta\tau_{\mathbf{q}\kappa\alpha}, \frac{\partial}{\partial \Delta\tau_{\mathbf{q}'\kappa'\alpha'}}\right] &= \frac{1}{N_p} \sum_{\mathbf{R_n},\mathbf{R_{n'}}\in \mathrm{L_f}} \mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R_n}} \mathrm{e}^{+\mathrm{i}\mathbf{q}'\mathbf{R_{n'}}} \left[\Delta\tau_{\mathbf{n}\kappa\alpha}, \frac{\partial}{\partial \Delta\tau_{\mathbf{n}'\kappa'\alpha'}}\right] \\ &= \frac{\delta_{\kappa,\kappa'}\delta_{\alpha,\alpha'}}{N_p} \sum_{\mathbf{R_n}\in \mathrm{L_f}} \mathrm{e}^{\mathrm{i}\mathrm{R}_\mathbf{n}\cdot(\mathbf{q}-\mathbf{q}')} \\ &= \delta_{\mathbf{q},\mathbf{q}'}\delta_{\kappa,\kappa'}\delta_{\alpha,\alpha'} \end{aligned} \tag{5.29}$$

Proof 5.17 Follows immediately from the linearity of the commutator, and the chain-rule for derivatives in Lemma 5.1.

Corollary 5.18 It follows for the Hamiltonian

$$T_{\text{ions}} + V_{\text{eff}}(\tau) \approx \sum_{\substack{\mathbf{q}\in \mathrm{Q} \\ \kappa\in\{1,\ldots,r\} \\ \alpha\in\{1,\ldots,d_\mathbf{x}\}}} -\frac{\hbar^2}{2M_\kappa} \frac{\partial}{\partial \Delta\tau_{-\mathbf{q}\kappa\alpha}} \frac{\partial}{\partial \Delta\tau_{\mathbf{q}\kappa\alpha}} + \frac{1}{2} \sum_{\substack{\mathbf{q}\in \mathrm{Q} \\ \kappa,\kappa'\in\{1,\ldots,r\} \\ \alpha,\alpha'\in\{1,\ldots,d_\mathbf{x}\}}} \Delta\tau_{-\mathbf{q}\kappa\alpha} \frac{D_{\kappa\alpha,\,\kappa'\alpha'}(\mathbf{q})}{\sqrt{M_\kappa M_{\kappa'}}^{-1}} \Delta\tau_{\mathbf{q}\kappa'\alpha'}.$$

Proof 5.18 See Appendix, p. 59.

Lemma 5.19 For each $\mathbf{q} \in \mathrm{Q}$ the matrix $D(\mathbf{q})$ with matrix entries $D_{\kappa\alpha,\kappa'\alpha'}(\mathbf{q})$ is hermitian.

Proof 5.19 See Appendix, p. 293.

Corollary 5.20 Since $D(\mathbf{q})$ is hermitian, there is a unitary matrix $U(\mathbf{q})$ consisting of orthonormal eigenvectors (as columns) that diagonalizes the dynamical matrix

$$U^\dagger(\mathbf{q}) D(\mathbf{q})\, U(\mathbf{q}) = \operatorname{diag}\left(\omega_1^2(\mathbf{q}), \ldots, \omega_{rd_\mathbf{x}}^2(\mathbf{q})\right). \tag{5.30}$$

Proof 5.20 This proof is carried out in any textbook on linear algebra.

Notation 5.21 Recall from any linear algebra course, that the matrix-entries of $U(\mathbf{q})$ are given by the orthonormal eigenvectors of the dynamical matrix $D(\mathbf{q})$. We introduce for each $\mathbf{q} \in \mathrm{Q}$ its eigenvectors as

$$\left\{\mathbf{e}_{:,1}(\mathbf{q}), \mathbf{e}_{:,2}(\mathbf{q}), \ldots, \mathbf{e}_{:,rd_{\mathbf{x}}}(\mathbf{q})\right\}.$$

This gives the equation for their components

$$\sum_{\kappa'\alpha'} D_{\kappa\alpha,\kappa'\alpha'}(\mathbf{q}) e_{\kappa'\alpha',\nu}(\mathbf{q}) = \omega_\nu^2(\mathbf{q}) e_{\kappa\alpha,\nu}(\mathbf{q}) \quad \Longleftrightarrow \quad D(\mathbf{q})\mathbf{e}_{:,\nu}(\mathbf{q}) = \omega_\nu^2(\mathbf{q})\mathbf{e}_{:,\nu}(\mathbf{q}) \tag{5.31}$$

Lemma 5.22 The unitarity of the transformation-matrix $U(\mathbf{q})$ implies the relations

$$\begin{aligned} \sum_{\kappa\alpha} e^*_{\kappa\alpha,\nu}(\mathbf{q}) e_{\kappa\alpha,\nu'}(\mathbf{q}) &= \delta_{\nu\nu'} \\ \sum_{\nu} e^*_{\kappa\alpha,\nu}(\mathbf{q}) e_{\kappa'\alpha',\nu}(\mathbf{q}) &= \delta_{\kappa',\kappa}\delta_{\alpha',\alpha} \end{aligned} \tag{5.32}$$

whereas the first relation merely resembles the orthonormality of the eigenvectors w.r.t. the standard scalar-product on the complex numbers, and the second equation is a version of the completeness-relation for the eigenvectors.

Proof 5.22 See any textbook on linear algebra.

Lemma 5.23 For the eigenvectors and eigenvalues of the dynamical matrix in reciprocal space one finds the relations

$$\omega^2_{-\mathbf{q}\nu} = \omega^2_{\mathbf{q}\nu}, \quad e_{\kappa\alpha,\nu}(-\mathbf{q}) = e^*_{\kappa\alpha,\nu}(\mathbf{q}). \tag{5.33}$$

Proof 5.23 See Appendix, p. 293.

Notation 5.24 For reasons that will become clear in a moment, we introduce coordinates according to

$$\Delta\tau_{\mathbf{q}\kappa\alpha} = \sum_{\nu'} e_{\kappa\alpha,\nu'}(\mathbf{q}) \frac{u_{\mathbf{q}\nu'}}{\sqrt{M_\kappa}}, \quad \Longleftrightarrow \quad u_{\mathbf{q}\nu} = \sum_{\kappa'\alpha'} e^*_{\kappa'\alpha',\nu}(\mathbf{q}) \sqrt{M_{\kappa'}} \Delta\tau_{\mathbf{q}\kappa'\alpha'} \tag{5.34}$$

Lemma 5.25 We will prove ...

- that the inverse-relation in Equation (5.34) is correct,
- the auxiliary-identity

$$\Delta\tau^\dagger_{\mathbf{q}\kappa\alpha} = \sum_{\nu'} e^*_{\kappa\alpha,\nu'}(\mathbf{q}) \frac{u^\dagger_{\mathbf{q}\nu'}}{\sqrt{M_\kappa}} = \Delta\tau_{-\mathbf{q}\kappa\alpha}, \tag{5.35}$$

- the derivatives change according to

$$\frac{\partial}{\partial \Delta\tau_{\mathbf{q}\kappa\alpha}} = \sum_{\nu'} e^*_{\kappa\alpha,\nu'}(\mathbf{q})\sqrt{M_\kappa}\frac{\partial}{\partial u_{\mathbf{q}\nu'}} \quad\Longleftrightarrow\quad \frac{\partial}{\partial u_{\mathbf{q}\nu}} = \sum_{\kappa'\alpha'} e_{\kappa'\alpha'\nu}(\mathbf{q})\frac{1}{\sqrt{M_{\kappa'}}}\frac{\partial}{\partial \Delta\tau_{\mathbf{q}\kappa'\alpha'}}. \tag{5.36}$$

- and this transformation preserves the CCR, i.e. one has

$$[u_{\mathbf{q}\nu}, \frac{\partial}{\partial u_{\mathbf{q}'\nu'}}] = -\delta_{\nu,\nu'}\delta_{\mathbf{q},\mathbf{q}'} \tag{5.37}$$

Proof 5.25 See Appendix, p. 294.

Lemma 5.26 Summarizing all the coordinate-changes, we obtain in total

$$\tau_{\mathbf{n}\kappa\alpha} = R_{\mathbf{n}\kappa\alpha} + \frac{1}{\sqrt{N_p}}\sum_{\substack{\mathbf{q}\in\mathrm{Q}\\ \nu'\in\{1,\dots,rd_{\mathbf{x}}\}}} e_{\kappa\alpha,\nu'}(\mathbf{q})\frac{\mathrm{e}^{\mathrm{i}\mathbf{q}\mathbf{R}_\mathbf{n}}}{\sqrt{M_\kappa}}u_{\mathbf{q}\nu'} \quad\Longleftrightarrow\quad u_{\mathbf{q}\nu} = \frac{1}{\sqrt{N_p}}\sum_{\substack{\mathbf{R}_\mathbf{n}\in\mathrm{L_f}\\ \kappa'\alpha'}} e^*_{\kappa'\alpha',\nu}(\mathbf{q})\sqrt{M_{\kappa'}}\mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R}_\mathbf{n}}\left(\tau_{\mathbf{n}\kappa'\alpha'} - R_{\mathbf{n}\kappa'\alpha'}\right) \tag{5.38}$$

which implies the equations

$$\frac{1}{\partial \tau_{\mathbf{n}\kappa\alpha}} = \frac{1}{\sqrt{N_p}}\sum_{\substack{\mathbf{q}\in\mathrm{Q}\\ \nu'\in\{1,\dots,rd_{\mathbf{x}}\}}} e^*_{\kappa'\alpha',\nu'}(\mathbf{q})\sqrt{M_\kappa}\mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R}_\mathbf{n}}\frac{\partial}{\partial u_{\mathbf{q}\nu'}} \quad\Longleftrightarrow\quad \frac{\partial}{\partial u_{\mathbf{q}\nu}} = \frac{1}{\sqrt{N_p}}\sum_{\substack{\mathbf{R}_\mathbf{n}\in\mathrm{L_f}\\ \kappa'\alpha'}} e_{\kappa'\alpha',\nu}(\mathbf{q})\frac{\mathrm{e}^{+\mathrm{i}\mathbf{q}\mathbf{R}_\mathbf{n}}}{\sqrt{M_{\kappa'}}}\frac{\partial}{\partial \tau_{\mathbf{n}\kappa'\alpha'}}$$

Proof 5.26 See Appendix, p. 295.

Corollary 5.27 It follows

$$T_{\mathrm{ions}} + V_{\mathrm{eff}}(\tau) \approx \sum_{\substack{\mathbf{q}\in\mathrm{Q}\\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} -\frac{\hbar^2}{2}\frac{\partial}{\partial u_{-\mathbf{q}\nu}}\frac{\partial}{\partial u_{\mathbf{q}\nu}} + \frac{\omega_\nu^2}{2}(\mathbf{q})\, u_{-\mathbf{q}\nu}u_{\mathbf{q}\nu}.$$

Proof 5.27 See Appendix, p. 297.

Definition 5.28 We introduce the ladder-operators

$$a_{\mathbf{q}\nu} = \sqrt{\frac{\omega_\nu(\mathbf{q})}{2\hbar}}\left(u_{\mathbf{q}\nu} + \frac{\hbar}{\omega_\nu(\mathbf{q})}\frac{\partial}{\partial u_{-\mathbf{q}\nu}}\right), \qquad a^\dagger_{\mathbf{q}\nu} = \sqrt{\frac{\omega_\nu(\mathbf{q})}{2\hbar}}\left(u_{-\mathbf{q}\nu} - \frac{\hbar}{\omega_\nu(\mathbf{q})}\frac{\partial}{\partial u_{+\mathbf{q}\nu}}\right) \tag{5.39}$$

Remark 5.28.1 Although this is simply a neat rewriting of operators, one oftentimes calls this step *canonical quantization* of position operators or *second quantization*.

Lemma 5.29 The ladder-operators fulfill the CCR

$$[a_{\mathbf{q}\nu},\, a^{\dagger}_{\mathbf{q}'\nu'}] = \delta_{\nu,\nu'}\delta_{\mathbf{q},\mathbf{q}'}, \qquad [a_{\mathbf{q}\nu},\, a_{\mathbf{q}'\nu'}] = [a^{\dagger}_{\mathbf{q}\nu},\, a^{\dagger}_{\mathbf{q}'\nu'}] = 0 \tag{5.40}$$

and one finds the inversion formulae

$$u_{\mathbf{q}\nu} = \sqrt{\frac{\hbar}{2\omega_\nu(\mathbf{q})}}\left(a_{\mathbf{q}\nu} + a^{\dagger}_{-\mathbf{q}\nu}\right), \qquad \frac{\partial}{\partial u_{\mathbf{q}\nu}} = \sqrt{\frac{\omega_\nu(\mathbf{q})}{2\hbar}}\left(a_{-\mathbf{q}\nu} - a^{\dagger}_{\mathbf{q}\nu}\right) \tag{5.41}$$

Proof 5.29 See Appendix, p. 299.

Corollary 5.30 Inserting this in the Hamiltonian above yields the final expression

$$H = \sum_{\substack{\mathbf{q}\in\mathrm{Q} \\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}}} \hbar\omega_\nu(\mathbf{q})\left(a^{\dagger}_{\mathbf{q}\nu}a_{\mathbf{q}\nu} + \frac{1}{2}\right). \tag{5.42}$$

Proof 5.30 See Appendix, p. 300.

Remark 5.30.1 Notice that we used $\omega_\nu(-\mathbf{q}) = \omega_\nu(+\mathbf{q})$ within the proof.

Lemma 5.31 Notice the important formula that is an integral component on any derivation of electron-phonon-couplings:

$$\Delta\tau_{\mathbf{n}\kappa\alpha} = \frac{1}{\sqrt{N_p}} \sum_{\substack{\mathbf{q}\in\mathrm{Q} \\ \nu'\in\{1,\ldots,rd_{\mathbf{x}}\}}} \sqrt{\frac{\hbar}{2M_\kappa\omega_{\nu'}(\mathbf{q})}}\, e_{\kappa\alpha,\nu'}(\mathbf{q})\, \mathrm{e}^{\mathrm{i}\mathbf{q}\mathbf{R}_\mathbf{n}}\left(a_{\mathbf{q}\nu'} + a^{\dagger}_{-\mathbf{q}\nu'}\right) \tag{5.43}$$

Proof 5.31 See Appendix, p. 301.

5.2. Electron-Phonon-Coupling

Cf. Waste-Folder 6, topic 10: For Felicianos derivation of electron-phonon-coupling.

5.3. Summary

The Main Tricks

By using a Taylor-expansion around the crystal-coordinates one has to simply diagonalize a matrix in order to solve the Schrödinger-equation (within the harmonic approximation). By going to reciprocal space, we have reduced the problem

$$\text{“Diagonalizing a symmetric } rd_{\mathbf{x}}N_p \times rd_{\mathbf{x}}N_p \text{ -matrix”} \tag{5.44}$$

with $N_p \in [10^{20},\ 10^{24}]$ to the *equivalent* problem

$$\text{“Diagonalizing } N_p \text{ hermitian } rd_{\mathbf{x}} \times rd_{\mathbf{x}} \text{ -matrices”} \tag{5.45}$$

with $rd_{\mathbf{x}} \leq 100$ (in general...).

Thus the problem of

"Solving the effective Schrödinger-equation for the wavefunctions of the ions with an effective potential that is the result of the solution to the electronic problem for different ion configurations"

turned out to be equivalent[3] (within the validity of the harmonic-approximation) to the problem

"Solving the Schrödinger-equation of harmonic-oscillators and simultaneously diagonalizing $rd_{\mathbf{x}} \times rd_{\mathbf{x}}$-matrices for different values of $\mathbf{q} \in \mathrm{Q}$.

Eigen-energies and eigenstates of the ionic Schrödinger-equation

Let us now introduce an enumeration of the cross-product

$$\left\{\lambda_1, \lambda_2, \dots, \lambda_{rd_{\mathbf{x}}N_p}\right\} \equiv \{1, 2, \dots, rd_{\mathbf{x}}\} \times \mathrm{Q}$$

such that $\lambda \equiv (j, \mathbf{q})$. Now each harmonic oscillator $H_{\lambda_\bullet}$ appearing in the harmonic-approximation can be diagonalized individually with standard-techniques, such that the total Hamiltonian

$$H \approx H_{\substack{\text{ionic}\\ \text{harmonic}}} = \sum_{\substack{\mathbf{q}\in \mathrm{Q}\\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} \hbar\omega_\nu(\mathbf{q}) \left(a^\dagger_{\mathbf{q}\nu} a_{\mathbf{q}\nu} + \frac{1}{2}\right) \equiv \sum_{\lambda\in\{1,2,\dots,rd_{\mathbf{x}}N_p\}} \hbar\omega_\lambda \left(a^\dagger_\lambda a_\lambda + \frac{1}{2}\right). \tag{5.46}$$

can be diagonalized with a product-ansatz. This yields a basis for the underlying Hilbert-space

$$\bigotimes_{j=1}^{rd_{\mathbf{x}}N_p} L^2(\mathbb{R}) \simeq L^2(\mathbb{R}^{rd_{\mathbf{x}}N_p}) \equiv \mathscr{H}$$

a basis by products of wavefunctions

[3]In the sense: If we have found the solutions, i.e. wavefunctions, to the latter problem, then we can find solutions to the prior problem by simply changing the coordinates and expanding w.r.t. these wavefunctions.

$$\mathscr{B} = \left\{ |\Psi_{n_{\lambda_1}, n_{\lambda_2}, \ldots, n_{\lambda_{rd_{\mathbf{x}} N_p}}}\rangle \equiv |n_{\lambda_1}, n_{\lambda_2}, \ldots, n_{\lambda_{rd_{\mathbf{x}} N_p}}\rangle \mid n_{\lambda_\bullet} \in \{0, 1, 2, \ldots, \infty\} \right\} \tag{5.47}$$

We thus conclude that the total energies that the system can have (corresponding to the individual eigenstates that the system can be in) are

$$E_{n_{\lambda_1}, n_{\lambda_2}, \ldots, n_{\lambda_{rd_{\mathbf{x}} N_p}}} = \sum_{j=1}^{rd_{\mathbf{x}} N_p} \hbar\omega_{\lambda_j} \left(n_{\lambda_j} + \frac{1}{2} \right) \equiv \sum_{\mathbf{q} \in \mathrm{Q}} \sum_{\nu=1}^{rd_{\mathbf{x}}} \hbar\omega_{\nu\mathbf{q}} \left(n_{\nu\mathbf{q}} + \frac{1}{2} \right). \tag{5.48}$$

Remark 5.31.1 Notice that each of these wavefunctions has $rd_{\mathbf{x}} N_p$-positional-arguments! However... Usually one does not work so much about the ionic wavefunctions themselves anymore in solid-state-theory. It will turn out that the energies are quite enough to get a good picture of what is happening inside a solid in a lot of situations.

Definition of Phonons

Definition 5.32 For each $\nu \in \{1, 2, \ldots, rd_{\mathbf{x}}\}$ one categorizes the *phonon-branches* as mappings

$$\mathrm{Q} \ni \mathbf{q} \mapsto \omega_\nu(\mathbf{q}) \in \mathbb{R}_{>0}. \tag{5.49}$$

Remark 5.32.1 A phonon branch ω_ν is called *acoustical* (or *optical*), if

$$\lim_{\mathbf{q} \to 0} \omega_\nu(\mathbf{q}) = 0 \qquad \left(\text{or} \quad \lim_{\mathbf{q} \to 0} \omega_\nu(\mathbf{q}) \neq 0 \right). \tag{5.50}$$

Fact 5.33 For a crystal, there are $d_{\mathbf{x}}$ acoustical phonon-branches and $rd_{\mathbf{x}} - d_{\mathbf{x}}$ optical branches. (Without Proof.)

Notation 5.34 Without loss of generality we enumerate the phonon-branches

$$\{\omega_1, \ldots, \omega_{d_{\mathbf{x}}}, \omega_{d_{\mathbf{x}}} + 1, \ldots, \omega_{rd_{\mathbf{x}}}\} \tag{5.51}$$

in such a way, that

- the first $d_{\mathbf{x}}$ branches are acoustical,
- the remaining $rd_{\mathbf{x}} - d_{\mathbf{x}}$ branches are optical.

Interpretation of Phonons as Quasiparticles

For this we borrow the words from Alexander Altland in his book about condensed matter field theory:

Consider a standard harmonic oscillator (HO) Hamiltonian [...]. The HO has, of course, the status of a single-particle problem. However, the equidistance of its energy levels suggests an alternative interpretation. One can think of a given energy state ϵ_n *as an accumulation of n elementary entities, or quasi-particles, each having energy* ω*. What can be said about the features of these new objects? First, they are structureless, i.e. the only "quantum number" identifying the quasi-particles is their energy* ω *(otherwise n-particle states formed of the quasi-particles would not be equidistant). This implies that the quasi-particles must be bosons. (The same state* ω *can be occupied by more than one particle [...].*

About the rewritten form of the HO-Hamiltonian

$$H_{\mathrm{HO}} = \frac{p^2}{2m} + \frac{m\omega^2}{2}x^2 = \omega\left(a^\dagger a + \frac{1}{2}\right) \tag{5.52}$$

[...] its "real" advantage is that it naturally affords a many-particle interpretation. To this end, let us declare $|0\rangle$ *to represent a "vacuum" state, i.e. a state with zero particles present. Next, imagine that* $a^\dagger|0\rangle$ *is a state with a single featureless particle [...] of energy* ω*. Similarly* $\left(a^\dagger\right)^n|0\rangle$ *is considered as a many-body state with n particles, i.e. within the new picture [...]. [...] While, at first sight, this may look unfamiliar, the new interpretation is internally consistent. Moreover, it achieves what we had asked for above, i.e. it allows an interpretation of the HO states as a superposition of independent structureless entities. Physically, the quasi-particles [...] are identified with the phonon modes of the solid.*

6. The Electronic Problem

To Be Done... Bloch-Theorem and reference to Slater-determinants solving the non-interacting problem.

6.1. The Many-Electron Problem

6.2. The One-Electron Problem (Bloch-Theorem)

Part II.

Green-Function-Techniques

Motivation and Overview

After this part of the notes you will understand ...

- what Hans Bethe meant, when he said[1] that

  > "*[The fact that] Particles can be created and annihilated, is the bread and butter of todays physics and has remained so.*"

  and why this is usually visualized through pictures like the one in Figure 6.1.

- How Green-Functions (in this work) are defined by combining two big pillars of modern physics, namely *Quantum Many Body Physics* and *Equilibrium Statistical Physics*.

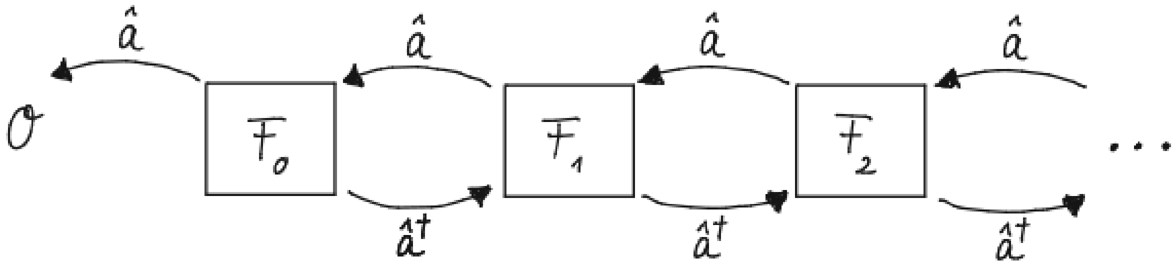
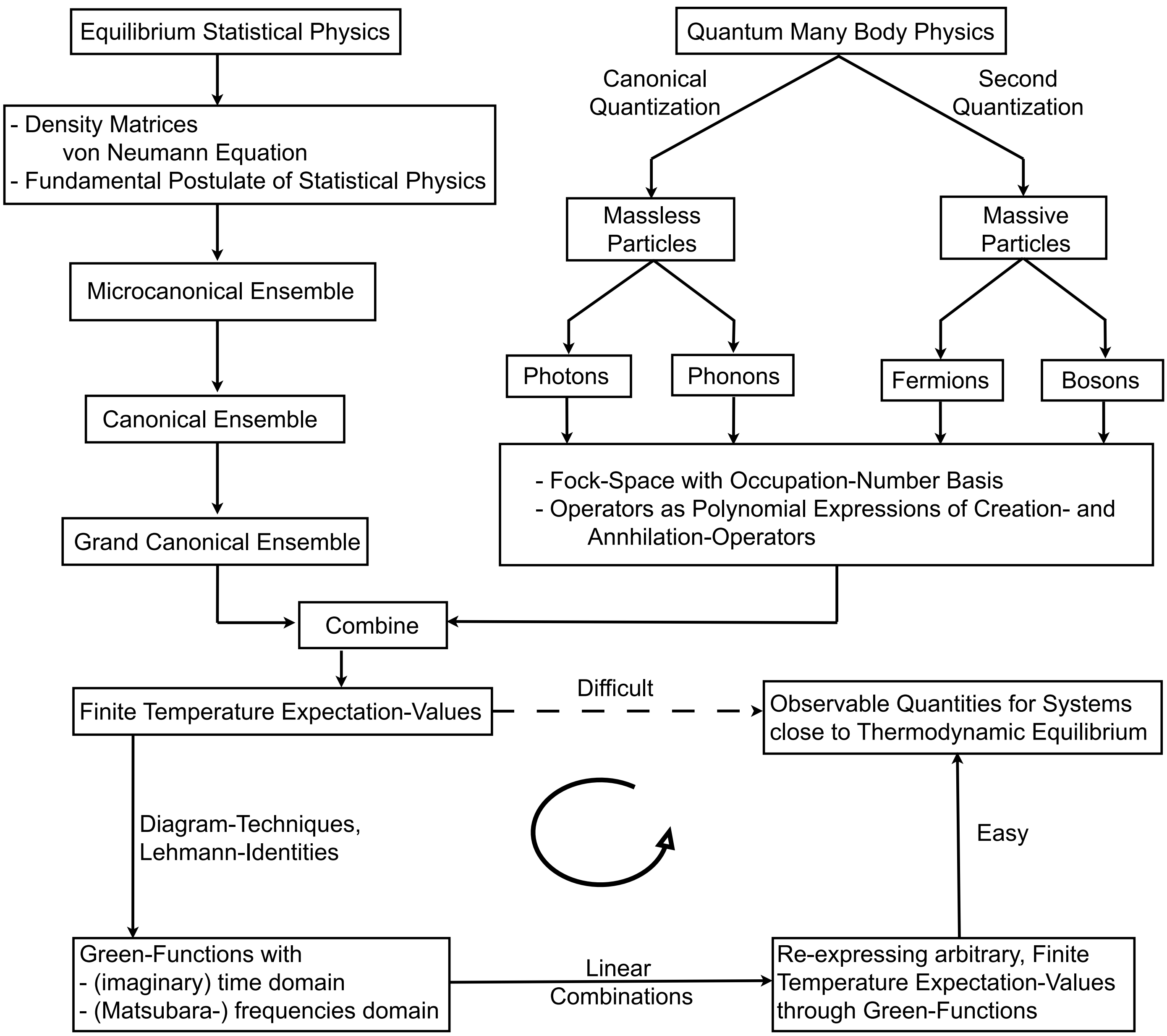


Figure 6.1.: Particles being createaed and annihilated, mapping from one sub-Hilbert-space to another.

[1] Hans Bethe lecture held at MIT Nov. 21 1977 `https://www.youtube.com/watch?v=E61UR4Lbifo`, timestamp: 01h 11min

Outline of the Presentation



7. Fundamentals I - Hilbert-Spaces, States and Operators

7.1. The Many-Body-Formalism for Distinguishable Objects

Fact 7.1 Let $\mathcal{H}_1, \dots, \mathcal{H}_n$ denote the single-particle-Hilbert-spaces with respective scalar-products of n (pairwise) distinguishable *objects*[1]. Then the Hilbert-space $\mathscr{H}$ of the total system is given as tensor product

$$\mathscr{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \cdots \otimes \mathcal{H}_n \equiv \bigotimes_{j=1}^{n} \mathcal{H}_j \tag{7.1}$$

with sesqui-linear scalar-product defined on product-states as

$$\langle \psi_1 \otimes \cdots \otimes \psi_n, \psi_1' \otimes \cdots \otimes \psi_n' \rangle_{\mathscr{H}} := \langle \psi_1, \psi_1' \rangle_{\mathcal{H}_1} \cdots \langle \psi_n, \psi_n' \rangle_{\mathcal{H}_n} \equiv \prod_{j=1}^{n} \langle \psi_j, \psi_j' \rangle_{\mathcal{H}_j}. \tag{7.2}$$

which is extended by linearity on the entire tensor-product.

Remark 7.1.1 Notice also that if $e_1^i, e_2^i, \dots, e_{\mathrm{d}_i}^i$ denote bases of $\mathcal{H}_i$, then the products

$$\left\{ e_{l_1}^1 \otimes \cdots \otimes e_{l_n}^n \quad \middle| \quad \left(l_1, \dots, l_n\right) \in \bigtimes_{j=1}^{n} \mathbb{N}_{\mathrm{d}_j} \right\} \tag{7.3}$$

form a basis of $\mathscr{H}$.

Fact 7.2 Phonons, photons and electrons are pairwise distinguishable from another.

[1] We are using the term "object", in order to represent a "wave" or a "particle" object.

7.2. The Many-Body-Formalism for Indistinguishable Particles (Second Quantization, Algebraic Formalism)

7.2.1. Algebras in Many Particle Physics

Definition 7.3 Let $\mathbb{K}$ be a field and $\mathcal{A}$ be a vector space over $\mathbb{K}$ equipped with an additional binary operation

$$\begin{aligned} \odot : \mathcal{A} \times \mathcal{A} &\to \mathcal{A} \\ (x\,,\, x') &\mapsto x \odot x'. \end{aligned} \tag{7.4}$$

Then $\mathcal{A}$ is an *associative algebra over* $\mathbb{K}$, if for all $x, y, z \in \mathcal{A}$ and $a, b \in \mathbb{K}$ the following identities hold

1. Right distributivity: $(x + y) \odot z = x \odot z + y \odot z$,
2. Left distributivity: $z \odot (x + y) = z \odot x + z \odot y$,
3. Compatibility with scalars: $(ax) \odot (by) = (ab)(x \odot y)$,
4. Associativity[2]: $(x \odot y) \odot z = x \odot (y \odot z)$.

Remark 7.3.1 For $x, y \in \mathcal{A}$ one calls $x \odot y$ the product of x and y and the binary operation is oftentimes referred to as *multiplication*. An associative algebra over a field $\mathbb{K}$ is sometimes called a *associative* $\mathbb{K}$*-algebra*, and $\mathbb{K}$ is called the *base-field*.

Remark 7.3.2 The algebra $\mathcal{A}$ is called *unitary*, if there exists an element $1 \in \mathcal{A}$ such that

$$\forall x \in \mathcal{A} \,:\, 1 \odot x = x \odot 1 = x. \tag{7.5}$$

Remark 7.3.3 When we construct the Fock-space in Definition 7.6, the product will be denoted by the so-called "zeta-product" $\odot \equiv \textcircled{\zeta}$. This will stress the interpretation of objects in the Fock-space as wavefunctions with coordinate-representations. In the constructions of the so-called "ζ-algebra" and the "polynomial algebra" in Definitions 7.15, 16.1 we will use the usual product symbol.

Definition 7.4 Let $\mathcal{A}$ be a vector-space over the field $\mathbb{K}$ with a basis. We define[3] the *associative, unitary algebra generated by* $\mathbb{K} \oplus \mathcal{A}$, as all formal polynomial expressions that one can build with the elements (or, equivalently, the basis elements) of $\mathcal{A}$, given coefficients from the field $\mathbb{K}$ and obeying the relations in Definition 7.3.

The initial situation (which one encounters quite naturally for particle-like objects in physics) is as follows.

[2]This requirement is not unique in the literature. There are non-associative algebras. We will however not be concerned with them.
[3]We don't give this object a symbol, because we will make use of this object three times. Depending on the situation it will be referenced with a different symbol. In order to avoid confusion, we leave out a symbol in this definition.

Notation 7.5 Let $\mathcal{H} = \mathrm{span}_{\mathbb{C}}\left\{|1\rangle, |2\rangle, \dots |\mathrm{d}_{\mathcal{H}}\rangle\right\}$ be a single-particle Hilbert-space with (not neccessarily, but oftentimes orthonormal) basis, corresponding to wavefunctions

$$\langle\, x \,|\, i\,\rangle \equiv \psi_i(x). \tag{7.6}$$

Definition 7.6 We define the *Fock-space* $\mathscr{F}_\zeta \equiv \mathscr{F}_\zeta(\mathcal{H})$ of the complex vector space $\mathcal{H}$ as the associative, unitary algebra generated by $\mathbb{C} \oplus \mathcal{H}$ with binary operation $\textcircled{\zeta}$ which we call ζ*-product* and relations

$$\forall |\psi\rangle, |\psi'\rangle \in \mathcal{H}: \quad |\psi\rangle \textcircled{\zeta} |\psi'\rangle - \zeta |\psi'\rangle \textcircled{\zeta} |\psi\rangle = 0. \tag{7.7}$$

Remark 7.6.1 One distinguishes two cases:

- For $\zeta = -1$, the Fock-space is called the *exterior algebra* and it is commonly denoted with $\wedge(\mathcal{H})$.
- For $\zeta = +1$, the Fock-space is called the *symmetric algebra* and commonly denoted with $\mathrm{S}(\mathcal{H})$.

Remark 7.6.2 We will occasioanlly denote the vectors and the algebra-product in a different way (without the Bra-Ket-Notation, or with adapted Bra-Ket-Notation) as

$$|\psi\rangle \textcircled{\zeta} |\psi'\rangle \equiv |\psi \textcircled{\zeta} \psi'\rangle = \psi \textcircled{\zeta} \psi'. \tag{7.8}$$

We do this out of notational convenience, for example if double-bra-kets appear otherwise, as in Equations (7.15), (7.30). The meaning of the notation will be clear from the respective context.

Theorem 7.7 The Fock space factorizes into subspaces s.t.

$$\mathscr{F}_\zeta(\mathcal{H}) = \mathscr{F}^0_\zeta(\mathcal{H}) \oplus \mathscr{F}^1_\zeta(\mathcal{H}) \oplus \mathscr{F}^2_\zeta(\mathcal{H}) \oplus \dots \equiv \begin{cases} \bigoplus\limits_{n=0}^{\mathrm{d}_{\mathcal{H}}} \wedge^n(\mathcal{H}) & \text{for } \zeta = -1 \\ \bigoplus\limits_{n=0}^{\infty} \mathrm{S}^n(\mathcal{H}) & \text{for } \zeta = +1 \end{cases} \tag{7.9}$$

with convention $\mathscr{F}^0_\zeta(\mathcal{H}) \equiv \mathbb{C}$ and dimensions

$$\dim_{\mathbb{C}} \mathscr{F}^n_\zeta(\mathcal{H}) = \begin{cases} \begin{pmatrix} \mathrm{d}_{\mathcal{H}} \\ n \end{pmatrix} & \text{for } \zeta = -1 \\ \begin{pmatrix} \mathrm{d}_{\mathcal{H}} + n - 1 \\ n \end{pmatrix} & \text{for } \zeta = +1 \end{cases} \tag{7.10}$$

Proof 7.7 See Appendix, Proofs, page 381.

Remark 7.7.1 Note further the identity for fermions

$$\dim_{\mathbb{C}} \mathscr{F}^{n}_{-1}(\mathcal{H}) = \binom{\mathrm{d}_{\mathcal{H}}}{n} = \binom{\mathrm{d}_{\mathcal{H}}}{\mathrm{d}_{\mathcal{H}} - n} = \dim_{\mathbb{C}} \mathscr{F}^{\mathrm{d}_{\mathcal{H}} - n}_{-1}(\mathcal{H}) \tag{7.11}$$

Corollary 7.8 One finds the asymptotic behaviour

$$\dim_{\mathbb{C}} \mathscr{F}^{n}_{\zeta}(\mathcal{H}) \overset{1 \ll \mathrm{d}_{\mathcal{H}}}{\approx} \frac{1}{n!} \left(\dim_{\mathbb{C}} \mathcal{H}\right)^{n}. \tag{7.12}$$

Proof 7.8 Follows by expanding the binomial coefficients, cancelling terms and using the asymptotic behaviour of the factorials.[4]

Lemma 7.9 It follows for the dimension of the Fock-space

$$\dim_{\mathbb{C}} \mathscr{F}_{\zeta}(\mathcal{H}) = \begin{cases} 2^{\mathrm{d}_{\mathcal{H}}} & \text{for } \zeta = -1, \\ \infty & \text{for } \zeta = +1. \end{cases} \tag{7.13}$$

Proof 7.9 The bosonic case is trivial. For the dimension of the fermionic Fock-space one finds

$$\dim_{\mathbb{C}} \Lambda(\mathcal{H}) = \sum_{n=0}^{\mathrm{d}_{\mathcal{H}}} \binom{\mathrm{d}_{\mathcal{H}}}{n} = 2^{\mathrm{d}_{\mathcal{H}}}. \tag{7.14}$$

Fact 7.10 By specifiying the product of the algebra, we can interprete it's elements as many-particle-wavefunctions. As the single-particle-wavefunctions are already defined by assumption, this procedure happens inductively.

Definition 7.11 Consider two elements $\psi \in \mathscr{F}^{p}_{\zeta}(\mathcal{H})$ and $\phi \in \mathscr{F}^{q}_{\zeta}(\mathcal{H})$ of the corresponding degree subspaces in the algebra. We define the *coordinate-representation* of the corresponding wave-function for $p+q$ particles as

$$\begin{aligned} \left\langle x_1, \dots, x_{p+q} \,|\, \psi \,ⓩ\, \phi \right\rangle &\equiv \left(\psi \,ⓩ\, \phi\right)(x_1, \dots, x_{p+q}) \\ &:= \frac{1}{p!q!} \sum_{\pi \in \mathrm{S}_{p+q}} \operatorname{sign}(\pi)^{1-\theta(\zeta)} \psi(x_{\pi(1)}, \dots, x_{\pi(p)}) \phi(x_{\pi(p+1)}, \dots, x_{\pi(p+q)}). \end{aligned} \tag{7.15}$$

Remark 7.11.1 Notice that, depending on the nature of the single-particle wavefunctions (scalar-wavefunctions that solve a Schrödinger-equation or spinor-wavefunctions that solve a Pauli-equation) the concrete meaning of the "coordinates" $x_{\bullet}$ is being specified from the context.

[4]Cf. Zirnbauer, statistical physics, page 20.

Corollary 7.12 For $\psi_i \in \mathscr{F}_\zeta^k(\mathcal{H})$, $\phi \in \mathscr{F}_\zeta^l(\mathcal{H})$, $\chi \in \mathscr{F}_\zeta^m(\mathcal{H})$ one finds for the ζ-product the following rules for the wavefunctions w.r.t. their coordinate representation:

1. $\left(\psi_1 + \psi_2\right) ⓩ \phi = \psi_1 ⓩ \phi + \psi_2 ⓩ \phi$ (linearity),
2. $(\psi ⓩ \phi) ⓩ \chi = \psi ⓩ (\phi ⓩ \chi)$ (associativity),
3. $\psi ⓩ \phi = \zeta^{kl} \phi ⓩ \psi$ (Koszul-sign-rule).

Proof 7.12 See Appendix, Proofs, page 381.

Remark 7.12.1 Notice that the third equation gives the compatibility of the ζ-product with eq. (7.7).

Definition 7.13 On the direct sum $\mathcal{H} \oplus \mathcal{H}^*$ we define the *ζ-bilinearform B_ζ* as mapping

$$\begin{aligned} B_\zeta : \mathcal{H} \oplus \mathcal{H}^* \times \mathcal{H} \oplus \mathcal{H}^* &\to \mathbb{C} \\ (|v\rangle + \langle\phi| \,,\, |v'\rangle + \langle\phi'|) &\mapsto B_\zeta(|v\rangle + \langle\phi| \,,\, |v'\rangle + \langle\phi'|) \equiv \langle\phi|v'\rangle - \zeta\langle\phi'|v\rangle \end{aligned}$$

Lemma 7.14 Notice that this bilinearform is (anti-)symmetric depending on the statistics of the system:

$$\forall u, u' \in \mathcal{H} \oplus \mathcal{H}^* : \quad B_\zeta(u, u') + \zeta B_\zeta(u', u) = 0 \tag{7.16}$$

Proof 7.14 See Appendix, Proofs, page 312.

Definition 7.15 We define the *ζ-algebra $\mathscr{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta)$* as the associative algebra generated by $\mathbb{C} \oplus \mathcal{H} \oplus \mathcal{H}^*$ with relations

$$\forall u, u' \in \mathcal{H} \oplus \mathcal{H}^* : \quad uu' - \zeta u'u = B_\zeta(u, u') \tag{7.17}$$

Remark 7.15.1 Again, one distinguishes two cases:

- For $\zeta = -1$, the underlying algebra is a so-called *Clifford-algebra*.
- For $\zeta = +1$, the underlying algebra is a so-called *Weyl-algebra*.

Definition 7.16 We define the *statistical commutator*, or *ζ-commutator* as

$$\left[A, B\right]_\zeta \equiv AB - \zeta BA. \tag{7.18}$$

Remark 7.16.1 This definition is a little bit tricky since it is a magnet for confusion and errors. We are following the prefactor-convention from [Altland p. 44, Negele, Orland p. 13], in contrast to the prefactor-convention from [Coleman p. 43, Negele, Orland p. 15][5].

Remark 7.16.2 Notice that this definition becomes generalized later on in Definition 8.12.

Lemma 7.17 A useful identity relates the ordinary commutator to statistical commutators

$$[A, BC] = [A, B]_\zeta C + \zeta B[A, C]_\zeta. \tag{7.19}$$

Proof 7.17 See Appendix, Proofs, page 312.

7.2.2. Representation and Scalar-Product on the Fock-Space

In order to define operators on the Fock-space, we will define a representation

$$\begin{aligned} \mathscr{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta) \times \mathscr{F}_\zeta \quad &\longrightarrow \mathscr{F}_\zeta \\ (x\,,\,\xi) \qquad &\longmapsto x \cdot \xi. \end{aligned} \tag{7.20}$$

We will do so inductively.

Fact 7.18 Since $\mathscr{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta)$ is generated by $\mathbb{C} \oplus \mathcal{H} \oplus \mathcal{H}^*$, it suffices to specify the *action / multiplication* in Equation (7.20) for

$$\text{(1) scalars:} \quad x \qquad \equiv k \quad \in \mathbb{C} \;\subset\; \mathscr{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta) \tag{7.21}$$

$$\text{(2) vectors:} \quad x = |\psi\rangle \equiv c^+_{|\psi\rangle} \in \mathcal{H} \;\subset\; \mathscr{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta) \tag{7.22}$$

$$\text{(3) dual-vectors:} \quad x = \langle\psi| \equiv c_{\langle\psi|} \in \mathcal{H}^* \subset\; \mathscr{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta) \tag{7.23}$$

whereas *multiplication by scalars* is given by the structure of a complex vector space.

Definition 7.19 *Multiplication by vectors* is defined by the statistical product computed inside the algebra

$$c^+_{|\psi\rangle} \cdot |\xi\rangle \equiv c^+_{|\psi\rangle}\,|\xi\rangle := |\,\psi\,\rangle \,ⓩ\, |\,\xi\,\rangle \equiv |\,\psi \,ⓩ\, \xi\,\rangle. \tag{7.24}$$

Remark 7.19.1 The notation $c^+_{|\psi\rangle}$ is used in order to dinstinguish the algebra element $|\psi\rangle \equiv |\psi\rangle + 0 \in \mathscr{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta)$ from its concrete realization as a linear operator $c^+_{|\psi\rangle}$ on $\mathscr{F}_\zeta$.

Remark 7.19.2 Note also that this multiplication increases the degree of a state in a statistical sense (in physics language: the number of particles):

[5] Notice that there is an inconsistency within the book from Negele and Orland!

- For $\zeta = +1$ we obtain a mapping $c^+_{|\psi\rangle} :\ \mathrm{S}^l(\mathcal{H}) \to \mathrm{S}^{l+1}(\mathcal{H})$.
- For $\zeta = -1$ we obtain a mapping $c^+_{|\psi\rangle} :\ \Lambda^l(\mathcal{H}) \to \Lambda^{l+1}(\mathcal{H}) \cup \{0\}$, that either increases the degree of a state, or maps it to zero.

Remark 7.19.3 Notice further, that one immediately finds the linearity

$$c^+_{\lambda|\psi\rangle+\lambda'|\psi'\rangle} = \lambda c^+_{|\psi\rangle} + \lambda' c^+_{|\psi'\rangle}. \tag{7.25}$$

for $\lambda, \lambda' \in \mathbb{C}$.

Definition 7.20 *Multiplication by co-vectors* is defined recursively and denoted as

$$\begin{aligned} c_{\langle\phi|} \cdot 1 \quad &:= \ 0 \\ c_{\langle\phi|} \cdot |\psi\rangle \quad &:= \ B_\zeta(\langle\phi|, |\psi\rangle) = \langle\phi|\psi\rangle \qquad\qquad \text{for } |\psi\rangle \in \mathcal{H} \subset \mathscr{F}_\zeta \\ c_{\langle\phi|} \cdot |\xi \textcircled{\zeta} \eta\rangle \quad &:= \ \left(c_{\langle\phi|} \cdot |\xi\rangle\right) \textcircled{\zeta} |\eta\rangle + \zeta^l |\xi\rangle \textcircled{\zeta} \left(c_{\langle\phi|} \cdot |\eta\rangle\right) \qquad \text{for } |\xi\rangle, |\eta\rangle \in \mathscr{F}_\zeta \end{aligned} \tag{7.26}$$

whereas $|\,\xi\,\rangle \in \mathscr{F}^l_\zeta(\mathcal{H})$ in the last equation.

Remark 7.20.1 Here, again, the notation $c_{\langle\phi|}$ is used in order to dinstinguish the algebra element $\langle\phi| \equiv 0 + \langle\phi| \in \mathscr{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta)$ from its concrete realization as a linear operator $c_{\langle\phi|}$ on $\mathscr{F}_\zeta$.

Remark 7.20.2 Notice that due to linearity, we obtain the action

$$\left(k + c^+_{|\psi\rangle} + c_{\langle\phi|}\right) \cdot |\xi\rangle \equiv k|\xi\rangle + c^+_{|\psi\rangle}\,|\xi\rangle + c_{\langle\phi|}|\xi\rangle \tag{7.27}$$

Remark 7.20.3 Notice that the commutation relations in Equation (7.17) imply the relations

$$\begin{aligned} 1_{\mathscr{F}_\zeta}\langle\phi|\psi\rangle &= c_{\langle\phi|}c^+_{|\psi\rangle} - \zeta c^+_{|\psi\rangle}c_{\langle\phi|} \\ 0 &= c_{\langle\phi|}c_{\langle\phi'|} - \zeta c_{\langle\phi'|}c_{\langle\phi|} = c^+_{|\upsilon\rangle}c^+_{|\upsilon'\rangle} - \zeta c^+_{|\psi'\rangle}c^+_{|\psi\rangle} \end{aligned}$$

Notation 7.21 Let us fix an orthonormal basis $\mathcal{H} = \mathrm{span}_\mathbb{C}\left\{|1\rangle, |2\rangle, \dots |\mathrm{d}_\mathcal{H}\rangle\right\}$ of the single-particle Hilbert-space with dual basis $\mathcal{H}^* = \mathrm{span}_\mathbb{C}\left\{\langle 1|, \langle 2|, \dots \langle \mathrm{d}_\mathcal{H}|\right\}$, and let us write $\langle i| \equiv |c_{\langle i|} \equiv c_i$ and $|i\rangle \equiv c^+_{|i\rangle} \equiv c^+_i$. Using the duality-relation $\langle i|j\rangle = \delta_{ij}$ we obtain the relations (7.17) in the form

$$[c_i, c^+_j]_\zeta = \delta_{ij}, \quad [c_i, c_j]_\zeta = [c^+_i, c^+_j]_\zeta = 0 \tag{7.28}$$

Proof 7.21 See Appendix, Proofs, page 312.

Remark 7.21.1 In the next subsection, we will elucidate how c_i and c_i^+ will be acting as linear operators on the Fock-space with its canonical Hermitian scalar-product. In that setting, the operator $c_i^+ \equiv c_i^\dagger$ will turn out to be the „Hermitian adjoint" of c_i.

Theorem 7.22 Equations (7.26), (7.24) define a representation of the algebra $\mathscr{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta)$ on the complex vector-space $\mathscr{F}_\zeta$.

Proof 7.22 See Appendix, Proofs, page 383.

Theorem 7.23 One obtains the identities

$$
\begin{aligned}
c_j^\dagger \cdot \left(c_{i_1}^\dagger c_{i_2}^\dagger \cdots c_{i_l}^\dagger | 0 \rangle \right) &= c_j^\dagger c_{i_1}^\dagger c_{i_2}^\dagger \cdots c_{i_l}^\dagger | 0 \rangle \\
c_j \cdot \left(c_{i_1}^\dagger c_{i_2}^\dagger \cdots c_{i_l}^\dagger | 0 \rangle \right) &= \delta_{ji_1} c_{i_2}^\dagger c_{i_3}^\dagger \cdots c_{i_l}^\dagger | 0 \rangle \quad + \quad \zeta \delta_{ji_2} c_{i_1}^\dagger c_{i_3}^\dagger \cdots c_{i_l}^\dagger | 0 \rangle \quad + \quad \ldots \quad + \quad \zeta^{l-1} \delta_{ji_l} c_{i_1}^\dagger c_{i_2}^\dagger \cdots c_{i_{l-1}}^\dagger | 0 \rangle \\
&= \sum_{k=1}^{l} \zeta^{k+1} \delta_{j,i_k} c_{i_1}^\dagger \cdots c_{i_{k-1}}^\dagger c_{i_{k+1}}^\dagger \cdots c_{i_l}^\dagger | 0 \rangle \\
&= \sum_{k=1}^{l} \zeta^{k+1} \delta_{j,i_k} c_{i_1}^\dagger \cdots \widehat{c_{i_k}^\dagger} \cdots c_{i_l}^\dagger | 0 \rangle .
\end{aligned}
$$

Proof 7.23 See Appendix, Proofs, page 386.

Definition 7.24 We introduce a mapping on the subspace

$$
\langle \cdot, \cdot \rangle_{\mathscr{F}_\zeta^n} : \mathscr{F}_\zeta^n \times \mathscr{F}_\zeta^n \longrightarrow \mathbb{C} \tag{7.29}
$$

by defining it through permanents ($\zeta = +1$) and determinants ($\zeta = -1$) as

$$
\begin{aligned}
\left\langle \psi_1 ⓩ \ldots ⓩ \psi_n, \psi_1' ⓩ \ldots ⓩ \psi_n' \right\rangle_{\mathscr{F}_\zeta(\mathcal{H})} &:= \sum_{\pi \in \mathrm{S}_n} \mathrm{sign}(\pi)^{1-\theta(\zeta)} \prod_{j=1}^{n} \left\langle \psi_n, \psi_{\pi(n)}' \right\rangle_{\mathcal{H}} \\
&= \left| \begin{matrix} \langle \psi_1, \psi_1' \rangle_{\mathcal{H}} & \cdots & \langle \psi_1, \psi_n' \rangle_{\mathcal{H}} \\ \vdots & \ddots & \vdots \\ \langle \psi_n, \psi_1' \rangle_{\mathcal{H}} & \cdots & \langle \psi_n, \psi_n' \rangle_{\mathcal{H}} \end{matrix} \right|_\zeta
\end{aligned} \tag{7.30}
$$

and sesqui-linear continuation.

Corollary 7.25 This mapping is a well-defined hermitian scalar-product on $\mathscr{F}_\zeta^n$, meaning that it is sesqui-linear, positive definite.

Proof 7.25 See Appendix, Proofs, page 422.

Remark 7.25.1 We continue this hermitian scalar-product on $\mathscr{F}_\zeta^n$ in the canonical way on the entire Fock-space $\mathscr{F}_\zeta$, by defining the scalar-product of different particle-number subspaces to vanish. Equipped with a scalar-product, the Fock-space is a Hilbert-space.

7.2.3. Occupation-Number States and Bra-Ket-Isomorphism

Definition 7.26 We introduce the *statistical occupation-numbers* as set

$$\mathbb{N}_\zeta = \begin{cases} \{0,1\} & \text{for } \zeta = -1, \\ \{0,1,2,\ \dots, \infty\} & \text{for } \zeta = +1. \end{cases} \tag{7.31}$$

Definition 7.27 We define the *occupation-number-states* as elements of the Fock-space

$$\mathscr{B}_\zeta = \left\{ \frac{\left(c_1^\dagger\right)^{n_1}}{\sqrt{n_1!}} \cdots \frac{\left(c_{\mathrm{d}_\mathcal{H}}^\dagger\right)^{n_{\mathrm{d}_\mathcal{H}}}}{\sqrt{n_{\mathrm{d}_\mathcal{H}}!}} |\, 0\, \rangle =: |n_1, \dots, n_{\mathrm{d}_\mathcal{H}}\rangle \;\middle|\; n_1, \dots, n_{\mathrm{d}_\mathcal{H}} \in \mathbb{N}_\zeta \right\} \quad \subset \quad \mathscr{F}_\zeta. \tag{7.32}$$

with the so-called *vacuum-state* a.k.a *the empty state*

$$|\, 0\, \rangle \equiv |\mathrm{vac.}\rangle \equiv 1 \in \mathbb{C}. \tag{7.33}$$

Remark 7.27.1 In the *physics-jargon*, one usually says that such a state possesses n_1 particles in the first state, n_2 particles in the second state, etc.

Remark 7.27.2 Notice that this definition comes with different prefactor-conventions in the literature. We are following the convention of [Zirnbauer], in contrast to, for example, the convention of [Stefanucci, Leuween].

Recall now that in a complex vector space with sesqui-linear scalar-product there is a canonically defined operation of taking the hermitian adjoint[6] of an operator, denoted by the symbol †.

Corollary 7.28 One has

$$\left(c_i^+\right)^\dagger = c_i \Longleftrightarrow c_i^+ = c_i^\dagger. \tag{7.35}$$

[6]The hermitian adjoint of an operator A w.r.t. a scalar-product is defined such that the following equation

$$TBD \tag{7.34}$$

is fulfilled.

Proof 7.28 See Appendix, Proofs, page 426.

Fact 7.29 The Hermitian scalar-product on the single-particle-Hilbert-space $\langle\cdot,\cdot\rangle : \mathcal{H}\times\mathcal{H}\longrightarrow\mathbb{C}$ determines a one-to-one-correspondence between the Hilbert-space and the dual-space known as *ket-bra-bijection* a.k.a. *Riesz-isomorphism*

$$\begin{aligned} \mathrm{J}_{\mathcal{H}} :& \mathcal{H}\longrightarrow\mathcal{H}^* \\ & \lambda|\,\psi\,\rangle \longmapsto \mathrm{J}_{\mathcal{H}}(\lambda\psi) \equiv \lambda^*\langle\,\psi\,| \equiv \lambda^*\langle\,\psi\,|\bullet\rangle \end{aligned}$$

for $\lambda\in\mathbb{C}$.

We will introduce this one-to-one-correspondence as well for the Fock-space.

Lemma 7.30 It holds $\mathscr{F}_\zeta(\mathcal{H})^*\simeq\mathscr{F}_\zeta(\mathcal{H}^*)$ as a vector-space.

Proof 7.30 See Appendix, Proofs, page 426.

Definition 7.31 We introduce the *ket-bra-correspondence of the Fock-space* as

$$\begin{aligned} \mathrm{J}_{\mathscr{F}} :& \mathscr{F}_\zeta(\mathcal{H})\longrightarrow\mathscr{F}_\zeta(\mathcal{H}^*) \\ & \lambda|\,\psi_1\,\rangle \circledS \dots \circledS |\,\psi_n\,\rangle \longmapsto \lambda^*\,\mathrm{J}_{\mathcal{H}}(\psi_1)\circledS \dots \circledS \mathrm{J}_{\mathcal{H}}(\psi_n) = \lambda^*\langle\,\psi_1\,|\circledS \dots \circledS\langle\,\psi_n\,| \end{aligned}$$

with $\lambda\in\mathbb{C}$.

Corollary 7.32 In *physics-notation*, this implies the ket-bra-correspondence

$$\mathscr{F}(\mathcal{H})\ni\quad \left(c_1^\dagger\right)^{n_1}\dots\left(c_{\mathrm{d}_\mathcal{H}}^\dagger\right)^{n_{\mathrm{d}_\mathcal{H}}}|0\rangle \quad\longmapsto\quad \mathrm{J}_{\mathscr{F}}\left(\left(c_1^\dagger\right)^{n_1}\dots\left(c_{\mathrm{d}_\mathcal{H}}^\dagger\right)^{n_{\mathrm{d}_\mathcal{H}}}|0\rangle\right) = \langle 0|c_{\mathrm{d}_\mathcal{H}}^{n_{\mathrm{d}_\mathcal{H}}}\cdots c_1^{n_1} \quad\in\mathscr{F}(\mathcal{H}^*) \tag{7.36}$$

with $\lambda\in\mathbb{C}$.

Proof 7.32 TBD.

Theorem 7.33 The states defined in Equation (7.32) are an orthonormal basis $\mathscr{F}_\zeta(\mathcal{H})$

Proof 7.33 See Appendix, Proofs, page 425.

7.2.4. Operators on the Fock-Space

One-Body-Operators

Notation 7.34 Let $L \in \mathrm{End}(\mathcal{H})$ be a linear operator on the single-particle-Hilbert-space $\mathcal{H}$.

Definition 7.35 *The second quantized version of* L is given as element of the ζ-algebra (which we denote by the same symbol for notational simplicity):

$$\mathrm{End}(\mathcal{H}) \ni \quad L \quad \mapsto \quad L^{2\mathrm{Q}} \equiv \sum_{j=1}^{\mathrm{d}_{\mathcal{H}}} c^{+}_{L|j\rangle}\, c_j \quad \in \mathscr{A}_{\zeta}(\mathcal{H} \oplus \mathcal{H}^{*}, B_{\zeta}) \tag{7.37}$$

Remark 7.35.1 For notational simplicity, one oftentimes simply drops the upper index and writes $L^{2\mathrm{Q}} \equiv L$ as it is clear from the context what is meant. The same hodls true for other operators in second quantization.

Corollary 7.36 On the one-particle-subspace of the Fock-space, this mapping acts precisely as its first quantized version, i.e. TBMathematised.

Proof 7.36 See Appendix, Proofs, page 427.

Remark 7.36.1 This implies, that the second-quantized-version of L is mathematically considered to be a so-called *derivation*[7].

Lemma 7.37 This is an hermitian operator on the Fock-space.

Proof 7.37 See Appendix, Proofs, page 427.

Lemma 7.38 Switching to standard-physics-notation, we obtain the second-quantized version of the one-body-operator L as

$$\mathrm{End}(\mathcal{H}) \ni \quad L \quad \mapsto \quad L^{2\mathrm{Q}} \equiv \sum_{i,j=1}^{\mathrm{d}_{\mathcal{H}}} \langle i|L|j\rangle\, c_i^{\dagger} c_j \quad \in \mathscr{A}_{\zeta}(\mathcal{H} \oplus \mathcal{H}^{*}, B_{\zeta}). \tag{7.38}$$

Proof 7.38 See Appendix, Proofs, page 428.

[7] TBElaborated?

Definition 7.39 We define the *number-operator* as the second-quantized version of the identity-operator on the single-particle-Hilbert-space, i.e.

$$\mathrm{End}(\mathcal{H}) \ni \quad \mathbb{1} \quad \mapsto \quad \mathbb{1}^{2\mathrm{Q}} \equiv \sum_{j=1}^{\mathrm{d}_{\mathcal{H}}} c_j^\dagger c_j \equiv N \quad \in \mathscr{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta). \tag{7.39}$$

Corollary 7.40 For basis-states as in equation (7.32) we have

$$\hat{N}|n_1, \dots, n_{\mathrm{d}_{\mathcal{H}}}\rangle = \sum_{j=1}^{\mathrm{d}_{\mathcal{H}}} n_j |n_1, \dots, n_{\mathrm{d}_{\mathcal{H}}}\rangle. \tag{7.40}$$

Proof 7.40 TBD

Lemma 7.41 For fermionic or bosonic operators, one finds in both cases the identity

$$[a_j^\dagger a_k, a_l^\dagger a_l] = a_j^\dagger a_l \delta_{k,l} - a_l^\dagger a_k \delta_{l,j} \tag{7.41}$$

Proof 7.41 Cf. Gross, AQM2013, Sheet 5, Exercise 1 a), c). (TBDigitalized)

Two-Body-Operators

Notation 7.42 Let $V \in \mathrm{End}(\mathcal{H} \otimes \mathcal{H})$ be a linear operator on the tensor product of two single-particle Hilbert-spaces.

Definition 7.43 The *second quantized version of* V is given as an element of the ζ-algebra

$$\mathrm{End}(\mathcal{H} \otimes \mathcal{H}) \ni \quad V \quad \mapsto \quad V^{2\mathrm{Q}} \equiv \frac{1}{2} \sum_{ijkl} \langle i, j|L|k, l\rangle \, c_i^\dagger c_j^\dagger c_l c_k \quad \in \mathscr{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta). \tag{7.42}$$

Remark 7.43.1 Notice also the peculiarity that the two latest indices in the expression are contra-symmetrically exchanged in order to be a derivation of its first-quantized version.

Corollary 7.44 On the two-particle-subspace the second-quantized-version of V acts precisely as its first-quantized-version. TBMathematised.

Proof 7.44 See Appendix, Proofs, page 429.

Remark 7.44.1 This again implies, that the second-quantized-version of V is mathematically considered to be a so-called derivation.

Lemma 7.45 This is an hermitian operator on the Fock-space.

Proof 7.45 See Appendix, Proofs, page 314.

n-Body-Operators

Definition 7.46 TBD Extension to higher-order-cases, even if they are not so relevant, to give an interpretation for SC.

Theorem 7.47 An Hermitian n-body-operator on $\bigotimes_{j=1}^{n} \mathcal{H}$ will be also Hermitian in its second quantized form as an operator on $\mathscr{F}_\zeta(\mathcal{H})$.

Proof 7.47 See Appendix, Proofs, page 314.

7.2.5. Useful Formulae

| ============= | Begin: To Be Updated / Corrected for n-body operators | ============= |

The template was the clean derivation of the Landauer-formula; Cf. also Zirnbauer p. 67, 68.

Probably this holds for Fermions *and* Bosons... One step needs to be filled.

Corollary 7.48 Let A and B be linear operators on $\mathcal{H}$. Denote by $[A, B] = AB - BA$ their commutator. Then

$$[A, B]^{2Q} = [A^{2Q}, B^{2Q}] \tag{7.43}$$

$$e^{A^{2Q}} B^{2Q} e^{-A^{2Q}} = \left(e^A B e^{-A}\right)^{2Q}, \tag{7.44}$$

$$\tag{7.45}$$

Proof 7.48 The first identity is easily proven using the anticommutation relations

$$\begin{aligned}
[A^{2Q}, B^{2Q}] &= \sum_{k,j,m,n} \langle e_k, Ae_j\rangle\langle e_m, Be_n\rangle[a_k^\dagger a_j, a_m^\dagger a_n] \\
&= \sum_{k,j,m,n} \langle e_k, Ae_j\rangle\langle e_m, Be_n\rangle \left(a_k^\dagger a_m \delta_{j,n} - a_m^\dagger a_j \delta_{k,n}\right) \\
&= \text{TBD} \\
&= \sum_{k,j} \langle e_k, (AB - BA)e_j\rangle a_k^\dagger a_j = [A, B]^{2Q},
\end{aligned}$$

while the second one is implied by the first equality, the Baker–Campbell–Hausdorff formula

$$e^{A^{2Q}} B^{2Q} e^{-A^{2Q}} = B^{2Q} + [A^{2Q}, B^{2Q}] + \frac{1}{2}[B^{2Q}, [B^{2Q}, A^{2Q}]] + \cdots,$$

and the linearity of the scalar-product.

| ============= | End: To Be Updated / Corrected for n-body operators | ============= |

8. Fundamentals II - Many-Particle Hilbert-Spaces, Time-Dependencies and Examples

8.1. Summary of the Many-Particle Hilbert-Spaces - Pure and Mixed Systems

Notation 8.1 When we speak about "Green-functions" (to be introduced later), we consider one of the following Hilbert-spaces:

$$
\begin{aligned}
&(1) \quad \mathscr{H} \equiv \mathscr{H}_{m=0} = \operatorname{span}_{\mathbb{C}} \left\{ \prod_{j=1}^{\mathrm{d}} \frac{1}{\sqrt{n_j!}} \left(c_j^\dagger \right)^{n_j} |\,\mathrm{vac.}\,\rangle \equiv |n_1, n_2, \dots, n_{\mathrm{d}}\rangle \,\middle|\, n_1, n_2, \dots, n_{\mathrm{d}} \in \mathbb{N}_\zeta \right\} \quad \text{(massless particles)},\\
&(2) \quad \mathscr{H} \equiv \mathscr{H}_{m>0} = \operatorname{span}_{\mathbb{C}} \left\{ \prod_{j=1}^{\mathrm{d}_{\mathcal{H}}} \frac{1}{\sqrt{n_j!}} \left(c_j^\dagger \right)^{n_j} |\,\mathrm{vac.}\,\rangle \equiv |n_1, n_2, \dots, n_{\mathrm{d}_{\mathcal{H}}}\rangle \,\middle|\, n_1, n_2, \dots, n_{\mathrm{d}_{\mathcal{H}}} \in \mathbb{N}_\zeta \right\} \quad \text{(massive particles)},\\
&(3) \quad \mathscr{H} = \mathscr{H}_{m>0} \otimes \mathscr{H}_{m=0} \quad \text{(massive-massless-mixtures)},\\
&(4) \quad \mathscr{H} = \mathscr{H}_{m_1>0} \otimes \mathscr{H}_{m_2=0} \otimes \mathscr{H}_{m_3=0} \quad \text{(massive-massless-massless-mixtures)},\\
&(5) \quad \mathscr{H} = \mathscr{H}_{m_1>0} \otimes \mathscr{H}_{m_2>0} \otimes \mathscr{H}_{m_3=0} \quad \text{(massive-massive-massless-mixtures)},\\
&\quad \vdots
\end{aligned}
$$

Remark 8.1.1 Notice that in every case except the first two cases, every Hilbert-space is composed of tensor-products of the particle-like or wave-like systems. In those cases, a basis of the total Hilbert-space is given by taking tensor-products of the respective occupation-number basis-elements. In turn, the basis of the total Hilbert-space is again given as an occupation-number basis (with a more tedious enumeration).

Remark 8.1.2 In the first two cases (i.e. in the case of exclusively massless or massive particles in the system), we will be speaking of a *pure system*.

Definition 8.2 Depending on the mass of the underlying particles in a system, we introduce a number η called *nature* as

$$
\eta = \begin{cases} 0 & \text{for massless particles} \\ 1 & \text{for massive particles} \end{cases} \tag{8.1}
$$

Particle / System (Example)	**Nature** η	**Statistics** ζ
Electron-Gas	1	-1
Phonon- & Photon-Gas	0	$+1$
Rubidium atoms / Bose-Einstein-Condensate	1	$+1$
XXZ-Model + Jordan-Wigner-Transformation (???)	0	-1

Remark 8.2.1 The table summarizes some examples of pure systems with their associated statistics ζ, nature η.

Definition 8.3 We introduce the so-called *ensemble Hamiltonian* Ξ as

$$\Xi = \begin{cases} H & \text{for } \eta = 0 \text{ (massless particles)}, \\ H - \mu N & \text{for } \eta = 1 \text{ (massive particles)}, \\ H_{m>0} - \mu N_{m>0} + H_{m=0} + V_{\text{int}} & \text{for massive-massless-mixtures}, \\ \vdots \end{cases} \tag{8.2}$$

whereas the role of the so-called "chemical potential" μ will become clear when we introduce the different ensembles that one uses.

Remark 8.3.1 Notice that one can rewrite the equation above in the concise form

$$\Xi = \begin{cases} H_\eta - \eta\mu N & \text{for a pure system with nature } \eta, \\ H_{m>0} - \mu N_{m>0} + H_{m=0} + V_{\text{int}} & \text{for massive-massless-mixtures}. \\ \vdots \end{cases} \tag{8.3}$$

Remark 8.3.2 For other cases, such as massive-massive-mixtures, one has to account for several chemical potentials.

Fact 8.4 The Hamiltonian of the system is usually expressed as a combination of a well-understood reference term, H_0, and a perturbation term, V, whereas the perturbation may represent a superposition of one-body interactions (such as external potentials or impurities in a crystal) and two-body interactions (such as Coulomb interactions)... Hence we concretely want to cover the cases ...

1. systems with massless particles

$$\begin{aligned} H &= \sum_{i,j=1}^{\mathrm{d}} H_{i,j} c_i^\dagger c_j &&+ \sum_{i,j,k=1}^{\mathrm{d}} \left(H_{i,j;k} c_i^\dagger c_j^\dagger c_k + H_{i;j,k} c_i^\dagger c_j c_k \right) + \dots, \\ &= H_0 &&+ V \end{aligned} \tag{8.4}$$

2. systems with massive particles

$$\begin{aligned} H &= \sum_{i,j=1}^{\mathrm{d}} H_{i,j} c_i^\dagger c_j &&+ \frac{1}{2} \sum_{i,j,k,l=1}^{\mathrm{d}} H_{i,j;kl} c_i^\dagger c_j^\dagger c_l c_k + \dots, \\ &= H_0 &&+ V \end{aligned} \tag{8.5}$$

whereas we emphasize the reversed order in the last two indices,

3. and for the interactions of, for example, massive-massless-systems

$$V_{\text{int}} = \sum_{i,j=1}^{d_{m>0}} \sum_{k=1}^{d_{m=0}} \left(V_{\substack{\text{int} \\ i,j;k}} c_i^\dagger c_j c_k^\dagger + \widetilde{V}_{\substack{\text{int} \\ i,j;k}} c_i^\dagger c_j c_k \right) + \dots , \tag{8.6}$$

with

$$H_0 \equiv H_{\substack{0 \\ m=0}} + H_{\substack{0 \\ m>0}} , \qquad V \equiv V_{m=0} + V_{m>0} + V_{\text{int}} \tag{8.7}$$

4. here and for the remaining cases, we consider similar Hamiltonians ...

Remark 8.4.1 Notice that for systems with massive particles the Hamiltonians are always particle-number conserving, as they emerge from a second quantization scheme.

8.2. Time-Dependencies in Many-Body Quantum Mechanics

8.2.1. Real Time

Notation 8.5 Let $\mathscr{H}$ a many-particle Hilbert-space of a quantum system with a (possibly spin-dependent) time-independent ensemble Hamiltonian Ξ (in the Schrödinger-picture) and consider the time-evolution in the Schrödinger- / Pauli-equation

$$\mathrm{i}\hbar\partial_t |\psi(t)\rangle = \Xi |\psi(t)\rangle .$$

Definition 8.6 For arbitrary times $t_\mathrm{i}, t_\mathrm{f} \in \mathbb{R}$ we define the *ensemble-time-evolution operator* $U_\Xi(t_\mathrm{f}, t_\mathrm{i})$ of the system as

$$U_\Xi(t_\mathrm{f}, t_\mathrm{i}) = \exp - \frac{\mathrm{i}}{\hbar} \left(t_\mathrm{f} - t_\mathrm{i} \right) \Xi .$$

Remark 8.6.1 Notice that one has the identity

$$U_\Xi(t_\mathrm{f}, t_\mathrm{i})^\dagger = U_\Xi(t_\mathrm{i}, t_\mathrm{f}). \tag{8.8}$$

Remark 8.6.2 For notational convenience we will oftentimes drop the index in the time-evolution-operator as $U_\Xi \equiv U$.

Theorem 8.7 The time-evolution operator defined above gives a formal solution to a Schrödinger- / Pauli-equation (with ensemble-Hamiltonian) for a given initial-condition.

Proof 8.7 By taking the derivative, one sees that one indeeds obtains a solution for a given initial condition.

Definition 8.8 Let $X \equiv X_\mathrm{S} \equiv X_\mathrm{S}(t)$ be a possibly time-dependent operator in the [S]chrödinger-picture, then one defines the corresponding *ensemble-[H]eisenberg-representation* as

$$X_\mathrm{H}(t) := U(t,0)^\dagger\, X\, U(t,0).$$

Theorem 8.9 One immediately concludes the *ensemble-Heisenberg-"equation of motion"*

$$\frac{\mathrm{d}}{\mathrm{d}t} X_\mathrm{H}(t) = -\frac{\mathrm{i}}{\hbar}[X, \Xi]_\mathrm{H}(t) + \left(\frac{\partial X}{\partial t}\right)_\mathrm{H}(t) \tag{8.9}$$

Proof 8.9 See Appendix, Proofs, page 314.

Remark 8.9.1 Most of the operators we consider are creation- and annihilation-operators c, $c^\dagger$, which are time-independent in the Schrödinger-picture. But one oftentimes simply drops the index, even if they are written in the Heisenberg-picture. This doesn't lead to any confusion since they acquire a time-argument then $c_\mathrm{H}(t) \equiv c(t)$, etc.

Definition 8.10 We consider a monomial in fermionic ($\zeta_\bullet \equiv -1$) and / or bosonic ($\zeta_\bullet \equiv +1$) creation and annihilation-operators

$$X \equiv c^\dagger_{\alpha_1} \cdots c^\dagger_{\alpha_n} c_{\alpha'_1} \cdots c_{\alpha'_m}, \tag{8.10}$$

with statistics $\zeta_{\alpha_1}, \ldots, \zeta_{\alpha_n}, \zeta_{\alpha'_1}, \ldots, \zeta_{\alpha'_m} \in \{-1, +1\}$ then we define the *statistics of the operator X* as a sign

$$\mathrm{S}(X) := \prod_{i=1}^{n} \zeta_{\alpha_i} \cdot \prod_{j=1}^{m} \zeta_{\alpha'_j} \in \{-1, +1\}. \tag{8.11}$$

Remark 8.10.1 Analogous one defines the *statistics of a polynomial expression in creation- and annihilation-operators*, if every monomial has the same statistics. For a polynomial expression in creation- and annihilation-operators with monomials that differ (individually) in their statistics, there is no reasonable definition of the statistics.

Remark 8.10.2 Notice that one finds for creation- and annihilation-operators $\mathrm{S}(c_\bullet) = \mathrm{S}(c^\dagger_\bullet) = \zeta$.

Definition 8.11 For a pair of operators X, X', we define the *pair-statistics* $\mathrm{S} \equiv \mathrm{S}(X, X') \in \{-1, +1\}$ as

$$\mathrm{S}(X, X') \equiv \mathrm{S}_{X,X'} := \begin{cases} -1 & \text{if } \mathrm{S}(X) = \mathrm{S}(X') = -1, \\ +1 & \text{otherwise.} \end{cases} \tag{8.12}$$

Remark 8.11.1 Notice that the only case, in which the pair-statistics is equal to minus one, is if X and X' are (sums of) odd monomial expressions in fermionic operators.

Remark 8.11.2 Note further that $\mathrm{S}(c_\bullet, c_\bullet^\dagger) = \zeta$.

Definition 8.12 For a pair of operators X, X', we define the *pair-commutator* as

$$\left[X, X'\right]_\mathrm{S} \equiv X X' - \mathrm{S}_{X,X'}\, X' X. \tag{8.13}$$

Definition 8.13 We consider a pair of operators X, X'. Then we define the *time-ordered-pair-product* of their ensemble-Heisenberg-representation as

$$\mathrm{T}_t \left[X_\mathrm{H}(t)\, X'_\mathrm{H}\left(t'\right) \right] \equiv \theta(t - t')\, X_\mathrm{H}(t)\, X'_\mathrm{H}\left(t'\right) + \mathrm{S}_{X,X'}\, \theta(t' - t) X'_\mathrm{H}\left(t'\right) X_\mathrm{H}(t) \tag{8.14}$$

Remark 8.13.1 The definition of the pair-commutator and time-ordered-pair-product seem suspicious at first, but they will be integral in the formulation of Green-functions.

Remark 8.13.2 For X, X' being creation and annihilation-operators of the same statistics one has $\mathrm{S}_{X,X'} \equiv \zeta$ and it follows

$$\mathrm{T}_t\, c_\alpha(t)\, c^\dagger_{\alpha'}(t') \equiv \theta(t - t')\, c_\alpha(t)\, c^\dagger_{\alpha'}(t') + \zeta\, \theta(t' - t)\, c^\dagger_{\alpha'}(t')\, c_\alpha(t). \tag{8.15}$$

Remark 8.13.3 Analogously one also introduces an *anti-time-ordering-operator* $\overline{\mathrm{T}}_t$; For creation and annihilation-operators it follows

$$\overline{\mathrm{T}}_t \left[X_\mathrm{H}(t)\, X'_\mathrm{H}\left(t'\right) \right] \equiv \theta(t' - t)\, X_\mathrm{H}(t)\, X'_\mathrm{H}\left(t'\right) + \mathrm{S}_{X,X'}\, \theta(t - t')\, X'_\mathrm{H}\left(t'\right) X_\mathrm{H}(t)\,. \tag{8.16}$$

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Definition 8.14 We consider a family of statistical operators $\left\{X_\alpha\right\}_{\alpha\in\mathscr{A}}$ with statistics $\left\{\mathrm{S}_\alpha \equiv \mathrm{S}(X_\alpha)\right\}_{\alpha\in\mathscr{A}}$. Then we define the *time-ordered-product* of their ensemble-Heisenberg-representation as

$$\mathrm{T}_t\left[X_{\alpha_1}\left(t_1\right)X_{\alpha_2}\left(t_2\right)\cdots X_{\alpha_n}(t_n)\right] \equiv \sum_{\pi\in S_n}\chi_\pi\,\theta_\pi(t_1,t_2,\ldots,t_n)\,X_{\alpha_{\pi(1)}}\left(t_{\pi(1)}\right)X_{\alpha_{\pi(2)}}\left(t_{\pi(2)}\right)\cdots X_{\alpha_{\pi(n)}}\left(t_{\pi(n)}\right) \tag{8.17}$$

with a product of Heaviside-functions

$$\theta_\pi(t_1,t_2,\ldots,t_n) = \prod_{j=1}^{n-1}\theta(t_{\pi(j)} - t_{\pi(j+1)}) \tag{8.18}$$

enforcing the time-ordering $t_{\pi(1)} \geq t_{\pi(2)} \geq \ldots \geq t_{\pi(n)}$, and *Koszul-sign*

$$\chi_\pi = \sum_{\substack{1\leq i<j\leq n\\ \pi(i)>\pi(j)}}\mathrm{S}_{\alpha_{\pi(i)}}\,\mathrm{S}_{\alpha_{\pi(j)}} \tag{8.19}$$

counting the number of statistical transpositions required to bring the product into the order specified by the permutation π.

Lemma 8.15 If all operators have the same statistics $\mathrm{S} = \mathrm{S}_\alpha$ for all $\alpha\in\mathscr{A}$, then one obtains the simple formula

$$\mathrm{T}_t\left[X_{\alpha_1}\left(t_1\right)X_{\alpha_2}\left(t_2\right)\cdots X_{\alpha_n}(t_n)\right] \equiv \mathrm{sign}(\pi)^{1-\theta(\mathrm{S})}X_{\alpha_{\pi(1)}}\left(t_{\pi(1)}\right)X_{\alpha_{\pi(2)}}\left(t_{\pi(2)}\right)\cdots X_{\alpha_{\pi(n)}}\left(t_{\pi(n)}\right) \tag{8.20}$$

where

$$t_{\pi(1)} > t_{\pi(2)} > \ldots > t_{\pi(n)}. \tag{8.21}$$

Remark 8.15.1 When it cannot lead to any confusion, it is convenient to omit the brackets, with the notation that the time-ordering-operator acts (as a derivative) to everything that is „positioned right to it“

| ============= | End: To Be Updated / Corrected | ============= |

8.2.2. „Imaginary“ Time

For reasons that are difficult to motivate several chapters in advance, we introduce another kind of time-evolution-operator.

Definition 8.16 For arbitrary values $\tau_{\mathrm{i}}, \tau_{\mathrm{f}} \in (-\hbar\beta, +\hbar\beta)$ we define the *ensemble-imaginary-time-evolution operator* $U_{\Xi}(\tau_{\mathrm{f}}, \tau_{\mathrm{i}})$ of the system as

$$U_{\Xi}(\tau_{\mathrm{f}}, \tau_{\mathrm{i}}) = \exp - \frac{1}{\hbar}\left(\tau_{\mathrm{f}} - \tau_{\mathrm{i}}\right) \Xi.$$

Remark 8.16.1 Notice that the phraseology „imaginary“ time is a little bit misleading: τ defines a real variable. The thing that changed w.r.t. the previous subsection is the evolution-operator, not the nature of the variables.

Remark 8.16.2 Notice further that we have, in contrast to the real-time case

$$U_{\Xi}(\tau_{\mathrm{f}}, \tau_{\mathrm{i}}) \neq U_{\Xi}(\tau_{\mathrm{i}}, \tau_{\mathrm{f}})^{\dagger}. \tag{8.22}$$

Remark 8.16.3 Just as in the real-time case, we usually drop the index $U_{\Xi} \equiv U$.

Definition 8.17 Let $X \equiv X_{\mathrm{S}} \equiv X_{\mathrm{S}}(\tau)$ be the Wick-rotated version of a possibly time-dependent operator. Then one defines the corresponding *ensemble-Heisenberg-representation in imaginary time* as

$$X_{\mathrm{H}}(\tau) := U(0, \tau)\, X\, U(\tau, 0).$$

Theorem 8.18 One immediately concludes the *ensemble-Heisenberg-"equation of motion" in imaginary time*

$$\frac{\mathrm{d}}{\mathrm{d}\tau} X_{\mathrm{H}}(\tau) = -\frac{1}{\hbar}[X, \Xi]_{\mathrm{H}}(\tau) + \left(\frac{\partial X}{\partial \tau}\right)_{\mathrm{H}}(\tau) \tag{8.23}$$

Proof 8.18 See Appendix, Proofs, page 314.

Remark 8.18.1 Accordingly one introduces also the *imaginary-time-ordering operator* T_{τ}, analog to Definition 8.13.

8.3. Examples - Pure Systems

To remain as general as possible (and simultaneously reasonable) we specifiy neither the statistics nor the nature of the systems in this section.

8.3.1. Non-Interacting Cases

Example: A Quantum-Dot with Nature η and Statistics ζ

Definition 8.19 A *quantum-dot* is a system described by the Hamiltonian

$$H_0 = \epsilon c^\dagger c$$

and Fock-space

$$\mathscr{F}_\zeta \equiv \operatorname{span}_{\mathbb{C}} \left\{ |n\rangle \,\middle|\, n \in \mathbb{N}_\zeta \right\}.$$

Remark 8.19.1 One finds for the Heisenberg-time-evolution in the grand canonical ensemble as a special case of Lemma 8.22.

Example: A Two-Level-System with Nature η and Statistics ζ

Definition 8.20 A *two-level-system* is a system described by the Hamiltonian

$$H_0 = \mathbf{c}^\dagger\, \epsilon\, \mathbf{c} \equiv \begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix}^\dagger \begin{pmatrix} \epsilon_0 + \epsilon^z & -\mathrm{i}\epsilon^y + \epsilon^x \\ +\mathrm{i}\epsilon^y + \epsilon^x & \epsilon_0 - \epsilon^z \end{pmatrix} \begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix}$$

with

$$\epsilon \equiv \operatorname{mat} \epsilon = \epsilon_0 \sigma_0 + \boldsymbol{\epsilon} \cdot \boldsymbol{\sigma}.$$

Remark 8.20.1 Notice that a *spin-$\frac{1}{2}$-particle*, described through the Zeeman-contribution to the Pauli-equation leads (after going over to the second quantized formalism) to a two-level-system.

Remark 8.20.2 Computing the Heisenberg-time-evolution for the creation- and annihilation-operators analogously to Lemma 8.22 requires in this case a basis-change. This is done in Lemma TBReferenced.

Example: A Diagonalized, Non-Interacting Gas with Nature η and Statistics ζ

Definition 8.21 A *diagonalized, non-interacting "gas"*[1] is a system described by the Hamiltonian

$$H_0 = \sum_{\alpha} \epsilon_\alpha c^\dagger_\alpha c_\alpha$$

and Fock-space $\mathscr{F}_\zeta$ spanned by occupation-number-states as usual

$$\mathscr{F}_\zeta = \operatorname{span}_{\mathbb{C}} \left\{ |n_{\alpha_1}, n_{\alpha_2}, \dots, n_{\alpha_{\mathrm{d}_{\mathcal{H}}}}\rangle \,\middle|\, n_{\alpha_1}, n_{\alpha_2}, \dots, n_{\alpha_{\mathrm{d}_{\mathcal{H}}}} \in \mathbb{N}_\zeta \right\} \tag{8.24}$$

whereas the state with $n_{\alpha_1} = n_{\alpha_2} = \dots = n_{\alpha_{\mathrm{d}_{\mathcal{H}}}} = 0$ represents the *vacuum-state* a.k.a *the empty state*.

Lemma 8.22 One finds for the ensemble-Heisenberg-time-evolutions the identities

$$\begin{aligned} c_\bullet(t) &= \mathrm{e}^{-\mathrm{i}t(\epsilon_\bullet - \eta\mu)/\hbar} c_\bullet, \qquad & c^\dagger_\bullet(t) &= \mathrm{e}^{\mathrm{i}t(\epsilon_\bullet - \eta\mu)/\hbar} c^\dagger_\bullet \\ c_\bullet(\tau) &= \mathrm{e}^{-\tau(\epsilon_\bullet - \eta\mu)/\hbar} c_\bullet, \qquad & c^\dagger_\bullet(\tau) &= \mathrm{e}^{\tau(\epsilon_\bullet - \eta\mu)/\hbar} c^\dagger_\bullet \end{aligned} \tag{8.25}$$

Proof 8.22 See Appendix, Proofs, page 315.

Remark 8.22.1 A fermionic, particle-like example of this Hamiltonian is the *free electron gas in the LDA-approximation* in a crystal

$$H_0 = \sum_{\substack{\mathbf{k}\in \mathrm{Q} \\ \sigma\in\{\downarrow,\uparrow\} \\ n\in\{1,\dots,\infty\}}} \epsilon_{n\sigma}(\mathbf{k})\, c^\dagger_{n\sigma}(\mathbf{k}) c_{n\sigma}(\mathbf{k}).$$

The states in the Fock-space $\mathscr{F}$ are then written as

$$\mathscr{F} = \operatorname{span}_{\mathbb{C}} \left\{ |n_{\alpha_1}, n_{\alpha_2}, \dots, n_{\alpha_{\mathrm{d}_{\mathcal{H}}}}\rangle \,\middle|\, n_{\alpha_\bullet} \in \{0,1\},\ \alpha_i \equiv (m_i, \sigma_i, \mathbf{k}_i) \in \{1,\dots,\infty\} \times \{\downarrow,\uparrow\} \times \mathrm{Q} \right\} \tag{8.26}$$

Notice that the dimension $\mathrm{d}_{\mathcal{H}}$ of the single-particle Hilbert-space is given as

$$\mathrm{d}_{\mathcal{H}} = \dim \mathcal{H} = \Big| \{1,\dots,\infty\} \times \{\downarrow,\uparrow\} \times \mathrm{Q} \Big| = \infty \tag{8.27}$$

Usually one introduces a cutoff for the bands, and one considers crystals that are „large“, but finite size, such that one can work with a finite-dimensional basis.

[1] Note that this unfortunately a poor convention regarding the phraseology. Systems that one describes with this formalism are not neccessarily gases in the traditional sense of having a gas-*phase* (like steam). One treats electrons and phonons in solid materials with this formalism.

Remark 8.22.2 A bosonic example of this Hamiltonian is the *free phonon gas* in a $d_{\mathbf{x}}$-dimensional crystal (within the harmonic approximation)

$$H_0 = \sum_{\substack{\mathbf{q}\in \mathrm{Q} \\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}}} \hbar\omega_\nu(\mathbf{q})\, c^\dagger_\nu(\mathbf{q}) c_\nu(\mathbf{q}) \tag{8.28}$$

The Hilbert-space $\mathscr{H}$ of the system reads

$$\mathscr{H} = \mathrm{span}_{\mathbb{C}} \left\{ |n_{\alpha_1}, n_{\alpha_2}, \ldots, n_{\alpha_{d_{\mathcal{H}}}}\rangle \,\middle|\, n_{\alpha_\bullet} \in \{0,1,2,\ldots\},\ \alpha_i \equiv (\nu_i, \mathbf{q}_i) \in \{1,\ldots,rd_{\mathbf{x}}\} \times \mathrm{Q} \right\} \tag{8.29}$$

Example: A Non-Interacting Gas with Nature η and Statistics ζ

Definition 8.23 A *quadratic, non-interacting gas* is a system described by the Hamiltonian

$$H_0 = \sum_{\alpha,\beta} H_{0\,\alpha,\beta}\, c^\dagger_\alpha c_\beta \equiv \mathbf{c}^\dagger \cdot \mathbf{H}_0 \cdot \mathbf{c}$$

and Fock-space $\mathscr{F}_\zeta$ spanned by occupation-number-states as usual

$$\mathscr{F}_\zeta = \mathrm{span}_{\mathbb{C}} \left\{ |n_{\alpha_1}, n_{\alpha_2}, \ldots, n_{\alpha_{\mathrm{d}_{\mathcal{H}}}}\rangle \,\middle|\, n_{\alpha_1}, n_{\alpha_2}, \ldots, n_{\alpha_{\mathrm{d}_{\mathcal{H}}}} \in \mathbb{N}_\zeta \right\} \tag{8.30}$$

whereas the state with $n_{\alpha_1} = n_{\alpha_2} = \ldots = n_{\alpha_{\mathrm{d}_{\mathcal{H}}}} = 0$ represents the vacuum-state.

Remark 8.23.1 Computing the Heisenberg-time-evolution for the creation- and annihilation-operators analogously to Lemma 8.22 requires in this case a basis-change. We will not elaborate on this.

8.3.2. Interacting Cases

Definition 8.24 The Hamiltonian of *a diagonal gas with quadratic interactions* reads

$$\begin{aligned} H \equiv{}& \quad H_0 \qquad + \quad V \\ \equiv{}& \sum_{\alpha\in\mathscr{A}} \epsilon_\alpha c^\dagger_\alpha c_\alpha + \sum_{\alpha_1,\alpha_2\in\mathscr{A}} V_{\alpha_1\alpha_2} c^\dagger_{\alpha_1} c_{\alpha_2}. \end{aligned}$$

Remark 8.24.1 A typical example of this model is a free electron gas in the LDA-approximation with non-local potentials (e.g. defects in the crystal).

Definition 8.25 The Hamiltonian of *a diagonal gas with cubic interactions* reads

$$\begin{aligned} H \equiv\ & H_0 + V \\ \equiv\ & \sum_{\alpha \in \mathscr{A}} \epsilon_\alpha c^\dagger_\alpha c_\alpha + \sum_{\substack{n_1,n_2,n_3 \in \mathbb{N}_\zeta \\ \overline{n_1},\overline{n_2},\overline{n_3} \in \mathbb{N}_\zeta}} \delta_{n_1+n_2+n_3+\overline{n_1}+\overline{n_2}+\overline{n_3}=3} \sum_{\alpha_1,\alpha_2,\alpha_3 \in \mathscr{A}} V_{\substack{\alpha_1\alpha_2\alpha_3 \\ n_1 n_2 n_3 \\ \overline{n_1 n_2 n_3}}} \left(c^\dagger_{\alpha_1}\right)^{\overline{n_1}} \left(c^\dagger_{\alpha_2}\right)^{\overline{n_2}} \left(c^\dagger_{\alpha_3}\right)^{\overline{n_3}} c^{n_3}_{\alpha_3} c^{n_2}_{\alpha_2} c^{n_1}_{\alpha_1}. \end{aligned}$$

Remark 8.25.1 Recall from the scheme of second quantization that cubic interactions are relevant only for systems with nature $\eta = 1$, since the Hamiltonian of a system with nature $\eta = 0$ must be even in the creation- and annihilation-operators (i.e. particle number conserving).

Remark 8.25.2 A typical example of this model is a free phonon gas in a crystal with anharmonic interactions, i.e. the *phono3py-model*[2]

$$= \overbrace{\sum_{\substack{\mathbf{q}\in Q \\ \nu\in\{1,\ldots,rd_\mathbf{x}\}}} \hbar\omega_\nu(\mathbf{q})\, c^\dagger_\nu(\mathbf{q}) c_\nu(\mathbf{q})}^{H_0} + \overbrace{\sum_{\substack{\mathbf{q},\mathbf{q}',\mathbf{q}''\in Q \\ \nu,\nu',\nu''\in\{1,\ldots,rd_\mathbf{x}\}}} \Phi_{\nu\nu'\nu''}(\mathbf{q},\mathbf{q}',\mathbf{q}'') \left(c_\nu(-\mathbf{q}) + c^\dagger_\nu(\mathbf{q})\right) \left(c_{\nu'}(\mathbf{q}') + c^\dagger_{\nu'}(-\mathbf{q}')\right) \left(c_{\nu''}(\mathbf{q}'') + c^\dagger_{\nu''}(-\mathbf{q}'')\right)}^{V}. \tag{8.31}$$

Definition 8.26 The Hamiltonian of *a diagonal gas in a crystal with quartic interactions* reads

$$\begin{aligned} H \equiv\ & H_0 + V \\ \equiv\ & \sum_{\alpha \in \mathscr{A}} \epsilon_\alpha c^\dagger_\alpha c_\alpha + \frac{1}{2} \sum_{\alpha_1,\alpha_2,\alpha_3,\alpha_4 \in \mathscr{A}} V_{\alpha_1\alpha_2\alpha_3\alpha_4} c^\dagger_{\alpha_1} c^\dagger_{\alpha_2} c_{\alpha_4} c_{\alpha_3} \end{aligned}$$

with the additional property

$$V_{\alpha_1\alpha_2\alpha_3\alpha_4} = V_{\alpha_2\alpha_1\alpha_4\alpha_3} \quad \text{for all } \alpha_1,\alpha_2,\alpha_3,\alpha_4 \in \mathscr{A}. \tag{8.32}$$

Remark 8.26.1 A typical example of this model is the "GW-system"[3] for electrons in a crystal with screened Coulomb-interactions.

[2] Cf. phono3py-paper.
[3] Cf. TBReferenced

Definition 8.27 The Hamiltonian of *a non-diagonal gas in a crystal with quartic interactions* reads

$$\begin{aligned} H \equiv \quad & H_0 \qquad\qquad + \quad V \\ \equiv & \sum_{\alpha,\beta\in\mathscr{A}} H_{0\,\alpha,\beta}\, c^\dagger_\alpha c_\beta + \frac{1}{2}\sum_{\alpha_1,\alpha_2,\alpha_3,\alpha_4\in\mathscr{A}} V_{\alpha_1\alpha_2\alpha_3\alpha_4} c^\dagger_{\alpha_1} c^\dagger_{\alpha_2} c_{\alpha_4} c_{\alpha_3} \end{aligned}$$

with the additional property

$$V_{\alpha_1\alpha_2\alpha_3\alpha_4} = V_{\alpha_2\alpha_1\alpha_4\alpha_3} \quad \text{for all } \alpha_1,\alpha_2,\alpha_3,\alpha_4 \in \mathscr{A}. \tag{8.33}$$

Remark 8.27.1 A typical example of this model is a tight-binding model for electrons in a crystal with Hubbard-interactions.

8.4. Examples - Mixed Systems

8.4.1. Non-Interacting Cases

Definition 8.28 We consider a spin unpolarized, diagonal electron-gas and a diagonal phonon-gas that do not interact with each other

$$\begin{aligned} H = & \qquad\qquad\qquad H_0 \\ = & \qquad H_{0,\mathrm{el}} \qquad\qquad + \qquad H_{0,\mathrm{ph}} \\ = & \sum_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\ldots,\infty\}}} \epsilon_{m\mathbf{k}\sigma} c^\dagger_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma} + \sum_{\substack{\mathbf{q}\in\mathrm{Q}\\ \nu\in\{1,\ldots,rd_\mathbf{x}\}}} \Omega_{\mathbf{q},\nu} b^\dagger_{\mathbf{q}\nu} b_{\mathbf{q}\nu} \end{aligned} \tag{8.34}$$

with $\epsilon_{m\mathbf{k}\uparrow} = \epsilon_{m\mathbf{k}\downarrow}$.

8.4.2. Interacting Cases

Definition 8.29 We consider the *EPW-system*, i.e. a spin unpolarized, diagonal electron-gas and a diagonal phonon-gas interacting with each other over a spin-independent-interaction as

$$\begin{aligned} H = & \qquad\qquad\qquad H_0 \qquad\qquad\qquad + \qquad V \\ = & \sum_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\ldots,\infty\}}} \epsilon_{m\mathbf{k}\sigma} c^\dagger_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma} + \sum_{\substack{\mathbf{q}\in\mathrm{Q}\\ \nu\in\{1,\ldots,rd_\mathbf{x}\}}} \hbar\omega_{\mathbf{q},\nu} b^\dagger_{\mathbf{q}\nu} b_{\mathbf{q}\nu} + \sum_{\substack{\mathbf{k},\mathbf{q}\in\mathrm{Q}\\ \sigma\in\{\uparrow,\downarrow\}\\ \nu\in\{1,\ldots,rd_\mathbf{x}\}\\ m,n\in\{1,\ldots,\infty\}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})}{\sqrt{N_p}} c^\dagger_{\mathbf{k}+\mathbf{q}m\sigma} c_{\mathbf{k}n\sigma} \left(b_{\mathbf{q}\nu} + b^\dagger_{-\mathbf{q}\nu}\right) \end{aligned}$$

with $\epsilon_{m\mathbf{k}\uparrow} = \epsilon_{m\mathbf{k}\downarrow}$.

Lemma 8.30 The hermiticity of the Hamiltonian implies for the coupling

$$g_{mn\nu}(\mathbf{k},\mathbf{q}) = g^*_{nm\nu}(\mathbf{k}+\mathbf{q},-\mathbf{q}). \tag{8.35}$$

Proof 8.30 See Appendix, Proofs, page 316.

Remark 8.30.1 Notice that this property implies that $g_{nn\nu}(\mathbf{k},0) = g^*_{nn\nu}(\mathbf{k},0) \in \mathbb{R}$, i.e. the diagonal coupling at zero phonon-momentum is real-valued.

Lemma 8.31 Notice that for inversion-symmetric crystals one has the additional symmetry

$$g_{mn\nu}(\mathbf{k},\mathbf{q}) = g^*_{mn\nu}(-\mathbf{k},-\mathbf{q}). \tag{8.36}$$

Proof 8.31 ToBeDigitalized from Waste-Folder 1.

Definition 8.32 We consider the *EPy3W-system*, i.e. a spin unpolarized, diagonal electron-gas and an interacting phonon-gas interacting with each other over a spin-independent-interaction as

$$\begin{aligned} H &= \qquad H_0 \qquad + \qquad V \\ &= \sum_{\substack{\mathbf{k}\in Q \\ \sigma\in\{\uparrow,\downarrow\} \\ m\in\{1,\ldots,\infty\}}} \epsilon_{m\mathbf{k}\sigma} c^\dagger_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma} + \sum_{\substack{\mathbf{q}\in Q \\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}}} \hbar\omega_{\mathbf{q},\nu} b^\dagger_{\mathbf{q}\nu} b_{\mathbf{q}\nu} + \quad V_{\text{el-ph}} + V_{\text{ph-ph}} \end{aligned}$$

with $\epsilon_{m\mathbf{k}\uparrow} = \epsilon_{m\mathbf{k}\downarrow}$ and

$$\begin{aligned} V_{\text{el-ph}} &= \sum_{\substack{\mathbf{k},\mathbf{q}\in Q \\ \sigma\in\{\uparrow,\downarrow\} \\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\} \\ m,n\in\{1,\ldots,\infty\}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})}{\sqrt{N_p}} c^\dagger_{\mathbf{k}+\mathbf{q}m\sigma} c_{\mathbf{k}n\sigma} \left(b_{\mathbf{q}\nu} + b^\dagger_{-\mathbf{q}\nu}\right) \\ V_{\text{ph-ph}} &= \sum_{\substack{\mathbf{q},\mathbf{q}',\mathbf{q}''\in Q \\ \nu,\nu',\nu''\in\{1,\ldots,rd_{\mathbf{x}}\}}} \Phi_{\nu\nu'\nu''}(\mathbf{q},\mathbf{q}',\mathbf{q}'') \left(c_\nu(-\mathbf{q}) + c^\dagger_\nu(\mathbf{q})\right) \left(c_{\nu'}(\mathbf{q}') + c^\dagger_{\nu'}(-\mathbf{q}')\right) \left(c_{\nu''}(\mathbf{q}'') + c^\dagger_{\nu''}(-\mathbf{q}'')\right). \end{aligned} \tag{8.37}$$

9. Fundamentals III - Quantum Statistical Mechanics - Pure Systems

Quoting from [Negele, Orland, p. 47]:

> *In quantum statistical mechanics, the thermodynamic properties of a system in equilibrium are specified by the assumption of the equal occupation of accessible states. The physical problem is idealized by considering an appropriate ensemble of identical systems, and the probability of observing a particular state is given by the ratio of the number of systems in the ensemble in that state to the total number of systems in the ensemble. Three ensembles are commonly used. The microcanonical ensemble describes an isolated system, which has fixed energy E and fixed particle number. The probability of obsenring a state of energy $E' \neq E$ is zero and that of observing a state of energy $E' = E$ is constant, equal to the inverse of the total number of states of energy E. The canonical ensemble describes a system of fixed particle number in equilibrium with a thermal reservoir with which it may exchange energy at a specified temperature. By enumerating the number of states in which the system and the reservoir combined have a fixed total energy, it follows that the probability of observing this system alone with energy E is proportional to $\mathrm{e}^{-E/k_BT} \equiv \mathrm{e}^{-\beta E}$ where k_B is Boltzmann's constant. The average energy of the system is fixed and is, controlled by the temperature T. The grand canonical ensemble describes a system in equilibrium with a particle reservoir, with which it may change particles, as well as a thermal reservoir. Enumeration of the states in which the system and reservoir have fixed total energy and particle number leads to the result that the probability of observing the system alone with energy E and particle number N is proportional to $\mathrm{e}^{-\beta(E-\mu N)}$. In addition to the average energy of the system, the average particle number is now fixed and is controlled by the chemical potential. The thermodynamic limit for a physical system of N particles contained in a volume V is defined by taking the limit as N and V go to infinity such that the ratio $\rho = N/V$ remains constant. An essential result of thermodynamics is that the three ensembles are equivalent for describing the macroscopic properties of most systems in the thermodynamic limit. One is then free to select the ensemble on the basis of formal convenience, and as in the case of classical statistical mechanics, the grand canonical ensemble offers the greatest formal simplicity.*

9.1. Linear Mappings

In this section, we will carefully explain how expressions of the type "ket-bra" are to be understood.

Definition 9.1 Let V,W denote vector-spaces over a field $\mathbb{K}$. Let us write further Hom(V, W) for the $\mathbb{K}$-vector space of linear mappings from V to W, and V^* for the dual vector space of V.

Theorem 9.2 The mapping

$$\begin{aligned} I : \; W \otimes V^\star &\longrightarrow \mathrm{Hom}(V, W) \\ w \otimes f \quad &\mapsto I(w \otimes f) \end{aligned}$$

with

$$\forall v \in V \; : \; I(w \otimes f)(v) := f(v)w \tag{9.1}$$

and linear extension over the remaining tensor-product, is an isomorphism.

Proof 9.2 The linearity is given by construction:

$$I(w_1 \otimes f_1 + w_2 \otimes f_2) = I(w_1 \otimes f_1) + I(w_2 \otimes f_2) \tag{9.2}$$

For the linearity w.r.t. $\mathbb{K}$-multiplication one finds for $\lambda \in \mathbb{K}$

$$I((\lambda w) \otimes f)(v) = f(v) \cdot \lambda \cdot w = I(w \otimes (\lambda f))(v) = (\lambda \cdot f)(v) \cdot w = \lambda \cdot f(v) \cdot w \tag{9.3}$$

For the injectivity, let us assume that $I(w \otimes f) = 0$, meaning that

$$\forall v \in V \; : \; I(w \otimes f)(v) = f(v)w = 0. \tag{9.4}$$

There is two solutions for this: Either $\forall v \in V \; : \; f(v) = 0$ or $w = 0$. Since f is a linear map, both cases imply that $w \otimes f = 0$, that only the zero gets mapped to zero.

For the surjectivity, we take $A \in \mathrm{Hom}(V, W)$ and prove that there is a decomposition, such that

$$\forall v \in V \; : \; Av = I\left(\sum_i w_i \otimes f_i\right)(v) \tag{9.5}$$

For this, we choose a basis $e_1, \ldots, e_n$ of V with dual basis $e^1, \ldots, e^n$ of V^*, such that $e^i(e_j) = \delta_{i,j}$. Next, we decompose v w.r.t. this basis, i.e.

$$v = \sum_{i=1}^{n} e^i(v) e_i \quad \left(\equiv \sum_{i=1}^{n} v_i e_i\right) \tag{9.6}$$

This yields, using the linearity of A,

$$Av = \sum_{i=1}^{n} e^i(v)Ae_i = \dots \tag{9.7}$$

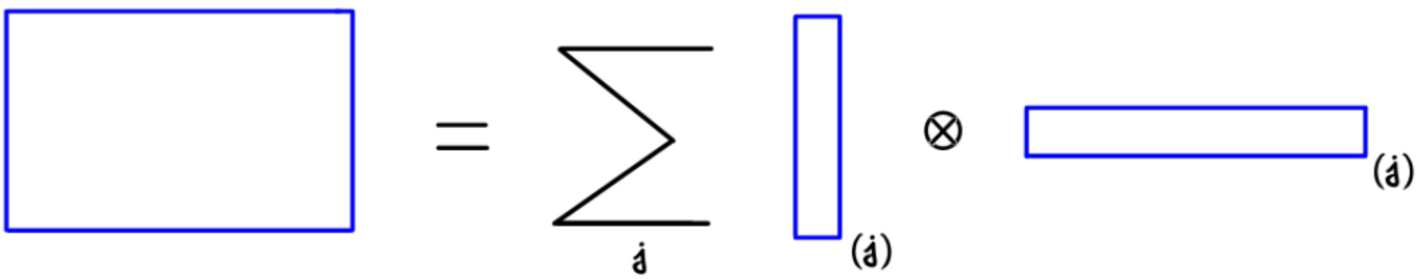
Since $Ae_i \in W$, and $e^i \in V^*$ we can rewrite this expression as

$$\dots = \sum_{i=1}^{n} I\left(Ae_i \otimes e^i\right)(v) = I\left(\sum_{i=1}^{n} Ae_i \otimes e^i\right)(v) \tag{9.8}$$

proving the surjectivity.

Remark 9.2.1 Since, after referring to a basis, vectors are usually interpreted as standing vectors, and dual-vectors are usually interpreted as lyings-vectors, one sees that each matrix of size $m \times n$ can be decomposed uniquely (up to an ambiguity due to scalar factors moving in the tensor product) as a sum of terms each of which is the tensor product (sometimes called a 'dyadic' product in physics textbooks) of an m-component column vector with an n-component row vector.

Remark 9.2.2 The figure elucidates the isomorphism.



Notation 9.3 Switching to the physics-conventions, we realize that in the case of $V = W \equiv \mathscr{H}$ for some (single- or many-particle) Hilbert-space, $V^* \equiv \mathscr{H}^*$ its dual space, and End($\mathscr{H}$) the set of all endomorphisms on the Hilbert-space, one can, for arbitrary $\psi, \chi \in \mathscr{H}$, interprete expressions of the type

$$|\psi\rangle\langle\chi| \tag{9.9}$$

as linear mappings, and by referring to a basis, their underlying matrix can be determined from a dyadic-product.

9.2. Density Matrices

Motivation: The need for a new Technique

Consider a Hilbert-space $\mathscr{H}$ with scalar-product $\langle\ ,\ \rangle$. So far, we have only considered state-vectors in this Hilbert-space, so called *pure states*. But with this idea, one can not tackle all physically relevant situations:

Let us assume that $\{|\psi_i\rangle\}_{i=1,\dots,N} \subset \mathscr{H}$ is an arbitrary series of pure states, and let us assume a situation, in which a pure state $|\psi_i\rangle$ is created with probability $q_i \in [0,1]$, s.t. $\sum_i q_i = 1$. Then, common-sense tells us, that we have for the expectation-value of an observable A the following equation

$$\langle A\rangle = \sum_i q_i \langle\psi_i|A|\psi_i\rangle. \tag{9.10}$$

Now let's rewrite the r.h.s.:

$$\langle A\rangle = \sum_i q_i\langle\psi_i|A|\psi_i\rangle = \sum_i q_i \,\mathrm{Tr}\left(|\psi_i\rangle\langle\psi_i|A\right) = \mathrm{Tr}\left(\sum_i q_i|\psi_i\rangle\langle\psi_i|A\right) =: \mathrm{Tr}\,(\rho A) \tag{9.11}$$

Remark 9.3.1 In this example it immediately follows that $\mathrm{Tr}\,\rho = 1$.

Remark 9.3.2 One calls such an operator ρ, that describes the systems state, a *density(-operator)*. We will introduce them now more carefully.

The Formalism of Density Matrices

Quantum mechanics distinguishes between two types of states that physical systems can be in: "pure" and "mixed states". In this section we will introduce both.

Definition 9.4 A **pure state** is a state that can be described by a ket vector $|\psi\rangle$ in a Hilbert-space $\mathscr{H}$.

Remark 9.4.1 Notice that this Hilbert-space can be either a single-particle Hilbert-space or a many-particle Hilbert-space.

Definition 9.5 We assume that the system is with probabilities

$$p_i \in [0,1) \quad \text{and} \quad \sum_{i=1}^{n} p_i = 1 \tag{9.12}$$

in orthonormalized pure states

$$|\psi_i\rangle \quad \text{with} \quad \langle\psi_i|\psi_j\rangle = \delta_{ij}. \tag{9.13}$$

The system is then described by the *density matrix*, or *density* for short,

$$\begin{aligned} \rho : \mathscr{H} &\longrightarrow \mathscr{H} \\ |\chi\rangle &\longmapsto \rho|\chi\rangle \end{aligned}$$

which is defined as a sum of *projectors*

$$\hat{\rho} = \sum_{i=1}^{n} p_i |\psi_i\rangle\langle\psi_i|. \tag{9.14}$$

Remark 9.5.1 For $n = 1$, the density matrix still represents a pure state, for $n > 1$, we shall refer to it as describing a *mixed state*.

Remark 9.5.2 One can also introduce densities for states that are not orthonormalized. We will however not do so, since we are not interested in these cases.

Lemma 9.6 We have the following properties for the density matrix $\hat{\rho}$:

1. $\hat{\rho}$ is hermitian, i.e. $\hat{\rho} = \hat{\rho}^\dagger$.
2. $\hat{\rho}$ has the eigenvalues p_i with eigenstates $|\psi_i\rangle$.
3. $\hat{\rho}$ is positive, i.e. $\langle\chi|\hat{\rho}|\chi\rangle \geq 0$ for all $|\chi\rangle$.
4. $\hat{\rho}$ is *normalized*, i.e. $\mathrm{Tr}(\hat{\rho}) = 1$.

Proof 9.6 TBDigitalized from Nolting p. 106 f.

Theorem 9.7 For time-dependent states $|\psi_i\rangle \equiv |\psi_i(t)\rangle$ that obey the Schrödinger-equation

$$\mathrm{i}\hbar\frac{d}{dt}|\psi_i(t)\rangle = \hat{H}|\psi_i(t)\rangle \tag{9.15}$$

we find that $\hat{\rho} \equiv |\psi_i\rangle\langle\psi_i|$ obeys the von Neumann-equation

$$\frac{d}{dt}\hat{\rho} = -\frac{\mathrm{i}}{\hbar}[\hat{H}, \hat{\rho}]. \tag{9.16}$$

Proof 9.7 TBDigitalized from Nolting p. 108

Remark 9.7.1 We call a solution a *stationary state* if $\hat{\rho}$ is time-independent.

Definition 9.8 We define the expectation value of an operator $\hat{A}$ in the state $\hat{\rho}$ as

$$\langle \hat{A} \rangle := \mathrm{Tr}(\hat{\rho}\hat{A}). \tag{9.17}$$

Theorem 9.9 The expectation value of an operator $\hat{A}$ in the state $\hat{\rho}$ is given by

$$\langle \hat{A} \rangle = \sum_{i=1}^{n} p_i \langle \psi_i | \hat{A} | \psi_i \rangle. \tag{9.18}$$

Proof 9.9 TBDigitalized from Nolting p. 109

Remark 9.9.1 Notice that this expression is completely intuitive, given the situation from Definition 9.5.

Definition 9.10 Given a state ρ, we define the *entropy* as

$$S(\hat{\rho}) = -k_{\mathrm{B}}\,\mathrm{Tr}(\hat{\rho}\ln\hat{\rho}). \tag{9.19}$$

Remark 9.10.1 Notice that in the template[1] it is also proven that this identity holds for all ensembles.

The subsequent sections will become a bit messy with notation, so we believe that it is appropriate to end this section with the following quote.

> *"Zudem ist es ein Irrtum zu glauben, dass die Strenge in der Beweisführung die Feindin der Einfachheit wäre."*
>
> David Hilbert in a lecture given at the International Congress of Mathematicians in Paris 1900

9.3. The Microcanonical Ensemble

Fact 9.11 For macroscopically large systems, it is impossible to determine the exact state of the system by solving the time-dependent Schrödinger-equation for some given initial value. In statistical physics, it is then common to "guess" / postulate the state of the system, by considering a more or less reasonable density matrix $\hat{\rho}$.

Definition 9.12 We define the *natural variables of the microcanonical ensemble* as the triplet $(E, V, N) \in (-\infty, +\infty) \times (0, +\infty) \times \mathbb{N}_0$, whereas ...

- $E \in (-\infty, +\infty)$ is the energy (relative to some reference-energy) of the system,
- $V \in (0, +\infty)$ is the volume of the system, and

[1] Cf. TBDigitalized from Nolting p. 151.

- $N \in \mathbb{N}_0 = \{0, 1, 2, \dots, \infty\}$ is the number of particles.

Notation 9.13 For a fixed particle number N and a fixed volume V we enumerate the set of eigen-energies $\{E_1, E_2, \dots\}$ and orthonormal eigenstates $\{|E_1\rangle, |E_2\rangle, \dots\}$ of the Hamilton-operator of the system

$$\hat{H}|E_i\rangle = E_i|E_i\rangle \quad \text{with} \quad \langle E_i|E_j\rangle = \delta_{ij} \quad \text{and} \quad \mathscr{H}_N = \text{span}_{\mathbb{C}}\left\{|E_1\rangle, |E_2\rangle, \dots\right\}, \tag{9.20}$$

whereas $\mathscr{H}_N$ represents the underlying N-particle Hilbert-space.

Remark 9.13.1 Notice that the volume influences the eigen-energies, and that the eigen-states above are such that they correspond to N-particle states

$$\langle E_i|\hat{N}|E_i\rangle = N. \tag{9.21}$$

with the particle-number operator $\hat{N}$ from Definition 7.39. Thus, the eigenenergies actually depend on the volume V and the particle number N as

$$E_i \equiv E_i(V, N). \tag{9.22}$$

However, we will suppress this dependence for notational convenience.

Definition 9.14 For an uncertainty Δ, that corresponds to the energy-uncertainty in the measuring-process, we define the subset

$$\mathscr{E} = \left\{E_\bullet \in \{E_1, E_2, \dots\} \,\middle|\, E_\bullet \in [E - \Delta, E]\right\}. \tag{9.23}$$

The corresponding states $\left\{|E_\bullet\rangle \,\middle|\, E_\bullet \in \mathscr{E}\right\}$ is called *the set of microstates corresponding to the triplet* (E, V, N). Further we introduce the *total number of accessible microstates in the microcanonical ensemble* $\Omega_\Delta(E, V, N)$ as the cardinal number of $\mathscr{E}$, i.e.

$$\Omega_\Delta(E, V, N) = \sum_{E_\bullet \in \mathscr{E}} 1 = |\mathscr{E}|. \tag{9.24}$$

Remark 9.14.1 When one explicitly computes the total number of microstates for some simple toy-models, one finds that Ω depends only lightly on Δ. One usually then *postulates* that this is also the case for more complicated systems and drops the index $\Omega_\Delta(E, V, N) \equiv \Omega(E, V, N)$.

Hypothesis 9.15 We *postulate* that the system is in a state that is described by the density matrix

$$\hat{\rho}_{\text{MCE}} = \sum_{m=1,2,\dots} p_m^{\text{MCE}}|E_m\rangle\langle E_m| \tag{9.25}$$

with the probabilities

$$p_m^{\mathrm{MCE}} = \begin{cases} \Omega(E,V,N)^{-1} & \text{for} \quad E_m \in \mathscr{E}. \\ 0 & \text{otherwise.} \end{cases} \tag{9.26}$$

Remark 9.15.1 This assumption is sometimes referred to as the *principle of equal a priori probabilities* or *the fundamental postulate of statistical mechanics*. It is oftentimes formulated in prose as

> *A closed system in equilibrium is equally probable in each of its accessible microstates.*

We further say, that the system is in the *thermal state* if the system is properly described by the density above.

Remark 9.15.2 The fact that this is the correct state to describe the system is proven a posteriori by the fact that it leads to the correct experimental results. Thinking about it longer, it is a rather unreasonable assumption.

Corollary 9.16 The following statements are immediate

- $\hat{\rho}$ is a density matrix in the sense of Definition TBReferenced.
- $\hat{\rho}$ solves the stationary von Neumann equation (TBReferenced).
- $\hat{\rho}$ is describes a system with energy E, i.e. $\langle \hat{H} \rangle \approx E$

Proof 9.16 TBDigitalized from Nolting p. 113 The only non-trivial thing is the normalization, i.e. $\mathrm{Tr}(\hat{\rho}) = 1$. For this we ...

Definition 9.17 We define the *Boltzmann-entropy* in the microcanonical ensemble as

$$S_{\mathrm{MCE}}(E,V,N) = k_{\mathrm{B}} \ln \Omega(E,V,N). \tag{9.27}$$

and the *temperature* $T \equiv T(E,V,N)$ as

$$\frac{1}{T(E,V,N)} = \left(\frac{\partial S_{\mathrm{MCE}}}{\partial E} \right)_{V,N} (E,V,N), \tag{9.28}$$

and the *pressure* $P \equiv P(E,V,N)$ as

$$P(E,V,N) = T(E,V,N) \cdot \left(\frac{\partial S_{\mathrm{MCE}}}{\partial V} \right)_{E,N} (E,V,N), \tag{9.29}$$

and the *chemical potential* $\mu \equiv \mu(E,V,N)$ as

$$\mu(E,V,N) = -T(E,V,N) \cdot \left(\frac{\partial S_{\mathrm{MCE}}}{\partial N} \right)_{E,V} (E,V,N). \tag{9.30}$$

Remark 9.17.1 The entropy is one of several so-called *state functions* or *thermodynamic potentials*. The temperature, pressure and chemical potential are then the corresponding *conjugate variables* to the energy, volume and particle number.

Remark 9.17.2 The appearance of the temperature in the definition of the pressure and chemical potential might seem to complicate matters at first glance. This will however turn out to be a very neat definition, as one can attribute physical interpretations to the pressure and the chemical potential, as we will see in the examples and in the following sections.

Remark 9.17.3 The chemical potential has, in physical terms, the meaning of the energy increase when one particle is added (at constant entropy and constant volume). This will become clear in Corollary 9.33.

Remark 9.17.4

Notice that one can rewrite the partial derivatives as

$$\left(\frac{\partial S_{\text{MCE}}}{\partial V}\right)_{E,N}(E,V,N)=\frac{P(E,V,N)}{T(E,V,N)},\qquad \left(\frac{\partial S_{\text{MCE}}}{\partial N}\right)_{E,V}(E,V,N)=-\frac{\mu(E,V,N)}{T(E,V,N)},\tag{9.31}$$

such that we can write for the differential of the entropy

$$\mathrm{d}S_{\text{MCE}}=\frac{1}{T}\mathrm{d}E+\frac{P}{T}\mathrm{d}V-\frac{\mu}{T}\mathrm{d}N\quad\Longleftrightarrow\quad \mathrm{d}E=T\mathrm{d}S_{\text{MCE}}-P\mathrm{d}V+\mu\mathrm{d}N\tag{9.32}$$

Definition 9.18 Definition of two systems being in thermal equilibrium.

Theorem 9.19 Zustandssumme des idealen Gases: Fließbach, Gleichung (6.20).

Proof 9.19 TBDigitalized.

9.4. The Canonical Ensemble

To introduce the canonical ensemble, we start from a description that uses the microcanonical ensemble.

Notation 9.20 We consider a isolated system $\Sigma_{I\cup II}\equiv\Sigma_I\cup\Sigma_{II}$ that consists of two parts Σ_I and Σ_{II}.

Here are some further assumptions and notations:

1. The Hilbert-space of the total system is given as a tensor-product $\mathscr{H}_{N_{I\cup II}}=\mathscr{H}_{N_I}\otimes\mathscr{H}_{N_{II}}$.

2. The Hamilton-operator of the total system is given by

$$H_{I\cup II} = H_I + H_{II} + H_{I\cap II}, \tag{9.33}$$

   where H_I and H_{II} are the Hamilton-operators of Σ_I and Σ_{II} and $H_{I\cap II}$ is the interaction[2] Hamilton-operator.

3. The total system has an energy, volume and particle-number $E_{I\cup II}$, $V_{I\cup II}$ and $N_{I\cup II}$, respectively.

4. The subsystems Σ_I and Σ_{II} have energies E_1 and E_2 and volumes V_1 and V_2 and particle-numbers N_1 and N_2, such that

$$\begin{aligned} E_I + E_{II} &= E_{I\cup II}, \\ V_I + V_{II} &= V_{I\cup II}, \\ N_I + N_{II} &= N_{I\cup II}. \end{aligned}$$

5. This total system $\Sigma_{I\cup II}$ is assumed to be properly described in the microcanonical ensemble, such that entropy $S_{I\cup II}$ and a temperature $T_{I\cup II}$ can be defined.

6. We further assume that Σ_I and Σ_{II} are in thermal equilibrium with each other *and* have the same temperature T as the total system $T \equiv T_{I\cup II} \equiv T_I \equiv T_{II}$.

7. For reasons that will become clear later, we call the second subsystem Σ_{II} the *reservoir* or *heat bath*.

The first subsystem $\Sigma_I \equiv \Sigma$ with Hilbert-space $\mathscr{H}_{N_I} \equiv \mathscr{H}_N$ and Hamilton-operator $H_I \equiv H$ is the system of interest, which we ultimately seek to describe. Thus we write for its volume $V_I \equiv V$ and particle number $N_1 \equiv N$. The system Σ is considered to be a "very small", yet "macroscopic", part of the "very large" system $\Sigma_{1\cup 2}$.

Hypothesis 9.21 We assume that the interaction between Σ_I and Σ_{II} is so weak that the energy exchange between the two systems is negligible $H_{I\cap II} \approx 0$ in the following discussions.

Definition 9.22 We define the *natural variables of the canonical ensemble* as the triplet (T, V, N), where T is the temperature (of a so-called "reservoir", see below), V is the volume and N is the number of particles.

Notation 9.23 Just as in Notation 9.13: For a fixed particle number N and a fixed volume V we enumerate the set of eigen-energies $\{E_1, E_2, \ldots\}$ and orthonormal eigenstates $\{|E_1\rangle, |E_2\rangle, \ldots\}$ of the Hamilton-operator H of the system Σ as

$$\hat{H}|E_i\rangle = E_i|E_i\rangle \quad \text{with} \quad \langle E_i|E_j\rangle = \delta_{ij} \quad \text{and} \quad \mathscr{H}_N = \mathrm{span}_{\mathbb{C}}\left\{|E_1\rangle, |E_2\rangle, \ldots\right\}, \tag{9.34}$$

whereas $\mathscr{H}_N$ represents the underlying N-particle Hilbert-space.

Hypothesis 9.24 We assume, that the energies $E_\bullet$ of the system are so small, such that the following approximation is valid

$$\text{TBDone.} \tag{9.35}$$

[2] Notice that this interaction in the Hamiltonian is neccessary for the two systems to reach thermal equilibrium.

Fact 9.25 The number of states of the total system at energy $E \approx E_{II} \gg E_\bullet$, surpressing the volumes and particle-numbers for notational convenience, is given by

$$\Omega_{I\cup II}(E) = \sum_{E_\bullet \in \{E_1, E_2, \ldots\}} \Omega_I(E_\bullet)\Omega_{II}(E - E_\bullet) \tag{9.36}$$

whereas we identify

$\Omega_I(E_\bullet)\Omega_{II}(E - E_\bullet) =$ "number of ways to configure the system + reservoir with the system in microstate $|E_\bullet\rangle$".

Thus we obtain in the expansion of the density matrix in the canonical ensemble

$$\hat{\rho}_{\mathrm{CE}} = \sum_{\bullet=1,2,\ldots} p_\bullet^{\mathrm{CE}} |E_\bullet\rangle\langle E_\bullet|. \tag{9.37}$$

the ansatz

$$p_\bullet^{\mathrm{CE}} = \frac{\Omega_I(E_\bullet)\Omega_{II}(E - E_\bullet)}{\Omega_{I\cup II}(E)} \propto \Omega_{II}(E - E_\bullet). \tag{9.38}$$

Theorem 9.26 One finds for the density $\rho \equiv \rho(T, V, N)$ in the canonical ensemble

$$\hat{\rho}_{\mathrm{CE}} = \frac{1}{Z} \sum_{\bullet=1,2,\ldots} \mathrm{e}^{-\beta E_\bullet} |E_\bullet\rangle\langle E_\bullet| \tag{9.39}$$

with

$$Z = \sum_{\bullet=1,2,\ldots} \mathrm{e}^{-\beta E_\bullet} \tag{9.40}$$

where $\beta = 1/k_{\mathrm{B}}T$ is the inverse temperature.

Proof 9.26 TBDigitalized.

Remark 9.26.1 Notice that we can rewrite the density as

$$\hat{\rho}_{\mathrm{CE}} = \frac{1}{Z} \sum_{\bullet=1,2,\ldots} \mathrm{e}^{-\beta E_\bullet} |E_\bullet\rangle\langle E_\bullet| = \frac{1}{Z} \mathrm{e}^{-\beta \hat{H}} \tag{9.41}$$

Corollary 9.27 The following statements are immediate

- $\hat{\rho}$ is a density matrix in the sense of Definition TBReferenced.

- $\hat{\rho}$ solves the stationary von Neumann equation (TBReferenced).

Proof 9.27 The normalization is trivial. The fact that it solves the von Neumann equation as well.

Theorem 9.28 The (Helmholtz) free energy is minimal in the canonical ensemble.

Proof 9.28 03 page 28 for minimality of (Helmholtz) free energy in the canonical ensemble.

9.5. The Grand Canonical Ensemble

TBD See also 03 page 35 for a derivation.

Minimality of the grand canonical potential a.k.a. Landau-potential : Nolting page 157.

Ehrenfest Categorization of Phase Transitions: Some book page 314.

Theorem 9.29 For fixed (T, V), the function

$$\mu \mapsto N(\mu) \equiv -\left(\frac{\partial \Omega}{\partial \mu}\right)(\mu, T, V) \tag{9.42}$$

is monotonically increasing.

Proof 9.29 See [Salmhofer, page 118].

9.6. Equivalence of Ensembles

9.6.1. Equivalence of the Microcanonical and Canonical Ensemble

TBDigitalized from Nolting p. 74.

9.6.2. Equivalence of the Microcanonical and Canonical Ensemble

TBDigitalized from Nolting p. 88 f. (partly).

9.7. Legendre Transformations

TBCopied from... Was it Schwabl that did a great job on this?

9.8. Thermodynamics

9.8.1. Deductions in the Microcanonical Ensemble

Notation 9.30 We consider a system that is properly described in the microcanonical ensemble by the natural variables (E, V, N), with entropy-function S_{MCE} as in Definition 9.17.

Remark 9.30.1 For notational convenience, we will in the following suppress the index "MCE" in $S_{\text{MCE}} \equiv S$.

Corollary 9.31 Inverting the entropy S w.r.t. the energy E we can conclude that there exists an energy-function E such that

$$(S, V, N) \mapsto E(S, V, N) \in \mathbb{R} \tag{9.43}$$

with differential as in Equation (9.32) as

$$\mathrm{d}E = T\mathrm{d}S - P\mathrm{d}V + \mu\mathrm{d}N \tag{9.44}$$

Proof 9.31 Follows from the theorem of local invertibility.

Hypothesis 9.32 We keep in mind all possible ways of supplying energy to a system. This can occur through ...

1. through contact with other bodies from which heat is transferred,
2. the performance of work,
3. through the addition of material (i.e. increasing the particle number).

The total change in energy therefore takes the form

$$\mathrm{d}E = \delta \mathrm{Q} + \delta W + \delta E_N \tag{9.45}$$

with ...

1. *heat supply* $\delta\mathrm{Q}$,
2. *mechanical work* δW,
3. *energy increase due to material supply* δE_N.

Remark 9.32.1 It is Equation (9.45), that is usually referred to as the *first law of thermodynamics*. Notice however that Equation (9.44) is equivalent to it.

Remark 9.32.2 One usually says that the energy of a system changes according to Equation (9.45) when passing from one equilibrium state to another, infinitesimally close one.

Corollary 9.33 By comparing these two expressions for $\mathrm{d}E$ we find the identifications ...

1. $\delta Q = T\,\mathrm{d}S$ (the change of energy due to heat supply),
2. $\delta W = -P\,\mathrm{d}V$ (the work performed on the system),
3. $\delta E_N = \mu\,\mathrm{d}N$ (the change of the energy when the particle number increases).

Proof 9.33 We interpret the individual terms in Equation (9.44) as in Equation (9.45).

Remark 9.33.1 The relationship

$$\delta Q = T\mathrm{d}S \tag{9.46}$$

is called *the second law of thermodynamics for reversible changes*. Later, we shall formulate the second law in its general form.

9.8.2. Transitioning from Microcanonical to Canonical Ensemble

Legendretransformation to the free energy etc...

Definition 9.34 We define the *heat capacity* of a system in the canonical ensemble as derivative

$$C_{V,N}(T) = T\left(\frac{\partial S}{\partial T}\right)_{V,N}(T) \tag{9.47}$$

Lemma 9.35 The following identities are immediate

$$\begin{aligned} &\bullet \quad C_{V,N}(T) = \partial_T \langle H\rangle_{V,N}(T) \\ &\bullet \quad C_{V,N}(T) = -T\left(\partial_T^2 F\right)(N,T,V) \\ &\bullet \quad C_{V,N}(T) = k_{\mathrm{B}}\beta^2\left(\langle H^2\rangle_{N,V}(T) - \langle H\rangle_{N,V}^2(T)\right) = k_{\mathrm{B}}\beta^2\left\langle (H-\langle H\rangle)^2\right\rangle_{N,V,T} \geq 0 \end{aligned} \tag{9.48}$$

Proof 9.35 TBD; Schwabl for the first two, Nolting page 73 for the third one.

Exception for positivity of the heat capacity: Black holes. Cf. Tong Lecture notes, under Equation (1.11).

Notation 9.36 For a crystal[3] at hand we introduce ...

- the *total mass of the crystal* m (with unit kg)
- the *molar mass of the crystal* M (with unit kg/mol)
- the *number of moles of the crystal* n (with unit mol)

[3]Notice that one can also introduce these definitions for materials that are not crystalline. However, in this document, we will not be concerned with those materials.

Fact 9.37 One has for the number of moles the identity

$$n = \frac{m}{M} \tag{9.49}$$

Notation 9.38 We enumerate the masses of the r atoms in the unit cell as $m_1, m_2, \dots, m_r$.

Fact 9.39 The total mass m of the crystal is given as

$$m = N_x N_y N_z \cdot \sum_{j=1}^{r} m_j. \tag{9.50}$$

Example: A monoatomic crystal, e.g. lead (Pb) in an fcc-lattice with a single atom per unit cell.

- The molar mass M of lead is

$$M_{\text{Pb}} = 207.2\,\text{g/mol} = 207.2 \cdot 10^{-3}\,\text{kg/mol} \tag{9.51}$$

- The total mass m of a lead-crystal (with monoatomic basis in an fcc-lattice) is given as the total number of cells multiplied with the number of lead-atoms per cell (i.e. one) multiplied with the mass of a lead atom i.e.

$$m_{\text{Pb}} = N_x N_y N_z \cdot m_{\text{Pb-atom}} \tag{9.52}$$

with

$$m_{\text{Pb-atom}} = 207.2\,\text{u} = 207.2 \cdot 1.66 \cdot 10^{-27}\,\text{kg} \tag{9.53}$$

Combining these two equations yields for the number of moles n in a crystal with mass m the equation

$$n = \frac{m_{\text{Pb}}}{M_{\text{Pb}}} = N_x N_y N_z \cdot \frac{1.66 \cdot 10^{-27}}{10^{-3}}\text{mol} = N_x N_y N_z \cdot 1.66 \cdot 10^{-24}\,\text{mol} \tag{9.54}$$

Definition 9.40 We define the *molar heat capacity* $C_{V,N,\text{molar}}(T)$ as the amount of heat required to raise the temperature of one mole of a substance by one Kelvin, i.e.

$$C_{V,N,\text{molar}}(T) = \frac{C_{V,N}(T)}{n} \tag{9.55}$$

whereas n is the amount of substance in moles.

Remark 9.40.1 Identities analogous to Lemma 9.35 follow immediately.

Theorem 9.41 TBD

See 03 page 41, 42 for a nice summary of potentials and partition functions.

TBD: 02: page 129 (useful formulae).

TBD: 03: page 22 (equilibrium is found by maximizing the entropy).

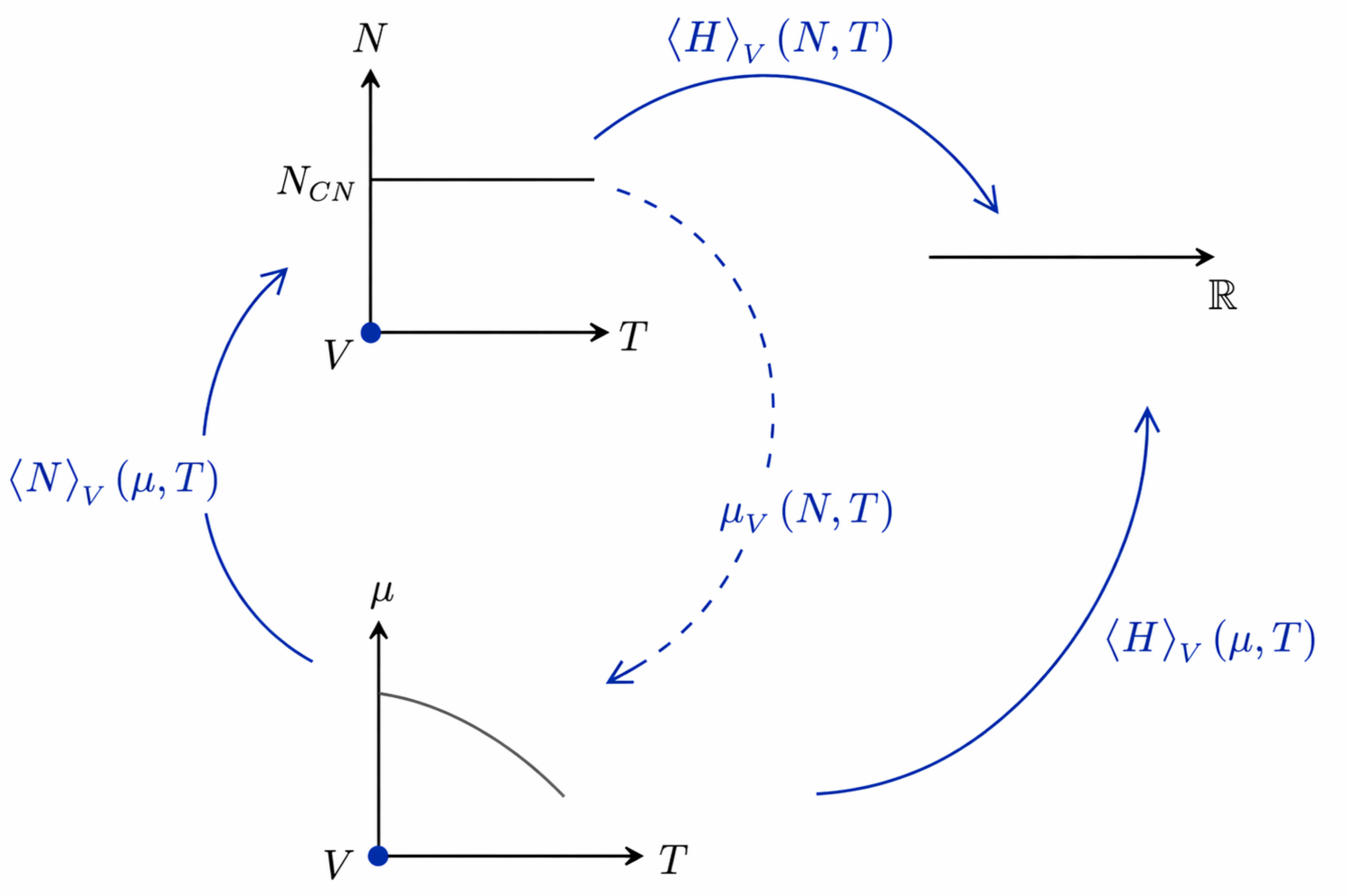
9.8.3. Transitioning from Canonical Ensemble to the Grand Canonical Ensemble

TBElaborated.

Corollary 9.42 Computing the specific heat capacity in the grand canonical ensemble one finds

$$\begin{aligned} C_{N,V}(T) &= \partial_T \langle H \rangle (N,T,V) \\ &= \left(\partial_T \langle H \rangle\right)(\mu(T,V),T,V) + \left(\partial_\mu \langle H \rangle\right)(\mu(T,V),T,V) \cdot \left(\partial_T \mu\right)(N,V,T) \end{aligned} \tag{9.56}$$

whereas the particle number N is kept constant.



Proof 9.42 By using the chain-rule. Cf. TBDigitalized for the details.

Might be Interesting later:

- Callen - Thermodynamics and an Introduction to Thermostatistics, Wiley Sons
- Texier Roux - Physique Statistique, Dunod
- Diu, Guthmann, Lederer, Roulet - Physique Statistique, Hermann
- Jancovici - Statistical Physics and Thermodynamics, McGraw-Hill
- Landau Lifschitz - Statistical Physics, Pergamon Press
- K. Huang - Statistical Mechanics, Wiley
- Barrat Hansen - Basic Concepts for Simple and Complex Fluids, Cambridge University Press

- Somewhere page 13, 14: Stirlings approximation.
- Somewhere page 19, 20, 22: Entropy and Temperature for two combined systems.
- Somewhere page 27, 28: Chapter 4 Canonical Ensemble; Same book page 35, 36 Chapter 5 Grand canonical ensemble.
- Somewhere pdfpage44: "Ueberlässt man ein abgeschlossenes Vielteilchensystem sich selbst, so streben die makroskopisch messbaren Groeszen gegen zeitlich konstante Werte. Dies ist ein Erfahrungssatz."
- Somewhere page 39: "Die Energie E kann nur mit einer endlichen Genauigkeit δE bestimmt werden (...)".
- Page 157 in the template: minimization of the grand canonical potential.

9.9. Auxiliary Identities

For later use we introduce the following notation

Notation 9.43 We introduce analoga for the [F]ermi- / [B]ose- / [ζ]-functions that take dimensionless arguments as

$$\mathrm{F}(x) = \frac{1}{\mathrm{e}^x + 1} \qquad / \qquad \mathrm{B}(x) = \frac{1}{\mathrm{e}^x - 1} \qquad / \qquad \mathrm{n}^\zeta(x) = \frac{1}{\mathrm{e}^x - \zeta} \tag{9.57}$$

with $\zeta \in \{-1, +1\}$ as usual.

Lemma 9.44 One obtains the identities

$$\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{n}^\zeta(x) = -\mathrm{n}^\zeta(x)\left(1 + \zeta \mathrm{n}^\zeta(x)\right) \equiv -\frac{1}{4}\zeta \mathrm{ch}^2(x/2) \equiv \begin{cases} -\frac{1}{4}\,\mathrm{sech}^2\,(x/2) & \text{for } \zeta = -1 \\ -\frac{1}{4}\,\mathrm{csch}^2\,(x/2) & \text{for } \zeta = +1 \end{cases} \tag{9.58}$$

Proof 9.44 See Appendix, Proofs, page 317.

Notation 9.45 For a temperature $\beta = \frac{1}{k_\mathrm{B}T}$ we write for the [F]ermi- / [B]ose- / [ζ]-functions

$$\mathrm{F}_{\hbar\beta}(\omega) = \frac{1}{\mathrm{e}^{\hbar\beta\omega} + 1} \qquad / \qquad \mathrm{B}_{\hbar\beta}(\omega) = \frac{1}{\mathrm{e}^{\hbar\beta\omega} - 1} \qquad / \qquad \mathrm{n}^\zeta_{\hbar\beta}(\omega) = \frac{1}{\mathrm{e}^{\hbar\beta\omega} - \zeta} \tag{9.59}$$

with $\zeta \in \{-1, +1\}$ as usual.

Remark 9.45.1 Notice that one has the relation $\mathrm{n}^\zeta_{\hbar\beta}(\omega) = \mathrm{n}^\zeta(\beta\hbar\omega)$, etc.

For reasons that are difficult to motivate in advance[4], we will need the following two identities.

Lemma 9.46 For $\omega_n \in \Omega_{-1}$ one has the identities

$$\mathrm{F}_{\hbar\beta}(\bullet + \mathrm{i}\omega_n) = -\,\mathrm{B}_{\hbar\beta}(\bullet), \qquad \mathrm{F}_{\hbar\beta}(\mathrm{i}\omega_n - \bullet) = \mathrm{B}_{\hbar\beta}(\bullet) + 1 \tag{9.60}$$

further one has

$$\mathrm{F}_{\hbar\beta}(-\bullet) = 1 - \mathrm{F}_{\hbar\beta}(\bullet). \tag{9.61}$$

Proof 9.46 See Appendix, Proofs, page 317.

[4] These identities will be needed in different situations, for example to compute the self-energy or the heat capacity of an interacting electron-phonon-system.

Remark 9.46.1 Notice further that one also has the trivial identities for $\omega_n \in \Omega_{+1}$

$$\mathrm{F}_{\hbar\beta}(\bullet \pm \mathrm{i}\omega_n) = \mathrm{F}_{\hbar\beta}(\bullet) \tag{9.62}$$

Lemma 9.47 One has the derivatives

$$\frac{\mathrm{d}}{\mathrm{d}T}\,\mathrm{n}^{\zeta}_{\beta}(\hbar\omega) = \frac{\hbar\omega}{k_{\mathrm{B}}T^2}\,\mathrm{n}^{\zeta}_{\beta}(\hbar\omega)\left(1+\zeta\,\mathrm{n}^{\zeta}_{\beta}(\hbar\omega)\right) = \begin{cases} \dfrac{\hbar\omega}{4k_{\mathrm{B}}T^2}\,\mathrm{sech}^2\left(\dfrac{\hbar\omega}{2k_{\mathrm{B}}T}\right) & \text{for } \zeta = -1 \\ \dfrac{\hbar\omega}{4k_{\mathrm{B}}T^2}\,\mathrm{csch}^2\left(\dfrac{\hbar\omega}{2k_{\mathrm{B}}T}\right) & \text{for } \zeta = +1 \end{cases} \tag{9.63}$$

Proof 9.47 See Appendix, Proofs, page 317.

Corollary 9.48 For $z \in \mathbb{C}$ we find the identity

$$\frac{\mathrm{d}}{\mathrm{d}z}\mathrm{n}^{\zeta}_{\hbar\beta}(z) = -\hbar\beta\mathrm{n}^{\zeta}_{\hbar\beta}(z)\left(1+\zeta\mathrm{n}^{\zeta}_{\hbar\beta}(z)\right) \equiv -\frac{\hbar\beta}{4}\zeta\mathrm{ch}^2(\hbar\beta z/2) \equiv \begin{cases} -\dfrac{\hbar\beta}{4}\,\mathrm{sech}^2\left(\hbar\beta z/2\right) & \text{for } \zeta = -1 \\ -\dfrac{\hbar\beta}{4}\,\mathrm{csch}^2\left(\hbar\beta z/2\right) & \text{for } \zeta = +1 \end{cases} \tag{9.64}$$

Proof 9.48 Follows directly from Lemma 9.44 with the chain-rule. Alternatively, by direct computation, see Appendix, Proofs, page 318.

10. Teaser for Equilibrium Green-Functions

"The Green's function method is one of the most powerful and versatile formalisms in physics"

Stefanucci / Leuween in their book [TBReferenced properly]

The ultimate goal of equilibrium Green-function-techniques is to *describe* (in the widest sense of the word) systems as in Notation 8.1 in the canonical ensemble, or in the grand canonical ensemble, or in hybrid ensembles.

10.1. General Setup - Pure Systems

Ingredient I: Many-Particle-Hilbert-Space, States and Operators

Notation 10.1 We consider a system with statistics ζ and nature η, and an underlying many-particle Hilbert-space $\mathscr{H} \equiv \mathscr{H}_\eta$ that is generated by creation- and annihilation-operators

$$\mathscr{H} = \mathrm{span}_{\mathbb{C}} \left\{ \prod_{j=1}^{\mathrm{d}_\eta} \frac{1}{\sqrt{n_j!}} \left(c_j^\dagger\right)^{n_j} |\,\mathrm{vac.}\rangle \equiv |n_1, n_2, \dots, n_{\mathrm{d}_\eta}\rangle \,\middle|\, n_1, n_2, \dots, n_{\mathrm{d}_\eta} \in \mathbb{N}_\zeta \right\} \tag{10.1}$$

with a number of states to occupy

$$\mathrm{d}_\eta = \begin{cases} \mathrm{d}_{m=0} & \text{for } \eta = 0, \\ \mathrm{d}_{\mathcal{H}} & \text{for } \eta = 1, \end{cases} \tag{10.2}$$

generated through creation- and annihilation-operators acting on a vacuum. The Hamiltonian of the system is then given by a contribution H_0 that we believe to understand and perturbations V as

$$\begin{aligned} H &= H_0 && + \quad V \\ &= \sum_{i,j} \underset{ij}{H_0} c_i^\dagger c_j && + \quad V(\mathbf{c}^\dagger, \mathbf{c}) \end{aligned} \tag{10.3}$$

with creation- and annihilation-operators $\left\{ c_1, c_1^\dagger, c_2, c_2^\dagger, \dots, c_{\mathrm{d}_\eta}, c_{\mathrm{d}_\eta}^\dagger \right\}$ obey the statistical commutation-relations

$$\left[c_i, c_j\right]_\zeta = \left[c_i^\dagger, c_j^\dagger\right]_\zeta = 0, \qquad \left[c_i, c_j^\dagger\right]_\zeta = \delta_{i,j} \tag{10.4}$$

Remark 10.1.1 Usually one switches the notation to a more *generic case*, and just writes generalized (system-specific) indices $\left\{ c_1, c_1^\dagger, c_2, c_2^\dagger, \dots, c_{\mathrm{d}_\eta}, c_{\mathrm{d}_\eta}^\dagger \right\} \equiv \bigcup_{\alpha \in \mathscr{A}} \left\{ c_\alpha, c_\alpha^\dagger \right\}$ with some index-set $\mathscr{A}$.

Ingredient II: The Natural Ensemble for Pure Systems

For systems with nature $\eta = 0$, we want to use the canonical ensemble and ignore the particle-number dependence, which is (on an enumerational level) equivalent to the grand canonical ensemble with $\mu = 0$. For systems with nature $\eta = 1$, we want to use the grand canonical ensemble with μ dictating the particle number. We combine these two cases into a single notation, by introducing the "natural ensemble".

Notation 10.2 We consider a system with nature $\eta \in \{0, 1\}$ in the grand canonical ensemble, with a volume V, temperature T and chemical potential

$$\begin{cases} \mu = 0 & \text{for } \eta = 0 \\ \mu \underset{\text{in general}}{\neq} 0 & \text{for } \eta = 1. \end{cases} \tag{10.5}$$

This gives us $2 + \eta$ natural variables.

Fact 10.3 In the absence of „spontaneous symmetry breaking“[1] expectation-values in the grand canonical ensemble are evaluated w.r.t. the density

$$\rho = \frac{\mathrm{e}^{-\beta(H-\eta\mu N)}}{\mathrm{Tr}_{\mathscr{H}}\, \mathrm{e}^{-\beta(H-\eta\mu N)}} \equiv \frac{1}{Z_\eta}\mathrm{e}^{-\beta(H-\eta\mu N)} \equiv \begin{cases} \dfrac{1}{Z}\mathrm{e}^{-\beta H} & \text{for } \eta = 0 \\ \dfrac{1}{Z}\mathrm{e}^{-\beta(H-\mu N)} & \text{for } \eta = 1. \end{cases}, \tag{10.6}$$

whereas we introduced the *natural partition-function*

$$Z_\eta \equiv \begin{cases} Z(T, V) & \text{for } \eta = 0 \\ Z(\mu, T, V) & \text{for } \eta = 1. \end{cases} \tag{10.7}$$

Remark 10.3.1 Recall that the state above represents a solution to the stationary von Neumann equation.

Remark 10.3.2 Although in the case of „spontaneous symmetry breaking“, the evaluation of observables w.r.t. the state above yields incorrect results, one can oftentimes however use the state above, and break the underlying symmetry „by hand“, in order to recover the „correct“ physical (ground) state (or, in the case that we will encounter later, the correct physical „field theory“) again.

Fact 10.4 The thermodynamic potential[2] in the natural ensemble is given by the so-called *natural energy*

$$\Omega_\eta \equiv \begin{cases} \Omega(T, V) = -\dfrac{\mathrm{Log}\, Z(T, V)}{\beta} & \text{for } \eta = 0 \\ \Omega(\mu, T, V) = -\dfrac{\mathrm{Log}\, Z(\mu, T, V)}{\beta} & \text{for } \eta = 1, \end{cases} \tag{10.8}$$

[1] We will introduce this concept in the chapter about superconductivity.

[2] Recall that the thermodynamic potential is a scalar-quantity which is used to represent the thermodynamic state of a system.

from which one can conclude the usual relations for the entropy, pressure and, in case of $\eta = 1$, the conjugated variable of N as

$$\frac{\partial \Omega_\eta}{\partial T} = -S, \qquad \frac{\partial \Omega_\eta}{\partial V} = -P, \quad \text{and if } \eta = 1 : \qquad \frac{\partial \Omega_\eta}{\partial \mu} = -N \tag{10.9}$$

whereas both sides of these respective equations depend on the $2 + \eta$ variables, which has been omitted for notational clarity.

Remark 10.4.1 Notice that the thermodynamic potential reduces to the Helmholtz free energy F for $\eta = 0$, and to the Landau-potential Ω for $\eta = 1$.

10.2. General Setup - Mixed Systems

We discuss the case of mixed systems at the example of mixture with massive and massless particles. The generalizations to other mixtures is straight-forward.

Ingredient I: Hilbert-Space, States and Operators

Notation 10.5 We consider a system, which is made of two subsystems, one with Fock-space $\mathscr{H}_{m>0} \equiv \mathscr{F}_\zeta$ with generators $\left\{ c_{\alpha_1}, c^\dagger_{\alpha_1}, c_{\alpha_2}, c^\dagger_{\alpha_2}, \ldots, c_{\alpha_{d_{m>0}}}, c^\dagger_{\alpha_{d_{m>0}}} \right\}$ and statistical operators, and another one with Hilbert-space $\mathscr{H}_{m=0}$ with bosonic generators $\left\{ c_{\beta_1}, c^\dagger_{\beta_1}, c_{\beta_2}, c^\dagger_{\beta_2}, \ldots, c_{\beta_{\mathrm{d}_{m=0}}}, c^\dagger_{\beta_{\mathrm{d}_{m=0}}} \right\}$. The total Hilbert-space $\mathscr{H}$ of the system is given as tensor-product

$$\mathscr{H} = \mathscr{H}_{m>0} \otimes \mathscr{H}_{m=0}$$

with basis[3] and scalar-product as usual. The generic Hamiltonian reads

$$\begin{aligned} H_{\text{tot}} &\equiv \qquad H_{m>0} && + \qquad H_{m=0} && + V_{\text{int}} \\ &\equiv \ \underset{m>0}{H_{\ 0}} + V_{m>0} && + \ \underset{m=0}{H_{\ 0}} + V_{m=0} && + V_{\text{int}} \\ &\equiv \sum_{i,j} \underset{m>0}{H_{0,ij}}\, c^\dagger_{\alpha_i} c_{\alpha_j} + V_{m>0}(\mathbf{c}^\dagger_\alpha, \mathbf{c}_\alpha) && + \sum_{i,j} \underset{m=0}{H_{0,ij}}\, c^\dagger_{\beta_i} b_{\beta_j} + V_{m=0}(\mathbf{c}^\dagger_\beta, \mathbf{c}_\beta) && + V_{\text{int}}(\mathbf{c}^\dagger, \mathbf{c}) \end{aligned} \tag{10.10}$$

with interactions as in Fact 8.4.

Ingredient II: The Hybrid Ensemble

Notation 10.6 We further restrict our attention to the case in which the above mentioned system is made up of two sub-systems which have a mutual volume V and temperature T and a chemical potential μ for the massive particles. This gives us the three natural variables (μ, T, V).

[3] Cf. Flensberg page 276.

Fact 10.7 In the absence of „spontaneous symmetry breaking" (to be introduced in a later chapter) expectation-values in the hybrid ensemble are evaluated w.r.t. the *hybrid density*

$$\rho = \frac{\mathrm{e}^{-\beta\left(H_{m>0}-\mu N_{m>0}+H_{\mathrm{w}}+V_{\mathrm{int}}\right)}}{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta\left(H_{\mathrm{p}}-\mu N+H_{\mathrm{w}}+V_{\mathrm{int}}\right)}} \equiv \frac{1}{Z}\mathrm{e}^{-\beta\left(H_{\mathrm{p}}-\mu N+H_{\mathrm{w}}+V_{\mathrm{int}}\right)}, \tag{10.11}$$

whereas we introduced the *mixed partition-function* $Z \equiv Z(N, \mu, T, V)$.

Remark 10.7.1 Recall that the state above represents a solution to the stationary von Neumann equation.

Remark 10.7.2 Although in the case of „spontaneous symmetry breaking", the evaluation of observables w.r.t. the state above yields incorrect results, one can oftentimes however use the state above, and break the underlying symmetry „by hand", in order to recover the „correct" physical state (or, in the case that we will encounter later, the correct physical „field theory") again.

Remark 10.7.3 Thermodynamic Relations?

10.3. Applications (Teaser)

In [Negele and Orland]:

> "*The ultimate objective of the quantum theory of many-particle systems is to understand experimentally observable properties of a diverse range of physical systems.*"

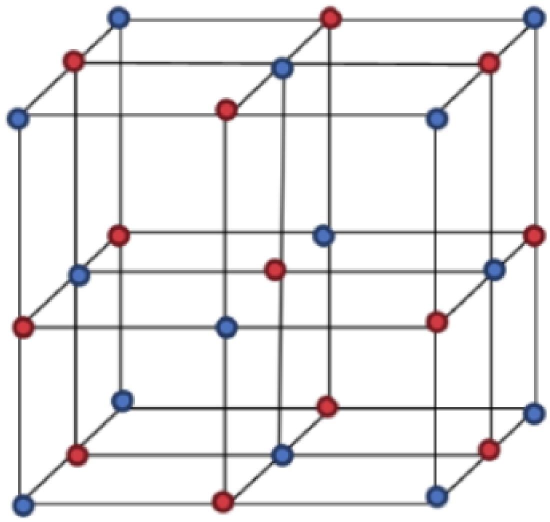
This quote from Negele and Orland is hitting very hard, because most of the theory and formalism presented in this work applies to *any crystalline system* (there are more than 200.000 known crystals in the Inorganic Crystal Structure Database[4]). Lying out the tools to study all of these systems is apparently a very powerful formalism.

It is a standard-fact, that in many experimental setups that are in some sense "close" to thermal equilibrium, observable quantities can be expressed as „equilibrium Green-functions“ (will be introduced later). We list some examples whereas the meaning of the individual symbols will become clear throughout this document.

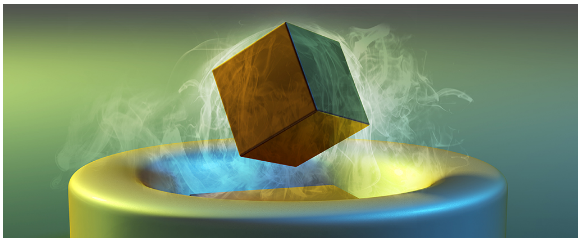
[4]`https://icsd.fiz-karlsruhe.de/`

System („Cartoon“):

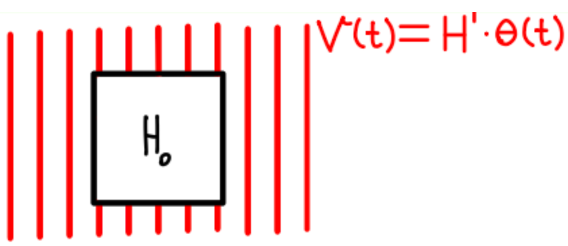
- **Any arbitrary Crystal:**



- **Superconductor**:



- **An *arbitrary* observable A of a System subjected to a „weak“ Perturbation H':**



Observable(s):

- **Electronic Density of States / Band-Structure:**

$$\rho_{n\mathbf{k}}(\omega) = -\frac{\hbar}{\pi}\,\mathrm{Im}\lim_{\epsilon\to 0} G^{\mathrm{ret}}_{\substack{n,n\\ \mathbf{k},\mathbf{k}}}(\omega + \mathrm{i}\epsilon)$$

- **Magnetization[5]:**

$$\langle M\rangle \propto \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in \mathrm{Q}\\ n\in\{1,2,\dots\}\\ \omega\in(-\infty,+\infty)}} G^{<}_{\substack{n,n\\ \mathbf{k},\mathbf{k}\\ \uparrow,\uparrow}}(\omega) - G^{<}_{\substack{n,n\\ \mathbf{k},\mathbf{k}\\ \downarrow,\downarrow}}(\omega)$$

- **Electronic / Phononic / Total Heat Capacity:**

$$C_{N,V}(T) = \partial_T\langle H\rangle_{N,T} = \text{"sum of Green-functions"}^{6}$$

- **Electronic Conductivity Tensor:**

$$\sigma(\omega) = \text{"Green-function with current-operators"}$$

- **Thermal Conductivity Tensor:**

$$\kappa = \text{"Green-function with energy-current-operators"}$$

- **Order Parameter[7] (in Eliashberg-Theory):**

$$\Delta_n(k) \propto \mathscr{G}^{\;n}_{\substack{1,2\\ \text{Gorkov}}}(k)$$

- **Kubo-Formula:**

$$\langle A\rangle(t) = \langle A\rangle(t=-\infty) + \int \mathrm{d}t'\, G^{\mathrm{ret}}_{AH'}(t,t') + \mathcal{O}(H')^2$$

[5]Note that for this to be unequal to zero one has to have spin orbit coupling or spontaneous symmetry breaking.
[6]Depending on the model, there can be a different number and different types of Green-functions.
[7]Here again, one has to ("spontaneously") break a symmetry in order to get non-zero values.

11. Green-Functions at Zero Temperature

Can be obtained from Green-Functions at Finite-Temperature (see next chapter) and taking the limit $T \to 0$, as elucidated in the Appendix, Section 33.3. A derivation of some standard-identities that one oftentimes encounters in literature on many-body-Green-function-techniques is also provided in the Appendix.

12. Real-Time-Two-Operator-Green-Functions at Finite Temperature

12.1. Two-Operator Green-Functions without Periodicity

For reasons that are impossible to motivate in advance, we introduce the following notations and functions.

Definition 12.1 For a pair of two operators (X, X') and times $t,\ t' \in \mathbb{R}$ we define the *real-time-two-operator-Green-function* in its various notations as *expectation-value*

$$
\begin{aligned}
G^{++}_{XX'}(t,t') \equiv G^{\mathbb{T}}_{XX'}(t,t') \equiv G_{XX'}(t,t') \\
&\equiv -\frac{\mathrm{i}}{\hbar}\left\langle \mathrm{T}_t\, X\,(t)\, X'\left(t'\right)\right\rangle \\
&\equiv -\frac{\mathrm{i}}{\hbar}\frac{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta\Xi}\,\mathrm{T}_t\, X\,(t)\, X'\,(t')}{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta\Xi}} \\
&\equiv -\frac{\mathrm{i}}{\hbar}\frac{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta\Xi}\left[\theta(t-t')\,X\,(t)\,X'\,(t') + \mathrm{S}_{X,X'}\,\theta(t'-t)X'\,(t')\,X\,(t)\right]}{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta\Xi}}
\end{aligned}
\tag{12.1}
$$

with time-ordering-operator and ensemble-Heisenberg-time-evolution as usual.

Definition 12.2 We introduce various *Green-functions corresponding to the operator-pair* (X, X') as

- $G_{X,X'}(t,t') \equiv G^{++}_{X,X'}(t,t') \equiv G^{\mathbb{T}}_{X,X'}(t,t') \equiv -\frac{\mathrm{i}}{\hbar}\langle \mathrm{T}_t\, X(t)X'(t')\rangle$
- $G^{+-}_{X,X'}(t,t') \equiv G^{<}_{X,X'}(t,t') \equiv -\frac{\mathrm{i}}{\hbar}S_{X,X'}\langle X'(t')X(t)\rangle$
- $G^{-+}_{X,X'}(t,t') \equiv G^{>}_{X,X'}(t,t') \equiv -\frac{\mathrm{i}}{\hbar}\langle X(t)X'(t)\rangle$
- $G^{--}_{X,X'}(t,t') \equiv G^{\overline{\mathbb{T}}}_{X,X'}(t,t') \equiv -\frac{\mathrm{i}}{\hbar}\langle \overline{\mathrm{T}}_t X(t)X'(t')\rangle$
- $G^{\mathrm{ret}}_{X,X'}(t,t') = -\frac{\mathrm{i}}{\hbar}\theta(t-t')\langle\left[X(t),\ X'(t')\right]_{\mathrm{S}}\rangle$
- $G^{\mathrm{adv}}_{X,X'}(t,t') = +\frac{\mathrm{i}}{\hbar}\theta(t'-t)\langle\left[X(t),\ X'(t')\right]_{\mathrm{S}}\rangle$

Lemma 12.3 The Green-functions defined above only depend on the time-difference, hence one can write

$$G^{\blacksquare}_{X,X'}(t,t') \equiv G^{\blacksquare}_{X,X'}(t-t'). \tag{12.2}$$

Proof 12.3 See Appendix, Proofs, page 318.

Remark 12.3.1 For the discussion of Feynman-diagrams, it is fruitful to introduce this redundant, extra time-argument.

Lemma 12.4 The following identities are immediate, if one neglects boundary-conditions[1]

- $G^{\mathbb{T}}_{X,X'}(t,t') = +\theta(t-t')\, G^{>}_{X,X'}(t,t') + \theta(t'-t)\, G^{<}_{X,X'}(t,t')$,
- $G^{\overline{\mathbb{T}}}_{X,X'}(t,t') = +\theta(t'-t)\, G^{>}_{X,X'}(t,t') + \theta(t-t')\, G^{<}_{X,X'}(t,t')$,
- $G^{\mathrm{ret}}_{X,X'}(t,t') = \left(G^{++}_{X,X'} - G^{+-}_{X,X'}\right)(t,t') = \left(G^{-+}_{X,X'} - G^{--}_{X,X'}\right)(t,t') = \theta(t-t')\left(G^{-+}_{X,X'} - G^{+-}_{X,X'}\right)(t,t')$,
- $G^{\mathrm{adv}}_{X,X'}(t,t') = \left(G^{++}_{X,X'} - G^{-+}_{X,X'}\right)(t,t') = \left(G^{+-}_{X,X'} - G^{--}_{X,X'}\right)(t,t') = \theta(t'-t)\left(G^{+-}_{X,X'} - G^{-+}_{X,X'}\right)(t,t')$,
- $G^{\mathbb{T}}_{X,X'}(t,t') + G^{\overline{\mathbb{T}}}_{X,X'}(t,t') = G^{>}_{X,X'}(A,A') + G^{<}_{X,X'}(t,t')$,
- $G^{\mathrm{ret}}_{X,X'}(t,t') - G^{\mathrm{adv}}_{X,X'}(t,t') = G^{>}_{X,X'}(t,t') - G^{<}_{X,X'}(t,t')$,

and if one has $X^\dagger = X'$ one also finds

- $G^{\mathrm{ret}}_{X,X'}(t,t') = G^{\mathrm{adv}}_{X,X'}(t',t)^\dagger$.

whereas the last equation is to be understood as a generalized matrix.

Proof 12.4 See Appendix, Proofs, page 318.

Remark 12.4.1 We emphasize a comment from Kamenev [TBReferenced properly]:

> "*I can see how one can be overwhelmed by a number of different Green-functions, rules to follow and by the length of the calculations.*"

[1]This is what one usually does in thermal quantum field theory. There are some comments about this in [Mahan].

12.2. Pure One-Body Green-Functions without Periodicity

Definition 12.5 We consider $X = c_\alpha$ and $X' = c^\dagger_{\alpha'}$ for multi-indices $A \equiv (\alpha, t) \in \mathscr{A} \times \mathbb{R}$, with

$$\left[c_\alpha, c^\dagger_{\alpha'}\right]_\zeta = \delta_{\alpha,\alpha'}, \qquad \mathrm{S}(c_\alpha) = \mathrm{S}(c^\dagger_{\alpha'}) = \zeta.$$

Then we write for the *real-time-one-body-Green-function*[2], *the lesser, greater, anti-time-ordered, retarded and advanced Green's-functions* in their various shapes as *expectation-value*

- $G(A, A') \equiv G^{++}(A, A') \equiv G^{\mathbb{T}}(A, A') \equiv -\frac{\mathrm{i}}{\hbar} \left\langle \mathrm{T}_t\, c_\alpha(t) c^\dagger_{\alpha'}(t') \right\rangle,$
- $G^{+-}(A, A') \equiv G^{<}(A, A') \equiv -\frac{\mathrm{i}}{\hbar}\zeta \left\langle c^\dagger_{\alpha'}(t') c_\alpha(t) \right\rangle,$
- $G^{-+}(A, A') \equiv G^{>}(A, A') \equiv -\frac{\mathrm{i}}{\hbar} \left\langle c_\alpha(t) c^\dagger_{\alpha'}(t') \right\rangle,$
- $G^{--}(A, A') \equiv G^{\overline{\mathbb{T}}}(A, A') \equiv -\frac{\mathrm{i}}{\hbar} \left\langle \overline{\mathrm{T}}_t c_\alpha(t) c^\dagger_{\alpha'}(t') \right\rangle,$
- $G^{\mathrm{ret}}(A, A') = -\frac{\mathrm{i}}{\hbar}\theta(t - t') \left\langle \left[c_\alpha(t),\, c^\dagger_{\alpha'}(t')\right]_\zeta \right\rangle,$
- $G^{\mathrm{adv}}(A, A') = +\frac{\mathrm{i}}{\hbar}\theta(t' - t) \left\langle \left[c_\alpha(t),\, c^\dagger_{\alpha'}(t')\right]_\zeta \right\rangle.$

with time-ordering-operator and ensemble-Heisenberg-time-evolution as usual.

Remark 12.5.1 Among physicists, the usual *interpretation* of the causal Green-function is as follows[3]

> *The one-particle Green's function describes a Gedankenexperiment which can never happen in practice. One creates a particle at time t' in the state α' and then (if $t > t'$) annihilates another particle at time t in the state α.*

Lemma 12.6 We obtain for the time-ordered, the anti-time-ordered, the retarded and the advanced Green-functions the equations of motion, and write them in wise foresight in sets as

$$\partial_t G^{\mathbb{T}}_{\alpha,\alpha'}(t) = -\frac{\mathrm{i}}{\hbar}\delta(t)\delta_{\alpha,\alpha'} \quad - \quad \frac{1}{\hbar}\left\langle \mathrm{T}_t \left[c_\alpha, \frac{\Xi}{\hbar}\right]_{\mathrm{H}}(t) c^\dagger_{\alpha'}(0) \right\rangle, \tag{12.3}$$

$$\partial_t G^{\mathrm{ret}}_{\alpha,\alpha'}(t) = -\frac{\mathrm{i}}{\hbar}\delta(t)\delta_{\alpha,\alpha'} \quad - \quad \frac{1}{\hbar}\theta(t)\left\langle \left[\left[c_\alpha, \frac{\Xi}{\hbar}\right]_{\mathrm{H}}(t), c^\dagger_{\alpha'}(0)\right]_\zeta \right\rangle, \tag{12.4}$$

and as

$$\partial_t G^{\overline{\mathbb{T}}}_{\alpha,\alpha'}(t) = \frac{\mathrm{i}}{\hbar}\delta(t)\delta_{\alpha,\alpha'} \quad - \quad \frac{1}{\hbar}\left\langle \overline{\mathrm{T}}_t \left[c_\alpha, \frac{\Xi}{\hbar}\right]_{\mathrm{H}}(t) c^\dagger_{\alpha'}(0) \right\rangle, \tag{12.5}$$

[2]Other names of this function are *time-ordered Green-function*, or *causal Green-function*. One also calls this Green-function the *propagator* of the underlying physical system.

[3]From [2], p. 167, with adapted notation to fit our purposes.

$$\partial_t G^{\mathrm{adv}}_{\alpha,\alpha'}(t) = -\frac{\mathrm{i}}{\hbar}\delta(-t)\delta_{\alpha,\alpha'} \quad + \quad \frac{1}{\hbar}\theta(-t)\left\langle \left[\left[c_\alpha, \frac{\Xi}{\hbar}\right]_{\mathrm{H}}(t), c^\dagger_{\alpha'}(0)\right]_\zeta \right\rangle . \tag{12.6}$$

Proof 12.6 See Appendix, Proofs, page 320.

Remark 12.6.1 Notice the symmetric behaviour of total minus-signs in these expressions.

Example: Pure, Non-Interacting Systems

Example: A Quantum Dot with Nature η and Statistics ζ

Can be inferred as a special case of the diagonal gas.

Example: A Diagonal Gas with Nature η and Statistics ζ

Theorem 12.7 Introducing the variables $\xi_\alpha \equiv \epsilon_\alpha - \eta\mu$ it follows

$$\begin{aligned}
G^{\mathbb{T}}_{\substack{0\\ \alpha,\alpha'}}(t) &= -\frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}t\xi_\alpha/\hbar}\delta_{\alpha,\alpha'}\left[\theta(t)\left(1+\zeta\,\mathrm{n}^\zeta_\beta(\xi_\alpha)\right)+\theta(-t)\zeta\mathrm{n}^\zeta_\beta(\xi_\alpha)\right], & G^{\mathbb{T}}_{\substack{0\\ \alpha,\alpha'}}(\omega) &= +\frac{1}{\hbar}\delta_{\alpha,\alpha'}\lim_{\epsilon\to 0}\frac{1+\zeta\mathrm{n}^\zeta_{\hbar\beta}(\omega)}{\left(\omega+\mathrm{i}\epsilon-\xi_\alpha/\hbar\right)}-\frac{\zeta\mathrm{n}^\zeta_{\hbar\beta}(\omega)}{\left(\omega-\mathrm{i}\epsilon-\xi_\alpha/\hbar\right)},\\
G^{<}_{\substack{0\\ \alpha,\alpha'}}(t) &= -\frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}t\xi_\alpha/\hbar}\delta_{\alpha,\alpha'}\zeta\mathrm{n}^\zeta_\beta(\xi_\alpha), & G^{<}_{\substack{0\\ \alpha,\alpha'}}(\omega) &= -\frac{\mathrm{i}}{\hbar}\delta_{\alpha,\alpha'}\zeta\mathrm{n}^\zeta_\beta(\xi_\alpha)2\pi\delta(\omega-\xi_\alpha/\hbar),\\
G^{>}_{\substack{0\\ \alpha,\alpha'}}(t) &= -\frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}t\xi_\alpha/\hbar}\delta_{\alpha,\alpha'}\left(1+\zeta\,\mathrm{n}^\zeta_\beta(\xi_\alpha)\right), & G^{>}_{\substack{0\\ \alpha,\alpha'}}(\omega) &= -\frac{\mathrm{i}}{\hbar}\delta_{\alpha,\alpha'}\left(1+\zeta\,\mathrm{n}^\zeta_\beta(\xi_\alpha)\right)2\pi\delta(\omega-\xi_\alpha/\hbar),\\
G^{\overline{\mathbb{T}}}_{\substack{0\\ \alpha,\alpha'}}(t) &= -\frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}t\xi_\alpha/\hbar}\delta_{\alpha,\alpha'}\left[\theta(-t)\left(1+\zeta\,\mathrm{n}^\zeta_\beta(\xi_\alpha)\right)+\theta(+t)\zeta\mathrm{n}^\zeta_\beta(\xi_\alpha)\right], & G^{\overline{\mathbb{T}}}_{\substack{0\\ \alpha,\alpha'}}(\omega) &= +\frac{1}{\hbar}\delta_{\alpha,\alpha'}\lim_{\epsilon\to 0}-\frac{1+\zeta\mathrm{n}^\zeta_{\hbar\beta}(\omega)}{\left(\omega-\mathrm{i}\epsilon-\xi_\alpha/\hbar\right)}+\frac{\zeta\mathrm{n}^\zeta_{\hbar\beta}(\omega)}{\left(\omega+\mathrm{i}\epsilon-\xi_\alpha/\hbar\right)},\\
G^{\mathrm{ret}}_{\substack{0\\ \alpha,\alpha'}}(t) &= -\frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}t\xi_\alpha/\hbar}\delta_{\alpha,\alpha'}\theta(t), & G^{\mathrm{ret}}_{\substack{0\\ \alpha,\alpha'}}(\omega+\mathrm{i}\epsilon) &= +\frac{1}{\hbar}\delta_{\alpha,\alpha'}\frac{1}{\omega+\mathrm{i}\epsilon-\xi_\alpha/\hbar},\\
G^{\mathrm{adv}}_{\substack{0\\ \alpha,\alpha'}}(t) &= -\frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}t\xi_\alpha/\hbar}\delta_{\alpha,\alpha'}\theta(-t), & G^{\mathrm{adv}}_{\substack{0\\ \alpha,\alpha'}}(\omega-\mathrm{i}\epsilon) &= +\frac{1}{\hbar}\delta_{\alpha,\alpha'}\frac{1}{\omega-\mathrm{i}\epsilon-\xi_\alpha/\hbar},
\end{aligned} \tag{12.7}$$

Proof 12.7 See Appendix, Proofs, page 322.

Remark 12.7.1 Notice that we can conclude the expectation-value for the number-operator from the lesser Green-function as

$$\langle c^\dagger_\alpha c_\alpha\rangle = -\frac{\hbar}{\mathrm{i}}\zeta G^{<}_{\alpha,\alpha}(t=0) = \mathrm{n}^\zeta_\beta(\xi_\alpha). \tag{12.8}$$

Example: A Non-Diagonal Gas with Nature η and Statistics ζ

A direct computation of all the individual Green-functions from the definition is (to the authors knowledge) impossible. One can however determine the individual Green-functions by using Theorem TBReferenced for the imaginary time retarded Green-function and subsequently using Green's chart (Figure TBReferenced) to determine the individual Green-functions.

Fermionic Examples

Example: A Spin-$\frac{1}{2}$-Particle

Lemma 12.8 One finds the expectation-values

$$\begin{aligned}
\langle d_\uparrow^\dagger d_\uparrow\rangle &= \frac{\mathrm{e}^{-\beta\xi_\uparrow}+\mathrm{e}^{2\mu\beta}}{1+\mathrm{e}^{-\beta\xi_\uparrow}+\mathrm{e}^{-\beta\xi_\downarrow}+\mathrm{e}^{2\mu\beta}}, &\qquad \langle d_\uparrow d_\uparrow^\dagger\rangle &= 1-\langle d_\uparrow^\dagger d_\uparrow\rangle,\\
\langle d_\downarrow^\dagger d_\downarrow\rangle &= \frac{\mathrm{e}^{-\beta\xi_\downarrow}+\mathrm{e}^{2\mu\beta}}{1+\mathrm{e}^{-\beta\xi_\uparrow}+\mathrm{e}^{-\beta\xi_\downarrow}+\mathrm{e}^{2\mu\beta}}, &\qquad \langle d_\downarrow d_\downarrow^\dagger\rangle &= 1-\langle d_\downarrow^\dagger d_\downarrow\rangle.
\end{aligned}$$

Proof 12.8 See Appendix, TBReferenced.

Theorem 12.9 It follows

$$\begin{aligned}
G^{\mathbb{T}}(t) &= \text{cf. Corollary 28.11}, & G^{\mathbb{T}}(\omega+\mathrm{i}\epsilon) &= \text{TBD},\\
G^{<}(t) &= \text{cf. Lemma 28.9.} & G^{<}(\omega) &= \text{TBD},\\
G^{>}(t) &= \text{cf. Lemma 28.10} & G^{>}(\omega) &= \text{TBD},\\
G^{\overline{\mathbb{T}}}(t) &= \text{TBD}, & G^{\overline{\mathbb{T}}}(\omega+\mathrm{i}\epsilon) &= \text{TBD},\\
G^{\mathrm{ret}}(t) &= \text{TBD}, & G^{\mathrm{ret}}(\omega+\mathrm{i}\epsilon) &= \frac{1}{\hbar}\frac{\left(\omega+\mathrm{i}\epsilon-\frac{\epsilon_0-\mu}{\hbar}\right)\sigma_0+\left(\frac{\vec{\epsilon}_1}{\hbar}\right)\cdot\vec{\sigma}}{\left(\omega+\mathrm{i}\epsilon-\frac{\epsilon_0-\mu}{\hbar}\right)^2-\left(\frac{\vec{\epsilon}_1}{\hbar}\right)^2},\\
G^{\mathrm{adv}}(t) &= \text{cf. Remark \textbf{??}}, & G^{\mathrm{adv}}(\omega-\mathrm{i}\epsilon) &= \text{TBD}.
\end{aligned}$$

Proof 12.9 See Appendix, Proofs, page 322.

12.3. Phononic Displacement-Green-Functions without Periodicity

Definition 12.10 For $\nu \in \{1, \dots, rd_{\mathbf{x}}\}$ and $\mathbf{q} \in \mathrm{Q}$, we introduce the *phononic-displacement-operator* $A_\nu(\mathbf{q})$ as

$$A_\nu(\mathbf{q}) = c_\nu(\mathbf{q}) + c_\nu^\dagger(-\mathbf{q}). \tag{12.9}$$

Remark 12.10.1 Notice that $A_\nu(\mathbf{q}) = A_\nu^\dagger(-\mathbf{q}) \Longleftrightarrow A_\nu(-\mathbf{q}) = A_\nu^\dagger(\mathbf{q})$

Definition 12.11 We consider $X = A_\nu(\mathbf{q})$ and $X' = A_{\nu'}^\dagger(\mathbf{q}')$ for multi-indices

$$(\alpha, t) \equiv (\mathbf{q}, \nu, t) \in \mathrm{Q} \times \{1, \dots, rd_{\mathbf{x}}\} \times \mathbb{R} \tag{12.10}$$

we introduce the different *real-time-[D]isplacement-Green-functions*[4]

- $D(A, A') \equiv D^{++}(A, A') \equiv D^{\mathbb{T}}(A, A') \equiv -\frac{\mathrm{i}}{\hbar} \left\langle \mathrm{T}_t\, A_\alpha(t) A_{\alpha'}^\dagger(t') \right\rangle,$
- $D^{+-}(A, A') \equiv D^{<}(A, A') \equiv -\frac{\mathrm{i}}{\hbar} \left\langle A_{\alpha'}^\dagger(t') A_\alpha(t) \right\rangle,$
- $D^{-+}(A, A') \equiv D^{>}(A, A') \equiv -\frac{\mathrm{i}}{\hbar} \left\langle A_\alpha(t) A_{\alpha'}^\dagger(t') \right\rangle,$
- $D^{--}(A, A') \equiv D^{\overline{\mathbb{T}}}(A, A') \equiv -\frac{\mathrm{i}}{\hbar} \left\langle \overline{\mathrm{T}}_t A_\alpha(t) A_{\alpha'}^\dagger(t') \right\rangle,$
- $D^{\mathrm{ret}}(A, A') = -\frac{\mathrm{i}}{\hbar}\theta(t - t') \left\langle \left[A_\alpha(t),\ A_{\alpha'}^\dagger(t') \right] \right\rangle,$
- $D^{\mathrm{adv}}(A, A') = +\frac{\mathrm{i}}{\hbar}\theta(t' - t) \left\langle \left[A_\alpha(t),\ A_{\alpha'}^\dagger(t') \right] \right\rangle,$

with canonical time-evolution-operator as usual.

Theorem 12.12 TBD: Differential Equations of phononic Green-functions. Cf. Rammer 01 p. 66.

[4]Cf. Cuevas p. 233 vs Mahan p. 99 and Negele p. 245, whereas there is a typo (missing minus-sign) in the first literature.

Example: A Diagonal Phonon Gas

Theorem 12.13 It follows[5]

$$D_{\mathbf{q}\nu}(t) = -\frac{\mathrm{i}}{\hbar}\sum_{\sigma\in\{+,-\}}\theta(\sigma t)\Big[\left(1+\mathrm{B}_{\hbar\beta}(\omega_{\mathbf{q}\nu})\right)\mathrm{e}^{-\sigma\omega_{\mathbf{q}\nu}\mathrm{i}t} + \mathrm{B}_{\hbar\beta}(\omega_{\mathbf{q}\nu})\mathrm{e}^{+\sigma\omega_{\mathbf{q}\nu}\mathrm{i}t}\Big] \tag{12.11}$$

$$D^{<}_{\mathbf{q}\nu}(t) = TBD \tag{12.12}$$

$$D^{>}_{\mathbf{q}\nu}(t) = TBD \tag{12.13}$$

$$D^{\mathrm{ret}}_{\mathbf{q}\nu}(t-t') = -\frac{\mathrm{i}}{\hbar}\theta(t-t')\left(\mathrm{e}^{-\mathrm{i}\omega_\nu(\mathbf{q})(t-t')} - \mathrm{e}^{+\mathrm{i}\omega_\nu(\mathbf{q})(t-t')}\right) \tag{12.14}$$

Cf. Mahan page 99 (might be contradicting these results here...

Proof 12.13 See Appendix, Proofs, page 359. TBD.

Theorem 12.14 We find for the frequency-space versions

$$D_{\mathbf{q}\nu}(\omega) = \lim_{\delta\to 0} TBD. \tag{12.15}$$

$$D^{<}_{\mathbf{q}\nu}(\omega) = -\frac{2\pi\mathrm{i}}{\hbar}\Big[\left(\mathrm{B}_{\hbar\beta}(\omega_{\mathbf{q}\nu})+1\right)\delta(\omega+\omega_{\mathbf{q}\nu}) + \mathrm{B}_{\hbar\beta}(\omega_{\mathbf{q}\nu})\delta(\omega-\omega_{\mathbf{q}\nu})\Big] \tag{12.16}$$

$$D^{>}_{\mathbf{q}\nu}(\omega) = -\frac{2\pi\mathrm{i}}{\hbar}\Big[\left(\mathrm{B}_{\hbar\beta}(\omega_{\mathbf{q}\nu})+1\right)\delta(\omega-\omega_{\mathbf{q}\nu}) + \mathrm{B}_{\hbar\beta}(\omega_{\mathbf{q}\nu})\delta(\omega+\omega_{\mathbf{q}\nu})\Big] \tag{12.17}$$

$$D^{\mathrm{ret}}_{\mathbf{q}\nu}(\omega+\mathrm{i}\epsilon) = \frac{1}{\hbar}\frac{1}{\omega+\mathrm{i}\epsilon-\omega_\nu(\mathbf{q})} - \frac{1}{\hbar}\frac{1}{\omega+\mathrm{i}\epsilon+\omega_\nu(\mathbf{q})} \tag{12.18}$$

- More identities: Rammer 01 p. 69

[5] For D^{ret} and $D^{<}$: Jauho page 73 and also Mahan page 116 with Wick-transformation.

13. Imaginary-Time-Two-Operator-Green-Functions at Finite Temperature

13.1. Two-Operator Green-Functions with Periodicity

Definition 13.1 For a pair of two operators (X, X') and some imaginary time $\tau \in \mathbb{R}$ we define the *imaginary-time two-operator Green-function* in its various shapes

$$\begin{aligned}\mathscr{G}_{X,X'}(\tau,\tau') \equiv \mathscr{G}_{X,X'}(\tau,\tau') &\equiv -\frac{1}{\hbar}\left\langle \mathrm{T}_\tau\, X(\tau)\, X'(\tau')\right\rangle \\ &\equiv -\frac{1}{\hbar}\frac{\mathrm{Tr}_{\mathscr{H}}\, \mathrm{e}^{-\beta\Xi}\, \mathrm{T}_\tau\, X(\tau)\, X'(\tau')}{\mathrm{Tr}_{\mathscr{H}}\, \mathrm{e}^{-\beta\Xi}} \\ &\equiv -\frac{1}{\hbar}\frac{\mathrm{Tr}_{\mathscr{H}}\, \mathrm{e}^{-\beta\Xi}\Big[\theta(\tau-\tau')\, X(\tau)\, X'(\tau') + S_{X,X'}\, \theta(\tau'-\tau)\, X'(\tau')\, X(\tau)\Big]}{\mathrm{Tr}_{\mathscr{H}}\, \mathrm{e}^{-\beta\Xi}}\end{aligned}$$

with imaginary-time-ordering-operator and ensemble-Heisenberg-imaginary-time-evolution as usual.

Theorem 13.2 It follows

$$\mathscr{G}_{X,X'}(\tau,\tau') \equiv \mathscr{G}_{X,X'}(\tau-\tau',0) \equiv \mathscr{G}_{X,X'}(\tau-\tau') \equiv S_{X,X'}\mathscr{G}_{X,X'}(\tau-\tau'+\hbar\beta), \tag{13.1}$$

which implies that the function

$$(-\hbar\beta,\, \hbar\beta) \ni \tau \mapsto \mathscr{G}_{X,X'}(\tau) \in \mathbb{C} \tag{13.2}$$

is periodic.

Proof 13.2 See Appendix, Proofs, page 332

Remark 13.2.1 Notice that this implies, with Lemma 1.11, that there exists a Matsubara-Fourier-transformation of the Green-function

$$\mathrm{i}\Omega_{S_{X,X'}} \ni \mathrm{i}\omega_n \mapsto \mathscr{G}_{X,X'}(\mathrm{i}\omega_n) \equiv \int_0^{\hbar\beta} \mathrm{d}\tau\, \mathscr{G}_{X,X'}(\tau)\, \mathrm{e}^{\mathrm{i}\omega_n\tau} \in \mathbb{C}. \tag{13.3}$$

13.2. Pure One-Body Green-Functions with Periodicity

Definition 13.3 We consider $X = c_\alpha$ and $X' = c^\dagger_{\alpha'}$ for multi-indices $A \equiv (\alpha, t) \in \mathscr{A} \times \mathbb{R}$, with

$$\left[c_\alpha, c^\dagger_{\alpha'}\right]_\zeta = \delta_{\alpha,\alpha'}, \qquad \mathrm{S}(c_\alpha) = \mathrm{S}(c^\dagger_{\alpha'}) = \zeta.$$

Then we write for the *imaginary-time one-body Green-function* in its various shapes

$$\mathscr{G}(A, A') \equiv \mathscr{G}_{\alpha,\alpha'}(\tau, \tau') \equiv -\frac{1}{\hbar} \frac{\mathrm{Tr}_{\mathscr{H}}\, \mathrm{e}^{-\beta\Xi} \left[\theta(\tau - \tau')\, c_\alpha(\tau)\, c^\dagger_{\alpha'}(\tau') + \zeta\, \theta(\tau' - \tau)\, c^\dagger_{\alpha'}(\tau')\, c_\alpha(\tau)\right]}{\mathrm{Tr}_{\mathscr{H}}\, \mathrm{e}^{-\beta\Xi}} \tag{13.4}$$

with *imaginary-time*-ordering-operator and ensemble-Heisenberg-imaginary-time-evolution as usual[1]

Theorem 13.4 It follows

$$\mathscr{G}_{\alpha,\alpha'}(\tau, \tau') \equiv \mathscr{G}_{\alpha,\alpha'}(\tau - \tau', 0) \equiv \mathscr{G}_{\alpha,\alpha'}(\tau - \tau') \equiv \zeta \mathscr{G}_{\alpha,\alpha'}(\tau - \tau' + \hbar\beta), \tag{13.5}$$

which implies that the function

$$(-\hbar\beta,\, \hbar\beta) \ni \tau \mapsto \mathscr{G}_{\alpha,\alpha'}(\tau) \in \mathbb{C} \tag{13.6}$$

is periodic.

Proof 13.4 See Appendix, Proofs, page 332

Lemma 13.5 An useful identity is the so-called *equation of motion*

$$\partial_\tau \mathscr{G}_{\alpha\alpha'}(\tau) = -\frac{1}{\hbar}\delta(\tau)\delta_{\alpha,\alpha'} \quad + \quad \frac{1}{\hbar}\left\langle \mathrm{T}_\tau \left[c_\alpha, \frac{\Xi}{\hbar}\right]_{\mathrm{H}}(\tau) c^\dagger_{\alpha'}(0) \right\rangle. \tag{13.7}$$

Proof 13.5 See Appendix, Proofs, page 333

[1] Notice that in the imaginary-time case, the creation- and annihilation-operators are no longer complex-conjugates to each other:

$$c_\bullet(\tau) = \mathrm{e}^{\tau(H-\eta\mu N)/\hbar} c_\bullet \mathrm{e}^{-\tau(H-\eta\mu N)/\hbar}, \qquad c^\dagger_\bullet(\tau) = \mathrm{e}^{\tau(H-\eta\mu N)/\hbar} c^\dagger_\bullet \mathrm{e}^{-\tau(H-\eta\mu N)/\hbar}.$$

Example: Pure, Non-Interacting Systems

Example: A Quantum Dot with Nature η and Statistics ζ

Can be inferred as a special case of the diagonal gas.

Example: A Diagonal Gas with Nature η and Statistics ζ

Theorem 13.6 Omitting convergence-generating factors, and introducing the variable $\xi_\alpha \equiv \epsilon_\alpha - \eta\mu$, it follows

$$\begin{aligned}\mathscr{G}_{\substack{0\\ \alpha,\alpha'}}(\mathrm{i}\omega_n) &= \frac{1}{\hbar}\frac{\delta_{\alpha,\alpha'}}{\mathrm{i}\omega_n - \xi_\alpha/\hbar},\\ \mathscr{G}_{\substack{0\\ \alpha,\alpha'}}(\tau) &= -\frac{1}{\hbar}\mathrm{e}^{-\tau\xi_\alpha/\hbar}\delta_{\alpha,\alpha'}\Big[\theta(\tau)\left(1+\zeta\,\mathrm{n}^\zeta_\beta(\xi_\alpha)\right)+\theta(-\tau)\zeta\mathrm{n}^\zeta_\beta(\xi_\alpha)\Big].\end{aligned} \tag{13.8}$$

Proof 13.6 See Appendix, Proofs, page 333

Example: A Non-Interacting Gas with Nature η and Statistics ζ

Theorem 13.7 Omitting convergence-generating factors it follows

$$\mathscr{G}_{\substack{0\\ \alpha,\alpha'}}(\mathrm{i}\omega_n) = \frac{1}{\hbar}\left(\left[\mathrm{i}\omega_n - \frac{H_0-\eta\mu}{\hbar}\right]^{\mathrm{inv}}\right)_{\alpha,\alpha'}, \qquad \mathscr{G}_{\substack{0\\ \alpha,\alpha'}}(\tau) = \frac{1}{\hbar\beta}\sum_{\omega_n\in\Omega_\zeta}\frac{1}{\hbar}\left(\frac{\mathrm{e}^{-\mathrm{i}\omega_n\tau}}{\mathrm{i}\omega_n - \frac{H_0-\eta\mu}{\hbar}}\right)_{\alpha,\alpha'}. \tag{13.9}$$

Proof 13.7 See Appendix, Proofs, page 334

Remark 13.7.1 Notice that we can write this in matrix-notation as

$$\mathscr{G}_0(\mathrm{i}\omega_n) = \frac{1}{\hbar}\frac{1}{\mathrm{i}\omega_n - \left(H_0-\eta\mu\right)/\hbar} \tag{13.10}$$

Remark 13.7.2 Note that within the proof of this we found the solution of the differential equation[2]

$$\left(\partial_\tau + \frac{H-\eta\mu}{\hbar}\right)\mathscr{G}(\tau) = -\delta(\tau)\,\delta_{\alpha,\alpha'}/\hbar \tag{13.11}$$

with $\mathscr{G}$ defined on the interval $(-\hbar\beta, +\hbar\beta)$, subject to the boundary-condition

$$\mathscr{G}(\tau+\hbar\beta) = -\mathscr{G}(\tau). \tag{13.12}$$

[2]Cf. [3], p. 73 and [4] p. Exercises for Chapter 9

Fermionic Systems

Example: A Spin-$\frac{1}{2}$-Particle

Theorem 13.8 It follows

$$\mathscr{G}(\tau) = \text{TBD}, \qquad \mathscr{G}(\mathrm{i}\omega_n) = \frac{1}{\hbar}\frac{\left(\mathrm{i}\omega_n - \frac{\epsilon_0-\mu}{\hbar}\right)\sigma_0 + (\boldsymbol{\epsilon}/\hbar)\cdot\boldsymbol{\sigma}}{\left(\mathrm{i}\omega_n - \omega_+\right)\left(\mathrm{i}\omega_n - \omega_-\right)},$$

with

$$\omega_\xi \equiv \frac{\xi\epsilon_1 + \epsilon_0 - \mu}{\hbar} \qquad \text{for } \xi \in \{-,+\}.$$

Proof 13.8 See Appendix, Proofs, page 335

13.3. Phononic Displacement-Green-Functions with Periodicity

Definition 13.9 We consider $X = A_\nu(\mathbf{q})$ and $X' = A^\dagger_{\nu'}(\mathbf{q}')$ for multi-indices

$$(\alpha, \tau) \equiv (\mathbf{q}, \nu, \tau) \in \mathrm{Q}\times\{1, \ldots, rd_{\mathbf{x}}\} \times \mathbb{R} \tag{13.13}$$

we introduce the *imaginary-time-[D]isplacement-Green-function* as

$$\mathscr{D}_{\alpha,\alpha'}(\tau, \tau') = -\frac{1}{\hbar}\left\langle \mathrm{T}_\tau\, A_\alpha(\tau) A^\dagger_{\alpha'}(\tau') \right\rangle \quad \Longleftrightarrow \quad \mathscr{D}_{\substack{\mathbf{q},\mathbf{q}'\\ \nu,\nu'}}(\tau, \tau') = -\frac{1}{\hbar}\left\langle \mathrm{T}_\tau\, A_\nu(\tau, \mathbf{q}) A^\dagger_{\nu'}(\tau', \mathbf{q}') \right\rangle . \tag{13.14}$$

Remark 13.9.1 It is far from obvious that this definition is a reasonable choice for a Green-function. It will turn out however, that this Green-function appears quite naturally in the functional integral formalism of the theory, after one has performed a suitable change of variables resulting in Theorem 33.74. It will also become useful in the discussion of Feynman-diagrams, cf. the proof of Theorem 20.3.

Theorem 13.10 Being a superposition of periodic functions that only depend on time-differences, it follows

$$\mathscr{D}_{\alpha,\alpha'}(\tau, \tau') \equiv \mathscr{D}_{\alpha,\alpha'}(\tau - \tau', 0) \equiv \mathscr{D}_{\alpha,\alpha'}(\tau - \tau') = +\mathscr{D}_{\alpha,\alpha'}(\tau - \tau' + \hbar\beta) \tag{13.15}$$

which implies that the function

$$(0, \hbar\beta) \ni \tau \mapsto \mathscr{D}_{\alpha,\alpha'}(\tau) \in \mathbb{C} \tag{13.16}$$

is periodic.

Proof 13.10 TBD

Example: A d-dimensional Phononic Gas

Theorem 13.11 It follows

$$\begin{aligned}\mathscr{D}_{\substack{0\\\alpha,\alpha'}}(\tau) &= -\frac{1}{\hbar}\delta_{\alpha,\alpha'}\sum_{\nu\in\{+,-\}}\theta(\nu\tau)\Big[\left(1+\mathrm{B}_{\hbar\beta}(\omega_\alpha)\right)\,\mathrm{e}^{-\nu\,\omega_\alpha\tau}+\mathrm{B}_{\hbar\beta}(\omega_\alpha)\,\mathrm{e}^{\nu\,\omega_\alpha\tau}\Big]\\ \mathscr{D}_{\substack{0\\\alpha\alpha'}}(\mathrm{i}\omega_n) &= -\frac{1}{\hbar}\frac{2\omega_\alpha\delta_{\alpha,\alpha'}}{\omega_n^2+\omega_\alpha^2}\end{aligned} \tag{13.17}$$

Proof 13.11 See Appendix, Proofs, page 336

Remark 13.11.1 Notice also, that in the non-interacting case, the periodic standard-continuation of the function is symmetric, i.e. satisfies the equation

$$\mathscr{D}_{\substack{0\\\alpha\alpha'}}(\tau) = \mathscr{D}_{\substack{0\\\alpha\alpha'}}(-\tau). \tag{13.18}$$

This relation is integral, when one writes down and computes Feynman-diagrams related to electron-phonon theories.

Remark 13.11.2 We now recall that we can rewrite the bare phonon-propagator

$$\mathscr{D}_{0,\nu}(\mathrm{i}\omega_n,\mathbf{q}) = \frac{1}{\hbar}\frac{2\omega_\nu(\mathbf{q})}{(\mathrm{i}\omega_n)^2-\omega_\nu^2(\mathbf{q})} = \frac{1}{\hbar}\int_0^\infty \mathrm{d}\omega\frac{2\omega}{(\mathrm{i}\omega_n)^2-\omega^2}\delta\left(\omega-\omega_\nu(\mathbf{q})\right) \tag{13.19}$$

and therefore conclude, that the electron-phonon coupling-strength contains the phonon-propagator as a factor of the integrand.

Remark 13.11.3 Notice further the decomposition

$$\begin{aligned}\mathscr{D}_{0,\nu}(\mathrm{i}\omega_n,\mathbf{q}) &= -\frac{1}{\hbar}\left(\frac{1}{\mathrm{i}\omega_n-\omega_\nu(\mathbf{q})}-\frac{1}{\mathrm{i}\omega_n+\omega_\nu(\mathbf{q})}\right)\\ &= -\frac{1}{\hbar}\left(\frac{-\mathrm{i}\omega_n-\omega_\nu(\mathbf{q})}{(\mathrm{i}\omega_n)^2-\omega_\nu^2(\mathbf{q})}-\frac{-\mathrm{i}\omega_n+\omega_\nu(\mathbf{q})}{(\mathrm{i}\omega_n)^2-\omega_\nu^2(\mathbf{q})}\right)\\ &= -\frac{1}{\hbar}\frac{2\omega_\nu(\mathbf{q})}{\omega_n^2+\omega_\nu^2(\mathbf{q})}\end{aligned}$$

which is useful when one has the intention of continuing this expression analytically to the real axis.

Theorem 13.12 The imaginary-time-displacement-Green-function of a phononic gas solves the differential equation

$$\left(\partial_\tau^2 - \omega_\alpha^2\right) \mathscr{D}^{\,0}_{\alpha,\alpha'}(\tau) = 2\frac{\omega_\alpha}{\hbar}\delta(\tau)\,\delta_{\alpha,\alpha'} \tag{13.20}$$

with $\mathscr{D}$ defined on the interval $(0, +\hbar\beta)$, subject to the boundary-condition

$$\mathscr{D}(\tau + \hbar\beta) = +\mathscr{D}(\tau). \tag{13.21}$$

Proof 13.12 See Appendix, Proofs, page 337

14. Lehmann-Representation and more Green-Functions

Notation 14.1 The *Lehmann-representation*, refers to the following algebraic trick: We introduce the exact eigenbasis of the Hamiltonian

$$\mathscr{B} \equiv \{|\,0\,\rangle, |\,1\,\rangle, |\,2\,\rangle, \dots\} \quad \text{s.t.} \quad \mathscr{H} = \operatorname{span}_{\mathbb{C}} \mathscr{B} \tag{14.1}$$

with eigenvalues

$$\xi_n|\,n\,\rangle \equiv \Xi|\,n\,\rangle \equiv \begin{cases} (H - \eta\mu N)\,|\,n\,\rangle & \text{for systems with nature } \eta \\ \left(H_{\mathrm{p}} - \mu N + H_{\mathrm{w}} + V_{\mathrm{int}}\right)|\,n\,\rangle & \text{for wave-particle-mixtures} \\ \quad\vdots \end{cases} \tag{14.2}$$

Remark 14.1.1 The existence of this basis is guaranteed, if we introduce some cutoff for our basis, since then we have an hermitian operator on a finite dimensional complex vector space.

Remark 14.1.2 Notice that we capture all three cases from 8.1.

With this eigenbasis, we can now formally derive integral identities that connect the various Green-functions.

Fact 14.2 Recall also the *Sokhotski-Plemelj-theorem* for integrals over the frequency ω

$$\frac{1}{\omega \pm \mathrm{i}\epsilon + \text{const.}} = \mathrm{P}\,\frac{1}{\omega + \text{const.}} \mp \mathrm{i}\pi\delta(\omega + \text{const.}) \tag{14.3}$$

in the distributional sense for $\epsilon \to 0$.

Proof 14.2 TBReferenced.

14.1. Identities for Two-Operator-Green-Functions

Definition 14.3 For reasons that are difficult to motivate in advance, we introduce the *spectral function corresponding to the operator pair* (X, X') as

$$A_{X,X'}(\omega) \equiv -\frac{\hbar}{\pi} \operatorname{Im} \lim_{\epsilon \to 0} G^{\text{ret}}_{X,X'}(\omega + \mathrm{i}\epsilon).$$

Whereas this limit is to be understood in the distributional sense[1].

Remark 14.3.1 Notice that we defined these functions by their frequency-dependencies, and not by their time-dependencies (in constrast to the Green-functions).

Lemma 14.4 With the Lehmann-representation one finds the expansion

$$\mathscr{G}_{X,X'}(\mathrm{i}\omega_l) = \frac{1}{Z} \sum_{(n,m)\in\mathbb{N}^2} \langle n|X|m\rangle\langle m|X'|n\rangle \, \frac{e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}}{\hbar} \frac{1}{\mathrm{i}\omega_l + (\xi_n - \xi_m)/\hbar} \tag{14.4}$$

and similar identities for all the other Green-functions (and the spectral function).

Proof 14.4 See Appendix, Proofs, page 339

Theorem 14.5 The Matsubara-Green-function can be analytically continued down to the real axis to the retarded / advanced Green-functions as

$$\begin{aligned}
\mathrm{i}\omega_n &\mapsto \omega + \mathrm{i}\epsilon \quad &&\Rightarrow \quad \mathscr{G}_{X,X'}(\mathrm{i}\omega_n) \mapsto G^{\text{ret}}_{X,X'}(\omega + \mathrm{i}\epsilon),\\
\mathrm{i}\omega_n &\mapsto \omega - \mathrm{i}\epsilon \quad &&\Rightarrow \quad \mathscr{G}_{X,X'}(\mathrm{i}\omega_n) \mapsto G^{\text{adv}}_{X,X'}(\omega - \mathrm{i}\epsilon).
\end{aligned}$$

Proof 14.5 Follows immediately from the Lehmann-representations in Eqs. (28.65)-(28.72) and the Carlson-theorem[2].

Theorem 14.6 One also obtains an adapted version of the *Fluctuation-Dissipation-Theorem*:

$$\begin{aligned}
G^{<}_{X,X'}(\omega) &= 2\mathrm{i}\ \mathrm{S}_{X,X'}\, \mathrm{n}^{\mathrm{S}_{X,X'}}_{\hbar\beta}(\omega) \operatorname{Im} \lim_{\epsilon\to 0} G^{\text{ret}}_{X,X'}(\omega + \mathrm{i}\epsilon),\\
G^{>}_{X,X'}(\omega) &= 2\mathrm{i} \left(1 + \mathrm{S}_{X,X'}\, \mathrm{n}^{\mathrm{S}_{X,X'}}_{\hbar\beta}(\omega)\right) \operatorname{Im} \lim_{\epsilon\to 0} G^{\text{ret}}_{X,X'}(\omega + \mathrm{i}\epsilon).
\end{aligned} \tag{14.5}$$

[1] This means, that this object needs to be integrated over.
[2] Cf. [3].

Proof 14.6 See Appendix, Proofs, page 349

Theorem 14.7 One finds the *Lehmann-identities*

$$\begin{aligned}
\operatorname{Re} G_{X,X'}(\omega) &= +\operatorname{Re} G^{\mathrm{ret}}_{X,X'}(\omega) && = +\operatorname{Re} G^{\mathrm{adv}}_{X,X'}(\omega), \\
\operatorname{Im} G_{X,X'}(\omega) &= +\operatorname{Im} G^{\mathrm{ret}}_{X,X'}(\omega) \cdot \left(\tanh \frac{\beta\hbar\omega}{2}\right)^{\mathrm{s}_{X,X'}} && = -\operatorname{Im} G^{\mathrm{adv}}_{X,X'}(\omega) \cdot \left(\tanh \frac{\beta\hbar\omega}{2}\right)^{\mathrm{s}_{X,X'}},
\end{aligned} \tag{14.6}$$

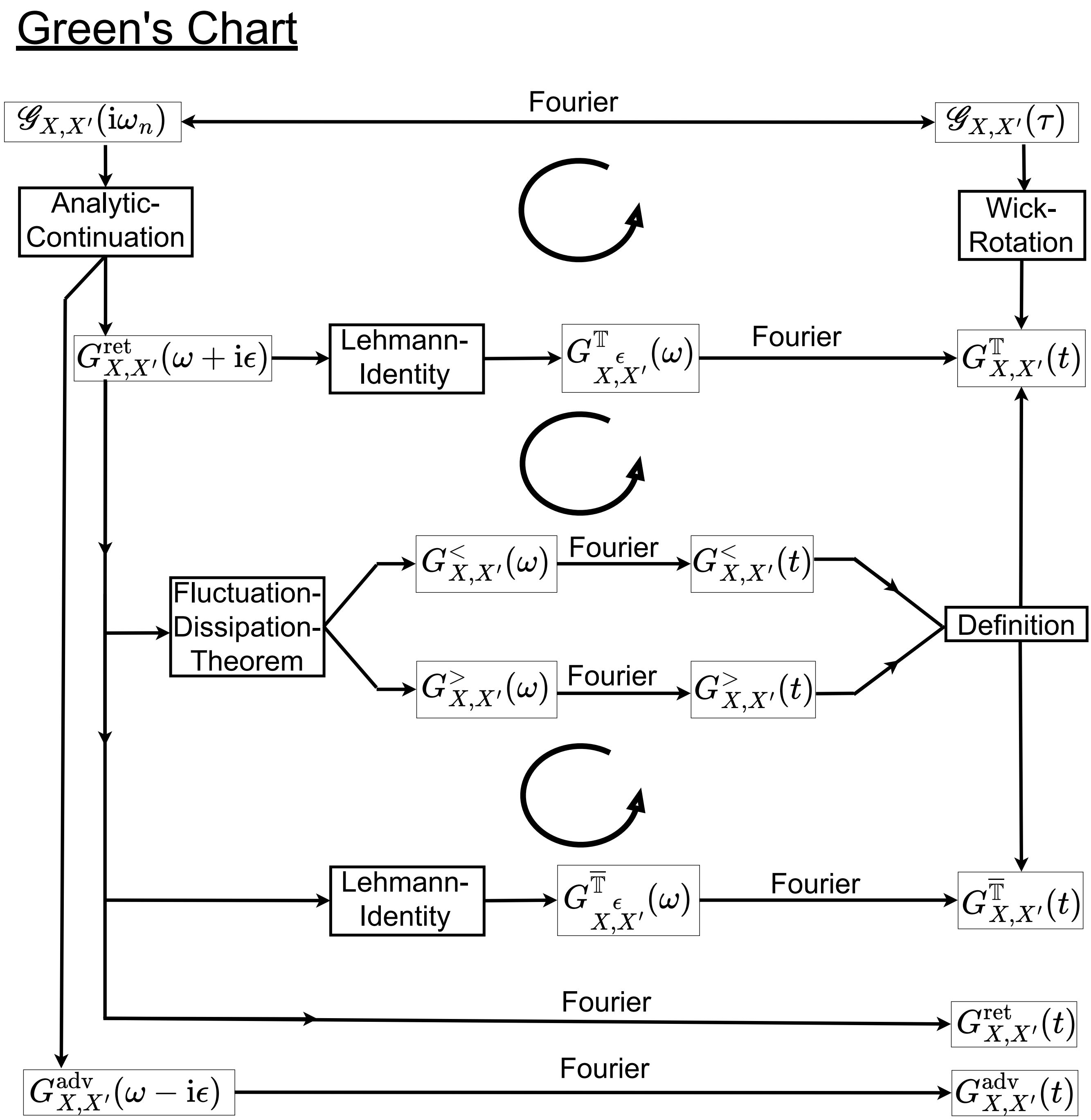
Proof 14.7 See Appendix, Proofs, page 351

Theorem 14.8 The so-called *Kramers-Kronig-relations*[3]:

$$\begin{aligned}
\operatorname{Re} G^{\mathrm{ret}}_{X,X'}(\omega) &= \frac{1}{\pi} \int \mathrm{d}\omega' \; \operatorname{Im} G^{\mathrm{ret}}_{X,X'}(\omega')\, \mathrm{P}\, \frac{1}{\omega' - \omega} \\
\operatorname{Im} G^{\mathrm{ret}}_{X,X'}(\omega) &= \frac{1}{\pi} \int \mathrm{d}\omega' \; \operatorname{Re} G^{\mathrm{ret}}_{X,X'}(\omega')\, \mathrm{P}\, \frac{1}{\omega' - \omega}
\end{aligned}$$

We summarize the different identities in *Green's Chart* which graphically depicts how one can determine every two-operator Green-function from the imaginary-time or (inclusive) Matsubara-Green-function.

[3] Also known as dispersion-relations



Theorem 14.9 More identities:

- See Schwabl 2 page 84 for more identities.
- Mahan Eqs. (3.326) - (3.328), NE, Equation (5.42).

Proof 14.9 TBDone.

14.2. Identities for Pure One-Body-Green-Functions

In this section, we consider $X = c_\alpha$ and $X' = c^\dagger_{\alpha'}$ for multi-indices $A \equiv (\alpha, t) \in \mathscr{A} \times \mathbb{R}$, with

$$\left[c_\alpha, c^\dagger_{\alpha'}\right]_\zeta = \delta_{\alpha,\alpha'}, \qquad \mathrm{S}(c_\alpha) = \mathrm{S}(c^\dagger_{\alpha'}) = \zeta.$$

Definition 14.10 For reasons that are difficult to motivate in advance, we introduce the *spectral matrix* or the *generalized spectral functions* $A_{\alpha\alpha'}(\omega)$ defined by

$$A_{\alpha\alpha'}(\omega) := -\frac{\hbar}{\pi} \operatorname{Im} \lim_{\epsilon \to 0} G^{\mathrm{ret}}_{\alpha\alpha'}(\omega + \mathrm{i}\epsilon).$$

Whereas this limit is to be understood in the distributional sense[4].

Remark 14.10.1 Notice that we defined these functions by their frequency-dependencies, and not by their time-dependencies (in constrast to the Green-functions).

Lemma 14.11 With the Lehmann-representation one finds the expansion

$$\mathscr{G}_{\alpha,\alpha'}(\mathrm{i}\omega_l) = \frac{1}{Z} \sum_{(n,m)\in\mathbb{N}^2} \langle n|c_\alpha|m\rangle\langle m|c^\dagger_{\alpha'}|n\rangle \frac{e^{-\beta\xi_n} - \zeta e^{-\beta\xi_m}}{\hbar} \frac{1}{\mathrm{i}\omega_l + (\xi_n - \xi_m)/\hbar} \tag{14.7}$$

and similar identities for all the other Green-functions (and the generalized spectral functions).

Proof 14.11 See Appendix, Proofs, page 339

Remark 14.11.1 In the derivation of these identities, it also becomes clear why there will always be the need to introduce convergence-generating factors for some of the Green-functions.

Theorem 14.12 The Matsubara-Green-function can be analytically continued down to the real axis to the retarded / advanced Green-functions as

$$\begin{aligned}
\mathrm{i}\omega_n \mapsto \omega + \mathrm{i}\epsilon \quad &\Rightarrow \quad \mathscr{G}_{\alpha,\alpha'}(\mathrm{i}\omega_n) \mapsto G^{\mathrm{ret}}_{\alpha,\alpha'}(\omega + \mathrm{i}\epsilon) \\
\mathrm{i}\omega_n \mapsto \omega - \mathrm{i}\epsilon \quad &\Rightarrow \quad \mathscr{G}_{\alpha,\alpha'}(\mathrm{i}\omega_n) \mapsto G^{\mathrm{adv}}_{\alpha,\alpha'}(\omega - \mathrm{i}\epsilon)
\end{aligned}$$

Proof 14.12 Follows immediately from Theorem 14.5.

[4]As explained earlier.

Remark 14.12.1 Notice however, that one has to be a little bit careful when one wants to perform the analytic continuation. The function has to be rewritten w.r.t. simple poles, or functions that decay „fast enough“ at infinity.

Theorem 14.13 The *fluctuation-dissipation-theorem*[5] states the two *Lehmann-identities*

$$\begin{aligned} G^{<}_{\alpha\alpha'}(\omega) &= 2\mathrm{i}\,\zeta\,\mathrm{n}^{\zeta}_{\hbar\beta}(\omega)\,\mathrm{Im}\lim_{\epsilon\to 0} G^{\mathrm{ret}}_{\alpha\alpha'}(\omega+\mathrm{i}\epsilon), \\ G^{>}_{\alpha\alpha'}(\omega) &= 2\mathrm{i}\left(1+\zeta\,\mathrm{n}^{\zeta}_{\hbar\beta}(\omega)\right)\mathrm{Im}\lim_{\epsilon\to 0} G^{\mathrm{ret}}_{\alpha\alpha'}(\omega+\mathrm{i}\epsilon). \end{aligned} \tag{14.8}$$

Proof 14.13 Follows from Theorem 14.6.

Lemma 14.14 For a given chemical potential μ, we can compute the average number of particles as

$$\langle N\rangle = \frac{-1}{\pi}\lim_{\epsilon\to 0} \sum\!\!\!\!\!\!\int_{\substack{\alpha\in\mathscr{A}\\ E\in(-\infty,+\infty)}} \mathrm{n}^{\zeta}_{\beta}(E)\ \mathrm{Im}\ G^{\mathrm{ret}}_{\alpha,\alpha}(E+\mathrm{i}\epsilon)$$

Proof 14.14 See Appendix, Proofs, page 352

Remark 14.14.1 Analogously one can compute the expectation-value for the particle numbers associated to a certain type of state, for example for electrons in a crystal, one can compute the number of spin-up or spin-down electrons. One can also compute the expectation-value of the holes.

Lemma 14.15 The spectral function fulfills the following identities

- $\delta_{\alpha,\alpha'} = \displaystyle\int \mathrm{d}\omega\, A_{\alpha,\alpha'}(\omega)$
- $\mathrm{d} = \displaystyle\int \mathrm{d}\omega\,\mathrm{Tr}\,A(\omega) \equiv \int \mathrm{d}\omega\,\rho(\omega)$
- For fermions ($\zeta = -1$) one finds the density

$$0 < \frac{\mathrm{d}N_{\mathrm{states}}}{\mathrm{d}\omega} \equiv \rho(\omega) \equiv \mathrm{Tr}\,A(\omega) = -\frac{\hbar}{\pi}\,\mathrm{Im}\lim_{\epsilon\to 0}\mathrm{Tr}\,G^{\mathrm{ret}}(\omega+\mathrm{i}\epsilon) \quad \text{for all } \omega\in\mathbb{R} \tag{14.9}$$

Proof 14.15 See Appendix, Proofs, page 352

Remark 14.15.1 Noting that the spectral function is in the fermionic case positive definite, one usually[6] proposes the interpretation of the spectral function as a *density (of states)* or (after renormalization) *probability-density* by generalizing from the non-interacting case. Although these are, strictly speaking, just single-particle-states, and not many-particle-states, that are being counted when one integrates this object.

[5] In [**?**], p. 46 the origin of this name is motivated.
[6] Cf. [Mahan, p. 123]

Remark 14.15.2 For systems, where the index can be decomposed one can also introduce specific densities. One has additional indices $i \in I$, for example ...

- electronic spin-up and spin-down-component $\mathscr{A} \ni \alpha \equiv (i, \sigma) \in I \times \{\downarrow, \uparrow\}$,
- electronic Wannier-states $\mathscr{A} \ni \alpha \equiv (\mathbf{R}_j, n, i) \in \mathrm{L_f} \times \{1, 2, 3, \dots\} \times I$,
- electronic Bloch-states $\mathscr{A} \ni \alpha \equiv (\mathbf{k}, n, i) \in \mathrm{Q} \times \{1, 2, 3, \dots\} \times I$.

Corollary 14.16 Notice that this interpretation allows us, in the fermionic case, to write the expectation-value of the particle-number in the intuitive form

$$\langle N \rangle = \int_{-\infty}^{+\infty} \mathrm{d}E \, \mathrm{F}_\beta(E) \, \rho(E) \tag{14.10}$$

Proof 14.16 Follows immediately from combining the lemmata.

Theorem 14.17 Another useful set of *Lehmann-identities* is

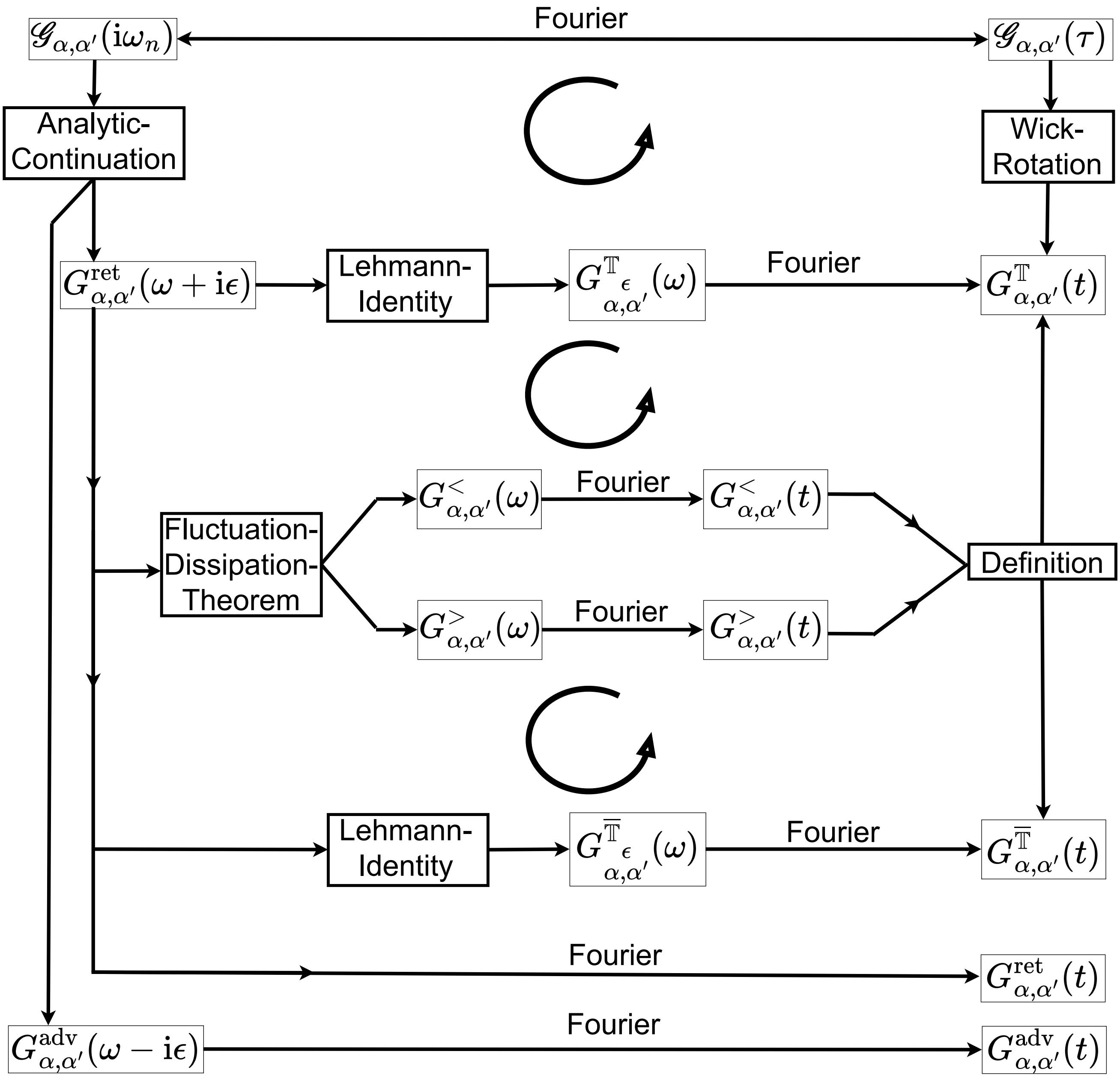
$$\begin{aligned}
\operatorname{Re} G^{\ \epsilon}_{\alpha,\alpha'}(\omega) &= + \operatorname{Re} G^{\mathrm{ret}}_{\alpha,\alpha'}(\omega + \mathrm{i}\epsilon) && = + \operatorname{Re} G^{\mathrm{adv}}_{\alpha,\alpha'}(\omega - \mathrm{i}\epsilon), \\
\operatorname{Im} G^{\ \epsilon}_{\alpha,\alpha'}(\omega) &= + \operatorname{Im} G^{\mathrm{ret}}_{\alpha,\alpha'}(\omega + \mathrm{i}\epsilon) \cdot \left(\coth \frac{\beta\hbar\omega}{2} \right)^{\zeta} && = - \operatorname{Im} G^{\mathrm{adv}}_{\alpha,\alpha'}(\omega - \mathrm{i}\epsilon) \cdot \left(\coth \frac{\beta\hbar\omega}{2} \right)^{\zeta},
\end{aligned}$$

in the limit $\epsilon \to 0$.

Proof 14.17 Follows from Theorem 14.7.

We summarize the different identities in *Green's Chart* which graphically depicts how one can determine every one-body Green-function from the imaginary-time or (inclusive) Matsubara-Green-function.

Green's Chart



14.3. Identities for Phononic Displacement-Green-Functions

+ ================ +
| Begin: To Be Implemented |
+ ================ +

Definition 14.18 We define the *generalized spectral function* $B(\mathbf{q}, \omega)$ as

$$B(\mathbf{q}, \omega) = -2\,\mathrm{Im}\, D^{\mathrm{ret}}_{\mathbf{q}}(\omega) \tag{14.11}$$

Lemma 14.19 It follows *for the non-interacting case (!!!)*

$$2\langle a^{\dagger}_{\mathbf{q}} a_{\mathbf{q}} \rangle + 1 = \langle A^{\dagger}_{\mathbf{q}} A_{\mathbf{q}} \rangle = \int_{-\infty}^{+\infty} \frac{\mathrm{d}\omega}{2\pi}\, \mathrm{B}_{\hbar\beta}(\omega) B(\mathbf{q}, \omega). \tag{14.12}$$

Proof 14.19 See Appendix, Proofs, page 353

Remark 14.19.1 Following Mahan, p. 125: The factor $\mathrm{B}_{\hbar\beta}(\omega) B(\mathbf{q}, \omega)$ is for $\omega > 0$ (!!!)[7] always positive and serves as the *temperature-dependent probability of having phonons with crystal-momentum* $\mathbf{q}$ *and frequency* ω.

- Generalizations for several branches?
- TBD; Cf. Mahan p. 116
- See Waste-Folder 1, topic 9.
- TBD; Cf. Mahan p. 117
- See also the footnote[8]
- Identity with Polarization: cf. Mahan p. 125.
- Cf. Mahan page 122 onwards for these and more identities.

| ================ |
| End: To Be Implemented |
| ================ |

[7] Notice that this detail is missing in the book from Mahan.
[8] `https://physics.stackexchange.com/questions/202672/free-phonon-propagator-in-imaginary-time?rq=1`

Definition 14.20 For reasons that are difficult to motivate in advance, we introduce the *phononic spectral matrix* or the *phononic generalized spectral functions* $B_{\alpha,\alpha'}(\omega)$ defined by

$$B_{\alpha,\alpha'}(\omega) := -\frac{\hbar}{\pi}\operatorname{Im}\lim_{\epsilon\to 0} D^{\text{ret}}_{\alpha,\alpha'}(\omega + \mathrm{i}\epsilon).$$

Whereas this limit is to be understood in the distributional sense[9].

Remark 14.20.1 Notice that we defined these functions by their frequency-dependencies, and not by their time-dependencies (in constrast to the Green-functions).

Theorem 14.21 The Matsubara-Green-function can be analytically continued down to the real axis to the retarded / advanced Green-functions as

$$\begin{aligned}
\mathrm{i}\omega_n \mapsto \omega + \mathrm{i}\epsilon \quad &\Rightarrow \quad \mathscr{D}_{\alpha,\alpha'}(\mathrm{i}\omega_n) \mapsto D^{\text{ret}}_{\alpha,\alpha'}(\omega + \mathrm{i}\epsilon)\\
\mathrm{i}\omega_n \mapsto \omega - \mathrm{i}\epsilon \quad &\Rightarrow \quad \mathscr{D}_{\alpha,\alpha'}(\mathrm{i}\omega_n) \mapsto D^{\text{adv}}_{\alpha,\alpha'}(\omega - \mathrm{i}\epsilon)
\end{aligned}$$

Proof 14.21 Follows immediately from Theorem 14.5.

Remark 14.21.1 Notice however, that one has to be a little bit careful when one wants to perform the analytic continuation. The function has to be rewritten w.r.t. simple poles, or functions that decay „fast enough" at infinity.

Theorem 14.22 The *fluctuation-dissipation-theorem*[10] states the two *Lehmann-identities*

$$\begin{aligned}
D^{<}_{\alpha\alpha'}(\omega) &= 2\mathrm{i}\mathrm{B}_{\hbar\beta}(\omega)\operatorname{Im}\lim_{\epsilon\to 0} D^{\text{ret}}_{\alpha\alpha'}(\omega + \mathrm{i}\epsilon),\\
D^{>}_{\alpha\alpha'}(\omega) &= 2\mathrm{i}\left(1 + \mathrm{B}_{\hbar\beta}(\omega)\right)\operatorname{Im}\lim_{\epsilon\to 0} D^{\text{ret}}_{\alpha\alpha'}(\omega + \mathrm{i}\epsilon).
\end{aligned} \tag{14.13}$$

Proof 14.22 Follows from Theorem 14.6.

Theorem 14.23 Another useful set of *Lehmann-identities* is

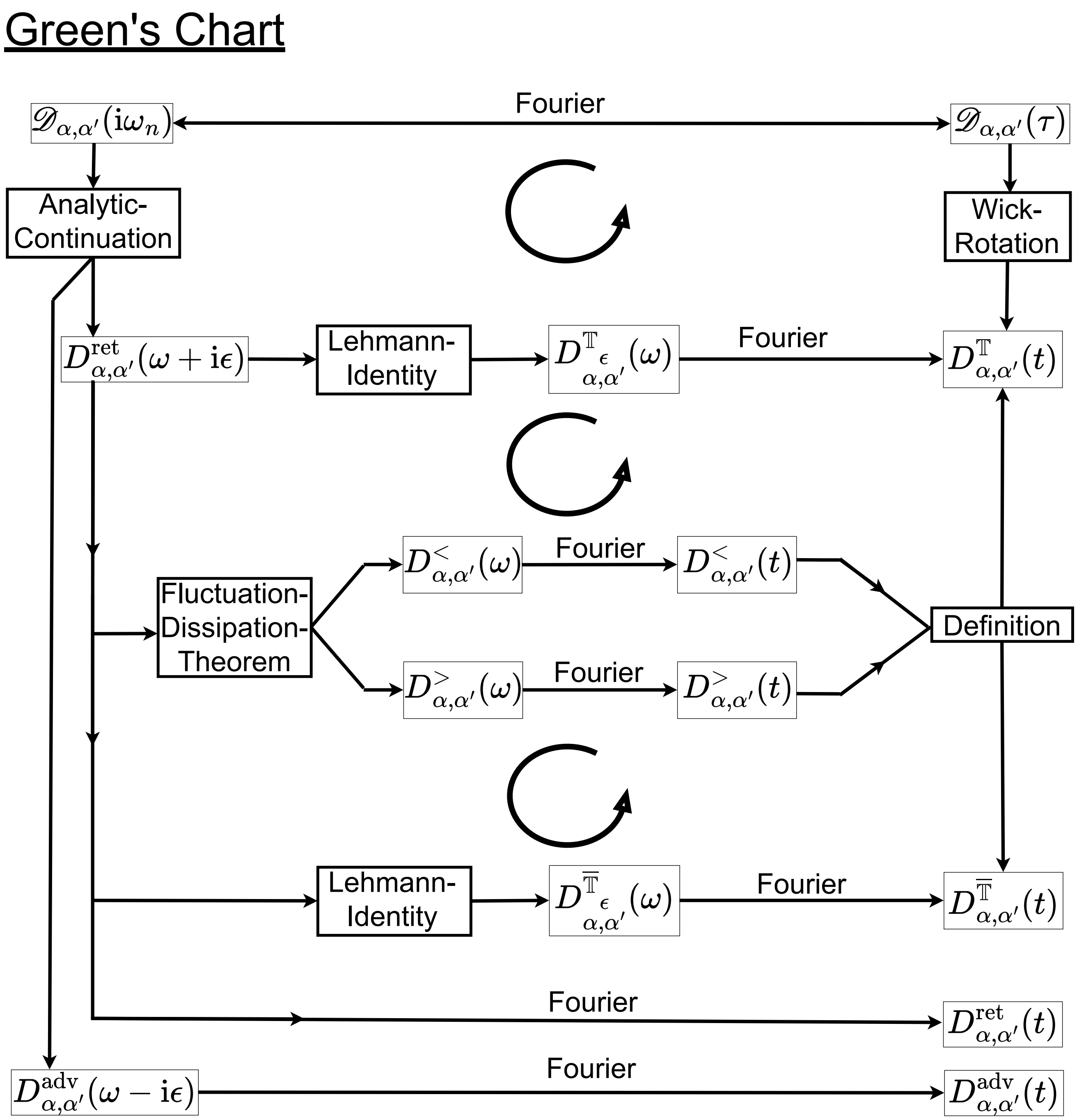
$$\begin{aligned}
\operatorname{Re} D^{\epsilon}_{\alpha,\alpha'}(\omega) &= +\operatorname{Re} D^{\text{ret}}_{\alpha,\alpha'}(\omega + \mathrm{i}\epsilon) &&= +\operatorname{Re} D^{\text{adv}}_{\alpha,\alpha'}(\omega - \mathrm{i}\epsilon),\\
\operatorname{Im} D^{\epsilon}_{\alpha,\alpha'}(\omega) &= +\operatorname{Im} D^{\text{ret}}_{\alpha,\alpha'}(\omega + \mathrm{i}\epsilon)\cdot\coth\frac{\beta\hbar\omega}{2} &&= -\operatorname{Im} D^{\text{adv}}_{\alpha,\alpha'}(\omega - \mathrm{i}\epsilon)\cdot\coth\frac{\beta\hbar\omega}{2},
\end{aligned}$$

[9] As explained earlier.
[10] In [?], p. 46 the origin of this name is motivated.

in the limit $\epsilon \to 0$.

Proof 14.23 Follows from Theorem 14.7.

We summarize the different identities in *Green's Chart* which graphically depicts how one can determine every one-body Green-function from the imaginary-time or (inclusive) Matsubara-Green-function.



14.4. Definitions of Pure n-Body Green-Functions

For reasons that are difficult to motivate in advance, we introduce generalize the one-body Green-functions from Equation (13.4) to n-body Green-functions as follows.

Definition 14.24 We define *real- and imaginary-time n-body Green-functions* as

$$G^n(A_1, \dots A_n \mid A'_n, \dots A'_1) \equiv \left(-\frac{\mathrm{i}}{\hbar}\right)^n \langle \mathrm{T}_t \, c_{\alpha_1}(t_1) \dots c_{\alpha_n}(t_n) \, c^\dagger_{\alpha'_1}(t'_1) \dots c^\dagger_{\alpha'_n}(t'_n) \rangle, \tag{14.14}$$

$$\mathscr{G}^n(A_1, \dots A_n \mid A'_n, \dots A'_1) \equiv \left(-\frac{1}{\hbar}\right)^n \langle \mathrm{T}_\tau \, c_{\alpha_1}(\tau_1) \dots c_{\alpha_n}(\tau_n) \, c^\dagger_{\alpha'_1}(\tau'_1) \dots c^\dagger_{\alpha'_n}(\tau'_n) \rangle. \tag{14.15}$$

Remark 14.24.1 Notice that we use another sign-convention here than Negele and Orland on page 53.

15. Systematic Calculation of Imaginary-Time and Matsubara-Green-Functions for Interacting Systems

15.1. S-Matrix-Expansion for the Two-Operator Green-Functions

Definition 15.1 We consider a system described by the Hamiltonian in Equation (**??**) and we assume that the the one-particle Green-functions of the system described by H_0 are known. Then we introduce the abbreviations

$$\Xi \quad = \quad \Xi_0 \quad + \quad V \tag{15.1}$$

to define the *S-matrix* as

$$S(\tau) \equiv \mathrm{e}^{\tau \Xi_0/\hbar} \mathrm{e}^{-\tau \Xi/\hbar}. \tag{15.2}$$

Lemma 15.2 It follows

$$S(\tau) = T_\tau \exp \frac{-1}{\hbar} \int_0^\tau \mathrm{d}\tau' \, V_{\Xi_0}(\tau'), \qquad \text{with} \qquad V_{\Xi_0}(\tau) \equiv \mathrm{e}^{\tau\Xi_0/\hbar} \, V \mathrm{e}^{-\tau\Xi_0/\hbar}$$

Proof 15.2 See Appendix, Proofs, page **??**

Theorem 15.3 One finds the *S-matrix-expansion of the Green-functions*

$$\begin{aligned} \mathscr{G}_{X,X'}(\tau,\tau') &= -\frac{1}{\hbar} \frac{\left\langle \mathrm{T}_\tau \, S(\beta) X_{\Xi_0}(\tau) X'_{\Xi_0}(\tau') \right\rangle_0}{\langle S(\beta) \rangle_0} \\ \mathscr{G}_{\alpha\alpha'}(\tau,\tau') &= -\frac{1}{\hbar} \frac{\left\langle \mathrm{T}_\tau \, S(\beta) c_{\alpha\Xi_0}(\tau) c^\dagger_{\alpha'\Xi_0}(\tau') \right\rangle_0}{\langle S(\beta) \rangle_0} \\ \mathscr{D}_{\alpha\alpha'}(\tau,\tau') &= -\frac{1}{\hbar} \frac{\left\langle \mathrm{T}_\tau \, S(\beta) \hat{A}_{\alpha\Xi_0}(\tau) \hat{A}^\dagger_{\alpha'\Xi_0}(\tau') \right\rangle_0}{\langle S(\beta) \rangle_0} \end{aligned} \tag{15.3}$$

with S-matrix defined as in Equation (15.2) and free-time-evolution defined as TBDigitalized and surpressed nature-index.

Proof 15.3 See Appendix, Proofs, page **??**

Remark 15.3.1 The S-matrix-expansion of the Green-functions can be derived in a much more straightforward way, by referring to the quantum field theory of the underlying system. We will present this in Equation (16.34).

Theorem 15.4 *Wick-Theorem from an operator-formalism.* [...]

Proof 15.4 TBDigitalized from [Flensberg].

Theorem 15.5 The formulae in eq. (15.3) can be simplified to

$$\mathscr{G}_{X,X'}(\tau,\tau') = -\sum_{n=0}^{\infty}\left(\frac{-1}{\hbar}\right)^n \int_0^{\hbar\beta} \mathrm{d}\tau_1 \cdots \int_0^{\hbar\beta} \mathrm{d}\tau_n \left\langle \mathrm{T}_\tau\, X_{\Xi_0}(\tau) X'_{\Xi_0}(0) V_{\Xi_0}(\tau_1)\cdots V_{\Xi_0}(\tau_n)\right\rangle_0^{\mathrm{DCD}}$$

$$\mathscr{G}_{\alpha\alpha'}(\tau,\tau') = -\sum_{n=0}^{\infty}\left(\frac{-1}{\hbar}\right)^n \int_0^{\hbar\beta} \mathrm{d}\tau_1 \cdots \int_0^{\hbar\beta} \mathrm{d}\tau_n \left\langle \mathrm{T}_\tau\, c_{\alpha\Xi_0}(\tau) c^\dagger_{\alpha'\Xi_0}(0) V_{\Xi_0}(\tau_1)\cdots V_{\Xi_0}(\tau_n)\right\rangle_0^{\mathrm{DCD}}$$

$$\mathscr{D}_{\alpha,\alpha'}(\tau,\tau') = -\sum_{n=0}^{\infty}\left(\frac{-1}{\hbar}\right)^n \int_0^{\hbar\beta} \mathrm{d}\tau_1 \cdots \int_0^{\hbar\beta} \mathrm{d}\tau_n \left\langle \mathrm{T}_\tau\, A_{\alpha\Xi_0}(\tau) A^\dagger_{\alpha'\Xi_0}(\tau') V_{\Xi_0}(\tau_1)\cdots V_{\Xi_0}(\tau_n)\right\rangle_0^{\mathrm{DCD}}$$

where only DCD $\equiv$ [D]ifferent [C]onnected [D]iagrams are evaluated[1].

Proof 15.5 See Appendix, Proofs, page **??**

To Be Proven with Graph-Theory. Cf. Renormalization-book from Salmhofer. Maybe here is a prefactor-error.

Remark 15.5.1 This theorem, that simplifies the computation of Green-functions drastically, is called the *linked cluster theorem*. It is basically never proven in the generality of cases in which people claim it's circle of applicability. For example in [Altland] this theorem is being introduced for a scalar field theory and it is only proven by an example. In [Salmhofer] this theorem *is* proven but only for a scalar field theory again. In [Mahan] this theorem is just stated as a fact without any proofs. In [Flensberg] this theorem is only proven for the case of electron-electron interactions. In [Negele-Orland] it is also only proven for electron-electron-interactions. All of the proofs (except) for the one from [Salmhofer] are imho very vaguely formulated.

Remark 15.5.2 The straight-forward evaluation of these infinite series is cumbersome and basically never done in practice. There is a much more elegant way to compute Green-functions by identifying proper kernels in the so-called "Dyson-equation" (to be introduced below). This process involves the identification of "good" Green-functions, which can be linked to many observable quantities one the one hand, and possess a kernel that is easy to compute on the other hand.

[1] Cf. Mahan p. 132.

15.2. The Dyson-Equation for Pure One-Body Green-Functions

Notation 15.6 For reasons that will become clear in a later chapter[2] we assume the existence of a function[3]

$$\begin{aligned}\Sigma :\ \mathrm{i}\Omega_\xi &\longrightarrow \mathrm{Mat}_{\mathrm{d}\times\mathrm{d}}(\mathbb{C})\\ \mathrm{i}\omega_n &\longmapsto \Sigma(\mathrm{i}\omega_n)\end{aligned}$$

which we call the *(statistical) self-energy* s.t. the *Dyson-equation* for the one-body Green-function is fulfilled

$$\mathscr{G}(\mathrm{i}\omega_n) = \mathscr{G}_0(\mathrm{i}\omega_n) + \left(\mathscr{G}_0 \cdot \Sigma \cdot \mathscr{G}\right)(\mathrm{i}\omega_n),$$

where the dot inside the parenthesis corresponds to matrix-multiplication.

Remark 15.6.1 It should be noted, that there are different sign-conventions for the self-energy in the literature. We are following the one from [Altland-Book, Thomale-Paper] in contrast to [Negele-Orland, Kopietz]. We will however motivate the here chosen convention below.

Remark 15.6.2 Recall from Lemma TBReferenced from B-Sides that this implies the existence of a corresponding Matsubara-Fourier-transformed version of this equation[4] and that the Matsubara-Fourier-transformed self-energy does not have the unit of an energy.

Remark 15.6.3 The justification for the name „self-energy“ can be seen in the following way: A formal inversion of the equation above yields

$$\begin{aligned}\mathscr{G}(\mathrm{i}\omega_n) = \left(1-\mathscr{G}_0\cdot\Sigma\right)^{-1}(\mathrm{i}\omega_n)\cdot\mathscr{G}_0(\mathrm{i}\omega_n) = \frac{1}{\left(\mathscr{G}_0^{-1}-\Sigma\right)(\mathrm{i}\omega_n)} &= \frac{1}{\hbar}\frac{1}{\mathrm{i}\omega_n - \left(H_0+\Sigma(\mathrm{i}\omega_n)-\eta\mu\right)/\hbar}\\ &= \frac{1}{\mathrm{i}\hbar\omega_n - \left(H_0+\Sigma(\mathrm{i}\omega_n)-\eta\mu\right)}\end{aligned} \tag{15.4}$$

The self-energy is therefore a term, that is formally being added on top of the Hamiltonian-matrix.

[2] When we have introduced the effective action formalism, it will become clear that one can conclude the existence and shape of this function from symmetry-considerations for *most* fermionic and bosonic systems. For phonons however one can not conclude the existence of a self-energy. Phonons will be discussed in the next section.

[3] Note that the self-energy depends on the statistics of the system $\Sigma_\zeta \equiv \Sigma$.

[4] More concretely

$$\begin{aligned}\Sigma :\ (0,\hbar\beta)^2 &\longrightarrow \mathrm{Mat}_{\mathrm{d}_{\mathcal{H}}\times\mathrm{d}_{\mathcal{H}}}(\mathbb{C})\\ (\tau,\tau') &\longmapsto \Sigma(\tau,\tau')\end{aligned}$$

s.t. the Dyson-equation in (imaginary-)time-space the (connected or unconnected) one-body Green-function is fulfilled

$$\mathscr{G}(\tau,\tau') = \mathscr{G}_0(\tau,\tau') + \left(\mathscr{G}_0\cdot\Sigma\cdot\mathscr{G}\right)(\tau,\tau'),$$

where the dot inside the parenthesis is to be understood as a generalized product as in TBReferenced from B-Sides.

Remark 15.6.4 Notice that in the equation-system of the last remark we found the identity

$$\mathscr{G}^{-1}(\mathrm{i}\omega_n) = \mathscr{G}_0^{-1}(\mathrm{i}\omega_n) - \Sigma(\mathrm{i}\omega_n) \tag{15.5}$$

If we now generalize the one-body Green-function to possess two Matsubara-frequencies, i.e.

$$\mathscr{G}\left(A|A'\right) \equiv \mathscr{G}\left(\mathrm{i}\omega_n, \alpha|\mathrm{i}\omega_{n'}, \alpha'\right) \equiv \mathscr{G}_{\substack{\alpha,\alpha' \\ \mathrm{i}\omega_n,\mathrm{i}\omega_{n'}}} \equiv \delta_{\omega_n,\omega_{n'}}\mathscr{G}_{\alpha,\alpha'}(\mathrm{i}\omega_n), \qquad \text{etc.} \tag{15.6}$$

then we see, that we can write the *generalized Dyson-equation*

$$\mathscr{G}^{-1} = \mathscr{G}_0^{-1} - \Sigma. \tag{15.7}$$

Corollary 15.7 With the von Neumann-series, one concludes

$$\mathscr{G}(\mathrm{i}\omega_n) = \left(1 - \mathscr{G}_0 \cdot \Sigma\right)^{-1}(\mathrm{i}\omega_n) \cdot \mathscr{G}_0(\mathrm{i}\omega_n) = \sum_{n=0}^{\infty} \left(\left[\mathscr{G}_0 \cdot \Sigma\right]^n \cdot \mathscr{G}_0\right)(\mathrm{i}\omega_n),$$

Proof 15.7 Follows immediately from using the von Neumann-series.

Remark 15.7.1 It will turn out, that depending on the situation, it is advantegous to make use of the first or second equation-sign.

Remark 15.7.2 It will also turn out, that the self-energy of the system, which seems like a very abstract concept here, can be calculated quite straight-forward. The self-energy Σ of a particle will turn out to be the sum of „all the connected irreducible diagrams“, whereas we will carefully explain what is meant by this statement in the following subsections. We will need all the identities presented here for the computation of the self-energy.

15.3. The Dyson-Equation for Phononic Displacement-Green-Functions

Notation 15.8 For reasons that will become clear in a later chapter we assume the existence of a function

$$\begin{aligned}\Pi : \mathrm{i}\Omega_{+1} &\longrightarrow \mathrm{Mat}_{d_{\max}\times d_{\max}}(\mathbb{C}) \\ \mathrm{i}\omega_n &\longmapsto \Pi(\mathrm{i}\omega_n)\end{aligned}$$

with $d_{\max} = rd_{\mathbf{x}}N_p$, which we call the *polarization* s.t. the *Dyson-equation* of the phononic displacement-Green-function is fulfilled

$$\mathscr{D}(\mathrm{i}\omega_n) = \mathscr{D}_0(\mathrm{i}\omega_n) + \left(\mathscr{D}_0 \cdot \Pi \cdot \mathscr{D}\right)(\mathrm{i}\omega_n),$$

where the dot inside the parenthesis corresponds to matrix-multiplication.

Remark 15.8.1 In contrast to its electronic counterpart, this equation is unique within the literature.

Remark 15.8.2 Recall from Lemma TBReferenced from B-Sides that this implies the existence of a corresponding Matsubara-Fourier-transformed version of this equation[5] which again has not the unit of an energy.

Remark 15.8.3 Notice that a formal inversion of the equation above yields[6]

$$\begin{aligned} &\mathscr{D}(\mathrm{i}\omega_n) = \left(1 - \mathscr{D}_0 \cdot \Pi\right)^{-1}(\mathrm{i}\omega_n) \cdot \mathscr{D}_0(\mathrm{i}\omega_n) = \frac{1}{\left(\mathscr{D}_0^{-1} - \Pi\right)(\mathrm{i}\omega_n)} \\ \iff &\mathscr{D}^{-1}_{\nu,\nu'}(\mathbf{q}, \mathrm{i}\omega_n) = \mathscr{D}^{-1}_{\substack{0\\ \nu,\nu'}}(\mathbf{q}, \mathrm{i}\omega_n) - \Pi_{\nu,\nu'}(\mathbf{q}, \mathrm{i}\omega_n) \end{aligned}$$

Remark 15.8.4 For a system with a single branch one obtains

$$\mathscr{D}_\bullet(\mathbf{q}, \mathrm{i}\omega_n) = \frac{-2\omega_\bullet(\mathbf{q})}{\omega_n^2 + \omega_\bullet^2(\mathbf{q}) + 2\omega_\bullet(\mathbf{q})\Pi_\bullet(\mathbf{q}, \mathrm{i}\omega_n)}. \tag{15.8}$$

Corollary 15.9 With the von Neumann-series, one concludes

$$\mathscr{D}(\mathrm{i}\omega_n) = \left(1 - \mathscr{D}_0 \cdot \Pi\right)^{-1}(\mathrm{i}\omega_n) \cdot \mathscr{D}_0(\mathrm{i}\omega_n) = \sum_{n=0}^{\infty} \left(\left[\mathscr{D}_0 \cdot \Pi\right]^n \cdot \mathscr{D}_0\right)(\mathrm{i}\omega_n),$$

Proof 15.9 Follows from a formal inversion with the von Neumann-series.

[5] More concretely

$$\begin{aligned} \Pi :\ (0, \hbar\beta)^2 &\longrightarrow \mathrm{Mat}_{d_{\max}\times d_{\max}}(\mathbb{C}) \\ (\tau, \tau') &\mapsto \Pi(\tau, \tau') \end{aligned}$$

s.t. the Dyson-equation in (imaginary-)time for phononic displacement-Green-function is fulfilled

$$\mathscr{D}(\tau, \tau') = \mathscr{D}_0(\tau, \tau') + \left(\mathscr{D}_0 \cdot \Pi \cdot \mathscr{D}\right)(\tau, \tau'),$$

where the dot inside the parenthesis is to be understood as a generalized product as in TBReferenced from B-Sides.

[6] Cf. [01 Allens Long Article.pdf (with underlines)].

Remark 15.9.1 It will turn out, that depending on the situation, it is advantegous to make use of the first or second equation-sign.

Remark 15.9.2 Just as for the self-energy of the system, the polarization can be calculated quite straight-forward.

15.4. The Bethe-Salpeter-Equation for Two-Body Green-Functions of Particle-Nature

For pedagogical reasons (we just presented a formula with which one can compute the one-body Green-function of an interacting system; Now we should present the formula for the two-body Green-function of an interacting system) we will now present the Bethe-Salpeter-Equation for the two-body Green-function and refer to the proof that is being carried out several chapters later.

Theorem 15.10 Under the proper circumstances, we can conclude the existence of a so-called *effective interaction* Vert which solves the equation

$$\begin{aligned}\mathscr{G}(i_1, i_2 \mid i_1', i_2') &= \mathscr{G}(i_1 \mid i_1')\mathscr{G}(i_2 \mid i_2') + \zeta\mathscr{G}(i_1 \mid i_2')\mathscr{G}(i_2 \mid i_1') \\ &- \sum\!\!\!\!\!\!\int_{j_1,j_2,j_1',j_2'} \mathscr{G}(i_1 \mid j_1)\mathscr{G}(i_2 \mid j_2)\ \mathrm{Vert}(j_1, j_2 \mid j_1', j_2')\mathscr{G}(j_1' \mid i_1')\mathscr{G}(j_2' \mid i_2')\end{aligned} \tag{15.9}$$

Proof 15.10 Can be concluded from Theorem 16.46.

Remark 15.10.1 Just like the self-energy and the polarization are not quantities that are "easy to identify", the effective interaction Vert is another quantity that needs to be computed. We will learn later how to do it.

15.5. Matsubara-Summations

Without any motivation, we will elucidate how to compute summations of the type

$$S_\zeta = \sum_{\omega_n \in \Omega_\zeta} h(\omega_n). \tag{15.10}$$

since we will encounter summations of this type quite often in the following chapters.

Notation 15.11 For reasons, that will become clear soon, we assume that the function

$$\mathbb{C} - \mathrm{Z} \ni z \mapsto h(-\mathrm{i}z) \in \mathbb{C} \tag{15.11}$$

is complex differentiable in the entire complex plane, with the exception of a finite number of poles of finite order which are no Matsubara-frequencies, which we denote with

$$\mathrm{Z} \equiv \left\{ z_1, \ldots , z_N \right\} . \tag{15.12}$$

Lemma 15.12 The complex function

$$\mathbb{C} - \mathrm{i}\Omega_\zeta \ni z \mapsto g_\zeta(z) = \frac{\hbar\beta}{\mathrm{e}^{\hbar\beta z} - \zeta} \in \mathbb{C} \tag{15.13}$$

has simple poles as

$$\forall \mathrm{i}\omega_n \in \mathrm{i}\Omega_\zeta \, : \, \lim_{z \to \mathrm{i}\omega_n} (z - \mathrm{i}\omega_n) g_\zeta(z) = \zeta . \tag{15.14}$$

Proof 15.12 See Appendix, Proofs, page 355

Remark 15.12.1 Notice that now the function

$$\mathbb{C} - \mathrm{P} \ni z \mapsto g_\zeta(z)\, h(-\mathrm{i}z) \equiv f_\zeta(z) \in \mathbb{C} \tag{15.15}$$

is complex differentiable as a product of complex differentiable functions, with the pole-set $\mathrm{P} = \mathrm{i}\Omega_\zeta \cup \mathrm{Z}$.

Remark 15.12.2 Notice further the identity

$$g_\zeta(z) = \frac{\hbar\beta}{\mathrm{e}^{\hbar\beta z} - \zeta} = \hbar\beta \mathrm{n}^{\zeta}_{\hbar\beta}(z) \tag{15.16}$$

We will now quote (without proof) and use the Residue-theorem as in the following form.

Theorem 15.13 The *Residue-theorem* in the form we will use it:

Consider a complex function $f : \mathbb{C} \to \mathbb{C}$, which has poles (defined below) at the points $z_1, \ldots, z_N$, and is *holomorphic* (= complex differentiable) on $\mathbb{C}\backslash\{z_1, \ldots, z_N\}$.
Let γ be a closed contour in the complex plane. Then the residue theorem states that

$$\oint_\gamma dz f(z) = 2\pi i \sum_{z_n} I(\gamma, z_n)\,\mathrm{Res}(f, z_n)\,, \text{ where}$$

$$I(\gamma, z) = \begin{cases} 0 & \text{if } z \text{ is not enclosed by } \gamma \\ +n & \text{if } z \text{ is enclosed } n \text{ times counter clockwise by } \gamma \\ -n & \text{if } z \text{ is enclosed } n \text{ times clockwise by } \gamma, \end{cases}$$

and the *residue* $\mathrm{Res}(f, z_0)$ of the function f at point z_0 is defined as follows. If f has pole of order n at z_0, meaning that $f(z) = \frac{g(z)}{(z-z_0)^n}$, where $g(z)$ is holomorphic in the vicinity of z_0, then:

$$\mathrm{Res}(f, z_0) = \frac{1}{(n-1)!}\frac{d^{n-1}}{dz^{n-1}}\left((z-z_0)^n f(z)\right)|_{z=z_0}.$$

Proof 15.13 See in the literature, for example TBReferenced.

Theorem 15.14 If f decays sufficiently fast[7] at complex infinity, then it follows

$$S_\zeta = \sum_{\omega_n \in \Omega_\zeta} h(\omega_n) \overset{\text{Claim}}{=} -\zeta \sum_{k=1}^{N} \mathrm{Res}\left(f_\zeta(z),\, z = z_k\right) \tag{15.17}$$

Proof 15.14 See Appendix, Proofs, page 355

Lemma 15.15 In the case that the complex function can be written as a product of simple poles

$$\mathbb{C} - \mathrm{Z} \ni z \mapsto h(-\mathrm{i}z) \equiv \text{const.} \prod_{k=1}^{N} \frac{1}{\left(z - z_k\right)} \in \mathbb{C} \tag{15.18}$$

one has

$$S_\zeta = \sum_{\omega_n \in \Omega_\zeta} h(\omega_n) \overset{\text{Claim}}{=} -\zeta\hbar\beta\,\text{const} \sum_{k=1}^{N} \mathrm{n}^{\zeta}_{\hbar\beta}(z_k) \prod_{\substack{l=1 \\ l\neq k}}^{N} \frac{1}{z_k - z_l} \tag{15.19}$$

[7]This means that ...

whereas an empty product is as usual defined to be equal to one.

Proof 15.15 Follows immediately by insertion.

Corollary 15.16 For

$$h(\omega_n) = \frac{1}{\hbar\beta}\frac{1}{\mathrm{i}\omega_n - C} \tag{15.20}$$

one finds the identity

$$S_\zeta = \sum_{\omega_n \in \Omega_\zeta} h(\omega_n) \stackrel{\text{Claim}}{=} -\zeta \mathrm{n}^{\zeta}_{\hbar\beta}(C). \tag{15.21}$$

Proof 15.16 See Appendix, Proofs, page 355

Remark 15.16.1 Notice that this is exactly the type of summation that appears when evaluating Matsubara-sums of the type

$$S_\zeta = \frac{1}{\beta}\sum_{\omega_n \in \Omega_\zeta} \underline{\mathscr{G}_0}\,(\mathrm{i}\omega_n). \tag{15.22}$$

Lemma 15.17 For $\mathrm{i}\omega_n \in \mathrm{i}\Omega_{-1}$ we notice the auxiliary-identities

$$\mathrm{F}_{\hbar\beta}(\bullet + \mathrm{i}\omega_n) = -\mathrm{B}_{\hbar\beta}(\bullet) \qquad \mathrm{F}_{\hbar\beta}(\mathrm{i}\omega_n - \bullet) = \mathrm{B}_{\hbar\beta}(\bullet) + 1$$

Proof 15.17 See Appendix, Proofs, page 356

Corollary 15.18 For $\omega_m \in \Omega_{-1}$ and summation-function

$$h(\omega_n) = \frac{1}{\hbar^2\beta}\frac{1}{\mathrm{i}\omega_n - C_1}\frac{-2C_2}{\left(\omega_m - \omega_n\right)^2 + C_2^2} \tag{15.23}$$

one finds the identity

$$S_{-1} = \sum_{\omega_n \in \Omega_{-1}} h(\omega_n) \stackrel{\text{Claim}}{=} \frac{-1}{\hbar}\left[\frac{\mathrm{B}_{\hbar\beta}(C_2) + \mathrm{F}_{\hbar\beta}(C_1)}{\mathrm{i}\omega_m - C_1 + C_2} + \frac{\mathrm{B}_{\hbar\beta}(C_2) + 1 - \mathrm{F}_{\hbar\beta}(C_1)}{\mathrm{i}\omega_m - C_1 - C_2}\right] \tag{15.24}$$

Proof 15.18 See Appendix, Proofs, page 356

Remark 15.18.1 Notice that this is exactly the type of summation that appears when evaluating Matsubara-sums of the type

$$\frac{1}{\beta} \sum_{\omega_{n_2} \in \Omega_{-1}} \mathscr{G}_{0\,\blacksquare_1}(\mathrm{i}\omega_{n_2})\, \mathscr{D}_{0\,\blacksquare_2}(\mathrm{i}(\omega_n - \omega_{n_2})). \tag{15.25}$$

Part III.

(Thermal) Quantum Field Theory

16. Algebraic Concepts in Quantum Field Theory

> *"Quantum field theory (QFT) provides an extremely powerful set of computational methods that have yet to find any fundamental limitations. It has led to the most fantastic agreement between theoretical predictions and experimental data in the history of science. It provides deep and profound insights into the nature of our universe, and into the nature of other possible self-consistent universes. On the other hand, the subject is a mess."*

M. D. Schwartz in [**?**], p. xv

> *"Go for the messes - that's where the action is."*

Steven Weinberg in [TBReferenced]

16.1. Quantum Field Theory - Pure Systems

> *"The formal calculus in Graßmann algebra, which was developed by Berezin in his first book Method of the Second Quantization, made him sure that a nontrivial analogue of analysis exists in which anticommuting variables appear on the same footing as the usual ones."*

Felix Berezin in [Introduction to Superanalysis]

Supermathematics is a branch of mathematical physics, which (among many other applications) applies algebraic concepts to the study of bosonic and in particular fermionic systems. The mathematical rigorous introduction of these concepts is (at least for the purposes that we have in mind) not very complicated, but it is a lot of simple definitions and theorems.

Since there exists already so much great introductory literature to these concepts we will not prove all the standard-theorems but just list those that we need, referring to the literature[1] for the proofs.

16.1.1. Fundamentals

Definition 16.1 We associate generators $U \cup \overline{U} = \left\{\psi_1, \overline{\psi}_1, \ldots, \psi_{\mathrm{D}}, \overline{\psi}_{\mathrm{D}}\right\}$ with annihilation-operators (ψ) and creation-operators ($\overline{\psi}$). Then we define the *algebra of statistical polynomials* $\mathscr{P}_\zeta$ as the associative algebra generated by $\mathbb{C} \oplus U \cup \overline{U}$ with relations

$$\psi_i \psi_j = \zeta \psi_j \psi_i, \qquad \overline{\psi}_i \psi_j = \zeta \psi_j \overline{\psi}_i, \qquad \overline{\psi}_i \overline{\psi}_j = \zeta \overline{\psi}_j \overline{\psi}_i. \tag{16.1}$$

for $i, j \in \{1, \ldots, \mathrm{D}\}$.

[1] Cf. [Altland][Zirnbauer][Zinn-Justin][Math-Physics-Article] etc.

Remark 16.1.1 One distinguishes between two cases:

- For $\zeta = -1$ this algebra is called *Graßmann-algebra* and oftentimes denotes as $\mathscr{P}_{-1} \equiv \mathscr{G}$. The generators $\{\psi_i, \overline{\psi}_i\}_{i\in\{1,\ldots,\mathrm{D}\}}$ are referred to as *Graßmann-numbers*.
- For $\zeta = +1$ this algebra is called *symmetric algebra*. The generators $\{\psi_i, \overline{\psi}_i\}_{i\in\{1,\ldots,\mathrm{D}\}}$ are referred to as *complex-numbers*.

Remark 16.1.2 Notice that these commutation-relations are the same as in Equation (7.7).

Remark 16.1.3 For the moment, we focus on the case $\mathrm{D} = \mathrm{d}_\eta$, such that for each creation- and annihilation-operator there is a Graßmann-variable associated. In the later application, it will be the case that $\mathrm{D} \equiv \mathrm{d}_\eta N$ with $N \to \infty$, such that for each creation- and annihilation-operator, there will be infinitely many Graßmann-variables associated. Typically one has a notation as $\psi_{n\alpha}$ etc. with $\alpha \in \mathscr{A}$ and $n \in \{1, 2, \ldots, N\}$. The index n is either associated with an imaginary time $\tau_n \in (0, \hbar\beta)$ (such that $\psi_{n\alpha} \equiv \psi_\alpha(\tau_n)$) or a Matsubara-frequency $\mathrm{i}\omega_n \in \mathrm{i}\Omega_\zeta$ (such that $\psi_{n\alpha} \equiv \psi_\alpha(\mathrm{i}\omega_n)$). When we consider different Graßmann-variables $\left\{\psi_1, \ldots, \psi_{\mathrm{d}_\eta}\right\}$ and $\left\{\phi_1, \ldots, \phi_{\mathrm{d}_\eta}\right\}$ that anticommute with each other (and correspond to different „timestamps“ or „Matsubara-frequencies“), it is this situation just elucidated that we have in mind.

Lemma 16.2 A basis of the algebra $\mathscr{P}_\zeta$ is given by the monomials

$$\left\{\psi_1^{n_1} \cdots \psi_D^{n_D} \overline{\psi}_1^{\overline{n}_1} \cdots \overline{\psi}_D^{\overline{n}_D} \;\middle|\; n_1, \overline{n}_1 \ldots, n_\mathrm{D}, \overline{n}_\mathrm{D} \in \mathbb{N}_\zeta\right\} \quad \subset \quad \mathscr{P}_\zeta \tag{16.2}$$

Proof 16.2 Follows from introductory courses on linear algebra.

Corollary 16.3 Every element $f(\psi, \overline{\psi}) \equiv f \in \mathscr{P}_\zeta$ with $\psi \equiv (\psi_1, \ldots, \psi_\mathrm{D})$ can be written as uniquely as

$$f(\psi, \overline{\psi}) = \sum_{\substack{n_1,\ldots,n_D \in \mathbb{N}_\zeta \\ \overline{n}_1,\ldots,\overline{n}_D \in \mathbb{N}_\zeta}} c_{\substack{n_1,\ldots,n_D \\ \overline{n}_1,\ldots,\overline{n}_D}} \psi_1^{n_1} \cdots \psi_D^{n_D} \overline{\psi}_1^{\overline{n}_1} \cdots \overline{\psi}_D^{\overline{n}_D} \tag{16.3}$$

with complex numbers $c_{\substack{n_1,\ldots,n_D \\ \overline{n}_1,\ldots,\overline{n}_D}} \in \mathbb{C}$.

Proof 16.3 Follows from introductory courses on linear algebra.

Remark 16.3.1 We decompose f as

$$\begin{aligned} f(\psi, \overline{\psi}) &= c_{\substack{0,\ldots,0 \\ 0,\ldots,0}} &&+ \left(f(\psi, \overline{\psi}) - c_{\substack{0,\ldots,0 \\ 0,\ldots,0}}\right) \\ &\equiv \mathrm{body}_f &&+ \mathrm{soul}_f(\psi, \overline{\psi}) \\ &\equiv f_0 &&+ f_{\mathrm{nil}}(\psi, \overline{\psi}) \end{aligned} \tag{16.4}$$

whereas $c_{\substack{0,\ldots,0\\0,\ldots,0}} \equiv f_0 \equiv \text{body}_f \in \mathbb{C}$ is called the *body* and $\text{soul}_f \equiv f_{\text{nil}}$ the *soul* of f.

Lemma 16.4 The soul of f is *nilpotent*, i.e.

$$\exists N_f \in \mathbb{N} : f_{\text{nil}}^{N_f}(\psi, \overline{\psi}) = 0. \tag{16.5}$$

whereas we define N_f as the *nilpotency index* of f.

Proof 16.4 The fact that f_{nil} is nilpotent is easily seen by writing down the polynomial expansion and taking the square... The exact value of N_f depends on the structure of the soul.

Remark 16.4.1 Notice that $N_f \leq 2D$.

Definition 16.5 We call a polynomial $f \in \mathscr{P}_\zeta$ *invertible*, if there is another element $f^{-1} \equiv \frac{1}{f} \in \mathscr{P}_\zeta$ such that

$$f f^{-1} = f^{-1} f = 1. \tag{16.6}$$

Remark 16.5.1 We call f^{-1} the *inverse element* of f.

Lemma 16.6 A polynomial f is invertible, iff $f_0 \neq 0$ and then the inverse element is given by

$$f^{-1}(\psi, \overline{\psi}) = \frac{1}{f_0} \sum_{n=0}^{N_f - 1} \left(- \frac{f_{\text{nil}}(\psi, \overline{\psi})}{f_0} \right)^n. \tag{16.7}$$

Proof 16.6 See Appendix, Proofs, page 359[2].

Remark 16.6.1 Notice further, that this is "simply" a Graßmann-version of the geometric series, formally

$$f^{-1}(\psi, \overline{\psi}) \equiv \frac{1}{f_0 + f_{\text{nil}}(\psi, \overline{\psi})} \equiv \frac{1}{f_0} \sum_{n=0}^{\infty} \left(- \frac{f_{\text{nil}}(\psi, \overline{\psi})}{f_0} \right)^n. \tag{16.8}$$

Remark 16.6.2 Notice that the inverse element is also again a polynomial.

[2] Cf. TBDigitalized TBD3; Waste-Folder 1, topic 14 We prove that Equation (16.7) is indeed the inverse element of f. TBD.

Definition 16.7 We define derivatives for $\zeta = +1$ in the usual way. For $\zeta = -1$ we define the *left-derivative* for each $j \in \{1, \dots, \mathrm{D}\}$ as

$$f \equiv f_0 + \psi_j f_1 \quad \Rightarrow \quad \frac{\partial}{\partial \psi_j} f := f_1 \tag{16.9}$$

whereas f_1 is independent of ψ_j.

Remark 16.7.1 Something peculiar about Graßmann-derivatives is that one can also define a *right-derivative* as

$$f \equiv f_0 + f_1 \psi_j \quad \Rightarrow \quad \left(\partial f \middle/ \partial \psi_j \right) \equiv f \frac{\overleftarrow{\partial}}{\partial \psi_j} := f_1 \tag{16.10}$$

Remark 16.7.2 The easiest way to see why one can do this, is to realize that derivatives are basically just linear maps on $\mathscr{P}_\zeta$. For $\zeta = 1$ this map is the same, no matter if one defines it as a left- or right-derivative. For $\zeta = -1$ one can define two different maps, which are related to each other.

Definition 16.8 For a monomial expression

$$m \equiv \psi_1^{n_1} \cdots \psi_D^{n_D} \overline{\psi}_1^{\overline{n}_1} \cdots \overline{\psi}_D^{\overline{n}_D} \tag{16.11}$$

we define the *degree* of m as

$$\deg(m) := \sum_{j=1}^{D} \left(n_j + \overline{n}_j \right) \tag{16.12}$$

Theorem 16.9 Let m_1 and m_2 be two monomial expressions in $\mathscr{P}_\zeta$. Then the *graded chain-rule* reads

$$\frac{\partial}{\partial \psi_j} \left(m_1 m_2 \right) (\psi, \overline{\psi}) = \left[\left(\frac{\partial m_1}{\partial \psi_j} \right) m_2 + \zeta^{\deg(m_1)} m_1 \left(\frac{\partial m_2}{\partial \psi_j} \right) \right] (\psi, \overline{\psi}) \tag{16.13}$$

Proof 16.9 TBDone.

Lemma 16.10 The *statistics neutral commutation relations* read

$$\left[\frac{\partial}{\partial \psi_i}, \psi_j \right]_\zeta = \delta_{i,j}, \qquad \left[\frac{\partial}{\partial \psi_i}, \overline{\psi}_j \right]_\zeta = 0 = \left[\frac{\partial}{\partial \overline{\psi}_i}, \psi_j \right]_\zeta \tag{16.14}$$

Proof 16.10 See Appendix, Proofs, page **??**.

Lemma 16.11 The *statistics neutral Schwartz-rule* reads

$$0 = \left[\frac{\partial}{\partial\psi_i}, \frac{\partial}{\partial\psi_j}\right]_\zeta = \left[\frac{\partial}{\partial\psi_i}, \frac{\partial}{\partial\overline{\psi}_j}\right]_\zeta = \left[\frac{\partial}{\partial\overline{\psi}_i}, \frac{\partial}{\partial\overline{\psi}_j}\right]_\zeta \tag{16.15}$$

Proof 16.11 See Appendix, Proofs, page **??**.

Remark 16.11.1 Notice that these commutation-relations are the same as in Notation 7.21. In fact, one can introduce the statistical algebra of polynomials and their derivatives as ζ-algebras quite straightforwardly by following the same steps that we did in the previous chapters. It becomes though a little bit difficult to follow on the notation (one needs, for the purposes that we have in mind, eight[3] „copies“ of generators that correspond to creation- and annihilation-operators), and therefore we will not do so here.

Notation 16.12 By inclusion of another set of generators $\left\{j_1, \overline{j}_1, \ldots, j_\mathrm{D}, \overline{j}_\mathrm{D}\right\}$, which are commonly referred to as *source-terms* we can consider combined polynomial expressions in even another polynomial algebra, which we don't give a name for notational simplicity.

Remark 16.12.1 The introduction of these so-called source-terms is the main trick when it comes to the computation of Green-functions within the framework of TQFT.

Remark 16.12.2 We introduce the notation

$$\overline{\psi}^\mathrm{T} \equiv \psi^\dagger \quad \text{and} \quad j^\dagger\psi \equiv \sum_{k=1}^{D} \overline{j}_k\psi_k \tag{16.16}$$

Lemma 16.13 The derivative of a source-term reads

$$\begin{aligned}
\frac{\partial}{\partial\overline{j}_i}\exp\sum_k \overline{j}_k\psi_k &= \psi_i \exp\sum_k \overline{j}_k\psi_k \\
\frac{\partial}{\partial j_{i'}}\exp\sum_k \overline{\psi}_k j_k &= \zeta\,\overline{\psi}_{i'} \exp\sum_k \overline{\psi}_k j_k
\end{aligned}$$

Proof 16.13 See Appendix, Proofs, page 358.

Corollary 16.14 It holds that

[3] Two copies for polynomials in variables and their conjugates. Two more for their derivatives. And again four more for „source-terms“ and their derivatives (to be introduced soon).

$$\frac{\partial^n}{\partial \bar{j}_{i_1} \cdots \partial \bar{j}_{i_n}} \exp \sum_k \bar{j}_k \psi_k = \psi_{i_1} \cdots \psi_{i_n} \exp \sum_k \bar{j}_k \psi_k$$
$$\frac{\partial^{n'}}{\partial j_{i'_1} \cdots \partial j_{i'_{n'}}} \exp \sum_k \overline{\psi}_k j_k = \zeta^n \overline{\psi}_{i'_1} \cdots \overline{\psi}_{i'_{n'}} \exp \sum_k \overline{\psi}_k j_k$$

Proof 16.14 Using the Lemma above proves the corollary.

Remark 16.14.1 Notice that the commutation-relations imply

$$\frac{\partial^{n+n'}}{\partial \bar{j}_{i_1} \cdots \partial \bar{j}_{i_n} \partial j_{i'_1} \cdots \partial j_{i'_{n'}}} \mathrm{e}^{\sum_k \bar{j}_k \psi_k + \overline{\psi}_k j_k} = \zeta^n \psi_{i_1} \cdots \psi_{i_n} \overline{\psi}_{i'_1} \cdots \overline{\psi}_{i'_{n'}} \mathrm{e}^{\sum_k \bar{j}_k \psi_k + \overline{\psi}_k j_k} \tag{16.17}$$

Definition 16.15 For $\zeta = +1$ we define integration as the usual Riemann- / Lebesgue-integral. For $\zeta = -1$ we define for a function as in Equation (16.3) the *Berezin-integral* as

$$\int \mathrm{d}_\zeta(\psi, \overline{\psi}) f(\psi, \overline{\psi}) := \frac{\partial^{2D}}{\partial \psi_1 \cdots \partial \psi_D \partial \overline{\psi}_1 \cdots \partial \overline{\psi}_D} f \tag{16.18}$$

with integral-measure[4]

$$\mathrm{d}_\zeta(\overline{\psi}, \psi) \equiv \prod_{j=1}^{D} \frac{\mathrm{d}\overline{\psi}_j \mathrm{d}\psi_j}{(2\pi \mathrm{i})^{\theta(\zeta)}} \tag{16.20}$$

16.1.2. Partition-Functions and Expectation-Values for Non-Interacting Models

Notation 16.16 Let $A \in \mathrm{GL}_n(\mathbb{C})$ be an invertible, quadratic matrix. Then we consider *the partition-function corresponding to A with statistics ζ* as

$$Z_0 := \int \mathrm{d}_\zeta(\overline{\psi}, \psi) \exp - \psi^\dagger A \psi \equiv \int \mathrm{d}_\zeta(\overline{\psi}, \psi) \exp - S_0[\psi, \overline{\psi}] \tag{16.21}$$

for complex- or Graßmann-variables.

Remark 16.16.1 Equivalently, we also speak about the *partition-function w.r.t. the action S*.

[4] Notice that in the case $\zeta = +1$, one has

$$\frac{\mathrm{d}\overline{\psi}_j \mathrm{d}\psi_j}{2\pi \mathrm{i}} \equiv \frac{\mathrm{d}\,\mathrm{Re}\,\psi_j\, \mathrm{d}\,\mathrm{Im}\,\psi_j}{\pi} \tag{16.19}$$

Remark 16.16.2 Note that we are, following the convention for notational simplicity, surpressing the dependence on ζ in the symbol Z. We will oftentimes surpress this dependence also on other quantities throughout this section (and work).

Lemma 16.17 The partition function can be calculated to

$$Z_0 = \frac{1}{\mathrm{Det}\, A^{\zeta}}. \tag{16.22}$$

Proof 16.17 See Appendix, Proofs, page **??**[5].

Corollary 16.18 It follows also

$$Z_0 = \exp\left(-\zeta \,\mathrm{Tr}\,\mathrm{Log}\, A\right) \tag{16.23}$$

Proof 16.18 Follows from the identity Tr Log • = Log Det •.

Definition 16.19 We consider a monomial expression in the integration-variables, and define its *expectation value w.r.t. A*

$$\left\langle \psi_{i_1} \cdots \psi_{i_n} \overline{\psi}_{i'_1} \cdots \overline{\psi}_{i'_m} \right\rangle_0 := \frac{1}{Z} \int \mathrm{d}_{\zeta}(\overline{\psi}, \psi)\, \psi_{i_1} \cdots \psi_{i_n} \overline{\psi}_{i'_1} \cdots \overline{\psi}_{i'_m} \exp - \psi^{\dagger} A\, \psi. \tag{16.24}$$

Remark 16.19.1 By $\mathbb{C}$-linearity, one defines *expectation-values w.r.t. A* for polynomial expressions in the integration-variables $\mathrm{P}(\overline{\psi}, \psi)$.

Definition 16.20 We define the *n-body Green-function* $\mathcal{G}_0$ *w.r.t. A* as expectation-value

$$\mathcal{G}_0(i_1, \ldots, i_n \,|\, i'_n, \ldots, i'_1) = \left(-\frac{1}{c}\right)^n \left\langle \psi_{i_1} \cdots \psi_{i_n} \overline{\psi}_{i'_1} \cdots \overline{\psi}_{i'_n} \right\rangle_0 \tag{16.25}$$

Remark 16.20.1 The constant c is arbitrary. For the setting that we have in mind later, it is *convention* to set it to $c \equiv \hbar$, such that the dimension of the Green-function becomes $1/$Energy.

Remark 16.20.2 It is a priori completely unclear, that this definition of the Green-function will turn out to have anything to do with the Green-functions that we defined in earlier chapters. We will however prove in the next

[5] For fermions, see Zirnbauers QFT1 2020 Lecture 18. for bosons, see Altland-book.

chapter, that the Green-functions defined in Equation (14.15) can be re-expressed as correlation-functions of integrals of the type introduced here.

Theorem 16.21 For monomial expressions, which contain the same number of ψ-variables as $\overline{\psi}$-variables, one finds the *Wick-theorem*

$$\left\langle \psi_{i_1} \cdots \psi_{i_n} \overline{\psi}_{i'_n} \cdots \overline{\psi}_{i'_1} \right\rangle_0 = \sum_{\pi \in \mathrm{S}_n} \mathrm{sign}(\pi)^{1-\theta(\zeta)} \prod_{k=1}^{n} \left\langle \psi_i \overline{\psi}_{i'_{\pi(k)}} \right\rangle_0 = \sum_{\pi \in \mathrm{S}_n} \mathrm{sign}(\pi)^{1-\theta(\zeta)} \prod_{k=1}^{n} A^{-1}_{i_k i'_{\pi(k)}} \tag{16.26}$$

Proof 16.21 See Appendix, Proofs, page **??**[6].

Remark 16.21.1 Notice the special case for $m = n = 1$

$$-c\mathscr{G}_0(i \mid i') = \left\langle \psi_i \overline{\psi}_{i'} \right\rangle_0 = A^{-1}_{i,i'} \quad \Longleftrightarrow \quad \mathscr{G}_0(i \mid i') = -\frac{1}{c} A^{-1}_{i,i'} \quad \Longleftrightarrow \quad A_{i,i'} = -\frac{1}{c}\mathscr{G}_0^{-1}(i \mid i') \tag{16.27}$$

which allows us to write formally in the non-interacting case

$$Z_0 = \int \mathrm{d}_\zeta(\overline{\psi}, \psi) \exp - \psi^\dagger \left(-\frac{1}{c}\mathscr{G}_0^{-1} \right) \psi \equiv \int \mathrm{d}_\zeta(\overline{\psi}, \psi) \exp - S_0[\psi, \overline{\psi}] \tag{16.28}$$

Corollary 16.22 The above theorem implies an identity for the calculation of Green-functions as

$$\mathscr{G}_0(i_1, \ldots, i_n \mid i'_1, \ldots, i'_n) = \left(-\frac{1}{c} \right)^n \left\langle \psi_{i_1} \cdots \psi_{i_n} \overline{\psi}_{i'_n} \cdots \overline{\psi}_{i'_1} \right\rangle_0 = \sum_{\pi \in \mathrm{S}_n} \mathrm{sign}(\pi)^{1-\theta(\zeta)} \prod_{k=1}^{n} \mathscr{G}_0 \left(i_k \mid i'_{\pi(k)} \right) \tag{16.29}$$

Proof 16.22 Using the definition, apparently gives the desired result.

Remark 16.22.1 Notice the reversed order in the second argument-series, in contrast to Eq. (16.25).

[6] ToBeDigitalized, cf. for example Zirnbauer QFT1 2014, page 46.

16.1.3. Partition-Functions and Expectation-Values for Interacting Models

Notation 16.23 Let $A \in \mathrm{GL}_n(\mathbb{C})$ be an invertible, quadratic matrix, and $\mathrm{P}[\overline{\psi}, \psi]$ an arbitrary[7] polynomial expression in the integration-variables. For reasons that are impossible to clarify in advance, we consider the quantity

$$
\begin{aligned}
Z &\equiv \int \mathrm{d}_\zeta(\overline{\psi}, \psi) \exp - \left(\psi^\dagger A \psi + \mathrm{P}[\psi, \overline{\psi}]\right) \\
&\equiv \int \mathrm{d}_\zeta(\overline{\psi}, \psi) \exp - \left(S_0[\psi, \overline{\psi}] + S_{\mathrm{int}}[\psi, \overline{\psi}]\right) \\
&\equiv \int \mathrm{d}_\zeta(\overline{\psi}, \psi) \exp - S[\psi, \overline{\psi}]
\end{aligned}
$$

for complex- or Graßmann-variables.

Notation 16.24 We consider a monomial expression in the integration-variables, and define its *expectation value w.r.t.* S as

$$\left\langle \psi_{i_1} \cdots \psi_{i_n} \overline{\psi}_{i'_1} \cdots \overline{\psi}_{i'_m} \right\rangle := \frac{1}{Z} \int \mathrm{d}_\zeta(\overline{\psi}, \psi)\, \psi_{i_1} \cdots \psi_{i_n} \overline{\psi}_{i'_1} \cdots \overline{\psi}_{i'_m} \exp - S[\psi, \overline{\psi}] \tag{16.30}$$

Remark 16.24.1 Accordingly, one also introduces the *n-body-Green-function w.r.t.* S as expectation-value

$$\mathscr{G}(i_1, \ldots, i_n \mid i'_n, \ldots, i'_1) = \left(-\frac{1}{c}\right)^n \left\langle \psi_{i_1} \cdots \psi_{i_n} \overline{\psi}_{i'_1} \cdots \overline{\psi}_{i'_n} \right\rangle. \tag{16.31}$$

Remark 16.24.2 Notice that this allows us to write down a *formal* equation that is *exclusive* for one-body Green-functions

$$\mathscr{G}(i \mid i') \equiv -\frac{1}{c} \frac{1}{Z} \int \mathrm{d}_\zeta(\overline{\psi}, \psi)\, \psi_i \overline{\psi}_{i'} \exp - \psi^\dagger \left(-\frac{1}{c} \mathscr{G}^{-1}\right) \psi. \tag{16.32}$$

Remark 16.24.3 Combining Equations (16.61) and Equation (16.32) yields that we can rewrite correlation functions as functional integrals over the so-called "quadratic part of the effective action $\mathtt{Vert}(i, i') \equiv \mathtt{Vert}$" (which will be introduced later) as

$$\mathscr{G}(i, i') = -\frac{1}{c} \frac{\displaystyle\int \mathscr{D}(\eta, \overline{\eta})\, \eta(i) \overline{\eta}(i') \exp\, - \eta^\dagger \left(-\frac{1}{c} \mathtt{Vert}(:, :)\right) \eta}{\displaystyle\int \mathscr{D}(\eta, \overline{\eta}) \exp\, - \eta^\dagger \left(-\frac{1}{c} \mathtt{Vert}(:, :)\right) \eta} \tag{16.33}$$

[7]Exception: For systems with fermionic statistics (i.e. $\zeta = -1$) we demand that P is even in the fields $\overline{\psi}$ and ψ, meaning that it only contains polynomials with equal number of Graßmann-variables and conjugated Graßmann-variables.

Fact 16.25 In general, there is no possibility to calculate this partition-function or these expectation-values w.r.t. S in closed-form. But it will turn out to be exactly these expectation-values that we need to compute in order to make predictions about the behaviour of non-interacting systems. There are several standard-techniques to evaluate integrals of this kind, for example

- Expanding in the interaction and using the so-called "Wick-theorem" (see below),
- performing a so-called "Hubbard-Stratanovich-transformation" (to be introduced later),
- applying a so-called "saddle-point approximation" (to be introduced later, works only for for $\zeta = 1$).

Assuming strong-convergence[8] one obtains for the expectation-value

$$\left\langle \psi_{i_1} \cdots \psi_{i_n} \overline{\psi}_{i'_1} \cdots \overline{\psi}_{i'_m} \right\rangle = \frac{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \left\langle \psi_{i_1} \cdots \psi_{i_n} \overline{\psi}_{i'_1} \cdots \overline{\psi}_{i'_m} \prod_{l=0}^{k} S_{\text{int}}[\psi, \overline{\psi}] \right\rangle_0}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \left\langle \prod_{l=0}^{k} S_{\text{int}}[\psi, \overline{\psi}] \right\rangle_0} \tag{16.34}$$

Remark 16.25.1 For monomial expressions with $n = m = 1$, we introduced the self-energy-concept in order to simplify the computation of one-body Green-functions. For monomial expressions with $n = m = 2$, one can introduce the so-called "Bethe-Salpeter-equation" in order to simplify the computation of two-body Green-functions. For general cases, one can simplify this expression with the so-called linked-cluster-theorem, which leads to the expression in Theorem (15.5).

We conclude with a remark from the literature[9]

> “*One often needs to sum the perturbation expansion up to infinite order. This is done by identifying the most important subset of diagrams and including them in a self-energy. A warning is in place: the correct identification of the most important diagrams can be very tricky, even in the presence of a small parameter. The reason is that some high-order diagrams, which are not included in a particular model, even though formally smaller, may occur in such large numbers that they yield a contribution that must be included. Any perturbation theory may be just an asymptotic series and not converging in the strict mathematical sense. Finally, it is by no means clear that a "good" self-energy always even exists. Strongly interacting systems of this kind need a nonperturbative approach.*”

[8] In the sense, that integral and infinite series can be exchanged without altering the result.
[9] Cf. Jauho, p. 52.

16.1.4. External Sources and Connected Green-Functions

Notation 16.26 We equip the partition-function Z in Notation 16.23 with external sources $j, \bar{j}$, by making the replacement[10]

$$Z \mapsto Z[j, \bar{j}] \equiv \int \mathrm{d}_\zeta(\overline{\psi}, \psi) \exp\Big[- \big(S_0[\psi, \overline{\psi}] + S_{\mathrm{int}}[\psi, \overline{\psi}] \big) + j^\dagger \psi + \psi^\dagger j \Big] \tag{16.35}$$

Theorem 16.27 One finds in the non-interacting case

$$Z_0[j, \bar{j}] = \int \mathrm{d}_\zeta(\overline{\psi}, \psi) \exp\left(-\psi^\dagger A\, \psi + j^\dagger \psi + \psi^\dagger j \right) = \frac{\exp\left(-j^\dagger A^{-1} j \right)}{\mathrm{Det}\, A^\zeta} \tag{16.36}$$

Proof 16.27 See Appendix, Proofs, page **??**[11].

Remark 16.27.1 Notice, that this identity can be combined with Equation (16.27) such that one obtains

$$\begin{aligned} \int \mathrm{d}_\zeta(\overline{\psi}, \psi) \exp\left(-\psi^\dagger \left(-\frac{1}{c} \mathscr{G}_0^{-1} \right) \psi + j^\dagger \psi + \psi^\dagger j \right) &= \mathrm{Det}\left(-\frac{\mathscr{G}_0^\zeta}{c^{-1}} \right) \exp\left(-\frac{j^\dagger \mathscr{G}_0 j}{c^{-1}} \right) \\ &= \exp\left(-\zeta\, \mathrm{Tr}\, \mathrm{Log}\left(-\frac{1}{c} \mathscr{G}_0^{-1} \right) \quad - \quad \frac{j^\dagger \mathscr{G}_0 j}{c^{-1}} \right) \end{aligned}$$

whereas the individual Equation-signs can be advantegous in different situations.

Definition 16.28 We introduce the generating functional

$$\frac{Z[\bar{j}, j]}{Z} = \frac{1}{Z} \int \mathscr{D}_\zeta(\psi, \overline{\psi})\, \mathrm{e}^{-S[\psi, \overline{\psi}] + j^\dagger \psi + \psi^\dagger j} = \left\langle \exp\left[j^\dagger \psi + \psi^\dagger j \right] \right\rangle \tag{16.37}$$

Definition 16.29 We also introduce the *analogue of the Helmholtz free energy / Landau-potential*[12][13]

$$W[\bar{j}, j] = + \log Z[\bar{j}, j] \tag{16.38}$$

[10] Warning: Signs are incosistent **within** Negele, Negele page 70, 75, 76, 105, 107.

[11] NO p. 22 / A p. 104 (bosons), NO p. 34 / A p. 165 (fermions).

[12] Correspondence to the free Energy (TBD), cf. Negele p. 107 following.

[13] In the literature [?], [?] this quantity is known as "generating functional for connected correlation functions". In the following we will understand why.

Remark 16.29.1 Notice that this is at zero external sources proportional (with proportionality constant $k_B T$) to the grand canonical potential of a system; Cf. Lemma TBReferenced.

Notation 16.30 We introduce the *field expectation values*

$$\begin{aligned} \phi[\bullet, j, \bar{j}] &= \langle \psi(\bullet) \rangle_{j, \bar{j}} &= 1 \frac{\partial}{\partial \bar{j}(\bullet)} W[j, \bar{j}], \\ \bar{\phi}[\bullet, j, \bar{j}] &= \langle \bar{\psi}(\bullet) \rangle_{j, \bar{j}} &= \zeta \frac{\partial}{\partial j(\bullet)} W[j, \bar{j}]. \end{aligned} \tag{16.39}$$

Remark 16.30.1 For $j, \bar{j} \neq 0$ it follows $\phi, \bar{\phi} \overset{\text{in general}}{\neq} 0$.

Lemma 16.31 The field-expectation values $\phi, \bar{\phi}$ are functionals of the sources $j, \bar{j}$, and they are made up of monomials that are *odd* in the sources $j, \bar{j}$.

Proof 16.31 TBDigitalized.[14]

Lemma 16.32 The mapping $(j, \bar{j}) \mapsto (\phi, \bar{\phi})$ is *invertible*, in the sense that we can find expansions

$$j[\bullet, \phi', \overline{\phi'}], \quad \bar{j}[\bullet, \phi', \overline{\phi'}] \tag{16.40}$$

such that

$$\phi\Big[\bullet, j[\bullet, \phi', \overline{\phi'}], \bar{j}[\bullet, \phi', \overline{\phi'}]\Big] = \phi', \qquad \bar{\phi}\Big[\bullet, j[\bullet, \phi', \overline{\phi'}], \bar{j}[\bullet, \phi', \overline{\phi'}]\Big] = \overline{\phi'}. \tag{16.41}$$

Proof 16.32 Iteratively we can construct the desired expansion TBDigitalized.

Remark 16.32.1 Notice that in the construction of the inverse, we realized that the expansions for the sources is *odd* in the fields ϕ and $\bar{\phi}$, which implies that the derivatives (w.r.t. to the ϕ-fields) are even in the fields and commute with the fields. This is a crucial observation for the effective action.

[14] Cf. TBDigitalized TBD4.

Intermezzo: Vectors with Graßmann-Entries and Chain-Rule

Remark 16.32.2

- Definitions ... of a module over a Graßmann-algebra $V_{\mathscr{P}_\zeta} = V \otimes \mathscr{P}_\zeta$
- Derivatives ... at a point $X \in V_{\mathscr{P}_\zeta}$ are defined as linear maps $dF_X : V_{\mathscr{P}_\zeta} \to V_{\mathscr{P}_\zeta}$, so derivatives live in $\mathrm{Hom}_{\mathscr{P}_\zeta}(V_{\mathscr{P}_\zeta}, V_{\mathscr{P}_\zeta})$
- TBD ...

Notation 16.33 With the assumption that the mapping $(j, \overline{j}) \mapsto (\phi, \overline{\phi})$ is invertible, and with suppression of the dependencies $j \equiv j[\bullet, \phi, \overline{\phi}]$, we write the fermionic effective action (being the analog of the Gibbs-potential) as a Legendre-transformation of W

$$\Gamma[\phi, \overline{\phi}] = -W[j, \overline{j}] \quad + \quad \phi^\dagger j + j^\dagger \phi. \tag{16.42}$$

Remark 16.33.1 For the Legendre-transformation, there is a diversity in signs throughout the literature... Basically every author seems to be using a different convention.

Lemma 16.34 One immediately concludes that

$$\begin{aligned}\frac{\partial}{\partial \overline{\phi}(\bullet)} \Gamma[\phi, \overline{\phi}] &= 1 j[\bullet, \phi, \overline{\phi}],\\ \frac{\partial}{\partial \phi(\bullet)} \Gamma[\phi, \overline{\phi}] &= \zeta \overline{j}[\bullet, \phi, \overline{\phi}].\end{aligned}$$

Proof 16.34.0 See Appendix, Proofs, page 358.

16.1.5. Series-Expansions

Notation 16.35 After one has established an order relation[15] for the generalized indices, one introduces the series-expansions for the functionals

$$\begin{aligned}F[\pi, \overline{\pi}] &\equiv \sum_{n,m=0}^{\infty} \sum_{\substack{i_1 < \ldots < i_n \\ i'_1 < i'_2 < \ldots < i'_m}} \overline{\pi}(i_1) \cdots \overline{\pi}(i_n) f(i_1, \ldots, i_n, i'_1, \ldots i'_m)\, \pi(i'_m) \cdots \pi(i'_1)\\ &= \sum_{n,m=0}^{\infty} \frac{1}{n!\, m!} \sum_{\substack{i_1, \ldots, i_n \\ i'_1, i'_2, \ldots, i'_m}} \overline{\pi}(i_1) \cdots \overline{\pi}(i_n) f(i_1, \ldots, i_n, i'_1, \ldots i'_m)\, \pi(i'_m) \cdots \pi(i'_1)\end{aligned} \tag{16.43}$$

with complex-valued coefficients $f(i_1, \ldots, i_n, i'_1, \ldots i'_m) \in \mathbb{C}$, such that

$$f(i_{\pi(1)}, \ldots, i_{\pi(n)}, i'_{\sigma(1)}, \ldots, i'_{\sigma(m)}) = \operatorname{sgn}(\pi)^{1-\theta(\zeta)} \operatorname{sgn}(\sigma)^{1-\theta(\zeta)} f(i_1, \ldots, i_n, i'_1, \ldots, i'_m)$$

for $(\pi, \sigma) \in \mathrm{S}_n \times \mathrm{S}_m$.

Remark 16.35.1 Sometimes[16] one might find oneself in the situation that one has access to a series-expansion of the type

[15] To avoid double counting.
[16] For example, in the later chapter on electron-phonon-systems.

$$F[\pi,\overline{\pi}] \equiv \sum_{n,m=0}^{\infty} \sum_{\substack{i_1,\dots,i_n \\ i'_1,i'_2,\dots,i'_m}} \overline{\pi}(i_1)\cdots\overline{\pi}(i_n)\widetilde{f}(i_1,\dots,i_n,i'_1,\dots i'_m)\,\pi(i'_m)\cdots\pi(i'_1) \tag{16.44}$$

with complex-valued coefficients $\widetilde{f}(i_1,\dots,i_n,i'_1,\dots i'_m) \in \mathbb{C}$ that do not obey any symmetry properties. In this case, one has to identify the coefficients f in the expansion in Equation (16.43) as the properly antisymmetrized contributions.

Remark 16.35.2 Sometimes (but not in this work ;-)) the prefactors $(n!m!)^{-1}$ are included in the vertex-functions, such that the definition changes slightly.

Remark 16.35.3 One notates the individual coefficients depending on the functional of consideration, whereas the exact meaning of the word "generate" will be elucidated below.

Functional F	**Fields** $\pi,\overline{\pi}$	**Coefficients** f	**Purpose**
$S[\psi,\overline{\psi}]$	$\psi,\overline{\psi}$	s	Defines the System
$Z[j,\overline{j}]/Z$	$j,\overline{j}$	z	"Generates" Green-functions
$W[j,\overline{j}]$	$j,\overline{j}$	w	"Generates" connected Green-functions
$\Gamma[\phi,\overline{\phi}]$	$\phi,\overline{\phi}$	γ	"Generates" vertex-functions

Table 16.1.: Summary of the functional expansions and their roles.

Lemma 16.36 For $n = m = 2$ and $\zeta = -1$ one obtains f from $\widetilde{f}$ as

$$f_{1,2,3,4} = \frac{1}{4}\left(\widetilde{f}_{1,2,3,4} - \widetilde{f}_{2,1,3,4} - \widetilde{f}_{1,2,4,3} + \widetilde{f}_{2,1,4,3}\right) \tag{16.45}$$

Proof 16.36 Cf. TBDigitalized TBD8.

Lemma 16.37 Notice that we obtain the identity

$$f(i_1,\dots,i_n\,|\,i'_1,\dots,i'_m) = \zeta^{nm+n(n-1)/2+m(m-1)/2}\frac{\partial^{n+m}}{\partial\overline{\pi}(i_1)\cdots\partial\overline{\pi}(i_n)\partial\pi(i'_m)\cdots\partial\pi(i'_1)}F[\overline{\pi},\pi]\Big|_{\overline{\pi}=\pi=0} \tag{16.46}$$

Proof 16.37 We compute the derivatives explicitly and find the identity

$$
\begin{aligned}
&\frac{\partial^{n+m}}{\partial\overline{\pi}(i_1)\cdots\partial\overline{\pi}(i_n)\partial\pi(i'_m)\cdots\partial\pi(i'_1)}F[\overline{\pi},\pi]\Big|_{\overline{\pi}=\pi=0}\\
&=\frac{\partial^{n+m}}{\partial\overline{\pi}(i_1)\cdots\partial\overline{\pi}(i_n)\partial\pi(i'_m)\cdots\partial\pi(i'_1)}\sum_{n,m=0}^{\infty}\sum_{\substack{i_1<\ldots<i_n\\ i'_1<i'_2<\ldots<i'_m}}\overline{\pi}(i_1)\cdots\overline{\pi}(i_n)f(i_1,\ldots,i_n,i'_1,\ldots i'_m)\,\pi(i'_m)\cdots\pi(i'_1)\\
&=\zeta^{nm}\sum_{n,m=0}^{\infty}\sum_{\substack{i_1<\ldots<i_n\\ i'_1<i'_2<\ldots<i'_n}}\frac{\partial^{n}}{\partial\overline{\pi}(i_1)\cdots\partial\overline{\pi}(i_n)}\overline{\pi}(i_1)\cdots\overline{\pi}(i_n)f(i_1,\ldots,i_n,i'_1,\ldots i'_m)\frac{\partial^{n}}{\partial\pi(i'_m)\cdots\partial\pi(i'_1)}\pi(i'_m)\cdots\pi(i'_1)\\
&=\zeta^{nm+n(n-1)/2+m(m-1)/2}f(i_1,\ldots,i_n,i'_1,\ldots i'_m)
\end{aligned}
\tag{16.47}
$$

Remark 16.37.1 Notice that in the special case that $m=n$ one obtains

$$
\zeta^{nm+n(n-1)/2+m(m-1)/2}=\zeta^{n^2+n(n-1)}=\zeta^{n^2}=\zeta^{n} \tag{16.48}
$$

whereas we used in the second last step that the number $n(n-1)$ is even and in the last step, we used that n is even / odd if n^2 is even / odd, which is something that one learns in the first semester of mathematics (usually when one proves that $\sqrt{2}$ is irrational). Thus one finds

$$
f(i_1,\ldots,i_n\mid i'_1,\ldots,i'_n)=\zeta^{n}\frac{\partial^{2n}}{\partial\overline{\pi}(i_1)\cdots\partial\overline{\pi}(i_n)\partial\pi(i'_n)\cdots\partial\pi(i'_1)}F[\overline{\pi},\pi]\Big|_{\overline{\pi}=\pi=0} \tag{16.49}
$$

Lemma 16.38 The n-body Green-function as functional derivative

$$
\mathscr{G}(i_1,\ldots,i_n\mid i'_1,\ldots,i'_n)=\left(-\frac{\zeta}{c}\right)^{n}\frac{\partial^{2n}}{\partial\overline{j}(i_1)\cdots\partial\overline{j}(i_n)\partial j(i'_n)\cdots\partial j(i'_1)}\frac{Z[\overline{j},j]}{Z}\Big|_{\overline{j}=j=0} \tag{16.50}
$$

Proof 16.38 See Appendix, Proofs, page 359.

Remark 16.38.1 One obtains thus

$$
\mathscr{G}(i_1,\ldots,i_n\mid i'_1,\ldots,i'_n)=\left(-\frac{1}{c}\right)^{n}z(i_1,\ldots,i_n\mid i'_1,\ldots,i'_n) \tag{16.51}
$$

and one says that the functional $Z[\overline{j},j]/Z$ *generates* the Green-functions, since it's expansion coefficients are equal to the Green-functions.

Definition 16.39 We introduce the *[c]onnected Green-functions* as

$$\mathscr{G}_{\mathrm{c}}(i_1,\ldots,i_n\,|\,i'_1,\ldots,i'_n) := \left(-\frac{\zeta}{\hbar}\right)^n \frac{\partial^{2n}}{\partial\bar{j}(i_1)\cdots\partial\bar{j}(i_n)\partial j(i'_n)\cdots\partial j(i'_1)} W[\bar{j},j]\Big|_{\bar{j}=j=0} \tag{16.52}$$

Remark 16.39.1 The fact that this functional generates all connected diagrams is far from trivial. We will come back to this statement in a later chapter.[17].

Remark 16.39.2 One obtains thus

$$\mathscr{G}_{\mathrm{c}}(i_1,\ldots,i_n\,|\,i'_1,\ldots,i'_n) = \left(-\frac{1}{\hbar}\right)^n w(i_1,\ldots,i_n\,|\,i'_1,\ldots,i'_n) \tag{16.53}$$

and one says that the functional $W[\bar{j},j]$ *generates* the *connect* Green-functions, since it's expansion coefficients are equal to the Green-functions.

Definition 16.40 We introduce the *vertex-functions* Vert by taking successive derivatives of Γ at the point $\phi = \bar{\phi} = 0$ as

$$\mathrm{Vert}(i_1,\ldots,i_n,i'_1,\ldots,i'_n) := \left(-\frac{\hbar}{\zeta}\right)^n \frac{\partial^{2n}}{\partial\bar{\phi}(i_1)\cdots\partial\bar{\phi}(i_n)\partial\phi(i'_n)\cdots\partial\phi(i'_1)} \Gamma[\bar{\phi},\phi]\Big|_{\bar{\phi}=\phi=0} \tag{16.54}$$

Remark 16.40.1 Notice that this implies for the coefficients

$$\mathrm{Vert}(i_1,\ldots,i_n,i'_1,\ldots,i'_n) = (-\hbar)^n\,\gamma(i_1,\ldots,i_n,i'_1,\ldots,i'_n) \tag{16.55}$$

Remark 16.40.2 The particular vertex-function $\gamma(i_1,i_2,i'_1,i'_2)$ is oftentimes [Kopietz][Thomale(?)][Altland] referred to as the *effective interaction*. It is said to ...

> *„describe the true interaction between two particles in the many-body system"*.

This is however irritating because it doesn't neccessarily have the unit of an energy. Only after appropriate rescaling / change of basis can this vertex-function be interpreted as an interaction (with unit of energy). We will come back to this point in a later chapter.

Remark 16.40.3 One also says that the functional $\Gamma[\phi,\bar{\phi}]$ *generates* the vertex-functions.

17

- TBD: Altland p. 215 with instructions-manual.
- TBD: Negele Orland: p. 79 for control-result in first order.
- TBD: Negele Orland: p. 97 for the same statement.
- Cf. Negele, page 107.

Corollary 16.41 It holds

$$\begin{aligned}\mathscr{G}_{\mathrm{c}}(i_1 \mid i_1') &= \mathscr{G}(i_1 \mid i_1') \\ \mathscr{G}_{\mathrm{c}}(i_1, i_2 \mid i_1', i_2') &= \mathscr{G}(i_1, i_2 \mid i_1', i_2') - \left[\mathscr{G}(i_1 \mid i_1')\mathscr{G}(i_2 \mid i_2') + \zeta\mathscr{G}(i_1 \mid i_2')\mathscr{G}(i_2 \mid i_1')\right]\end{aligned} \tag{16.56}$$

Proof 16.41 See Appendix, Proofs, page 360.

Lemma 16.42 One finds for a functional $F \equiv F[\overline{j}, j]$ the chain-rule

$$\begin{aligned}\frac{\partial}{\partial\phi(\bullet)}F[\overline{j}, j] &= \sum_i \zeta\frac{\partial F}{\partial\overline{j}(i)}\frac{\partial^2\Gamma}{\partial\phi(\bullet)\partial\phi(i)} + \frac{\partial F}{\partial j(i)}\frac{\partial^2\Gamma}{\partial\phi(\bullet)\partial\overline{\phi}(i)} \\ \frac{\partial}{\partial\overline{\phi}(\bullet)}F[\overline{j}, j] &= \sum_i \zeta\frac{\partial F}{\partial\overline{j}(i)}\frac{\partial^2\Gamma}{\partial\overline{\phi}(\bullet)\partial\phi(i)} + \frac{\partial F}{\partial j(i)}\frac{\partial^2\Gamma}{\partial\overline{\phi}(\bullet)\partial\overline{\phi}(i)}\end{aligned} \tag{16.57}$$

Proof 16.42 See Appendix, Proofs, page 360.

Corollary 16.43 It follows that

$$\sum_{i_2}\begin{pmatrix}\dfrac{\partial^2 W}{\partial\overline{j}(i_3)\partial j(i_2)} & \zeta\dfrac{\partial^2 W}{\partial\overline{j}(i_3)\partial\overline{j}(i_2)} \\ \zeta\dfrac{\partial^2 W}{\partial j(i_3)\partial j(i_2)} & \dfrac{\partial^2 W}{\partial j(i_3)\partial\overline{j}(i_2)}\end{pmatrix}\begin{pmatrix}\dfrac{\partial^2\Gamma}{\partial\overline{\phi}(i_2)\partial\phi(i_1)} & \dfrac{\partial^2\Gamma}{\partial\overline{\phi}(i_2)\partial\overline{\phi}(i_1)} \\ \dfrac{\partial^2\Gamma}{\partial\phi(i_2)\partial\phi(i_1)} & \dfrac{\partial^2\Gamma}{\partial\phi(i_2)\partial\overline{\phi}(i_1)}\end{pmatrix} = \delta_{i_1,i_3}\begin{pmatrix}1 & 0 \\ 0 & 1\end{pmatrix} \tag{16.58}$$

Proof 16.43 See Appendix, Proofs, page 361.

Lemma 16.44 In the absense of "spontaneous symmetry breaking"[18], we can write down the quadratic part of the effective action $\mathrm{Vert}(:,:)$ by simple inversion of a matrix:

$$\mathrm{Vert}(:,:) = \mathscr{G}_{\mathrm{c}}^{-1}(:,:) = \mathscr{G}^{-1}(:,:) \tag{16.59}$$

Proof 16.44 Using that the effective action has the same symmetries as the original action (cf. TBReferenced) we obtain from Equation (29.9) the simplified identity

$$\delta_{1,3} = \sum_2\left(-\frac{c}{\zeta}\mathscr{G}(2|3)\left(-\frac{\zeta}{c}\frac{\partial^2\Gamma}{\partial\overline{\phi}(1)\partial\phi(2)}\right)\right) = \sum_2\mathscr{G}(2|3)\,\mathrm{Vert}(1|2) = \sum_2\mathrm{Vert}(1|2)\mathscr{G}(2|3) \tag{16.60}$$

[18]This will be introduced later.

Thus we obtain

$$\Big[\text{Vert}(:,:)\cdot\mathcal{G}(:,:)\Big]_{n,m}=\delta_{n,m}\quad\Longleftrightarrow\quad\text{Vert}=\mathcal{G}^{-1} \tag{16.61}$$

which yields in matrix-notation the desired result.

Remark 16.44.1 Notice that from here, we can easily obtain the generic version of the Dyson-equation (cf. Equation (15.7))

$$\mathcal{G}^{-1}=\mathcal{G}_0^{-1}-\Sigma \tag{16.62}$$

by introducing the self-energy Σ as the difference between the inverse of the free Green-function and the inverse of the full Green-function. This result alone is already quite nice, but we will see (in the next chapter) that one can conclude the explicit shape of the self-energy as well from symmetry-principles which will be quite a revealing moment for the interested reader.

Remark 16.44.2 One can[19] also rewrite this equation in a more general way for system that have a broken symmetry, but we will not do this here for pedagogical purposes.

Theorem 16.45 In the absence of spontaneous symmetry breaking, another useful identity[20]. is

$$\mathcal{G}_c(i_1,i_2\,|\,i_1',i_2')=-\sum_{l_1,l_2,l_1',l_2'}\mathcal{G}_c(i_1\,|\,l_1)\mathcal{G}_c(i_2\,|\,l_2)\ \text{Vert}(l_1,l_2\,|\,l_1',l_2')\mathcal{G}_c(l_1'\,|\,i_1')\mathcal{G}_c(l_2'\,|\,i_2') \tag{16.63}$$

Proof 16.45 See Appendix, Proofs, page 361.

Remark 16.45.1 In a later chapter we will see how this can be rewritten diagrammatically[21].

Theorem 16.46 In the absence of spontaneous symmetry breaking, one concludes the *Bethe-Salpeter-equation for the two-body Green-function* as

$$\begin{aligned}\mathcal{G}(i_1,i_2\,|\,i_1',i_2')&=\mathcal{G}(i_1\,|\,i_1')\mathcal{G}(i_2\,|\,i_2')+\zeta\mathcal{G}(i_1\,|\,i_2')\mathcal{G}(i_2\,|\,i_1')\\&\quad-\sum_{l_1,l_2,l_1',l_2'}\mathcal{G}(i_1\,|\,l_1)\mathcal{G}(i_2\,|\,l_2)\ \text{Vert}(l_1,l_2\,|\,l_1',l_2')\mathcal{G}(l_1'\,|\,i_1')\mathcal{G}(l_2'\,|\,i_2')\end{aligned} \tag{16.64}$$

[19] Cf. Negele, Orland eq. (7.52).

[20] Other identities that might be included:

- Cf. Negele p. 108.
- cf. Negele, Orland eq. (6.79 a,b) Identity for quartic green-function as correlation-functions
- For the relations between $\mathcal{G}^{(6)}$ and $\Gamma^{(6)}$: see Salmhofers Functional renormalization group approach to correlated fermion systems, eq. (12)

[21] Cf. Negele, p. 118.

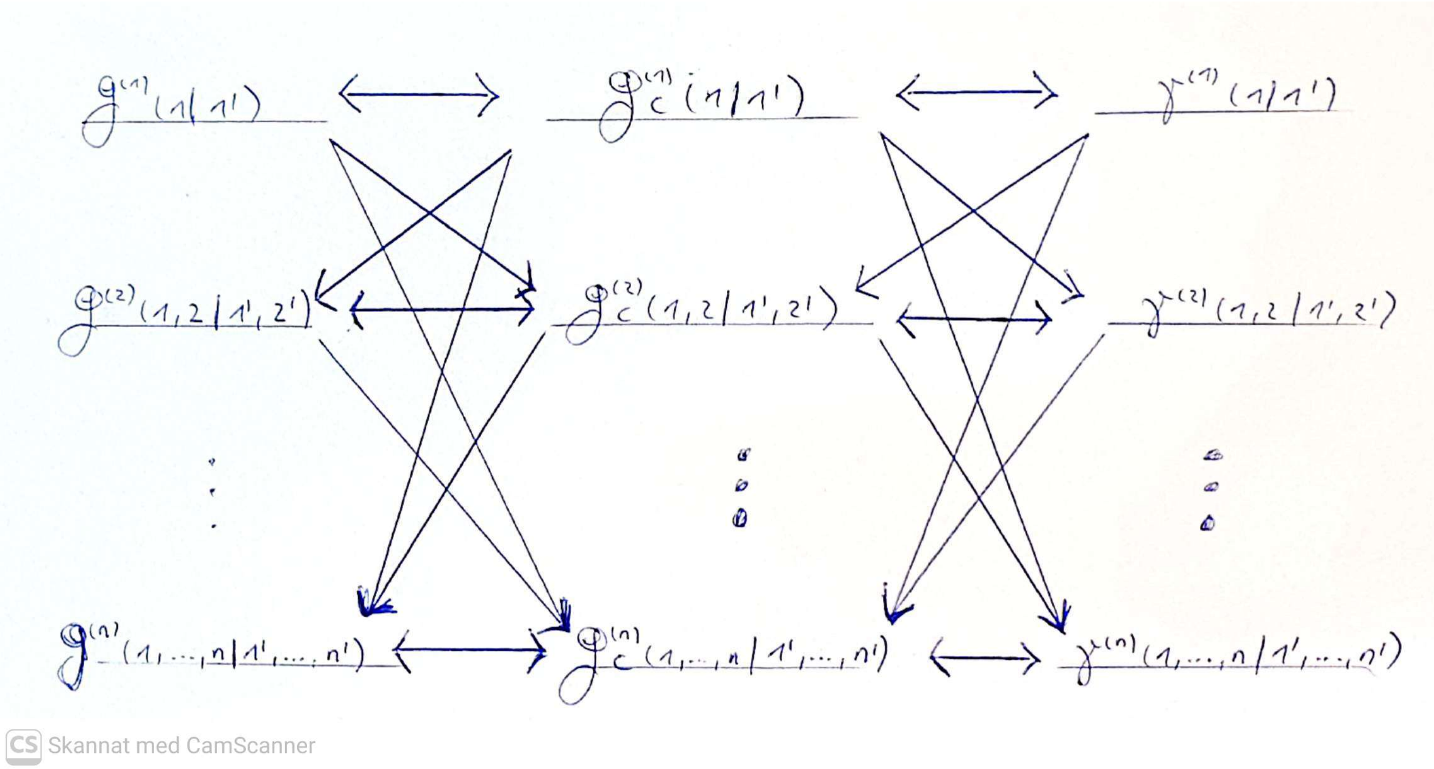
Proof 16.46 Combining Corollary 16.41 and Theorem 16.45 and rearranging terms gives the desired result.

Hypothesis 16.47 With induction, one sees that the following three problems are equivalent, in the sense that if one of them is solved, the other two are solved as well:

- *Finding all many-body Green-Functions* $\mathscr{G}^{(1)}(i_1, i_1'), \quad \mathscr{G}^{(2)}(i_1, i_2, i_1', i_2'), \ \dots, \quad \mathscr{G}^{(n)}(i_1, \dots, i_n, i_1', \dots, i_n')$
- *Finding all connected many-body Green-Functions* $\mathscr{G}_{\mathrm{c}}^{(1)}(i_1, i_1'), \quad \mathscr{G}_{\mathrm{c}}^{(2)}(i_1, i_2, i_1', i_2'), \ \dots, \quad \mathscr{G}_{\mathrm{c}}^{(n)}(i_1, \dots, i_n, i_1', \dots, i_n')$
- *Finding all many-body vertex-Functions* $\gamma^{(1)}(i_1, i_1'), \quad \gamma^{(2)}(i_1, i_2, i_1', i_2'), \ \dots, \quad \gamma^{(n)}(i_1, \dots, i_n, i_1', \dots, i_n')$

Remark 16.47.1 This is being concluded from Equations (TBReferenced 1, 2, 3) and "probably" one can show with induction somehow that this is a general statement.

Remark 16.47.2 The picture visualizes the equivalence of the three problems.



16.1.6. Change of Coordinates

Notation 16.48 We consider an invertible linear mapping M (with complex conjugate $\overline{M}$) between integration-variables („coordinate-change“)

$$\psi_i \mapsto (M \cdot \psi)_i \equiv \sum_{i'} M_{i,i'} \psi_{i'}, \qquad \overline{\psi}_i \mapsto (\overline{M} \cdot \overline{\psi})_i \equiv \sum_{i'} \overline{M}_{i,i'} \overline{\psi}_{i'}$$

Remark 16.48.1 Notice that in the notation of physicists, this would be written as

$$\psi'_i = \psi'_i(\psi) = \sum_i M_{i',i} \psi_i, \qquad \overline{\psi}_i{}' = \overline{\psi}_i{}'(\psi) = \sum_i \overline{M}_{i',i} \overline{\psi}_i \tag{16.65}$$

Remark 16.48.2 Notice that this mapping changes a monomial expression in the integration-variables according to

$$\psi_{i_1} \cdots \psi_{i_n} \overline{\psi}_{i'_1} \cdots \overline{\psi}_{i'_m} \mapsto \sum_{\substack{j_1,\ldots,j_n \\ j'_1,j'_2,\ldots,j'_{m-1},j'_m}} \left(\prod_{l=1}^n M_{i_l,j_l} \right) \left(\prod_{l=1}^m \overline{M}_{i'_l j'_l} \right) \psi_{j_1} \cdots \psi_{j_n} \overline{\psi}_{j'_1} \cdots \overline{\psi}_{j'_m} \tag{16.66}$$

Remark 16.48.3 The most important transformation will turn out to be the Fourier-series. It removes the derivatives (if one concatenates the variable-transformation with a standard-approximation) s.t. we can simply invert the matrix appearing in the action.

Lemma 16.49 Changing the coordinates according to Equation (16.65) yields for the derivatives

$$\frac{\partial}{\partial \psi'_i} = \sum_i \frac{\partial \psi_i}{\partial \psi'_i} \frac{\partial}{\partial \psi_i}, \qquad \frac{\partial}{\partial \overline{\psi}_i{}'} = \sum_i \frac{\partial \overline{\psi}_i}{\partial \overline{\psi}_i{}'} \frac{\partial}{\partial \overline{\psi}_i} \tag{16.67}$$

Proof 16.49 See Appendix, Proofs, page **??**[22].

Lemma 16.50 Changing the coordinates according to Equation (16.65) changes the integration-measure as

$$\mathrm{d}_\zeta(\overline{\psi}', \psi') \equiv \mathrm{Det} \left(\frac{\partial \psi'_i}{\partial \psi_j} \right)^\zeta \mathrm{Det} \left(\frac{\partial \overline{\psi}_i{}'}{\partial \overline{\psi}_j} \right)^\zeta \mathrm{d}_\zeta(\overline{\psi}, \psi) \tag{16.68}$$

Corollary 16.51 Emphasizing the role of the matrices A and $M(A) \equiv M^\dagger A M$ into the notation, we obtain the important result

$$\langle \psi_{i_1} \cdots \psi_{i_n} \overline{\psi}_{i'_1} \cdots \overline{\psi}_{i'_m} \rangle_\zeta [A] = \sum_{\substack{j_1,\ldots,j_n \\ j'_1,j'_2,\ldots,j'_{m-1},j'_m}} \left(\prod_{l=1}^n M_{i_l,j_l} \right) \left(\prod_{l=1}^m \overline{M}_{i'_l j'_l} \right) \langle \psi_{j_1} \cdots \psi_{j_n} \overline{\psi}_{j'_1} \cdots \overline{\psi}_{j'_m} \rangle_\zeta [M(A)]$$

[22]TBD. Cf. Zinn-Justin page 11.

Proof 16.51 See Appendix, Proofs, page **??**.

Using this mapping as transformation-matrix, we obtain for the partition-function the scaling-factor

$$Z_\zeta[A] \equiv \int \mathrm{d}_\zeta(\overline{\psi}, \psi)\, \mathrm{e}^{-\psi^\dagger A \psi} \equiv \mathrm{Det}\, M^{-\zeta} \int \mathrm{d}_\zeta(\overline{\psi}, \psi)\, \mathrm{e}^{-\psi^\dagger M^\dagger A M \psi} \equiv \mathrm{Det}\, M^{-\zeta} \cdot Z_\zeta[\widetilde{A}]$$

and equally we obtain another scaling-factor for the numerator in the expression of the expectation-value.

16.1.7. The Tree-Level-Approximation for the Effective Action

Approximation 16.52 In the so-called Tree-Level-Approximation, one obtains for the fermionic effective action

$$\Gamma[\phi, \overline{\phi}] \quad \approx \quad S[\phi, \overline{\phi}]. \tag{16.69}$$

Proof 16.52 See Appendix, Proofs, page 367.

Remark 16.52.1 This approximation is usually depicted with Feynman-diagrams as in the figure

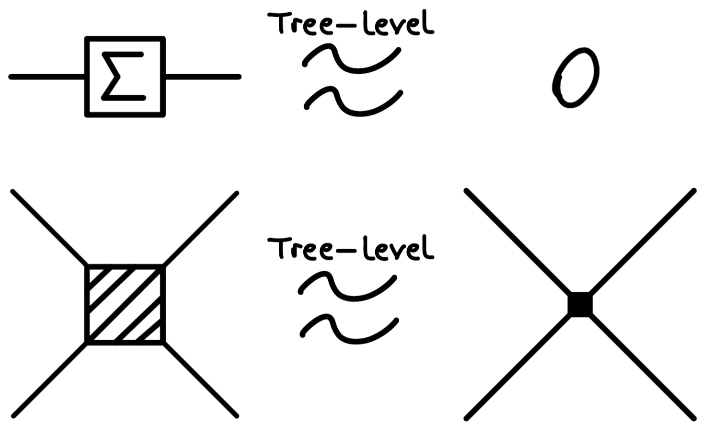


Figure 16.1.: Feynman-Diagrams (schematically) for the Tree-Level-Approximation.

Corollary 16.53 Within the Tree-Level-Approximation, equations (16.42) and (16.69) can be combined, which gives

$$\frac{\partial}{\partial \phi} S[\phi, \overline{\phi}] = \overline{j} \qquad \text{and} \qquad \frac{\partial}{\partial \overline{\phi}} S[\phi, \overline{\phi}] = -j. \tag{16.70}$$

Proof 16.53 Follows immediately from the approximation and Remark 16.34.

16.1.8. The One-Loop-Approximation for the Effective Action

Approximation 16.54 In the so-called One-Loop-Approximation, one obtains for the fermionic effective action

$$\Gamma[\phi,\overline{\phi}] \approx S[\phi,\overline{\phi}] - \frac{1}{2}\,\mathrm{Tr}\log\begin{pmatrix} \frac{\partial^2}{\partial\phi\,\partial\phi}S[\phi,\overline{\phi}] & \frac{\partial^2}{\partial\phi\,\partial\overline{\phi}}S[\phi,\overline{\phi}] \\ \frac{\partial^2}{\partial\overline{\phi}\,\partial\phi}S[\phi,\overline{\phi}] & \frac{\partial^2}{\partial\overline{\phi}\,\partial\overline{\phi}}S[\phi,\overline{\phi}] \end{pmatrix}, \tag{16.71}$$

where

$$\left(\frac{\partial^2}{\partial\phi\,\partial\phi}S[\phi,\overline{\phi}]\right) \equiv \left(\frac{\partial^2}{\partial\phi\,\partial\phi}S[\phi,\overline{\phi}]\right)(x,y) \equiv \frac{\partial^2}{\partial\phi(i)\,\partial\phi(j)}S[\phi,\overline{\phi}], \qquad \text{etc.}$$

Proof 16.54 See Appendix, Proofs, page 369.

Notation 16.55 From now on we consider the special case[23]

$$S[\psi,\overline{\psi}] = S_0[\psi,\overline{\psi}] + S_{\mathrm{int}}[\psi,\overline{\psi}], \tag{16.72}$$

with inverse propagator $G_0^{-1} \equiv G_0^{-1}(i,j)$ in the action

$$S_0[\psi,\overline{\psi}] = \sum_{i,j} \overline{\psi}(i)\, G_0^{-1}(i,j)\, \psi(j)$$

and lastly we assume that the interaction S_{int} possesses an U(1)-invariance.

Lemma 16.56 For actions of the type in Notation 16.55 the following identities hold

This needs to be updated cause it doesn't hold for arbitrary operators.

$$\frac{\delta^2}{\delta\overline{\psi}(x_1)\delta\psi(x_2)} = -\frac{\vec{\delta}^2}{\delta\overline{\psi}(x_1)\delta\psi(x_2)} = -\frac{\delta^2}{\delta\psi(x_2)\delta\overline{\psi}(x_1)}. \tag{16.73}$$

Proof 16.56 See Appendix, Proofs, page 370.

Approximation 16.57 Expanding the r.h.s. of eq. (16.71) up to quartic order in the Graßmann-fields and neglecting higher order terms yields

[23] It is exactly this case which we will encounter in the later application; many problems of physical relevance are of this type, such as electrons in a box with periodic boundary conditions with Coulomb-interaction, or with local, attractive interaction (BCS-theory) [1].

$$
\begin{aligned}
\Gamma[\phi, \overline{\phi}] \quad \approx \quad S[\phi, \overline{\phi}] \quad - \quad & \operatorname{Tr}\left[G_0 \cdot \frac{\partial^2}{\partial \overline{\phi}\, \partial \phi} S_{\text{int}}[\phi, \overline{\phi}]\right] \\
- \quad \frac{1}{2} \quad & \operatorname{Tr}\left[G_0 \cdot \frac{\partial^2}{\partial \overline{\phi}\, \partial \overline{\phi}} S_{\text{int}}[\phi, \overline{\phi}] \cdot G_0^{\mathrm{T}} \cdot \frac{\partial^2}{\partial \phi\, \partial \phi} S_{\text{int}}[\phi, \overline{\phi}]\right] \\
+ \quad \frac{1}{2} \quad & \operatorname{Tr}\left[G_0 \cdot \frac{\partial^2}{\partial \overline{\phi}\, \partial \phi} S_{\text{int}}[\phi, \overline{\phi}] \cdot G_0 \cdot \frac{\partial^2}{\partial \overline{\phi}\, \partial \phi} S_{\text{int}}[\phi, \overline{\phi}]\right]
\end{aligned}
\tag{16.74}
$$

Proof 16.57 See Appendix, Proofs, page 371.

Remark 16.57.1 This approximation is usually depicted with Feynman-diagrams as in the figure. The corrections to the Tree-Level contribution of the effective interaction $\gamma(x_1, x_2, x_3, x_4)$ are the particle-particle- and particle-hole-diagrams respectively.

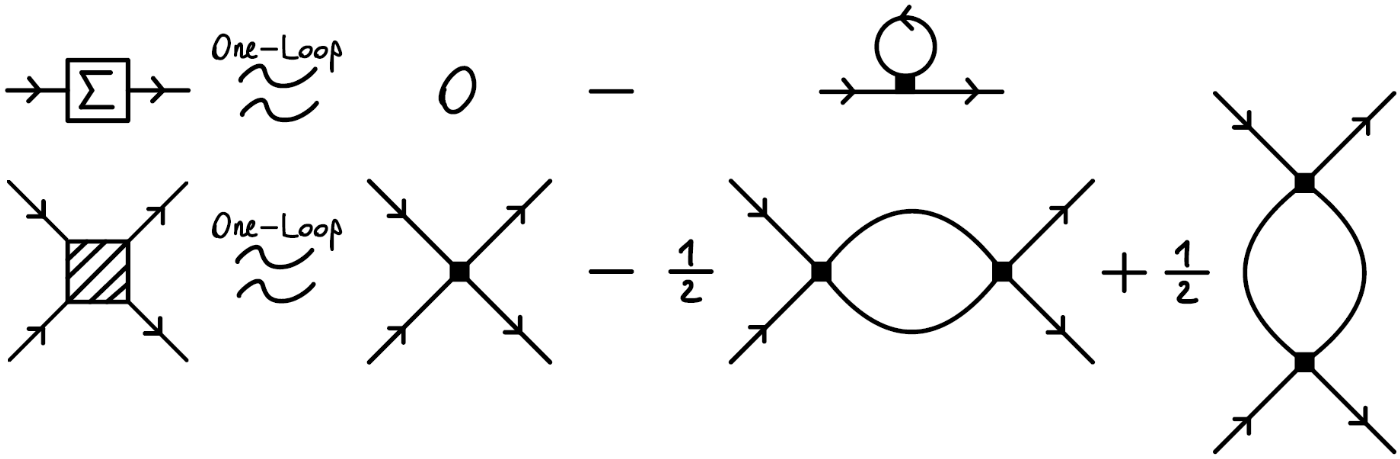


Figure 16.2.: Feynman-Diagrams (schematically) for the One-Loop-Approximation.

16.2. Quantum Field Theory - Mixed Systems

16.2.1. Fundamentals

TBCopied (a lot from Zinn-Justin).

16.2.2. Partition-Functions and Expectation-Values for Non-Interacting Models

TBD

16.2.3. Partition-Functions and Expectation-Values for Interacting Models

TBD

16.2.4. Gauss-Identities

TBD

16.2.5. External Sources, Connected Green-Functions and Vertex-Functions

TBD

17. Symmetry-Transformations

17.1. Pure Systems

17.1.1. Introduction

In this section we will see how symmetry considerations will constrain the shape of the coefficients γ.

Definition 17.1 We call an invertible linear mapping M (with complex conjugate $\overline{M}$) between Graßmann-fields

$$\psi(x) \mapsto (M \cdot \psi)(i) \equiv \sum_y M(i,j)\,\psi(j), \qquad \overline{\psi}(i) \mapsto (\overline{M} \cdot \overline{\psi})(i) \equiv \sum_y \overline{M}(i,j)\,\overline{\psi}(j)$$

a *symmetry transformation* if both, the integration measure and the action remain unchanged under the action of the mapping M (and $\overline{M}$):

$$\mathscr{D}(\psi, \overline{\psi}) = \mathscr{D}(M \cdot \psi, \overline{M} \cdot \overline{\psi}) \quad \text{and} \quad S[\psi, \overline{\psi}] = S[M \cdot \psi, \overline{M} \cdot \overline{\psi}]. \tag{17.1}$$

Remark 17.1.1 The set of all symmetry transformations, equipped with composition as binary-operation, is a group. For each action and integration measure one refers to this group as the "symmetry group" of the problem [**?**].

Lemma 17.2 For a symmetry transformation M one has

$$W[j, \overline{j}] = W[\overline{M}^{\mathrm{T}} \cdot j, M^{\mathrm{T}} \cdot \overline{j}]. \tag{17.2}$$

implying the equations

$$\begin{aligned} \phi[\bullet, \overline{M}^{\mathrm{T}} \cdot j, M^{\mathrm{T}} \cdot \overline{j}] &= (M^{-1} \cdot \phi)[\bullet, j, \overline{j}], \\ \overline{\phi}[\bullet, \overline{M}^{\mathrm{T}} \cdot j, M^{\mathrm{T}} \cdot \overline{j}] &= (\overline{M}^{-1} \cdot \overline{\phi})[\bullet, j, \overline{j}]. \end{aligned}$$

Proof 17.2 See Appendix, Proofs, page 366.

Theorem 17.3 With the above preparations one obtains for a symmetry transformation M the equation

$$\Gamma[\phi, \overline{\phi}] = \Gamma[M^{-1} \cdot \phi, \overline{M}^{-1} \cdot \overline{\phi}]. \tag{17.3}$$

Proof 17.3 See Appendix, Proofs, page 367.

Remark 17.3.1 Thus, by equation (17.3) we obtain for each symmetry transformation M (respectively its inverse M^{-1}), by comparing coefficients in eq. (**??**), restricting conditions for the vertex-functions:

$$\gamma(i_1, \dots, i_n, j_1, \dots, j_m) = \sum_{\substack{\tilde{i}_1, \dots, \tilde{i}_n \\ \tilde{j}_1, \tilde{j}_2, \dots, \tilde{j}_{m-1}, \tilde{j}_m}} \left(\prod_{l=1}^{n} \overline{M}(i_l, \tilde{i}_l) \right) \left(\prod_{l=1}^{m} M(j_l, \tilde{j}_l) \right) \gamma(\tilde{i}_1, \dots, \tilde{i}_n, \tilde{j}_1, \dots, \tilde{j}_m)$$

The shape of the coefficients is therefore dictated by the symmetries of the underlying physical system. This realization will simplify concrete calculations later.

Remark 17.3.2 The above results can be memorized by the following sentence from [**?**]:

> *The effective action Γ shares all the symmetries of the action S.*

Remark 17.3.3 Equation (17.2) gives analogous results for the free energy W.

Remark 17.3.4 The figure summarizes the results pictographically.

$$\left.\begin{array}{l} S[\psi, \overline{\psi}] = S[M \cdot \psi, \overline{M} \cdot \overline{\psi}] \\ \mathcal{D}(\psi, \overline{\psi}) = \mathcal{D}(M \cdot \psi, \overline{M} \cdot \overline{\psi}) \end{array}\right\} \Longleftrightarrow W[j, \overline{j}] = W[\overline{M}^T \cdot j, M^T \cdot \overline{j}] \Longleftrightarrow \Gamma[\phi, \overline{\phi}] = \Gamma[M^{-1} \cdot \phi, \overline{M}^{-1} \cdot \overline{\phi}]$$

Figure 17.1.: Summary for Symmetry Transformations.

17.1.2. Implications

TBD.

17.2. Mixed Systems

TBD.

17.2.1. Introduction

TBD.

17.2.2. Implications

TBD.

18. Feynman-Diagramology

In quantum field theory, there are *four* famous theorems about "Feynman-diagrams", which are never proven in the generalaity in which one claims their applicability. Those are …

- The partition-function is the sum of all "diagrams" without "external vertices".
- The free energy is the sum of all "connected diagrams" without "external vertices".
- The self-energy is the sum of all "one-particle-irreducible two-point[1] diagrams".
- The n-body Green-functions are computed as sums of all "connected diagrams" with n "external vertices" (a.k.a. "Linked Cluster Theorem").

18.1. Fundamentals on Graph Theory

18.2. Pure Systems

18.3. Mixed Systems

[1] I.e. diagrams with two external legs.

19. Thermal Quantum Field Theory

19.1. Many-Body Quantum Field Theory - Pure Systems

19.1.1. Coherent States and Resolutions of Identity

In the previous chapter we learned many techniques on how to evaluate (functional) integrals. In this chapter, we will provide the bridge between the many-body-concepts developed earlier, and the (functional) integral formalism. For this we still require some tools.

Definition 19.1 We consider a polynomial algebra $\mathscr{P}_\zeta$ as in Definition 16.1 and a Fock-space $\mathscr{H}_\eta$ that is generated by creation and annihilation-operators, as in Equation (10.4). We then construct[1] a [G]eneralized [H]ilbert-space $\mathscr{G}\mathscr{H}_{\eta,\zeta}$ as the set of linear combinations of states of the Hilbert-space $\mathscr{H}_\eta$ with coefficients in the polynomial algebra $\mathscr{P}_\zeta$.

Remark 19.1.1 Notice that this implies, that any vector $|\psi\rangle$ in the generalized Hilbert-space can be expanded as

$$|\psi\rangle = \sum_\alpha \chi_\alpha |\phi_\alpha\rangle \tag{19.1}$$

where the χ_α are elements of the polynomial algebra and the $|\phi_\alpha\rangle$ vectors of the Hilbert-space.

In order to treat expressions containing combinations of polynomial-algebra-generators and creation- and annihilation-operators, it is necessary to specify commutation-relations between the ψ's and c's.

Definition 19.2 In the generalized Hilbert-space, we define commutation-relations as

$$[\psi_i, c_j]_\zeta := 0 \tag{19.2}$$

etc. for the other combinations.

Definition 19.3 We define a *coherent state* $|\psi\rangle \in \mathscr{G}\mathscr{H}_{\eta,\zeta}$ as an eigenstate of the statistical annihilation-operator, i.e. fulfilling the equation

$$c_j|\psi\rangle = \psi_j|\psi\rangle \tag{19.3}$$

with $\psi_j \in \mathscr{P}_\zeta$.

[1]Cf. Negele Orland p. 29 for this part, or Marcos Marinos QFT-Skript p. 64: enlarged Fock-space for fermionic coherent states.

Theorem 19.4 The state $|\psi\rangle$ defined through

$$|\psi\rangle = \exp\left(\sum_{j=1}^{d_\eta} c_j^\dagger \psi_j\right)|0\rangle \tag{19.4}$$

is a *coherent state with respect to generators* $\{c_1, c_1^\dagger, \dots, c_{d_\eta}, c_{d_\eta}^\dagger\}$ of $\mathscr{H}$ *and* $\{\psi_1, \overline{\psi}_1, \dots, \psi_{d_\eta}, \overline{\psi}_{d_\eta}\}$ of $\mathscr{P}_\zeta$, i.e. it is an eigenstate of the statistical annihilation-operators fulfilling

$$c_j|\psi\rangle = \psi_j|\psi\rangle \quad \text{for all } j \in \{1, \dots, d_\eta\} \tag{19.5}$$

Proof 19.4 See Appendix, Proofs, page 372.

Remark 19.4.1 Notice that we have $D = d_\eta$ in the sense of Notations TBReferenced1 and TBReferenced2.

Corollary 19.5 By taking the hermitian conjugate of Equation (19.4) we find that the „bra“ $\langle\psi|$ associated with the „ket“ $|\psi\rangle$ is a left-eigenstate of the set of creation-operators, i.e. for

$$\langle\psi| = \langle 0|\exp\left(\sum_{j=1}^{d_\eta} \overline{\psi}_j c_j\right) \tag{19.6}$$

it holds

$$\langle\psi|c_j^\dagger = \langle\psi|\overline{\psi}_j \tag{19.7}$$

Proof 19.5 See Appendix, Proofs, page 372.

Lemma 19.6 One finds the *derivative-identities* for the coherent states as

- $c_j^\dagger|\psi\rangle = \zeta\dfrac{\partial}{\partial\psi_j}|\psi\rangle$,
- $\langle\psi|c_j = \dfrac{\partial}{\partial\overline{\psi}_j}\langle\psi|$

Proof 19.6 See Appendix, Proofs, page 372.

Theorem 19.7 One finds for the overlap between two coherent states

$$\langle\psi'|\psi\rangle = \exp\left(\sum_{j=1}^{d_\eta} \overline{\psi}'_j \psi_j\right) \tag{19.8}$$

Proof 19.7 See Appendix, Proofs, page **??**.

Theorem 19.8 Taking $\mathrm{D} = \mathrm{d}_\eta$ in Definition 16.1, we obtain the *completeness-relation*

$$\mathbb{1}_{\mathscr{H}} = \int \mathrm{d}_\zeta(\overline{\psi}, \psi)\, \mathrm{e}^{-\sum_{j=1}^{\mathrm{d}_\eta} \overline{\psi}_j \psi_j} |\psi\rangle\langle\psi| \tag{19.9}$$

with integration-measure as in Definition 16.15

$$\mathrm{d}_\zeta(\overline{\psi}, \psi) \equiv \prod_{j=1}^{\mathrm{d}_\eta} \frac{\mathrm{d}\overline{\psi}_j \mathrm{d}\psi_j}{(2\pi\mathrm{i})^{\theta(\zeta)}} \tag{19.10}$$

Proof 19.8 See Appendix, Proofs, page 372.

Lemma 19.9 The Hamiltonian in Equation (TBReferenced) is block-diagonal w.r.t. the number of particles.

Proof 19.9 See Appendix, Proofs, page 373.

Remark 19.9.1 Notice that this theorem implies, that $\langle N|H|N'\rangle = 0$ if $|N\rangle$ and $|N'\rangle$ represents occupation number states with different numbers of particles.

Lemma 19.10 Let $|n\rangle$ and $|n'\rangle$ denote two N-particle states in their occupation-number representation. Then it follows that

$$\langle n|\psi\rangle\langle\psi|n'\rangle = \langle\zeta\psi|n'\rangle\langle n|\psi\rangle. \tag{19.11}$$

Proof 19.10 See Appendix, Proofs, page 373.

Lemma 19.11 For an operator X that is given as a polynomial expression in ladder-operators $X \equiv X(c, c^\dagger)$, we write for the normalized overlap between coherent states

$$\frac{\langle\psi|X|\psi'\rangle}{\langle\psi|\psi'\rangle} = X(\overline{\psi}, \psi'), \tag{19.12}$$

i.e. the normalized overlap is *the* polynomial expression in Graßmann-variables, in which one replaced each ladder-operator by its Graßmann-counterpart.

Proof 19.11 See Appendix, Proofs, page 373.

Remark 19.11.1 Notice that for the Hamiltonian-operators in Fact 8.4, one obtains the identities

$$\frac{\langle\psi|H|\psi'\rangle}{\langle\psi|\psi'\rangle} \equiv H(\overline{\psi},\psi') \equiv \begin{cases} \sum_{i,j=1}^{\mathrm{d}} H_{i,j}\overline{\psi}_i\psi_j + \sum_{i,j,k=1}^{\mathrm{d}} \left(H_{i,j;k}\overline{\psi}_i\overline{\psi}_j\psi_k + H_{i;j,k}\overline{\psi}_i\psi_j\psi_k\right) + \ldots & \text{for } \eta = 0 \\ \sum_{i,j=1}^{\mathrm{d}} H_{i,j}\overline{\psi}_i\psi_j + \frac{1}{2}\sum_{i,j,k,l=1}^{\mathrm{d}} H_{i,j;kl}\overline{\psi}_i\overline{\psi}_j\psi_l\psi_k + \ldots & \text{for } \eta = 1 \end{cases} \tag{19.13}$$

and the interacting part will be considered in the next subsection.

Lemma 19.12 One finds for the trace of an operator $X \equiv X(c, c^\dagger)$ that is block-diagonal in the number of fermionic particle-number (and arbitrary in the bosonic particle-number)

$$\mathrm{Tr}_{\mathscr{F}_\zeta} X(c, c^\dagger) = \int \mathrm{d}_\zeta(\overline{\psi},\psi)\, \mathrm{e}^{-\sum_{j=1}^{\mathrm{d}_\eta}\overline{\psi}_j\psi_j}\, \langle\zeta\psi|X|\psi\rangle\,. \tag{19.14}$$

with

$$\langle\zeta\psi| \equiv \langle 0|\exp\left(\zeta\sum_{j=1}^{\mathrm{d}_\eta}\overline{\psi}_j c_j\right) \tag{19.15}$$

Proof 19.12 See Appendix, Proofs, page 373.

19.1.2. The Functional Field Integral of the Partition Function

Theorem 19.13 One can rewrite the grand canonical partition function Z as a functional field integral

$$Z = \mathrm{Tr}_{\mathscr{H}_\eta}\, \mathrm{e}^{-\beta(H-\eta\mu N)} = \int \mathscr{D}_\zeta(\psi,\overline{\psi})\, \mathrm{e}^{-S_\eta[\psi,\overline{\psi}]}$$

where the action is written in its *continuum-notation*

$$\begin{aligned} S_\eta[\psi,\overline{\psi}] &\equiv \int_0^{\hbar\beta} \mathrm{d}\tau \left[\overline{\psi}\cdot\partial_\tau\psi + \frac{H(\overline{\psi},\psi) - \eta\mu N(\overline{\psi},\psi)}{\hbar}\right] \\ &\equiv \int_0^{\hbar\beta} \mathrm{d}\tau \left[\overline{\psi}\cdot\partial_\tau\psi + \frac{H_0(\overline{\psi},\psi) - \eta\mu N(\overline{\psi},\psi)}{\hbar}\right] + \int_0^{\hbar\beta} \mathrm{d}\tau\, \frac{V(\overline{\psi}(\tau),\psi(\tau))}{\hbar} \end{aligned} \tag{19.16}$$

and the other symbols will be introduced and explained in the course of the text.

Proof 19.13 See Appendix, Proofs, page 373.

Remark 19.13.1 Notice that the continuum-notation is a bit misleading, as we are strictly speaking dealing with finite sums. However, the charm of the continuum-notation has prevailed.

Remark 19.13.2 We notice, that the field integral above is *almost* in the form of the generic field integral that was introduced in Chapter TBReferenced. The only difference is that the action has a continous variable, whereas we based our formalism only on the discrete case. To this end, we need to translate the action into a discrete form.

Remark 19.13.3 Notice that for the generic Hamilton-operators in Fact 8.4, one obtains the identities

$$S_\eta[\psi,\overline{\psi}] = \begin{cases} \displaystyle\oint\!\!\!\!\!\!\sum \overline{\psi}_\alpha(\tau)\Big[\partial_\tau\delta_{\alpha\alpha'} + H^{0}_{\alpha\alpha'}/\hbar\Big]\psi_{\alpha'}(\tau) \quad + \oint\!\!\!\!\!\!\sum \frac{V_{\alpha,\beta;\gamma}}{\hbar}\overline{\psi}_\alpha(\tau)\overline{\psi}_\beta(\tau)\psi_\gamma(\tau) + \frac{V_{\alpha;\beta,\gamma}}{\hbar}\overline{\psi}_\alpha(\tau)\psi_\beta(\tau)\psi_\gamma(\tau) + \dots & \text{for } \eta = 0 \\ \displaystyle\oint\!\!\!\!\!\!\sum \overline{\psi}_\alpha(\tau)\Big[\left(\partial_\tau - \frac{\mu}{\hbar}\right)\delta_{\alpha\alpha'} + H^{0}_{\alpha\alpha'}/\hbar\Big]\psi_{\alpha'}(\tau) + \frac{1}{2}\oint\!\!\!\!\!\!\sum \frac{V_{\alpha_1\alpha_2\alpha_3\alpha_4}}{\hbar}\overline{\psi}_{\alpha_1}(\tau)\overline{\psi}_{\alpha_2}(\tau)\psi_{\alpha_4}(\tau)\psi_{\alpha_3}(\tau) \quad + \dots & \text{for } \eta = 1 \end{cases} \tag{19.17}$$

Lemma 19.14 The following transformation is unitary up to a (diverging) scaling-factor

$$\begin{aligned} \psi(\tau) &= \frac{1}{\sqrt{\hbar\beta}}\sum_{\omega_n\in\Omega_\zeta}\psi(\mathrm{i}\omega_n)\,\mathrm{e}^{-\mathrm{i}\omega_n\tau}, \qquad & \psi(\mathrm{i}\omega_n) &= \frac{1}{\sqrt{\hbar\beta}}\int_0^{\hbar\beta}\mathrm{d}\tau\,\psi(\tau)\,\mathrm{e}^{\mathrm{i}\omega_n\tau} \\ \overline{\psi}(\tau) &= \frac{1}{\sqrt{\hbar\beta}}\sum_{\omega_n\in\Omega_\zeta}\overline{\psi}(\mathrm{i}\omega_n)\,\mathrm{e}^{\mathrm{i}\omega_n\tau}, \qquad & \overline{\psi}(\mathrm{i}\omega_n) &= \frac{1}{\sqrt{\hbar\beta}}\int_0^{\hbar\beta}\mathrm{d}\tau\,\overline{\psi}(\tau)\,\mathrm{e}^{-\mathrm{i}\omega_n\tau} \end{aligned}$$

where

$$\mathbb{Z} \ni n \mapsto \omega_n = \frac{2n+1-\theta(\zeta)}{\hbar\beta}\pi = \begin{cases} 2n\pi/\hbar\beta & \text{for bosons} \\ (2n+1)\pi/\hbar\beta & \text{for fermions} \end{cases}$$

Proof 19.14 See Appendix, Proofs, page 374.

Remark 19.14.1 Notice that this is a change of coordinates in the sense of Equation (16.65) with Jacobian-matrix

$$\mathrm{Det}\left(\frac{\partial\psi(\tau_i)}{\partial\psi(\mathrm{i}\omega_j)}\right) = \mathrm{Det}\left(\frac{\partial\overline{\psi}(\tau_i)}{\partial\overline{\psi}(\mathrm{i}\omega_j)}\right) = \text{TBCalculated} \tag{19.18}$$

Remark 19.14.2 This diverging factor is mostly irrelevant. There are though situations, where it becomes important to keep the diverging factor inside the Equations. Examples are when one computes the natural energy for a non-interacting system in Lemma TBReferenced, or when one considers corrections to the partition-function in the superconducting state.

Remark 19.14.3 Notice that we found in the proof of this theorem the identity

$$\text{Kronecker-Delta relations for zeta-transformations.} \tag{19.19}$$

Corollary 19.15 Transforming to Matsubara-frequencies gives in the non-interacting case ($V \equiv 0$)

$$S[\psi, \overline{\psi}] = \sum_{\substack{\omega_n \in \Omega_\zeta \\ \alpha, \alpha' \in \mathscr{A}}} \overline{\psi}_\alpha(\mathrm{i}\omega_n) \Big[\Big(-\mathrm{i}\omega_n - \frac{\eta\mu}{\hbar} \Big) \delta_{\alpha\alpha'} + H_{\substack{0 \\ \alpha\alpha'}} / \hbar \Big] \psi_{\alpha'}(\mathrm{i}\omega_n) \tag{19.20}$$

and for the Hamiltonians in Fact 8.4 ...

- in the wave-case

$$\begin{aligned} S[\psi, \overline{\psi}] = &\sum_{\substack{\omega_n \in \Omega_\zeta \\ \alpha, \alpha' \in \mathscr{A}}} \overline{\psi}_\alpha(\mathrm{i}\omega_n) \Big[-\mathrm{i}\omega_n \delta_{\alpha\alpha'} + H_{\substack{0 \\ \alpha\alpha'}} / \hbar \Big] \psi_{\alpha'}(\mathrm{i}\omega_n) \\ &+ \text{TBD} \end{aligned} \tag{19.21}$$

- in the particle-case

$$\begin{aligned} S[\psi, \overline{\psi}] = &\sum_{\substack{\omega_n \in \Omega_\zeta \\ \alpha, \alpha' \in \mathscr{A}}} \overline{\psi}_\alpha(\mathrm{i}\omega_n) \Big[\Big(-\mathrm{i}\omega_n - \frac{\mu}{\hbar} \Big) \delta_{\alpha\alpha'} + H_{\substack{0 \\ \alpha\alpha'}} / \hbar \Big] \psi_{\alpha'}(\mathrm{i}\omega_n) \\ &+ \frac{1}{2} \frac{1}{\hbar\beta} \sum_{\substack{\alpha_1\alpha_2\alpha_3\alpha_4 \in \mathscr{A} \\ \omega_{n_1}\omega_{n_2}\omega_{n_3}\omega_{n_4} \in \Omega_\zeta}} \frac{V_{\alpha_1\alpha_2\alpha_3\alpha_4} \delta_{n_1+n_2,\, n_3+n_4}}{\hbar} \overline{\psi}_{\alpha_1}(\mathrm{i}\omega_{n_1}) \overline{\psi}_{\alpha_2}(\mathrm{i}\omega_{n_2}) \psi_{\alpha_4}(\mathrm{i}\omega_{n_4}) \psi_{\alpha_3}(\mathrm{i}\omega_{n_3}) \end{aligned} \tag{19.22}$$

with rescaled[2] partition-function.

Proof 19.15 See Appendix, Proofs, page 374.

Lemma 19.16 The grand canonical potential from Equation (TBReferenced) is ...

- in the non-interacting case $V \equiv 0$ given as

$$\Omega_0 = \frac{\zeta}{\beta} \sum_{\alpha \in \mathscr{A}} \log \left(1 - \zeta \mathrm{e}^{-\beta \xi_\alpha} \right) \tag{19.23}$$

  whereas $\xi_{\alpha_1}, \xi_{\alpha_2}, \dots, \xi_{\alpha_{d_\mathcal{H}}}$ denote the eigenvalues of $H_0 - \eta\mu$.

- in the interacting case $V \neq 0$ given as

$$\Omega = \Omega_0 - \frac{1}{\beta} \sum \text{"all connected graphs"} \tag{19.24}$$

Proof 19.16 See Appendix, Proofs, page 375.

[2]Notice that this does not change the expectation-value, as was elaborated in Theorem TBD.

19.1.3. Imaginary-Time Generalized Green-Functions

From the proof of Theorem TBReferenced we see that expectation-values of operators can be computed within the functional integral formalism only if they are normal ordered. Therefore one has to re-express X and X' through their normal-ordered expressions. TBD / TBInvestigated.

19.1.4. Imaginary-Time Pure n-Body Green-Functions

Theorem 19.17 For $\tau_1, \dots, \tau_n, \tau'_1, \dots, \tau'_n \in (0, \hbar\beta)$ one finds for the imaginary-time n-body Green-Functions

$$\begin{aligned}\mathscr{G}(A_1, \dots, A_n \,|\, A'_n, \dots, A'_1) &= \left(-\frac{1}{\hbar}\right)^n \frac{1}{Z} \int \mathscr{D}_\zeta(\psi, \overline{\psi})\, \mathrm{e}^{-S[\psi,\overline{\psi}]}\, \psi_{\alpha_1}(\tau_1) \cdots \psi_{\alpha_n}(\tau_n)\, \overline{\psi}_{\alpha'_1}(\tau'_1) \cdots \overline{\psi}_{\alpha'_n}(\tau'_n) \\ &\equiv \left(-\frac{1}{\hbar}\right)^n \left\langle\, \psi(A_1) \cdots \psi(A_n)\, \overline{\psi}(A'_1) \cdots \overline{\psi}(A'_n) \,\right\rangle .\end{aligned}$$

Proof 19.17 See Appendix, Proofs, page 375.

Theorem 19.18 For the one-body-Green-function of a non-interacting (fermionic or bosonic) gas one finds[3]

$$\mathscr{G}_{\substack{0\\ \alpha,\alpha'}}(\tau) = \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} \mathscr{G}_{\substack{0\\ \alpha,\alpha'}}(\mathrm{i}\omega_n)\, \mathrm{e}^{-\mathrm{i}\omega_n \tau} \tag{19.26}$$

with

$$\mathscr{G}_{\substack{0\\ \alpha,\alpha'}}(\mathrm{i}\omega_n) = \frac{1}{\hbar} \left(\left[\mathrm{i}\omega_n - \frac{H_0 - \eta\mu}{\hbar}\right]^{\mathrm{inv}} \right)_{\alpha,\alpha'} \quad \Longleftrightarrow: \quad \mathscr{G}_0(\mathrm{i}\omega_n) \equiv \frac{1}{\mathrm{i}\hbar\omega_n - (H_0 - \eta\mu)} \tag{19.27}$$

Proof 19.18 See Appendix, Proofs, page 376.

[3]Notice that we are dropping the infinitesimal phases for notational convenience. Keeping track of them yields

$$\mathscr{G}_{\substack{0\\ \alpha,\alpha'}}(\mathrm{i}\omega_n) = \frac{1}{\hbar} \left(\left[\mathrm{i}\omega_n - \frac{H_0 - \eta\mu}{\hbar} \mathrm{e}^{\mathrm{i}\omega_n \delta}\right]^{\mathrm{inv}} \right)_{\alpha,\alpha'} \tag{19.25}$$

19.1.5. Intermezzo: The Continuum-Notation with it's Perks and Problems

We introduced the mathematical framework of quantum field theory in Chapter TBReferenced based on discrete indices. In the derivation of the field-integral however, we noticed that the resulting expressions are more conveniently written in a continuum-notation. This is a re-occuring phenomenon in physics. For example the action in Equation TBReferenced can also be re-expressed with field operators (which are introduced in Appendix TBReferenced) in a continuum-notation over space, such that the entire action becomes an integral over four variables.

Definition 19.19 We equip our partition-function with external sources and introduce the generating functional with continuum-notation

$$\frac{Z[\bar{j},j]}{Z} = \frac{1}{Z}\int \mathscr{D}_\zeta(\psi,\overline{\psi})\,\mathrm{e}^{-S[\psi,\overline{\psi}]+\sum\!\!\!\!\!\int j^\dagger\psi+\psi^\dagger j} = \left\langle \exp \sum\!\!\!\!\!\!\int j^\dagger\psi + \psi^\dagger j \right\rangle \tag{19.28}$$

whereas the source-terms are given in their continuum-notation as

$$\bullet \sum\!\!\!\!\!\!\int j^\dagger\psi \equiv \int_0^{\hbar\beta} \mathrm{d}\tau\, j^\dagger(\tau)\psi(\tau) \equiv \int_0^{\hbar\beta} \mathrm{d}\tau \sum_{\alpha\in\mathscr{A}} \bar{j}_\alpha(\tau)\psi_\alpha(\tau) \equiv \Delta\tau \sum_{\substack{\alpha\in\mathscr{A}\\ n=1,\dots,N_\tau}} \bar{j}_\alpha(\tau_n)\psi_\alpha(\tau_n) \tag{19.29}$$

Lemma 19.20 One concludes, that one can determine the n-body Green-function as functional derivative

$$\mathscr{G}(A_1,\dots,A_n\,|\,A'_1,\dots,A'_n) = \left(-\frac{\zeta}{\hbar}\right)^n \frac{\delta^{2n}}{\delta\bar{j}(A_1)\cdots\delta\bar{j}(A_n)\delta j(A'_n)\cdots\delta j(A'_1)} \frac{Z[\bar{j},j]}{Z}\bigg|_{\bar{j}=j=0} \tag{19.30}$$

Proof 19.20 See Appendix, Proofs, page 359.

Definition 19.21 We also introduce the analogue of the Gibbs free energy / grand canonical potential[4]

$$W[\bar{j},j] = +\log Z[\bar{j},j] \tag{19.31}$$

Theorem 19.22 We find for the [c]onnected Green-functions

$$\mathscr{G}^{\mathrm{c}}(A_1,\dots,A_n\,|\,A'_1,\dots,A'_n) = \left(-\frac{\zeta}{\hbar}\right)^n \frac{\delta^{2n}}{\delta\bar{j}(A_1)\cdots\delta\bar{j}(A_n)\delta j(A'_n)\cdots\delta j(A'_1)} W[\bar{j},j]\bigg|_{\bar{j}=j=0} \tag{19.32}$$

Proof 19.22 See Appendix, Proofs, page **??**[5].

[4]
- Warning for myself: A Sign-Convention-Difference relative to the Master-Thesis.
- Correspondence to the free Energy (TBD), cf. Negele p. 107 following.

[5] Cf. Negele, page 107.

Theorem 19.23 One has the identity

$$\left\langle c^{\dagger}_{\alpha'}(0)\, c_{\alpha}(0)\right\rangle = \frac{\delta^2}{\delta\bar{j}(\tau=0-,\alpha)\delta j(\tau'=0,\alpha')}\frac{Z[\bar{j},j]}{Z}\Big|_{\bar{j}=j=0} \tag{19.33}$$

Proof 19.23 Recall first that from the definition of the Green-function

$$\mathscr{G}(A,A') \equiv \mathscr{G}_{\alpha,\alpha'}(\tau,\tau') \equiv -\frac{1}{\hbar}\frac{\mathrm{Tr}_{\mathscr{F}_\zeta}\, \mathrm{e}^{-\beta(H-\mu N)}\left[\theta(\tau-\tau')\, c_{\alpha}(\tau)\, c^{\dagger}_{\alpha'}(\tau') + \zeta\,\theta(\tau'-\tau)\, c^{\dagger}_{\alpha'}(\tau')\, c_{\alpha}(\tau)\right]}{\mathrm{Tr}_{\mathscr{F}_\zeta}\, \mathrm{e}^{-\beta(H-\mu N)}}$$

one finds the identity

$$\left\langle c^{\dagger}_{\alpha'}(0)\, c_{\alpha}(0)\right\rangle = -\frac{\hbar}{\zeta}\mathscr{G}_{\alpha,\alpha'}(\tau=0-,\tau'=0) \tag{19.34}$$

This then implies with Lemma 16.38

$$\begin{aligned}\left\langle c^{\dagger}_{\alpha'}(0)\, c_{\alpha}(0)\right\rangle &= -\frac{\hbar}{\zeta}\mathscr{G}_{\alpha,\alpha'}(\tau=0-,\tau'=0)\\ &= -\frac{\hbar}{\zeta}\left(-\frac{\zeta}{\hbar}\right)^{1}\frac{\delta^2}{\delta\bar{j}(\tau=0-,\alpha)\delta j(\tau'=0,\alpha')}\frac{Z[\bar{j},j]}{Z}\Big|_{\bar{j}=j=0}\\ &= \frac{\delta^2}{\delta\bar{j}(\tau=0-,\alpha)\delta j(\tau'=0,\alpha')}\frac{Z[\bar{j},j]}{Z}\Big|_{\bar{j}=j=0}\end{aligned} \tag{19.35}$$

Lemma 19.24 One can rewrite the source-term derivatives as

$$\frac{\delta}{\delta\bar{j}(\tau=0-,\alpha)} = \frac{1}{\sqrt{\hbar\beta}}\sum_{\omega_n\in\Omega_\zeta}\mathrm{e}^{+\mathrm{i}\omega_n 0^+}\frac{\partial}{\partial\bar{j}(\mathrm{i}\omega_n,\alpha)} \tag{19.36}$$

Proof 19.24 TBD.

19.2. Many-Body Quantum Field Theory - Mixed Systems

We present the formalism at the example of **massive-massless-particles-mixtures**. The generalization to other mixtures is straight-forward.

19.2.1. Coherent States and Resolutions of Identity

In one to one correspondence to the respective creation- and annihilation-operators we again introduce Graßmann-numbers $\left\{\psi_1, \overline{\psi}_1, \dots, \psi_{\mathrm{d}_{m>0}}, \overline{\psi}_{\mathrm{d}_{m>0}}\right\}$ and complex-numbers $\left\{\phi_1, \overline{\phi}_1, \dots, \phi_{\mathrm{d}_{m=0}}, \overline{\phi}_{\mathrm{d}_{m=0}}\right\}$ to form polynomial expressions with them and creation- and annihilation-operators.

Theorem 19.25 The bra- and ket-states $\langle\, \chi \,|$ and $|\, \chi \,\rangle$ as

$$\langle\, \chi \,| \equiv \langle\, \text{vac.} \,| \exp\left(+ \sum_{j=1}^{\mathrm{d}_{m=0}} b_j \overline{\phi}_j \quad - \quad \sum_{j=1}^{\mathrm{d}_{m>0}} c_j \overline{\psi}_j \right), \qquad |\, \chi \,\rangle \equiv \exp\left(+ \sum_{j=1}^{\mathrm{d}_{m=0}} \phi_j b_j^\dagger \quad - \quad \sum_{j=1}^{\mathrm{d}_{m>0}} \psi_j c_j^\dagger \right) |\, \text{vac.} \,\rangle,$$

and are left- and right-eigenstates of creation- and annihilation-operators, implying

$$\langle\, \chi \,| b_\beta^\dagger = \langle\, \chi \,| \overline{\phi}_\beta, \quad \langle\, \chi \,| c_\alpha^\dagger = \langle\, \chi \,| \overline{\psi}_\alpha, \qquad b_\beta |\, \chi \,\rangle = \phi_\beta |\, \chi \,\rangle \quad \text{and} \quad c_\alpha |\, \chi \,\rangle = \psi_\alpha |\, \chi \,\rangle, \tag{19.37}$$

for all $\alpha \in \{1, \dots \mathrm{d}_{m>0}\}$, $\beta \in \{1, \dots, \mathrm{d}_{m=0}\}$.

Proof 19.25 TBD.

Theorem 19.26 It follows

- Overlap
- Completeness Relation for the total Hilbert-space.

Proof 19.26 See Appendix, Proofs, page **??**.

19.2.2. The Functional Field Integral of the Partition Function

Theorem 19.27 One can rewrite the mixed partition function Z as a functional field integral

$$Z = \mathrm{Tr}_{\mathscr{H}} \exp -\beta \left(H_{m>0} - \mu N_{m>0} + H_{m=0} \right) = \int \mathscr{D}(\chi, \overline{\chi})\, \mathrm{e}^{-S[\chi, \overline{\chi}]}$$

where

$$
\begin{aligned}
S[\chi,\overline{\chi}] \equiv \quad & S_0[\chi,\overline{\chi}] \quad + \quad S_{\substack{m>0\\ \text{int}}}[\psi,\overline{\psi}] + S_{\substack{m=0\\ \text{int}}}[\phi,\overline{\phi}] + S_{\text{int}}[\chi,\overline{\chi}] \\
= & \int_0^{\hbar\beta} \mathrm{d}\tau \begin{pmatrix} \overline{\boldsymbol{\psi}} \\ \overline{\boldsymbol{\phi}} \end{pmatrix}^{\mathrm{T}} \cdot \begin{pmatrix} \partial_\tau + \frac{H_{\substack{0\\ m>0}} - \mu}{\hbar} & 0 \\ 0 & \partial_\tau + \frac{H_{\substack{0\\ m=0}}}{\hbar} \end{pmatrix} \cdot \begin{pmatrix} \boldsymbol{\psi} \\ \boldsymbol{\phi} \end{pmatrix} \quad + \quad \int_0^{\hbar\beta} \mathrm{d}\tau \, \frac{V_{m>0}(\overline{\boldsymbol{\psi}},\boldsymbol{\psi}) + V_{m=0}(\overline{\boldsymbol{\phi}},\boldsymbol{\phi}) + V_{\text{int}}(\overline{\chi},\chi)}{\hbar}
\end{aligned}
$$

Proof 19.27 See Appendix, Proofs, page **??**.

Remark 19.27.1 Going over to Matsubara-frequencies turns this expression into a real mess, as one has to account for the different statistics within the respective summations. It is however common to use Matsubara-frequencies, as it is still the most tractable way to deal with these integrals.

19.2.3. Pure Imaginary-Time Generalized Green-Functions

TBDigitalized for interaction-expectation-values.

19.2.4. Pure Imaginary-Time n-Body Green-Functions

20. Examples of Self-Energies and Polarizations for Interacting Systems

In this whole section, we consider a situation as in Equation (TBReferenced), where the total system is written as

$$H \quad = \quad H_0 \quad + \quad V \tag{20.1}$$

whereas H_0 is a Hamiltonian of which the Green-functions are accessible, and V is a perturbation of some kind. The electronic self-energy is the main object of interest. It will be always written in a series

$$\Sigma(\mathrm{i}\omega_n) \quad = \quad \sum_{j=1}^{\infty} \Sigma^{(j)}(\mathrm{i}\omega_n) \quad = \quad \Sigma^{(1)}(\mathrm{i}\omega_n) + \Sigma^{(2)}(\mathrm{i}\omega_n) + \Sigma^{(3)}(\mathrm{i}\omega_n) + \mathcal{O}(\,V^4\,) \tag{20.2}$$

in the interaction-vertices.

20.1. Example: A Diagonalized Gas with Single-Particle-Interactions

Theorem 20.1 The self-energy reads

$$\Sigma_{\beta\beta'}(\mathrm{i}\omega_n) = V_{\beta\beta'} = \text{„frequency-independent“} . \tag{20.3}$$

Proof 20.1 See Appendix, Proofs, page 431.

Remark 20.1.1 Notice that with the Dyson-equation (TBReferenced) one can conclude that this is actually the only term that appears in the self-energy, i.e. $\Sigma^{(1)}_{\beta\beta'} \equiv \Sigma_{\beta\beta'}$ and $\Sigma^{(1)}_{\beta\beta'} = \Sigma^{(2)}_{\beta,\beta'} = \ldots = 0$. When dealing with quadratic perturbations, self-energy concepts are basically useless. It's only when one has other, higher order perturbations on top of it, where it becomes reasonable to compute self-energies.

20.2. Example: A Diagonalized Gas with Quartic Interactions

Theorem 20.2 The electronic self-energy of this system reads can be approximated in the series expansion as[1]

$$\Sigma^{(1)}_{\alpha,\alpha'}(\mathrm{i}\omega_n) \equiv \zeta \sum_{\beta\in\mathscr{A}} \left(V_{\alpha\beta\beta\alpha'} - V_{\alpha\beta\alpha'\beta}\right) \mathrm{n}^{\zeta}_{\hbar\beta}(\epsilon_\beta - \mu) = \text{„frequency-independent“} . \tag{20.4}$$

Proof 20.2 See Appendix, Proofs, page 434.

[1] Cf. this to Henk p. 167 - 169.

20.3. Example: A Spin-Independent-Interaction between an Electron-Gas and a Phonon-Gas

Theorem 20.3 The electronic self-energy of this system reads can be approximated in the series expansion as

$$\Sigma^{(1)}_{\substack{\mathbf{k},\mathbf{k}'\\ m,m'\\ \mu,\mu'}}(\mathrm{i}\omega_n) = 0, \tag{20.5}$$

$$\Sigma^{(2)}_{\substack{\mathbf{k},\mathbf{k}'\\ m,m'\\ \mu,\mu'}}(\mathrm{i}\omega_n) = \Sigma^{\mathrm{Hartree}}_{\substack{\mathbf{k},\mathbf{k}'\\ m,m'\\ \mu,\mu'}}(\mathrm{i}\omega_n) + \Sigma^{\mathrm{Fock}}_{\substack{\mathbf{k},\mathbf{k}'\\ m,m'\\ \mu,\mu'}}(\mathrm{i}\omega_n), \tag{20.6}$$

with Hartree-contribution

$$\begin{aligned}\Sigma^{\mathrm{Hartree}}_{\substack{\mathbf{k},\mathbf{k}'\\ m,m'\\ \mu,\mu'}}(\mathrm{i}\omega_n) &= \delta_{\mathbf{k},\mathbf{k}'}\delta_{\mu,\mu'} \sum_{\substack{\mathbf{q}\in\mathrm{Q}\\ \sigma\in\{\downarrow,\uparrow\}\\ n\in\{1,2,\dots\}\\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \frac{g_{mm'\nu}(\mathbf{k},0)g_{nn\nu}(\mathbf{q},0)}{N_p}\mathscr{D}_{\substack{0\\ \mathbf{0},\mathbf{0}\\ \nu,\nu}}(0)\,\mathrm{F}_\beta(\xi_{n\sigma\mathbf{q}})\\ &= -\delta_{\mathbf{k},\mathbf{k}'}\delta_{\mu,\mu'} \sum_{\substack{\mathbf{q}\in\mathrm{Q}\\ \sigma\in\{\downarrow,\uparrow\}\\ n\in\{1,2,\dots\}\\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \frac{g_{mm'\nu}(\mathbf{k},0)g_{nn\nu}(\mathbf{q},0)}{N_p}\frac{\mathrm{F}_\beta(\xi_{n\sigma\mathbf{q}})}{\Omega_\nu(\mathbf{q}'=0)}\end{aligned} \tag{20.7}$$

and Fock-contribution

$$\Sigma^{\mathrm{Fock}}_{\substack{\mathbf{k},\mathbf{k}'\\ m,m'\\ \mu,\mu'}}(\mathrm{i}\omega_n) = \delta_{\mathbf{k},\mathbf{k}'}\delta_{\mu,\mu'} \sum_{\substack{\mathbf{q}\in\mathrm{Q}\\ n\in\{1,2,\dots\}\\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \frac{g^*_{nm\nu}(\mathbf{k},\mathbf{q})g_{nm'\nu'}(\mathbf{k},\mathbf{q})}{N_p}\left[\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))+\mathrm{F}_\beta(\xi_{\mathbf{q}+\mathbf{k}n\mu})}{\mathrm{i}\hbar\omega_n-\xi_{\mathbf{q}+\mathbf{k}n\mu}+\Omega_\nu(\mathbf{q})}+\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))+1-\mathrm{F}_\beta(\xi_{\mathbf{q}+\mathbf{k}n\mu})}{\mathrm{i}\hbar\omega_n-\xi_{\mathbf{q}+\mathbf{k}n\mu}-\Omega_\nu(\mathbf{q})}\right] \tag{20.8}$$

Proof 20.3 See Appendix, Proofs, page 439.

Remark 20.3.1 In the *diagonal band approximation*, one assumes that the self-energy is diagonal in the band indices, i.e.

$$\Sigma_{\substack{\mathbf{k},\mathbf{k}'\\ m,m'\\ \mu,\mu'}}(\mathrm{i}\omega_n) \approx \delta_{m,m'}\delta_{\mu,\mu'}\Sigma_{\mathbf{k}m\sigma}(\mathrm{i}\omega_n) \qquad \text{with} \qquad \Sigma_{\mathbf{k}m\sigma}(\mathrm{i}\omega_n) \approx \Sigma^{\mathrm{Hartree}}_{\mathbf{k}m\sigma}(\mathrm{i}\omega_n) + \Sigma^{\mathrm{Fock}}_{\mathbf{k}m\sigma}(\mathrm{i}\omega_n), \tag{20.9}$$

and with Hartree-contribution

$$\Sigma^{\mathrm{Hartree}}_{m\sigma\mathbf{k}}(\bullet) = -\sum_{\substack{\mathbf{q}\in\mathrm{Q}\\ \sigma'\in\{\downarrow,\uparrow\}\\ n\in\{1,2,\dots\}\\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \frac{g_{mm\nu}(\mathbf{k},0)g_{nn\nu}(\mathbf{q},0)}{N_p}\frac{\mathrm{F}_\beta(\epsilon_{n\sigma'\mathbf{q}}-\mu)}{\Omega_\nu(\mathbf{q}'=0)} \quad \in \quad \mathbb{R} \tag{20.10}$$

and Fock-contribution

$$\Sigma^{\mathrm{Fock}}_{m\sigma\mathbf{k}}(\mathrm{i}\omega_n) = \sum_{\substack{\mathbf{q}\in \mathrm{Q} \\ n\in\{1,2,\ldots\} \\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\}}} \frac{|g_{nm\nu}(\mathbf{k},\mathbf{q})|^2}{N_p}\left[\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) + \mathrm{F}_\beta(\xi_{\mathbf{q}+\mathbf{k}n\sigma})}{\mathrm{i}\omega_n - \xi_{\mathbf{q}+\mathbf{k}n\sigma} + \Omega_\nu(\mathbf{q})} + \frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) + 1 - \mathrm{F}_\beta(\xi_{\mathbf{q}+\mathbf{k}n\sigma})}{\mathrm{i}\omega_n - \xi_{\mathbf{q}+\mathbf{k}n\sigma} - \Omega_\nu(\mathbf{q})}\right] \tag{20.11}$$

which is usually motivated by looking at the complex numbers appearing in Equations (20.7), (20.8) and using Remark 8.30.1.

Remark 20.3.2 Another common approximation is the *Migdal approximation*, which neglects the Hartree-term, which is good, when the change of the chemical potential as a function of temperature is small enough.

The Polarization

Theorem 20.4 The polarization of the system is given as

$$\Pi = 2\mathrm{i}\int \frac{\mathrm{d}2}{(2\pi)^4}|g(1,2)|^2 G_0(1)G_0(2) \tag{20.12}$$

or, in coordinates

$$\Pi_{\substack{\mathbf{q},\mathbf{q}' \\ \nu,\nu'}}(\mathrm{i}\omega_n) = 2\delta_{\mathbf{q},\mathbf{q}'}\delta_{\nu,\nu'}\int \frac{\mathrm{d}\mathbf{q}}{\Omega_{\mathrm{BZ}}}|g_{mn\nu}(\mathbf{k},\mathbf{q})|^2 \frac{\mathrm{F}_\beta(\epsilon_{m\mathbf{k}}) - \mathrm{F}_\beta(\epsilon_{m\mathbf{k}+\mathbf{q}})}{\epsilon_{m\mathbf{k}+\mathbf{q}} - \epsilon_{m\mathbf{k}} - \mathrm{i}\omega_n} \tag{20.13}$$

Proof 20.4 See Appendix, Proofs, page 448.

21. Quasi-Equilibrium-Formalism (Kubo-Formalism)

21.1. Kubo-Formula for arbitrary Perturbations

This argument is evidently quite general, and a large class of experiments in which a system is subjected to a weak, controllable external probe is characterized by simple correlation functions of operators at different times (...) - Negele Orland page 51.

Notation 21.1 We consider a time-dependent Hamiltonian $H(t)$ as

$$H(t) = H_0 + V(t) = H_0 + H'(t)\theta(t - t_0) \tag{21.1}$$

and some observable of interest A, given in second quantization as

$$A \equiv A(c, c^\dagger). \tag{21.2}$$

Remark 21.1.1 Recall that the expectation value of an operator A is (in the absence of spontaneous symmetry breaking) given by

$$\langle A(t)\rangle = \mathrm{Tr}_{\mathscr{H}}\left(\rho(t)A\right). \tag{21.3}$$

Theorem 21.2 The *Kubo-formula* gives an expression for the change of the expectation value of an operator A due to a perturbation $V(t)$ as

$$\langle A(t)\rangle = \langle A\rangle_0(t) + \int_{t_0}^{\infty} \mathrm{d}t' G^{\mathrm{ret}}_{\substack{0\\A,H'(t')}}(t,t') + \mathcal{O}(V^2) \tag{21.4}$$

with

$$G^{\mathrm{ret}}_{\substack{0\\A,H'(t')}}(t,t') = -\frac{\mathrm{i}}{\hbar}\theta(t-t')\langle[A(t), H'(t')]\rangle_0 = -\frac{\mathrm{i}}{\hbar}\theta(t-t')\,\mathrm{Tr}_{\mathscr{H}}\left[\rho_0[A(t), H'(t')]\right]. \tag{21.5}$$

and we use the notation $\rho_0 \equiv \rho(t_0)$.

Remark 21.2.1 Cf. also Schwabl 2 page 88. „dynamic susceptibility“.

Proof 21.2 First we note that with the Lehmann-representation we obtain

$$\rho \equiv \rho(t_0) = \frac{1}{Z}\mathrm{e}^{-\beta(H-\mu N)} = \frac{1}{Z}\sum_n |n\rangle\langle n|\mathrm{e}^{-\beta\xi_n} \tag{21.6}$$

whereas $|n\rangle \equiv |n(t_0)\rangle$. Now, applying the time-evolution of the von Neumann equation onto the density-matrix ρ we obtain

$$\rho(t) = \frac{1}{Z}\sum_n |n(t)\rangle\langle n(t)|\mathrm{e}^{-\beta\xi_n} \tag{21.7}$$

Now we rewrite the ket-vectors $|n(t)\rangle$ in the interaction picture as

$$|n(t)\rangle = \mathrm{e}^{-\mathrm{i}H_0(t-t_0)/\hbar}\hat{U}(t,t_0)|n(t_0)\rangle \tag{21.8}$$

with

$$\hat{U}(t,0) = \mathrm{e}^{\mathrm{i}H_0(t-t_0)/\hbar}U(t,0) \tag{21.9}$$

Before we continue, note the general expression for the time-evolution-operator from Theorem 2.13. Therefore, using now the identity

$$\partial_t U(t,0) = H(t)U(t,0). \tag{21.10}$$

This yields therefore

$$\begin{aligned}
\partial_t\hat{U}(t,0) &= \mathrm{i}\mathrm{e}^{\mathrm{i}H_0(t-t_0)/\hbar}\left(H_0 - H(t)\right)U(t,0) \\
&= -\frac{\mathrm{i}}{\hbar}\mathrm{e}^{\mathrm{i}H_0(t-t_0)/\hbar}V(t)U(t,0) \\
&= -\frac{\mathrm{i}}{\hbar}\mathrm{e}^{\mathrm{i}H_0(t-t_0)/\hbar}V(t)\mathrm{e}^{-\mathrm{i}H_0(t-t_0)/\hbar}\hat{U}(t,0) \\
&= -\frac{\mathrm{i}}{\hbar}V_{H_0}(t)\hat{U}(t,0)
\end{aligned} \tag{21.11}$$

and with this we obtain

$$\begin{aligned}
\hat{U}(t,0) - \hat{U}(0,0) \quad &= -\frac{\mathrm{i}}{\hbar}\int_{t_0}^{t}\mathrm{d}t' V_{H_0}(t')\hat{U}(t',0) \\
\Leftrightarrow \hat{U}(t,0) \quad &= 1 - \frac{\mathrm{i}}{\hbar}\int_{t_0}^{t}\mathrm{d}t' V_{H_0}(t')\hat{U}(t',0) \\
\Leftrightarrow \hat{U}(t,0) \quad &= 1 - \frac{\mathrm{i}}{\hbar}\int_{t_0}^{t}\mathrm{d}t' V_{H_0}(t') + \mathcal{O}(V^2)
\end{aligned} \tag{21.12}$$

this gives us then

$$|n(t)\rangle = \mathrm{e}^{-\mathrm{i}H_0(t-t_0)/\hbar}\left(1 - \frac{\mathrm{i}}{\hbar}\int_{t_0}^{t}\mathrm{d}t' V_{H_0}(t') + \mathcal{O}(V^2)\right)|n(t_0)\rangle \tag{21.13}$$

This gives us, inserting it into the definition of the expectation value,

$$\begin{aligned}\langle A(t)\rangle &= \mathrm{Tr}_{\mathscr{H}}\left(\rho(t)A\right)\\ &= \frac{1}{Z}\sum_n \mathrm{e}^{-\beta\xi_n}\,\mathrm{Tr}_{\mathscr{H}}\left(|n(t)\rangle\langle n(t)|A\right)\\ &= \dots\end{aligned} \tag{21.14}$$

now we compute the trace in the last line as

$$\begin{aligned}&\mathrm{Tr}_{\mathscr{H}}\left(|n(t)\rangle\langle n(t)|A\right)\\ &= \mathrm{Tr}_{\mathscr{H}}\left(\mathrm{e}^{-\mathrm{i}H_0(t-t_0)/\hbar}\left(1-\frac{\mathrm{i}}{\hbar}\int_{t_0}^{t}\mathrm{d}t'\mathrm{d}t' V_{H_0}(t')\right)|n(t_0)\rangle\langle n(t_0)|\left(1+\frac{\mathrm{i}}{\hbar}\int_0^t \mathrm{d}t'' V_{H_0}(t'')\right)\mathrm{e}^{+\mathrm{i}H_0(t-t_0)/\hbar}\right)+\mathcal{O}(V^3)\\ &= \mathrm{Tr}_{\mathscr{H}}\left(\mathrm{e}^{-\mathrm{i}H_0(t-t_0)/\hbar}|n(t_0)\rangle\langle n(t_0)|\mathrm{e}^{+\mathrm{i}H_0(t-t_0)/\hbar}A\right)\\ &-\frac{\mathrm{i}}{\hbar}\int_{t_0}^{t}\mathrm{d}t'\,\mathrm{Tr}_{\mathscr{H}}\left(\mathrm{e}^{-\mathrm{i}H_0(t-t_0)/\hbar}V_{H_0}(t')|n(t_0)\rangle\langle n(t_0)|\mathrm{e}^{+\mathrm{i}H_0(t-t_0)/\hbar}A-\mathrm{e}^{-\mathrm{i}H_0(t-t_0)/\hbar}|n(t_0)\rangle\langle n(t_0)|V_{H_0}(t')\mathrm{e}^{+\mathrm{i}H_0(t-t_0)/\hbar}A\right)\\ &= \mathrm{Tr}_{\mathscr{H}}\left(\mathrm{e}^{-\mathrm{i}H_0(t-t_0)/\hbar}|n(t_0)\rangle\langle n(t_0)|\mathrm{e}^{+\mathrm{i}H_0(t-t_0)/\hbar}A\right)\\ &-\frac{\mathrm{i}}{\hbar}\int_{t_0}^{t}\mathrm{d}t'\langle n(t_0)|A_{H_0}(t)V_{H_0}(t')-V_{H_0}(t')A_{H_0}(t)|n(t_0)\rangle\end{aligned} \tag{21.15}$$

inserting this above yields

$$\begin{aligned}\dots &= \langle A\rangle_0(t-t_0)\\ &-\frac{\mathrm{i}}{\hbar}\frac{1}{Z}\sum_n \mathrm{e}^{-\beta\xi_n}\int_{t_0}^{t}\mathrm{d}t'\langle n(t_0)|A_{H_0}(t)V_{H_0}(t')-V_{H_0}(t')A_{H_0}(t)|n(t_0)\rangle+\mathcal{O}(V^2)\\ &= \langle A\rangle_0(t-t_0)-\frac{\mathrm{i}}{\hbar}\int_{t_0}^{t}\mathrm{d}t'\langle[A_{H_0}(t),V_{H_0}(t')]\rangle_0+\mathcal{O}(V^2)\\ &= \langle A\rangle_0(t-t_0)-\frac{\mathrm{i}}{\hbar}\int_{t_0}^{t}\mathrm{d}t'\theta(t'-t_0)\langle[A_{H_0}(t),H'_{H_0}(t')]\rangle_0+\mathcal{O}(V^2)\\ &= \dots\end{aligned} \tag{21.16}$$

now we use

$$\theta(t'-t_0)=\begin{cases}1 & t'>t_0\\ 0 & t'<t_0\end{cases} \tag{21.17}$$

which is, within the integral, always fulfilled, and this gives

$$\begin{aligned}\dots &= \langle A\rangle_0(t-t_0)-\frac{\mathrm{i}}{\hbar}\int_{t_0}^{t}\mathrm{d}t'\langle[A_{H_0}(t),H'_{H_0}(t')]\rangle_0+\mathcal{O}(V^2)\\ &= \dots\end{aligned} \tag{21.18}$$

now we will extend the integral further, by introducing the theta-function as

$$\theta(t-t') = \begin{cases} 1 & t > t' \\ 0 & t < t' \end{cases} \tag{21.19}$$

to rewrite the expression above as

$$\begin{aligned} \ldots &= \langle A \rangle_0 (t-t_0) - \frac{\mathrm{i}}{\hbar} \int_{t_0}^{\infty} \mathrm{d}t' \theta(t-t') \langle [A_{H_0}(t), H'_{H_0}(t')] \rangle_0 + \mathcal{O}(V^2) \\ &= \langle A \rangle_0 (t-t_0) - \int_{t_0}^{\infty} \mathrm{d}t' G^{\mathrm{ret}}{}_{A,H'(t')}^{\;\;0}(t,t') + \mathcal{O}(V^2) \end{aligned} \tag{21.20}$$

Part IV.

Applications to Real Materials

22. Thermal Quantum Field Theories of Real Materials

22.1. Non-Interacting Systems

22.1.1. Electrons

We consider the system from Remark 8.22.1.

Lemma 22.1 For a crystal with time-reversal-symmetry one obtains $\epsilon_{m,\uparrow}(+\mathbf{k}) = \epsilon_{m,\downarrow}(-\mathbf{k})$.

Proof 22.1 TBReferenced

Lemma 22.2 The inversion-symmetry of the crystal implies for the bands $\epsilon_{m,\sigma}(+\mathbf{k}) = \epsilon_{m,\sigma}(-\mathbf{k})$.

Proof 22.2 TBReferenced.

Fact 22.3 In the derivation of the Eliashberg-equations, it is used that one has a *two-fold Kramers-degeneracy of the bands*, i.e. $\epsilon_m(\mathbf{k}) \equiv \epsilon_{m,\uparrow}(\mathbf{k}) = \epsilon_{m,\downarrow}(\mathbf{k})$, which holds for systems that are time-reversal symmetric *and* inversion-symmetric [][1].

Fact 22.4 Recall that the retarded Green-function for the electrons is given by analytical continuation of the Matsubara-Green-function according to Theorem 14.21 which is determined as

$$\mathscr{G}_{\substack{m,m' \\ \sigma,\sigma'}}(\mathrm{i}\omega_n, \mathbf{k}) = \frac{\delta_{\sigma,\sigma'}}{\hbar} \left(\frac{1}{\mathrm{i}\omega_n - \left(H_{0,\sigma}(\mathbf{k}) - \mu\right)/\hbar} \right)_{m,m'} = \frac{\delta_{\sigma,\sigma'}}{\mathrm{i}\hbar\omega_n - \left(\epsilon_{m\sigma}(\mathbf{k}) - \mu\right)} \tag{22.1}$$

with non-interacting band-dispersion-matrix $H_{0,\sigma}(\mathbf{k}) \equiv \mathrm{diag}(\epsilon_{1,\sigma}(\mathbf{k}), \ldots, \epsilon_{n_{\mathrm{bnds}},\sigma}(\mathbf{k}))$, whereas n_{bnds} represents the number of bands that are under consideration[2].

[1] Cf. Waste-Folder 1 book reference; Cf. also Lemmas 25.19, 25.22 where we used $\omega(+\mathbf{q}) = \omega(-\mathbf{q})$

[2] Usually this is the number of bands were are reproduced in the Wannierization-procedure.

Field Integral Formulation

Fact 22.5 Going over to the field-integral-description of the problem gives the action

$$S[\psi,\overline{\psi}] \equiv S_{\text{el}}[\psi,\overline{\psi}] \tag{22.2}$$

with

- $S_{\text{el}}[\psi,\overline{\psi}] = \displaystyle\sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\downarrow,\uparrow\}\\ \tau\in(0,\hbar\beta)\\ m\in\{1,2,\ldots,\infty\}}} \overline{\psi}_{m\sigma}(\tau,\mathbf{k})\left(\partial_\tau + \frac{\epsilon_{m\sigma}(\mathbf{k})-\mu}{\hbar}\right)\psi_{m\sigma}(\tau,\mathbf{k})$

Notation 22.6 We implement external sources into the partition-function as

$$Z \mapsto Z[j_{\text{el}},\overline{j}_{\text{el}}] \equiv \int \mathscr{D}(\overline{\psi},\psi)\,\exp\left[-S_{\text{el}}[\psi,\overline{\psi}] + \overline{j}_{\text{el}}\psi + \overline{\psi} j_{\text{el}}\right] \tag{22.3}$$

with abbreviation

- $\overline{j}_{\text{el}}\psi \equiv \displaystyle\sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\downarrow,\uparrow\}\\ \tau\in(0,\hbar\beta)\\ m\in\{1,2,\ldots,\infty\}}} \overline{j}_{m\sigma}(\tau,\mathbf{k})\psi_{m\sigma}(\tau,\mathbf{k})$

Lemma 22.7 Going over to Matsubara-frequencies and introducing the four-momenta $k \equiv (\mathrm{i}\omega_n, \mathbf{k}) \in \mathrm{i}\Omega_{-1} \times \mathrm{Q}$, we obtain

$$S_{[j,\overline{j}]}[\psi,\overline{\psi}] = S_{\text{el}}[\psi,\overline{\psi}] - \left(\overline{j}_{\text{el}}\psi + \overline{\psi} j_{\text{el}}\right)$$

with

- $S_{\text{el}}[\psi,\overline{\psi}] = \displaystyle\sum_{\substack{\sigma\in\{\downarrow,\uparrow\}\\ k\in\mathrm{i}\Omega_{-1}\times\mathrm{Q}\\ m\in\{1,2,\ldots,\infty\}}} \overline{\psi}_{m\sigma}(k)\left(-\mathrm{i}\omega_k + \frac{\epsilon_{m\sigma}(\mathbf{k})-\mu}{\hbar}\right)\psi_{m\sigma}(k) \equiv \sum_{\substack{\sigma\in\{\downarrow,\uparrow\}\\ k\in\mathrm{i}\Omega_{-1}\times\mathrm{Q}\\ m\in\{1,2,\ldots,\infty\}}} \overline{\psi}_{m\sigma}(k)\left(-\frac{1}{\hbar}\mathscr{G}^{-\text{el}}_{\substack{0\\ m,\sigma}}(k)\right)\psi_{m\sigma}(k)$

with negative four-momentum $-q \equiv (-\mathrm{i}\omega_m, -\mathbf{q})$.

Proof 22.7 Follows from Corollary 19.15.

22.1.2. Phonons

Lemma 22.8 The time-reversal-symmetry of the crystal implies for the dispersions $\omega_\nu(+\mathbf{q}) = \omega_\nu(-\mathbf{q})$.

Proof 22.8 TBReferenced.

Field Integral Formulation

We consider the system from Remark 8.22.2.

Fact 22.9 Going over to the field-integral-description of the problem gives the action

$$S[\phi,\overline{\phi}] = S_{\text{ph}}[\phi,\overline{\phi}] \tag{22.4}$$

with

- $S_{\text{ph}}[\phi,\overline{\phi}] = \displaystyle\sum\!\!\!\!\!\!\int_{\substack{\mathbf{q}\in\mathrm{Q}\\ \tau\in(0,\hbar\beta)\\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\}}} \overline{\phi}_\nu(\tau,\mathbf{q})\left(\partial_\tau + \omega_\nu(\mathbf{q})\right)\phi_\nu(\tau,\mathbf{q})$

Notation 22.10 We implement external sources into the partition-function as

$$Z \mapsto Z[j_{\text{ph}},\overline{j}_{\text{ph}}] \equiv \int \mathscr{D}(\overline{\phi},\phi)\, \exp\Big[-S_{\text{ph}}[\phi,\overline{\phi}] + \sum_{\bullet\in\{\text{el},\text{ph}\}} \overline{j}_{\text{ph}}\phi_{\bullet} + \overline{\phi}j_{\text{ph}}\Big] \tag{22.5}$$

with abbreviation

- $\overline{j}_{\text{ph}}\phi \equiv \displaystyle\sum\!\!\!\!\!\!\int_{\substack{\mathbf{q}\in\mathrm{Q}\\ \tau\in(0,\hbar\beta)\\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\}}} \overline{j}_\nu(\tau,\mathbf{q})\phi_\nu(\tau,\mathbf{q})$

Lemma 22.11 Going over to Matsubara-frequencies and introducing the four-momenta $q \equiv (\mathrm{i}\omega_m,\mathbf{q}) \in \mathrm{i}\Omega_{+1}\times\mathrm{Q}$ and $k \equiv (\mathrm{i}\omega_n,\mathbf{k}) \in \mathrm{i}\Omega_{-1}\times\mathrm{Q}$, we obtain

$$S_{[j,\overline{j}]}[\phi,\overline{\phi}] = S_{\text{ph}}[\phi,\overline{\phi}] - \left(\overline{j}_{\text{ph}}\phi_{\text{ph}} + \overline{\phi}_{\text{ph}}j_{\text{ph}}\right)$$

with

- $S_{\text{ph}}[\phi,\overline{\phi}] = \displaystyle\sum_{\substack{q\in\mathrm{i}\Omega_{+1}\times\mathrm{Q}\\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\}}} \overline{\phi}_\nu(q)\left(-\mathrm{i}\omega_q + \omega_\nu(\mathbf{q})\right)\phi_\nu(q) \equiv \sum_{\substack{q\in\mathrm{i}\Omega_{+1}\times\mathrm{Q}\\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\}}} \overline{\phi}_\nu(q)\left(-\frac{1}{\hbar}\mathscr{G}_{0,\nu}^{-\text{ph}}(q)\right)\phi_\nu(q)$

with negative four-momentum $-q \equiv (-\mathrm{i}\omega_m,-\mathbf{q})$.

Proof 22.11 Follows from Corollary 19.15.

Notice that in the case of phonons, one can also simplify the action by changing the field-variables. This is elaborated in Section 33.6.

22.1.3. Total Crystal

We consider the system from Definition 8.28.

Fact 22.12 Going over to the field-integral-description of the problem gives the action

$$S[\chi,\overline{\chi}] = S_{\mathrm{el}}[\psi,\overline{\psi}] + S_{\mathrm{ph}}[\phi,\overline{\phi}] \tag{22.6}$$

with

- $S_{\mathrm{el}}[\psi,\overline{\psi}] = \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\downarrow,\uparrow\}\\ \tau\in(0,\hbar\beta)\\ m\in\{1,2,\dots,\infty\}}} \overline{\psi}_{m\sigma}(\tau,\mathbf{k})\left(\partial_\tau + \frac{\epsilon_{m\sigma}(\mathbf{k})-\mu}{\hbar}\right)\psi_{m\sigma}(\tau,\mathbf{k})$
- $S_{\mathrm{ph}}[\phi,\overline{\phi}] = \sum\!\!\!\!\!\!\int_{\substack{\mathbf{q}\in\mathrm{Q}\\ \tau\in(0,\hbar\beta)\\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \overline{\phi}_\nu(\tau,\mathbf{q})\left(\partial_\tau + \omega_\nu(\mathbf{q})\right)\phi_\nu(\tau,\mathbf{q})$

Notation 22.13 We implement external sources into the partition-function as

$$Z \mapsto Z[j_{\mathrm{el}},\overline{j}_{\mathrm{el}},j_{\mathrm{ph}},\overline{j}_{\mathrm{ph}}] \equiv \int \mathscr{D}(\overline{\chi},\chi)\,\exp\left[-\left(S_{\mathrm{el}}[\psi,\overline{\psi}] + S_{\mathrm{ph}}[\phi,\overline{\phi}]\right) + \sum_{\bullet\in\{\mathrm{el},\mathrm{ph}\}} \overline{j}_\bullet\chi_\bullet + \overline{\chi}_\bullet j_\bullet\right] \tag{22.7}$$

with abbreviations

- $\overline{j}_{\mathrm{el}}\chi_{\mathrm{el}} \equiv \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\downarrow,\uparrow\}\\ \tau\in(0,\hbar\beta)\\ m\in\{1,2,\dots,\infty\}}} \overline{j}_{m\sigma}(\tau,\mathbf{k})\psi_{m\sigma}(\tau,\mathbf{k})$
- $\overline{j}_{\mathrm{ph}}\chi_{\mathrm{ph}} \equiv \sum\!\!\!\!\!\!\int_{\substack{\mathbf{q}\in\mathrm{Q}\\ \tau\in(0,\hbar\beta)\\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \overline{j}_\nu(\tau,\mathbf{q})\phi_\nu(\tau,\mathbf{q})$
- etc...

Lemma 22.14 Going over to Matsubara-frequencies and introducing the four-momenta $q \equiv (\mathrm{i}\omega_m,\mathbf{q}) \in \mathrm{i}\Omega_{+1}\times\mathrm{Q}$ and $k \equiv (\mathrm{i}\omega_n,\mathbf{k}) \in \mathrm{i}\Omega_{-1}\times\mathrm{Q}$, we obtain

$$S_{[j,\overline{j}]}[\chi,\overline{\chi}] = S_{\mathrm{el}}[\psi,\overline{\psi}] + S_{\mathrm{ph}}[\phi,\overline{\phi}] - \sum_{\bullet\in\{\mathrm{el},\mathrm{ph}\}}\left(\overline{j}_\bullet\chi_\bullet + \overline{\chi}_\bullet j_\bullet\right)$$

with

- $S_{\mathrm{el}}[\psi,\overline{\psi}] = \sum_{\substack{\sigma\in\{\downarrow,\uparrow\}\\ k\in\mathrm{i}\Omega_{-1}\times\mathrm{Q}\\ m\in\{1,2,\dots,\infty\}}} \overline{\psi}_{m\sigma}(k)\left(-\mathrm{i}\omega_k + \frac{\epsilon_{m\sigma}(\mathbf{k})-\mu}{\hbar}\right)\psi_{m\sigma}(k) \equiv \sum_{\substack{\sigma\in\{\downarrow,\uparrow\}\\ k\in\mathrm{i}\Omega_{-1}\times\mathrm{Q}\\ m\in\{1,2,\dots,\infty\}}} \overline{\psi}_{m\sigma}(k)\left(-\frac{1}{\hbar}\mathscr{G}^{-\mathrm{el}}_{\substack{0\\ m,\sigma}}(k)\right)\psi_{m\sigma}(k)$
- $S_{\mathrm{ph}}[\phi,\overline{\phi}] = \sum_{\substack{q\in\mathrm{i}\Omega_{+1}\times\mathrm{Q}\\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \overline{\phi}_\nu(q)\left(-\mathrm{i}\omega_q + \omega_\nu(\mathbf{q})\right)\phi_\nu(q) \equiv \sum_{\substack{q\in\mathrm{i}\Omega_{+1}\times\mathrm{Q}\\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \overline{\phi}_\nu(q)\left(-\frac{1}{\hbar}\mathscr{G}^{-\mathrm{ph}}_{0,\nu}(q)\right)\phi_\nu(q)$

with negative four-momentum $-q \equiv (-\mathrm{i}\omega_m, -\mathbf{q})$.

Proof 22.14 See Appendix, Proofs, page 306.

22.2. Interacting Systems

We consider the system from Definition 8.29.

22.2.1. Introduction

Fact 22.15 Recall that the retarded Green-function for the electrons is given by analytical continuation of the Matsubara-Green-function according to Theorem 14.21 which is determined by solving the Dyson-equation (15.4) as

$$\underset{\sigma,\sigma'}{\mathscr{G}_{m,m'}}(\mathrm{i}\omega_n, \mathbf{k}) = \frac{\delta_{\sigma,\sigma'}}{\hbar}\left(\frac{1}{\mathrm{i}\omega_n - \left(H_{0,\sigma}(\mathbf{k}) + \Sigma_\sigma(\mathbf{k}, \mathrm{i}\omega_n) - \mu\right)/\hbar}\right)_{m,m'} \tag{22.8}$$

with non-interacting band-dispersion-matrix $H_{0,\sigma}(\mathbf{k}) \equiv \mathrm{diag}(\epsilon_{1,\sigma}(\mathbf{k}), \dots, \epsilon_{n_{\mathrm{bnds}},\sigma}(\mathbf{k}))$, whereas n_{bnds} represents the number of bands that are under consideration[3].

Remark 22.15.1 Notice further that we have determined the self-energy in this system up to second / third order perturbation theory in Theorem 20.3.

Remark 22.15.2 Usually, we enumerate the indices in such a way, that the matrix-representation of the Green-function becomes

$$\mathscr{G}(\mathrm{i}\omega_n, \mathbf{k}) = \begin{pmatrix} \mathscr{G}_{\uparrow,\uparrow}(\mathrm{i}\omega_n, \mathbf{k}) & \mathbf{0} \\ \mathbf{0} & \mathscr{G}_{\downarrow,\downarrow}(\mathrm{i}\omega_n, \mathbf{k}) \end{pmatrix}.$$

with individual blocks for spin-up and spin-down electrons. More explicitly, this means

$$\mathscr{G}(\mathrm{i}\omega_n, \mathbf{k}) = \begin{pmatrix} \underset{\uparrow,\uparrow}{\mathscr{G}_{1,1}} & \underset{\uparrow,\uparrow}{\mathscr{G}_{1,2}} & \cdots & \underset{\uparrow,\uparrow}{\mathscr{G}_{1,n_{\mathrm{bnds}}}} & 0 & 0 & \cdots & 0 \\ \underset{\uparrow,\uparrow}{\mathscr{G}_{2,1}} & \underset{\uparrow,\uparrow}{\mathscr{G}_{2,2}} & \cdots & \underset{\uparrow,\uparrow}{\mathscr{G}_{2,n_{\mathrm{bnds}}}} & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \underset{\uparrow,\uparrow}{\mathscr{G}_{n_{\mathrm{bnds}},1}} & \underset{\uparrow,\uparrow}{\mathscr{G}_{n_{\mathrm{bnds}},2}} & \cdots & \underset{\uparrow,\uparrow}{\mathscr{G}_{n_{\mathrm{bnds}},n_{\mathrm{bnds}}}} & 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & \underset{\downarrow,\downarrow}{\mathscr{G}_{1,1}} & \underset{\downarrow,\downarrow}{\mathscr{G}_{1,2}} & \cdots & \underset{\downarrow,\downarrow}{\mathscr{G}_{1,n_{\mathrm{bnds}}}} \\ 0 & 0 & \cdots & 0 & \underset{\downarrow,\downarrow}{\mathscr{G}_{2,1}} & \underset{\downarrow,\downarrow}{\mathscr{G}_{2,2}} & \cdots & \underset{\downarrow,\downarrow}{\mathscr{G}_{2,n_{\mathrm{bnds}}}} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \underset{\downarrow,\downarrow}{\mathscr{G}_{n_{\mathrm{bnds}},1}} & \underset{\downarrow,\downarrow}{\mathscr{G}_{n_{\mathrm{bnds}},2}} & \cdots & \underset{\downarrow,\downarrow}{\mathscr{G}_{n_{\mathrm{bnds}},n_{\mathrm{bnds}}}} \end{pmatrix}(\mathrm{i}\omega_n, \mathbf{k}).$$

[3]Usually this is the number of bands were are reproduced in the Wannierization-procedure.

Notation 22.16 We simplify the notation further, by recalling that the self-energy is, within the diagonal-band-approximation (cf. Remark 20.3.1), diagonal in spin and wave-vector indices, such that we can write

$$G^{\text{ret}}_{\substack{m,m'\\ \sigma,\sigma'\\ \mathbf{k},\mathbf{k}'}}(E+\mathrm{i}\delta) \equiv \delta_{\mathbf{k},\mathbf{k}'}\delta_{\sigma,\sigma'}G^{\text{ret}}_{\substack{m,m'\\ \sigma,\sigma'}}(E+\mathrm{i}\delta,\mathbf{k}) \tag{22.9}$$

whereas we keep the double spin-indices explicit for reasons that will become clear in the chapter about superconductivity.

22.2.2. Field Integral Formulation

Fact 22.17 Going over to the field-integral-description of the problem gives the action

$$S[\chi,\overline{\chi}] = S_{\text{el}}[\psi,\overline{\psi}] + S_{\text{ph}}[\phi,\overline{\phi}] + S_{\text{el-ph-int}}[\chi,\overline{\chi}] \tag{22.10}$$

with

- $S_{\text{el}}[\psi,\overline{\psi}] = \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\downarrow,\uparrow\}\\ \tau\in(0,\hbar\beta)\\ m\in\{1,2,\ldots,\infty\}}} \overline{\psi}_{m\sigma}(\tau,\mathbf{k})\left(\partial_\tau + \dfrac{\epsilon_{m\sigma}(\mathbf{k})-\mu}{\hbar}\right)\psi_{m\sigma}(\tau,\mathbf{k})$
- $S_{\text{ph}}[\phi,\overline{\phi}] = \sum\!\!\!\!\!\!\int_{\substack{\mathbf{q}\in Q\\ \tau\in(0,\hbar\beta)\\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\}}} \overline{\phi}_\nu(\tau,\mathbf{q})\left(\partial_\tau + \omega_\nu(\mathbf{q})\right)\phi_\nu(\tau,\mathbf{q})$
- $S_{\text{el-ph-int}}[\chi,\overline{\chi}] = \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k},\mathbf{q}\in Q\\ \tau\in(0,\hbar\beta)\\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\}\\ m,n\in\{1,2,\ldots,\infty\}}} \dfrac{g_{mn\nu}(\mathbf{k},\mathbf{q})}{\hbar\sqrt{N_p}}\left(\phi_\nu(\tau,\mathbf{q}) + \overline{\phi}_\nu(\tau,-\mathbf{q})\right)\overline{\psi}_{m\sigma}(\tau,\mathbf{k}+\mathbf{q})\psi_{n\sigma}(\tau,\mathbf{k})$

Notation 22.18 We implement external sources into the partition-function as

$$Z \mapsto Z[j_{\text{el}},\overline{j}_{\text{el}},j_{\text{ph}},\overline{j}_{\text{ph}}] \equiv \int \mathscr{D}(\overline{\chi},\chi)\,\exp\left[-\left(S_{\text{el}}[\psi,\overline{\psi}] + S_{\text{ph}}[\phi,\overline{\phi}] + S_{\text{el-ph-int}}[\chi,\overline{\chi}]\right) + \sum_{\bullet\in\{\text{el},\text{ph}\}} \overline{j}_\bullet\chi_\bullet + \overline{\chi}_\bullet j_\bullet\right] \tag{22.11}$$

with abbreviations

- $\overline{j}_{\text{el}}\chi_{\text{el}} \equiv \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\downarrow,\uparrow\}\\ \tau\in(0,\hbar\beta)\\ m\in\{1,2,\ldots,\infty\}}} \overline{j}_{m\sigma}(\tau,\mathbf{k})\psi_{m\sigma}(\tau,\mathbf{k})$
- $\overline{j}_{\text{ph}}\chi_{\text{ph}} \equiv \sum\!\!\!\!\!\!\int_{\substack{\mathbf{q}\in Q\\ \tau\in(0,\hbar\beta)\\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\}}} \overline{j}_\nu(\tau,\mathbf{q})\phi_\nu(\tau,\mathbf{q})$
- etc...

Lemma 22.19 Going over to Matsubara-frequencies and introducing the four-momenta $q \equiv (\mathrm{i}\omega_m, \mathbf{q}) \in \mathrm{i}\Omega_{+1} \times \mathrm{Q}$ and $k \equiv (\mathrm{i}\omega_n, \mathbf{k}) \in \mathrm{i}\Omega_{-1} \times \mathrm{Q}$, we obtain

$$S_{[j,\bar{j}]}[\chi, \overline{\chi}] = S_{\mathrm{el}}[\psi, \overline{\psi}] + S_{\mathrm{ph}}[\phi, \overline{\phi}] + S_{\text{el-ph-int}}[\chi, \overline{\chi}] - \sum_{\bullet \in \{\mathrm{el},\mathrm{ph}\}} \left(\bar{j}_{\bullet} \chi_{\bullet} + \overline{\chi}_{\bullet} j_{\bullet} \right)$$

with

- $S_{\mathrm{el}}[\psi, \overline{\psi}] = \sum_{\substack{\sigma \in \{\downarrow,\uparrow\} \\ k \in \mathrm{i}\Omega_{-1}\times \mathrm{Q} \\ m \in \{1,2,\ldots,\infty\}}} \overline{\psi}_{m\sigma}(k) \left(-\mathrm{i}\omega_k + \frac{\epsilon_{m\sigma}(\mathbf{k}) - \mu}{\hbar} \right) \psi_{m\sigma}(k) \equiv \sum_{\substack{\sigma \in \{\downarrow,\uparrow\} \\ k \in \mathrm{i}\Omega_{-1}\times \mathrm{Q} \\ m \in \{1,2,\ldots,\infty\}}} \overline{\psi}_{m\sigma}(k) \left(-\frac{1}{\hbar} \mathscr{G}^{-\mathrm{el}}_{\substack{0 \\ m,\sigma}}(k) \right) \psi_{m\sigma}(k)$
- $S_{\mathrm{ph}}[\phi, \overline{\phi}] = \sum_{\substack{q \in \mathrm{i}\Omega_{+1}\times \mathrm{Q} \\ \nu \in \{1,2,\ldots,rd_{\mathbf{x}}\}}} \overline{\phi}_\nu(q) \left(-\mathrm{i}\omega_q + \omega_\nu(\mathbf{q}) \right) \phi_\nu(q) \equiv \sum_{\substack{q \in \mathrm{i}\Omega_{+1}\times \mathrm{Q} \\ \nu \in \{1,2,\ldots,rd_{\mathbf{x}}\}}} \overline{\phi}_\nu(q) \left(-\frac{1}{\hbar} \mathscr{G}^{-\mathrm{ph}}_{0,\nu}(q) \right) \phi_\nu(q)$
- $S_{\text{el-ph-int}}[\chi, \overline{\chi}] = \frac{1}{\sqrt{\hbar\beta}} \sum_{\substack{k \in \mathrm{i}\Omega_{-1}\times \mathrm{Q} \\ q \in \mathrm{i}\Omega_{+1}\times \mathrm{Q} \\ \nu \in \{1,2,\ldots,rd_{\mathbf{x}}\} \\ m,n \in \{1,2,\ldots,\infty\}}} \frac{g_{mn\nu}(\mathbf{k}, \mathbf{q})}{\hbar\sqrt{N_p}} \left(\phi_\nu(q) + \overline{\phi}_\nu(-q) \right) \overline{\psi}_{m\sigma}(q+k) \psi_{n\sigma}(k)$

with negative four-momentum $-q \equiv (-\mathrm{i}\omega_m, -\mathbf{q})$.

Proof 22.19 See Appendix, Proofs, page 306.

Lemma 22.20 We can rewrite the interaction-term as

$$\begin{aligned} \underset{[j_{\mathrm{ph}},\bar{j}_{\mathrm{ph}}]}{S_{\text{el-ph-int}}}[\chi, \overline{\chi}] &= \left(S_{\text{el-ph-int}} + \underset{[j_{\mathrm{ph}},\bar{j}_{\mathrm{ph}}]}{S_{\text{ph-sources}}} \right) [\chi, \overline{\chi}] \\ &= \sum_{\substack{q \in \mathrm{i}\Omega_{+1}\times \mathrm{Q} \\ \nu \in \{1,\ldots,rd_{\mathbf{x}}\}}} \phi_\nu(q) \left(\rho_\nu(q) - \bar{j}_\nu(q) \right) + \overline{\phi}_\nu(q) \left(\rho_\nu(-q) - j_\nu(q) \right) \end{aligned}$$

with density-like-interaction

$$\rho_\nu(q) = \frac{1}{\sqrt{\hbar\beta N_p}} \sum_{\substack{\sigma \in \{\downarrow,\uparrow\} \\ k \in \mathrm{i}\Omega_{-1}\times \mathrm{Q} \\ n,m \in \{1,\ldots,\infty\}}} \frac{g_{mn\nu}(\mathbf{k}, \mathbf{q})}{\hbar} \overline{\psi}_{m\sigma}(k+q) \psi_{n\sigma}(k) \tag{22.12}$$

Proof 22.20 See Appendix, Proofs, page 381.

Corollary 22.21 One finds

$$\overline{\rho_\nu(-q)} = \rho_\nu(q) \tag{22.13}$$

Proof 22.21 TBDigitalized from SC-notes and TB-brought in harmony with WF1-topic-32.

22.2.3. Integrating out the Phonons

Lemma 22.22 Changing variables in a standard way allows us to integrate out the phonons, such that we obtain the effective fermionic action

$$\begin{aligned} S_{[j,\bar{j}]}[\psi,\overline{\psi}] = & \quad S_{\mathrm{el}}[\psi,\overline{\psi}] && + \underset{[j_{\mathrm{ph}},\bar{j}_{\mathrm{ph}}]}{S_{\text{el-el-int}}}[\psi,\overline{\psi}] - \left(\bar{j}_{\mathrm{el}}\psi + \overline{\psi} j_{\mathrm{el}}\right) \\ = & \sum_{\substack{\sigma\in\{\downarrow,\uparrow\} \\ k\in \mathrm{i}\Omega_{-1}\times \mathrm{Q} \\ m\in\{1,2,\ldots,\infty\}}} \overline{\psi}_{m\sigma}(k)\left(-\frac{1}{\hbar}\mathscr{G}^{-\mathrm{el}}_{\substack{0\\ m,\sigma}}(k)\right)\psi_{m\sigma}(k) && + \underset{[j_{\mathrm{ph}},\bar{j}_{\mathrm{ph}}]}{S_{\text{el-el-int}}}[\psi,\overline{\psi}] - \left(\bar{j}_{\mathrm{el}}\psi + \overline{\psi} j_{\mathrm{el}}\right) \end{aligned} \tag{22.14}$$

giving four terms, whereas the term that is quartic in the fermionic fields can be simplified to

$$\begin{aligned} \underset{[j_{\mathrm{ph}},\bar{j}_{\mathrm{ph}}]}{S_{\text{el-el-int}}}[\psi,\overline{\psi}] = & -\sum_{\substack{q\in \mathrm{i}\Omega_{+1}\times \mathrm{Q} \\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}}} \frac{\omega_\nu(\mathbf{q})}{\omega_m^2+\omega_\nu^2(\mathbf{q})}\rho_\nu(q)\rho_\nu(-q) \\ & -\hbar \sum_{\substack{q\in \mathrm{i}\Omega_{+1}\times \mathrm{Q} \\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}}} \rho_\nu(q)\mathscr{G}_{0,\nu}(q) j_\nu(q) - \hbar \sum_{\substack{q\in \mathrm{i}\Omega_{+1}\times \mathrm{Q} \\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}}} \bar{j}_\nu(q)\mathscr{G}_{0,\nu}(q)\rho_\nu(-q) \\ & +\hbar \sum_{\substack{q\in \mathrm{i}\Omega_{+1}\times \mathrm{Q} \\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}}} \bar{j}_\nu(q)\mathscr{G}_{0,\nu}(q) j_\nu(q) \end{aligned} \tag{22.15}$$

Proof 22.22 See Appendix, Proofs, page 381.

Lemma 22.23 Introducing the multi-index

$$\alpha \equiv (m_\alpha, \sigma_\alpha, \mathrm{i}\omega_\alpha, \mathbf{k}_\alpha) \in \{1,2,\ldots\}\times\{\downarrow,\uparrow\}\times \mathrm{i}\Omega_{-1}\times \mathrm{Q} \equiv \Xi, \tag{22.16}$$

one can rewrite the interaction-term-contribution that is quartic in the Graßmann-variables as generalised bilinear-form

$$S_{\text{el-el-int}}[\psi,\overline{\psi}] = \sum_{\alpha_1,\ldots,\alpha_4\in\Xi} \overline{\psi}_{\alpha_1}\overline{\psi}_{\alpha_2} V_{\alpha_1,\alpha_2;\alpha_3,\alpha_4}\psi_{\alpha_4}\psi_{\alpha_3}$$

with

$$\begin{aligned}
&V_{\alpha_1,\alpha_2;\alpha_3,\alpha_4}\\
&= \frac{1}{\hbar\beta N_p}\sum_{\nu\in\{1,\ldots,rd_x\}}\frac{\omega_\nu(\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4})}{\omega^2_{\alpha_1-\alpha_4}+\omega^2_\nu(\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4})}\frac{g_{m_{\alpha_1}m_{\alpha_4}\nu}(\mathbf{k}_{\alpha_4},\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4})g^*_{m_{\alpha_3}m_{\alpha_2}\nu}(\mathbf{k}_{\alpha_2},\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4})}{\hbar^2}\delta_{k_{\alpha_3}+k_{\alpha_4},k_{\alpha_2}+k_{\alpha_1}}\delta_{\sigma_{\alpha_1},\sigma_{\alpha_4}}\delta_{\sigma_{\alpha_2},\sigma_{\alpha_3}}\\
&= \frac{1}{\hbar\beta N_p}\sum_{\nu\in\{1,\ldots,rd_x\}}\frac{\omega_\nu(\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4})}{\omega^2_{\alpha_1-\alpha_4}+\omega^2_\nu(\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4})}\frac{g_{m_{\alpha_1}m_{\alpha_4}\nu}(\mathbf{k}_{\alpha_4},\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4})g_{m_{\alpha_2}m_{\alpha_3}\nu}(\mathbf{k}_{\alpha_2}+\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4},-\mathbf{k}_{\alpha_1}+\mathbf{k}_{\alpha_4})}{\hbar^2}\delta_{k_{\alpha_3}+k_{\alpha_4},k_{\alpha_2}+k_{\alpha_1}}\delta_{\sigma_{\alpha_1},\sigma_{\alpha_4}}\delta_{\sigma_{\alpha_2},\sigma_{\alpha_3}}
\end{aligned} \tag{22.17}$$

Proof 22.23 See Appendix, Proofs, page 383.

Corollary 22.24 This[4] action is invariant under conjugation.

Proof 22.24 See Appendix, Proofs, page 386.

22.2.4. Expectation-Values as Source-Derivatives

Notation 22.25 We assume that we have a partition-function with ($Z[j_{\text{el}},\bar{j}_{\text{el}},j_{\text{ph}},\bar{j}_{\text{ph}}]$) and without ($Z$) electronic and phononic source-terms consider the ratio as generating functional

$$\frac{Z[j_{\text{el}},\bar{j}_{\text{el}},j_{\text{ph}},\bar{j}_{\text{ph}}]}{Z}\equiv\frac{Z[j,\bar{j}]}{Z} \tag{22.18}$$

whereas the external sources enter the partition function as in Notation 16.26. After obtaining a final expression for the partition function, we compute various quantities by taking functional derivatives.

Remark 22.25.1 Notice that the partition function can be simplified, by integrating out the phonons. Then one obtains Equation (22.14), which we rewrite here symbolically as

$$\begin{aligned}
Z[j,\bar{j}] &\equiv \int\mathscr{D}(\overline{\chi},\chi)\,\exp\left[-\left(S_{\text{el}}[\psi,\overline{\psi}]+S_{\text{ph}}[\phi,\overline{\phi}]+S_{\text{el-ph-int}}[\chi,\overline{\chi}]\right)\quad+\sum_{\bullet\in\{\text{el},\text{ph}\}}\bar{j}.\chi.+\overline{\chi}.j.\right]\\
&=\int\mathscr{D}(\overline{\psi},\psi)\,\exp-S_{[j,\bar{j}]}[\psi,\overline{\psi}]\\
&=\int\mathscr{D}(\overline{\psi},\psi)\,\exp-\left(S_{\text{el}}[\psi,\overline{\psi}]+\underset{[j_{\text{ph}},\bar{j}_{\text{ph}}]}{S_{\text{el-el-int}}}[\psi,\overline{\psi}]-\left(\bar{j}_{\text{el}}\psi+\overline{\psi}j_{\text{el}}\right)\right)\\
&=\int\mathscr{D}(\overline{\psi},\psi)\,\exp\left(\bar{j}_{\text{el}}\psi+\overline{\psi}j_{\text{el}}\right)\exp-\left(S_{\text{el}}+\underset{[j_{\text{ph}},\bar{j}_{\text{ph}}]}{S_{\text{el-el-int}}}\right)[\psi,\overline{\psi}]
\end{aligned} \tag{22.19}$$

[4] This is actually a much more general theorem that should hold IN GENERAL. This should be implemented somewhere in the text...

Corollary 22.26 Under the assumption, that the Matsubara-Green-function is diagonal w.r.t. the Matsubara-frequencies, we can re-express the following quantities as functional derivatives of the partition function:

Electronic Observables:

- The expectation-value of an occupation-number-operator w.r.t. the electronic state $(n, \sigma, \mathbf{k})$

$$\langle n_{n\sigma}(\mathbf{k})\rangle = \frac{1}{\hbar\beta} \sum_{\omega_m \in \Omega_{-1}} \mathrm{e}^{+\mathrm{i}\omega_m 0^+} \frac{\partial^2}{\partial \bar{j}_{n,\sigma}(\mathrm{i}\omega_m, \mathbf{k}) \partial j_{n,\sigma}(\mathrm{i}\omega_m, \mathbf{k})} \frac{Z[j_{\mathrm{el}}, \bar{j}_{\mathrm{el}}, j_{\mathrm{ph}}, \bar{j}_{\mathrm{ph}}]}{Z}\Big|_{\bar{j}=j=0}, \tag{22.20}$$

- the total particle number

$$\langle N\rangle = \frac{1}{\hbar\beta} \sum_{\substack{\sigma\in\{\downarrow,\uparrow\} \\ k\in \mathrm{i}\Omega_{-1}\times Q \\ n\in\{1,2,\ldots,\infty\}}} \mathrm{e}^{+\mathrm{i}\omega_k 0^+} \frac{\partial^2}{\partial \bar{j}_{n,\sigma}(k) \partial j_{n,\sigma}(k)} \frac{Z[j_{\mathrm{el}}, \bar{j}_{\mathrm{el}}, j_{\mathrm{ph}}, \bar{j}_{\mathrm{ph}}]}{Z}\Big|_{\bar{j}=j=0}, \tag{22.21}$$

- the electronic contribution to the total energy we have

$$\langle H_{\mathrm{el}}\rangle = \frac{1}{\hbar\beta} \sum_{\substack{\sigma\in\{\downarrow,\uparrow\} \\ k\in \mathrm{i}\Omega_{-1}\times Q \\ n\in\{1,2,\ldots,\infty\}}} \mathrm{e}^{+\mathrm{i}\omega_k 0^+} \epsilon_{n\sigma}(\mathbf{k}) \frac{\partial^2}{\partial \bar{j}_{n,\sigma}(k) \partial j_{n,\sigma}(k)} \frac{Z[j_{\mathrm{el}}, \bar{j}_{\mathrm{el}}, j_{\mathrm{ph}}, \bar{j}_{\mathrm{ph}}]}{Z}\Big|_{\bar{j}=j=0}, \tag{22.22}$$

Phononic Observables:

- the phononic occupation-number-operator

$$\langle n_\nu(\mathbf{q})\rangle = \frac{1}{\hbar\beta} \sum_{\omega_m \in \Omega_{+1}} \mathrm{e}^{+\mathrm{i}\omega_m 0^+} \frac{\partial^2}{\partial \bar{j}_{\nu}(\mathrm{i}\omega_m, \mathbf{q}) \partial j_{\nu}(\mathrm{i}\omega_m, \mathbf{q})} \frac{Z[j_{\mathrm{el}}, \bar{j}_{\mathrm{el}}, j_{\mathrm{ph}}, \bar{j}_{\mathrm{ph}}]}{Z}\Big|_{\bar{j}=j=0},$$

- the phononic contribution to the total energy we have

$$\left\langle H_{\mathrm{ph}}\right\rangle = \frac{1}{\hbar\beta} \sum_{\substack{q\in \mathrm{i}\Omega_{+1}\times Q \\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\}}} \mathrm{e}^{+\mathrm{i}\omega_q 0^+} \Omega_\nu(\mathbf{q}) \frac{\partial^2}{\partial \bar{j}_{\nu}(q) \partial j_{\nu}(q)} \frac{Z[j_{\mathrm{el}}, \bar{j}_{\mathrm{el}}, j_{\mathrm{ph}}, \bar{j}_{\mathrm{ph}}]}{Z}\Big|_{\bar{j}=j=0}. \tag{22.23}$$

Interaction Observables:

- the electron-phonon-interaction energy is given by

$$\left\langle c^\dagger_{\mathbf{k}+\mathbf{q}m\sigma} c_{\mathbf{k}n\sigma}\left(b_{\mathbf{q}\nu} + b^\dagger_{-\mathbf{q}\nu}\right)\right\rangle = \frac{1}{\sqrt{\hbar\beta}^3} \sum_{\substack{\omega_m\in\Omega_{+1} \\ \omega_n,\omega_{n'}\in\Omega_{-1}}} \mathrm{e}^{\mathrm{i}\omega_n 0^+} \frac{\partial^2}{\partial \bar{j}_{n\sigma\mathbf{k}}(\mathrm{i}\omega_n) \partial j_{m\sigma\mathbf{k}+\mathbf{q}}(\mathrm{i}\omega_{n'})} \left(\frac{\partial}{\partial \bar{j}_\nu(\mathrm{i}\omega_m, \mathbf{q})} + \frac{\partial}{\partial j_\nu(-\mathrm{i}\omega_m, -\mathbf{q})}\right) \frac{Z[j_{\mathrm{el}}, \bar{j}_{\mathrm{el}}, j_{\mathrm{ph}}, \bar{j}_{\mathrm{ph}}]}{Z}\Big|_{\bar{j}=j=0}. \tag{22.24}$$

Proof 22.26 See Appendix, Proofs, page 387.

Remark 22.26.1 Equations (22.20), (22.20) and (22.22) are merely listed for completeness. The Green-function-techniques developed in the earlier chapters allow us a much more elegant computation of the internal energy, as will become clear in Theorem 24.19.

23. Density of States, Thermodynamic Potential & Conjugated Variables

<u>**Motivation and Overview**</u>

After this part of the notes you will understand how to calculate for arbitrary materials and for arbitrary temperatures the Band-Structure and [D]ensity [O]f [S]tates (DOS) ...

- ...for a typical non-interacting model, resulting in a plot as in Figure 23.1.
- ...for a typical interacting model, resulting in a plot as in Figure [TBCreated].

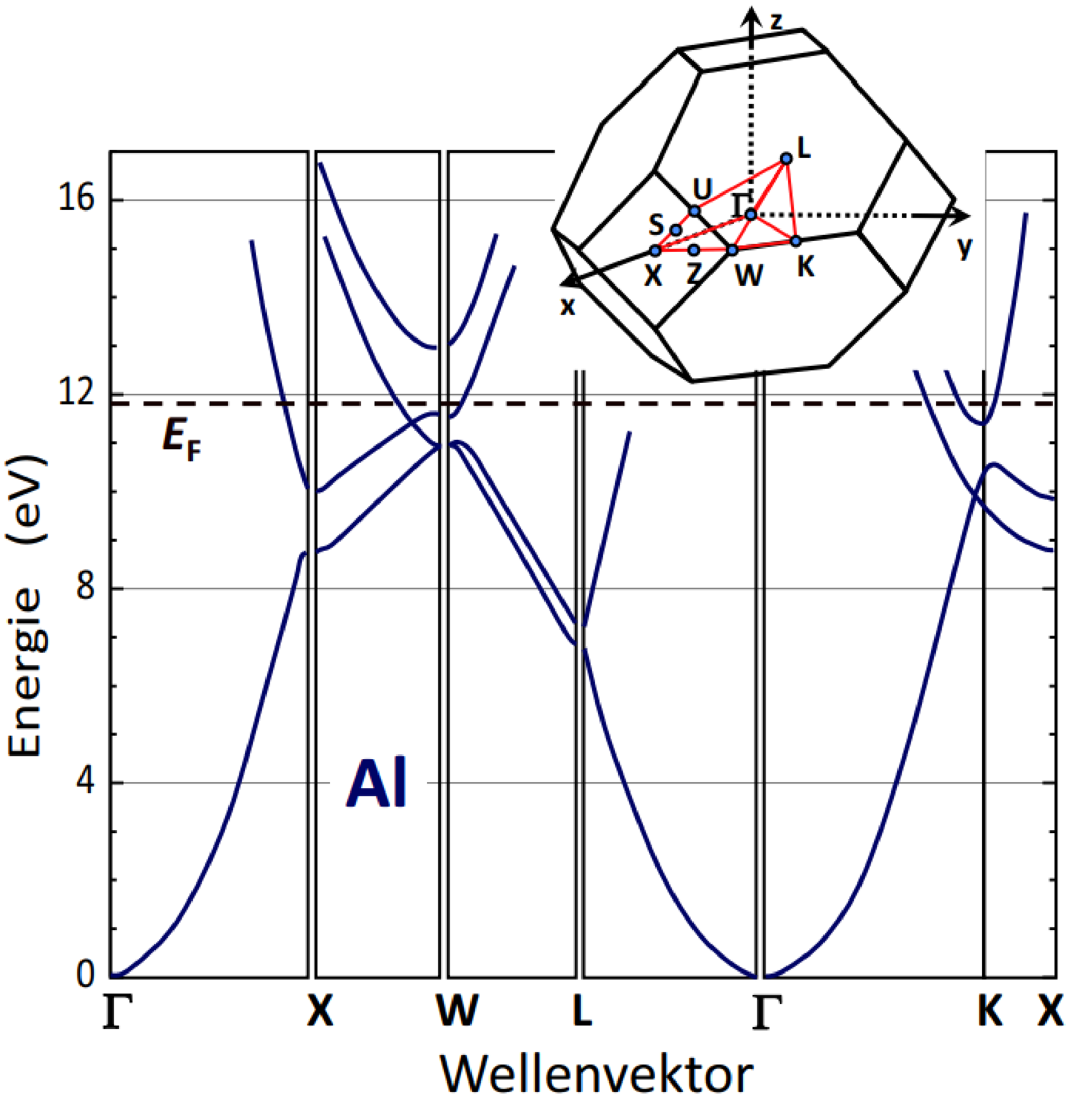


Figure 23.1.: The Band-Structure of Aluminium along the high-symmetry **k**-points. Taken from [GM].

23.1. Non-Interacting Gas-Models

23.1.1. Electrons

Electronic Density of States

Notation 23.1 Within usual approximations, we describe the non-interacting electron gas as

$$H_{\mathrm{el}} = \sum_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\ldots,\infty\}}} \epsilon_{m\mathbf{k}\sigma} c^{\dagger}_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma} \tag{23.1}$$

Fact 23.2 Recall[1] that using Equation (14.9) we obtain the electronic density of states $\rho_0(E)$ for the system as

$$\rho_0(E) = \sum_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\ldots,\infty\}}} \delta(E - (\epsilon_{m\mathbf{k}\sigma} - \mu)) \tag{23.2}$$

Proof 23.2 Follows immediately from the Sokhotski-Plemelj-Theorem in Equation (14.3).

Remark 23.2.1 Notice that this density of states is formally just a sum of delta-peaks. However, when physicists depict the density of states, they recall that this model is just an approximation to a more realistic model which includes interactions. Since interactions usually imply the presence of smearing, one usually "artificially" introduces smearing in the non-interacting density of states by replacing the delta-peaks with Lorentzians or Gaussians.

Remark 23.2.2 Regularizing the delta-functions with Gaussians converges much faster (in momenta) than regularizing the delta-functions with Lorentzians.

Notation 23.3 Notice that this quantity is (after introducing cutoffs for the number of bands) normalized to

$$\int \mathrm{d}E \rho_0(E) = N_x N_y N_z \cdot N_{\mathrm{band}} \cdot 2 \tag{23.3}$$

whereas the density of states, as computed in [Q]uantum [E]spresso (QE) within the [L]ocal [D]ensity [A]pproximation (LDA), $\rho_{\mathrm{QE}}(E)$ is normalized to the number of bands, i.e.

$$\frac{1}{N_x N_y N_z}\rho_0(E-\mu) = \frac{1}{N_x N_y N_z} \sum_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\downarrow,\uparrow\}\\ n\in\{1,2,\ldots\}}} \delta(E-\epsilon_{n\mathbf{k}}) \mapsto \frac{1}{N_x N_y N_z} \sum_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\downarrow,\uparrow\}\\ n\in\{1,2,\ldots,N_{\mathrm{band}}\}}} \delta(E-\epsilon_{n\mathbf{k}}) \equiv \rho_{\mathrm{QE}}(E) \tag{23.4}$$

[1] Cf. Lemma 14.15.

such that it is normalized to

$$\int \mathrm{d}E \rho_{\mathrm{QE}}(E) = 2N_{\mathrm{band}} \tag{23.5}$$

Electronic Thermodynamic Potential - Landau Potential

Lemma 23.4 The grand canonical potential $\Omega_{0,\mathrm{el}}(\mu, T, V)$ for the electronic system is given as

$$\Omega_{0,\mathrm{el}}(\mu, T, V) = \frac{-1}{\beta} \sum_{\substack{\mathbf{k}\in \mathrm{Q} \\ \sigma\in\{\uparrow,\downarrow\} \\ m\in\{1,\ldots,\infty\}}} \mathrm{Log}\left(1 + \mathrm{e}^{-\beta(\epsilon_{m\mathbf{k}\sigma}-\mu)}\right) = \frac{-1}{\beta}\int \mathrm{d}E\,\mathrm{Log}\left(1+\mathrm{e}^{-\beta E}\right)\rho_0(E) \tag{23.6}$$

Proof 23.4 We use Lemma 19.16 and obtain the first equality. For the second equation, we notice that the expression can be rewritten with respect to the density of states as

$$\begin{aligned}
\Omega_{0,\mathrm{el}}(\mu, T, V) &= \frac{-1}{\beta} \sum_{\substack{\mathbf{k}\in \mathrm{Q} \\ \sigma\in\{\uparrow,\downarrow\} \\ m\in\{1,\ldots,\infty\}}} \mathrm{Log}\left(1 + \mathrm{e}^{-\beta(\epsilon_{m\mathbf{k}\sigma}-\mu)}\right) \\
&= \frac{-1}{\beta}\int \mathrm{d}E \sum_{\substack{\mathbf{k}\in \mathrm{Q} \\ \sigma\in\{\uparrow,\downarrow\} \\ m\in\{1,\ldots,\infty\}}} \mathrm{Log}\left(1 + \mathrm{e}^{-\beta(E-\mu)}\right)\delta(E-\epsilon_{m\mathbf{k}\sigma}) \\
&= \frac{-1}{\beta}\int \mathrm{d}E\,\mathrm{Log}\left(1+\mathrm{e}^{-\beta(E-\mu)}\right)\rho_0(E-\mu) \\
&= \frac{-1}{\beta}\int \mathrm{d}E\,\mathrm{Log}\left(1+\mathrm{e}^{-\beta E}\right)\rho_0(E)
\end{aligned}$$

Electronic Thermodynamic Conjugated Variable - Chemical Potential

Fact 23.5 The chemical potential has to be computed by constraining the number of particles according to Lemma 14.14 as

$$\langle N\rangle_0 = \sum_{\substack{\mathbf{k}\in \mathrm{Q} \\ \sigma\in\{\uparrow,\downarrow\} \\ m\in\{1,\ldots,\infty\}}} \mathrm{F}_\beta(\epsilon_{m\mathbf{k}\sigma}-\mu) = \int_{-\infty}^{+\infty}\mathrm{d}E\,\mathrm{F}_\beta(E)\,\rho_0(E) \iff \frac{\langle N\rangle_0}{N_xN_yN_z} = \int_{-\infty}^{+\infty}\mathrm{d}E\,\mathrm{F}_\beta(E-\mu)\,\rho_{\mathrm{QE,\,LDA}}(E) \tag{23.7}$$

Remark 23.5.1 Notice that we implicitly assumed that the number of bands is large enough to accommodate all electrons in the system.

Fact 23.6 We first notice that dictating the particle number (for example by requiring [C]harge [N]eutrality of the crystal) through Equation (23.7) can be rewritten as

$$\begin{aligned} F_0(\mu, T) \equiv \quad & \langle N \rangle_0(\mu, T) & - N_{\mathrm{CN}} \\ = & \left[\int_{-\infty}^{+\infty} \mathrm{d}E \, \mathrm{F}_\beta(E) \, \rho_0(E) \right] & - N_{\mathrm{CN}} \\ = & \left[\int_{-\infty}^{+\infty} \mathrm{d}E \, \mathrm{F}_\beta(E - \mu) \, \rho_{\mathrm{QE,\,LDA}}(E) \right] N_x N_y N_z & - N_{\mathrm{CN}} = 0. \end{aligned} \tag{23.8}$$

Remark 23.6.1 Notice that the chemical potential that is defined through this Equation is unique due to Theorem 9.29.

Remark 23.6.2 Notice further that the partial derivatives of the function F_0 are given as the partial derivatives of the expected particle number $\langle N \rangle_0$, i.e.

$$\left(\partial_T F_0\right)(\mu, T) = \partial_T \langle N \rangle_0(\mu, T), \qquad \left(\partial_\mu F_0\right)(\mu, T) = \partial_\mu \langle N \rangle_0(\mu, T). \tag{23.9}$$

Lemma 23.7 The temperature-derivative of the function F_0 can be computed with Lemma 9.44 and Equation (23.4) as

$$\begin{aligned} \left(\partial_T F_0\right)(\mu, T) = \partial_T \langle N \rangle_0(\mu, T) = & \quad \frac{\beta}{4T} \sum_{\substack{\mathbf{k} \in \mathrm{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \ldots, \infty\}}} \operatorname{sech}^2\left(\frac{\beta \xi_{m\mathbf{k}\sigma}}{2}\right) \quad \xi_{n\mathbf{k}\sigma} \\ = & \quad \frac{\beta}{4T} \int_{-\infty}^{+\infty} \mathrm{d}E \quad \operatorname{sech}^2\left(\frac{\beta E}{2}\right) \quad E \quad \rho_0(E) \\ = N_x N_y N_z & \quad \frac{\beta}{4T} \int_{-\infty}^{+\infty} \mathrm{d}E \quad \operatorname{sech}^2\left(\beta \frac{E - \mu}{2}\right) \quad (E - \mu) \quad \rho_{\mathrm{QE}}(E) \end{aligned} \tag{23.10}$$

Proof 23.7 TBD.

Remark 23.7.1 Notice the identity that we obtain through Lemma (TBReferenced)

$$\frac{\beta}{T} \int_{-\infty}^{+\infty} \mathrm{d}E \, \mathrm{F}_\beta(E) \left(1 - \mathrm{F}_\beta(E)\right) \rho_0(E) E = \frac{\beta}{4T} \sum_{\substack{\mathbf{k} \in \mathrm{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \ldots, \infty\}}} \xi_{n\mathbf{k}\sigma} \operatorname{sech}^2\left(\frac{\beta \xi_{m\mathbf{k}\sigma}}{2}\right) \tag{23.11}$$

Lemma 23.8 The chemical-potential-derivative of the function F can be computed as ...

$$
\begin{aligned}
\left(\partial_\mu F_0\right)(\mu,T) = \partial_\mu \langle N\rangle_0(\mu,T) &= \frac{\beta}{4} \sum_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\dots,\infty\}}} \operatorname{sech}^2\left(\frac{\xi_{m\mathbf{k}\sigma}}{2k_BT}\right)\\
&= \frac{\beta}{4}\int_{-\infty}^{+\infty} \mathrm{d}E \quad \operatorname{sech}^2\left(\frac{\beta E}{2}\right)\rho_0(E)\\
&= N_xN_yN_z\ \frac{\beta}{4}\int_{-\infty}^{+\infty} \mathrm{d}E \quad \operatorname{sech}^2\left(\beta\frac{E-\mu}{2}\right)\rho_{\mathrm{QE}}(E)
\end{aligned}
\tag{23.12}
$$

Proof 23.8 TBDigitalized 1 / TBChecked. We compute the derivative of the Fermi-function w.r.t. the chemical potential as

$$
\mathrm{F}_\beta(\epsilon_{n\mathbf{k}\sigma}-\mu) \equiv \mathrm{n}^{-1}(\beta(\epsilon_{n\mathbf{k}\sigma}-\mu)) \tag{23.13}
$$

which implies with Lemma 9.44 that

$$
\partial_\mu\, \mathrm{n}^{-1}(\beta(\epsilon_{n\mathbf{k}\sigma}-\mu)) = -\frac{1}{4}\operatorname{sech}^2\left(\beta\xi_{n\mathbf{k}\sigma}/2\right)\cdot(-\beta) = \frac{\beta}{4}\operatorname{sech}^2\left(\beta\xi_{n\mathbf{k}\sigma}/2\right) \tag{23.14}
$$

Corollary 23.9 Assuming now we have access[2] to the values $\mu(T)$ that fulfill Equation (23.8), we can compute the temperature derivative of the chemical potential as

$$
\left(\frac{\partial\mu}{\partial T}\right)(T) = -\left(\frac{\partial_T F_0}{\partial_\mu F_0}\right)(\mu(T),T) = -\left(\frac{\partial_T\langle N\rangle_0}{\partial_\mu\langle N\rangle_0}\right)(\mu(T),T)\,. \tag{23.15}
$$

Proof 23.9 Follows from the Theorem of implicit functions with Equation (23.8).

Remark 23.9.1 Notice that this equation can be rewritten in the physics-notation as

$$
\frac{\partial\mu}{\partial T} = -\frac{\partial N}{\partial T}\frac{\partial\mu}{\partial N} \quad\Longleftrightarrow\quad \frac{\partial\mu}{\partial T}\frac{\partial T}{\partial N}\frac{\partial N}{\partial\mu} = -1 \tag{23.16}
$$

Corollary 23.10 Notice that this implies for Equation (23.15) in the non-interacting case the formula

$$
\left(\frac{\partial\mu}{\partial T}\right)(T) = -\frac{1}{T}\frac{\int_{-\infty}^{+\infty}\mathrm{d}E\operatorname{sech}^2\left(\beta\frac{E-\mu(T)}{2}\right)\rho_{\mathrm{QE}}(E)\left[E-\mu(T)\right]}{\int_{-\infty}^{+\infty}\mathrm{d}E\operatorname{sech}^2\left(\beta\frac{E-\mu(T)}{2}\right)\rho_{\mathrm{QE}}(E)} \tag{23.17}
$$

Proof 23.10 Follows immediately from Lemmas 24.22 and 23.8.

[2]Notice that this is the case, as we compute the chemical potential numerically.

Electronic Thermodynamic Conjugated Variable - Pressure

Fact 23.11 Notice that one can compute the pressure $P \equiv P(\mu, T, V)$ as

$$-\frac{\Omega_0(\mu, T, V + \Delta V) - \Omega_0(\mu, T, V)}{\Delta V} \tag{23.18}$$

with $V \equiv N_p V_{\text{UC}}$, $N_p \equiv N_x N_y N_z$ and unit-cell volume

$$V_{\text{UC}} = \int_{\text{span}_{[0,1]}\{\mathbf{a}_1,\mathbf{a}_2,\mathbf{a}_3\}} 1\mathrm{d}^3x = \det(\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3) \tag{23.19}$$

Electronic Thermodynamic Conjugated Variable - Entropy

TBD

23.1.2. Phonons

Phononic Density of States

Notation 23.12 Within usual approximations, we describe the non-interacting phonon gas as

$$H_{\text{ph}} = \sum_{\substack{\mathbf{q}\in Q \\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\}}} \Omega_\nu(\mathbf{q}) b^\dagger_{\mathbf{q}\nu} b_{\mathbf{q}\nu} \tag{23.20}$$

Definition 23.13 We introduce the *phononic density of states* ρ_{ph} as

$$\rho_{\text{ph}}(E) = \frac{1}{N_x N_y N_z} \sum_{\substack{\mathbf{q}\in Q \\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\}}} \delta(E - \Omega_\nu(\mathbf{q})) = \frac{1}{N_x N_y N_z} \frac{-1}{\pi} \sum_{\substack{\mathbf{q}\in Q \\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\}}} \lim_{\eta\to 0^+} \operatorname{Im} \frac{1}{E + \mathrm{i}\eta - \Omega_\nu(\mathbf{q})} \tag{23.21}$$

Phononic Thermodynamic Potential - Helmholtz Free Energy

TBD

Phononic Thermodynamic Conjugated Variable - Chemical Potential

This is not zero! TBElaborated.

Phononic Thermodynamic Conjugated Variable - Pressure

TBD

Phononic Thermodynamic Conjugated Variable - Entropy

TBD

23.1.3. Total Crystal

Total Crystal Thermodynamic Potential

Fact 23.14 The grand canonical potential $\Omega_0(\mu, T, V)$ is given as

$$\Omega_0(\mu, T, V) = \Omega_{0,\mathrm{el}}(\mu, T, V) \qquad + \Omega_{0,\mathrm{ph}}(\mu, T, V) \tag{23.22}$$
$$= \frac{-1}{\beta} \sum_{\substack{\mathbf{k}\in Q \\ \sigma\in\{\uparrow,\downarrow\} \\ m\in\{1,\ldots,\infty\}}} \mathrm{Log}\left(1 + \mathrm{e}^{-\beta(\epsilon_{m\sigma}(\mathbf{k})-\mu)}\right) + \frac{1}{\beta} \sum_{\substack{\mathbf{q}\in Q \\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}}} \mathrm{Log}\left(1 - \mathrm{e}^{-\beta\Omega_\nu(\mathbf{q})}\right) \tag{23.23}$$

23.2. Interacting Gas-Model

TBDone

23.2.1. Electrons

Fact 23.15 We compute the chemical potential, by constraining the number of particles according to Lemma 14.14 as

$$\langle N\rangle = \sum_{\substack{\mathbf{k}\in Q \\ \sigma\in\{\uparrow,\downarrow\} \\ m\in\{1,\ldots,\infty\}}} \langle n_{m\mathbf{k}\sigma}\rangle = \frac{-1}{\pi} \lim_{\delta\to 0} \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in Q \\ \sigma\in\{\uparrow,\downarrow\} \\ m\in\{1,\ldots,\infty\} \\ E\in(-\infty,+\infty)}} \mathrm{F}_\beta(E)\, \mathrm{Im}\, G^{\mathrm{ret}}_{\substack{m,m \\ \sigma,\sigma \\ \mathbf{k},\mathbf{k}}}(E + \mathrm{i}\delta) \tag{23.24}$$

TBDone

23.2.2. Phonons

TBDone

23.2.3. Total Crystal

Theorem 23.16 For the grand canonical potential of the interacting system we have in the lowest order of perturbation theory

$$\text{TBDigitalized from Pinboard, Topic 17} \tag{23.25}$$

Proof 23.16 See Appendix, Proofs, page 402.

Remark 23.16.1 Notice further, that one can compute the pressure $P \equiv P(\mu, T, V)$ as

$$-\frac{\Omega(\mu, T, V + \Delta V) - \Omega(\mu, T, V)}{\Delta V} \tag{23.26}$$

with $V \equiv N_p V_{\mathrm{UC}}$, $N_p \equiv N_x N_y N_z$ and unit-cell volume

$$V_{\mathrm{UC}} = \int_{\mathrm{span}_{[0,1]}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}} 1 \mathrm{d}^3 x = \det(\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3) \tag{23.27}$$

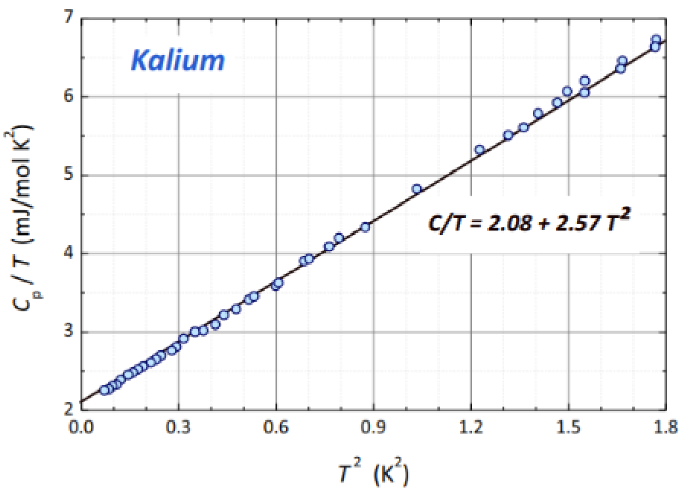
24. Internal Energy & Specific Heat (Normal State)

Motivation and Overview

After this part of the notes you will understand ...

- how you calculate the specific heat for arbitrary materials in a typical non-interacting model at low temperatures as in Figures 24.1 and 24.2.

$$C_p = \gamma \cdot T + A \cdot T^3 \,.$$



Metall	γ_{exp} $(10^{-3}$ J/mol K)	γ_{theor} $(10^{-3}$ J/mol K)	$\gamma_{\mathrm{exp}}/\gamma_{\mathrm{theor}}$
Li	1.63	0.749	2.18
Na	1.38	1.094	1.26
K	2.08	1.668	1.25
Rb	2.41	1.911	1.26
Cs	3.20	2.238	1.43
Fe	4.98	0.498	10
Co	4.98	0.483	10.3
Ni	7.02	0.458	15.3
Cu	0.695	0.505	1.38
Ag	0.646	0.645	1.00
Au	0.729	0.642	1.14
Sn	1.78	1.41	1.26
Pb	2.98	1.509	1.97

Figure 24.1.: Specific heat of various materials at low temperatures.

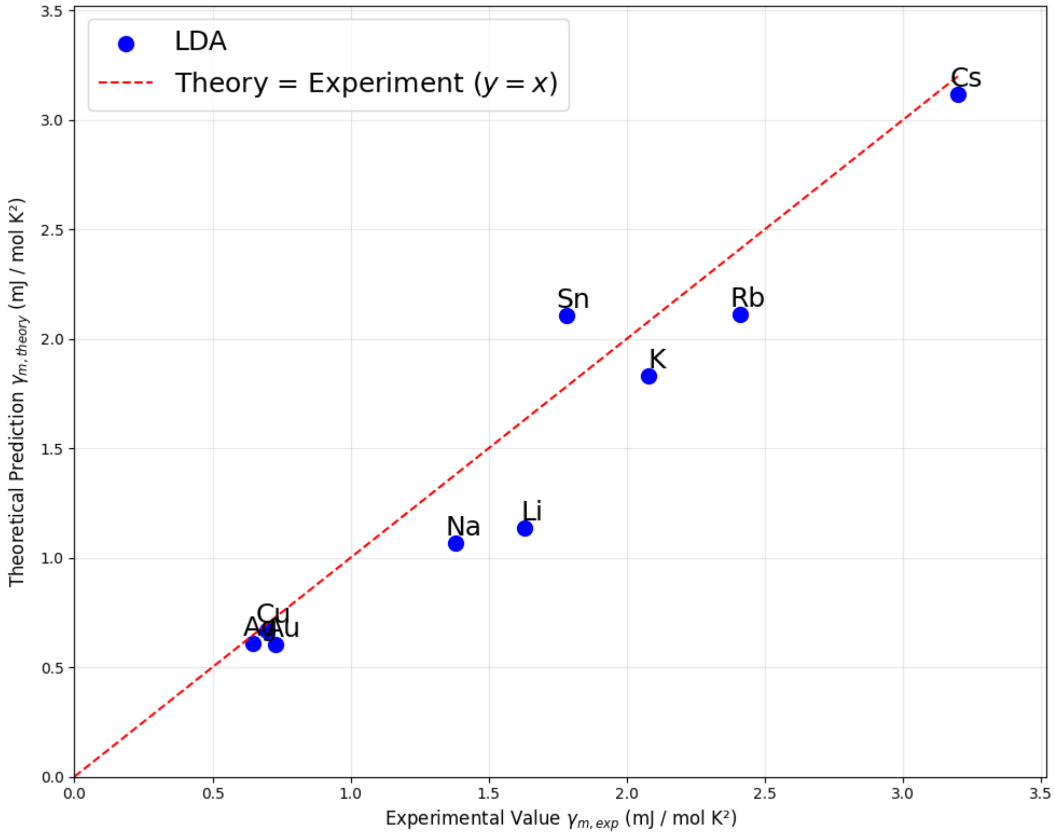


Figure 24.2.: Specific heat of various materials contrasting LDA and Experimental Results.

24.1. Non-Interacting Gas-Models

24.1.1. Electrons

Theorem 24.1 The expectation-value of the electronic internal energy can be computed as

$$\langle H_{\mathrm{el}} \rangle_0 = \sum_{\substack{\mathbf{k} \in \mathrm{Q} \\ \sigma \in \{\uparrow,\downarrow\} \\ m \in \{1,\dots,\infty\}}} \epsilon_{m\mathbf{k}\sigma} \left\langle c^\dagger_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma} \right\rangle_0 = \sum_{\substack{\mathbf{k} \in \mathrm{Q} \\ \sigma \in \{\uparrow,\downarrow\} \\ m \in \{1,\dots,\infty\}}} \epsilon_{m\mathbf{k}\sigma} \, \mathrm{F}_\beta(\epsilon_{m\mathbf{k}\sigma} - \mu) \tag{24.1}$$

Corollary 24.2 Notice, that in the non-interacting case, this formula simplifies to

$$\langle H_{\mathrm{el}} \rangle_0 = \int \mathrm{d}E \; E \; \mathrm{F}_\beta(E) \rho_0(E) \quad \Longleftrightarrow \quad \frac{\langle H_{\mathrm{el}} \rangle_0}{N_x N_y N_z} = \int \mathrm{d}E \, (E - \mu) \; \mathrm{F}_\beta(E - \mu) \rho_{\mathrm{QE}}(E) \tag{24.2}$$

Proof 24.2 Follows with the Sokhotski-Plemelj theorem from Equation (24.14).

Lemma 24.3 In the absence of interactions one obtains the simple formula

$$\begin{aligned} \partial_T \frac{\langle H_{\mathrm{el}} \rangle_0}{N_x N_y N_z} &= \frac{1}{N_x N_y N_z} \sum_{\substack{\mathbf{k} \in \mathrm{Q} \\ \sigma \in \{\uparrow,\downarrow\} \\ m \in \{1,\dots,\infty\}}} \frac{\epsilon_{m\mathbf{k}\sigma}}{4} \; \mathrm{sech}^2\left(\frac{\epsilon_{m\mathbf{k}\sigma} - \mu(T)}{2k_B T}\right) \left[\frac{\epsilon_{m\mathbf{k}\sigma} - \mu(T)}{k_B T^2}\right] \\ &= \int \mathrm{d}E \; \frac{E}{4} \; \mathrm{sech}^2\left(\frac{E - \mu(T)}{2k_B T}\right) \left[\frac{E - \mu(T)}{k_B T^2}\right] \rho_{\mathrm{QE}}(E) \end{aligned} \tag{24.3}$$

Proof 24.3 See Appendix, Proofs, page 379.

Lemma 24.4 In the absence of interactions one obtains the simple formula

$$\begin{aligned} \partial_\mu \frac{\langle H_{\mathrm{el}} \rangle_0}{N_x N_y N_z} &= \frac{1}{N_x N_y N_z} \frac{\beta}{4} \sum_{\substack{\mathbf{k} \in \mathrm{Q} \\ \sigma \in \{\uparrow,\downarrow\} \\ m \in \{1,\dots,\infty\}}} \epsilon_{m\mathbf{k}\sigma} \; \mathrm{sech}^2\left(\frac{\epsilon_{m\mathbf{k}\sigma} - \mu}{2k_B T}\right) \\ &= \frac{\beta}{4} \int \mathrm{d}E \; E \; \mathrm{sech}^2\left(\frac{E - \mu}{2k_B T}\right) \rho_{\mathrm{QE}}(E) \end{aligned} \tag{24.4}$$

Proof 24.4 See Appendix, Proofs, page 380.

24.1.2. Phonons

Notation 24.5 Within usual approximations, we describe the non-interacting phonon gas as

$$H_{\text{ph}} = \sum_{\substack{\mathbf{q}\in \mathrm{Q} \\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \Omega_\nu(\mathbf{q}) b^\dagger_{\mathbf{q}\nu} b_{\mathbf{q}\nu} \tag{24.5}$$

Fact 24.6 The expectation-value of the phononic internal energy can be computed as

$$\langle H_{\text{ph}}\rangle_0 = \sum_{\substack{\mathbf{q}\in \mathrm{Q} \\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \Omega_\nu(\mathbf{q})\, \mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) \tag{24.6}$$

Lemma 24.7 The temperature-derivatives of the non-interacting phononic contribution to the internal energy can be computed as

$$\partial_T \langle H_{\text{ph}}\rangle_0 = \frac{\beta}{4T} \sum_{\substack{\mathbf{q}\in \mathrm{Q} \\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \Big[\Omega_\nu(\mathbf{q})\,\mathrm{csch}(\beta\Omega_\nu(\mathbf{q})/2)\Big]^2 \tag{24.7}$$

Proof 24.7 See Appendix, Proofs, page 401.

Remark 24.7.1 Notice that this allows us to rewrite the temperature-derivative of the phononic internal energy as

$$\begin{aligned}\partial_T \langle H_{\text{ph}}\rangle_0 &= \frac{\beta}{4T}\int \mathrm{d}E\,\Big[E\,\mathrm{csch}(\beta E/2)\Big]^2 \rho_{\text{ph}}(E) \\ &= \frac{\beta}{4T}\lim_{\eta\to 0^+}\int \mathrm{d}E\,\Big[E\,\mathrm{csch}(\beta E/2)\Big]^2 \left(-\frac{1}{\pi}\right) \sum_{\substack{\mathbf{q}\in \mathrm{Q} \\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \mathrm{Im}\,\frac{1}{E+\mathrm{i}\eta-\Omega_\nu(\mathbf{q})}\end{aligned} \tag{24.8}$$

24.1.3. Total Crystal

Notation 24.8 Within usual approximations, we describe the union of two non-interacting gases

$$H = H_{\text{el}} + H_{\text{ph}} = \sum_{\substack{\mathbf{k}\in \mathrm{Q} \\ \sigma\in\{\uparrow,\downarrow\} \\ m\in\{1,\dots,\infty\}}} \epsilon_{m\mathbf{k}\sigma} c^\dagger_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma} + \sum_{\substack{\mathbf{q}\in \mathrm{Q} \\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} \Omega_{\mathbf{q},\nu} b^\dagger_{\mathbf{q}\nu} b_{\mathbf{q}\nu} \tag{24.9}$$

Fact 24.9 The expectation-value of the total internal energy can be computed as

$$\begin{aligned}\langle H\rangle_0 = &\quad \langle H_{\text{el}}\rangle_0 &&+ \quad \langle H_{\text{ph}}\rangle_0 \\ = &\sum_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\dots,\infty\}}} \epsilon_{m\mathbf{k}\sigma}\,\mathrm{F}_\beta(\epsilon_{m\mathbf{k}\sigma}-\mu) &&+ \sum_{\substack{\mathbf{q}\in Q\\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} \Omega_\nu(\mathbf{q})\,\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))\end{aligned} \tag{24.10}$$

Remark 24.9.1 Notice that these identities can be further simplified by using the density of states for the electronic and / or the phononic system.

Fact 24.10 The basic formula from is according to Equation (9.56)

$$\begin{aligned}c_{N,V}(T) = \frac{C_{N,V}(T)}{n} &= \frac{1}{n}\partial_T\,\langle H\rangle_{0,N,T,V}\\ &= \frac{1}{n}\Big[\big(\partial_T\langle H\rangle_0\big)(\mu,T,V) + \big(\partial_\mu\langle H\rangle_0\big)(\mu,T,V)\cdot\big(\partial_T\mu\big)(N(\mu),V,T)\Big]\end{aligned} \tag{24.11}$$

The Hamiltonian is made up of two contributions

$$H_{\text{tot}} = H_{\text{el}} + H_{\text{ph}} \tag{24.12}$$

Therefore, also for the specific heat, we have contributions from the electronic part and from the phononic part. We further need to account for the temperature derivative of the chemical potential. We consider all three quantities separately and then merge the results.

24.2. Interacting Gas-Model

Notation 24.11 We consider the interacting system that is described by the Hamiltonian from Definition 8.29 as

$$H = \sum_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\dots,\infty\}}} \epsilon_{m\mathbf{k}\sigma}c^\dagger_{m\mathbf{k}\sigma}c_{m\mathbf{k}\sigma} + \sum_{\substack{\mathbf{q}\in Q\\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} \hbar\omega_{\mathbf{q},\nu}b^\dagger_{\mathbf{q}\nu}b_{\mathbf{q}\nu} + \sum_{\substack{\mathbf{k},\mathbf{q}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}\\ m,n\in\{1,\dots,\infty\}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})}{\sqrt{N_p}}c^\dagger_{\mathbf{k}+\mathbf{q}m\sigma}c_{\mathbf{k}n\sigma}\left(b_{\mathbf{q}\nu}+b^\dagger_{-\mathbf{q}\nu}\right) \tag{24.13}$$

24.2.1. Electrons

Theorem 24.12 One finds the identity

$$\langle H_{\text{el}}\rangle = \sum_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\dots,\infty\}}} \epsilon_{m\mathbf{k}\sigma}\langle n_{m\mathbf{k}\sigma}\rangle = \frac{-1}{\pi}\lim_{\delta\to 0}\sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\dots,\infty\}\\ E\in(-\infty,+\infty)}} \epsilon_{m\mathbf{k}\sigma}\,\mathrm{F}_\beta(E)\,\mathrm{Im}\,G^{\text{ret}}_{\substack{m,m\\ \sigma,\sigma}}(E+\mathrm{i}\delta,\mathbf{k}) \tag{24.14}$$

Proof 24.12 See Appendix, Proofs, page 389.

The Electronic Contribution

Theorem 24.13 One finds for the interacting system two equivalent identities

$$\frac{\partial^2}{\partial \bar{j}_{n,\sigma}(k)\partial j_{n,\sigma}(k)} \frac{Z[j_{\rm el}, \bar{j}_{\rm el}, j_{\rm ph}, \bar{j}_{\rm ph}]}{Z}\Big|_{\bar{j}=j=0} = +\hbar \mathscr{G}_{\substack{n,n\\ \sigma,\sigma}}(k,k) \tag{24.15}$$

$$\begin{aligned}\frac{\partial^2}{\partial \bar{j}_{n,\sigma}(k)\partial j_{n,\sigma}(k)} \frac{Z[j_{\rm el}, \bar{j}_{\rm el}, j_{\rm ph}, \bar{j}_{\rm ph}]}{Z}\Big|_{\bar{j}=j=0} &= +\hbar \mathscr{G}_{\substack{0\\ n,n\\ \sigma,\sigma}}(k,k) \\ &- \frac{\hbar}{\beta N_p} \sum_{\substack{k\in \mathrm{i}\Omega_{-1}\times \mathrm{Q}\\ n'\in\{1,\dots,\infty\}\\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} |g_{nn'\nu}(\mathbf{k},\mathbf{k}'-\mathbf{k})|^2 \mathscr{G}^2_{0,n,\sigma}(k)\mathscr{G}_{0,n',\sigma}(k')\mathscr{D}_{0,\nu}(k'-k) \\ &+ \mathcal{O}(g^4)\end{aligned} \tag{24.16}$$

Proof 24.13 See Appendix, Proofs, page 388.

Remark 24.13.1 We only mention the second identity for completeness; In practice, it is difficult to converge this expression. The second identity can be simplified further by evaluating the occuring Matsuara-sums (cf. B-sides). The first identity is the one that is used in practice, since it can be simplified with the fluctuation-dissipation-theorem.

24.2.2. Phonons

The Phononic Contribution

We first focus on the term in Equation (22.23).

Theorem 24.14 One finds for the interacting system the identity

$$\begin{aligned}\frac{\partial^2}{\partial \bar{j}_{\nu}(q)\partial j_{\nu}(q)} \frac{Z[j_{\rm el}, \bar{j}_{\rm el}, j_{\rm ph}, \bar{j}_{\rm ph}]}{Z}\Big|_{\bar{j}=j=0} &\approx -\hbar \mathscr{G}_{0,\nu}(q) \\ &+ \frac{\hbar}{\beta N_p}\mathscr{G}^2_{0,\nu}(q)\delta_{q,0} \sum_{\substack{k,k'\in \mathrm{i}\Omega_{-1}\times \mathrm{Q}\\ n,n'\in\{1,\dots,\infty\}\\ \sigma,\sigma'\in\{\downarrow,\uparrow\}}} g_{nn\nu}(\mathbf{k},0)g_{n'n'\nu}(\mathbf{k}',0)\mathscr{G}_{0,n,\sigma}(k)\mathscr{G}_{0,n',\sigma'}(k') \\ &- \frac{\hbar}{\beta N_p}\mathscr{G}^2_{0,\nu}(q) \sum_{\substack{k\in \mathrm{i}\Omega_{-1}\times \mathrm{Q}\\ n,n'\in\{1,\dots,\infty\}\\ \sigma\in\{\downarrow,\uparrow\}}} |g_{n'n\nu}(\mathbf{k},\mathbf{q})|^2 \mathscr{G}_{0,n,\sigma}(k)\mathscr{G}_{0,n',\sigma}(k+q) \quad + \quad \mathcal{O}(g^4)\end{aligned} \tag{24.17}$$

whereas the terms of order $\mathcal{O}(g^4)$ are those that are being neglected when one approximates the Bethe-Salpeter-equation.

Proof 24.14 See Appendix, Proofs, page 389.

Lemma 24.15 For the phononic occupation numbers of the interacting system we have the identity

$$\begin{aligned}\langle H_{\mathrm{ph}}\rangle \approx &-\frac{1}{\beta}\sum_{\substack{q\,\in\,\mathrm{i}\Omega_{+1}\times\mathrm{Q}\\ \nu\,\in\,\{1,2,\ldots,rd_{\mathbf{x}}\}}} \mathrm{e}^{+\mathrm{i}\omega_q 0^+}\Omega_\nu(\mathbf{q})\mathscr{G}_{0,\nu}(q)\\ &+\frac{1}{\beta^2 N_p}\sum_{\substack{\sigma,\sigma'\,\in\,\{\downarrow,\uparrow\}\\ k,k'\,\in\,\mathrm{i}\Omega_{-1}\times\mathrm{Q}\\ n,n'\,\in\,\{1,\ldots,\infty\}\\ \nu\,\in\,\{1,2,\ldots,rd_{\mathbf{x}}\}}} \Omega_\nu(\mathbf{q}=\mathbf{0})g_{nn\nu}(\mathbf{k},0)g_{n'n'\nu}(\mathbf{k}',0)\mathscr{G}^2_{0,\nu}(q=0)\mathscr{G}_{0,n,\sigma}(k)\mathscr{G}_{0,n',\sigma'}(k')\\ &-\frac{1}{\beta^2 N_p}\sum_{\substack{k\,\in\,\mathrm{i}\Omega_{-1}\times\mathrm{Q}\\ q\,\in\,\mathrm{i}\Omega_{+1}\times\mathrm{Q}\\ \nu\,\in\,\{1,2,\ldots,rd_{\mathbf{x}}\}\\ n,n'\,\in\,\{1,\ldots,\infty\}\\ \sigma\,\in\,\{\downarrow,\uparrow\}}} \Omega_\nu(\mathbf{q})|g_{n'n\nu}(\mathbf{k},\mathbf{q})|^2\mathscr{G}^2_{0,\nu}(q)\mathscr{G}_{0,n,\sigma}(k)\mathscr{G}_{0,n',\sigma}(k+q) \quad + \quad \mathcal{O}(g^4)\end{aligned} \tag{24.18}$$

whereas the terms of order $\mathcal{O}(g^4)$ are those that are being neglected when one approximates the Bethe-Salpeter-equation.

Proof 24.15 See Appendix, Proofs, page 391.

Remark 24.15.1 Notice that the individual terms have the unit of an energy and are hence consistent.

Corollary 24.16 Evaluating the Matsubara-sums simplifies the individual terms as

$$\begin{aligned}\langle H_{\mathrm{ph}}\rangle \approx &\sum_{\substack{\mathbf{q}\,\in\,\mathrm{Q}\\ \nu\,\in\,\{1,2,\ldots,rd_{\mathbf{x}}\}}} \Omega_\nu(\mathbf{q})\,\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) \quad + \sum_{\nu\,\in\,\{1,2,\ldots,rd_{\mathbf{x}}\}} A_\nu B_\nu^2\\ &-\sum_{\substack{\mathbf{k},\mathbf{q}\,\in\,\mathrm{Q}\\ \sigma\,\in\,\{\downarrow,\uparrow\}\\ \nu\,\in\,\{1,2,\ldots,rd_{\mathbf{x}}\}\\ n,n'\,\in\,\{1,\ldots,\infty\}}} \frac{\Omega_\nu(\mathbf{q})}{N_p}|g_{n'n\nu}(\mathbf{k},\mathbf{q})|^2\left(\mathrm{F}_\beta(\xi_{n,\sigma,\mathbf{k}})-\mathrm{F}_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}})\right)\left[\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))-\mathrm{B}_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}-\xi_{n,\sigma,\mathbf{k}})}{\left(\Omega_\nu(\mathbf{q})+\xi_{n,\sigma,\mathbf{k}}-\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}\right)^2}+\frac{1}{4}\frac{\beta\,\mathrm{csch}(\beta\Omega_\nu(\mathbf{q})/2)^2}{\Omega_\nu(\mathbf{q})+\xi_{n,\sigma,\mathbf{k}}-\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}}\right]\\ &+\mathcal{O}(g^4)\end{aligned}$$

with real coefficients A_ν and B_ν defined as

$$A_\nu = \Omega_\nu(\mathbf{q}=\mathbf{0})\mathscr{G}^2_{0,\nu}(q=0) = \Omega_\nu^{-1}(\mathbf{q}=\mathbf{0}), \qquad B_\nu = \frac{1}{\sqrt{N_p}}\sum_{\substack{\mathbf{k}\,\in\,\mathrm{Q}\\ \sigma\,\in\,\{\downarrow,\uparrow\}\\ n\,\in\,\{1,\ldots,\infty\}}} g_{nn\nu}(\mathbf{k},0)\,\mathrm{F}_\beta(\xi_{n\mathbf{k}\sigma}) \tag{24.19}$$

whereas the terms of order $\mathcal{O}(g^4)$ are those that are being neglected when one approximates the Bethe-Salpeter-equation.

Proof 24.16 See Appendix, Proofs, page 391.

24.2.3. Total Crystal

The Interaction Contribution

Lemma 24.17 One obtains for the electron-phonon-interaction energy the identity

$$\begin{aligned}\langle H_{\text{int}}\rangle &= \sum_{\substack{\mathbf{k},\mathbf{q}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}\\ m,n\in\{1,\dots,\infty\}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})}{\sqrt{N_p}} \left\langle c^{\dagger}_{\mathbf{k}+\mathbf{q}m\sigma} c_{\mathbf{k}n\sigma}\left(b_{\mathbf{q}\nu}+b^{\dagger}_{-\mathbf{q}\nu}\right)\right\rangle \\ &= \sum_{\substack{k\in \mathrm{i}\Omega_{-1}\times Q\\ q\in\mathrm{i}\Omega_{+1}\times Q\\ \sigma\in\{\downarrow,\uparrow\}\\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}\\ n,m\in\{1,\dots,\infty\}\\ \omega_{n'}\in\Omega_{-1}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})\hbar}{\sqrt{N_p}\sqrt{\hbar\beta}^3}\left\langle \overline{\psi}_{m\sigma}(\mathrm{i}\omega_{n'},\mathbf{k}+\mathbf{q})\psi_{n\sigma}(\mathrm{i}\omega_n,\mathbf{k})\left[\rho_\nu(q)\mathscr{G}_{0,\nu}(q)+\mathscr{G}_{0,\nu}(-q)\rho_\nu(q)\right]\right\rangle\end{aligned} \tag{24.20}$$

Proof 24.17 See Appendix, Proofs, page 394.

Corollary 24.18 Simplifying the above expression with an approximation of the Bethe-Salpeter-equation, and evaluating the Matsubara-sums gives the two contributions

$$\langle H_{\text{int}}\rangle = \mathrm{I}+\mathrm{II} \tag{24.21}$$

Proof 24.18 See Appendix, Proofs, page 395.

Collecting all Contributions

Fact 24.19 For the electronic and phononic occupation numbers of the interacting system we have the identity

$$\langle H_{\text{el}}\rangle = \sum_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\dots,\infty\}}} \epsilon_{m\mathbf{k}\sigma}\langle n_{m\mathbf{k}\sigma}\rangle = \frac{-1}{\pi}\lim_{\delta\to 0} \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\dots,\infty\}\\ E\in(-\infty,+\infty)}} \epsilon_{m\mathbf{k}\sigma}\,\mathrm{F}_\beta(E)\,\mathrm{Im}\,G^{\text{ret}}_{\substack{m,m\\ \sigma,\sigma}}(E+\mathrm{i}\delta,\mathbf{k}) \tag{24.22}$$

$$
\begin{aligned}
\langle H_{\mathrm{ph}} \rangle = &\sum_{\substack{\mathbf{q} \in \mathrm{Q} \\ \nu \in \{1,2,\ldots,rd_{\mathbf{x}}\}}} \Omega_\nu(\mathbf{q}) \langle n_\nu(\mathbf{q}) \rangle \\
\approx &\sum_{\substack{\mathbf{q} \in \mathrm{Q} \\ \nu \in \{1,2,\ldots,rd_{\mathbf{x}}\}}} \Omega_\nu(\mathbf{q})\, \mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) \quad + \sum_{\nu \in \{1,2,\ldots,rd_{\mathbf{x}}\}} A_\nu B_\nu^2 \\
- &\sum_{\substack{\mathbf{k},\mathbf{q} \in \mathrm{Q} \\ \nu \in \{1,2,\ldots,rd_{\mathbf{x}}\} \\ n,n' \in \{1,\ldots,\infty\} \\ \sigma \in \{\downarrow,\uparrow\}}} \frac{\Omega_\nu(\mathbf{q})}{N_p} |g_{n'n\nu}(\mathbf{k},\mathbf{q})|^2 \left(\mathrm{F}_\beta(\xi_{n,\sigma,\mathbf{k}}) - \mathrm{F}_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}})\right) \left[\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) - \mathrm{B}_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}} - \xi_{n,\sigma,\mathbf{k}})}{\left(\Omega_\nu(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}\right)^2} + \frac{1}{4} \frac{\beta \operatorname{csch}(\beta\Omega_\nu(\mathbf{q})/2)^2}{\Omega_\nu(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}}\right] \\
+ &\,\mathcal{O}(g^4)
\end{aligned}
\tag{24.23}
$$

$$
\langle H_{\mathrm{int}} \rangle = \text{TBDigitalized from PP3.} \tag{24.24}
$$

Proof 24.19 See Appendix, Proofs, page 397.

The Molar Heat Capacity of the Interacting System

We combine Equations (9.55) and (9.48)

$$
c_{N,T,V} = \frac{C_{N,V}(T)}{n} = \frac{1}{n} \partial_T \langle H \rangle_{N,T,V}\,. \tag{24.25}
$$

Using now Equation (9.56) yields

$$
\begin{aligned}
C_{N,V}(T) &= \partial_T \langle H \rangle (N,T,V) \\
&= \left(\partial_T \langle H \rangle\right)(\mu(T,V),T,V) + \left(\partial_\mu \langle H \rangle\right)(\mu(T,V),T,V) \cdot \left(\partial_T \mu\right)(N,V,T)
\end{aligned}
\tag{24.26}
$$

whereas the Hamiltonian is made up of three contributions

$$
H = H_{\mathrm{el}} + H_{\mathrm{ph}} + H_{\mathrm{int}}. \tag{24.27}
$$

Therefore, also for the molar heat capacity, we have contributions from the electronic part, from the phononic part, and from the interaction part. We further need to account for the temperature derivative of the chemical potential. We consider all four quantities separately and then merge the results.

24.2.3.1. Temperature-Derivative of the Chemical Potential

Fact 24.20 We first notice that dictating the particle number (for example by requiring [C]harge [N]eutrality of the crystal) through Equation (23.24) can be rewritten as

$$
F(\mu,T) \equiv \langle N \rangle(\mu,T) - N_{\mathrm{CN}} = 0. \tag{24.28}
$$

Remark 24.20.1 Notice that the chemical potential that is defined through this Equation is unique due to Theorem 9.29.

Corollary 24.21 Assuming now we have access[1] to the values $\mu(T)$ that fulfill Equation (24.28), we can compute the temperature derivative of the chemical potential as

$$\left(\frac{\partial\mu}{\partial T}\right)(T) = -\left(\frac{\partial_T F}{\partial_\mu F}\right)(\mu(T), T)\,. \tag{24.29}$$

Proof 24.21 Follows from the Theorem of implicit functions.

Lemma 24.22 The temperature-derivative of the function F can be computed as

$$\begin{aligned}\left(\partial_T F\right)(\mu, T) = \frac{-1}{\pi}\frac{\beta}{T}\lim_{\delta\to 0}\mathop{\sum\!\!\!\!\!\!\int}_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\dots,\infty\}\\ E\in(-\infty,+\infty)}} \mathrm{F}_\beta(E)\,\mathrm{Im}\Bigg[&\left(1-\mathrm{F}_\beta(E)\right)E \quad \frac{1}{E+\mathrm{i}\delta-(\xi_{m\mathbf{k}\sigma}+\Sigma^{\mathrm{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta))}\\ &+\quad \frac{k_B T^2\partial_T\Sigma_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta)}{\left(E+\mathrm{i}\delta-(\xi_{m\mathbf{k}\sigma}+\Sigma^{\mathrm{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta))\right)^2}\Bigg]\end{aligned} \tag{24.30}$$

with partial derivative of the retarded self-energy

$$k_B T^2\partial_T\Sigma^{\mathrm{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta) \quad = \quad k_B T^2\partial_T\Sigma^{\mathrm{ret,\,H}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta) \quad + \quad k_B T^2\partial_T\Sigma^{\mathrm{ret,\,F}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta) \tag{24.31}$$

and invididual contributions

$$k_B T^2\partial_T\Sigma^{\mathrm{ret,\,H}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta) = \text{TBDone / TBDigitalized} \tag{24.32}$$

and

$$\begin{aligned}k_B T^2\partial_T\Sigma^{\mathrm{ret,\,F}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta) = \sum_{\substack{\mathbf{q}\in Q\\ n\in\{1,2,\dots,\infty\}\\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \frac{|g_{nm\nu}(\mathbf{k},\mathbf{q})|^2}{4N_p}\Bigg[&\frac{\Omega_\nu(\mathbf{q})\,\mathrm{csch}^2(\beta\Omega_\nu(\mathbf{q})/2)+\xi_{n,\sigma,\mathbf{k+q}}\,\mathrm{sech}^2(\beta\xi_{n,\sigma,\mathbf{k+q}}/2)}{E+\mathrm{i}\delta-\xi_{n,\sigma,\mathbf{k+q}}+\Omega_\nu(\mathbf{q})}\\ &+\frac{\Omega_\nu(\mathbf{q})\,\mathrm{csch}^2(\beta\Omega_\nu(\mathbf{q})/2)-\xi_{n,\sigma,\mathbf{k+q}}\,\mathrm{sech}^2(\beta\xi_{n,\sigma,\mathbf{k+q}}/2)}{E+\mathrm{i}\delta-\xi_{n,\sigma,\mathbf{k+q}}-\Omega_\nu(\mathbf{q})}\Bigg]\end{aligned} \tag{24.33}$$

Proof 24.22 See Appendix, Proofs, page 398.

[1]Notice that this is the case, as we compute the chemical potential numerically.

Lemma 24.23 The chemical-potential-derivative of the function F can be computed as

$$\left(\partial_\mu F\right)(\mu,T) = \frac{-1}{\pi}\frac{\beta}{4}\lim_{\delta\to 0}\sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\dots,\infty\}\\ E\in(-\infty,+\infty)}} \mathrm{Im}\left[\mathrm{sech}^2\left(\frac{\beta E}{2}\right)\frac{1}{E+\mathrm{i}\delta-(\xi_{m\mathbf{k}\sigma}+\Sigma^{\mathrm{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta))}\right.$$
$$\left.+\mathrm{F}_\beta(E)\cdot\frac{\sigma_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta)}{\left(E+\mathrm{i}\delta-(\xi_{m\mathbf{k}\sigma}+\Sigma^{\mathrm{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta))\right)^2}\right] \tag{24.34}$$

with auxiliary function

$$\sigma_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta) = \sigma^{\mathrm{Hartree}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta)+\sigma^{\mathrm{Fock}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta), \tag{24.35}$$

with the Hartree-contribution

$$\sigma^{\mathrm{Hartree}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta) = -\sum_{\substack{\mathbf{q}\in Q\\ \sigma'\in\{\downarrow,\uparrow\}\\ n\in\{1,2,\dots\}\\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}}\frac{g_{mm\nu}(\mathbf{k},0)g_{nn\nu}(\mathbf{q},0)}{N_p}\frac{1}{\Omega_\nu(\mathbf{q}'=0)}\,\mathrm{sech}^2\left(\beta\xi_{n\sigma'\mathbf{q}}/2\right) \tag{24.36}$$

and the Fock-contribution

$$\sigma^{\mathrm{Fock}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta) = \sum_{\substack{\mathbf{q}\in Q\\ n\in\{1,2,\dots\}\\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}}\frac{|g_{nm\nu}(\mathbf{k},\mathbf{q})|^2}{N_p}\mathrm{sech}^2(\beta\xi_{\mathbf{q}+\mathbf{k}n\sigma}/2)\left[\frac{1}{E+\mathrm{i}\delta-\xi_{\mathbf{q}+\mathbf{k}n\sigma}+\Omega_\nu(\mathbf{q})}-\frac{1}{E+\mathrm{i}\delta-\xi_{\mathbf{q}+\mathbf{k}n\sigma}-\Omega_\nu(\mathbf{q})}\right] \tag{24.37}$$

Proof 24.23 See Appendix, Proofs, page 398.

24.2.3.2. Temperature-Derivative of the Electronic Energy

Lemma 24.24 The temperature-derivatives of the electronic contribution to the internal energy can be computed as

$$\partial_T\langle H_{\mathrm{el}}\rangle = \frac{-1}{\pi}\frac{1}{k_B T^2}\lim_{\delta\to 0}\sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\dots,\infty\}\\ E\in(-\infty,+\infty)}}\epsilon_{m\mathbf{k}\sigma}\,\mathrm{F}_\beta(E)\,\mathrm{Im}\left[\left(1-\mathrm{F}_\beta(E)\right)E\frac{1}{E+\mathrm{i}\delta-(\xi_{m\mathbf{k}\sigma}+\Sigma^{\mathrm{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta))}\right.$$
$$\left.+\frac{k_B T^2\partial_T\Sigma^{\mathrm{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta)}{\left(E+\mathrm{i}\delta-(\xi_{m\mathbf{k}\sigma}+\Sigma^{\mathrm{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta,\mathbf{k}))\right)^2}\right] \tag{24.38}$$

with partial derivative of the retarded self-energy as in Equation (24.33)

Proof 24.24 See Appendix, Proofs, page 401.

Remark 24.24.1 Notice that this is the very same Equation as (24.30), except for the factor of $\epsilon_{m\mathbf{k}\sigma}$ in the integrand-summand.

Lemma 24.25 The chemical-potential-derivative of the electronic contribution to the internal energy can be computed as

$$\partial_\mu \langle H_{\mathrm{el}} \rangle = \frac{-1}{\pi} \frac{\beta}{4} \lim_{\delta \to 0} \mathop{\Large\Sigma\!\!\!\!\int}_{\substack{\mathbf{k} \in \mathrm{Q} \\ \sigma \in \{\uparrow,\downarrow\} \\ m \in \{1,\ldots,\infty\} \\ E \in (-\infty,+\infty)}} \epsilon_{m\mathbf{k}\sigma} \,\mathrm{Im}\Bigg[\mathrm{sech}^2\left(\frac{\beta E}{2}\right) \frac{1}{E + \mathrm{i}\delta - (\xi_{m\mathbf{k}\sigma} + \Sigma^{\mathrm{ret}}_{m\sigma\mathbf{k}}(E + \mathrm{i}\delta))} + \mathrm{F}_\beta(E) \cdot \frac{\sigma_{m\sigma\mathbf{k}}(E + \mathrm{i}\delta)}{\big(E + \mathrm{i}\delta - (\xi_{m\mathbf{k}\sigma} + \Sigma^{\mathrm{ret}}_{m\sigma\mathbf{k}}(E + \mathrm{i}\delta))\big)^2} \Bigg] \tag{24.39}$$

with auxiliary function

$$\sigma_{m\sigma\mathbf{k}}(E + \mathrm{i}\delta) = \sigma^{\mathrm{Hartree}}_{m\sigma\mathbf{k}}(E + \mathrm{i}\delta) + \sigma^{\mathrm{Fock}}_{m\sigma\mathbf{k}}(E + \mathrm{i}\delta), \tag{24.40}$$

as above.

Proof 24.25 Completely analogous to the proof of Lemma 24.23.

24.2.3.3. Temperature-Derivative of the Phononic Energy

Lemma 24.26 The temperature-derivatives of the non-interacting phononic contribution to the internal energy can be computed as

$$\partial_T \langle H_{\mathrm{ph}} \rangle_0 = \frac{1}{4 k_B T^2} \sum_{\substack{\mathbf{q} \in \mathrm{Q} \\ \nu \in \{1,2,\ldots,r d_{\mathbf{x}}\}}} \Big[\Omega_\nu(\mathbf{q}) \,\mathrm{csch}(\beta \Omega_\nu(\mathbf{q})/2) \Big]^2 \tag{24.41}$$

and for the interacting phononic correction contribution

$$
\begin{aligned}
&\partial_T \delta\langle H_{\mathrm{ph}}\rangle \\
&\approx 2 \sum_{\nu\in\{1,2,\ldots,rd_{\mathbf{x}}\}} A_\nu B_\nu \partial_T B_\nu \\
&+ \frac{-1}{N_p} \sum_{\substack{\mathbf{k},\mathbf{q}\in\mathrm{Q} \\ n,n'\in\{1,\ldots,\infty\} \\ \sigma\in\{\downarrow,\uparrow\} \\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\}}} \frac{\Omega_\nu(\mathbf{q})|g_{n'n\nu}(\mathbf{k},\mathbf{q})|^2}{4k_BT^2} \left(\mathrm{F}_\beta(\xi_{n,\sigma,\mathbf{k}}) - \mathrm{F}_\beta(\xi_{n',\sigma,\mathbf{k+q}})\right) \frac{\operatorname{csch}^2(\beta\Omega_\nu(\mathbf{q})/2)}{\Omega_\nu(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k+q}}} \left(\beta\Omega_\nu(\mathbf{q})\coth\left(\beta\Omega_\nu(\mathbf{q})/2\right) - 1\right) \\
&+ \frac{-1}{N_p} \sum_{\substack{\mathbf{k},\mathbf{q}\in\mathrm{Q} \\ n,n'\in\{1,\ldots,\infty\} \\ \sigma\in\{\downarrow,\uparrow\} \\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\}}} \frac{\Omega_\nu(\mathbf{q})|g_{n'n\nu}(\mathbf{k},\mathbf{q})|^2}{4k_BT^2} \left(\mathrm{F}_\beta(\xi_{n,\sigma,\mathbf{k}}) - \mathrm{F}_\beta(\xi_{n',\sigma,\mathbf{k+q}})\right) \left[\frac{\Omega_\nu(\mathbf{q})\operatorname{csch}^2\left(\beta\Omega_\nu(\mathbf{q})/2\right) - (\xi_{n',\sigma,\mathbf{k+q}} - \xi_{n,\sigma,\mathbf{k}})\operatorname{csch}^2\left(\frac{\xi_{n',\sigma,\mathbf{k+q}}-\xi_{n,\sigma,\mathbf{k}}}{2k_BT}\right)}{\left(\Omega_\nu(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k+q}}\right)^2}\right] \\
&+ \frac{-1}{N_p} \sum_{\substack{\mathbf{k},\mathbf{q}\in\mathrm{Q} \\ n,n'\in\{1,\ldots,\infty\} \\ \sigma\in\{\downarrow,\uparrow\} \\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\}}} \frac{\Omega_\nu(\mathbf{q})|g_{n'n\nu}(\mathbf{k},\mathbf{q})|^2}{4k_BT^2} \left(\xi_{n,\sigma,\mathbf{k}}\operatorname{sech}^2\left(\beta\xi_{n,\sigma,\mathbf{k}}/2\right) - \xi_{n',\sigma,\mathbf{k+q}}\operatorname{sech}^2\left(\beta\xi_{n',\sigma,\mathbf{k+q}}/2\right)\right) \left[\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) - \mathrm{B}_\beta(\xi_{n',\sigma,\mathbf{k+q}} - \xi_{n,\sigma,\mathbf{k}})}{\left(\Omega_\nu(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k+q}}\right)^2} + \frac{1}{4}\frac{\beta\operatorname{csch}(\beta\Omega_\nu(\mathbf{q})/2)^2}{\Omega_\nu(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k+q}}}\right]
\end{aligned}
\tag{24.42}
$$

with partial derivative

$$
\partial_T B_\nu = \frac{1}{\sqrt{N_p}} \frac{\xi_{n,\sigma,\mathbf{k}}}{4k_BT^2} \sum_{\substack{\mathbf{k}\in\mathrm{Q} \\ n\in\{1,2,\ldots,\infty\} \\ \sigma\in\{\downarrow,\uparrow\}}} g_{nn\nu}(\mathbf{k},0)\operatorname{sech}^2\left(\beta\xi_{n,\sigma,\mathbf{k}}/2\right) \tag{24.43}
$$

Proof 24.26 See Appendix, Proofs, page 401.

Lemma 24.27 For the chemical potential-derivative of the phononic correction-term we obtain

$$
\begin{aligned}
&\partial_\mu \delta\langle H_{\mathrm{ph}}\rangle \\
&\approx 2 \sum_{\nu\in\{1,2,\ldots,rd_{\mathbf{x}}\}} A_\nu B_\nu \partial_\mu B_\nu \\
&- \sum_{\substack{\mathbf{k},\mathbf{q}\in\mathrm{Q} \\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\} \\ n,n'\in\{1,\ldots,\infty\} \\ \sigma\in\{\downarrow,\uparrow\}}} \frac{\beta}{4}\frac{\Omega_\nu(\mathbf{q})}{N_p}|g_{n'n\nu}(\mathbf{k},\mathbf{q})|^2 \left(\operatorname{sech}^2(\beta\xi_{n,\sigma,\mathbf{k}}/2) - \operatorname{sech}^2(\beta\xi_{n',\sigma,\mathbf{k+q}}/2)\right) \left[\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) - \mathrm{B}_\beta(\xi_{n',\sigma,\mathbf{k+q}} - \xi_{n,\sigma,\mathbf{k}})}{\left(\Omega_\nu(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k+q}}\right)^2} + \frac{1}{4}\frac{\beta\operatorname{csch}(\beta\Omega_\nu(\mathbf{q})/2)^2}{\Omega_\nu(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k+q}}}\right] \\
&+ \mathcal{O}(g^4)
\end{aligned}
$$

and partial derivative

$$
\partial_\mu B_\nu = \frac{1}{\sqrt{N_p}} \frac{\beta}{4} \sum_{\substack{\mathbf{k}\in\mathrm{Q} \\ n\in\{1,2,\ldots,\infty\} \\ \sigma\in\{\downarrow,\uparrow\}}} g_{nn\nu}(\mathbf{k},0)\operatorname{sech}^2\left(\beta\xi_{n,\sigma,\mathbf{k}}/2\right) \tag{24.44}
$$

Proof 24.27 We use Equation (24.23) and note that we obtain with Lemma (9.44) and the chain-rule the identity

$$
\partial_\mu \mathrm{F}_\beta(\xi_{n,\sigma,\bullet}) = \frac{\beta}{4}\operatorname{sech}^2\left(\beta\xi_{n,\sigma,\bullet}/2\right) \tag{24.45}
$$

24.2.3.4. Temperature-Derivative of the Interaction Energy

TBDigitalized... (Only contribute for crystals with more than one atom per unit cell.)

25. Superconductivity

General Remarks

Apart from the toy-model for BCS-theory in section 34.2, we will introduce two formalisms for superconductivity.

1. The Eliashberg-Theory (ET for short),
2. The Path-Integral-Formalism for ([G]eneralized) [E]liashberg-[T]heory (GET / ET for short).

The strength and beauty in the path-integral-approach lies in the fact that it is carried out analogously to BCS-theory and reproduces the already established behavior for the order-parameters as a function of temperature. Further it reproduces and even opens the possibility to generalize Eliashberg-Theory in the appropriate limits. The path-integral-formalism can be considered as a generalization / unification of these theories. Another folklore-statement that is being reproduced is that superconductivity originates from an attractive interaction[1] that is mediated through phonons.

25.1. Eliashberg-Theory in the Operator-Formalism

25.1.1. Preliminaries

We consider the Hamiltonian from Definition 8.29

$$H = \sum_{\substack{\mathbf{k}\in Q \\ \sigma\in\{\uparrow,\downarrow\} \\ m\in\{1,\dots,\infty\}}} \epsilon_{m\sigma}(\mathbf{k}) c^\dagger_{m\sigma}(\mathbf{k}) c_{m\sigma}(\mathbf{k}) + \sum_{\substack{\mathbf{q}\in Q \\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} \hbar\omega_\nu(\mathbf{q}) b^\dagger_\nu(\mathbf{q}) b_\nu(\mathbf{q}) + \sum_{\substack{\mathbf{k},\mathbf{q}\in Q \\ \sigma\in\{\uparrow,\downarrow\} \\ \nu\in\{1,\dots,rd_{\mathbf{x}}\} \\ m,n\in\{1,\dots,\infty\}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})}{\sqrt{N_p}} c^\dagger_{m\sigma}(\mathbf{k}+\mathbf{q}) c_{n\sigma}(\mathbf{k}) \left(b_\nu(\mathbf{q}) + b^\dagger_\nu(-\mathbf{q})\right) \tag{25.1}$$

[1] Note however that contrary to the folklore, this interaction is attractive *not only* close to the Fermi-surface, but for all energies. But it will turn later out, that only those electronic energies close to the Fermi-surface and hence only those interactions will contribute to the emergence of superconductivity, cf. Theorem 25.46.

25.1.2. Derivation of the Eliashberg-Equations

Imarginary Time Formalism

Definition 25.1 We define[2] the *single-band Gorkov-Green-function* as

$$\widehat{\mathscr{G}}_{\underset{\text{Gorkov}}{m}}(\tau,\mathbf{k}) = \begin{pmatrix} \mathscr{G}_{\underset{\text{Gorkov}}{m}}(\tau,\mathbf{k}) & \mathscr{F}_m(\tau,\mathbf{k}) \\ \mathscr{F}_m^\dagger(\tau,\mathbf{k}) & -\mathscr{G}_{\underset{\text{Gorkov}}{m}}(-\tau,-\mathbf{k}) \end{pmatrix}$$
$$= -\frac{1}{\hbar}\begin{pmatrix} \left\langle \mathrm{T}_\tau\, c_{m,\uparrow}(\tau,\mathbf{k}) c^\dagger_{m,\uparrow}(0,\mathbf{k})\right\rangle_{\mathrm{SC}} & \left\langle \mathrm{T}_\tau\, c_{m,\uparrow}(\tau,\mathbf{k}) c_{m,\downarrow}(0,-\mathbf{k})\right\rangle_{\mathrm{SC}} \\ \left\langle \mathrm{T}_\tau\, c^\dagger_{m,\downarrow}(\tau,-\mathbf{k}) c^\dagger_{m,\uparrow}(0,\mathbf{k})\right\rangle_{\mathrm{SC}} & \left\langle \mathrm{T}_\tau\, c^\dagger_{m,\downarrow}(\tau,-\mathbf{k}) c_{m,\downarrow}(0,-\mathbf{k})\right\rangle_{\mathrm{SC}} \end{pmatrix}$$

whereas the average is with respect to the action in the state in which „the U(1) symmetry is spontaneously broken“, which we call the *[S]uper-[C]onducting state*.

Remark 25.1.1 We don't have the tools yet to clearly state how the expectation-value is to be understood, and what exactly we mean with „the state in which the U(1) symmetry is spontaneously broken“. For the moment we will leave it with this vague definition since it will allow us to derive the Eliashberg-equations in the "traditional way"[3]. The only thing that we demand from this expectation-value is linearity and compatibality with the Bethe-Salpeter-equation.

Remark 25.1.2 Notice that inside the time-ordering-operator, the exchange of two fermionic operators yields a minus sign.

Remark 25.1.3 The off-diagonal components are usually called *Gorkov's anomalous Green-functions*.

Remark 25.1.4 Gorkov's anomalous Green-functions are only non-zero in the superconducting phase[4]. Evaluating these expressions w.r.t. the thermal state yields zero[5].

Fact 25.2 It holds for all wavevectors $\mathbf{k} \in \mathrm{Q}$ and bands $m \in \{1, 2, 3, \dots, \infty\}$ the identity

[2]Notice that we are following the prefactor-convention of 03-Marsiglio, Eqs. (43-44), and Coleman page 519 Equations (14.158), or Polaron-Physics 07, p. 16 in contrast to Mahan3rd page 635, Eq. (10.31). Notice further that our Equations are in harmony with Allens long article on pages 8 and 33 and Equation (7.2) in particular.

[3]In the following sections we will rederive these equations in a more rigorous and comprehensive way, namely through the path integral formalism.

[4]Quoting Mahan p. 635:

> *A basic feature of the BCS ground state (...) is that it is composed of a superposition of electronic states containing a different number of electrons. Such a formulation is possible in a grand canonical ensemble.*

[5]Easy to prove, but also quoting Negele and Orland p. 77:

> *The expectation value of two creation or annihilation operators is zero in a state of definite particle number.*

$$\forall \tau \in (-\hbar\beta, 0): \quad \underset{\text{Gorkov}}{\widehat{\mathscr{G}}_m}(\tau + \hbar\beta, \mathbf{k}) = -\underset{\text{Gorkov}}{\widehat{\mathscr{G}}_m}(\tau, \mathbf{k}) \tag{25.2}$$

Proof 25.2 In the superconducting state, it is a priori not clear whether the Green-functions are periodic or antiperiodic. As we shall see later on in the functional integral formalism, these Green-functions are also antiperiodic, and it can be shown be introducing a "Hubbard-Stratanovich-transformation" in an appropriate channel. We will use this fact here, but we will prove it later on in section TBReferenced. Strictly speaking this assumption was a leap of faith when Eliashberg derived his equations (because the functional integral formalism was not invented back then).

Matsubara-Formalism

Lemma 25.3 In the Matsubara-representation one has the equation

$$\underset{\text{Gorkov}}{\widehat{\mathscr{G}}_m}(\mathrm{i}\omega_n, \mathbf{k}) = \begin{pmatrix} \underset{\text{Gorkov}}{\mathscr{G}_m}(\mathrm{i}\omega_n, \mathbf{k}) & \mathscr{F}_m(\mathrm{i}\omega_n, \mathbf{k}) \\ \mathscr{F}_m^*(\mathrm{i}\omega_n, \mathbf{k}) & -\underset{\text{Gorkov}}{\mathscr{G}_m}(-\mathrm{i}\omega_n, -\mathbf{k}) \end{pmatrix} \tag{25.3}$$

Proof 25.3 See Appendix, Proofs, page 409.

Lemma 25.4 Assuming a spin-degenerate and inversion-symmetric system, we obtain for the Green-function in the non-superconducting state

$$\begin{aligned} \underset{\text{Gorkov}}{\widehat{\mathscr{G}}^{-1}_{0,m}}(\mathrm{i}\omega_n, \mathbf{k}) &= \mathrm{i}\hbar\omega_n\sigma_0 - \big(\epsilon_m(\mathbf{k}) - \mu\big)\,\sigma_3 \\ \Longleftrightarrow \quad \underset{\text{Gorkov}}{\widehat{\mathscr{G}}_{0,m}}(\mathrm{i}\omega_n, \mathbf{k}) &= \begin{pmatrix} \dfrac{1}{\mathrm{i}\hbar\omega_n - \big(\epsilon_m(\mathbf{k}) - \mu\big)} & 0 \\ 0 & \dfrac{1}{\mathrm{i}\hbar\omega_n + \big(\epsilon_m(\mathbf{k}) - \mu\big)} \end{pmatrix}, \end{aligned} \tag{25.4}$$

whereas we introduced $\epsilon_m(\mathbf{k}) = \epsilon_{m,\uparrow}(\pm\mathbf{k}) = \epsilon_{m,\downarrow}(\pm\mathbf{k})$.

Proof 25.4 See Appendix, Proofs, page 409.

Notation 25.5 We write for *the Dyson-equation (cf. Remark 16.44.1) of the Gorkov-Green-function in the superconducting state*

$$\underset{\text{Gorkov}}{\widehat{\mathscr{G}}^{-1}_m}(\mathrm{i}\omega_n, \mathbf{k}) = \underset{\text{Gorkov}}{\widehat{\mathscr{G}}^{-1}_{0m}}(\mathrm{i}\omega_n, \mathbf{k}) - \underset{\text{Gorkov}}{\widehat{\Sigma}_m}(\mathrm{i}\omega_n, \mathbf{k}) \tag{25.5}$$

whereas $\underset{\text{Gorkov}}{\widehat{\mathscr{G}}_{0m}}$ represents the Gorkov-Green-function in the non-superconducting state and absence of electron-phonon-interactions.

Remark 25.5.1 Notice further, that this equation can be rewritten in several ways, e.g. as

$$\widehat{\mathscr{G}}^{-1} = \widehat{\mathscr{G}}_0^{-1} - \widehat{\Sigma} \Longleftrightarrow \widehat{\mathscr{G}}_0^{-1} = \widehat{\mathscr{G}}^{-1} + \widehat{\Sigma} \Longleftrightarrow \widehat{\mathscr{G}}_0^{-1}\widehat{\mathscr{G}} = 1 + \widehat{\Sigma}\widehat{\mathscr{G}} \Longleftrightarrow \widehat{\mathscr{G}} = \widehat{\mathscr{G}}_0 + \widehat{\mathscr{G}}_0\widehat{\Sigma}\widehat{\mathscr{G}} \tag{25.6}$$

whereas we lifted the multitude of arguments to lighten the notation.

Remark 25.5.2 Notice that just as the anti-periodicity of the Gorkov-Green-function, the existence (not to mention the computability) of the self-energy is a priori not clear.

Corollary 25.6 This gives for the Dyson-equation in the superconducting state the set of equations (for each component)

$$\begin{aligned} \mathscr{G}_{\substack{m\\ \text{Gorkov}}}(k) = +\mathscr{G}_{\substack{0m\\ \text{Gorkov}}}(k) \quad & +\mathscr{G}_{\substack{0m\\ \text{Gorkov}}}(k) && \left[\Sigma_{\substack{m\\ 11}}(k)\mathscr{G}_{\substack{m\\ \text{Gorkov}}}(k) \quad + \Sigma_{\substack{m\\ 12}}(k)\mathscr{F}_m^*(k)\right] \\ \mathscr{F}_m(k) = \quad & +\mathscr{G}_{\substack{0m\\ \text{Gorkov}}}(k) && \left[\Sigma_{\substack{m\\ 11}}(k)\mathscr{F}_m(k) \quad - \Sigma_{\substack{m\\ 12}}(k)\mathscr{G}_{\substack{m\\ \text{Gorkov}}}(-k)\right] \\ \mathscr{F}_m^*(k) = \quad & -\mathscr{G}_{\substack{0m\\ \text{Gorkov}}}(-k) && \left[\Sigma_{\substack{m\\ 21}}(k)\mathscr{G}_{\substack{m\\ \text{Gorkov}}}(k) \quad - \Sigma_{\substack{m\\ 22}}(k)\mathscr{F}_m^*(k)\right] \\ -\mathscr{G}_{\substack{m\\ \text{Gorkov}}}(-k) = -\mathscr{G}_{\substack{0m\\ \text{Gorkov}}}(-k) \quad & -\mathscr{G}_{\substack{0m\\ \text{Gorkov}}}(-k) && \left[\Sigma_{\substack{m\\ 21}}(k)\mathscr{F}_m(k) \quad - \Sigma_{\substack{m\\ 22}}(k)\mathscr{G}_{\substack{m\\ \text{Gorkov}}}(-k)\right] \end{aligned} \tag{25.7}$$

Proof 25.6 See Appendix, Proofs, page 410.

Notation 25.7 For the self-energy we introduce abbreviations as

$$\Sigma_m(\mathrm{i}\omega_n, \mathbf{k}) = \mathrm{i}\hbar\omega_n\left[1 - Z_m(\mathrm{i}\omega_n, \mathbf{k})\right]\sigma_0 + \phi_m(\mathrm{i}\omega_n, \mathbf{k})\sigma_1 + \chi_m(\mathrm{i}\omega_n, \mathbf{k})\sigma_3 \overset{\star\star\star}{=} \begin{pmatrix} \Sigma_m(\mathrm{i}\omega_n, \mathbf{k}) & \Xi_m(\mathrm{i}\omega_n, \mathbf{k}) \\ \Xi_m^*(\mathrm{i}\omega_n, \mathbf{k}) & -\Sigma_m(-\mathrm{i}\omega_n, -\mathbf{k}) \end{pmatrix} \tag{25.8}$$

whereas $\star\star\star$ only holds in the case that ϕ is real. (Is this the case???)

Corollary 25.8 Combining the Dyson-equation (25.5) with Eq. (25.8) for the self-energy yields for the full Green-function

$$\mathscr{G}_{\substack{m\\ \text{Gorkov}}}(\mathrm{i}\omega_n, \mathbf{k}) = \frac{-1}{\theta_m(\mathrm{i}\omega_n, \mathbf{k})}\left[\hbar\mathrm{i}\omega_n Z_m(\mathrm{i}\omega_n, \mathbf{k})\sigma_0 + \phi_m(\mathrm{i}\omega_n, \mathbf{k})\sigma_1 + \left(\chi_m(\mathrm{i}\omega_n, \mathbf{k}) + (\epsilon_m(\mathbf{k}) - \mu)\right)\sigma_3\right] \tag{25.9}$$

with

$$\theta_m(\mathrm{i}\omega_n, \mathbf{k}) = \hbar^2\omega_n^2 Z_m(\mathbf{k}, \mathrm{i}\omega_n)^2 + \left[\chi_m(\mathbf{k}, \mathrm{i}\omega_n) + (\epsilon_m(\mathbf{k}) - \mu)\right]^2 + \phi_m(\mathrm{i}\omega_n, \mathbf{k})^2 \tag{25.10}$$

Proof 25.8 See Appendix, Proofs, page 410.

==

Everything from here onwards has to be double-checked. Especially Thm 3.13

Theorem 25.9 The self-energies for electron-phonon- and electron-electron-interactions are in first (non-vanishing) order perturbation theory given by

$$\Sigma_{\text{Gorkov}}{}_n(\mathrm{i}\omega_n, \mathbf{k}) = \Sigma^{\text{ep}}_n(\mathrm{i}\omega_n, \mathbf{k}) + \Sigma^{\text{c}}_n(\mathrm{i}\omega_n, \mathbf{k}) \tag{25.11}$$

with *electron-phonon* and *Coulomb self energies*

$$\Sigma^{\text{ep}}_n(\mathrm{i}\omega_n, \mathbf{k}) = -T \sum_{\substack{\mathbf{q}\in\mathrm{Q} \\ \omega_{n'}\in\Omega_{-1} \\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\} \\ m\in\{1,\ldots,\infty\}}} \sigma_3 \mathscr{G}_{\text{Gorkov}}{}_m(\mathrm{i}\omega_{n'}, \mathbf{q}+\mathbf{k})\sigma_3 |g_{mn\nu}(\mathbf{k},\mathbf{q})|^2 \mathscr{D}_\nu(\mathrm{i}\omega_n - \mathrm{i}\omega_{n'}, \mathbf{q}) \tag{25.12}$$

$$\Sigma^{\text{c}}_n(\mathrm{i}\omega_n, \mathbf{k}) = -T \sum_{\substack{\mathbf{q}\in\mathrm{Q} \\ \omega_{n'}\in\Omega_{-1} \\ m\in\{1,\ldots,\infty\}}} \sigma_3 \mathscr{G}^{\text{od}}_{\text{Gorkov}}{}_m(\mathrm{i}\omega_{n'}, \mathbf{q})\sigma_3 V_{n,m}(\mathbf{k}, \mathbf{k}+\mathbf{q}, \mathrm{i}\omega_n - \mathrm{i}\omega_{n'}) \tag{25.13}$$

Proof 25.9 These equations will be proven in section 25.1.4.

Lemma 25.10 Writing out these equations gives for the electron-phonon self-energy

$$\Sigma^{\text{ep}}_m(\mathrm{i}\omega_n, \mathbf{k}) = -T \sum_{\substack{\mathbf{q}\in\mathrm{Q} \\ \omega_{n'}\in\Omega_{-1} \\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\} \\ n\in\{1,\ldots,\infty\}}} \begin{pmatrix} \mathscr{G}_{\text{Gorkov}}{}_{n\uparrow}(\bullet) & -\mathscr{F}_n(\bullet) \\ -\mathscr{F}^*_n(\bullet) & -\mathscr{G}_{\text{Gorkov}}{}_{n\downarrow}(-\bullet) \end{pmatrix}_{\bullet=(\mathrm{i}\omega_{n'},\mathbf{q}+\mathbf{k})} |g_{mn\nu}(\mathbf{k},\mathbf{q})|^2 \mathscr{D}_\nu(\mathrm{i}\omega_n - \mathrm{i}\omega_{n'}, \mathbf{q}) \tag{25.14}$$

Proof 25.10 See Appendix, Proofs, page 410.

Definition 25.11 We introduce the *density of states per spin at the Fermi-level* as

$$N_{\text{F}} = \text{TBD.} \tag{25.15}$$

Definition 25.12 We introduce[6] the *anisotropic electron-phonon coupling strength*

[6]Cf. [06 Superhydrides] in "Success of Eliashberg".

$$
\begin{aligned}
\lambda_{\mathbf{k},\mathbf{k}+\mathbf{q}}^{n,m}(\omega_n) &= \frac{N_\mathrm{F}}{\hbar} \sum\!\!\!\!\!\!\int_{\substack{\omega\in(0,\infty)\\ \nu\in\{1,\ldots,rd_\mathrm{x}\}}} \frac{2\omega}{\omega_n^2+\omega^2} |g_{mn\nu}(\mathbf{k},\mathbf{q})|^2 \delta\left(\omega-\omega_\nu(\mathbf{q})\right) \\
&= \frac{N_\mathrm{F}}{\hbar} \sum_{\nu\in\{1,\ldots,rd_\mathrm{x}\}} \frac{2\omega_\nu(\mathbf{q})}{\omega_n^2+\omega_\nu^2(\mathbf{q})} |g_{mn\nu}(\mathbf{k},\mathbf{q})|^2 \\
&= -N_\mathrm{F} \sum_{\nu\in\{1,\ldots,rd_\mathrm{x}\}} \mathscr{D}_{0,\nu}(\mathrm{i}\omega_n,\mathbf{q}) |g_{mn\nu}(\mathbf{k},\mathbf{q})|^2
\end{aligned}
\tag{25.16}
$$

Remark 25.12.1 Notice that the electron-phonon coupling strength has the unit of an energy.

Corollary 25.13 This definition allows us to rewrite the electron-phonon self-energy as

$$
\Sigma_n^{\mathrm{ep}}(\mathrm{i}\omega_n,\mathbf{k}) = \frac{T}{\hbar N_F} \sum_{\substack{\mathbf{q}\in\mathrm{Q}\\ \omega_{n'}\in\Omega_{-1}\\ m\in\{1,\ldots,\infty\}}} \sigma_3 \mathscr{G}_{\mathrm{Gorkov}}^{m}(\mathrm{i}\omega_{n'},\mathbf{q}+\mathbf{k}) \sigma_3 \lambda_{\mathbf{k},\mathbf{k}+\mathbf{q}}^{n,m}(\omega_n-\omega_{n'})
\tag{25.17}
$$

This expression has the incorrect unit.

Proof 25.13 Follows from TBD.

Theorem 25.14 Assuming all expressions are exact, lets us conclude *the Eliashberg-equations*

$$
Z_n(\mathrm{i}\omega_n,\mathbf{k}) = 1 + \frac{T}{\omega_n} \sum_{\substack{\mathbf{q}\in\mathrm{Q}\\ \omega_{n'}\in\Omega_{-1}\\ m\in\{1,\ldots,\infty\}}} \frac{Z_m(\mathrm{i}\omega_{n'},\mathbf{k}+\mathbf{q})}{\theta_m(\mathrm{i}\omega_{n'},\mathbf{k}+\mathbf{q})} \frac{\lambda_{n,m}(\mathrm{i}\omega_n-\mathrm{i}\omega_{n'};\mathbf{k},\mathbf{k}+\mathbf{q})}{N_\mathrm{F}}
\tag{25.18}
$$

$$
\chi_n(\mathrm{i}\omega_n,\mathbf{k}) = -T \sum_{\substack{\mathbf{q}\in\mathrm{Q}\\ \omega_{n'}\in\Omega_{-1}\\ m\in\{1,\ldots,\infty\}}} \frac{\epsilon_m(\mathbf{k}+\mathbf{q})-\mu+\chi_m(\mathrm{i}\omega_{n'},\mathbf{k}+\mathbf{q})}{\theta_m(\mathrm{i}\omega_{n'},\mathbf{k}+\mathbf{q})} \frac{\lambda_{n,m}(\mathrm{i}\omega_n-\mathrm{i}\omega_{n'};\mathbf{k},\mathbf{k}+\mathbf{q})}{N_\mathrm{F}}
\tag{25.19}
$$

$$
\phi_n(\mathrm{i}\omega_n,\mathbf{k}) = T \sum_{\substack{\mathbf{q}\in\mathrm{Q}\\ \omega_{n'}\in\Omega_{-1}\\ m\in\{1,\ldots,\infty\}}} \frac{\phi_m(\mathrm{i}\omega_{n'},\mathbf{k}+\mathbf{q})}{\theta_m(\mathrm{i}\omega_{n'},\mathbf{k}+\mathbf{q})} \left[\frac{\lambda_{n,m}(\mathrm{i}\omega_n-\mathrm{i}\omega_{n'};\mathbf{k},\mathbf{k}+\mathbf{q})}{N_\mathrm{F}} + V_{n,m}(\mathrm{i}\omega_n-\mathrm{i}\omega_{n'};\mathbf{k},\mathbf{k}+\mathbf{q})\right]
\tag{25.20}
$$

Proof 25.14 See Appendix, Proofs, page 411.

Lemma 25.15 Fixing the number of particles n by demanding charge-neutrality yields the additional equation

$$n = 1 - 2T \sum_{\substack{\mathbf{q} \in \mathrm{Q} \\ \omega_{n'} \in \Omega_{-1} \\ m \in \{1,\dots,\infty\}}} \frac{\epsilon_m(\mathbf{k}+\mathbf{q}) - \mu + \chi_m(\mathrm{i}\omega_{n'}\mathbf{k}+\mathbf{q})}{\theta_m(\mathrm{i}\omega_{n'},\mathbf{k}+\mathbf{q})} \tag{25.21}$$

Proof 25.15 See Appendix, Proofs, page 412.

From here onwards everything should be fine again.

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25.1.3. Simplified Equations

For reasons that will be clear in the subsequent chapters, we consider the following limiting case.

Corollary 25.16 In the limiting case $Z \equiv 1$, $\chi \equiv 0$ we obtain, changing the notation $\phi \equiv \Delta$, the system of gap equations

$$\overline{\Delta}_n(k') = \sum_{\substack{k \in \Omega_{-1} \times \mathrm{Q} \\ \nu \in \{1,\ldots,rd_{\mathbf{x}}\} \\ m \in \{1,2,3,\ldots,\infty\}}} \frac{\overline{\Delta}_m(k)}{\omega_k^2 + \frac{\left(\epsilon_m(\mathbf{k}) - \mu\right)^2}{\hbar^2} + |\Delta_m(k)|^2} \frac{2\omega_\nu(\mathbf{k}-\mathbf{k}')}{\omega_{k-k'}^2 + \omega_\nu^2(\mathbf{k}-\mathbf{k}')} \frac{|g_{mn\nu}(\mathbf{k}', \mathbf{k}-\mathbf{k}')|^2}{\hbar^3 \beta N_p}$$

TBD: Multiplying with Plancks-constant such that the equations can be expressed on both sides in eV / meV.

Proof 25.16 See Appendix, Proofs, page 412.

Remark 25.16.1 Notice that we can get an upper bound for gap-function by introducing the maximal value

$$|\Delta_{\max}| = \max \left\{ |\Delta_m(k)| \quad | \quad k \in \Omega_{-1} \times \mathrm{Q} \wedge m \in \{1, 2, 3, \ldots, \infty\} \right\} \tag{25.22}$$

which allows us to estimate

$$\begin{aligned} 0 \leq |\Delta_m(k)| &\leq \sum_{\substack{k \in \Omega_{-1} \times \mathrm{Q} \\ \nu \in \{1,\ldots,rd_{\mathbf{x}}\} \\ m \in \{1,2,3,\ldots,\infty\}}} \frac{|\overline{\Delta}_m(k)|}{\omega_k^2 + \frac{\left(\epsilon_m(\mathbf{k}) - \mu\right)^2}{\hbar^2} + |\Delta_m(k)|^2} \frac{2\omega_\nu(\mathbf{k}-\mathbf{k}')}{\omega_{k-k'}^2 + \omega_\nu^2(\mathbf{k}-\mathbf{k}')} \frac{|g_{mn\nu}(\mathbf{k}', \mathbf{k}-\mathbf{k}')|^2}{\hbar^3 \beta N_p} \\ &\leq |\Delta_{\max}| \sum_{\substack{k \in \Omega_{-1} \times \mathrm{Q} \\ \nu \in \{1,\ldots,rd_{\mathbf{x}}\} \\ m \in \{1,2,3,\ldots,\infty\}}} \frac{1}{\omega_k^2 + \frac{\left(\epsilon_m(\mathbf{k}) - \mu\right)^2}{\hbar^2}} \frac{2\omega_\nu(\mathbf{k}-\mathbf{k}')}{\omega_{k-k'}^2 + \omega_\nu^2(\mathbf{k}-\mathbf{k}')} \frac{|g_{mn\nu}(\mathbf{k}', \mathbf{k}-\mathbf{k}')|^2}{\hbar^3 \beta N_p} \end{aligned} \tag{25.23}$$

which can be evaluated further with the Residue-theorem TBDone.

25.1.4. Derivation of the Self-Energy-Expressions

The main-result of this subsection is the following Theorem.

Theorem 25.17 With the equation of motion technique, one obtains the individual components of the Dyson-equation as

$$\mathscr{G}^{\;m}_{\text{Gorkov}}(k) = \mathscr{G}^{\;0m}_{\text{Gorkov}}(k) + \mathscr{G}^{\;0m}_{\text{Gorkov}}(k)\Bigg[-T \sum_{\substack{\mathbf{q}\in\mathrm{Q}\\ \omega_{n'}\in\Omega_{-1}\\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}\\ n\in\{1,\ldots,\infty\}}} \mathscr{G}^{\;n}_{\text{Gorkov}}(\mathrm{i}\omega_{n'},\mathbf{q}+\mathbf{k})|g_{mn\nu}(\mathbf{k},\mathbf{q})|^2\mathscr{D}_\nu(\mathrm{i}\omega_n-\mathrm{i}\omega_{n'},\mathbf{q})\mathscr{G}^{\;m}_{\text{Gorkov}}(k)$$
$$+T \sum_{\substack{\mathbf{q}\in\mathrm{Q}\\ \omega_{n'}\in\Omega_{-1}\\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}\\ n\in\{1,\ldots,\infty\}}} \mathscr{F}_n(\mathrm{i}\omega_{n'},\mathbf{q}+\mathbf{k})|g_{mn\nu}(\mathbf{k},\mathbf{q})|^2\mathscr{D}_\nu(\mathrm{i}\omega_n-\mathrm{i}\omega_{n'},\mathbf{q})\mathscr{F}^*_m(k)\Bigg] \tag{25.24}$$

$$\text{TBD} \tag{25.25}$$

$$\text{TBD} \tag{25.26}$$

$$\text{TBD} \tag{25.27}$$

Proof 25.17 Combining Equations (25.7) and (25.14) yields the desired result.

Remark 25.17.1 These equations are not implemented like this in the EPW-code, but are used in order to identify the self-energy in the next section.

Preparation 0/4: Commutators and Equations of Motion for Creation- and Annihilation-Operators

Lemma 25.18 One finds for the imaginary-time-evolution of an operator X the Heisenberg-equation of motion

$$\partial_\tau X_{\mathrm{H}}(\xi\tau) = -\frac{\xi}{\hbar}\left[H-\mu N, X\right]_{\mathrm{H}}(\xi\tau) + \xi\left(\partial_\tau X\right)_{\mathrm{H}}(\xi\tau) \tag{25.28}$$

Proof 25.18 See Appendix, Proofs, page 413.

Lemma 25.19 One finds the identities

$$\partial_\tau b_\nu(\xi\tau,\bullet) = -\xi\omega_\nu(\bullet)b_\nu(\xi\tau,\bullet) - \xi \sum_{\substack{\mathbf{k}\in \mathrm{Q}\\ \sigma\in\{\uparrow,\downarrow\}\\ m,n\in\{1,\ldots,\infty\}}} \frac{g_{mn\nu}(\mathbf{k},-\bullet)}{\hbar\sqrt{N_p}} c^\dagger_{m\sigma}(\xi\tau,\mathbf{k}-\bullet)c_{n\sigma}(\xi\tau,\mathbf{k})$$

$$\partial_\tau b^\dagger_\nu(\xi\tau,-\bullet) = +\xi\omega_\nu(-\bullet)b^\dagger_\nu(\xi\tau,-\bullet) + \xi \sum_{\substack{\mathbf{k}\in \mathrm{Q}\\ \sigma\in\{\uparrow,\downarrow\}\\ m,n\in\{1,\ldots,\infty\}}} \frac{g_{mn\nu}(\mathbf{k},-\bullet)}{\hbar\sqrt{N_p}} c^\dagger_{m\sigma}(\xi\tau,\mathbf{k}-\bullet)c_{n\sigma}(\xi\tau,\mathbf{k})$$

for $\xi \in \{-1,+1\}$.

Proof 25.19 See Appendix, Proofs, page 413.

Remark 25.19.1 Notice that we used in the proof that $\omega_\nu(\mathbf{q}) = \omega_\nu(-\mathbf{q})$, and hence used the time-reversal-symmetry of the crystal.

Remark 25.19.2 Notice further, that the $\xi\tau$ dependence of the operators on the right-hand side of the equations can be simplified to refer to the operator-*pair* $c^\dagger c$.

Preparation 1/4: Equations of Motion for the normal, spin-up Green-Function

Lemma 25.20 One finds for the Gorkov-Green-function

$$\left(\partial_\tau + \frac{\epsilon_n(\mathbf{k})-\mu}{\hbar}\right)\mathscr{G}_{\substack{n\\ \text{Gorkov}}}(\tau,\mathbf{k})$$
$$= -\frac{1}{\hbar}\delta(\tau) + \sum_{\substack{\mathbf{k}'\in \mathrm{Q}\\ n'\in\{1,\ldots,\infty\}\\ \nu'\in\{1,\ldots,rd_{\mathbf{x}}\}}} \frac{g_{nn'\nu'}(\mathbf{k}',\mathbf{k}-\mathbf{k}')}{\hbar^2\sqrt{N_p}} \left\langle \mathrm{T}_\tau\, c_{n',\uparrow}(\tau,\mathbf{k}')c^\dagger_{n,\uparrow}(\mathbf{k},0)\left(b_{\nu'}(\tau,\mathbf{k}-\mathbf{k}') + b^\dagger_{\nu'}(\tau,\mathbf{k}'-\mathbf{k})\right)\right\rangle_{\text{SC}} \tag{25.29}$$

Proof 25.20 See Appendix, Proofs, page 415.

Definition 25.21 We define *the auxiliary-Green-function* for $\tau,\tau' \in (0,\hbar\beta)$

$$\mathscr{G}_{\substack{\text{aux}\\ n,n',\nu'}}(\tau,\tau',\mathbf{k},\mathbf{k}') = -\frac{1}{\hbar}\left\langle \mathrm{T}_\tau\, c_{n',\uparrow}(\tau',\mathbf{k}')c^\dagger_{n,\uparrow}(0,\mathbf{k})\left(b_{\nu'}(\tau,\mathbf{k}-\mathbf{k}') + b^\dagger_{\nu'}(\tau,\mathbf{k}'-\mathbf{k})\right)\right\rangle_{\text{SC}} \tag{25.30}$$

Remark 25.21.1 Notice that this Green-function has, just like a "regular" Green-function the unit of $1/\text{Energy} \times \text{Time}$.

Remark 25.21.2 Notice that Equation (25.29) can now be rewritten as

$$\left(\partial_\tau + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar}\right) \mathscr{G}^{\,n}_{\text{Gorkov}}(\tau, \mathbf{k}) = -\frac{1}{\hbar}\delta(\tau) - \sum_{\substack{\mathbf{k}' \in \mathrm{Q} \\ n' \in \{1,\ldots,\infty\} \\ \nu \in \{1,\ldots,rd_{\mathbf{x}}\}}} \frac{g_{nn'\nu}(\mathbf{k}', \mathbf{k} - \mathbf{k}')}{\hbar\sqrt{N_p}} \mathscr{G}^{\text{aux}}_{n,n',\nu}(\tau, \tau, \mathbf{k}, \mathbf{k}') \tag{25.31}$$

Lemma 25.22 For $\mathbf{k} \neq \mathbf{k}'$ we obtain the equations for the first partial derivative

$$\partial_\tau \mathscr{G}^{\text{aux}}_{n,n',\nu'}(\tau, \tau', \mathbf{k}, \mathbf{k}') = +\frac{\omega_{\nu'}(\mathbf{k} - \mathbf{k}')}{\hbar} \left\langle \mathrm{T}_\tau \left(b_{\nu'}(\tau, \mathbf{k} - \mathbf{k}') - b^\dagger_{\nu'}(\tau, \mathbf{k}' - \mathbf{k})\right) c_{n',\uparrow}(\tau', \mathbf{k}') c^\dagger_{n,\uparrow}(0, \mathbf{k}) \right\rangle_{\text{SC}} \tag{25.32}$$

and for the second partial derivative[7]

$$\begin{aligned} &\left(\partial^2_\tau - \omega^2_{\nu'}(\mathbf{k} - \mathbf{k}')\right) \mathscr{G}^{\text{aux}}_{n,n',\nu'}(\tau, \tau', \mathbf{k}, \mathbf{k}') \\ &= \frac{-2\omega_{\nu'}(\mathbf{k} - \mathbf{k}')}{\hbar} \sum_{\substack{\mathbf{q} \in \mathrm{Q} \\ m,m' \in \{1,\ldots,\infty\} \\ \sigma \in \{\uparrow,\downarrow\}}} \frac{g_{mm'\nu'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar\sqrt{N_p}} \left\langle \mathrm{T}_\tau\, c^\dagger_{m,\sigma}(\tau, \mathbf{q} + \mathbf{k}' - \mathbf{k}) c_{m',\sigma}(\tau, \mathbf{q}) c_{n',\uparrow}(\tau', \mathbf{k}') c^\dagger_{n,\uparrow}(0, \mathbf{k}) \right\rangle_{\text{SC}} \end{aligned} \tag{25.33}$$

Proof 25.22 See Appendix, Proofs, page 416.

Remark 25.22.1 Notice that we used $\omega_\nu(\mathbf{q}) = \omega_\nu(-\mathbf{q})$, and hence used the time-reversal-symmetry of the crystal.

Remark 25.22.2 We now exchange[8] the roles of τ and τ' in the above equations. This yields for the two equations

$$\partial_{\tau'} \mathscr{G}^{\text{aux}}_{\nu',n,n'}(\tau', \tau, \mathbf{k}, \mathbf{k}') = +\frac{\omega_{\nu'}(\mathbf{k} - \mathbf{k}')}{\hbar} \left\langle \mathrm{T}_\tau \left(b_{\nu'}(\tau', \mathbf{k} - \mathbf{k}') - b^\dagger_{\nu'}(\tau', \mathbf{k}' - \mathbf{k})\right) c_{n',\uparrow}(\tau, \mathbf{k}') c^\dagger_{n,\uparrow}(0, \mathbf{k}) \right\rangle_{\text{SC}}$$

$$\begin{aligned} &\left(\partial^2_{\tau'} - \omega^2_{\nu'}(\mathbf{k} - \mathbf{k}')\right) \mathscr{G}^{\text{aux}}_{\nu',n,n'}(\tau', \tau, \mathbf{k}, \mathbf{k}') \\ &= \frac{-2\omega_{\nu'}(\mathbf{k} - \mathbf{k}')}{\hbar} \sum_{\substack{\mathbf{q} \in \mathrm{Q} \\ m,m' \in \{1,\ldots,\infty\} \\ \sigma \in \{\uparrow,\downarrow\}}} \frac{g_{mm'\nu'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar\sqrt{N_p}} \left\langle \mathrm{T}_\tau\, c^\dagger_{m,\sigma}(\tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}) c_{m',\sigma}(\tau', \mathbf{q}) c_{n',\uparrow}(\tau, \mathbf{k}') c^\dagger_{n,\uparrow}(0, \mathbf{k}) \right\rangle_{\text{SC}} \end{aligned} \tag{25.34}$$

[7] Note that this is in perfect harmony with Marsiglios article, except that he has typos in ω and g.
[8] Notice that this has no influence on the time-ordering-operator which has the unconvenient notation of a τ in its index.

Corollary 25.23 It follows for the equal-time Green-function[9] the equation

$$\begin{aligned}
&\mathscr{G}^{\text{aux}}_{n,n',\nu'}(\tau,\tau,\mathbf{k},\mathbf{k}') \\
&= \sum\!\!\!\!\!\!\int_{\substack{\mathbf{q}\in Q \\ \tau'\in[0,\hbar\beta] \\ m,m'\in\{1,\ldots,\infty\} \\ \sigma\in\{\uparrow,\downarrow\}}} -1\cdot \mathscr{D}_{\substack{0,\nu \\ \mathbf{k}-\mathbf{k}'}}(\tau-\tau')\frac{g_{mm'\nu'}(\mathbf{q},\mathbf{k}'-\mathbf{k})}{\hbar\sqrt{N_p}}\left\langle \mathrm{T}_\tau\, c^\dagger_{m,\sigma}(\tau',\mathbf{q}+\mathbf{k}'-\mathbf{k})c_{m',\sigma}(\tau',\mathbf{q})c_{n',\uparrow}(\tau,\mathbf{k}')c^\dagger_{n,\uparrow}(0,\mathbf{k})\right\rangle_{\text{SC}}
\end{aligned} \tag{25.35}$$

Proof 25.23 See Appendix, Proofs, page 418.

Remark 25.23.1 Notice that in this equation all the units are correct.

Remark 25.23.2 Notice that now the equation of motion (25.31) can be rewritten as

$$\begin{aligned}
&\left(\partial_\tau+\frac{\epsilon_n(\mathbf{k})-\mu}{\hbar}\right)\mathscr{G}^{\;n}_{\text{Gorkov}}(\tau,\mathbf{k}) \\
&= -\frac{1}{\hbar}\delta(\tau) \\
&+ \sum\!\!\!\!\!\!\int_{\substack{\mathbf{q},\mathbf{k}'\in Q \\ \tau'\in(0,\hbar\beta) \\ \nu'\in\{1,\ldots,rd_{\mathbf{x}}\} \\ m,m',n'\in\{1,\ldots,\infty\} \\ \sigma\in\{\uparrow,\downarrow\}}} \frac{g_{nn'\nu'}(\mathbf{k}',\mathbf{k}-\mathbf{k}')g_{mm'\nu'}(\mathbf{q},\mathbf{k}'-\mathbf{k})}{\hbar^2 N_p}\mathscr{D}_{\substack{0,\nu' \\ \mathbf{k}-\mathbf{k}'}}(\tau-\tau')\left\langle \mathrm{T}_\tau\, c^\dagger_{m,\sigma}(\tau',\mathbf{q}+\mathbf{k}'-\mathbf{k})c_{m',\sigma}(\tau',\mathbf{q})c_{n',\uparrow}(\tau,\mathbf{k}')c^\dagger_{n,\uparrow}(0,\mathbf{k})\right\rangle
\end{aligned} \tag{25.36}$$

Fact 25.24 The Bethe-Salpeter-equation (15.9), in which one neglects the vertex-correction, reads

$$\mathscr{G}(i_1,i_2|i_1',i_2') \approx \mathscr{G}(i_1|i_1')\mathscr{G}(i_2|i_2') + \zeta\mathscr{G}(i_1|i_2')\mathscr{G}(i_2|i_1') \tag{25.37}$$

The left hand side of this equation is

$$\mathscr{G}(i_1,i_2|i_1',i_2') = \left(-\frac{1}{\hbar}\right)^2\left\langle \mathrm{T}_\tau\, c_{i_1}c_{i_2}c^\dagger_{i_2'}c^\dagger_{i_1'}\right\rangle \tag{25.38}$$

Now, following Definition 25.38, we introduce the following notation

$$c^\dagger_{m\sigma}(\tau',\mathbf{q}+\mathbf{k}'-\mathbf{k}) = \begin{cases}\eta_{m,2}(\tau',-\mathbf{q}-\mathbf{k}'+\mathbf{k}) & \text{if } \sigma=\downarrow \\ \overline{\eta}_{m,1}(\tau',\mathbf{q}+\mathbf{k}'-\mathbf{k}) & \text{if } \sigma=\uparrow\end{cases} \tag{25.39}$$

and

[9] Notice that Marsiglio has exchanged values for the couplings and he has no global minus-sign, which is because we defined our auxiliary-Green-function with a global minus-sign, which is more consistent with the usual definition of Green-functions.

$$c_{m'\sigma}(\tau,\mathbf{q}) = \begin{cases} \overline{\eta}_{m',2}(\tau,-\mathbf{q}) & \text{if } \sigma = \downarrow \\ \eta_{m',1}(\tau,\mathbf{q}) & \text{if } \sigma = \uparrow \end{cases} \tag{25.40}$$

and hence also $c_{n'\uparrow}(\tau,\mathbf{k}') \equiv \eta_{n',1}(\tau,\mathbf{k}')$, and $c^\dagger_{n\uparrow}(0,\mathbf{k}) \equiv \overline{\eta}_{n,1}(0,\mathbf{k})$.

This gives thus for the sum from Equation (25.36)

$$\begin{aligned}
&\sum_{\sigma\in\{\downarrow,\uparrow\}} \Big\langle \mathrm{T}_\tau\, c^\dagger_{m,\sigma}(\tau',\mathbf{q}+\mathbf{k}'-\mathbf{k}) c_{m',\sigma}(\tau',\mathbf{q}) c_{n',\uparrow}(\tau,\mathbf{k}') c^\dagger_{n,\uparrow}(0,\mathbf{k}) \Big\rangle \\
&= \left\langle \mathrm{T}_\tau\, \eta_{m,2}(\tau',-\mathbf{q}-\mathbf{k}'+\mathbf{k}) \overline{\eta}_{m',2}(\tau',-\mathbf{q}) \eta_{n',1}(\tau,\mathbf{k}') \overline{\eta}_{n,1}(0,\mathbf{k}) \right\rangle \\
&\quad + \left\langle \mathrm{T}_\tau\, \overline{\eta}_{m,1}(\tau',+\mathbf{q}+\mathbf{k}'-\mathbf{k}) \eta_{m',1}(\tau',+\mathbf{q}) \eta_{n',1}(\tau,\mathbf{k}') \overline{\eta}_{n,1}(0,\mathbf{k}) \right\rangle \\
&= \left\langle \mathrm{T}_\tau\, \eta_{n',1}(\tau,\mathbf{k}') \eta_{m,2}(\tau',-\mathbf{q}-\mathbf{k}'+\mathbf{k}) \overline{\eta}_{m',2}(\tau',-\mathbf{q}) \overline{\eta}_{n,1}(0,\mathbf{k}) \right\rangle \\
&\quad + \left\langle \mathrm{T}_\tau\, \eta_{m',1}(\tau',+\mathbf{q}) \eta_{n',1}(\tau,\mathbf{k}') \overline{\eta}_{m,1}(\tau',+\mathbf{q}+\mathbf{k}'-\mathbf{k}) \overline{\eta}_{n,1}(0,\mathbf{k}) \right\rangle \\
&\equiv I + II
\end{aligned} \tag{25.41}$$

Now we use the decoupling for the $\sigma = \downarrow$ contribution (for the first of the two consecutive lines)

$$\begin{aligned}
\mathrm{I} &= \hbar^2 \mathscr{G}\left(i_1 = [n',1,\tau,\mathbf{k}'], \quad i_2 = [m,2,\tau',-\mathbf{q}-\mathbf{k}'+\mathbf{k}] \quad | \quad i'_1 = [n,1,0,\mathbf{k}], \quad i'_2 = [m',2,\tau',-\mathbf{q}] \right) \\
&\approx \hbar^2 \mathscr{G}_{\substack{n',n\\1,1}}(\tau-0;\mathbf{k}',\mathbf{k}) \mathscr{G}_{\substack{m,m'\\2,2}}(0;-\mathbf{q}-\mathbf{k}'+\mathbf{k},-\mathbf{q}) \quad - \quad \hbar^2 \mathscr{G}_{\substack{n',m'\\1,2}}(\tau-\tau';\mathbf{k}',-\mathbf{q}) \mathscr{G}_{\substack{m,n\\2,1}}(\tau';-\mathbf{q}-\mathbf{k}'+\mathbf{k},\mathbf{k}) \\
&= \hbar^2 \delta_{n,n'}\delta_{m,m'}\delta_{\mathbf{k},\mathbf{k}'} \mathscr{G}_{\substack{n\\1,1}}(\tau,\mathbf{k}) \mathscr{G}_{\substack{m\\2,2}}(0,-\mathbf{q}) \quad - \quad \hbar^2 \delta_{n',m'}\delta_{m,n}\delta_{\mathbf{k}',-\mathbf{q}} \mathscr{G}_{\substack{n'\\1,2}}(\tau-\tau',\mathbf{k}') \mathscr{G}_{\substack{m\\2,1}}(\tau',\mathbf{k}) \\
&= \hbar^2 - \delta_{n,n'}\delta_{m,m'}\delta_{\mathbf{k},\mathbf{k}'} \mathscr{G}_{\substack{n\\\text{Gorkov}}}(\tau,\mathbf{k}) \mathscr{G}_{\substack{m\\\text{Gorkov}}}(0,+\mathbf{q}) \quad - \quad \hbar^2 \delta_{n',m'}\delta_{m,n}\delta_{\mathbf{k}',-\mathbf{q}} \mathscr{F}_{n'}(\tau-\tau',\mathbf{k}') \mathscr{F}^\dagger_n(\tau',\mathbf{k})
\end{aligned}$$

whereas we used Definition 25.1 in the last equation. Now inserting this into the r.h.s. of Equation (25.36) yields (in addition to the delta-functions) two terms

$$\begin{aligned}
(I,1) &= -1 \cdot \sum\!\!\!\!\!\!\int_{\substack{\tau'\in(0,\hbar\beta)\\ \mathbf{q},\mathbf{k}'\in \mathrm{Q}\\ m,m',n'\in\{1,\dots,\infty\}\\ \nu'\in\{1,\dots,rd_{\mathbf{x}}\}}} \frac{g_{nn'\nu'}(\mathbf{k}',\mathbf{k}-\mathbf{k}') g_{mm'\nu'}(\mathbf{q},\mathbf{k}'-\mathbf{k})}{\hbar^2 N_p} \mathscr{D}_{\substack{0,\nu'\\ \mathbf{k}-\mathbf{k}'}}(\tau-\tau') \delta_{n,n'}\delta_{m,m'}\delta_{\mathbf{k},\mathbf{k}'} \mathscr{G}_{\substack{n\\\text{Gorkov}}}(\tau,\mathbf{k}) \mathscr{G}_{\substack{m\\\text{Gorkov}}}(0,+\mathbf{q})\hbar^2 \\
&= -1 \cdot \sum\!\!\!\!\!\!\int_{\substack{\tau'\in(0,\hbar\beta)\\ \mathbf{q}\in \mathrm{Q}\\ m\in\{1,\dots,\infty\}\\ \nu'\in\{1,\dots,rd_{\mathbf{x}}\}}} \frac{g_{nn\nu'}(\mathbf{k}',0) g_{mm\nu'}(\mathbf{q},0)}{N_p \hbar^2} \mathscr{D}_{\substack{0,\nu'\\ \mathbf{0}}}(\tau-\tau') \mathscr{G}_{\substack{n\\\text{Gorkov}}}(\tau,\mathbf{k}) \mathscr{G}_{\substack{m\\\text{Gorkov}}}(0,+\mathbf{q})\hbar^2
\end{aligned} \tag{25.42}$$

and the second term can be treated similarly, yielding

$$
\begin{aligned}
(I,2) &= -1 \cdot \sum\!\!\!\!\!\!\int_{\substack{\tau' \in (0,\hbar\beta)\\ \mathbf{q},\mathbf{k}' \in \mathrm{Q}\\ m,m',n' \in \{1,\ldots,\infty\}\\ \nu' \in \{1,\ldots,rd_{\mathbf{x}}\}}} \frac{g_{nn'\nu'}(\mathbf{k}', \mathbf{k}-\mathbf{k}') g_{mm'\nu'}(\mathbf{q}, \mathbf{k}'-\mathbf{k})}{\hbar^2 N_p} \mathscr{D}_{\substack{0,\nu'\\ \mathbf{k}-\mathbf{k}'}}(\tau-\tau') \delta_{n',m'} \delta_{m,n} \delta_{\mathbf{k}',-\mathbf{q}} \mathscr{F}_{n'}(\tau-\tau', \mathbf{k}') \mathscr{F}^{\dagger}_{n}(\tau', \mathbf{k}) \hbar^2 \\
&= -1 \cdot \sum\!\!\!\!\!\!\int_{\substack{\tau' \in (0,\hbar\beta)\\ \mathbf{q} \in \mathrm{Q}\\ n' \in \{1,\ldots,\infty\}\\ \nu' \in \{1,\ldots,rd_{\mathbf{x}}\}}} \frac{g_{nn'\nu'}(-\mathbf{q}, \mathbf{k}+\mathbf{q}) g_{nn'\nu'}(\mathbf{q}, -\mathbf{q}-\mathbf{k})}{N_p \hbar^2} \mathscr{D}_{\substack{0,\nu'\\ \mathbf{k}+\mathbf{q}}}(\tau-\tau') \mathscr{F}_{n'}(\tau-\tau', -\mathbf{q}) \mathscr{F}^{\dagger}_{n}(\tau', \mathbf{k}) \hbar^2 \\
&= -1 \cdot \sum\!\!\!\!\!\!\int_{\substack{\tau' \in (0,\hbar\beta)\\ \mathbf{q} \in \mathrm{Q}\\ n' \in \{1,\ldots,\infty\}\\ \nu' \in \{1,\ldots,rd_{\mathbf{x}}\}}} \frac{|g_{nn'\nu'}(\mathbf{q}, -\mathbf{q}-\mathbf{k})|^2}{N_p} \mathscr{D}_{\substack{0,\nu'\\ \mathbf{k}+\mathbf{q}}}(\tau-\tau') \mathscr{F}_{n'}(\tau-\tau', -\mathbf{q}) \mathscr{F}^{\dagger}_{n}(\tau', \mathbf{k})
\end{aligned}
\tag{25.43}
$$

whereas we terminated the summations over m and m' in the second last equation-sign and we used the inversion-symmetry for the couplings in the last equation-sign. Now we do the same procedure for the other term in Equation (25.41), i.e. the second line, giving

$$
\begin{aligned}
\mathrm{II} &= \hbar^2 \mathscr{G}\left(i_1 = \left[m', 1, \tau', \mathbf{q}\right], \quad i_2 = \left[n', 1, \tau, \mathbf{k}'\right] \quad | \quad i'_1 = \left[n, 1, 0, \mathbf{k}\right], \quad i'_2 = \left[m, 1, \tau', +\mathbf{q}+\mathbf{k}'-\mathbf{k}\right]\right) \\
&\approx \hbar^2 \mathscr{G}_{\substack{m',n\\ 1,1\\ \mathbf{q},\mathbf{k}}}(\tau'-0) \mathscr{G}_{\substack{n',m\\ 1,1\\ \mathbf{k}',+\mathbf{q}}}(\tau-\tau') \quad - \quad \hbar^2 \mathscr{G}_{\substack{m',m\\ 1,1\\ \mathbf{q},+\mathbf{q}}}(0) \mathscr{G}_{\substack{n',n\\ 1,1\\ \mathbf{k}',\mathbf{k}}}(\tau-0) \\
&= \hbar^2 \delta_{m',n} \delta_{n',m} \delta_{\mathbf{q},\mathbf{k}} \delta_{\mathbf{q},\mathbf{k}'} \mathscr{G}_{\substack{n\\ 1,1\\ \mathbf{k}}}(\tau') \mathscr{G}_{\substack{m\\ 1,1\\ +\mathbf{k}'}}(\tau-\tau') \quad - \quad \hbar^2 \delta_{m',m} \delta_{n',n} \delta_{\mathbf{q},+\mathbf{q}} \delta_{\mathbf{k}',\mathbf{k}} \mathscr{G}_{\substack{m\\ 1,1\\ +\mathbf{q}}}(0) \mathscr{G}_{\substack{n\\ 1,1\\ \mathbf{k}}}(\tau-0) \\
&= \hbar^2 \delta_{m',n} \delta_{n',m} \delta_{\mathbf{q},\mathbf{k}} \delta_{\mathbf{q},\mathbf{k}'} \mathscr{G}^{\;n}_{\mathrm{Gorkov}}(\tau', \mathbf{k}) \mathscr{G}^{\;m}_{\mathrm{Gorkov}}(\tau-\tau', +\mathbf{k}') \quad - \quad \hbar^2 \delta_{m',m} \delta_{n',n} \delta_{\mathbf{q},+\mathbf{q}} \delta_{\mathbf{k}',\mathbf{k}} \mathscr{G}^{\;m}_{\mathrm{Gorkov}}(0, +\mathbf{q}) \mathscr{G}^{\;n}_{\mathrm{Gorkov}}(\tau, \mathbf{k})
\end{aligned}
$$

whereas we used Definition 25.1 in the last equation.

Now we insert this into Equation (25.36), giving two additional terms

$$
\begin{aligned}
(II,3) &= \sum\!\!\!\!\!\!\int_{\substack{\tau' \in (0,\hbar\beta)\\ \mathbf{q},\mathbf{k}' \in \mathrm{Q}\\ m,m',n' \in \{1,\ldots,\infty\}\\ \nu' \in \{1,\ldots,rd_{\mathbf{x}}\}}} \frac{g_{nn'\nu'}(\mathbf{k}', \mathbf{k}-\mathbf{k}') g_{mm'\nu'}(\mathbf{q}, \mathbf{k}'-\mathbf{k})}{\hbar^2 N_p} \mathscr{D}_{\substack{0,\nu'\\ \mathbf{k}-\mathbf{k}'}}(\tau-\tau') \delta_{m',n} \delta_{n',m} \delta_{\mathbf{q},\mathbf{k}} \delta_{\mathbf{q},\mathbf{k}'} \mathscr{G}^{\;n}_{\mathrm{Gorkov}}(\tau', \mathbf{k}) \mathscr{G}^{\;m}_{\mathrm{Gorkov}}(\tau-\tau', +\mathbf{k}' \\
&= \sum\!\!\!\!\!\!\int_{\substack{\tau' \in (0,\hbar\beta)\\ \mathbf{k}' \in \mathrm{Q}\\ n' \in \{1,\ldots,\infty\}\\ \nu' \in \{1,\ldots,rd_{\mathbf{x}}\}}} \frac{g_{nn'\nu'}(\mathbf{k}', \mathbf{k}-\mathbf{k}') g_{n'n\nu'}(\mathbf{k}, \mathbf{k}'-\mathbf{k})}{N_p} \mathscr{D}_{\substack{0,\nu'\\ \mathbf{k}-\mathbf{k}'}}(\tau-\tau') \mathscr{G}^{\;n}_{\mathrm{Gorkov}}(\tau', \mathbf{k}) \mathscr{G}^{\;n'}_{\mathrm{Gorkov}}(\tau-\tau', +\mathbf{k}') \\
&= \sum\!\!\!\!\!\!\int_{\substack{\tau' \in (0,\hbar\beta)\\ \mathbf{k}' \in \mathrm{Q}\\ n' \in \{1,\ldots,\infty\}\\ \nu' \in \{1,\ldots,rd_{\mathbf{x}}\}}} \frac{|g_{n'n\nu'}(\mathbf{k}, \mathbf{k}'-\mathbf{k})|^2}{N_p} \mathscr{D}_{\substack{0,\nu'\\ \mathbf{k}-\mathbf{k}'}}(\tau-\tau') \mathscr{G}^{\;n}_{\mathrm{Gorkov}}(\tau', \mathbf{k}) \mathscr{G}^{\;n'}_{\mathrm{Gorkov}}(\tau-\tau', +\mathbf{k}')
\end{aligned}
\tag{25.44}
$$

whereas in the second last equation, we terminated the summations over m and m' and $\mathbf{q}$, and in the last equation we used the hermiticity of the couplings and something went wrong because we lost a Kronecker-delta. . And for the other term we obtain

$$
\begin{aligned}
(II,4) &= -1\cdot \sum\!\!\!\!\!\!\int_{\substack{\tau'\in(0,\hbar\beta)\\ \mathbf{q},\mathbf{k}'\in Q\\ m,m',n'\in\{1,\ldots,\infty\}\\ \nu'\in\{1,\ldots,rd_{\mathbf{x}}\}}} \frac{g_{nn'\nu'}(\mathbf{k}',\mathbf{k}-\mathbf{k}')g_{mm'\nu'}(\mathbf{q},\mathbf{k}'-\mathbf{k})}{\hbar^2 N_p}\mathscr{D}_{\substack{0,\nu'\\ \mathbf{k}-\mathbf{k}'}}(\tau-\tau')\delta_{m',m}\delta_{n',n}\delta_{\mathbf{q},+\mathbf{q}}\delta_{\mathbf{k}',\mathbf{k}}\mathscr{G}_{\substack{m\\ \text{Gorkov}}}(0,+\mathbf{q})\mathscr{G}_{\substack{n\\ \text{Gorkov}}}(\tau,\mathbf{k}\\
&= -1\cdot \sum\!\!\!\!\!\!\int_{\substack{\tau'\in(0,\hbar\beta)\\ \mathbf{q}\in Q\\ m\in\{1,\ldots,\infty\}\\ \nu'\in\{1,\ldots,rd_{\mathbf{x}}\}}} \frac{g_{nn\nu'}(\mathbf{k},0)g_{mm\nu'}(\mathbf{q},0)}{N_p}\mathscr{D}_{\substack{0,\nu'\\ 0}}(\tau-\tau')\mathscr{G}_{\substack{m\\ \text{Gorkov}}}(0,+\mathbf{q})\mathscr{G}_{\substack{n\\ \text{Gorkov}}}(\tau,\mathbf{k})
\end{aligned}
\tag{25.45}
$$

whereas we terminated the summations over m', n' and $\mathbf{k}'$ in the last equation.

Fact 25.25 Standard Eliashberg-theory now assumes $g(\mathbf{k},0) = 0$, i.e. restricting attention to systems with only one atom per unit cell.

Corollary 25.26 Combining the approximation of the Bethe-Salpether-equation with Remark 25.23.2 yields the equation of motion for the Green-function in imaginary time as

$$
\begin{aligned}
&\left(\partial_\tau + \frac{\epsilon_n(\mathbf{k})-\mu}{\hbar}\right)\mathscr{G}_{\substack{n\\ \text{Gorkov}}}(\tau,\mathbf{k})\\
&= -\frac{1}{\hbar}\delta(\tau)\\
&+ \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}'\in Q\\ \nu'\in\{1,\ldots,rd_{\mathbf{x}}\}\\ n'\in\{1,\ldots,\infty\}\\ \tau'\in(0,\hbar\beta)}} \frac{|g_{nn'\nu'}(\mathbf{k}',\mathbf{k}-\mathbf{k}')|^2}{N_p\hbar^2}\mathscr{D}_{\substack{0,\nu'\\ \mathbf{k}-\mathbf{k}'}}(\tau-\tau')\mathscr{G}_{\substack{n'\\ \text{Gorkov}}}(\tau-\tau',\mathbf{k}')\mathscr{G}_{\substack{n\\ \text{Gorkov}}}(\tau',\mathbf{k})\hbar^2\\
&- \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}'\in Q\\ \nu'\in\{1,\ldots,rd_{\mathbf{x}}\}\\ n'\in\{1,\ldots,\infty\}\\ \tau'\in(0,\hbar\beta)}} \frac{|g_{nn'\nu'}(\mathbf{k}',\mathbf{k}-\mathbf{k}')|^2}{N_p\hbar^2}\mathscr{D}_{\substack{0,\nu'\\ \mathbf{k}-\mathbf{k}'}}(\tau-\tau')\mathscr{F}_{n'}(\tau-\tau',\mathbf{k}')\mathscr{F}_n^\dagger(\tau',\mathbf{k})\hbar^2
\end{aligned}
\tag{25.46}
$$

Proof 25.26 See Appendix, Proofs, page 418.

Remark 25.26.1 Notice that the terms on the r.h.s. of this Equation are (II,3) and (I,2).

Remark 25.26.2 Notice that both sides of this Equation have the (correct) unit $\text{Energy}^{-1}\times\text{Time}^{-2}$.

Theorem 25.27

We collect the individual contributions and obtain

$$\left(\partial_\tau + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar}\right) \mathscr{G}^{\,n}_{\text{Gorkov}}(\tau, \mathbf{k}) = -\frac{1}{\hbar}\delta(\tau)$$
$$- \sum\!\!\!\!\!\!\int_{\substack{\mathbf{q}\in Q \\ \nu'\in\{1,\dots,rd_{\mathbf{x}}\} \\ n'\in\{1,\dots,\infty\} \\ \tau'\in(0,\hbar\beta)}} \left(|g_{nn'\nu'}(-\mathbf{q}, \mathbf{k}+\mathbf{q})|^2/N_p\hbar\right) \mathscr{D}_{\substack{0,\nu' \\ \mathbf{k}+\mathbf{q}}}(\tau-\tau')\mathscr{F}_{n'}(\tau-\tau', -\mathbf{q})\mathscr{F}^\dagger_n(\tau', \mathbf{k})\hbar^2$$
$$+ \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}'\in Q \\ \nu'\in\{1,\dots,rd_{\mathbf{x}}\} \\ n'\in\{1,\dots,\infty\} \\ \tau'\in(0,\hbar\beta)}} \left(|g_{n'n\nu'}(\mathbf{k}, \mathbf{k}'-\mathbf{k})|^2/N_p\hbar\right) \mathscr{D}_{\substack{0,\nu' \\ \mathbf{k}-\mathbf{k}'}}(\tau-\tau')\mathscr{G}^{\,n}_{\text{Gorkov}}(\tau', \mathbf{k})\mathscr{G}^{\,n'}_{\text{Gorkov}}(\tau-\tau', \mathbf{k}')\hbar^2$$
$$= \dots \tag{25.47}$$

(Notice that those were terms "2" and "4" in the enumeration earlier.) Now we replace $\mathbf{q} \mapsto -\mathbf{k}'$ in the first interaction-term and obtain

$$\dots = -\frac{1}{\hbar}\delta(\tau) + \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}'\in Q \\ \nu'\in\{1,\dots,rd_{\mathbf{x}}\} \\ n'\in\{1,\dots,\infty\} \\ \tau'\in(0,\hbar\beta)}} \hbar^2 \Big[|g_{nn'\nu'}(\mathbf{k}', \mathbf{k}-\mathbf{k}')|^2/(N_p\hbar)\mathscr{D}_{\substack{0,\nu' \\ \mathbf{k}-\mathbf{k}'}}(\tau-\tau')(-1)\mathscr{F}_{n'}(\tau-\tau', \mathbf{k}')\mathscr{F}^\dagger_n(\tau', \mathbf{k})$$
$$+ |g_{n'n\nu'}(\mathbf{k}, \mathbf{k}'-\mathbf{k})|^2/(N_p\hbar)\mathscr{D}_{\substack{0,\nu' \\ \mathbf{k}-\mathbf{k}'}}(\tau-\tau')\mathscr{G}^{\,n}_{\text{Gorkov}}(\tau', \mathbf{k})\mathscr{G}^{\,n'}_{\text{Gorkov}}(\tau-\tau', \mathbf{k}')\Big]$$
$$= \dots \tag{25.48}$$

and then we obtain

$$= -\frac{1}{\hbar}\delta(\tau)$$
$$+ \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}'\in Q \\ \nu'\in\{1,\dots,rd_{\mathbf{x}}\} \\ n'\in\{1,\dots,\infty\} \\ \tau'\in(0,\hbar\beta)}} \hbar^2 \left(|g_{nn'\nu'}(\mathbf{k}', \mathbf{k}-\mathbf{k}')/(N_p\hbar)\right) \mathscr{D}_{\substack{0,\nu' \\ \mathbf{k}-\mathbf{k}'}}(\tau-\tau')\Big[\mathscr{G}^{\,n}_{\text{Gorkov}}(\tau', \mathbf{k})\mathscr{G}^{\,n'}_{\text{Gorkov}}(\tau-\tau', \mathbf{k}') - \mathscr{F}_{n'}(\tau-\tau', \mathbf{k}')\mathscr{F}^\dagger_n(\tau', \mathbf{k}$$
$$\tag{25.49}$$

Going to Matsubara-space gives the equation of motion for the Green-function as

$$\left(-\hbar\mathscr{G}^{\,0,n}_{\text{Gorkov}}(\mathrm{i}\omega_{n_1}, \mathbf{k})\right)^{-1}\mathscr{G}^{\,n\uparrow}_{\text{Gorkov}}(\mathrm{i}\omega_{n_1}, \mathbf{k})$$
$$= -\frac{1}{\hbar} + \sum_{\substack{\omega_{n_2}\in\Omega_{-1} \\ \mathbf{k}'\in Q \\ n'\in\{1,\dots,\infty\} \\ \nu'\in\{1,\dots,rd_{\mathbf{x}}\}}} \hbar^2\frac{|g_{nn'\nu'}(\mathbf{k}', \mathbf{k}-\mathbf{k}')|^2}{N_p}\left(+\frac{1}{\beta}\right)\mathscr{D}_{\substack{0,\nu' \\ \mathbf{k}-\mathbf{k}'}}(\mathrm{i}(\omega_{n_1}-\omega_{n_2}))\Big[\quad \mathscr{G}^{\,n'}_{\text{Gorkov}}(\mathrm{i}\omega_{n_2}, \mathbf{k}')\mathscr{G}^{\,n}_{\text{Gorkov}}(\mathrm{i}\omega_{n_1}, \mathbf{k})$$
$$- \mathscr{F}_{n'}(\mathrm{i}\omega_{n_2}, \mathbf{k}')\mathscr{F}^\dagger_n(\mathrm{i}\omega_{n_1}, \mathbf{k})\Big]$$

Proof 25.27 See Appendix, Proofs, page 419.

Comparing this with Equation (25.7) as

$$\mathscr{G}_{n\mathbf{k}} = \mathscr{G}_{0,n\mathbf{k}} + \mathscr{G}_{0,n\mathbf{k}} \left[\underset{n\mathbf{k}}{\Sigma_{11}} \mathscr{G}_{n\mathbf{k}} + \underset{n\mathbf{k}}{\Sigma_{12}} \mathscr{F}^{\dagger}_{n\mathbf{k}} \right] \tag{25.50}$$

yields the components Σ_{11} and Σ_{12}

Preparation 2/4: Equations of Motion for the normal, spin-down Green-Function

To be done.

Preparation 3/4: Equations of Motion for the first anomalous Green-Function

Fact 25.28 The equation of motion for the anomalous Green-function in imaginary time reads

$$\partial_\tau \overline{\mathscr{F}}_n(\tau, \mathbf{k}) = TBD \tag{25.51}$$

Proof 25.28 TBD

Preparation 4/4: Equations of Motion for the other anomalous Green-Function

Fact 25.29 The equation of motion for the anomalous Green-function in imaginary time reads

$$\partial_\tau \mathscr{F}_n(\tau, \mathbf{k}) = TBD \tag{25.52}$$

Proof 25.29 TBD

25.2. Eliashberg-Theory in the Field Integral Formalism

We divide the whole procedure into several steps. All this time we will have a mean-field-theory for the bosonic field in mind (in analogy to Altlands description of superconductivity). The second quantized Hamiltonian of the underlying system is as in section 25.1.1.

25.2.1. The Electron-Electron-Interactions originating from the Phonons

Theorem 25.30 The effective electron-electron interaction can be decomposed into so-called *Spin-Singlet*-contributions („↓↑“) and two *Spin-Triplet*-contributions („↓↓“, „↑↑“) as

$$
\begin{aligned}
S_{\text{el-el-int}}[\psi,\overline{\psi}] &= \sum_{\alpha_1,\dots,\alpha_4\in\Xi} \overline{\psi}_{\alpha_1}\overline{\psi}_{\alpha_2} V_{\alpha_1,\alpha_2;\alpha_3,\alpha_4} \psi_{\alpha_4}\psi_{\alpha_3} \\
&= \sum_{\substack{q,k,k'\in Q\times\Omega_{-1}\\ m_{\alpha_1},\dots,m_{\alpha_4}\in\{1,2,3,\dots\}}} \overline{\psi}_{m_{\alpha_1}\uparrow}(q+k)\overline{\psi}_{m_{\alpha_2}\downarrow}(-k) V^{\downarrow\uparrow}_{m_{\alpha_1},m_{\alpha_2};m_{\alpha_3},m_{\alpha_4}}(q,k,k')\psi_{m_{\alpha_3}\downarrow}(q-k')\psi_{m_{\alpha_4}\uparrow}(k') \\
&+ \sum_{\substack{q,k,k'\in Q\times\Omega_{-1}\\ m_{\alpha_1},\dots,m_{\alpha_4}\in\{1,2,3,\dots\}}} \overline{\psi}_{m_{\alpha_1}\uparrow}(q+k)\overline{\psi}_{m_{\alpha_2}\uparrow}(-k) V^{\uparrow\uparrow}_{m_{\alpha_1},m_{\alpha_2};m_{\alpha_3},m_{\alpha_4}}(q,k,k')\psi_{m_{\alpha_3}\uparrow}(q-k')\psi_{m_{\alpha_4}\uparrow}(k') \\
&+ \sum_{\substack{q,k,k'\in Q\times\Omega_{-1}\\ m_{\alpha_1},\dots,m_{\alpha_4}\in\{1,2,3,\dots\}}} \overline{\psi}_{m_{\alpha_1}\downarrow}(q+k)\overline{\psi}_{m_{\alpha_2}\downarrow}(-k) V^{\downarrow\downarrow}_{m_{\alpha_1},m_{\alpha_2};m_{\alpha_3},m_{\alpha_4}}(q,k,k')\psi_{m_{\alpha_3}\downarrow}(q-k')\psi_{m_{\alpha_4}\downarrow}(k') \\
&= S_{\downarrow\uparrow}[\psi,\overline{\psi}] + S_{\uparrow\uparrow}[\psi,\overline{\psi}] + S_{\downarrow\downarrow}[\psi,\overline{\psi}]
\end{aligned}
$$

with spin-triplet vertices as above and spin-singlet matrix-elements (after adding up and collecting the individual contributions) given as

$$
\begin{aligned}
&2V^{\sigma\sigma}_{m_{\alpha_1},m_{\alpha_2};m_{\alpha_3},m_{\alpha_4}}(q,k,k') \\
&= V^{\downarrow\uparrow}_{m_{\alpha_1},m_{\alpha_2};m_{\alpha_3},m_{\alpha_4}}(q,k,k') \\
&= \frac{-1}{\hbar\beta N_p}\sum_{\nu\in\{1,\dots,rd_{\mathbf{x}}\}} \frac{2\omega_\nu(\mathbf{k}+\mathbf{q}-\mathbf{k}')}{\omega^2_{k+q-k'}+\omega^2_\nu(\mathbf{k}+\mathbf{q}-\mathbf{k}')} \frac{g_{m_{\alpha_1}m_{\alpha_4}\nu}(\mathbf{k}',\mathbf{k}+\mathbf{q}-\mathbf{k}')g^*_{m_{\alpha_3}m_{\alpha_2}\nu}(-\mathbf{k},\mathbf{k}+\mathbf{q}-\mathbf{k}')}{\hbar^2} \\
&= \frac{-1}{\hbar\beta N_p}\sum_{\nu\in\{1,\dots,rd_{\mathbf{x}}\}} \frac{2\omega_\nu(\mathbf{k}+\mathbf{q}-\mathbf{k}')}{\omega^2_{k+q-k'}+\omega^2_\nu(\mathbf{k}+\mathbf{q}-\mathbf{k}')} \frac{g_{m_{\alpha_1}m_{\alpha_4}\nu}(\mathbf{k}',\mathbf{k}+\mathbf{q}-\mathbf{k}')g_{m_{\alpha_2}m_{\alpha_3}\nu}(\mathbf{q}-\mathbf{k}',-(\mathbf{k}+\mathbf{q}-\mathbf{k}'))}{\hbar^2}
\end{aligned}
\tag{25.53}
$$

Proof 25.30 See Appendix, Proofs, page 422.

Lemma 25.31 The vertices in Equation (25.53) are lesser than zero iff

- either $\mathbf{k}=-\mathbf{k}'$ AND $m_{\alpha_1}=m_{\alpha_3}$ AND $m_{\alpha_4}=m_{\alpha_2}$ (upper equation - „[N]on-[E]liashberg-[T]heory“ cases),
- or $\mathbf{q}=0$ AND $m_{\alpha_1}=m_{\alpha_2}$ AND $m_{\alpha_4}=m_{\alpha_3}$ (lower equation - „[E]liashberg-[T]heory“ cases).

Proof 25.31 Referring to Equations (8.35), (8.36) proves the theorem.

Remark 25.31.1 Notice that the BCS-action in Equation (34.4) turns out to be a special case of the spin-singlet contributions in the limit of constant vertices.

Theorem 25.32 We decompose the spin-singlet contributions as

$$S[\psi,\overline{\psi}] \equiv S^{\mathrm{ET}}[\psi,\overline{\psi}] + S^{\mathrm{NET}}[\psi,\overline{\psi}] + S^{\mathrm{RPC}}[\psi,\overline{\psi}] \tag{25.54}$$

whereas for the ([N]on-)[E]liashberg-[T]heory contribution

$$\begin{aligned} S^{\mathrm{ET}}[\psi,\overline{\psi}] = & \sum_{\substack{k,k' \in \Omega_{-1}\times \mathrm{Q} \\ m,n \in \{1,2,3\ldots\}}} \overline{\psi}_{m\uparrow}(k)\overline{\psi}_{m\downarrow}(-k) V^{\mathrm{ET}}_{\substack{\downarrow\uparrow \\ m,n}}(k,k')\psi_{n\downarrow}(-k')\psi_{n\uparrow}(k') \\ & + \sum_{\substack{\sigma\in\{\downarrow,\uparrow\} \\ k,k' \in \Omega_{-1}\times \mathrm{Q} \\ m,n \in \{1,2,3\ldots\}}} \overline{\psi}_{m\sigma}(k)\overline{\psi}_{m\sigma}(-k) V^{\mathrm{ET}}_{\substack{\sigma\sigma \\ m,n}}(k,k')\psi_{n\sigma}(-k')\psi_{n\sigma}(k') \end{aligned}$$

whereas the first term produces the off-diagonal contributions to the Gorkov-Green-functions.

$$\begin{aligned} S^{\mathrm{NET}}_{\uparrow\downarrow}[\psi,\overline{\psi}] = & \sum_{\substack{q,k \in \mathrm{i}\Omega_{-1}\times \mathrm{Q} \\ m,n \in \{1,2,3\ldots\}}} \overline{\psi}_{m\uparrow}(q)\psi_{m\downarrow}(q) V^{\mathrm{NET}}_{\substack{\downarrow\uparrow \\ m,n}}(q+k,-k)\psi_{n\uparrow}(k)\overline{\psi}_{n\downarrow}(k) \\ & + \sum_{\substack{\sigma\in\{\downarrow,\uparrow\} \\ q,k \in \mathrm{i}\Omega_{-1}\times \mathrm{Q} \\ m,n \in \{1,2,3\ldots\}}} \overline{\psi}_{m\sigma}(q)\psi_{m\sigma}(q) V^{\mathrm{NET}}_{\substack{\sigma\sigma \\ m,n}}(q-k,k)\psi_{n\sigma}(-k)\overline{\psi}_{n\sigma}(-k) \end{aligned}$$

whereas the terms in the second line should be able to produce the diagonal contributions to the Gorkov-Green-functions. The remaining terms are the [R]andom [P]hase [C]ancellations. (We assume them to be random phases that don't contribute to the physical observables.) Then we have, when we take the double contributions into account, the vertices

$$2V^{\mathrm{ET}}_{\substack{\sigma\sigma \\ m,n}}(k,k') = V^{\mathrm{ET}}_{\substack{\downarrow\uparrow \\ m,n}}(k,k') = \frac{-2}{\hbar\beta N_p} \sum_{\nu\in\{1,\ldots,rd_{\mathbf{x}}\}} \frac{\omega_\nu(\mathbf{k}-\mathbf{k}')}{\omega^2_{k-k'} + \omega^2_\nu(\mathbf{k}-\mathbf{k}')} \frac{|g_{mn\nu}(\mathbf{k}',\mathbf{k}-\mathbf{k}')|^2}{\hbar^2} \tag{25.55}$$

$$2V^{\mathrm{NET}}_{\substack{\sigma\sigma \\ m,n}}(q+k,-k) = V^{\mathrm{NET}}_{\substack{\downarrow\uparrow \\ m,n}}(q+k,-k) = \frac{-2}{\hbar\beta N_p} \sum_{\nu\in\{1,\ldots,rd_{\mathbf{x}}\}} \frac{\omega_\nu(-\mathbf{k}+\mathbf{q})}{\omega^2_{-k+q} + \omega^2_\nu(-\mathbf{k}+\mathbf{q})} \frac{|g_{mn\nu}(+\mathbf{k},-\mathbf{k}+\mathbf{q})|^2}{\hbar^2} \tag{25.56}$$

Proof 25.32 See Appendix, Proofs, page 425.

25.2.2. Electron-Electron-Interactions for Spin-Singlet Eliashberg-Theory

In order to prove the validity of our approach, we derive the [E]liashberg-[T]heory-equations from a limiting case of our theory.

Approximation 25.33 We approximate the interactions by focusing on the ET-↓↑-contributions and neglecting the other terms as

$$S_{\text{el-el-int}}[\psi,\overline{\psi}] = S^{\text{ET}}_{\downarrow\uparrow}[\psi,\overline{\psi}] + S_{\Delta V}[\psi,\overline{\psi}] \approx S^{\text{ET}}_{\downarrow\uparrow}[\psi,\overline{\psi}]$$

with the Eliashberg contribution

$$S^{\text{ET}}_{\downarrow\uparrow}[\psi,\overline{\psi}] = \sum_{\substack{k,k' \in \mathrm{i}\Omega_{-1}\times\mathrm{Q} \\ m,n \in \{1,2,3\dots\}}} \overline{\psi}_{m\uparrow}(k)\overline{\psi}_{m\downarrow}(-k) \underset{m,n}{V^{\text{ET}}_{\downarrow\uparrow}}(k,k')\psi_{n\downarrow}(-k')\psi_{n\uparrow}(k')$$

and the vertex

$$\underset{m,n}{V^{\text{ET}}_{\downarrow\uparrow}}(k,k') = \frac{-2}{\hbar\beta N_p} \sum_{\nu\in\{1,\dots,rd\}} \frac{\omega_\nu(\mathbf{k}-\mathbf{k}')}{\omega^2_{k-k'} + \omega^2_\nu(\mathbf{k}-\mathbf{k}')} \frac{|g_{mn\nu}(\mathbf{k}',\mathbf{k}-\mathbf{k}')|^2}{\hbar^2} \tag{25.57}$$

Notation 25.34 We condense the notation even further, by introducing the multi-index

$$A \equiv (k,m,\downarrow,\uparrow) \in \Pi, \quad B \equiv (k',n;\downarrow,\uparrow) \in \Pi \quad \text{with} \quad \Pi \equiv \mathrm{i}\Omega_{-1} \times \mathrm{Q} \times \{1,2,3,\dots\} \times \{\downarrow\} \times \{\uparrow\} \tag{25.58}$$

and density-like terms

$$\overline{\rho}(A) \equiv \overline{\psi}_{m\uparrow}(k)\overline{\psi}_{m\downarrow}(-k), \qquad \rho(B) \equiv \psi_{n\downarrow}(-k')\psi_{n\uparrow}(k') \tag{25.59}$$

with $\rho(A) = \psi_{m\downarrow}(-k)\psi_{m\uparrow}(k)$.

Remark 25.34.1 Notice that with this, we can rewrite the Eliashberg-interaction term as a bilinear

$$S^{\text{ET}}_{\downarrow\uparrow}[\psi,\overline{\psi}] = \sum_{A,B\in\Pi} \overline{\rho}(A) V^{\text{ET}}_{\downarrow\uparrow}(A,B)\rho(B) \tag{25.60}$$

25.2.3. Hubbard-Stratanovich-Transformation

We now exchange the quartic interaction for a quadratic-part through the Cooper-channel. We therefore have to find the inverse of $V^{\text{ET}}_{\downarrow\uparrow}$.

Hypothesis 25.35 We assume, the existence of another generalised matrix $\left(V^{\text{ET}}_{\downarrow\uparrow}\right)^{-1} \equiv \Lambda^{\text{ET}}_{\downarrow\uparrow}$ such that the inverse relation holds

$$\left(V^{\text{ET}}_{\downarrow\uparrow} \cdot \Lambda^{\text{ET}}_{\downarrow\uparrow}\right)(A, A') = \sum_{B} V^{\text{ET}}_{\downarrow\uparrow}(A, B)\Lambda^{\text{ET}}_{\downarrow\uparrow}(B, A') = \delta(A, A'). \tag{25.61}$$

Remark 25.35.1 For the moment, we will not bother with the concrete shape of this inverse matrix. We simply assume its existence and give it a name to perform further mathematical operations.

Lemma 25.36 Performing a Hubbard-Stratanovich-transformation of the Eliashberg-interaction term through the Cooper-channel yields

$$\begin{aligned}
\exp -S^{\text{ET}}_{\downarrow\uparrow}[\psi, \overline{\psi}] &= \exp - \sum_{A,B\,\in\,\Pi^2} \overline{\rho}(A)V^{\text{ET}}_{\downarrow\uparrow}(A, B)\rho(B) \\
&= \int \mathscr{D}(\Delta_{\downarrow\uparrow}, \overline{\Delta}_{\downarrow\uparrow}) \exp + \left[\sum_{A,B\,\in\,\Pi^2} \overline{\Delta}(A)\Lambda^{\text{ET}}_{\downarrow\uparrow}(A, B)\Delta(B) - \sum_{A\,\in\,\Pi} \left(\overline{\Delta}(A)\rho(A) + \overline{\rho}(A)\Delta(A)\right)\right] \\
&= \int \mathscr{D}(\Delta_{\downarrow\uparrow}, \overline{\Delta}_{\downarrow\uparrow}) \exp - \left[S_{\text{aux.}}[\Delta_{\downarrow\uparrow}, \overline{\Delta}_{\downarrow\uparrow}] + S_{\text{int}}[\Delta_{\downarrow\uparrow}, \overline{\Delta}_{\downarrow\uparrow}, \psi, \overline{\psi}]\right].
\end{aligned}$$

Proof 25.36 Shifting variables according to

$$\Delta(A) \mapsto \Delta(A) + \sum_{B\,\in\,\Pi} V^{\text{ET}}_{\downarrow\uparrow}(A, B)\rho(B), \qquad \overline{\Delta}(A) \mapsto \overline{\Delta}(A) + \sum_{B\,\in\,\Pi} \overline{\rho}(B)V^{\text{ET}}_{\downarrow\uparrow}(B, A) \tag{25.62}$$

proves the Lemma.

Fact 25.37 The full action, in absense of external sources, now reads

$$S_{\text{tot}}[\Delta_{\downarrow\uparrow}, \overline{\Delta}_{\downarrow\uparrow}, \psi, \overline{\psi}] = S_{\text{el}}[\psi, \overline{\psi}] + S_{\text{aux}}[\Delta_{\downarrow\uparrow}, \overline{\Delta}_{\downarrow\uparrow}] + S_{\text{int}}[\Delta_{\downarrow\uparrow}, \overline{\Delta}_{\downarrow\uparrow}, \psi, \overline{\psi}] \tag{25.63}$$

with

- $S_{\text{el}}[\psi, \overline{\psi}] \equiv \sum_{\substack{\sigma\,\in\,\{\downarrow,\uparrow\} \\ k\,\in\,\mathrm{i}\Omega_{-1}\times\mathrm{Q} \\ m\,\in\,\{1,2,3,\ldots\}}} \overline{\psi}_{m\sigma}(k) \left(-\frac{1}{\hbar}\mathscr{G}^{-\mathrm{F}}_{\substack{0 \\ m,\sigma}}(k)\right) \psi_{m\sigma}(k)$
- $S_{\text{aux}}[\Delta_{\downarrow\uparrow}, \overline{\Delta}_{\downarrow\uparrow}] = - \sum_{A,B\,\in\,\Pi^2} \overline{\Delta}(A)\Lambda^{\text{ET}}_{\downarrow\uparrow}(A, B)\Delta(B)$

- $S_{\mathrm{int}}[\Delta_{\downarrow\uparrow}, \overline{\Delta}_{\downarrow\uparrow}, \psi, \overline{\psi}] = \sum_{\substack{k \in \mathrm{i}\Omega_{-1} \times \mathrm{Q} \\ m \in \{1,2,3,\dots\}}} \overline{\Delta}_{m\downarrow\uparrow}(k)\psi_{m\downarrow}(-k)\psi_{m\uparrow}(k) + \overline{\psi}_{m\uparrow}(k)\overline{\psi}_{m\downarrow}(-k)\Delta_{m\downarrow\uparrow}(k)$

Definition 25.38 We now introduce Nambu-spinors as

$$\eta^{\dagger}_{m}(k) = \begin{pmatrix} \overline{\psi}_{m,\uparrow}(k) & \psi_{m,\downarrow}(-k) \end{pmatrix}, \qquad \eta_{m'}(k') = \begin{pmatrix} \psi_{m'\uparrow}(k') \\ \overline{\psi}_{m'\downarrow}(-k') \end{pmatrix} \tag{25.64}$$

Remark 25.38.1 Notice that this allows us to rewrite the electronic contributions as

$$\begin{aligned} S_{\mathrm{el}}[\psi, \overline{\psi}] + S_{\mathrm{int}}[\Delta_{\downarrow\uparrow}, \overline{\Delta}_{\downarrow\uparrow}, \psi, \overline{\psi}] &= \sum_{\substack{k \in \mathrm{i}\Omega_{-1} \times \mathrm{Q} \\ m \in \{1,2,\dots\}}} \begin{pmatrix} \overline{\psi}_{m,\uparrow}(k) \\ \psi_{m,\downarrow}(-k) \end{pmatrix}^{\mathrm{T}} \begin{pmatrix} -\frac{1}{\hbar}\mathscr{G}^{-\mathrm{F}}_{\substack{0 \\ m\uparrow}}(k) & \Delta_{m\downarrow\uparrow}(k) \\ \overline{\Delta}_{m\downarrow\uparrow}(k) & +\frac{1}{\hbar}\mathscr{G}^{-\mathrm{F}}_{\substack{0 \\ m\downarrow}}(-k) \end{pmatrix} \begin{pmatrix} \psi_{m\uparrow}(k) \\ \overline{\psi}_{m\downarrow}(-k) \end{pmatrix} \\ &= \sum_{\substack{k \in \mathrm{i}\Omega_{-1} \times \mathrm{Q} \\ m \in \{1,2,\dots\}}} \overline{\eta}^{\mathrm{T}}_{m}(k) \left[-\frac{1}{\hbar} \mathscr{G}^{-1}_{\substack{m \\ [\Delta_{\downarrow\uparrow}, \overline{\Delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}(k) \right] \eta_{m}(k) \end{aligned}$$

Lemma 25.39 One finds for the imaginary time components of the Nambu-spinors the identity in imaginary time space

$$\overline{\eta}^{\mathrm{T}}_{m}(\tau, \mathbf{k}) = \begin{pmatrix} \overline{\psi}_{m\uparrow}(\tau, \mathbf{k}) & \psi_{m\downarrow}(\tau, -\mathbf{k}) \end{pmatrix} \qquad \eta_{m}(\tau, \mathbf{k}) = \begin{pmatrix} \psi_{m\uparrow}(\tau, \mathbf{k}) \\ \overline{\psi}_{m\downarrow}(\tau, -\mathbf{k}) \end{pmatrix} \tag{25.65}$$

Proof 25.39 See Appendix, Proofs, page 426.

Corollary 25.40 We note the correlators

$$\left\langle \eta_{m1}(\tau,\mathbf{k})\overline{\eta}_{m,1}(\tau',\mathbf{k})\right\rangle = \left\langle \psi_{m\uparrow}(\tau,\mathbf{k})\overline{\psi}_{m,\uparrow}(\tau',\mathbf{k})\right\rangle = -\hbar \mathscr{G}^{\ \ m}_{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov},11}}(\tau-\tau',\mathbf{k}) = -\hbar \mathscr{G}^{\ \ m\uparrow}_{\text{Gorkov}}(\tau-\tau',\mathbf{k}) \tag{25.66}$$

$$\left\langle \eta_{m1}(\tau,\mathbf{k})\overline{\eta}_{m,2}(\tau',\mathbf{k})\right\rangle = \left\langle \psi_{m\uparrow}(\tau,\mathbf{k})\psi_{m,\uparrow}(\tau',-\mathbf{k})\right\rangle = -\hbar \mathscr{G}^{\ \ m}_{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov},12}}(\tau-\tau',\mathbf{k}) = -\hbar \mathscr{F}_m(\tau-\tau',\mathbf{k}) \tag{25.67}$$

$$\left\langle \eta_{m2}(\tau,\mathbf{k})\overline{\eta}_{m,1}(\tau',\mathbf{k})\right\rangle = \left\langle \overline{\psi}_{m\downarrow}(\tau,-\mathbf{k})\overline{\psi}_{m,\uparrow}(\tau',\mathbf{k})\right\rangle = -\hbar \mathscr{G}^{\ \ m}_{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov},21}}(\tau-\tau',\mathbf{k}) = -\hbar \mathscr{F}^{\dagger}_m(\tau-\tau',\mathbf{k}) \tag{25.68}$$

$$\left\langle \eta_{m2}(\tau,\mathbf{k})\overline{\eta}_{m,2}(\tau',\mathbf{k})\right\rangle = \left\langle \overline{\psi}_{m\downarrow}(\tau,-\mathbf{k})\psi_{m,\downarrow}(\tau',-\mathbf{k})\right\rangle = -\hbar \mathscr{G}^{\ \ m}_{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov},22}}(\tau-\tau',\mathbf{k}) = -\hbar\left(-\mathscr{G}^{\ \ m\downarrow}_{\text{Gorkov}}(\tau'-\tau,-\mathbf{k})\right) \tag{25.69}$$

Proof 25.40 See Appendix, Proofs, page 426.

Lemma 25.41 Integrating out the fermions gives us an effective bosonic theory

$$S_{\text{tot}}[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}] \quad = \quad S_{\text{aux.}}[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}] \quad - \quad \operatorname{Tr}\operatorname{Log}-\frac{1}{\hbar}\mathscr{G}^{-1}_{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov}}} \tag{25.70}$$

Proof 25.41 By using standard-identities about Gauss-Berezin-integrals and using the standard-identity Tr Log $\bullet$ = Log Det $\bullet$ gives us the desired identity.

25.2.4. Spontaneous Symmetry Breaking and Gap-Equations

We now break the U(1) symmetry by applying a saddle-point approximation and keeping only one contribution, analogous to section 34.2.5.

Lemma 25.42 We note the identity

$$\frac{\delta}{\delta\Delta(\bullet)}\operatorname{Tr}\operatorname{Log}-\frac{1}{\hbar}\mathscr{G}^{-1}_{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov}}} = +\operatorname{Tr}\mathscr{G}_{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov}}}\frac{\delta}{\delta\Delta(\bullet)}\mathscr{G}^{-1}_{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov}}} \tag{25.71}$$

Proof 25.42 With Altlands prescription: We define $f(x) = \operatorname{Log}\left(-\frac{1}{\hbar}x\right)$, such that we obtain $f'(x) = \frac{1}{x}$. This yields for a x-dependent matrix $A \equiv A(x)$ the identity

$$\partial_x \operatorname{Tr} f(A) = \operatorname{Tr}\left(A^{-1}\partial_x A\right). \tag{25.72}$$

Corollary 25.43 The stationary points of the action are determined by the set of coupled equations

$$\overline{\Delta}_{n\downarrow\uparrow}(k') = + \sum_{\substack{k\in \mathrm{i}\Omega_{-1}\times \mathrm{Q} \\ \nu\in\{1,\ldots,rd\} \\ m\in\{1,2,3,\ldots\}}} \mathrm{Tr}_{2\times 2}\left(\mathscr{G}_{\substack{m \\ [\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}} \frac{\delta}{\delta \Delta_{m\downarrow\uparrow}(k)} \mathscr{G}^{-1}_{\substack{m \\ [\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}} \right) \frac{2\omega_\nu(\mathbf{k}-\mathbf{k}')}{\omega_{k-k'}^2 + \omega_\nu^2(\mathbf{k}-\mathbf{k}')} \frac{|g_{mn\nu}(\mathbf{k}',\mathbf{k}-\mathbf{k}')|^2}{\hbar^3 \beta N_p}$$

Proof 25.43 See Appendix, Proofs, page 427.

Lemma 25.44 One has the identities

$$\frac{\delta}{\delta \Delta_{m\downarrow\uparrow}(k)} \mathscr{G}^{-1}_{\substack{m \\ [\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}} = \begin{pmatrix} 0 & -\hbar \\ 0 & 0 \end{pmatrix} \tag{25.73}$$

and

$$\mathscr{G}_{\substack{m \\ [\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}(k) = -\frac{1}{\hbar} \frac{\begin{pmatrix} +\mathrm{i}\omega_k + (\epsilon_m(-\mathbf{k}) - \mu)/\hbar & +\Delta_{m\downarrow\uparrow}(k) \\ +\overline{\Delta}_{m\downarrow\uparrow}(k) & +\mathrm{i}\omega_k - (\epsilon_m(\mathbf{k}) - \mu)/\hbar \end{pmatrix}}{+\omega_k^2 + \dfrac{\left(\epsilon_m(\mathbf{k}) - \mu\right)^2}{\hbar^2} + |\Delta_{m\downarrow\uparrow}(k)|^2} \tag{25.74}$$

Proof 25.44 See Appendix, Proofs, page 427.

Notice that we are now in the position to prove

Lemma 25.45 Remark **??** is true, i.e. it holds that

$$\mathscr{G}_{\substack{m,12 \\ [\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}(A, A')^\dagger = \mathscr{G}_{\substack{m,21 \\ [\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}(A, A') \tag{25.75}$$

Proof 25.45 See Appendix, Proofs, page 428.

Theorem 25.46 The gap equations become

$$\overline{\Delta}_{n\downarrow\uparrow}(k') = + \sum_{\substack{k\in \mathrm{i}\Omega_{-1}\times \mathrm{Q} \\ \nu\in\{1,\ldots,rd\} \\ m\in\{1,2,3,\ldots\}}} \frac{\overline{\Delta}_{m\downarrow\uparrow}(k)}{\omega_k^2 + \dfrac{\left(\epsilon_m(\mathbf{k}) - \mu\right)^2}{\hbar^2} + |\Delta_{m\downarrow\uparrow}(k)|^2} \frac{2\omega_\nu(\mathbf{k}-\mathbf{k}')}{\omega_{k-k'}^2 + \omega_\nu^2(\mathbf{k}-\mathbf{k}')} \frac{|g_{mn\nu}(\mathbf{k}',\mathbf{k}-\mathbf{k}')|^2}{\hbar^3 \beta N_p}.$$

Proof 25.46 See Appendix, Proofs, page 429.

Remark 25.46.1 This result coincides perfectly with the limiting equations from the EPW-code in Equation (TBReferenced).

26. Material Analysis with Quantum Espresso & Wannier90 & Wannier-Tools & EPW & GRIT

We consider only stochiometric crystals, i.e. crystals with a fixed number of atoms per unit cell and a fixed number of valence electrons per atom.

Introduction

Fact 26.1 For fixed (averaged) particle-number and volume we describe a crystal as a system of interacting phonons (massless particles) and electrons (massive particles). We work in the grand canonical ensemble with natural variables depending on the nature of the subsystem of interest, meaning that …

- *for the phononic part of the system*, we are working in the grand canonical ensemble with natural variables (T, V),
- *for the electronic part of the system*, we are working in the grand canonical ensemble with natural variables (μ, T, V).

Fact 26.2 Notice first that for spin-unpolarised systems (i.e. no spin-dependent exchange-interaction or spin-orbit-coupling) one finds the inequality

$$\text{"number of at least partially occupied bands"} \times 2 \quad \geq \quad \text{"number of electrons per unit cell".}$$

whereas spin-degenerate bands are counted once. In the more general notation (i.e. counting two degenerate bands as band nr. 1 and band nr. 2) or in the case of spin-polarised systems, one finds the inequality

$$\text{"number of at least partially occupied bands"} \quad \geq \quad \text{"number of electrons per unit cell".}$$

Fact 26.3 One has to count the number of expected electrons $\langle N \rangle$. For a crystal containing r atoms in the basis and $Z_1, Z_2, \dots, Z_r$ valence electrons per atom, the number of expected electrons is given by

$$\begin{aligned} \langle N \rangle = \text{"number of cells"} &\quad \times \quad \text{"the total number of valence electrons per unit-cell"} \\ = \quad N_x N_y N_z &\quad \cdot \quad \left(Z_1 + Z_2 + \dots + Z_r\right) \\ \equiv \quad N_x N_y N_z &\quad \cdot \quad n_{\mathrm{UC}} \end{aligned}$$

26.1. Materials for a Specific Heat Analysis

26.1.1. Rb Crystal - Spacegroup Im3m (Rubidium)

Quantum Espresso Inputs

Crystal Structure from CIF-File

We have the following crystal-structure and atomic basis vectors from the cif-file:

```
&SYSTEM
  A           = 5.6568708402
  (...)
/

CELL_PARAMETERS {alat}
 -0.500000000000000   0.500000000000000   0.500000000000000
  0.500000000000000  -0.500000000000000   0.500000000000000
  0.500000000000000   0.500000000000000  -0.500000000000000

ATOMIC_POSITIONS {crystal}
Rb   0.000000000000000   0.000000000000000   0.000000000000000
```

Computational Input-Parameters

Excerpt from the python-code in which all the parameters are defined:

```
TBD
```

Step 0: vc-relax Calculation

We obtain the following final coordinates.

```
Begin final coordinates
     new unit-cell volume =    611.01184 a.u.^3 (    90.54261 Ang^3 )
     density =      1.56747 g/cm^3

CELL_PARAMETERS (alat= 10.55412041)
  -0.506494229   0.506494229   0.506494229
   0.506494229  -0.506494229   0.506494229
   0.506494229   0.506494229  -0.506494229

ATOMIC_POSITIONS (crystal)
Rb            0.0000000000        0.0000000000        0.0000000000
End final coordinates
```

such that the new input-file for the subsequent calculations reads

```
&SYSTEM
  A           = 5.6575405376
  (...)
```

```
/

CELL_PARAMETERS {alat}
 -0.500000000000000   0.500000000000000   0.500000000000000
  0.500000000000000  -0.500000000000000   0.500000000000000
  0.500000000000000   0.500000000000000  -0.500000000000000

ATOMIC_POSITIONS {crystal}
Rb   0.000000000000000   0.000000000000000   0.000000000000000
```

Step 1: SCF, Band-Structure and (Partial) Density of States

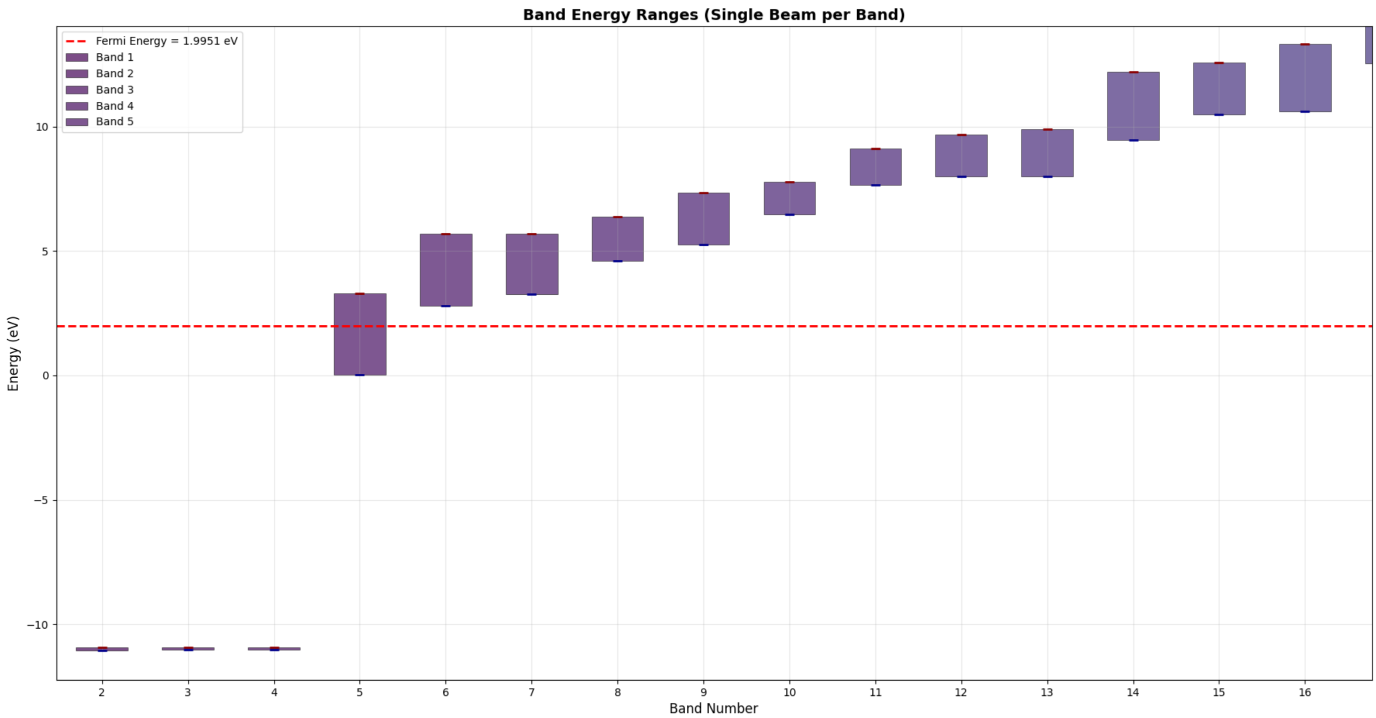


Figure 26.1.: Band-ranges and Band-indices.

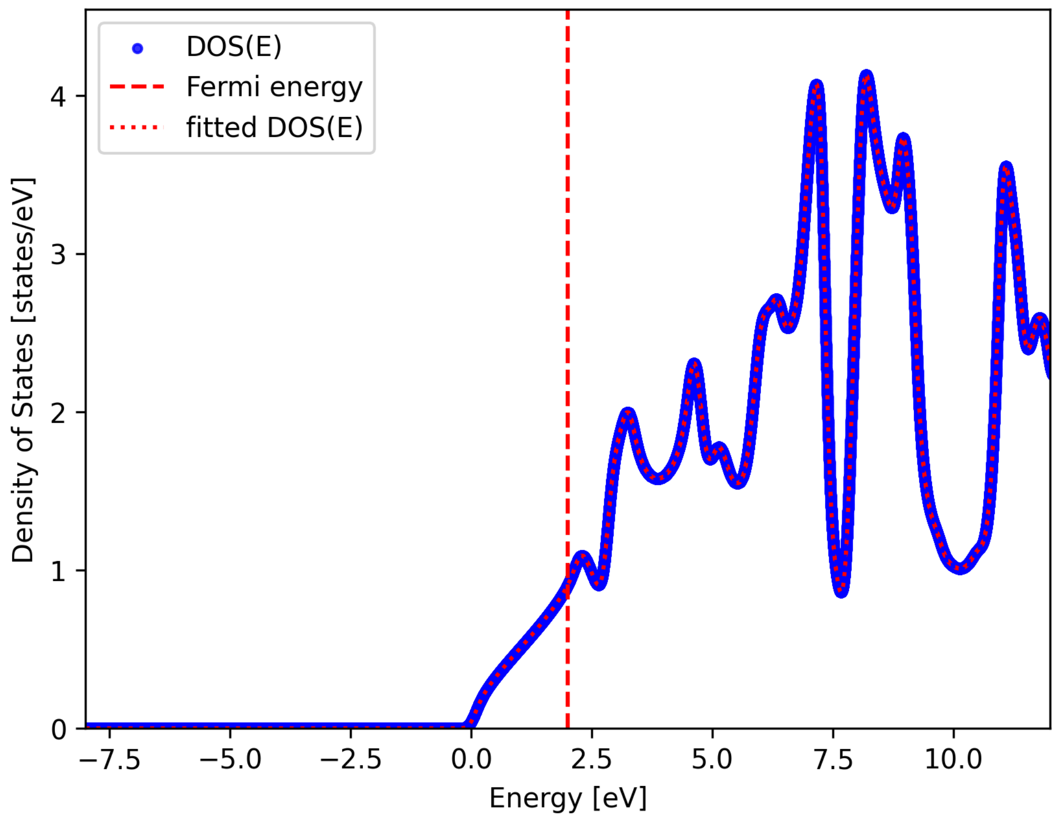


Figure 26.2.: Density of States in the LDA-Approximation.

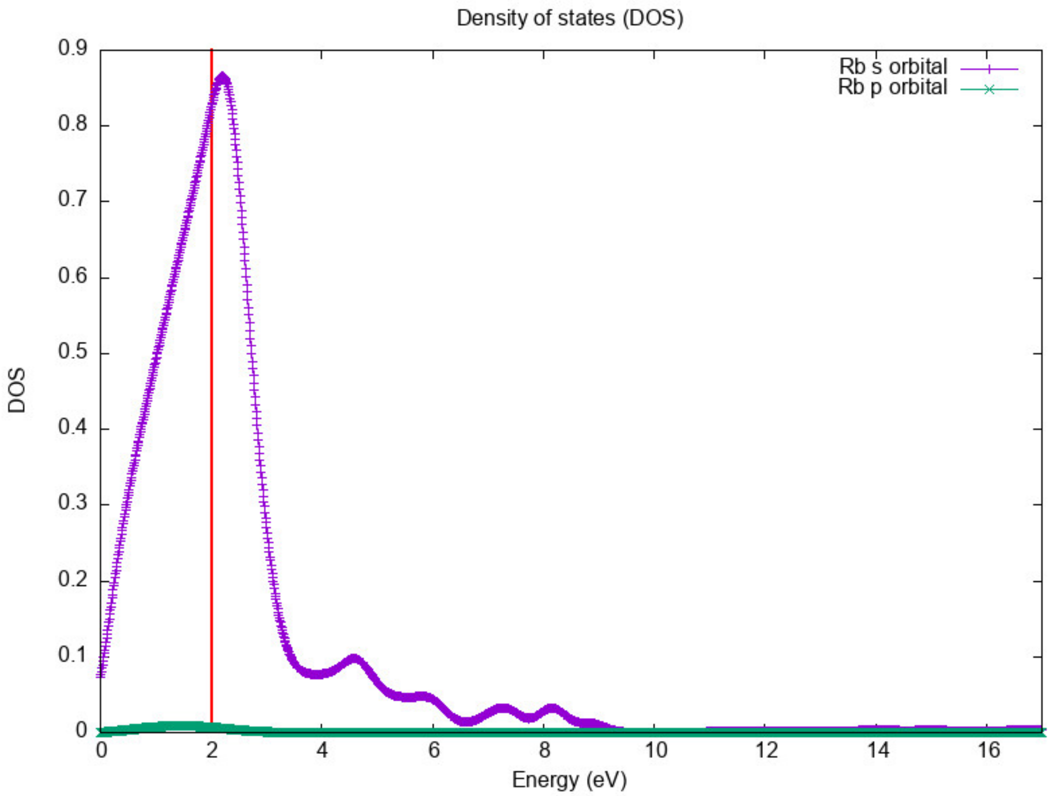


Figure 26.3.: Partial Density of States in the LDA-Approximation.

Notice that the projections give zero overlap for some highest energy, and that we don't see any d-states which is a consequence from the fact that we are using a pseudopotential.

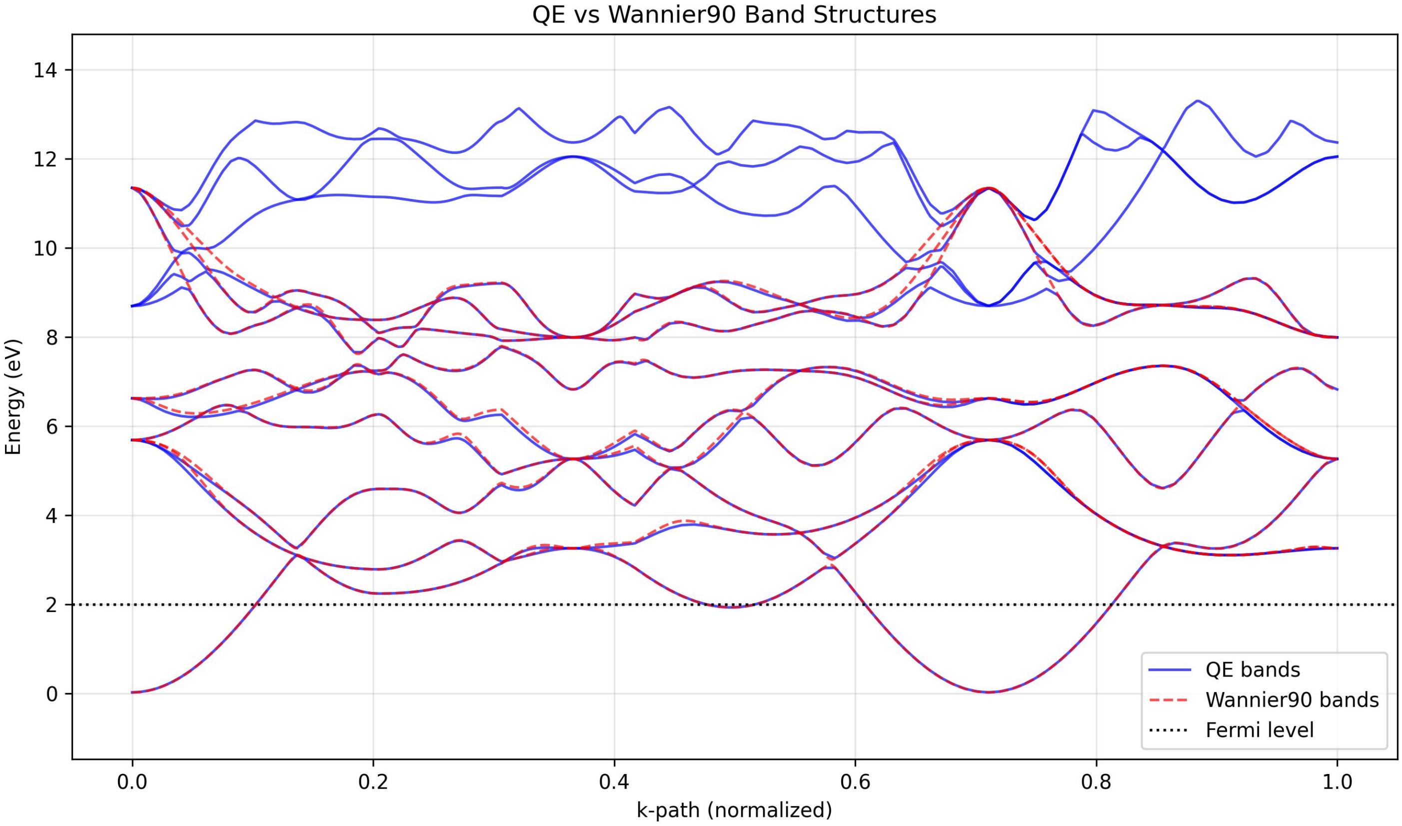


Figure 26.4.: Quantum Espresso vs. Wannier90 Bands.

TBContinued.

26.1.2. Au Crystal - Spacegroup Fm3m (Gold)

26.1.3. Ag Crystal - Spacegroup Fm3m (Silver)

26.1.4. K Crystal - Spacegroup Im3m (Potassium)

26.1.5. Cs Crystal - Spacegroup Im3m (Caesium)

26.1.6. Cu Crystal - Spacegroup Fm3m (Copper)

26.1.7. Si Crystal - Spacegroup Fd3m (Silicon)

26.1.8. C Crystal - Spacegroup Fd3m (Diamond)

26.1.9. KCl Crystal - Spacegroup Fm3m (Potassium Chloride)

26.2. Materials for a Weyl-Point Analysis

26.2.1. TaAs Crystal - Spacegroup I41md (Tantalum Arsenide)

Quantum Espresso Inputs

Crystal Structure from CIF-File

We have the following crystal-structure and atomic basis vectors from the cif-file:

```
&SYSTEM
  A          = 3.43700
  (...)
/

CELL_PARAMETERS {alat}
  1.000000000000000   0.000000000000000   0.000000000000000
  0.000000000000000   1.000000000000000   0.000000000000000
  0.500000000000000   0.500000000000000   1.695664823974397

ATOMIC_SPECIES
  As   74.92100  As.upf
  Ta  180.94700  Ta.upf

ATOMIC_POSITIONS {crystal}
Ta   0.000000000000000   0.000000000000000   0.000000000000000
Ta   0.750000000000000   0.250000000000000   0.500000000000000
As   0.583000000000000   0.583000000000000   0.834000000000000
As   0.333000000000000   0.833000000000000   0.334000000000000
```

Computational Input-Parameters

Excerpt from the python-code in which all the parameters are defined:

```
TBD
```

Step 0: vc-relax Calculation

We didn't do a relaxation for this material, as the automation is not yet implemented.

Step 1: SCF, Band-Structure and (Partial) Density of States

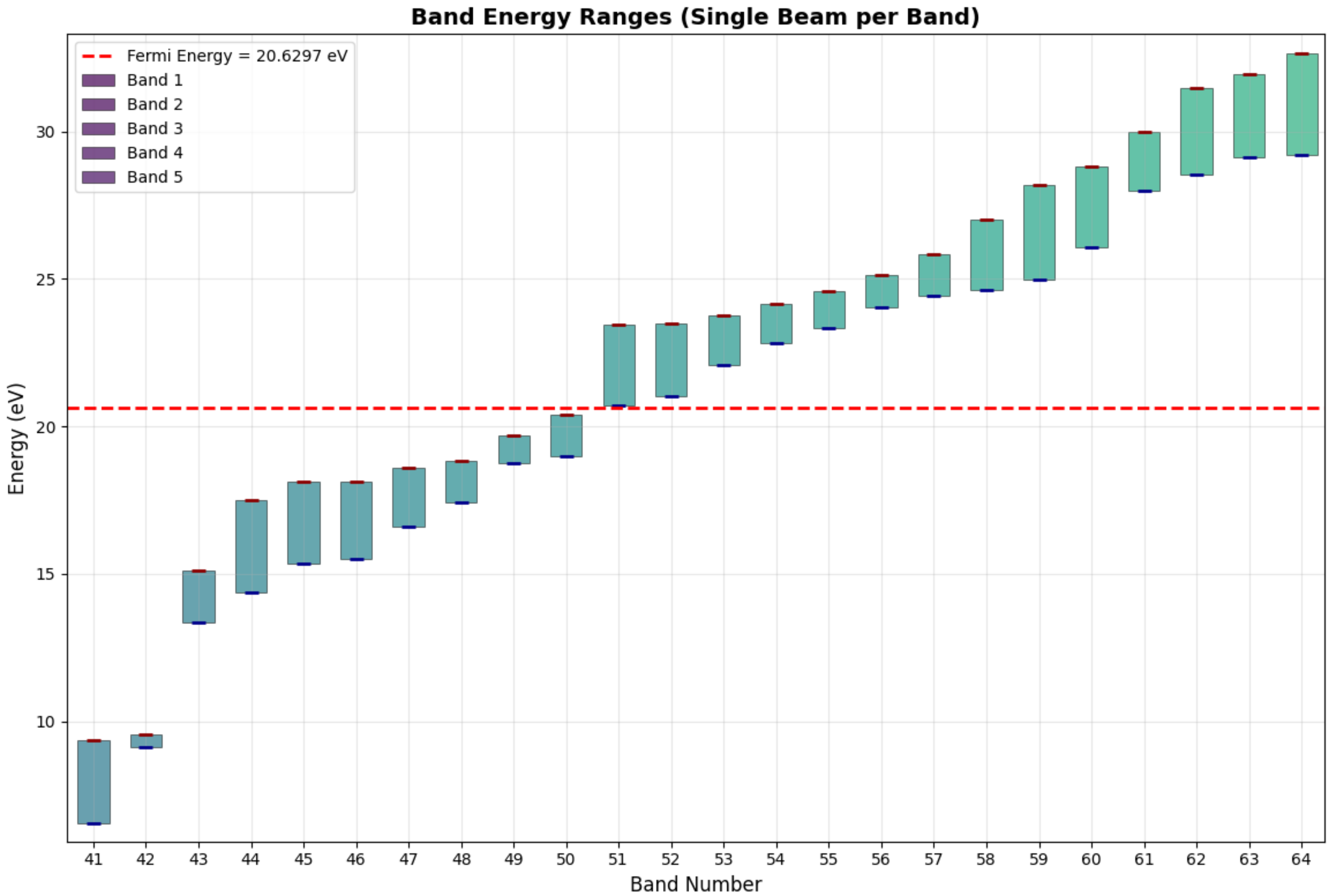


Figure 26.5.: Band-ranges and Band-indices.

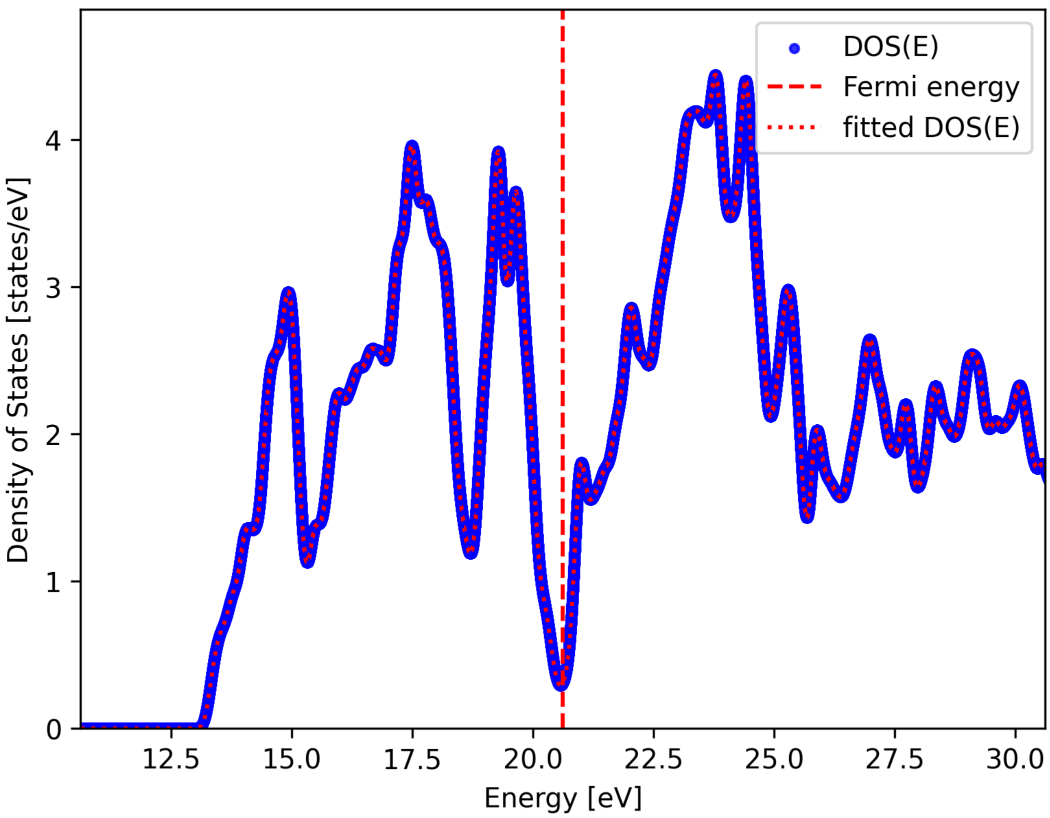


Figure 26.6.: Density of States in the LDA-Approximation.

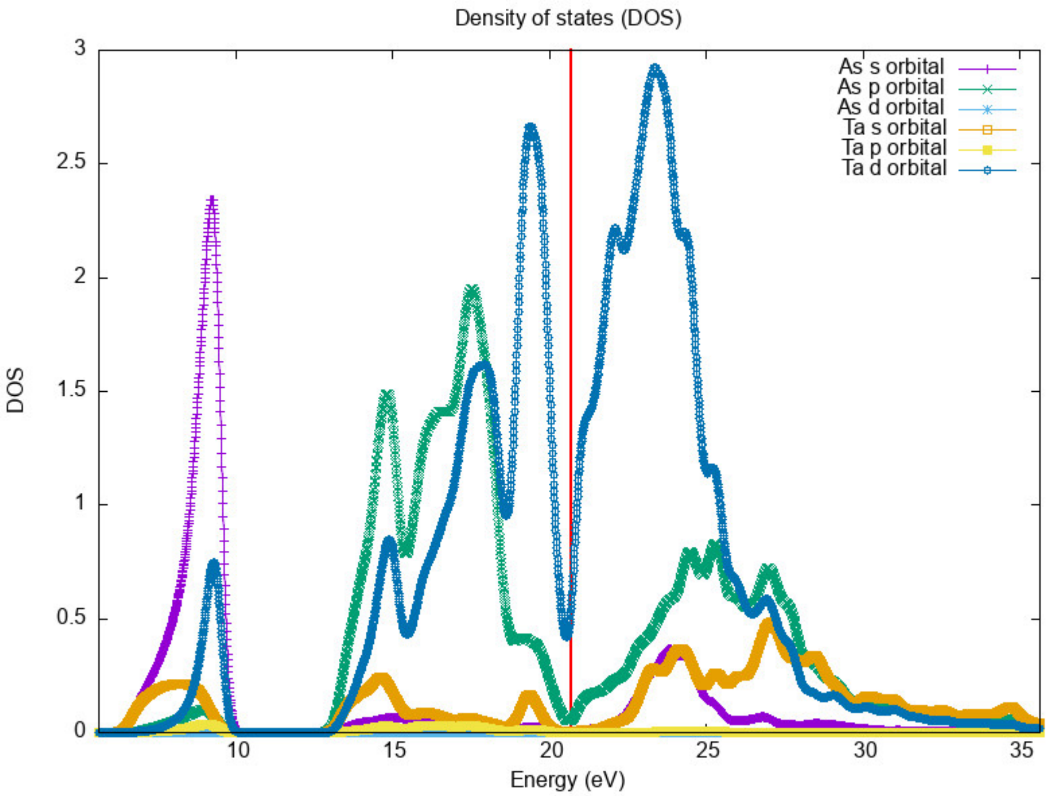


Figure 26.7.: Partial Density of States in the LDA-Approximation.

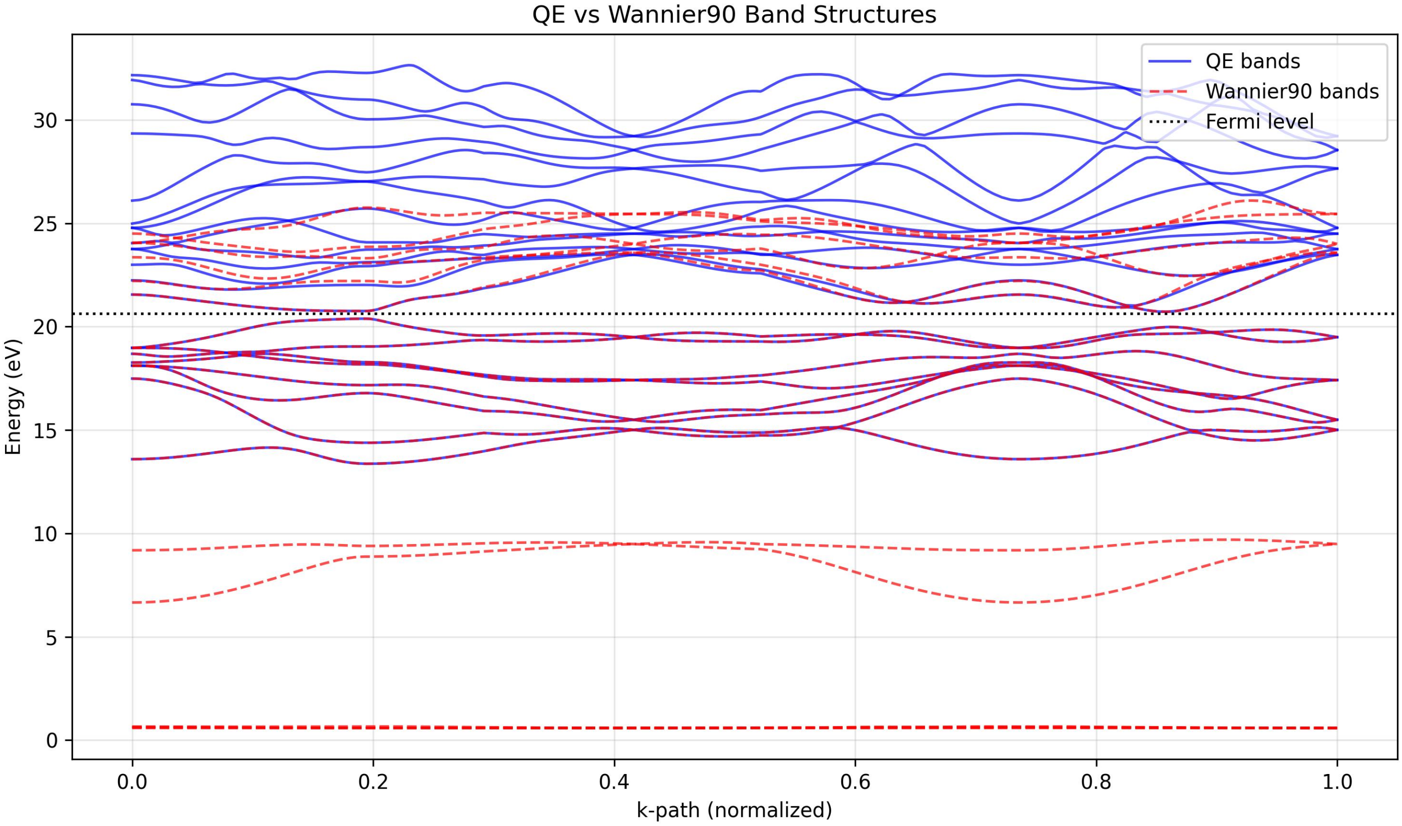


Figure 26.8.: Quantum Espresso vs. Wannier90 Bands.

26.2.2. MnTe Crystal - Spacegroup P63mmc (Manganese Telluride)

Quantum Espresso Inputs

Crystal Structure from CIF-File

We have the following crystal-structure and atomic basis vectors from the cif-file:

```
&SYSTEM
  A           = 4.15020
  (...)
/

CELL_PARAMETERS {alat}
  0.866025403784439  -0.500000000000000   0.000000000000000
  0.000000000000000   1.000000000000000   0.000000000000000
  0.000000000000000   0.000000000000000   1.619030408173100

ATOMIC_SPECIES
  Te  127.60000  Te.upf
  Mn   54.93800  Mn.upf

  ATOMIC_POSITIONS {crystal}
Te   0.333333333333333   0.666666666666667   0.250000000000000
Te   0.666666666666667   0.333333333333333   0.750000000000000
Mn   0.000000000000000   0.000000000000000   0.000000000000000
Mn   0.000000000000000   0.000000000000000   0.500000000000000
```

Computational Input-Parameters

Excerpt from the python-code in which all the parameters are defined:

```
TBD
```

Step 0: vc-relax Calculation

We didn't do a relaxation for this material, as the automation is not yet implemented.

Step 1: SCF, Band-Structure and (Partial) Density of States

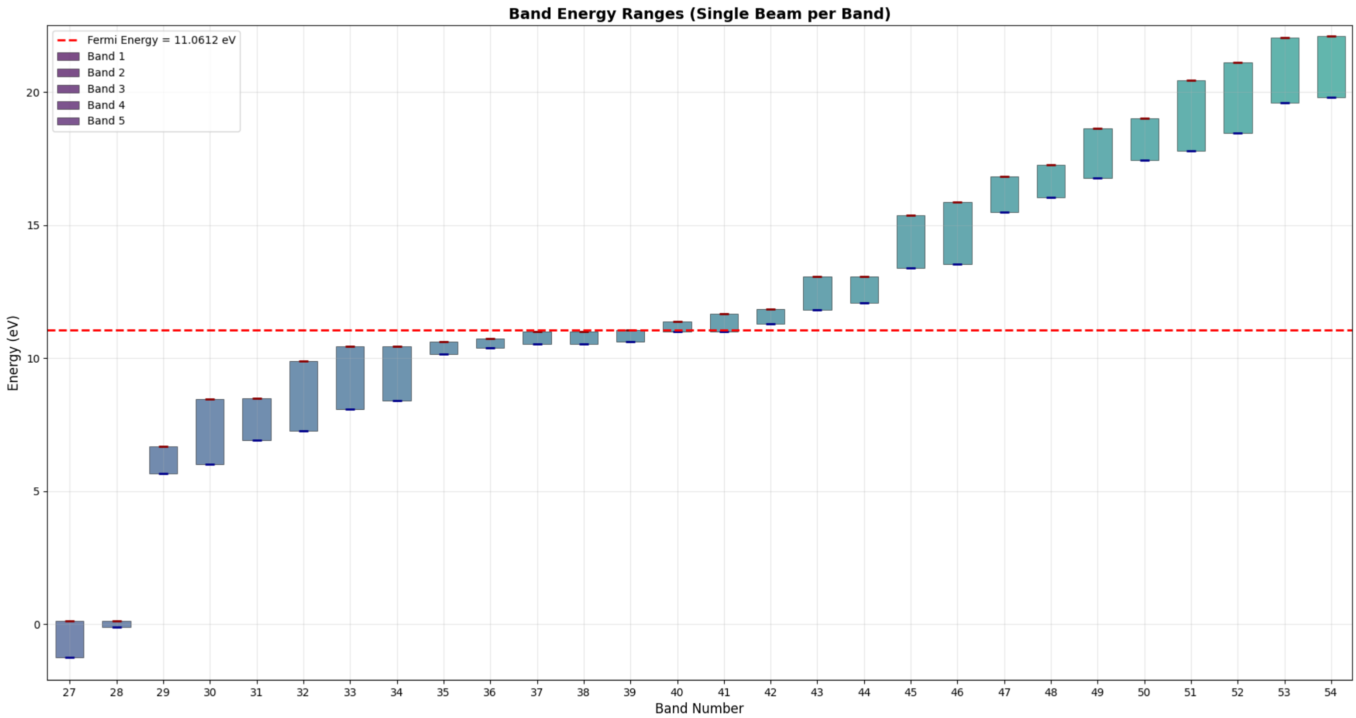


Figure 26.9.: Band-ranges and Band-indices.

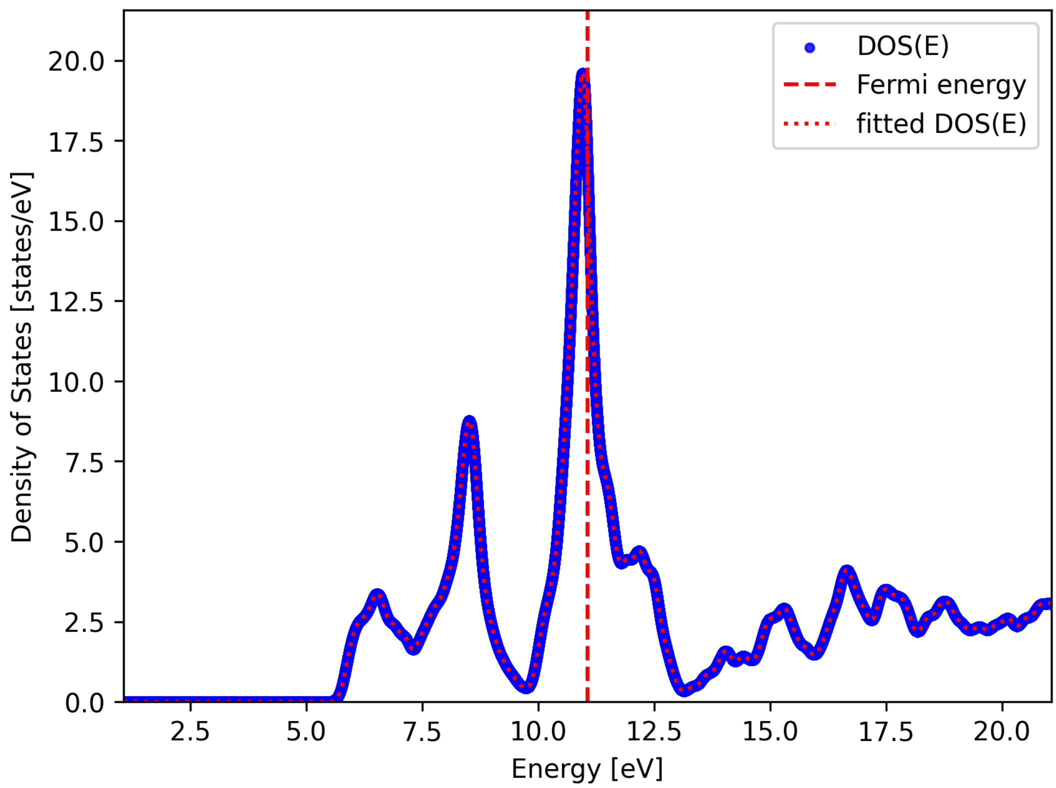


Figure 26.10.: Density of States in the LDA-Approximation.

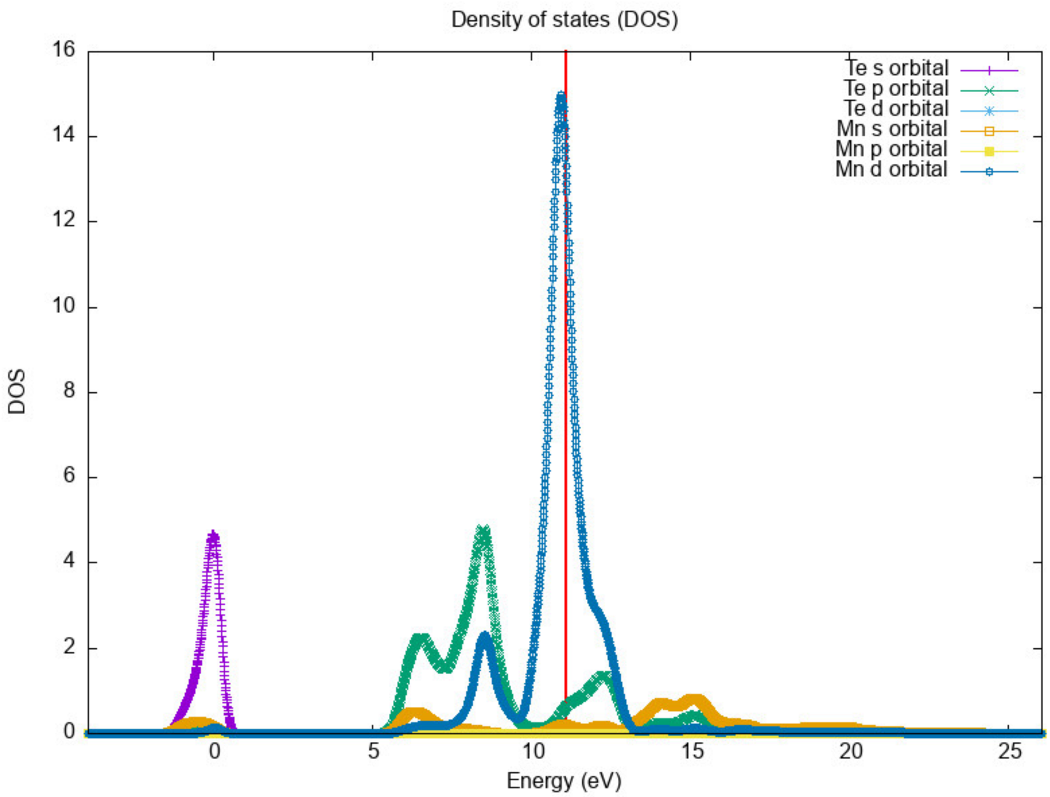


Figure 26.11.: Partial Density of States in the LDA-Approximation.

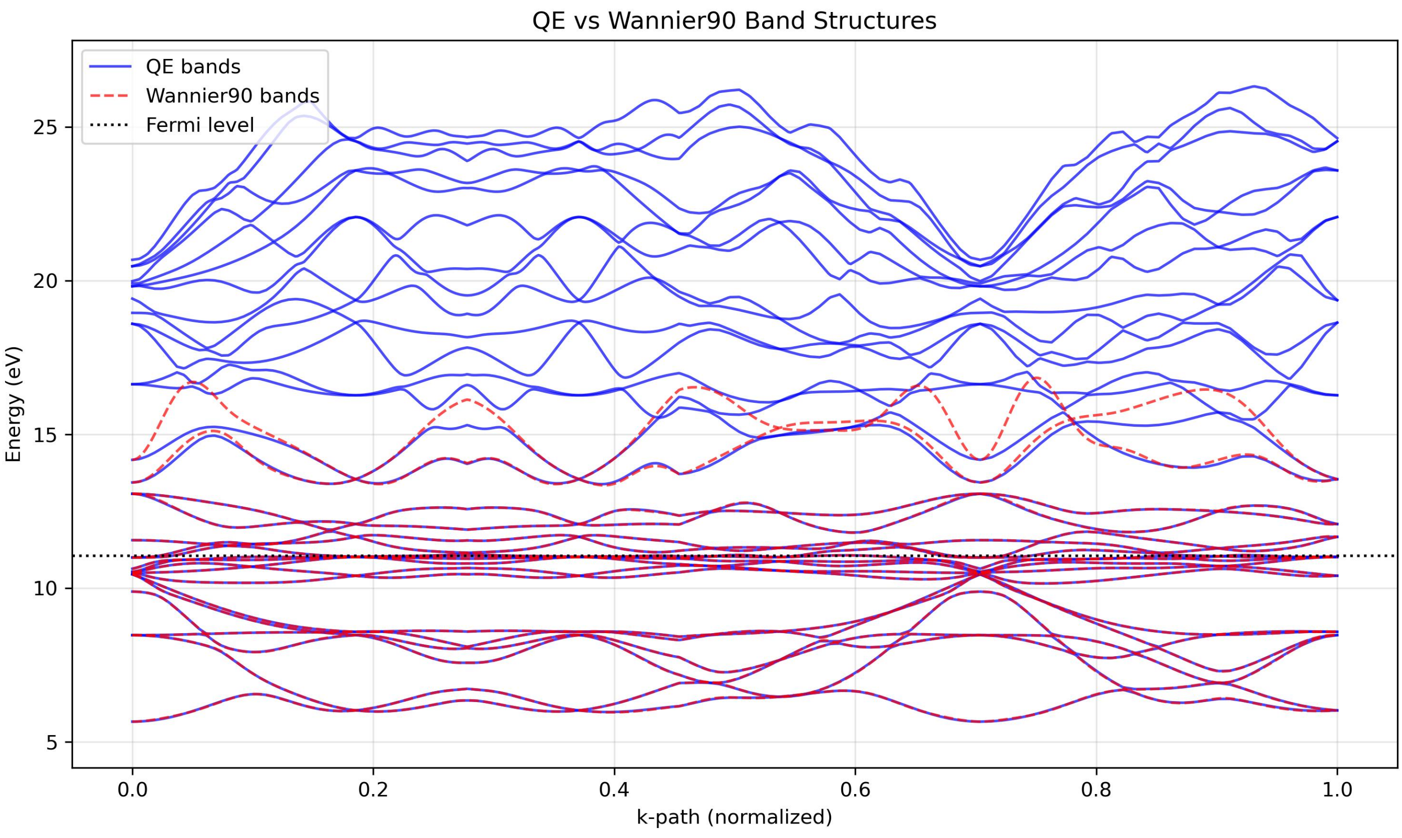


Figure 26.12.: Quantum Espresso vs. Wannier90 Bands.

26.2.3. WTe2 Crystal - Spacegroup ??? (Tungsten Ditelluride)

Quantum Espresso Inputs

Crystal Structure

We have the following crystal-structure and atomic basis vectors from the quantum-espresso input-file:

```
&SYSTEM
  A             = 3.47700
  (...)
/

CELL_PARAMETERS {alat}
  1.000000000000000   0.000000000000000   0.000000000000000
  0.000000000000000   1.797238999137187   0.000000000000000
  0.000000000000000   0.000000000000000   4.031636468219730

ATOMIC_SPECIES
  W   183.84000   W.upf
  Te  127.60000  Te.upf

ATOMIC_POSITIONS {crystal}
 W    0.000000000000000   0.600620000000000   0.764525000000000
 W    0.000000000000000   0.039800000000000   0.279745000000000
 W    0.500000000000000   0.399380000000000   0.264525000000000
 W    0.500000000000000   0.960200000000000   0.779745000000000
Te    0.000000000000000   0.857610000000000   0.919775000000000
Te    0.500000000000000   0.142390000000000   0.419775000000000
Te    0.000000000000000   0.646310000000000   0.375645000000000
Te    0.500000000000000   0.353690000000000   0.875645000000000
Te    0.000000000000000   0.298450000000000   0.124355000000000
Te    0.500000000000000   0.701550000000000   0.624355000000000
Te    0.000000000000000   0.207220000000000   0.668395000000000
Te    0.500000000000000   0.792780000000000   0.168395000000000
```

Computational Input-Parameters

Excerpt from the python-code in which all the parameters are defined:

```
TBD
```

Step 0: vc-relax Calculation

We didn't do a relaxation for this material, as the automation is not yet implemented.

Step 1: SCF, Band-Structure and (Partial) Density of States

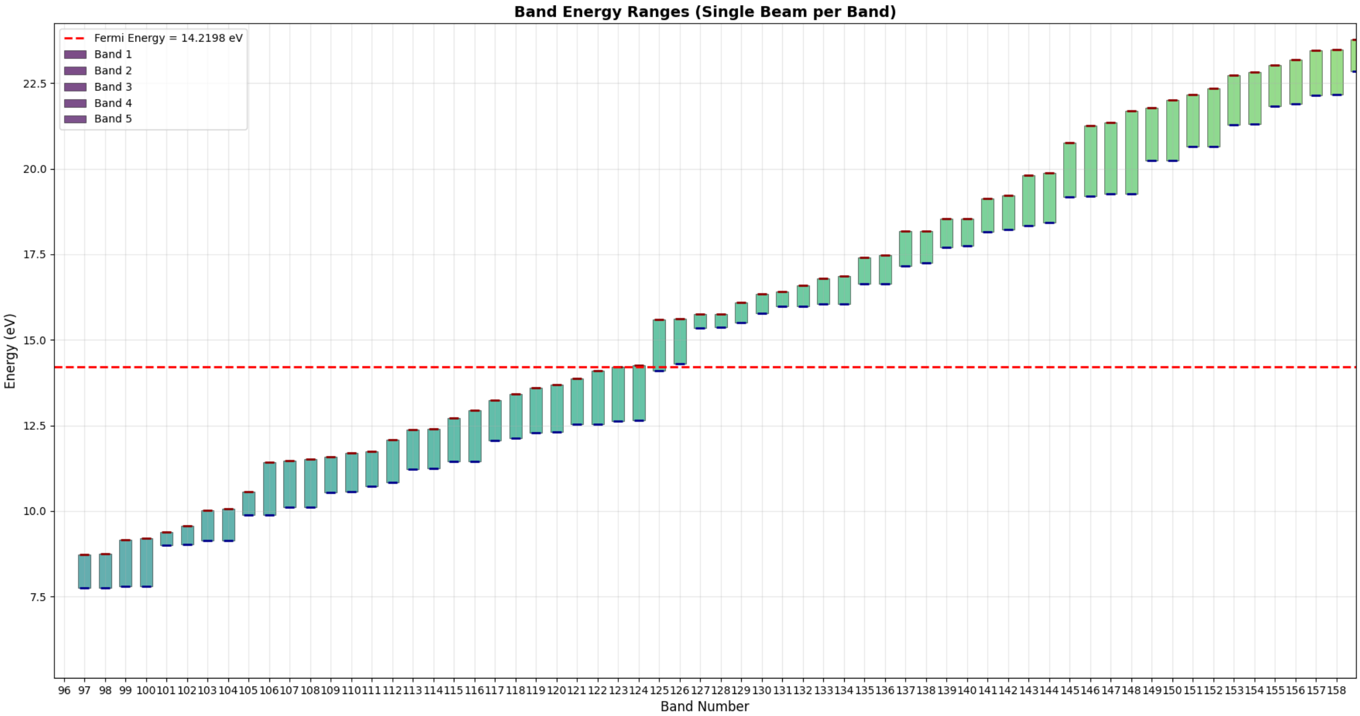


Figure 26.13.: Band-ranges and Band-indices.

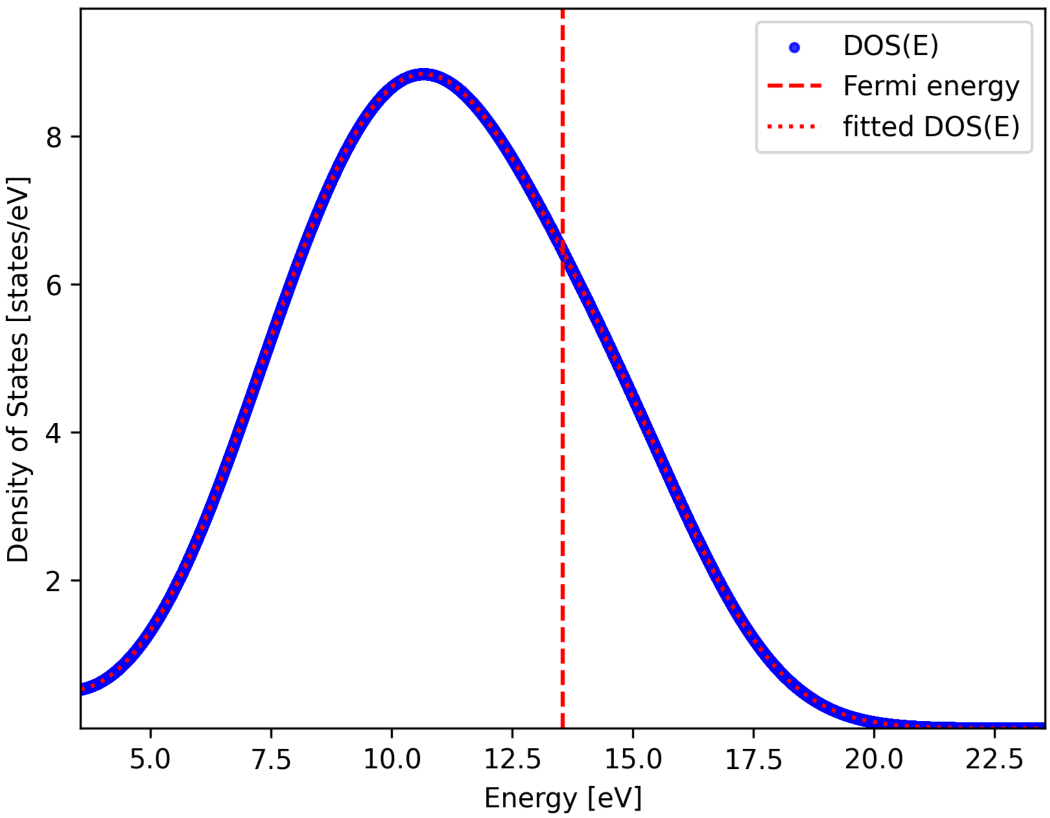


Figure 26.14.: Density of States in the LDA-Approximation.

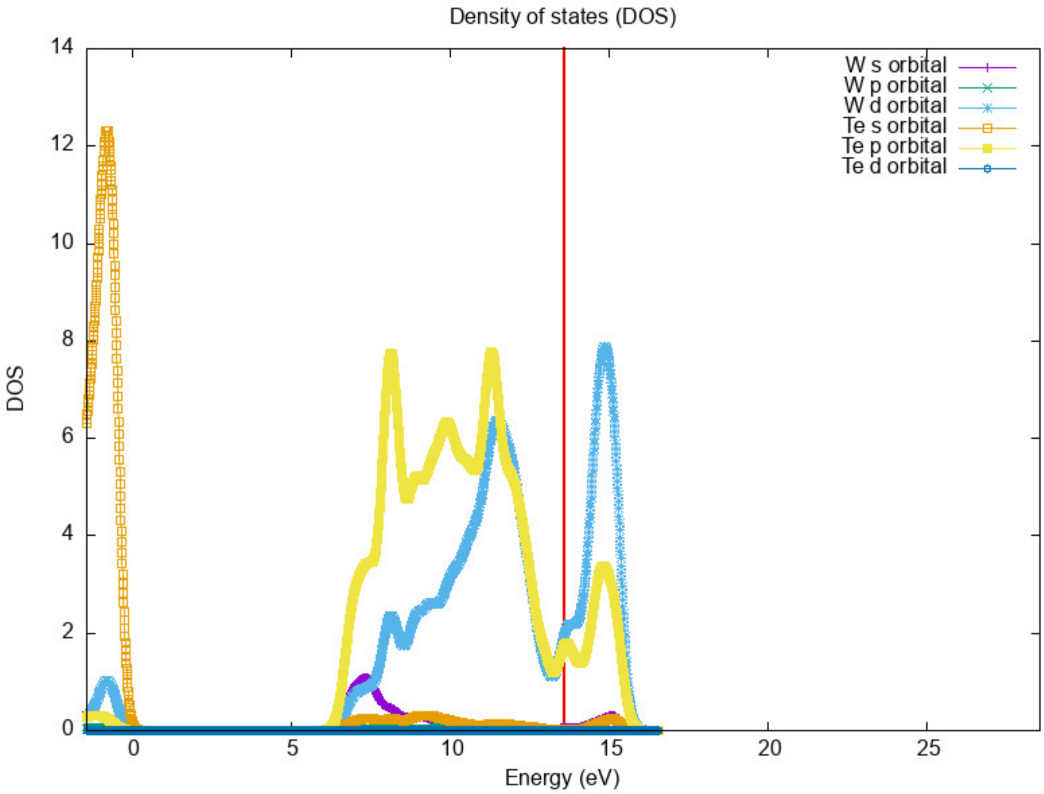


Figure 26.15.: Partial Density of States in the LDA-Approximation.

Notice that the projections give zero overlap for some highest energy which is a consequence from the fact that we are using a pseudopotential.

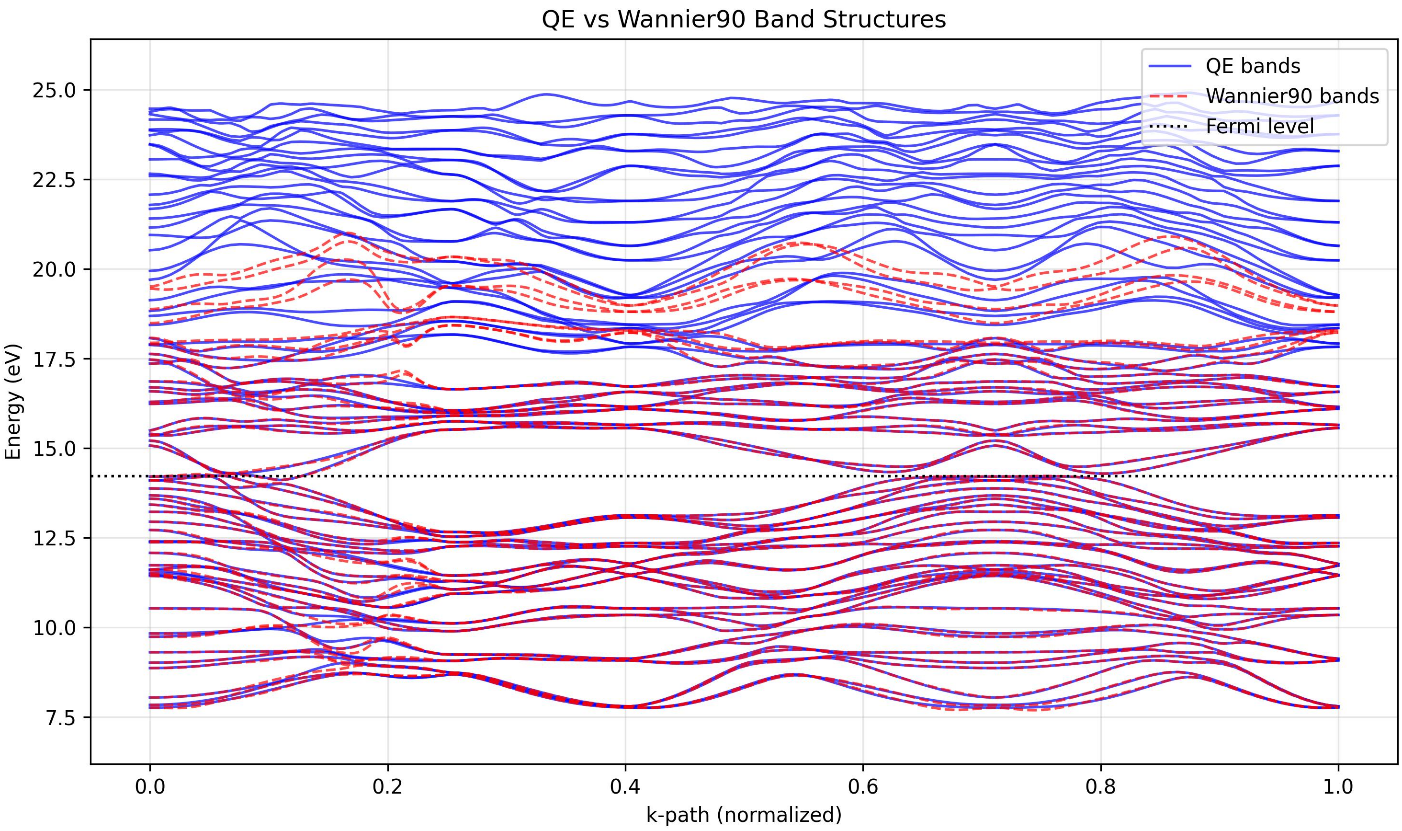


Figure 26.16.: Quantum Espresso vs. Wannier90 Bands.

26.3. Materials for a Superconductivity Analysis

26.3.1. Al Crystal - Spacegroup Fm3m (Aluminium)

26.3.2. Sn Crystal - Spacegroup I41/amd (Tin)

26.3.3. Hg Crystal - Spacegroup R-3m (Mercury)

26.3.4. C2Na Crystal - Spacegroup ??? (Sodium Carbide)

26.3.5. V Crystal - Spacegroup Im3m (Vanadium)

26.3.6. C6Yb Crystal - Spacegroup ??? (Ytterbium Graphite Intercalation)

26.3.7. Pb Crystal - Spacegroup Fm3m (Lead)

26.3.8. Tc Crystal - Spacegroup P63/mmc (Technetium)

26.3.9. MoTe2 Crystal - Spacegroup P63/mmc (Molybdenum Ditelluride)

26.3.10. Nb Crystal - Spacegroup Im3m (Niobium)

26.3.11. ZrN Crystal - Spacegroup Fm3m (Zirconium Nitride)

26.3.12. TaC Crystal - Spacegroup Fm3m (Tantalum Carbide)

26.3.13. CaC6 Crystal - Spacegroup R-3m (Calcium Graphite Intercalation)

26.3.14. Nb Crystal - Spacegroup Im3m (Niobium)

26.3.15. V3 Crystal - Spacegroup ??? (Vanadium Phase)

26.3.16. Nb3Al Crystal - Spacegroup Pm3n (Niobium Aluminide)

26.3.17. Nb3Sn Crystal - Spacegroup Pm3n (Niobium-Tin)

26.3.18. Nb3Ga Crystal - Spacegroup Pm3n (Niobium Gallide)

26.3.19. Nb3Ge Crystal - Spacegroup Pm3n (Niobium Germanide)

26.3.20. MgB2 Crystal - Spacegroup P6/mmm (Magnesium Diboride)

26.3.21. H2S Crystal - Spacegroup ??? (Hydrogen Sulfide)

26.3.22. YH6 Crystal - Spacegroup ??? (Yttrium Hexahydride)

26.3.23. LaH10 Crystal - Spacegroup ??? (Lanthanum Decahydride)

Part V.

Appendices

27. Appendix - Proofs for Part I

Proof 27.1 of Lemma 2.11 To prove the canonical commutation relations:

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij},$$

we start with the definitions of the position and momentum operators:

$$\hat{x}_i = x_i, \quad \hat{p}_j = -i\hbar\frac{\partial}{\partial x_j}.$$

The commutator is defined as:

$$[\hat{x}_i, \hat{p}_j] = \hat{x}_i\hat{p}_j - \hat{p}_j\hat{x}_i.$$

Applying this to a test function $f(x)$, we have:

$$[\hat{x}_i, \hat{p}_j]f(x) = x_i\left(-i\hbar\frac{\partial f}{\partial x_j}\right) - \left(-i\hbar\frac{\partial}{\partial x_j}(x_i f(x))\right).$$

Expanding the derivative $\frac{\partial}{\partial x_j}(x_i f(x))$ using the product rule:

$$\frac{\partial}{\partial x_j}(x_i f(x)) = \delta_{ij}f(x) + x_i\frac{\partial f}{\partial x_j}.$$

Substituting back, we get:

$$[\hat{x}_i, \hat{p}_j]f(x) = -i\hbar x_i\frac{\partial f}{\partial x_j} + i\hbar\left(\delta_{ij}f(x) + x_i\frac{\partial f}{\partial x_j}\right).$$

Simplifying:

$$[\hat{x}_i, \hat{p}_j]f(x) = i\hbar\delta_{ij}f(x).$$

Thus, we conclude:

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}.$$

Proof 27.2 of Theorem 2.15 For concreteness we consider a single particle with Hamilton-function given as $H = \frac{p^2}{2m} + V(x)$. The proof can be straightfowardly generalized to many-particle systems with more complicated

Hamilton-functions. Substituting the product ansatz $\psi(\mathbf{x}, t) = \psi(\mathbf{x})\phi(t)$ into the time-dependent Schrödinger equation:

$$i\hbar \frac{\partial \psi(\mathbf{x}, t)}{\partial t} = \left(-\frac{\hbar^2}{2m} \nabla_{\mathbf{x}}^2 + V(\mathbf{x}) \right) \psi(\mathbf{x}, t), \tag{27.1}$$

we obtain

$$i\hbar \frac{\partial}{\partial t} (\psi(\mathbf{x})\phi(t)) = \psi(\mathbf{x}) \left(-\frac{\hbar^2}{2m} \nabla_{\mathbf{x}}^2 \psi(\mathbf{x}) + V(\mathbf{x})\psi(\mathbf{x}) \right). \tag{27.2}$$

Using the product rule for differentiation on the left-hand side gives:

$$i\hbar \psi(\mathbf{x}) \frac{d\phi(t)}{dt} = \phi(t) \left(-\frac{\hbar^2}{2m} \nabla_{\mathbf{x}}^2 \psi(\mathbf{x}) + V(\mathbf{x})\psi(\mathbf{x}) \right). \tag{27.3}$$

Dividing through by $\psi(\mathbf{x})\phi(t)$, we get:

$$i\hbar \frac{1}{\phi(t)} \frac{d\phi(t)}{dt} = \frac{-\frac{\hbar^2}{2m} \nabla_{\mathbf{x}}^2 \psi(\mathbf{x}) + V(\mathbf{x})\psi(\mathbf{x})}{\psi(\mathbf{x})}. \tag{27.4}$$

The left-hand side is a function of t only, while the right-hand side is a function of $\mathbf{x}$ only. Hence, both must equal a constant, say E. This gives two separate equations:

$$i\hbar \frac{1}{\phi(t)} \frac{d\phi(t)}{dt} = E, \tag{27.5}$$

$$-\frac{\hbar^2}{2m} \nabla_{\mathbf{x}}^2 \psi(\mathbf{x}) + V(\mathbf{x})\psi(\mathbf{x}) = E\psi(\mathbf{x}). \tag{27.6}$$

The first equation simplifies to

$$\frac{d\phi(t)}{dt} = -\frac{iE}{\hbar} \phi(t), \tag{27.7}$$

which has the solution

$$\phi(t) = e^{-iEt/\hbar}. \tag{27.8}$$

The second equation is the time-independent Schrödinger equation:

$$E\psi(\mathbf{x}) = H\psi(\mathbf{x}). \tag{27.9}$$

Thus, the theorem is proven.

Proof 27.3 of Corollary 2.16 A stationary state is defined as a quantum state for which the expectation value of any observable $\hat{O}$ is constant in time. The expectation value is given by

$$\langle \hat{O} \rangle(t) = \int \psi^*(\mathbf{x}, t) \hat{O} \psi(\mathbf{x}, t)\, d^{d_{\mathrm{x}}}. \tag{27.10}$$

Substituting $\psi(\mathbf{x}, t) = \psi(\mathbf{x})\phi(t)$ and $\phi(t) = e^{-\mathrm{i}Et/\hbar}$ gives

$$\psi^*(\mathbf{x}, t) = \psi^*(\mathbf{x}) e^{\mathrm{i}Et/\hbar}, \quad \psi(\mathbf{x}, t) = \psi(\mathbf{x}) e^{-\mathrm{i}Et/\hbar}. \tag{27.11}$$

The time-dependent factors cancel out:

$$\langle \hat{O} \rangle(t) = \int \psi^*(\mathbf{x}) \hat{O} \psi(\mathbf{x})\, d^{d_{\mathrm{x}}}. \tag{27.12}$$

Since $\psi(\mathbf{x})$ is time-independent, the expectation value $\langle \hat{O} \rangle(t)$ is constant in time. Therefore, the state is stationary, and the corollary is proven.

Proof 27.4 of Theorem 2.25

We first note the auxiliary identities

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} U(t,0) &= -\frac{i}{\hbar} H U(t,0) \\
\frac{\mathrm{d}}{\mathrm{d}t} U(t,0)^\dagger &= \frac{\mathrm{d}}{\mathrm{d}t} U(0,t) = \frac{i}{\hbar} U(0,t) H = \frac{i}{\hbar} U(t,0)^\dagger H
\end{aligned}$$

Then it follows

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} X_{\mathrm{H}}(t) &= \frac{\mathrm{d}}{\mathrm{d}t} U(t,0)^\dagger X(t) U(t,0) \\
&= \left(\frac{\mathrm{d}}{\mathrm{d}t} U(t,0)^\dagger \right) X(t) U(t,0) + U(t,0)^\dagger \left(\frac{\mathrm{d}}{\mathrm{d}t} X(t) \right) U(t,0) + U(t,0)^\dagger X(t) \left(\frac{\mathrm{d}}{\mathrm{d}t} U(t,0) \right) \\
&= \frac{i}{\hbar} U(t,0)^\dagger H X(t) U(t,0) + U(t,0)^\dagger \left(\frac{\mathrm{d}}{\mathrm{d}t} X(t) \right) U(t,0) + U(t,0)^\dagger X(t) \left(-\frac{i}{\hbar} H U(t,0) \right) \\
&= \frac{i}{\hbar} U(t,0)^\dagger \left(H X(t) - X(t) H \right) U(t,0) + U(t,0)^\dagger \left(\frac{\mathrm{d}}{\mathrm{d}t} X(t) \right) U(t,0) \\
&= \frac{i}{\hbar} U(t,0)^\dagger \left[H, X(t) \right] U(t,0) + U(t,0)^\dagger \left(\frac{\mathrm{d}}{\mathrm{d}t} X(t) \right) U(t,0) \\
&= -\frac{i}{\hbar} \left[X, H \right]_{\mathrm{H}} (t) + U(t,0)^\dagger \left(\frac{\mathrm{d}}{\mathrm{d}t} X(t) \right) U(t,0)
\end{aligned}$$

Proof 27.5 of Corollary 3.13

To be done: Needs to be updated such that the details are fine.

- Cf. the weblink for the details of the proof:

  https://en.wikipedia.org/wiki/Fourier_series#Fourier_series_of_a_Bravais-lattice-periodi
  function

- TBD: Fixing the details of the proof with the notes. Cf. Waste-Folder-1

We provide the proof for the three-dimensional case[1]. The one- and two-dimensional cases can be proven analogously.[2]

We define a new function g as,

$$g(x_1, x_2, x_3) := f\left(x_1\frac{\mathbf{a}_1}{a_1} + x_2\frac{\mathbf{a}_2}{a_2} + x_3\frac{\mathbf{a}_3}{a_3}\right).$$

This new function, $g(x_1, x_2, x_3)$, is now a function of three variables, each of which has periodicity a_1, a_2, and a_3 respectively:

$$g(x_1, x_2, x_3) = g(x_1 + a_1, x_2, x_3) = g(x_1, x_2 + a_2, x_3) = g(x_1, x_2, x_3 + a_3).$$

This enables us to build up a set of Fourier coefficients, each being indexed by three independent integers m_1, m_2, m_3. If we write a series for g on the interval $[0, a_1]$ for x_1, we can define the following:

$$h^{\text{one}}(m_1, x_2, x_3) \equiv \frac{1}{a_1}\int_0^{a_1} g(x_1, x_2, x_3) \cdot e^{-i2\pi\frac{m_1}{a_1}x_1}\, dx_1.$$

And then we can write:

$$g(x_1, x_2, x_3) = \sum_{m_1=-\infty}^{\infty} h^{\text{one}}(m_1, x_2, x_3) \cdot e^{\mathrm{i}2\pi\frac{m_1}{a_1}x_1}.$$

Further defining:

$$\begin{aligned}
h^{\text{two}}(m_1, m_2, x_3) &\equiv \frac{1}{a_2}\int_0^{a_2} h^{\text{one}}(m_1, x_2, x_3) \cdot e^{-\mathrm{i}2\pi\frac{m_2}{a_2}x_2}\, dx_2 \\
&= \frac{1}{a_2}\int_0^{a_2} dx_2 \frac{1}{a_1}\int_0^{a_1} dx_1 g(x_1, x_2, x_3) \cdot e^{-\mathrm{i}2\pi\frac{m_1}{a_1}x_1} \cdot e^{-\mathrm{i}2\pi\frac{m_2}{a_2}x_2}
\end{aligned}$$

We can write g once again as:

$$g(x_1, x_2, x_3) = \sum_{m_1=-\infty}^{\infty}\sum_{m_2=-\infty}^{\infty} h^{\text{two}}(m_1, m_2, x_3) \cdot e^{\mathrm{i}2\pi\frac{m_1}{a_1}x_1} \cdot e^{\mathrm{i}2\pi\frac{m_2}{a_2}x_2}$$

[1] Cf. [85] in Waste-Folder 3.

[2] TBD: Add the proof for the one- and two-dimensional cases.

Finally, applying the same for the third coordinate, we define:

$$
\begin{aligned}
h^{\text{three}}(m_1, m_2, m_3) &\equiv \frac{1}{a_3}\int_0^{a_3} h^{\text{two}}(m_1, m_2, x_3)\cdot e^{-\mathrm{i}2\pi\frac{m_3}{a_3}x_3}\,dx_3 \\
&= \frac{1}{a_3}\int_0^{a_3} dx_3 \frac{1}{a_2}\int_0^{a_2} dx_2 \frac{1}{a_1}\int_0^{a_1} dx_1 g(x_1, x_2, x_3)\cdot e^{-\mathrm{i}2\pi\frac{m_1}{a_1}x_1}\cdot e^{-\mathrm{i}2\pi\frac{m_2}{a_2}x_2}\cdot e^{-\mathrm{i}2\pi\frac{m_3}{a_3}x_3}
\end{aligned}
$$

We write g as:

$$
\begin{aligned}
g(x_1, x_2, x_3) &= \sum_{m_1=-\infty}^{\infty}\sum_{m_2=-\infty}^{\infty}\sum_{m_3=-\infty}^{\infty} h^{\text{three}}(m_1, m_2, m_3)\cdot e^{\mathrm{i}2\pi\frac{m_1}{a_1}x_1}\cdot e^{\mathrm{i}2\pi\frac{m_2}{a_2}x_2}\cdot e^{\mathrm{i}2\pi\frac{m_3}{a_3}x_3}. \\
&= \sum_{m_1, m_2, m_3\in\mathbb{Z}} h^{\text{three}}(m_1, m_2, m_3)\cdot e^{i2\pi\left(\frac{m_1}{a_1}x_1+\frac{m_2}{a_2}x_2+\frac{m_3}{a_3}x_3\right)}.
\end{aligned}
$$

Now, every reciprocal lattice vector can be written as

$\mathbf{G} = m_1\mathbf{b}_1 + m_2\mathbf{b}_2 + m_3\mathbf{b}_3$, where m_i are integers and $\mathbf{b}_i$ are reciprocal lattice vectors to satisfy $\mathbf{b}_i \cdot \mathbf{a}_j = 2\pi\delta_{ij}$ ($\delta_{ij} = 1$ for $i = j$, and $\delta_{ij} = 0$ for $i \neq j$). Then for any arbitrary reciprocal lattice vector $\mathbf{G}$ and arbitrary position vector $\mathbf{r} = \left(x_1\frac{\mathbf{a}_1}{a_1} + x_2\frac{\mathbf{a}_2}{a_2} + x_3\frac{\mathbf{a}_3}{a_3}\right)$ in the original Bravais lattice space, their scalar product is:

$$
\mathbf{G}\cdot\mathbf{r} = (m_1\mathbf{g}_1 + m_2\mathbf{g}_2 + m_3\mathbf{g}_3)\cdot\left(x_1\frac{\mathbf{a}_1}{a_1} + x_2\frac{\mathbf{a}_2}{a_2} + x_3\frac{\mathbf{a}_3}{a_3}\right) = 2\pi\left(x_1\frac{m_1}{a_1} + x_2\frac{m_2}{a_2} + x_3\frac{m_3}{a_3}\right).
$$

So it is clear that in our expansion of $g(x_1, x_2, x_3) = f(\mathbf{r})$, the sum is actually over reciprocal lattice vectors:

$$
f(\mathbf{r}) = \sum_{\mathbf{G}} h(\mathbf{G})\cdot e^{i\mathbf{G}\cdot\mathbf{r}},
$$

where

$$
h(\mathbf{G}) = \frac{1}{a_3}\int_0^{a_3} dx_3\,\frac{1}{a_2}\int_0^{a_2} dx_2\,\frac{1}{a_1}\int_0^{a_1} dx_1\, f\left(x_1\frac{\mathbf{a}_1}{a_1} + x_2\frac{\mathbf{a}_2}{a_2} + x_3\frac{\mathbf{a}_3}{a_3}\right)\cdot e^{-i\mathbf{G}\cdot\mathbf{r}}.
$$

Assuming

$$
\mathbf{r} = (x, y, z) = a_1x_1 + a_2x_2 + a_3x_3,
$$

we can solve this system of three linear equations for x, y, z in terms of x_1, x_2, x_3 in order to calculate the volume element in the original Cartesian coordinate system. Once we have x, y, z in terms of x_1, x_2, x_3, we can calculate the Jacobian determinant:

$$
\begin{vmatrix}
\frac{\partial x_1}{\partial x} & \frac{\partial x_1}{\partial y} & \frac{\partial x_1}{\partial z} \\
\frac{\partial x_2}{\partial x} & \frac{\partial x_2}{\partial y} & \frac{\partial x_2}{\partial z} \\
\frac{\partial x_3}{\partial x} & \frac{\partial x_3}{\partial y} & \frac{\partial x_3}{\partial z}
\end{vmatrix}
$$

which after some calculation and applying some non-trivial cross-product identities (TBDone) can be shown to be equal to:

$$\frac{a_1 a_2 a_3}{a_1 \cdot (a_2 \times a_3)}.$$

(It may be advantageous for the sake of simplifying calculations to work in such a Cartesian coordinate system in which it just so happens that a_1 is parallel to the x-axis, a_2 lies in the xy-plane, and a_3 has components of all three axes). The denominator is exactly the volume of the primitive unit cell which is enclosed by the three primitive vectors a_1, a_2, a_3. In particular, we now know that:

$$dx_1 dx_2 dx_3 = \frac{a_1 a_2 a_3}{a_1 \cdot (a_2 \times a_3)} \, dx \, dy \, dz.$$

We can write now $\hat{h}(\mathbf{G})$ as an integral with the traditional coordinate system over the volume of the primitive cell, instead of with the x_1, x_2, x_3 variables:

$$\hat{h}(\mathbf{G}) = \frac{1}{a_1 \cdot (a_2 \times a_3)} \int_C e^{-i\mathbf{G}\cdot\mathbf{r}} f(\mathbf{r}) \, d^3 r,$$

writing $d^3 r$ for the volume element $dx\, dy\, dz$; and where C is the primitive unit cell, thus, $a_1 \cdot (a_2 \times a_3)$ is the volume of the primitive unit cell.

Proof 27.6 of Theorem 4.8 We compute

$$
\begin{aligned}
0 &= (H - E)\,\psi(\mathbf{r}, \boldsymbol{\tau}) \\
&= \sum_{\alpha \in \mathscr{A}} \left(H_0 + T_\mathrm{a} - E\right) \chi_\alpha(\boldsymbol{\tau}) \phi_{\alpha,\tau}(\mathbf{r}) \\
&= \sum_{\alpha \in \mathscr{A}} \left(\epsilon_{\alpha,\tau} + T_\mathrm{a} - E\right) \chi_\alpha(\boldsymbol{\tau}) \phi_{\alpha,\tau}(\mathbf{r}) \\
&= \dots
\end{aligned}
$$

We now use the general Leibniz-rule and compute

$$
\Delta_{\tau_i}\left[\chi_\alpha(\boldsymbol{\tau})\phi_{\alpha,\tau}(\mathbf{r})\right] \quad = \quad \left[\Delta_{\tau_i}\chi_\alpha(\boldsymbol{\tau})\right]\phi_{\alpha,\tau}(\mathbf{r}) + \left[\nabla_{\tau_i}\chi_\alpha(\boldsymbol{\tau})\right]\cdot\left[\nabla_{\tau_i}\phi_{\alpha,\tau}(\mathbf{r})\right] + \chi_\alpha(\boldsymbol{\tau})\left[\Delta_{\tau_i}\phi_{\alpha,\tau}(\mathbf{r})\right] \tag{27.13}
$$

and therefore obtain

$$
\begin{aligned}
\mathrm{T}_a\, \chi_\alpha(\boldsymbol{\tau})\phi_{\alpha,\tau}(\mathbf{r}) &= \sum_{i=1}^{N_a} -\frac{\hbar^2}{2M_i}\left[\Delta_{\tau_i}\chi_\alpha(\boldsymbol{\tau})\right]\phi_{\alpha,\tau}(\mathbf{r}) \quad + \quad \text{remaining terms which will be neglected} \\
&\approx \sum_{i=1}^{N_a} -\frac{\hbar^2}{2M_i}\left[\Delta_{\tau_i}\chi_\alpha(\boldsymbol{\tau})\right]\phi_{\alpha,\tau}(\mathbf{r})
\end{aligned} \tag{27.14}
$$

Inserting this in the expression above yields

$$
\dots = \sum_{\alpha \in \mathscr{A}} \left[\left(\epsilon_{\alpha,\tau} + T_\mathrm{a} - E\right) \chi_\alpha(\boldsymbol{\tau})\right]\phi_{\alpha,\tau}(\mathbf{r}) \tag{27.15}
$$

multiplying with $\phi^*_{\beta,\tau}(\mathbf{r})$ and integrating over the electronic coordinates yields

$$
\sum_{\alpha \in \mathscr{A}} \left[\left(\epsilon_{\alpha,\tau} + T_\mathrm{a} - E\right) \chi_\alpha(\boldsymbol{\tau})\right] \int \mathrm{d}\mathbf{r}\phi^*_{\beta,\tau}(\mathbf{r})\phi_{\alpha,\tau}(\mathbf{r}) = \left(\epsilon_{\beta,\tau} + T_\mathrm{a} - E\right)\chi_\beta(\boldsymbol{\tau}) = 0 \Longleftrightarrow \left(T_\mathrm{a} + \epsilon_{\beta,\tau}\right)\chi_\beta(\boldsymbol{\tau}) = E\chi_\beta(\boldsymbol{\tau}) \tag{27.16}
$$

Proof 27.7 of Corollary 5.18 The kinetic energy term is straightforward from Lemma 5.17. The potential energy term is a bit more involved, since we have to use the different expansion formulae Equations (5.25) and (5.27) for the coordinates. However, by direct calculation one has

$$
\begin{aligned}
V_{\mathrm{eff}}(\tau) &\approx \frac{1}{2} \sum_{\substack{\mathbf{R_n},\mathbf{R_{n'}}\in \mathrm{L} \\ \kappa,\kappa'\in\{1,\ldots,r\} \\ \alpha,\alpha'\in\{1,2,3\}}} \Delta\tau_{\mathbf{n}\kappa\alpha} \left(\frac{\partial^2 V_{\mathrm{eff}}}{\partial\tau_{\mathbf{n}\kappa\alpha}\partial\tau_{\mathbf{n'}\kappa'\alpha'}}\right)(\mathbf{R})\, \Delta\tau_{\mathbf{n'}\kappa'\alpha'} \\
&= \frac{1}{2} \sum_{\substack{\kappa,\kappa'\in\{1,\ldots,r\} \\ \alpha,\alpha'\in\{1,2,3\}}} \sqrt{M_\kappa M_{\kappa'}} \sum_{\mathbf{R_n},\mathbf{R_{n'}}\in \mathrm{L}} \Delta\tau_{\mathbf{n}\kappa\alpha}\, D_{\kappa\alpha,\,\kappa'\alpha'}(\mathbf{R_n}-\mathbf{R_{n'}})\, \Delta\tau_{\mathbf{n'}\kappa'\alpha'} \\
&= \frac{1}{2}\frac{1}{N_p^2} \sum_{\substack{\kappa,\kappa'\in\{1,\ldots,r\} \\ \alpha,\alpha'\in\{1,2,3\}}} \sqrt{M_\kappa M_{\kappa'}} \sum_{\mathbf{q}_1,\mathbf{q}_2,\mathbf{q}_3\in \mathrm{Q}} \Delta\tau_{-\mathbf{q}_1\kappa\alpha}\, D_{\kappa\alpha,\,\kappa'\alpha'}(\mathbf{q}_2)\, \Delta\tau_{\mathbf{q}_3\kappa'\alpha'} \sum_{\mathbf{R_n},\mathbf{R_{n'}}\in \mathrm{L}} \mathrm{e}^{\mathrm{i}\mathbf{R_n}(\mathbf{q}_1-\mathbf{q}_2)}\mathrm{e}^{-\mathrm{i}\mathbf{R_{n'}}(+\mathbf{q}_3-\mathbf{q}_2)} \\
&= \frac{1}{2} \sum_{\substack{\kappa,\kappa'\in\{1,\ldots,r\} \\ \alpha,\alpha'\in\{1,2,3\}}} \sqrt{M_\kappa M_{\kappa'}} \sum_{\mathbf{q}\in \mathrm{Q}} \Delta\tau_{-\mathbf{q}\kappa\alpha}\, D_{\kappa\alpha,\,\kappa'\alpha'}(\mathbf{q})\, \Delta\tau_{\mathbf{q}\kappa'\alpha'} \\
&= \frac{1}{2} \sum_{\mathbf{q}\in \mathrm{Q}} \sum_{\substack{\kappa,\kappa'\in\{1,\ldots,r\} \\ \alpha,\alpha'\in\{1,2,3\}}} \Delta\tau_{-\mathbf{q}\kappa\alpha} \frac{D_{\kappa\alpha,\,\kappa'\alpha'}(\mathbf{q})}{\sqrt{M_\kappa M_{\kappa'}}^{-1}} \Delta\tau_{\mathbf{q}\kappa'\alpha'}
\end{aligned}
$$

proving the claim.

Proof 27.8 of Lemma 5.19 By direct calculation one obtains with the symmetry of the real-space-dynamical-matrix from Remark 5.6.1, together with the translation invariance from Theorem 5.15, through a long chain of equations

$$
\begin{aligned}
\left(D^{\mathrm{T}}(\mathbf{q})\right)_{\kappa\alpha,\kappa'\alpha'} &= D_{\kappa'\alpha',\kappa\alpha}(\mathbf{q}) = \sum_{\mathbf{R_n}\in \mathrm{L_f}} D_{\kappa'\alpha',\kappa\alpha}(\mathbf{R_n})\mathrm{e}^{+\mathrm{i}\mathbf{q}\cdot\mathbf{R_n}} = \sum_{\mathbf{R_n}\in \mathrm{L_f}} D_{\mathbf{n}\kappa'\alpha',\mathbf{0}\kappa\alpha}\mathrm{e}^{+\mathrm{i}\mathbf{q}\cdot\mathbf{R_n}} = \sum_{\mathbf{R_n}\in \mathrm{L_f}} D_{\mathbf{0}\kappa'\alpha',-\mathbf{n}\kappa\alpha}\mathrm{e}^{+\mathrm{i}\mathbf{q}\cdot\mathbf{R_n}} = \ldots \\
\ldots &\overset{\mathbf{R_n}\mapsto-\mathbf{R_n}}{=} \sum_{\mathbf{R_n}\in \mathrm{L_f}} D_{\mathbf{0}\kappa'\alpha',\mathbf{n}\kappa\alpha}\mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot\mathbf{R_n}} \overset{D \text{ symmetric}}{=} \sum_{\mathbf{R_n}\in \mathrm{L_f}} D_{\mathbf{n}\kappa\alpha,\mathbf{0}\kappa'\alpha'}\mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot\mathbf{R_n}} \\
&= \sum_{\mathbf{R_n}\in \mathrm{L_f}} D_{\kappa\alpha,\kappa'\alpha'}(\mathbf{R_n})\mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot\mathbf{R_n}} = \left(D_{\kappa\alpha,\kappa'\alpha'}(\mathbf{q})\right)^*
\end{aligned}
$$

Proof 27.9 of Lemma 5.23

$D_{\kappa\alpha,\,\kappa'\alpha'}(\mathbf{q})$ is hermitian, according to Lemma 5.19. We obtain further with Equation (5.25) and the fact that the dynamical matrix is real-valued in real-space as follows from Equation (5.2)

$$
D_{\kappa\alpha,\,\kappa'\alpha'}(\mathbf{q}) \equiv \sum_{\mathbf{R_n}\in \mathrm{L_f}} D_{\kappa\alpha,\,\kappa'\alpha'}(\mathbf{R_n})\mathrm{e}^{\mathrm{i}\mathbf{q}\mathbf{R}_n} \quad \Rightarrow \quad D_{\kappa\alpha,\,\kappa'\alpha'}(-\mathbf{q}) = D^*_{\kappa\alpha,\,\kappa'\alpha'}(\mathbf{q}) \tag{27.17}
$$

Thus we obtain for $\nu \in \{1,\ldots,rd_x\}$ the equations

$$
D(\mathbf{q})e_{:\nu}(\mathbf{q}) = \omega_\nu^2(\mathbf{q})e_{:\nu}(\mathbf{q}) \quad \Longleftrightarrow \quad D(-\mathbf{q})e^*_{:\nu}(\mathbf{q}) = \omega_\nu^2(\mathbf{q})e^*_{:\nu}(\mathbf{q}) \tag{27.18}
$$

which lets us conclude that the eigenvectors $\{\mathbf{e}_{:,1}(-\mathbf{q}), \mathbf{e}_{:,2}(-\mathbf{q}), \dots, \mathbf{e}_{:,rd_x}(-\mathbf{q})\}$ of $D(-\mathbf{q})$ are given by the complex conjugates of the eigenvectors of $D(\mathbf{q})$, and that $\omega_\nu^2(-\mathbf{q}) = \omega_\nu^2(\mathbf{q})$, proving the theorem.

Proof 27.10 of Lemma 5.25

- For the inverse-relation, we find with Lemma 5.22 that

$$\Delta\boldsymbol{\tau}_{\mathbf{q}\kappa\alpha} = \sum_{\nu'} e_{\kappa\alpha,\nu'}(\mathbf{q}) u_{\mathbf{q}\nu'} \frac{1}{\sqrt{M_\kappa}} \tag{27.19}$$

$$= \sum_{\kappa'\alpha'} \underbrace{\sum_{\nu'} e^*_{\kappa'\alpha',\nu'}(\mathbf{q}) e_{\kappa\alpha,\nu'}(\mathbf{q})}_{=\delta_{\kappa'\kappa}\delta_{\alpha'\alpha}} \sqrt{\frac{M_{\kappa'}}{M_\kappa}} \Delta\boldsymbol{\tau}_{\mathbf{q}\kappa'\alpha'} \tag{27.20}$$

$$= \sum_{\kappa'\alpha'} \delta_{\kappa'\kappa}\delta_{\alpha'\alpha} \sqrt{\frac{M_{\kappa'}}{M_\kappa}} \Delta\boldsymbol{\tau}_{\mathbf{q}\kappa'\alpha'} \tag{27.21}$$

$$= \Delta\boldsymbol{\tau}_{\mathbf{q}\kappa\alpha} \tag{27.22}$$

- The second auxiliary identity is evident from Remark 5.16.2.
- For the derivative-relations, we find that

$$u_{\mathbf{q}\nu} = \sum_{\kappa'\alpha'} e^*_{\kappa'\alpha',\nu}(\mathbf{q}) \sqrt{M_{\kappa'}} \Delta\boldsymbol{\tau}_{\mathbf{q}\kappa'\alpha'} \tag{27.23}$$

  which implies

$$\frac{\partial}{\partial \Delta\boldsymbol{\tau}_{\mathbf{q}\kappa\alpha}} = \sum_{\nu'} \frac{\partial u_{\mathbf{q}\nu'}}{\partial \Delta\boldsymbol{\tau}_{\mathbf{q}\kappa\alpha}} \frac{\partial}{\partial u_{\mathbf{q}\nu'}} = \sum_{\nu'} e^*_{\kappa\alpha,\nu'}(\mathbf{q}) \sqrt{M_\kappa} \frac{\partial}{\partial u_{\mathbf{q}\nu'}} \tag{27.24}$$

  which proves the left side of the identity. Now we come to the right side, and compute straight forwardly as well

$$\Delta\boldsymbol{\tau}_{\mathbf{q}\kappa\alpha} = \sum_{\nu'} e_{\kappa\alpha,\nu'}(\mathbf{q}) u_{\mathbf{q}\nu'} \frac{1}{\sqrt{M_\kappa}} \tag{27.25}$$

  which implies

$$\frac{\partial}{\partial u_{\mathbf{q}\nu}} = \sum_{\kappa'\alpha'} \frac{\partial \Delta\boldsymbol{\tau}_{\mathbf{q}\kappa'\alpha'}}{\partial u_{\mathbf{q}\nu}} \frac{\partial}{\partial \Delta\boldsymbol{\tau}_{\mathbf{q}\kappa'\alpha'}} = \sum_{\kappa'\alpha'} e_{\kappa'\alpha',\nu}(\mathbf{q}) \frac{1}{\sqrt{M_{\kappa'}}} \frac{\partial}{\partial \Delta\boldsymbol{\tau}_{\mathbf{q}\kappa'\alpha'}} \tag{27.26}$$

- For the preservation of the CCR, we fist note that

$$\left[\tau_{\mathbf{n}\kappa\alpha}, -\mathrm{i}\hbar \frac{\partial}{\partial \tau_{\mathbf{n}'\kappa'\alpha'}}\right] = +\mathrm{i}\hbar \delta_{\mathbf{n}\mathbf{n}'} \delta_{\kappa\kappa'} \delta_{\alpha\alpha'}, \tag{27.27}$$

it follows

$$\left[u_{\mathbf{q}\nu}, -\mathrm{i}\hbar\frac{\partial}{\partial u_{\mathbf{q}'\nu'}}\right] = \frac{1}{N_p}\sum_{\substack{nn'\\ \kappa\kappa'\\ \alpha\alpha'}} e^*_{\kappa\alpha,\nu}(\mathbf{q})\sqrt{M_\kappa}\mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R}_n}\left[\left(\tau_{n\kappa\alpha} - R_{n\kappa\alpha}\right), \quad e_{\kappa'\alpha',\nu'}(\mathbf{q}')\,\mathrm{e}^{\mathrm{i}\mathbf{q}'\mathbf{R}_{\mathbf{n}'}}\frac{1}{\sqrt{M_{\kappa'}}}\frac{\partial}{\partial\tau_{\mathbf{n}'\kappa'\alpha'}}(-\mathrm{i}\hbar)\right] \tag{27.28}$$

$$= \frac{1}{N_p}\sum_{\substack{\mathbf{n},\mathbf{n}'\\ \kappa,\kappa'\\ \alpha,\alpha'}} e^*_{\kappa\alpha,\nu}(\mathbf{q})\sqrt{\frac{M_\kappa}{M_{\kappa'}}}\mathrm{e}^{\mathrm{i}\left(\mathbf{q}'\mathbf{R}_{\mathbf{n}'}-\mathbf{q}\mathbf{R}_{\mathbf{n}}\right)}\underbrace{\left[\tau_{\mathbf{n}\kappa\alpha}, \quad -\mathrm{i}\hbar\frac{\partial}{\partial\tau_{\mathbf{n}'\kappa'\alpha'}}\right]}_{=\delta_{\mathbf{n},\mathbf{n}'}\delta_{\kappa,\kappa'}\delta_{\alpha,\alpha'}\mathrm{i}\hbar} e_{\kappa'\alpha',\nu'}(\mathbf{q}') \tag{27.29}$$

$$= \frac{1}{N_p}\sum_{\substack{\mathbf{n}\\ \kappa,\alpha}} e^*_{\kappa\alpha,\nu}(\mathbf{q})\sqrt{\frac{M_\kappa}{M_{\kappa'}}}\mathrm{e}^{\mathrm{i}\mathbf{R}_n(\mathbf{q}'-\mathbf{q})}e_{\kappa\alpha,\nu'}(\mathbf{q}')\mathrm{i}\hbar \tag{27.30}$$

$$= \underbrace{\frac{1}{N_p}\sum_{\kappa\alpha} e^*_{\kappa\alpha,\nu}(\mathbf{q})e_{\kappa\alpha,\nu'}(\mathbf{q}')}_{=\delta_{\nu\nu'}}\,\delta_{\mathbf{q}\mathbf{q}'}\mathrm{i}\hbar \tag{27.31}$$

$$= \delta_{\nu\nu'}\delta_{\mathbf{q}\mathbf{q}'}\mathrm{i}\hbar \tag{27.32}$$

Implying

$$-\mathrm{i}\hbar\left[u_{\mathbf{q}\nu}, \frac{\partial}{\partial u_{\mathbf{q}'\nu'}}\right] = \mathrm{i}\hbar\delta_{\nu\nu'}\delta_{\mathbf{q}\mathbf{q}'} \quad\Longleftrightarrow\quad \left[u_{\mathbf{q}\nu}, \frac{\partial}{\partial u_{\mathbf{q}'\nu'}}\right] = -\delta_{\nu\nu'}\delta_{\mathbf{q}\mathbf{q}'} \tag{27.33}$$

Proof 27.11 of Lemma 5.26

- For the identity $\tau_{\mathbf{n}\kappa\alpha} = \ldots$:

  We simply have to summarize all the coordinate-changes. Those were ...

$$\begin{aligned}
\tau_{\mathbf{n}\kappa\alpha} \quad &\overset{\text{Translation}}{\longrightarrow} \quad \Delta\tau_{\mathbf{n}\kappa\alpha} := \tau_{\mathbf{n}\kappa\alpha} - R_{\mathbf{n}\kappa\alpha}\\
&\overset{\text{Fourier}}{\longrightarrow} \quad \Delta\tau_{\mathbf{q}\kappa\alpha} := \frac{1}{\sqrt{N_p}}\sum_{\mathbf{n}}\Delta\tau_{\mathbf{n}\kappa\alpha}\mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R}_n}\\
&\overset{\text{Eigenmodes and rescaling}}{\longrightarrow} \quad u_{\mathbf{q}\nu} := \sum_{\kappa\alpha} e^*_{\kappa\alpha,\nu}(\mathbf{q})\sqrt{M_\kappa}\Delta\tau_{\mathbf{q}\kappa\alpha}
\end{aligned}$$

  Now combining the equations gives

$$\tau_{\mathbf{n}\kappa\alpha} \equiv R_{\mathbf{n}\kappa\alpha} + \Delta\tau_{\mathbf{n}\kappa\alpha} \tag{27.34}$$

$$= R_{\mathbf{n}\kappa\alpha} + \frac{1}{\sqrt{N_p}} \sum_{\mathbf{q}} \underbrace{\Delta\tau_{\mathbf{q}\kappa\alpha}}_{=\sum_{\nu'} e_{\kappa\alpha,\nu'}(\mathbf{q}) u_{\mathbf{q}\nu'} \frac{1}{\sqrt{M_\kappa}}} \mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R}_\mathbf{n}} \tag{27.35}$$

$$= R_{\mathbf{n}\kappa\alpha} + \frac{1}{\sqrt{N_p}} \sum_{\substack{\mathbf{q}\\ \nu'}} e_{\kappa\alpha,\nu'}(\mathbf{q}) \mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R}_n} \frac{1}{\sqrt{M_\kappa}} u_{\mathbf{q}\nu'} \tag{27.36}$$

which is the left hand side of Equation (5.38). Combining equations from the other side, gives the right hand side of the equation.

- For the identity $u_{\mathbf{q}\nu} = \ldots$: Taking the individual inverses one gets the relation

$$u_{\mathbf{q}\nu} = \sum_{\kappa'\alpha'} e^*_{\kappa'\alpha',\nu}(\mathbf{q}) \sqrt{M_{\kappa'}} \Delta\boldsymbol{\tau}_{\mathbf{q}\kappa'\alpha'} \tag{27.37}$$

$$= \sum_{\substack{\kappa'\alpha'\\ \mathbf{n}}} e^*_{\kappa'\alpha',\nu}(\mathbf{q}) \sqrt{M_{\kappa'}} \mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R}_n} \frac{1}{\sqrt{N_p}} \underbrace{\Delta\boldsymbol{\tau}_{n\kappa'\alpha'}}_{=(\boldsymbol{\tau}_{n\kappa'\alpha'} - \mathbf{R}_{\mathbf{n}\kappa'\alpha'})} \tag{27.38}$$

$$= \frac{1}{\sqrt{N_p}} \sum_{\substack{\mathbf{R_n}\in L\\ \kappa'\alpha'}} e^*_{\kappa'\alpha',\nu}(\mathbf{q}) \sqrt{M_{\kappa'}} \mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R}_n} \left(\boldsymbol{\tau}_{n\kappa'\alpha'} - \mathbf{R}_{\mathbf{n}\kappa'\alpha'}\right) \tag{27.39}$$

We can also simply insert the equation and see that it describely corrects the inverse relation:

$$u_{\mathbf{q}\nu} = \sum_{\kappa'\alpha'} e^*_{\kappa'\alpha',\nu}(\mathbf{q}) \sqrt{M_{\kappa'}} \Delta\tau_{\mathbf{q}\kappa'\alpha'} \tag{27.40}$$

$$= \sum_{\kappa'\alpha'} e^*_{\kappa'\alpha',\nu}(\mathbf{q}) \sqrt{M_{\kappa'}} \sum_{\nu''} e_{\kappa'\alpha',\nu''}(\mathbf{q}) u_{\mathbf{q}\nu''} \frac{1}{\sqrt{M_{\kappa'}}} \tag{27.41}$$

$$= \sum_{\nu''} u_{\mathbf{q}\nu''} \sqrt{\frac{M_{\kappa'}}{M_{\kappa'}}} \underbrace{\sum_{\kappa'\alpha'} \mathrm{e}^*_{\kappa'\alpha',\nu}(\mathbf{q}) \mathrm{e}_{\kappa'\alpha',\nu''}(\mathbf{q})}_{\delta_{\nu\nu''}} \tag{27.42}$$

$$= u_{\mathbf{q}\nu} \tag{27.43}$$

- For the identity $\frac{1}{\partial\tau_{\mathbf{n}\kappa\alpha}} = \ldots$:

  Formula: If $x_i = x_i\left(x'_1, \ldots, x'_n\right)$ then

$$\frac{\partial}{\partial x'_i} = \sum_j \frac{\partial x_j}{\partial x'_i} \frac{\partial}{\partial x_j}$$

  Therefore

$$\frac{\partial}{\partial \tau_{n\kappa\alpha}} = \frac{\partial}{\partial\left(\Delta\tau_{\mathbf{n}\kappa\alpha}\right)} \underbrace{\frac{\partial\left(\Delta\tau_{\mathbf{n}\kappa\alpha}\right)}{\partial \boldsymbol{\tau}_{\mathbf{n}\kappa\alpha}}}_{=1} = \frac{\partial}{\partial\left(\Delta\tau_{\mathbf{n}\kappa\alpha}\right)} \tag{27.44}$$

$$\frac{\partial}{\partial\left(\Delta\tau_{\mathbf{n}\kappa\alpha}\right)} = \sum_{\mathbf{q}} \frac{\partial}{\partial\left(\Delta\tau_{\mathbf{q}\kappa\alpha}\right)} \underbrace{\frac{\partial\left(\Delta\tau_{\mathbf{q}\kappa\alpha}\right)}{\partial\left(\Delta\tau_{\mathbf{n}\kappa\alpha}\right)}}_{=\mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R}_\mathbf{n}}\frac{1}{\sqrt{N_p}}} = \frac{1}{\sqrt{N_p}} \sum_{\mathbf{q}} \mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R}_\mathbf{n}} \frac{\partial}{\partial\left(\Delta\tau_{\mathbf{q}\kappa\alpha}\right)} \tag{27.45}$$

$$\frac{\partial}{\partial\left(\Delta\tau_{\mathbf{q}\kappa\alpha}\right)} = \sum_{\nu'} \frac{\partial}{\partial u_{\mathbf{q}\nu'}} \underbrace{\frac{\partial u_{\mathbf{q}\nu'}}{\partial\left(\Delta\tau_{\mathbf{q}\kappa\alpha}\right)}}_{=\sqrt{M_\kappa}e^*_{\kappa\alpha,\nu'}(\mathbf{q})} = \sum_{\nu'} \sqrt{M_\kappa} e^*_{\kappa\alpha,\nu'}(\mathbf{q}) \frac{\partial}{\partial u_{\mathbf{q}\nu'}} \tag{27.46}$$

Combining these equations yields the desired result as

$$\frac{\partial}{\partial \tau_{n\kappa\alpha}} = \frac{\partial}{\partial\left(\Delta\tau_{n\kappa\alpha}\right)} \tag{27.47}$$

$$= \frac{1}{\sqrt{N_p}} \sum_{\mathbf{q}} \mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R}_n} \underbrace{\frac{\partial}{\partial\left(\Delta\tau_{\mathbf{q}\kappa\alpha}\right)}}_{=\sum_\mu \sqrt{M_\kappa} e^*_{\kappa\alpha,\mu}(\mathbf{q}) \frac{\partial}{\partial u_{\mathbf{q}\mu}}} \tag{27.48}$$

$$= \frac{\sqrt{M_\kappa}}{\sqrt{N_p}} \sum_{\substack{\mathbf{q}\\ \mu}} \mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R}_n} e^*_{\kappa\alpha,\mu}(\mathbf{q}) \frac{\partial}{\partial u_{\mathbf{q}\mu}} \tag{27.49}$$

- For the identity $\frac{\partial}{\partial u_{\mathbf{q}\nu}} = \ldots$:

$$\frac{\partial}{\partial u_{\mathbf{q}\nu}} = \sum_{\substack{\mathbf{R_n}\in \mathrm{L_f}\\ \kappa',\alpha'}} \frac{\partial}{\partial \tau_{\mathbf{n}\kappa'\alpha'}} \underbrace{\frac{\partial \tau_{\mathbf{n}\kappa'\alpha'}}{\partial u_{\mathbf{q}\nu}}}_{=\frac{1}{\sqrt{N_p}} e^*_{\kappa'\alpha',\nu}(\mathbf{q}) \frac{1}{\sqrt{M_{\kappa'}}} \mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R_n}}} \tag{27.50}$$

$$= \frac{1}{\sqrt{N_p}} \sum_{\substack{\mathbf{R_n}\in L\\ \kappa'\alpha'}} e_{\kappa'\alpha',\nu}(\mathbf{q}) \mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R_n}} \frac{1}{\sqrt{M_{\kappa'}}} \frac{\partial}{\partial \tau_{\mathbf{n}\kappa'\alpha'}} \tag{27.51}$$

Proof 27.12 of Corollary 5.27

We can the terms appearing in the kinetic and potential energy expression through their new coordinates as

$$\Delta\tau^*_{\mathbf{q}\kappa\alpha} = \sum_\nu e^*_{\kappa\alpha,\nu}(\mathbf{q}) u^*_{\mathbf{q}\nu} \frac{1}{\sqrt{M_\kappa}}, \qquad \Delta\tau_{\mathbf{q}\kappa'\alpha'} = \sum_\nu e_{\kappa'\alpha',\nu}(\mathbf{q}) u_{\mathbf{q}\nu} \frac{1}{\sqrt{M_{\kappa'}}} \tag{27.52}$$

with $u_{\mathbf{q}\nu}$ as new coordinates.

Potential Energy Term:

Inserting this into the expression for the potential energy gives

$$\frac{1}{2}\sum_{\mathbf{q}}\sum_{\substack{\kappa\kappa'\\ \alpha\alpha'}}\Delta\tau^*_{\mathbf{q}\kappa\alpha}\frac{D_{\kappa\alpha,\kappa'\alpha'}(\mathbf{q})}{\sqrt{M_\kappa M_{\kappa'}}^{-1}}\Delta\tau_{\mathbf{q}\kappa'\alpha'} = \frac{1}{2}\sum_{\mathbf{q}}\sum_{\substack{\kappa\kappa'\\ \alpha\alpha'\\ \nu\nu'}} e^*_{\kappa\alpha,\nu}(\mathbf{q})u^*_{\mathbf{q}\nu}D_{\kappa\alpha,\kappa'\alpha'}(\mathbf{q})e_{\kappa'\alpha',\nu'}(\mathbf{q})u_{\mathbf{q}\nu'} \tag{27.53}$$

$$= \dots \tag{27.54}$$

Performing the sum over $\kappa'\alpha'$ (i.e. using Equation (5.31)) gives

$$\dots = \frac{1}{2}\sum_{\mathbf{q}}\sum_{\substack{\kappa\alpha\\ \nu\nu'}} e^*_{\kappa\alpha,\nu}(\mathbf{q})u^*_{\mathbf{q}\nu}\omega^2_\nu(\mathbf{q})e_{\kappa\alpha,\nu'}(\mathbf{q})u_{\mathbf{q}\nu'} = \dots \tag{27.55}$$

Now we use Equation (5.32)

$$\sum_{\kappa\alpha} e^*_{\kappa\alpha,\nu}(\mathbf{q})e_{\kappa\alpha,\nu'}(\mathbf{q}) = \delta_{\nu,\nu'} \qquad (**) \tag{27.56}$$

such that we obtain

$$\dots = \frac{1}{2}\sum_{\mathbf{q}\nu} u^*_{\mathbf{q}\nu}\omega^2_\nu(\mathbf{q})u_{\mathbf{q}\nu} = \frac{1}{2}\sum_{\mathbf{q}\nu}\omega^2_\nu(\mathbf{q})u_{-\mathbf{q}\nu}u_{\mathbf{q}\nu} \tag{27.57}$$

Kinetic Energy Term:

Before we proceed let's pay attention to the Momentum again:

$$T_e \equiv \sum_{\substack{\mathbf{R_n}\in L\\ \kappa\alpha}} -\frac{\hbar^2}{2M_\kappa}\frac{\partial^2}{\partial\tau^2_{\mathbf{n}\kappa\alpha}} \tag{27.58}$$

$$\equiv \sum_{\mathbf{n}\kappa\alpha}\frac{\partial}{\partial\tau_{\mathbf{n}\kappa\alpha}}\left(-\frac{\hbar^2}{2M_\kappa}\right)\frac{\partial}{\partial\tau_{\mathbf{n}\kappa\alpha}} \tag{27.59}$$

Then it follows

$$T_e = \sum_{n\kappa\alpha}\frac{\partial}{\partial\tau_{n\kappa\alpha}}\left(-\frac{\hbar^2}{2M_\kappa}\right)\frac{\partial}{\partial\tau_{n\kappa\alpha}} \tag{27.60}$$

With $\sum_{\kappa\alpha} \equiv \sum_{\nu}$

$$T_e = \sum_{\substack{\mathbf{q}\mathbf{q}' \\ \mu\mu' \\ n\nu}} \left(-\frac{\hbar^2}{2M_\kappa}\right)\left(\frac{M_\kappa}{N_p}\right) \mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R}_n} e^*_{\nu\mu}(\mathbf{q}) \frac{\partial}{\partial u_{\mathbf{q}\mu}} \mathrm{e}^{-\mathrm{i}\mathbf{q}'\mathbf{R}_n} e^*_{\nu\mu'}\left(\mathbf{q}'\right) \frac{\partial}{\partial u_{\mathbf{q}'\mu}} \tag{27.61}$$

$$= \sum_{\substack{\mathbf{q}\mathbf{q}' \\ \mu\mu' \\ \nu}} \underbrace{\sum_n \mathrm{e}^{-\mathrm{i}\mathbf{R}_n(\mathbf{q}+\mathbf{q}')}}_{=N_p\delta_{\mathrm{q}+\mathrm{q}',0}} \left(-\frac{\hbar^2}{2N_p}\right) e^*_{\nu\mu}(\mathbf{q}) \frac{\partial}{\partial u_{\mathbf{q}\mu}} e^*_{\nu\mu'}\left(\mathbf{q}'\right) \frac{\partial}{\partial u_{\mathbf{q}'\mu}} \tag{27.62}$$

$$= \sum_{\substack{\mathbf{q} \\ \mu\mu' \\ \nu}} -\frac{\hbar^2}{2} e^*_{\nu\mu}(\mathbf{q}) \underbrace{e^*_{\nu\mu'}(-\mathbf{q})}_{=e_{\nu\mu'}(\mathbf{q})} \frac{\partial}{\partial u_{\mathbf{q}\mu}} \frac{\partial}{\partial u_{-\mathbf{q}\mu}} \tag{27.63}$$

$$= \sum_{\substack{\mathbf{q} \\ \mu\mu'}} -\frac{\hbar^2}{2} \underbrace{\sum_\nu e^*_{\nu\mu}(\mathbf{q})\, e_{\nu\mu'}(\mathbf{q})}_{=\delta_{\mu\mu'}} \frac{\partial}{\partial u_{\mathbf{q}\mu}} \frac{\partial}{\partial u_{-\mathbf{q}\mu}} \tag{27.64}$$

$$= \sum_{\mathbf{q}\mu} -\frac{\hbar^2}{2} \frac{\partial}{\partial u_{\mathbf{q}\mu}} \frac{\partial}{\partial u_{-\mathbf{q}\mu}} \tag{27.65}$$

Proof 27.13 of Lemma 5.29

Proof of CCR:

$$\left[a_{\mathbf{q}\mu}, a^\dagger_{\mathbf{q}\mu}\right] = \sqrt{\frac{\omega_\mu(\mathbf{q})\,\omega_{\mu'}(\mathbf{q}')}{(2\hbar)^2}} \left(\frac{\hbar}{\omega_\mu(\mathbf{q})} \underbrace{\left[\frac{\partial}{\partial u_{-\mathbf{q}\mu}}, u_{-\mathbf{q}'\mu'}\right]}_{=\,\delta_{\mu\mu'}\delta_{\mathbf{q}\mathbf{q}'}} - \frac{\hbar}{\omega_{\mu'}(\mathbf{q}')} \left[u_{\mathbf{q}\mu}, \frac{\partial}{\partial u_{\mathbf{q}\mu}}\right] \right) \tag{27.66}$$

$$= \sqrt{\frac{\omega_\mu(\mathbf{q})\,\omega_{\mu'}(\mathbf{q}')}{(2\hbar)^2}}\,\hbar \left(\frac{\delta_{\mathbf{q}\mathbf{q}'}\delta_{\mu\mu'}}{\omega_\mu(\mathbf{q})} + \frac{\delta_{\mathbf{q}\mathbf{q}'}\delta_{\mu\mu'}}{\omega_{\mu'}(\mathbf{q}')}\right) \tag{27.67}$$

$$= \delta_{\mathbf{q}\mathbf{q}'}\delta_{\mu\mu'} \tag{27.68}$$

Proof of the inversion Formulae:

$$u_{\mathbf{q}\mu} = \left(a_{\mathbf{q}\mu} + a^\dagger_{-\mathbf{q}\mu}\right)\sqrt{\frac{\hbar}{2\omega_\mu(\mathbf{q})}} \tag{27.69}$$

Proof:

$$
\begin{aligned}
&\left(a_{\mathbf{q}\mu} + a^{\dagger}_{-\mathbf{q}\mu}\right)\sqrt{\frac{\hbar}{2\omega_{\mu}(\mathbf{q})}} && (27.70)\\
=&\left(\sqrt{\frac{\omega_{\mu}(\mathbf{q})}{2\hbar}}\left(u_{\mathbf{q}\mu} + \frac{\hbar}{\omega_{\mu}(\mathbf{q})}\frac{\partial}{\partial u_{-\mathbf{q}\mu}}\right) + \sqrt{\frac{\omega_{\mu}(-\mathbf{q})}{2\hbar}}\left(u_{\mathbf{q}\mu} - \frac{\hbar}{\omega_{\mu}(-\mathbf{q})}\frac{\partial}{\partial u_{-\mathbf{q}\mu}}\right)\right)\sqrt{\frac{\hbar}{2\omega_{\mu}(\mathbf{q})}} && (27.71)\\
\overset{\text{symmetry of }\omega_{\mu}}{=}&\sqrt{\frac{\omega_{\mu}(\mathbf{q})}{2\hbar}}\left(\left(u_{\mathbf{q}\mu} + \frac{\hbar}{\omega_{\mu}(\mathbf{q})}\frac{\partial}{\partial u_{-\mathbf{q}\mu}}\right) + \left(u_{\mathbf{q}\mu} - \frac{\hbar}{\omega_{\mu}(\mathbf{q})}\frac{\partial}{\partial u_{-\mathbf{q}\mu}}\right)\right)\sqrt{\frac{\hbar}{2\omega_{\mu}(\mathbf{q})}} && (27.72)\\
=&\sqrt{\frac{\omega_{\mu}(\mathbf{q})}{2\hbar}}\left(u_{\mathbf{q}\mu} + u_{\mathbf{q}\mu}\right)\sqrt{\frac{\hbar}{2\omega_{\mu}(\mathbf{q})}} && (27.73)\\
=&u_{\mathbf{q}\mu} && (27.74)
\end{aligned}
$$

Now the other identity:

$$
\frac{\partial}{\partial u_{\mathbf{q}\mu}} = \left(a_{-\mathbf{q}\mu} - a^{\dagger}_{\mathbf{q}\mu}\right)\sqrt{\frac{\omega_{\mu}(\mathbf{q})}{2\hbar}} \tag{27.75}
$$

By straight forward computation we find

$$
\begin{aligned}
&\left(a_{-\mathbf{q}\mu} - a^{\dagger}_{\mathbf{q}\mu}\right)\sqrt{\frac{\omega_{\mu}(\mathbf{q})}{2\hbar}} && (27.76)\\
=&\left(\sqrt{\frac{\omega_{\mu}(-\mathbf{q})}{2\hbar}}\left(u_{-\mathbf{q}\mu} + \frac{\hbar}{\omega_{\mu}(-\mathbf{q})}\frac{\partial}{\partial u_{\mathbf{q}\mu}}\right) - \sqrt{\frac{\omega_{\mu}(\mathbf{q})}{2\hbar}}\left(u_{-\mathbf{q}\mu} - \frac{\hbar}{\omega_{\mu}(\mathbf{q})}\frac{\partial}{\partial u_{\mathbf{q}\mu}}\right)\right)\sqrt{\frac{\omega_{\mu}(\mathbf{q})}{2\hbar}} && (27.77)\\
\overset{\text{symmetry of }\omega_{\mu}}{=}&\sqrt{\frac{\omega_{\mu}(\mathbf{q})}{2\hbar}}\left(\left(u_{-\mathbf{q}\mu} + \frac{\hbar}{\omega_{\mu}(\mathbf{q})}\frac{\partial}{\partial u_{\mathbf{q}\mu}}\right) - \left(u_{-\mathbf{q}\mu} - \frac{\hbar}{\omega_{\mu}(\mathbf{q})}\frac{\partial}{\partial u_{\mathbf{q}\mu}}\right)\right)\sqrt{\frac{\omega_{\mu}(\mathbf{q})}{2\hbar}} && (27.78)\\
=&\sqrt{\frac{\omega_{\mu}(\mathbf{q})}{2\hbar}}\left(\frac{2\hbar}{\omega_{\mu}(\mathbf{q})}\frac{\partial}{\partial u_{\mathbf{q}\mu}}\right)\sqrt{\frac{\omega_{\mu}(\mathbf{q})}{2\hbar}} && (27.79)\\
=&\frac{\partial}{\partial u_{\mathbf{q}\mu}} && (27.80)
\end{aligned}
$$

Proof 27.14 of Corollary 5.30

$$H \approx \sum_{q\mu}\left(-\frac{\hbar^2}{2}\frac{\partial}{\partial u_{\mathbf{q}\mu}}\frac{\partial}{\partial u_{-\mathbf{q}\mu}}+\frac{1}{2}\omega_\mu^2(q)\,u_{\mathbf{q}\mu}u_{-\mathbf{q}\mu}\right) \tag{27.81}$$

$$= \sum_{q\mu} -\frac{\hbar^2}{2}\left(a_{-\mathbf{q}\mu}-a^\dagger_{\mathbf{q}\mu}\right)\sqrt{\frac{\omega_\mu(\mathbf{q})}{2\hbar}}\left(a_{\mathbf{q}\mu}-a^\dagger_{-\mathbf{q}\mu}\right)\sqrt{\frac{\omega_\mu(-\mathbf{q})}{2\hbar}} \tag{27.82}$$

$$+\frac{1}{2}\omega_\mu^2(q)\left(a_{\mathbf{q}\mu}+a^\dagger_{-\mathbf{q}\mu}\right)\sqrt{\frac{\hbar}{2\omega_\mu(\mathbf{q})}}\left(a_{-\mathbf{q}\mu}+a^\dagger_{\mathbf{q}\mu}\right)\sqrt{\frac{\hbar}{2\omega_\mu(-\mathbf{q})}} \tag{27.83}$$

$$= \sum_{q\mu} -\frac{\hbar\omega_\mu(q)}{4}\left(a_{-\mathbf{q}\mu}-a^\dagger_{\mathbf{q}\mu}\right)\left(a_{\mathbf{q}\mu}-a^\dagger_{-\mathbf{q}\mu}\right) \tag{27.84}$$

$$+\frac{\hbar\omega_\mu(q)}{4}\left(a_{\mathbf{q}\mu}+a^\dagger_{-\mathbf{q}\mu}\right)\left(a_{-\mathbf{q}\mu}+a^\dagger_{\mathbf{q}\mu}\right) \tag{27.85}$$

$$= \sum_{q\mu}\frac{\hbar\omega_\mu(q)}{4}\left(-a_{-\mathbf{q}\mu}a_{\mathbf{q}\mu}+a_{-\mathbf{q}\mu}a^\dagger_{-\mathbf{q}\mu}+a^\dagger_{\mathbf{q}\mu}a_{\mathbf{q}\mu}-a^\dagger_{\mathbf{q}\mu}a^\dagger_{-\mathbf{q}\mu}+a_{\mathbf{q}\mu}a_{-\mathbf{q}\mu}+a_{\mathbf{q}\mu}a^\dagger_{\mathbf{q}\mu}+a^\dagger_{-\mathbf{q}\mu}a_{-\mathbf{q}\mu}+a^\dagger_{-\mathbf{q}\mu}a^\dagger_{\mathbf{q}\mu}\right) \tag{27.86}$$

$$= \sum_{q\mu}\frac{\hbar\omega_\mu(q)}{4}\left(\underbrace{a_{-\mathbf{q}\mu}a^\dagger_{-\mathbf{q}\mu}}_{=1+a^\dagger_{-\mathbf{q}\mu}a_{-\mathbf{q}\mu}}+a^\dagger_{\mathbf{q}\mu}a_{\mathbf{q}\mu}\underbrace{+a_{\mathbf{q}\mu}a^\dagger_{\mathbf{q}\mu}}_{=1+a^\dagger_{\mathbf{q}\mu}a_{\mathbf{q}\mu}}+a^\dagger_{-\mathbf{q}\mu}a_{-\mathbf{q}\mu}\right) \tag{27.87}$$

$$= \sum_{q\mu}\frac{\hbar\omega_\mu(q)}{4}\left(2+4a^\dagger_{\mathbf{q}\mu}a_{\mathbf{q}\mu}\right) \tag{27.88}$$

$$= \sum_{q\mu}\hbar\omega_\mu(q)\left(a^\dagger_{\mathbf{q}\mu}a_{\mathbf{q}\mu}+\frac{1}{2}\right) \tag{27.89}$$

Proof 27.15 of Lemma 5.31

Combining the derived equations we obtain

$$\Delta\boldsymbol{\tau}_{\mathbf{n}\nu} = \frac{1}{\sqrt{N_p}}\sum_{\substack{\mathbf{q}\\ \nu'}} e_{\nu\nu'}(\mathbf{q})\mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R}_n}\frac{1}{\sqrt{M_\nu}}\underbrace{u_{\mathbf{q}\nu'}}_{=\left(a_{\mathbf{q}\nu'}+a^\dagger_{-\mathbf{q}\nu'}\right)\sqrt{\frac{\hbar}{2\omega_{\nu'}(\mathbf{q})}}} \tag{27.90}$$

$$= \frac{1}{\sqrt{N_p}}\sum_{\substack{\mathbf{q}\\ \nu'}}\sqrt{\frac{\hbar}{2M_\nu\omega_{\nu'}(\mathbf{q})}}e_{\nu\nu'}(\mathbf{q})\mathrm{e}^{-\mathrm{i}\mathbf{q}\mathbf{R}_n}\left(a_{\mathbf{q}\nu'}+a^\dagger_{-\mathbf{q}\nu'}\right) \tag{27.91}$$

Proof 27.16 of Lemma 35.5 As long as $m_{1,2,3} \neq \infty$ (or, equivalently $h, k, l \neq 0$) we can consider the three special vectors

$$\mathbf{T}_1 = m_1\mathbf{a}_1 - m_2\mathbf{a}_2, \quad \mathbf{T}_2 = m_2\mathbf{a}_2 - m_3\mathbf{a}_3, \quad \mathbf{T}_3 = m_3\mathbf{a}_3 - m_1\mathbf{a}_1. \tag{27.92}$$

According to the mnemonic

$$\text{Plane} = \text{Support-Vector} + c_1\text{Directional-Vector-1} + c_2\text{Directional-Vector-2} \quad c_i \in \mathbb{R} \tag{27.93}$$

we have to show that the scalar-product $\mathbf{T}_i \cdot \mathbf{G}$ vanishes.

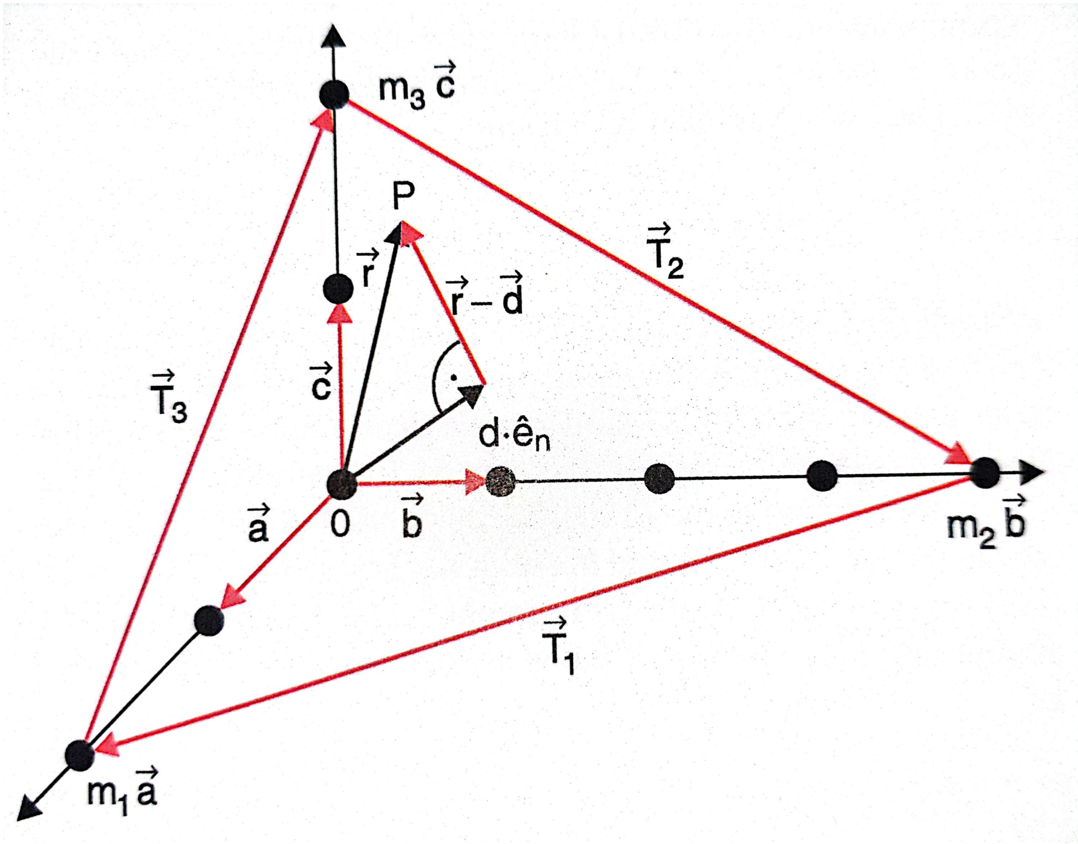


Figure 27.1.: Excerpt from Wolfang Demtröders Experimentalphysik 3.

And indeed, a straight-forward calculation yields

$$\begin{aligned}\mathbf{T}_1 \cdot \mathbf{G} &= \left(m_1\mathbf{a}_1 - m_2\mathbf{a}_2\right) \cdot \left(h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3\right) \\ &= 2\pi\left(m_1 h - m_2 k\right) \\ &= 0\end{aligned}$$

and analog for $\mathbf{T}_{1,2}$.

Now we consider the case that at least one of the m_i is equal to infinity. Without loss of generality, we consider $m_1 = \infty$ (equivalently one has $h = 0$). Then we consider $\widetilde{\mathbf{T}}_1 = \mathbf{a}_1$ and $\mathbf{T}_2 = m_2\mathbf{a}_2 - m_3\mathbf{a}_3$ as directional-vectors of the planes and the scalar-product vanishes again.

Now we consider the case that two of the m_i are equal to infinity. Without loss of generality, we consider $m_1 = m_2 = \infty$ (equivalently one has $h = k = 0$). Then we consider $\widetilde{\mathbf{T}}_1 = \mathbf{a}_1$ and $\widetilde{\mathbf{T}}_2 = \mathbf{a}_2$ as directional-vectors of the planes and the scalar-product vanishes again.

Adding Pictures.

Proof 27.17 of Corollary 33.8 We compute

$$
\begin{aligned}
\left(\mathrm{P}_\zeta^2\,\psi\right)(x_1,\dots,x_n) &= \mathrm{P}_\zeta \frac{1}{N!} \sum_{\pi\in \mathrm{S}_N} \operatorname{sign}(\pi)^{1-\theta(\zeta)} \psi(x_{\pi(1)},\dots,x_{\pi(N)}) \\
&= \frac{1}{N!^2} \sum_{\pi,\sigma\in \mathrm{S}_N} \operatorname{sign}(\pi)^{1-\theta(\zeta)} \operatorname{sign}(\sigma)^{1-\theta(\zeta)} \psi(x_{\sigma(\pi(1))},\dots,x_{\sigma(\pi(N))}) \\
&= \frac{1}{N!^2} \sum_{\pi,\sigma\in \mathrm{S}_N} \operatorname{sign}(\sigma\circ\pi)^{1-\theta(\zeta)} \psi(x_{\sigma(\pi(1))},\dots,x_{\sigma(\pi(N))}) \\
&= \dots
\end{aligned}
$$

Now we use, that for each $\pi \in \mathrm{S}_N$ one has $\mathrm{S}_N = \mathrm{S}_N \circ\pi$ (as a set). Hence we can write

$$
\begin{aligned}
\dots &= \frac{1}{N!^2} \sum_{\pi\in \mathrm{S}_N} \sum_{\tilde\sigma\in \mathrm{S}_N} \operatorname{sign}(\tilde\sigma)^{1-\theta(\zeta)} \psi(x_{\tilde\sigma(1)},\dots,x_{\tilde\sigma(N)}) \\
&= \frac{1}{N!} \sum_{\sigma\in \mathrm{S}_N} \operatorname{sign}(\sigma)^{1-\theta(\zeta)} \psi(x_{\sigma(1)},\dots,x_{\sigma(N)}) \\
&= \left(\mathrm{P}_\zeta\,\psi\right)(x_1,\dots,x_n)
\end{aligned}
$$

Proof 27.18 of Lemma 33.11

We have to show that the two expressions coincide. The wavefunction associated with Equation (33.10) is given as

$$
(x_1,\dots,x_N|\psi\rangle = \prod_{j=1}^{N} \langle x_i|\psi_j\rangle \equiv \psi(x_1,\dots,x_N) \tag{27.94}
$$

Therefore we obtain on the one hand

$$
\begin{aligned}
\left(\mathrm{P}_\zeta\,\psi\right)(x_1,\dots,x_N) &= \frac{1}{N!} \sum_{\pi\in \mathrm{S}_N} \operatorname{sign}(\pi)^{1-\theta(\zeta)} \psi(x_{\pi(1)},\dots,x_{\pi(N)}) \\
&= \frac{1}{N!} \sum_{\pi\in \mathrm{S}_N} \operatorname{sign}(\pi)^{1-\theta(\zeta)} \prod_{j=1}^{N} \langle x_{\pi(j)}|\psi_j\rangle \\
&= \frac{1}{N!} \sum_{\pi\in \mathrm{S}_N} \operatorname{sign}(\pi)^{1-\theta(\zeta)} \prod_{j=1}^{N} \psi_j(x_{\pi(j)})
\end{aligned}
$$

and we obtain on the other hand with Equation (33.11)

$$
(x_1,\dots,x_N|\,\mathrm{P}_\zeta\,|\psi\rangle = \frac{1}{N!} \sum_{\pi\in \mathrm{S}_N} \operatorname{sign}(\pi)^{1-\theta(\zeta)} \prod_{j=1}^{N} \psi_{\pi(j)}(x_j)
$$

the two expressions coincide which can be seen by reordering the products and using $\operatorname{sign}(\sigma) = \operatorname{sign}(\sigma^{-1})$.

Proof 27.19 of Lemma 33.14 We have to show, that

$$\forall \psi, \chi \in \mathscr{H}_N \,:\, \langle \psi |\, \mathrm{P}_\zeta\, \chi \rangle = \langle \mathrm{P}_\zeta\, \psi | \chi \rangle \tag{27.95}$$

It suffices to prove this identity for basis-elements. Therefore we compute on the one hand

$$\begin{aligned}
(i_1, \dots, i_N |\, \mathrm{P}_\zeta\, | j_1, \dots, j_N) &= \frac{1}{N!} \sum_{\pi \in \mathrm{S}_N} \operatorname{sign}(\pi)^{1-\theta(\zeta)} (i_1, \dots, i_N | j_{\pi(1)}, \dots, j_{\pi(N)}) \\
&= \frac{1}{N!} \sum_{\pi \in \mathrm{S}_N} \operatorname{sign}(\pi)^{1-\theta(\zeta)} \prod_{l=1}^{N} \langle i_l | j_{\pi(l)} \rangle \\
&= \frac{1}{N!} \sum_{\pi \in \mathrm{S}_N} \operatorname{sign}(\pi)^{1-\theta(\zeta)} \prod_{l=1}^{N} \delta_{i_l, j_{\pi(l)}}
\end{aligned}$$

and we obtain on the other hand

$$(i_1, \dots, i_N |\, \mathrm{P}_\zeta^\dagger\, | j_1, \dots, j_N) = \frac{1}{N!} \sum_{\pi \in \mathrm{S}_N} \operatorname{sign}(\pi)^{1-\theta(\zeta)} \prod_{l=1}^{N} \delta_{i_{\pi(l)}, j_l}$$

Just as in the prove of the Lemma before, the two expressions coincide, which can also be seen by expanding the products reordering the appearing factors and using $\operatorname{sign}(\sigma) = \operatorname{sign}(\sigma^{-1})$.

Proof 27.20 of Theorem 33.57 We adopt the more general notation from the earlier section. This yields for the Hamiltonian in the harmonic approximation

$$H \equiv \sum_{\lambda \in \mathscr{A}} \hbar\omega_\lambda b_\lambda^\dagger b_\lambda \tag{27.96}$$

in the grand-canonical Ensemble, we obtain, referring to an enumeration $\lambda_1, \lambda_2, \ldots, \lambda_{rd_\mathrm{x} N_p}$ of $\mathscr{A}$, the identities

$$\langle b_\lambda^\dagger b_\lambda \rangle = \frac{\mathrm{Tr}_\mathscr{H}\left(\mathrm{e}^{-\beta H} b_\lambda^\dagger b_\lambda\right)}{\mathrm{Tr}_\mathscr{H}\left(\mathrm{e}^{-\beta H}\right)} = \frac{\sum_{n_{\lambda_1}=0}^\infty \sum_{n_{\lambda_2}=0}^\infty \cdots \sum_{n_{\lambda_{rd_\mathrm{x} N_p}}=0}^\infty \left\langle n_{\lambda_1}, \ldots, n_{\lambda_{rd_\mathrm{x} N_p}} \mid \mathrm{e}^{-\beta H} b_\lambda^\dagger b_\lambda \mid n_{\lambda_1}, \ldots, n_{\lambda_{rd_\mathrm{x} N_p}} \right\rangle}{\sum_{n_{\lambda_1}=0}^\infty \sum_{n_{\lambda_2}=0}^\infty \cdots \sum_{n_{\lambda_{rd_\mathrm{x} N_p}}=0}^\infty \left\langle n_{\lambda_1}, \ldots, n_{\lambda_{rd_\mathrm{x} N_p}} \mid \mathrm{e}^{-\beta H} \mid n_{\lambda_1}, \ldots, n_{\lambda_{rd_\mathrm{x} N_p}} \right\rangle} = \star$$

using now the hermiticity of the Hamiltonian, we see that we can replace $b_{\lambda_i}^\dagger b_{\lambda_i}$ by the numbers n_{λ_i} and factorize the expression, such that everything except for one pair of sums cancels out and we obtain

$$\star = \frac{\sum_{n_\lambda=0}^\infty \mathrm{e}^{-\beta\hbar\omega_\lambda n_\lambda} n_\lambda}{\sum_{n_\lambda=0}^\infty \mathrm{e}^{-\beta n_\lambda \hbar\omega_\lambda}} \equiv -\left(\frac{f'(x)}{f(x)}\right)_{|x=\beta\hbar\omega_\lambda}$$

We therefore examine the function

$$f(x) = \sum_{n_\lambda=0}^\infty \mathrm{e}^{-n_\lambda x} = \frac{1}{1-\mathrm{e}^{-x}}$$

and obtain the derivative

$$f'(x) = -\frac{\mathrm{e}^{-x}}{(1-\mathrm{e}^{-x})^2}$$

such that we obtain

$$\frac{-f'(x)}{f(x)} = \frac{\mathrm{e}^{-x}}{(1-\mathrm{e}^{-x})^2} \cdot \frac{1-\mathrm{e}^{-x}}{1} = \frac{\mathrm{e}^{-x}}{1-\mathrm{e}^{-x}} = \frac{1}{\mathrm{e}^{+x}-1} = \mathrm{B}_1(x)$$

28. Appendix - Proofs for Part II

Proof 28.1 of Lemma 1.11[1]

Using the Fourier-expansion in Equation (1.7) we obtain[2]

$$\mathscr{G}_\zeta(\tau) = \frac{1}{\hbar\beta} \sum_{n=-\infty}^{+\infty} \mathscr{G}_\zeta(\mathrm{i}\omega_n)\, \mathrm{e}^{-\mathrm{i}n\pi\tau/\hbar\beta} \tag{28.1}$$

with coefficients

$$\mathscr{G}_\zeta(\mathrm{i}\omega_n) = \frac{1}{2} \int_{-\hbar\beta}^{+\hbar\beta} \mathrm{d}\tau\, \mathscr{G}_\zeta(\tau) \mathrm{e}^{\mathrm{i}n\pi\tau/\hbar\beta} \tag{28.2}$$

Bosons: $\zeta = +1$

We use that

$$-\hbar\beta < \tau < 0 \qquad \Rightarrow \qquad \mathscr{G}_\zeta(\tau) = \mathscr{G}_\zeta(\tau + \hbar\beta) \tag{28.3}$$

and therefore we obtain

$$\mathscr{G}_\zeta(\mathrm{i}\omega_n) = \frac{1}{2} \left[\int_0^{\hbar\beta} \mathrm{d}\tau\, \mathscr{G}_\zeta(\tau) \mathrm{e}^{\mathrm{i}n\pi\tau/\hbar\beta} + \int_{-\hbar\beta}^{0} \mathrm{d}\tau\, \mathscr{G}_\zeta(\tau) \mathrm{e}^{\mathrm{i}n\pi\tau/\hbar\beta} \right] \tag{28.4}$$

For the second term we change variables from $\tau \mapsto \tau + \hbar\beta$ which gives

$$\mathscr{G}_\zeta(\mathrm{i}\omega_n) = \frac{1}{2} \left(1 + \mathrm{e}^{\mathrm{i}n\pi}\right) \int_0^{\hbar\beta} \mathrm{d}\tau\, \mathscr{G}_\zeta(\tau) \mathrm{e}^{\mathrm{i}n\pi\tau/\hbar\beta} \tag{28.5}$$

This expression has the feature that $\mathscr{G}_\zeta(\mathrm{i}\omega_n) = 0$ whenever n is an odd integer. For bosons we hence obtain

$$\begin{aligned}
\mathscr{G}_\zeta(\tau) &= \sum_{\omega_n \in \Omega_{+1}} \mathrm{e}^{-\mathrm{i}\omega_n\tau} \mathscr{G}_\zeta(\mathrm{i}\omega_n), \\
\mathscr{G}_\zeta(\mathrm{i}\omega_n) &= \int_0^{\hbar\beta} \mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}\omega_n\tau} \mathscr{G}_\zeta(\tau), \\
\Omega_{+1} &= \left\{ \omega_n = 2n\pi/\hbar\beta \quad | \quad n \in \mathbb{Z} \right\}.
\end{aligned}$$

[1] Reference: [2] p. 110 onwards.

[2] Notice that we have a different-prefactor-convention w.r.t. the equation before. This is *the* prefactor-convention for Matsubara-Green-functions.

Fermions: $\zeta = -1$

We use that

$$-\hbar\beta < \tau < 0 \quad \Rightarrow \quad \mathscr{G}_\zeta(\tau) = -\mathscr{G}_\zeta(\tau + \hbar\beta) \tag{28.6}$$

and therefore we obtain

$$\mathscr{G}_\zeta(\mathrm{i}\omega_n) = \frac{1}{2}\left[\int_0^{\hbar\beta} \mathrm{d}\tau\, \mathscr{G}_\zeta(\tau)\mathrm{e}^{\mathrm{i}n\pi\tau/\hbar\beta} + \int_{-\hbar\beta}^{0} \mathrm{d}\tau\, \mathscr{G}_\zeta(\tau)\mathrm{e}^{\mathrm{i}n\pi\tau/\hbar\beta}\right] \tag{28.7}$$

For the second term we obtain in the same manner

$$\mathscr{G}_\zeta(\mathrm{i}\omega_n) = \frac{1}{2}\left(1 - \mathrm{e}^{\mathrm{i}n\pi}\right)\int_0^{\hbar\beta} \mathrm{d}\tau\, \mathscr{G}_\zeta(\tau)\mathrm{e}^{\mathrm{i}n\pi\tau/\hbar\beta} \tag{28.8}$$

This expression has the feature that $\mathscr{G}_\zeta(\mathrm{i}\omega_n) = 0$ whenever n is an even integer. For fermions we hence obtain

$$\begin{aligned}
\mathscr{G}_\zeta(\tau) &= \sum_{\omega_n \in \Omega_{-1}} \mathrm{e}^{-\mathrm{i}\omega_n\tau}\mathscr{G}_\zeta(\mathrm{i}\omega_n), \\
\mathscr{G}_\zeta(\mathrm{i}\omega_n) &= \int_0^{\hbar\beta} \mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}\omega_n\tau}\mathscr{G}_\zeta(\tau), \\
\Omega_{+1} &= \left\{\omega_n = (2n+1)\pi/\hbar\beta \quad | \quad n \in \mathbb{Z}\right\}.
\end{aligned}$$

Proof 28.2 of Corollary 1.12

By straight-forward integration we obtain

$$\int_0^{\hbar\beta} \mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}\tau\omega_m} = \begin{cases} \dfrac{1}{\mathrm{i}\omega_m}\left(\mathrm{e}^{\mathrm{i}\tau\omega_m} - 1\right) & \text{for } \omega_m \neq 0 \\ \hbar\beta & \text{for } \omega_m = 0 \end{cases} \tag{28.9}$$

hence we have proven the identity.

Proof 28.3 of Lemma 1.15

We obtain in usual Dirac-notation

$$\begin{aligned}
\left\langle\, \tau \,|\, \mathscr{G}_\zeta \,\right\rangle &= \mathscr{G}_\zeta(\tau) \\
&= \sum_{\omega_n \in \Omega_\zeta} \langle\, \tau \,|\, \mathrm{i}\omega_n \,\rangle\langle\, \mathrm{i}\omega_n \,|\, \mathscr{G}_\zeta \,\rangle \\
&= \frac{1}{\hbar\beta}\sum_{\omega_n \in \Omega_\zeta} \mathscr{G}_\zeta(\mathrm{i}\omega_n)\mathrm{e}^{-\mathrm{i}\omega_n\tau}
\end{aligned}$$

and also

$$\begin{aligned}\langle\, \tau \mid \tau' \,\rangle &= \delta(\tau-\tau') \\ &= \frac{1}{\hbar\beta}\sum_{\omega_n\in\Omega_\zeta} \mathrm{e}^{-\mathrm{i}\omega_n\tau+\mathrm{i}\omega_n\tau'} \\ &= \frac{1}{\hbar\beta}\sum_{\omega_n\in\Omega_\zeta} \mathrm{e}^{\mathrm{i}\omega_n(\tau-\tau')}\end{aligned}$$

Proof 28.4 of Theorem 7.7[3]

The factorization of the Fock-space into polynomial expressions of different orders is a direct consequence of the contruction.

The fermionic ($\zeta = -1$) case. One way to find out the number of linearly independent elements in $\Lambda^n(\mathcal{H})$ is by direct counting. Linearity allows any element in $\omega = \Lambda^n(\mathcal{H})$ in the following way

$$\omega = \sum_{1\le i_1,\ldots,i_n\le \mathrm{d}_\mathcal{H}} \omega_{i_1,\ldots,i_n} |i_1 ⓩ \cdots ⓩ i_n\rangle \tag{28.10}$$

Due to the defining relation $vv' + v'v = 0$ for all $v, v' \in \mathcal{H}$, any two elements $|i_1 ⓩ \cdots ⓩ i_n\rangle$, $|i'_1 ⓩ \cdots ⓩ i'_n\rangle$ are equal to zero, if one basis element appears more than one time $|i_j ⓩ \cdots ⓩ i_j\rangle = 0$. Thus, the number of possible non-zero combinations for $|i_1 ⓩ \cdots ⓩ i_n\rangle$ is

$$\begin{aligned}\#\left\{ |i_1 ⓩ \cdots ⓩ i_n\rangle \,\middle|\, i_1,\ldots,i_n \in \{1,\ldots,\mathrm{d}_\mathcal{H}\} \wedge \nexists j,j' \in \{1,\ldots,n\} \,:\, i_j = i_{j'} \right\} &= \mathrm{d}_\mathcal{H}\cdot(\mathrm{d}_\mathcal{H}-1)\cdot\ldots\cdot(\mathrm{d}_\mathcal{H}-n+1) \\ &= \frac{\mathrm{d}_\mathcal{H}!}{(\mathrm{d}_\mathcal{H}-n)!}.\end{aligned}$$

We now notice the following things:

- Every element $|i_1 ⓩ \cdots ⓩ i_n\rangle$ can be re-ordered as $|\tilde{i}_1 ⓩ \cdots ⓩ \tilde{i}_n\rangle$ such that $\tilde{i}_1 < \ldots < \tilde{i}_n$.
- The number of ways to reorder elements as $|i_1 ⓩ \cdots ⓩ i_n\rangle$ is $n!$ (due to the dimension of the symmetric group).

Hence $\omega = \Lambda^n(\mathcal{H})$ can be rewritten as

$$\omega = \sum_{1\le i_1<\ldots<i_n\le \mathrm{d}_\mathcal{H}} \tilde{\omega}_{i_1,\ldots,i_n} |i_1 ⓩ \cdots ⓩ i_n\rangle \tag{28.11}$$

Hence we obtain

$$\#\left\{ |i_1 ⓩ \cdots ⓩ i_n\rangle \,\middle|\, 1 \le i_1 < \ldots < i_n \le \mathrm{d}_\mathcal{H} \right\} = \binom{\mathrm{d}_\mathcal{H}}{n} = \dim_\mathbb{C} \Lambda^n(\mathcal{H}).$$

[3] References: For the Bosonic part: See Zirnbauer 2018 / 2019 Sheet 09 Exercise 2 a). For the fermionic part: See Zirnbauer SS 2020, Sheet 6 Exercise 3 a).

The bosonic ($\zeta = +1$) case.

Notice first, that the identity to prove can be written as

$$\dim_{\mathbb{C}} \mathrm{S}^n(\mathcal{H}) = \binom{\mathrm{d}_{\mathcal{H}} + n - 1}{n} = \#\left\{ (x_1, \dots, x_{\mathrm{d}_{\mathcal{H}}}) \quad \middle| \quad \sum_{i=1}^{\mathrm{d}_{\mathcal{H}}} x_i = n \quad \wedge \quad x_1, \dots, x_{\mathrm{d}_{\mathcal{H}}} \geq 0 \right\} \tag{28.12}$$

by induction. Since we have two integers $\mathrm{d}_{\mathcal{H}}$, n to vary, we need to perform the induction-start on the lines

$$\left\{ (n, \mathrm{d}_{\mathcal{H}}) \in \mathbb{N}_+^2 \quad | \quad n \in \{1, \dots, \infty\} \wedge \mathrm{d}_{\mathcal{H}} = 1 \right\} \tag{28.13}$$

and

$$\left\{ (n, \mathrm{d}_{\mathcal{H}}) \in \mathbb{N}_+^2 \quad | \quad n = 1 \wedge \mathrm{d}_{\mathcal{H}} \in \{1, \dots, \infty\} \right\} \tag{28.14}$$

This is fulfilled

Now we prove that

$$\#\left\{ (x_1, \dots, x_{\mathrm{d}_{\mathcal{H}}}, x_{\mathrm{d}_{\mathcal{H}}+1}) \quad \middle| \quad \sum_{i=1}^{\mathrm{d}_{\mathcal{H}}+1} x_i = n + 1 \quad \wedge \quad x_1, \dots, x_{\mathrm{d}_{\mathcal{H}}}, x_{\mathrm{d}_{\mathcal{H}}+1} \geq 0 \right\} = \binom{\mathrm{d}_{\mathcal{H}} + n + 1}{n + 1} \tag{28.15}$$

under the assumption that Equation (28.12) is true for $(n + 1, \mathrm{d}_{\mathcal{H}})$ and $(n, \mathrm{d}_{\mathcal{H}} + 1)$.

TBD: Adding sketch.

For this, we decompose the set

$$\#\left\{ (x_1, \dots, x_{\mathrm{d}_{\mathcal{H}}}, x_{\mathrm{d}_{\mathcal{H}}+1}) \quad \middle| \quad \sum_{i=1}^{\mathrm{d}_{\mathcal{H}}+1} x_i = n + 1 \quad \wedge \quad x_1, \dots, x_{\mathrm{d}_{\mathcal{H}}}, x_{\mathrm{d}_{\mathcal{H}}+1} \geq 0 \right\} \tag{28.16}$$

into the two cases $x_1 = 0$ and $x_1 \geq 1$.

- Case 1: We consider the set

$$\#\left\{ (x_2, \dots, x_{\mathrm{d}_{\mathcal{H}}}, x_{\mathrm{d}_{\mathcal{H}}+1}) \quad \middle| \quad \sum_{i=2}^{\mathrm{d}_{\mathcal{H}}+1} x_i = n + 1 \quad \wedge \quad x_2, \dots, x_{\mathrm{d}_{\mathcal{H}}}, x_{\mathrm{d}_{\mathcal{H}}+1} \geq 0 \right\} = \binom{\mathrm{d}_{\mathcal{H}} + n}{n + 1} \tag{28.17}$$

- Case 2: We have now that $y_1 \equiv x_1 - 1 \geq 0$ We consider the set

$$\#\left\{ (y_1, x_2, \dots, x_{\mathrm{d}_{\mathcal{H}}}, x_{\mathrm{d}_{\mathcal{H}}+1}) \quad \middle| \quad y_1 + \sum_{i=2}^{\mathrm{d}_{\mathcal{H}}+1} x_i = n \quad \wedge \quad y_1, x_2, \dots, x_{\mathrm{d}_{\mathcal{H}}}, x_{\mathrm{d}_{\mathcal{H}}+1} \geq 0 \right\} = \binom{\mathrm{d}_{\mathcal{H}} + n}{n} \tag{28.18}$$

Thus we obtain in total

$$\#\left\{(x_1,\ldots,x_{\mathrm{d}_\mathcal{H}},x_{\mathrm{d}_\mathcal{H}+1}) \;\middle|\; \sum_{i=1}^{\mathrm{d}_\mathcal{H}+1} x_i = n+1 \quad\wedge\quad x_1,\ldots,x_{\mathrm{d}_\mathcal{H}},x_{\mathrm{d}_\mathcal{H}+1}\geq 0\right\} = \binom{\mathrm{d}_\mathcal{H}+n}{n+1}+\binom{\mathrm{d}_\mathcal{H}+n}{n}$$
$$= \binom{\mathrm{d}_\mathcal{H}+n+1}{n+1}$$

Proving the theorem for the bosonic case.

Proof 28.5 of Corollary 7.12

We have to show, that taking two elements $v \equiv \psi$ and $v' \equiv \phi$ (which are then interpreted as many-particle wavefunctions), the condition in Equation (7.7) is still satisfied. This is done by straightforward computation

$$\left(\psi \,\textcircled{\scriptsize$\zeta$}\, \phi\right)(x_1,x_2) - \zeta\left(\phi \,\textcircled{\scriptsize$\zeta$}\, \psi\right)(x_1,x_2) = \sum_{\pi\in \mathrm{S}_2} \mathrm{sign}(\pi)^{1-\theta(\zeta)}\psi(x_{\pi(1)})\phi(x_{\pi(2)}) - \zeta\sum_{\pi\in \mathrm{S}_2} \mathrm{sign}(\pi)^{1-\theta(\zeta)}\phi(x_{\pi(1)})\psi(x_{\pi(2)}) = 0$$

Maybe we need to do more here...

whereas in the last line we used that $\sum_{\sigma\circ\pi\,\in\,\mathrm{S}_{p+q}} = \sum_{\pi\,\in\,\mathrm{S}_{p+q}}$ for each $\sigma \in \mathrm{S}_{p+q}$. Hence we can choose σ such that

$$\begin{pmatrix} 1 & 2 & \dots & q & q+1 & \dots & p+q \\ p+1 & p+2 & \dots & p+q & 1 & \dots & p \end{pmatrix} \tag{28.19}$$

and the contributions cancel.

It remains to prove the associativity. This is proven more or less straight-forwardly[4]

[4]Reference: Sweers, Analysis 3, Satz 12.10.

Proof 28.6 of Lemma 7.14

Introducing $u \equiv v + \phi$ and $u' \equiv v' + \phi'$ we obtain by direct computation

$$
\begin{aligned}
&B_\zeta(u, u') + \zeta B_\zeta(u', u) \\
&= B_\zeta(v + \phi, v' + \phi') + \zeta B_\zeta(v' + \phi', v + \phi) \\
&= \phi(v') - \zeta\phi'(v) \quad + \quad \zeta\left(\phi'(v) - \zeta\phi(v')\right) \\
&= 0
\end{aligned}
$$

Proof 28.7 of Lemma 7.17

Expanding the r.h.s. of the identity yields

$$
\begin{aligned}
[A, B]_\zeta C + \zeta B[A, C]_\zeta &= (AB - \zeta BA)\, C + \zeta B\, (AC - \zeta CA) \\
&= ABC - \zeta^2 BCA \\
&= ABC - BCA \\
&= [A, BC]
\end{aligned}
$$

Proof 28.8 of Notation 7.21

One fixes the elements $u \equiv |i\rangle + 0$ and $u' \equiv 0 + \langle j|$ and immediately obtains the relations by considering their commutators.

Proof 28.9 of Theorem 7.22 Recall that a mapping

$$
\begin{aligned}
\Phi :\ \mathscr{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta) \times \mathscr{F}_\zeta &\longrightarrow \mathscr{F}_\zeta \\
(g, v) &\longmapsto \Phi(g, v) \equiv g \cdot v
\end{aligned}
$$

qualifies as a representation, iff

1. for all $g \in \mathscr{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta)$ the map

$$
\begin{aligned}
\Phi(g) :\ \mathscr{F}_\zeta &\mapsto \mathscr{F}_\zeta \\
v &\longmapsto \Phi(g, v)
\end{aligned}
$$

   is linear over $\mathbb{C}$,

2. for all $g_1, g_2 \in \mathscr{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta)$ and $v \in \mathscr{F}_\zeta$ one has

$$
\begin{aligned}
e \cdot v &= v \\
g_1 \cdot (g_2 \cdot v) &= (g_1 g_2) \cdot v
\end{aligned}
$$

   whereas $e \equiv 1 \equiv 1_\mathbb{C}$ is the identity in the algebra.

The first property is obviously fulfilled, as we defined our representation on the basis-elements and extended the definition by linearity. The identity condition is also fulfilled by applying the definition for multiplication with scalars. For the third condition there are two non-trivial cases:

1. $g_1 \equiv w + 0$, $g_2 \equiv 0 + \phi$
2. $g_1 \equiv 0 + \phi$, $g_2 \equiv w + 0$

and we have to check in each case whether $v \in \mathbb{C}$, $v \in \mathcal{H}$ or $v \in \mathscr{F}_\zeta^n$ for $n \geq 2$.

Case 1:

In this case we obtain

$$\begin{aligned} g_1 g_2 w &= (w + 0)\,(0 + \phi) \cdot v \\ &\equiv \iota(w)\epsilon(\phi) \cdot v = \dots \end{aligned}$$

which gives for each type of v:

- $v \in \mathbb{C}$: $\dots = 0$.
- $v \in \mathcal{H}$: $\dots = \iota(w) \cdot \phi(v) = w \cdot_\zeta \phi(v)$
- $v \equiv v_1 \cdot_\zeta v_2 \in \mathscr{F}_\zeta^n$ for $n \geq 2$: $\dots =$

TBContinued.

Proof 28.10 of Theorem 7.23 TBD.

Proof 28.11 of Corollary 7.25 TBD. With computation-rules of determinants / permanents.

Proof 28.12 of Corollary 7.28 TBD. Recall that the hermitian adjoint of an operator w.r.t. a scalar-product is defined as TBWritten.

Proof 28.13 of Theorem 7.33 TBD. The states form obviously a basis, since every element in the Fock-space can be expanded w.r.t. the states. Now to the orthonormality: TBDone.

Proof 28.14 of Corollary 7.36 TBD.

Proof 28.15 of Lemma 7.37 TBD.

Proof 28.16 of Lemma 7.38 TBD.

Proof 28.17 of Corollary 7.36 TBD.

Proof 28.18 of Corollary 7.44 TBD.

Proof 28.19 of Corollary 7.45 TBD.

Proof 28.20 of Theorem 7.47 TBD.

Proof 28.21 of Theorem 8.9

TBUpdated w.r.t. the ensemble Hamiltonian.

We first note the auxiliary identities

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}U(t,0) &= -\frac{i}{\hbar}HU(t,0) \\
\frac{\mathrm{d}}{\mathrm{d}t}U(t,0)^\dagger &= \frac{\mathrm{d}}{\mathrm{d}t}U(0,t) = \frac{i}{\hbar}U(0,t)H = \frac{i}{\hbar}U(t,0)^\dagger H
\end{aligned}$$

Then it follows

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}X_{\mathrm{H}}(t) &= \frac{\mathrm{d}}{\mathrm{d}t}U(t,0)^\dagger X(t)U(t,0) \\
&= \left(\frac{\mathrm{d}}{\mathrm{d}t}U(t,0)^\dagger\right)X(t)U(t,0) + U(t,0)^\dagger\left(\frac{\mathrm{d}}{\mathrm{d}t}X(t)\right)U(t,0) + U(t,0)^\dagger X(t)\left(\frac{\mathrm{d}}{\mathrm{d}t}U(t,0)\right) \\
&= \frac{i}{\hbar}U(t,0)^\dagger HX(t)U(t,0) + U(t,0)^\dagger\left(\frac{\mathrm{d}}{\mathrm{d}t}X(t)\right)U(t,0) + U(t,0)^\dagger X(t)\left(-\frac{i}{\hbar}HU(t,0)\right) \\
&= \frac{i}{\hbar}U(t,0)^\dagger\left(HX(t) - X(t)H\right)U(t,0) + U(t,0)^\dagger\left(\frac{\mathrm{d}}{\mathrm{d}t}X(t)\right)U(t,0) \\
&= \frac{i}{\hbar}U(t,0)^\dagger\left[H,X(t)\right]U(t,0) + U(t,0)^\dagger\left(\frac{\mathrm{d}}{\mathrm{d}t}X(t)\right)U(t,0) \\
&= -\frac{i}{\hbar}\left[X,H\right]_{\mathrm{H}}(t) + U(t,0)^\dagger\left(\frac{\mathrm{d}}{\mathrm{d}t}X(t)\right)U(t,0)
\end{aligned}$$

Proof 28.22 of Theorem 8.18

TBUpdated w.r.t. the ensemble Hamiltonian.

Analogously to the real-time theorem we first note the auxiliary identities

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}\tau}U(\tau,0) &= -\frac{1}{\hbar}HU(\tau,0) \\
\frac{\mathrm{d}}{\mathrm{d}\tau}U(0,\tau) &= \frac{1}{\hbar}U(0,\tau)H
\end{aligned}$$

Then it follows

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}\tau}X_{\mathrm{H}}(\tau) &= \frac{\mathrm{d}}{\mathrm{d}t}U(0,\tau)X(\tau)U(\tau,0) \\
&= \left(\frac{\mathrm{d}}{\mathrm{d}\tau}U(0,\tau)\right)X(\tau)U(\tau,0) + U(0,\tau)\left(\frac{\mathrm{d}}{\mathrm{d}\tau}X(\tau)\right)U(\tau,0) + U(0,\tau)X(\tau)\left(\frac{\mathrm{d}}{\mathrm{d}\tau}U(\tau,0)\right) \\
&= \frac{1}{\hbar}U(0,\tau)HX(\tau)U(\tau,0) + U(0,\tau)\left(\frac{\mathrm{d}}{\mathrm{d}\tau}X(\tau)\right)U(\tau,0) + U(0,\tau)X(\tau)\left(-\frac{1}{\hbar}HU(\tau,0)\right) \\
&= \frac{1}{\hbar}U(0,\tau)\left(HX(\tau) - X(\tau)H\right)U(\tau,0) + U(0,\tau)\left(\frac{\mathrm{d}}{\mathrm{d}t}X(\tau)\right)U(\tau,0) \\
&= \frac{1}{\hbar}U(0,\tau)\left[H, X(\tau)\right]U(\tau,0) + U(0,\tau)\left(\frac{\mathrm{d}}{\mathrm{d}\tau}X(\tau)\right)U(\tau,0) \\
&= -\frac{1}{\hbar}\left[X,H\right]_{\mathrm{H}}(\tau) + U(0,\tau)\left(\frac{\mathrm{d}}{\mathrm{d}\tau}X(\tau)\right)U(\tau,0)
\end{aligned}$$

Proof 28.23 of Lemma 8.22

There is at least two possibilities to prove the lemma.

1. With the Equation of motion-method.
2. With the Baker-Campbell-Hausdorff-series[5].

TBElaborated with Mahan eq. (3.44) and BCH-series TBElaborated with Mahan eq. (3.67) and BCH-series

The Equation of motion-method

We compute with the Heisenberg-equation-of-motion in Equation TBReferenced, and with the identity for statistical commutators

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}c_{\bullet}(t) &= -\frac{\mathrm{i}}{\hbar}\sum_{\alpha\in\mathscr{A}}(\epsilon_\alpha-\mu)[c_\bullet, c_\alpha^\dagger c_\alpha]_{\mathrm{H}}(t) \\
&= -\frac{\mathrm{i}}{\hbar}\sum_{\alpha\in\mathscr{A}}(\epsilon_\alpha-\mu)\left([c_\bullet, c_\alpha^\dagger]_\zeta c_\alpha + \zeta c_\alpha^\dagger [c_\bullet, c_\alpha]_\zeta\right)_{\mathrm{H}}(t) \\
&= \frac{\mathrm{i}}{\hbar}(\epsilon_\bullet-\mu)c_\bullet(t)
\end{aligned}$$

with the initial conditions $c_\alpha(t=0) = c_\alpha$ we see, that

$$c_\bullet(t) = \mathrm{e}^{-\mathrm{i}(\epsilon_\bullet-\mu)t/\hbar}c_\bullet \tag{28.20}$$

is a solution of the differential equation. Taking the hermitian conjugate proves the other identity.

The Baker-Campbell-Hausdorff-method

TBD.

Other references related to this[6]

[5] TBElaborated with Mahan eq. (3.44) and Zirnbauers exercise-sheet

[6] TBCopied: cf. Zirnbauer AQM, 2018, 2019, Sheet 10 Exercise 1b).

Proof 28.24 of Lemma 8.30

We compute the hermitian conjugate of the interaction-term as

$$
\begin{aligned}
\left(H_{\text{el-ph}}\right)^{\dagger} &= \sum_{\substack{\mathbf{k},\mathbf{q}\in \mathrm{Q} \\ \sigma\in\{\uparrow,\downarrow\} \\ \nu\in\{1,\dots,rd_{\mathbf{x}}\} \\ n,m\in\{1,\dots,\infty\}}} \frac{g^{*}_{mn\nu}(\mathbf{k},\mathbf{q})}{\sqrt{N_p}} c^{\dagger}_{n\sigma}(\mathbf{k}) c_{m\sigma}(\mathbf{k}+\mathbf{q}) \left(b_{\nu}(-\mathbf{q}) + b^{\dagger}_{\nu}(+\mathbf{q})\right) \\
&= \sum_{\substack{\mathbf{k},\mathbf{q}\in \mathrm{Q} \\ \sigma\in\{\uparrow,\downarrow\} \\ \nu\in\{1,\dots,rd_{\mathbf{x}}\} \\ n,m\in\{1,\dots,\infty\}}} \frac{g^{*}_{mn\nu}(\mathbf{k},-\mathbf{q})}{\sqrt{N_p}} c^{\dagger}_{n\sigma}(\mathbf{k}) c_{m\sigma}(\mathbf{k}-\mathbf{q}) \left(b_{\nu}(+\mathbf{q}) + b^{\dagger}_{\nu}(-\mathbf{q})\right) \\
&= \sum_{\substack{\mathbf{k},\mathbf{q}\in \mathrm{Q} \\ \sigma\in\{\uparrow,\downarrow\} \\ \nu\in\{1,\dots,rd_{\mathbf{x}}\} \\ n,m\in\{1,\dots,\infty\}}} \frac{g^{*}_{nm\nu}(\mathbf{k}+\mathbf{q},-\mathbf{q})}{\sqrt{N_p}} c^{\dagger}_{m\sigma}(\mathbf{k}+\mathbf{q}) c_{n\sigma}(\mathbf{k}) \left(b_{\nu}(\mathbf{q}) + b^{\dagger}_{\nu}(-\mathbf{q})\right)
\end{aligned}
$$

whereas we used $\mathbf{q} \mapsto -\mathbf{q}$, in the second equation sign and $\mathbf{k} \mapsto \mathbf{k}+\mathbf{q}$ and $m \leftrightarrow n$ in the third equation sign.

Therefore we obtain

$$
g_{mn\nu}(\mathbf{k},\mathbf{q}) = g^{*}_{nm\nu}(\mathbf{k}+\mathbf{q},-\mathbf{q})
$$

Proof 28.25 of Lemma 9.46

We first rewrite

$$\mathrm{n}^{\zeta}(x) = \frac{1}{\mathrm{e}^{x} - \zeta} = (\mathrm{e}^{x} - \zeta)^{-1} \tag{28.21}$$

and then compute for the first identity

$$\frac{\mathrm{d}}{\mathrm{d}x}\,\mathrm{n}^{\zeta}(x) = -1 \cdot \frac{1}{(\mathrm{e}^{x} - \zeta)^{2}} \cdot \mathrm{e}^{x} = \frac{-1}{4} \cdot \frac{2\mathrm{e}^{x/2}}{\mathrm{e}^{x} - \zeta} \cdot \frac{2\mathrm{e}^{x/2}}{\mathrm{e}^{x} - \zeta} = \frac{-1}{4}\zeta\,\mathrm{ch}^{2}\left(\frac{x}{2}\right) \tag{28.22}$$

For the second identity we simplify the appearing expressions in another way as

$$\frac{\mathrm{d}}{\mathrm{d}x}\,\mathrm{n}^{\zeta}(x) = -1 \cdot \frac{1}{(\mathrm{e}^{x} - \zeta)^{2}} \cdot \mathrm{e}^{x} = - \cdot \frac{1}{\mathrm{e}^{x} - \zeta} \cdot \frac{\mathrm{e}^{x} - \zeta + \zeta}{\mathrm{e}^{x} - \zeta} = -\,\mathrm{n}^{\zeta}(x) \cdot \big(1 + \zeta\,\mathrm{n}^{\zeta}(x)\big) \tag{28.23}$$

Proof 28.26 of Lemma 9.46

We compute

$$\mathrm{F}_{\hbar\beta}(\bullet + \mathrm{i}\omega_{n}) = \frac{1}{\mathrm{e}^{\hbar\beta(\bullet + \mathrm{i}\omega_{n})} + 1} = \frac{1}{-\mathrm{e}^{\hbar\beta\bullet} + 1} = -\,\mathrm{B}_{\hbar\beta}(\bullet)$$

and

$$\mathrm{F}_{\hbar\beta}(\mathrm{i}\omega_{n} - \bullet) - \mathrm{B}_{\hbar\beta}(\bullet) = \frac{1}{\mathrm{e}^{\hbar\beta(\mathrm{i}\omega_{n} - \bullet)} + 1} - \frac{1}{\mathrm{e}^{\hbar\beta\bullet} - 1} = \frac{1}{1 - \mathrm{e}^{-\hbar\beta\bullet}} - \frac{1}{\mathrm{e}^{\hbar\beta\bullet} - 1} = \frac{\mathrm{e}^{-\hbar\beta\bullet} - 1}{\mathrm{e}^{-\hbar\beta\bullet} - 1} = 1$$

For the third identity we still have to digitalize / create the proof.

Proof 28.27 of Lemma 9.47

We define $x = \hbar\beta\omega = \frac{\hbar\omega}{k_{\mathrm{B}}T}$ such that we obtain

$$\frac{\mathrm{d}x}{\mathrm{d}T} = \hbar\omega\,\frac{\mathrm{d}\beta}{\mathrm{d}T} = \hbar\omega\,\frac{\mathrm{d}}{\mathrm{d}T}\left(\frac{1}{k_{\mathrm{B}}T}\right) = -\frac{\hbar\omega}{k_{\mathrm{B}}T^{2}}. \tag{28.24}$$

and

$$\mathrm{n}^{\zeta}_{\hbar\beta}(\omega) = \mathrm{n}^{\zeta}(x) \tag{28.25}$$

and therefore

$$\frac{\mathrm{d}}{\mathrm{d}T}\,\mathrm{n}^{\zeta}_{\hbar\beta}(\omega) = \frac{\mathrm{d}}{\mathrm{d}T}\,\mathrm{n}^{\zeta}(x) = \frac{\mathrm{d}}{\mathrm{d}x}\,\mathrm{n}^{\zeta}(x) \cdot \frac{\mathrm{d}x}{\mathrm{d}T} \tag{28.26}$$

Combining this with Lemma 9.44 yields the desired result.

Proof 28.28 of Corollary 9.48 We recall the definition of the Bose-function

$$\mathrm{B}_{\hbar\beta}(z) = \frac{1}{\mathrm{e}^{\hbar\beta z} - 1} = \left(\mathrm{e}^{\hbar\beta z} - 1\right)^{-1} \tag{28.27}$$

and compute straightforwardly

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}z}\,\mathrm{B}_{\hbar\beta}(z) &= -\left(\mathrm{e}^{\hbar\beta z} - 1\right)^{-2} \cdot \underbrace{\frac{\mathrm{d}}{\mathrm{d}z}\mathrm{e}^{\hbar\beta z}}_{\hbar\beta\mathrm{e}^{\hbar\beta z}} \\ &= -\hbar\beta\frac{\mathrm{e}^{\hbar\beta z} - 1 + 1}{(\mathrm{e}^{\hbar\beta z} - 1)^2} \\ &= -\hbar\beta\Big[\,\mathrm{B}_{\hbar\beta}(z) + \mathrm{B}^2_{\hbar\beta}(z\Big] \\ &= -\hbar\beta\,\mathrm{B}_{\hbar\beta}(z)\Big[1 + \mathrm{B}_{\hbar\beta}(z)\Big] \end{aligned} \tag{28.28}$$

For the Fermi-function the computation is analogous.

Proof 28.29 of Lemma 12.3 With the cyclicity of the trace, and the Baker-Campbell-Hausdorff-formula, the proof is straightforward:

TBElaborated.

Proof 28.30 of Lemma 12.4

See Waste-Folder 1, topic 10.

The proofs are more or less straightforward by simply writing out the underlying equations. We obtain for the individual Green-functions

$$\begin{aligned} G(A,A') &= -\frac{\mathrm{i}}{\hbar}\langle\theta(t-t')c_\alpha(t)c^\dagger_{\alpha'}(t') + \zeta\theta(t'-t)c^\dagger_{\alpha'}(t')c_\alpha(t)\rangle \\ &= \theta(t-t')G^>(A,A') + \theta(t'-t)G^<(A,A') \end{aligned}$$

$$\begin{aligned} G^{\mathbb{T}}(A,A') &= -\frac{\mathrm{i}}{\hbar}\langle\theta(t'-t)c_\alpha(t)c^\dagger_{\alpha'}(t') + \zeta\theta(t-t')c^\dagger_{\alpha'}(t')c_\alpha(t)\rangle \\ &= \theta(t'-t)G^>(A,A') + \theta(t-t')G^<(A,A') \end{aligned}$$

$$G^{\mathrm{ret}}(A,A') = -\frac{\mathrm{i}}{\hbar}\theta(t-t')\langle c_\alpha(t)c^\dagger_{\alpha'}(t') - \zeta c^\dagger_{\alpha'}(t')c_\alpha(t)\rangle$$

$$G^{\mathrm{adv}}(A,A') = +\frac{\mathrm{i}}{\hbar}\theta(t'-t)\langle c_\alpha(t)c^\dagger_{\alpha'}(t') - \zeta c^\dagger_{\alpha'}(t')c_\alpha(t)\rangle$$

(i) We obtain for the retarded Green-function

$$G^{\text{ret}}_{\alpha,\alpha'}(t,t') = -\frac{\mathrm{i}}{\hbar}\theta(t-t')\langle c_\alpha(t)c^\dagger_{\alpha'}(t') - \zeta c^\dagger_{\alpha'}(t')c_\alpha(t)\rangle \tag{28.29}$$

and for the advanced Green-function

$$G^{\text{adv}}_{\alpha,\alpha'}(t,t') = +\frac{\mathrm{i}}{\hbar}\theta(t'-t)\langle c_\alpha(t)c^\dagger_{\alpha'}(t') - \zeta c^\dagger_{\alpha'}(t')c_\alpha(t)\rangle \tag{28.30}$$

and hence

$$\begin{aligned}\left(G^{\text{adv}}_{\alpha',\alpha}(t',t)\right)^\dagger &= \left(+\frac{\mathrm{i}}{\hbar}\theta(t-t')\langle c_{\alpha'}(t')c^\dagger_\alpha(t) + c^\dagger_\alpha(t)c_{\alpha'}(t')\rangle\right)^\dagger \\ &= -\frac{\mathrm{i}}{\hbar}\theta(t-t')\langle c_\alpha(t)c^\dagger_{\alpha'}(t') + c^\dagger_{\alpha'}(t')c_\alpha(t)\rangle = G^{\text{ret}}_{\alpha,\alpha'}(t,t')\end{aligned}$$

Therefore we obtain

$$G^{\text{ret}}_{\alpha,\alpha'}(t,t') = \left(G^{\text{adv}}_{\alpha',\alpha}(t',t)\right)^\dagger \iff G^{\text{ret}}(A,A') = \left(G^{\text{adv}}(A',A)\right)^\dagger \tag{28.31}$$

(ii) We obtain further

$$\begin{aligned}\left(G^{++} - G^{-+}\right)(A,A') &= -\frac{\mathrm{i}}{\hbar}\langle\left(\theta(t-t') - 1\right)c_\alpha(t)c_{\alpha'}(t') + \zeta\theta(t'-t)c_{\alpha'}(t')c_\alpha(t)\rangle \\ &= -\frac{\mathrm{i}}{\hbar}\langle -\theta(t'-t)c_\alpha(t)c_{\alpha'}(t') + \zeta\theta(t'-t)c_{\alpha'}(t')c_\alpha(t)\rangle \\ &= \frac{\mathrm{i}}{\hbar}\theta(t'-t)\langle c_\alpha(t)c_{\alpha'}(t') - \zeta c_{\alpha'}(t')c_\alpha(t)\rangle\end{aligned}$$

(iii) We obtain further

$$\begin{aligned}\left(G^{++} - G^{--}\right)(A,A') &= -\frac{\mathrm{i}}{\hbar}\langle -\theta(t'-t)c_\alpha(t)c_{\alpha'}(t') + \left(-\zeta\theta(t-t') + \zeta\right)c_{\alpha'}(t')c_\alpha(t)\rangle \\ &= -\frac{\mathrm{i}}{\hbar}\langle -\theta(t'-t)c_\alpha(t)c_{\alpha'}(t') + \zeta\theta(t'-t)c_{\alpha'}(t')c_\alpha(t)\rangle \\ &= +\frac{\mathrm{i}}{\hbar}\theta(t'-t)\langle c_\alpha(t)c_{\alpha'}(t') - \zeta c_{\alpha'}(t')c_\alpha(t)\rangle\end{aligned}$$

(iv) We obtain further

$$\begin{aligned}\left(G^{++} - G^{-+}\right)(A,A') &= -\frac{\mathrm{i}}{\hbar}\langle \zeta c_{\alpha'}(t')c_\alpha(t) - c_\alpha(t)c_{\alpha'}(t')\rangle \\ &= \frac{\mathrm{i}}{\hbar}\langle c_\alpha(t)c_{\alpha'}(t') - \zeta\theta(t'-t)c_{\alpha'}(t')c_\alpha(t)\rangle\end{aligned}$$

(v) We obtain further

$$\begin{aligned}\left(G^{++}-G^{+-}\right)(A,A') &= -\frac{\mathrm{i}}{\hbar}\langle\theta(t-t')c_\alpha(t)c_{\alpha'}(t')+\zeta\left(\theta(t'-t)-1\right)c_{\alpha'}(t')c_\alpha(t)\rangle\\ &= -\frac{\mathrm{i}}{\hbar}\theta(t-t')\langle c_\alpha(t)c_{\alpha'}(t')-\zeta c_{\alpha'}(t')c_\alpha(t)\rangle\end{aligned}$$

(vi) We obtain further

$$\begin{aligned}\left(G^{-+}-G^{--}\right)(A,A') &= -\frac{\mathrm{i}}{\hbar}\langle\left(1-\theta(t'-t)\right)c_\alpha(t)c_{\alpha'}(t')-\zeta\theta(t-t')c_{\alpha'}(t')c_\alpha(t)\rangle\\ &= -\frac{\mathrm{i}}{\hbar}\theta(t-t')\langle c_\alpha(t)c_{\alpha'}(t')-\zeta c_{\alpha'}(t')c_\alpha(t)\rangle\end{aligned}$$

and therefore we obtain

$$\theta(t-t')\left(G^{-+}-G^{+-}\right)(A,A') = -\frac{\mathrm{i}}{\hbar}\theta(t-t')\langle c_\alpha(t)c_{\alpha'}(t')-\zeta c_{\alpha'}(t')c_\alpha(t)\rangle$$

ToBeSorted. The last two identities follow from combining the results we just obtained TBDone.

Proof 28.31 of Lemma 12.6

The time-ordered Green-function

With the equation of motion technique, we consider the definition

$$\mathbb{R}\ni t\mapsto G^{\mathbb{T}}_{\alpha,\alpha'}(t) = -\frac{\mathrm{i}}{\hbar}\frac{\mathrm{Tr}_{\mathscr{F}_\zeta}\,\mathrm{e}^{-\beta(H-\mu N)}\left[\theta(t)\,c_\alpha(t)\,c^\dagger_{\alpha'}(0)+\zeta\,\theta(-t)\,c^\dagger_{\alpha'}(0)\,c_\alpha(t)\right]}{\mathrm{Tr}_{\mathscr{F}_\zeta}\,\mathrm{e}^{-\beta(H-\mu N)}}\in\mathbb{C}$$

and want to calculate its derivative. We therefore calculate first

$$\begin{aligned}\partial_t\,\mathrm{T}_t\,c_\alpha(t)\,c^\dagger_{\alpha'}(0) \propto{}& \partial_t\left[\theta(t)\,c_\alpha(t)\,c^\dagger_{\alpha'}(0)+\zeta\,\theta(-t)\,c^\dagger_{\alpha'}(0)\,c_\alpha(t)\right]\\ ={}& \delta(t)c_\alpha(t)\,c^\dagger_{\alpha'}(0)-\zeta\delta(-t)c^\dagger_{\alpha'}(0)\,c_\alpha(t)-\frac{\mathrm{i}}{\hbar}\theta(t)\left[c_\alpha,H-\mu\right]_{\mathrm{H}}(t)c^\dagger_{\alpha'}(0)-\frac{\mathrm{i}}{\hbar}\theta(-t)\zeta c^\dagger_{\alpha'}(0)\left[c_\alpha,H-\mu\right]_{\mathrm{H}}(t)\\ ={}& \delta(t)\left[c_\alpha(t)\,,c^\dagger_{\alpha'}(0)\right]_\zeta-\frac{\mathrm{i}}{\hbar}\,\mathrm{T}_t\left(\left[c_\alpha,H-\mu\right]_{\mathrm{H}}(t)\,c^\dagger_{\alpha'}(0)\right)\end{aligned}$$

whereas we used the statistical commutator from Definition TBD and the Heisenberg-equation-of-motions from Lemma TBD. This gives the equation of motion

$$\partial_t G^{\mathbb{T}}_{\alpha,\alpha'}(t) = -\frac{\mathrm{i}}{\hbar}\delta(t)\delta_{\alpha,\alpha'} \quad - \quad \frac{1}{\hbar}\left\langle \mathrm{T}_t\left[c_\alpha, \frac{H-\mu N}{\hbar}\right]_{\mathrm{H}}(t)c^\dagger_{\alpha'}(0)\right\rangle \tag{28.32}$$

The anti-time-ordered Green-function

With the equation of motion technique, we consider the definition

$$\mathbb{R} \ni t \mapsto G^{\overline{\mathbb{T}}}_{\alpha,\alpha'}(t) = -\frac{\mathrm{i}}{\hbar}\frac{\mathrm{Tr}_{\mathscr{F}_\zeta}\,\mathrm{e}^{-\beta(H-\mu N)}\left[\theta(-t)\,c_\alpha(t)\,c^\dagger_{\alpha'}(0) + \zeta\,\theta(+t)\,c^\dagger_{\alpha'}(0)\,c_\alpha(t)\right]}{\mathrm{Tr}_{\mathscr{F}_\zeta}\,\mathrm{e}^{-\beta(H-\mu N)}} \in \mathbb{C}$$

and want to calculate its derivative. We therefore calculate first

$$\begin{aligned}
\partial_t\overline{\mathrm{T}}_t c_\alpha(t)\,c^\dagger_{\alpha'}(0) &\propto \partial_t\left[\theta(-t)\,c_\alpha(t)\,c^\dagger_{\alpha'}(0) + \zeta\,\theta(+t)\,c^\dagger_{\alpha'}(0)\,c_\alpha(t)\right]\\
&= -\delta(-t)c_\alpha(t)\,c^\dagger_{\alpha'}(0) + \zeta\delta(t)c^\dagger_{\alpha'}(0)\,c_\alpha(t) - \frac{\mathrm{i}}{\hbar}\theta(-t)\left[c_\alpha, H-\mu\right]_{\mathrm{H}}(t)c^\dagger_{\alpha'}(0) - \frac{\mathrm{i}}{\hbar}\theta(t)\zeta c^\dagger_{\alpha'}(0)\left[c_\alpha, H-\mu\right]_{\mathrm{H}}(t)\\
&= -\delta(-t)\left[c_\alpha(t)\,,c^\dagger_{\alpha'}(0)\right]_\zeta - \frac{\mathrm{i}}{\hbar}\overline{\mathrm{T}}_t\left(\left[c_\alpha, H-\mu\right]_{\mathrm{H}}(t)\,c^\dagger_{\alpha'}(0)\right)
\end{aligned}$$

whereas we used the statistical commutator from Definition TBD and the Heisenberg-equation-of-motions from Lemma TBD. This gives the equation of motion

$$\partial_t G^{\overline{\mathbb{T}}}_{\alpha,\alpha'}(t) = \frac{\mathrm{i}}{\hbar}\delta(t)\delta_{\alpha,\alpha'} \quad - \quad \frac{1}{\hbar}\left\langle \overline{\mathrm{T}}_t\left[c_\alpha, \frac{H-\mu N}{\hbar}\right]_{\mathrm{H}}(t)c^\dagger_{\alpha'}(0)\right\rangle \tag{28.33}$$

The retarded Green-function

With the equation of motion technique, we consider the definition

$$\mathbb{R} \ni t \mapsto G^{\mathrm{ret}}_{\alpha,\alpha'}(t) = -\frac{\mathrm{i}}{\hbar}\theta(t)\frac{\mathrm{Tr}_{\mathscr{F}_\zeta}\,\mathrm{e}^{-\beta(H-\mu N)}\left[c_\alpha(t)\,c^\dagger_{\alpha'}(0) - \zeta\,c^\dagger_{\alpha'}(0)\,c_\alpha(t)\right]}{\mathrm{Tr}_{\mathscr{F}_\zeta}\,\mathrm{e}^{-\beta(H-\mu N)}} \in \mathbb{C}$$

and want to calculate its derivative. We therefore calculate first

$$\begin{aligned}
\partial_t\theta(t)\left[c_\alpha(t),\,c^\dagger_{\alpha'}(0)\right]_\zeta &= \delta(t)\left[c_\alpha(t),\,c^\dagger_{\alpha'}(0)\right]_\zeta + \theta(t)\left[\partial_t c_\alpha(t),\,c^\dagger_{\alpha'}(0)\right]_\zeta\\
&= \delta(t)\delta_{\alpha,\alpha'} - \frac{\mathrm{i}}{\hbar}\theta(t)\left[\left[c_\alpha, H-\mu\right]_{\mathrm{H}}(t), c^\dagger_{\alpha'}(0)\right]_\zeta
\end{aligned}$$

whereas we used the statistical commutator from Definition TBD and the Heisenberg-equation-of-motions from Lemma TBD. This gives the equation of motion

$$\partial_t G^{\mathrm{ret}}_{\alpha,\alpha'}(t) = -\frac{\mathrm{i}}{\hbar}\delta(t)\delta_{\alpha,\alpha'} \quad - \quad \frac{1}{\hbar}\theta(t)\left\langle \left[\left[c_\alpha, H-\mu\right]_{\mathrm{H}}(t), c^\dagger_{\alpha'}(0)\right]_\zeta \right\rangle \tag{28.34}$$

The advanced Green-function

With the equation of motion technique, we consider the definition

$$\mathbb{R} \ni t \mapsto G^{\mathrm{adv}}_{\alpha,\alpha'}(t) = \frac{\mathrm{i}}{\hbar}\theta(-t)\frac{\mathrm{Tr}_{\mathscr{F}_\zeta}\, \mathrm{e}^{-\beta(H-\mu N)}\left[c_\alpha(t)\, c^\dagger_{\alpha'}(0) - \zeta\, c^\dagger_{\alpha'}(0)\, c_\alpha(t)\right]}{\mathrm{Tr}_{\mathscr{F}_\zeta}\, \mathrm{e}^{-\beta(H-\mu N)}} \in \mathbb{C}$$

and want to calculate its derivative. We therefore calculate first

$$\begin{aligned} \partial_t\theta(-t)\left[c_\alpha(t),\, c^\dagger_{\alpha'}(0)\right]_\zeta &= -\delta(-t)\left[c_\alpha(t),\, c^\dagger_{\alpha'}(0)\right]_\zeta + \theta(-t)\left[\partial_t c_\alpha(t),\, c^\dagger_{\alpha'}(0)\right]_\zeta \\ &= -\delta(-t)\delta_{\alpha,\alpha'} - \frac{\mathrm{i}}{\hbar}\theta(-t)\left[\left[c_\alpha, H-\mu\right]_{\mathrm{H}}(t), c^\dagger_{\alpha'}(0)\right]_\zeta \end{aligned}$$

whereas we used the statistical commutator from Definition TBD and the Heisenberg-equation-of-motions from Lemma TBD. This gives the equation of motion

$$\partial_t G^{\mathrm{adv}}_{\alpha,\alpha'}(t) = -\frac{\mathrm{i}}{\hbar}\delta(-t)\delta_{\alpha,\alpha'} \quad + \quad \frac{1}{\hbar}\theta(-t)\left\langle \left[\left[c_\alpha, H-\mu\right]_{\mathrm{H}}(t), c^\dagger_{\alpha'}(0)\right]_\zeta \right\rangle \tag{28.35}$$

Proof 28.32 of Theorem 12.7

See Waste-Folder 1, topic 17.

Proof 28.33 of Theorem 12.9

This is one of the more difficult proofs and it involves several steps that need to be made. The different steps are …

1. Finding the eigenvalues.
2. Finding the eigenvectors.
3. Expanding the creation- and annihilation-operators w.r.t. these vectors.
4. Calculating the different Green-functions by their definition.

Lemma 28.1 In order to find the eigenvalues of

$$H = c^\dagger\, \epsilon\, c \equiv \begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix}^\dagger \begin{pmatrix} \epsilon_0 + \epsilon_1^z & -\mathrm{i}\epsilon_1^y + \epsilon_1^x \\ +\mathrm{i}\epsilon_1^y + \epsilon_1^x & \epsilon_0 - \epsilon_1^z \end{pmatrix} \begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix} \tag{28.36}$$

it suffices to examine the matrix

$$h = \begin{pmatrix} n_z & -\mathrm{i}n_y + n_x \\ +\mathrm{i}n_y + n_x & -n_z \end{pmatrix}, \quad \text{with } |\vec{n}| = 1. \tag{28.37}$$

Proof 28.1 We consider the unit-vector

$$\frac{\boldsymbol{\epsilon}}{\epsilon} \equiv \mathbf{n} \equiv n_x \mathbf{e}_x + n_y \mathbf{e}_y + n_z \mathbf{e}_z \in \mathrm{S}^2 \subset \mathbb{R}^3$$

and find that

$$H \equiv H(\mathbf{n}) := \epsilon_0 \sigma_0 + \epsilon \mathbf{n} \cdot \boldsymbol{\sigma} = \epsilon_0 \sigma_0 + \epsilon \left(n_x \sigma_x + n_y \sigma_y + n_z \sigma_z \right).$$

Therefore, if $|\lambda\rangle$ is an eigenstate of $\mathbf{n} \cdot \boldsymbol{\sigma}$ with eigenvalue λ then it follows

$$\epsilon_0 \sigma_0 + \epsilon \left(n_x \sigma_x + n_y \sigma_y + n_z \sigma_z \right) |\lambda\rangle = \left(\epsilon_0 \sigma_0 + \epsilon \lambda \right) |\lambda\rangle$$

therefore $|\lambda\rangle$ is also ein eigenvalue of H with eigenvalue $\left(\epsilon_0 \sigma_0 + \epsilon \lambda \right)$.

Notation 28.2 The unit-vector $\mathbf{n}$ can be expressed in spherical coordinates $(r, \phi, \theta) \in 1 \times (0, 2\pi) \times (0, \pi)$ as:

$$\mathbf{n} = \begin{pmatrix} \cos(\phi) \sin(\theta) \\ \sin(\phi) \sin(\theta) \\ \cos(\theta) \end{pmatrix}.$$

Lemma 28.3 The matrix

$$h = \begin{pmatrix} n_z & -\mathrm{i}n_y + n_x \\ +\mathrm{i}n_y + n_x & -n_z \end{pmatrix}, \quad \text{with } |\vec{n}| = 1. \tag{28.38}$$

possesses the eigenvalues ± 1 with eigenvectors

$$\begin{aligned} |\uparrow_n\rangle &= \cos\frac{\theta}{2} |\uparrow\rangle + \sin\frac{\theta}{2} \mathrm{e}^{\mathrm{i}\phi} |\downarrow\rangle \\ |\downarrow_n\rangle &= \sin\frac{\theta}{2} |\uparrow\rangle - \cos\frac{\theta}{2} \mathrm{e}^{\mathrm{i}\phi} |\downarrow\rangle \end{aligned}$$

whereas

$$\left\{ |\uparrow\rangle \equiv \mathbf{e}_x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |\downarrow\rangle \equiv \mathbf{e}_y = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\} \subset \mathbb{R}^2$$

represent the eigenstates of the third Pauli-matrix.

Proof 28.3 We compute the eigenvalues of the 2×2-matrix:

$$\mathbf{n} \cdot \boldsymbol{\sigma} = \begin{pmatrix} n_z & -\mathrm{i}n_y + n_x \\ +\mathrm{i}n_y + n_x & -n_z \end{pmatrix}$$

by solving the characteristic equation

$$\mathrm{Det}\left(\mathbf{n} \cdot \boldsymbol{\sigma} - \lambda \cdot \mathbf{1}_{2\times 2}\right) = \mathrm{Det}\begin{pmatrix} n_z - \lambda & -\mathrm{i}n_y + n_x \\ +\mathrm{i}n_y + n_x & -n_z - \lambda \end{pmatrix} \overset{\text{Binomial Formulae}}{=} \lambda^2 - n_z^2 - (n_x^2 + n_y^2) = \lambda^2 - \mathbf{n}^2 = 0$$
$$\iff \lambda^2 = \mathbf{n}^2 = 1 \iff \lambda_\pm = \pm 1$$

Now we compute the eigenstates of $\mathbf{n} \cdot \boldsymbol{\sigma}$:

$$\begin{aligned} \mathrm{Kern}\left(\mathbf{n} \cdot \boldsymbol{\sigma} - (\pm 1) \cdot \mathbf{1}_{2\times 2}\right) &= \mathrm{Kern}\begin{pmatrix} n_z \mp 1 & -\mathrm{i}n_y + n_x \\ 0 & 0 \end{pmatrix} \\ &= \left\{ \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \in \mathbb{C}^2 \mid (n_z \mp 1)v_1 + (-\mathrm{i}n_y + n_x)v_2 = 0 \right\} \\ &= \mathrm{Span}_{\alpha \in \mathbb{C}} \left\{ \begin{pmatrix} \alpha \\ \frac{-(n_z \mp 1)\alpha}{n_x - \mathrm{i}n_y} \end{pmatrix} \right\} \qquad (\star) \end{aligned}$$

The eigenstate to the eigenvalue $+1$:

We multiply a "clever one"

$$\frac{1}{n_x - \mathrm{i}n_y} = \frac{n_x + \mathrm{i}n_y}{n_x^2 + n_y^2} = \frac{(\cos(\phi) + \mathrm{i}\sin(\phi))\sin(\theta)}{\left(\cos^2(\phi) + \sin^2(\phi)\right)\sin^2(\theta)} = \frac{\mathrm{e}^{\mathrm{i}\phi}}{\sin(\theta)}$$

In order to reproduce the control result, we consider[7] the eigenvector in $(\star)$ with $\alpha = \cos\frac{\theta}{2}$. Then it follows for the eigenvector

$$\begin{aligned} |\uparrow_{\mathbf{n}}\rangle &= \cos\frac{\theta}{2}|\uparrow\rangle + \frac{(1 - n_z)\cos\frac{\theta}{2}}{n_x - \mathrm{i}n_y}|\downarrow\rangle \\ &= \cos\frac{\theta}{2}|\uparrow\rangle + \frac{(1 - \cos(\theta))\cos(\frac{\theta}{2})}{\sin(\theta)}\mathrm{e}^{\mathrm{i}\phi}|\downarrow\rangle \\ &= \cos\frac{\theta}{2}|\uparrow\rangle + \sin\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi}|\downarrow\rangle \end{aligned}$$

whereas in the last equation-sign we used this ...

`https://www.wolframalpha.com/input?i=%281+-+cos%28theta%29%29*cos%28theta+%2F+2%29+%2F+sin%28theta%29++%3D+sin%28theta+%2F+2%29`

... identity.

---

[7] Recall that a complex eigenvector is only unique up to a complex phase.

This eigenstate is also normalized as one finds

$$\langle \uparrow_{\mathbf{n}} \mid \uparrow_{\mathbf{n}} \rangle = \left(\cos\frac{\theta}{2}\langle \uparrow \mid + \sin\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}\langle \downarrow \mid\right)\left(\cos\frac{\theta}{2}\mid \uparrow \rangle + \sin\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi}\mid \downarrow \rangle\right) = \cos^2\frac{\theta}{2} + \sin^2\frac{\theta}{2} = 1.$$

The eigenstate to the eigenvalue -1:

This eigenstate can be concluded by demanding orthogonality to $\mid \uparrow_{\mathbf{n}} \rangle$. One finds that

$$\mid \downarrow_{\mathbf{n}} \rangle = \sin\frac{\theta}{2}\mid \uparrow \rangle - \cos\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi}\mid \downarrow \rangle$$

is orthogonal since

$$\langle \downarrow_{\mathbf{n}} \mid \uparrow_{\mathbf{n}} \rangle = \left(\sin\frac{\theta}{2}\langle \uparrow \mid - \cos\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}\langle \downarrow \mid\right)\left(\cos\frac{\theta}{2}\mid \uparrow \rangle + \sin\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi}\mid \downarrow \rangle\right) = \sin\frac{\theta}{2}\cos\frac{\theta}{2} - \cos\frac{\theta}{2}\sin\frac{\theta}{2}\mathrm{e}^{0} = 0$$

Analogously one finds that $\mid \downarrow_{\mathbf{n}} \rangle$ is normalized.

Remark: In the limit $\mathbf{n} \to \mathbf{e}_z$ one obtains states, that are in the same equivalence-class of states as the usual ones of the third Pauli-matrix.

Corollary 28.4 It follows

$$
\begin{aligned}
|\uparrow\rangle &= \cos\frac{\theta}{2}|\uparrow_n\rangle + \sin\frac{\theta}{2}|\downarrow_n\rangle \\
|\downarrow\rangle &= \sin\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}|\uparrow_n\rangle - \cos\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}|\downarrow_n\rangle
\end{aligned}
$$

Proof 28.4

The equations

$$
\begin{aligned}
|\uparrow_n\rangle &= \cos\frac{\theta}{2}|\uparrow\rangle + \sin\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi}|\downarrow\rangle \\
|\downarrow_n\rangle &= \sin\frac{\theta}{2}|\uparrow\rangle - \cos\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi}|\downarrow\rangle
\end{aligned}
$$

can be rewritten in matrix-form as

$$
\begin{aligned}
\begin{pmatrix} |\uparrow_n\rangle \\ |\downarrow_n\rangle \end{pmatrix} &= \begin{pmatrix} \cos\frac{\theta}{2} & \sin\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi} \\ \sin\frac{\theta}{2} & -\cos\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi} \end{pmatrix} \begin{pmatrix} |\uparrow\rangle \\ |\downarrow\rangle \end{pmatrix} \\
\Longleftrightarrow \begin{pmatrix} |\uparrow\rangle \\ |\downarrow\rangle \end{pmatrix} &= \begin{pmatrix} \cos\frac{\theta}{2} & \sin\frac{\theta}{2} \\ \sin\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi} & -\cos\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi} \end{pmatrix} \begin{pmatrix} |\uparrow_n\rangle \\ |\downarrow_n\rangle \end{pmatrix}
\end{aligned}
$$

since the determinant of the first matrix is equal to $-\mathrm{e}^{\mathrm{i}\phi}$, and therefore we have

$$
\begin{aligned}
|\uparrow\rangle &= \cos\frac{\theta}{2}|\uparrow_n\rangle + \sin\frac{\theta}{2}|\downarrow_n\rangle \\
|\downarrow\rangle &= \sin\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}|\uparrow_n\rangle - \cos\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}|\downarrow_n\rangle
\end{aligned}
$$

Remark 28.4.1 Notice that we obtain for the corresponding creation- (and annihilation-) operators

$$
\begin{aligned}
c_\uparrow &= \cos\frac{\theta}{2}d_\uparrow + \sin\frac{\theta}{2}d_\downarrow \\
c_\downarrow &= \sin\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}d_\uparrow - \cos\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}d_\downarrow
\end{aligned}
$$

whereas we denote $c_{\uparrow_n} \equiv d_\uparrow$ etc for notational convenience.

Notation 28.5 We notate the annihilation- and creation-operators corresponding to these states with $d_\bullet$ and $d_\bullet^\dagger$ with $\bullet \in \{\downarrow, \uparrow\}$.

Remark 28.5.1 Since this is a unitary transformation, these new operators also obey the CAR.

Remark 28.5.2 We have now found the spectral decomposition of the Hamiltonian

$$H = \sum_{\star \in \{\uparrow,\downarrow\}} E_\star d_\star^\dagger d_\star \tag{28.39}$$

with

$$E_\star \equiv \epsilon_0 \pm \epsilon. \tag{28.40}$$

Notation 28.6 We write $\xi_\star \equiv E_\star - \mu$.

Remark 28.6.1 It follows for the grand-canonical-time-evolution

$$\begin{aligned} d_\star(t) &= \mathrm{e}^{-\mathrm{i}\xi_\star t/\hbar} d_\star \\ d_\star^\dagger(t) &= \mathrm{e}^{+\mathrm{i}\xi_\star t/\hbar} d_\star^\dagger \end{aligned}$$

Proof 28.6.1 [Siehe altes Übungsblatt vom Zirnbauer für eine Anleitung.]

Corollary 28.7 It follows for the time-evolution

$$\begin{aligned} c_\uparrow(t) &= \cos\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\xi_\uparrow t/\hbar} d_\uparrow + \sin\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\xi_\downarrow t/\hbar} d_\downarrow \\ c_\downarrow(t) &= \sin\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi}\mathrm{e}^{-\mathrm{i}\xi_\uparrow t/\hbar} d_\uparrow - \cos\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi}\mathrm{e}^{-\mathrm{i}\xi_\downarrow t/\hbar} d_\downarrow \\ c_\uparrow^\dagger(t) &= \cos\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\xi_\uparrow t/\hbar} d_\uparrow^\dagger + \sin\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\xi_\downarrow t/\hbar} d_\downarrow^\dagger \\ c_\downarrow^\dagger(t) &= \sin\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}\mathrm{e}^{\mathrm{i}\xi_\uparrow t/\hbar} d_\uparrow^\dagger - \cos\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}\mathrm{e}^{\mathrm{i}\xi_\downarrow t/\hbar} d_\downarrow^\dagger \end{aligned}$$

Proof 28.7 Follows by straightforward application fo the time-evolution operator and Lemma TBReferenced.

Lemma 28.8 It holds

$$\begin{aligned}
\langle d_\uparrow^\dagger d_\uparrow \rangle &= \frac{\mathrm{e}^{-\beta\xi_\uparrow} + \mathrm{e}^{2\mu\beta}}{1 + \mathrm{e}^{-\beta\xi_\uparrow} + \mathrm{e}^{-\beta\xi_\downarrow} + \mathrm{e}^{2\mu\beta}}, & \langle d_\uparrow d_\uparrow^\dagger \rangle &= 1 - \langle d_\uparrow^\dagger d_\uparrow \rangle, \\
\langle d_\downarrow^\dagger d_\downarrow \rangle &= \frac{\mathrm{e}^{-\beta\xi_\downarrow} + \mathrm{e}^{2\mu\beta}}{1 + \mathrm{e}^{-\beta\xi_\uparrow} + \mathrm{e}^{-\beta\xi_\downarrow} + \mathrm{e}^{2\mu\beta}}, & \langle d_\downarrow d_\downarrow^\dagger \rangle &= 1 - \langle d_\downarrow^\dagger d_\downarrow \rangle, \\
\langle d_\uparrow^\dagger d_\downarrow \rangle &= 0, & \langle d_\downarrow d_\uparrow^\dagger \rangle &= 0, \\
\langle d_\downarrow^\dagger d_\uparrow \rangle &= 0, & \langle d_\uparrow d_\downarrow^\dagger \rangle &= 0.
\end{aligned}$$

Proof 28.8 We compute

$$\langle d_\uparrow^\dagger d_\uparrow \rangle = \frac{1}{Z} \sum_{\psi \in \mathbb{B}} \langle \psi | \mathrm{e}^{-\beta(H-\mu N)} d_\uparrow^\dagger d_\uparrow | \psi \rangle = \frac{\mathrm{e}^{-\beta(E_\uparrow - \mu)} + \mathrm{e}^{-\beta(-2\mu)}}{1 + \mathrm{e}^{-\beta(E_\uparrow-\mu)} + \mathrm{e}^{-\beta(E_\downarrow - \mu)} + \mathrm{e}^{2\beta\mu}}$$

$$\langle d_\downarrow^\dagger d_\downarrow \rangle = \frac{1}{Z} \sum_{\psi \in \mathbb{B}} \langle \psi | \mathrm{e}^{-\beta(H-\mu N)} d_\downarrow^\dagger d_\downarrow | \psi \rangle = \frac{\mathrm{e}^{-\beta(E_\downarrow - \mu)} + \mathrm{e}^{-\beta(-2\mu)}}{1 + \mathrm{e}^{-\beta(E_\uparrow-\mu)} + \mathrm{e}^{-\beta(E_\downarrow - \mu)} + \mathrm{e}^{2\beta\mu}}$$

$$\langle d_\uparrow^\dagger d_\downarrow \rangle = \langle d_\downarrow^\dagger d_\uparrow \rangle = 0$$

and the other identities follow from the CCR.

Lemma 28.9 It follows

$$\begin{aligned}
G^<_{\uparrow\uparrow}(t) &= \frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}\xi_\uparrow t/\hbar} \cos^2\frac{\theta}{2} \langle d_\uparrow^\dagger d_\uparrow \rangle & &+ \frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}\xi_\downarrow t/\hbar} \sin^2\frac{\theta}{2} \langle d_\downarrow^\dagger d_\downarrow \rangle \\
G^<_{\uparrow\downarrow}(t) &= \frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}\xi_\uparrow t/\hbar} \frac{\sin\theta}{2} \mathrm{e}^{-\mathrm{i}\phi} \langle d_\uparrow^\dagger d_\uparrow \rangle & &- \frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}\xi_\downarrow t/\hbar} \frac{\sin\theta}{2} \mathrm{e}^{-\mathrm{i}\phi} \langle d_\downarrow^\dagger d_\downarrow \rangle \\
G^<_{\downarrow\uparrow}(t) &= \frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}\xi_\uparrow t/\hbar} \frac{\sin\theta}{2} \mathrm{e}^{\mathrm{i}\phi} \langle d_\uparrow^\dagger d_\uparrow \rangle & &- \frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}\xi_\downarrow t/\hbar} \frac{\sin\theta}{2} \mathrm{e}^{\mathrm{i}\phi} \langle d_\downarrow^\dagger d_\downarrow \rangle \\
G^<_{\downarrow\downarrow}(t) &= \frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}\xi_\uparrow t/\hbar} \sin^2\frac{\theta}{2} \langle d_\uparrow^\dagger d_\uparrow \rangle & &+ \frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}\xi_\downarrow t/\hbar} \cos^2\frac{\theta}{2} \langle d_\downarrow^\dagger d_\downarrow \rangle
\end{aligned}$$

Proof 28.9 We compute straightforwardly

$$\begin{aligned}
G^<_{\uparrow\uparrow}(t) &= \frac{\mathrm{i}}{\hbar} \langle c_\uparrow^\dagger c_\uparrow(t) \rangle \\
&= \frac{\mathrm{i}}{\hbar} \left\langle \left(\cos\frac{\theta}{2} d_\uparrow^\dagger + \sin\frac{\theta}{2} d_\downarrow^\dagger \right) \left(\cos\frac{\theta}{2} d_\uparrow(t) + \sin\frac{\theta}{2} d_\downarrow(t) \right) \right\rangle \\
&= \cos^2\frac{\theta}{2} \frac{\mathrm{i}}{\hbar} \left\langle d_\uparrow^\dagger d_\uparrow(t) \right\rangle + \sin^2\frac{\theta}{2} \frac{\mathrm{i}}{\hbar} \left\langle d_\downarrow^\dagger d_\downarrow(t) \right\rangle \\
&= \frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}\xi_\uparrow t/\hbar} \cos^2\frac{\theta}{2} \left\langle d_\uparrow^\dagger d_\uparrow \right\rangle + \frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}\xi_\downarrow t/\hbar} \sin^2\frac{\theta}{2} \left\langle d_\downarrow^\dagger d_\downarrow \right\rangle
\end{aligned}$$

proving the first identity.

$$\begin{aligned}
G^<_{\downarrow\downarrow}(t) &= \frac{\mathrm{i}}{\hbar}\left\langle c^\dagger_\downarrow c_\downarrow(t)\right\rangle \\
&= \frac{\mathrm{i}}{\hbar}\left\langle\left(\sin\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}d^\dagger_\uparrow - \cos\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}d^\dagger_\downarrow\right)\left(\sin\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi}d_\uparrow(t) - \cos\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi}d_\downarrow(t)\right)\right\rangle \\
&= \frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}\xi_\uparrow t/\hbar}\sin^2\frac{\theta}{2}\left\langle d^\dagger_\uparrow d_\uparrow\right\rangle + \frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}\xi_\downarrow t/\hbar}\cos^2\frac{\theta}{2}\left\langle d^\dagger_\downarrow d_\downarrow\right\rangle
\end{aligned}$$

Proving the fourth identity. For the other identities we obtain

$$\begin{aligned}
G^<_{\uparrow\downarrow}(t) &= \frac{\mathrm{i}}{\hbar}\left\langle c^\dagger_\downarrow c_\uparrow(t)\right\rangle \\
&= \frac{\mathrm{i}}{\hbar}\left\langle\left(\sin\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}d^\dagger_\uparrow - \cos\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}d^\dagger_\downarrow\right)\left(\cos\frac{\theta}{2}d_\uparrow(t) + \sin\frac{\theta}{2}d_\downarrow(t)\right)\right\rangle \\
&= \frac{\mathrm{i}}{\hbar}\sin\frac{\theta}{2}\cos\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}\left\langle d^\dagger_\uparrow d_\uparrow(t)\right\rangle - \frac{\mathrm{i}}{\hbar}\sin\frac{\theta}{2}\cos\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}\left\langle d^\dagger_\downarrow d_\downarrow(t)\right\rangle
\end{aligned}$$

proving the second identity. For the last identity we obtain

$$\begin{aligned}
G^<_{\downarrow\uparrow}(t) &= \frac{\mathrm{i}}{\hbar}\left\langle c^\dagger_\uparrow c_\downarrow(t)\right\rangle \\
&= \frac{\mathrm{i}}{\hbar}\left\langle\left(\cos\frac{\theta}{2}d^\dagger_\uparrow + \sin\frac{\theta}{2}d^\dagger_\downarrow\right)\left(\sin\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi}d_\uparrow(t) - \cos\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi}d_\downarrow(t)\right)\right\rangle \\
&= \sin\frac{\theta}{2}\cos\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi}\frac{\mathrm{i}}{\hbar}\left\langle d^\dagger_\uparrow d_\uparrow(t)\right\rangle - \sin\frac{\theta}{2}\cos\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi}\frac{\mathrm{i}}{\hbar}\left\langle d^\dagger_\downarrow d_\downarrow(t)\right\rangle
\end{aligned}$$

proving the last identity.

Lemma 28.10 It follows

$$\begin{aligned}
G^>_{\uparrow\uparrow}(t) &= \mathrm{i}\mathrm{e}^{-\frac{\mathrm{i}}{\hbar}\xi_\uparrow t/\hbar}\cos^2\frac{\theta}{2}\left(\langle d^\dagger_\uparrow d_\uparrow\rangle - 1\right) \quad + \frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}\xi_\downarrow t/\hbar}\sin^2\frac{\theta}{2}\left(\langle d^\dagger_\downarrow d_\downarrow\rangle - 1\right) \\
G^>_{\uparrow\downarrow}(t) &= \frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}\xi_\uparrow t/\hbar}\frac{\sin\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}\left(\langle d^\dagger_\uparrow d_\uparrow\rangle - 1\right) - \frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}\xi_\downarrow t/\hbar}\frac{\sin\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}\left(\langle d^\dagger_\downarrow d_\downarrow\rangle - 1\right) \\
G^>_{\downarrow\uparrow}(t) &= \frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}\xi_\uparrow t/\hbar}\frac{\sin\theta}{2}\mathrm{e}^{\mathrm{i}\phi}\left(\langle d^\dagger_\uparrow d_\uparrow\rangle - 1\right) \quad - \frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}\xi_\downarrow t/\hbar}\frac{\sin\theta}{2}\mathrm{e}^{\mathrm{i}\phi}\left(\langle d^\dagger_\downarrow d_\downarrow\rangle - 1\right) \\
G^>_{\downarrow\downarrow}(t) &= \frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}\xi_\uparrow t/\hbar}\sin^2\frac{\theta}{2}\left(\langle d^\dagger_\uparrow d_\uparrow\rangle - 1\right) \quad + \frac{\mathrm{i}}{\hbar}\mathrm{e}^{-\mathrm{i}\xi_\downarrow t/\hbar}\cos^2\frac{\theta}{2}\left(\langle d^\dagger_\downarrow d_\downarrow\rangle - 1\right)
\end{aligned}$$

Proof 28.10 We compute

$$\begin{aligned}
G^>_{\uparrow\uparrow}(t) &= -\frac{\mathrm{i}}{\hbar}\langle c_\uparrow(t)c^\dagger_\uparrow\rangle \\
&= -\frac{\mathrm{i}}{\hbar}\langle(\cos\frac{\theta}{2}d_\uparrow(t) + \sin\frac{\theta}{2}d_\downarrow(t))(\cos\frac{\theta}{2}d^\dagger_\uparrow + \sin\frac{\theta}{2}d^\dagger_\downarrow)\rangle \\
&= -\frac{\mathrm{i}}{\hbar}\cos^2\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\xi_\uparrow t/\hbar}\langle d_\uparrow d^\dagger_\uparrow\rangle - \frac{\mathrm{i}}{\hbar}\sin^2\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\xi_\downarrow t/\hbar}\langle d_\downarrow d^\dagger_\downarrow\rangle
\end{aligned}$$

proving the first identity. For the second identity we find

$$
\begin{aligned}
G^{>}_{\downarrow\downarrow}(t) &= -\frac{\mathrm{i}}{\hbar}\langle c_\downarrow(t)c^\dagger_\downarrow\rangle \\
&= -\frac{\mathrm{i}}{\hbar}\left\langle\left(\sin\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi}d_\uparrow(t)-\cos\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi}d_\downarrow(t)\right)\left(\sin\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}d^\dagger_\uparrow-\cos\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}d^\dagger_\downarrow\right)\right\rangle \\
&= -\frac{\mathrm{i}}{\hbar}\sin^2\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\xi_\uparrow t/\hbar}\langle d_\uparrow d^\dagger_\uparrow\rangle-\frac{\mathrm{i}}{\hbar}\cos^2\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\xi_\downarrow t/\hbar}\langle d_\downarrow d^\dagger_\downarrow\rangle
\end{aligned}
$$

proving the fourth identity.

$$
\begin{aligned}
G^{>}_{\uparrow\downarrow}(t) &= -\frac{\mathrm{i}}{\hbar}\langle c_\uparrow(t)c^\dagger_\downarrow\rangle \\
&= -\frac{\mathrm{i}}{\hbar}\left\langle\left(\cos\frac{\theta}{2}d_\uparrow(t)+\sin\frac{\theta}{2}d_\downarrow(t)\right)\left(\sin\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}d^\dagger_\uparrow-\cos\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}d^\dagger_\downarrow\right)\right\rangle \\
&= -\frac{\mathrm{i}}{\hbar}\cos\frac{\theta}{2}\sin\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}\mathrm{e}^{-\xi_\uparrow t/\hbar}\langle d_\uparrow d^\dagger_\uparrow\rangle+\frac{\mathrm{i}}{\hbar}\cos\frac{\theta}{2}\sin\frac{\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}\mathrm{e}^{-\xi_\downarrow t/\hbar}\langle d_\downarrow d^\dagger_\downarrow\rangle
\end{aligned}
$$

For the last identity we find

$$
\begin{aligned}
G^{<}_{\downarrow\uparrow}(t) &= -\frac{\mathrm{i}}{\hbar}\langle c_\downarrow(t)c^\dagger_\uparrow\rangle \\
&= -\frac{\mathrm{i}}{\hbar}\left\langle\left(\sin\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi}d_\uparrow(t)-\cos\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi}d_\downarrow(t)\right)\left(\cos\frac{\theta}{2}d^\dagger_\uparrow+\sin\frac{\theta}{2}d^\dagger_\downarrow\right)\right\rangle \\
&= -\frac{\mathrm{i}}{\hbar}\sin\frac{\theta}{2}\cos\frac{\theta}{2}\mathrm{e}^{\mathrm{i}\phi}\left(\mathrm{e}^{\mathrm{i}\xi_\uparrow t/\hbar}\langle d_\uparrow d^\dagger_\uparrow\rangle-\mathrm{e}^{\mathrm{i}\xi_\downarrow t/\hbar}\langle d_\downarrow d^\dagger_\downarrow\rangle\right)
\end{aligned}
$$

proving the last identity after using $\sin\frac{\theta}{2}\cos\frac{\theta}{2}=\frac{1}{2}\sin\theta$.

Corollary 28.11 Recalling Remark **??**

$$G(t)=\theta(t)\,G^{>}(t)+\theta(-t)\,G^{<}(t)$$

we have determined the time-ordered-Green-function, although we do not want to write it down at this moment.

Corollary 28.12 Recalling another identity from Remark **??**, $G^{\mathrm{r}}(t)=\theta(t)\cdot(G^{>}(t)-G^{<}(t))$, we obtain the retarded Green-function as a function of time:

$$
\begin{aligned}
G^{\mathrm{r}}_{\uparrow\uparrow}(t) &= -\frac{\mathrm{i}}{\hbar}\theta(t)\mathrm{e}^{-\mathrm{i}\xi_\uparrow t/\hbar}\cos^2\frac{\theta}{2} &&-\frac{\mathrm{i}}{\hbar}\theta(t)\mathrm{e}^{-\mathrm{i}\xi_\downarrow t/\hbar}\sin^2\frac{\theta}{2} \\
G^{\mathrm{r}}_{\uparrow\downarrow}(t) &= -\frac{\mathrm{i}}{\hbar}\theta(t)\mathrm{e}^{-\mathrm{i}\xi_\uparrow t/\hbar}\frac{\sin\theta}{2}\mathrm{e}^{-\mathrm{i}\phi} &&+\frac{\mathrm{i}}{\hbar}\theta(t)\mathrm{e}^{-\mathrm{i}\xi_\downarrow t/\hbar}\frac{\sin\theta}{2}\mathrm{e}^{-\mathrm{i}\phi} \\
G^{\mathrm{r}}_{\downarrow\uparrow}(t) &= -\frac{\mathrm{i}}{\hbar}\theta(t)\mathrm{e}^{-\mathrm{i}\xi_\uparrow t/\hbar}\frac{\sin\theta}{2}\mathrm{e}^{\mathrm{i}\phi} &&+\frac{\mathrm{i}}{\hbar}\theta(t)\mathrm{e}^{-\mathrm{i}\xi_\downarrow t/\hbar}\frac{\sin\theta}{2}\mathrm{e}^{\mathrm{i}\phi} \\
G^{\mathrm{r}}_{\downarrow\downarrow}(t) &= -\frac{\mathrm{i}}{\hbar}\theta(t)\mathrm{e}^{-\mathrm{i}\xi_\uparrow t/\hbar}\sin^2\frac{\theta}{2} &&-\frac{\mathrm{i}}{\hbar}\theta(t)\mathrm{e}^{-\mathrm{i}\xi_\downarrow t/\hbar}\cos^2\frac{\theta}{2}
\end{aligned}
$$

Proof 28.12 Follows immediately by insertion.

Corollary 28.13 With the Fourier-transformation we obtain the retarded Green-function as a function of frequency:

$$
\begin{aligned}
G^{\mathrm{r}}_{\uparrow\uparrow}(\omega+\mathrm{i}\eta) &= \frac{1}{\hbar}\frac{1}{\omega+\mathrm{i}\eta-\xi_{\uparrow}/\hbar}\cos^2\frac{\theta}{2} &&+ \frac{1}{\hbar}\frac{1}{\omega+\mathrm{i}\eta-\xi_{\downarrow}/\hbar}\sin^2\frac{\theta}{2}\\
G^{\mathrm{r}}_{\uparrow\downarrow}(\omega+\mathrm{i}\eta) &= \frac{1}{\hbar}\frac{1}{\omega+\mathrm{i}\eta-\xi_{\uparrow}/\hbar}\frac{\sin\theta}{2}\mathrm{e}^{-\mathrm{i}\phi} &&- \frac{1}{\hbar}\frac{1}{\omega+\mathrm{i}\eta-\xi_{\downarrow}/\hbar}\frac{\sin\theta}{2}\mathrm{e}^{-\mathrm{i}\phi}\\
G^{\mathrm{r}}_{\downarrow\uparrow}(\omega+\mathrm{i}\eta) &= \frac{1}{\hbar}\frac{1}{\omega+\mathrm{i}\eta-\xi_{\uparrow}/\hbar}\frac{\sin\theta}{2}\mathrm{e}^{\mathrm{i}\phi} &&- \frac{1}{\hbar}\frac{1}{\omega+\mathrm{i}\eta-\xi_{\downarrow}/\hbar}\frac{\sin\theta}{2}\mathrm{e}^{\mathrm{i}\phi}\\
G^{\mathrm{r}}_{\downarrow\downarrow}(\omega+\mathrm{i}\eta) &= \frac{1}{\hbar}\frac{1}{\omega+\mathrm{i}\eta-\xi_{\uparrow}/\hbar}\sin^2\frac{\theta}{2} &&+ \frac{1}{\hbar}\frac{1}{\omega+\mathrm{i}\eta-\xi_{\downarrow}/\hbar}\cos^2\frac{\theta}{2}
\end{aligned}
$$

Proof 28.13 One finds

$$
\begin{aligned}
(\omega-\xi_{\downarrow})\cos^2\frac{\theta}{2}+(\omega-\xi_{\uparrow})\sin^2\frac{\theta}{2} &= \omega-(\epsilon_0-\epsilon_z)\cos^2\frac{\theta}{2}-(\epsilon_0+\epsilon_z)\sin^2\frac{\theta}{2}\\
&= \omega-\epsilon_0+\epsilon_z(\cos^2\frac{\theta}{2}-\sin^2\frac{\theta}{2})\\
&= \omega-\epsilon_0+\epsilon_z
\end{aligned}
$$

whereas we used $\cos^2\frac{\theta}{2}-\sin^2\frac{\theta}{2}=\cos\theta$, etc. for the other entries.

Theorem 28.14 It follows

$$
G^{\mathrm{r}}(\omega+\mathrm{i}\eta) = \frac{1}{\hbar}\frac{\left(\omega+\mathrm{i}\eta-\frac{\epsilon_0-\mu}{\hbar}\right)\sigma_0+\left(\frac{\vec{\epsilon}_1}{\hbar}\right)\cdot\vec{\sigma}}{\left(\omega+\mathrm{i}\eta-\frac{\epsilon_0-\mu}{\hbar}\right)^2-\left(\frac{\vec{\epsilon}_1}{\hbar}\right)^2} \tag{28.41}
$$

Proof 28.14 Follows immediately by insertion.

Remark 28.14.1 Note that the same result can be obtained by matrix-inversion and analytic continuation, as will be elaborated below.

Proof 28.34 of Theorem 13.4 We compute

$$\begin{aligned}\mathscr{G}_{X,X'}(\tau,\tau') &= -\frac{1}{\hbar}\frac{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta(H-\eta\mu N)}\Big[\theta(\tau-\tau')\,X(\tau)\,X'(\tau') + \mathrm{S}_{X,X'}\,\theta(\tau'-\tau)\,X'(\tau')\,X(\tau)\Big]}{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta(H-\eta\mu N)}}\\ &= -\frac{1}{\hbar}\frac{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta(H-\eta\mu N)}\Big[\theta(\tau-\tau')\,X(\tau-\tau')\,X'(0) + \mathrm{S}_{X,X'}\,\theta(\tau'-\tau)\,X'(0)\,X(\tau-\tau')\Big]}{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta(H-\eta\mu N)}}\end{aligned}$$

Therefore the Green-function depends only on time-differences and we can write

$$\mathscr{G}_{X,X'}(\tau,\tau') \equiv \mathscr{G}_{X,X'}(\tau-\tau'). \tag{28.42}$$

With the periodic standard-continuation of the θ-function, we obtain, using the cyclicity property twice, for $-\hbar\beta < \tau < 0$ on the one hand

$$\begin{aligned}\mathscr{G}_{X,X'}(\tau) &= -\frac{\mathrm{S}_{X,X'}}{\hbar}\frac{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta(H-\eta\mu N)}X'(0)\,X(\tau)}{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta(H-\eta\mu N)}}\\ &= -\frac{\mathrm{S}_{X,X'}}{\hbar}\frac{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta(H-\eta\mu N)}X'\,\mathrm{e}^{+\tau(H-\eta\mu N)/\hbar}X\mathrm{e}^{-\tau(H-\eta\mu N)/\hbar}}{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta(H-\mu N)}}\\ &= -\frac{\mathrm{S}_{X,X'}}{\hbar}\frac{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{+\tau(H-\eta\mu N)/\hbar}X\mathrm{e}^{-(\beta+\tau/\hbar)(H-\mu N)}X'}{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta(H-\eta\mu N)}}\\ &= \dots\end{aligned}$$

now we multiply with one: $1 = \mathrm{e}^{\pm\beta(H-\eta\mu N)}$ in order to get

$$\dots = -\frac{\mathrm{S}_{X,X'}}{\hbar}\frac{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta(H-\eta\mu N)}\mathrm{e}^{+(\beta+\tau/\hbar)(H-\eta\mu N)}X\mathrm{e}^{-(\beta+\tau/\hbar)(H-\eta\mu N)}X'}{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta(H-\eta\mu N)}}$$

and we ultimately compute on the other hand (again for $-\hbar\beta < \tau < 0$)

$$\begin{aligned}\mathscr{G}_{X,X'}(\tau+\hbar\beta) &= -\frac{1}{\hbar}\frac{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta(H-\eta\mu N)}X(\tau+\hbar\beta)\,X'(0)}{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta(H-\eta\mu N)}}\\ &= -\frac{1}{\hbar}\frac{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta(H-\eta\mu N)}\mathrm{e}^{+(\tau/\hbar+\beta)(H-\eta\mu N)}X\,\mathrm{e}^{+(\tau/\hbar+\beta)(H-\eta\mu N)}X'}{\mathrm{Tr}_{\mathscr{H}}\,\mathrm{e}^{-\beta(H-\eta\mu N)}}\end{aligned}$$

and therefore

$$\mathscr{G}_{X,X'}(\tau+\hbar\beta) = \mathrm{S}_{X,X'}\mathscr{G}_{X,X'}(\tau). \tag{28.43}$$

and hence there is a periodic standard-continuation, which is unique, and therefore the originally defined function.

Alternative proof: With the Lehmann-representation. Cf. Equation (TBReferenced).

Proof 28.35 of Lemma 13.5

We consider the definition

$$(-\hbar\beta, +\hbar\beta) \ni \tau \mapsto \mathscr{G}_{\alpha,\alpha'}(\tau) = -\frac{1}{\hbar} \frac{\mathrm{Tr}_{\mathscr{F}_\zeta}\, \mathrm{e}^{-\beta(H-\mu N)} \left[\theta(\tau)\, c_\alpha(\tau)\, c^\dagger_{\alpha'}(0) + \zeta\, \theta(-\tau)\, c^\dagger_{\alpha'}(0)\, c_\alpha(\tau)\right]}{\mathrm{Tr}_{\mathscr{F}_\zeta}\, \mathrm{e}^{-\beta(H-\mu N)}} \in \mathbb{C}$$

and want to calculate its derivative. We therefore calculate first

$$\begin{aligned}
\partial_\tau\, \mathrm{T}_\tau\, c_\alpha(\tau)\, c^\dagger_{\alpha'}(0) &\propto \partial_\tau \left[\theta(\tau)\, c_\alpha(\tau)\, c^\dagger_{\alpha'}(0) + \zeta\, \theta(-\tau)\, c^\dagger_{\alpha'}(0)\, c_\alpha(\tau)\right] \\
&= \delta(\tau) c_\alpha(\tau)\, c^\dagger_{\alpha'}(0) - \zeta \delta(-\tau) c^\dagger_{\alpha'}(0)\, c_\alpha(\tau) - \frac{1}{\hbar}\theta(\tau)\left[c_\alpha, H-\mu\right]_{\mathrm{H}}(\tau) c^\dagger_{\alpha'}(0) - \frac{\zeta}{\hbar}\theta(-\tau) c^\dagger_{\alpha'}(0)\left[c_\alpha, H-\mu\right]_{\mathrm{H}}(\tau) \\
&= \delta(\tau)\left[c_\alpha(\tau)\,, c^\dagger_{\alpha'}(0)\right]_\zeta - \frac{1}{\hbar}\, \mathrm{T}_\tau \left(\left[c_\alpha, H-\mu\right]_{\mathrm{H}}(\tau)\, c^\dagger_{\alpha'}(0)\right)
\end{aligned}$$

whereas we used the statistical commutator from Definition TBD and the imaginary-time Heisenberg-equation-of-motions from Lemma TBD and the fact. This gives the equation of motion

$$\partial_\tau \mathscr{G}_{\alpha\alpha'}(\tau) = -\frac{1}{\hbar}\delta(\tau)\delta_{\alpha,\alpha'} \quad + \quad \frac{1}{\hbar}\left\langle \mathrm{T}_\tau \left[c_\alpha, \frac{H-\mu N}{\hbar}\right]_{\mathrm{H}}(\tau) c^\dagger_{\alpha'}(0)\right\rangle \tag{28.44}$$

Proof 28.36 of Theorem 13.6

For Fermions: See Waste-Folder 1, Topic 4, page 8 and Mahan p. 115

For Bosons: See Waste-Folder 1, topic 9.

For both: See Flensberg, page 164f. , Negele and Orland, page 73.

Fermions

For the imaginary-time Green-function one obtains

$$\mathscr{G}(\tau) = -\frac{1}{\hbar}\theta(\tau)\mathrm{e}^{-\xi\tau/\hbar}\left(1 - \mathrm{F}_\beta(\xi)\right) + \frac{1}{\hbar}\theta(-\tau)\mathrm{e}^{-\xi\tau/\hbar}\,\mathrm{F}_\beta(\xi) \tag{28.45}$$

and for the coefficients of the Fourier-series one has

$$\begin{aligned}
\mathscr{G}(\mathrm{i}\omega_n) &= \int_0^{\hbar\beta} \mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}\omega_n\tau} \mathscr{G}(\tau) \\
&= -\frac{1}{\hbar}\left(1 - \mathrm{F}_\beta(\xi)\right) \int_0^{\hbar\beta} \mathrm{d}\tau\, \mathrm{e}^{\tau(\mathrm{i}\omega_n - \xi/\hbar)} = \dots
\end{aligned}$$

Now we compute

$$\int_0^{\hbar\beta} \mathrm{d}\tau\, \mathrm{e}^{\tau(\mathrm{i}\omega_n - \xi/\hbar)} = \left(-\mathrm{e}^{\beta\xi} - 1\right) \frac{1}{\mathrm{i}\omega_n - \xi/\hbar} \\ = -(1 + \mathrm{e}^{\beta\xi}) \frac{1}{\mathrm{i}\omega_n - \xi/\hbar}$$

which yields

$$\ldots = \frac{1}{\hbar} \frac{(1 - \mathrm{F}_\beta(\xi))(1 + \mathrm{e}^{\beta\xi})}{\mathrm{i}\omega_n - \xi/\hbar} \\ = \frac{1}{\hbar} \frac{1}{\mathrm{i}\omega_n - \xi/\hbar}$$

whereas the last equation-sign is mentioned in Mahan on page 115.

Proof 28.37 of Theorem 13.7

We will prove the theorem in two steps. First we will consider the equation of motion of the function to conclude its form. Then we will verify that the function has the correct periodicity. Fundamental theorems on the uniqueness of differential equations yields then that we have obtained the correct solution.

We now evaluate

$$\begin{aligned} \left[a_\alpha, H - \mu\right] &= \sum_{\beta,\beta'} H_{\beta,\beta'} \left[a_\alpha, a_\beta^\dagger a_{\beta'}\right] \quad + \quad \mu \sum_\beta \left[a_\alpha, a_\beta^\dagger a_\beta\right] \\ &= \sum_{\beta,\beta'} H_{\beta,\beta'} \left[a_\alpha, a_\beta^\dagger\right]_\zeta a_{\beta'} \quad + \quad \mu \sum_\beta \left[a_\alpha, a_\beta^\dagger\right]_\zeta a_\beta \\ &= \sum_{\beta'} H_{\alpha,\beta'} a_{\beta'} \quad + \quad \mu a_\alpha \end{aligned}$$

Therefore we obtain

$$\ldots = \delta(\tau)\delta_{\alpha,\alpha'} - \sum_\beta \frac{H_{\alpha,\beta} - \delta_{\alpha,\beta}\mu}{\hbar} \mathrm{T}_\tau \left(a_\beta(\tau) a_{\alpha'}^\dagger(0)\right)$$

which gives combined in total

$$\begin{aligned} \partial_\tau \mathscr{G}_{\alpha,\alpha'}(\tau) = -\delta(\tau)/\hbar\delta_{\alpha,\alpha'} - \sum_\beta \frac{H_{\alpha,\beta} - \delta_{\alpha,\beta}\mu}{\hbar} \mathscr{G}_{\beta,\alpha'}(\tau) \quad &\Longleftrightarrow \quad \partial_\tau \mathscr{G}(\tau) = -\delta(\tau)/\hbar - \frac{H-\mu}{\hbar}\mathscr{G}(\tau) \\ &\Longleftrightarrow \quad \left(\partial_\tau + \frac{H-\mu}{\hbar}\right)\mathscr{G}(\tau) = -\delta(\tau)/\hbar \\ &\Longleftrightarrow \ldots \end{aligned}$$

whereas the r.h.sides are to be understood as matrix-identities. Solving this differential equation w.r.t. the boundary-conditions $\mathscr{G}(\hbar\beta) = \zeta\mathscr{G}(0)$ yields the desired result.

We will now use the Dirac-identity for (anti-)periodic functions as[8]

$$\delta(\tau) = \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} \mathrm{e}^{-\mathrm{i}\omega_n \tau} \tag{28.46}$$

and use the expansion from Lemma TBD to obtain

$$\left(\partial_\tau + \frac{H-\mu}{\hbar}\right)\mathscr{G}(\tau) = -\delta(\tau)/\hbar \Longleftrightarrow \frac{1}{\hbar}\sum_{\omega_n \in \Omega_\zeta} \mathrm{e}^{-\mathrm{i}\omega_n \tau}\left(-\mathrm{i}\omega_n + \frac{H-\mu}{\hbar}\right)\mathscr{G}(\mathrm{i}\omega_n) = \frac{1}{\hbar}\sum_{\omega_n \in \Omega_\zeta} \mathrm{e}^{-\mathrm{i}\omega_n \tau}(-1/\hbar)$$

using now that the coefficients of a Fourier-series are unique, one obtains by comparing coefficients

$$\left(-\mathrm{i}\omega_n + \frac{H-\mu}{\hbar}\right)\mathscr{G}(\mathrm{i}\omega_n) = -\frac{1}{\hbar} \quad \Longleftrightarrow \quad \mathscr{G}(\mathrm{i}\omega_n) = \frac{1}{\hbar}\left(\frac{1}{\mathrm{i}\omega_n - \frac{H-\mu}{\hbar}}\right)$$

Notice that we can't expand the r.h. sides w.r.t. Matsubara-frequencies according to Lemma TBD since we cant use the usual Kronecker-delta-relation since we are only integrating from 0 to $\hbar\beta$.

Equation of Motion Technique for Hybridized Systems

Equations of motion for generalized time-ordered, retarded and advanced Green-functions: Elk-Gasser p. 25

With the equation-of-motion-technique; See Appendix (ToBeReferenced.)

Proof 28.38 of Theorem 13.8

See Waste-Folder 2, topic 4.

Proof 28.39 of Theorem ??

We compute

$$\mathscr{G}_{\mathbf{q},\mathbf{q}'}(\tau, 0) = -\theta(\tau)\langle b_{\mathbf{q}}(\tau) b^\dagger_{\mathbf{q}'}(0)\rangle - \theta(-\tau)\langle b^\dagger_{\mathbf{q}'}(0) b_{\mathbf{q}}(\tau)\rangle = \dots$$

Now we use

$$b_{\mathbf{q}}(\tau) = \mathrm{e}^{\tau H/\hbar} b_{\mathbf{q}} \mathrm{e}^{-\tau H/\hbar} = \mathrm{e}^{-\tau\omega_{\mathbf{q}}} b_{\mathbf{q}} \tag{28.47}$$

and

[8] Cf. Waste-Folder 1, topic 34.

$$b_{\mathbf{q}}^{\dagger}(\tau) = \mathrm{e}^{\tau H/\hbar} b_{\mathbf{q}} \mathrm{e}^{-\tau H/\hbar} = \mathrm{e}^{+\tau\omega_{\mathbf{q}}} b_{\mathbf{q}}^{\dagger} \tag{28.48}$$

and therefore we obtain

$$-\langle b_{\mathbf{q}}(\tau) b_{\mathbf{q}'}^{\dagger}(0)\rangle = -\mathrm{e}^{-\tau\omega_{\mathbf{q}}} \langle b_{\mathbf{q}} b_{\mathbf{q}'}^{\dagger}\rangle = -\delta_{\mathbf{q},\mathbf{q}'} \mathrm{e}^{-\tau\omega_{\mathbf{q}}} \left(\mathrm{B}_{\beta}(\hbar\omega_{\mathbf{q}}) + 1\right) \tag{28.49}$$

and

$$-\langle b_{\mathbf{q}'}^{\dagger}(0) b_{\mathbf{q}}(\tau)\rangle = -\mathrm{e}^{-\tau\omega_{\mathbf{q}}} \langle b_{\mathbf{q}'}^{\dagger} b_{\mathbf{q}}\rangle = -\delta_{\mathbf{q},\mathbf{q}'} \mathrm{e}^{-\tau\omega_{\mathbf{q}}} \mathrm{B}_{\beta}(\hbar\omega_{\mathbf{q}}) \tag{28.50}$$

and therefore we obtain

$$\begin{aligned}
\mathscr{G}_{\mathbf{q},\mathbf{q}'}(\tau,0) &= \dots \\
&= -\theta(\tau)\delta_{\mathbf{q},\mathbf{q}'} \mathrm{e}^{-\tau\omega_{\mathbf{q}}} \left(\mathrm{B}_{\beta}(\hbar\omega_{\mathbf{q}}) + 1\right) - \theta(-\tau)\delta_{\mathbf{q},\mathbf{q}'} \mathrm{e}^{-\tau\omega_{\mathbf{q}}} \mathrm{B}_{\beta}(\hbar\omega_{\mathbf{q}})
\end{aligned}$$

Proofs to Section 7.2

Proof 28.40 of Theorem 13.11 By direct computation we obtain

$$\begin{aligned}
\mathscr{D}_{\alpha,\alpha'}(\tau) &= \frac{-\theta(\tau)}{\hbar} \langle A_{\alpha}(\tau)\, A_{\alpha'}^{\dagger}(0)\rangle + \theta(-\tau)\, \langle A_{\alpha'}^{\dagger}(0)\, A_{\alpha}(\tau)\rangle \\
&= \frac{-\theta(\tau)}{\hbar}\Big[\langle a_{\mathbf{q}l}(\tau) a_{\mathbf{q}'l'}^{\dagger}(0)\rangle + \langle a_{-\mathbf{q}l}^{\dagger}(\tau) a_{-\mathbf{q}'l'}(0)\rangle\Big] \quad + \quad \frac{-\theta(-\tau)}{\hbar}\Big[\langle a_{-\mathbf{q}'l'}(0) a_{-\mathbf{q}l}^{\dagger}(\tau)\rangle + \langle a_{\mathbf{q}'l'}^{\dagger}(0) a_{\mathbf{q}l}(\tau)\rangle\Big] \\
&= \dots
\end{aligned}$$

Now we use the identities in Equation (8.25)

$$\begin{aligned}
a_{\mathbf{q}l}(\tau) &= a_{\mathbf{q}l}(0)\mathrm{e}^{-\tau\omega_{\mathbf{q}l}} = a_{\mathbf{q}l}\mathrm{e}^{-\tau\omega_{\mathbf{q}l}} \qquad & \langle a_{\alpha}^{\dagger} a_{\beta}\rangle &= \delta_{\alpha,\beta}\, \mathrm{B}_{\hbar\beta}(\omega_{\alpha}) \\
a_{\mathbf{q}l}^{\dagger}(\tau) &= a_{\mathbf{q}l}^{\dagger}(0)\mathrm{e}^{+\tau\omega_{\mathbf{q}l}} = a_{\mathbf{q}l}^{\dagger}\mathrm{e}^{+\tau\omega_{\mathbf{q}l}} \qquad & \langle a_{\beta} a_{\alpha}^{\dagger}\rangle &= \langle a_{\alpha}^{\dagger} a_{\beta}\rangle + \delta_{\alpha,\beta} \equiv \delta_{\alpha,\beta}\, \mathrm{B}_{\hbar\beta}(\omega_{\alpha}) + \delta_{\alpha,\beta}
\end{aligned}$$

This gives (together with Remark 12.7.1)

$$\begin{aligned}
\dots &= \frac{-\theta(\tau)}{\hbar}\Big[\mathrm{e}^{-\tau\omega_{\mathbf{q}l}} \langle a_{\mathbf{q}l} a_{\mathbf{q}'l'}^{\dagger}\rangle + \mathrm{e}^{\tau\omega_{\mathbf{q}l}} \langle a_{-\mathbf{q}l}^{\dagger} a_{-\mathbf{q}'l'}\rangle\Big] \qquad + \quad \frac{-\theta(-\tau)}{\hbar}\Big[\mathrm{e}^{\tau\omega_{\mathbf{q}l}} \langle a_{-\mathbf{q}'l'} a_{-\mathbf{q}l}^{\dagger}\rangle + \mathrm{e}^{-\tau\omega_{\mathbf{q}l}} \langle a_{\mathbf{q}'l'}^{\dagger} a_{\mathbf{q}l}\rangle\Big] \\
&= \delta_{\mathbf{q},\mathbf{q}'}\delta_{l,l'} \frac{-\theta(\tau)}{\hbar}\Big[\mathrm{e}^{-\tau\omega_{\mathbf{q}l}} \left(\mathrm{B}_{\hbar\beta}(\omega_{\alpha}) + 1\right) + \mathrm{e}^{+\tau\omega_{\mathbf{q}l}} \mathrm{B}_{\hbar\beta}(\omega_{\alpha})\Big] + \delta_{\mathbf{q},\mathbf{q}'}\delta_{l,l'} \frac{-\theta(-\tau)}{\hbar}\Big[\mathrm{e}^{+\tau\omega_{\mathbf{q}l}} \left(\mathrm{B}_{\hbar\beta}(\omega_{\alpha}) + 1\right) + \mathrm{e}^{-\tau\omega_{\mathbf{q}l}} \mathrm{B}_{\hbar\beta}(\omega_{\alpha})\Big] \\
&= \dots
\end{aligned}$$

which can be summarized as

$$\ldots = -\delta_{\alpha,\alpha'}\frac{1}{\hbar}\sum_{\nu\in\{+,-\}}\theta(\nu\tau)\Big[\left(1+\mathrm{B}_{\hbar\beta}(\omega_\alpha)\right)\,\mathrm{e}^{-\nu\,\omega_\alpha\tau}+\mathrm{B}_{\hbar\beta}(\omega_\alpha)\,\mathrm{e}^{\nu\,\omega_\alpha\tau}\Big] \tag{28.51}$$

For the Matsubara-Fourier-transform: Cf. the literature (around equation (3.72).

$$\mathscr{D}_{\alpha\alpha'}(\mathrm{i}\omega_n)=\int_0^{\hbar\beta}\mathrm{d}\tau\,\mathscr{D}_{\alpha\alpha'}(\tau)=(...TBD...)=-\frac{1}{\hbar}\frac{2\omega_\alpha\delta_{\alpha,\alpha'}}{\omega_n^2+\omega_\alpha^2} \tag{28.52}$$

Proof 28.41 of Theorem 13.12

Expanding the time-ordering operator yields the equation

$$\begin{aligned}\mathscr{D}_{\substack{0\\\alpha\alpha'}}(\tau)&=-\frac{1}{\hbar}\langle T_\tau A_\alpha(\tau)A^\dagger_{\alpha'}(0)\rangle_0\\&=-\frac{1}{\hbar}\langle\theta(\tau)A_\alpha(\tau)A^\dagger_{\alpha'}(0)+\theta(-\tau)A^\dagger_{\alpha'}(0)A_\alpha(\tau)\rangle_0\end{aligned}$$

Hence we obtain for the time-derivative

$$\begin{aligned}\partial_\tau\mathscr{D}_{\substack{0\\\alpha\alpha'}}(\tau)&=-\frac{1}{\hbar}\langle\mathrm{T}_\tau\,\partial_\tau A_\alpha(\tau)A^\dagger_{\alpha'}(0)\rangle_0-\frac{1}{\hbar}\delta(\tau)\langle A_\alpha(0)A^\dagger_{\alpha'}(0)\rangle_0+\frac{1}{\hbar}\delta(-\tau)\langle A^\dagger_{\alpha'}(0)A_\alpha(0)\rangle_0\\&=\ldots\end{aligned}$$

for the first term we have with Heisenbergs equation of motion

$$\partial_\tau A_\alpha(\tau)=-\frac{1}{\hbar}[A_\alpha,H_0]_{\mathrm{H}}(\tau) \tag{28.53}$$

Now we compute the commutators

$$\begin{aligned}[a_\alpha,a^\dagger_\beta a_\beta]&=\delta_{\alpha,\beta}a_\beta\\ [a^\dagger_\alpha,a^\dagger_\beta a_\beta]&=-\delta_{\alpha,\beta}a^\dagger_\beta\end{aligned}$$

and obtain

$$\begin{aligned}[A_\nu(\mathbf{k}),H_0]&=\sum_{\mathbf{k}',\nu'}\hbar\omega_\nu(\mathbf{k}')\left([a_\nu(\mathbf{k}),a^\dagger_{\nu'}(\mathbf{k}')a_{\nu'}(\mathbf{k}')]+[a^\dagger_\nu(-\mathbf{k}),a^\dagger_{\nu'}(\mathbf{k}')a_{\nu'}(\mathbf{k}')]\right)\\&=\hbar\omega_\nu(\mathbf{k})\left(a_\nu(\mathbf{k})-a^\dagger_\nu(-\mathbf{k})\right)\\&\equiv\hbar\omega_\nu(\mathbf{k})P_\nu(\mathbf{k})\end{aligned}$$

hence we obtain, summarizing the terms with delta-functions with the distributivity-law

$$\ldots = -\frac{1}{\hbar}\left(\frac{-1}{\hbar}\omega_\nu(\mathbf{k})\langle \mathrm{T}_\tau\, P_\nu(\mathbf{k})A^\dagger_{\nu'}(\mathbf{k}')(0)\rangle_0 + \delta(\tau)\Big[\langle A_\nu(\mathbf{k})A^\dagger_{\nu'}(\mathbf{k}')\rangle_0 - \langle A^\dagger_{\nu'}(\mathbf{k}')A_\nu(\mathbf{k})\rangle_0\Big]\right)$$

Now we compute the δ-functions

$$\begin{aligned}
&\langle A_\nu(\mathbf{k})A^\dagger_{\nu'}(\mathbf{k}')\rangle_0 - \langle A^\dagger_{\nu'}(\mathbf{k}')A_\nu(\mathbf{k})\rangle_0\\
&= \delta_{\nu,\nu'}\delta_{\mathbf{k},\mathbf{k}'}\left(1-\mathrm{B}_{\hbar\beta}(\omega_\nu(\mathbf{k}))+0+0+\mathrm{B}_{\hbar\beta}(\omega_\nu(-\mathbf{k}))-\left[\mathrm{B}_{\hbar\beta}(\omega_\nu(\mathbf{k}))+0+0+1-\mathrm{B}_{\hbar\beta}(\omega_\nu(-\mathbf{k}))\right]\right)\\
&= 0
\end{aligned}$$

in total we obtain thus

$$\partial_\tau \mathscr{D}^{\,0}_{\alpha\alpha'}(\tau) = \frac{+\omega_\nu(\mathbf{k})}{\hbar}\langle \mathrm{T}_\tau\, P_\nu(\mathbf{k},\tau)A^\dagger_{\nu'}(\mathbf{k}',0)\rangle_0 \tag{28.54}$$

Hence we obtain for the second derivative analogously

$$\begin{aligned}
\partial^2_\tau \mathscr{D}^{\,0}_{\alpha\alpha'}(\tau) &= \frac{+\omega_\nu(\mathbf{k})}{\hbar}\left(\langle \mathrm{T}_\tau\, \partial_\tau P_\nu(\mathbf{k},\tau)A^\dagger_{\nu'}(\mathbf{k}',0)\rangle_0 + \delta(\tau)\langle P_\nu(\mathbf{k})A^\dagger_{\nu'}(\mathbf{k}')\rangle_0 - \delta(-\tau)\langle A^\dagger_{\nu'}(\mathbf{k}')P_\nu(\mathbf{k})\rangle_0\right)\\
&= \ldots
\end{aligned}$$

The first term becomes a commutator with Heisenbergs equation of motion

$$\partial_\tau P_\nu(\mathbf{k},\tau) = -\frac{1}{\hbar}[P_\nu(\mathbf{k}),H_0]_{\mathrm{H}}(\tau). \tag{28.55}$$

Analogously to the previous computation we obtain

$$[P_\nu(\mathbf{k}),H_0] = \hbar\omega_\nu(\mathbf{k})A_\nu(\mathbf{k})$$

and for the other term we compute

$$\begin{aligned}
&\langle P_\nu(\mathbf{k})A^\dagger_{\nu'}(\mathbf{k}')\rangle_0 - \langle A^\dagger_{\nu'}(\mathbf{k}')P_\nu(\mathbf{k})\rangle_0\\
&= \delta_{\nu,\nu'}\delta_{\mathbf{k},\mathbf{k}'}\left(1-\mathrm{B}_{\hbar\beta}(\omega_\nu(\mathbf{k}))+0+0-\mathrm{B}_{\hbar\beta}(\omega_\nu(-\mathbf{k}))-\left[\mathrm{B}_{\hbar\beta}(\omega_\nu(\mathbf{k}))+0+0-(1-\mathrm{B}_{\hbar\beta}(\omega_\nu(-\mathbf{k})))\right]\right)\\
&= 2\delta_{\nu,\nu'}\delta_{\mathbf{k},\mathbf{k}'}
\end{aligned}$$

Hence we obtain in total

$$\begin{aligned}
\partial^2_\tau \mathscr{D}^{\,0}_{\alpha\alpha'}(\tau) &= \frac{+\omega_\nu(\mathbf{k})}{\hbar}\left(-\omega_\nu(\mathbf{k})\langle \mathrm{T}_\tau\, A_\nu(\mathbf{k},\tau)A^\dagger_{\nu'}(\mathbf{k}',0)\rangle_0 + 2\delta(\tau)\delta_{\nu,\nu'}\delta_{\mathbf{k},\mathbf{k}'}\right)\\
\Longleftrightarrow \left(\partial^2_\tau - \omega_\nu(\mathbf{k})^2\right)\mathscr{D}^{\,0}_{\alpha\alpha'}(\tau) &= 2\omega_\nu(\mathbf{k})\delta(\tau)\delta_{\nu,\nu'}\delta_{\mathbf{k},\mathbf{k}'}
\end{aligned}$$

Proof 28.42 of Lemma 14.4

We complete the proof, by sorting the Green-functions into four groups.

- **Group 1:** Time- and anti-time-ordered Green-function $G^{\mathbb{T}}$, $G^{\overline{\mathbb{T}}}$.
- **Group 2:** Retarded and advanced Green-function G^{ret}, G^{adv}.
- **Group 3:** Lesser and greater Green-function $G^{<}$, $G^{>}$ and the spectral function A.
- **Group 4:** Imaginary time Green-function $\mathscr{G}$.

For each group, we will prove some Lemmata in order to complete the main proof. Ultimately we want to compute the Fourier-expansion coefficients for each Green-function. In each case we set, without loss of generality, the second time argument to zero $t' = \tau' = 0$ in order to simplify the notation.

Lemmata for the first two groups

In the first two groups we are confronted with the obstacle, that the usual Fourier-transform will turn out to be undefined in the functional sense[9]. The Fourier-transform only exists in the distributional sense. We therefore introduce the auxiliary functions

$$G^{\blacksquare}_{\substack{\epsilon\\X,X'}}(t) = G^{\blacksquare}_{X,X'}(t)\,\mathrm{e}^{-\epsilon|t|} \qquad \text{for } \blacksquare \in \{\mathbb{T}, \overline{\mathbb{T}}, \mathrm{ret}, \mathrm{adv}\} \tag{28.56}$$

then one has

$$G^{\blacksquare}_{\substack{\epsilon\\X,X'}}(\omega) = \int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} G^{\blacksquare}_{\substack{\epsilon\\X,X'}}(t) = \int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} G^{\blacksquare}_{X,X'}(t)\,\mathrm{e}^{-\epsilon|t|} \tag{28.57}$$

For reasons that are impossible to understand in advance, we compute the following integrals.

Lemma 28.15 Let $\xi_{\bullet} \in \mathbb{R}$. Then we obtain

$$\frac{\mathrm{i}}{\hbar}\int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}\, \mathrm{e}^{-\epsilon|t|}\, \mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar}\, \theta(t) = \frac{-1}{\hbar}\frac{1}{\omega + \mathrm{i}\epsilon + (\xi_n - \xi_m)/\hbar} \tag{28.58}$$

Proof 28.15 We compute straightforwardly

$$\begin{aligned}\frac{\mathrm{i}}{\hbar}\int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}\, \mathrm{e}^{-\epsilon|t|}\, \mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar}\, \theta(t) &= \frac{\mathrm{i}}{\hbar}\int_0^\infty \mathrm{d}t\, \mathrm{e}^{t\left(-\epsilon+\mathrm{i}\left[\omega+(\xi_n-\xi_m)/\hbar\right]\right)}\\ &= \frac{\mathrm{i}}{\hbar}\frac{-1}{-\epsilon+\mathrm{i}\left[\omega+(\xi_n-\xi_m)/\hbar\right]}\\ &= \frac{-1}{\hbar}\frac{1}{\omega+\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar}\end{aligned}$$

Lemma 28.16 Let $\xi_{\bullet} \in \mathbb{R}$. Then we obtain

[9] Cf. [Kaul, Fischer].

$$\frac{\mathrm{i}}{\hbar}\int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}\, \mathrm{e}^{-\epsilon|t|}\, \mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar}\, \theta(-t) = \frac{1}{\hbar}\frac{1}{\omega - \mathrm{i}\epsilon + (\xi_n - \xi_m)/\hbar} \tag{28.59}$$

Proof 28.16 We compute straightforwardly

$$\begin{aligned}\frac{\mathrm{i}}{\hbar}\int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}\, \mathrm{e}^{-\epsilon|t|}\, \mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar}\, \theta(-t) &= \frac{\mathrm{i}}{\hbar}\int_{-\infty}^{0} \mathrm{d}t\, \mathrm{e}^{t\left(+\epsilon+\mathrm{i}\left[\omega+(\xi_n-\xi_m)/\hbar\right]\right)} \\ &= \frac{\mathrm{i}}{\hbar}\frac{1}{+\epsilon+\mathrm{i}\left[\omega+(\xi_n-\xi_m)/\hbar\right]} \\ &= \frac{1}{\hbar}\frac{1}{\omega-\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar}\end{aligned}$$

Fact 28.17 Notice that these results can be summarized by the identity

$$\frac{\mathrm{i}}{\hbar}\int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}\, \mathrm{e}^{-\epsilon|t|}\, \mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar}\, \theta(\pm t) = \frac{\mp 1}{\hbar}\frac{1}{\omega \pm \mathrm{i}\epsilon + (\xi_n - \xi_m)/\hbar} \tag{28.60}$$

Group 1

The time-ordered Green-function

For the time-ordered Green-function we obtain

$$\begin{aligned}G^{\mathbb{T}}_{X,X'}(t) &= G^{\mathbb{T}}_{X,X'}(t,t'=0) \\ &= -\frac{\mathrm{i}}{\hbar}\langle \theta(t)X(t)X'(0) + S_{X,X'}\theta(-t)X'(0)X(t)\rangle \\ &= -\frac{\mathrm{i}}{\hbar}\frac{1}{Z}\sum_{n,m}\theta(t)\langle n|\mathrm{e}^{-\beta\Xi}X(t)|m\rangle\langle m|X'|n\rangle + S_{X,X'}\theta(-t)\langle n|\mathrm{e}^{-\beta\Xi}X'|m\rangle\langle m|X(t)|n\rangle \\ &= -\frac{\mathrm{i}}{\hbar}\frac{1}{Z}\sum_{n,m}\theta(t)\mathrm{e}^{-\beta\xi_n}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar}\langle n|X|m\rangle\langle m|X'|n\rangle + S_{X,X'}\theta(-t)\mathrm{e}^{-\beta\xi_n}\mathrm{e}^{\mathrm{i}t(\xi_m-\xi_n)/\hbar}\langle n|X'|m\rangle\langle m|X|n\rangle \\ &= -\frac{\mathrm{i}}{\hbar}\frac{1}{Z}\sum_{n,m}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\theta(t)\mathrm{e}^{-\beta\xi_n} + S_{X,X'}\theta(-t)\mathrm{e}^{-\beta\xi_m}\right)\end{aligned}$$

We therefore see, that the Fourier-transform of the time-ordered Green-function is strictly speaking ill defined. We therefore introduce for each „small“ $\epsilon > 0$, the regularized time-ordered Green-function

$$\begin{aligned}G^{\mathbb{T}}_{\substack{\epsilon\\X,X'}}(t) &= G^{\mathbb{T}}_{X,X'}(t)\,\mathrm{e}^{-\epsilon|t|} \\ &= -\frac{\mathrm{i}}{\hbar}\frac{1}{Z}\sum_{n,m}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar-\epsilon|t|}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\theta(t)\mathrm{e}^{-\beta\xi_n} + S_{X,X'}\theta(-t)\mathrm{e}^{-\beta\xi_m}\right)\end{aligned}$$

which gives for the Fourier-transform then

$$
\begin{aligned}
G^{\mathbb{T}}{}_{\substack{\epsilon \\ X,X'}}(\omega) &= \int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} G^{\mathbb{T}}{}_{\substack{\epsilon \\ X,X'}}(t) \\
&= -\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle \mathrm{e}^{-\beta\xi_n}\frac{\mathrm{i}}{\hbar}\int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar-\epsilon|t|}\theta(t) \\
&\quad - S_{X,X'}\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle \mathrm{e}^{-\beta\xi_m}\frac{\mathrm{i}}{\hbar}\int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar-\epsilon|t|}\theta(-t) \\
&= -\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle \mathrm{e}^{-\beta\xi_n}\frac{-1}{\hbar}\frac{1}{\omega+\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar} \\
&\quad - S_{X,X'}\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle \mathrm{e}^{-\beta\xi_m}\frac{1}{\hbar}\frac{1}{\omega-\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar} \\
&= \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\frac{1}{\hbar}\left(\frac{\mathrm{e}^{-\beta\xi_n}}{\omega+\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar} - S_{X,X'}\frac{\mathrm{e}^{-\beta\xi_m}}{\omega-\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar}\right)
\end{aligned}
$$

The anti-time-ordered Green-function

For the anti-time-ordered Green-function we obtain

$$
\begin{aligned}
G^{\overline{\mathbb{T}}}_{X,X'}(t) &= G^{\overline{\mathbb{T}}}_{X,X'}(t,t'=0) \\
&= -\frac{\mathrm{i}}{\hbar}\langle\theta(-t)X(t)X'(0)+S_{X,X'}\theta(+t)X'(0)X(t)\rangle \\
&= -\frac{\mathrm{i}}{\hbar}\frac{1}{Z}\sum_{n,m}\theta(-t)\langle n|\mathrm{e}^{-\beta\Xi}X(t)|m\rangle\langle m|X'|n\rangle + S_{X,X'}\theta(+t)\langle n|\mathrm{e}^{-\beta\Xi}X'|m\rangle\langle m|X(t)|n\rangle \\
&= -\frac{\mathrm{i}}{\hbar}\frac{1}{Z}\sum_{n,m}\theta(-t)\mathrm{e}^{-\beta\xi_n}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar}\langle n|X|m\rangle\langle m|X'|n\rangle + S_{X,X'}\theta(+t)\mathrm{e}^{-\beta\xi_n}\mathrm{e}^{\mathrm{i}t(\xi_m-\xi_n)/\hbar}\langle n|X'|m\rangle\langle m|X|n\rangle \\
&= -\frac{\mathrm{i}}{\hbar}\frac{1}{Z}\sum_{n,m}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\theta(-t)\mathrm{e}^{-\beta\xi_n}+S_{X,X'}\theta(+t)\mathrm{e}^{-\beta\xi_m}\right)
\end{aligned}
$$

We therefore see, that the Fourier-transform of the anti-time-ordered Green-function is strictly speaking ill defined. We therefore introduce for each „small“ $\epsilon > 0$, the regularized anti time-ordered Green-function

$$
\begin{aligned}
G^{\overline{\mathbb{T}}}{}_{\substack{\epsilon \\ X,X'}}(t) &= G^{\overline{\mathbb{T}}}_{X,X'}(t)\,\mathrm{e}^{-\epsilon|t|} \\
&= -\frac{\mathrm{i}}{\hbar}\frac{1}{Z}\sum_{n,m}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar-\epsilon|t|}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\theta(-t)\mathrm{e}^{-\beta\xi_n}+S_{X,X'}\theta(+t)\mathrm{e}^{-\beta\xi_m}\right)
\end{aligned}
$$

which gives for the Fourier-transform then

$$
\begin{aligned}
G^{\overline{\mathbb{T}}}_{\substack{\epsilon \\ X,X'}}(\omega) &= \int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} G^{\overline{\mathbb{T}}}_{\substack{\epsilon \\ X,X'}}(t) \\
&= -\frac{1}{Z}\sum_{n,m} \langle n|X|m\rangle\langle m|X'|n\rangle \mathrm{e}^{-\beta\xi_n} \frac{\mathrm{i}}{\hbar}\int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar-\epsilon|t|}\theta(-t) \\
&- S_{X,X'}\frac{1}{Z}\sum_{n,m} \langle n|X|m\rangle\langle m|X'|n\rangle \mathrm{e}^{-\beta\xi_m} \frac{\mathrm{i}}{\hbar}\int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar-\epsilon|t|}\theta(+t) \\
&= -\frac{1}{Z}\sum_{n,m} \langle n|X|m\rangle\langle m|X'|n\rangle \mathrm{e}^{-\beta\xi_n} \frac{+1}{\hbar}\frac{1}{\omega-\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar} \\
&- S_{X,X'}\frac{1}{Z}\sum_{n,m} \langle n|X|m\rangle\langle m|X'|n\rangle \mathrm{e}^{-\beta\xi_m} \frac{-1}{\hbar}\frac{1}{\omega+\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar} \\
&= \frac{1}{Z}\sum_{n,m} \langle n|X|m\rangle\langle m|X'|n\rangle \frac{1}{\hbar}\left(\frac{-\mathrm{e}^{-\beta\xi_n}}{\omega-\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar} + S_{X,X'}\frac{\mathrm{e}^{-\beta\xi_m}}{\omega+\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar}\right)
\end{aligned}
$$

Group 2

The retarded Green-function

For the retarded Green-function we obtain

$$
\begin{aligned}
G^{\mathrm{ret}}_{X,X'}(t) &= G^{\mathrm{ret}}_{X,X'}(t,t'=0) \\
&= -\frac{\mathrm{i}}{\hbar}\theta(t)\langle X(t)X'(0) - S_{X,X'}X'(0)X(t)\rangle \\
&= -\frac{\mathrm{i}}{\hbar}\theta(t)\frac{1}{Z}\sum_{n,m}\langle n|\mathrm{e}^{-\beta\Xi}X(t)|m\rangle\langle m|X'|n\rangle - S_{X,X'}\langle n|\mathrm{e}^{-\beta\Xi}X'|m\rangle\langle m|X(t)|n\rangle \\
&= -\frac{\mathrm{i}}{\hbar}\theta(t)\frac{1}{Z}\sum_{n,m}\mathrm{e}^{-\beta\xi_n}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar}\langle n|X|m\rangle\langle m|X'|n\rangle - S_{X,X'}\mathrm{e}^{-\beta\xi_n}\mathrm{e}^{\mathrm{i}t(\xi_m-\xi_n)/\hbar}\langle n|X'|m\rangle\langle m|X|n\rangle \\
&= -\frac{\mathrm{i}}{\hbar}\frac{1}{Z}\theta(t)\sum_{n,m}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)
\end{aligned}
$$

We therefore see, that the Fourier-transform of the retarded Green-function is strictly speaking ill defined. We therefore introduce for each „small“ $\epsilon > 0$, the regularized retarded Green-function

$$
\begin{aligned}
G^{\mathrm{ret}}_{\substack{\epsilon \\ X,X'}}(t) &= G^{\mathrm{ret}}_{X,X'}(t)\,\mathrm{e}^{-\epsilon|t|} \\
&= -\frac{\mathrm{i}}{\hbar}\frac{1}{Z}\theta(t)\sum_{n,m}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar-\epsilon|t|}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)
\end{aligned}
$$

which gives for the Fourier-transform then

$$
\begin{aligned}
G^{\text{ret}}_{\substack{\epsilon\\X,X'}}(\omega) &= \int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} G^{\text{ret}}_{\substack{\epsilon\\X,X'}}(t) \\
&= -\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle \left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right) \frac{\mathrm{i}}{\hbar}\int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar-\epsilon|t|}\theta(t) \\
&= -\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle \left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right) \frac{-1}{\hbar}\frac{1}{\omega+\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar} \\
&= \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle \left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right) \frac{1}{\hbar}\frac{1}{\omega+\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar}
\end{aligned}
$$

Lastly we switch to the usual notation found in physics-textbooks and note the formal identity

$$
\begin{aligned}
G^{\text{ret}}_{\substack{\epsilon\\X,X'}}(\omega) &= \int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} G^{\text{ret}}_{\substack{\epsilon\\X,X'}}(t) \\
&= -\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle \left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right) \frac{\mathrm{i}}{\hbar}\int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar-\epsilon|t|}\theta(t) \\
&= -\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle \left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right) \frac{\mathrm{i}}{\hbar}\int_0^\infty \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar-\epsilon t} \\
&= -\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle \left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right) \frac{\mathrm{i}}{\hbar}\int_0^\infty \mathrm{d}t\, \mathrm{e}^{\mathrm{i}(\omega+\mathrm{i}\epsilon)t}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar} \\
&= -\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle \left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right) \frac{\mathrm{i}}{\hbar}\int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}(\omega+\mathrm{i}\epsilon)t}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar}\theta(t) \\
&= \int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}(\omega+\mathrm{i}\epsilon)t} G^{\text{ret}}_{X,X'}(t) \\
&= G^{\text{ret}}_{X,X'}(\omega+\mathrm{i}\epsilon)
\end{aligned}
$$

Hence, for future reference, we have

$$
G^{\text{ret}}_{X,X'}(\omega+\mathrm{i}\epsilon) = \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle \left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right) \frac{1}{\hbar}\frac{1}{\omega+\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar}
$$

The advanced Green-function

For the advanced Green-function we obtain

$$
\begin{aligned}
G^{\mathrm{adv}}_{X,X'}(t) &= G^{\mathrm{adv}}_{X,X'}(t, t'=0) \\
&= +\frac{\mathrm{i}}{\hbar}\theta(-t)\langle X(t)X'(0) - S_{X,X'}X'(0)X(t)\rangle \\
&= +\frac{\mathrm{i}}{\hbar}\theta(-t)\frac{1}{Z}\sum_{n,m}\langle n|\mathrm{e}^{-\beta\Xi}X(t)|m\rangle\langle m|X'|n\rangle - S_{X,X'}\langle n|\mathrm{e}^{-\beta\Xi}X'|m\rangle\langle m|X(t)|n\rangle \\
&= +\frac{\mathrm{i}}{\hbar}\theta(-t)\frac{1}{Z}\sum_{n,m}\mathrm{e}^{-\beta\xi_n}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar}\langle n|X|m\rangle\langle m|X'|n\rangle - S_{X,X'}\mathrm{e}^{-\beta\xi_n}\mathrm{e}^{\mathrm{i}t(\xi_m-\xi_n)/\hbar}\langle n|X'|m\rangle\langle m|X|n\rangle \\
&= +\frac{\mathrm{i}}{\hbar}\frac{1}{Z}\theta(-t)\sum_{n,m}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)
\end{aligned}
$$

We therefore see, that the Fourier-transform of the advanced Green-function is strictly speaking ill defined. We therefore introduce for each „small“ $\epsilon > 0$, the regularized advanced Green-function

$$
\begin{aligned}
G^{\mathrm{adv}}_{\substack{\epsilon\\X,X'}}(t) &= G^{\mathrm{adv}}_{X,X'}(t)\,\mathrm{e}^{-\epsilon|t|} \\
&= +\frac{\mathrm{i}}{\hbar}\frac{1}{Z}\theta(-t)\sum_{n,m}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar-\epsilon|t|}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)
\end{aligned}
$$

which gives for the Fourier-transform then

$$
\begin{aligned}
G^{\mathrm{adv}}_{\substack{\epsilon\\X,X'}}(\omega) &= \int \mathrm{d}t\,\mathrm{e}^{\mathrm{i}\omega t}G^{\mathrm{adv}}_{\substack{\epsilon\\X,X'}}(t) \\
&= +\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)\frac{\mathrm{i}}{\hbar}\int \mathrm{d}t\,\mathrm{e}^{\mathrm{i}\omega t}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar-\epsilon|t|}\theta(-t) \\
&= +\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)\frac{1}{\hbar}\frac{1}{\omega - \mathrm{i}\epsilon + (\xi_n-\xi_m)/\hbar} \\
&= \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)\frac{1}{\hbar}\frac{1}{\omega - \mathrm{i}\epsilon + (\xi_n-\xi_m)/\hbar}
\end{aligned}
$$

Lastly we switch to the usual notation found in physics-textbooks and note the formal identity

$$
\begin{aligned}
G^{\mathrm{adv}}_{\substack{\epsilon\\X,X'}}(\omega) &= \int \mathrm{d}t\,\mathrm{e}^{\mathrm{i}\omega t}G^{\mathrm{adv}}_{\substack{\epsilon\\X,X'}}(t) \\
&= +\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)\frac{\mathrm{i}}{\hbar}\int \mathrm{d}t\,\mathrm{e}^{\mathrm{i}\omega t}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar-\epsilon|t|}\theta(-t) \\
&= +\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)\frac{\mathrm{i}}{\hbar}\int_{-\infty}^{0} \mathrm{d}t\,\mathrm{e}^{\mathrm{i}\omega t}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar+\epsilon t}
\end{aligned}
$$

$$
\begin{aligned}
&= +\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)\frac{\mathrm{i}}{\hbar}\int_{-\infty}^{0}\mathrm{d}t\,\mathrm{e}^{\mathrm{i}(\omega-\mathrm{i}\epsilon)t}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar} \\
&= +\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)\frac{\mathrm{i}}{\hbar}\int\mathrm{d}t\,\mathrm{e}^{\mathrm{i}(\omega-\mathrm{i}\epsilon)t}\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar}\theta(-t) \\
&= \int\mathrm{d}t\,\mathrm{e}^{\mathrm{i}(\omega-\mathrm{i}\epsilon)t}G^{\mathrm{adv}}_{X,X'}(t) \\
&= G^{\mathrm{adv}}_{X,X'}(\omega-\mathrm{i}\epsilon)
\end{aligned}
$$

Hence, for future reference, we have

$$
G^{\mathrm{adv}}_{X,X'}(\omega-\mathrm{i}\epsilon) = \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)\frac{1}{\hbar}\frac{1}{\omega-\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar}
$$

Group 3

Lemmata for the third group

In the third group we are in the comfortable situation that we can refer to standard-theorems about distribution-theory. We therefore note the standard-identities for test-functions that decay sufficiently fast at infinity

$$
\delta(\omega) = \frac{1}{2\pi}\int\mathrm{d}t\,\mathrm{e}^{\mathrm{i}\omega t} \tag{28.61}
$$

and (in wise foresight)

$$
\frac{1}{\omega+\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar} = \mathrm{P}\,\frac{1}{\omega+(\xi_n-\xi_m)/\hbar} - \mathrm{i}\pi\delta(\omega+(\xi_n-\xi_m)/\hbar) \tag{28.62}
$$

The lesser Green-function

For the lesser Green-function we obtain

$$
\begin{aligned}
G^{<}_{X,X'}(t) &= G^{<}_{X,X'}(t,t'=0) \\
&= -\frac{\mathrm{i}}{\hbar}S_{X,X'}\langle X'(0)X(t)\rangle \\
&= -\frac{\mathrm{i}}{\hbar}S_{X,X'}\frac{1}{Z}\sum_{n,m}\langle n|\mathrm{e}^{-\beta\Xi}X'|m\rangle\langle m|X(t)|n\rangle \\
&= -\frac{\mathrm{i}}{\hbar}S_{X,X'}\frac{1}{Z}\sum_{n,m}\mathrm{e}^{-\beta\xi_n}\mathrm{e}^{\mathrm{i}t(\xi_m-\xi_n)/\hbar}\langle n|X'|m\rangle\langle m|X|n\rangle \\
&= \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\mathrm{e}^{-\beta\xi_m}\left(-\frac{\mathrm{i}}{\hbar}S_{X,X'}\right)\mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar}
\end{aligned}
$$

We therefore see, that the Fourier-transform of the lesser Green-function is simply the usual delta-distribution. This gives for the Fourier-transform then

$$
\begin{aligned}
G^{<}_{X,X'}(\omega) &= \int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} G^{<}_{X,X'}(t) \\
&= \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle\langle m|X'|n\rangle \left(-\frac{\mathrm{i}}{\hbar} S_{X,X'}\right) \mathrm{e}^{-\beta\xi_m} \int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} \mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar} \\
&= -\frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle\langle m|X'|n\rangle \left(-2\pi\frac{\mathrm{i}}{\hbar} S_{X,X'}\right) \mathrm{e}^{-\beta\xi_m} \delta(\omega + (\xi_n - \xi_m)/\hbar)
\end{aligned}
$$

The greater Green-function

For the greater Green-function we obtain

$$
\begin{aligned}
G^{>}_{X,X'}(t) &= G^{>}_{X,X'}(t, t' = 0) \\
&= -\frac{\mathrm{i}}{\hbar} \langle X(t) X'(0)\rangle \\
&= -\frac{\mathrm{i}}{\hbar} \frac{1}{Z} \sum_{n,m} \langle n|\mathrm{e}^{-\beta\Xi} X(t)|m\rangle\langle m|X'|n\rangle \\
&= -\frac{\mathrm{i}}{\hbar} \frac{1}{Z} \sum_{n,m} \mathrm{e}^{-\beta\xi_n} \mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar} \langle n|X|m\rangle\langle m|X'|n\rangle \\
&= \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle\langle m|X'|n\rangle \mathrm{e}^{-\beta\xi_n} \left(-\frac{\mathrm{i}}{\hbar}\right) \mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar}
\end{aligned}
$$

We therefore see, that the Fourier-transform of the greater Green-function is simply the usual delta-distribution. This gives for the Fourier-transform then

$$
\begin{aligned}
G^{>}_{X,X'}(\omega) &= \int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} G^{>}_{X,X'}(t) \\
&= \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle\langle m|X'|n\rangle \left(-\frac{\mathrm{i}}{\hbar}\right) \mathrm{e}^{-\beta\xi_n} \int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} \mathrm{e}^{\mathrm{i}t(\xi_n-\xi_m)/\hbar} \\
&= \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle\langle m|X'|n\rangle \left(-2\pi\frac{\mathrm{i}}{\hbar}\right) \mathrm{e}^{-\beta\xi_n} \delta(\omega + (\xi_n - \xi_m)/\hbar)
\end{aligned}
$$

The spectral function

We have for the spectral function

$$
\begin{aligned}
A_{X,X'}(\omega) &= -\frac{\hbar}{\pi}\,\mathrm{Im}\lim_{\epsilon\to 0} G^{\mathrm{ret}}_{X,X'}(\omega+\mathrm{i}\epsilon)\\
&= -\frac{\hbar}{\pi}\,\mathrm{Im}\lim_{\epsilon\to 0}\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\mathrm{e}^{-\beta\xi_n}-S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)\frac{1}{\hbar}\frac{1}{\omega+\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar}\\
&= -\frac{\hbar}{\pi}\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\mathrm{e}^{-\beta\xi_n}-S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)\frac{1}{\hbar}\,\mathrm{Im}\left[\mathrm{P}\frac{1}{\omega+(\xi_n-\xi_m)/\hbar}-\mathrm{i}\pi\delta(\omega+(\xi_n-\xi_m)/\hbar)\right]\\
&= \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\mathrm{e}^{-\beta\xi_n}-S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)\delta(\omega+(\xi_n-\xi_m)/\hbar)\\
&= \dots
\end{aligned}
$$

Notice, for later reference, that this can be simplified further as[10]

$$
\begin{aligned}
&= \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\mathrm{e}^{-\beta\xi_n}\left(1-S_{X,X'}\mathrm{e}^{-\beta(\xi_m-\xi_n)}\right)\delta(\omega+(\xi_n-\xi_m)/\hbar)\\
&= \frac{1}{Z}\left(1-S_{X,X'}\mathrm{e}^{-\beta\omega}\right)\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\mathrm{e}^{-\beta\xi_n}\delta(\omega+(\xi_n-\xi_m)/\hbar)
\end{aligned}
$$

Group 4 (The imaginary-time Green-function)

Lemma for the four group

In the third group we are in the comfortable situation that we are dealing with a periodic functions, which are particularly simple to handle due to the vast literature about them. We only need to prove one Lemma to perform the subsequent calculations.

Lemma 28.18 For a Matsubara-frequency $\omega_n \in \Omega_{S_{X,X'}}$ it holds

$$
\frac{1}{\hbar}\int_0^{\hbar\beta}\mathrm{d}\tau\,\mathrm{e}^{\mathrm{i}\omega_n\tau}\mathrm{e}^{\tau\left(\xi_n-\xi_m\right)/\hbar} = \frac{1}{\hbar}\frac{S_{X,X'}\mathrm{e}^{\beta\left(\xi_n-\xi_m\right)}-1}{\mathrm{i}\omega_n+\left(\xi_n-\xi_m\right)/\hbar} \tag{28.63}
$$

Proof 28.18 With the definition of the Matsubara-frequencies in Equation TBReferenced, we remark the trivial identity

$$
\mathrm{e}^{\mathrm{i}\hbar\omega_n\beta} = S_{X,X'} \tag{28.64}
$$

and then we compute straightforwardly

[10]Cf. Mahan page 122.

$$
\begin{aligned}
\frac{1}{\hbar}\int_0^{\hbar\beta} \mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}\omega_n\tau}\mathrm{e}^{\tau\left(\xi_n-\xi_m\right)/\hbar} &= \frac{1}{\hbar}\int_0^{\hbar\beta} \mathrm{d}\tau\, \mathrm{e}^{\tau\left[\mathrm{i}\omega_n+\left(\xi_n-\xi_m\right)/\hbar\right]} \\
&= \frac{1}{\hbar}\frac{1}{\mathrm{i}\omega_n+\left(\xi_n-\xi_m\right)/\hbar}\left[\mathrm{e}^{\tau\left[\mathrm{i}\omega_n+\left(\xi_n-\xi_m\right)/\hbar\right]}\right]_0^{\hbar\beta} \\
&= \frac{1}{\hbar}\frac{1}{\mathrm{i}\omega_n+\left(\xi_n-\xi_m\right)/\hbar}\left[S_{X,X'}\mathrm{e}^{\beta\left(\xi_n-\xi_m\right)}-1\right] \\
&= \frac{1}{\hbar}\frac{S_{X,X'}\mathrm{e}^{\beta\left(\xi_n-\xi_m\right)}-1}{\mathrm{i}\omega_n+\left(\xi_n-\xi_m\right)/\hbar}
\end{aligned}
$$

The imaginary-time-ordered Green-function

For the imaginary-time-ordered Green-function we obtain

$$
\begin{aligned}
\mathscr{G}_{X,X'}(\tau) &= \mathscr{G}_{X,X'}(\tau,\tau'=0) \\
&= -\frac{1}{\hbar}\langle\theta(\tau)X(\tau)X'(0)+S_{X,X'}\theta(-\tau)X'(0)X(\tau)\rangle \\
&= -\frac{1}{\hbar}\frac{1}{Z}\sum_{n,m}\theta(\tau)\langle n|\mathrm{e}^{-\beta\Xi}X(\tau)|m\rangle\langle m|X'|n\rangle+S_{X,X'}\theta(-\tau)\langle n|\mathrm{e}^{-\beta\Xi}X'|m\rangle\langle m|X(\tau)|n\rangle \\
&= -\frac{1}{\hbar}\frac{1}{Z}\sum_{n,m}\theta(\tau)\mathrm{e}^{-\beta\xi_n}\mathrm{e}^{\tau(\xi_n-\xi_m)/\hbar}\langle n|X|m\rangle\langle m|X'|n\rangle+S_{X,X'}\theta(-\tau)\mathrm{e}^{-\beta\xi_n}\mathrm{e}^{\tau(\xi_m-\xi_n)/\hbar}\langle n|X'|m\rangle\langle m|X|n\rangle \\
&= -\frac{1}{\hbar}\frac{1}{Z}\sum_{n,m}\mathrm{e}^{\tau(\xi_n-\xi_m)/\hbar}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\theta(\tau)\mathrm{e}^{-\beta\xi_n}+S_{X,X'}\theta(-\tau)\mathrm{e}^{-\beta\xi_m}\right)
\end{aligned}
$$

We now compute the Matsubara-coefficients of these functions.

$$
\begin{aligned}
\mathscr{G}_{X,X'}(\mathrm{i}\omega_n) &= \int_0^{\hbar\beta}\mathrm{d}\tau\,\mathrm{e}^{\mathrm{i}\omega_\tau}\mathscr{G}_{X,X'}(\tau) \\
&= -\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\mathrm{e}^{-\beta\xi_n}\frac{1}{\hbar}\int_0^{\hbar\beta}\mathrm{d}\tau\,\mathrm{e}^{\mathrm{i}\omega_n\tau}\mathrm{e}^{\tau(\xi_n-\xi_m)/\hbar} \\
&= -\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\mathrm{e}^{-\beta\xi_n}\frac{1}{\hbar}\frac{S_{X,X'}\mathrm{e}^{\beta\left(\xi_n-\xi_m\right)}-1}{\mathrm{i}\omega_n+\left(\xi_n-\xi_m\right)/\hbar} \\
&= \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\frac{1}{\hbar}\frac{\mathrm{e}^{-\beta\xi_n}-S_{X,X'}\mathrm{e}^{-\beta\xi_m}}{\mathrm{i}\omega_n+\left(\xi_n-\xi_m\right)/\hbar}
\end{aligned}
$$

Summarizing the results, one finds for the different Green-functions (and the spectral function)

$$G^{\mathbb{T}_\epsilon}_{X,X'}(\omega) = \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\frac{1}{\hbar}\left(\frac{\mathrm{e}^{-\beta\xi_n}}{\omega+\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar} - S_{X,X'}\frac{\mathrm{e}^{-\beta\xi_m}}{\omega-\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar}\right) \tag{28.65}$$

$$G^{<}_{X,X'}(\omega) = \frac{1}{Z}\sum_{(n,m)\in\mathbb{N}^2}\langle n|X|m\rangle\langle m|X'|n\rangle\,\frac{-2\pi\mathrm{i}S_{X,X'}e^{-\beta\xi_m}}{\hbar}\delta(\omega+(\xi_n-\xi_m)/\hbar) \tag{28.66}$$

$$G^{>}_{X,X'}(\omega) = \frac{1}{Z}\sum_{(n,m)\in\mathbb{N}^2}\langle n|X|m\rangle\langle m|X'|n\rangle\,\frac{2\pi\mathrm{i}e^{-\beta\xi_n}}{\hbar}\delta(\omega+(\xi_n-\xi_m)/\hbar) \tag{28.67}$$

$$G^{\overline{\mathbb{T}}_\epsilon}_{X,X'}(\omega) = \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\frac{1}{\hbar}\left(\frac{-\mathrm{e}^{-\beta\xi_n}}{\omega-\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar} + S_{X,X'}\frac{\mathrm{e}^{-\beta\xi_m}}{\omega+\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar}\right) \tag{28.68}$$

$$G^{\mathrm{ret}}_{X,X'}(\omega+\mathrm{i}\epsilon) = \frac{1}{Z}\sum_{(n,m)\in\mathbb{N}^2}\langle n|X|m\rangle\langle m|X'|n\rangle\,\frac{e^{-\beta\xi_n}-S_{X,X'}e^{-\beta\xi_m}}{\hbar}\frac{1}{\omega+\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar} \tag{28.69}$$

$$G^{\mathrm{adv}}_{X,X'}(\omega-\mathrm{i}\epsilon) = \frac{1}{Z}\sum_{(n,m)\in\mathbb{N}^2}\langle n|X|m\rangle\langle m|X'|n\rangle\,\frac{e^{-\beta\xi_n}-S_{X,X'}e^{-\beta\xi_m}}{\hbar}\frac{1}{\omega-\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar} \tag{28.70}$$

$$\mathscr{G}_{X,X'}(\mathrm{i}\omega_l) = \frac{1}{Z}\sum_{(n,m)\in\mathbb{N}^2}\langle n|X|m\rangle\langle m|X'|n\rangle\,\frac{e^{-\beta\xi_n}-S_{X,X'}e^{-\beta\xi_m}}{\hbar}\frac{1}{\mathrm{i}\omega_l+(\xi_n-\xi_m)/\hbar} \tag{28.71}$$

$$A_{X,X'}(\omega) = \frac{1}{Z}\sum_{(n,m)\in\mathbb{N}^2}\langle n|X|m\rangle\langle m|X'|n\rangle\left(e^{-\beta\xi_n}-S_{X,X'}e^{-\beta\xi_m}\right)\delta(\omega+(\xi_n-\xi_m)/\hbar) \tag{28.72}$$

Proof 28.43 of Theorem 14.6 We divide the proof into the two equation signs.

The spectral function for the lesser Green-function

We recall from Equation (28.72) that the spectral function

$$\begin{aligned}
A_{X,X'}(\omega) &= \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\mathrm{e}^{-\beta\xi_n}-S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)\delta(\omega+(\xi_n-\xi_m)/\hbar)\\
&= \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\mathrm{e}^{-\beta\xi_m}\left(\mathrm{e}^{\beta(\xi_m-\xi_n)}-S_{X,X'}\right)\delta(\omega+(\xi_n-\xi_m)/\hbar)\\
&= \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\mathrm{e}^{-\beta\xi_m}\left(\mathrm{e}^{\beta\hbar\omega}-S_{X,X'}\right)\delta(\omega+(\xi_n-\xi_m)/\hbar)\\
&= \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\mathrm{e}^{-\beta\xi_m}\left(\mathrm{n}^{S_{X,X'}}_{\hbar\beta}(\omega)\right)^{-1}\delta(\omega+(\xi_n-\xi_m)/\hbar)
\end{aligned}$$

With the the ζ-distribution from Equation (TBReferenced). Now comparing the Lehmann-representation of the spectral-function (with a proper prefactor) with the one for the lesser-Green-function gives

$$
\begin{aligned}
-2\pi\frac{\mathrm{i}}{\hbar}S_{X,X'}\,\mathrm{n}_{\hbar\beta}^{S_{X,X'}}(\omega)A_{X,X'}(\omega) &= 2\mathrm{i}S_{X,X'}\,\mathrm{n}_{\hbar\beta}^{S_{X,X'}}(\omega)\,\mathrm{Im}\lim_{\epsilon\to 0}G^{\mathrm{ret}}_{X,X'}(\omega+\mathrm{i}\epsilon)\\
&= -2\pi\frac{\mathrm{i}}{\hbar}S_{X,X'}\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle \mathrm{e}^{-\beta\xi_m}\delta(\omega+(\xi_n-\xi_m)/\hbar)\\
&= G^{<}_{X,X'}(\omega)
\end{aligned}
$$

whereas in the last equation we used Equation (28.66).

The spectral function for the greater Green-function

We recall from Equation (28.72) that the spectral function

$$
\begin{aligned}
A_{X,X'}(\omega) &= \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\left(\mathrm{e}^{-\beta\xi_n}-S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)\delta(\omega+(\xi_n-\xi_m)/\hbar)\\
&= \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle \mathrm{e}^{-\beta\xi_n}\left(1-S_{X,X'}\mathrm{e}^{-\beta(\xi_m-\xi_n)}\right)\delta(\omega+(\xi_n-\xi_m)/\hbar)\\
&= \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle \mathrm{e}^{-\beta\xi_n}\left(1-S_{X,X'}\mathrm{e}^{-\beta\hbar\omega}\right)\delta(\omega+(\xi_n-\xi_m)/\hbar)\\
&= \dots
\end{aligned}
$$

Now we compute with the ζ-distribution

$$
\begin{aligned}
&1+S_{X,X'}\,\mathrm{n}_{\hbar\beta}^{S_{X,X'}}(\omega) = 1+\frac{S_{X,X'}}{\mathrm{e}^{\hbar\beta\omega}-S_{X,X'}} = \frac{\mathrm{e}^{\hbar\beta\omega}-S_{X,X'}+S_{X,X'}}{\mathrm{e}^{\hbar\beta\omega}-S_{X,X'}} = \frac{1}{1-S_{X,X'}\mathrm{e}^{-\hbar\beta\omega}} = \left(1-S_{X,X'}\mathrm{e}^{-\hbar\beta\omega}\right)^{-1}\\
\iff &\left(1+S_{X,X'}\,\mathrm{n}_{\hbar\beta}^{S_{X,X'}}(\omega)\right)^{-1} = 1-S_{X,X'}\mathrm{e}^{-\hbar\beta\omega}
\end{aligned}
$$

and inserting this above allows us to rewrite the expression above to the form

$$
A_{X,X'}(\omega) = \dots = \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle \mathrm{e}^{-\beta\xi_n}\left(1+S_{X,X'}\,\mathrm{n}_{\hbar\beta}^{S_{X,X'}}(\omega)\right)^{-1}\delta(\omega+(\xi_n-\xi_m)/\hbar)
$$

This yields then

$$
\begin{aligned}
-\frac{\mathrm{i}}{\hbar}2\pi\left(1+S_{X,X'}\,\mathrm{n}_{\hbar\beta}^{S_{X,X'}}(\omega)\right)A_{X,X'}(\omega) &= 2\mathrm{i}\left(1+S_{X,X'}\,\mathrm{n}_{\hbar\beta}^{S_{X,X'}}(\omega)\right)\mathrm{Im}\lim_{\epsilon\to 0}G^{\mathrm{ret}}_{X,X'}(\omega+\mathrm{i}\epsilon)\\
&= \frac{-\mathrm{i}}{\hbar}2\pi\frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle \mathrm{e}^{-\beta\xi_n}\delta(\omega+(\xi_n-\xi_m)/\hbar)\\
&= G^{>}_{X,X'}(\omega)
\end{aligned}
$$

whereas we used Equation (28.67) in the last step.

Proof 28.44 of Theorem 14.7

We first prove the identities relating the time-ordered and the retarded Green-function. For this we recall Equations (28.65), (28.69) and the Sokhotski-Plemelj-theorem in Equation (14.3)

$$\frac{1}{\omega + \mathrm{i}\epsilon + \text{const.}} = \mathrm{P}\,\frac{1}{\omega + \text{const.}} - \mathrm{i}\pi\delta(\omega + \text{const.}), \tag{28.73}$$

$$\frac{1}{\omega - \mathrm{i}\epsilon + \text{const.}} = \mathrm{P}\,\frac{1}{\omega + \text{const.}} + \mathrm{i}\pi\delta(\omega + \text{const.}), \tag{28.74}$$

such that we obtain the identities

$$G^{\mathbb{T}}_{\epsilon\,X,X'}(\omega) = \frac{1}{Z}\sum_{n,m}\langle n|X|m\rangle\langle m|X'|n\rangle\frac{1}{\hbar}\left(\frac{\mathrm{e}^{-\beta\xi_n}}{\omega + \mathrm{i}\epsilon + (\xi_n - \xi_m)/\hbar} - S_{X,X'}\frac{\mathrm{e}^{-\beta\xi_m}}{\omega - \mathrm{i}\epsilon + (\xi_n - \xi_m)/\hbar}\right) \tag{28.75}$$

$$G^{\mathrm{ret}}_{X,X'}(\omega + \mathrm{i}\epsilon) = \frac{1}{Z}\sum_{(n,m)\in\mathbb{N}^2}\langle n|X|m\rangle\langle m|X'|n\rangle\,\frac{e^{-\beta\xi_n} - S_{X,X'}e^{-\beta\xi_m}}{\hbar}\,\frac{1}{\omega + \mathrm{i}\epsilon + (\xi_n - \xi_m)/\hbar} \tag{28.76}$$

We will prove the identity for each term in the individual summations, which will hence prove the theorem.

The time-ordered Green-function $G^{\mathbb{T}}$

For the time-ordered Green-function we obtain on the one hand

$$\begin{aligned}\frac{\mathrm{e}^{-\beta\xi_n}}{\omega + \mathrm{i}\epsilon + (\xi_n - \xi_m)/\hbar} - S_{X,X'}\frac{\mathrm{e}^{-\beta\xi_m}}{\omega - \mathrm{i}\epsilon + (\xi_n - \xi_m)/\hbar} &= \left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)\mathrm{P}\,\frac{1}{\omega + (\xi_n - \xi_m)/\hbar}\\ &\quad + \left(-\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)\mathrm{i}\pi\delta(\omega + (\xi_n - \xi_m)/\hbar)\end{aligned}$$

and hence we obtain

$$\mathrm{Re}\left(\frac{\mathrm{e}^{-\beta\xi_n}}{\omega + \mathrm{i}\epsilon + (\xi_n - \xi_m)/\hbar} - S_{X,X'}\frac{\mathrm{e}^{-\beta\xi_m}}{\omega - \mathrm{i}\epsilon + (\xi_n - \xi_m)/\hbar}\right) = \left(\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)\mathrm{P}\,\frac{1}{\omega + (\xi_n - \xi_m)/\hbar}$$

$$\mathrm{Im}\left(\frac{\mathrm{e}^{-\beta\xi_n}}{\omega + \mathrm{i}\epsilon + (\xi_n - \xi_m)/\hbar} - S_{X,X'}\frac{\mathrm{e}^{-\beta\xi_m}}{\omega - \mathrm{i}\epsilon + (\xi_n - \xi_m)/\hbar}\right) = \left(-\mathrm{e}^{-\beta\xi_n} - S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)\pi\delta(\omega + (\xi_n - \xi_m)/\hbar)$$

The retarded Green-function G^{ret}

For the retarded Green-function we obtain on the other hand

$$\left(e^{-\beta\xi_n} - S_{X,X'}e^{-\beta\xi_m}\right)\frac{1}{\omega + \mathrm{i}\epsilon + (\xi_n - \xi_m)/\hbar} = \left(e^{-\beta\xi_n} - S_{X,X'}e^{-\beta\xi_m}\right)\left(\mathrm{P}\,\frac{1}{\omega + (\xi_n - \xi_m)/\hbar} - \mathrm{i}\pi\delta(\omega + (\xi_n - \xi_m)/\hbar)\right)$$

and hence

$$\mathrm{Re}\left(e^{-\beta\xi_n}-S_{X,X'}e^{-\beta\xi_m}\right)\frac{1}{\omega+\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar}=\left(e^{-\beta\xi_n}-S_{X,X'}e^{-\beta\xi_m}\right)\mathrm{P}\,\frac{1}{\omega+(\xi_n-\xi_m)/\hbar}$$

$$\mathrm{Im}\left(e^{-\beta\xi_n}-S_{X,X'}e^{-\beta\xi_m}\right)\frac{1}{\omega+\mathrm{i}\epsilon+(\xi_n-\xi_m)/\hbar}=-\left(e^{-\beta\xi_n}-S_{X,X'}e^{-\beta\xi_m}\right)\pi\delta(\omega+(\xi_n-\xi_m)/\hbar)$$

Comparing the Results

The real parts obviously coincide. For the imaginary part we obtain, using the delta-distribution

$$\begin{aligned}&-\left(e^{-\beta\xi_n}-S_{X,X'}e^{-\beta\xi_m}\right)\cdot\left(\coth\frac{\beta\hbar\omega}{2}\right)^{S_{X,X'}}\overset{!}{=}-\left(\mathrm{e}^{-\beta\xi_n}+S_{X,X'}\mathrm{e}^{-\beta\xi_m}\right)\\
\Longleftrightarrow&\left(\coth\frac{\beta\hbar\omega}{2}\right)^{S_{X,X'}}\overset{!}{=}\frac{\mathrm{e}^{-\beta\xi_n}+S_{X,X'}\mathrm{e}^{-\beta\xi_m}}{e^{-\beta\xi_n}-S_{X,X'}e^{-\beta\xi_m}}\equiv\frac{\mathrm{e}^{-\beta\xi_n}+S_{X,X'}\mathrm{e}^{-\beta(\xi_n+\hbar\omega)}}{e^{-\beta\xi_n}-S_{X,X'}e^{-\beta(\xi_n+\hbar\omega)}}=\frac{1+S_{X,X'}\mathrm{e}^{-\beta\hbar\omega}}{1-S_{X,X'}e^{-\beta\hbar\omega}}=\frac{\mathrm{e}^{\beta\hbar\omega/2}+S_{X,X'}\mathrm{e}^{-\beta\hbar\omega/2}}{\mathrm{e}^{\beta\hbar\omega/2}-S_{X,X'}e^{-\beta\hbar\omega/2}}\end{aligned}$$

Recalling now the definition of the cotangens-hyperbolicus we have proven the theorem. For the advanced Green-function the theorem is proven analogously TBD.

Proof 28.45 of Lemma 14.14

We re-express the average particle number as lesser Green-functions and then use the fluctuation-dissipation-theorem

$$\begin{aligned}\langle N\rangle&=\sum_{\alpha\in\mathscr{A}}\langle c_\alpha^\dagger c_\alpha\rangle\\
&=-\zeta\frac{\hbar}{\mathrm{i}}\sum_{\alpha\in\mathscr{A}}G^<_{\alpha,\alpha}(t=0)\\
&=-\zeta\frac{1}{2\pi\mathrm{i}}\mathop{\sum\!\!\!\!\!\!\int}_{\substack{\alpha\in\mathscr{A}\\E\in(-\infty,+\infty)}}G^<_{\alpha,\alpha}(E)\\
&=\frac{-1}{\pi}\lim_{\eta\to0}\mathop{\sum\!\!\!\!\!\!\int}_{\substack{\alpha\in\mathscr{A}\\E\in(-\infty,+\infty)}}\mathrm{n}^\zeta_\beta(E)\ \mathrm{Im}\ G^{\mathrm{ret}}_{\alpha,\alpha}(E+\mathrm{i}\eta)\end{aligned}\tag{28.77}$$

Proof 28.46 of Lemma 14.15

For the first identity we compute

$$
\begin{aligned}
\delta_{\alpha,\alpha'} &= \langle \delta_{\alpha,\alpha'} \rangle \\
&= \langle [c_\alpha, c^\dagger_{\alpha'}]_\zeta \rangle \\
&= \langle c_\alpha c^\dagger_{\alpha'} \rangle - \zeta \langle c^\dagger_{\alpha'} c_\alpha \rangle \\
&= -\frac{\hbar}{\mathrm{i}} \left(G^>_{\alpha,\alpha'}(t=0) - G^<_{\alpha,\alpha'}(t=0) \right) \\
&= -\frac{\hbar}{\mathrm{i}} \int \frac{\mathrm{d}\omega}{2\pi} \left(G^>_{\alpha,\alpha'}(\omega) - G^<_{\alpha,\alpha'}(\omega) \right) \\
&= -\frac{\hbar}{\mathrm{i}} \lim_{\eta \to 0} \int \frac{\mathrm{d}\omega}{2\pi} 2\mathrm{i} \,\mathrm{Im}\, G^{\mathrm{ret}}_{\alpha,\alpha'}(\omega + \mathrm{i}\eta) \\
&= \lim_{\eta \to 0} -\frac{\hbar}{\pi} \int \mathrm{d}\omega \; \mathrm{Im}\, G^{\mathrm{ret}}_{\alpha,\alpha'}(\omega + \mathrm{i}\eta) \\
&\equiv \int \mathrm{d}\omega \; A_{\alpha,\alpha'}(\omega)
\end{aligned}
$$

The second identity follows from the first identity with

$$
\mathrm{d} = \sum_{\alpha \in \mathscr{A}} \delta_{\alpha,\alpha} = \ldots = \int \mathrm{d}\omega \sum_{\alpha \in \mathscr{A}} \delta_{\alpha,\alpha} \, A_{\alpha,\alpha}(\omega) \tag{28.78}
$$

The third identity follows immediately from Equation (28.72).

Proof 28.47 of Lemma 14.19

TBDone: Proving this and reexpressing this w.r.t. the lesser GF.

Proofs to Section 9.1

Proof 28.48 of Lemma 15.2

TBDone.

Proof 28.49 of Theorem 15.3

TBDone.

Proof 28.50 of Theorem ?? (Linked-Cluster-Theorem)

TBUpdated: Ordering of the indices.

We rewrite the partition-function in its symmetrized form

With the expansion

$$\begin{aligned}\exp - S_{\text{int}}[\xi] &= \sum_{n=0}^{\infty} \frac{(-1/\hbar)^n}{n!} \prod_{\bullet=1}^{n} \int_0^{\hbar\beta} \mathrm{d}\tau_\bullet\, V(\xi(\tau_\bullet)) \\ &= \sum_{n=0}^{\infty} \frac{(-1/\hbar)^n}{n!} \prod_{\bullet=1}^{n} \sumint V_{\alpha_1^\bullet,\dots,\alpha_m^\bullet}\, \xi_{\alpha_1^\bullet}(\tau_\bullet) \cdots \xi_{\alpha_m^\bullet}(\tau_\bullet)\end{aligned}$$

one finds

$$\begin{aligned}\mathscr{G}_{\alpha,\alpha'}(\tau) &= \frac{-1}{\hbar} \langle \xi_\alpha(\tau)\xi_{\alpha'}(0)\rangle \\ &= \frac{-1}{\hbar} \frac{\sum_{n=0}^{\infty} \frac{(-1/\hbar)^n}{n!} \langle \xi_\alpha(\tau)\xi_{\alpha'}(0) \prod_{\bullet=1}^{n} \sumint V_{\alpha_1^\bullet,\dots,\alpha_m^\bullet}\, \xi_{\alpha_1^\bullet}(\tau_\bullet) \cdots \xi_{\alpha_m^\bullet}(\tau_\bullet)\rangle_0}{\sum_{n=0}^{\infty} \frac{(-1/\hbar)^n}{n!} \langle \prod_{\bullet=1}^{n} \sumint V_{\alpha_1^\bullet,\dots,\alpha_m^\bullet}\, \xi_{\alpha_1^\bullet}(\tau_\bullet) \cdots \xi_{\alpha_m^\bullet}(\tau_\bullet)\rangle_0}\end{aligned}$$

We will now examine the numerator of this expression and see that it factorizes into two factors, with one being equal to the denominator; This is what Alexander Altland called a „miraculous cancellation mechanism“. It follows by combinatorical counting

$$\begin{aligned}&\text{numerator} \\ &= \sum_{n=0}^{\infty} \frac{(-1/\hbar)^n}{n!} \langle \xi_\alpha(\tau)\xi_{\alpha'}(0) \prod_{\bullet=1}^{n} \sumint V_{\alpha_1^\bullet,\dots,\alpha_m^\bullet}\, \xi_{\alpha_1^\bullet}(\tau_\bullet) \cdots \xi_{\alpha_m^\bullet}(\tau_\bullet)\rangle_0 \\ &= \sum_{n=0}^{\infty} \sum_{p=0}^{n} \frac{(-1/\hbar)^n}{(n-p)!p!} \langle \xi_\alpha(\tau)\xi_{\alpha'}(0) \prod_{\bullet=1}^{n-p} \sumint V_{\alpha_1^\bullet,\dots,\alpha_m^\bullet}\, \xi_{\alpha_1^\bullet}(\tau_\bullet) \cdots \xi_{\alpha_m^\bullet}(\tau_\bullet)\rangle_0^{\text{n.v.}} \langle \prod_{\bullet=1}^{p} \sumint V_{\alpha_1^\bullet,\dots,\alpha_m^\bullet}\, \xi_{\alpha_1^\bullet}(\tau_\bullet) \cdots \xi_{\alpha_m^\bullet}(\tau_\bullet)\rangle_0 \\ &= \sum_{n=0}^{\infty} \frac{(-1/\hbar)^n}{n!} \langle \xi_\alpha(\tau)\xi_{\alpha'}(0) \prod_{\bullet=1}^{n} \sumint V_{\alpha_1^\bullet,\dots,\alpha_m^\bullet}\, \xi_{\alpha_1^\bullet}(\tau_\bullet) \cdots \xi_{\alpha_m^\bullet}(\tau_\bullet)\rangle_0^{\text{n.v.}} \sum_{p=0}^{\infty} \frac{(-1/\hbar)^p}{p!} \langle \prod_{\bullet=1}^{p} \sumint V_{\alpha_1^\bullet,\dots,\alpha_m^\bullet}\, \xi_{\alpha_1^\bullet}(\tau_\bullet) \cdots \xi_{\alpha_m^\bullet}(\tau_\bullet)\rangle_0\end{aligned}$$

and it follows that

$$\mathscr{G}_{\alpha,\alpha'}(\tau) = -\frac{1}{\hbar} \sum_{n=0}^{\infty} \frac{(-1/\hbar)^n}{n!} \langle \xi_\alpha(\tau)\xi_{\alpha'}(0) \prod_{\bullet=1}^{n} \sumint V_{\alpha_1^\bullet,\dots,\alpha_m^\bullet}\, \xi_{\alpha_1^\bullet}(\tau_\bullet) \cdots \xi_{\alpha_m^\bullet}(\tau_\bullet)\rangle_0^{\text{n.v.}} \tag{28.79}$$

ToBeRedone with Graph Theory, Cf. the book from Salmhofer

Proofs to Section 9.3

Proof 28.51 of Theorem 15.10

TBDone.

Proofs to Section 9.5

Proof 28.52 of Lemma 15.12 We write $z = \mathrm{i}\omega_n + \Delta z$ and consider

$$\begin{aligned}
\lim_{\Delta z \to 0} \Delta z \, g_\zeta(\mathrm{i}\omega_n + \Delta z) &= \lim_{\Delta z \to 0} \Delta z \frac{\hbar\beta}{\mathrm{e}^{\mathrm{i}\omega_n \hbar\beta} \mathrm{e}^{\hbar\beta\Delta z} - \zeta} \\
&= \lim_{\Delta z \to 0} \Delta z \frac{\hbar\beta}{\mathrm{e}^{\hbar\beta\Delta z} - 1} \zeta \\
&= \lim_{\Delta z \to 0} \frac{\hbar\beta\Delta z}{\hbar\beta\Delta z + \frac{1}{2}(\hbar\beta\Delta z)^2 + \dots} \zeta \\
&= \zeta .
\end{aligned}$$

Proof 28.53 of Theorem 15.14

We compute

$$\operatorname{Res}\left(f_\zeta(z),\ z = \mathrm{i}\omega_n\right) = \lim_{z \to \mathrm{i}\omega_n} (z - \mathrm{i}\omega_n) g_\zeta(z) h(-\mathrm{i}z) = \zeta h(\omega_n)$$

and obtain therefore

$$\begin{aligned}
0 = \frac{1}{2\pi \mathrm{i}} \oint_\gamma \mathrm{d}z \, f_\zeta(z) \quad &\Longleftrightarrow \quad 0 = \sum_{z_\star \in \mathrm{P}} \operatorname{Res}\left(f(z),\ z = z_\star\right) \\
&\Longleftrightarrow \quad 0 = \sum_{z_\star \in \mathrm{Z}} \operatorname{Res}\left(f(z),\ z = z_\star\right) + \sum_{\mathrm{i}\omega_n \in \Omega_\zeta} \operatorname{Res}\left(f(z),\ z = \mathrm{i}\omega_n\right) \\
&\Longleftrightarrow S = -\zeta \sum_{z_\star \in \mathrm{Z}} \operatorname{Res}\left(f(z),\ z = z_\star\right)
\end{aligned}$$

Proof 28.54 of Corollary 15.16

One obtains the function

$$\mathbb{C} - \{\mathrm{C}\} \ni z \mapsto h(-\mathrm{i}z) \equiv \frac{1}{\hbar\beta} \frac{1}{(z - C)} \in \mathbb{C} \tag{28.80}$$

Therefore, with the Lemma above, we obtain

$$\sum_{\omega_n \in \Omega_\zeta} \frac{1}{\hbar} \frac{1}{\mathrm{i}\omega_n - C} = -\frac{\zeta}{\hbar\beta} g_\zeta(C) = -\zeta \mathrm{n}^\zeta_{\hbar\beta}(C) = -\zeta \mathrm{n}^\zeta_{\hbar\beta}(C) \tag{28.81}$$

Proof 28.55 of Lemma 15.17 TBDigitalized from .[11]

Proof 28.56 of Corollary 15.18

One obtains the function

$$\mathbb{C} - \{z_1, z_2, z_3\} \ni z \mapsto h(-\mathrm{i}z) \equiv \frac{1}{\hbar^2\beta} \frac{1}{\left(z - C_1\right)} \frac{-2C_2}{\left(\omega_m + \mathrm{i}z\right)^2 + C_2^2} \in \mathbb{C} \tag{28.82}$$

The first pole of this function is $z_1 = C_1$. The other two poles are determined by the equation

$$\begin{aligned} 0 = \left(\omega_m + \mathrm{i}z\right)^2 + C_2^2 &\iff \pm\mathrm{i}C_2 = \omega_m + \mathrm{i}z_\pm \\ &\iff \mathrm{i}z_\pm = \pm\mathrm{i}C_2 - \omega_m \\ &\iff z_\pm \equiv z_{2,3} = \mp C_2 + \mathrm{i}\omega_m \end{aligned}$$

Hence

$$z_1 = C_1, \qquad z_2 = -C_2 + \mathrm{i}\omega_m, \qquad z_3 = +C_2 + \mathrm{i}\omega_m \tag{28.83}$$

and since

$$\begin{aligned} (z - z_2)(z - z_3) &= (z + C_2 - \mathrm{i}\omega_m)(z - C_2 - \mathrm{i}\omega_m) \\ &= (z - \mathrm{i}\omega_m)^2 - C_2^2 \\ &= -\mathrm{i}^2 \left((z - \mathrm{i}\omega_m)^2 - C_2^2\right) \\ &= -\left((\omega_m + \mathrm{i}z)^2 + C_2^2\right) \end{aligned}$$

we conclude that

$$\text{const.} \equiv \frac{2C_2}{\hbar^2\beta} \tag{28.84}$$

Therefore we can now write down the products appearing in Equation (15.19) to evaluate the Matsubara-summation

$$k = 1: \qquad \mathrm{n}^\zeta_{\hbar\beta}(z_1) = \mathrm{F}_{\hbar\beta}(C_1), \qquad z_1 - z_2 = C_1 + C_2 - \mathrm{i}\omega_m, \quad z_1 - z_3 = C_1 - C_2 - \mathrm{i}\omega_m$$

[11]Cf. Mahan Eqs. (3.236), (3.237).

$$k = 2: \qquad \mathrm{n}^{\zeta}_{\hbar\beta}(z_1) = \mathrm{F}_{\hbar\beta}(-C_2 + \mathrm{i}\omega_m), \qquad z_2 - z_1 = -C_2 + \mathrm{i}\omega_m - C_1, \quad z_2 - z_3 = -C_2 + \mathrm{i}\omega_m - C_2 - \mathrm{i}\omega_m = -2C_2$$

$$k = 3: \qquad \mathrm{n}^{\zeta}_{\hbar\beta}(z_1) = \mathrm{F}_{\hbar\beta}(+C_2 + \mathrm{i}\omega_m), \qquad z_3 - z_1 = C_2 + \mathrm{i}\omega_m - C_1, \quad z_3 - z_2 = C_2 + \mathrm{i}\omega_m + C_2 - \mathrm{i}\omega_m = 2C_2$$

Using the identities in Lemma 15.17 we obtain therefore

$$k = 1: \qquad \mathrm{n}^{\zeta}_{\hbar\beta}(z_1) = \mathrm{F}_{\hbar\beta}(C_1), \qquad z_1 - z_2 = C_1 + C_2 - \mathrm{i}\omega_m, \quad z_1 - z_3 = C_1 - C_2 - \mathrm{i}\omega_m$$

$$k = 2: \qquad \mathrm{n}^{\zeta}_{\hbar\beta}(z_1) = \mathrm{B}_{\hbar\beta}(C_2) + 1, \qquad z_2 - z_1 = -C_2 + \mathrm{i}\omega_m - C_1, \quad z_2 - z_3 = -2C_2$$

$$k = 3: \qquad \mathrm{n}^{\zeta}_{\hbar\beta}(z_1) = -\mathrm{B}_{\hbar\beta}(C_2), \qquad z_3 - z_1 = C_2 + \mathrm{i}\omega_m - C_1, \quad z_3 - z_2 = 2C_2$$

This gives with Equation (15.19) the intermediate result

$$\begin{aligned}
S_{-1} &= \sum_{\omega_n \in \Omega_{-1}} h(\omega_n) \\
&= +1\hbar\beta \,\mathrm{const} \sum_{k=1}^{N} \mathrm{n}^{\zeta}_{\hbar\beta}(z_k) \prod_{\substack{l=1 \\ l \neq k}}^{N} \frac{1}{z_k - z_l} \\
&= \frac{2C_2}{\hbar} \left[\frac{\mathrm{F}_{\hbar\beta}(C_1)}{(C_1 + C_2 - \mathrm{i}\omega_m)(C_1 - C_2 - \mathrm{i}\omega_m)} + \frac{\mathrm{B}_{\hbar\beta}(C_2) + 1}{(-C_2 + \mathrm{i}\omega_m - C_1)(-2C_2)} + \frac{-\mathrm{B}_{\hbar\beta}(C_2)}{(C_2 + \mathrm{i}\omega_m - C_1)(2C_2)} \right] \\
&= \frac{1}{\hbar} \left[\frac{2C_2 \mathrm{F}_{\hbar\beta}(C_1)}{(\mathrm{i}\omega_m - C_1 - C_2)(\mathrm{i}\omega_m - C_1 + C_2)} - \frac{\mathrm{B}_{\hbar\beta}(C_2) + 1}{-C_2 + \mathrm{i}\omega_m - C_1} - \frac{\mathrm{B}_{\hbar\beta}(C_2)}{C_2 + \mathrm{i}\omega_m - C_1} \right] \\
&= \star\star\star
\end{aligned}$$

We now use the partial fraction expansion (cf. WolframAlpha)

$$\frac{2C_2}{(x + C_2)(x - C_2)} = -\frac{1}{x + C_2} + \frac{1}{x - C_2} \tag{28.85}$$

and obtain

$$\begin{aligned}
\star\star\star &= \frac{1}{\hbar} \left[\frac{-\mathrm{F}_{\hbar\beta}(C_1)}{\mathrm{i}\omega_m - C_1 + C_2} + \frac{\mathrm{F}_{\hbar\beta}(C_1)}{\mathrm{i}\omega_m - C_1 - C_2} - \frac{\mathrm{B}_{\hbar\beta}(C_2) + 1}{-C_2 + \mathrm{i}\omega_m - C_1} - \frac{\mathrm{B}_{\hbar\beta}(C_2)}{C_2 + \mathrm{i}\omega_m - C_1} \right] \\
&= \frac{-1}{\hbar} \left[\frac{\mathrm{B}_{\hbar\beta}(C_2) + \mathrm{F}_{\hbar\beta}(C_1)}{\mathrm{i}\omega_m - C_1 + C_2} + \frac{\mathrm{B}_{\hbar\beta}(C_2) + 1 - \mathrm{F}_{\hbar\beta}(C_1)}{\mathrm{i}\omega_m - C_1 - C_2} \right]
\end{aligned}$$

29. Appendix - Proofs for Part III

Proof 29.1 of Lemma 16.13 The bosonic case is trivial. In the fermionic case we obtain by direct computation for the first identity

$$\exp \sum_k \bar{j}_k \psi_k = \prod_k \exp \bar{j}_k \psi_k = \exp \bar{j}_i \psi_i \prod_{\substack{k \\ k \neq i}} \exp \bar{j}_k \psi_k \tag{29.1}$$

which can be combined with

$$\frac{\partial}{\partial \bar{j}_i} \exp \bar{j}_i \psi_i = \frac{\partial}{\partial \bar{j}_i} \left(1 + \bar{j}_i \psi_i\right) = \psi_i = \psi_i \left(1 + \bar{j}_i \psi_i\right) \tag{29.2}$$

and

$$\frac{\partial}{\partial j_{i'}} \exp \overline{\psi}_{i'} j_{i'} = \frac{\partial}{\partial j_{i'}} \left(1 + \overline{\psi}_{i'} j_{i'}\right) = -\overline{\psi}_{i'} = -\overline{\psi}_{i'} \left(1 + \overline{\psi}_{i'} j_{i'}\right) \tag{29.3}$$

Proof 29.2 of Lemma 16.34. One has[1]

$$\begin{aligned}
\frac{\partial}{\partial \phi(\bullet)} \Gamma[\phi, \bar{\phi}] &= -\sum_x \left(\frac{\partial j(x)}{\partial \phi(\bullet)}, \frac{\partial \bar{j}(x)}{\partial \phi(\bullet)} \right) \cdot \begin{pmatrix} \frac{\partial}{\partial j(x)} W[j, \bar{j}] \\ \frac{\partial}{\partial \bar{j}(x)} W[j, \bar{j}] \end{pmatrix} + \zeta \bar{j}(\bullet) + \sum_x \left(\frac{\partial \bar{j}(x)}{\partial \phi(\bullet)} \phi(x) + \zeta \bar{\phi}(x) \frac{\partial j(x)}{\partial \phi(\bullet)} \right) \\
&= -\sum_x \left(\frac{\partial j(x)}{\partial \phi(\bullet)}, \frac{\partial \bar{j}(x)}{\partial \phi(\bullet)} \right) \cdot \begin{pmatrix} \zeta \bar{\phi}(x) \\ \phi(x) \end{pmatrix} + \zeta \bar{j}(\bullet) + \sum_x \left(\frac{\partial \bar{j}(x)}{\partial \phi(\bullet)} \phi(x) + \zeta \bar{\phi}(x) \frac{\partial j(x)}{\partial \phi(\bullet)} \right) \\
&= -\sum_x \left(\frac{\partial j(x)}{\partial \phi(\bullet)} \zeta \bar{\phi}(x) + \frac{\partial \bar{j}(x)}{\partial \phi(\bullet)} \phi(x) \right) + \zeta \bar{j}(\bullet) + \sum_x \left(\frac{\partial \bar{j}(x)}{\partial \phi(\bullet)} \phi(x) + \zeta \bar{\phi}(x) \frac{\partial j(x)}{\partial \phi(\bullet)} \right) \\
&= -\sum_x \left(\frac{\partial j(x)}{\partial \phi(\bullet)} \zeta \bar{\phi}(x) \right) + \zeta \bar{j}(\bullet) + \sum_x \left(\zeta \bar{\phi}(x) \frac{\partial j(x)}{\partial \phi(\bullet)} \right) \\
&= \zeta \bar{j}(\bullet),
\end{aligned}$$

and

[1]Cf. the derivative-rule for Grassmann-vector-valued functions. Cf. the differentiation-rule from Zinn-Justin on page 8!

$$\begin{aligned}\frac{\partial}{\partial\overline{\phi}(\bullet)}\Gamma[\phi,\overline{\phi}] &= -\sum_x\left(\frac{\partial j(x)}{\partial\overline{\phi}(\bullet)},\frac{\partial\overline{j}(x)}{\partial\overline{\phi}(\bullet)}\right)\cdot\begin{pmatrix}\frac{\partial}{\partial j(x)}W[j,\overline{j}]\\ \frac{\partial}{\partial\overline{j}(x)}W[j,\overline{j}]\end{pmatrix}+j(\bullet)+\sum_x\left(\frac{\partial\overline{j}(x)}{\partial\overline{\phi}(\bullet)}\phi(x)+\zeta\overline{\phi}(x)\frac{\partial j(x)}{\partial\overline{\phi}(\bullet)}\right)\\
&= -\sum_x\left(\frac{\partial j(x)}{\partial\overline{\phi}(\bullet)},\frac{\partial\overline{j}(x)}{\partial\overline{\phi}(\bullet)}\right)\cdot\begin{pmatrix}\zeta\overline{\phi}(x)\\ \phi(x)\end{pmatrix}+j(\bullet)+\sum_x\left(\frac{\partial\overline{j}(x)}{\partial\overline{\phi}(\bullet)}\phi(x)+\zeta\overline{\phi}(x)\frac{\partial j(x)}{\partial\overline{\phi}(\bullet)}\right)\\
&= +j(\bullet).\end{aligned}$$

Proof 29.3 of Lemma 16.38

We use Corollary 16.14 and find

$$\begin{aligned}\frac{\partial^n}{\partial\overline{j}(i_1)\cdots\partial\overline{j}(i_n)}\frac{\partial^n}{\partial j(i_n')\cdots\partial j(i_1')}\frac{Z[\overline{j},j]}{Z} &= \frac{\partial^n}{\partial\overline{j}(i_1)\cdots\partial\overline{j}(i_n)}\zeta^n\left\langle\overline{\psi}(i_n')\cdots\overline{\psi}(i_1')\exp\left[j^\dagger\psi+\psi^\dagger j\right]\right\rangle\\
&= \pm\zeta^n\left\langle\overline{\psi}(i_n')\cdots\overline{\psi}(i_1')\frac{\partial^n}{\partial\overline{j}(i_1)\cdots\partial\overline{j}(i_n)}\exp\left[j^\dagger\psi+\psi^\dagger j\right]\right\rangle\\
&= \pm\zeta^n\left\langle\overline{\psi}(i_n')\cdots\overline{\psi}(i_1')\psi(i_1)\cdots\psi(i_n)\exp\left[j^\dagger\psi+\psi^\dagger j\right]\right\rangle\\
&= +\zeta^n\left\langle\psi(i_1)\cdots\psi(i_n)\overline{\psi}(i_n')\cdots\overline{\psi}(i_1')\exp\left[j^\dagger\psi+\psi^\dagger j\right]\right\rangle\end{aligned}$$

Therefore we obtain

$$\begin{aligned}(-\zeta/c)^n\frac{\partial^n}{\partial\overline{j}(i_1)\cdots\partial\overline{j}(i_n)}\frac{\partial^n}{\partial j(i_n')\cdots\partial j(i_1')}\frac{Z[\overline{j},j]}{Z} &= +(-1/c)^n\left\langle\psi(i_1)\cdots\psi(i_n)\overline{\psi}(i_n')\cdots\overline{\psi}(i_1')\exp\left[j^\dagger\psi+\psi^\dagger j\right]\right\rangle\\
&= \dots\end{aligned}$$

comparing this now with Equation (16.31) proves the Lemma

$$\dots = \mathscr{G}(i_1,\dots,i_n\,|\,i_1',\dots,i_n'). \tag{29.4}$$

Proof 29.4 of Lemma 16.41

We compute

$$
\begin{aligned}
\mathscr{G}_c(i_1|i_1') &= -\left(\frac{\zeta}{c}\right)\left(\frac{\partial^2}{\partial\bar{j}(i_1)\partial j(i_1')}W\right)[j,\bar{j}]_{|j=\bar{j}=0} \\
&= -\left(\frac{\zeta}{c}\right)\partial_{\bar{j}(i_1)}\partial_{j(i_1')}\log\left\langle\exp\left[j^\dagger\psi+\psi^\dagger j\right]\right\rangle_{|j=\bar{j}=0} \\
&= -\left(\frac{\zeta}{c}\right)\partial_{\bar{j}(i_1)}\left\langle\exp\left[j^\dagger\psi+\psi^\dagger j\right]\right\rangle^{-1}\left\langle\zeta\overline{\psi}(i_1')\exp\left[j^\dagger\psi+\psi^\dagger j\right]\right\rangle_{|j=\bar{j}=0} \\
&= -\left(\frac{1}{c}\right)\partial_{\bar{j}(i_1)}\left\langle\exp\left[j^\dagger\psi+\psi^\dagger j\right]\right\rangle^{-1}\left\langle\overline{\psi}(i_1')\exp\left[j^\dagger\psi+\psi^\dagger j\right]\right\rangle_{|j=\bar{j}=0} \\
&= -\left(\frac{1}{c}\right)\Bigg[\left\langle\exp\left[j^\dagger\psi+\psi^\dagger j\right]\right\rangle^{-2}\left\langle\psi(i_1)\exp\left[j^\dagger\psi+\psi^\dagger j\right]\right\rangle\left\langle\overline{\psi}(i_1')\exp\left[j^\dagger\psi+\psi^\dagger j\right]\right\rangle \\
&\qquad\qquad+\left\langle\exp\left[j^\dagger\psi+\psi^\dagger j\right]\right\rangle^{-2}\left\langle\zeta\overline{\psi}(i_1')\psi(i_1)\exp\left[j^\dagger\psi+\psi^\dagger j\right]\right\rangle\Bigg]_{|j=\bar{j}=0} \\
&= -\frac{\zeta}{c}\langle\overline{\psi}(i_1')\psi(i_1)\rangle \\
&= -\frac{1}{c}\langle\psi(i_1)\overline{\psi}(i_1')\rangle \\
&= \mathscr{G}(i_1|i_1')
\end{aligned}
\tag{29.5}
$$

For the other formula: Cf. Negele and Orland somewhere around page 100. TBD.

Proof 29.5 of Lemma 16.42 We compute straight-forwardly

$$
\begin{aligned}
\frac{\partial}{\partial\phi(\bullet)}F[\bar{j},j] &= \sum_x\left(\frac{\partial F}{\partial\bar{j}(x)},\frac{\partial F}{\partial j(x)},\right)\cdot\begin{pmatrix}\frac{\partial\bar{j}(x)}{\partial\phi(\bullet)}\\ \frac{\partial j(x)}{\partial\phi(\bullet)}\end{pmatrix} \\
&= \sum_x\frac{\partial F}{\partial\bar{j}(x)}\frac{\partial\bar{j}(x)}{\partial\phi(\bullet)}+\frac{\partial F}{\partial j(x)}\frac{\partial j(x)}{\partial\phi(\bullet)} \\
&= \sum_x\zeta\frac{\partial F}{\partial\bar{j}(x)}\frac{\partial^2\Gamma}{\partial\phi(\bullet)\partial\phi(x)}+\frac{\partial F}{\partial j(x)}\frac{\partial^2\Gamma}{\partial\phi(\bullet)\partial\overline{\phi}(x)}
\end{aligned}
\tag{29.6}
$$

and

$$
\begin{aligned}
\frac{\partial}{\partial\overline{\phi}(\bullet)}F[\bar{j},j] &= \sum_x\left(\frac{\partial F}{\partial\bar{j}(x)},\frac{\partial F}{\partial j(x)},\right)\cdot\begin{pmatrix}\frac{\partial\bar{j}(x)}{\partial\overline{\phi}(\bullet)}\\ \frac{\partial j(x)}{\partial\overline{\phi}(\bullet)}\end{pmatrix} \\
&= \sum_x\frac{\partial F}{\partial\bar{j}(x)}\frac{\partial\bar{j}(x)}{\partial\overline{\phi}(\bullet)}+\frac{\partial F}{\partial j(x)}\frac{\partial j(x)}{\partial\overline{\phi}(\bullet)} \\
&= \sum_x\zeta\frac{\partial F}{\partial\bar{j}(x)}\frac{\partial^2\Gamma}{\partial\overline{\phi}(\bullet)\partial\phi(x)}+\frac{\partial F}{\partial j(x)}\frac{\partial^2\Gamma}{\partial\overline{\phi}(\bullet)\partial\overline{\phi}(x)}
\end{aligned}
\tag{29.7}
$$

Proof 29.6 of Corollary 16.43 We use the Lemma and compute ...

The first equation:

$$
\begin{aligned}
\delta_{1,3} = \frac{\partial \phi(3)}{\partial \phi(1)} &= \frac{\partial}{\partial \phi(1)} \frac{\partial W}{\partial \bar{j}(3)} \\
&= \sum_2 \zeta \left(\frac{\partial^2 W}{\partial \bar{j}(2) \partial \bar{j}(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \phi(1) \partial \phi(2)} \right) + \left(\frac{\partial^2 W}{\partial j(2) \partial \bar{j}(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \phi(1) \partial \bar{\phi}(2)} \right)
\end{aligned}
\tag{29.8}
$$

The second equation:

$$
\begin{aligned}
\delta_{1,3} = \frac{\partial \bar{\phi}(3)}{\partial \bar{\phi}(1)} &= \zeta \frac{\partial}{\partial \phi(1)} \frac{\partial W}{\partial j(3)} \\
&= \zeta \sum_2 \zeta \left(\frac{\partial^2 W}{\partial \bar{j}(2) \partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1) \partial \phi(2)} \right) + \left(\frac{\partial^2 W}{\partial j(2) \partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1) \partial \bar{\phi}(2)} \right) \\
&= \sum_2 \left(\frac{\partial^2 W}{\partial \bar{j}(2) \partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1) \partial \phi(2)} \right) + \zeta \left(\frac{\partial^2 W}{\partial j(2) \partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1) \partial \bar{\phi}(2)} \right)
\end{aligned}
\tag{29.9}
$$

The third equation:

$$
\begin{aligned}
0 = \frac{\partial \phi(3)}{\partial \bar{\phi}(1)} &= \frac{\partial}{\partial \bar{\phi}(1)} \frac{\partial W}{\partial \bar{j}(3)} \\
&= \sum_2 \zeta \left(\frac{\partial^2 W}{\partial \bar{j}(2) \partial \bar{j}(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1) \partial \phi(2)} \right) + \left(\frac{\partial^2 W}{\partial j(2) \partial \bar{j}(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1) \partial \bar{\phi}(2)} \right)
\end{aligned}
\tag{29.10}
$$

The fourth equation:

$$
\begin{aligned}
0 = \frac{\partial \bar{\phi}(3)}{\partial \phi(1)} &= \zeta \frac{\partial}{\partial \phi(1)} \frac{\partial W}{\partial j(3)} \\
&= \sum_2 \left(\frac{\partial^2 W}{\partial \bar{j}(2) \partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \phi(1) \partial \phi(2)} \right) + \zeta \left(\frac{\partial^2 W}{\partial j(2) \partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \phi(1) \partial \bar{\phi}(2)} \right)
\end{aligned}
\tag{29.11}
$$

Now summarizing[2] the equations we find

$$
\sum_{i_2} \begin{pmatrix} \frac{\partial^2 W}{\partial \bar{j}(i_3) \partial j(i_2)} & \zeta \frac{\partial^2 W}{\partial \bar{j}(i_3) \partial \bar{j}(i_2)} \\ \zeta \frac{\partial^2 W}{\partial j(i_3) \partial j(i_2)} & \frac{\partial^2 W}{\partial j(i_3) \partial \bar{j}(i_2)} \end{pmatrix} \begin{pmatrix} \frac{\partial^2 \Gamma}{\partial \bar{\phi}(i_2) \partial \phi(i_1)} & \frac{\partial^2 \Gamma}{\partial \bar{\phi}(i_2) \partial \bar{\phi}(i_1)} \\ \frac{\partial^2 \Gamma}{\partial \phi(i_2) \partial \phi(i_1)} & \frac{\partial^2 \Gamma}{\partial \phi(i_2) \partial \bar{\phi}(i_1)} \end{pmatrix} = \delta_{i_1, i_3} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}
\tag{29.12}
$$

Proof 29.7 of Theorem 16.45

We start with Equation (29.9)

[2]The first equation gives the 1,1 component, the second to the 2,2 component, the third for the 1,2 and the fourth to the 2,1 component.

$$\delta_{1,3} = \sum_2 \left(\frac{\partial^2 W}{\partial \bar{j}(2)\partial j(3)}\right)\left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1)\partial \phi(2)}\right) + \zeta \left(\frac{\partial^2 W}{\partial j(2)\partial j(3)}\right)\left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1)\partial \bar{\phi}(2)}\right) \tag{29.13}$$

and now we take the derivative w.r.t. $\phi(4)$ on both sides

$$\begin{aligned} 0 = \partial_{\phi(4)}\delta_{1,3} = &\sum_2 \left(\partial_{\phi(4)}\frac{\partial^2 W}{\partial \bar{j}(2)\partial j(3)}\right)\left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1)\partial \phi(2)}\right) + \left(\frac{\partial^2 W}{\partial \bar{j}(2)\partial j(3)}\right)\left(\partial_{\phi(4)}\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1)\partial \phi(2)}\right) \\ &+ \zeta \left(\partial_{\phi(4)}\frac{\partial^2 W}{\partial j(2)\partial j(3)}\right)\left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1)\partial \bar{\phi}(2)}\right) + \zeta \left(\frac{\partial^2 W}{\partial j(2)\partial j(3)}\right)\left(\partial_{\phi(4)}\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1)\partial \bar{\phi}(2)}\right) \\ = &\ldots \end{aligned} \tag{29.14}$$

now we consider the following two terms individually:

$$\partial_{\phi(4)}\frac{\partial^2 W}{\partial \bar{j}(2)\partial j(3)} = \sum_5 \zeta \left(\frac{\partial^3 W}{\partial \bar{j}(5)\partial \bar{j}(2)\partial j(3)}\right)\left(\frac{\partial^2 \Gamma}{\partial \phi(4)\partial \phi(5)}\right) + \left(\frac{\partial^3 W}{\partial j(5)\partial \bar{j}(2)\partial j(3)}\right)\left(\frac{\partial^2 \Gamma}{\partial \phi(4)\partial \bar{\phi}(5)}\right) \tag{29.15}$$

and

$$\partial_{\phi(4)}\frac{\partial^2 W}{\partial j(2)\partial j(3)} = \sum_5 \zeta \left(\frac{\partial^3 W}{\partial \bar{j}(5)\partial j(2)\partial j(3)}\right)\left(\frac{\partial^2 \Gamma}{\partial \phi(4)\partial \phi(5)}\right) + \left(\frac{\partial^3 W}{\partial j(5)\partial j(2)\partial j(3)}\right)\left(\frac{\partial^2 \Gamma}{\partial \phi(4)\partial \bar{\phi}(5)}\right) \tag{29.16}$$

inserting this above yields

$$\begin{aligned} 0 = \ldots = \sum_2 \Bigg[&\left(\frac{\partial^2 W}{\partial \bar{j}(2)\partial j(3)}\right)\left(\frac{\partial^3 \Gamma}{\partial \phi(4)\partial \bar{\phi}(1)\partial \bar{\phi}(2)}\right) + \zeta \left(\frac{\partial^2 W}{\partial j(2)\partial j(3)}\right)\left(\frac{\partial^3 \Gamma}{\partial \phi(4)\partial \bar{\phi}(1)\partial \bar{\phi}(2)}\right) \\ &+ \sum_5 \left[\zeta \frac{\partial^3 W}{\partial \bar{j}(5)\partial \bar{j}(2)\partial j(3)} \cdot \frac{\partial^2 \Gamma}{\partial \phi(4)\partial \bar{\phi}(5)} + \frac{\partial^3 W}{\partial j(5)\partial \bar{j}(2)\partial j(3)} \cdot \frac{\partial^2 \Gamma}{\partial \phi(4)\partial \bar{\phi}(5)}\right] \cdot \frac{\partial^2 \Gamma}{\partial \bar{\phi}(1)\partial \phi(2)} \\ &+ \sum_2 \left[\frac{\partial^3 W}{\partial \bar{j}(5)\partial j(2)\partial j(3)} \frac{\partial^2 \Gamma}{\partial \phi(4)\partial \phi(5)} + \zeta \frac{\partial^3 W}{\partial j(5)\partial j(2)\partial j(3)} \frac{\partial^2 \Gamma}{\partial \phi(4)\partial \bar{\phi}(5)}\right] \cdot \frac{\partial^2 \Gamma}{\partial \bar{\phi}(1)\partial \bar{\phi}(2)} \Bigg] \end{aligned} \tag{29.17}$$

Now we take the derivative w.r.t. $\bar{\phi}(6)$. We consider each of the six terms (seperated by plus-symbols respectively) separately:

1. We compute

$$\begin{aligned} &\partial_{\bar{\phi}(6)} \sum_2 \left(\frac{\partial^2 W}{\partial \bar{j}(2)\partial j(3)}\right)\left(\frac{\partial^3 \Gamma}{\partial \phi(4)\partial \bar{\phi}(1)\partial \phi(2)}\right) \\ &= \sum_2 \left(\partial_{\bar{\phi}(6)}\frac{\partial^2 W}{\partial \bar{j}(2)\partial j(3)}\right)\left(\frac{\partial^3 \Gamma}{\partial \phi(4)\partial \bar{\phi}(1)\partial \phi(2)}\right) + \left(\frac{\partial^2 W}{\partial \bar{j}(2)\partial j(3)}\right)\left(\frac{\partial^4 \Gamma}{\partial \bar{\phi}(6)\partial \phi(4)\partial \bar{\phi}(1)\partial \phi(2)}\right) \\ &\equiv \sum_2 0 + \left(\frac{\partial^2 W}{\partial \bar{j}(2)\partial j(3)}\right)\left(\frac{\partial^4 \Gamma}{\partial \bar{\phi}(6)\partial \phi(4)\partial \bar{\phi}(1)\partial \phi(2)}\right) \end{aligned} \tag{29.18}$$

2. This term doesn't contribute because of $\partial_{j(2)}\partial_{j(3)}$
3. This term doesn't contribute because of $\partial_{\phi(4)}\partial_{\phi(5)}$
4. We compute straight-forwardly

$$\begin{aligned}
&\partial_{\overline{\phi}(6)}\sum_{2,5}\left(\frac{\partial^3 W}{\partial j(5)\partial\overline{j}(2)\partial j(3)}\right)\left(\frac{\partial^2\Gamma}{\partial\phi(4)\partial\overline{\phi}(5)}\right)\left(\frac{\partial^2\Gamma}{\partial\overline{\phi}(1)\partial\phi(2)}\right)\\
&=\sum_{2,5}\left(\partial_{\overline{\phi}(6)}\frac{\partial^3 W}{\partial j(5)\partial\overline{j}(2)\partial j(3)}\right)\left(\frac{\partial^2\Gamma}{\partial\phi(4)\partial\overline{\phi}(5)}\right)\left(\frac{\partial^2\Gamma}{\partial\overline{\phi}(1)\partial\phi(2)}\right)\\
&+\left(\frac{\partial^3 W}{\partial j(5)\partial\overline{j}(2)\partial j(3)}\right)\left[\frac{\partial^3\Gamma}{\partial\overline{\phi}(6)\partial\phi(4)\partial\overline{\phi}(5)}\cdot\frac{\partial^2\Gamma}{\partial\overline{\phi}(1)\partial\phi(2)}+\frac{\partial^2\Gamma}{\partial\phi(4)\partial\overline{\phi}(5)}\cdot\frac{\partial^3\Gamma}{\partial\overline{\phi}(6)\partial\overline{\phi}(1)\partial\phi(2)}\right]\\
&\equiv\dots
\end{aligned}\tag{29.19}$$

now we will simplify this, using that the terms in the second line dont contribute in the absence of external sources and spontaneous symmetry breaking, implying that one obtains

$$\dots=\sum_{2,5,7}\left[\zeta\frac{\partial^4 W}{\partial\overline{j}(7)\partial j(5)\partial\overline{j}(2)\partial j(3)}\cdot\frac{\partial^2\Gamma}{\partial\overline{\phi}(6)\partial\phi(7)}+\cdots\right]\cdot\frac{\partial^2\Gamma}{\partial\phi(4)\partial\overline{\phi}(5)}\cdot\frac{\partial^2\Gamma}{\partial\overline{\phi}(1)\partial\phi(2)}\tag{29.20}$$

5. This term doesn't contribute because of $\partial_{\overline{\phi}(1)}\partial_{\overline{\phi}(2)}$
6. This term doesn't contribute because of $\partial_{\overline{\phi}(1)}\partial_{\overline{\phi}(2)}$

Now we collect the individual (non-vanishing) contributions and obtain

$$\begin{aligned}
0&=\sum_{2}\left(\frac{\partial^2 W}{\partial\overline{j}(2)\partial j(3)}\right)\left(\frac{\partial^4\Gamma}{\partial\overline{\phi}(6)\partial\phi(4)\partial\overline{\phi}(1)\partial\phi(2)}\right)\\
&+\zeta\sum_{2,5,7}\left(\frac{\partial^4 W}{\partial\overline{j}(7)\partial j(5)\partial\overline{j}(2)\partial j(3)}\right)\left(\frac{\partial^2\Gamma}{\partial\phi(4)\partial\overline{\phi}(5)}\right)\left(\frac{\partial^2\Gamma}{\partial\overline{\phi}(1)\partial\phi(2)}\right)\left(\frac{\partial^2\Gamma}{\partial\overline{\phi}(6)\partial\phi(7)}\right)\\
&=\sum_{2}\left(\frac{\partial^2 W}{\partial\overline{j}(2)\partial j(3)}\right)\left(\frac{\zeta\partial^4\Gamma}{\partial\overline{\phi}(6)\partial\overline{\phi}(1)\partial\phi(4)\partial\phi(2)}\right)\\
&+\zeta\sum_{2,5,7}\left(\frac{\zeta\partial^4 W}{\partial\overline{j}(7)\partial\overline{j}(2)\partial j(5)\partial j(3)}\right)\left(\frac{\zeta\partial^2\Gamma}{\partial\overline{\phi}(5)\partial\phi(4)}\right)\left(\frac{\partial^2\Gamma}{\partial\overline{\phi}(1)\partial\phi(2)}\right)\left(\frac{\partial^2\Gamma}{\partial\overline{\phi}(6)\partial\phi(7)}\right)\\
&=\sum_{2}\left(-\frac{\hbar}{\zeta}\mathscr{G}_c(2,3)\right)\zeta\left(-\frac{\zeta}{\hbar}\right)^2 V(6,1|2,4)\\
&+\zeta\sum_{2,5,7}\zeta\left(-\frac{\hbar}{\zeta}\right)^2\mathscr{G}_c(7,2|3,5)\zeta\left(-\frac{\zeta}{\hbar}V(5,4)\right)\left(-\frac{\zeta}{\hbar}V(1,2)\right)\left(-\frac{\zeta}{\hbar}V(6,7)\right)\\
&=-\frac{1}{\hbar}\sum_{2}\mathscr{G}_c(2,3)V(6,1|2,4)\\
&-\frac{1}{\hbar}\sum_{2,5,7}\mathscr{G}_c(7,2|3,5)V(5,4)V(1,2)V(6,7)
\end{aligned}\tag{29.21}$$

rearranging terms and multiplying both sides with $\hbar$ gives therefore

$$\sum_{2} \mathcal{G}_c(2,3)V(6,1|2,4) = -\sum_{2,5,7} \mathcal{G}_c(7,2|3,5)V(5,4)V(1,2)V(6,7) \tag{29.22}$$

Notice that this equation holds for all 1, 3, 4, 6. Now we make the replacements

- $1 \mapsto \tilde{1}$
- $3 \mapsto \tilde{2}$
- $4 \mapsto \tilde{3}$
- $6 \mapsto \tilde{4}$

and obtain therefore

$$\sum_{2} \mathcal{G}_c(2,\tilde{2})V(\tilde{4},\tilde{1}|2,\tilde{3}) = -\sum_{2,5,7} \mathcal{G}_c(7,2|\tilde{2},5)V(5,\tilde{3})V(\tilde{1},2)V(\tilde{4},7) \tag{29.23}$$

or, equivalently

$$\sum_{2} \mathcal{G}_c(2,i_2)V(i_4,i_1|2,i_3) = -\sum_{2,5,7} \mathcal{G}_c(7,2|i_2,5)V(5,i_3)V(i_1,2)V(i_4,7) \tag{29.24}$$

Now we multiply both sides with $\mathcal{G}_c(i_3|x_1)$ and sum over i_3, whereas x_1 still has to be determined, and we obtain

$$\begin{aligned}\sum_{2,i_3} \mathcal{G}_c(2,i_2)V(i_4,i_1|2,i_3)\mathcal{G}_c(i_3|x_1) &= -\sum_{2,5,7} \mathcal{G}_c(7,2|i_2,5)V(i_1,2)V(i_4,7)\sum_{i_3} V(5,i_3)\mathcal{G}_c(i_3|x_1) \\ &= -\sum_{2,5,7} \mathcal{G}_c(7,2|i_2,5)V(i_1,2)V(i_4,7)\delta_{5,x_1} \\ &= -\sum_{2,7} \mathcal{G}_c(7,2|i_2,x_1)V(i_1,2)V(i_4,7)\end{aligned} \tag{29.25}$$

Now we multiply both sides with $\mathcal{G}_c(x_2|i_1)\mathcal{G}_c(x_3|i_4)$ and sum over i_1, whereas i_4 still has to be determined, and we obtain

$$\begin{aligned}\sum_{2,i_1,i_3,i_4} \mathcal{G}_c(2,i_2)V(i_4,i_1|2,i_3)\mathcal{G}_c(i_3|x_1)\mathcal{G}_c(x_2|i_1)\mathcal{G}_c(x_3|i_4) &= -\sum_{2,7} \mathcal{G}_c(7,2|i_2,x_1)\sum_{i_1} \mathcal{G}_c(x_2|i_1)V(i_1,2)\sum_{i_4} \mathcal{G}_c(x_3|i_4)V(i_4,7) \\ &= -\sum_{2,7} \mathcal{G}_c(7,2|i_2,x_1)\delta_{x_2,2}\delta_{x_3,7} \\ &= -\mathcal{G}_c(x_3,x_2|i_2,x_1)\end{aligned} \tag{29.26}$$

Now we replace

1. $x_3 \mapsto y_1$
2. $x_2 \mapsto y_2$
3. $i_2 \mapsto y_3$
4. $x_1 \mapsto y_4$

and we obtain

$$\begin{aligned}\mathscr{G}_c(y_1, y_2|y_3, y_4) &= -\sum_{2,i_1,i_3,i_4} \mathscr{G}_c(2, y_3)V(i_4, i_1|2, i_3)\mathscr{G}_c(i_3|y_4)\mathscr{G}_c(y_2|i_1)\mathscr{G}_c(y_1|i_4) \\ &= -\sum_{2,i_1,i_3,i_4} \mathscr{G}_c(y_1|i_4)\mathscr{G}_c(y_2|i_1)V(i_4, i_1|2, i_3)\mathscr{G}_c(2, y_3)\mathscr{G}_c(i_3|y_4) \\ &= \dots\end{aligned} \tag{29.27}$$

and now, we replace finally

1. $i_4 \mapsto l_1$
2. $i_1 \mapsto l_2$
3. $2 \mapsto l'_1$
4. $i_3 \mapsto l'_2$

and obtain

$$\mathscr{G}_c(y_1, y_2|y_3, y_4) = -\sum_{l_1,l_2,l'_1,l'_2} \mathscr{G}_c(y_1|l_1)\mathscr{G}_c(y_2|l_2)V(l_1, l_2|l'_1, l'_2)\mathscr{G}_c(l'_1, y_3)\mathscr{G}_c(l'_2|y_4) \tag{29.28}$$

proving the theorem.

This entire section of proofs needs to be revised, cause we adjusted the prefactor convention for the functional derivati

Proof 29.8 of Lemma 17.2.

For the first equation we compute

$$
\begin{aligned}
W[j,\bar{j}] &= -\log \int \mathscr{D}(\psi,\overline{\psi}) \exp\left(-S[\psi,\overline{\psi}] + \sum\!\!\!\!\!\!\int (\bar{j}\psi + \overline{\psi} j)\right) \\
&= -\log \int \mathscr{D}(M\cdot\psi, \overline{M}\cdot\overline{\psi}) \exp\left(-S[M\cdot\psi, \overline{M}\cdot\overline{\psi}] + \sum\!\!\!\!\!\!\int_x \bar{j}(x)\,(M\cdot\psi)(x) + (\overline{M}\cdot\overline{\psi})(x)\,j(x)\right) \\
&= \dots
\end{aligned}
$$

using now

$$
\sum\!\!\!\!\!\!\int_x \bar{j}(x)\,(M\cdot\psi)(x) = \sum\!\!\!\!\!\!\int_x \sum\!\!\!\!\!\!\int_y \bar{j}(x)\,M(x,y)\,\psi(y) = \sum\!\!\!\!\!\!\int_y \sum\!\!\!\!\!\!\int_x M^{\mathrm{T}}(y,x)\bar{j}(x)\,\psi(y) = \sum\!\!\!\!\!\!\int_y (M^{\mathrm{T}}\cdot\bar{j})(y)\,\psi(y)
$$

and analogously one has

$$
\sum\!\!\!\!\!\!\int_x (\overline{M}\cdot\overline{\psi})(x)\,j(x) = \sum\!\!\!\!\!\!\int_y \overline{\psi}(y)\,(\overline{M}^{\mathrm{T}}\cdot j)(y).
$$

Then, it follows (together with Definition 17.1)

$$
\begin{aligned}
\dots &= -\log \int \mathscr{D}(\psi,\overline{\psi}) \exp\left(-S[\psi,\overline{\psi}] + \sum\!\!\!\!\!\!\int_y (M^{\mathrm{T}}\cdot\bar{j})(y)\,\psi(y) + \overline{\psi}(y)\,(\overline{M}^{\mathrm{T}}\cdot j)(y)\right) \\
&= W[\overline{M}^{\mathrm{T}}\cdot j, M^{\mathrm{T}}\cdot\bar{j}].
\end{aligned}
$$

For the second equation one has

$$
\begin{aligned}
\phi[\bullet, \overline{M}^{\mathrm{T}}\cdot j, M^{\mathrm{T}}\cdot\bar{j}] &= -\frac{\vec{\delta}}{\delta(M^{\mathrm{T}}\cdot\bar{j})(\bullet)} W[\overline{M}^{\mathrm{T}}\cdot j, M^{\mathrm{T}}\cdot\bar{j}] \\
&= -\frac{\vec{\delta}}{\delta(M^{\mathrm{T}}\cdot\bar{j})(\bullet)} W[j,\bar{j}] \\
&= \dots
\end{aligned}
$$

Now one calculates

$$
\bar{j}(x) = (M^{-\mathrm{T}}\cdot M^{\mathrm{T}}\cdot\bar{j})(x) = \sum\!\!\!\!\!\!\int_y M^{-\mathrm{T}}(x,y)\,(M^{\mathrm{T}}\cdot\bar{j})(y)
$$

and therefore

$$\frac{\vec{\delta}}{\delta(M^{\mathrm{T}}\cdot\bar{j})(\bullet)} = \sumint_x \left(\frac{\vec{\delta}}{\delta(M^{\mathrm{T}}\cdot\bar{j})(\bullet)}\bar{j}(x)\right)\frac{\vec{\delta}}{\delta\bar{j}(x)} = \sumint_x M^{-\mathrm{T}}(x,\bullet)\frac{\vec{\delta}}{\delta\bar{j}(x)} = \sumint_x M^{-1}(\bullet,x)\frac{\vec{\delta}}{\delta\bar{j}(x)}.$$

It then follows with Notation 16.30

$$\ldots = -\sumint_x M^{-1}(\bullet,x)\left(\frac{\vec{\delta}}{\delta\bar{j}(x)}W[j,\bar{j}]\right) = +\sumint_x M^{-1}(\bullet,x)\,\phi[x,j,\bar{j}] = \left(M^{-1}\cdot\phi\right)[\bullet,j,\bar{j}],$$

Analogously, one obtains

$$\bar{\phi}[\bullet,\overline{M}^{\mathrm{T}}\cdot j, M^{\mathrm{T}}\cdot\bar{j}] = \frac{\vec{\delta}}{\delta(\overline{M}^{\mathrm{T}}\cdot j)(\bullet)}W[j,\bar{j}] = \sumint_x \overline{M}^{-1}(\bullet,x)\left(\frac{\vec{\delta}}{\delta j(x)}W[j,\bar{j}]\right) = \left(\overline{M}^{-1}\cdot\bar{\phi}\right)[\bullet,j,\bar{j}].$$

Proof 29.9 of Theorem 17.3.

With Lemma 17.2 one can prove the theorem very easily:

$$\begin{aligned}\Gamma[M^{-1}\cdot\phi,\overline{M}^{-1}\cdot\bar{\phi}] &= W[\overline{M}^{\mathrm{T}}\cdot j, M^{\mathrm{T}}\cdot\bar{j}] + \sumint_x (M^{\mathrm{T}}\cdot\bar{j})(x)\,(M^{-1}\cdot\phi)(x) + (\overline{M}^{-1}\cdot\bar{\phi})(x)\,(\overline{M}^{\mathrm{T}}\cdot j)(x)\\ &= W[j,\bar{j}] + \sumint_x \bar{j}(x)\,\phi(x) + \bar{\phi}(x)\,j(x)\\ &= \Gamma[\phi,\bar{\phi}].\end{aligned}$$

Proof 29.10 of Approximation 16.52.

Starting with eq. (16.42):

$$\begin{aligned}\Gamma[\phi,\bar{\phi}] &= W[j,\bar{j}] + \sumint \bar{j}\phi + \bar{\phi}j\\ &= -\log Z[j,\bar{j}] + \sumint \bar{j}\phi + \bar{\phi}j,\end{aligned}$$

We do the following things:

1. We multiply this equation by -1 and apply the exponential-function to both sides,
2. we insert the definition of the source-dependent partition function $Z[j,\bar{j}]$ from eq. (**??**),
3. we use the invariance of the functional integral by Graßmann-valued translations $\psi-\phi\to\Delta$, etc.

This gives in equations

$$\begin{aligned}\exp\left(-\Gamma[\phi,\overline{\phi}]\right) &= Z[j,\overline{j}]\exp\left(-\sum\!\!\!\!\!\!\int \overline{j}\phi+\overline{\phi}j\right)\\ &= \int \mathscr{D}(\psi,\overline{\psi})\exp\left(-S[\psi,\overline{\psi}] + \sum\!\!\!\!\!\!\int \overline{j}(\psi-\phi)+(\overline{\psi}-\overline{\phi})j\right)\\ &= \int \mathscr{D}(\Delta,\overline{\Delta})\exp\left(-S[\phi+\Delta,\overline{\phi}+\overline{\Delta}] + \sum\!\!\!\!\!\!\int \overline{j}\Delta+\overline{\Delta}j\right).\end{aligned}$$

The action S can now be expanded analogously to Notation **??** (again, in schematic notation) in the variation

$$\begin{aligned}S[\phi+\Delta,\overline{\phi}+\overline{\Delta}] = {}& S[\phi,\overline{\phi}]\\ &+ \sum\!\!\!\!\!\!\int \left(\frac{\delta}{\delta\phi}S[\phi,\overline{\phi}], \frac{\delta}{\delta\overline{\phi}}S[\phi,\overline{\phi}]\right)\cdot\begin{pmatrix}\Delta\\ \overline{\Delta}\end{pmatrix}\\ &+ \frac{1}{2!}\sum\!\!\!\!\!\!\int\sum\!\!\!\!\!\!\int \begin{pmatrix}\Delta\\ \overline{\Delta}\end{pmatrix}^{\mathrm{T}}\cdot\begin{pmatrix}\frac{\delta^2}{\delta\phi\,\delta\phi}S[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\phi\,\delta\overline{\phi}}S[\phi,\overline{\phi}]\\ \frac{\delta^2}{\delta\overline{\phi}\,\delta\phi}S[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\overline{\phi}\,\delta\overline{\phi}}S[\phi,\overline{\phi}]\end{pmatrix}\cdot\begin{pmatrix}\Delta\\ \overline{\Delta}\end{pmatrix}\\ &+\mathcal{O}\left(\Delta^3,\Delta^2\overline{\Delta}^1,\Delta^1\overline{\Delta}^2,\overline{\Delta}^3\right).\end{aligned}$$

With this we get

$$\exp\left(-\Gamma[\phi,\overline{\phi}]\right) = \exp\left(-S[\phi,\overline{\phi}]\right)\int \mathscr{D}(\Delta,\overline{\Delta})\ \exp \mathrm{P}(\Delta,\overline{\Delta},\phi,\overline{\phi}) \tag{29.29}$$

with a polynomial P in the integration-variables and in the classical fields. In the so-called “Tree-Level-approximation” one now approximates[3]

$$\int \mathscr{D}(\Delta,\overline{\Delta})\ \exp \mathrm{P}(\Delta,\overline{\Delta},\phi,\overline{\phi}) \quad\approx\quad \mathcal{O}(1).$$

So, one assumes that the functional integral (which, when evaluated, is again a polynomial expression in the classical fields) gives essentially ϕ-independent contributions.

After applying the logarithm and using the inverse property (also for Graßmann-valued polynomial expressions?!) we thus obtain

$$\Gamma[\phi,\overline{\phi}] \quad\approx\quad S[\phi,\overline{\phi}] + \text{const}.$$

Remembering now that the physically relevant part of the effective action is in the numerical values of the non-trivial correlators (and thus in the coefficients of the non-constant monomials in Γ (cf. the series-expansion in eq. (**??**) and section **??**), then, one usually simply notes for convenience:

$$\Gamma[\phi,\overline{\phi}] \quad\approx\quad S[\phi,\overline{\phi}].$$

[3]The meaning of the symbol “$\approx$” is discussed in section **??**.

Proof 29.11 of Approximation 16.54.

The "One-Loop-approximation" builds on the Tree-Level-approximation. If we use Corollary 16.53, then (in the context of the Tree-Level-approximation) in eq. (29.29) the terms linear in Δ and $\overline{\Delta}$ cancel out. Then we can use this to connect to eq. (29.29), by neglecting the contributions to the polynomial P of order $\Delta^n\,\overline{\Delta}^m$ with $n+m \geq 3$:

$$\begin{aligned}\exp\left(-\Gamma[\phi,\overline{\phi}]\right) &\approx \exp\left(-S[\phi,\overline{\phi}]\right)\int \mathscr{D}(\Delta,\overline{\Delta})\,\exp\,-\frac{1}{2}\sum\!\!\!\!\!\!\int\sum\!\!\!\!\!\!\int \begin{pmatrix}\Delta\\ \overline{\Delta}\end{pmatrix}^{\mathrm{T}}\cdot\begin{pmatrix}\frac{\delta^2}{\delta\phi\,\delta\phi}S[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\phi\,\delta\overline{\phi}}S[\phi,\overline{\phi}]\\ \frac{\delta^2}{\delta\overline{\phi}\,\delta\phi}S[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\overline{\phi}\,\delta\overline{\phi}}S[\phi,\overline{\phi}]\end{pmatrix}\cdot\begin{pmatrix}\Delta\\ \overline{\Delta}\end{pmatrix}\\ &= \exp\left(-S[\phi,\overline{\phi}]\right)\mathrm{Pf}\begin{pmatrix}\frac{\delta^2}{\delta\phi\,\delta\phi}S[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\phi\,\delta\overline{\phi}}S[\phi,\overline{\phi}]\\ \frac{\delta^2}{\delta\overline{\phi}\,\delta\phi}S[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\overline{\phi}\,\delta\overline{\phi}}S[\phi,\overline{\phi}]\end{pmatrix}.\end{aligned}$$

If we now take the logarithm on both sides and use the inversion-property between exponential-function and logarithm (?!) and fundamental identities for the matrix logarithm (also for Graßmann-valued matrices?!), we get

$$\begin{aligned}-\Gamma[\phi,\overline{\phi}] &\approx -S[\phi,\overline{\phi}] + \log\mathrm{Pf}\begin{pmatrix}\frac{\delta^2}{\delta\phi\,\delta\phi}S[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\phi\,\delta\overline{\phi}}S[\phi,\overline{\phi}]\\ \frac{\delta^2}{\delta\overline{\phi}\,\delta\phi}S[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\overline{\phi}\,\delta\overline{\phi}}S[\phi,\overline{\phi}]\end{pmatrix}\\ &\overset{?!}{=} -S[\phi,\overline{\phi}] + \frac{1}{2}\log\mathrm{Det}\begin{pmatrix}\frac{\delta^2}{\delta\phi\,\delta\phi}S[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\phi\,\delta\overline{\phi}}S[\phi,\overline{\phi}]\\ \frac{\delta^2}{\delta\overline{\phi}\,\delta\phi}S[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\overline{\phi}\,\delta\overline{\phi}}S[\phi,\overline{\phi}]\end{pmatrix}\\ &\overset{?!}{=} -S[\phi,\overline{\phi}] + \frac{1}{2}\,\mathrm{Tr}\,\log\begin{pmatrix}\frac{\delta^2}{\delta\phi\,\delta\phi}S[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\phi\,\delta\overline{\phi}}S[\phi,\overline{\phi}]\\ \frac{\delta^2}{\delta\overline{\phi}\,\delta\phi}S[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\overline{\phi}\,\delta\overline{\phi}}S[\phi,\overline{\phi}]\end{pmatrix}\\ \Longleftrightarrow +\Gamma[\phi,\overline{\phi}] &\approx +S[\phi,\overline{\phi}] - \frac{1}{2}\,\mathrm{Tr}\,\log\begin{pmatrix}\frac{\delta^2}{\delta\phi\,\delta\phi}S[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\phi\,\delta\overline{\phi}}S[\phi,\overline{\phi}]\\ \frac{\delta^2}{\delta\overline{\phi}\,\delta\phi}S[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\overline{\phi}\,\delta\overline{\phi}}S[\phi,\overline{\phi}]\end{pmatrix}.\end{aligned}$$

Proof 29.12 of Lemma 16.56.

First Identity

One has not only for the quadratic sector

$$\frac{\vec{\delta}^2}{\delta\overline{\psi}(y_1)\delta\psi(y_2)}\overline{\psi}(x_1)\,\psi(x_2) = -\delta(x_2-y_2)\delta(x_1-y_1) = -\frac{\delta^2}{\delta\overline{\psi}(y_1)\delta\psi(y_2)}\overline{\psi}(x_1)\,\psi(x_2),$$

but also for the quartic sector

$$\begin{aligned}-\frac{\vec{\delta}^2}{\delta\overline{\psi}(y_1)\delta\psi(y_2)}\overline{\psi}(x_1)\,\overline{\psi}(x_2)\,\psi(x_3)\,\psi(x_4) &= (\delta(x_1-y_1)\overline{\psi}(x_2)-\delta(x_2-y_1)\overline{\psi}(x_1))(\delta(x_4-y_2)\psi(x_3)-\delta(x_3-y_2)\psi(x_4))\\ &= \frac{\delta^2}{\delta\overline{\psi}(y_1)\delta\psi(y_2)}\overline{\psi}(x_1)\,\overline{\psi}(x_2)\,\psi(x_3)\,\psi(x_4),\end{aligned}$$

where the Delta-symbols denote a multiplication of a Kronecker-Delta and Dirac-Delta for the discrete and continuous part of the generalized indices.

Second Identity

Here, one also has for the quadratic sector

$$\frac{\vec{\delta}^2}{\delta\overline{\psi}(y_1)\delta\psi(y_2)}\overline{\psi}(x_1)\,\psi(x_2) = -\delta(x_2-y_2)\delta(x_1-y_1) = \frac{\delta^2}{\delta\psi(y_2)\delta\overline{\psi}(y_1)}\overline{\psi}(x_1)\,\psi(x_2),$$

but also in the quartic sector

$$\begin{aligned}\frac{\vec{\delta}^2}{\delta\overline{\psi}(y_1)\delta\psi(y_2)}\overline{\psi}(x_1)\,\overline{\psi}(x_2)\,\psi(x_3)\,\psi(x_4) &= (\delta(x_1-y_1)\overline{\psi}(x_2)-\delta(x_2-y_1)\overline{\psi}(x_1))(\delta(x_3-y_2)\psi(x_4)-\delta(x_4-y_2)\psi(x_3))\\ &= \frac{\delta^2}{\delta\psi(y_2)\delta\overline{\psi}(y_1)}\overline{\psi}(x_1)\,\overline{\psi}(x_2)\,\psi(x_3)\,\psi(x_4).\end{aligned}$$

The proof is finished by using the U(1) invariance of the action, implying[4] that in a series expansion of the fields each monomial possesses equally many ψ-variables as $\overline{\psi}$-variables. Then one considers a monomial of this type

$$\overline{\psi}(x_1)\cdots\overline{\psi}(x_n)\,\psi(\tilde{x}_1)\cdots\psi(\tilde{x}_n)$$

and proves the identities by induction[5] over the number $n \in \mathbb{N}_{>1}$. Noting the linearity of the derivates completes the proof.

[4]See, for example, [**?**].

[5]As this is a little bit lengthy (but very easy), it is not documented in this thesis.

Proof 29.13 of Approximation 16.57.

One separates the second variation according to

$$\begin{aligned}\begin{pmatrix}\frac{\delta^2}{\delta\phi\,\delta\phi}S[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\phi\,\delta\overline{\phi}}S[\phi,\overline{\phi}]\\ \frac{\delta^2}{\delta\overline{\phi}\,\delta\phi}S[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\overline{\phi}\,\delta\overline{\phi}}S[\phi,\overline{\phi}]\end{pmatrix} &= \begin{pmatrix}0 & -G_0^{-\mathrm{T}}\\ G_0^{-1} & 0\end{pmatrix} + \begin{pmatrix}\frac{\delta^2}{\delta\phi\,\delta\phi}S_{\mathrm{int}}[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\phi\,\delta\overline{\phi}}S_{\mathrm{int}}[\phi,\overline{\phi}]\\ \frac{\delta^2}{\delta\overline{\phi}\,\delta\phi}S_{\mathrm{int}}[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\overline{\phi}\,\delta\overline{\phi}}S_{\mathrm{int}}[\phi,\overline{\phi}]\end{pmatrix}\\ &\equiv \qquad \mathcal{G}_0^{-1} \qquad + \qquad \mathcal{V}[\phi,\overline{\phi}].\end{aligned}$$

Using now that

$$\mathcal{G}_0 = \begin{pmatrix}0 & -G_0^{-\mathrm{T}}\\ G_0^{-1} & 0\end{pmatrix}^{-1} = \begin{pmatrix}0 & G_0\\ -G_0^{\mathrm{T}} & 0\end{pmatrix},$$

it follows, after simplifying the notation $\mathcal{V}[\phi,\overline{\phi}] \equiv \mathcal{V}$, that

$$\begin{aligned}\mathrm{Tr}\log\left[\mathcal{G}_0^{-1} + \mathcal{V}\right] &= \mathrm{Tr}\log\left[\mathcal{G}_0^{-1}\cdot\left(1+\mathcal{G}_0\mathcal{V}\right)\right]\\ &\overset{?!}{=} \mathrm{Tr}\log\left[\mathcal{G}_0^{-1}\right] + \mathrm{Tr}\log\left[1+\mathcal{G}_0\cdot\mathcal{V}\right]\\ &= \mathcal{O}(1) + \mathrm{Tr}\log\left[1+\mathcal{G}_0\cdot\mathcal{V}\right].\end{aligned}$$

The first term again provides a (in ϕ and $\overline{\phi}$) constant (and thus physically irrelevant) contribution. The second term provides (physically relevant) corrections. Therefore, we focus on the second term and expand the logarithm in its series representation for "small" corrections to the identity:

$$\begin{aligned}\mathrm{Tr}\log\left[1+\mathcal{G}_0\cdot\mathcal{V}\right] &\approx \mathrm{Tr}\left[\mathcal{G}_0\cdot\mathcal{V} - \frac{(\mathcal{G}_0\cdot\mathcal{V})^2}{2} \overset{+}{-} \cdots\right]\\ &= \mathrm{Tr}\left[\mathcal{G}_0\cdot\mathcal{V}\right] - \frac{1}{2}\mathrm{Tr}\left[\mathcal{G}_0\cdot\mathcal{V}\cdot\mathcal{G}_0\cdot\mathcal{V}\right] \overset{+}{-} \cdots\end{aligned}$$

First summand. One computes

$$\begin{aligned}\mathcal{G}_0\mathcal{V} &= \begin{pmatrix}0 & G_0\\ -G_0^{\mathrm{T}} & 0\end{pmatrix}\cdot\begin{pmatrix}\frac{\delta^2}{\delta\phi\,\delta\phi}S_{\mathrm{int}}[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\phi\,\delta\overline{\phi}}S_{\mathrm{int}}[\phi,\overline{\phi}]\\ \frac{\delta^2}{\delta\overline{\phi}\,\delta\phi}S_{\mathrm{int}}[\phi,\overline{\phi}] & \frac{\delta^2}{\delta\overline{\phi}\,\delta\overline{\phi}}S_{\mathrm{int}}[\phi,\overline{\phi}]\end{pmatrix}\\ &= \begin{pmatrix}G_0\cdot\frac{\delta^2}{\delta\overline{\phi}\,\delta\phi}S_{\mathrm{int}}[\phi,\overline{\phi}] & G_0\cdot\frac{\delta^2}{\delta\overline{\phi}\,\delta\overline{\phi}}S_{\mathrm{int}}[\phi,\overline{\phi}]\\ -G_0^{\mathrm{T}}\cdot\frac{\delta^2}{\delta\phi\,\delta\phi}S_{\mathrm{int}}[\phi,\overline{\phi}] & -G_0^{\mathrm{T}}\cdot\frac{\delta^2}{\delta\phi\,\delta\overline{\phi}}S_{\mathrm{int}}[\phi,\overline{\phi}]\end{pmatrix},\end{aligned}$$

where the formal products are to be evaluated by insertion of corresponding "unities".

Because of Lemma 16.56 one easily concludes that

$$\frac{\delta^2}{\delta\phi\,\delta\overline{\phi}}S_{\mathrm{int}}[\phi,\overline{\phi}] = -\left(\frac{\delta^2}{\delta\overline{\phi}\,\delta\phi}S_{\mathrm{int}}[\phi,\overline{\phi}]\right)^{\mathrm{T}}, \qquad \text{and} \qquad B^{\mathrm{T}}A^{\mathrm{T}} = (AB)^{\mathrm{T}}. \tag{29.30}$$

Due to the cyclicity of the trace we arrive at the first term in eq. (16.74).

Second summand. We write

$$
\begin{aligned}
\mathcal{G}_0 \mathcal{V}\, \mathcal{G}_0 \mathcal{V} &= \begin{pmatrix} G_0 \cdot \frac{\delta^2}{\delta\overline{\phi}\,\delta\phi} S_{\text{int}}[\phi, \overline{\phi}] & G_0 \cdot \frac{\delta^2}{\delta\overline{\phi}\,\delta\overline{\phi}} S_{\text{int}}[\phi, \overline{\phi}] \\ -G_0^{\text{T}} \cdot \frac{\delta^2}{\delta\phi\,\delta\phi} S_{\text{int}}[\phi, \overline{\phi}] & -G_0^{\text{T}} \cdot \frac{\delta^2}{\delta\phi\,\delta\overline{\phi}} S_{\text{int}}[\phi, \overline{\phi}] \end{pmatrix} \cdot \begin{pmatrix} G_0 \cdot \frac{\delta^2}{\delta\overline{\phi}\delta\phi} S_{\text{int}}[\phi, \overline{\phi}] & G_0 \cdot \frac{\delta^2}{\delta\overline{\phi}\,\delta\overline{\phi}} S_{\text{int}}[\phi, \overline{\phi}] \\ -G_0^{\text{T}} \cdot \frac{\delta^2}{\delta\phi\,\delta\phi} S_{\text{int}}[\phi, \overline{\phi}] & -G_0^{\text{T}} \cdot \frac{\delta^2}{\delta\phi\,\delta\overline{\phi}} S_{\text{int}}[\phi, \overline{\phi}] \end{pmatrix} \\
&\equiv \begin{pmatrix} A + B & \cdots \\ \cdots & C + D \end{pmatrix},
\end{aligned}
$$

(the secondary diagonals are irrelevant if one takes the trace and only provide corrections on higher orders of perturbation theory) with

$$
\begin{aligned}
A &= G_0 \cdot \frac{\delta^2}{\delta\overline{\phi}\,\delta\phi} S_{\text{int}}[\phi, \overline{\phi}] \cdot G_0 \cdot \frac{\delta^2}{\delta\overline{\phi}\,\delta\phi} S_{\text{int}}[\phi, \overline{\phi}], \\
B &= -G_0 \cdot \frac{\delta^2}{\delta\overline{\phi}\,\delta\overline{\phi}} S_{\text{int}}[\phi, \overline{\phi}] \cdot G_0^{\text{T}} \cdot \frac{\delta^2}{\delta\phi\,\delta\phi} S_{\text{int}}[\phi, \overline{\phi}], \\
C &= -G_0^{\text{T}} \cdot \frac{\delta^2}{\delta\phi\,\delta\phi} S_{\text{int}}[\phi, \overline{\phi}] \cdot G_0 \cdot \frac{\delta^2}{\delta\overline{\phi}\,\delta\overline{\phi}} S_{\text{int}}[\phi, \overline{\phi}], \\
D &= G_0^{\text{T}} \cdot \frac{\delta^2}{\delta\phi\,\delta\overline{\phi}} S_{\text{int}}[\phi, \overline{\phi}] \cdot G_0^{\text{T}} \cdot \frac{\delta^2}{\delta\phi\,\delta\overline{\phi}} S_{\text{int}}[\phi, \overline{\phi}].
\end{aligned}
$$

Now, one has $\text{Tr}\, B = \text{Tr}\, C$. Due to the cyclicity of the trace and $\text{Tr}\, A = \text{Tr}\, D$ because of eq. (29.30). Eq. (16.74) is now proven.

Proof 29.14 of Theorem 19.4

TBDigitalized. (Negele, Orland p. 29, Altland p.163 for fermions, and Altland p. 159 for bosons.)

Proof 29.15 of Theorem 19.5

Proof 29.16 of Lemma 19.6

TBDigitalized, cf. Negele Orland page 30or Altland page 181 for fermions, Negele Orland page 21 or Altland p. 159 for bosons.

Proof 29.17 of Theorem 19.7

TBD.

Proof 29.18 of Theorem 19.8

- NO page 22, Altland page 160 (bosons)

- NO page 39, Altland page 181 f (fermions)

Proof 29.19 of Lemma 19.9

TBD

Proof 29.20 of Lemma 19.10

We obtain by direct computation ...

- Altland page 182.

Proof 29.21 of Lemma 19.11

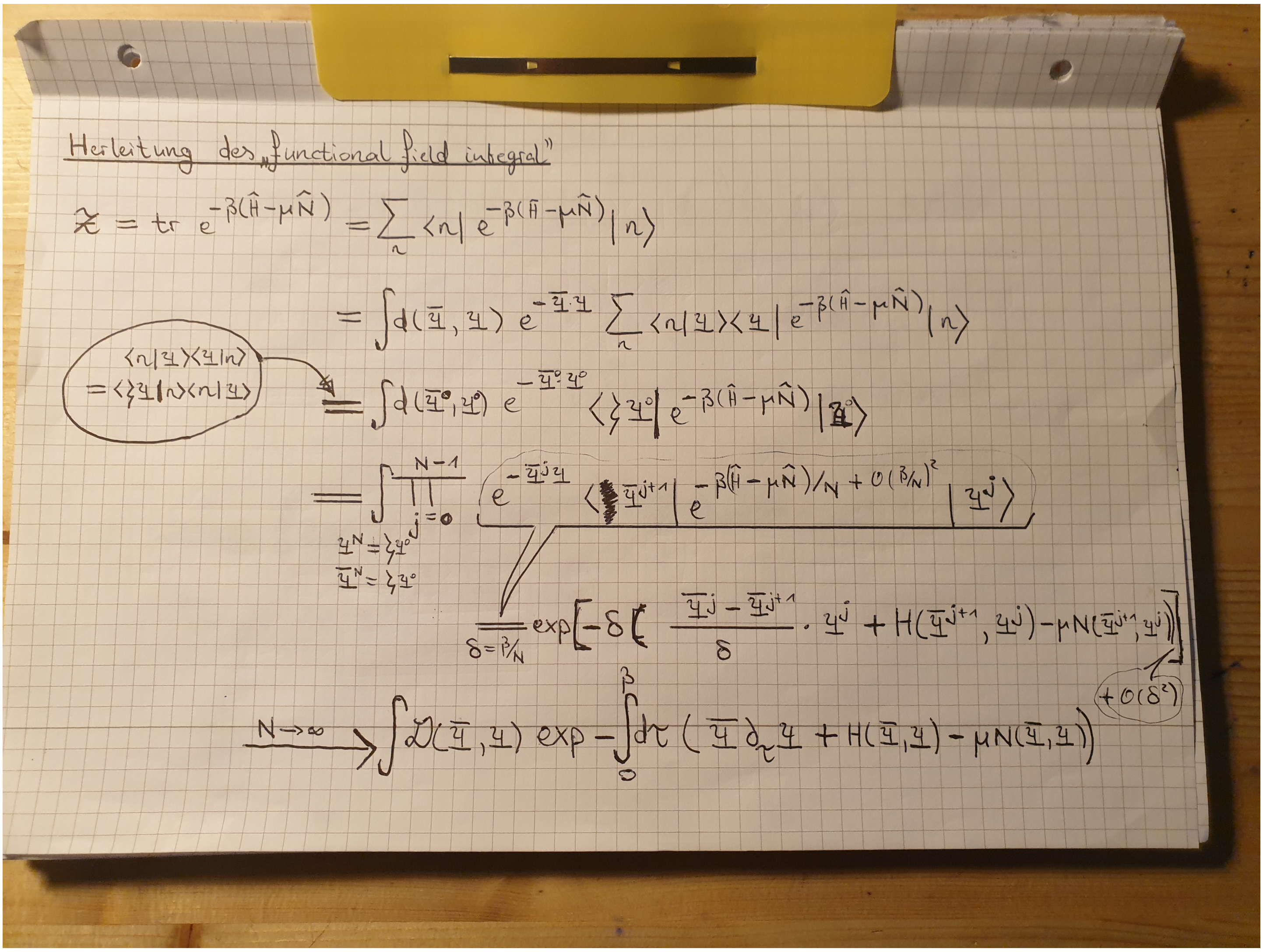
TBDIgitalized, cf. Negele Orland page 24 f.

Proof 29.22 of Lemma 19.12

By a direct computation we obtain

- TBDigitalized.

Proof 29.23 of Theorem 19.13



Proof 29.24 of Lemma 19.14

TBDigitalized.

Proof 29.25 of Corollary 19.15

The discrete version of the partition-function reads

$$Z = \lim_{N\to\infty} \int_{\substack{\psi^N=\zeta\psi^0 \\ \overline{\psi}^N=\zeta\overline{\psi}^0}} \prod_{j=0}^{N} \mathrm{d}_\zeta(\overline{\psi}^j, \psi^j) \exp -\delta \sum_{j=0}^{N-1} -\frac{\overline{\psi}^{j+1} - \overline{\psi}^j}{\delta} \cdot \psi^j + H(\overline{\psi}^{j+1}, \psi^j) - \mu N(\overline{\psi}^{j+1}, \psi^j)$$

Now we examine each of the terms in the exponents individually, when we transform coordinates according to

$$\psi^j = \frac{1}{\hbar\beta} \sum_{\omega_n} \psi_{\omega_n} \mathrm{e}^{-\mathrm{i}\omega_n\tau_j}, \qquad \overline{\psi}^j = \frac{1}{\hbar\beta} \sum_{\omega_n} \overline{\psi}_{\omega_n} \mathrm{e}^{+\mathrm{i}\omega_n\tau_j} \tag{29.31}$$

for

$$n \in \left\{-\frac{N-1}{2}, \dots, -1, 0, +1, \dots, \frac{N-1}{2}\right\} \tag{29.32}$$

We also notice that

$$\sum_{j=0}^{N-1} \mathrm{e}^{\mathrm{i}\tau_j(\omega_n-\omega_m)} = N\delta_{n,m} \tag{29.33}$$

The Derivative

Noticing that $1 \equiv \mathrm{e}^{\mathrm{i}\omega_n\tau_0}$ and remembering that $\delta = \hbar\beta/N$ we obtain

$$\begin{aligned}
&\delta \sum_{j=0}^{N-1} -\frac{1}{\delta}\left(\frac{1}{\sqrt{\hbar\beta}}\sum_{\omega_n}\overline{\psi}_{\omega_n}\mathrm{e}^{\mathrm{i}\omega_n\tau_{j+1}} - \overline{\psi}_{\omega_n}\mathrm{e}^{\mathrm{i}\omega_n\tau_j}\right)\frac{1}{\sqrt{\hbar\beta}}\sum_{\omega_m}\psi_{\omega_m}\mathrm{e}^{-\mathrm{i}\omega_m\tau_j} \\
&= \delta \sum_{j=0}^{N-1} -\frac{1}{\delta\hbar\beta}\sum_{\omega_n}\overline{\psi}_{\omega_n}\left(\mathrm{e}^{\mathrm{i}\omega_n\tau_1}-1\right)\mathrm{e}^{\mathrm{i}\omega_n\tau_j}\sum_{\omega_m}\psi_{\omega_m}\mathrm{e}^{-\mathrm{i}\omega_m\tau_j} \\
&\approx \delta \sum_{j=0}^{N-1} -\frac{1}{\delta\hbar\beta}\sum_{\omega_n}\overline{\psi}_{\omega_n}\mathrm{i}\omega_n\delta\mathrm{e}^{\mathrm{i}\omega_n\tau_j}\sum_{\omega_m}\psi_{\omega_m}\mathrm{e}^{-\mathrm{i}\omega_m\tau_j} \\
&= \frac{\delta}{\hbar\beta}\sum_{\omega_n,\omega_m} -\mathrm{i}\omega_n\overline{\psi}_{\omega_n}\psi_{\omega_m}\sum_{j=0}^{N-1}\mathrm{e}^{\mathrm{i}\tau_j(\omega_n-\omega_m)} \\
&= \sum_{\omega_n} -\mathrm{i}\omega_n\overline{\psi}_{\omega_n}\psi_{\omega_n}
\end{aligned}$$

The other Quadratic Terms

$$\begin{aligned}
\delta\sum_{j=0}^{N-1}\overline{\psi}(\tau_{j+1})\psi(\tau_j) &= \frac{\delta}{\hbar\beta}\sum_{j=0}^{N-1}\sum_{\omega_n,\omega_m}\overline{\psi}_{\omega_n}\mathrm{e}^{\mathrm{i}\omega_n\tau_{j+1}}\psi_{\omega_m}\mathrm{e}^{-\mathrm{i}\omega_m\tau_j} \\
&= \sum_{\omega_n,\omega_m}\overline{\psi}_{\omega_n}\psi_{\omega_m}\frac{\delta}{\hbar\beta}\left(\sum_{j=0}^{N-1}\mathrm{e}^{\mathrm{i}(\omega_n-\omega_m)\tau_j}\right)\mathrm{e}^{\mathrm{i}\omega_n\tau_1} \\
&= \sum_{\omega_n}\overline{\psi}_{\omega_n}\psi_{\omega_n}\mathrm{e}^{\mathrm{i}\omega_n\delta}
\end{aligned}$$

Proof 29.26 of Lemma 19.16

TBDigitalized from the Altland-book chapter 4 and Henk chapter 7.2 p. 139, 145, 149. Residue theorem etc. from Altland-book and NO eq. (2.137b) for the non-interacting case, Negele Orland p. something something for the interacting case. Cf. also Fetter, Walecka p. 37, 38.

Proof 29.27 of Theorem 19.17

- Cf. Negele p. 70 following.
- Cf. Diehl QFT-1-Script 20, 21, page 98
- Cf. Coleman p. 428.

Proof 29.28 of Theorem 19.18

We compute straightforwardly

$$\begin{aligned}\mathscr{G}_{\substack{0\\ \alpha,\alpha'}}(\tau) &= -\frac{1}{\hbar}\langle\psi_\alpha(\tau)\overline{\psi}_{\alpha'}(0)\rangle \\ &= -\frac{1}{\hbar}\frac{1}{\hbar\beta}\sum_{\omega_n,\omega_m\in\Omega_\zeta}\langle\psi_\alpha(\omega_n)\overline{\psi}_{\alpha'}(\omega_m)\rangle \mathrm{e}^{-\mathrm{i}\omega_n\tau} \\ &= \dots\end{aligned}$$

now we use that the action is diagonal w.r.t. Matsubara-frequencies s.t. we obtain

$$\langle\psi_\alpha(\omega)\overline{\psi}_{\alpha'}(\omega_m)\rangle = \delta_{n,m}\langle\psi_\alpha(\omega_n)\overline{\psi}_{\alpha'}(\omega_n)\rangle \tag{29.34}$$

and therefore we obtain

$$\dots = \frac{1}{\hbar\beta}\sum_{\omega_n\in\Omega_\zeta}\langle\psi_\alpha(\omega_n)\overline{\psi}_{\alpha'}(\omega_m)\rangle\mathrm{e}^{-\mathrm{i}\omega_n\tau} \equiv \frac{1}{\hbar\beta}\sum_{\omega_n\in\Omega_\zeta}\mathscr{G}_{\substack{0\\ \alpha,\alpha'}}(\mathrm{i}\omega_n)\mathrm{e}^{-\mathrm{i}\omega_n\tau} \tag{29.35}$$

Therefore we obtain for the non-interacting Green-function

$$\mathscr{G}_{\substack{0\\ \alpha,\alpha'}}(\mathrm{i}\omega_n) \equiv -\frac{1}{\hbar}\langle\psi_\alpha(\omega_n)\overline{\psi}_{\alpha'}(\omega_n)\rangle_0 = -\frac{1}{\hbar}\left(\left[-\mathrm{i}\omega_n + \frac{H-\mu}{\hbar}\right]^{\mathrm{inv}}\right)_{\alpha,\alpha'} = \frac{1}{\hbar}\left(\left[\mathrm{i}\omega_n - \frac{H-\mu}{\hbar}\right]^{\mathrm{inv}}\right)_{\alpha,\alpha'} \tag{29.36}$$

which completes the proof.

Proof 29.29 of Theorem 33.74

We rewrite the action w.r.t. to the new fields

$$
\begin{aligned}
S_{\mathrm{ph}}[\phi,\overline{\phi}] &= \sum \overline{\phi}_\nu(q)\left(-\mathrm{i}\omega_n+\omega_{\mathbf{q},\nu}\right)\phi_\nu(q)\\
&= \frac{1}{4}\sum_q \left(A_\nu-\Pi_\nu\right)(-q)\left[-\mathrm{i}\omega_n+\omega_\nu(\mathbf{q})\right]\left(A_\nu+\Pi_\nu\right)(q)\\
&= \frac{1}{4}\sum_q A_\nu(-q)\left[-\mathrm{i}\omega_n+\omega_\nu(\mathbf{q})\right]A_\nu(q)\\
&- \frac{1}{4}\sum_q \Pi_\nu(-q)\left[-\mathrm{i}\omega_n+\omega_\nu(\mathbf{q})\right]\Pi_\nu(+q)\\
&+ \frac{1}{4}\sum_q A_\nu(-q)\left[-\mathrm{i}\omega_n+\omega_\nu(\mathbf{q})\right]\Pi_\nu(q)\\
&- \frac{1}{4}\sum_q \Pi_\nu(-q)\left[-\mathrm{i}\omega_n+\omega_\nu(\mathbf{q})\right]A_\nu(q)\\
&\equiv \mathrm{I}+\mathrm{II}+\mathrm{III}+\mathrm{IV}\\
&= \dots
\end{aligned}
$$

Now we will examine each term individually.

$$
\mathrm{IV} = -\frac{1}{4}\sum_q A_\nu(-q)\left[+\mathrm{i}\omega_n+\omega_\nu(\mathbf{q})\right]\Pi_\nu(+q) \tag{29.37}
$$

Therefore the terms proportional to $\omega_\nu(\mathbf{q})$ in III and IV will cancel out. Due to the symmetries $q \leftrightarrow -q$ there are more cancellations happening. This gives

$$
\begin{aligned}
\dots &= \frac{1}{4}\sum_q A_\nu(-q)\left[\omega_\nu(\mathbf{q})\right]A_\nu(q)\\
&- \frac{1}{4}\sum_q \Pi_\nu(-q)\left[\omega_\nu(\mathbf{q})\right]\Pi_\nu(+q)\\
&+ \frac{1}{4}\sum_q A_\nu(-q)\left[-\mathrm{i}\omega_n\right]\Pi_\nu(q)\\
&- \frac{1}{4}\sum_q A_\nu(-q)\left[\mathrm{i}\omega_n\right]\Pi_\nu(+q)\\
&\equiv \widetilde{\mathrm{I}}+\widetilde{\mathrm{II}}+\widetilde{\mathrm{III}}+\widetilde{\mathrm{IV}}\\
&= \dots
\end{aligned}
$$

Each term gets evaluated

$$
\widetilde{\mathrm{I}} = \frac{1}{2}\sum_{q\in\Omega_{+1}\times\mathrm{Q}/\sim} A_\nu(-q)\left[\omega_\nu(\mathbf{q})\right]A_\nu(q) \tag{29.38}
$$

$$
\widetilde{\mathrm{II}} = -\frac{1}{2}\sum_{q\in\Omega_{+1}\times\mathrm{Q}/\sim} \Pi_\nu(-q)\left[\omega_\nu(\mathbf{q})\right]\Pi_\nu(+q) \tag{29.39}
$$

$$\widetilde{\mathrm{III}} + \widetilde{\mathrm{IV}} = -\frac{1}{2} \sum_q A_\nu(-q) \left[\mathrm{i}\omega_n\right] \Pi_\nu(q)$$
$$= -\frac{1}{2} \sum_{q \in \Omega_{+1} \times \mathrm{Q}/\sim} A_\nu(-q) \left[\mathrm{i}\omega_n\right] \Pi_\nu(q) + A_\nu(+q) \left[-\mathrm{i}\omega_n\right] \Pi_\nu(-q)$$

Now we focus on the terms that are proportional to Π and find by performing the integral

$$\widetilde{\mathrm{II}} + \widetilde{\mathrm{III}} + \widetilde{\mathrm{IV}} = \sum_{q \in \Omega_{+1} \times \mathrm{Q}/\sim} \left\{ \Pi_\nu(-q) \frac{-\omega_\nu(\mathbf{q})}{2} \Pi_\nu(+q) + A_\nu(-q) \frac{-\mathrm{i}\omega_n}{2} \Pi_\nu(q) + A_\nu(+q) \frac{+\mathrm{i}\omega_n}{2} \Pi_\nu(-q) \right\}$$
$$\equiv \sum_{q \in \Omega_{+1} \times \mathrm{Q}/\sim} -A_\nu(-q) \frac{-\mathrm{i}\omega_n}{2} \frac{2}{-\omega_\nu(\mathbf{q})} \frac{+\mathrm{i}\omega_n}{2} A_\nu(+q)$$
$$\equiv \sum_{q \in \Omega_{+1} \times \mathrm{Q}/\sim} A_\nu(-q) \frac{\omega_n^2}{2\omega_\nu(\mathbf{q})} A_\nu(+q)$$

and therefore

$$\ldots = \sum_{q \in \Omega_{+1} \times \mathrm{Q}/\sim} A_\nu(-q) \left[\omega_\nu(\mathbf{q}) + \frac{\omega_n^2}{2\omega_\nu(\mathbf{q})} \right] A_\nu(+q)$$
$$= \sum_{q \in \Omega_{+1} \times \mathrm{Q}/\sim} A_\nu(-q) \left[\frac{\omega_n^2 + \omega_\nu^2(\mathbf{q})}{2\omega_\nu(\mathbf{q})} \right] A_\nu(+q)$$
$$= \sum_{q \in \Omega_{+1} \times \mathrm{Q}/\sim} A_\nu(-q) \left[-\frac{1}{\hbar} \mathscr{D}_{0,\nu}^{-1}(q) \right] A_\nu(+q)$$

30. Appendix - Proofs for Part IV

Proof 30.1 of Lemma 24.3

We recall

$$\left\langle H_{0,\mathrm{el}}\right\rangle_0(\mu,T,V)=\sum_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,2,\ldots,\infty\}}}\epsilon_{m\mathbf{k}\sigma}\,\mathrm{F}_\beta(\epsilon_{m,\sigma,\mathbf{k}}-\mu) \tag{30.1}$$

and then compute the temperature-derivative as

$$\partial_T\left\langle H_{0,\mathrm{el}}\right\rangle_0(\mu,T,V)=\sum_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,2,\ldots,\infty\}}}\epsilon_{m\mathbf{k}\sigma}\partial_T\,\mathrm{F}_1\left(\beta(\epsilon_{m,\sigma,\mathbf{k}}-\mu)\right)=\ldots \tag{30.2}$$

Now we use chain and product rule to obtain

$$\begin{aligned}\partial_T\,\mathrm{F}_1\left(\beta(\epsilon_{m,\sigma,\mathbf{k}}-\mu)\right)&=\left(\partial_T\,\mathrm{F}_1\right)\left(\beta(\epsilon_{m,\sigma,\mathbf{k}}-\mu)\right) &&\cdot\,\partial_T\beta\left(\epsilon_{m,\sigma,\mathbf{k}}-\mu\right)\\ &\overset{\text{Lemma 9.44}}{=}-\frac{1}{4}\operatorname{sech}^2\left(\frac{\epsilon_{m,\sigma,\mathbf{k}}-\mu}{2k_BT}\right) &&\cdot\left(\partial_T\beta\right)(\epsilon_{m,\sigma,\mathbf{k}}-\mu)\\ &=-\frac{1}{4}\operatorname{sech}^2\left(\frac{\epsilon_{m,\sigma,\mathbf{k}}-\mu}{2k_BT}\right) &&\cdot\left(\frac{-1}{k_BT^2}\right)(\epsilon_{m,\sigma,\mathbf{k}}-\mu)\end{aligned} \tag{30.3}$$

which allows us to write the semi-final result as

$$\partial_T\langle H_{0,\mathrm{el}}\rangle_0(\mu,T,V)=\ldots=\sum_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\ldots,\infty\}}}\frac{\epsilon_{m\mathbf{k}\sigma}}{4}\operatorname{sech}^2\left(\frac{\epsilon_{m\mathbf{k}\sigma}-\mu}{2k_BT}\right)\left[\frac{\epsilon_{m\mathbf{k}\sigma}-\mu}{k_BT^2}\right]=\star\star\star \tag{30.4}$$

Now we introduce

$$\rho(E)=\sum_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,2,\ldots,\infty\}}}\delta(E-\xi_{m\mathbf{k}\sigma})=\sum_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,2,\ldots,\infty\}}}\delta(E+\mu-\epsilon_{m\mathbf{k}\sigma}) \tag{30.5}$$

and

$$\rho_{\mathrm{aux}}(E)=\sum_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,2,\ldots,\infty\}}}\delta(E-\epsilon_{m\mathbf{k}\sigma}) \tag{30.6}$$

which implies $\rho_{\text{aux}}(E+\mu) = \rho(E)$. This allows us to rewrite the expression as

$$\begin{aligned}\partial_T \langle H_{0,\text{el}}\rangle_0 = \star &= \int \mathrm{d}E\, \rho(E) \frac{E+\mu}{4} \operatorname{sech}^2\left(\frac{E}{2k_B T}\right) \quad \left[\frac{E}{k_B T^2}\right] \\ &= \int \mathrm{d}E\, \rho_{\text{aux}}(E) \frac{E}{4} \operatorname{sech}^2\left(\frac{E-\mu}{2k_B T}\right) \quad \left[\frac{E-\mu}{k_B T^2}\right] \\ = \star\star\end{aligned} \tag{30.7}$$

We also recall the definition of the density of states from quantum espresso as

$$\rho_{\text{QE}}(E) = \frac{1}{N_p} \sum_{\substack{\mathbf{k}\in \mathrm{Q} \\ \sigma\in\{\uparrow,\downarrow\} \\ m\in\{1,2,\ldots,\infty\}}} \delta(E - \epsilon_{m\mathbf{k}\sigma}) = \frac{1}{N_p} \rho_{\text{aux}}(E) \tag{30.8}$$

such that we obtain the final result

$$\frac{1}{N_x N_y N_z} \partial_T \langle H_{0,\text{el}}\rangle_0(\mu, T, V) = \int \mathrm{d}E\, \rho_{\text{QE}}(E) \frac{E}{4} \operatorname{sech}^2\left(\frac{E-\mu}{2k_B T}\right) \left[\frac{E-\mu(T)}{k_B T^2}\right] \tag{30.9}$$

Proof 30.2 of Lemma 24.4 We compute the derivative w.r.t. the chemical potential and obtain the desired result with Lemma 9.44 and the chain-rule.

Proof 30.3 of Lemma ??

The only non-trivial identity is the interaction-term.

$$\begin{aligned}S_{\text{el-ph-int}}[\chi,\overline{\chi}] &= \sum\!\!\!\!\!\!\!\!\int_{\substack{\mathbf{k},\mathbf{q}\in \mathrm{Q} \\ \tau\in(0,\hbar\beta) \\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\} \\ m,n\in\{1,2,3,\ldots\}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})}{\hbar\sqrt{N_p}} \left(\phi_\nu(\tau,\mathbf{q}) + \overline{\phi}_\nu(\tau,-\mathbf{q})\right) \overline{\psi}_{m\sigma}(\tau,\mathbf{k}+\mathbf{q}) \psi_{n\sigma}(\tau,\mathbf{k}) \\ &= \sum\!\!\!\!\!\!\!\!\int_{\substack{\mathbf{k},\mathbf{q}\in \mathrm{Q} \\ \tau\in(0,\hbar\beta) \\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\} \\ m,n\in\{1,2,3,\ldots\}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})}{\hbar\sqrt{N_p}} \phi_\nu(\tau,\mathbf{q}) \overline{\psi}_{m\sigma}(\tau,\mathbf{k}+\mathbf{q}) \psi_{n\sigma}(\tau,\mathbf{k}) \\ &+ \sum\!\!\!\!\!\!\!\!\int_{\substack{\mathbf{k},\mathbf{q}\in \mathrm{Q} \\ \tau\in(0,\hbar\beta) \\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\} \\ m,n\in\{1,2,3,\ldots\}}} \frac{g_{mn\nu}(\mathbf{k},-\mathbf{q})}{\hbar\sqrt{N_p}} \overline{\phi}_\nu(\tau,\mathbf{q}) \overline{\psi}_{m\sigma}(\tau,\mathbf{k}-\mathbf{q}) \psi_{n\sigma}(\tau,\mathbf{k}) \\ &= \frac{1}{(\hbar\beta)^{3/2}} \sum_{\substack{\omega_q\in\Omega_{+1} \\ \omega_n,\omega_k\in \mathrm{i}\Omega_{-1}}} \sum_{\substack{\mathbf{k},\mathbf{q}\in \mathrm{Q} \\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\} \\ m,n\in\{1,2,3,\ldots\}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})}{\hbar\sqrt{N_p}} \phi_\nu(\mathrm{i}\omega_q,\mathbf{q}) \overline{\psi}_{m\sigma}(\mathrm{i}\omega_n,\mathbf{k}+\mathbf{q}) \psi_{n\sigma}(\mathrm{i}\omega_k,\mathbf{k}) \int_0^{\hbar\beta} \mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}\tau\left(-\omega_q+\omega_n-\omega_k\right)} \\ &+ \frac{1}{(\hbar\beta)^{3/2}} \sum_{\substack{\omega_q\in\Omega_{+1} \\ \omega_n,\omega_k\in \mathrm{i}\Omega_{-1}}} \sum_{\substack{\mathbf{k},\mathbf{q}\in \mathrm{Q} \\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\} \\ m,n\in\{1,2,3,\ldots\}}} \frac{g_{mn\nu}(\mathbf{k},-\mathbf{q})}{\hbar\sqrt{N_p}} \overline{\phi}_\nu(\mathrm{i}\omega_q,\mathbf{q}) \overline{\psi}_{m\sigma}(\mathrm{i}\omega_n,\mathbf{k}-\mathbf{q}) \psi_{n\sigma}(\mathrm{i}\omega_k,\mathbf{k}) \int_0^{\hbar\beta} \mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}\tau\left(+\omega_q+\omega_n-\omega_k\right)}\end{aligned}$$

$$= \frac{1}{\sqrt{\hbar\beta}} \sum_{\substack{\omega_q \in \Omega_{+1} \\ \omega_n,\omega_k \in \mathrm{i}\Omega_{-1}}} \sum_{\substack{\mathbf{k},\mathbf{q} \in \mathrm{Q} \\ \nu \in \{1,2,\ldots,rd_{\mathbf{x}}\} \\ m,n \in \{1,2,3,\ldots\}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})}{\hbar\sqrt{N_p}} \phi_\nu(\mathrm{i}\omega_q,\mathbf{q}) \overline{\psi}_{m\sigma}(\mathrm{i}\omega_n,\mathbf{k}+\mathbf{q}) \psi_{n\sigma}(\mathrm{i}\omega_k,\mathbf{k})\, \delta_{\omega_n,\omega_k+\omega_q}$$

$$+ \frac{1}{\sqrt{\hbar\beta}} \sum_{\substack{\omega_q \in \Omega_{+1} \\ \omega_n,\omega_k \in \mathrm{i}\Omega_{-1}}} \sum_{\substack{\mathbf{k},\mathbf{q} \in \mathrm{Q} \\ \nu \in \{1,2,\ldots,rd_{\mathbf{x}}\} \\ m,n \in \{1,2,3,\ldots\}}} \frac{g_{mn\nu}(\mathbf{k},-\mathbf{q})}{\hbar\sqrt{N_p}} \overline{\phi}_\nu(\mathrm{i}\omega_q,\mathbf{q}) \overline{\psi}_{m\sigma}(\mathrm{i}\omega_n,\mathbf{k}-\mathbf{q}) \psi_{n\sigma}(\mathrm{i}\omega_k,\mathbf{k})\, \delta_{\omega_n,\omega_k-\omega_q}$$

$$= \frac{1}{\sqrt{\hbar\beta}} \sum_{\substack{k \in \mathrm{Q}\times\Omega_{-1} \\ q \in \mathrm{Q}\times\Omega_{+1} \\ \nu \in \{1,2,\ldots,rd_{\mathbf{x}}\} \\ m,n \in \{1,2,3,\ldots\}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})}{\hbar\sqrt{N_p}} \left(\phi_\nu(q) + \overline{\phi}_\nu(-q)\right) \overline{\psi}_{m\sigma}(q+k) \psi_{n\sigma}(k)$$

Proof 30.4 of Lemma 22.20

By straightforward computation we obtain

$$\begin{aligned} S_{\text{el-ph-int}}[\chi,\overline{\chi}] &= \frac{1}{\sqrt{\hbar\beta}} \sum_{\substack{\omega_q \in \Omega_{+1} \\ \omega_k \in \mathrm{i}\Omega_{-1}}} \sum \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})}{\hbar\sqrt{N_p}} \phi_\nu(\mathrm{i}\omega_q,\mathbf{q}) \overline{\psi}_{m\sigma}(\mathrm{i}\left(\omega_k+\omega_q\right),\mathbf{k}+\mathbf{q}) \psi_{n\sigma}(\mathrm{i}\omega_k,\mathbf{k}) \\ &+ \frac{1}{\sqrt{\hbar\beta}} \sum_{\substack{\omega_q \in \Omega_{+1} \\ \omega_k \in \mathrm{i}\Omega_{-1}}} \sum \frac{g_{mn\nu}(\mathbf{k},-\mathbf{q})}{\hbar\sqrt{N_p}} \overline{\phi}_\nu(\mathbf{q},\mathrm{i}\omega_q) \overline{\psi}_{m\sigma}(\mathbf{k}-\mathbf{q},\mathrm{i}\left(\omega_k-\omega_q\right)) \psi_{n\sigma}(\mathbf{k},\mathrm{i}\omega_k) \\ &= \sum_{\substack{\mathbf{q} \in \mathrm{Q} \\ \omega_q \in \Omega_{+1} \\ \nu \in \{1,\ldots,rd_{\mathbf{x}}\}}} \phi_\nu(\mathrm{i}\omega_q,\mathbf{q}) \rho_\nu(\mathrm{i}\omega_q,\mathbf{q}) \quad + \quad \overline{\phi}_\nu(\mathrm{i}\omega_q,\mathbf{q}) \rho_\nu(-\mathrm{i}\omega_q,-\mathbf{q}) \end{aligned}$$

with density-like-interaction

$$\rho_\nu(\mathrm{i}\omega_q,\mathbf{q}) = \frac{1}{\sqrt{\hbar\beta N_p}} \sum_{\substack{\mathbf{k} \in \mathrm{Q} \\ \sigma \in \{\downarrow,\uparrow\} \\ \omega_k \in \mathrm{i}\Omega_{-1} \\ n,m \in \{1,\ldots,\infty\}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})}{\hbar} \overline{\psi}_{m\sigma}(\mathrm{i}\left(\omega_k+\omega_q\right),\mathbf{k}+\mathbf{q}) \psi_{n\sigma}(\mathrm{i}\omega_k,\mathbf{k}) \tag{30.10}$$

Proof 30.5 of Lemma 22.22

We keep in mind that

$$S_{\text{ph}}[\phi,\overline{\phi}] = \sum_{\substack{q \in \mathrm{i}\Omega_{+1}\times\mathrm{Q} \\ \nu \in \{1,2,\ldots,rd_{\mathbf{x}}\}}} \overline{\phi}_\nu(q) \left(-\frac{1}{\hbar} \mathscr{G}_{\substack{0 \\ \nu}}^{-\mathrm{B}}(q)\right) \phi_\nu(q) \tag{30.11}$$

and

$$\underset{[j_{\mathrm{B}},\overline{j}_{\mathrm{B}}]}{S_{\text{el-ph-int}}}[\chi,\overline{\chi}] = \sum_{\substack{q \in \mathrm{i}\Omega_{+1}\times\mathrm{Q} \\ \nu \in \{1,\ldots,rd_{\mathbf{x}}\}}} \phi_\nu(q) \left(\rho_\nu(q) - \overline{j}_{\substack{\mathrm{B} \\ \nu}}(q)\right) + \overline{\phi}_\nu(q) \left(\rho_\nu(-q) - j_{\substack{\mathrm{B} \\ \nu}}(q)\right)$$

Now we change the variables according to

$$
\begin{aligned}
\phi_\nu(q) &\mapsto \phi_\nu(q) + \hbar \underset{\nu}{\mathscr{G}_0^{\mathrm{B}}}(q) \left(\rho_\nu(-q) - \underset{\nu}{j_{\mathrm{B}}}(q) \right) \\
\overline{\phi}_\nu(q) &\mapsto \overline{\phi}_\nu(q) + \hbar \underset{\nu}{\mathscr{G}_0^{\mathrm{B}}}(q) \left(\rho_\nu(q) - \underset{\nu}{\overline{j}_{\mathrm{B}}}(q) \right)
\end{aligned}
$$

This yields

$$
\begin{aligned}
\left(S_{\mathrm{ph}} + S_{\text{el-ph}} \right) [\overline{\chi}, \chi] = & \sum_{\substack{q \in \mathrm{i}\Omega_{+1} \times \mathrm{Q} \\ \nu \in \{1,\dots,rd_{\mathbf{x}}\}}} \left[\overline{\phi}_\nu(q) + \hbar \underset{\nu}{\mathscr{G}_0^{\mathrm{B}}}(q) \left(\rho_\nu(q) - \underset{\nu}{\overline{j}_{\mathrm{B}}}(q) \right) \right] \left(-\frac{1}{\hbar} \underset{\nu}{\mathscr{G}_0^{-\mathrm{B}}}(q) \right) \left[\phi_\nu(q) + \hbar \underset{\nu}{\mathscr{G}_0^{\mathrm{B}}}(q) \left(\rho_\nu(-q) - \underset{\nu}{j_{\mathrm{B}}}(q) \right) \right. \\
& + \sum_{\substack{q \in \mathrm{i}\Omega_{+1} \times \mathrm{Q} \\ \nu \in \{1,\dots,rd_{\mathbf{x}}\}}} \left[\phi_\nu(q) + \hbar \underset{\nu}{\mathscr{G}_0^{\mathrm{B}}}(q) \left(\rho_\nu(-q) - \underset{\nu}{j_{\mathrm{B}}}(q) \right) \right] \left(\rho_\nu(q) - \underset{\nu}{\overline{j}_{\mathrm{B}}}(q) \right) \\
& + \sum_{\substack{q \in \mathrm{i}\Omega_{+1} \times \mathrm{Q} \\ \nu \in \{1,\dots,rd_{\mathbf{x}}\}}} \left[\overline{\phi}_\nu(q) + \hbar \underset{\nu}{\mathscr{G}_0^{\mathrm{B}}}(q) \left(\rho_\nu(q) - \underset{\nu}{\overline{j}_{\mathrm{B}}}(q) \right) \right] \left(\rho_\nu(-q) - \underset{\nu}{j_{\mathrm{B}}}(q) \right)
\end{aligned}
$$

Now we obtain 8 terms.

1.
$$
\sum_{\substack{q \in \mathrm{i}\Omega_{+1} \times \mathrm{Q} \\ \nu \in \{1,\dots,rd_{\mathbf{x}}\}}} \overline{\phi}_\nu(q) \left(-\frac{1}{\hbar} \underset{\nu}{\mathscr{G}_0^{-\mathrm{B}}}(q) \right) \phi_\nu(q) \tag{30.12}
$$

2.
$$
-\hbar \sum_{\substack{q \in \mathrm{i}\Omega_{+1} \times \mathrm{Q} \\ \nu \in \{1,\dots,rd_{\mathbf{x}}\}}} \left(\rho_\nu(q) - \underset{\nu}{\overline{j}_{\mathrm{B}}}(q) \right) \underset{\nu}{\mathscr{G}_0^{\mathrm{B}}}(q) \left(\rho_\nu(-q) - \underset{\nu}{j_{\mathrm{B}}}(q) \right) \tag{30.13}
$$

3.
$$
- \sum_{\substack{q \in \mathrm{i}\Omega_{+1} \times \mathrm{Q} \\ \nu \in \{1,\dots,rd_{\mathbf{x}}\}}} \left(\rho_\nu(q) - \underset{\nu}{\overline{j}_{\mathrm{B}}}(q) \right) \phi_\nu(q) \tag{30.14}
$$

4.
$$
- \sum_{\substack{q \in \mathrm{i}\Omega_{+1} \times \mathrm{Q} \\ \nu \in \{1,\dots,rd_{\mathbf{x}}\}}} \overline{\phi}_\nu(q) \left(\rho_\nu(-q) - \underset{\nu}{j_{\mathrm{B}}}(q) \right)
$$

5.
$$
\sum_{\substack{q \in \mathrm{i}\Omega_{+1} \times \mathrm{Q} \\ \nu \in \{1,\dots,rd_{\mathbf{x}}\}}} \phi_\nu(q) \left(\rho_\nu(q) - \underset{\nu}{\overline{j}_{\mathrm{B}}}(q) \right) \tag{30.15}
$$

6.
$$
\hbar \sum_{\substack{q \in \mathrm{i}\Omega_{+1} \times \mathrm{Q} \\ \nu \in \{1,\dots,rd_{\mathbf{x}}\}}} \left(\rho_\nu(-q) - \underset{\nu}{j_{\mathrm{B}}}(q) \right) \underset{\nu}{\mathscr{G}_0^{\mathrm{B}}}(q) \left(\rho_\nu(q) - \underset{\nu}{\overline{j}_{\mathrm{B}}}(q) \right) \tag{30.16}
$$

7.

$$\sum_{\substack{q\in \mathrm{i}\Omega_{+1}\times \mathrm{Q}\\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} \overline{\phi}_\nu(q)\left(\rho_\nu(-q)-\underset{\nu}{j_{\mathrm{B}}}(q)\right) \tag{30.17}$$

8.

$$\hbar\sum_{\substack{q\in \mathrm{i}\Omega_{+1}\times \mathrm{Q}\\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} \left(\rho_\nu(q)-\underset{\nu}{\overline{j}_{\mathrm{B}}}(q)\right)\underset{\nu}{\mathscr{G}_0^{\mathrm{B}}}(q)\left(\rho_\nu(-q)-\underset{\nu}{j_{\mathrm{B}}}(q)\right) \tag{30.18}$$

Now we obtain

$$\text{"}2+6=0=3+5=4+7\text{"} \tag{30.19}$$

such that only 1 and 8 contribute. Then, when one integrates out the phonons, one obtains the effective electron-electron interaction, giving us

$$\underset{[j_{\mathrm{B}},\overline{j}_{\mathrm{B}}]}{S_{\text{el-el-int}}}[\psi,\overline{\psi}]=+\hbar\sum_{\substack{q\in \mathrm{i}\Omega_{+1}\times \mathrm{Q}\\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} \left(\rho_\nu(q)-\underset{\nu}{\overline{j}_{\mathrm{B}}}(q)\right)\underset{\nu}{\mathscr{G}_0^{\mathrm{B}}}(q)\left(\rho_\nu(-q)-\underset{\nu}{j_{\mathrm{B}}}(q)\right)$$

For the latter identity: We compute straightforwardly

$$\begin{aligned}
S_{\text{el-el-int}}[\psi,\overline{\psi}] &= +\hbar\sum_{\substack{\mathbf{q}\in \mathrm{Q}\\ \omega_q\in\Omega_{+1}\\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} \rho_\nu(\mathbf{q},\mathrm{i}\omega_q)\underset{\nu}{\mathscr{G}_0^{\mathrm{B}}}(\mathbf{q},\mathrm{i}\omega_q)\rho_\nu(-\mathbf{q},-\mathrm{i}\omega_q)\\
&= +\sum_{\substack{\mathbf{q}\in \mathrm{Q}\\ \omega_q\in\Omega_{+1}\\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} \frac{\rho_\nu(\mathbf{q},\mathrm{i}\omega_q)\rho_\nu(-\mathbf{q},-\mathrm{i}\omega_q)}{\mathrm{i}\omega_q-\omega_\nu(\mathbf{q})}\\
&= -\sum_{\substack{\mathbf{q}\in \mathrm{Q}\\ \omega_q\in\Omega_{+1}\\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} \frac{\rho_\nu(\mathbf{q},\mathrm{i}\omega_q)\rho_\nu(-\mathbf{q},-\mathrm{i}\omega_q)}{\omega_q^2+\omega_\nu^2(\mathbf{q})}\left(\mathrm{i}\omega_q+\omega_\nu(\mathbf{q})\right)\\
&= -\sum_{\substack{\mathbf{q}\in \mathrm{Q}\\ \omega_q\in\Omega_{+1}\\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} \frac{\omega_\nu(\mathbf{q})}{\omega_q^2+\omega_\nu^2(\mathbf{q})}\rho_\nu(\mathbf{q},\mathrm{i}\omega_q)\rho_\nu(-\mathbf{q},-\mathrm{i}\omega_q)
\end{aligned}$$

Proof 30.6 of Lemma 22.23

Now we simplify the interaction-term

$$\rho_\nu(\mathbf{q},\mathrm{i}\omega_q)\rho_\nu(-\mathbf{q},-\mathrm{i}\omega_q)=\frac{1}{\hbar\beta N_p}\sum_{\substack{\mathbf{k},\mathbf{k}'\in \mathrm{Q}\\ \omega_k,\omega_{k'}\in\mathrm{i}\Omega_{-1}\\ \sigma,\sigma'\in\{\downarrow,\uparrow\}\\ n,n',m,m'\in\{1,\dots,\infty\}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})g_{m'n'\nu}(\mathbf{k}',-\mathbf{q})}{\hbar^2}\overline{\psi}_{m\sigma}(\mathbf{k}+\mathbf{q},\mathrm{i}\left(\omega_k+\omega_q\right))\psi_{n\sigma}(\mathbf{k},\mathrm{i}\omega_k)\overline{\psi}_{m'\sigma'}(\mathbf{k}'-\mathbf{q},\mathrm{i}\left(\omega_{k'}-\omega_q\right))\psi_{n'\sigma'}(\mathbf{k}',\mathrm{i}\omega_{k'}) \tag{30.20}$$

introducing the four-momentum we can write this shorter as

$$
\begin{aligned}
\rho_\nu(q)\rho_\nu(-q) &= \frac{1}{\hbar\beta N_p} \sum_{\substack{k,k'\in Q\times\Omega_{-1}\\ \sigma,\sigma'\in\{\downarrow,\uparrow\}\\ n,n',m,m'\in\{1,\ldots,\infty\}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})g_{m'n'\nu}(\mathbf{k}',-\mathbf{q})}{\hbar^2}\overline{\psi}_{m\sigma}(k+q)\psi_{n\sigma}(k)\overline{\psi}_{m'\sigma'}(k'-q)\psi_{n'\sigma'}(k') \\
&= \frac{-1}{\hbar\beta N_p} \sum_{\substack{k,k'\in Q\times\Omega_{-1}\\ \sigma,\sigma'\in\{\downarrow,\uparrow\}\\ n,n',m,m'\in\{1,\ldots,\infty\}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})g_{m'n'\nu}(\mathbf{k}',-\mathbf{q})}{\hbar^2}\overline{\psi}_{m\sigma}(k+q)\overline{\psi}_{m'\sigma'}(k'-q)\psi_{n\sigma}(k)\psi_{n'\sigma'}(k')
\end{aligned}
$$

Therefore we obtain for the interaction-term

$$
\begin{aligned}
S_{\text{el-el-int}}[\psi,\overline{\psi}] &= \frac{+1}{\hbar\beta N_p} \sum_{\substack{\mathbf{q}\in Q\\ \omega_q\in\Omega_{+1}\\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}\\ k,k'\in Q\times\Omega_{-1}\\ \sigma,\sigma'\in\{\downarrow,\uparrow\}\\ n,n',m,m'\in\{1,\ldots,\infty\}}} \frac{\omega_\nu(\mathbf{q})}{\omega_q^2+\omega_\nu^2(\mathbf{q})}\frac{g_{mn\nu}(\mathbf{k},\mathbf{q})g_{m'n'\nu}(\mathbf{k}',-\mathbf{q})}{\hbar^2}\overline{\psi}_{m\sigma}(k+q)\overline{\psi}_{m'\sigma'}(k'-q)\psi_{n\sigma}(k)\psi_{n'\sigma'}(k') \\
&= \sum_{\substack{\mathbf{q}\in Q\\ \omega_q\in\Omega_{+1}\\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}\\ k,k'\in Q\times\Omega_{-1}\\ \sigma,\sigma'\in\{\downarrow,\uparrow\}\\ n,n',m,m'\in\{1,\ldots,\infty\}}} \overline{\psi}_{m\sigma}(k+q)\overline{\psi}_{m'\sigma'}(k'-q)\frac{+1}{\hbar\beta N_p}\frac{\omega_\nu(\mathbf{q})}{\omega_q^2+\omega_\nu^2(\mathbf{q})}\frac{g_{mn\nu}(\mathbf{k},\mathbf{q})g_{m'n'\nu}(\mathbf{k}',-\mathbf{q})}{\hbar^2}\psi_{n\sigma}(k)\psi_{n'\sigma'}(k') \\
&= \sum_{\substack{\mathbf{q}\in Q\\ \omega_q\in\Omega_{+1}\\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}\\ k,k'\in Q\times\Omega_{-1}\\ \sigma,\sigma'\in\{\downarrow,\uparrow\}\\ n,n',m,m'\in\{1,\ldots,\infty\}}} \overline{\psi}_{m\sigma}(k+q)\overline{\psi}_{m'\sigma'}(k')\frac{-1}{\hbar\beta N_p}\frac{\omega_\nu(\mathbf{q})}{\omega_q^2+\omega_\nu^2(\mathbf{q})}\frac{g_{mn\nu}(\mathbf{k},\mathbf{q})g_{m'n'\nu}(\mathbf{k}'+\mathbf{q},-\mathbf{q})}{\hbar^2}\psi_{n'\sigma'}(k'+q)\psi_{n\sigma}(k) \\
&= \sum_{\substack{\mathbf{q}\in Q\\ \omega_q\in\Omega_{+1}\\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}\\ k,k'\in Q\times\Omega_{-1}\\ \sigma,\sigma'\in\{\downarrow,\uparrow\}\\ n,n',m,m'\in\{1,\ldots,\infty\}}} \overline{\psi}_{m\sigma}(k+q)\overline{\psi}_{m'\sigma'}(k')\frac{-1}{\hbar\beta N_p}\frac{\omega_\nu(\mathbf{q})}{\omega_q^2+\omega_\nu^2(\mathbf{q})}\frac{g_{mn\nu}(\mathbf{k},\mathbf{q})g^*_{n'm'\nu}(\mathbf{k}',\mathbf{q})}{\hbar^2}\psi_{n'\sigma'}(k'+q)\psi_{n\sigma}(k)
\end{aligned}
$$

Now we introduce[1] the new fermionic Matsubara summation-variable $p \equiv k+q$ in order to replace the summation over q. This yields

[1]More concretely: For each k we introduce a p such that the equation $p \equiv k+q$ is fulfilled.

$$
\begin{aligned}
&S_{\text{el-el-int}}[\psi, \overline{\psi}] \\
&= \sum_{\substack{\nu \in \{1,\dots,rd_{\mathbf{x}}\} \\ p,k,k' \in Q\times\Omega_{-1} \\ \sigma,\sigma' \in \{\downarrow,\uparrow\} \\ n,n',m,m' \in \{1,\dots,\infty\}}} \overline{\psi}_{m\sigma}(p)\overline{\psi}_{m'\sigma'}(k')\frac{-1}{\hbar\beta N_p}\frac{\omega_\nu(\mathbf{p}-\mathbf{k})}{\omega^2_{p_0-k_0}+\omega^2_\nu(\mathbf{p}-\mathbf{k})}\frac{g_{mn\nu}(\mathbf{k},\mathbf{p}-\mathbf{k})g^*_{n'm'\nu}(\mathbf{k}',\mathbf{p}-\mathbf{k})}{\hbar^2}\psi_{n'\sigma'}(k'+p-k)\psi_{n\sigma}(k) \\
&= \sum_{\substack{p,k,k' \in Q\times\Omega_{-1} \\ \sigma,\sigma' \in \{\downarrow,\uparrow\} \\ n,n',m,m' \in \{1,\dots,\infty\}}} \overline{\psi}_{m\sigma}(p)\overline{\psi}_{m'\sigma'}(k')\frac{-1}{\hbar\beta N_p}\sum_{\nu \in \{1,\dots,rd_{\mathbf{x}}\}}\frac{\omega_\nu(\mathbf{p}-\mathbf{k})}{\omega^2_{p_0-k_0}+\omega^2_\nu(\mathbf{p}-\mathbf{k})}\frac{g_{mn\nu}(\mathbf{k},\mathbf{p}-\mathbf{k})g^*_{n'm'\nu}(\mathbf{k}',\mathbf{p}-\mathbf{k})}{\hbar^2}\psi_{n'\sigma'}(k'+p-k)\psi_{n\sigma}(k) \\
&= \sum_{\substack{p,p',k,k' \in Q\times\Omega_{-1} \\ \sigma,\sigma' \in \{\downarrow,\uparrow\} \\ n,n',m,m' \in \{1,\dots,\infty\}}} \overline{\psi}_{m\sigma}(p)\overline{\psi}_{m'\sigma'}(k')V_{m,m';n',n}(p,k';p',k)\psi_{n'\sigma'}(p')\psi_{n\sigma}(k)
\end{aligned} \tag{30.21}
$$

with

$$
V_{m,m';n',n}(p,k';p',k) = \frac{-1}{\hbar\beta N_p}\sum_{\nu \in \{1,\dots,rd_{\mathbf{x}}\}}\frac{\omega_\nu(\mathbf{p}-\mathbf{k})}{\omega^2_{p_0-k_0}+\omega^2_\nu(\mathbf{p}-\mathbf{k})}\frac{g_{mn\nu}(\mathbf{k},\mathbf{p}-\mathbf{k})g^*_{n'm'\nu}(\mathbf{k}',\mathbf{p}-\mathbf{k})}{\hbar^2}\delta_{p',k'+p-k} \tag{30.22}
$$

Now we will simplify this expression, by relabelling the indices. This yields

$$
\begin{aligned}
&S_{\text{el-el-int}}[\psi, \overline{\psi}] \\
&= \sum_{\substack{p,k,k' \in Q\times\Omega_{-1} \\ \sigma,\sigma' \in \{\downarrow,\uparrow\} \\ n,n',m,m' \in \{1,\dots,\infty\}}} \overline{\psi}_{m\sigma}(p)\overline{\psi}_{m'\sigma'}(p')V_{m,m';n',n}(p,p';k',k)\psi_{n'\sigma'}(k')\psi_{n\sigma}(k)
\end{aligned}
$$

with

$$
V_{m,m';n',n}(p,p';k',k) = \frac{-1}{\hbar\beta N_p}\sum_{\nu \in \{1,\dots,rd_{\mathbf{x}}\}}\frac{\omega_\nu(\mathbf{p}-\mathbf{k})}{\omega^2_{p_0-k_0}+\omega^2_\nu(\mathbf{p}-\mathbf{k})}\frac{g_{mn\nu}(\mathbf{k},\mathbf{p}-\mathbf{k})g^*_{n'm'\nu}(\mathbf{p}',\mathbf{p}-\mathbf{k})}{\hbar^2}\delta_{k',p'+p-k} \tag{30.23}
$$

Now we rewrite this even further to find a bilinearform

$$
\begin{aligned}
&S_{\text{el-el-int}}[\psi, \overline{\psi}] \\
&= \sum_{\substack{p,k,k' \in Q\times\Omega_{-1} \\ \sigma,\sigma',\pi,\pi' \in \{\downarrow,\uparrow\} \\ n,n',m,m' \in \{1,\dots,\infty\}}} \overline{\psi}_{m\sigma}(p)\overline{\psi}_{m'\sigma'}(p')V^{\sigma,\sigma',\pi',\pi}_{m,m';n',n}(p,p';k',k)\psi_{n'\pi'}(k')\psi_{n\pi}(k)
\end{aligned}
$$

with

$$V^{\sigma,\sigma',\pi',\pi}_{m,m';n',n}(p,p';k',k) = \frac{-1}{\hbar\beta N_p}\sum_{\nu\in\{1,\dots,rd_{\mathbf{x}}\}}\frac{\omega_\nu(\mathbf{p}-\mathbf{k})}{\omega^2_{p_0-k_0}+\omega^2_\nu(\mathbf{p}-\mathbf{k})}\frac{g_{mn\nu}(\mathbf{k},\mathbf{p}-\mathbf{k})g^*_{n'm'\nu}(\mathbf{p}',\mathbf{p}-\mathbf{k})}{\hbar^2}\delta_{k'+k,p'+p}\delta_{\sigma,\pi}\delta_{\sigma',\pi'} \tag{30.24}$$

Now we condense the notation even further, by introducing the multi-index

$$\alpha \equiv (m_\alpha,\sigma_\alpha,\omega_\alpha,\mathbf{k}_\alpha)\in\{1,2,\dots\}\times\{\downarrow,\uparrow\}\times\Omega_{-1}\times\mathrm{Q}\equiv\Xi, \tag{30.25}$$

which then yields (notice the exchanged order of the Grassmann-fields)

$$S_{\text{el-el-int}}[\psi,\overline{\psi}] = \sum_{\alpha_1,\dots,\alpha_4\in\Xi}\overline{\psi}_{\alpha_1}\overline{\psi}_{\alpha_2}V_{\alpha_1,\alpha_2;\alpha_3,\alpha_4}\psi_{\alpha_4}\psi_{\alpha_3}$$

with Lemma 8.30

$$\begin{aligned}&V_{\alpha_1,\alpha_2;\alpha_3,\alpha_4}\\ &=\frac{1}{\hbar\beta N_p}\sum_{\nu\in\{1,\dots,rd_{\mathbf{x}}\}}\frac{\omega_\nu(\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4})}{\omega^2_{\alpha_1-\alpha_4}+\omega^2_\nu(\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4})}\frac{g_{m_{\alpha_1}m_{\alpha_4}\nu}(\mathbf{k}_{\alpha_4},\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4})g^*_{m_{\alpha_3}m_{\alpha_2}\nu}(\mathbf{k}_{\alpha_2},\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4})}{\hbar^2}\delta_{k_{\alpha_3}+k_{\alpha_4},k_{\alpha_2}+k_{\alpha_1}}\delta_{\sigma_{\alpha_1},\sigma_{\alpha_4}}\delta_{\sigma_{\alpha_2},\sigma_{\alpha_3}}\\ &=\frac{1}{\hbar\beta N_p}\sum_{\nu\in\{1,\dots,rd_{\mathbf{x}}\}}\frac{\omega_\nu(\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4})}{\omega^2_{\alpha_1-\alpha_4}+\omega^2_\nu(\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4})}\frac{g_{m_{\alpha_1}m_{\alpha_4}\nu}(\mathbf{k}_{\alpha_4},\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4})g_{m_{\alpha_2}m_{\alpha_3}\nu}(\mathbf{k}_{\alpha_2}+\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4},-\mathbf{k}_{\alpha_1}+\mathbf{k}_{\alpha_4})}{\hbar^2}\delta_{k_{\alpha_3}+k_{\alpha_4},k_{\alpha_2}+k_{\alpha_1}}\delta_{\sigma_{\alpha_1},\sigma_{\alpha_4}}\delta_{\sigma_{\alpha_2},\sigma_{\alpha_3}}\end{aligned} \tag{30.26}$$

Proof 30.7 of Corollary 22.24 We compute

$$\left(\sum_{\alpha_i}\psi^\dagger_{\alpha_1}\psi^\dagger_{\alpha_2}V_{\alpha_1,\alpha_2;\alpha_3,\alpha_4}\psi_{\alpha_4}\psi_{\alpha_3}\right)^\dagger = \sum_{\alpha_i}\psi^\dagger_{\alpha_3}\psi^\dagger_{\alpha_4}V^*_{\alpha_1,\alpha_2;\alpha_3,\alpha_4}\psi_{\alpha_2}\psi_{\alpha_1} = \sum_{\alpha_i}\psi^\dagger_{\alpha_1}\psi^\dagger_{\alpha_2}V_{\alpha_4,\alpha_3;\alpha_2,\alpha_1}\psi_{\alpha_4}\psi_{\alpha_3}$$

Therefore this is hermitian, if the following equation is fulfilled[2]

$$\begin{aligned}&\sum_{\nu\in\{1,\dots,rd_{\mathbf{x}}\}}\frac{\omega_\nu(\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4})}{\omega^2_{\alpha_1-\alpha_4}+\omega^2_\nu(\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4})}\frac{g_{m_{\alpha_1}m_{\alpha_4}\nu}(\mathbf{k}_{\alpha_4},\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4})g^*_{m_{\alpha_3}m_{\alpha_2}\nu}(\mathbf{k}_{\alpha_2},\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4})}{\hbar^3\beta N_p}\\ \stackrel{!}{=}&\sum_{\nu\in\{1,\dots,rd_{\mathbf{x}}\}}\frac{\omega_\nu(\mathbf{k}_{\alpha_4}-\mathbf{k}_{\alpha_1})}{\omega^2_{\alpha_4-\alpha_1}+\omega^2_\nu(\mathbf{k}_{\alpha_4}-\mathbf{k}_{\alpha_1})}\frac{g^*_{m_{\alpha_4}m_{\alpha_1}\nu}(\mathbf{k}_{\alpha_1},\mathbf{k}_{\alpha_4}-\mathbf{k}_{\alpha_1})g_{m_{\alpha_2}m_{\alpha_3}\nu}(\mathbf{k}_{\alpha_3},\mathbf{k}_{\alpha_4}-\mathbf{k}_{\alpha_1})}{\hbar^3\beta N_p}\end{aligned}$$

Now since

[2] Maybe we need to go through this proof again, since we symmetrized the expression.

$$g_{m_{\alpha_2}m_{\alpha_3}\nu}(\mathbf{k}_{\alpha_3},\mathbf{k}_{\alpha_4}-\mathbf{k}_{\alpha_1}) = g^*_{m_{\alpha_3}m_{\alpha_2}\nu}(\mathbf{k}_{\alpha_3}+\mathbf{k}_{\alpha_4}-\mathbf{k}_{\alpha_1},\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4}) = g^*_{m_{\alpha_3}m_{\alpha_2}\nu}(\mathbf{k}_{\alpha_2},\mathbf{k}_{\alpha_1}-\mathbf{k}_{\alpha_4})$$

etc, which proves the corollary.

Proof 30.8 of Corollary 22.26

Electrons:

One obtains through direct computation using Equation (19.34) and using further the identity relating Green-functions to functional derivatives

$$\begin{aligned}
\langle n_{n\sigma}(\mathbf{k})\rangle &= \langle c^\dagger_{n\sigma}(\mathbf{k})c_{n\sigma}(\mathbf{k})\rangle \\
&= \frac{\delta^2}{\delta\bar{j}_{n,\sigma}(\tau=0-,\mathbf{k})\delta j_{n,\sigma}(\tau'=0,\mathbf{k})}\frac{Z[j_{\mathrm{F}},\bar{j}_{\mathrm{F}},j_{\mathrm{B}},\bar{j}_{\mathrm{B}}]}{Z}\Big|_{\bar{j}=j=0} \\
&= \frac{1}{\hbar\beta}\lim_{\delta\downarrow 0}\sum_{\omega_m\in\Omega_{-1}} \mathrm{e}^{+\mathrm{i}\omega_m\delta}\frac{\partial^2}{\partial\bar{j}_{n,\sigma}(\mathrm{i}\omega_m,\mathbf{k})\partial j_{n,\sigma}(\mathrm{i}\omega_m,\mathbf{k})}\frac{Z[j_{\mathrm{F}},\bar{j}_{\mathrm{F}},j_{\mathrm{B}},\bar{j}_{\mathrm{B}}]}{Z}\Big|_{\bar{j}=j=0}
\end{aligned}$$

whereas we used that the partition-function is diagonal w.r.t. the Matsubara-frequencies. After recalling the definition of the total electronic number operator as

$$\langle N\rangle = \sum_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\downarrow,\uparrow\}\\ n\in\{1,2,\dots,\infty\}}} \langle n_{n\sigma}(\mathbf{k})\rangle.$$

Follows from using the Lemma **??** as

$$\begin{aligned}
\langle H_{\mathrm{el}}\rangle &= \sum_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\uparrow,\downarrow\}\\ n\in\{1,\dots,\infty\}}} \epsilon_{n\sigma}(\mathbf{k})\left\langle c^\dagger_{n\sigma}(\mathbf{k})c_{n\sigma}(\mathbf{k})\right\rangle \\
&= \frac{1}{\hbar\beta}\lim_{\delta\downarrow 0}\sum_{\substack{\sigma\in\{\downarrow,\uparrow\}\\ k\in\mathrm{i}\Omega_{-1}\times\mathrm{Q}\\ n\in\{1,2,\dots,\infty\}}} \epsilon_{n\sigma}(\mathbf{k})\mathrm{e}^{+\mathrm{i}\omega_k\delta}\frac{\partial^2}{\partial\bar{j}_{n,\sigma}(k)\partial j_{n,\sigma}(k)}\frac{Z[j_{\mathrm{F}},\bar{j}_{\mathrm{F}},j_{\mathrm{B}},\bar{j}_{\mathrm{B}}]}{Z}\Big|_{\bar{j}=j=0}
\end{aligned}$$

Phonons:

The equations for the phonons are derived analogously.

Interactions:

We first rewrite the interaction term as

$$\left\langle c^\dagger_{\mathbf{k}+\mathbf{q}m\sigma} c_{\mathbf{k}n\sigma} \left(b_{\mathbf{q}\nu} + b^\dagger_{-\mathbf{q}\nu} \right) \right\rangle = \left\langle \overline{\psi}_{\mathbf{k}+\mathbf{q}m\sigma}(0) \psi_{\mathbf{k}n\sigma}(0) \left(\phi_{\mathbf{q}\nu}(0) + \overline{\phi}_{-\mathbf{q}\nu}(0) \right) \right\rangle$$
$$= \frac{\delta^2}{\delta \overline{j}(\tau = 0-, \mathbf{k}n\sigma) \delta j(\tau = 0, \mathbf{k}+\mathbf{q}m\sigma)} \left(\frac{\delta}{\delta \overline{j}(\tau = 0-, \mathbf{q}\nu) + \delta j(\tau = 0, -\mathbf{q}\nu)} \right) \frac{Z[j,\overline{j}]}{Z} \Big|_{j=\overline{j}=0} \tag{30.27}$$

Now we compute functional derivatives again. First we note that the functional derivatives can be expanded as

$$\frac{\delta}{\delta \overline{j}(\tau = 0, \mathbf{q}\nu)} = \frac{1}{\sqrt{\hbar\beta}} \sum_{\omega_m \in \Omega_{+1}} \frac{\partial}{\partial \overline{j}(\mathrm{i}\omega_m, \mathbf{q}\nu)}, \tag{30.28}$$

and

$$\frac{\delta}{\delta j(\tau = 0, -\mathbf{q}\nu)} = \frac{1}{\sqrt{\hbar\beta}} \sum_{\omega_m \in \Omega_{+1}} \frac{\partial}{\partial j(-\mathrm{i}\omega_m, -\mathbf{q}\nu)}. \tag{30.29}$$

We also expand the other functional derivatives as

$$\frac{\delta^2}{\delta \overline{j}(\tau = 0-, \mathbf{k}n\sigma) \delta j(\tau = 0, \mathbf{k}+\mathbf{q}m\sigma)} = \frac{1}{\hbar\beta} \lim_{\delta \downarrow 0} \sum_{\omega_{n,n'} \in \Omega_{-1}} \mathrm{e}^{\mathrm{i}\omega_n \delta} \frac{\partial^2}{\partial \overline{j}_{n,\sigma}(\mathrm{i}\omega_n, \mathbf{k}) \partial j_{m,\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k}+\mathbf{q})} \tag{30.30}$$

which gives the desired result.

Proof 30.9 of Theorem 24.13

We use Equation (22.19) and compute

$$\frac{\partial}{\partial j_{n\sigma}(k)} Z[j,\overline{j}] = -\int \mathscr{D}(\overline{\psi},\psi)\, \overline{\psi}_{n\sigma}(k) \exp\left(\overline{j}_{\mathrm{el}}\psi + \overline{\psi} j_{\mathrm{el}}\right) \exp -\left(S_{\mathrm{el}} + \underset{[j_{\mathrm{ph}},\overline{j}_{\mathrm{ph}}]}{S_{\text{el-el-int}}} \right)[\psi,\overline{\psi}] \tag{30.31}$$

and this yields for the double-derivative

$$\frac{\partial^2}{\partial \overline{j}_{n\sigma}(k) \partial j_{n\sigma}(k)} Z[j,\overline{j}] = -\int \mathscr{D}(\overline{\psi},\psi)\, \psi_{n\sigma}(k) \overline{\psi}_{n\sigma}(k) \exp\left(\overline{j}_{\mathrm{el}}\psi + \overline{\psi} j_{\mathrm{el}}\right) \exp -\left(S_{\mathrm{el}} + \underset{[j_{\mathrm{ph}},\overline{j}_{\mathrm{ph}}]}{S_{\text{el-el-int}}} \right)[\psi,\overline{\psi}] \tag{30.32}$$

hence we obtain

$$\frac{\partial^2}{\partial \overline{j}_{n\sigma}(k) \partial j_{n\sigma}(k)} \frac{Z[j,\overline{j}]}{Z} \Big|_{j=\overline{j}=0} = -\langle \psi_{n\sigma}(k) \overline{\psi}_{n\sigma}(k) \rangle = +\underset{\sigma,\sigma}{\mathscr{G}_{n,n}}(k,k) \tag{30.33}$$

whereas in the last step we used Equation TBReferenced.

I think the other identity is proven in Waste-Folder 1

Proof 30.10 of Theorem 24.12

We perform a calculation analogously to Equation (28.77)

$$\begin{aligned}\langle H_{\text{el}}\rangle &= \sum_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\ldots,\infty\}}} \epsilon_{m\mathbf{k}\sigma}\langle n_{m\mathbf{k}\sigma}\rangle \\ &= \frac{\hbar}{\mathrm{i}} \sum_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\ldots,\infty\}}} \epsilon_{m\mathbf{k}\sigma} G^{<}_{\substack{m,m\\ \mathbf{k},\mathbf{k}\\ \sigma,\sigma}}(t=0) \\ &= \frac{1}{2\pi\mathrm{i}} \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\ldots,\infty\}\\ E\in(-\infty,+\infty)}} \epsilon_{m\mathbf{k}\sigma} G^{<}_{\substack{m,m\\ \mathbf{k},\mathbf{k}\\ \sigma,\sigma}}(E) \\ &= \frac{-1}{\pi}\lim_{\eta\to 0} \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\ldots,\infty\}\\ E\in(-\infty,+\infty)}} \epsilon_{m\mathbf{k}\sigma}\, \mathrm{F}_{\beta}(E)\, \mathrm{Im}\; G^{\text{ret}}_{\substack{m,m\\ \mathbf{k},\mathbf{k}\\ \sigma,\sigma}}(E+\mathrm{i}\eta).\end{aligned} \tag{30.34}$$

Proof 30.11 of Theorem 24.14

We divide the proof in consecutive steps.

1. We first compute the first derivative as

$$\frac{\partial}{\partial j_{\nu}(q)} Z[j,\bar{j}] = \int \mathscr{D}(\overline{\psi},\psi) \left(\frac{-\partial}{\partial j_{\nu}(q)} \underset{[j,\bar{j}]}{S_{\text{el-el-int}}}[\psi,\overline{\psi}]\right) \exp - S_{[j,\bar{j}]}[\psi,\overline{\psi}] \tag{30.35}$$

2. We then compute the second derivative with Equation (22.19) as

$$\frac{\partial^2}{\partial \bar{j}_{\nu}(q)\partial j_{\nu}(q)} Z[j,\bar{j}] = \int \mathscr{D}(\overline{\psi},\psi) \left[\frac{-\partial^2}{\partial \bar{j}_{\nu}(q)\partial j_{\nu}(q)} \underset{[j,\bar{j}]}{S_{\text{el-el-int}}}[\psi,\overline{\psi}] + \left(\frac{-\partial}{\partial j_{\nu}(q)} \underset{[j,\bar{j}]}{S_{\text{el-el-int}}}[\psi,\overline{\psi}]\right)\left(\frac{-\partial}{\partial \bar{j}_{\nu}(q)} \underset{[j,\bar{j}]}{S_{\text{el-el-int}}}[\psi,\overline{\psi}]\right)\right] \exp - S_{[j,\bar{j}]}[\psi,\overline{\psi}] \tag{30.36}$$

3. We then compute the individual terms with Equation (22.15)

$$\frac{\partial}{\partial j_{\nu}(q)} \underset{[j,\bar{j}]}{S_{\text{el-el-int}}}[\psi,\overline{\psi}] = \hbar \bar{j}_{\nu}(q)\mathscr{G}_{0,\nu}(q) - \hbar\rho_{\nu}(q)\mathscr{G}_{0,\nu}(q) \tag{30.37}$$

4. and for the other derivative we have with Equation (22.15)

$$\frac{\partial}{\partial \bar{j}_{\nu}(q)} \underset{[j,\bar{j}]}{S_{\text{el-el-int}}}[\psi,\overline{\psi}] = \hbar\mathscr{G}_{0,\nu}(q) j_{\nu}(q) - \hbar\mathscr{G}_{0,\nu}(q)\rho_{\nu}(-q) \tag{30.38}$$

5. and for the double-derivative we have

$$\frac{\partial^2}{\partial \bar{j}_{\nu}(q)\partial j_{\nu}(q)} \underset{[j,\bar{j}]}{S_{\text{el-el-int}}}[\psi,\overline{\psi}] = \hbar\mathscr{G}_{0,\nu}(q) \tag{30.39}$$

Combining these results, we obtain

$$\frac{\partial^2}{\partial \bar{j}_\nu(q)\partial j_\nu(q)} \frac{Z[j,\bar{j}]}{Z}\bigg|_{j=\bar{j}=0} = -\hbar\mathscr{G}_{0,\nu}(q) + \mathscr{G}^2_{0,\nu}(q)\frac{1}{Z}\int \mathscr{D}(\overline{\psi},\psi)\,\left(\hbar^2\rho_\nu(q)\rho_\nu(-q)\right)\exp -S_{[j,\bar{j}]}[\psi,\overline{\psi}]$$
$$= -\hbar\mathscr{G}_{0,\nu}(q) + \mathscr{G}^2_{0,\nu}(q)\hbar^2\left\langle \rho_\nu(q)\rho_\nu(-q)\right\rangle$$
$$= \star\star\star \tag{30.40}$$

Now we focus on the second term, which can be expanded using

$$\rho_\nu(q) = \frac{1}{\sqrt{\hbar\beta N_p}} \sum_{\substack{\sigma\in\{\downarrow,\uparrow\} \\ k\in \mathrm{i}\Omega_{-1}\times \mathrm{Q} \\ n,m\in\{1,\ldots,\infty\}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})}{\hbar}\overline{\psi}_{m\sigma}(k+q)\psi_{n\sigma}(k) \tag{30.41}$$

which implies for the product

$$\mathscr{G}^2_{0,\nu}(q)\hbar^2\left\langle\rho_\nu(q)\rho_\nu(-q)\right\rangle = \frac{\mathscr{G}^2_{0,\nu}(q)}{\hbar\beta N_p}\sum_{\substack{\sigma,\sigma'\in\{\downarrow,\uparrow\} \\ k,k'\in \mathrm{i}\Omega_{-1}\times \mathrm{Q} \\ n,n',m,m'\in\{1,\ldots,\infty\}}} g_{mn\nu}(\mathbf{k},\mathbf{q})g_{m'n'\nu}(\mathbf{k}',-\mathbf{q})\left\langle\overline{\psi}_{m\sigma}(k+q)\psi_{n\sigma}(k)\overline{\psi}_{m'\sigma'}(k'-q)\psi_{n'\sigma'}(k')\right\rangle$$
$$= \frac{\mathscr{G}^2_{0,\nu}(q)}{\hbar\beta N_p}\sum_{\substack{\sigma,\sigma'\in\{\downarrow,\uparrow\} \\ k,k'\in \mathrm{i}\Omega_{-1}\times \mathrm{Q} \\ n,n',m,m'\in\{1,\ldots,\infty\}}} g_{mn\nu}(\mathbf{k},\mathbf{q})g_{m'n'\nu}(\mathbf{k}',-\mathbf{q})\left\langle\psi_{n\sigma}(k)\psi_{n'\sigma'}(k')\overline{\psi}_{m'\sigma'}(k'-q)\overline{\psi}_{m\sigma}(k+q)\right\rangle \tag{30.42}$$

Now we use the Bethe-Salpeter-equation TBReferenced and Equation Exp. Value as GF to approximate[3] the expectation-value as

$$\left\langle\psi_{n\sigma}(k)\psi_{n'\sigma'}(k')\overline{\psi}_{m'\sigma'}(k'-q)\overline{\psi}_{m\sigma}(k+q)\right\rangle$$
$$\approx \hbar^2\left(\mathscr{G}_{\substack{n,m\\ \sigma,\sigma}}(k,k+q)\mathscr{G}_{\substack{n',m'\\ \sigma',\sigma'}}(k',k'-q) - \mathscr{G}_{\substack{n,m'\\ \sigma,\sigma'}}(k,k'-q)\mathscr{G}_{\substack{n',m\\ \sigma',\sigma}}(k',k+q)\right) + \mathcal{O}(g^2)$$
$$= \hbar^2\left(\delta_{q,0}\delta_{n,m}\delta_{n',m'}\mathscr{G}_{n,\sigma}(k)\mathscr{G}_{n',\sigma'}(k') - \delta_{k+q,k'}\delta_{n,m'}\delta_{n',m}\delta_{\sigma,\sigma'}\mathscr{G}_{n,\sigma}(k)\mathscr{G}_{n',\sigma'}(k')\right) + \mathcal{O}(g^2) \tag{30.43}$$

Plugging this into the equation above yields the final results

[3] In order to go beyond this approximation: See TBDigitalized, TBD6.

$$
\begin{aligned}
\star\star\star = -\hbar\mathscr{G}_{0,\nu}(q) \quad &+ \quad \frac{\hbar}{\beta N_p}\mathscr{G}^2_{0,\nu}(q)\delta_{q,0} \sum_{\substack{k,k' \in \mathrm{i}\Omega_{-1}\times \mathrm{Q} \\ n,n' \in \{1,\dots,\infty\} \\ \sigma,\sigma' \in \{\downarrow,\uparrow\}}} g_{nn\nu}(\mathbf{k},0)g_{n'n'\nu}(\mathbf{k}',0)\mathscr{G}_{n,\sigma}(k)\mathscr{G}_{n',\sigma'}(k') \\
&- \frac{\hbar}{\beta N_p}\mathscr{G}^2_{0,\nu}(q) \sum_{\substack{k \in \mathrm{i}\Omega_{-1}\times \mathrm{Q} \\ n,n' \in \{1,\dots,\infty\} \\ \sigma \in \{\downarrow,\uparrow\}}} g_{n'n\nu}(\mathbf{k},\mathbf{q})g_{nn'\nu}(\mathbf{k}+\mathbf{q},-\mathbf{q})\mathscr{G}_{n,\sigma}(k)\mathscr{G}_{n',\sigma}(k+q) \quad + \quad \mathcal{O}(g^4) \\
= -\hbar\mathscr{G}_{0,\nu}(q) \quad &+ \quad \frac{\hbar}{\beta N_p}\mathscr{G}^2_{0,\nu}(q)\delta_{q,0} \sum_{\substack{k,k' \in \mathrm{i}\Omega_{-1}\times \mathrm{Q} \\ n,n' \in \{1,\dots,\infty\} \\ \sigma,\sigma' \in \{\downarrow,\uparrow\}}} g_{nn\nu}(\mathbf{k},0)g_{n'n'\nu}(\mathbf{k}',0)\mathscr{G}_{n,\sigma}(k)\mathscr{G}_{n',\sigma'}(k') \\
&- \frac{\hbar}{\beta N_p}\mathscr{G}^2_{0,\nu}(q) \sum_{\substack{k \in \mathrm{i}\Omega_{-1}\times \mathrm{Q} \\ n,n' \in \{1,\dots,\infty\} \\ \sigma \in \{\downarrow,\uparrow\}}} |g_{n'n\nu}(\mathbf{k},\mathbf{q})|^2\mathscr{G}_{n,\sigma}(k)\mathscr{G}_{n',\sigma}(k+q) \quad + \quad \mathcal{O}(g^4)
\end{aligned}
\tag{30.44}
$$

since we can now replace $\mathscr{G}$ with $\mathscr{G}_0$ since the corrections will only contribute to higher orders in g, which proves the theorem.

Proof 30.12 of Lemma 24.15

We combine Theorem 24.14 with Equation (22.23) to obtain

$$
\begin{aligned}
\langle H_{\mathrm{ph}}\rangle \approx &-\frac{1}{\hbar\beta} \sum_{\substack{q \in \mathrm{i}\Omega_{+1}\times \mathrm{Q} \\ \nu \in \{1,2,\dots,rd_{\mathbf{x}}\}}} \mathrm{e}^{+\mathrm{i}\omega_q 0^+}\Omega_\nu(\mathbf{q})\hbar\mathscr{G}_{0,\nu}(q) \\
&+ \frac{1}{\hbar\beta} \sum_{\substack{q \in \mathrm{i}\Omega_{+1}\times \mathrm{Q} \\ \nu \in \{1,2,\dots,rd_{\mathbf{x}}\}}} \mathrm{e}^{+\mathrm{i}\omega_q 0^+}\Omega_\nu(\mathbf{q})\frac{\hbar}{\beta N_p}\mathscr{G}^2_{0,\nu}(q)\delta_{q,0} \sum_{\substack{k,k' \in \mathrm{i}\Omega_{-1}\times \mathrm{Q} \\ n,n' \in \{1,\dots,\infty\} \\ \sigma,\sigma' \in \{\downarrow,\uparrow\}}} g_{nn\nu}(\mathbf{k},0)g_{n'n'\nu}(\mathbf{k}',0)\mathscr{G}_{n,\sigma}(k)\mathscr{G}_{n',\sigma'}(k') \\
&- \frac{1}{\hbar\beta} \sum_{\substack{q \in \mathrm{i}\Omega_{+1}\times \mathrm{Q} \\ \nu \in \{1,2,\dots,rd_{\mathbf{x}}\}}} \mathrm{e}^{+\mathrm{i}\omega_q 0^+}\Omega_\nu(\mathbf{q})\frac{\hbar}{\beta N_p}\mathscr{G}^2_{0,\nu}(q) \sum_{\substack{k \in \mathrm{i}\Omega_{-1}\times \mathrm{Q} \\ n,n' \in \{1,\dots,\infty\} \\ \sigma \in \{\downarrow,\uparrow\}}} |g_{n'n\nu}(\mathbf{k},\mathbf{q})|^2\mathscr{G}_{n,\sigma}(k)\mathscr{G}_{n',\sigma}(k+q)
\end{aligned}
\tag{30.45}
$$

Now simplifying the appearing terms and neglecting terms of the order $\mathcal{O}(g^4)$ and rearranging the terms yields the desired result.

Proof 30.13 of Corollary 24.16

First Term We compute the phononic, non-interacting contribution and introduce $q \equiv (\mathrm{i}\omega_n, \mathbf{q})$

$$
\begin{aligned}
\frac{1}{\hbar\beta}\sum_{\mathrm{i}\omega_n \in \mathrm{i}\Omega_{+1}} \left(-\hbar\mathscr{G}_{0,\nu}(q)\right) &= \frac{-1}{\hbar\beta}\sum_{\mathrm{i}\omega_n \in \mathrm{i}\Omega_{+1}} \frac{1}{\mathrm{i}\omega_n - \omega_\nu(\mathbf{q})} \\
&\overset{\text{Corollary 15.16}}{=} \mathrm{B}_{\hbar\beta}(\omega_\nu(\mathbf{q}))
\end{aligned}
\tag{30.46}
$$

Second Term For the mix-terms we compute

$$\frac{1}{\beta^2}\sum_{\substack{\omega_k\in\Omega_{-1}\\ \omega_q\in\Omega_{+1}}}\mathscr{G}^2_{0,\nu}(q)\mathscr{G}_{0,n,\sigma}(k)\mathscr{G}_{0,n',\sigma}(k+q) = \frac{1}{\beta^2\hbar^4}\sum_{\substack{\omega_k\in\Omega_{-1}\\ \omega_q\in\Omega_{+1}}}\frac{1}{\left(\mathrm{i}\omega_q-\omega_\nu(\mathbf{q})\right)^2}\cdot\frac{1}{\mathrm{i}\omega_k-\xi_{n,\sigma,\mathbf{k}}/\hbar}\cdot\frac{1}{\mathrm{i}\omega_{k+q}-\xi_{n',\sigma,\mathbf{k+q}}/\hbar}$$
$$= \dots \tag{30.47}$$

We now decompose $\Omega_{-1}\ni\omega_{q+k}\equiv\omega_q+\omega_k$, then we can evaluate the sum over ω_k first as

$$\dots = \frac{1}{\hbar^3\beta}\sum_{\omega_q\in\Omega_{+1}}\frac{1}{\left(\mathrm{i}\omega_q-\omega_\nu(\mathbf{q})\right)^2}\cdot\underbrace{\frac{1}{\hbar\beta}\sum_{\omega_k\in\Omega_{-1}}\frac{1}{\mathrm{i}\omega_k-\xi_{n,\sigma,\mathbf{k}}/\hbar}\cdot\frac{1}{\mathrm{i}\omega_k+\mathrm{i}\omega_q-\xi_{n',\sigma,\mathbf{k+q}}/\hbar}}_{=S_{-1}\equiv\sum_{\omega_k\in\Omega_{-1}}h(\omega_k)} \tag{30.48}$$
$$= \dots$$

We now consider the function

$$\begin{aligned}\mathbb{C}-Z\ni z\mapsto h(-\mathrm{i}z) &= \frac{1}{\hbar\beta}\cdot\frac{1}{z-\xi_{n,\sigma,\mathbf{k}}/\hbar}\cdot\frac{1}{z+\mathrm{i}\omega_q-\xi_{n',\sigma,\mathbf{k+q}}/\hbar}\\ &= \frac{1}{\hbar\beta}\cdot\frac{1}{z-z_1}\cdot\frac{1}{z-z_2}\end{aligned} \tag{30.49}$$

with first order poles

$$z_1=\xi_{n,\sigma,\mathbf{k}}/\hbar,\qquad z_2=\xi_{n',\sigma,\mathbf{k+q}}/\hbar-\mathrm{i}\omega_q \tag{30.50}$$

This gives with Theorem 15.14 (a version of the Residue-theorem)

$$S_{-1}=\sum_{\omega_k\in\Omega_{-1}}h(\omega_k)=-\zeta\sum_{l=1}^{2}\mathrm{Res}(f_\zeta(z),z=z_l)|_{\zeta=-1} \tag{30.51}$$

with auxiliary function

$$f_\zeta(z)|_{\zeta=-1}=g_\zeta(z)h(-\mathrm{i}z)|_{\zeta=-1}=\hbar\beta n^\zeta_{\hbar\beta}(z)h(-\mathrm{i}z)|_{\zeta=-1} \tag{30.52}$$

This gives for the first residue

$$\begin{aligned}\mathrm{Res}(f_\zeta(z),z=z_1)|_{\zeta=-1} &= \lim_{z\to z_1}(z-z_1)f_\zeta(z)|_{\zeta=-1}\\ &= \lim_{z\to z_1}(z-z_2)^{-1}\,\mathrm{F}_{\hbar\beta}(z)\\ &= \frac{\mathrm{F}_\beta(\xi_{n,\sigma,\mathbf{k}})}{\xi_{n,\sigma,\mathbf{k}}/\hbar-\xi_{n',\sigma,\mathbf{k+q}}/\hbar+\mathrm{i}\omega_q}\end{aligned} \tag{30.53}$$

For the second residue we obtain likewise

$$
\begin{aligned}
\operatorname{Res}(f_\zeta(z), z = z_2)|_{\zeta=-1} &= \lim_{z \to z_2} (z - z_2) f_\zeta(z)|_{\zeta=-1} \\
&= \lim_{z \to z_2} (z - z_1)^{-1} \, \mathrm{F}_{\hbar\beta}(z) \\
&= \frac{\mathrm{F}_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}} - \mathrm{i}\hbar\omega_q)}{\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar - \xi_{n,\sigma,\mathbf{k}}/\hbar - \mathrm{i}\omega_q} \\
&\overset{\star}{=} \frac{\mathrm{F}_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}})}{\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar - \xi_{n,\sigma,\mathbf{k}}/\hbar - \mathrm{i}\omega_q}
\end{aligned} \tag{30.54}
$$

whereas in $\star$ we used Remark 9.46.1. Summing up the residues yields then

$$
\begin{aligned}
\ldots &= \frac{\mathrm{F}_\beta(\xi_{n,\sigma,\mathbf{k}}) - \mathrm{F}_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}})}{\hbar^2} \underbrace{\frac{1}{\hbar\beta} \sum_{\omega_q \in \Omega_{+1}} \frac{1}{\left(\mathrm{i}\omega_q - \omega_\nu(\mathbf{q})\right)^2} \cdot \frac{1}{\mathrm{i}\omega_q + \xi_{n,\sigma,\mathbf{k}}/\hbar - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar}}_{=S_{+1} \equiv \sum_{\omega_q \in \Omega_{+1}} h(\omega_q)} \\
&= \ldots
\end{aligned} \tag{30.55}
$$

We now, again, consider the function

$$
\begin{aligned}
\mathbb{C} - Z \ni z \mapsto h(-\mathrm{i}z) &= \frac{1}{\hbar\beta} \cdot \frac{1}{(z - \omega_\nu(\mathbf{q}))^2} \cdot \frac{1}{z + \xi_{n,\sigma,\mathbf{k}}/\hbar - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar} \\
&= \frac{1}{\hbar\beta} \cdot \frac{1}{(z - z_2)^2} \cdot \frac{1}{z - z_1}
\end{aligned} \tag{30.56}
$$

with poles of different order

- $z_1 = \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar - \xi_{n,\sigma,\mathbf{k}}/\hbar$ (first order pole)
- $z_2 = \omega_\nu(\mathbf{q})$ (second order pole)

This gives with Theorem 15.14

$$
S_{+1} = \sum_{\omega_q \in \Omega_{+1}} h(\omega_q) = -\sum_{l=1}^{2} \operatorname{Res}(f_{+1}(z), z = z_l) = \star\star\star \tag{30.57}
$$

with auxiliary function

$$
f_{+1}(z) = \hbar\beta \, \mathrm{B}_{\hbar\beta}(z) h(-\mathrm{i}z) = \hbar\beta \, \mathrm{B}_{\hbar\beta}(z) \cdot \frac{1}{\hbar\beta} \cdot \frac{1}{(z - z_2)^2} \cdot \frac{1}{z - z_1} = \mathrm{B}_{\hbar\beta}(z) \cdot \frac{1}{(z - z_2)^2} \cdot \frac{1}{z - z_1} \tag{30.58}
$$

This gives for the first residue

$$
-\operatorname{Res}(f_{+1}(z), z = z_1) = -\lim_{z \to z_1} (z - z_2)^{-2} \, \mathrm{B}_{\hbar\beta}(z) = \frac{-\mathrm{B}_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}} - \xi_{n,\sigma,\mathbf{k}})}{\left(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar - \xi_{n,\sigma,\mathbf{k}}/\hbar - \omega_\nu(\mathbf{q})\right)^2} \tag{30.59}
$$

Now we compute the second residue

$$
-\operatorname{Res}(f_{+1}(z), z = z_2) = -\lim_{z \to z_2} \frac{\mathrm{d}}{\mathrm{d}z} \left[(z - z_1)^{-1} \, \mathrm{B}_{\hbar\beta}(z)\right] = \square\square\square \tag{30.60}
$$

which gives with Lemma 9.58 the result

$$\square\square\square = \lim_{z\to z_2} -\Big[-(z-z_1)^{-2}\,\mathrm{B}_{\hbar\beta}(z) + (z-z_1)^{-1}\cdot(-\hbar\beta)\,\mathrm{B}_{\hbar\beta}(z)\left(1+\mathrm{B}_{\hbar\beta}(z)\right)\Big]$$
$$= \frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))}{\left(\omega_\nu(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}}/\hbar - \xi_{n',\sigma,\mathbf{k+q}}/\hbar\right)^2} + \frac{\hbar\beta\,\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))\left(1+\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))\right)}{\omega_\nu(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}}/\hbar - \xi_{n',\sigma,\mathbf{k+q}}/\hbar} \tag{30.61}$$

This gives the result

$$\star\star\star = \text{TBD}. \tag{30.62}$$

Using now Equation (TBReferenced) we obtain the final expression as

$$\frac{1}{\beta^2}\sum_{\substack{\omega_k\in\Omega_{-1}\\ \omega_q\in\Omega_{+1}}} \mathscr{G}^2_{0,\nu}(q)\mathscr{G}_{0,n,\sigma}(k)\mathscr{G}_{0,n',\sigma}(k+q)$$
$$= \dots \tag{30.63}$$
$$= \frac{\mathrm{F}_\beta(\xi_{n,\sigma,\mathbf{k}}) - \mathrm{F}_\beta(\xi_{n',\sigma,\mathbf{k+q}})}{\hbar^2}\left[\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) - \mathrm{B}_\beta(\xi_{n',\sigma,\mathbf{k+q}} - \xi_{n,\sigma,\mathbf{k}})}{\left(\omega_\nu(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}}/\hbar - \xi_{n',\sigma,\mathbf{k+q}}/\hbar\right)^2} + \frac{\hbar\beta\,\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))\left(1+\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))\right)}{\omega_\nu(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}}/\hbar - \xi_{n',\sigma,\mathbf{k+q}}/\hbar}\right]$$

now we use Lemma 9.47 to obtain

$$\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))\left(1+\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))\right) = 4^{-1}\,\mathrm{csch}^2\left(\frac{\beta\Omega_\nu(\mathbf{q})}{2}\right) \tag{30.64}$$

which proves the corollary. Notice that Equation (30.63) has the right unit, namely Energy^{-2}.

Proof 30.14 of Lemma 24.17

We first simplify Equation (22.24): We use Equation (22.19) to integrate out the phonons, such that we can obtain

$$\frac{\partial}{\partial\bar{j}(\mathrm{i}\omega_m,\mathbf{q}\nu)}Z[j,\bar{j}] = \int \mathscr{D}(\overline{\psi},\psi)\left(\frac{-\partial}{\partial\bar{j}(\mathrm{i}\omega_m,\mathbf{q}\nu)}\underset{[j_{\mathrm{ph}},\bar{j}_{\mathrm{ph}}]}{S_{\text{el-el-int}}}[\psi,\overline{\psi}]\right)\exp - S_{j,\bar{j}}[\psi,\overline{\psi}] \tag{30.65}$$

and

$$\frac{\partial}{\partial j(-\mathrm{i}\omega_m,-\mathbf{q}\nu)}Z[j,\bar{j}] = \int \mathscr{D}(\overline{\psi},\psi)\left(\frac{-\partial}{\partial j(-\mathrm{i}\omega_m,-\mathbf{q}\nu)}\underset{[j_{\mathrm{ph}},\bar{j}_{\mathrm{ph}}]}{S_{\text{el-el-int}}}[\psi,\overline{\psi}]\right)\exp - S_{j,\bar{j}}[\psi,\overline{\psi}] \tag{30.66}$$

Now we use Equation (22.15) to obtain the explicit form of these derivatives as

$$\frac{-\partial}{\partial j(\mathrm{i}\omega_m,\mathbf{q}\nu)}\underset{[j_{\mathrm{ph}},\bar{j}_{\mathrm{ph}}]}{S_{\text{el-el-int}}}[\psi,\overline{\psi}] = -\hbar\bar{j}_\nu(q)\mathscr{G}_{0,\nu}(q) + \hbar\rho_\nu(q)\mathscr{G}_{0,\nu}(q) \tag{30.67}$$

and

$$\frac{-\partial}{\partial \overline{j}(\mathrm{i}\omega_m, \mathbf{q}\nu)} S_{\text{el-el-int}}\underset{[j_{\text{ph}}, \overline{j}_{\text{ph}}]}{[\psi, \overline{\psi}]} = -\hbar \mathscr{G}_{0,\nu}(-q) j_\nu(-q) + \hbar \mathscr{G}_{0,\nu}(-q) \rho_\nu(q) \tag{30.68}$$

Now we can use Equation (22.19) again to obtain

$$\frac{\partial}{\partial j_{m,\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k}+\mathbf{q})} Z[j, \overline{j}] = \int \mathscr{D}(\overline{\psi}, \psi) \underbrace{\left(\frac{\partial}{\partial j_{m,\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k}+\mathbf{q})} \sum \overline{\psi} j_{\text{el}} \right)}_{=-\psi_{m\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k}+\mathbf{q})} \exp - S_{[j,\overline{j}]}[\psi, \overline{\psi}] \tag{30.69}$$

This implies that

$$\frac{\partial^2}{\partial \overline{j}_{n,\sigma}(\mathrm{i}\omega_n, \mathbf{k}) \partial j_{m,\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k}+\mathbf{q})} \frac{Z[j, \overline{j}]}{Z} \Big|_{j=\overline{j}=0} = \left\langle \overline{\psi}_{m\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k}+\mathbf{q}) \psi_{n\sigma}(\mathrm{i}\omega_n, \mathbf{k}) \right\rangle \tag{30.70}$$

Now we combine this with Equation (24.13) to obtain the desired result:

$$\begin{aligned}
\langle H_{\text{int}} \rangle &= \sum_{\substack{\mathbf{k},\mathbf{q} \in \mathrm{Q} \\ \sigma \in \{\downarrow,\uparrow\} \\ \nu \in \{1,2,\ldots,rd_{\mathbf{x}}\} \\ n,m \in \{1,\ldots,\infty\} \\ \omega_m \in \Omega_{+1} \\ \omega_n, \omega_{n'} \in \Omega_{-1}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})\hbar}{\sqrt{N_p}\sqrt{\hbar\beta}^3} \left\langle \overline{\psi}_{m\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k}+\mathbf{q}) \psi_{n\sigma}(\mathrm{i}\omega_n, \mathbf{k}) \Big[\rho_\nu(\mathrm{i}\omega_m, \mathbf{q}) \mathscr{G}_{0,\nu}(\mathrm{i}\omega_m, \mathbf{q}) + \mathscr{G}_{0,\nu}(-\mathrm{i}\omega_m, -\mathbf{q}) \rho_\nu(\mathrm{i}\omega_m, \mathbf{q}) \Big] \right\rangle \\
&= \sum_{\substack{k \in \mathrm{i}\Omega_{-1} \times \mathrm{Q} \\ q \in \mathrm{i}\Omega_{+1} \times \mathrm{Q} \\ \sigma \in \{\downarrow,\uparrow\} \\ \nu \in \{1,2,\ldots,rd_{\mathbf{x}}\} \\ n,m \in \{1,\ldots,\infty\} \\ \omega_{n'} \in \Omega_{-1}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})\hbar}{\sqrt{N_p}\sqrt{\hbar\beta}^3} \left\langle \overline{\psi}_{m\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k}+\mathbf{q}) \psi_{n\sigma}(\mathrm{i}\omega_n, \mathbf{k}) \Big[\rho_\nu(q) \mathscr{G}_{0,\nu}(q) + \mathscr{G}_{0,\nu}(-q) \rho_\nu(q) \Big] \right\rangle
\end{aligned} \tag{30.71}$$

Proof 30.15 of Corollary 24.18

We decompose Equation (24.20) as

$$\begin{aligned}
\langle H_{\text{int}} \rangle &= \sum_{\substack{k \in \mathrm{i}\Omega_{-1} \times \mathrm{Q} \\ q \in \mathrm{i}\Omega_{+1} \times \mathrm{Q} \\ \sigma \in \{\downarrow,\uparrow\} \\ \nu \in \{1,2,\ldots,rd_{\mathbf{x}}\} \\ n,m \in \{1,\ldots,\infty\} \\ \omega_{n'} \in \Omega_{-1}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})\hbar}{\sqrt{N_p}\sqrt{\hbar\beta}^3} \left\langle \overline{\psi}_{m\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k}+\mathbf{q}) \psi_{n\sigma}(\mathrm{i}\omega_n, \mathbf{k}) \Big[\rho_\nu(q) \mathscr{G}_{0,\nu}(q) + \mathscr{G}_{0,\nu}(-q) \rho_\nu(q) \Big] \right\rangle \\
&\equiv \qquad\qquad\qquad\qquad \mathrm{I} \qquad + \qquad \mathrm{II}
\end{aligned} \tag{30.72}$$

We focus on the first term I and compute

$$\begin{aligned}
\mathrm{I} &= \sum_{\substack{k,\widetilde{k} \in \mathrm{i}\Omega_{-1} \times \mathrm{Q} \\ q \in \mathrm{i}\Omega_{+1} \times \mathrm{Q} \\ \sigma,\widetilde{\sigma} \in \{\downarrow,\uparrow\} \\ \nu \in \{1,2,\ldots,rd_{\mathbf{x}}\} \\ n,\widetilde{n},m,\widetilde{m} \in \{1,\ldots,\infty\} \\ \omega_{n'} \in \Omega_{-1}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q}) g_{\widetilde{m}\widetilde{n}\nu}(\widetilde{\mathbf{k}},\mathbf{q})}{N_p (\hbar\beta)^2} \left\langle \overline{\psi}_{m\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k}+\mathbf{q}) \psi_{n\sigma}(k) \overline{\psi}_{\widetilde{m}\widetilde{\sigma}}(\widetilde{k}+q) \psi_{\widetilde{n}\widetilde{\sigma}}(\widetilde{k}) \right\rangle \mathscr{G}_{0,\nu}(q) \\
&= \ldots
\end{aligned} \tag{30.73}$$

Now we simplify the correlator as

$$\begin{aligned}
&\left\langle \overline{\psi}_{m\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k}+\mathbf{q})\psi_{n\sigma}(k)\overline{\psi}_{\widetilde{m}\widetilde{\sigma}}(\widetilde{k}+q)\psi_{\widetilde{n}\widetilde{\sigma}}(\widetilde{k})\right\rangle \\
&= -\left\langle \psi_{n\sigma}(k)\psi_{\widetilde{n}\widetilde{\sigma}}(\widetilde{k})\overline{\psi}_{m\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k}+\mathbf{q})\overline{\psi}_{\widetilde{m}\widetilde{\sigma}}(\widetilde{k}+q)\right\rangle \\
&= -\Big[\left\langle \psi_{n\sigma}(k)\overline{\psi}_{\widetilde{m}\widetilde{\sigma}}(\widetilde{k}+q)\right\rangle_0 \left\langle \psi_{\widetilde{n}\widetilde{\sigma}}(\widetilde{k})\overline{\psi}_{m\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k}+\mathbf{q})\right\rangle_0 \quad - \left\langle \psi_{n\sigma}(k)\overline{\psi}_{m\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k}+\mathbf{q})\right\rangle_0 \left\langle \psi_{\widetilde{n}\widetilde{\sigma}}(\widetilde{k})\overline{\psi}_{\widetilde{m}\widetilde{\sigma}}(\widetilde{k}+q)\right\rangle_0 + \mathcal{O}(g^2)\Big] \\
&= -\hbar^2\Big[\mathcal{G}_{\substack{0\\ n,\widetilde{m}\\ \sigma,\widetilde{\sigma}}}(k,\widetilde{k}+q)\mathcal{G}_{\substack{0\\ \widetilde{n},m\\ \widetilde{\sigma},\sigma}}\begin{pmatrix}\mathrm{i}\omega_{\widetilde{k}}, \mathrm{i}\omega_{n'}\\ \widetilde{\mathbf{k}}, \mathbf{k}+\mathbf{q}\end{pmatrix} \quad - \mathcal{G}_{\substack{0\\ n,m\\ \sigma,\sigma}}\begin{pmatrix}\mathrm{i}\omega_{k}, \mathrm{i}\omega_{n'}\\ \mathbf{k}, \mathbf{k}+\mathbf{q}\end{pmatrix}\mathcal{G}_{\substack{0\\ \widetilde{n},\widetilde{m}\\ \widetilde{\sigma},\widetilde{\sigma}}}(\widetilde{k},\widetilde{k}+q)\Big]
\end{aligned} \tag{30.74}$$

This gives a natural decomposition for

$$\mathrm{I} = \ldots = \mathrm{I1} + \mathrm{I2} \tag{30.75}$$

Now we simplify

$$\begin{aligned}
I1 = &\sum_{\substack{k,\tilde{k}\in \mathrm{i}\Omega_{-1}\times \mathrm{Q}\\ (\mathrm{i}\omega_q,\mathbf{q})\equiv q\in \mathrm{i}\Omega_{+1}\times \mathrm{Q}\\ \sigma,\widetilde{\sigma}\in\{\downarrow,\uparrow\}\\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\}\\ n,\tilde{n},m,\tilde{m}\in\{1,\ldots,\infty\}\\ \omega_{n'}\in\Omega_{-1}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})g_{\widetilde{m}\widetilde{n}\nu}(\widetilde{\mathbf{k}},\mathbf{q})}{N_p\beta^2}\cdot(-1)\cdot\Big[\delta_{n,\widetilde{m}}\delta_{\sigma,\widetilde{\sigma}}\delta_{k-q,\widetilde{k}}\mathcal{G}_{0,n,\sigma}(k)\delta_{\widetilde{n},m}\delta_{\omega_{\widetilde{k}},\omega_{n'}}\delta_{\widetilde{\mathbf{k}},\mathbf{k}+\mathbf{q}}\mathcal{G}_{0,m,\sigma}(\mathrm{i}\omega_k,\mathbf{k})\Big] \\
&= \star\star\star
\end{aligned} \tag{30.76}$$

Now we use the individual delta-functions to ...

1. kill $\tilde{m}$,
2. kill $\widetilde{\sigma}$,
3. kill $\widetilde{k}$,
4. kill $\widetilde{n}$,
5. kill $\omega_{n'}$ as $\tilde{k}$ is already killed,
6. kill $\mathbf{q}$ as $\widetilde{k}$ is already killed.

This allows us to simplify

$$
\begin{aligned}
\star\star\star &= \sum_{\substack{k\in \mathrm{i}\Omega_{-1}\times \mathrm{Q}\\ \mathrm{i}\omega_q\in\Omega_{+1}\\ \sigma\in\{\downarrow,\uparrow\}\\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}\\ n,m\in\{1,\dots,\infty\}}} \frac{g_{mn\nu}(\mathbf{k},0)g_{nm\nu}(\mathbf{k},0)}{N_p\beta^2}\cdot(-1)\cdot\mathscr{G}_{0,n,\sigma}(k)\mathscr{G}_{0,m,\sigma}(\mathrm{i}\omega_{k-q},\mathbf{k}) \\
&= \sum_{\substack{\mathbf{k}\in \mathrm{Q}\\ \sigma\in\{\downarrow,\uparrow\}\\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}\\ n,m\in\{1,\dots,\infty\}}} \frac{-|g_{mn\nu}(\mathbf{k},0)|^2}{N_p\beta^2\hbar^2}\sum_{\substack{\mathrm{i}\omega_k\in\Omega_{-1}\\ \mathrm{i}\omega_q\in\Omega_{+1}}}\frac{1}{\mathrm{i}\omega_k-\xi_{n,\sigma}/\hbar}\cdot\frac{1}{\mathrm{i}\omega_k-\mathrm{i}\omega_q-\xi_{m,\sigma}/\hbar} \\
&= \sum_{\substack{\mathbf{k}\in \mathrm{Q}\\ \sigma\in\{\downarrow,\uparrow\}\\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}\\ n,m\in\{1,\dots,\infty\}}} \frac{-|g_{mn\nu}(\mathbf{k},0)|^2}{N_p}\frac{1}{\hbar\beta}\sum_{\omega_k\in\Omega_{-1}}\frac{1}{\mathrm{i}\omega_k-\xi_{n,\sigma}/\hbar}\cdot\underbrace{\frac{1}{\hbar\beta}\sum_{\omega_q\in\Omega_{+1}}\frac{1}{\mathrm{i}\omega_q-\mathrm{i}\omega_k+\xi_{m,\sigma}/\hbar}}_{=-\mathrm{B}_\beta(\xi_{m,\sigma}-\mathrm{i}\hbar\omega_k)}
\end{aligned}
\tag{30.77}
$$

whereas we used corollary 16.15 (or something).

Intermezzo: We now simplify $\mathrm{B}_\beta(\bullet-\mathrm{i}\hbar\omega_k)$ ... TBContinued.

Proof 30.16 of Theorem 24.19

For **the electronic contribution** of the expectation-value we simply refer to Theorem 24.12.

For **the phononic contribution**, we simplify refer to Corollary 24.16.

For **the interaction contribution**, we refer to Theorem TBCreated after PP3 has been digitalized.

Proof 30.17 of Lemma 24.22 We compute the derivative of Equation (23.24) as

$$
\begin{aligned}
\left(\partial_T F\right)(\mu,T) &= \langle N\rangle(\mu,T) \\
&= \partial_T \frac{-1}{\pi} \lim_{\delta\to 0} \sumint_{\substack{\mathbf{k}\in \mathrm{Q} \\ \sigma\in\{\uparrow,\downarrow\} \\ m\in\{1,\dots,\infty\} \\ E\in(-\infty,+\infty)}} \mathrm{F}_\beta(E)\, \mathrm{Im}\, G^{\mathrm{ret}}_{\substack{m,m \\ \sigma,\sigma}}(E+\mathrm{i}\delta,\mathbf{k}) \\
&= \frac{-1}{\pi} \lim_{\delta\to 0} \sumint_{\substack{\mathbf{k}\in \mathrm{Q} \\ \sigma\in\{\uparrow,\downarrow\} \\ m\in\{1,\dots,\infty\} \\ E\in(-\infty,+\infty)}} \left[\left(\partial_T \mathrm{F}_\beta(E)\right) \mathrm{Im}\, G^{\mathrm{ret}}_{\substack{m,m \\ \sigma,\sigma}}(E+\mathrm{i}\delta,\mathbf{k}) + \mathrm{F}_\beta(E)\, \mathrm{Im}\, \partial_T G^{\mathrm{ret}}_{\substack{m,m \\ \sigma,\sigma}}(E+\mathrm{i}\delta,\mathbf{k}) \right] \\
&= \dots
\end{aligned}
\tag{30.78}
$$

Now we use Lemma 9.47, i.e. we use the Equation

$$
\partial_T \mathrm{F}_\beta(E) = \frac{E}{k_B T^2} \mathrm{F}_\beta(E) \left(1 - \mathrm{F}_\beta(E)\right) \tag{30.79}
$$

and for the other derivative we have

$$
\partial_T G^{\mathrm{ret}}_{m,\sigma,\mathbf{k}}(E+\mathrm{i}\delta) = \partial_T \frac{1}{E+\mathrm{i}\delta - \left(\epsilon_{m\mathbf{k}\sigma} + \Sigma^{\mathrm{ret}}_{m,\sigma,\mathbf{k}}(E+\mathrm{i}\delta) - \mu\right)} = \frac{\partial_T \Sigma^{\mathrm{ret}}_{m,\sigma,\mathbf{k}}(E+\mathrm{i}\delta)}{\left(E+\mathrm{i}\delta - \left(\epsilon_{m\mathbf{k}\sigma} + \Sigma^{\mathrm{ret}}_{m,\sigma,\mathbf{k}}(E+\mathrm{i}\delta) - \mu\right)\right)^2} \tag{30.80}
$$

Now we compute the derivative and use the diagonal bands approximation for the self-energy and use Lemma 9.47

$$
\begin{aligned}
\partial_T \Sigma^{\mathrm{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta) &= \sum_{\substack{\mathbf{q}\in \mathrm{Q} \\ n\in\{1,\dots,\infty\} \\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} \frac{|g_{nm\nu}(\mathbf{k},\mathbf{q})|^2}{N_p} \left[\partial_T \frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) + \mathrm{F}_\beta(\xi_{n\sigma\mathbf{k}+\mathbf{q}})}{E+\mathrm{i}\delta - \xi_{n\sigma\mathbf{k}+\mathbf{q}} + \Omega_\nu(\mathbf{q})} + \partial_T \frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) + 1 - \mathrm{F}_\beta(\xi_{n\sigma\mathbf{k}+\mathbf{q}})}{E+\mathrm{i}\delta - \xi_{n\sigma\mathbf{k}+\mathbf{q}} - \Omega_\nu(\mathbf{q})} \right] \\
&= \sum_{\substack{\mathbf{q}\in \mathrm{Q} \\ n\in\{1,\dots,\infty\} \\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} \frac{|g_{nm\nu}(\mathbf{k},\mathbf{q})|^2}{4k_B T^2 N_p} \left[\frac{\Omega_\nu(\mathbf{q})\,\mathrm{csch}^2(\beta\Omega_\nu(\mathbf{q})/2) + \xi_{n\sigma\mathbf{k}+\mathbf{q}}\,\mathrm{sech}^2(\beta\xi_{n\sigma\mathbf{k}+\mathbf{q}}/2)}{E+\mathrm{i}\delta - \xi_{n\sigma\mathbf{k}+\mathbf{q}} + \Omega_\nu(\mathbf{q})} + \frac{\Omega_\nu(\mathbf{q})\,\mathrm{csch}^2(\beta\Omega_\nu(\mathbf{q})/2) - \xi_{n\sigma\mathbf{k}+\mathbf{q}}\,\mathrm{sech}^2(\beta\xi_{n\sigma\mathbf{k}+\mathbf{q}}/2)}{E+\mathrm{i}\delta - \xi_{n\sigma\mathbf{k}+\mathbf{q}} - \Omega_\nu(\mathbf{q})} \right]
\end{aligned}
\tag{30.81}
$$

now we multiply both sides of this equation with $k_B T^2$.

Proof 30.18 of Lemma 24.23 We rewrite the particle-number equation as

$$
\langle N\rangle(\mu,T) = \frac{-1}{\pi} \lim_{\delta\to 0} \sumint_{\substack{\mathbf{k}\in \mathrm{Q} \\ \sigma\in\{\uparrow,\downarrow\} \\ m\in\{1,\dots,\infty\} \\ E\in(-\infty,+\infty)}} \mathrm{F}_\beta(E)\, \mathrm{Im} \frac{1}{E+\mu+\mathrm{i}\delta - \left(\epsilon_{m\sigma\mathbf{k}} + \widetilde{\Sigma}^{\mathrm{ret}}_{m\sigma\mathbf{k}}(E+\mu+\mathrm{i}\delta)\right)} = \dots \tag{30.82}
$$

whereas we introduced

$$
\widetilde{\Sigma}^{\mathrm{ret}}_{m\sigma\mathbf{k}}(E+\mu+\mathrm{i}\delta) = \widetilde{\Sigma}^{\mathrm{Hartree}}_{m\sigma\mathbf{k}}(E+\mu+\mathrm{i}\delta) + \widetilde{\Sigma}^{\mathrm{Fock}}_{m\sigma\mathbf{k}}(E+\mu+\mathrm{i}\delta) \tag{30.83}
$$

with Hartree-contribution

$$\widetilde{\Sigma}^{\text{Hartree}}_{m\sigma\mathbf{k}}(\bullet) = - \sum_{\substack{\mathbf{q}\in \mathrm{Q} \\ \sigma' \in \{\downarrow,\uparrow\} \\ n\in\{1,2,\dots\} \\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \frac{g_{mm\nu}(\mathbf{k},0) g_{nn\nu}(\mathbf{q},0)}{N_p} \frac{\mathrm{F}_\beta(\epsilon_{n\sigma'\mathbf{q}} - \mu)}{\Omega_\nu(\mathbf{q}' = 0)} \tag{30.84}$$

and Fock-contribution

$$\widetilde{\Sigma}^{\text{Fock}}_{m\sigma\mathbf{k}}(E + \mu + \mathrm{i}\delta) = \sum_{\substack{\mathbf{q}\in \mathrm{Q} \\ n\in\{1,2,\dots\} \\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \frac{|g_{nm\nu}(\mathbf{k},\mathbf{q})|^2}{N_p} \left[\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) + \mathrm{F}_\beta(\xi_{\mathbf{q}+\mathbf{k}n\sigma})}{E + \mu + \mathrm{i}\delta - \epsilon_{\mathbf{q}+\mathbf{k}n\sigma} + \Omega_\nu(\mathbf{q})} + \frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) + 1 - \mathrm{F}_\beta(\xi_{\mathbf{q}+\mathbf{k}n\sigma})}{E + \mu + \mathrm{i}\delta - \epsilon_{\mathbf{q}+\mathbf{k}n\sigma} - \Omega_\nu(\mathbf{q})} \right] \tag{30.85}$$

now, using the substitution-rule allows us to rewrite the above Equation as

$$\langle N \rangle (\mu, T) = \ldots = \frac{-1}{\pi} \lim_{\delta\to 0} \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in \mathrm{Q} \\ \sigma\in\{\uparrow,\downarrow\} \\ m\in\{1,\dots,\infty\} \\ E\in(-\infty,+\infty)}} \mathrm{F}_\beta(E-\mu) \,\mathrm{Im}\, \frac{1}{E + \mathrm{i}\delta - \left(\epsilon_{m\sigma\mathbf{k}} + \widetilde{\Sigma}^{\text{ret}}_{m\sigma\mathbf{k}}(E + \mathrm{i}\delta)\right)} \tag{30.86}$$

Now we use Equation (TBReferenced) together with Lemma TBReferenced, and introduce the function

$$g(\mu) = \beta(E - \mu) \quad \text{which implies} \quad \left(\frac{\mathrm{d}g}{\mathrm{d}\mu}\right)(\mu) = -\beta \tag{30.87}$$

such that we obtain

$$\mathrm{F}_\beta(E - \mu) = \mathrm{F}(g(\mu)) \tag{30.88}$$

which implies with the chain-rule

$$\partial_\mu \left(\mathrm{F} \circ g\right)(\mu) = \left(\partial_\mu \mathrm{F}\right)(g(\mu)) \cdot \left(\frac{\mathrm{d}g}{\mathrm{d}\mu}\right)(\mu) = \frac{1}{4}\,\mathrm{sech}^2\left(\beta \frac{E-\mu}{2}\right) \cdot \beta \tag{30.89}$$

Analogously we obtain

$$\partial_\mu \widetilde{\Sigma}^{\text{ret}}_{m\sigma\mathbf{k}}(E + \mathrm{i}\delta) = \partial_\mu \widetilde{\Sigma}^{\text{Hartree}}_{m\sigma\mathbf{k}}(E + \mathrm{i}\delta) + \partial_\mu \widetilde{\Sigma}^{\text{Fock}}_{m\sigma\mathbf{k}}(E + \mathrm{i}\delta) \tag{30.90}$$

with Hartree-contribution

$$\partial_\mu \widetilde{\Sigma}^{\text{Hartree}}_{m\sigma\mathbf{k}}(\bullet) = - \sum_{\substack{\mathbf{q}\in \mathrm{Q} \\ \sigma' \in \{\downarrow,\uparrow\} \\ n\in\{1,2,\dots\} \\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \frac{g_{mm\nu}(\mathbf{k},0) g_{nn\nu}(\mathbf{q},0)}{4N_p} \frac{\beta}{\Omega_\nu(\mathbf{q}' = 0)} \,\mathrm{sech}^2\left(\beta \xi_{n\sigma'\mathbf{q}}/2\right) \tag{30.91}$$

and Fock-contribution

$$\partial_\mu \widetilde{\Sigma}^{\text{Fock}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta) = \sum_{\substack{\mathbf{q}\in Q\\ n\in\{1,2,\dots\}\\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \frac{|g_{nm\nu}(\mathbf{k},\mathbf{q})|^2}{4N_p k_B T}\operatorname{sech}^2(\beta\xi_{\mathbf{q}+\mathbf{k}n\sigma}/2)\left[\frac{1}{E+\mathrm{i}\delta-\epsilon_{\mathbf{q}+\mathbf{k}n\sigma}+\Omega_\nu(\mathbf{q})}-\frac{1}{E+\mathrm{i}\delta-\epsilon_{\mathbf{q}+\mathbf{k}n\sigma}-\Omega_\nu(\mathbf{q})}\right] \tag{30.92}$$

Now we can compute the derivative as

$$\langle N\rangle(\mu,T) = \frac{-1}{\pi}\lim_{\delta\to 0}\sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\dots,\infty\}\\ E\in(-\infty,+\infty)}} \mathrm{F}_\beta(E-\mu)\,\mathrm{Im}\left[E+\mathrm{i}\delta-\left(\epsilon_{m\sigma\mathbf{k}}+\widetilde{\Sigma}^{\text{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta)\right)\right]^{-1} \tag{30.93}$$

which implies

$$\begin{aligned}
\partial_\mu\langle N\rangle(\mu,T) &= \frac{-1}{\pi}\lim_{\delta\to 0}\sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\dots,\infty\}\\ E\in(-\infty,+\infty)}} \frac{\beta}{4}\operatorname{sech}^2\left(\beta\frac{E-\mu}{2}\right)\mathrm{Im}\left[E+\mathrm{i}\delta-\left(\epsilon_{m\sigma\mathbf{k}}+\widetilde{\Sigma}^{\text{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta)\right)\right]^{-1}\\
&+\frac{-1}{\pi}\lim_{\delta\to 0}\sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\dots,\infty\}\\ E\in(-\infty,+\infty)}} \mathrm{F}_\beta(E-\mu)\cdot(-1)^2\cdot\mathrm{Im}\frac{\partial_\mu\widetilde{\Sigma}^{\text{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta)}{\left(E+\mathrm{i}\delta-\left(\epsilon_{m\sigma\mathbf{k}}+\widetilde{\Sigma}^{\text{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta)\right)\right)^2}\\
&= \frac{-1}{\pi}\lim_{\delta\to 0}\sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\dots,\infty\}\\ E\in(-\infty,+\infty)}} \frac{\beta}{4}\left[\operatorname{sech}^2(\beta E/2)\,\mathrm{Im}\frac{1}{E+\mathrm{i}\delta-\left(\xi_{m\sigma\mathbf{k}}+\Sigma^{\text{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta)\right)}\right.\\
&\qquad\left.+\mathrm{F}_\beta(E)\,\mathrm{Im}\frac{\sigma_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta)}{\left[E+\mathrm{i}\delta-\left(\xi_{m\sigma\mathbf{k}}+\Sigma^{\text{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta)\right)\right]^2}\right]
\end{aligned} \tag{30.94}$$

whereas we introduced the notation

$$\sigma_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta) = \sigma^{\text{Hartree}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta)+\sigma^{\text{Fock}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta), \tag{30.95}$$

with the Hartree-contribution

$$\sigma^{\text{Hartree}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta) = -\sum_{\substack{\mathbf{q}\in Q\\ \sigma'\in\{\downarrow,\uparrow\}\\ n\in\{1,2,\dots\}\\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \frac{g_{mm\nu}(\mathbf{k},0)g_{nn\nu}(\mathbf{q},0)}{N_p}\frac{1}{\Omega_\nu(\mathbf{q}'=0)}\operatorname{sech}^2\left(\beta\xi_{n\sigma'\mathbf{q}}/2\right) \tag{30.96}$$

and the Fock-contribution

$$\sigma^{\text{Fock}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta) = \sum_{\substack{\mathbf{q}\in Q\\ n\in\{1,2,\dots\}\\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \frac{|g_{nm\nu}(\mathbf{k},\mathbf{q})|^2}{N_p}\operatorname{sech}^2(\beta\xi_{\mathbf{q}+\mathbf{k}n\sigma}/2)\left[\frac{1}{E+\mathrm{i}\delta-\xi_{\mathbf{q}+\mathbf{k}n\sigma}+\Omega_\nu(\mathbf{q})}-\frac{1}{E+\mathrm{i}\delta-\xi_{\mathbf{q}+\mathbf{k}n\sigma}-\Omega_\nu(\mathbf{q})}\right] \tag{30.97}$$

Proof 30.19 of Lemma 24.24

We compute the derivative of Equation (24.22) as

$$\begin{aligned}
\partial_T \langle H_{\mathrm{el}} \rangle &= \partial_T \frac{-1}{\pi} \lim_{\delta\to 0} \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\ldots,\infty\}\\ E\in(-\infty,+\infty)}} \epsilon_{m\mathbf{k}\sigma}\, \mathrm{F}_\beta(E)\, \mathrm{Im}\, G^{\mathrm{ret}}_{\substack{m,m\\ \sigma,\sigma}}(E+\mathrm{i}\delta,\mathbf{k}) \\
&= \frac{-1}{\pi} \lim_{\delta\to 0} \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in Q\\ \sigma\in\{\uparrow,\downarrow\}\\ m\in\{1,\ldots,\infty\}\\ E\in(-\infty,+\infty)}} \epsilon_{m\mathbf{k}\sigma} \Big[\big(\partial_T \mathrm{F}_\beta(E)\big)\, \mathrm{Im}\, G^{\mathrm{ret}}_{\substack{m,m\\ \sigma,\sigma}}(E+\mathrm{i}\delta,\mathbf{k}) + \mathrm{F}_\beta(E)\, \mathrm{Im}\, \partial_T G^{\mathrm{ret}}_{\substack{m,m\\ \sigma,\sigma}}(E+\mathrm{i}\delta,\mathbf{k}) \Big] \\
&= \ldots
\end{aligned} \tag{30.98}$$

Now we use Lemma 9.47, i.e. we use the Equation

$$\mathrm{d}_T \mathrm{F}_\beta(E) = \frac{E}{k_B T^2} \mathrm{F}_\beta(E) \big(1 - \mathrm{F}_\beta(E)\big) \tag{30.99}$$

and for the other derivative we have

$$\partial_T G^{\mathrm{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta) = \partial_T \frac{1}{E+\mathrm{i}\delta - \big(\epsilon_{m\mathbf{k}\sigma} + \Sigma^{\mathrm{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta) - \mu\big)} = \frac{\partial_T \Sigma^{\mathrm{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta)}{E+\mathrm{i}\delta - \big(\epsilon_{m\mathbf{k}\sigma} + \Sigma^{\mathrm{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta) - \mu\big)^2} \tag{30.100}$$

Now we compute the derivative and use the diagonal bands approximation for the self-energy and use Lemma 9.47

$$\begin{aligned}
\partial_T \Sigma^{\mathrm{ret}}_{m\sigma\mathbf{k}}(E+\mathrm{i}\delta) &= \sum_{\substack{\mathbf{q}\in Q\\ n\in\{1,\ldots,\infty\}\\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}}} \frac{|g_{nm\nu}(\mathbf{k},\mathbf{q})|^2}{N_p} \left[\partial_T \frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) + \mathrm{F}_\beta(\xi_{n\sigma\mathbf{k+q}})}{E+\mathrm{i}\delta - \xi_{n\sigma\mathbf{k+q}} + \Omega_\nu(\mathbf{q})} + \partial_T \frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) + 1 - \mathrm{F}_\beta(\xi_{n\sigma\mathbf{k+q}})}{E+\mathrm{i}\delta - \xi_{n\sigma\mathbf{k+q}} - \Omega_\nu(\mathbf{q})} \right] \\
&= \sum_{\substack{\mathbf{q}\in Q\\ n\in\{1,\ldots,\infty\}\\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}}} \frac{|g_{nm\nu}(\mathbf{k},\mathbf{q})|^2}{4 k_B T^2 N_p} \left[\frac{\Omega_\nu(\mathbf{q}) \operatorname{csch}^2(\beta\Omega_\nu(\mathbf{q})/2) + \xi_{n\sigma\mathbf{k+q}} \operatorname{sech}^2(\beta\xi_{n\sigma\mathbf{k+q}}/2)}{E+\mathrm{i}\delta - \xi_{n\sigma\mathbf{k+q}} + \Omega_\nu(\mathbf{q})} + \frac{\Omega_\nu(\mathbf{q}) \operatorname{csch}^2(\beta\Omega_\nu(\mathbf{q})/2) - \xi_{n\sigma\mathbf{k+q}} \operatorname{sech}^2(\beta\xi_{n\sigma\mathbf{k+q}}/2)}{E+\mathrm{i}\delta - \xi_{n\sigma\mathbf{k+q}} - \Omega_\nu(\mathbf{q})} \right]
\end{aligned} \tag{30.101}$$

Proof 30.20 of Lemma 24.26

Taking the temperature-derivative of

$$\langle H_{\mathrm{ph}} \rangle_0 = \sum_{\substack{\mathbf{q}\in Q\\ \nu\in\{1,2,\ldots,rd_{\mathbf{x}}\}}} \Omega_\nu(\mathbf{q})\, \mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) \tag{30.102}$$

and using Lemma 9.47 we obtain the desired result.

Now we take care of the corrections. We compute the derivative

$$
\begin{aligned}
\mathrm{d}_T\langle \delta H_{\mathrm{ph}}\rangle \approx & \sum_{\nu\in\{1,2,\dots,rd_x\}} A_\nu \mathrm{d}_T\left(B_\nu\right)^2 \\
& - \mathrm{d}_T \sum_{\substack{\mathbf{k},\mathbf{q}\in Q\\ \nu\in\{1,2,\dots,rd_x\}\\ n,n'\in\{1,\dots,\infty\}\\ \sigma\in\{\downarrow,\uparrow\}}} \frac{\Omega_\nu(\mathbf{q})}{N_p}|g_{n'n\nu}(\mathbf{k},\mathbf{q})|^2\left(\mathrm{F}_\beta(\xi_{n,\sigma,\mathbf{k}})-\mathrm{F}_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}})\right)\left[\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))-\mathrm{B}_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}-\xi_{n,\sigma,\mathbf{k}})}{\left(\Omega_\nu(\mathbf{q})+\xi_{n,\sigma,\mathbf{k}}-\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}\right)^2}+\frac{1}{4}\frac{\beta\,\mathrm{csch}(\beta\Omega_\nu(\mathbf{q})/2)^2}{\Omega_\nu(\mathbf{q})+\xi_{n,\sigma,\mathbf{k}}-\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}}\right]
\end{aligned}
$$

we neglect the first term for the beginning, since it only contributes for crystals with more than one atom in the UC and focus on the second term, which can be computed as

$$
\begin{aligned}
& - \mathrm{d}_T \sum_{\substack{\mathbf{k},\mathbf{q}\in Q\\ \nu\in\{1,2,\dots,rd_x\}\\ n,n'\in\{1,\dots,\infty\}\\ \sigma\in\{\downarrow,\uparrow\}}} \frac{\Omega_\nu(\mathbf{q})}{N_p}|g_{n'n\nu}(\mathbf{k},\mathbf{q})|^2\left(\mathrm{F}_\beta(\xi_{n,\sigma,\mathbf{k}})-\mathrm{F}_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}})\right)\left[\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))-\mathrm{B}_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}-\xi_{n,\sigma,\mathbf{k}})}{\left(\Omega_\nu(\mathbf{q})+\xi_{n,\sigma,\mathbf{k}}-\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}\right)^2}+\frac{1}{4}\frac{\beta\,\mathrm{csch}(\beta\Omega_\nu(\mathbf{q})/2)^2}{\Omega_\nu(\mathbf{q})+\xi_{n,\sigma,\mathbf{k}}-\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}}\right] \\
= & - \sum_{\substack{\mathbf{k},\mathbf{q}\in Q\\ \nu\in\{1,2,\dots,rd_x\}\\ n,n'\in\{1,\dots,\infty\}\\ \sigma\in\{\downarrow,\uparrow\}}} \frac{\Omega_\nu(\mathbf{q})}{N_p}|g_{n'n\nu}(\mathbf{k},\mathbf{q})|^2\left[\mathrm{d}_T\left(\mathrm{F}_\beta(\xi_{n,\sigma,\mathbf{k}})-\mathrm{F}_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}})\right)\right]\left[\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))-\mathrm{B}_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}-\xi_{n,\sigma,\mathbf{k}})}{\left(\Omega_\nu(\mathbf{q})+\xi_{n,\sigma,\mathbf{k}}-\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}\right)^2}+\frac{1}{4}\frac{\beta\,\mathrm{csch}(\beta\Omega_\nu(\mathbf{q})/2)^2}{\Omega_\nu(\mathbf{q})+\xi_{n,\sigma,\mathbf{k}}-\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}}\right] \\
& - \sum_{\substack{\mathbf{k},\mathbf{q}\in Q\\ \nu\in\{1,2,\dots,rd_x\}\\ n,n'\in\{1,\dots,\infty\}\\ \sigma\in\{\downarrow,\uparrow\}}} \frac{\Omega_\nu(\mathbf{q})}{N_p}|g_{n'n\nu}(\mathbf{k},\mathbf{q})|^2\left(\mathrm{F}_\beta(\xi_{n,\sigma,\mathbf{k}})-\mathrm{F}_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}})\right)\left[\mathrm{d}_T\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))-\mathrm{B}_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}-\xi_{n,\sigma,\mathbf{k}})}{\left(\Omega_\nu(\mathbf{q})+\xi_{n,\sigma,\mathbf{k}}-\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}\right)^2}+\frac{1}{4}\mathrm{d}_T\frac{\beta\,\mathrm{csch}(\beta\Omega_\nu(\mathbf{q})/2)^2}{\Omega_\nu(\mathbf{q})+\xi_{n,\sigma,\mathbf{k}}-\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}}\right]
\end{aligned}
$$

Using now again Lemma 9.47 we obtain the stated result for the first two terms. For the last term, we note the identity

$$
\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{csch}(x) = -\coth(x)\,\mathrm{csch}(x) \tag{30.103}
$$

which gives

$$
\begin{aligned}
& \mathrm{d}_T\left(\frac{1}{k_BT}\cdot\mathrm{csch}^2\left(\frac{\Omega_\nu(\mathbf{q})}{2k_BT}\right)\right) \\
= & \frac{-1}{k_BT^2}\mathrm{csch}^2\left(\frac{\Omega_\nu(\mathbf{q})}{2k_BT}\right)+\frac{1}{k_BT}\cdot 2\cdot\mathrm{csch}\left(\frac{\Omega_\nu(\mathbf{q})}{2k_BT}\right)\cdot\left(-\coth\left(\frac{\Omega_\nu(\mathbf{q})}{2k_BT}\right)\cdot\mathrm{csch}\left(\frac{\Omega_\nu(\mathbf{q})}{2k_BT}\right)\cdot\left(\frac{-\Omega_\nu(\mathbf{q})}{2k_BT^2}\right)\right) \\
= & \frac{-1}{k_BT^2}\mathrm{csch}^2\left(\frac{\Omega_\nu(\mathbf{q})}{2k_BT}\right)\cdot\left[1-\frac{\Omega_\nu(\mathbf{q})}{k_BT}\coth\left(\frac{\Omega_\nu(\mathbf{q})}{2k_BT}\right)\right]
\end{aligned} \tag{30.104}
$$

Proof 30.21 of Theorem 23.16

The grand canonical potential $\Omega(\mu,T,V)$ obeys the equation[4]

$$
\Omega-\Omega_0 = -\frac{1}{\beta}\,\mathrm{Log}\,\frac{Z}{Z_0} = -\frac{1}{\beta}\times\text{„all connected diagrams“} \tag{30.105}
$$

Thus we compute the connected diagrams of the system.

[4] See also Mahan 3.6.1 for an expansion of the thermodynamic potential.

Computing the connected Diagrams through a Series-Expansion

We can therefore simply compute the partition-function in a perturbative expansion, take from the appearing Feynman-diagrams the connected ones and add the expression for the non-interacting grand canonical potential on top of it.

$$
\begin{aligned}
&\frac{Z}{Z_0}\\
&= \langle \exp - S_{\text{int}}[\overline{\chi}, \chi] \rangle_0\\
&= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \langle S_{\text{int}}^n[\overline{\chi}, \chi] \rangle_0\\
&= 1 + 0\\
&+ \frac{1}{2} \sum\!\!\!\!\!\!\int_{\substack{\tau_1,\tau_2 \in [0,\hbar\beta]\\ \mu,\mu' \in \{\downarrow,\uparrow\}\\ \mathbf{k},\mathbf{k}',\mathbf{q},\mathbf{q}' \in \mathrm{Q}\\ m,m',n,n' \in \{1,2,\dots\}\\ \nu,\nu' \in \{1,2,\dots,rd_{\mathbf{x}}\}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q}) g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{\hbar^2 N_p} \left\langle \left(\phi_{\mathbf{q}\nu}(\tau_1) + \overline{\phi}_{-\mathbf{q}\nu}(\tau_1)\right) \overline{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \psi_{\mathbf{k}n\mu}(\tau_1) \left(\phi_{\mathbf{q}'\nu'}(\tau_2) + \overline{\phi}_{-\mathbf{q}'\nu'}(\tau_2)\right) \overline{\psi}_{\mathbf{k}'+\mathbf{q}'m'}\right.\\
&= \dots
\end{aligned}
$$

We now factorize and compute the correlators

$$
\begin{aligned}
&\left\langle \left(\phi_{\mathbf{q}\nu}(\tau_1) + \overline{\phi}_{-\mathbf{q}\nu}(\tau_1)\right) \overline{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \psi_{\mathbf{k}n\mu}(\tau_1) \left(\phi_{\mathbf{q}'\nu'}(\tau_2) + \overline{\phi}_{-\mathbf{q}'\nu'}(\tau_2)\right) \overline{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2) \psi_{\mathbf{k}'n'\mu'}(\tau_2) \right\rangle_0\\
&= \left\langle \overline{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \psi_{\mathbf{k}n\mu}(\tau_1) \overline{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2) \psi_{\mathbf{k}'n'\mu'}(\tau_2) \right\rangle_0 \left\langle \left(\phi_{\mathbf{q}\nu}(\tau_1) + \overline{\phi}_{-\mathbf{q}\nu}(\tau_1)\right) \left(\phi_{\mathbf{q}'\nu'}(\tau_2) + \overline{\phi}_{-\mathbf{q}'\nu'}(\tau_2)\right) \right\rangle_0\\
&= -\hbar \left\langle \overline{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \psi_{\mathbf{k}n\mu}(\tau_1) \overline{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2) \psi_{\mathbf{k}'n'\mu'}(\tau_2) \right\rangle_0 \mathscr{D}^{\,0}_{\substack{\mathbf{q},-\mathbf{q}'\\ \nu,\nu'}}(\tau_1 - \tau_2)\\
&= +\hbar \left\langle \psi_{\mathbf{k}n\mu}(\tau_1) \psi_{\mathbf{k}'n'\mu'}(\tau_2) \overline{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \overline{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2) \right\rangle_0 \mathscr{D}^{\,0}_{\substack{\mathbf{q},-\mathbf{q}'\\ \nu,\nu'}}(\tau_1 - \tau_2)\\
&= +\hbar^3 \left(\mathscr{G}^{\,0}_{\substack{\mathbf{k},\mathbf{k}'+\mathbf{q}'\\ n,m'\\ \mu,\mu'}}(\tau_1 - \tau_2) \mathscr{G}^{\,0}_{\substack{\mathbf{k}',\mathbf{k}+\mathbf{q}\\ n',m\\ \mu',\mu}}(\tau_2 - \tau_1) - \mathscr{G}^{\,0}_{\substack{\mathbf{k},\mathbf{k}+\mathbf{q}\\ n,m\\ \mu,\mu}}(\tau_1 - \tau_1) \mathscr{G}^{\,0}_{\substack{\mathbf{k}',\mathbf{k}'+\mathbf{q}'\\ n',m'\\ \mu',\mu'}}(\tau_2 - \tau_2) \right) \mathscr{D}^{\,0}_{\substack{\mathbf{q},-\mathbf{q}'\\ \nu,\nu'}}(\tau_1 - \tau_2)
\end{aligned}
$$

Reinserting this in the equation above yields

$$
\begin{aligned}
\dots &= 1\\
&+ \frac{\hbar}{2N_p} \sum\!\!\!\!\!\!\int_{\substack{\tau_1,\tau_2 \in [0,\hbar\beta]\\ \mu \in \{\downarrow,\uparrow\}\\ \mathbf{k},\mathbf{k}',\mathbf{q} \in \mathrm{Q}\\ m,n \in \{1,2,\dots\}\\ \nu \in \{1,2,\dots,rd_{\mathbf{x}}\}}} g_{mn\nu}(\mathbf{k},\mathbf{q}) g_{nm\nu}(\mathbf{k}',-\mathbf{q}) \mathscr{G}^{\,0}_{\substack{\mathbf{k},\mathbf{k}'-\mathbf{q}\\ n,n\\ \mu,\mu}}(\tau_1 - \tau_2) \mathscr{G}^{\,0}_{\substack{\mathbf{k}',\mathbf{k}+\mathbf{q}\\ m,m\\ \mu,\mu}}(\tau_2 - \tau_1) \mathscr{D}_{0\,\mathbf{q}\nu}(\tau_1 - \tau_2)\\
&- \frac{\hbar}{2N_p} \sum\!\!\!\!\!\!\int_{\substack{\tau_1,\tau_2 \in [0,\hbar\beta]\\ \mu,\mu' \in \{\downarrow,\uparrow\}\\ \mathbf{k},\mathbf{k}',\mathbf{q} \in \mathrm{Q}\\ m,m' \in \{1,2,\dots\}\\ \nu,\nu' \in \{1,2,\dots,rd_{\mathbf{x}}\}}} g_{mm\nu}(\mathbf{k},\mathbf{q}) g_{m'm'\nu'}(\mathbf{k}',-\mathbf{q}) \mathscr{G}^{\,0}_{\substack{\mathbf{k},\mathbf{k}+\mathbf{q}\\ m,m\\ \mu,\mu}}(\tau_1 - \tau_1) \mathscr{G}^{\,0}_{\substack{\mathbf{k}',\mathbf{k}'-\mathbf{q}\\ m',m'\\ \mu',\mu'}}(\tau_2 - \tau_2) \mathscr{D}_{0\,\mathbf{q},\nu}(\tau_1 - \tau_2)
\end{aligned}
$$

This can be simplified by using the Kronecker-relation for the delta-functions

$$\dots = 1 + \frac{\hbar}{2N_p} \sum\!\!\!\!\!\!\int_{\substack{\tau_1,\tau_2 \in [0,\hbar\beta] \\ \mu \in \{\downarrow,\uparrow\} \\ \mathbf{k}',\mathbf{q} \in \mathrm{Q} \\ m,n \in \{1,2,\dots\} \\ \nu \in \{1,2,\dots,rd_{\mathbf{x}}\}}} g_{mn\nu}(\mathbf{k}'-\mathbf{q},\mathbf{q}) g^*_{mn\nu}(\mathbf{k}'-\mathbf{q},+\mathbf{q}) \mathscr{G}_{\substack{\mathbf{k}'-\mathbf{q},\mathbf{k}'-\mathbf{q} \\ n,n \\ \mu,\mu}}(\tau_1-\tau_2) \mathscr{G}_{\substack{\mathbf{k}',\mathbf{k}' \\ m,m \\ \mu,\mu}}(\tau_2-\tau_1) \mathscr{D}_{\substack{0 \\ \mathbf{q}\nu}}(\tau_1-\tau_2)$$

$$- \frac{\hbar}{2N_p} \sum\!\!\!\!\!\!\int_{\substack{\tau_1,\tau_2 \in [0,\hbar\beta] \\ \mu,\mu' \in \{\downarrow,\uparrow\} \\ \mathbf{k},\mathbf{k}' \in \mathrm{Q} \\ m,m' \in \{1,2,\dots\} \\ \nu,\nu' \in \{1,2,\dots,rd_{\mathbf{x}}\}}} g_{mm\nu}(\mathbf{k},0) g_{m'm'\nu'}(\mathbf{k}',0) \mathscr{G}_{\substack{\mathbf{k},\mathbf{k} \\ m,m \\ \mu,\mu}}(\tau_1-\tau_1) \mathscr{G}_{\substack{\mathbf{k}',\mathbf{k}' \\ m',m' \\ \mu',\mu'}}(\tau_2-\tau_2) \mathscr{D}_{\substack{0 \\ \mathbf{q}=0,\nu}}(\tau_1-\tau_2) = \dots$$

Now using the hermiticity of g and changing summation $\mathbf{q} \mapsto -\mathbf{q}$ and simplifying the notation we obtain

$$\dots = 1 + \frac{\hbar}{2N_p} \sum\!\!\!\!\!\!\int_{\substack{\tau_1,\tau_2 \in [0,\hbar\beta] \\ \mu \in \{\downarrow,\uparrow\} \\ \mathbf{k}',\mathbf{q} \in \mathrm{Q} \\ m,n \in \{1,2,\dots\} \\ \nu \in \{1,2,\dots,rd_{\mathbf{x}}\}}} |g_{mn\nu}(\mathbf{k}'+\mathbf{q},-\mathbf{q})|^2 \mathscr{G}_{\substack{0 \\ n,\mu}}(\mathbf{k}'+\mathbf{q},\tau_1-\tau_2) \mathscr{G}_{\substack{0 \\ m,\mu}}(\mathbf{k}',\tau_2-\tau_1) \mathscr{D}_{\substack{0 \\ \nu}}(\mathbf{q},\tau_1-\tau_2)$$

$$- \frac{\hbar}{2N_p} \sum\!\!\!\!\!\!\int_{\substack{\tau_1,\tau_2 \in [0,\hbar\beta] \\ \mu,\mu' \in \{\downarrow,\uparrow\} \\ \mathbf{k},\mathbf{k}' \in \mathrm{Q} \\ m,m' \in \{1,2,\dots\} \\ \nu,\nu' \in \{1,2,\dots,rd_{\mathbf{x}}\}}} g_{mm\nu}(\mathbf{k},0) g_{m'm'\nu'}(\mathbf{k}',0) \mathscr{G}_{\substack{0 \\ m,\mu}}(\mathbf{k},\tau_1-\tau_1) \mathscr{G}_{\substack{0 \\ m',\mu'}}(\mathbf{k}',\tau_2-\tau_2) \mathscr{D}_{\substack{0 \\ \nu}}(\mathbf{q}=0,\tau_1-\tau_2) = \dots$$

ToBeUpdated, because one of the terms doesn't neccessarily vanish. [5]

[See Waste-Folder 1, topic 18 for the final expressions.]

Remark: Notice that one could also use now

$$|g_{mn\nu}(\mathbf{k}'+\mathbf{q},-\mathbf{q})| = |g_{nm\nu}(\mathbf{k}',\mathbf{q})| \tag{30.106}$$

[5] Cf. the GPAW-documentation:
For the three translational modes at $\mathbf{q} = 0$ the matrix elements $g_{mn\nu}(\mathbf{k},\mathbf{q}) = 0$, as a consequence of the acoustic sum rule.
See `https://gpaw.readthedocs.io/documentation/elph/elph.html`
See further Felicianos article:
In the case of the three translational modes at $|\mathbf{q}| = 0$ we set the matrix elements $g_{mn\nu}(\mathbf{k},\mathbf{q})$ to zero, as a consequence of the acoustic sum rule (...).

31. Appendix - Proofs for Part V

Proof 31.1 of Lemma 34.4

We compute the matrix-elements to be

$$
\begin{aligned}
\langle \mathbf{k}_1\sigma_1, \mathbf{k}_2\sigma_2|V|\mathbf{k}_3\sigma_3, \mathbf{k}_4\sigma_4\rangle &= -|g|\delta_{\sigma_3\neq\sigma_4}\delta_{\sigma_1,\sigma_3}\delta_{\sigma_2,\sigma_4}\int \mathrm{d}^3\mathbf{r}\mathrm{d}^3\mathbf{r}'\delta(\mathbf{r}-\mathbf{r}')\psi^*_{\mathbf{k}_1}(\mathbf{r})\psi^*_{\mathbf{k}_2}(\mathbf{r}')\psi_{\mathbf{k}_3}(\mathbf{r})\psi_{\mathbf{k}_4}(\mathbf{r}') \\
&= -|g|\delta_{\sigma_3\neq\sigma_4}\delta_{\sigma_1,\sigma_3}\delta_{\sigma_2,\sigma_4}\int \mathrm{d}^3\mathbf{r}\psi^*_{\mathbf{k}_1}(\mathbf{r})\psi^*_{\mathbf{k}_2}(\mathbf{r})\psi_{\mathbf{k}_3}(\mathbf{r})\psi_{\mathbf{k}_4}(\mathbf{r}') \\
&= -\frac{|g|}{L^{d_\mathbf{x}}}\delta_{\sigma_3\neq\sigma_4}\delta_{\sigma_1,\sigma_3}\delta_{\sigma_2,\sigma_4}\delta_{\mathbf{k}_1+\mathbf{k}_2,\mathbf{k}_3+\mathbf{k}_4}
\end{aligned}
$$

Proof 31.2 of Theorem 34.5 We have computed the matrix-elements and will now insert them into the definition of the second quantized version of V. This yields

$$
\begin{aligned}
V^{2\mathrm{Q}} &= \frac{1}{2}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4}\sum_{\sigma_1,\sigma_2,\sigma_3,\sigma_4}\frac{-|g|}{L^{d_\mathbf{x}}}\delta_{\sigma_3\neq\sigma_4}\delta_{\sigma_1,\sigma_3}\delta_{\sigma_2,\sigma_4}\delta_{\mathbf{k}_1+\mathbf{k}_2,\mathbf{k}_3+\mathbf{k}_4}c^\dagger_{\mathbf{k}_1\sigma_1}c^\dagger_{\mathbf{k}_2\sigma_2}c_{\mathbf{k}_4\sigma_4}c_{\mathbf{k}_3\sigma_3} \\
&= \frac{-|g|}{2L^{d_\mathbf{x}}}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4}\sum_{\sigma_1,\sigma_2}\delta_{\sigma_1\neq\sigma_2}\delta_{\mathbf{k}_1+\mathbf{k}_2,\mathbf{k}_3+\mathbf{k}_4}c^\dagger_{\mathbf{k}_1\sigma_1}c^\dagger_{\mathbf{k}_2\sigma_2}c_{\mathbf{k}_4\sigma_2}c_{\mathbf{k}_3\sigma_1} \\
&= \frac{-|g|}{2L^{d_\mathbf{x}}}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4}\delta_{\mathbf{k}_1+\mathbf{k}_2,\mathbf{k}_3+\mathbf{k}_4}\left(c^\dagger_{\mathbf{k}_1\uparrow}c^\dagger_{\mathbf{k}_2\downarrow}c_{\mathbf{k}_4\downarrow}c_{\mathbf{k}_3\uparrow} + c^\dagger_{\mathbf{k}_1\downarrow}c^\dagger_{\mathbf{k}_2\uparrow}c_{\mathbf{k}_4\uparrow}c_{\mathbf{k}_3\downarrow}\right) \\
&= \frac{-|g|}{L^{d_\mathbf{x}}}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4}\delta_{\mathbf{k}_1+\mathbf{k}_2,\mathbf{k}_3+\mathbf{k}_4}c^\dagger_{\mathbf{k}_1\uparrow}c^\dagger_{\mathbf{k}_2\downarrow}c_{\mathbf{k}_4\downarrow}c_{\mathbf{k}_3\uparrow}
\end{aligned}
$$

Lastly we need to relabel the summation-indices to obtain the desired form of the Hamiltonian. For this we do the following manipulations

$$
\begin{aligned}
V^{2\mathrm{Q}} &= \frac{-|g|}{L^{d_{\mathbf{x}}}} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} \delta_{\mathbf{k}_1+\mathbf{k}_2,\mathbf{k}_3+\mathbf{k}_4} c^\dagger_{\mathbf{k}_1\uparrow} c^\dagger_{\mathbf{k}_2\downarrow} c_{\mathbf{k}_4\downarrow} c_{\mathbf{k}_3\uparrow} \\
&\overset{k_3\leftrightarrow k_4}{=} \frac{-|g|}{L^{d_{\mathbf{x}}}} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} \delta_{\mathbf{k}_1+\mathbf{k}_2,\mathbf{k}_3+\mathbf{k}_4} c^\dagger_{\mathbf{k}_1\uparrow} c^\dagger_{\mathbf{k}_2\downarrow} c_{\mathbf{k}_3\downarrow} c_{\mathbf{k}_4\uparrow} \\
&= \frac{-|g|}{L^{d_{\mathbf{x}}}} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_4} c^\dagger_{\mathbf{k}_1\uparrow} c^\dagger_{\mathbf{k}_2\downarrow} c_{\mathbf{k}_1+\mathbf{k}_2-\mathbf{k}_4\downarrow} c_{\mathbf{k}_4\uparrow} \\
&\overset{k_2\to -k_2}{=} \frac{-|g|}{L^{d_{\mathbf{x}}}} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_4} c^\dagger_{\mathbf{k}_1\uparrow} c^\dagger_{-\mathbf{k}_2\downarrow} c_{\mathbf{k}_1-\mathbf{k}_2-\mathbf{k}_4\downarrow} c_{\mathbf{k}_4\uparrow} \\
&\overset{k_2\equiv k, k_4\equiv k'}{=} \frac{-|g|}{L^{d_{\mathbf{x}}}} \sum_{\mathbf{k}_1,\mathbf{k},\mathbf{k}'} c^\dagger_{\mathbf{k}_1\uparrow} c^\dagger_{-\mathbf{k}\downarrow} c_{\mathbf{k}_1-\mathbf{k}-\mathbf{k}'\downarrow} c_{\mathbf{k}'\uparrow} \\
&\overset{k_1\equiv q}{=} \frac{-|g|}{L^{d_{\mathbf{x}}}} \sum_{\mathbf{q},\mathbf{k},\mathbf{k}'} c^\dagger_{\mathbf{q}\uparrow} c^\dagger_{-\mathbf{k}\downarrow} c_{\mathbf{q}-\mathbf{k}-\mathbf{k}'\downarrow} c_{\mathbf{k}'\uparrow} \\
&\overset{q\to q+k}{=} \frac{-|g|}{L^{d_{\mathbf{x}}}} \sum_{\mathbf{q},\mathbf{k},\mathbf{k}'} c^\dagger_{\mathbf{q}+\mathbf{k}\uparrow} c^\dagger_{-\mathbf{k}\downarrow} c_{\mathbf{q}-\mathbf{k}'\downarrow} c_{\mathbf{k}'\uparrow}
\end{aligned}
$$

proving the theorem.

Proof 31.3 of Lemma 34.7

See Draft-Summary.

Proof 31.4 of Corollary 34.8

The First Identity

We have the notation

$$
\langle \mathbf{x}|\mathbf{q}\rangle \equiv \frac{1}{\sqrt{L^{d_{\mathbf{x}}}}} \mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{x}} \equiv \psi_{\mathbf{q}}(\mathbf{x}) \tag{31.1}
$$

for the spatial part of wavefunctions. The only term that was not proven in the [Draft Summary] is the interaction-term.

Hence we compute

$$
\begin{aligned}
\int \mathrm{d}\mathbf{r}\, c^\dagger_\uparrow(\mathbf{r}) c^\dagger_\downarrow(\mathbf{r}) c_\downarrow(\mathbf{r}) c_\uparrow(\mathbf{r}) &= \sum_{\mathbf{k}_1,\ldots,\mathbf{k}_4} \int \mathrm{d}\mathbf{r}\, \overline{\psi}_{\mathbf{k}_1\uparrow}(\mathbf{r}) \overline{\psi}_{\mathbf{k}_2\downarrow}(\mathbf{r}) \psi_{\mathbf{k}_3\downarrow}(\mathbf{r}) \psi_{\mathbf{k}_4\uparrow}(\mathbf{r}) c^\dagger_\uparrow(\mathbf{k}_1) c^\dagger_\downarrow(\mathbf{k}_2) c_\downarrow(\mathbf{k}_3) c_\uparrow(\mathbf{k}_4) \\
&= \ldots
\end{aligned}
$$

the integral factor can[1] be simplified to

[1]Cf. page 64 in Supermathematics.

$$\int \mathrm{d}\mathbf{r}\, \overline{\psi}_{\mathbf{k}_1\uparrow}(\mathbf{r})\overline{\psi}_{\mathbf{k}_2\downarrow}(\mathbf{r})\psi_{\mathbf{k}_3\downarrow}(\mathbf{r})\psi_{\mathbf{k}_4\uparrow}(\mathbf{r}) = \frac{1}{L^{2d_\mathrm{x}}}\int \mathrm{d}\mathbf{r}\, \mathrm{e}^{\mathrm{i}(\mathbf{k}_3+\mathbf{k}_4-(\mathbf{k}_1+\mathbf{k}_2))} = \frac{1}{L^{d_\mathrm{x}}}\delta_{\mathbf{k}_1+\mathbf{k}_2,\mathbf{k}_3+\mathbf{k}_4} \tag{31.2}$$

hence we obtain

$$\begin{aligned}
\ldots &= \frac{1}{L^{d_\mathrm{x}}}\sum_{\mathbf{k}_1,\ldots,\mathbf{k}_3} c_\uparrow^\dagger(\mathbf{k}_1)c_\downarrow^\dagger(\mathbf{k}_2)c_\downarrow(\mathbf{k}_3)c_\uparrow(\mathbf{k}_1+\mathbf{k}_2-\mathbf{k}_3)\\
&\overset{\mathbf{k}_2\to-\mathbf{k}_2}{=} \frac{1}{L^{d_\mathrm{x}}}\sum_{\mathbf{k}_1,\ldots,\mathbf{k}_3} c_\uparrow^\dagger(\mathbf{k}_1)c_\downarrow^\dagger(-\mathbf{k}_2)c_\downarrow(\mathbf{k}_3)c_\uparrow(\mathbf{k}_1-(\mathbf{k}_2+\mathbf{k}_3))\\
&\overset{\mathbf{k}_2\to\mathbf{k}}{=} \frac{1}{L^{d_\mathrm{x}}}\sum_{\mathbf{k}_1,\mathbf{k},\mathbf{k}_3} c_\uparrow^\dagger(\mathbf{k}_1)c_\downarrow^\dagger(-\mathbf{k})c_\downarrow(\mathbf{k}_3)c_\uparrow(\mathbf{k}_1-(\mathbf{k}+\mathbf{k}_3))\\
&\overset{\mathbf{k}_1\to\mathbf{k}_1+\mathbf{k}}{=} \frac{1}{L^{d_\mathrm{x}}}\sum_{\mathbf{k}_1,\mathbf{k},\mathbf{k}_3} c_\uparrow^\dagger(\mathbf{k}_1+\mathbf{k})c_\downarrow^\dagger(-\mathbf{k})c_\downarrow(\mathbf{k}_3)c_\uparrow(\mathbf{k}_1-\mathbf{k}_3)\\
&\overset{\mathbf{k}_1\equiv\mathbf{q}}{=} \frac{1}{L^{d_\mathrm{x}}}\sum_{\mathbf{q},\mathbf{k},\mathbf{k}_3} c_\uparrow^\dagger(\mathbf{k}+\mathbf{q})c_\downarrow^\dagger(-\mathbf{k})c_\downarrow(\mathbf{k}_3)c_\uparrow(\mathbf{q}-\mathbf{k}_3)\\
&\overset{\mathbf{k}_3\to-\mathbf{k}_3+\mathbf{q}}{=} \frac{1}{L^{d_\mathrm{x}}}\sum_{\mathbf{q},\mathbf{k},\mathbf{k}_3} c_\uparrow^\dagger(\mathbf{k}+\mathbf{q})c_\downarrow^\dagger(-\mathbf{k})c_\downarrow(\mathbf{q}-\mathbf{k}_3)c_\uparrow(\mathbf{k}_3)\\
&\overset{\mathbf{k}_3\equiv\mathbf{k}'}{=} \frac{1}{L^{d_\mathrm{x}}}\sum_{\mathbf{q},\mathbf{k},\mathbf{k}'} c_\uparrow^\dagger(\mathbf{k}+\mathbf{q})c_\downarrow^\dagger(-\mathbf{k})c_\downarrow(\mathbf{q}-\mathbf{k}')c_\uparrow(\mathbf{k}')
\end{aligned}$$

The Second Identity

TBD.

Proof 31.5 of Corollary 34.12

See Waste-Folder 6, Topic 3 for a proof.

We compute with results from [Draft Summary]

$$\begin{aligned}
S_\mathrm{int}[\overline{\psi},\psi] &\equiv \sum_{\omega_{n_1},\omega_{n_2},\omega_{n_3},\omega_{n_4}} \delta_{\omega_{n_1}+\omega_{n_2},\omega_{n_3}+\omega_{n_4}}\overline{\psi}(\omega_{n_1})\overline{\psi}(\omega_{n_2})\psi(\omega_{n_3})\psi(\omega_{n_4})\\
&= \ldots
\end{aligned}$$

now we change the enumeration of the second sum as $\omega_{n_2} \mapsto -\omega_{n_2}$ and obtain hence $\omega_{n_1} = \omega_{n_3} + \omega_{n_4} + \omega_{n_2}$ which gives

$$\begin{aligned}
\ldots &= \sum_{\omega_{n_1},\omega_{n_2},\omega_{n_3},\omega_{n_4}} \delta_{\omega_{n_1}+\omega_{n_2},\omega_{n_3}+\omega_{n_4}}\overline{\psi}(\omega_{n_2}+\omega_{n_3}+\omega_{n_4})\overline{\psi}(\omega_{n_2})\psi(\omega_{n_3})\psi(\omega_{n_4})\\
&= \ldots
\end{aligned}$$

Now we shift the third summation according to $\omega_{n_3} \mapsto \omega_{n_3} - \omega_{n_4}$ and obtain finally

$$\dots = \sum_{\omega_{n_1},\omega_{n_2},\omega_{n_3},\omega_{n_4}} \delta_{\omega_{n_1}+\omega_{n_2},\omega_{n_3}+\omega_{n_4}} \overline{\psi}(\omega_{n_2}+\omega_{n_3})\overline{\psi}(\omega_{n_2})\psi(\omega_{n_3}-\omega_{n_4})\psi(\omega_{n_4})$$

Proof 31.6 of Lemma 34.14 See Supermathematics, equation (3.31) and See Waste-Folder 6, topic 5, page 6 for the boundary-conditions.

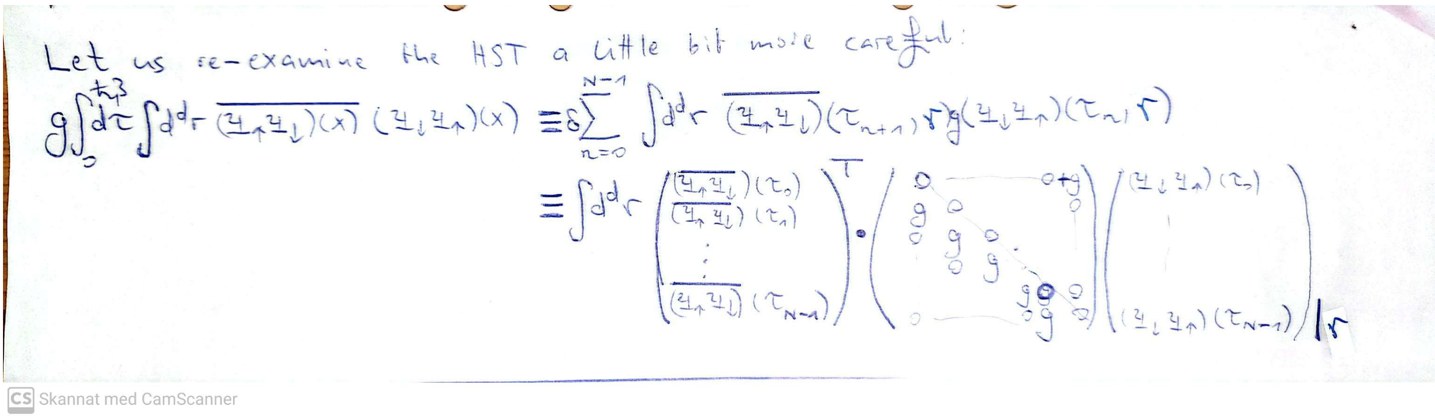


Figure 31.1.: Notes to clarify why periodic boundary-conditions emerge.

Proof 31.7 of Lemma 34.18

TBD

Proof 31.8 of Lemma 34.29

By direct computation, using the von Neumann series

$$\begin{aligned}\operatorname{Tr}\operatorname{Log}\mathscr{G}^{-1}_{\text{Gorkov}} &\equiv \operatorname{Tr}\operatorname{Log}\mathscr{G}^{-1}_{\substack{0\\ \text{Gorkov}}} + \operatorname{Tr}\operatorname{Log}\left(1+\mathscr{G}_{\substack{0\\ \text{Gorkov}}}V\right)^{-1}\\ &\equiv \text{const.} \quad + \sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{k}\operatorname{Tr}\left(\mathscr{G}_{\substack{0\\ \text{Gorkov}}}V\right)^{k}.\end{aligned}$$

Proof 31.9 of Theorem 34.30

TBD

Proof 31.10 of Corollary 34.22

TBD

Proof 31.11 of Lemma 25.3

Starting from the definition

$$\widehat{\mathscr{G}}_{\substack{m\\ \text{Gorkov}}}(\tau,\mathbf{k}) = \begin{pmatrix} \mathscr{G}_m(\tau,\mathbf{k}) & \mathscr{F}_m(\tau,\mathbf{k}) \\ \mathscr{F}^{\dagger}_m(\tau,\mathbf{k}) & -\mathscr{G}_m(-\tau,-\mathbf{k}) \end{pmatrix}$$

we compute the Matsubara-component of this function. For reasons that are difficult to motivate in advance, it is advantegous to use the definition from Theorem 1.7 in this calculation

$$\widehat{\mathscr{G}}_{\substack{m\\ \text{Gorkov}}}(\mathrm{i}\omega_n,\mathbf{k}) = \frac{1}{2}\int_{-\hbar\beta}^{\hbar\beta} \mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}\omega_n\tau} \begin{pmatrix} \mathscr{G}_m(\tau,\mathbf{k}) & \mathscr{F}_m(\tau,\mathbf{k}) \\ \mathscr{F}^{\dagger}_m(\tau,\mathbf{k}) & -\mathscr{G}_m(-\tau,-\mathbf{k}) \end{pmatrix}$$

Now the only component that requires further calculations is the lower right component of this matrix. By changing variables as $\tau \longmapsto -\tau$ we compute

$$\begin{aligned} -\left(\widehat{\mathscr{G}}_{\substack{m\\ \text{Gorkov}}}(\mathrm{i}\omega_n,\mathbf{k})\right)_{2,2} &= \frac{1}{2}\int_{-\hbar\beta}^{\hbar\beta} \mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}\omega_n\tau}\mathscr{G}_m(-\tau,-\mathbf{k}) \\ &= \frac{1}{2}\int_{+\hbar\beta}^{-\hbar\beta} \mathrm{d}\tau\, \mathrm{e}^{-\mathrm{i}\omega_n\tau}\mathscr{G}_m(+\tau,-\mathbf{k})\cdot(-1) \\ &= \frac{1}{2}\int_{-\hbar\beta}^{+\hbar\beta} \mathrm{d}\tau\, \mathrm{e}^{-\mathrm{i}\omega_n\tau}\mathscr{G}_m(+\tau,-\mathbf{k}) \\ &= \mathscr{G}_m(-\mathrm{i}\omega_n,-\mathbf{k}) \end{aligned}$$

Combining this with the very definition, we obtain the desired equation.

Proof 31.12 of Lemma 25.4 Recall from Theorem 13.6 that

$$\mathscr{G}_m(\mathrm{i}\omega_n,\mathbf{k}) = \frac{1}{\hbar}\frac{1}{\mathrm{i}\omega_n - \left(\epsilon_m(\mathbf{k})-\mu\right)/\hbar}$$

and

$$\begin{aligned} -\mathscr{G}_m(-\mathrm{i}\omega_n,-\mathbf{k}) &= \frac{-1}{\hbar}\frac{1}{-\mathrm{i}\omega_n - \left(\epsilon_m(-\mathbf{k})-\mu\right)/\hbar} \\ &= \frac{1}{\hbar}\frac{1}{\mathrm{i}\omega_n + \left(\epsilon_m(-\mathbf{k})-\mu\right)/\hbar} \end{aligned}$$

Thus we obtain

$$\mathscr{G}^{-1}_m(\mathrm{i}\omega_n,\mathbf{k}) = \hbar\left(\mathrm{i}\omega_n - \left(\epsilon_m(\mathbf{k})-\mu\right)/\hbar\right)$$

and

$$-\widehat{\mathscr{G}}_m^{-1}(-\mathrm{i}\omega_n, -\mathbf{k}) = \hbar\left(\mathrm{i}\omega_n + \left(\epsilon_m(-\mathbf{k}) - \mu\right)/\hbar\right)$$

and using the Pauli-matrices and the theorem about inversion-symmetric and spin-degenerate band-structures we obtain

$$\widehat{\mathscr{G}}^{-1}_{\substack{0,m\\ \text{Gorkov}}}(\mathrm{i}\omega_n, \mathbf{k}) = \mathrm{i}\hbar\omega_n\sigma_0 - \left(\epsilon_m(\mathbf{k}) - \mu\right)\sigma_3 \tag{31.3}$$

proving the lemma.

Proof 31.13 of Corollary 25.6 We use Equation (25.6) and introduce the self-energy

$$\widehat{\Sigma}_{\substack{m\\ \text{Gorkov}}}(k) = \begin{pmatrix} \Sigma_{\substack{m\\11}}(k) & \Sigma_{\substack{m\\12}}(k) \\ \Sigma_{\substack{m\\21}}(k) & \Sigma_{\substack{m\\22}}(k) \end{pmatrix} \tag{31.4}$$

This yields for the product

$$\begin{aligned}
\left(\widehat{\mathscr{G}}_0\widehat{\Sigma}\widehat{\mathscr{G}}\right)(k) &= \widehat{\mathscr{G}}_0(k)\begin{pmatrix} \Sigma_{\substack{m\\11}}(k) & \Sigma_{\substack{m\\12}}(k) \\ \Sigma_{\substack{m\\21}}(k) & \Sigma_{\substack{m\\22}}(k) \end{pmatrix}\begin{pmatrix} \mathscr{G}_{\substack{m\\ \text{Gorkov}}}(k) & \mathscr{F}_m(k) \\ \mathscr{F}^*_m(k) & -\mathscr{G}_{\substack{m\\ \text{Gorkov}}}(-k) \end{pmatrix} \\
&= \widehat{\mathscr{G}}_0(k)\begin{pmatrix} \Sigma_{\substack{m\\11}}(k)\mathscr{G}_{\substack{m\\ \text{Gorkov}}}(k) + \Sigma_{\substack{m\\12}}(k)\mathscr{F}^*_m(k) & \Sigma_{\substack{m\\11}}(k)\mathscr{F}_m(k) - \Sigma_{\substack{m\\12}}(k)\mathscr{G}_{\substack{m\\ \text{Gorkov}}}(-k) \\ \Sigma_{\substack{m\\21}}(k)\mathscr{G}_{\substack{m\\ \text{Gorkov}}}(k) + \Sigma_{\substack{m\\22}}(k)\mathscr{F}^*_m(k) & \Sigma_{\substack{m\\21}}(k)\mathscr{F}_m(k) - \Sigma_{\substack{m\\22}}(k)\mathscr{G}_{\substack{m\\ \text{Gorkov}}}(-k) \end{pmatrix} \\
&= \begin{pmatrix} \mathscr{G}_0(k) & 0 \\ 0 & -\mathscr{G}_0(-k) \end{pmatrix}\begin{pmatrix} \Sigma_{\substack{m\\11}}(k)\mathscr{G}_{\substack{m\\ \text{Gorkov}}}(k) + \Sigma_{\substack{m\\12}}(k)\mathscr{F}^*_m(k) & \Sigma_{\substack{m\\11}}(k)\mathscr{F}_m(k) - \Sigma_{\substack{m\\12}}(k)\mathscr{G}_{\substack{m\\ \text{Gorkov}}}(-k) \\ \Sigma_{\substack{m\\21}}(k)\mathscr{G}_{\substack{m\\ \text{Gorkov}}}(k) + \Sigma_{\substack{m\\22}}(k)\mathscr{F}^*_m(k) & \Sigma_{\substack{m\\21}}(k)\mathscr{F}_m(k) - \Sigma_{\substack{m\\22}}(k)\mathscr{G}_{\substack{m\\ \text{Gorkov}}}(-k) \end{pmatrix}
\end{aligned}$$

whereas we used Equation (25.4). Using now Equation (25.6) yields the desired result.

Proof 31.14 of Corollary 25.8

We use the decompositions in Equation (25.8) for self-energy and Equation (25.4) for the non-interacting Gorkov-Green-Function and insert them into the Dyson-equation (25.5) to obtain

$$\widehat{\mathscr{G}}^{-1}_{\substack{m\\ \text{Gorkov}}}(\mathrm{i}\omega_n, \mathbf{k}) = \hbar\mathrm{i}\omega_n Z_m(\mathrm{i}\omega_n, \mathbf{k})\sigma_0 - \phi_m(\mathrm{i}\omega_n, \mathbf{k})\sigma_1 - \left(\chi_m(\mathrm{i}\omega_n, \mathbf{k}) + (\epsilon_m(\mathbf{k}) - \mu)\right)\sigma_3 \tag{31.5}$$

which can be inverted with the inversion-formula for 2x2-matrices[2] to

$$\widehat{\mathscr{G}}_{\substack{m\\ \text{Gorkov}}}(\mathrm{i}\omega_n, \mathbf{k}) = \frac{\hbar\mathrm{i}\omega_n Z_m(\mathrm{i}\omega_n, \mathbf{k})\sigma_0 + \phi_m(\mathrm{i}\omega_n, \mathbf{k})\sigma_1 + \left(\chi_m(\mathrm{i}\omega_n, \mathbf{k}) + (\epsilon_m(\mathbf{k}) - \mu)\right)\sigma_3}{\mathrm{Det}\,\widehat{\mathscr{G}}^{-1}_{\substack{m\\ \text{Gorkov}}}(\mathrm{i}\omega_n, \mathbf{k})} \tag{31.6}$$

[2]Cf. Waste-Folder 1 Topic 22: For $A = a_0\sigma_0 + \mathbf{a}\cdot\boldsymbol{\sigma}$

Noticing

$$\begin{aligned}
\operatorname{Det} \underset{\text{Gorkov}}{\widehat{\mathscr{G}}^{-1}_{m}}(\mathrm{i}\omega_n, \mathbf{k}) &= -\hbar^2\omega_n^2 Z_m(\mathbf{k}, \mathrm{i}\omega_n)^2 - \left(\chi_m(\mathbf{k}, \mathrm{i}\omega_n) + (\epsilon_m(\mathbf{k}) - \mu)\right)^2 - \left(\phi_m(\mathbf{k}, \mathrm{i}\omega_n)\right)^2 \\
&= -\left(\hbar^2\omega_n^2 Z_m(\mathbf{k}, \mathrm{i}\omega_n)^2 + \left(\chi_m(\mathbf{k}, \mathrm{i}\omega_n) + (\epsilon_m(\mathbf{k}) - \mu)\right)^2 + \phi_m(\mathbf{k}, \mathrm{i}\omega_n)^2\right) \\
&\equiv -\theta_m(\mathrm{i}\omega_n, \mathbf{k})
\end{aligned}$$

Inserting this above yields the desired result

$$\underset{\text{Gorkov}}{\widehat{\mathscr{G}}_{m}}(\mathrm{i}\omega_n, \mathbf{k}) = -\frac{\hbar\mathrm{i}\omega_n Z_m(\mathrm{i}\omega_n, \mathbf{k})\sigma_0 + \phi_m(\mathrm{i}\omega_n, \mathbf{k})\sigma_1 + \left(\chi_m(\mathrm{i}\omega_n, \mathbf{k}) + (\epsilon_m(\mathbf{k}) - \mu)\right)\sigma_3}{\theta_m(\mathrm{i}\omega_n, \mathbf{k})} \tag{31.7}$$

Proof 31.15 of Lemma 25.10 We need to compute

$$\sigma_3 \underset{\text{Gorkov}}{\widehat{\mathscr{G}}_{m}}(\mathrm{i}\omega_{n'}, \mathbf{q}+\mathbf{k})\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \underset{\text{Gorkov}}{\mathscr{G}_{m}}(\bullet) & \mathscr{F}_m(\bullet) \\ \mathscr{F}^*_m(\bullet) & -\underset{\text{Gorkov}}{\mathscr{G}_{m}}(-\bullet) \end{pmatrix}_{\bullet=(\mathrm{i}\omega_{n'}, \mathbf{q}+\mathbf{k})} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{31.8}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \underset{\text{Gorkov}}{\mathscr{G}_{m}}(\bullet) & -\mathscr{F}_m(\bullet) \\ \mathscr{F}^*_m(\bullet) & \underset{\text{Gorkov}}{\mathscr{G}_{m}}(-\bullet) \end{pmatrix}_{\bullet=(\mathrm{i}\omega_{n'}, \mathbf{q}+\mathbf{k})} \tag{31.9}$$

$$= \begin{pmatrix} \underset{\text{Gorkov}}{\mathscr{G}_{m}}(\bullet) & -\mathscr{F}_m(\bullet) \\ -\mathscr{F}^*_m(\bullet) & -\underset{\text{Gorkov}}{\mathscr{G}_{m}}(-\bullet) \end{pmatrix}_{\bullet=(\mathrm{i}\omega_{n'}, \mathbf{q}+\mathbf{k})} \tag{31.10}$$

Proof 31.16 of Theorem 25.14

Inserting Equation (25.9) for the Green-function into Equation (25.12) for the self-energy, and comparing coefficients with Equation (25.8) for the self-energy, we obtain the Eliashberg-equations

$$\begin{aligned}
\underset{n,m}{\widehat{\Sigma}_{\text{ep}}}(\mathrm{i}\omega_n, \mathbf{k}) = T \sum_{\substack{\mathbf{k}' \in \mathrm{Q} \\ \omega_{n'} \in \Omega_{-1}}} \hbar\mathrm{i}\omega_{n'} \frac{Z_m(\mathrm{i}\omega_{n'}, \mathbf{k}')}{\theta_m(\mathrm{i}\omega_{n'}, \mathbf{k}')} \sigma_3\sigma_0\sigma_3 \sum_{\nu} |g_{mn\nu}(\mathbf{k}, \mathbf{k}')|^2 \mathscr{D}_\nu(\mathrm{i}\omega_n - \mathrm{i}\omega_{n'}, \mathbf{k} - \mathbf{k}') \\
- T \sum_{\substack{\mathbf{k}' \in \mathrm{Q} \\ \omega_{n'} \in \Omega_{-1}}} \frac{\phi_m(\mathrm{i}\omega_{n'}, \mathbf{k}')}{\theta_m(\mathrm{i}\omega_{n'}, \mathbf{k}')} \sigma_1 \sum_{\nu} |g_{mn\nu}(\mathbf{k}, \mathbf{k}')|^2 \mathscr{D}_\nu(\mathrm{i}\omega_n - \mathrm{i}\omega_{n'}, \mathbf{k} - \mathbf{k}') \\
+ T \sum_{\substack{\mathbf{k}' \in \mathrm{Q} \\ \omega_{n'} \in \Omega_{-1}}} \frac{\chi_m(\mathrm{i}\omega_{n'}, \mathbf{k}') + (\epsilon_m(\mathbf{k}) - \mu)}{\theta_m(\mathrm{i}\omega_{n'}, \mathbf{k}')} \sigma_3 \sum_{\nu} |g_{mn\nu}(\mathbf{k}, \mathbf{k}')|^2 \mathscr{D}_\nu(\mathrm{i}\omega_n - \mathrm{i}\omega_{n'}, \mathbf{k} - \mathbf{k}') \\
\overset{!!!}{=} \mathrm{i}\omega_n \left(1 - Z_m(\mathrm{i}\omega_n, \mathbf{k})\right)\sigma_0 + \phi_m(\mathrm{i}\omega_n, \mathbf{k})\sigma_1 + \chi_m(\mathrm{i}\omega_n, \mathbf{k})\sigma_3
\end{aligned}$$

In this equation, all of the momentum-indices are messed up!

Notice further, that we used the following identities for the Pauli-matrices ToBeWrittenDown.

Proof 31.17 of Lemma 25.15 Cf. 06 Superhydrides Success of Eliashberg. ToBeDone. -

Proof 31.18 of Corollary 25.16

The Eliashberg-Equations from 03-EPW reads for ϕ:

$$\overline{\phi}_n(\mathrm{i}\omega_j, \mathbf{k}) = \frac{T}{N_\mathrm{F}} \sum_{\substack{\mathbf{q}\in \mathrm{Q} \\ \omega_{j'}\in\Omega_{-1} \\ m\in\{1,\ldots,\infty\}}} \frac{\overline{\phi}_m(\mathrm{i}\omega_{j'}, \mathbf{k}+\mathbf{q})}{\theta_m(\mathrm{i}\omega_{j'}, \mathbf{k}+\mathbf{q})} \lambda_{n,m}(\mathrm{i}\omega_j - \mathrm{i}\omega_{j'}; \mathbf{k}, \mathbf{k}+\mathbf{q}) \tag{31.11}$$

with

$$\theta_m(\mathrm{i}\omega_n, \mathbf{k}) = \hbar^2\omega_n^2 Z_m(\mathbf{k}, \mathrm{i}\omega_n)^2 + \left(\chi_m(\mathbf{k}, \mathrm{i}\omega_n) + (\epsilon_m(\mathbf{k}) - \mu)\right)^2 + \phi_m(\mathrm{i}\omega_n, \mathbf{k})^2 \tag{31.12}$$

and

$$\lambda_{\substack{n,m \\ \mathbf{k},\mathbf{k}+\mathbf{q}}}(\omega_n) = N_\mathrm{F} \sum\!\!\!\!\!\!\int_{\substack{\omega\in(0,\infty) \\ \nu\in\{1,\ldots,rd_\mathrm{x}\}}} \frac{2\omega}{\omega_n^2+\omega^2} |g_{mn\nu}(\mathbf{k},\mathbf{q})|^2 \delta\left(\omega - \omega_\nu(\mathbf{q})\right) \tag{31.13}$$

Considering now the limiting case $Z \equiv 1$, $\chi \equiv 0$ we obtain

$$\begin{aligned}
\overline{\phi}_n(\mathrm{i}\omega_j, \mathbf{k}) &= \sum_{\substack{\mathbf{q}\in \mathrm{Q} \\ \omega_{j'}\in\Omega_{-1} \\ m\in\{1,\ldots,\infty\}}} \frac{\overline{\phi}_m(\mathrm{i}\omega_{j'}, \mathbf{k}+\mathbf{q})}{\hbar^2\omega_{j'}^2 + \left(\epsilon_m(\mathbf{k}+\mathbf{q}) - \mu\right)^2 + |\phi_m(\mathrm{i}\omega_{j'}, \mathbf{k}+\mathbf{q})|^2} |g_{mn\nu}(\mathbf{k},\mathbf{q})|^2 \frac{2\omega_\nu(\mathbf{q})}{\left(\omega_j - \omega_{j'}\right)^2 + \omega_\nu(\mathbf{q})^2} \frac{1}{\hbar\beta} \\
&\overset{\mathbf{q}\mapsto\mathbf{q}-\mathbf{k}}{=} \sum_{\substack{\mathbf{q}\in \mathrm{Q} \\ \omega_{j'}\in\Omega_{-1} \\ m\in\{1,\ldots,\infty\}}} \frac{\overline{\phi}_m(\mathrm{i}\omega_{j'}, \mathbf{q})}{\hbar^2\omega_{j'}^2 + \left(\epsilon_m(\mathbf{q}) - \mu\right)^2 + |\phi_m(\mathrm{i}\omega_{j'}, \mathbf{q})|^2} |g_{mn\nu}(\mathbf{k},\mathbf{q}-\mathbf{k})|^2 \frac{2\omega_\nu(\mathbf{q}-\mathbf{k})}{\left(\omega_j - \omega_{j'}\right)^2 + \omega_\nu(\mathbf{q}-\mathbf{k})^2} \frac{1}{\hbar\beta} \\
&\overset{q\equiv(\mathbf{q},\mathrm{i}\omega_{j'}),k\equiv(\mathbf{k},\mathrm{i}\omega_j)}{=} \sum_{\substack{q\in \mathrm{Q}\times\Omega_{-1} \\ m\in\{1,\ldots,\infty\}}} \frac{\overline{\phi}_m(q)}{\hbar^2\omega_q^2 + \left(\epsilon_m(q) - \mu\right)^2 + |\phi_m(q)|^2} |g_{mn\nu}(\mathbf{k},\mathbf{q}-\mathbf{k})|^2 \frac{2\omega_\nu(\mathbf{q}-\mathbf{k})}{\omega_{q-k}^2 + \omega_\nu(\mathbf{q}-\mathbf{k})^2} \frac{1}{\hbar\beta}
\end{aligned}$$

which can be summarized as

$$\overline{\phi}_n(k) = \sum_{\substack{q\in \mathrm{Q}\times\Omega_{-1} \\ m\in\{1,\ldots,\infty\}}} \frac{\overline{\phi}_m(q)}{\hbar^2\omega_q^2 + \left(\epsilon_m(q) - \mu\right)^2 + |\phi_m(q)|^2} |g_{mn\nu}(\mathbf{k},\mathbf{q}-\mathbf{k})|^2 \frac{2\omega_\nu(\mathbf{q}-\mathbf{k})}{\omega_{q-k}^2 + \omega_\nu(\mathbf{q}-\mathbf{k})^2} \frac{1}{\hbar\beta}$$

Note: The template for this was 03-EPW.

Proof 31.19 of Lemma 25.18

Recalling that $U(\xi\tau, 0) = \exp -\frac{1}{\hbar}\xi\tau\widetilde{H} = \exp -\xi(\tau - 0)\widetilde{H}/\hbar$, with $\widetilde{H} = H - \mu N$ we obtain analogously the auxiliary identities

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}\tau}U(\xi\tau, 0) &= -\frac{\xi}{\hbar}\widetilde{H}U(\xi\tau, 0)\\ \frac{\mathrm{d}}{\mathrm{d}\tau}U(0, \xi\tau) &= \frac{\xi}{\hbar}U(0, \xi\tau)\widetilde{H}\end{aligned}$$

Remark: There are more identities like this, but we only need these here. Then it follows

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}\tau}X_{\mathrm{H}}(\xi\tau) &= \frac{\mathrm{d}}{\mathrm{d}t}U(0, \xi\tau)X(\xi\tau)U(\xi\tau, 0)\\ &= \left(\frac{\mathrm{d}}{\mathrm{d}\tau}U(0, \xi\tau)\right)X(\xi\tau)U(\xi\tau, 0) + U(0, \xi\tau)\left(\frac{\mathrm{d}}{\mathrm{d}\tau}X(\xi\tau)\right)U(\xi\tau, 0) + U(0, \xi\tau)X(\xi\tau)\left(\frac{\mathrm{d}}{\mathrm{d}\tau}U(\xi\tau, 0)\right)\\ &= \frac{\xi}{\hbar}U(0, \xi\tau)\widetilde{H}X(\xi\tau)U(\xi\tau, 0) + U(0, \xi\tau)\left(\frac{\mathrm{d}}{\mathrm{d}\tau}X(\xi\tau)\right)U(\xi\tau, 0) + U(0, \xi\tau)X(\xi\tau)\left(-\frac{\xi}{\hbar}\widetilde{H}U(\xi\tau, 0)\right)\\ &= \frac{\xi}{\hbar}U(0, \xi\tau)\left(\widetilde{H}X(\xi\tau) - X(\xi\tau)\widetilde{H}\right)U(\xi\tau, 0) + U(0, \xi\tau)\left(\frac{\mathrm{d}}{\mathrm{d}t}X(\xi\tau)\right)U(\xi\tau, 0)\\ &= \frac{\xi}{\hbar}U(0, \xi\tau)\left[\widetilde{H}, X(\xi\tau)\right]U(\xi\tau, 0) + U(0, \xi\tau)\left(\frac{\mathrm{d}}{\mathrm{d}\tau}X(\xi\tau)\right)U(\xi\tau, 0)\\ &= -\frac{\xi}{\hbar}\left[X, \widetilde{H}\right]_{\mathrm{H}}(\xi\tau) + U(0, \xi\tau)\left(\frac{\mathrm{d}}{\mathrm{d}\tau}X(\xi\tau)\right)U(\xi\tau, 0)\end{aligned}$$

Proof 31.20 of Lemma 25.19

Removing the primes to lighten the notation.

We compute the equation of motion for the annihilation-operators using the Heisenberg-equation of motion (25.28) for arbitrary $\bullet \in \mathrm{Q}$ with H from Equation (25.1)

$$\begin{aligned}\partial_\tau b_\nu(\xi\tau, \bullet) &= -\frac{\xi}{\hbar}[b_\nu(\bullet), H - \mu N]_{\mathrm{H}}(\xi\tau)\\ &= -\xi \sum_{\substack{\mathbf{q}' \in \mathrm{Q}\\ \nu' \in \{1,\ldots,rd_{\mathbf{x}}\}}} \omega_{\nu'}(\mathbf{q}')[b_\nu(\bullet), b^\dagger_{\nu'}(\mathbf{q}')b_{\nu'}(\mathbf{q}')]_{\mathrm{H}}(\xi\tau) \quad + \quad 0\\ &\quad - \xi \sum_{\substack{\mathbf{k}',\mathbf{q}' \in \mathrm{Q}\\ \sigma' \in \{\uparrow,\downarrow\}\\ \nu' \in \{1,\ldots,rd_{\mathbf{x}}\}\\ m',n' \in \{1,\ldots,\infty\}}} \frac{g_{m'n'\nu'}(\mathbf{k}', \mathbf{q}')}{\hbar\sqrt{N_p}} c^\dagger_{m'\sigma'}(\xi\tau, \mathbf{k}' + \mathbf{q}')c_{n'\sigma'}(\xi\tau, \mathbf{k}')\left([b_\nu(\bullet), b^\dagger_{\nu'}(-\mathbf{q}')]_{\mathrm{H}}(\xi\tau) \quad + 0\right)\\ &= \ldots\end{aligned}$$

Now we use Equation (10.4) and Lemma 7.17, i.e.

$$[A, BC] = [A, B]_\zeta C + \zeta B[A, C]_\zeta \tag{31.14}$$

to simplify the second and third row as

$$[b_\nu(\bullet), b^\dagger_{\nu'}(\mathbf{q}')b_{\nu'}(\mathbf{q}')] = \delta_{\nu,\nu'}\delta_{\bullet,\mathbf{q}'}b_{\nu'}(\mathbf{q}')$$
$$[b_\nu(\bullet), b^\dagger_{\nu'}(-\mathbf{q}')] = \delta_{\nu,\nu'}\delta_{\bullet,-\mathbf{q}'}$$

such that we obtain

$$\ldots = -\xi\omega_\nu(\bullet)b_\nu(\xi\tau,\bullet) - \xi \sum_{\substack{\mathbf{k}'\in Q\\ \sigma'\in\{\uparrow,\downarrow\}\\ m',n'\in\{1,\ldots,\infty\}}} \frac{g_{m'n'\nu}(\mathbf{k}',-\bullet)}{\hbar\sqrt{N_p}} c^\dagger_{m'\sigma'}(\xi\tau,\mathbf{k}'-\bullet)c_{n'\sigma'}(\xi\tau,\mathbf{k}')$$

replacing now the summation indices (i.e. dropping the primes), we obtain the desired result.

In the same fashion we obtain

$$\begin{aligned}
\partial_\tau b^\dagger_\nu(\xi\tau,-\bullet) &= -\frac{\xi}{\hbar}[b^\dagger_\nu(-\bullet), H-\mu N]_{\mathrm{H}}(\xi\tau)\\
&= -\xi \sum_{\substack{\mathbf{q}'\in Q\\ \nu'\in\{1,\ldots,rd_{\mathbf{x}}\}}} \omega_{\nu'}(\mathbf{q}')[b^\dagger_\nu(-\bullet), b^\dagger_{\nu'}(\mathbf{q}')b_{\nu'}(\mathbf{q}')]_{\mathrm{H}}(\xi\tau) \quad + \quad 0\\
&\quad -\xi \sum_{\substack{\mathbf{k}',\mathbf{q}'\in Q\\ \sigma'\in\{\uparrow,\downarrow\}\\ \nu'\in\{1,\ldots,rd_{\mathbf{x}}\}\\ m',n'\in\{1,\ldots,\infty\}}} \frac{g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{\hbar\sqrt{N_p}} c^\dagger_{m'\sigma'}(\xi\tau,\mathbf{k}'+\mathbf{q}')c_{n'\sigma'}(\xi\tau,\mathbf{k}')\left([b^\dagger_\nu(-\bullet), b_{\nu'}(+\mathbf{q}')]_{\mathrm{H}}(\xi\tau) \quad + \quad 0\right)\\
&= \ldots
\end{aligned}$$

Now we use again Equation (10.4) and Lemma 7.17

$$[b^\dagger_\nu(-\bullet), b^\dagger_{\nu'}(\mathbf{q}')b_{\nu'}(\mathbf{q}')] = -\delta_{\nu,\nu'}\delta_{-\bullet,\mathbf{q}'}b^\dagger_{\nu'}(\mathbf{q}')$$
$$[b^\dagger_\nu(-\bullet), b_{\nu'}(\mathbf{q}')] = -\delta_{\nu,\nu'}\delta_{-\bullet,\mathbf{q}'}$$

(notice that the extra minus sign appears due to the commutation-relations), such that we obtain

$$\ldots = +\xi\omega_\nu(-\bullet)b^\dagger_\nu(\xi\tau,-\bullet) + \xi \sum_{\substack{\mathbf{k}'\in Q\\ \sigma'\in\{\uparrow,\downarrow\}\\ m',n'\in\{1,\ldots,\infty\}}} \frac{g_{m'n'\nu}(\mathbf{k}',-\bullet)}{\hbar\sqrt{N_p}} c^\dagger_{m'\sigma'}(\xi\tau,\mathbf{k}'-\bullet)c_{n'\sigma'}(\xi\tau,\mathbf{k}')$$

using now the inversion-symmetry of ω and dropping the primes we obtain the desired result.

Proof 31.21 of Lemma 25.20 Recall from Lemma 13.5 that the equation of motion for the Green-function in imaginary time reads

$$\partial_\tau \mathscr{G}_{\text{Gorkov}}^{\;n}(\tau,\mathbf{k}) = -\frac{1}{\hbar}\delta(\tau) - \frac{1}{\hbar}\left\langle \mathrm{T}_\tau \left[\frac{H-\mu N}{\hbar},\, c_{n,\uparrow}(\mathbf{k})\right]_{\mathrm{H}}(\tau) c^\dagger_{n,\uparrow}(\mathbf{k},0)\right\rangle_{\mathrm{SC}} \tag{31.15}$$

whereas N refers to the total electronic occupation-number operator. We now compute the commutator for the EPW-system as

$$\begin{aligned}
\left[\frac{H-\mu N}{\hbar}, c_{n\uparrow}(\mathbf{k})\right] = &\sum_{\substack{\mathbf{k}'\in\mathrm{Q}\\ \sigma'\in\{\uparrow,\downarrow\}\\ n'\in\{1,\ldots,\infty\}}} \frac{1}{\hbar}\left(\epsilon_{n'}(\mathbf{k}')-\mu\right)[c^\dagger_{n'\sigma'}(\mathbf{k}')c_{n'\sigma'}(\mathbf{k}'), c_{n\uparrow}(\mathbf{k})] \\
&+ \sum_{\substack{\mathbf{q}',\mathbf{k}'\in\mathrm{Q}\\ \sigma'\in\{\uparrow,\downarrow\}\\ \nu'\in\{1,\ldots,rd_{\mathbf{x}}\}\\ n',m'\in\{1,\ldots,\infty\}}} \frac{g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{\hbar\sqrt{N_p}}[c^\dagger_{m'\sigma'}(\mathbf{k}'+\mathbf{q}')c_{n'\sigma'}(\mathbf{k}')\left(b_{\nu'}(\mathbf{q}')+b^\dagger_{\nu'}(-\mathbf{q}')\right), c_{n\uparrow}(\mathbf{k})] \\
= &\ldots
\end{aligned}$$

Now we evaluate the individual commutators. For the first line we have

$$[c^\dagger_{n'\sigma'}(\mathbf{k}')c_{n'\sigma'}(\mathbf{k}'), c_{n\uparrow}(\mathbf{k})] = -[c_{n\uparrow}(\mathbf{k}), c^\dagger_{n'\sigma'}(\mathbf{k}')c_{n'\sigma'}(\mathbf{k}')] = -\delta_{n',n}\delta_{\sigma',\uparrow}\delta_{\mathbf{k}',\mathbf{k}}c_{n'\sigma'}(\mathbf{k}') \tag{31.16}$$

and

$$\begin{aligned}
[c^\dagger_{m'\sigma'}(\mathbf{k}'+\mathbf{q}')c_{n'\sigma'}(\mathbf{k}')\left(b_{\nu'}(\mathbf{q}')+b^\dagger_{\nu'}(-\mathbf{q}')\right), c_{n\uparrow}(\mathbf{k})] &= -[c_{n\uparrow}(\mathbf{k}), c^\dagger_{m'\sigma'}(\mathbf{k}'+\mathbf{q}')c_{n'\sigma'}(\mathbf{k}')]\left(b_{\nu'}(\mathbf{q}')+b^\dagger_{\nu'}(-\mathbf{q}')\right) \\
&= -\delta_{m',n}\delta_{\sigma',\uparrow}\delta_{\mathbf{k}'+\mathbf{q}',\mathbf{k}}c_{n'\sigma'}(\mathbf{k}')\left(b_{\nu'}(\mathbf{q}')+b^\dagger_{\nu'}(-\mathbf{q}')\right)
\end{aligned}$$

whereas we used Lemma 7.17 in both cases, and thus we obtain, after terminating the sums over $\mathbf{q}'$, m' and σ', the equation

$$\ldots = -\left(\frac{\epsilon_n(\mathbf{k})-\mu}{\hbar}\right)c_{n\uparrow}(\mathbf{k}) - \sum_{\substack{\mathbf{k}'\in\mathrm{Q}\\ \nu'\in\{1,\ldots,rd_{\mathbf{x}}\}\\ n'\in\{1,\ldots,\infty\}}} \frac{g_{nn'\nu'}(\mathbf{k}',\mathbf{k}-\mathbf{k}')}{\hbar\sqrt{N_p}}\left(b_{\nu'}(\mathbf{k}-\mathbf{k}')+b^\dagger_{\nu'}(\mathbf{k}'-\mathbf{k})\right)c_{n'\uparrow}(\mathbf{k}')$$

Plugging this into the equation of motion, Equation (31.15), yields

$$\begin{aligned}
\partial_\tau \mathscr{G}_{\text{Gorkov}}^{\;n}(\tau,\mathbf{k}) = &-\delta(\tau)/\hbar + \frac{\epsilon_n(\mathbf{k})-\mu}{\hbar^2}\left\langle \mathrm{T}_\tau\, c_{n\uparrow}(\tau,\mathbf{k})c^\dagger_{n\uparrow}(0,\mathbf{k})\right\rangle_{\mathrm{SC}} \\
&+ \sum_{\substack{\mathbf{k}'\in\mathrm{Q}\\ \nu'\in\{1,\ldots,rd_{\mathbf{x}}\}\\ n'\in\{1,\ldots,\infty\}}} \frac{g_{nn'\nu'}(\mathbf{k}',\mathbf{k}-\mathbf{k}')}{\hbar^2\sqrt{N_p}}\left\langle \mathrm{T}_\tau\left(b_{\nu'}(\tau,\mathbf{k}-\mathbf{k}')+b^\dagger_{\nu'}(\tau,\mathbf{k}'-\mathbf{k})\right)c_{n'\uparrow}(\tau,\mathbf{k}')c^\dagger_{n\uparrow}(\mathbf{k},0)\right\rangle_{\mathrm{SC}}
\end{aligned}$$

which can be simplified to

$$\left(\partial_\tau + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar}\right) \mathcal{G}_{\substack{n\\ \text{Gorkov}}}(\tau, \mathbf{k})$$
$$= -\delta(\tau)/\hbar + \sum_{\substack{\mathbf{k}' \in \mathrm{Q}\\ \nu' \in \{1,\ldots,rd_{\mathbf{x}}\}\\ n' \in \{1,\ldots,\infty\}}} \frac{g_{nn'\nu'}(\mathbf{k}', \mathbf{k}-\mathbf{k}')}{\hbar^2 \sqrt{N_p}} \left\langle \mathrm{T}_\tau \left(b_{\nu'}(\tau, \mathbf{k}-\mathbf{k}') + b^\dagger_{\nu'}(\tau, \mathbf{k}'-\mathbf{k}) \right) c_{n'\uparrow}(\tau, \mathbf{k}') c^\dagger_{n\uparrow}(\mathbf{k}, 0) \right\rangle_{\mathrm{SC}} \tag{31.17}$$

Proof 31.22 of Lemma 25.22 Notice first that Equation (25.30) can be expanded as

$$\begin{aligned} \mathcal{G}_{\substack{\text{aux}\\ \nu',n,n'}}(\tau, \tau', \mathbf{k}, \mathbf{k}') = -\frac{1}{\hbar}\Big[&+ \theta(\tau') \left\langle \left(b_{\nu'}(\tau, \mathbf{k}-\mathbf{k}') + b^\dagger_{\nu'}(\tau, \mathbf{k}'-\mathbf{k}) \right) c_{n',\uparrow}(\tau', \mathbf{k}') c^\dagger_{n,\uparrow}(0, \mathbf{k}) \right\rangle_{\mathrm{SC}} \\ &- \theta(-\tau') \left\langle \left(b_{\nu'}(\tau, \mathbf{k}-\mathbf{k}') + b^\dagger_{\nu'}(\tau, \mathbf{k}'-\mathbf{k}) \right) c^\dagger_{n,\uparrow}(0, \mathbf{k}) c_{n',\uparrow}(\tau', \mathbf{k}') \right\rangle_{\mathrm{SC}} \Big]. \end{aligned} \tag{31.18}$$

Since the time-ordering-operator decomposese into the bosonic and fermionic part and it is sufficient[3] to focus on the bosonic part. Thus taking the derivative w.r.t. τ yields doesn't involve any δ-functions. One finds by straightforward computation with Equation (31.18) that[4]

$$\begin{aligned} \partial_\tau \mathcal{G}_{\substack{\text{aux}\\ \nu',n,n'}}(\tau, \tau', \mathbf{k}, \mathbf{k}') = &-\frac{1}{\hbar} \left\langle \mathrm{T}_\tau \left(\partial_\tau b_{\nu'}(\tau, \mathbf{k}-\mathbf{k}') \right) c_{n',\uparrow}(\tau', \mathbf{k}') c^\dagger_{n,\uparrow}(0, \mathbf{k}) \right\rangle_{\mathrm{SC}} \\ &- \frac{1}{\hbar} \left\langle \mathrm{T}_\tau \left(\partial_\tau b^\dagger_{\nu'}(\tau, \mathbf{k}'-\mathbf{k}) \right) c_{n',\uparrow}(\tau' \pm \epsilon, \mathbf{k}') c^\dagger_{n,\uparrow}(0, \mathbf{k}) \right\rangle_{\mathrm{SC}} = \ldots \end{aligned}$$

and then by using Lemma 25.19 we obtain

$$\begin{aligned} &\ldots \\ &= -\frac{1}{\hbar} \left\langle \mathrm{T}_\tau\, c_{n',\uparrow}(\tau', \mathbf{k}') c^\dagger_{n,\uparrow}(0, \mathbf{k}) \Big[-\omega_{\nu'}(\mathbf{k}-\mathbf{k}') b_{\nu'}(\tau, \mathbf{k}-\mathbf{k}') - \sum_{\substack{\widetilde{\mathbf{k}} \in \mathrm{Q}\\ \widetilde{\sigma} \in \{\uparrow,\downarrow\}\\ \widetilde{m},\widetilde{n} \in \{1,\ldots,\infty\}}} \frac{g_{\widetilde{m}\widetilde{n}\nu'}(\widetilde{\mathbf{k}}, \mathbf{k}'-\mathbf{k})}{\hbar\sqrt{N_p}} c^\dagger_{\widetilde{m}\widetilde{\sigma}}(\tau, \widetilde{\mathbf{k}}+\mathbf{k}'-\mathbf{k}) c_{\widetilde{n}\widetilde{\sigma}}(\tau, \widetilde{\mathbf{k}}) \Big] \right\rangle \\ &- \frac{1}{\hbar} \left\langle \mathrm{T}_\tau\, c_{n',\uparrow}(\tau', \mathbf{k}') c^\dagger_{n,\uparrow}(0, \mathbf{k}) \Big[\omega_{\nu'}(\mathbf{k}'-\mathbf{k}) b^\dagger_{\nu'}(\tau, \mathbf{k}'-\mathbf{k}) + \sum_{\substack{\widetilde{\mathbf{k}} \in \mathrm{Q}\\ \widetilde{\sigma} \in \{\uparrow,\downarrow\}\\ \widetilde{m},\widetilde{n} \in \{1,\ldots,\infty\}}} \frac{g_{\widetilde{m}\widetilde{n}\nu'}(\widetilde{\mathbf{k}}, \mathbf{k}'-\mathbf{k})}{\hbar\sqrt{N_p}} c^\dagger_{\widetilde{m}\widetilde{\sigma}}(\tau, \widetilde{\mathbf{k}}+\mathbf{k}'-\mathbf{k}) c_{\widetilde{n}\widetilde{\sigma}}(\tau, \widetilde{\mathbf{k}}) \Big] \right\rangle \\ &= \ldots \end{aligned}$$

and see that the contributions containing the electron-phonon-coupling g cancel out and we obtain

[3] Maybe this should be added as a corollary. Cf. definition 8.14.

[4] Notice that the order doesn't matter cause within T_τ it has even statistics.

$$\begin{aligned}\partial_\tau \mathcal{G}^{\text{aux}}_{\nu',n,n'}(\tau,\tau',\mathbf{k},\mathbf{k}') &= +\frac{\omega_{\nu'}(\mathbf{k}-\mathbf{k}')}{\hbar}\left\langle \mathrm{T}_\tau\, b_{\nu'}(\tau,\mathbf{k}-\mathbf{k}')c_{n',\uparrow}(\tau',\mathbf{k}')c^\dagger_{n,\uparrow}(0,\mathbf{k})\right\rangle_{\text{SC}} \\ &\quad - \frac{\omega_{\nu'}(\mathbf{k}'-\mathbf{k})}{\hbar}\left\langle \mathrm{T}_\tau\, b^\dagger_{\nu'}(\tau,\mathbf{k}'-\mathbf{k})c_{n',\uparrow}(\tau',\mathbf{k}')c^\dagger_{n,\uparrow}(0,\mathbf{k})\right\rangle_{\text{SC}} \\ &= \frac{\omega_{\nu'}(\mathbf{k}-\mathbf{k}')}{\hbar}\left\langle \mathrm{T}_\tau \left(b_{\nu'}(\tau,\mathbf{k}-\mathbf{k}') - b^\dagger_{\nu'}(\tau,\mathbf{k}'-\mathbf{k})\right) c_{n',\uparrow}(\tau',\mathbf{k}')c^\dagger_{n,\uparrow}(0,\mathbf{k})\right\rangle_{\text{SC}}\end{aligned}$$

whereas we used the inversion-symmetry $\omega_\nu(+\bullet) = \omega_\nu(-\bullet)$ in the last equation-sign and hence we obtain the desired result. For the second derivative we obtain

$$\begin{aligned}\partial^2_\tau \mathcal{G}^{\text{aux}}_{\nu',n,n'}(\tau,\tau',\mathbf{k},\mathbf{k}') &= \frac{\omega_{\nu'}(\mathbf{k}-\mathbf{k}')}{\hbar}\left\langle \mathrm{T}_\tau\, \partial_\tau \left(b_{\nu'}(\tau,\mathbf{k}-\mathbf{k}') - b^\dagger_{\nu'}(\tau,\mathbf{k}'-\mathbf{k})\right) c_{n',\uparrow}(\tau',\mathbf{k}')c^\dagger_{n,\uparrow}(0,\mathbf{k})\right\rangle_{\text{SC}} \\ &= \mathrm{I} + \mathrm{II} \\ &= \ldots\end{aligned} \tag{31.19}$$

using now Lemma 25.19 we obtain for the individual contributions

$$\begin{aligned}\mathrm{I} = \frac{\omega_{\nu'}(\mathbf{k}-\mathbf{k}')}{\hbar}\Bigg\langle \mathrm{T}_\tau \Bigg[&-\omega_{\nu'}(\mathbf{k}-\mathbf{k}')b_{\nu'}(\tau,\mathbf{k}-\mathbf{k}') \\ &- \sum_{\substack{\widetilde{\mathbf{k}}\in Q \\ \widetilde{\sigma}\in\{\uparrow,\downarrow\} \\ \widetilde{m},\widetilde{n}\in\{1,\ldots,\infty\}}} \frac{g_{\widetilde{m}\widetilde{n}\nu'}(\widetilde{\mathbf{k}},\mathbf{k}'-\mathbf{k})}{\hbar\sqrt{N_p}} c^\dagger_{\widetilde{m}\widetilde{\sigma}}(\tau,\widetilde{\mathbf{k}}+\mathbf{k}'-\mathbf{k})c_{\widetilde{n}\widetilde{\sigma}}(\tau,\widetilde{\mathbf{k}})\Bigg] c_{n',\uparrow}(\tau',\mathbf{k}')c^\dagger_{n,\uparrow}(0,\mathbf{k})\Bigg\rangle\end{aligned}$$

and

$$\begin{aligned}\mathrm{II} = -\frac{\omega_{\nu'}(\mathbf{k}'-\mathbf{k})}{\hbar}\Bigg\langle \mathrm{T}_\tau \Bigg[&+\omega_{\nu'}(\mathbf{k}'-\mathbf{k})b^\dagger_{\nu'}(\tau,\mathbf{k}'-\mathbf{k}) \\ &+ \sum_{\substack{\widetilde{\mathbf{k}}\in Q \\ \widetilde{\sigma}\in\{\uparrow,\downarrow\} \\ \widetilde{m},\widetilde{n}\in\{1,\ldots,\infty\}}} \frac{g_{\widetilde{m}\widetilde{n}\nu'}(\widetilde{\mathbf{k}},\mathbf{k}'-\mathbf{k})}{\hbar\sqrt{N_p}} c^\dagger_{\widetilde{m}\widetilde{\sigma}}(\tau,\widetilde{\mathbf{k}}+\mathbf{k}'-\mathbf{k})c_{\widetilde{n}\widetilde{\sigma}}(\tau,\widetilde{\mathbf{k}})\Bigg] c_{n',\uparrow}(\tau',\mathbf{k}')c^\dagger_{n,\uparrow}(0,\mathbf{k})\Bigg\rangle\end{aligned}$$

and collecting the individual contributions we obtain

$$\begin{aligned}\partial^2_\tau \mathcal{G}^{\text{aux}}_{\nu',n,n'}(\tau,\tau',\mathbf{k},\mathbf{k}') &= \omega^2_{\nu'}(\mathbf{k}-\mathbf{k}')\mathcal{G}^{\text{aux}}_{\nu',n,n'}(\tau,\tau',\mathbf{k},\mathbf{k}') \\ &\quad - \frac{2\omega_{\nu'}(\mathbf{k}-\mathbf{k}')}{\hbar} \sum_{\substack{\widetilde{\mathbf{k}}\in Q \\ \widetilde{\sigma}\in\{\uparrow,\downarrow\} \\ \widetilde{m},\widetilde{n}\in\{1,\ldots,\infty\}}} \frac{g_{\widetilde{m}\widetilde{n}\nu'}(\widetilde{\mathbf{k}},\mathbf{k}'-\mathbf{k})}{\hbar\sqrt{N_p}} \left\langle \mathrm{T}_\tau\, c^\dagger_{\widetilde{m}\widetilde{\sigma}}(\tau,\widetilde{\mathbf{k}}+\mathbf{k}'-\mathbf{k})c_{\widetilde{n}\widetilde{\sigma}}(\tau,\widetilde{\mathbf{k}})c_{n',\uparrow}(\tau',\mathbf{k}')c^\dagger_{n,\uparrow}(0,\mathbf{k})\right\rangle\end{aligned} \tag{31.20}$$

Now relabelling the summation-indices we obtain

$$\begin{aligned}&\left(\partial_\tau^2-\omega_{\nu'}^2(\mathbf{k}-\mathbf{k}')\right)\mathscr{G}_{\substack{\text{aux}\\ \nu',n,n'}}(\tau,\tau',\mathbf{k},\mathbf{k}')\\ &=\frac{-2\omega_{\nu'}(\mathbf{k}-\mathbf{k}')}{\hbar}\sum_{\substack{\mathbf{q}\in\mathrm{Q}\\ mm'\in\{1,\ldots,\infty\}\\ \sigma\in\{\uparrow,\downarrow\}}}\frac{g_{mm'\nu'}(\mathbf{q},\mathbf{k}'-\mathbf{k})}{\hbar N_p}\left\langle \mathrm{T}_\tau\, c^\dagger_{m,\sigma}(\tau,\mathbf{q}+\mathbf{k}'-\mathbf{k})c_{m',\sigma}(\tau,\mathbf{q})c_{n',\uparrow}(\tau',\mathbf{k}')c^\dagger_{n,\uparrow}(0,\mathbf{k})\right\rangle\end{aligned}$$

Proof 31.23 of Corollary 25.23 Starting with equation (25.33) which is

$$\begin{aligned}&\left(\partial_{\tau'}^2-\omega_{\nu'}^2(\mathbf{k}-\mathbf{k}')\right)\mathscr{G}_{\substack{\text{aux}\\ \nu',n,n'}}(\tau',\tau,\mathbf{k},\mathbf{k}')\\ &=\frac{-2\omega_{\nu'}(\mathbf{k}-\mathbf{k}')}{\hbar}\sum_{\substack{\mathbf{q}\in\mathrm{Q}\\ m,m'\in\{1,\ldots,\infty\}\\ \sigma\in\{\uparrow,\downarrow\}}}\frac{g_{mm'\nu'}(\mathbf{q},\mathbf{k}'-\mathbf{k})}{\hbar\sqrt{N_p}}\left\langle \mathrm{T}_\tau\, c^\dagger_{m,\sigma}(\tau',\mathbf{q}+\mathbf{k}'-\mathbf{k})c_{m',\sigma}(\tau',\mathbf{q})c_{n',\uparrow}(\tau,\mathbf{k}')c^\dagger_{n,\uparrow}(0,\mathbf{k})\right\rangle\end{aligned}$$

we multiply both sides with $\mathscr{D}_{\substack{0\\ \nu',\nu'\\ \mathbf{k},\mathbf{k}'}}(\tau-\tau')$, integrate τ' from 0 to $\hbar\beta$, perform an integration by parts and use the identity

$$\left(\partial_\tau^2-\omega_{\nu'}^2(\mathbf{k}-\mathbf{k}')\right)\mathscr{D}_{\substack{0,\nu'\\ \mathbf{k}-\mathbf{k}'}}(\tau-\tau')=-2\omega_{\nu'}(\mathbf{k}-\mathbf{k}')\delta(\tau-\tau')/\hbar$$

(which is a special case of a more general identity in [Draft Summary]). This yields

$$\begin{aligned}&\int_0^{\hbar\beta}\mathrm{d}\tau'\,\mathscr{G}_{\substack{\text{aux}\\ \nu',n,n'}}(\tau',\tau,\mathbf{k},\mathbf{k}')\left(\partial_{\tau'}^2-\omega_{\nu'}^2(\mathbf{k}-\mathbf{k}')\right)\mathscr{D}_{\substack{0,\nu'\\ \mathbf{k}-\mathbf{k}'}}(\tau-\tau')\\ &=\frac{-2\omega_{\nu'}(\mathbf{k}-\mathbf{k}')}{\hbar}\mathscr{G}_{\substack{\text{aux}\\ \nu',n,n'}}(\tau,\tau,\mathbf{k},\mathbf{k})\\ &=\int_0^{\hbar\beta}\mathrm{d}\tau'\,\mathscr{D}_{\substack{0,\nu'\\ \mathbf{k}-\mathbf{k}'}}(\tau-\tau')\frac{-2\omega_{\nu'}(\mathbf{k}-\mathbf{k}')}{\hbar}\sum_{\substack{\mathbf{q}\in\mathrm{Q}\\ m,m'\in\{1,\ldots,\infty\}\\ \sigma\in\{\uparrow,\downarrow\}}}\frac{g_{mm'\nu'}(\mathbf{q},\mathbf{k}'-\mathbf{k})}{\hbar\sqrt{N_p}}\left\langle \mathrm{T}_\tau\, c^\dagger_{m,\sigma}(\tau',\mathbf{q}+\mathbf{k}'-\mathbf{k})c_{m',\sigma}(\tau',\mathbf{q})c_{n',\uparrow}(\tau,\mathbf{k}')c^\dagger_{n,\uparrow}(0,\mathbf{k})\right\rangle\end{aligned} \tag{31.21}$$

simplifying both sides gives then the final result

$$\mathscr{G}_{\substack{\text{aux}\\ \nu',n,n'}}(\tau,\tau,\mathbf{k},\mathbf{k}') \tag{31.22}$$

$$=\mathop{\Large\sum\!\!\!\!\!\!\int}_{\substack{\mathbf{q}\in\mathrm{Q}\\ \tau'\in[0,\hbar\beta]\\ m,m'\in\{1,\ldots,\infty\}\\ \sigma\in\{\uparrow,\downarrow\}}}\mathscr{D}_{\substack{0,\nu'\\ \mathbf{k}-\mathbf{k}'}}(\tau-\tau')\frac{g_{mm'\nu'}(\mathbf{q},\mathbf{k}'-\mathbf{k})}{\hbar\sqrt{N_p}}\left\langle \mathrm{T}_\tau\, c^\dagger_{m,\sigma}(\tau',\mathbf{q}+\mathbf{k}'-\mathbf{k})c_{m',\sigma}(\tau',\mathbf{q})c_{n',\uparrow}(\tau,\mathbf{k}')c^\dagger_{n,\uparrow}(0,\mathbf{k})\right\rangle \tag{31.23}$$

Proof 31.24 of Corollary 25.26

$$\left(\partial_\tau + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar}\right) \underset{\text{Gorkov}}{\mathscr{G}_{n\uparrow}}(\tau, \mathbf{k})$$
$$= -\frac{1}{\hbar}\delta(\tau)$$
$$- \sum\!\!\!\!\!\!\int_{\substack{\mathbf{q},\mathbf{k}' \in \mathrm{Q} \\ \tau' \in (0,\hbar\beta) \\ \nu' \in \{1,\ldots,rd_{\mathbf{x}}\} \\ m,m',n' \in \{1,\ldots,\infty\} \\ \sigma \in \{\uparrow,\downarrow\}}} \frac{g_{nn'\nu'}(\mathbf{k}', \mathbf{k} - \mathbf{k}') g_{mm'\nu'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar N_p} \underset{\mathbf{k}-\mathbf{k}'}{\mathscr{D}_{0,\nu'}}(\tau - \tau')\hbar^2 \Bigg[\delta_{\sigma,\uparrow}\delta_{m,n'}\delta_{n,m'}\delta_{\mathbf{q},\mathbf{k}}\delta_{\sigma,\uparrow} \underset{\text{Gorkov}}{\mathscr{G}_{m\uparrow}}(\tau' - \tau, \mathbf{k}') \underset{\text{Gorkov}}{\mathscr{G}_{n\uparrow}}(\tau', \mathbf{k})$$
$$- \delta_{m,n}\delta_{n',m'}\delta_{\sigma,\downarrow}\delta_{\mathbf{q},-\mathbf{k}'}\Delta_{m\downarrow\uparrow}(\tau', \mathbf{k})\overline{\Delta}_{m'\downarrow\uparrow}(\tau - \tau', \mathbf{k}')\Bigg]$$
$$= -\frac{1}{\hbar}\delta(\tau) \quad + \quad \mathrm{I} \quad + \quad \mathrm{II} \tag{31.24}$$

with

$$\mathrm{I} = - \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}' \in \mathrm{Q} \\ \nu' \in \{1,\ldots,rd_{\mathbf{x}}\} \\ m \in \{1,\ldots,\infty\} \\ \tau' \in (0,\hbar\beta)}} \hbar \frac{g_{nm\nu'}(\mathbf{k}', \mathbf{k} - \mathbf{k}') g_{mn\nu'}(\mathbf{k}, \mathbf{k}' - \mathbf{k})}{N_p} \underset{\mathbf{k}-\mathbf{k}'}{\mathscr{D}_{0,\nu'}}(\tau - \tau') \underset{\text{Gorkov}}{\mathscr{G}_{m\uparrow}}(\tau' - \tau, \mathbf{k}') \underset{\text{Gorkov}}{\mathscr{G}_{n\uparrow}}(\tau', \mathbf{k})$$
$$= - \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}' \in \mathrm{Q} \\ \nu' \in \{1,\ldots,rd_{\mathbf{x}}\} \\ m \in \{1,\ldots,\infty\} \\ \tau' \in (0,\hbar\beta)}} \hbar \frac{|g_{nm\nu'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{N_p} \underset{\mathbf{k}-\mathbf{k}'}{\mathscr{D}_{0,\nu'}}(\tau - \tau') \underset{\text{Gorkov}}{\mathscr{G}_{m\uparrow}}(\tau' - \tau, \mathbf{k}') \underset{\text{Gorkov}}{\mathscr{G}_{n\uparrow}}(\tau', \mathbf{k})$$

and

$$\mathrm{II} = + \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}' \in \mathrm{Q} \\ \nu' \in \{1,\ldots,rd_{\mathbf{x}}\} \\ n' \in \{1,\ldots,\infty\} \\ \tau' \in (0,\hbar\beta)}} \hbar \frac{g_{nn'\nu'}(\mathbf{k}', \mathbf{k} - \mathbf{k}') g_{nn'\nu'}(-\mathbf{k}', \mathbf{k}' - \mathbf{k})}{N_p} \underset{\mathbf{k}-\mathbf{k}'}{\mathscr{D}_{0,\nu'}}(\tau - \tau')\Delta_{n\downarrow\uparrow}(\tau', \mathbf{k})\overline{\Delta}_{n'\downarrow\uparrow}(\tau - \tau', \mathbf{k}')$$
$$= + \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}' \in \mathrm{Q} \\ \nu' \in \{1,\ldots,rd_{\mathbf{x}}\} \\ n' \in \{1,\ldots,\infty\} \\ \tau' \in (0,\hbar\beta)}} \hbar \frac{|g_{nn'\nu'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{N_p} \underset{\mathbf{k}-\mathbf{k}'}{\mathscr{D}_{0,\nu'}}(\tau - \tau')\Delta_{n\downarrow\uparrow}(\tau', \mathbf{k})\overline{\Delta}_{n'\downarrow\uparrow}(\tau - \tau', \mathbf{k}')$$

Proof 31.25 of Theorem 25.27 We rewrite the equation of motion, i.e. Equation (31.24) in Matsubara-space. We start with the l.h.s. and obtain

$$\left(\partial_\tau + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar}\right) \underset{\text{Gorkov}}{\mathscr{G}_{n\uparrow}}(\tau, \mathbf{k}) = \frac{1}{\hbar\beta} \sum_{\omega_{n_1} \in \Omega_{-1}} \left(-i\omega_{n_1} + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar}\right) \underset{\text{Gorkov}}{\mathscr{G}_{n\uparrow}}(\mathbf{k}, \omega_{n_1}) \mathrm{e}^{-i\omega_{n_1}\tau}$$
$$= \frac{1}{\hbar\beta} \sum_{\omega_{n_1} \in \Omega_{-1}} \left(-\hbar \underset{\text{Gorkov}}{\overset{0}{\mathscr{G}}_{n\uparrow}}(\mathrm{i}\omega_{n_1}, \mathbf{k})\right)^{-1} \underset{\text{Gorkov}}{\mathscr{G}_{n\uparrow}}(\mathrm{i}\omega_{n_1}, \mathbf{k}) \mathrm{e}^{-i\omega_{n_1}\tau}$$

On the r.h.s. we expand each term individually and obtain

$$-\frac{1}{\hbar}\delta(\tau) = \frac{1}{\hbar\beta}\sum_{\omega_{n_1}\in\Omega_{-1}} -\frac{1}{\hbar}\mathrm{e}^{-i\omega_{n_1}\tau} \tag{31.25}$$

for the delta-function and for the other term number 1:

$$\begin{aligned}\mathrm{I} = &-\frac{\hbar}{(\hbar\beta)^3}\sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}'\in \mathrm{Q}\\ \nu'\in\{1,\ldots,rd_{\mathbf{x}}\}\\ m\in\{1,\ldots,\infty\}\\ \tau'\in(0,\hbar\beta)}} \frac{|g_{nm\nu'}(\mathbf{k}',\mathbf{k}-\mathbf{k}')|^2}{N_p}\sum_{\substack{\omega_{n_1},\omega_{n_2}\in\Omega_{-1}\\ \omega_{n_3}\in\Omega_1}} \mathrm{e}^{-\mathrm{i}\omega_{n_3}(\tau-\tau')-\mathrm{i}\omega_{n_1}(\tau'-\tau)-\mathrm{i}\omega_{n_2}\tau'}\mathscr{D}_{\substack{0,\nu'\\ \mathbf{k}-\mathbf{k}'}}(\omega_{n_3})\mathscr{G}_{\substack{m\uparrow\\ \text{Gorkov}}}(\mathrm{i}\omega_{n_1},\mathbf{k}')\mathscr{G}_{\substack{n\uparrow\\ \text{Gorkov}}}(\mathrm{i}\omega_{n_2},\mathbf{k}\\ =&\ \ldots\end{aligned}$$

Now we collect the terms in τ' and perform the integration

$$\int_0^{\hbar\beta}\mathrm{d}\tau'\,\mathrm{e}^{\mathrm{i}\tau'(\omega_{n_3}-\omega_{n_1}-\omega_{n_2})} = \hbar\beta\delta_{\omega_{n_3},\omega_{n_1}+\omega_{n_2}} \tag{31.26}$$

Noting further that

$$\mathrm{e}^{\mathrm{i}\tau\left(-\omega_{n_3}+\omega_{n_1}\right)}\Big|_{\omega_{n_3}=\omega_{n_1}+\omega_{n_2}} = \mathrm{e}^{-\mathrm{i}\tau\omega_{n_2}} \tag{31.27}$$

This allows us to terminate the summation over ω_{n_3} and continue as

$$\mathrm{I} = \ldots = \frac{1}{\hbar\beta}\sum_{\omega_{n_1},\omega_{n_2}\in\Omega_{-1}}\left(\frac{|g_{nm\nu'}(\mathbf{k}',\mathbf{k}-\mathbf{k}')|^2}{N_p}\right)\left(-\frac{\hbar}{\hbar\beta}\right)\mathrm{e}^{-\mathrm{i}\omega_{n_2}\tau}\mathscr{D}_{\substack{0,\nu'\\ \mathbf{k}-\mathbf{k}'}}(\mathrm{i}\omega_{n_1}+\mathrm{i}\omega_{n_2})\mathscr{G}_{\substack{m\uparrow\\ \text{Gorkov}}}(\mathrm{i}\omega_{n_1},\mathbf{k}')\mathscr{G}_{\substack{n\uparrow\\ \text{Gorkov}}}(\mathrm{i}\omega_{n_2},\mathbf{k})$$

Now we shift our attention to the term II and obtain for the other term number 2:

$$\mathrm{II} = \frac{\hbar}{(\hbar\beta)^3}\sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}'\in \mathrm{Q}\\ \nu'\in\{1,\ldots,rd_{\mathbf{x}}\}\\ n'\in\{1,\ldots,\infty\}\\ \tau'\in(0,\hbar\beta)}} \frac{|g_{nn'\nu'}(\mathbf{k}',\mathbf{k}-\mathbf{k}')|^2}{N_p}\sum_{\substack{\omega_{n_1},\omega_{n_2}\in\Omega_{-1}\\ \omega_{n_3}\in\Omega_1}} \mathrm{e}^{-\mathrm{i}\omega_{n_3}(\tau-\tau')-\mathrm{i}\omega_{n_1}(\tau-\tau')-\mathrm{i}\omega_{n_2}\tau'}\mathscr{D}_{\substack{0,\nu'\\ \mathbf{k}-\mathbf{k}'}}(\omega_{n_3})\Delta_{n\downarrow\uparrow}(\mathbf{k},\omega_{n_2})\overline{\Delta}_{n'\downarrow\uparrow}(\mathbf{k}',\omega_{n_1})$$

Now we collect the terms in τ' and perform the integration

$$\int_0^{\hbar\beta}\mathrm{d}\tau'\,\mathrm{e}^{\mathrm{i}\tau'(\omega_{n_3}+\omega_{n_1}-\omega_{n_2})} = \hbar\beta\delta_{\omega_{n_3},\omega_{n_2}-\omega_{n_1}} \tag{31.28}$$

Noting further that

$$\mathrm{e}^{-\mathrm{i}\tau(-\omega_{n_1}+\omega_{n_2})-\omega_{n_1}\tau} = \mathrm{e}^{-\mathrm{i}\tau\omega_{n_2}}$$

which gives the result

$$\mathrm{II} = \ldots = \frac{1}{\hbar\beta} \sum_{\substack{\omega_{n_1}, \omega_{n_2} \in \Omega_{-1} \\ n' \in \{1,\ldots,\infty\} \\ \nu' \in \{1,\ldots,rd_{\mathbf{x}}\} \mathbf{k}' \in \mathrm{Q}}} \left(\frac{|g_{nn'\nu'}(\mathbf{k}', \mathbf{k}-\mathbf{k}')|^2}{N_p} \right) \left(\frac{\hbar}{\hbar\beta} \right) \mathrm{e}^{-\mathrm{i}\omega_{n_2}\tau} \mathscr{D}_{\substack{0,\nu' \\ \mathbf{k}-\mathbf{k}'}}(\omega_{n_1} - \omega_{n_2}) \Delta_{n\downarrow\uparrow}(\mathrm{i}\omega_{n_2}, \mathbf{k}) \overline{\Delta}_{n'\downarrow\uparrow}(\mathrm{i}\omega_{n_1}, \mathbf{k}')$$

Combining the results gives the final result

$$\begin{aligned}
&- \frac{1}{\hbar\beta} \sum_{\omega_{n_1} \in \Omega_{-1}} \left(-\hbar \mathscr{G}_{\substack{0 \\ n\uparrow \\ \text{Gorkov}}}(\mathrm{i}\omega_{n_1}, \mathbf{k}) \right)^{-1} \mathscr{G}_{\substack{n\uparrow \\ \text{Gorkov}}}(\mathrm{i}\omega_{n_1}, \mathbf{k}) \mathrm{e}^{-i\omega_{n_1}\tau} \\
&= \frac{1}{\hbar\beta} \sum_{\omega_{n_1} \in \Omega_{-1}} -\frac{1}{\hbar} \mathrm{e}^{-i\omega_{n_1}\tau} \\
&+ \sum_{\substack{\omega_{n_1}, \omega_{n_2} \in \Omega_{-1} \\ m \in \{1,\ldots,\infty\} \\ \nu' \in \{1,\ldots,rd_{\mathbf{x}}\} \\ \mathbf{k}' \in \mathrm{Q}}} \left(\frac{|g_{nm\nu'}(\mathbf{k}', \mathbf{k}-\mathbf{k}')|^2}{N_p} \right) \left(-\frac{1}{\beta} \right) \mathrm{e}^{-\mathrm{i}\omega_{n_2}\tau} \mathscr{D}_{\substack{0,\nu' \\ \mathbf{k}-\mathbf{k}'}}(\mathrm{i}\omega_{n_1} + \mathrm{i}\omega_{n_2}) \mathscr{G}_{\substack{m\uparrow \\ \text{Gorkov}}}(\mathrm{i}\omega_{n_1}, \mathbf{k}') \mathscr{G}_{\substack{n\uparrow \\ \text{Gorkov}}}(\mathrm{i}\omega_{n_2}, \mathbf{k}) \\
&+ \sum_{\substack{\omega_{n_1}, \omega_{n_2} \in \Omega_{-1} \\ n' \in \{1,\ldots,\infty\} \\ \nu' \in \{1,\ldots,rd_{\mathbf{x}}\} \\ \mathbf{k}' \in \mathrm{Q}}} \left(\frac{|g_{nn'\nu'}(\mathbf{k}', \mathbf{k}-\mathbf{k}')|^2}{N_p} \right) \left(\frac{1}{\beta} \right) \mathrm{e}^{-\mathrm{i}\omega_{n_2}\tau} \mathscr{D}_{\substack{0,\nu' \\ \mathbf{k}-\mathbf{k}'}}(\omega_{n_2} - \omega_{n_1}) \Delta_{n\downarrow\uparrow}(\mathrm{i}\omega_{n_2}, \mathbf{k}) \overline{\Delta}_{n'\downarrow\uparrow}(\mathrm{i}\omega_{n_1}, \mathbf{k}')
\end{aligned}$$

now comparing coefficients gives the desired result.

Proof 31.26 of Theorem 25.30

There are many couplings contributing. We will „divide and conquer“ the problem. The Kronecker-deltas for the spin yield the enumeration

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \sigma, k_{\alpha_1})$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \pi, k_{\alpha_2})$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \pi, k_{\alpha_3})$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \sigma, k_{\alpha_4})$

Spin-Singlet Contributions

Case 1: $\sigma =\uparrow$ and $\pi =\downarrow$. Then we obtain

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \uparrow, k_{\alpha_1})$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \downarrow, k_{\alpha_2})$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \downarrow, k_{\alpha_3})$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \uparrow, k_{\alpha_4})$

Then we can proceed by enumerating the momenta in the following way

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \uparrow, k_{\alpha_1})$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \downarrow, -k)$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \downarrow, k_{\alpha_3})$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \uparrow, k')$

Momentum conservation yields then

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \uparrow, k_{\alpha_1})$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \downarrow, -k)$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \downarrow, k_{\alpha_1} - k - k')$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \uparrow, k')$

Now we replace the summation over $k_{\alpha_1} \mapsto k_{\alpha_1} + k$.

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \uparrow, k_{\alpha_1} + k)$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \downarrow, -k)$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \downarrow, k_{\alpha_1} - k')$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \uparrow, k')$

And now we write $k_{\alpha_1} \equiv q$ and obtain finally

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \uparrow, q + k)$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \downarrow, -k)$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \downarrow, q - k')$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \uparrow, k')$

This gives for the matrix-element

$$
\begin{aligned}
&V_{\alpha_1,\alpha_2;\alpha_3,\alpha_4} \\
&= \frac{-1}{\hbar\beta N_p} \sum_{\nu\in\{1,\dots,rd_{\mathbf{x}}\}} \frac{\omega_\nu(\mathbf{k}+\mathbf{q}-\mathbf{k}')}{\omega^2_{k+q-k'}+\omega^2_\nu(\mathbf{k}+\mathbf{q}-\mathbf{k}')} \frac{g_{m_{\alpha_1}m_{\alpha_4}\nu}(\mathbf{k}',\mathbf{k}+\mathbf{q}-\mathbf{k}')g^*_{m_{\alpha_3}m_{\alpha_2}\nu}(-\mathbf{k},\mathbf{k}+\mathbf{q}-\mathbf{k}')}{\hbar^2} \\
&= \frac{-1}{\hbar\beta N_p} \sum_{\nu\in\{1,\dots,rd_{\mathbf{x}}\}} \frac{\omega_\nu(\mathbf{k}+\mathbf{q}-\mathbf{k}')}{\omega^2_{k+q-k'}+\omega^2_\nu(\mathbf{k}+\mathbf{q}-\mathbf{k}')} \frac{g_{m_{\alpha_1}m_{\alpha_4}\nu}(\mathbf{k}',\mathbf{k}+\mathbf{q}-\mathbf{k}')g_{m_{\alpha_2}m_{\alpha_3}\nu}(\mathbf{q}-\mathbf{k}',-(\mathbf{k}+\mathbf{q}-\mathbf{k}'))}{\hbar^2}
\end{aligned}
$$

Case 2: $\sigma =\downarrow$ and $\pi =\uparrow$. Then we obtain

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \downarrow, k_{\alpha_1})$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \uparrow, k_{\alpha_2})$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \uparrow, k_{\alpha_3})$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \downarrow, k_{\alpha_4})$

Then we can proceed by enumerating the momenta in the following way

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \downarrow, -k)$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \uparrow, k_{\alpha_2})$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \uparrow, k')$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \downarrow, k_{\alpha_4})$

Momentum conservation yields then

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \downarrow, -k)$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \uparrow, k_{\alpha_2})$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \uparrow, k')$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \downarrow, k_{\alpha_2} - k - k')$

Now we replace the summation over $k_{\alpha_2} \mapsto k_{\alpha_2} + k$.

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \downarrow, -k)$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \uparrow, k_{\alpha_2} + k)$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \uparrow, k')$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \downarrow, k_{\alpha_2} - k')$

And now we write $k_{\alpha_2} \equiv q$ and obtain

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \downarrow, -k)$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \uparrow, q + k)$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \uparrow, k')$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \downarrow, q - k')$

finally we exchange $m_{\alpha_1} \leftrightarrow m_{\alpha_2}$ and (simultaneously) $m_{\alpha_3} \leftrightarrow m_{\alpha_4}$ which leads to

- $\alpha_1 \equiv (m_{\alpha_2}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_2}, \downarrow, -k)$
- $\alpha_2 \equiv (m_{\alpha_1}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_1}, \uparrow, q+k)$
- $\alpha_3 \equiv (m_{\alpha_4}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_4}, \uparrow, k')$
- $\alpha_4 \equiv (m_{\alpha_3}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_3}, \downarrow, q-k')$

This gives for the matrix-element

$$
\begin{aligned}
&V_{\alpha_1,\alpha_2;\alpha_3,\alpha_4} \\
&= \frac{-1}{\hbar\beta N_p} \sum_{\nu\in\{1,\ldots,rd_{\mathbf{x}}\}} \frac{\omega_\nu(-\mathbf{k}-\mathbf{q}+\mathbf{k}')}{\omega^2_{-k-q+k'} + \omega^2_\nu(-\mathbf{k}-\mathbf{q}+\mathbf{k}')} \frac{g_{m_{\alpha_2} m_{\alpha_3}\nu}(\mathbf{q}-\mathbf{k}', -\mathbf{k}-\mathbf{q}+\mathbf{k}') g^*_{m_{\alpha_4} m_{\alpha_1}\nu}(\mathbf{q}+\mathbf{k}, -\mathbf{k}-\mathbf{q}+\mathbf{k}')}{\hbar^2} \\
&= \frac{-1}{\hbar\beta N_p} \sum_{\nu\in\{1,\ldots,rd_{\mathbf{x}}\}} \frac{\omega_\nu(-\mathbf{k}-\mathbf{q}+\mathbf{k}')}{\omega^2_{-k-q+k'} + \omega^2_\nu(-\mathbf{k}-\mathbf{q}+\mathbf{k}')} \frac{g_{m_{\alpha_2} m_{\alpha_3}\nu}(\mathbf{q}-\mathbf{k}', -\mathbf{k}-\mathbf{q}+\mathbf{k}') g_{m_{\alpha_1} m_{\alpha_4}\nu}(\mathbf{k}', +\mathbf{k}+\mathbf{q}-\mathbf{k}')}{\hbar^2}
\end{aligned}
$$

Spin-Triplet-Contributions

Case 1: $\sigma = \uparrow$ and $\pi = \uparrow$. Then we obtain by the exact same calculations as above

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \uparrow, q + k)$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \uparrow, -k)$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \uparrow, q - k')$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \uparrow, k')$

This gives for the matrix-element

$$\begin{aligned}
&V^{\uparrow\uparrow}_{\alpha_1,\alpha_2;\alpha_3,\alpha_4} \\
&= \frac{-1}{\hbar\beta N_p} \sum_{\nu\in\{1,\ldots,rd_{\mathbf{x}}\}} \frac{\omega_\nu(\mathbf{k}+\mathbf{q}-\mathbf{k}')}{\omega^2_{k+q-k'} + \omega^2_\nu(\mathbf{k}+\mathbf{q}-\mathbf{k}')} \frac{g_{m_{\alpha_1} m_{\alpha_4}\nu}(\mathbf{k}', \mathbf{k}+\mathbf{q}-\mathbf{k}') g^*_{m_{\alpha_3} m_{\alpha_2}\nu}(-\mathbf{k}, \mathbf{k}+\mathbf{q}-\mathbf{k}')}{\hbar^2} \\
&= \frac{-1}{\hbar\beta N_p} \sum_{\nu\in\{1,\ldots,rd_{\mathbf{x}}\}} \frac{\omega_\nu(\mathbf{k}+\mathbf{q}-\mathbf{k}')}{\omega^2_{k+q-k'} + \omega^2_\nu(\mathbf{k}+\mathbf{q}-\mathbf{k}')} \frac{g_{m_{\alpha_1} m_{\alpha_4}\nu}(\mathbf{k}', \mathbf{k}+\mathbf{q}-\mathbf{k}') g_{m_{\alpha_2} m_{\alpha_3}\nu}(\mathbf{q}-\mathbf{k}', -(\mathbf{k}+\mathbf{q}-\mathbf{k}'))}{\hbar^2}
\end{aligned}$$

Case 2: $\sigma = \downarrow$ and $\pi = \downarrow$. Then we obtain by the exact same calculations as above

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \downarrow, q + k)$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \downarrow, -k)$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \downarrow, q - k')$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \downarrow, k')$

This gives for the matrix-element

$$\begin{aligned}
&V^{\downarrow\downarrow}_{\alpha_1,\alpha_2;\alpha_3,\alpha_4} \\
&= \frac{-1}{\hbar\beta N_p} \sum_{\nu\in\{1,\ldots,rd_{\mathbf{x}}\}} \frac{\omega_\nu(\mathbf{k}+\mathbf{q}-\mathbf{k}')}{\omega^2_{k+q-k'} + \omega^2_\nu(\mathbf{k}+\mathbf{q}-\mathbf{k}')} \frac{g_{m_{\alpha_1} m_{\alpha_4}\nu}(\mathbf{k}', \mathbf{k}+\mathbf{q}-\mathbf{k}') g^*_{m_{\alpha_3} m_{\alpha_2}\nu}(-\mathbf{k}, \mathbf{k}+\mathbf{q}-\mathbf{k}')}{\hbar^2} \\
&= \frac{-1}{\hbar\beta N_p} \sum_{\nu\in\{1,\ldots,rd_{\mathbf{x}}\}} \frac{\omega_\nu(\mathbf{k}+\mathbf{q}-\mathbf{k}')}{\omega^2_{k+q-k'} + \omega^2_\nu(\mathbf{k}+\mathbf{q}-\mathbf{k}')} \frac{g_{m_{\alpha_1} m_{\alpha_4}\nu}(\mathbf{k}', \mathbf{k}+\mathbf{q}-\mathbf{k}') g_{m_{\alpha_2} m_{\alpha_3}\nu}(\mathbf{q}-\mathbf{k}', -(\mathbf{k}+\mathbf{q}-\mathbf{k}'))}{\hbar^2}
\end{aligned}$$

Proof 31.27 of Theorem 25.32 We decompose the spin-singlet contributions as

$$S[\psi, \overline{\psi}] \equiv S^{\mathrm{ET}}[\psi, \overline{\psi}] + S^{\mathrm{NET}}[\psi, \overline{\psi}] + S^{\mathrm{RPC}}[\psi, \overline{\psi}] \tag{31.29}$$

whereas for the ([N]on-)[E]liashberg-[T]heory contribution

$$S^{\mathrm{ET}}[\psi,\overline{\psi}] = \sum_{\substack{k,k'\in \mathrm{i}\Omega_{-1}\times \mathrm{Q}\\ m,n\in\{1,2,3\dots\}}} \overline{\psi}_{m\uparrow}(k)\overline{\psi}_{m\downarrow}(-k)V^{\mathrm{ET}}_{\substack{\downarrow\uparrow\\ m,n}}(k,k')\psi_{n\downarrow}(-k')\psi_{n\uparrow}(k') + \sum_{\substack{\sigma\in\{\downarrow,\uparrow\}\\ k,k'\in \mathrm{i}\Omega_{-1}\times \mathrm{Q}\\ m,n\in\{1,2,3\dots\}}} \overline{\psi}_{m\sigma}(k)\overline{\psi}_{m\sigma}(-k)V^{\mathrm{ET}}_{\substack{\sigma\sigma\\ m,n}}(k,k')\psi_{n\sigma}(-k')\psi_{n\sigma}(k')$$

$$S^{\mathrm{NET}}_{\downarrow\uparrow}[\psi,\overline{\psi}] = \sum_{\substack{q,k\in \mathrm{i}\Omega_{-1}\times \mathrm{Q}\\ m,n\in\{1,2,3\dots\}}} \overline{\psi}_{m\uparrow}(q+k)\overline{\psi}_{n\downarrow}(-k)V^{\mathrm{NET}}_{\substack{\downarrow\uparrow\\ m,n}}(q,k)\psi_{m\downarrow}(q+k)\psi_{n\uparrow}(-k) + \sum_{\substack{\sigma\in\{\downarrow,\uparrow\}\\ q,k\in \mathrm{i}\Omega_{-1}\times \mathrm{Q}\\ m,n\in\{1,2,3\dots\}}} \overline{\psi}_{m\sigma}(q+k)\overline{\psi}_{n\sigma}(-k)V^{\mathrm{NET}}_{\substack{\sigma\sigma\\ m,n}}(q,k)\psi_{m\sigma}(q+k)\psi_{n\sigma}(-k)$$

we have, when we take the double contributions into account, the vertices

$$2V^{\mathrm{ET}}_{\substack{\sigma\sigma\\ m,n}}(k,k') = V^{\mathrm{ET}}_{\substack{\downarrow\uparrow\\ m,n}}(k,k') = \frac{-2}{\hbar\beta N_p}\sum_{\nu\in\{1,\dots,rd_{\mathbf{x}}\}} \frac{\omega_\nu(\mathbf{k}-\mathbf{k}')}{\omega^2_{k-k'}+\omega^2_\nu(\mathbf{k}-\mathbf{k}')}\frac{|g_{mn\nu}(\mathbf{k}',\mathbf{k}-\mathbf{k}')|^2}{\hbar^2} \tag{31.30}$$

$$2V^{\mathrm{NET}}_{\substack{\sigma\sigma\\ m,n}}(q,k) = V^{\mathrm{NET}}_{\substack{\downarrow\uparrow\\ m,n}}(q,k) = \frac{-2}{\hbar\beta N_p}\sum_{\nu\in\{1,\dots,rd_{\mathbf{x}}\}} \frac{\omega_\nu(2\mathbf{k}+\mathbf{q})}{\omega^2_{2k+q}+\omega^2_\nu(2\mathbf{k}+\mathbf{q})}\frac{|g_{mn\nu}(-\mathbf{k},2\mathbf{k}+\mathbf{q})|^2}{\hbar^2} \tag{31.31}$$

and the [R]andom [P]hase [C]ancellations are all the remaining contributions (which have complex vertices).

Proof 31.28 of Lemma 25.39

We compute straightforwardly

$$\eta_m(\tau,\mathbf{k}) = \frac{1}{\sqrt{\hbar\beta}}\sum_{\omega_n\in \mathrm{i}\Omega_{-1}} \eta_m(\mathbf{k},\mathrm{i}\omega_n)\mathrm{e}^{-\mathrm{i}\omega_n\tau} = \frac{1}{\sqrt{\hbar\beta}}\sum_{\mathrm{i}\omega_n\in \mathrm{i}\Omega_{-1}} \begin{pmatrix}\psi_{m\uparrow}(\mathrm{i}\omega_n,\mathbf{k})\\ \overline{\psi}_{m\downarrow}(-\mathrm{i}\omega_n,-\mathbf{k})\end{pmatrix}\mathrm{e}^{-\mathrm{i}\omega_n\tau} = \begin{pmatrix}\psi_{m\uparrow}(\tau,\mathbf{k})\\ \overline{\psi}_{m\downarrow}(\tau,-\mathbf{k})\end{pmatrix}$$

whereas in the last step we replaced $\omega_n \mapsto -\omega_n$ in the second component. Likewise, one obtains also

$$\overline{\eta}^{\mathrm{T}}_m(\tau,\mathbf{k}) = \begin{pmatrix}\overline{\psi}_{m\uparrow}(\tau,\mathbf{k})\\ \psi_{m\downarrow}(\tau,-\mathbf{k})\end{pmatrix} \tag{31.32}$$

Proof 31.29 of Corollary 25.40 We note that the action can be rewritten in imaginary-time space as

$$S_{\text{int}}[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow},\psi,\overline{\psi}] = \sum_{\substack{k\in \mathrm{i}\Omega_{-1}\times \mathrm{Q}\\ m\in\{1,2,3,\dots\}}} \overline{\eta}^{\mathrm{T}}_{m}(k)\left(-\frac{1}{\hbar}\underset{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov}}}{\mathscr{G}^{-1}{}_{m}}(k)\right)\eta_m(k)$$
$$= \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in \mathrm{Q}\\ m\in\{1,2,3,\dots\}\\ \tau,\tau'\in(0,\beta\hbar)}} \overline{\eta}^{\mathrm{T}}_{m}(\mathbf{k},\tau)\left(-\frac{1}{\hbar}\frac{1}{\hbar\beta}\sum_{\mathrm{i}\omega_n\in \mathrm{i}\Omega_{-1}}\underset{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov}}}{\mathscr{G}^{-1}{}_{m}}(\mathrm{i}\omega_n,\mathbf{k})\,\mathrm{e}^{-\mathrm{i}\omega_n(\tau'-\tau)}\right)\eta_m(\tau',\mathbf{k})$$
$$\equiv \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in \mathrm{Q}\\ m\in\{1,2,3,\dots\}\\ \tau,\tau'\in(0,\beta\hbar)}} \overline{\eta}^{\mathrm{T}}_{m}(\tau,\mathbf{k})\left(-\frac{1}{\hbar}\underset{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov}}}{\mathscr{G}^{-1}{}_{m}}(\tau-\tau',\mathbf{k})\right)\eta_m(\tau',\mathbf{k})$$

Now the identities follow by simply using the usual relations from [Draft Summary].

Proof 31.30 of Corollary 25.43

Searching for stationary points yields the equation

$$0 = \left(\frac{\delta}{\delta\Delta(\Gamma)}S_{\text{tot}}\right)[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]$$
$$= -\sum_{A\in\Pi}\overline{\Delta}(A)\Lambda^{\mathrm{ET}}_{\downarrow\uparrow}(A,\Gamma) - \operatorname{Tr}\underset{\text{Gorkov}}{\mathscr{G}_{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]}}\frac{\delta}{\delta\Delta(\Gamma)}\underset{\text{Gorkov}}{\mathscr{G}^{-1}_{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]}}$$

Multiplying now from both sides with the inverse of $\Lambda^{\mathrm{ET}}_{\downarrow\uparrow}$ from the right yields the gap-equation

$$\overline{\Delta}(B) = -\sum_{B\in\Pi}\operatorname{Tr}\left(\underset{\text{Gorkov}}{\mathscr{G}_{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]}}\frac{\delta}{\delta\Delta(A)}\underset{\text{Gorkov}}{\mathscr{G}^{-1}_{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]}}\right)V^{\mathrm{ET}}_{\downarrow\uparrow}(A,B) \tag{31.33}$$

Referring now to the multi-indices in Equation (25.58) and the matrix-elements in Equation (25.57), and partially performing the trace over the non-vanishing contributions, we obtain for each band and each four-momentum the desired equations.

Proof 31.31 of Lemma 25.44

Recalling that the Gorkov-Green-function block-diagonal w.r.t. bands and wave-vectors with individual blocks

$$-\frac{1}{\hbar}\underset{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov}}}{\mathscr{G}^{-1}{}_{m}}(k) = \begin{pmatrix} -\frac{1}{\hbar}\underset{m\uparrow}{\mathscr{G}_0^{-\mathrm{F}}}(k) & \Delta_{m\downarrow\uparrow}(k)\\ \overline{\Delta}_{m\downarrow\uparrow}(k) & +\frac{1}{\hbar}\underset{m\downarrow}{\mathscr{G}_0^{-\mathrm{F}}}(-k)\end{pmatrix} \tag{31.34}$$

which allows us on the one hand to compute the derivative

$$\frac{\delta}{\delta\Delta_{m\downarrow\uparrow}(k)}\underset{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov}}}{\mathscr{G}^{-1}{}_{m}} = \begin{pmatrix}0 & -\hbar\\ 0 & 0\end{pmatrix} \tag{31.35}$$

and by inversion of the equation

$$-\hbar \mathscr{G}_{\substack{m \\ [\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}(k) = \begin{pmatrix} -\frac{1}{\hbar}\mathscr{G}^{-\mathrm{F}}_{\substack{0 \\ m\uparrow}}(k) & \Delta_{m\downarrow\uparrow}(k) \\ \overline{\Delta}_{m\downarrow\uparrow}(k) & +\frac{1}{\hbar}\mathscr{G}^{-\mathrm{F}}_{\substack{0 \\ m\downarrow}}(-k) \end{pmatrix}^{-1} \iff \mathscr{G}_{\substack{m \\ [\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}(k) = -\frac{1}{\hbar}\begin{pmatrix} -\frac{1}{\hbar}\mathscr{G}^{-\mathrm{F}}_{\substack{0 \\ m\uparrow}}(k) & \Delta_{m\downarrow\uparrow}(k) \\ \overline{\Delta}_{m\downarrow\uparrow}(k) & +\frac{1}{\hbar}\mathscr{G}^{-\mathrm{F}}_{\substack{0 \\ m\downarrow}}(-k) \end{pmatrix}^{-1} \tag{31.36}$$

Recalling that one has for the non-interacting Green-functions the relations (we are considering a spin-degenerate system)

$$\begin{aligned} -\frac{1}{\hbar}\mathscr{G}^{-\mathrm{F}}_{\substack{0 \\ m\uparrow}}(k) &= -\mathrm{i}\omega_n + (\epsilon_m(\mathbf{k}) - \mu)/\hbar \\ +\frac{1}{\hbar}\mathscr{G}^{-\mathrm{F}}_{\substack{0 \\ m\downarrow}}(-k) &= -\mathrm{i}\omega_n - (\epsilon_m(-\mathbf{k}) - \mu)/\hbar \end{aligned}$$

we obtain for the inverse with the rule for 2×2 matrices

$$\mathscr{G}_{\substack{m \\ [\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}(k) = -\frac{1}{\hbar}\frac{\begin{pmatrix} -\mathrm{i}\omega_k - (\epsilon_m(-\mathbf{k}) - \mu)/\hbar & -\Delta_{m\downarrow\uparrow}(k) \\ -\overline{\Delta}_{m\downarrow\uparrow}(k) & -\mathrm{i}\omega_k + (\epsilon_m(\mathbf{k}) - \mu)/\hbar \end{pmatrix}}{\mathrm{Det}\begin{pmatrix} -\frac{1}{\hbar}\mathscr{G}^{-\mathrm{F}}_{\substack{0 \\ m\uparrow}}(k) & \Delta_{m\downarrow\uparrow}(k) \\ \overline{\Delta}_{m\downarrow\uparrow}(k) & +\frac{1}{\hbar}\mathscr{G}^{-\mathrm{F}}_{\substack{0 \\ m\downarrow}}(-k) \end{pmatrix}} \tag{31.37}$$

we therefore obtain ONLY IN THE CASE OF INVERSION SYMMETRIC CRYSTALS[5]

$$\mathrm{Det}\begin{pmatrix} -\frac{1}{\hbar}\mathscr{G}^{-\mathrm{F}}_{\substack{0 \\ m\uparrow}}(k) & \Delta_{m\downarrow\uparrow}(k) \\ \overline{\Delta}_{m\downarrow\uparrow}(k) & +\frac{1}{\hbar}\mathscr{G}^{-\mathrm{F}}_{\substack{0 \\ m\downarrow}}(-k) \end{pmatrix} = -\omega_k^2 - \frac{\left(\epsilon_m(\mathbf{k}) - \mu\right)^2}{\hbar^2} - |\Delta_{m\downarrow\uparrow}(k)|^2 \tag{31.38}$$

which simplifies the expression above to

$$\mathscr{G}_{\substack{m \\ [\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}(k) = -\frac{1}{\hbar}\frac{\begin{pmatrix} +\mathrm{i}\omega_k + (\epsilon_m(-\mathbf{k}) - \mu)/\hbar & +\Delta_{m\downarrow\uparrow}(k) \\ +\overline{\Delta}_{m\downarrow\uparrow}(k) & +\mathrm{i}\omega_k - (\epsilon_m(\mathbf{k}) - \mu)/\hbar \end{pmatrix}}{+\omega_k^2 + \frac{\left(\epsilon_m(\mathbf{k}) - \mu\right)^2}{\hbar^2} + |\Delta_{m\downarrow\uparrow}(k)|^2} \tag{31.39}$$

Proof 31.32 of Lemma 25.45 We first notice that

$$\mathscr{G}_{\substack{m,12 \\ [\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}(k) = \mathscr{G}_{\substack{m,21 \\ [\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}(k)^* \tag{31.40}$$

[5]In this case one finds $\epsilon_m(-\mathbf{k}) = \epsilon_m(+\mathbf{k})$

and obtain therefore

$$\underset{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov}}}{\mathscr{G}_{m,12}}(\tau-\tau') = \frac{1}{\hbar\beta}\sum_{\mathrm{i}\omega_n\in\mathrm{i}\Omega_{-1}} \underset{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov}}}{\mathscr{G}_{m,12}}(\mathrm{i}\omega_n,\mathbf{k})\mathrm{e}^{-\mathrm{i}\omega_n(\tau-\tau')}$$
$$= \frac{1}{\hbar\beta}\sum_{\mathrm{i}\omega_n\in\mathrm{i}\Omega_{-1}} \frac{\Delta_{m\downarrow\uparrow}(k)}{\omega_n^2+\left(\epsilon_m(\mathbf{k})-\mu\right)^2/\hbar^2+|\Delta_{m\downarrow\uparrow}(k)|^2}\mathrm{e}^{-\mathrm{i}\omega_n(\tau-\tau')}$$

And on the other hand we obtain

$$\underset{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov}}}{\mathscr{G}_{m,21}}(\tau'-\tau) = \frac{1}{\hbar\beta}\sum_{\mathrm{i}\omega_n\in\mathrm{i}\Omega_{-1}} \underset{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov}}}{\mathscr{G}_{m,21}}(\mathrm{i}\omega_n,\mathbf{k})\mathrm{e}^{-\mathrm{i}\omega_n(\tau'-\tau)}$$
$$= \frac{1}{\hbar\beta}\sum_{\mathrm{i}\omega_n\in\mathrm{i}\Omega_{-1}} \frac{\overline{\Delta}_{m\downarrow\uparrow}(k)}{\omega_n^2+\left(\epsilon_m(\mathbf{k})-\mu\right)^2/\hbar^2+|\Delta_{m\downarrow\uparrow}(k)|^2}\mathrm{e}^{+\mathrm{i}\omega_n(\tau-\tau')}$$
$$= \underset{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov}}}{\mathscr{G}_{m,12}}(\tau-\tau')^*$$

Thus, we obtain in the sense of generalized matrices the desired identity.

Proof 31.33 of Theorem 25.46

Combining the results TBReferenced and TBReferenced gives us

$$\left(\underset{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov}}}{\mathscr{G}_{m}}\frac{\delta}{\delta\Delta_{m\downarrow\uparrow}(k)}\underset{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov}}}{\mathscr{G}^{-1}_{m}}\right) = -\frac{1}{\hbar}\frac{\begin{pmatrix}+\mathrm{i}\omega_k+(\epsilon_m(-\mathbf{k})-\mu)/\hbar & +\Delta_{m\downarrow\uparrow}(k)\\ +\overline{\Delta}_{m\downarrow\uparrow}(k) & +\mathrm{i}\omega_k-(\epsilon_m(\mathbf{k})-\mu)/\hbar\end{pmatrix}}{+\omega_k^2+\frac{\left(\epsilon_m(\mathbf{k})-\mu\right)^2}{\hbar^2}+|\Delta_{m\downarrow\uparrow}(k)|^2}\cdot\begin{pmatrix}0 & -\hbar\\ 0 & 0\end{pmatrix}$$
$$= +\frac{\begin{pmatrix}0 & +\mathrm{i}\omega_k+(\epsilon_m(-\mathbf{k})-\mu)/\hbar\\ 0 & \overline{\Delta}_{m\downarrow\uparrow}(k)\end{pmatrix}}{+\omega_k^2+\frac{\left(\epsilon_m(\mathbf{k})-\mu\right)^2}{\hbar^2}+|\Delta_{m\downarrow\uparrow}(k)|^2}$$

which gives for the trace

$$\mathrm{Tr}_{2\times 2}\left(\underset{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov}}}{\mathscr{G}_{m}}\frac{\delta}{\delta\Delta_{m\downarrow\uparrow}(k)}\underset{\substack{[\Delta_{\downarrow\uparrow},\overline{\Delta}_{\downarrow\uparrow}]\\ \text{Gorkov}}}{\mathscr{G}^{-1}_{m}}\right) = +\frac{\overline{\Delta}_{m\downarrow\uparrow}(k)}{\omega_k^2+\frac{\left(\epsilon_m(\mathbf{k})-\mu\right)^2}{\hbar^2}+|\Delta_{m\downarrow\uparrow}(k)|^2} \tag{31.41}$$

Now we arrive at the final form of the gap-equations

$$\overline{\Delta}_{n\downarrow\uparrow}(k') = + \sum_{\substack{k \in \mathrm{i}\Omega_{-1} \times \mathrm{Q} \\ \nu \in \{1,\ldots,rd\} \\ m \in \{1,2,3,\ldots\}}} \frac{\overline{\Delta}_{m\downarrow\uparrow}(k)}{\omega_k^2 + \frac{\left(\epsilon_m(\mathbf{k}) - \mu\right)^2}{\hbar^2} + |\Delta_{m\downarrow\uparrow}(k)|^2} \frac{2\omega_\nu(\mathbf{k} - \mathbf{k}')}{\omega_{k-k'}^2 + \omega_\nu^2(\mathbf{k} - \mathbf{k}')} \frac{|g_{mn\nu}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{\hbar^3 \beta N_p}$$

32. Appendix - Derivation of Self-Energies and Polarizations

32.1. Self Energy of a Diagonalized Gas with Single-Particle-Interactions

Proof 32.1 of Theorem 20.1

Computing the electronic Self-Energy

Preparations

We consider the system

$$\begin{aligned}\left(S_0 + S_{\text{int}}\right)[\psi, \overline{\psi}] &\equiv \sum_{\substack{\omega_n \in \Omega_\zeta \\ \alpha,\alpha' \in \mathscr{A}}} \overline{\psi}_\alpha(\mathrm{i}\omega_n) \left[\left(-\mathrm{i}\omega_n + \frac{\epsilon_\alpha - \mu}{\hbar} \right) \delta_{\alpha,\alpha'} + \frac{V_{\alpha\alpha'}}{\hbar} \right] \psi_{\alpha'}(\mathrm{i}\omega_n) \\ &\equiv \sum_{\substack{\omega_n \in \Omega_\zeta \\ \alpha,\alpha' \in \mathscr{A}}} \overline{\psi}_\alpha(\mathrm{i}\omega_n) \left(-\frac{1}{\hbar} \mathscr{G}^{-1}_{\alpha,\alpha'}(\mathrm{i}\omega_n) \right) \psi_{\alpha'}(\mathrm{i}\omega_n)\end{aligned}$$

and compute the self-energy using the standard-trick.

We will use the following trick: Inverting the Dyson-equation with the von Neumann series on the one hand, and re-expressing the imaginary-time-Green-function with help of the field integral on the other hand, gives the two equations

$$\mathscr{G}_{\alpha,\alpha'}(\mathrm{i}\omega_n) = \sum_{n=0}^{\infty} \left[\left(\mathscr{G}_0 \cdot \Sigma \right)^n (\mathrm{i}\omega_n) \cdot \mathscr{G}_0(\mathrm{i}\omega_n) \right]_{\alpha,\alpha'} \tag{32.1}$$

$$\mathscr{G}_{\alpha,\alpha'}(\mathrm{i}\omega_n) = \frac{-1}{\hbar} \left\langle \psi_\alpha(\mathrm{i}\omega_n) \overline{\psi}_{\alpha'}(\mathrm{i}\omega_n) \right\rangle \tag{32.2}$$

The self-energy of the electronic Green-function is known to be given as the sum of all the connected irreducible diagrams of the interaction vertex

$$\Sigma(\mathrm{i}\omega_n) \equiv \sum_{j=1}^{\infty} \Sigma^{(j)}(\mathrm{i}\omega_n) = \Sigma^{(1)}(\mathrm{i}\omega_n) + \Sigma^{(2)}(\mathrm{i}\omega_n) + \dots . \tag{32.3}$$

Inserting this in the upper-equation above and keeping the first non-vanishing terms yields the sets of equations

$$\mathscr{G}_{\alpha,\alpha'}(\mathrm{i}\omega_n) = \mathscr{G}_{\substack{0\\ \alpha,\alpha'}}(\mathrm{i}\omega_n) + \left(\mathscr{G}_0 \cdot \Sigma^{(1)} \cdot \mathscr{G}_0\right)_{\alpha,\alpha'}(\mathrm{i}\omega_n) + \left(\mathscr{G}_0 \cdot \Sigma^{(2)} \cdot \mathscr{G}_0\right)_{\alpha,\alpha'}(\mathrm{i}\omega_n) + \dots \tag{32.4}$$

$$\mathscr{G}_{\alpha,\alpha'}(\mathrm{i}\omega_n) = \frac{-1}{\hbar}\left\langle \psi_\alpha(\mathrm{i}\omega_n)\overline{\psi}_{\alpha'}(\mathrm{i}\omega_n)\right\rangle \tag{32.5}$$

The lower equation can be calculated explicitly with help of the Wick-theorem, linked-cluster-theorem, and the residue-theorem, and then, by comparing the first non-vanishing coefficients with the upper-equation, we can conclude what $\Sigma^{(2)}(\mathrm{i}\omega_n)$ must be.

Fact 1: The action, corresponding to this system, is given by the equation above.

Fact 2: The electronic Green-function is written in second order perturbation-theory as

$$\mathscr{G}_{\alpha\alpha'}(\mathrm{i}\omega_n) = \mathscr{G}^{(0)}_{\alpha\alpha'}(\mathrm{i}\omega_n) + \mathscr{G}^{(1)}_{\alpha\alpha'}(\mathrm{i}\omega_n) + \mathscr{G}^{(2)}_{\alpha\alpha'}(\mathrm{i}\omega_n) + \dots$$

The First Non-Vanishing Order

We therefore calculate

$$\mathscr{G}^{(1)}_{\alpha\alpha'}(\mathrm{i}\omega_n) = \frac{-1}{\hbar}\sum_{\substack{\beta,\beta'\in\mathscr{A}\\ \omega_m\in\Omega_\zeta}} \frac{-V_{\beta\beta'}}{\hbar}\left\langle \psi_\alpha(\mathrm{i}\omega_n)\overline{\psi}_{\alpha'}(\mathrm{i}\omega_n)\overline{\psi}_{\beta}(\mathrm{i}\omega_m)\psi_{\beta'}(\mathrm{i}\omega_m)\right\rangle_0^{\mathrm{CD}} \tag{32.6}$$

where the last expectation-value contains only [C]onnected [D]iagrams (linked-cluster-theorem). We can therefore simply calculate the free correlator and discard the contributions from unconnected diagrams.

We therefore consider the correlator

$$\begin{aligned}
\left\langle \psi_\alpha(\mathrm{i}\omega_n)\overline{\psi}_{\alpha'}(\mathrm{i}\omega_n)\overline{\psi}_{\beta}(\mathrm{i}\omega_m)\psi_{\beta'}(\mathrm{i}\omega_m)\right\rangle_0 &= \left\langle \psi_\alpha(\mathrm{i}\omega_n)\psi_{\beta'}(\mathrm{i}\omega_m)\overline{\psi}_{\alpha'}(\mathrm{i}\omega_n)\overline{\psi}_{\beta}(\mathrm{i}\omega_m)\right\rangle_0 \\
&= \left\langle \psi(A_1)\psi(A_2)\overline{\psi}(B_1)\overline{\psi}(B_2)\right\rangle_0 \\
&= -\left\langle \psi(A_1)\psi(A_2)\overline{\psi}(B_2)\overline{\psi}(B_1)\right\rangle_0 \\
&= -\sum_{\pi\in\mathrm{S}_2}\mathrm{sign}(\pi)^{1-\theta(\zeta)}\prod_{l=1}^{2}\left\langle \psi(A_l)\overline{\psi}(B_{\pi(l)})\right\rangle_0 \\
&= -\hbar^2\sum_{\pi\in\mathrm{S}_2}\mathrm{sign}(\pi)^{1-\theta(\zeta)}\prod_{l=1}^{2}\mathscr{G}_0(A_l, B_{\pi(l)})
\end{aligned}$$

The electronic correlator can now be evaluated with the Wick-theorem. In order to so, we will enumerate all the permutations of the group S_3.

$$\mathrm{S}_2 = \{\pi_1, \pi_2\} = \left\{\begin{pmatrix}1 & 2\\ 1 & 2\end{pmatrix}, \begin{pmatrix}1 & 2\\ 2 & 1\end{pmatrix}\right\} \quad \text{with signatures} \quad \mathrm{sgn}(\pi_1,\pi_2) = (+1,-1). \tag{32.7}$$

The first contribution is a vacuum diagram and does not contribute to the series expansion (linked cluster theorem). We therefore focus on the second term. We note that

$$A_1 = (\alpha, \mathrm{i}\omega_n), \qquad A_2 = (\beta, \mathrm{i}\omega_m) \tag{32.8}$$

$$B_1 = (\alpha', \mathrm{i}\omega_n), \qquad B_2 = (\beta, \mathrm{i}\omega_m) \tag{32.9}$$

and therefore we obtain for the non-vacuum contribution to the Green-function

$$\mathscr{G}^{(1)}_{\alpha\alpha'}(\mathrm{i}\omega_n) = + \sum_{\beta,\beta' \in \mathscr{A}} \mathscr{G}_{\underset{\alpha,\beta}{0}}(\mathrm{i}\omega_n)\, V_{\beta,\beta'}\, \mathscr{G}_{\underset{\beta'\alpha'}{0}}(\mathrm{i}\omega_n)$$

comparison of coefficients gives the first-order correction to the self-energy

$$\Sigma^{(1)}_{\beta\beta'}(\mathrm{i}\omega_n) = +V_{\beta,\beta'} = \text{„frequency-independent“} \tag{32.10}$$

32.2. Self Energy of a Diagonalized Gas with Quartic Interactions

Proof 32.2 of Theorem 20.2

Computing the electronic Self-Energy

Preparations

We consider the system

$$\left(S_0 + S_{\text{int}}\right)[\psi, \overline{\psi}] \equiv \sum_{\substack{\omega_n \in \Omega_\zeta \\ \alpha,\alpha' \in \mathscr{A}}} \overline{\psi}_\alpha(\mathrm{i}\omega_n) \left[\left(-\mathrm{i}\omega_n + \frac{\epsilon_\alpha - \mu}{\hbar} \right) \delta_{\alpha,\alpha'} \right] \psi_{\alpha'}(\mathrm{i}\omega_n)$$
$$+ \frac{1}{2} \sum_{\substack{\alpha_1,\alpha_2,\alpha_3,\alpha_4 \in \mathscr{A} \\ \omega_{n_1},\omega_{n_2},\omega_{n_3},\omega_{n_4} \in \Omega_\zeta}} \frac{-V_{\alpha_1\alpha_2\alpha_3\alpha_4} \delta_{\omega_1+\omega_2,\omega_3+\omega_4}}{\hbar^2 \beta} \overline{\psi}_{\alpha_1}(\mathrm{i}\omega_1) \overline{\psi}_{\alpha_2}(\mathrm{i}\omega_2) \psi_{\alpha_3}(\mathrm{i}\omega_3) \psi_{\alpha_4}(\mathrm{i}\omega_4)$$

and compute the self-energy using the standard-trick.

We will use the following trick: Inverting the Dyson-equation with the von Neumann series on the one hand, and re-expressing the imaginary-time-Green-function with help of the field integral on the other hand, gives the two equations

$$\mathscr{G}_{\alpha,\alpha'}(\mathrm{i}\omega_n) = \sum_{n=0}^{\infty} \left[\left(\mathscr{G}_0 \cdot \Sigma \right)^n (\mathrm{i}\omega_n) \cdot \mathscr{G}_0(\mathrm{i}\omega_n) \right]_{\alpha,\alpha'} \tag{32.11}$$
$$\mathscr{G}_{\alpha,\alpha'}(\mathrm{i}\omega_n) = \frac{-1}{\hbar} \left\langle \psi_\alpha(\mathrm{i}\omega_n) \overline{\psi}_{\alpha'}(\mathrm{i}\omega_n) \right\rangle \tag{32.12}$$

The self-energy of the electronic Green-function is known to be given as the sum of all the connected irreducible diagrams of the interaction vertex

$$\Sigma(\mathrm{i}\omega_n) \equiv \sum_{j=1}^{\infty} \Sigma^{(j)}(\mathrm{i}\omega_n) = \Sigma^{(1)}(\mathrm{i}\omega_n) + \Sigma^{(2)}(\mathrm{i}\omega_n) + \dots . \tag{32.13}$$

Inserting this in the upper-equation above and keeping the first non-vanishing terms yields the sets of equations

$$\mathscr{G}_{\alpha,\alpha'}(\mathrm{i}\omega_n) = \mathscr{G}_{\substack{0 \\ \alpha,\alpha'}}(\mathrm{i}\omega_n) + \left(\mathscr{G}_0 \cdot \Sigma^{(1)} \cdot \mathscr{G}_0 \right)_{\alpha,\alpha'} (\mathrm{i}\omega_n) + \left(\mathscr{G}_0 \cdot \Sigma^{(2)} \cdot \mathscr{G}_0 \right)_{\alpha,\alpha'} (\mathrm{i}\omega_n) + \dots \tag{32.14}$$
$$\mathscr{G}_{\alpha,\alpha'}(\mathrm{i}\omega_n) = \frac{-1}{\hbar} \left\langle \psi_\alpha(\mathrm{i}\omega_n) \overline{\psi}_{\alpha'}(\mathrm{i}\omega_n) \right\rangle \tag{32.15}$$

The lower equation can be calculated explicitly with help of the Wick-theorem, linked-cluster-theorem, and the residue-theorem, and then, by comparing the first non-vanishing coefficients with the upper-equation, we can conclude what $\Sigma^{(2)}(\mathrm{i}\omega_n)$ must be.

Fact 1: The action, corresponding to this system, is given by the equation above.

Fact 2: The electronic Green-function is written in second order perturbation-theory as

$$\mathscr{G}_{\alpha\alpha'}(\mathrm{i}\omega_n) = \mathscr{G}^{(0)}_{\alpha\alpha'}(\mathrm{i}\omega_n) + \mathscr{G}^{(1)}_{\alpha\alpha'}(\mathrm{i}\omega_n) + \mathscr{G}^{(2)}_{\alpha\alpha'}(\mathrm{i}\omega_n) + \dots$$

The First Non-Vanishing Order

We therefore calculate

$$\mathscr{G}^{(1)}_{\alpha\alpha'}(\mathrm{i}\omega_n) = \frac{-1}{\hbar}\frac{1}{2} \sum_{\substack{\alpha_1,\alpha_2,\alpha_3,\alpha_4 \in \mathscr{A} \\ \omega_{n_1},\omega_{n_2},\omega_{n_3},\omega_{n_4} \in \Omega_\zeta}} \frac{+V_{\alpha_1\alpha_2\alpha_3\alpha_4}\delta_{\omega_{n_1}+\omega_{n_2},\omega_{n_3}+\omega_{n_4}}}{\hbar^2\beta} \left\langle \psi_\alpha(\mathrm{i}\omega_n)\overline{\psi}_{\alpha'}(\mathrm{i}\omega_n)\overline{\psi}_{\alpha_1}(\mathrm{i}\omega_{n_1})\overline{\psi}_{\alpha_2}(\mathrm{i}\omega_{n_2})\psi_{\alpha_3}(\mathrm{i}\omega_{n_3})\psi_{\alpha_4}(\mathrm{i}\omega_{n_4})\right\rangle_0 \tag{32.16}$$

We now focus on the correlator

$$\begin{aligned}
\left\langle \psi_\alpha(\mathrm{i}\omega_n)\overline{\psi}_{\alpha'}(\mathrm{i}\omega_n)\overline{\psi}_{\alpha_1}(\mathrm{i}\omega_1)\overline{\psi}_{\alpha_2}(\mathrm{i}\omega_2)\psi_{\alpha_3}(\mathrm{i}\omega_3)\psi_{\alpha_4}(\mathrm{i}\omega_4)\right\rangle_0 &= \left\langle \psi_\alpha(\mathrm{i}\omega_n)\psi_{\alpha_3}(\mathrm{i}\omega_3)\psi_{\alpha_4}(\mathrm{i}\omega_4)\overline{\psi}_{\alpha'}(\mathrm{i}\omega_n)\overline{\psi}_{\alpha_1}(\mathrm{i}\omega_1)\overline{\psi}_{\alpha_2}(\mathrm{i}\omega_2)\right\rangle_0 \\
&= \left\langle \psi(A_1)\psi(A_2)\psi(A_3)\overline{\psi}(B_1)\overline{\psi}(B_2)\overline{\psi}(B_3)\right\rangle_0 \\
&= -\left\langle \psi(A_1)\psi(A_2)\psi(A_3)\overline{\psi}(B_3)\overline{\psi}(B_2)\overline{\psi}(B_1)\right\rangle_0 \\
&= -\sum_{\pi\in \mathrm{S}_3} \mathrm{sign}(\pi)^{1-\theta(\zeta)} \prod_{l=1}^{3} \left\langle \psi(A_l)\overline{\psi}(B_{\pi(l)})\right\rangle_0 \\
&= +\hbar^3 \sum_{\pi\in \mathrm{S}_3} \mathrm{sign}(\pi)^{1-\theta(\zeta)} \prod_{l=1}^{3} \mathscr{G}_0(A_l, B_{\pi(l)})
\end{aligned}$$

The electronic correlator can now be evaluated with the Wick-theorem. In order to so, we will enumerate all the permutations of the group S_3.

$$\begin{aligned}
\mathrm{S}_3 &= \{\pi_1,\pi_2,\pi_3,\pi_4,\pi_5,\pi_6\} \\
&= \left\{ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3\end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2\end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3\end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1\end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2\end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1\end{pmatrix} \right\}
\end{aligned}$$

with signatures

$$\mathrm{sgn}(\pi_1,\pi_2,\pi_3,\pi_4,\pi_5,\pi_6) = (+1,-1,-1,+1,+1,-1)$$

Facts (???):

- The contributions from π_1 and π_2 are vacuum diagrams (which cancel out due to the linked-cluster-theorem)
- The contributions from π_3 and π_6 are "Baloon"-diagrams / "Hartree"-diagrams.
- The contributions from π_4 and π_5 are "Sunset"-diagrams / "Fock"-diagrams.

We note that

$$\begin{aligned}
A_1 &= (\alpha, \mathrm{i}\omega_n), \qquad & A_2 &= (\alpha_3, \mathrm{i}\omega_{n_3}), \qquad & A_3 &= (\alpha_4, \mathrm{i}\omega_{n_4}) \\
B_1 &= (\alpha', \mathrm{i}\omega_n), \qquad & B_2 &= (\alpha_1, \mathrm{i}\omega_{n_1}), \qquad & B_3 &= (\alpha_2, \mathrm{i}\omega_{n_2})
\end{aligned}$$

and therefore we obtain

$$\begin{aligned}
&B_{\pi_3(1)} \equiv B_2 \equiv (\alpha_1, \mathrm{i}\omega_{n_1}), \quad B_{\pi_3(2)} \equiv B_1 \equiv (\alpha', \mathrm{i}\omega_n), \quad B_{\pi_3(3)} \equiv B_3 \equiv (\alpha_2, \mathrm{i}\omega_{n_2}), \\
&B_{\pi_5(1)} \equiv B_3 \equiv (\alpha_2, \mathrm{i}\omega_{n_2}), \quad B_{\pi_5(2)} \equiv B_1 \equiv (\alpha', \mathrm{i}\omega_n), \quad B_{\pi_5(3)} \equiv B_2 \equiv (\alpha_1, \mathrm{i}\omega_{n_1}), \\
&B_{\pi_6(1)} \equiv B_3 \equiv (\alpha_2, \mathrm{i}\omega_{n_2}), \quad B_{\pi_6(2)} \equiv B_2 \equiv (\alpha_1, \mathrm{i}\omega_{n_1}), \quad B_{\pi_6(3)} \equiv B_1 \equiv (\alpha', \mathrm{i}\omega_n), \\
&B_{\pi_4(1)} \equiv B_2 \equiv (\alpha_1, \mathrm{i}\omega_{n_1}), \quad B_{\pi_4(2)} \equiv B_3 \equiv (\alpha_2, \mathrm{i}\omega_{n_2}), \quad B_{\pi_4(3)} \equiv B_1 \equiv (\alpha', \mathrm{i}\omega_n).
\end{aligned}$$

Baloon-Diagrams

π_3**:**

This gives then for the first order correction of the Green-function

$$\begin{aligned}
\mathcal{G}^{\pi_3}_{\alpha\alpha'}(\mathrm{i}\omega_n) &= +\frac{1}{2} \sum_{\substack{\alpha_1,\alpha_2,\alpha_3,\alpha_4 \in \mathcal{A} \\ \omega_{n_1},\omega_{n_2},\omega_{n_3},\omega_{n_4} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_1\alpha_2\alpha_3\alpha_4} \delta_{\omega_{n_1}+\omega_{n_2},\omega_{n_3}+\omega_{n_4}} \mathcal{G}^{0}_{\alpha,\alpha_1}(\mathrm{i}\omega_n) \delta_{\omega_n,\omega_{n_1}} \mathcal{G}^{0}_{\alpha_3\alpha'}(\mathrm{i}\omega_n) \delta_{\omega_{n_3},\omega_n} \mathcal{G}^{0}_{\alpha_4\alpha_2}(\mathrm{i}\omega_{n_2}) \delta_{\omega_{n_2},\omega_{n_4}} \\
&= +\frac{1}{2} \sum_{\substack{\alpha_1,\alpha_2,\alpha_3,\alpha_4 \in \mathcal{A} \\ \omega_{n_1},\omega_{n_2},\omega_{n_3},\omega_{n_4} \in \Omega_\zeta}} \mathcal{G}^{0}_{\alpha,\alpha_1}(\mathrm{i}\omega_n) \delta_{\omega_n,\omega_{n_1}} \frac{1}{\beta} V_{\alpha_1\alpha_2\alpha_3\alpha_4} \delta_{\omega_{n_1}+\omega_{n_2},\omega_{n_3}+\omega_{n_4}} \mathcal{G}^{0}_{\alpha_4\alpha_2}(\mathrm{i}\omega_{n_2}) \delta_{\omega_{n_2},\omega_{n_4}} \mathcal{G}^{0}_{\alpha_3\alpha'}(\mathrm{i}\omega_n) \delta_{\omega_{n_3},\omega_n} \\
&= +\frac{1}{2} \sum_{\substack{\alpha_1,\alpha_2,\alpha_3,\alpha_4 \in \mathcal{A} \\ \omega_{n_1},\omega_{n_2} \in \Omega_\zeta}} \mathcal{G}^{0}_{\alpha,\alpha_1}(\mathrm{i}\omega_n) \delta_{\omega_n,\omega_{n_1}} \frac{1}{\beta} V_{\alpha_1\alpha_2\alpha_3\alpha_4} \delta_{\omega_{n_1}+\omega_{n_2},\omega_n+\omega_{n_2}} \mathcal{G}^{0}_{\alpha_4\alpha_2}(\mathrm{i}\omega_{n_2}) \mathcal{G}^{0}_{\alpha_3\alpha'}(\mathrm{i}\omega_n) \\
&= +\frac{1}{2} \sum_{\substack{\alpha_1,\alpha_2,\alpha_3,\alpha_4 \in \mathcal{A} \\ \omega_{n_2} \in \Omega_\zeta}} \mathcal{G}^{0}_{\alpha,\alpha_1}(\mathrm{i}\omega_n) \frac{1}{\beta} V_{\alpha_1\alpha_2\alpha_3\alpha_4} \mathcal{G}^{0}_{\alpha_4\alpha_2}(\mathrm{i}\omega_{n_2}) \mathcal{G}^{0}_{\alpha_3\alpha'}(\mathrm{i}\omega_n)
\end{aligned}$$

From this we can read off the self-energy

$$\Sigma^{\pi_3}_{\alpha_1\alpha_3}(\mathrm{i}\omega_n) = +\frac{1}{2} \sum_{\substack{\alpha_2 \in \mathcal{A} \\ \omega_{n_2} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_1\alpha_2\alpha_3\alpha_2} \mathcal{G}^{0}_{\alpha_2\alpha_2}(\mathrm{i}\omega_{n_2}) = +\frac{1}{2} \sum_{\alpha_2 \in \mathcal{A}} V_{\alpha_1\alpha_2\alpha_3\alpha_2} \mathrm{F}_{\hbar\beta}(\epsilon_{\alpha_2} - \mu) = -\zeta \frac{1}{2} \sum_{\alpha_2 \in \mathcal{A}} V_{\alpha_1\alpha_2\alpha_3\alpha_2} \mathrm{n}^{\zeta}_{\hbar\beta}(\epsilon_{\alpha_2} - \mu) \tag{32.17}$$

π_6**:**

This gives then for the first order correction of the Green-function

$$
\begin{aligned}
\mathscr{G}^{\pi_6}_{\alpha\alpha'}(\mathrm{i}\omega_n) &= +\frac{1}{2} \sum_{\substack{\alpha_1,\alpha_2,\alpha_3,\alpha_4 \in \mathscr{A} \\ \omega_{n_1},\omega_{n_2},\omega_{n_3},\omega_{n_4} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_1\alpha_2\alpha_3\alpha_4} \delta_{\omega_{n_1}+\omega_{n_2},\omega_{n_3}+\omega_{n_4}} \mathscr{G}^{\,0}_{\alpha,\alpha_2}(\mathrm{i}\omega_n) \delta_{\omega_n,\omega_{n_2}} \mathscr{G}^{\,0}_{\alpha_3,\alpha_1}(\mathrm{i}\omega_{n_3}) \delta_{\omega_{n_3},\omega_{n_1}} \mathscr{G}^{\,0}_{\alpha_4,\alpha'}(\mathrm{i}\omega_n) \delta_{\omega_{n_4},\omega_n} \\
&= +\frac{1}{2} \sum_{\substack{\alpha_1,\alpha_2,\alpha_3,\alpha_4 \in \mathscr{A} \\ \omega_{n_3} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_1\alpha_2\alpha_3\alpha_4} \delta_{\omega_{n_3}+\omega_n,\omega_{n_3}+\omega_n} \mathscr{G}^{\,0}_{\alpha,\alpha_2}(\mathrm{i}\omega_n) \mathscr{G}^{\,0}_{\alpha_3\alpha_1}(\mathrm{i}\omega_{n_3}) \mathscr{G}^{\,0}_{\alpha_4\alpha'}(\mathrm{i}\omega_n)
\end{aligned}
$$

From this we can read off the self-energy

$$
\Sigma^{\pi_6}_{\alpha_2\alpha_4}(\mathrm{i}\omega_n) = +\frac{1}{2} \sum_{\substack{\alpha_3 \in \mathscr{A} \\ \omega_{n_3} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_3\alpha_2\alpha_3\alpha_4} \mathscr{G}^{\,0}_{\alpha_3\alpha_3}(\mathrm{i}\omega_{n_3}) = +\frac{1}{2} \sum_{\alpha_3 \in \mathscr{A}} V_{\alpha_3\alpha_2\alpha_3\alpha_4} \mathrm{F}_{\hbar\beta}(\epsilon_{\alpha_3} - \mu) = -\zeta \frac{1}{2} \sum_{\alpha_3 \in \mathscr{A}} V_{\alpha_3\alpha_2\alpha_3\alpha_4} \mathrm{n}^{\zeta}_{\hbar\beta}(\epsilon_{\alpha_3} - \mu) \tag{32.18}
$$

π_4:

This gives then for the first order correction of the Green-function

$$
\begin{aligned}
\mathscr{G}^{\pi_4}_{\alpha\alpha'}(\mathrm{i}\omega_n) &= -\frac{1}{2} \sum_{\substack{\alpha_1,\alpha_2,\alpha_3,\alpha_4 \in \mathscr{A} \\ \omega_{n_1},\omega_{n_2},\omega_{n_3},\omega_{n_4} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_1\alpha_2\alpha_3\alpha_4} \delta_{\omega_{n_1}+\omega_{n_2},\omega_{n_3}+\omega_{n_4}} \mathscr{G}^{\,0}_{\alpha,\alpha_1}(\mathrm{i}\omega_n) \delta_{\omega_n,\omega_{n_1}} \mathscr{G}^{\,0}_{\alpha_3\alpha_2}(\mathrm{i}\omega_{n_3}) \delta_{\omega_{n_3},\omega_{n_2}} \mathscr{G}^{\,0}_{\alpha_4\alpha'}(\mathrm{i}\omega_n) \delta_{\omega_{n_4},\omega_n} \\
&= -\frac{1}{2} \sum_{\substack{\alpha_1,\alpha_2,\alpha_3,\alpha_4 \in \mathscr{A} \\ \omega_{n_3} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_1\alpha_2\alpha_3\alpha_4} \delta_{\omega_n+\omega_{n_3},\omega_{n_3}+\omega_n} \mathscr{G}^{\,0}_{\alpha,\alpha_1}(\mathrm{i}\omega_n) \mathscr{G}^{\,0}_{\alpha_3,\alpha_2}(\mathrm{i}\omega_{n_3}) \mathscr{G}^{\,0}_{\alpha_4,\alpha'}(\mathrm{i}\omega_n)
\end{aligned}
$$

From this we can read off the self-energy

$$
\Sigma^{\pi_4}_{\alpha_1\alpha_4}(\mathrm{i}\omega_n) = -\frac{1}{2} \sum_{\substack{\alpha_3 \in \mathscr{A} \\ \omega_{n_3} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_1\alpha_3\alpha_3\alpha_4} \mathscr{G}^{\,0}_{\alpha_3\alpha_3}(\mathrm{i}\omega_{n_3}) = -\frac{1}{2} \sum_{\alpha_3 \in \mathscr{A}} V_{\alpha_1\alpha_3\alpha_3\alpha_4} \mathrm{F}_{\hbar\beta}(\epsilon_{\alpha_3} - \mu) = +\zeta \frac{1}{2} \sum_{\alpha_3 \in \mathscr{A}} V_{\alpha_1\alpha_3\alpha_3\alpha_4} \mathrm{n}^{\zeta}_{\hbar\beta}(\epsilon_{\alpha_3} - \mu) \tag{32.19}
$$

π_5:

This gives then for the first order correction of the Green-function

$$
\begin{aligned}
\mathscr{G}^{\pi_5}_{\alpha\alpha'}(\mathrm{i}\omega_n) &= -\frac{1}{2} \sum_{\substack{\alpha_1,\alpha_2,\alpha_3,\alpha_4 \in \mathscr{A} \\ \omega_{n_1},\omega_{n_2},\omega_{n_3},\omega_{n_4} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_1\alpha_2\alpha_3\alpha_4} \delta_{\omega_{n_1}+\omega_{n_2},\omega_{n_3}+\omega_{n_4}} \mathscr{G}^{\,0}_{\alpha,\alpha_2}(\mathrm{i}\omega_n) \delta_{\omega_n,\omega_{n_2}} \mathscr{G}^{\,0}_{\alpha_3\alpha'}(\mathrm{i}\omega_n) \delta_{\omega_{n_3},\omega_n} \mathscr{G}^{\,0}_{\alpha_4\alpha_1}(\mathrm{i}\omega_{n_4}) \delta_{\omega_{n_4},\omega_{n_1}} \\
&= -\frac{1}{2} \sum_{\substack{\alpha_1,\alpha_2,\alpha_3,\alpha_4 \in \mathscr{A} \\ \omega_{n_4} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_1\alpha_2\alpha_3\alpha_4} \delta_{\omega_{n_4}+\omega_n,\omega_n+\omega_{n_4}} \mathscr{G}^{\,0}_{\alpha,\alpha_2}(\mathrm{i}\omega_n) \mathscr{G}^{\,0}_{\alpha_4\alpha_1}(\mathrm{i}\omega_{n_4}) \mathscr{G}^{\,0}_{\alpha_3\alpha'}(\mathrm{i}\omega_n)
\end{aligned}
$$

From this we can read off the self-energy

$$\Sigma^{\pi_5}_{\alpha_2\alpha_3}(\mathrm{i}\omega_n) = -\frac{1}{2}\sum_{\substack{\alpha_4\in\mathscr{A}\\ \omega_{n_4}\in\Omega_\zeta}}\frac{1}{\beta}V_{\alpha_4\alpha_3\alpha_3\alpha_4}\,\mathscr{G}^{\;0}_{\alpha_4\alpha_4}(\mathrm{i}\omega_{n_4}) = -\frac{1}{2}\sum_{\alpha_3\in\mathscr{A}}V_{\alpha_4\alpha_3\alpha_3\alpha_4}\,\mathrm{F}_{\hbar\beta}(\epsilon_{\alpha_4}-\mu) = +\zeta\frac{1}{2}\sum_{\alpha_3\in\mathscr{A}}V_{\alpha_4\alpha_3\alpha_3\alpha_4}\,\mathrm{n}^{\zeta}_{\hbar\beta}(\epsilon_{\alpha_4}-\mu) \tag{32.20}$$

32.3. Self Energy of a Spin-Independent-Interaction between an Electron-Gas and a Phonon-Gas

Proof 32.3 of Theorem 20.3

Notice first, that going over to the field-integral-description of the problem gives the action

$$\begin{aligned} S[\chi,\overline{\chi}] = \quad & S_{\text{el}}[\psi,\overline{\psi}] \quad + \quad S_{\text{ph}}[\phi,\overline{\phi}] \quad + \quad S_{\text{el-ph-int}}[\chi,\overline{\chi}] \\ = & \sum\!\!\!\!\!\!\int \overline{\psi}_{m\mathbf{k}\sigma}(\tau)\left(\partial_\tau + \frac{\epsilon_{m\mathbf{k}\sigma}-\mu}{\hbar}\right)\psi_{m\mathbf{k}\sigma}(\tau) + \sum\!\!\!\!\!\!\int \overline{\phi}_{\mathbf{q}\nu}(\tau)\left(\partial_\tau + \omega_{\mathbf{q},\nu}\right)\phi_{\mathbf{q}\nu}(\tau) + S_{\text{el-ph-int}}[\chi,\overline{\chi}] \end{aligned}$$

with interaction-term

$$S_{\text{el-ph-int}}[\chi,\overline{\chi}] = \sum\!\!\!\!\!\!\int \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})}{\hbar\sqrt{N_p}}\left(\phi_{\mathbf{q}\nu}(\tau)+\overline{\phi}_{-\mathbf{q}\nu}(\tau)\right)\overline{\psi}_{\mathbf{k}+\mathbf{q}m\sigma}(\tau)\psi_{\mathbf{k}n\sigma}(\tau)$$

We will now determine the self-energy of the electronic system.

We will use the following trick: Inverting the Dyson-equation with the von Neumann series on the one hand, and re-expressing the imaginary-time-Green-function with help of the field integral on the other hand, gives the two equations

$$\mathscr{G}_{\alpha,\alpha'}(\mathrm{i}\omega_n) = \sum_{n=0}^{\infty}\left[\left(\mathscr{G}_0\cdot\Sigma\right)^n(\mathrm{i}\omega_n)\cdot\mathscr{G}_0(\mathrm{i}\omega_n)\right]_{\alpha,\alpha'} \tag{32.21}$$

$$\mathscr{G}_{\alpha,\alpha'}(\mathrm{i}\omega_n) = \int_0^{\hbar\beta}\mathrm{d}\tau\,\mathrm{e}^{\mathrm{i}\omega_n\tau}\,\mathscr{G}_{\alpha,\alpha'}(\tau) = \frac{-1}{\hbar}\int_0^{\hbar\beta}\mathrm{d}\tau\,\mathrm{e}^{\mathrm{i}\omega_n\tau}\,\langle\psi_\alpha(\tau)\overline{\psi}_{\alpha'}(0)\rangle \tag{32.22}$$

The self-energy of the electronic Green-function is known to be given as the sum of all the connected irreducible diagrams of the interaction

$$\Sigma(\mathrm{i}\omega_n) = \sum_{j=0}^{\infty}\Sigma^{(j)}(\mathrm{i}\omega_n) \approx 0 + \Sigma^{(2)}(\mathrm{i}\omega_n) + 0 + \mathcal{O}(\,\mathrm{V}^4\,), \tag{32.23}$$

whereas the first contribution will turn out to vanish since the interaction can be expanded as a sum of monomials of creation- and annihilation-operators with uneven degree. Inserting this in the upper-equation above and keeping the first non-vanishing terms yields the sets of equations

$$\mathscr{G}_{\alpha,\alpha'}(\mathrm{i}\omega_n) = \mathscr{G}_{\underset{\alpha,\alpha'}{0}}(\mathrm{i}\omega_n) + \left(\mathscr{G}_0\cdot\Sigma^{(2)}\cdot\mathscr{G}_0\right)_{\alpha,\alpha'}(\mathrm{i}\omega_n) \tag{32.24}$$

$$\mathscr{G}_{\alpha,\alpha'}(\mathrm{i}\omega_n) = \frac{-1}{\hbar}\int_0^{\hbar\beta}\mathrm{d}\tau\,\mathrm{e}^{\mathrm{i}\omega_n\tau}\,\langle\psi_\alpha(\tau)\overline{\psi}_{\alpha'}(0)\rangle \tag{32.25}$$

The lower equation can be calculated explicitly with help of the Wick-theorem, linked-cluster-theorem, and the residue-theorem, and then, by comparing the first non-vanishing coefficients with the upper-equation, we can conclude what $\Sigma^{(2)}(\mathrm{i}\omega_n)$ must be.

Fact 1: The action, corresponding to this system, is given by the equation above.

Fact 2: The electronic Green-function is written in second order perturbation-theory as

$$\begin{aligned}
\mathscr{G}_{\substack{b,b'\\ \mathbf{p},\mathbf{p}'\\ \sigma,\sigma'}}(\tau) &= \mathscr{G}^{(0)}_{\substack{b,b'\\ \mathbf{p},\mathbf{p}'\\ \sigma,\sigma'}}(\tau) + \mathscr{G}^{(1)}_{\substack{b,b'\\ \mathbf{p},\mathbf{p}'\\ \sigma,\sigma'}}(\tau) + \mathscr{G}^{(2)}_{\substack{b,b'\\ \mathbf{p},\mathbf{p}'\\ \sigma,\sigma'}}(\tau) + \dots \\
&= \mathscr{G}_{\substack{0\\ b,b'\\ \mathbf{p},\mathbf{p}'\\ \sigma,\sigma'}}(\tau) + \quad 0 \quad + \mathscr{G}^{(2)}_{\substack{b,b'\\ \mathbf{p},\mathbf{p}'\\ \sigma,\sigma'}}(\tau) + \dots
\end{aligned}$$

The second contribution reads

$$\begin{aligned}
&\mathscr{G}^{(2)}_{\substack{b,b'\\ \mathbf{p},\mathbf{p}'\\ \sigma,\sigma'}}(\tau) \\
&= \frac{1}{2!} \sum\!\!\!\!\!\!\int_{\substack{\tau_1,\tau_2 \in [0,\hbar\beta]\\ \mu,\mu' \in \{\downarrow,\uparrow\}\\ \mathbf{k},\mathbf{k}',\mathbf{q},\mathbf{q}' \in \mathrm{Q}\\ m,m',n,n' \in \{1,2,\dots\}\\ \nu,\nu' \in \{1,2,\dots,rd\}}} \frac{-1}{\hbar} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q}) g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{\hbar^2 N_p} \langle \psi_{\mathbf{p}b\sigma}(\tau)\overline{\psi}_{\mathbf{p}'b'\sigma'}(0) \left(\phi_{\mathbf{q}\nu}(\tau_1) + \overline{\phi}_{-\mathbf{q}\nu}(\tau_1)\right) \overline{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1)\psi_{\mathbf{k}n\mu}(\tau_1) \left(\phi_{\mathbf{q}'\nu'}(\tau_2) + \overline{\phi}_{-\mathbf{q}'\nu'}(\tau_2)\right) \overline{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2)\psi_{\mathbf{k}'n'\mu'}(\tau_2)\rangle_0^{\mathrm{CD}}
\end{aligned} \tag{32.26}$$

where only [C]onnected [D]iagrams are summed over (linked cluster theorem). We can therefore simply sum over all diagrams and discard the disconnected diagrams.

Fact 3: The correlator in it's decomposition into it's fermionic and bosonic component reads

$$\begin{aligned}
&\frac{-1}{\hbar} \langle \psi_{\mathbf{p}b\sigma}(\tau)\overline{\psi}_{\mathbf{p}'b'\sigma'}(0) \left(\phi_{\mathbf{q}\nu}(\tau_1) + \overline{\phi}_{-\mathbf{q}\nu}(\tau_1)\right) \overline{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1)\psi_{\mathbf{k}n\mu}(\tau_1) \left(\phi_{\mathbf{q}'\nu'}(\tau_2) + \overline{\phi}_{-\mathbf{q}'\nu'}(\tau_2)\right) \overline{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2)\psi_{\mathbf{k}'n'\mu'}(\tau_2)\rangle_0^{\mathrm{CD}} \\
&= \frac{-1}{\hbar} \langle \psi_{\mathbf{p}b\sigma}(\tau)\overline{\psi}_{\mathbf{p}'b'\sigma'}(0)\overline{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1)\psi_{\mathbf{k}n\mu}(\tau_1)\overline{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2)\psi_{\mathbf{k}'n'\mu'}(\tau_2)\rangle_0^{\mathrm{CD}} \langle \left(\phi_{\mathbf{q}\nu}(\tau_1) + \overline{\phi}_{-\mathbf{q}\nu}(\tau_1)\right)\left(\phi_{\mathbf{q}'\nu'}(\tau_2) + \overline{\phi}_{-\mathbf{q}'\nu'}(\tau_2)\right)\rangle_0^{\mathrm{CD}} \\
&= -\langle \psi_{\mathbf{p}b\sigma}(\tau)\psi_{\mathbf{k}n\mu}(\tau_1)\psi_{\mathbf{k}'n'\mu'}(\tau_2)\overline{\psi}_{\mathbf{p}'b'\sigma'}(0)\overline{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1)\overline{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2)\rangle_0^{\mathrm{CD}} \cdot \mathscr{D}_{\substack{0\\ \mathbf{q},-\mathbf{q}'\\ \nu,\nu'}}(\tau_1 - \tau_2)
\end{aligned} \tag{32.27}$$

The electronic correlator can now be evaluated with the Wick-theorem. In order to so, we will enumerate all the permutations of the group S_3.

$$\begin{aligned}
\mathrm{S}_3 &= \{\pi_1, \pi_2, \pi_3, \pi_4, \pi_5, \pi_6\} \\
&= \left\{ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \right\}
\end{aligned}$$

with signatures

$$\mathrm{sgn}(\pi_1, \pi_2, \pi_3, \pi_4, \pi_5, \pi_6) = (+1, -1, -1, +1, +1, -1)$$

Now we write

$$
\begin{aligned}
&- \langle \psi_{\mathbf{p}b\sigma}(\tau)\psi_{\mathbf{k}n\mu}(\tau_1)\psi_{\mathbf{k}'n'\mu'}(\tau_2)\overline{\psi}_{\mathbf{p}'b'\sigma'}(0)\overline{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1)\overline{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2)\rangle_0^{\mathrm{CD}} \\
&\equiv -\langle \psi(A_1)\psi(A_2)\psi(A_3)\overline{\psi}(B_1)\overline{\psi}(B_2)\overline{\psi}(B_3)\rangle_0 \\
&\equiv +\langle \psi(A_1)\psi(A_2)\psi(A_3)\overline{\psi}(B_3)\overline{\psi}(B_2)\overline{\psi}(B_1)\rangle_0 \\
&= + \sum_{\pi \in S_3} \mathrm{sgn}(\pi) \prod_{l=1}^{3} \langle \psi(A_l)\,\overline{\psi}(B_{\pi(l)})\rangle_0 \\
&= -\hbar^3 \sum_{\pi \in S_3} \mathrm{sgn}(\pi) \prod_{l=1}^{3} \mathscr{G}_0(A_l, B_{\pi(l)})
\end{aligned}
$$

Facts:

- The contributions from π_1 and π_2 are vacuum diagrams (which cancel out due to the linked-cluster-theorem)
- The contributions from π_3 and π_6 are "Baloon"-diagrams / "Hartree"-diagrams.
- The contributions from π_4 and π_5 are "Sunset"-diagrams / "Fock"-diagrams.

Note that the two Hartree- and Fock-terms don't cancel out, but contribute equally to the series expansion (respectively) which cancels out the prefactor of $1/2!$ from the Taylor-expansion.

We note that

$$
\begin{aligned}
A_1 &= (\mathbf{p}, b, \sigma, \tau), & A_2 &= (\mathbf{k}, n, \mu, \tau_1), & A_3 &= (\mathbf{k}', n', \mu', \tau_2) \\
B_1 &= (\mathbf{p}', b', \sigma', 0), & B_2 &= (\mathbf{k}+\mathbf{q}, m, \mu, \tau_1), & B_3 &= (\mathbf{k}'+\mathbf{q}', m', \mu', \tau_2)
\end{aligned}
$$

and therefore we obtain

$$
\begin{aligned}
B_{\pi_3(1)} &\equiv B_2 \equiv (\mathbf{q}+\mathbf{k}, m, \mu, \tau_1), & B_{\pi_3(2)} &\equiv B_1 \equiv (\mathbf{p}', b', \sigma', 0), & B_{\pi_3(3)} &\equiv B_3 \equiv (\mathbf{k}'+\mathbf{q}', m', \mu', \tau_2), \\
B_{\pi_4(1)} &\equiv B_2 \equiv (\mathbf{k}+\mathbf{q}, m, \mu, \tau_1), & B_{\pi_4(2)} &\equiv B_3 \equiv (\mathbf{k}'+\mathbf{q}', m', \mu', \tau_2), & B_{\pi_4(3)} &\equiv B_1 \equiv (\mathbf{p}', b', \sigma', 0).
\end{aligned}
$$

This gives now for the second-order-contribution

$$
\begin{aligned}
\mathscr{G}^{(2)}_{\substack{b,b' \\ \mathbf{p},\mathbf{p}' \\ \sigma,\sigma'}}(\tau) &= -\hbar \sum\!\!\!\!\!\!\int_{\substack{\tau_1,\tau_2 \in [0,\hbar\beta] \\ \mu,\mu' \in \{\downarrow,\uparrow\} \\ \mathbf{k},\mathbf{k}',\mathbf{q},\mathbf{q}' \in Q \\ m,m',n,n' \in \{1,2,\ldots\} \\ \nu,\nu' \in \{1,2,\ldots,rd\}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{N_p} \mathscr{D}_{\substack{0 \\ \mathbf{q},-\mathbf{q}' \\ \nu,\nu'}}(\tau_1-\tau_2) \sum_{\pi \in \{\pi_3,\pi_4\}} \mathrm{sgn}(\pi) \prod_{l=1}^{3} \mathscr{G}_0(A_l, B_{\pi(l)}) \\
&= \hbar \sum\!\!\!\!\!\!\int_{\substack{\tau_1,\tau_2 \in [0,\hbar\beta] \\ \mu,\mu' \in \{\downarrow,\uparrow\} \\ \mathbf{k},\mathbf{k}',\mathbf{q},\mathbf{q}' \in Q \\ m,m',n,n' \in \{1,2,\ldots\} \\ \nu,\nu' \in \{1,2,\ldots,rd\}}} +\frac{g_{mn\nu}(\mathbf{k},\mathbf{q})g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{N_p} \mathscr{G}_{\substack{0 \\ \mathbf{p},\mathbf{k}+\mathbf{q} \\ b,m \\ \sigma,\mu}}(\tau-\tau_1)\, \mathscr{G}_{\substack{0 \\ \mathbf{k},\mathbf{p}' \\ n,b' \\ \mu,\sigma'}}(\tau_1-0)\, \mathscr{G}_{\substack{0 \\ \mathbf{k}',\mathbf{k}'+\mathbf{q}' \\ n',m' \\ \mu',\mu'}}(0)\, \mathscr{D}_{\substack{0 \\ \mathbf{q},-\mathbf{q}' \\ \nu,\nu'}}(\tau_1-\tau_2) \\
&+ \hbar \sum\!\!\!\!\!\!\int_{\substack{\tau_1,\tau_2 \in [0,\hbar\beta] \\ \mu,\mu' \in \{\downarrow,\uparrow\} \\ \mathbf{k},\mathbf{k}',\mathbf{q},\mathbf{q}' \in Q \\ m,m',n,n' \in \{1,2,\ldots\} \\ \nu,\nu' \in \{1,2,\ldots,rd\}}} -\frac{g_{mn\nu}(\mathbf{k},\mathbf{q})g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{N_p} \mathscr{G}_{\substack{0 \\ \mathbf{p},\mathbf{k}+\mathbf{q} \\ b,m \\ \sigma,\mu}}(\tau-\tau_1)\, \mathscr{G}_{\substack{0 \\ \mathbf{k},\mathbf{k}'+\mathbf{q}' \\ n,m' \\ \mu,\mu'}}(\tau_1-\tau_2)\, \mathscr{G}_{\substack{0 \\ \mathbf{k}',\mathbf{p}' \\ n',b' \\ \mu',\sigma'}}(\tau_2-0)\, \mathscr{D}_{\substack{0 \\ \mathbf{q},-\mathbf{q}' \\ \nu,\nu'}}(\tau_1-\tau_2)
\end{aligned}
$$

$$\equiv \mathscr{G}^{\text{Hartree}}_{\substack{b,b' \\ \mathbf{p},\mathbf{p}' \\ \sigma,\sigma'}}(\tau) \quad + \quad \mathscr{G}^{\text{Fock}}_{\substack{b,b' \\ \mathbf{p},\mathbf{p}' \\ \sigma,\sigma'}}(\tau)$$

In order to compute the self-energy, we compute the Fourier-Matsubara-transform of these functions. We therefore determine $\mathscr{G}(\mathrm{i}\omega_n)$.

Fourier-Matsubara-Component of the Hartree-Contribution

Lemma 32.1 It holds

$$\mathscr{G}^{\text{Hartree}}_{\sigma,\sigma'}(\mathrm{i}\omega_n) = \sum_{\substack{\mu,\mu' \in \{\downarrow,\uparrow\} \\ \mathbf{k},\mathbf{k}',\mathbf{q},\mathbf{q}' \in \mathrm{Q} \\ m,m',n,n' \in \{1,2,\dots\} \\ \nu,\nu' \in \{1,2,\dots,rd\}}} \mathscr{G}_{\substack{0 \\ \mathbf{p},\mathbf{k}+\mathbf{q} \\ b,m \\ \sigma,\mu}}(\mathrm{i}\omega_n) \left(\frac{1}{\beta} \sum_{\omega_m \in \Omega_{-1}} + \frac{g_{mn\nu}(\mathbf{k},\mathbf{q}) g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{N_p} \mathscr{G}_{\substack{0 \\ \mathbf{k}',\mathbf{k}'+\mathbf{q}' \\ n',m' \\ \mu',\mu'}}(\mathrm{i}\omega_m) \mathscr{D}_{\substack{0 \\ \mathbf{q},-\mathbf{q}' \\ \nu,\nu'}}(0) \right) \mathscr{G}_{\substack{0 \\ \mathbf{k},\mathbf{p}' \\ n,b' \\ \mu,\sigma'}}(\mathrm{i}\omega_n) \tag{32.28}$$

Proof 32.4

We will now focus on the temporal components of the Hartree-contribution to the Green-function. For this we consider the function

$$\widetilde{\mathscr{G}}^{\text{Hartree}}(\tau) = \iint_{[0,\hbar\beta]^2} \mathrm{d}(\tau_1,\tau_2)\, \underset{\mathrm{I}}{\mathscr{G}_0}(\tau-\tau_1) \underset{\mathrm{II}}{\mathscr{G}_0}(\tau_1 - 0) \underset{\mathrm{III}}{\mathscr{G}_0}(0) \underset{\mathrm{IV}}{\mathscr{D}_0}(\tau_1-\tau_2) \tag{32.29}$$

and this gives for the Matsubara-components

$$\begin{aligned}
\widetilde{\mathscr{G}}^{\text{Hartree}}(\mathrm{i}\omega_n) &= \int_0^{\hbar\beta} \mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}\omega_n\tau} \widetilde{\mathscr{G}}^{\text{Hartree}}(\tau) \\
&= \iiint_{[0,\hbar\beta]^3} \mathrm{d}(\tau,\tau_1,\tau_2)\, \mathrm{e}^{\mathrm{i}\omega_n\tau} \underset{\mathrm{I}}{\mathscr{G}_0}(\tau-\tau_1) \underset{\mathrm{II}}{\mathscr{G}_0}(\tau_1-0) \underset{\mathrm{III}}{\mathscr{G}_0}(0) \underset{\mathrm{IV}}{\mathscr{D}_0}(\tau_1-\tau_2) \\
&= \dots
\end{aligned}$$

now we expand the statistical functions according to Lemma TBReferenced and we obtain

$$\begin{aligned}
\dots &= \frac{1}{(\hbar\beta)^4} \sum_{\substack{\omega_{n_4} \in \Omega_{+1} \\ \omega_{n_1},\omega_{n_2},\omega_{n_3} \in \Omega_{-1}}} \iiint_{[0,\hbar\beta]^3} \mathrm{d}(\tau,\tau_1,\tau_2)\, \mathscr{G}_{\substack{0 \\ \alpha,\xi}}(\mathrm{i}\omega_{n_1}) \mathscr{G}_{\substack{0 \\ \xi',\alpha'}}(\mathrm{i}\omega_{n_2}) \mathscr{G}_{\substack{0 \\ \zeta',\zeta}}(\mathrm{i}\omega_{n_3}) \mathscr{D}_0(\mathrm{i}\omega_{n_4})\, \mathrm{e}^{\mathrm{i}\left(\omega_n\tau - \omega_{n_1}(\tau-\tau_1) - \omega_{n_2}\tau_1 - \omega_{n_4}(\tau_1-\tau_2)\right)} \\
&= \sum_{\substack{\omega_{n_4} \in \Omega_{+1} \\ \omega_{n_1,n_2,n_3} \in \Omega_{-1}}} \underset{\mathrm{I}}{\mathscr{G}_0}(\mathrm{i}\omega_{n_1}) \underset{\mathrm{II}}{\mathscr{G}_0}(\mathrm{i}\omega_{n_2}) \underset{\mathrm{III}}{\mathscr{G}_0}(\mathrm{i}\omega_{n_3}) \underset{\mathrm{IV}}{\mathscr{D}_0}(\mathrm{i}\omega_{n_4}) \frac{1}{(\hbar\beta)^4} \iiint_{[0,\hbar\beta]^3} \mathrm{d}(\tau,\tau_1,\tau_2) \mathrm{e}^{\mathrm{i}\left(\omega_n\tau - \omega_{n_1}(\tau-\tau_1) - \omega_{n_2}\tau_1 - \omega_{n_4}(\tau_1-\tau_2)\right)} \\
&= \dots
\end{aligned}$$

and we find for the integral-factor

$$\iiint\limits_{[0,\hbar\beta]^3} \mathrm{d}(\tau,\tau_1,\tau_2)\mathrm{e}^{\mathrm{i}\left(\omega_n\tau-\omega_{n_1}(\tau-\tau_1)-\omega_{n_2}\tau_1-\omega_{n_4}(\tau_1-\tau_2)\right)} = \int_0^{\hbar\beta}\mathrm{d}\tau\,\mathrm{e}^{\mathrm{i}\tau\left(\omega_n-\omega_{n_1}\right)}\int_0^{\hbar\beta}\mathrm{d}\tau_1\,\mathrm{e}^{\mathrm{i}\tau_1\left(\omega_{n_1}-\omega_{n_2}-\omega_{n_4}\right)}\int_0^{\hbar\beta}\mathrm{d}\tau_2\,\mathrm{e}^{\mathrm{i}\tau_2\omega_{n_4}}$$
$$= (\hbar\beta)^3\delta_{\omega_n,\omega_{n_1}}\delta_{\omega_{n_1},\omega_{n_2}}\delta_{\omega_{n_4},0}$$

We use the first, second and third delta-function in order to eliminate the summations over ω_{n_1}, ω_{n_2} and ω_{n_4} respectively. This yields then

$$\widetilde{\mathscr{G}}^{\text{Hartree}}(\mathrm{i}\omega_n) = \ldots = \sum_{\omega_{n_3}\in\Omega_{-1}}\frac{1}{\hbar\beta}\underset{\mathrm{I}}{\mathscr{G}_0}(\mathrm{i}\omega_n)\underset{\mathrm{II}}{\mathscr{G}_0}(\mathrm{i}\omega_n)\underset{\mathrm{III}}{\mathscr{G}_0}(\mathrm{i}\omega_{n_3})\underset{\mathrm{IV}}{\mathscr{D}_0}(0)$$
$$\equiv \underset{\mathrm{I}}{\mathscr{G}_0}(\mathrm{i}\omega_n)\left(\frac{1}{\hbar\beta}\sum_{\omega_{n_3}\in\Omega_{-1}}\underset{\mathrm{III}}{\mathscr{G}_0}(\mathrm{i}\omega_{n_3})\underset{\mathrm{IV}}{\mathscr{D}_0}(0)\right)\underset{\mathrm{II}}{\mathscr{G}_0}(\mathrm{i}\omega_n)$$

or in short:

$$\widetilde{\mathscr{G}}^{\text{Hartree}}(\tau) = \iint\limits_{[0,\hbar\beta]^2}\mathrm{d}(\tau_1,\tau_2)\underset{\mathrm{I}}{\mathscr{G}_0}(\tau-\tau_1)\underset{\mathrm{II}}{\mathscr{G}_0}(\tau_1-0)\underset{\mathrm{III}}{\mathscr{G}_0}(0)\underset{\mathrm{IV}}{\mathscr{D}_0}(\tau_1-\tau_2)$$
$$\Rightarrow \quad \widetilde{\mathscr{G}}^{\text{Hartree}}(\mathrm{i}\omega_n) = \underset{\mathrm{I}}{\mathscr{G}_0}(\mathrm{i}\omega_n)\left(\frac{1}{\hbar\beta}\sum_{\omega_{n_3}\in\Omega_{-1}}\underset{\mathrm{III}}{\mathscr{G}_0}(\mathrm{i}\omega_{n_3})\underset{\mathrm{IV}}{\mathscr{D}_0}(0)\right)\underset{\mathrm{II}}{\mathscr{G}_0}(\mathrm{i}\omega_n) \tag{32.30}$$

Using this identity for the Matsubara-component of a function proves the Lemma: We can compute the Fourier-Matsubara-component as

$$\mathscr{G}^{\text{Hartree}}_{\sigma,\sigma'}(\mathrm{i}\omega_n) = \int_0^{\hbar\beta}\mathrm{d}\tau\,\mathscr{G}^{\text{Hartree}}_{\sigma,\sigma'}(\tau)\,\mathrm{e}^{\mathrm{i}\omega_n\tau}$$
$$= \hbar \sum\!\!\!\!\!\!\!\int_{\substack{\tau,\tau_1,\tau_2\in[0,\hbar\beta]\\ \mu,\mu'\in\{\downarrow,\uparrow\}\\ \mathbf{k},\mathbf{k}',\mathbf{q},\mathbf{q}'\in Q\\ m,m',n,n'\in\{1,2,\ldots\}\\ \nu,\nu'\in\{1,2,\ldots,rd\}}} + \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{N_p}\mathscr{G}_{\substack{0\\ \mathbf{p},\mathbf{k}+\mathbf{q}\\ b,m\\ \sigma,\mu}}(\tau-\tau_1)\,\mathscr{G}_{\substack{0\\ \mathbf{k},\mathbf{p}'\\ n,b'\\ \mu,\sigma'}}(\tau_1-0)\,\mathscr{G}_{\substack{0\\ \mathbf{k}',\mathbf{k}'+\mathbf{q}'\\ n',m'\\ \mu',\mu'}}(0)\,\mathscr{D}_{\substack{0\\ \mathbf{q},-\mathbf{q}'\\ \nu,\nu'}}(\tau_1-\tau_2)\,\mathrm{e}^{\mathrm{i}\omega_n\tau}$$
$$= \sum_{\substack{\mu,\mu'\in\{\downarrow,\uparrow\}\\ \mathbf{k},\mathbf{k}',\mathbf{q},\mathbf{q}'\in Q\\ m,m',n,n'\in\{1,2,\ldots\}\\ \nu,\nu'\in\{1,2,\ldots,rd\}}}\mathscr{G}_{\substack{0\\ \mathbf{p},\mathbf{k}+\mathbf{q}\\ b,m\\ \sigma,\mu}}(\mathrm{i}\omega_n)\left(\frac{1}{\beta}\sum_{\omega_m\in\Omega_{-1}} + \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{N_p}\mathscr{G}_{\substack{0\\ \mathbf{k}',\mathbf{k}'+\mathbf{q}'\\ n',m'\\ \mu',\mu'}}(\mathrm{i}\omega_m)\mathscr{D}_{\substack{0\\ \mathbf{q},-\mathbf{q}'\\ \nu,\nu'}}(0)\right)\mathscr{G}_{\substack{0\\ \mathbf{k},\mathbf{p}'\\ n,b'\\ \mu,\sigma'}}(\mathrm{i}\omega_n)$$

Fourier-Matsubara-Component of the Fock-Contribution

Lemma 32.2 It holds

$$\mathscr{G}^{\text{Fock}}_{\sigma,\sigma'}(\mathrm{i}\omega_n) = \sum_{\substack{\mu,\mu' \in \{\downarrow,\uparrow\} \\ \mathbf{k},\mathbf{k}',\mathbf{q},\mathbf{q}' \in \mathrm{Q} \\ m,m',n,n' \in \{1,2,\ldots\} \\ \nu,\nu' \in \{1,2,\ldots,rd\}}} \mathscr{G}_{\substack{0 \\ \mathbf{p},\mathbf{k}+\mathbf{q} \\ b,m \\ \sigma,\mu}}(\mathrm{i}\omega_n) \left(\frac{1}{\beta} \sum_{\omega_{n_2} \in \Omega_{-1}} -\frac{g_{mn\nu}(\mathbf{k},\mathbf{q}) g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{N_p} \mathscr{G}_{\substack{0 \\ \mathbf{k},\mathbf{k}'+\mathbf{q}' \\ n,m' \\ \mu,\mu'}}(\mathrm{i}\omega_{n_2}) \mathscr{D}_{\substack{0 \\ \mathbf{q},-\mathbf{q}' \\ \nu,\nu'}}(\mathrm{i}(\omega_n - \omega_{n_2})) \right) \mathscr{G}_{\substack{0 \\ \mathbf{k}',\mathbf{p}' \\ n',b' \\ \mu',\sigma'}}(\mathrm{i}\omega_n) \tag{32.31}$$

Proof 32.5

We will now focus on the temporal components of the Fock-contribution to the Green-function. For this we consider the function

$$\widetilde{\mathscr{G}}^{\text{Fock}}(\tau) = \iint_{[0,\hbar\beta]^2} \mathrm{d}(\tau_1,\tau_2)\, \underset{\mathrm{I}}{\mathscr{G}_0}(\tau-\tau_1) \underset{\mathrm{II}}{\mathscr{G}_0}(\tau_1-\tau_2) \underset{\mathrm{III}}{\mathscr{G}_0}(\tau_2) \underset{\mathrm{IV}}{\mathscr{D}_0}(\tau_1-\tau_2) \tag{32.32}$$

and this gives for the Matsubara-components

$$\begin{aligned} \widetilde{\mathscr{G}}^{\text{Fock}}(\mathrm{i}\omega_n) &= \int_0^{\hbar\beta} \mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}\omega_n\tau} \widetilde{\mathscr{G}}^{\text{Fock}}(\tau) \\ &= \iiint_{[0,\hbar\beta]^3} \mathrm{d}(\tau,\tau_1,\tau_2)\, \mathrm{e}^{\mathrm{i}\omega_n\tau} \underset{\mathrm{I}}{\mathscr{G}_0}(\tau-\tau_1) \underset{\mathrm{II}}{\mathscr{G}_0}(\tau_1-\tau_2) \underset{\mathrm{III}}{\mathscr{G}_0}(\tau_2) \underset{\mathrm{IV}}{\mathscr{D}_0}(\tau_1-\tau_2) \\ &= \ldots \end{aligned}$$

now we expand the statistical functions according to Lemma TBReferenced and we obtain

$$\begin{aligned} \ldots &= \sum_{\substack{\omega_{n_1,n_2,n_3} \in \Omega_{-1} \\ \omega_{n_4} \in \Omega_{+1}}} \underset{\mathrm{I}}{\mathscr{G}_0}(\omega_{n_1}) \underset{\mathrm{II}}{\mathscr{G}_0}(\omega_{n_2}) \underset{\mathrm{III}}{\mathscr{G}_0}(\omega_{n_3}) \underset{\mathrm{IV}}{\mathscr{D}_0}(\omega_{n_4}) \frac{1}{(\hbar\beta)^4} \iiint_{[0,\hbar\beta]^3} \mathrm{d}(\tau,\tau_1,\tau_2) \mathrm{e}^{\mathrm{i}\left(\omega_n\tau - \omega_{n_1}(\tau-\tau_1) - \omega_{n_2}(\tau_1-\tau_2) - \omega_{n_3}\tau_2 - \omega_{n_4}(\tau_1-\tau_2)\right)} \\ &= \ldots \end{aligned}$$

and we find for the integral-factor

$$\begin{aligned} &\iiint_{[0,\hbar\beta]^3} \mathrm{d}(\tau,\tau_1,\tau_2) \mathrm{e}^{\mathrm{i}\left(\omega_n\tau - \omega_{n_1}(\tau-\tau_1) - \omega_{n_2}(\tau_1-\tau_2) - \omega_{n_3}\tau_2 - \omega_{n_4}(\tau_1-\tau_2)\right)} \\ &= \int_0^{\hbar\beta} \mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}\tau\left(\omega_n - \omega_{n_1}\right)} \int_0^{\hbar\beta} \mathrm{d}\tau_1\, \mathrm{e}^{\mathrm{i}\tau_1\left(\omega_{n_1} - \omega_{n_2} - \omega_{n_4}\right)} \int_0^{\hbar\beta} \mathrm{d}\tau_2\, \mathrm{e}^{\mathrm{i}\tau_2\left(\omega_{n_2} - \omega_{n_3} + \omega_{n_4}\right)} \\ &= (\hbar\beta)^3 \delta_{\omega_n,\omega_{n_1}} \delta_{\omega_{n_1},\omega_{n_2}+\omega_{n_4}} \delta_{\omega_{n_2}+\omega_{n_4},\omega_{n_3}} \end{aligned}$$

We use the first delta-function to eliminate the summation over ω_{n_1}. The second delta-function can be simplified according to

$$
\begin{aligned}
\delta_{\omega_{n_1},\omega_{n_2}+\omega_{n_4}} &= \delta_{\omega_{n_2},\omega_{n_1}-\omega_{n_4}} \\
&= \delta_{\omega_{n_2},\omega_{n_1}-\omega_{n_3}+\omega_{n_2}} \\
&= \delta_{\omega_{n_3},\omega_{n_1}}
\end{aligned}
$$

and will eliminate the summation over ω_{n_3}. For the third delta-relation we simplify it to

$$
\delta_{\omega_{n_2}+\omega_{n_4},\omega_{n_3}} = \delta_{\omega_{n_4},\omega_{n_3}-\omega_{n_2}}
$$

and we will use this to eliminate the summation over ω_{n_4}.

This yields then

$$
\begin{aligned}
\widetilde{\mathscr{G}}^{\text{Fock}}(\mathrm{i}\omega_n) = \ldots &= \sum_{\omega_{n_2}\in\Omega_{-1}} \frac{1}{\hbar\beta} \underset{\text{I}}{\mathscr{G}_0}(\omega_n) \underset{\text{II}}{\mathscr{G}_0}(\omega_{n_2}) \underset{\text{III}}{\mathscr{G}_0}(\omega_n) \underset{\text{IV}}{\mathscr{D}_0}(\mathrm{i}(\omega_n-\omega_{n_2})) \\
&\equiv \underset{\text{I}}{\mathscr{G}_0}(\omega_n) \left(\frac{1}{\hbar\beta} \sum_{\omega_{n_2}\in\Omega_{-1}} \underset{\text{II}}{\mathscr{G}_0}(\omega_{n_2}) \underset{\text{IV}}{\mathscr{D}_0}(\mathrm{i}(\omega_n-\omega_{n_2})) \right) \underset{\text{III}}{\mathscr{G}_0}(\omega_n)
\end{aligned}
$$

Using this identity for the Matsubara-component of a function proves the Lemma: We can compute the Fourier-Matsubara-component as

$$
\begin{aligned}
\mathscr{G}^{\text{Fock}}_{\sigma,\sigma'}(\mathrm{i}\omega_n) &= \int_0^{\hbar\beta} \mathrm{d}\tau\, \mathscr{G}^{\text{Fock}}_{\sigma,\sigma'}(\tau)\, \mathrm{e}^{\mathrm{i}\omega_n\tau} \\
&= \hbar \sum\!\!\!\!\!\!\int_{\substack{\tau,\tau_1,\tau_2\in[0,\hbar\beta] \\ \mu,\mu'\in\{\downarrow,\uparrow\} \\ \mathbf{k},\mathbf{k}',\mathbf{q},\mathbf{q}'\in Q \\ m,m',n,n'\in\{1,2,\ldots\} \\ \nu,\nu'\in\{1,2,\ldots,rd\}}} -\frac{g_{mn\nu}(\mathbf{k},\mathbf{q})g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{N_p} \mathscr{G}_{\substack{0 \\ \mathbf{p},\mathbf{k}+\mathbf{q} \\ b,m \\ \sigma,\mu}}(\tau-\tau_1) \mathscr{G}_{\substack{0 \\ \mathbf{k},\mathbf{k}'+\mathbf{q}' \\ n,m' \\ \mu,\mu'}}(\tau_1-\tau_2) \mathscr{G}_{\substack{0 \\ \mathbf{k}',\mathbf{p}' \\ n',b' \\ \mu',\sigma'}}(\tau_2-0)\, \mathscr{D}_{\substack{0 \\ \mathbf{q},-\mathbf{q}' \\ \nu,\nu'}}(\tau_1-\tau_2)\, \mathrm{e}^{\mathrm{i}\omega_n\tau} \\
&= \sum_{\substack{\mu,\mu'\in\{\downarrow,\uparrow\} \\ \mathbf{k},\mathbf{k}',\mathbf{q},\mathbf{q}'\in Q \\ m,m',n,n'\in\{1,2,\ldots\} \\ \nu,\nu'\in\{1,2,\ldots,rd\}}} \mathscr{G}_{\substack{0 \\ \mathbf{p},\mathbf{k}+\mathbf{q} \\ b,m \\ \sigma,\mu}}(\mathrm{i}\omega_n) \left(\frac{1}{\beta} \sum_{\omega_{n_2}\in\Omega_{-1}} -\frac{g_{mn\nu}(\mathbf{k},\mathbf{q})g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{N_p} \mathscr{G}_{\substack{0 \\ \mathbf{k},\mathbf{k}'+\mathbf{q}' \\ n,m' \\ \mu,\mu'}}(\mathrm{i}\omega_{n_2})\, \mathscr{D}_{\substack{0 \\ \mathbf{q},-\mathbf{q}' \\ \nu,\nu'}}(\mathrm{i}(\omega_n-\omega_{n_2})) \right) \mathscr{G}_{\substack{0 \\ \mathbf{k}',\mathbf{p}' \\ n',b' \\ \mu',\sigma'}}(\mathrm{i}\omega_n).
\end{aligned}
\tag{32.33}
$$

which proves the lemma.

Continuation of the Initial Proof

Continuing with the initial proof, we obtained in total

$$
\begin{aligned}
\mathscr{G}^{(2)}_{\substack{b,b'\\ \mathbf{p},\mathbf{p}'\\ \sigma,\sigma'}}(\mathrm{i}\omega_n) &= \mathscr{G}^{\mathrm{Hartree}}_{\sigma,\sigma'}(\mathrm{i}\omega_n) \quad + \quad \mathscr{G}^{\mathrm{Fock}}_{\sigma,\sigma'}(\mathrm{i}\omega_n) \\
&\equiv \sum_{\substack{\mu,\mu'\in\{\downarrow,\uparrow\}\\ \mathbf{k},\mathbf{k}',\mathbf{q},\mathbf{q}'\in Q\\ m,m',n,n'\in\{1,2,\dots\}\\ \nu,\nu'\in\{1,2,\dots,rd\}}} \mathscr{G}_{0\,\substack{\mathbf{p},\mathbf{k}+\mathbf{q}\\ b,m\\ \sigma,\mu}}(\mathrm{i}\omega_n)\left(\frac{1}{\beta}\sum_{\omega_m\in\Omega_{-1}} +\frac{g_{mn\nu}(\mathbf{k},\mathbf{q})g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{N_p}\mathscr{G}_{0\,\substack{\mathbf{k}',\mathbf{k}'+\mathbf{q}'\\ n',m'\\ \mu',\mu'}}(\mathrm{i}\omega_m)\mathscr{D}_{0\,\substack{\mathbf{q},-\mathbf{q}'\\ \nu,\nu'}}(0)\right)\mathscr{G}_{0\,\substack{\mathbf{k},\mathbf{p}'\\ n,b'\\ \mu,\sigma'}}(\mathrm{i}\omega_n) \\
&+ \sum_{\substack{\mu,\mu'\in\{\downarrow,\uparrow\}\\ \mathbf{k},\mathbf{k}',\mathbf{q},\mathbf{q}'\in Q\\ m,m',n,n'\in\{1,2,\dots\}\\ \nu,\nu'\in\{1,2,\dots,rd\}}} \mathscr{G}_{0\,\substack{\mathbf{p},\mathbf{k}+\mathbf{q}\\ b,m\\ \sigma,\mu}}(\mathrm{i}\omega_n)\left(\frac{1}{\beta}\sum_{\omega_{n_2}\in\Omega_{-1}} -\frac{g_{mn\nu}(\mathbf{k},\mathbf{q})g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{N_p}\mathscr{G}_{0\,\substack{\mathbf{k},\mathbf{k}'+\mathbf{q}'\\ n,m'\\ \mu,\mu'}}(\mathrm{i}\omega_{n_2})\mathscr{D}_{0\,\substack{\mathbf{q},-\mathbf{q}'\\ \nu,\nu'}}(\mathrm{i}(\omega_n-\omega_{n_2}))\right)\mathscr{G}_{0\,\substack{\mathbf{k}',\mathbf{p}'\\ n',b'\\ \mu',\sigma'}}(\mathrm{i}\omega_n) \\
&\overset{\mathbf{q}\mapsto\mathbf{q}-\mathbf{k}}{=} \sum_{\substack{\mathbf{k},\mathbf{q}\in Q\\ \mu\in\{\downarrow,\uparrow\}\\ m,n,\in\{1,2,\dots\}}} \mathscr{G}_{0\,\substack{\mathbf{p},\mathbf{q}\\ b,m\\ \sigma,\mu}}(\mathrm{i}\omega_n)\left(\frac{1}{\beta}\sum_{\substack{\mathbf{k}',\mathbf{q}'\in Q\\ \mu'\in\{\downarrow,\uparrow\}\\ m',n'\in\{1,2,\dots\}\\ \nu,\nu'\in\{1,2,\dots,rd\}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q}-\mathbf{k})g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{N_p}\mathscr{D}_{0\,\substack{\mathbf{q}-\mathbf{k},-\mathbf{q}'\\ \nu,\nu'}}(0)\sum_{\omega_m\in\Omega_{-1}}\mathscr{G}_{0\,\substack{\mathbf{k}',\mathbf{k}'+\mathbf{q}'\\ n',m'\\ \mu',\mu'}}(\mathrm{i}\omega_m)\right)\mathscr{G}_{0\,\substack{\mathbf{k},\mathbf{p}'\\ n,b'\\ \mu,\sigma'}}(\mathrm{i}\omega_n) \\
&+ \sum_{\substack{\mathbf{k}',\mathbf{q}\in Q\\ \mu,\mu'\in\{\downarrow,\uparrow\}\\ m,n'\in\{1,2,\dots\}}} \mathscr{G}_{0\,\substack{\mathbf{p},\mathbf{q}\\ b,m\\ \sigma,\mu}}(\mathrm{i}\omega_n)\left(\sum_{\substack{\mathbf{k},\mathbf{q}'\in Q\\ m',n\in\{1,2,\dots\}\\ \nu,\nu'\in\{1,2,\dots,rd\}}}\frac{1}{\beta}\sum_{\omega_{n_2}\in\Omega_{-1}} -\frac{g_{mn\nu}(\mathbf{k},\mathbf{q}-\mathbf{k})g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{N_p}\mathscr{G}_{0\,\substack{\mathbf{k},\mathbf{k}'+\mathbf{q}'\\ n,m'\\ \mu,\mu'}}(\mathrm{i}\omega_{n_2})\mathscr{D}_{0\,\substack{\mathbf{q}-\mathbf{k},-\mathbf{q}'\\ \nu,\nu'}}(\mathrm{i}(\omega_n-\omega_{n_2}))\right)\mathscr{G}_{0\,\substack{\mathbf{k}',\mathbf{p}'\\ n',b'\\ \mu',\sigma'}}(\mathrm{i}\omega_n) \\
&= \sum_{\substack{\mathbf{k},\mathbf{q}\in Q\\ \mu\in\{\downarrow,\uparrow\}\\ m,n,\in\{1,2,\dots\}}} \mathscr{G}_{0\,\substack{\mathbf{p},\mathbf{q}\\ b,m\\ \sigma,\mu}}(\mathrm{i}\omega_n)\;\Sigma^{\mathrm{Hartree}}_{\substack{\mathbf{q},\mathbf{k}\\ m,n\\ \mu,\mu}}(\mathrm{i}\omega_n)\;\mathscr{G}_{0\,\substack{\mathbf{k},\mathbf{p}'\\ n,b'\\ \mu,\sigma'}}(\mathrm{i}\omega_n) \quad + \sum_{\substack{\mathbf{k}',\mathbf{q}\in Q\\ \mu\in\{\downarrow,\uparrow\}\\ m,n,\in\{1,2,\dots\}}} \mathscr{G}_{0\,\substack{\mathbf{p},\mathbf{q}\\ b,m\\ \sigma,\mu}}(\mathrm{i}\omega_n)\;\Sigma^{\mathrm{Fock}}_{\substack{\mathbf{q},\mathbf{k}'\\ m,n'\\ \mu,\mu'}}(\mathrm{i}\omega_n)\;\mathscr{G}_{0\,\substack{\mathbf{k}',\mathbf{p}'\\ n',b'\\ \mu',\sigma'}}(\mathrm{i}\omega_n)
\end{aligned}
\tag{32.34}
$$

In the next step we will perform the Matsubara-summations and simplify the self-energy expressions.

<u>**The Hartree-Contribution**</u>

Now since $\mathscr{G}_0$ is diagonal (in bands and wave-vectors) we see that only the $\mathbf{q}'=0$ and $m'=n'$ summand contributes. Since $\mathscr{D}_0$ is diagonal in branches and wave-vectors only the $\nu=\nu'$ and $\mathbf{k}=\mathbf{q}$ summands contribute. This gives the simplification

$$
\Sigma^{\mathrm{Hartree}}_{\substack{\mathbf{q},\mathbf{k}\\ m,n\\ \mu,\mu'}}(\mathrm{i}\omega_n) = \delta_{\mu,\mu'}\frac{1}{\beta}\sum_{\substack{\mathbf{k}'\in Q\\ \mu'\in\{\downarrow,\uparrow\}\\ m'\in\{1,2,\dots\}\\ \nu\in\{1,2,\dots,rd\}}} +\frac{g_{mn\nu}(\mathbf{k},0)g_{m'm'\nu}(\mathbf{k}',0)}{N_p}\mathscr{D}_{0\,\substack{\mathbf{0},\mathbf{0}\\ \nu,\nu}}(0)\sum_{\omega_m\in\Omega_{-1}}\mathscr{G}_{0\,\substack{\mathbf{k}',\mathbf{k}'\\ m',m'\\ \mu',\mu'}}(\mathrm{i}\omega_m)\delta_{\mathbf{q},\mathbf{k}}
\tag{32.35}
$$

Therefore this contribution to the self-energy is diagonal in wave-vectors but not in the bands.

We now evaluate the summation over Matsubara-frequencies with the Residue-theorem. Omitting convergence-generating factors for notational simplicity, this gives with Corollary 15.16

$$
\frac{1}{\beta}\sum_{\omega_m\in\Omega_{-1}}\mathscr{G}_{0\,\substack{\mathbf{k}',\mathbf{k}'\\ m',m'\\ \mu',\mu'}}(\mathrm{i}\omega_m) = \frac{1}{\hbar\beta}\sum_{\omega_m\in\Omega_{-1}}\frac{1}{\mathrm{i}\omega_m-\left(\epsilon_{m'\mu'\mathbf{k}'}-\mu\right)/\hbar} = \mathrm{F}_\beta(\epsilon_{m'\mu'\mathbf{k}'}-\mu)
\tag{32.36}
$$

yielding as final expression

$$\Sigma^{\text{Hartree}}_{\substack{0\\ \mathbf{q},\mathbf{k}\\ m,n\\ \mu,\mu'}}(\mathrm{i}\omega_n) = \delta_{\mu,\mu'} \sum_{\substack{\mathbf{k}'\in \mathrm{Q}\\ \mu'\in\{\downarrow,\uparrow\}\\ m'\in\{1,2,\ldots\}\\ \nu\in\{1,2,\ldots,rd\}}} \frac{g_{mn\nu}(\mathbf{k},0)g_{m'm'\nu}(\mathbf{k}',0)}{N_p} \mathscr{D}_{\substack{0\\ \mathbf{0},\mathbf{0}\\ \nu,\nu}}(0)\,\mathrm{F}_\beta(\epsilon_{m'\mu'\mathbf{k}'}-\mu)\delta_{\mathbf{q},\mathbf{k}} \tag{32.37}$$

The Fock-Contribution

Using now again the diagonality of the unperturbed Green-functions we obtain for the Fock-contribution to the self-energy

$$\begin{aligned}
\Sigma^{\text{Fock}}_{\substack{\mathbf{q},\mathbf{k}'\\ m,n'\\ \mu,\mu'}}(\mathrm{i}\omega_n) &= \sum_{\substack{\mathbf{k},\mathbf{q}'\in\mathrm{Q}\\ m',n\in\{1,2,\ldots\}\\ \nu,\nu'\in\{1,2,\ldots,rd\}}} \frac{1}{\beta}\sum_{\omega_{n_2}\in\Omega_{-1}} -\frac{g_{mn\nu}(\mathbf{k},\mathbf{q}-\mathbf{k})g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{N_p} \mathscr{G}_{\substack{0\\ \mathbf{k},\mathbf{k}'+\mathbf{q}'\\ n,m'\\ \mu,\mu'}}(\mathrm{i}\omega_{n_2})\,\mathscr{D}_{\substack{0\\ \mathbf{q}-\mathbf{k},-\mathbf{q}'\\ \nu,\nu'}}(\mathrm{i}(\omega_n-\omega_{n_2}))\\
&= \sum_{\substack{\mathbf{q}'\in\mathrm{Q}\\ m',n\in\{1,2,\ldots\}\\ \nu\in\{1,2,\ldots,rd\}}} \frac{1}{\beta}\sum_{\omega_{n_2}\in\Omega_{-1}} -\frac{g_{mn\nu}(\mathbf{k}'+\mathbf{q}',\mathbf{q}-(\mathbf{k}'+\mathbf{q}'))g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{N_p} \mathscr{G}_{\substack{0\\ \mathbf{k}'+\mathbf{q}',\mathbf{k}'+\mathbf{q}'\\ n,m'\\ \mu,\mu'}}(\mathrm{i}\omega_{n_2})\,\mathscr{D}_{\substack{0\\ \mathbf{q}-(\mathbf{k}'+\mathbf{q}'),-\mathbf{q}'\\ \nu,\nu}}(\mathrm{i}(\omega_n-\omega_{n_2}))\\
&= \sum_{\substack{\mathbf{q}'\in\mathrm{Q}\\ m',n\in\{1,2,\ldots\}\\ \nu\in\{1,2,\ldots,rd\}}} \frac{1}{\beta}\sum_{\omega_{n_2}\in\Omega_{-1}} -\frac{g_{mn\nu}(\mathbf{k}'+\mathbf{q}',\mathbf{q}-(\mathbf{k}'+\mathbf{q}'))g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{N_p} \mathscr{G}_{\substack{0\\ \mathbf{k}'+\mathbf{q}',\mathbf{k}'+\mathbf{q}'\\ n,m'\\ \mu,\mu'}}(\mathrm{i}\omega_{n_2})\,\mathscr{D}_{\substack{0\\ -\mathbf{q}',-\mathbf{q}'\\ \nu,\nu}}(\mathrm{i}(\omega_n-\omega_{n_2}))\delta_{\mathbf{q}-(\mathbf{k}'+\mathbf{q}'),-\mathbf{q}'}\\
&= \sum_{\substack{\mathbf{q}'\in\mathrm{Q}\\ m',n\in\{1,2,\ldots\}\\ \nu\in\{1,2,\ldots,rd\}}} \frac{1}{\beta}\sum_{\omega_{n_2}\in\Omega_{-1}} -\frac{g_{mn\nu}(\mathbf{k}'+\mathbf{q}',-\mathbf{q}')g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{N_p} \mathscr{G}_{\substack{0\\ \mathbf{k}'+\mathbf{q}',\mathbf{k}'+\mathbf{q}'\\ n,m'\\ \mu,\mu'}}(\mathrm{i}\omega_{n_2})\,\mathscr{D}_{\substack{0\\ -\mathbf{q}',-\mathbf{q}'\\ \nu,\nu}}(\mathrm{i}(\omega_n-\omega_{n_2}))\delta_{\mathbf{q},\mathbf{k}'}\\
&= \sum_{\substack{\mathbf{q}'\in\mathrm{Q}\\ m',n\in\{1,2,\ldots\}\\ \nu\in\{1,2,\ldots,rd\}}} \frac{1}{\beta}\sum_{\omega_{n_2}\in\Omega_{-1}} -\frac{g^*_{nm\nu}(\mathbf{k}',\mathbf{q}')g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{N_p} \mathscr{G}_{\substack{0\\ \mathbf{k}'+\mathbf{q}',\mathbf{k}'+\mathbf{q}'\\ n,m'\\ \mu,\mu'}}(\mathrm{i}\omega_{n_2})\,\mathscr{D}_{\substack{0\\ -\mathbf{q}',-\mathbf{q}'\\ \nu,\nu}}(\mathrm{i}(\omega_n-\omega_{n_2}))\delta_{\mathbf{q},\mathbf{k}'}\\
&= \sum_{\substack{\mathbf{q}'\in\mathrm{Q}\\ n\in\{1,2,\ldots\}\\ \nu\in\{1,2,\ldots,rd\}}} \frac{1}{\beta}\sum_{\omega_{n_2}\in\Omega_{-1}} -\frac{g_{nn'\nu'}(\mathbf{k}',\mathbf{q}')g^*_{nm\nu}(\mathbf{k}',\mathbf{q}')}{N_p} \mathscr{G}_{\substack{0\\ \mathbf{k}'+\mathbf{q}',\mathbf{k}'+\mathbf{q}'\\ n,n\\ \mu,\mu'}}(\mathrm{i}\omega_{n_2})\,\mathscr{D}_{\substack{0\\ \mathbf{q}',\mathbf{q}'\\ \nu,\nu}}(\mathrm{i}(\omega_n-\omega_{n_2}))\delta_{\mathbf{q},\mathbf{k}'}\\
&= \sum_{\substack{\mathbf{q}'\in\mathrm{Q}\\ n\in\{1,2,\ldots\}\\ \nu\in\{1,2,\ldots,rd\}}} -\frac{g_{nn'\nu'}(\mathbf{k}',\mathbf{q}')g^*_{nm\nu}(\mathbf{k}',\mathbf{q}')}{N_p}\delta_{\mathbf{q},\mathbf{k}'}\delta_{\mu,\mu'}\frac{1}{\beta}\sum_{\omega_{n_2}\in\Omega_{-1}} \mathscr{G}_{\substack{0\\ \mathbf{k}'+\mathbf{q}',\mathbf{k}'+\mathbf{q}'\\ n,n\\ \mu,\mu}}(\mathrm{i}\omega_{n_2})\,\mathscr{D}_{\substack{0\\ \mathbf{q}',\mathbf{q}'\\ \nu,\nu}}(\mathrm{i}(\omega_n-\omega_{n_2}))
\end{aligned} \tag{32.38}$$

Note that we used $\omega_\nu(-\mathbf{q}) = \omega_\nu(\mathbf{q})$ in the last step. We now evaluate the summation over Matsubara-frequencies with the Residue-theorem. Omitting convergence-generating factors for notational simplicity, this confronts us with the task of evaluating the sum

$$\begin{aligned}
\frac{1}{\beta}\sum_{\omega_{n_2}\in\Omega_{-1}} \mathscr{G}_{\substack{0\\ \mathbf{q}'+\mathbf{k}',\mathbf{q}'+\mathbf{k}'\\ n,n\\ \mu,\mu}}(\mathrm{i}\omega_{n_2})\,\mathscr{D}_{\substack{0\\ \mathbf{q}',\mathbf{q}'\\ \nu,\nu}}(\mathrm{i}(\omega_n-\omega_{n_2})) &= \frac{1}{\hbar^2\beta}\sum_{\omega_{n_2}\in\Omega_{-1}} \frac{1}{\mathrm{i}\omega_{n_2}-(\epsilon_{n\mu\mathbf{q}'+\mathbf{k}'}-\mu)/\hbar}\,\frac{-2\omega_\nu(\mathbf{q}')}{(\omega_n-\omega_{n_2})^2+\omega_\nu^2(\mathbf{q}')}\\
&\overset{\text{Corollary 15.18}}{=} \frac{-1}{\hbar}\left[\frac{\mathrm{B}_{\hbar\beta}(\omega_\nu(\mathbf{q}'))+\mathrm{F}_\beta(\xi_{\mathbf{q}'+\mathbf{k}'n\mu})}{\mathrm{i}\omega_n-\xi_{\mathbf{q}'+\mathbf{k}'n\mu}/\hbar+\omega_\nu(\mathbf{q}')} + \frac{\mathrm{B}_{\hbar\beta}(\omega_\nu(\mathbf{q}'))+1-\mathrm{F}_\beta(\xi_{\mathbf{q}'+\mathbf{k}'n\mu})}{\mathrm{i}\omega_n-\xi_{\mathbf{q}'+\mathbf{k}'n\mu}/\hbar-\omega_\nu(\mathbf{q}')}\right]
\end{aligned} \tag{32.39}$$

whereas we introduced the variable $\xi_{\mathbf{q}'+\mathbf{k}'n\mu} = \epsilon_{\mathbf{q}'+\mathbf{k}'n\mu} - \mu$.

In total we obtain the final expression

$$\Sigma^{\text{Fock}}_{\substack{\mathbf{q},\mathbf{k}' \\ m,n' \\ \mu,\mu'}}(\mathrm{i}\omega_n) = \sum_{\substack{\mathbf{q}'\in \mathrm{Q} \\ n\in\{1,2,\dots\} \\ \nu\in\{1,2,\dots,rd\}}} \frac{g_{nn'\nu'}(\mathbf{k}',\mathbf{q}')g^*_{nm\nu}(\mathbf{k}',\mathbf{q}')}{N_p}\delta_{\mathbf{q},\mathbf{k}'}\delta_{\mu,\mu'}\left[\frac{\mathrm{B}_\beta(\hbar\omega_\nu(\mathbf{q}'))+\mathrm{F}_\beta(\xi_{\mathbf{q}'+\mathbf{k}'n\mu})}{\mathrm{i}\hbar\omega_n-\xi_{\mathbf{q}'+\mathbf{k}'n\mu}+\hbar\omega_\nu(\mathbf{q}')}+\frac{\mathrm{B}_\beta(\hbar\omega_\nu(\mathbf{q}'))+1-\mathrm{F}_\beta(\xi_{\mathbf{q}'+\mathbf{k}'n\mu})}{\mathrm{i}\hbar\omega_n-\xi_{\mathbf{q}'+\mathbf{k}'n\mu}-\hbar\omega_\nu(\mathbf{q}')}\right] \tag{32.40}$$

32.4. Polarization of a Spin-Independent-Interaction between an Electron-Gas and a Phonon-Gas

Proof 32.6 of Theorem 20.4 From PHYSICAL REVIEW B 76, 165108 2007 („03“ in the phonon-folder) and / or Computer Physics Communications 209 (2016) 116–133 („04“ in the EPW-folder.)

We will use the following trick: Inverting the Dyson-equation with the von Neumann series on the one hand, and re-expressing the imaginary-time-phononic-Green-function with help of the field integral on the other hand, gives the two equations

$$\mathscr{D}_{\substack{\mathbf{p},\mathbf{p}' \\ \pi,\pi'}}(\mathrm{i}\omega_n) = \sum_{n=0}^{\infty}\left[\left(\mathscr{D}_0\cdot\Pi\right)^n(\mathrm{i}\omega_n)\cdot\mathscr{D}_0(\mathrm{i}\omega_n)\right]_{\substack{\mathbf{p},\mathbf{p}' \\ \pi,\pi'}} \tag{32.41}$$

$$\mathscr{D}_{\substack{\mathbf{p},\mathbf{p}' \\ \pi,\pi'}}(\mathrm{i}\omega_n) = \int_0^{\hbar\beta}\mathrm{d}\tau\,\mathrm{e}^{\mathrm{i}\omega_n\tau}\,\mathscr{D}_{\substack{\mathbf{p},\mathbf{p}' \\ \pi,\pi'}}(\tau) = \frac{-1}{\hbar}\int_0^{\hbar\beta}\mathrm{d}\tau\,\mathrm{e}^{\mathrm{i}\omega_n\tau}\,\langle A_{\mathbf{p}\pi}(\tau)A_{-\mathbf{p}'\pi'}(0)\rangle \tag{32.42}$$

The polarization of the phononic Green-function will be a series in the interaction-parameters

$$\Pi(\mathrm{i}\omega_n) = \sum_{j=0}^{\infty}\Pi^{(j)}(\mathrm{i}\omega_n) \approx 0+\Pi^{(2)}(\mathrm{i}\omega_n)+0+\mathcal{O}(\,\mathrm{V}^4\,), \tag{32.43}$$

whereas the first contribution will turn out to vanish since the interaction can be expanded as a sum of monomials of creation- and annihilation-operators with uneven degree. Inserting this in the upper-equation above and keeping the first non-vanishing terms yields the sets of equations

$$\mathscr{D}_{\substack{\mathbf{p},\mathbf{p}' \\ \pi,\pi'}}(\mathrm{i}\omega_n) = \mathscr{D}_{\substack{0 \\ \mathbf{p},\mathbf{p}' \\ \pi,\pi'}}(\mathrm{i}\omega_n) + \left(\mathscr{D}_0\cdot\Pi^{(2)}\cdot\mathscr{D}_0\right)_{\substack{\mathbf{p},\mathbf{p}' \\ \pi,\pi'}}(\mathrm{i}\omega_n) \tag{32.44}$$

$$\mathscr{D}_{\substack{\mathbf{p},\mathbf{p}' \\ \pi,\pi'}}(\mathrm{i}\omega_n) = \frac{-1}{\hbar}\int_0^{\hbar\beta}\mathrm{d}\tau\,\mathrm{e}^{\mathrm{i}\omega_n\tau}\,\langle A_{\mathbf{p}\pi}(\tau)A_{-\mathbf{p}'\pi'}(0)\rangle \tag{32.45}$$

The lower equation can be calculated explicitly with help of the Wick-theorem, linked-cluster-theorem, and the residue-theorem, and then, by comparing the first non-vanishing coefficients with the upper-equation, we can conclude what $\Pi^{(2)}(\mathrm{i}\omega_n)$ must be.

Fact 1: The action, corresponding to this system, is given by the equation above.

Fact 2: The phononic Green-function is written in second order perturbation-theory as

$$\begin{aligned}\mathscr{D}_{\substack{\mathbf{p},\mathbf{p}'\\\pi,\pi'}}(\tau) &= \mathscr{D}^{(0)}_{\substack{\mathbf{p},\mathbf{p}'\\\pi,\pi'}}(\tau) + \mathscr{D}^{(1)}_{\substack{\mathbf{p},\mathbf{p}'\\\pi,\pi'}}(\tau) + \mathscr{D}^{(2)}_{\substack{\mathbf{p},\mathbf{p}'\\\pi,\pi'}}(\tau) + \dots\\ &= \mathscr{D}_{\substack{0\\\mathbf{p},\mathbf{p}'\\\pi,\pi'}}(\tau) + \quad 0 \quad + \mathscr{D}^{(2)}_{\substack{\mathbf{p},\mathbf{p}'\\\pi,\pi'}}(\tau) + \dots\end{aligned}$$

The second contribution reads

$$\frac{1}{2!}\oint\!\!\!\!\!\!\sum_{\substack{\tau_1,\tau_2\in[0,\hbar\beta]\\\mu,\mu'\in\{\downarrow,\uparrow\}\\\mathbf{k},\mathbf{k}',\mathbf{q},\mathbf{q}'\in Q\\m,m',n,n'\in\{1,2,\dots\}\\\nu,\nu'\in\{1,2,\dots,rd\}}} \frac{-1}{\hbar}\frac{g_{mn\nu}(\mathbf{k},\mathbf{q})g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{\hbar^2 N_p}\langle A_{\mathbf{p}\pi}(\tau)A_{-\mathbf{p}'\pi'}(0)A_{\mathbf{q}\nu}(\tau_1)\overline{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1)\psi_{\mathbf{k}n\mu}(\tau_1)A_{\mathbf{q}'\nu'}(\tau_2)\overline{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2)\psi_{\mathbf{k}'n'\mu'}(\tau_2$$

where only [C]onnected [D]iagrams are summed over (linked cluster theorem). We can therefore simply sum over all diagrams and discard the disconnected diagrams.

$$\begin{aligned}&\langle A_{\mathbf{p}\pi}(\tau)A_{-\mathbf{p}'\pi'}(0)A_{\mathbf{q}\nu}(\tau_1)\overline{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1)\psi_{\mathbf{k}n\mu}(\tau_1)A_{\mathbf{q}'\nu'}(\tau_2)\overline{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2)\psi_{\mathbf{k}'n'\mu'}(\tau_2)\rangle_0^{\mathrm{CD}}\\ &= \langle A_{\mathbf{p}\pi}(\tau)A_{-\mathbf{p}'\pi'}(0)A_{\mathbf{q}\nu}(\tau_1)A_{\mathbf{q}'\nu'}(\tau_2)\rangle_0^{\mathrm{CD}}\,\langle\overline{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1)\psi_{\mathbf{k}n\mu}(\tau_1)\overline{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2)\psi_{\mathbf{k}'n'\mu'}(\tau_2)\rangle_0^{\mathrm{CD}}\end{aligned}$$

Computing the correlator of the phononic Green-functions yields two non-trivial contributions. Computing the correlator of the electronic Green-functions yields two more non-trivial contributions.

We therefore consider the correlator

$$\begin{aligned}\langle\overline{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1)\psi_{\mathbf{k}n\mu}(\tau_1)\overline{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2)\psi_{\mathbf{k}'n'\mu'}(\tau_2)\rangle_0 &= -\langle\psi_{\mathbf{k}n\mu}(\tau_1)\psi_{\mathbf{k}'n'\mu'}(\tau_2)\overline{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1)\overline{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2)\rangle_0\\ &= -\langle\psi(A_1)\psi(A_2)\overline{\psi}(B_2)\overline{\psi}(B_1)\rangle_0\\ &= -\sum_{\pi\in S_2}\mathrm{sign}(\pi)\prod_{l=1}^{2}\langle\psi(A_l)\overline{\psi}(B_{\pi(l)})\rangle_0\\ &= -\hbar^2\sum_{\pi\in S_2}\mathrm{sign}(\pi)\prod_{l=1}^{2}\mathscr{G}_0(A_l,B_{\pi(l)})\\ &= \dots\end{aligned}$$

The electronic correlator can now be evaluated with the Wick-theorem. In order to so, we will enumerate all the permutations of the group S_3.

$$S_2 = \{\pi_1,\pi_2\} = \left\{\begin{pmatrix}1&2\\1&2\end{pmatrix},\begin{pmatrix}1&2\\2&1\end{pmatrix}\right\} \quad \text{with signatures} \quad \mathrm{sgn}(\pi_1,\pi_2) = (+1,-1). \tag{32.46}$$

this gives

$$\dots = -\hbar^2\left(\mathscr{G}_{\substack{0\\\mathbf{k},\mathbf{k}'+\mathbf{q}'\\n,m'\\\mu,\mu'}}(\tau_1-\tau_2)\mathscr{G}_{\substack{0\\\mathbf{k}',\mathbf{k}+\mathbf{q}\\n',m\\\mu',\mu}}(\tau_2-\tau_1) - \mathscr{G}_{\substack{0\\\mathbf{k},\mathbf{k}+\mathbf{q}\\n,m\\\mu,\mu}}(\tau_1-\tau_1)\mathscr{G}_{\substack{0\\\mathbf{k}',\mathbf{k}'+\mathbf{q}'\\n',m'\\\mu',\mu'}}(\tau_2-\tau_2)\right)$$

Now we turn to the phononic correlators

$$\begin{aligned}\langle A_{\mathbf{p}\pi}(\tau)A_{-\mathbf{p}'\pi'}(0)A_{\mathbf{q}\nu}(\tau_1)A_{\mathbf{q}'\nu'}(\tau_2)\rangle_0 &\equiv \langle A(x_1)A(x_2)A(x_3)A(x_4)\rangle_0 \\ &= \sum_{\substack{\text{Pairings of} \\ \{x_1,\dots,x_4\}}} \langle A(x_{k_1})A(x_{k_2})\rangle_0 \langle A(x_{k_3})A(x_{k_4})\rangle_0 \\ &= \dots\end{aligned}$$

which gives [cf. Altland]

$$\begin{aligned}\dots &= \langle A(x_1)A(x_2)\rangle_0\langle A(x_3)A(x_4)\rangle_0 + \langle A(x_1)A(x_3)\rangle_0\langle A(x_2)A(x_4)\rangle_0 + +\langle A(x_1)A(x_4)\rangle_0\langle A(x_2)A(x_3)\rangle_0 \\ &= \langle A_{\mathbf{p}\pi}(\tau)A_{-\mathbf{p}'\pi'}(0)\rangle_0\langle A_{\mathbf{q}\nu}(\tau_1)A_{\mathbf{q}'\nu'}(\tau_2)\rangle_0 + \langle A_{\mathbf{p}\pi}(\tau)A_{\mathbf{q}\nu}(\tau_1)\rangle_0\langle A_{-\mathbf{p}'\pi'}(0)A_{\mathbf{q}'\nu'}(\tau_2)\rangle_0 + +\langle A_{\mathbf{p}\pi}(\tau)A_{\mathbf{q}'\nu'}(\tau_2)\rangle_0\langle A_{-\mathbf{p}'\pi'}(0)A_{\mathbf{q}} \\ &= \hbar^2 \left(\mathscr{D}^{0}_{\substack{\pi,\pi' \\ \mathbf{p},\mathbf{p}'}}(\tau)\mathscr{D}^{0}_{\substack{\nu,\nu' \\ \mathbf{q},-\mathbf{q}'}}(\tau_1-\tau_2) + \mathscr{D}^{0}_{\substack{\pi,\nu \\ \mathbf{p},-\mathbf{q}}}(\tau-\tau_1)\mathscr{D}^{0}_{\substack{\pi',\nu' \\ -\mathbf{p},-\mathbf{q}'}}(-\tau_2) + \mathscr{D}^{0}_{\substack{\pi,\nu' \\ \mathbf{p},-\mathbf{q}'}}(\tau-\tau_2)\mathscr{D}^{0}_{\substack{\pi',\nu \\ -\mathbf{p}',-\mathbf{q}}}(-\tau_1) \right)\end{aligned}$$

whereas the first term will not contribute since it does not resemble a one particle irreducible diagram (or something like that... external vertices are not connected to inner ones...).

This gives us the intermediate result

$$\begin{aligned}\mathscr{D}^{(2)}_{\substack{\mathbf{p},\mathbf{p}' \\ \pi,\pi'}}(\tau) = \sum\!\!\!\!\!\!\int_{\substack{\tau_1,\tau_2\in[0,\hbar\beta] \\ \mu,\mu'\in\{\downarrow,\uparrow\} \\ \mathbf{k},\mathbf{k}',\mathbf{q},\mathbf{q}'\in \mathrm{Q} \\ m,m',n,n'\in\{1,2,\dots\} \\ \nu,\nu'\in\{1,2,\dots,rd\}}} \frac{g_{mn\nu}(\mathbf{k},\mathbf{q})g_{m'n'\nu'}(\mathbf{k}',\mathbf{q}')}{2\hbar^{-1}N_p} &\left(\mathscr{G}^{0}_{\substack{\mathbf{k},\mathbf{k}'+\mathbf{q}' \\ n,m' \\ \mu,\mu'}}(\tau_1-\tau_2)\mathscr{G}^{0}_{\substack{\mathbf{k}',\mathbf{k}+\mathbf{q} \\ n',m \\ \mu',\mu}}(\tau_2-\tau_1) - \mathscr{G}^{0}_{\substack{\mathbf{k},\mathbf{k}+\mathbf{q} \\ n,m \\ \mu,\mu}}(\tau_1-\tau_1)\mathscr{G}^{0}_{\substack{\mathbf{k}',\mathbf{k}'+\mathbf{q}' \\ n',m' \\ \mu',\mu'}}(\tau_2-\tau_2) \right) \\ &\times \left(\mathscr{D}^{0}_{\substack{\pi,\nu \\ \mathbf{p},-\mathbf{q}}}(\tau-\tau_1)\mathscr{D}^{0}_{\substack{\pi',\nu' \\ -\mathbf{p},-\mathbf{q}'}}(-\tau_2) + \mathscr{D}^{0}_{\substack{\pi,\nu' \\ \mathbf{p},-\mathbf{q}'}}(\tau-\tau_2)\mathscr{D}^{0}_{\substack{\pi',\nu \\ -\mathbf{p}',-\mathbf{q}}}(-\tau_1) \right)\end{aligned}$$

This yields four contributions to the phonon-propagator, which we enumerate with

- $(1)-(i)$
- $(1)-(ii)$
- $(2)-(i)$
- $(2)-(ii)$

We now consider a generic (periodic) function of the type

$$\mathscr{F}(\tau) = \iint_{[0,\hbar\beta]^2} \mathrm{d}(\tau_1,\tau_2)\, \mathscr{D}(\tau-\tau_{k_1})\mathscr{D}(\tau_{k_2})\mathscr{G}(\tau_{k_3}-\tau_{k_4})\mathscr{G}(\tau_{k_5}-\tau_{k_6}) \tag{32.47}$$

The Matsubara-component of this function reads

$$
\begin{aligned}
\mathscr{F}(\mathrm{i}\omega_n) &= \iiint\limits_{[0,\hbar\beta]^3} \mathrm{d}(\tau,\tau_1,\tau_2)\, \mathscr{D}(\tau-\tau_{k_1})\mathscr{D}(\tau_{k_2})\mathscr{G}(\tau_{k_3}-\tau_{k_4})\mathscr{G}(\tau_{k_5}-\tau_{k_6})\mathrm{e}^{\mathrm{i}\omega_n\tau} \\
&= \frac{1}{(\hbar\beta)^4} \sum_{\substack{\omega_{n_1},\omega_{n_2}\in\Omega_{+1} \\ \omega_{n_3},\omega_{n_4}\in\Omega_{-1}}} \mathscr{D}(\mathrm{i}\omega_{n_1})\mathscr{D}(\mathrm{i}\omega_{n_2})\mathscr{G}(\mathrm{i}\omega_{n_3})\mathscr{G}(\mathrm{i}\omega_{n_4}) \iiint\limits_{[0,\hbar\beta]^3} \mathrm{d}(\tau,\tau_1,\tau_2)\mathrm{e}^{\mathrm{i}\tau(\omega_n-\omega_{n_1})+\mathrm{i}\tau_{k_1}\omega_{n_1}-\mathrm{i}\tau_{k_2}\omega_{n_2}-\mathrm{i}(\tau_{k_3}-\tau_{k_4})\omega_{n_3}-\mathrm{i}\omega_{n_4}(\tau_{k_5}-\tau_{k_6})}
\end{aligned}
$$

We now compute the Matsubara-component of each contribution individually and sum up the results afterwards.

ToBeContinued.

33. Appendix - B-Sides

33.1. The Many-Body-Formalism for Indistinguishable Particles: Second Quantization - Operator Formalism

33.1.1. The Fock-Space

The construction of many-body Hilbert-spaces for indistinguishable particles is way more tedious. The initial situation (which one encounters quite naturally in typical materials) is as follows.

Notation 33.1 Let $\mathcal{H} = \text{span}_{\mathbb{C}}\{|1\rangle, |2\rangle, \dots |\text{d}_{\mathcal{H}}\rangle\}$ be a single-particle Hilbert-space with (not neccessarily, oftentimes orthonormal) basis and corresponding wavefunctions / state-vectors with coordinates x

$$\langle\, x \mid i\, \rangle \equiv \psi_i(x). \tag{33.1}$$

whereas $\langle\, ,\, \rangle$ refers to the scalarproduct on $\mathcal{H}$.

Notation 33.2 In this section we denote $N \in \mathbb{N}_0$ to be a natural number.

Definition 33.3 We define the *unsymmetrized (and unphysical) N-particle-Hilbert-space $\mathscr{H}_N$* as tensor product

$$\mathscr{H}_N = \mathcal{H} \otimes \mathcal{H} \otimes \cdots \otimes \mathcal{H}. \tag{33.2}$$

Remark 33.3.1 This is not the Hilbert-space that one is allowed to work with. Only a subspace of it (depending on the "statistics" of the particles at hand) is relevant for computations.

<u>**Important Spanning Set / Basis 1 (of $\mathscr{H}_N$)**</u>

Corollary 33.4 A basis of $\mathscr{H}_N$ is given by the set

$$\mathscr{B}_{\mathscr{H}_N} \equiv \left\{ |i_1, i_2, \dots, i_N) \equiv |i_1\rangle \otimes |i_2\rangle \otimes \cdots \otimes |i_N\rangle \mid i_j \in \{1, \dots, \text{d}_{\mathcal{H}}\} \right\} \tag{33.3}$$

Proof 33.4 See introductory lectures about tensor-products.

Remark 33.4.1 Recall that the *scalar-product* between two basis-vectors is defined as

$$
\begin{aligned}
(i_1, i_2, \dots, i_N \mid j_1, j_2, \dots, j_N) &\equiv \left(\langle i_1| \otimes \langle i_2| \otimes \dots \otimes \langle i_N|\right)\left(|j_1\rangle \otimes |j_2\rangle \otimes \dots \otimes |j_N\rangle\right) \\
&\equiv \langle i_1 \mid j_1\rangle\langle i_2 \mid j_2\rangle \cdots \langle i_N \mid j_N\rangle
\end{aligned}
$$

and by sesqui-linearity thus on the entire Hilbert-space.

Remark 33.4.2 Recall that the basis yields a *completeness relation* on $\mathscr{H}_N$ given through projectors as

$$
1_{\mathscr{H}_N} \equiv \sum_{i_1,i_2,\dots,i_N=1}^{\mathrm{d}_{\mathcal{H}}} |i_1, i_2, \dots, i_N)(i_1, i_2, \dots, i_N|. \tag{33.4}
$$

Notation 33.5 We write for the value of a wavefunction at some coordinate-tuple

$$
\begin{aligned}
\psi_{i_1,i_2,\dots,i_N}(x_1, x_2, \dots, x_N) &\equiv (x_1, x_2, \dots, x_N \mid i_1, i_2, \dots, i_N), \\
&\equiv \left(\langle x_1| \otimes \langle x_2| \otimes \dots \otimes \langle x_N|\right)\left(|i_1\rangle \otimes |i_2\rangle \otimes \dots \otimes |i_N\rangle\right), \\
&\equiv \psi_{i_1}(x_1)\psi_{i_2}(x_2) \cdots \psi_{i_N}(x_N).
\end{aligned}
$$

Definition 33.6 For $\zeta \in \{-1, +1\}$, and $N \in \mathbb{N}_0$ we define the *ζ-Fock-subspaces* $\mathscr{F}_\zeta^N$ as

$$
\begin{aligned}
&\mathscr{F}_\zeta^0 = \mathbb{C}, \\
&\mathscr{F}_\zeta^N = \left\{\psi \in \mathscr{H}_N \mid \forall \pi \in \mathrm{S}_N \,:\, \psi(x_{\pi(1)}, \dots, x_{\pi(N)}) = \mathrm{sign}(\pi)^{1-\theta(\zeta)}\psi(x_1, \dots, x_N)\right\} \quad \text{for } N \geq 1.
\end{aligned} \tag{33.5}
$$

Remark 33.6.1 It might seem redundant to consider a subspace with zero-particles. This has computational reasons. It will later turn out, that it is easier to compute observables, when one is working in the grand-canonical ensemble, which makes this construction (mathematically) neat.

Remark 33.6.2 We remark that the dimensions of the individual subspaces can be written explicitly in terms of N, ζ and $\mathrm{d}_{\mathcal{H}}$. We will compute them in Theorem 7.7. At this point we simply state the result that in the case of fermions (i.e. $\zeta = -1$) there is an $N_{\max} \equiv N_{\max}(\mathrm{d}_{\mathcal{H}}) \in \mathbb{N}$ such that $\mathscr{F}_\zeta^N = \{0\}$ for all $N > N_{\max}$.

Lemma 33.7 We define the *statistical-symmetrization-operator* as linear mapping

$$
\begin{aligned}
\mathrm{P}_\zeta :\ \mathscr{H}_N &\rightarrow \mathscr{F}_\zeta^N \\
\psi &\mapsto \mathrm{P}_\zeta\, \psi
\end{aligned}
$$

as

$$\left(\mathrm{P}_\zeta\, \psi\right)(x_1, x_2, \dots, x_N) := \frac{1}{N!} \sum_{\pi \in \mathrm{S}_N} \mathrm{sign}(\pi)^{1-\theta(\zeta)}\, \psi(x_{\pi(1)}, x_{\pi(2)}, \dots, x_{\pi(N)}). \tag{33.6}$$

Proof 33.7 We have to prove that

$$\left(\mathrm{P}_\zeta\, \psi\right)(x_{\sigma(1)}, \dots, x_{\sigma(N)}) = \mathrm{sign}(\sigma)^{1-\theta(\zeta)} \left(\mathrm{P}_\zeta\, \psi\right)(x_1, \dots, x_N) \tag{33.7}$$

which is easy to see after relabeling the summations and since $\mathrm{sign}(\sigma) = \mathrm{sign}(\sigma^{-1})$.

Corollary 33.8 The statistical-symmetrization-operator is a projector, i.e. it holds $\mathrm{P}_\zeta \circ \mathrm{P}_\zeta = \mathrm{P}_\zeta$.

Proof 33.8 See Appendix, p. 303.

Lemma 33.9 The statistical symmetrization-operator P_ζ acts as the identity on the subspace $\mathscr{F}_\zeta^N$, i.e.

$$\forall \psi \in \mathscr{F}_\zeta^N \,:\, \mathrm{P}_\zeta\, \psi \equiv 1_{\mathscr{F}_\zeta^N} \psi = \psi \tag{33.8}$$

Proof 33.9 Using the symmetry-property in Equation (33.5) we obtain a sum of $N!$ identical contributions which cancels out the prefactor in Equation (33.6).

Theorem 33.10 It follows

$$\mathrm{P}_{+1}\, \mathrm{P}_{-1} = \mathrm{P}_{-1}\, \mathrm{P}_{+1} = 0 \tag{33.9}$$

Proof 33.10 We compute for example

$$\begin{aligned}
\left(\mathrm{P}_{+1}\, \mathrm{P}_{-1}\, \psi\right)(x_1, \dots, x_N) &= \frac{1}{N!} \sum_{\pi \in \mathrm{S}_N} \left(\mathrm{P}_{-1}\, \psi\right)(x_{\pi(1)}, \dots, x_{\pi(N)}) \\
&= \frac{1}{N!} \left(\sum_{\pi \in \mathrm{S}_N} \mathrm{sign}(\pi) \right) \left(\mathrm{P}_{-1}\, \psi\right)(x_1, \dots, x_N) \\
&= 0
\end{aligned}$$

Lemma 33.11 For *pure tensor-product states* $|\psi\rangle$, i.e. states that can be written as

$$|\psi\rangle = \bigotimes_{j=1}^{N} |\psi_j\rangle \tag{33.10}$$

for some $|\psi_j\rangle \in \mathcal{H}$, it follows

$$\mathrm{P}_\zeta\,|\psi\rangle = \frac{1}{N!}\sum_{\pi\,\in\,\mathrm{S}_N} \mathrm{sign}(\pi)^{1-\theta(\zeta)} \bigotimes_{j=1}^{N} |\psi_{\pi(j)}\rangle \tag{33.11}$$

and one obtains *Slater-determinants* and *permanents* for the coordinate representations as

$$(x_1, x_2, \dots, x_N\,|\,\mathrm{P}_\zeta\,|\,\psi\,\rangle = \frac{1}{N!}\begin{vmatrix} \langle x_1|\psi_1\rangle_{\mathcal{H}} & \cdots & \langle x_1|\psi_N\rangle_{\mathcal{H}} \\ \vdots & \ddots & \vdots \\ \langle x_n|\psi_1\rangle_{\mathcal{H}} & \cdots & \langle x_n|\psi_n\rangle_{\mathcal{H}} \end{vmatrix}_\zeta . \tag{33.12}$$

Proof 33.11 See Appendix, p. 303.

Definition 33.12 For a permutation $\pi \in \mathrm{S}_N$ we define its *action on a product-state*

$$|\psi\rangle = \bigotimes_{j=1}^{N} |\psi_j\rangle \tag{33.13}$$

as

$$\pi|\psi\rangle := \bigotimes_{j=1}^{N} |\psi_{\pi(j)}\rangle \tag{33.14}$$

Lemma 33.13 Now the statistical symmetrization-operator can be written for pure states as

$$\mathrm{P}_\zeta := \frac{1}{N!}\sum_{\pi\,\in\,\mathrm{S}_N} \mathrm{sign}(\pi)^{1-\theta(\zeta)}\,\pi \tag{33.15}$$

Proof 33.13 Follows with Equation (33.11).

Remark 33.13.1 Notice that by linear extension this is equation can be used to act on any state in $\mathscr{H}_N$ (since every element can be expandend w.r.t. the basis in Equation (33.3).

Lemma 33.14 The statistical symmtrization-operator P_ζ is hermitian w.r.t. the scalar-product on $\mathscr{H}_N$.

Proof 33.14 See Appendix, p. 304.

Important Spanning Set / Basis 2 (of $\mathscr{F}_\zeta^N$)

Definition 33.15 For $i_1, \dots, i_N \in \{1, 2, \dots, \mathrm{d}_\mathcal{H}\}$, we define *auxiliary-states* as

$$\begin{aligned} |i_1, i_2, \dots, i_N\} &= \frac{\mathrm{P}_\zeta}{N!^{-1/2}} |i_1, i_2, \dots, i_N) \\ &= \frac{1}{\sqrt{N!}} \sum_{\pi \in \mathrm{S}_N} \mathrm{sign}(\pi)^{1-\theta(\zeta)} |i_{\pi(1)}\rangle \otimes |i_{\pi(2)}\rangle \otimes \cdots \otimes |i_{\pi(N)}\rangle. \end{aligned} \tag{33.16}$$

Remark 33.15.1 Notice that in the special case of $N = 1$ one finds $|i\} = |i)$.

Remark 33.15.2 Introducing these states is a bit problematic from a pedagogical perspective, since they **only** serve in order to introduce the so-called „occupation number states“ later on (which will turn out to be the basis of states that we are **actually** interested in working with).

Lemma 33.16 The auxiliary-states have the property

$$\forall \sigma \in \mathrm{S}_N \ : \ |i_{\sigma(1)}, i_{\sigma(2)}, \dots, i_{\sigma(N)}\} = \mathrm{sign}(\sigma)^{1-\theta(\zeta)} |i_1, i_2, \dots, i_N\} \tag{33.17}$$

Proof 33.16 We prove this theorem by referring to the coordinate representation of the auxiliary-states. We therefore consider $\sigma \in \mathrm{S}_N$ and we compute

$$\begin{aligned} (x_1, x_2, \dots, x_N | i_{\sigma(1)}, i_{\sigma(2)}, \dots, i_{\sigma(N)}\} &= \frac{1}{\sqrt{N!}} \sum_{\pi \in \mathrm{S}_N} \mathrm{sign}(\pi)^{1-\theta(\zeta)} \prod_{l=1}^{N} \langle x_l | i_{\pi(\sigma(l))}\rangle \\ &= \mathrm{sign}(\sigma)^{1-\theta(\zeta)} \frac{1}{\sqrt{N!}} \sum_{\pi \in \mathrm{S}_N} \mathrm{sign}(\pi \circ \sigma)^{1-\theta(\zeta)} \prod_{l=1}^{N} \langle x_l | i_{\pi(\sigma(l))}\rangle \\ &= \mathrm{sign}(\sigma)^{1-\theta(\zeta)} (x_1, x_2, \dots, x_N | i_1, i_2, \dots, i_N\} \end{aligned}$$

which proves the Lemma.

Remark 33.16.1 Notice that this implies in the fermionic case (i.e. $\zeta = -1$) that if $i_j = i_{j'}$ for some $j, j' \in \{1, \dots, N\}$, then $|i_1, i_2, \dots, i_N\} = 0$.

Corollary 33.17 One obtains the completeness relation on $\mathscr{F}_\zeta^N$ as

$$1_{\mathscr{F}_\zeta^N} \equiv \frac{1}{N!} \sum_{i_1,\ldots,i_N=1}^{\mathrm{d}_\mathcal{H}} |i_1,\ldots,i_N\}\{i_1,\ldots,i_N|. \tag{33.18}$$

Proof 33.17 We compute

$$\begin{aligned}
\frac{1}{N!} \sum_{i_1,\ldots,i_N=1}^{\mathrm{d}_\mathcal{H}} |i_1,\ldots,i_N\}\{i_1,\ldots,i_N| &= \frac{1}{N!} \sum_{i_1,\ldots,i_N=1}^{\mathrm{d}_\mathcal{H}} \sqrt{N!}\ \mathrm{P}_\zeta\ |i_1,\ldots,i_N)(i_1,\ldots,i_N|\ \sqrt{N!}\ \mathrm{P}_\zeta \\
&= \mathrm{P}_\zeta \left[\sum_{i_1,\ldots,i_N=1}^{\mathrm{d}_\mathcal{H}} |i_1,\ldots,i_N)(i_1,\ldots,i_N| \right] \mathrm{P}_\zeta \\
&= \mathrm{P}_\zeta\ 1_{\mathscr{H}_N}\ \mathrm{P}_\zeta \\
&= \mathrm{P}_\zeta \\
&= 1_{\mathscr{F}_\zeta^N}\ .
\end{aligned}$$

whereas we used Equation (33.4), the fact that P_ζ is a projector (cf. Corollary 33.8) and that it acts as the identity on $\mathscr{F}_\zeta^N$ (cf. Lemma 33.9).

Lemma 33.18 It follows

$$\{i_1,i_2,\ldots,i_N \mid j_1,j_2,\ldots,j_N\} = \sum_{\pi\in \mathrm{S}_N} \mathrm{sign}(\pi)^{1-\theta(\zeta)} \prod_{k=1}^{N} \langle i_k \mid j_{\pi(k)}\rangle \tag{33.19}$$

Proof 33.18 We compute straightforwardly

$$\begin{aligned}
\{i_1,i_2,\ldots,i_N \mid j_1,j_2,\ldots,j_N\} &= N!(i_1,i_2,\ldots,i_N|\,\mathrm{P}_\zeta^2\,|j_1,j_2,\ldots,j_N) \\
&= N!(i_1,i_2,\ldots,i_N|\,\mathrm{P}_\zeta\,|j_1,j_2,\ldots,j_N) \\
&= \sum_{\pi\in \mathrm{S}_N} \mathrm{sign}(\pi)^{1-\theta(\zeta)} (i_1,\ldots,i_N| \left(|j_{\pi(1)}\rangle \otimes |j_{\pi(2)}\rangle \otimes \cdots \otimes |j_{\pi(N)}\rangle) \right) \\
&= \sum_{\pi\in \mathrm{S}_N} \mathrm{sign}(\pi)^{1-\theta(\zeta)} \prod_{k=1}^{N} \langle i_k \mid j_{\pi(k)}\rangle \\
&= \sum_{\pi\in \mathrm{S}_N} \mathrm{sign}(\pi)^{1-\theta(\zeta)}\ \langle i_1 \mid j_{\pi(1)}\rangle\langle i_2 \mid j_{\pi(2)}\rangle \cdots \langle i_N \mid j_{\pi(N)}\rangle
\end{aligned} \tag{33.20}$$

Remark 33.18.1 Notice that the auxiliary states in Definition 33.15 are not a basis, because of Lemma 33.16, but they form an overcomplete set, because of Corollary 33.17. Because of Lemma 33.18 it is evident that these states form an *overcomplete set*.

Definition 33.19 For each $k \in \left\{1, 2, \ldots, \mathrm{d}_{\mathcal{H}}\right\}$, we introduce the *occupation-numbers of a state w.r.t. a basis* as functions

$$\begin{aligned} \mathrm{n}_k : \{1, \ldots, \mathrm{d}_{\mathcal{H}}\}^N &\longrightarrow \mathbb{N} \\ (i_1, \ldots, i_N) \equiv i \mapsto \mathrm{n}_k(i_1, \ldots, i_N) &:= \sum_{l=1}^{N} \delta_{i_l,k} \equiv \mathrm{n}_k, \end{aligned}$$

whereas we define $\mathrm{n}_k(\bullet) \equiv 0$ for $N = 0$.

Remark 33.19.1 Notice that this implies with Remark 33.16.1 that in the fermionic case (i.e. $\zeta = -1$) it holds $\mathrm{n}_k \in \{0, 1\}$ for all $k \in \{1, 2, \ldots, \mathrm{d}_{\mathcal{H}}\}$.

Corollary 33.20 The overlap between different vectors is given by

$$\{i_1, i_2, \ldots, i_N \mid j_1, j_2, \ldots, j_N\} = \begin{cases} \mathrm{sign}(\sigma)^{1-\theta(\zeta)} \prod_{k=1}^{\mathrm{d}_{\mathcal{H}}} \mathrm{n}_k! & \text{if } \exists\, \sigma \in \mathrm{S}_N \ : \ (i_1, i_2, \ldots, i_N) = (j_{\sigma(1)}, \ldots, j_{\sigma(N)}) \\ 0 & \text{otherwise.} \end{cases}$$

Proof 33.20 If the numbers $i_1, \ldots, i_N$ are *not* a permutation of the numbers $j_1, \ldots, j_N$, then there is in the right hand side of Equation (33.19) for each summand at least one factor which is zero, such that the entire sum vanishes. We divide the rest of the proof into two cases.

Case 1: $\zeta = -1$:

Then it follows with Remarks 33.19.1 and 33.16.1 that if the numbers are a permutation of each other, i.e. if there is some $\sigma \in \mathrm{S}_N$ such that $(i_1, i_2, \ldots, i_N) = (j_{\sigma(1)}, \ldots, j_{\sigma(N)})$, then only one summand contributes in Equation (33.19), yielding the result from Equation (33.18) as

$$\begin{aligned} \{i_1, i_2, \ldots, i_N \mid j_1, j_2, \ldots, j_N\} &= \sum_{\pi \in \mathrm{S}_N} \mathrm{sign}(\pi) \prod_{k=1}^{N} \langle i_k \mid j_{\pi(k)} \rangle \\ &= \sum_{\pi \in \mathrm{S}_N} \mathrm{sign}(\pi) \prod_{k=1}^{N} \langle j_{\sigma(k)} \mid j_{\pi(k)} \rangle \\ &= \mathrm{sign}(\sigma) \sum_{\pi \in \mathrm{S}_N} \mathrm{sign}(\sigma^{-1} \circ \pi) \prod_{k=1}^{N} \langle i_k \mid j_{(\sigma^{-1} \circ \pi)(k)} \rangle \\ &= \mathrm{sign}(\sigma) \end{aligned} \tag{33.21}$$

since there is only one permutation that contributes to the sum.

Case 2: $\zeta = +1$:

For bosons, we use Lemma 33.16 in order to write $\{i_1, i_2, \ldots, i_N \mid = \{j_1, j_2, \ldots, j_N \mid$ and we obtain with Equation (33.18) then the overlap as

$$\begin{aligned}\{i_1, i_2, \dots, i_N \,|\, j_1, j_2, \dots, j_N\} &= \{j_1, j_2, \dots, j_N \,|\, j_1, j_2, \dots, j_N\} \\ &= \sum_{\pi \in S_N} \prod_{k=1}^{N} \langle j_k \,|\, j_{\pi(k)} \rangle \\ &= \dots\end{aligned} \tag{33.22}$$

the only permutations that contribute are now those, where identical j_k are being mapped onto each other. For each $k \in \{1, 2, \dots, \mathrm{d}_{\mathcal{H}}\}$ there are $\mathrm{n}_k!$ such permutations, which yields the result.

Important Spanning Set / Basis 3 (of $\mathscr{F}_\zeta^N$)

Definition 33.21 For $i_1, \dots, i_N \in \{1, 2, \dots, \mathrm{d}_{\mathcal{H}}\}$, we define the *quasi-occupation-number-states* as

$$\begin{aligned}|\, i_1, i_2, \dots, i_N \rangle &\equiv \frac{1}{\sqrt{\prod_{k=1}^{\mathrm{d}_{\mathcal{H}}} \mathrm{n}_k!}} |i_1, i_2, \dots, i_N\} \\ &\equiv \frac{1}{\sqrt{N!\, \prod_{k=1}^{\mathrm{d}_{\mathcal{H}}} \mathrm{n}_k!}} \sum_{\pi \in S_N} \mathrm{sign}(\pi)^{1-\theta(\zeta)} \, |i_{\pi(1)}\rangle \otimes |i_{\pi(2)}\rangle \otimes \cdots \otimes |i_{\pi(N)}\rangle\end{aligned} \tag{33.23}$$

with occupation-numbers as $\forall k \in \left\{1, 2, \dots, \mathrm{d}_{\mathcal{H}}\right\} \;:\; \mathrm{n}_k \equiv \mathrm{n}_k(i_1, \dots, i_N)$.

Remark 33.21.1 Notice that these states give *Slater-determinants* and *permanents* in the coordinate representation, i.e.

Corollary 33.22 One obtains the completeness relation

$$1_{\mathscr{F}_\zeta^N} \equiv \sum_{i_1, i_2, \dots, i_N = 1}^{\mathrm{d}_{\mathcal{H}}} \frac{\prod_{k=1}^{\mathrm{d}_{\mathcal{H}}} \mathrm{n}_k!}{N!} |i_1, i_2, \dots, i_N\rangle\langle i_1, i_2, \dots, i_N| \tag{33.24}$$

with occupation-numbers as $\forall k \in \left\{1, 2, \dots, \mathrm{d}_{\mathcal{H}}\right\} \;:\; \mathrm{n}_k \equiv \mathrm{n}_k(i_1, \dots, i_N)$.

Proof 33.22 Follows from the completeness relation in Corollary 33.17.

TBD: Overlap between different quasi-occupation-number-states and relation to Slater-determinants and permanents.

33.1.2. Creation- and Annihilation-Operators

Definition 33.23 In the zero-particle subspace $\mathscr{F}_\zeta^0 = \mathbb{C}$, we introduce *the vacuum-state* as

$$|0\rangle \equiv |\mathrm{vac.}\rangle \equiv 1 \in \mathbb{C}.$$

Remark 33.23.1 Notice that with this supplement, we can extend the completeness-relation in Equation (33.24) to the entire Fock-space $\mathscr{F}_\zeta$ as

$$1_{\mathscr{F}_\zeta} = \sum_{N=0}^{\infty} \sum_{i_1,i_2,\dots,i_N=1}^{\mathrm{d}_\mathcal{H}} \frac{\prod_{k=1}^{\mathrm{d}_\mathcal{H}} \mathrm{n}_k!}{N!} |i_1, i_2, \dots, i_N\rangle\langle i_1, i_2, \dots, i_N|. \tag{33.25}$$

with occupation-numbers as $\forall k \in \left\{1, 2, \dots, \mathrm{d}_\mathcal{H}\right\} \;:\; \mathrm{n}_k \equiv \mathrm{n}_k(i_1, \dots, i_N)$.

Definition 33.24 For each $k \in \{1, \dots, \mathrm{d}_\mathcal{H}\}$ we introduce a mapping as the *(particle-)creation-operators*

$$\begin{aligned} c_k^\dagger : \mathscr{F}_\zeta^N &\longrightarrow \mathscr{F}_\zeta^{N+1} \\ |\psi\rangle &\mapsto c_k^\dagger|\psi\rangle \end{aligned}$$

by specifying its action on the overcomplete basis in Equation (33.15) and extending its definition by linearity on the entire Fock-space. We define for the states

$$\begin{aligned} c_k^\dagger|0\rangle &\equiv |k\}, \\ c_k^\dagger|i_1, i_2, \dots, i_N\} &\equiv |k, i_1, i_2, \dots, i_N\} \end{aligned} \tag{33.26}$$

Remark 33.24.1 Since these operators are mappings from the N-particle Hilbert-space to the $N+1$-particle Hilbert-space, the name "creation-operators" becomes evident.

Remark 33.24.2 Another thing to notice is that this definition / prefactor-convention is not unique within the literature. We are following the prefactor-convention from [Negele, Orland] in contrast to [TBD].

Corollary 33.25 It follows

$$c_k^\dagger|i_1, i_2, \dots, i_N\rangle = \sqrt{\mathrm{n}_k + 1}\,|k, i_1, i_2, \dots, i_N\rangle \tag{33.27}$$

with occupation-numbers as $\forall k \in \left\{1, 2, \dots, \mathrm{d}_\mathcal{H}\right\} \;:\; \mathrm{n}_k \equiv \mathrm{n}_k(i_1, \dots, i_N)$.

Proof 33.25 Follows immediately by considering the respective definitions: We have

$$\begin{aligned} c_k^\dagger|i_1, i_2, \dots, i_N\rangle &\overset{\text{Eq. (33.23)}}{=} \frac{1}{\sqrt{\prod_{k=1}^{\mathrm{d}_\mathcal{H}} \mathrm{n}_k!}} c_k^\dagger|i_1, i_2, \dots, i_N\} \\ &\overset{\text{Eq. (33.26)}}{=} \frac{1}{\sqrt{\prod_{k=1}^{\mathrm{d}_\mathcal{H}} \mathrm{n}_k!}} |k, i_1, i_2, \dots, i_N\} \\ &\overset{\text{Eq. (33.23)}}{=} \sqrt{\mathrm{n}_k + 1}\,|k, i_1, i_2, \dots, i_N\rangle \end{aligned} \tag{33.28}$$

with occupation-numbers $\mathrm{n}_k \equiv \mathrm{n}_k(i_1, \dots, i_N)$.

Remark 33.25.1 By repeated application of Corollary 33.25 one finds that any basis-vector can be rewritten as a product of particle creations on the vacuum-state as

$$|i_1, i_2, \dots, i_N\} = c^\dagger_{i_1} \cdots c^\dagger_{i_N} |0\rangle \quad \Longleftrightarrow \quad |i_1, i_2, \dots, i_N\rangle = \frac{1}{\sqrt{\prod_{k=1}^{\mathrm{d}_\mathcal{H}} \mathrm{n}_k!}} c^\dagger_{i_1} \cdots c^\dagger_{i_N} |0\rangle \tag{33.29}$$

with occupation-numbers $\mathrm{n}_k \equiv \mathrm{n}_k(i_1, \dots, i_N)$.

Corollary 33.26 One obtains the statistical commutation relations for the creation-operators

$$c^\dagger_\lambda c^\dagger_\mu - \zeta c^\dagger_\mu c^\dagger_\lambda \equiv [c^\dagger_\lambda, c^\dagger_\mu]_\zeta = 0 \tag{33.30}$$

Proof 33.26 We compute

$$\begin{aligned} c^\dagger_\lambda c^\dagger_\mu |i_1, i_2, \dots, i_N\} &= c^\dagger_\lambda |\mu, i_1, i_2, \dots, i_N\} = |\lambda, \mu, i_1, i_2, \dots, i_N\} \\ &= \zeta |\mu, \lambda, i_1, i_2, \dots, i_N\} = \zeta c^\dagger_\mu |\lambda, i_1, i_2, \dots, i_N\} = \zeta c^\dagger_\mu c^\dagger_\lambda |i_1, i_2, \dots, i_N\} \end{aligned} \tag{33.31}$$

Now rearranging terms yields the desired result.

Definition 33.27 We introduce the *particle-annihilation-operators* $\{c_1, \dots c_{\mathrm{d}_\mathcal{H}}\}$ as the hermitian-adjoints of the creation-operators $\{c^\dagger_1, \dots c^\dagger_{\mathrm{d}_\mathcal{H}}\}$ w.r.t. the scalar-product on the Fock-space.

Lemma 33.28 It follows

$$c_\lambda c_\mu - \zeta c_\mu c_\lambda \equiv [c_\lambda, c_\mu]_\zeta = 0 \tag{33.32}$$

Proof 33.28 Follows immediately from taking the hermitian adjoint of Equation (33.30).

Lemma 33.29 For the action of the hermitian adjoint it follows

$$c_j |i_1, i_2, \dots, i_N\} = \sum_{k=1}^{N} \zeta^{k-1} \delta_{j,i_k} |i_1, \dots i_{k-1}, i_{k+1}, \dots, i_N\} \equiv \sum_{k=1}^{N} \zeta^{k-1} \delta_{j,i_k} |i_1, \dots \hat{i}_k, \dots, i_N\}. \tag{33.33}$$

Proof 33.29

We prove this identity, by taking an arbitrary state $\{j_2, \ldots, j_n|$ and compute the overlap of the two two sides of the equation. This gives

$$\begin{aligned}
\{j_2, \ldots, j_n|c_j|i_1, i_2, \ldots, i_N\} &= \{j, j_2, \ldots, j_n|i_1, i_2, \ldots, i_N\} \\
&\overset{j\equiv j_1,\text{Lemma (33.18)}}{=} \sum_{\pi\in S_N} \operatorname{sign}(\pi)^{1-\theta(\zeta)} \prod_{k=1}^{N} \langle j_k|i_{\pi(k)}\rangle \\
&= \ldots
\end{aligned} \tag{33.34}$$

In order to proceed, we have to make some considerations about the permutations $\pi \in \mathrm{S}_N$:

Intermezzo

Every permutation in $\pi \in \mathrm{S}_N$ is uniquely determined after one specifies

1. the image of the first element, i.e. $\pi(1) = k$ for some $k \in \{1, 2, \ldots, N\}$ and
2. the permutation of the remaining $N - 1$ elements.

For some $\tau \in \mathrm{S}_{N-1}$ that is acting bijectively on the set $\{2, \ldots, N\}$, we define ...

- a *natural embedding* $\imath :\ \mathrm{S}_{N-1} \longrightarrow \mathrm{S}_N$ with $\imath(\tau) = \widetilde{\tau}$ as

$$\begin{aligned}
\widetilde{\tau}(1) &= 1 \qquad , \\
\widetilde{\tau}(i) &= \tau(i) \quad \text{if } i > 1,
\end{aligned} \tag{33.35}$$

- for each $k \in \{1, 2, \ldots, N\}$ a *canonical insertion map* $\imath_k :\ \mathrm{S}_{N-1} \longrightarrow \mathrm{S}_N$ with $\imath_k(\tau) = \pi$ as

$$\begin{aligned}
&\pi(1) = k, \\
&\pi(i) = \tau(i) - 1 \quad \text{if } i \geq 2 \wedge \tau(i) \leq k, \\
&\pi(i) = \tau(i) \qquad\quad \text{if } i \geq 2 \wedge \tau(i) > k.
\end{aligned} \tag{33.36}$$

Then it follows that

$$\pi = \imath_k(\tau) = (k\ \ k-1) \cdots (2\ \ 1) \circ \imath(\tau) \quad \text{and} \ \operatorname{sign}(\pi) = \operatorname{sign}(\imath_k(\tau)) = \operatorname{sign}(\tau) \cdot (-1)^{k-1}. \tag{33.37}$$

This gives a disjoint union of the symmetric group S_N as

$$S_N = \bigsqcup_{k=1}^{N} \imath_k(\mathrm{S}_{N-1}). \tag{33.38}$$

which allows us to reorder the sum in Equation (33.34) by summing over permutations $\tau \in \mathrm{S}_{N-1}$

$$\begin{aligned}
\ldots &= \sum_{k=1}^{N} \sum_{\tau \in \mathrm{S}_{N-1}} \mathrm{sign}(\iota_k(\tau))^{1-\theta(\zeta)} \prod_{l=1}^{N} \langle j_l | i_{\iota_k(\tau)(l)} \rangle \\
&= \sum_{k=1}^{N} \underbrace{(-1)^{(k-1)(1-\theta(\zeta))}}_{=\zeta^{k-1}} \sum_{\tau \in \mathrm{S}_{N-1}} \mathrm{sign}(\tau)^{1-\theta(\zeta)} \underbrace{\langle j_1 | i_{\iota_k(\tau)(1)} \rangle}_{=\langle j_1 | i_k \rangle = \delta_{j,i_k}} \prod_{l=2}^{N} \langle j_l | i_{\iota_k(\tau)(l)} \rangle \\
&= \sum_{k=1}^{N} \zeta^{k-1} \delta_{j,i_k} \sum_{\tau \in \mathrm{S}_{N-1}} \mathrm{sign}(\tau)^{1-\theta(\zeta)} \prod_{l=2}^{N} \langle j_l | i_{\tau(l)} \rangle \\
&= \sum_{k=1}^{N} \zeta^{k-1} \delta_{j,i_k} \{ j_2, \ldots, j_N | i_1, \ldots i_{k-1}, i_{k+1}, \ldots, i_N \} \\
&= \sum_{k=1}^{N} \zeta^{k-1} \delta_{j,i_k} \{ j_2, \ldots, j_N | i_1, \ldots \hat{i}_k, \ldots, i_N \}
\end{aligned} \tag{33.39}$$

Since this equation holds for any vector $\{j_2, \ldots, j_N\}$, we can conclude that the identity claimed in the Lemma is correct.

Corollary 33.30 Analogously one finds

$$c_j |i_1, i_2, \ldots, i_N\rangle = \frac{1}{\sqrt{n_j}} \sum_{k=1}^{N} \zeta^{k-1} \delta_{j,i_k} |i_1, \ldots i_{k-1}, i_{k+1}, \ldots, i_N\rangle = \frac{1}{\sqrt{n_j}} \sum_{k=1}^{N} \zeta^{k-1} \delta_{j,i_k} |i_1, \ldots \hat{i}_k, \ldots, i_N\rangle \tag{33.40}$$

Proof 33.30 Follows immediately with Equation (33.23) and the previous Lemma.

Theorem 33.31 In total, we arrive at creation- and annihilation-operators $\left\{ c_1, c_1^\dagger, c_2, c_2^\dagger, \ldots, c_{\mathrm{d}_\mathcal{H}}, c_{\mathrm{d}_\mathcal{H}}^\dagger \right\}$ obey the statistics-dependent commutation-relations

$$\left[c_i, c_j\right]_\zeta = \left[c_i^\dagger, c_j^\dagger\right]_\zeta = 0, \qquad \left[c_i, c_j^\dagger\right]_\zeta = \delta_{i,j}$$

Proof 33.31

- TBD, cf. Negele, Orland page 15.

TBD: Basis Change etc according to page 15...

Important Spanning Set / Basis 4 (of $\mathscr{F}_\zeta^N$)

Definition 33.32 We consider a state $|i_1, i_2, \dots, i_N\rangle$. We denote its occupation-numbers $\mathrm{n}_1, \mathrm{n}_2, \dots, \mathrm{n}_{\mathrm{d}_\mathcal{H}}$ with occupation-number-operators as usual $\forall k \in \left\{1, 2, \dots, \mathrm{d}_\mathcal{H}\right\} \ : \ \mathrm{n}_k \equiv \mathrm{n}_k(i_1, \dots, i_N)$. Then we introduce the *occupation-number-representation* of the state a.k.a. *occupation-number-states* as

$$|\,\mathrm{n}_1, \mathrm{n}_2, \dots, \mathrm{n}_{\mathrm{d}_\mathcal{H}}\rangle_{\mathrm{occ}} := |i_1, i_2, \dots, i_N\rangle \in \mathscr{F}_\zeta^N. \tag{33.41}$$

Remark 33.32.1 Usually one drops the index "occ" in the notation as it is clear from the context which states one is referring to, i.e.

$$|\,\mathrm{n}_1, \mathrm{n}_2, \dots, \mathrm{n}_{\mathrm{d}_\mathcal{H}}\rangle_{\mathrm{occ}} \equiv |\,\mathrm{n}_1, \mathrm{n}_2, \dots, \mathrm{n}_{\mathrm{d}_\mathcal{H}}\rangle. \tag{33.42}$$

Remark 33.32.2 Notice that this is a *many-to-one-correspondence*.

Corollary 33.33 One obtains the simple formulae,

$$\begin{aligned} c_j|\,\mathrm{n}_1, \mathrm{n}_2, \dots, \mathrm{n}_{\mathrm{d}_\mathcal{H}}\rangle &= \sqrt{\mathrm{n}_j} & &|\,\mathrm{n}_1, \mathrm{n}_2, \dots, \mathrm{n}_{j-1}, \mathrm{n}_j - 1, \mathrm{n}_{j+1}, \dots, \mathrm{n}_{\mathrm{d}_\mathcal{H}}\rangle \\ c_j^\dagger|\,\mathrm{n}_1, \mathrm{n}_2, \dots, \mathrm{n}_{\mathrm{d}_\mathcal{H}}\rangle &= \sqrt{\mathrm{n}_j + 1}\,\zeta^{\sum_{l=1}^{j-1}\mathrm{n}_l} & &|\,\mathrm{n}_1, \mathrm{n}_2, \dots, \mathrm{n}_{j-1}, \mathrm{n}_j + 1, \mathrm{n}_{j+1}, \dots, \mathrm{n}_{\mathrm{d}_\mathcal{H}}\rangle \end{aligned} \tag{33.43}$$

whereas the state on the r.h.s. is defined as 0 for $\mathrm{n}_j = 0$.

Proof 33.33

- TBD, cf. Negele, Orland page 14 and Altland page 44.

Theorem 33.34 The occupation number states are an orthonormal basis of the Fock-space, i.e. one has the orthonormality condition

$$\langle n_1, n_2, \dots, n_{\mathrm{d}_\mathcal{H}}|n_1', n_2', \dots, n_{\mathrm{d}_\mathcal{H}}'\rangle = \prod_{l=1}^{\mathrm{d}_\mathcal{H}} \delta_{n_l, n_l'} \tag{33.44}$$

and the resolution of identity

$$\sum_{n_1, \dots, n_{\mathrm{d}_\mathcal{H}} \in \mathbb{N}_\zeta} |n_1, n_2, \dots, n_{\mathrm{d}_\mathcal{H}}\rangle\langle n_1, n_2, \dots, n_{\mathrm{d}_\mathcal{H}}| = 1_{\mathscr{F}_\zeta} \tag{33.45}$$

Proof 33.34

- Cf. Negele, Orland page 22, Nolting 2 page 277, Schwabl 2 page 1 onwards.

The second identity (resolution of identity) follows from Equation (33.25) and the fact that each occupation-number-state appears $\frac{N!}{\prod_k n_k!}$ times in the sum.

33.1.3. Operators on the Fock-Space

33.1.3.1. 1-Body-Operators

Corollary 33.35 We define the number-operator $n_j \equiv a_j^\dagger a_j$, which counts the "number of particles in a given state"

$$n_j|n_1, n_2, \dots, n_{\mathrm{d}_\mathcal{H}}\rangle = n_j|n_1, n_2, \dots, n_{\mathrm{d}_\mathcal{H}}\rangle \tag{33.46}$$

whereas on the l.h.s. of the equation n_j represents an operator, and on the r.h.s. of the equation it represents a number.

Proof 33.35 TBD.

Remark 33.35.1 Analog one also introduces the operator for the total number of particles

$$N = \sum_{j=1}^{\mathrm{d}_\mathcal{H}} n_j. \tag{33.47}$$

Theorem 33.36 One can write for a one-body-operator U that acts on occupation-number-states

$$U = \frac{1}{1!}\sum_{i,j=1}^{\mathrm{d}_\mathcal{H}} \langle\, i \,|\, U \,|\, j \,\rangle\, c_i^\dagger c_j \tag{33.48}$$

Proof 33.36

- TBD, cf. Negele, Orland page 17.

33.1.3.2. 2-Body-Operators

Theorem 33.37 One can write for a two-body-operator V that acts on occupation-number-states

$$V = \frac{1}{2!} \sum_{i,j,k,l} \langle\, i\, , j \mid U \mid k\, , l\, \rangle\, c_i^\dagger c_j^\dagger c_l c_k \tag{33.49}$$

Proof 33.37

- TBD, cf. Negele, Orland page 18.

33.1.3.3. n-Body-Operators

Theorem 33.38 For an n-body-operator L we write

$$L = \sum_{\substack{i_1<\ldots<i_n \\ j_1<j_2<\ldots<j_{n-1}<j_n}} L^{i_1,\ldots,i_n}_{j_1,\ldots,j_n} a^\dagger_{i_1} \cdots a^\dagger_{i_n} a_{j_1} \cdots a_{j_n},$$

with $L^{i_1,\ldots,i_n}_{j_1,\ldots,j_n} \in \mathbb{C}$ being defined through the scalar-product (and omitting units for notational simplicity).

Proof 33.38

- TBD, cf. Negele, Orland page 19.

Definition 33.39 An operator R on the many-particle-Hilbert-space is called an m-body-operator if it can be written in the following way

$$R|i_1, i_2, \ldots, i_N) =: \frac{1}{m!} \sum_{1\leq n_1 \neq n_2 \neq \ldots \neq n_m \leq \mathrm{d}_{\mathcal{H}}} R_{n_1,n_2,\ldots,n_m} |i_1, i_2, \ldots, i_N) \tag{33.50}$$

Lemma 33.40 For any permutation $\pi \in \mathrm{S}_N$ one finds

$$\left(i_1, i_2, \ldots, i_N | R | i'_1, i'_2, \ldots, i'_N\right) = \left(i_{\pi(1)}, i_{\pi(2)}, \ldots, i_{\pi(N)} | R | i'_{\pi(1)}, i'_{\pi(2)}, \ldots, i'_{\pi(N)}\right) \tag{33.51}$$

Proof 33.40

- TBD, cf. Negele, Orland p. 10.

Corollary 33.41 Notice the identity

$$\frac{\left(i_1, i_2, \ldots, i_N | R | i'_1, i'_2, \ldots, i'_N\right)}{\left(i_1, i_2, \ldots, i_N | i'_1, i'_2, \ldots, i'_N\right)} = \frac{1}{m!} \sum_{1 \leq n_1 \neq n_2 \neq \ldots \neq n_m \leq \mathrm{d}_{\mathcal{H}}} \frac{\left(i_1, i_2, \ldots, i_N | R | i'_1, i'_2, \ldots, i'_N\right)}{\langle i_1 | i'_1 \rangle \langle i_2 | i'_2 \rangle \cdots \langle i_N | i'_N \rangle} \tag{33.52}$$

Proof 33.41

- TBD, cf. Negele, Orland p. 10.

33.2. The Reason why so few People in the DFT-Community use Second Quantization

Notation 33.42 Let $L \in \mathrm{End}(\mathcal{H})$ be a linear operator on the single-particle-Hilbert-space $\mathcal{H}$ with second-quantized version $L^{2\mathrm{Q}}$ acting on the Fock-space $\mathscr{F}_\zeta$ as

$$L^{2\mathrm{Q}} \equiv \sum_{ij} \langle i | L | j \rangle \, c_i^\dagger c_j \quad \in \mathscr{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta). \tag{33.53}$$

Notation 33.43 We write for the spectral decomposition of the Hamiltonian

$$H - \mu 1_{\mathcal{H}} = \sum_{n=0}^{\infty} \left(E_n - \mu\right) |n\rangle\langle n| \tag{33.54}$$

Definition 33.44 Finally, we define

$$\rho_{\mathcal{H}} \equiv \mathrm{F}_\beta(H - \mu 1_{\mathcal{H}}) := \sum_{n=0}^{\infty} \mathrm{F}_\beta \left(E_n - \mu\right) |n\rangle\langle n|$$

Theorem 33.45 One finds the identity

$$\mathrm{Tr}_{\mathscr{F}}(\rho_{\mathscr{F}} A^{2\mathrm{Q}}) = \mathrm{Tr}_{\mathcal{H}}(\rho_{\mathcal{H}} A). \tag{33.55}$$

Proof 33.45

We first note that the trace is invariant with respect to the basis used, so we may take the basis as the set of eigenvectors of H for the trace of the single-particle Hilbert space, and we consider the occupation number basis derived from it for the trace of the Fock-space. Then

$$Z = \prod_{n=0}^{\infty} (1 + e^{-\beta(E_n - \mu)}) \tag{33.56}$$

and we can write

$$\mathrm{Tr}_{\mathscr{F}}(\rho_{\mathscr{F}} A^{2\mathrm{Q}}) = \sum_{n_1, n_2, \ldots \in \{0,1\}} \langle \Psi_{n_1, n_2, \ldots}, \rho_{\mathscr{F}} A^{2\mathrm{Q}} \Psi_{n_1, n_2, \ldots} \rangle$$

Since

$$\rho_{\mathscr{F}} \Psi_{n_1, n_2, \ldots} = \frac{1}{Z} e^{-\beta \sum_{j=0}^{\infty} n_j (E_j - \mu)} \Psi_{n_1, n_2, \ldots} = \frac{1}{Z} \prod_{j=0}^{\infty} e^{-\beta n_j (E_j - \mu)} \Psi_{n_1, n_2, \ldots},$$

and

$$\langle \Psi_{n_1, n_2, \ldots}, A^{2\mathrm{Q}} \Psi_{n_1, n_2, \ldots} \rangle = \sum_{k,m} \langle e_k, A e_m \rangle \delta_{n_k, 1} \delta_{n_m, 1} \delta_{k,m},$$

we have

$$\mathrm{Tr}_{\mathscr{F}}(\rho_{\mathscr{F}} A^{2\mathrm{Q}}) = \sum_{k=0}^{\infty} \langle e_k, A e_k \rangle \frac{1}{Z} \sum_{n_1, n_2, \ldots \in \{0,1\}} \prod_{j=0}^{\infty} e^{-\beta n_j (E_j - \mu)} \delta_{n_k, 1}.$$

Note that

$$\sum_{n_1, n_2, \ldots \in \{0,1\}} \prod_{j=0}^{\infty} e^{-\beta n_j (E_j - \mu)} \delta_{n_k, 1} = (1 + e^{-\beta(E_1 - \mu)})(1 + e^{-\beta(E_2 - \mu)}) \cdots e^{-\beta(E_k - \mu)} \cdots (1 + e^{-\beta(E_r - \mu)}) \cdots,$$

hence

$$\frac{1}{Z} \sum_{n_1, n_2, \ldots \in \{0,1\}} \prod_{j=0}^{\infty} e^{-\beta n_j (E_j - \mu)} \delta_{n_k, 1} = \mathrm{F}_{\beta}(E_k - \mu),$$

and therefore

$$\mathrm{Tr}_{\mathscr{F}}(\rho_{\mathscr{F}} A^{2\mathrm{Q}}) = \sum_{k=0}^{\infty} \langle e_k, A e_k \rangle \, \mathrm{F}_{\beta}(E_k - \mu) = \mathrm{Tr}_{\mathcal{H}}(\rho_{\mathcal{H}} A),$$

proving the theorem.

33.3. The Zero-Temperature Limit

In this section, we show how one obtains the usual zero-temperature-expressions for Green-functions from the finite-temperature-expressions.

Notation 33.46 We write for the temperature-dependent expectation-value in the natural ensemble

$$\langle \cdots \rangle := \frac{1}{Z}\,\mathrm{Tr}_{\mathscr{H}}\left(\cdots\ \mathrm{e}^{-\beta\Xi}\right) \tag{33.57}$$

Case 1: The Non-Degenerate Case

Hypothesis 33.47 We assume, that we can enumerate (neglecting degeneracies) the ground-state, first excited state for the operator $\Xi = H - \eta\mu N$ as

$$|0\rangle,\ |1\rangle,\ |2\rangle, \ldots \tag{33.58}$$

with energies

$$E_{|0\rangle} < E_{|1\rangle} < E_{|2\rangle} < \ldots \quad \Longleftrightarrow: \quad E_0 < E_1 < E_2 < \ldots \tag{33.59}$$

We further assume, that these states form an orthonormal basis of the Hilbert-space[1], such that the basis-independence of the trace yields

$$\mathrm{Tr}_{\mathscr{H}} \bullet = \sum_{k=0}^{\infty} \langle\, k\,|\, \bullet \,|\, k\,\rangle \tag{33.60}$$

Corollary 33.48 It holds

$$\lim_{T\to 0}\langle\cdots\rangle = \langle\, 0\,|\ \cdots\ |\, 0\,\rangle \tag{33.61}$$

and therefore in particular

$$\lim_{T\to 0} G_{\alpha,\alpha'}(t,t') = -\frac{\mathrm{i}}{\hbar}\langle\, 0\,|\ \mathrm{T}_t\, a_\alpha(t)\, a^\dagger_{\alpha'}(t')\,|\, 0\,\rangle \tag{33.62}$$

etc for the other Green-functions / expectation-values.

Proof 33.48 We compute

[1]Notice that this is for example the case for usual Slater-determinants / permanents (in the case of an underlying Fock-space).

$$
\begin{aligned}
\langle \cdots \rangle &\equiv -\frac{\mathrm{i}}{\hbar} \frac{\mathrm{Tr}_{\mathscr{H}} \left(\cdots \, \mathrm{e}^{-\beta(H-\eta\mu N)} \right)}{\mathrm{Tr}_{\mathscr{H}} \, \mathrm{e}^{-\beta(H-\eta\mu N)}} \\
&= -\frac{\mathrm{i}}{\hbar} \frac{\mathrm{e}^{-\beta E_0} \langle \, 0 \, | \, \cdots \, | \, 0 \, \rangle + \mathrm{e}^{-\beta E_1} \langle \, 1 \, | \, \cdots \, | \, 1 \, \rangle + \mathrm{e}^{-\beta E_2} \langle \, 2 \, | \, \cdots \, | \, 2 \, \rangle + \ldots}{\mathrm{e}^{-\beta E_0} + \mathrm{e}^{-\beta E_1} + \mathrm{e}^{-\beta E_2} + \ldots} \\
&= -\frac{\mathrm{i}}{\hbar} \frac{\langle \, 0 \, | \, \cdots \, | \, 0 \, \rangle + \mathrm{e}^{-\beta(E_1-E_0)} \langle \, 1 \, | \, \cdots \, | \, 1 \, \rangle + \mathrm{e}^{-\beta(E_2-E_0)} \langle \, 2 \, | \, \cdots \, | \, 2 \, \rangle + \ldots}{1 + \mathrm{e}^{-\beta(E_1-E_0)} + \mathrm{e}^{-\beta(E_2-E_0)} + \ldots}
\end{aligned}
$$

Therefore one obtains

$$
\lim_{T\to 0} \langle \cdots \rangle = -\frac{\mathrm{i}}{\hbar} \langle \, 0 \, | \, \cdots \, | \, 0 \, \rangle
$$

This yields then for example for the time-ordered Green-function

$$
\begin{aligned}
G_{\alpha,\alpha'}(t,t') &\equiv -\frac{\mathrm{i}}{\hbar} \frac{\mathrm{Tr}_{\mathscr{H}} \, \mathrm{e}^{-\beta(H-\eta\mu N)} \, \mathrm{T}_t \, a_\alpha(t) \, a^\dagger_{\alpha'}(t')}{\mathrm{Tr}_{\mathscr{H}} \, \mathrm{e}^{-\beta(H-\eta\mu N)}} \\
&= -\frac{\mathrm{i}}{\hbar} \frac{\mathrm{e}^{-\beta E_0} \langle \, 0 \, | \, \mathrm{T}_t \, a_\alpha(t) \, a^\dagger_{\alpha'}(t') \, | \, 0 \, \rangle + \mathrm{e}^{-\beta E_1} \langle \, 1 \, | \, \mathrm{T}_t \, a_\alpha(t) \, a^\dagger_{\alpha'}(t') \, | \, 1 \, \rangle + \mathrm{e}^{-\beta E_2} \langle \, 2 \, | \, \mathrm{T}_t \, a_\alpha(t) \, a^\dagger_{\alpha'}(t') \, | \, 2 \, \rangle + \ldots}{\mathrm{e}^{-\beta E_0} + \mathrm{e}^{-\beta E_1} + \mathrm{e}^{-\beta E_2} + \ldots} \\
&= -\frac{\mathrm{i}}{\hbar} \frac{\langle \, 0 \, | \, \mathrm{T}_t \, a_\alpha(t) \, a^\dagger_{\alpha'}(t') \, | \, 0 \, \rangle + \mathrm{e}^{-\beta(E_1-E_0)} \langle \, 1 \, | \, \mathrm{T}_t \, a_\alpha(t) \, a^\dagger_{\alpha'}(t') \, | \, 1 \, \rangle + \mathrm{e}^{-\beta(E_2-E_0)} \langle \, 2 \, | \, \mathrm{T}_t \, a_\alpha(t) \, a^\dagger_{\alpha'}(t') \, | \, 2 \, \rangle + \ldots}{1 + \mathrm{e}^{-\beta(E_1-E_0)} + \mathrm{e}^{-\beta(E_2-E_0)} + \ldots}
\end{aligned}
$$

Therefore one obtains

$$
\lim_{T\to 0} G_{\alpha,\alpha'}(t,t') = -\frac{\mathrm{i}}{\hbar} \langle \, 0 \, | \, \mathrm{T}_t \, a_\alpha(t) \, a^\dagger_{\alpha'}(t') \, | \, 0 \, \rangle
$$

Case 2: The Degenerate Case

Corollary 33.49 In the case of degenerate states

$$
|0_1\rangle, |0_2\rangle, \ldots |0_{n_0}\rangle, \; |1_1\rangle, |1_2\rangle, \ldots |1_{n_1}\rangle, \; |2_1\rangle, |2_2\rangle, \ldots |2_{n_2}\rangle, \ldots \tag{33.63}
$$

with energies

$$
E_{|0_1\rangle} = E_{|0_2\rangle} = \ldots = E_{|0_{n_0}\rangle} \quad < \quad E_{|1_1\rangle} = E_{|1_2\rangle} = \ldots = E_{|1_{n_1}\rangle} \quad < \quad E_{|2_1\rangle} = E_{|2_2\rangle} = \ldots = E_{|2_{n_2}\rangle} < \ldots
$$

It holds

$$\lim_{T\to 0}\langle\cdots\rangle = \frac{1}{n_0}\sum_{j=1}^{n_0}\langle\, 0_j \mid \cdots \mid 0_j\,\rangle \tag{33.64}$$

and therefore in particular

$$\lim_{T\to 0} G_{\alpha,\alpha'}(t,t') = -\frac{\mathrm{i}}{\hbar}\sum_{j=1}^{n_0}\langle\, 0_j \mid \mathrm{T}_t\, a_\alpha(t)\, a^\dagger_{\alpha'}(t') \,|\, 0_j\,\rangle \tag{33.65}$$

etc for the other Green-functions / expectation-values.

Proof 33.49 By straightforward generalization of the computation before.

33.4. The Chemical Potential and the Fermi-Energy of (Non-)Interacting Electrons

Theorem 33.50 For a non-interacting system

$$H = \sum_{\substack{\mathbf{k}\in Q \\ m\in\{1,2,3,\dots\}}} \epsilon_{m,\mathbf{k}}\, c^{\dagger}_{m,\mathbf{k}} c_{m,\mathbf{k}} \tag{33.66}$$

without degeneracies one finds

$$|0\rangle = \prod_{j=1}^{N} c^{\dagger}_{m_j,\mathbf{k}_j} |\text{vac.}\rangle =: \prod_{\substack{\mathbf{k}\in Q \\ m\in\{1,2,3,\dots\} \\ \epsilon_{m,\mathbf{k}}<\epsilon_{\text{Fermi}}}} c^{\dagger}_{m,\mathbf{k}} |\text{vac.}\rangle \tag{33.67}$$

with an (a priori unknown) Fermi-energy ϵ_{Fermi} that yet has to be determined.

Proof 33.50 Non-interacting systems can be diagonalized with standard-techniques from linear algebra, such that one arrives at the diagonal form of the Hamiltonian in second quantization. The orthonormal Slater-determinants span the entire Fock-space. Since there are no degeneracies the ground state is unique. The desired particle number N therefore dictates the Fermi-energy.

Lemma 33.51 It holds in the case of non-interacting electrons in a crystal

$$\lim_{T\to 0} \mu(T) = \epsilon_{\text{Fermi}} \tag{33.68}$$

whereas ϵ_{Fermi} is the highest occupied state in the Slater-determinant $|0\rangle$.

Proof 33.51 We compute the average particle twice.

On the one hand we have

$$\langle N\rangle = \sum_{\substack{\mathbf{k}\in Q \\ m\in\{1,2,3,\dots\}}} \langle c^{\dagger}_{m,\mathbf{k}} c_{m,\mathbf{k}}\rangle = \frac{\hbar}{\mathrm{i}} \sum_{\substack{\mathbf{k}\in Q \\ m\in\{1,2,3,\dots\}}} G^{<}_{\substack{m,m \\ \mathbf{k},\mathbf{k}}}(0) = \sum_{\substack{\mathbf{k}\in Q \\ m\in\{1,2,3,\dots\}}} \mathrm{F}_{\beta}(\epsilon_{m,\mathbf{k}} - \mu) \tag{33.69}$$

on the other hand we have

$$\langle 0|N|0\rangle = \sum_{\substack{\mathbf{k}\in Q \\ m\in\{1,2,3,\dots\}}} \langle 0|c^{\dagger}_{m,\mathbf{k}} c_{m,\mathbf{k}}|0\rangle = \sum_{\substack{\mathbf{k}\in Q \\ m\in\{1,2,3,\dots\}}} \theta(\epsilon_{\text{Fermi}} - \epsilon_{m,\mathbf{k}}) \tag{33.70}$$

This gives in the zero-temperature-limit the equation

$$\sum_{\substack{\mathbf{k}\in \mathrm{Q} \\ m\in\{1,2,3,\dots\}}} \theta(\epsilon_{\text{Fermi}} - \epsilon_{m,\mathbf{k}}) = \langle 0|N|0\rangle = \lim_{T\to 0}\langle N\rangle(T) = \sum_{\substack{\mathbf{k}\in \mathrm{Q} \\ m\in\{1,2,3,\dots\}}} \theta(\,\mu(T=0)\,-\epsilon_{m,\mathbf{k}}) \tag{33.71}$$

direct comparison of the two expression yields $\mu(T=0) = \epsilon_{\text{Fermi}}$.

Interacting Electrons

Theorem 33.52 We compute the chemical potential, by constraining the number of particles

$$\langle N\rangle = \frac{-1}{\pi}\lim_{\delta\to 0} \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in \mathrm{Q} \\ m\in\{1,\dots,\infty\} \\ E\in(-\infty,+\infty)}} \mathrm{F}_\beta(E)\,\mathrm{Im}\,G^{\text{ret}}_{m,m}(E+\mathrm{i}\delta,\mathbf{k})$$

Proof 33.52

$$\begin{aligned}
\langle N\rangle &= \sum_{\substack{\mathbf{k}\in \mathrm{Q} \\ m\in\{1,2,3,\dots\}}} \langle c^\dagger_{m,\mathbf{k}} c_{m,\mathbf{k}}\rangle \\
&= \frac{\hbar}{\mathrm{i}} \sum_{\substack{\mathbf{k}\in \mathrm{Q} \\ m\in\{1,2,3,\dots\}}} G^<_{\substack{m,m\\ \mathbf{k},\mathbf{k}}}(0) \\
&= \frac{\hbar}{2\pi\mathrm{i}} \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in \mathrm{Q} \\ m\in\{1,2,3,\dots\} \\ \omega\in(-\infty,\infty)}} G^<_{\substack{m,m\\ \mathbf{k},\mathbf{k}}}(\omega) \\
&= \frac{-\hbar}{\pi} \sum\!\!\!\!\!\!\int_{\substack{\mathbf{k}\in \mathrm{Q} \\ m\in\{1,2,3,\dots\} \\ \omega\in(-\infty,\infty)}} \mathrm{F}_{\hbar\beta}(\omega)\, G^{\text{ret}}_{\substack{m,m\\ \mathbf{k},\mathbf{k}}}(\omega+\mathrm{i}\delta)
\end{aligned}$$

Remark 33.52.1 Recall that the retarded Green-function is given by analytical continuation of the Matsubara-Green-function which is determined by solving the Dyson-equation

$$\mathscr{G}(\mathbf{k},\mathrm{i}\omega_n) = \frac{1}{\hbar}\frac{1}{\mathrm{i}\omega_n - \big(H_0(\mathbf{k}) + \Sigma(\mathbf{k},\mathrm{i}\omega_n) - \mu\big)/\hbar} \tag{33.72}$$

with non-interacting band-dispersion-matrix $H_0(\mathbf{k}) \equiv \mathrm{diag}(\epsilon_1(\mathbf{k}),\dots,\epsilon_J(\mathbf{k}))$, whereas J represents the maximal number of bands that are being considered.

33.5. Specific Heat in the Sommerfeld-Approximation and the Dulong-Petit-Law

33.5.1. Intermezzo: The [C]anonical [E]nsemble

33.5.1.1. Formalism

Notation 33.53 We consider the case in which the above mentioned system has a volume V, and is being treated in the canonical ensemble with temperature T and particle number N and Hilbert-space $\mathscr{H} \equiv \mathscr{H}_N$. This gives us the three natural variables (N, T, V).

Fact 33.54 In the absence of „spontaneous symmetry breaking“[2] expectation-values in the canonical-ensemble are evaluated w.r.t. the state

$$\rho_{\mathrm{CE}} = \frac{\mathrm{e}^{-\beta H}}{\mathrm{Tr}_{\mathscr{H}_N}\, \mathrm{e}^{-\beta H}} \equiv \frac{1}{Z_{\mathrm{CE}}} \mathrm{e}^{-\beta H}, \tag{33.73}$$

whereas we introduced the canonical partition-function $Z_{\mathrm{CE}} \equiv Z_{\mathrm{CE}}(N, T, V)$.

Remark 33.54.1 Note that this is a solution to the stationary von Neumann-equation which represents a stationary-state.

Fact 33.55 Recall, that the thermodynamic potential[3] in the canonical ensemble is given by the so-called grand free energy

$$F \equiv F(N, T, V) = -k_{\mathrm{B}} T \log_{\mathrm{e}} Z_{\mathrm{CE}}(N, T, V) \tag{33.74}$$

from which one can conclude the usual relations for the conjugated variables chemical potential, entropy and pressure

$$\frac{\partial F}{\partial N} = \mu, \qquad \frac{\partial F}{\partial T} = -S, \qquad \frac{\partial F}{\partial V} = -P. \tag{33.75}$$

Fact 33.56 The ionic system of a crystal is usually modelled in the canonical ensemble.

Remark 33.56.1 Looking at the appearing equations, we will realize, that one is tempted (and it is quite usual to do so) to re-interprete the ionic system as a system in the so-called „grand canonical ensemble“ with the additional condition that one sets the chemical potential to zero. However, this is just a mnemonic[4]. A non-zero chemical potential can be defined also in the case of an ionic system.

[2]This subject goes far beyond the scope of this lecture.
[3]Recall that this is a scalar-quantity which is used to represent the thermodynamic state of a system.
[4]See attached excerpts at the end of this chapter.

33.5.1.2. Application to the Ionic System

Preparation:

Before we start calculating and predicting observable quantities for the ionic system, we collect all the things we need.

- Hilbert-space: $\mathscr{H} = \bigotimes_{j=1}^{rd_{\mathbf{x}}N_p} L^2(\mathbb{R}^1) \simeq L^2(\mathbb{R}^{rd_{\mathbf{x}}N_p})$
- Basis: $\mathscr{B} = \left\{ |N\rangle \equiv |n_{\lambda_1}, n_{\lambda_2}, \dots, n_{\lambda_{rd_{\mathbf{x}}N_p}}\rangle \equiv \prod_{j=1}^{rd_{\mathbf{x}}N_p} |n_{\lambda_j}\rangle \,|\, \forall j \in \{1, \dots, rd_{\mathbf{x}}N_p\} \,:\, n_{\lambda_j} \in \{0, 1, 2, \dots, \infty\} \right\}$
- Expectation-value: $\langle A \rangle = \dfrac{\mathrm{Tr}_{\mathscr{H}}\, A\, \mathrm{e}^{-\beta H_{\text{harm.}}}}{\mathrm{Tr}_{\mathscr{H}}\, \mathrm{e}^{-\beta H_{\text{harm.}}}} \equiv \dfrac{\sum_{N\in\mathscr{B}} \langle\, N \mid A\, \mathrm{e}^{-\beta H_{\text{harm.}}} \mid N\, \rangle}{\sum_{N\in\mathscr{B}} \langle\, N \mid \mathrm{e}^{-\beta H_{\text{harm.}}} \mid N\, \rangle}$

(33.76)

whereas A represents an observable (i.e. an hermitian operator, usually written in second quantized form).

Theorem 33.57 One finds for the average occupation-number

$$\langle n_{\nu\mathbf{q}} \rangle = \langle b^{\dagger}_{\nu\mathbf{q}} b_{\nu\mathbf{q}} \rangle = \mathrm{B}_{\beta}(\hbar\omega_{\nu\mathbf{q}}) \tag{33.77}$$

Proof 33.57 See Appendix, p. 305.

Corollary 33.58 It follows for the internal energy of the ionic system

$$\langle H_{\text{harm.}} \rangle (N, V, T) = \sum_{\mathbf{q}\in\mathrm{Q}} \sum_{\nu=1}^{rd_{\mathbf{x}}} \hbar\omega_{\nu\mathbf{q}} \left(\left\langle b^{\dagger}_{\nu\mathbf{q}} b_{\nu\mathbf{q}} \right\rangle + \frac{1}{2} \right) = \sum_{\mathbf{q}\in\mathrm{Q}} \sum_{\nu=1}^{rd_{\mathbf{x}}} \hbar\omega_{\nu\mathbf{q}} \left(\mathrm{B}_{\beta}(\hbar\omega_{\nu\mathbf{q}}) + \frac{1}{2} \right)$$

Proof 33.58 Follows immediately from the linearity of the expectation-value and Theorem 33.57.

33.5.1.3. The Low-Temperature Limit

Approximation 33.59 We assume that

$$k_{\mathrm{B}}T \ll \hbar\omega_{\text{opt,min}} \wedge k_{\mathrm{B}}T \ll \hbar\omega_{\text{ac,max}} \quad 1 \ll \beta\hbar\omega_{\text{opt,min}} \wedge 1 \ll \beta\hbar\omega_{\text{ac,max}} \tag{33.78}$$

Corollary 33.60 The contribution from the optical branches is negligible.

Proof 33.60 For an optical phonon branch $\nu \in \{d_{\mathbf{x}} + 1, \dots, rd_{\mathbf{x}}\}$ we obtain

$$0 \leq \sum_{\mathbf{q}\in Q} \hbar\omega_{\nu\mathbf{q}} \mathrm{B}_\beta(\hbar\omega_{\nu\mathbf{q}}) = \sum_{\mathbf{q}\in Q} \hbar\omega_{\nu\mathbf{q}} \frac{1}{\mathrm{e}^{\beta\hbar\omega_{\nu\mathbf{q}}} - 1} \leq \sum_{\mathbf{q}\in Q} \hbar\omega_{\nu\mathbf{q}} \frac{1}{\mathrm{e}^{\beta\hbar\omega_{\mathrm{opt,min}}} - 1} \approx \sum_{\mathbf{q}\in Q} \hbar\omega_{\nu\mathbf{q}} \mathrm{e}^{-\beta\hbar\omega_{\mathrm{opt,min}}} \stackrel{T\to 0}{\longrightarrow} 0 \tag{33.79}$$

Approximation 33.61 Recalling that

$$\hbar\omega_\nu(\mathbf{q}) \approx c_\nu q \quad \text{for } \nu \in \{1, \dots, d_{\mathbf{x}}\} \equiv \{\text{"indices for acoustic phonons"}\} \text{ and } q \approx 0 \tag{33.80}$$

we can also evaluate the contribution from the acoustic branches.

Corollary 33.62 In the low-temperature limit, the internal energy is dominated by the acoustic phonons, i.e.

$$E \stackrel{T\approx 0}{\equiv} \langle H_{\mathrm{harm.}}\rangle(N, V, T) \approx E_0 + \sum_{\nu=1}^{d_{\mathbf{x}}} \sum_{\mathbf{q}\in Q} c_\nu q \frac{1}{\mathrm{e}^{\beta c_\nu q} - 1} \tag{33.81}$$

Proof 33.62 ToBeFormulated Cleanly

Theorem 33.63 It follows

$$E - E_0 \approx \frac{\pi^2}{30} \frac{1}{\hbar^3 \overline{c^3}} N_p V_{\mathrm{uc}} k_{\mathrm{B}}^4 T^4 \tag{33.82}$$

Proof 33.63 TBDigitalized; Cf. Bulla Script.

33.5.1.4. The High-Temperature Limit

TBD.

33.5.1.5. Summary

Corollary 33.64 One finds for the internal energy of the system

$$\langle H_{\text{harm.}}\rangle(N,V,T) \equiv E(N,V,T)$$
$$= \begin{cases} E_0 + rd_{\mathbf{x}} N_p k_{\mathrm{B}} T & \text{if } \forall \underline{q} \in \Xi :\ k_{\mathrm{B}} T \gg \hbar\omega_{\underline{q}} \\ E_0 + \text{const.} \times V \times T^4 & \text{if both equations are fulfilled } \forall \underline{q} \in \Xi :\ k_{\mathrm{B}} T \ll \hbar\omega_{\underline{q}} \approx \hbar c q. \end{cases} \tag{33.83}$$

Proof 33.64 See below.

Remark 33.64.1 Notice that one gets for the specific heat capacity per volume

$$C_{N,V}(T) \equiv \partial_T \langle H_{\text{harm.}}\rangle(N,V,T)$$
$$\propto \begin{cases} \text{const.} & \text{if } \forall \underline{q} \in \Xi :\ k_{\mathrm{B}} T \gg \hbar\omega_{\underline{q}} \\ \text{const.} T^3 & \text{if both equations are fulfilled } \forall \underline{q} \in \Xi :\ k_{\mathrm{B}} T \ll \hbar\omega_{\underline{q}} \approx \hbar c q. \end{cases} \tag{33.84}$$

which reproduces the Dulong-Petit-law for high-temperatures and the experimental behaviour advertised in the beginning of the lecture for low-temperatures.

33.5.2. Intermezzo: The [G]rand [C]anonical [E]nsemble

33.5.2.1. Formalism

Notation 33.65 We further restrict our attention to the case in which the above mentioned system has a volume V, and is being treated in the grand canonical ensemble with temperature T and chemical potential μ that is being set such that the average particle number is fixed to be a constant (see below). We further assume the system to be described the Hilbert-space[5] $\mathscr{H}$. This gives us the three natural variables (μ, T, V).

Remark 33.65.1 Notice that, if we are for example treating a crystal, and we count the number of electrons that we would expect to be inside the crystal, then (in general) we have to *numerically* determine the chemical potential.

Fact 33.66 In the absence of „spontaneous symmetry breaking" expectation-values in the grand-canonical-ensemble are evaluated w.r.t. the state

$$\rho_{\mathrm{GCE}} = \frac{\mathrm{e}^{-\beta(H-\mu N)}}{\mathrm{Tr}_{\mathscr{H}}\, \mathrm{e}^{-\beta(H-\mu N)}} \equiv \frac{1}{Z_{\mathrm{GCE}}} \mathrm{e}^{-\beta(H-\mu N)}, \tag{33.85}$$

whereas we introduced the grand canonical partition-function $Z_{\mathrm{GCE}} \equiv Z_{\mathrm{GCE}}(\mu, T, V)$.

Remark 33.66.1 Recall that the state above represents a solution to the stationary von Neumann equation.

Remark 33.66.2 Although in the case of „spontaneous symmetry breaking", the evaluation of observables w.r.t. the state above yields incorrect results, one can oftentimes however use the state above, and break the underlying symmetry „by hand", in order to recover the „correct" physical state (or, in the case that we will encounter later, the correct physical „field theory") again.

Fact 33.67 Recall, that the thermodynamic potential in the grand canonical ensemble is given by the so-called grand canonical potential

$$\Omega \equiv \Omega(\mu, T, V) = -\frac{\mathrm{Log}_{\mathrm{e}}\, Z_{\mathrm{GCE}}(\mu, T, V)}{\beta} \tag{33.86}$$

from which one can conclude the usual relations for the conjugated variables particle-number, entropy and pressure

$$\frac{\partial \Omega}{\partial \mu} = -N, \qquad \frac{\partial \Omega}{\partial T} = -S, \qquad \frac{\partial \Omega}{\partial V} = -P. \tag{33.87}$$

[5] For indistinguishable particles, this Hilbert-space will be equal to a Fock-space $\mathscr{H} \equiv \mathscr{F}_{\zeta}$.

33.5.2.2. Application to the Electronic System

We treat the electronic system within the LDA-approximation of Quantum-Espresso[6].

Notation 33.68 We have the electronic density of states in a free electron gas given as obtained through quantum espresso as

$$\rho_{\text{aux}}(E) = \sum_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\downarrow,\uparrow\}\\ n\in\{1,2,\dots\}}} \delta(E - \epsilon_{n\mathbf{k}}) \tag{33.88}$$

Lemma 33.69 One obtains the formulae

$$\begin{aligned} \langle N\rangle &= \sum_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\downarrow,\uparrow\}\\ n\in\{1,2,\dots\}}} \mathrm{F}_\beta(\epsilon_{n\mathbf{k}\sigma} - \mu) &&= \int \mathrm{d}E\ \mathrm{F}_\beta(E-\mu)\rho_{\text{aux}}(E) \\ \langle H_{\text{el}}\rangle &= \sum_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\downarrow,\uparrow\}\\ n\in\{1,2,\dots\}}} \mathrm{F}_\beta(\epsilon_{n\mathbf{k}\sigma} - \mu)\epsilon_{n\mathbf{k}\sigma} &&= \int \mathrm{d}E\ \mathrm{F}_\beta(E-\mu)\rho_{\text{aux}}(E)E \end{aligned} \tag{33.89}$$

Proof 33.69 TBDigitalized.

Remark 33.69.1 Although the discrete formulae look very tempting and beautiful, they are oftentimes quite inconvenient to use in practice as they are difficult to use for reliable calculations. Therefore, it is oftentimes more convenient to instead use the integral formulae after one introduced a smearing in the density of states in Equation (33.88).

Remark 33.69.2 The smearing that one usually introduced can be explained on physical grounds. This becomes clear after we introduced the concept of self-energies for electron-phonon-systems.

Theorem 33.70 One finds for the heat-capacity $C_{V,N}(T)$ the equation

$$C_{V,N}(T) = \frac{\pi^2}{3}\rho_{\text{aux}}(E_{\text{F}})k_{\text{B}}^2 T \equiv \gamma T \tag{33.90}$$

with heat-capacity coefficient $\gamma = \frac{\pi^2}{3}\rho_{\text{aux}}(E_{\text{F}})k_{\text{B}}^2$.

Proof 33.70 TBDigitalized.

[6]The templates ...
- Cf. Schwabl page 186
- Bulla-Skript
- Czycholl-Book.

Corollary 33.71 Combining the results, we obtain an expression for the molar heat capacity

$$\begin{aligned}
C_{V,N,\text{molar}}(T) &= \frac{1}{n} \cdot C_{V,N}(T) \\
&= \frac{1}{N_x N_y N_z 1.66 \cdot 10^{-24}\,\text{mol}} \cdot \frac{\pi^2}{3} \rho_{\text{aux}}(0) k_{\text{B}}^2 T \\
&= \frac{\rho_{\text{aux}}(0)}{N_x N_y N_z} \cdot \frac{\pi^2}{3} \frac{k_{\text{B}}^2}{1.66 \cdot 10^{-24}\,\text{mol}} \cdot T \\
&= \rho_{\text{QE}}(E_{\text{F}}) \cdot \frac{\pi^2}{3} \frac{k_{\text{B}}^2}{1.66 \cdot 10^{-24}\,\text{mol}} \cdot T \\
&\equiv \gamma_m \cdot T
\end{aligned}$$

with the final expression for γ_m as

$$\gamma_m = \rho_{\text{QE}}(E_{\text{F}}) \cdot \frac{\pi^2}{3} \frac{k_{\text{B}}^2}{1.66 \cdot 10^{-24}\,\text{mol}} \tag{33.91}$$

with ρ_{QE} as in Equation (23.4).

Proof 33.71 Follows from combining Equation (33.90) with Equation (9.54).

33.5.3. Thermodynamics of a Crystal in the Non-Interacting Gases Approximation

$$\begin{aligned} H_{\text{tot}} &\equiv H_{\text{el}} + H_{\text{ph}} + H_{\text{el-ph-int}} \\ &\equiv \sum_{\substack{\mathbf{k}\in Q \\ \sigma\in\{\downarrow,\uparrow\} \\ n\in\{1,2,\dots\}}} \epsilon_{n\mathbf{k}} c^{\dagger}_{n\sigma\mathbf{k}} c_{n\sigma\mathbf{k}} + \sum_{\substack{\mathbf{k}\in Q \\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \hbar\omega_{\nu\mathbf{k}} b^{\dagger}_{\nu\mathbf{k}} b_{\nu\mathbf{k}} + 0 \end{aligned}$$

with allowed wave-vectors $Q \subset$ "First Brillouin Zone". Notice that the total Hilbert-space $\mathscr{H}$ of the system is given as tensor-product

$$\mathscr{H} = \mathscr{H}_{\text{F}} \otimes \mathscr{H}_{\text{B}}$$

with basis and scalar-product inherited from the respective many-identical-particle-Hilbert-spaces as usual. Modelling the system as being connected to a heat-reservoir (the environment dictating the temperature), we work in a hybrid ensemble with thermodynamic variables (T, V, μ, N_p), expectation-values are evaluated w.r.t. the stationary solution of the von Neumann equation. Concretely one has for an observable quantity (expressed as a polynomial expression in second quantized operators) A the expectation-value

$$\langle A\rangle \equiv \frac{\operatorname{Tr}_{\mathscr{H}} A\, \mathrm{e}^{-\beta\left(H_{\text{el}}-\mu N_{\text{el}}+H_{\text{ph}}\right)}}{\operatorname{Tr}_{\mathscr{H}} \mathrm{e}^{-\beta\left(H_{\text{el}}-\mu N_{\text{el}}+H_{\text{ph}}\right)}} \equiv \frac{\operatorname{Tr}_{\mathscr{H}} A\, \mathrm{e}^{-\beta\left(H_{\text{el}}-\mu N_{\text{el}}+H_{\text{ph}}\right)}}{Z}. \tag{33.92}$$

Other thermodynamic properties (such as pressure) are determined by derivatives of the hybrid partition-function.

Combining the results, we obtain for the heat capacity at low temperatures

$$C_{V,N,\text{molar}}(T) = \frac{1}{n} \cdot C_{V,N}(T) \approx \gamma_m \cdot T + A \cdot T^3$$

in excellent agreement with several experimental results.

The folklore statement that the chemical potential of phonons is zero is correct, if one treats the phononic system in the grand canonical ensmeble. Notice however that in our derivation it became evident that we are *actually* working in the canonical ensemble. There is a possibility to define a chemical potential for the phononic system, which does not neccessarily need to be zero. Treating the phononic system in the grand canonical ensemble and agreeing to set $\mu = 0$ is just a mnemonic, which is quite common in the literature, but it is not a fundamental statement.

Addendum: The Chemical Potential of Bosons.

Here is a (hopefully convincing) selection of excerpts supporting this claim.

als

$$H_{\text{harm}} = \sum_{\boldsymbol{q}} \sum_{j=1}^{dr} \hbar\omega_j(\boldsymbol{q}) \left(b^\dagger_{\boldsymbol{q}j} b_{\boldsymbol{q}j} + \frac{1}{2} \right) . \tag{4.41}$$

Hierbei erfüllen die Oszillator-Auf- und Absteige-Operatoren $b^{(\dagger)}_{\boldsymbol{q}j}$ gemäß (4.15) die Vertauschungsrelation

$$\left[b_{\boldsymbol{q}j}, b^\dagger_{\boldsymbol{q}'j'} \right] = \delta_{\boldsymbol{q}\boldsymbol{q}'}\delta_{jj'} , \quad \left[b^\dagger_{\boldsymbol{q}j}, b^\dagger_{\boldsymbol{q}'j'} \right] = \left[b_{\boldsymbol{q}j}, b_{\boldsymbol{q}'j'} \right] = 0 . \tag{4.42}$$

Dies sind die Vertauschungsrelationen für Bose[2]-Erzeugungs- und Vernichtungs-Operatoren. Die Elementaranregungen des Hamilton-Operators in harmonischer Näherung, also die quantisierten Normalschwingungen, verhalten sich also wie die eines Systems **wechselwirkungsfreier Bosonen**. Diese nennt man auch **Phononen**. Phononen sind keine wirklichen Teilchen sondern **Quasiteilchen**. Insbesondere gibt es keine Teilchenzahlerhaltung (und deshalb auch kein chemisches Potential). Auch in dieser Hinsicht verhalten sich Phononen analog wie Photonen (Quanten des elektromagnetischen Feldes). In der Tat weist auch die Thermodynamik der Phononen (quantisierten Gitterwellen) eine enge Analogie zur Thermodynamik des Photonengases (Hohlraum-Strahlung, quantisierte elektromagnetische Wellen) auf.

Figure 33.1.: An excerpt from the book from Czycholl.

Chemisches Potential: Das chemische Potential des Photongases kann aus der Gibbs-Duhem-Relation $E = TS - PV + \mu N$ berechnet werden, da es sich um ein homogenes System handelt:

206 4. Ideale Quanten-Gase

$$\mu = \frac{1}{N}(E - TS + PV) = \frac{1}{N}\left(4 - \frac{16}{3} + \frac{4}{3}\right)\frac{\sigma V T^3}{3c} \equiv 0 . \tag{4.5.22}$$

Das chemische Potential des Photonengases ist für alle Temperaturen identisch 0, da die Zahl der Photonen nicht fest ist, sondern sich an die Temperatur und das Volumen anpaßt. Photonen werden von der umgebenden Materie, den Wänden des Hohlraums, absorbiert und emittiert. Generell verschwindet das chemische Potential von Teilchen und von Quasiteilchen, wie z.B. Phononen, deren Teilchenzahl durch keinen Erhaltungssatz eingeschränkt ist. Betrachten wir nämlich die freie Energie für eine fiktive feste Zahl von Photonen (Phononen etc.) $F(T, V, N_{\text{Ph}})$. Da die Zahl der Photonen (Phononen) nicht eingeschränkt ist, adjustiert sie sich so, daß die freie Energie minimal wird $\left(\frac{\partial F}{\partial N_{\text{Ph}}}\right)_{T,V} = 0$. Das ist aber gerade der Ausdruck für das chemische Potential, welches somit verschwindet: $\mu = 0$. Ebenso hätten wir von der Maximalität der Entropie ausgehen können $\left(\frac{\partial S}{\partial N_{\text{Ph}}}\right)_{E,V} = -\frac{\mu}{T} = 0$.

Figure 33.2.: An excerpt from the book from Schwabl.

particle reservoirs, or a subsystem is considered. The system can exchange particles with the reservoirs which are described by their chemical potentials (assuming in general several particle species present). This feature is simply included by introducing Lagrange multipliers, i.e. tacitly understanding that single-particle energies are measured relative to their chemical potential, $H \to H - \sum_s \mu_s N_s$. In this case, the partition function instead of being a function of the number of particles, is a function of the chemical potentials for the species in question

$$Z_{\text{gr}}(T, \mu_s) = \text{Tr}\, e^{-(\mathcal{H} - \mu_s N_s)/kT} = e^{-\Omega(T,\mu_s)/kT} \tag{2.90}$$

specified by the average number of particles according to

$$\langle N_s \rangle = -\left(\frac{\partial \Omega}{\partial \mu_s}\right)_{T,V} . \tag{2.91}$$

For systems of particles where the total number of *particles* is not conserved, i.e. where the Hamiltonian and the total number operator do not commute, such as for phonons and photons, the chemical potential vanishes, and the grand canonical ensemble is employed. For degenerate fermions, such as electrons in a metal, the

Figure 33.3.: An excerpt from the book from Rammer.

In thermal equilibrium, the average number of particles in a state λ is given by the usual boson occupation factor:

$$\langle n_\lambda \rangle = \frac{1}{e^{\beta(\varepsilon_\lambda - \mu)} - 1} \equiv N_\lambda = n_B(\varepsilon_\lambda - \mu) \tag{1.118}$$

Now there is a chemical potential, μ, which can vary with temperature and concentration. It is absent in the phonon, and photon, cases because these excitations do not conserve particle number. One may make as many phonons or photons as one wishes. Another operator of interest is the density operator:

Figure 33.4.: An excerpt from the book from Mahan.

33.6. The Wave-like / Massless Nature of the Phononic Quantum Field Theory

Theorem 33.72 Going over to the field-integral-description of the problem gives the action

$$S_{\text{ph}}[\phi, \overline{\phi}] = \sum\!\!\!\!\!\!\int_{\substack{\mathbf{q}\in Q \\ \tau\in(0,\hbar\beta) \\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}}} \overline{\phi}_\nu(\tau, \mathbf{q}) \left(\partial_\tau + \omega_\nu(\mathbf{q})\right) \phi_\nu(\tau, \mathbf{q}) \tag{33.93}$$

and switching to the Matsubara-representation yields

$$S_{\text{ph}}[\phi, \overline{\phi}] = \sum\!\!\!\!\!\!\int_{\substack{\mathbf{q}\in Q \\ \omega_n\in\Omega_{+1} \\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}}} \overline{\phi}_\nu(\mathrm{i}\omega_n, \mathbf{q}) \left(-\mathrm{i}\omega_n + \omega_\nu(\mathbf{q})\right) \phi_\nu(\mathrm{i}\omega_n, \mathbf{q}) \tag{33.94}$$

$$\equiv \sum\!\!\!\!\!\!\int_{\substack{q\in \mathrm{i}\Omega_{+1}\times Q \\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}}} \overline{\phi}_\nu(q) \left(-\mathrm{i}\omega_n + \omega_\nu(\mathbf{q})\right) \phi_\nu(q) \tag{33.95}$$

whereas we introduced the four-momentum $q \equiv (\mathbf{q}, \mathrm{i}\omega_n)$

Proof 33.72 Follows from Corollary 19.15.

Lemma 33.73 We introduce now the auxiliary fields

$$A_\nu(q) = \phi_\nu(q) + \overline{\phi}_\nu(-q), \qquad \Pi_\nu(q) = \phi_\nu(q) - \overline{\phi}_\nu(-q)$$

with inverse transformation

$$\phi(q) = \frac{A_\nu(q) + \Pi_\nu(q)}{2}, \qquad \overline{\phi}(q) = \frac{A_\nu(-q) - \Pi_\nu(-q)}{2}$$

Proof 33.73 TBD.

Remark 33.73.1 Notice that the fields defined like this are not independent from each other; One finds symmetries

$$A_\nu(q) = \overline{A}_\nu(-q), \qquad \Pi_\nu(q) = -\overline{\Pi}_\nu(-q). \tag{33.96}$$

In order to be able to change coordinates without double-counting, we restrict attention to the variables

$$\left\{A_\nu(q), \Pi_\nu(q)\right\}_{\substack{q\in \mathrm{i}\Omega_{+1}\times \mathrm{Q}/\sim \\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}}} \quad \text{with} \quad q\sim q' \;:\Longleftrightarrow\; q=-q'. \tag{33.97}$$

Part 1: No External Sources

Theorem 33.74 Changing integration-variables according to Equations (TBD) and integrating out the Π-fields yields the action

$$S_{\mathrm{ph}}[A,\overline{A}] = \sum_{\substack{\nu\in\{1,\ldots,rd_{\mathbf{x}}\} \\ q\in \mathrm{i}\Omega_{+1}\times \mathrm{Q}/\sim}} \overline{A}_\nu(\mathrm{i}\omega_n,\mathbf{q}) \frac{\omega_n^2+\omega_\nu(\mathbf{q})^2}{2\omega_\nu(\mathbf{q})} A_\nu(\mathrm{i}\omega_n,\mathbf{q}) \tag{33.98}$$

Proof 33.74 See Appendix, Proofs, page 376.

Remark 33.74.1 It will turn out, that all the interacting theories that one encounters in materials theory are independent of the Π-fields, such that integrating them out is a legitimate step.

Corollary 33.75 It follows

$$\mathscr{D}_{\nu\nu'}(\mathrm{i}\omega_n,\mathbf{q}) = -\frac{1}{\hbar}\frac{2\omega_\alpha\delta_{\alpha,\alpha'}}{\omega_n^2+\omega_\alpha^2} \tag{33.99}$$

from matrix-inversion.

Proof 33.75 Follows from the identity in Theorem TBD.

Remark 33.75.1 Notice that we derived this identity earlier by straight-forward computation of expectation-values, in Theorem TBReferenced.

TBElaborated.

Part 2: Incorporating External Sources

Theorem 33.76 The action in presence of external sources reads

$$\underset{[j,\bar{j}]}{S_{\text{ph}}}[\phi,\overline{\phi}] = \sum\!\!\!\!\!\!\int_{\substack{\mathbf{q}\in \mathrm{Q} \\ \tau\in(0,\hbar\beta) \\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} \overline{\phi}_\nu(\tau,\mathbf{q})\left(\partial_\tau + \omega_\nu(\mathbf{q})\right)\phi_\nu(\tau,\mathbf{q}) \quad + \quad \sum\!\!\!\!\!\!\int_{\substack{\mathbf{q}\in \mathrm{Q} \\ \tau\in(0,\hbar\beta) \\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} \left[\overline{\phi}_\nu(\tau,\mathbf{q})j(\tau,\mathbf{q}) + \bar{j}(\tau,\mathbf{q})\phi_\nu(\tau,\mathbf{q})\right] \tag{33.100}$$

and switching to the Matsubara-representation yields

$$S_{\text{ph}}[\phi,\overline{\phi}] = \sum\!\!\!\!\!\!\int_{\substack{q\in \mathrm{i}\Omega_{+1}\times\mathrm{Q} \\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} \overline{\phi}_\nu(q)\left(-\mathrm{i}\omega_n + \omega_\nu(\mathbf{q})\right)\phi_\nu(q) \quad + \quad \sum\!\!\!\!\!\!\int_{\substack{q\in \mathrm{i}\Omega_{+1}\times\mathrm{Q} \\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}}} \left[\overline{\phi}_\nu(q)j(q) + \bar{j}(q)\phi_\nu(q)\right] \tag{33.101}$$

whereas we introduced the four-momentum $q \equiv (\mathbf{q}, \mathrm{i}\omega_n)$

33.7. The Equation-of-Motion-Technique as double-edged sword

Proof 33.1 of Lemma ??

First note that

$$\mathrm{i}\partial_t G_{\alpha,\alpha'}(t,t') = \partial_t \Big[\theta(t-t')a_\alpha(t)a^\dagger_{\alpha'}(t') - \theta(t'-t)a^\dagger_{\alpha'}(t')a_\alpha(t)\Big] \tag{33.102}$$

Now using the CAR and Heisenbergs equation of motion (cf. eq. (8.9)) and the definition of the time-ordering-operator (backwards) gives the desired result.

Corollary 33.77 For non-interacting-systems (i.e. quadratic Hamiltonians as in eq. (**??**)) one finds the equation of motion

$$\begin{aligned}
(\omega+\mathrm{i}\delta)\ G^{\mathrm{r}}_{\alpha\alpha'}(\omega+\mathrm{i}\delta) &= \delta_{\alpha\alpha'} + \sum_\beta \frac{h_{\alpha\beta}-\mu\delta_{\alpha\beta}}{\hbar} G^{\mathrm{r}}_{\beta\alpha'}(\omega+\mathrm{i}\delta)\\
\Longleftrightarrow (\omega+\mathrm{i}\delta)\ \mathbf{G}^{\mathrm{r}}(\omega+\mathrm{i}\delta) &= \mathbb{1} + \frac{\mathbf{H}-\mu\mathbb{1}}{\hbar}\mathbf{G}^{\mathrm{r}}(\omega+\mathrm{i}\delta)\\
\Longleftrightarrow \mathbf{G}^{\mathrm{r}}(\omega+\mathrm{i}\delta) &= \left[\omega+\mathrm{i}\delta - \frac{\mathbf{H}-\mu\mathbb{1}}{\hbar}\right]^{\mathrm{inv}}
\end{aligned}$$

Proof 33.77 TBD.

Remark 33.77.1 By straight forward calculation, one also sees, that the retarded ("+") and advanced ("−") Green-functions obey the very same equation of motion in eq. (**??**), after replacing $\omega \mapsto \omega \pm \mathrm{i}\delta$ one finds similar solutions as in Corollary 33.77.

Proof 33.77.1 TBD.

Remark 33.77.2 The fact, that three Green-functions obey the same equation of motion makes me (personally) a little bit uncomfortable; In [Francesco 04] one comments this in the following way (with adapted notation for better understanding)

> *(...) On the other hand, the detailed time-dependence of $G^r(t)$, $G^a(t)$ and $G(t)$ is obviously very different. From the definition of these quantities follows for example that $G^r(t<0)\equiv 0$, while $G^a(t>0)\equiv 0$. Thus, if one wants to determine the correct solution of the equation of motion, one must incorporate those boundary conditions appropriately. This implies that the Fourier-transformation in eq. (...) has to be performed with some care.*

And nonetheless, despite these warnings, one usually determines the retarded- and advanced-Green-functions by the equation-of-motion-technique.

Update: It can be proven (from the functional integral formalism) that the equation-of-motion-technique yields the correct result for the retarded- and advanced-Green-functions. But **not** for the time-ordered Green-function.

33.8. The Curious Formalism of Position Eigenstates

In materials theory, it is unfortunately common to invoke the concept of so-called position eigenstates. We will elucidate the concept here, and then make comments about their usefulness.

Notation 33.78 We consider the situation in Notation 10.1. Let

$$\mathcal{H} = \mathrm{span}_{\mathbb{C}}\{|\alpha\rangle \quad | \quad \alpha \in \mathscr{A}\} \tag{33.103}$$

be a single-particle Hilbert-space with orthonormal basis. Let $\{c_\alpha, c_\alpha^\dagger\}_{\alpha\in\mathscr{A}}$ denote the corresponding creation- and annihilation-operators obeying the statistical commutation-relations.

Definition 33.79 One introduces the so-called *position eigenstates*

$$\begin{aligned}\psi(x) &= \sum_{\alpha\in\mathscr{A}} \langle\, x \,|\, \alpha \,\rangle c_\alpha = \sum_{\alpha\in\mathscr{A}} \phi_\alpha(x) c_\alpha,\\ \psi^\dagger(x) &= \sum_{\alpha\in\mathscr{A}} \langle\, \alpha \,|\, x \,\rangle c_\alpha^\dagger = \sum_{\alpha\in\mathscr{A}} \phi_\alpha^*(x) c_\alpha^\dagger.\end{aligned}$$

Remark 33.79.1 The usual interpretation is that these operators create or annihilate a particle in a position-/spin-state x.

Lemma 33.80 One finds the commutation-relations

$$[\psi(x),\, \psi(x')]_\zeta = 0 = [\psi^\dagger(x),\, \psi^\dagger(x')]_\zeta, \qquad [\psi(x),\, \psi^\dagger(x')]_\zeta = \delta(x-x') \tag{33.104}$$

Proof 33.80 We compute straightforwardly

$$\begin{aligned}[\psi(x),\, \psi^\dagger(x')]_\zeta &= \sum_{\alpha,\alpha'\in\mathscr{A}} \phi_\alpha(x)\phi_{\alpha'}^*(x)[c_\alpha, c_{\alpha'}^\dagger]_\zeta\\ &= \sum_{\alpha,\alpha'\in\mathscr{A}} \phi_\alpha(x)\phi_{\alpha'}^*(x)\delta_{\alpha,\alpha'}\\ &= \sum_{\alpha\in\mathscr{A}} \phi_\alpha(x)\phi_\alpha^*(x)\\ &= \langle\, x \,|\, \sum_{\alpha\in\mathscr{A}} |\, \alpha \,\rangle\langle\, \alpha \,|\,|\, x' \,\rangle\\ &= \delta(x-x')\end{aligned}$$

whereas we used the completeness-relation of the basis.

Lemma 33.81 The inverse transformation reads

$$c_\alpha^\dagger = \int_y \phi_\alpha(y)\,\psi^\dagger(y), \qquad c_\alpha = \int_y \phi_\alpha^*(y)\,\psi(y) \tag{33.105}$$

Proof 33.81 We compute straightforwardly

$$\psi^\dagger(x) = \sum_{\alpha\in\mathscr{A}} \phi_\alpha^*(x) c_\alpha^\dagger = \sum_{\alpha\in\mathscr{A}} \int_y \phi_\alpha^*(x)\phi_\alpha(y)\psi^\dagger(y) = \int_y \sum_{\alpha\in\mathscr{A}} \langle y|\alpha\rangle\langle\alpha|x\rangle\psi^\dagger(y) = \int_y \delta(y-x)\psi^\dagger(y) = \psi^\dagger(x)$$

Lemma 33.82 If $x \equiv (\mathbf{x}, s)$ represents a combination of a spatial- and a spin-coordinate, then the kinetic energy operator $T \equiv -\frac{\hbar^2}{2m}\Delta$ and a local one-body potential operator $\mathbf{x} \mapsto U(\mathbf{x})$ and a two-body operator V can be written in second quantization as

- $T = \frac{\hbar^2}{2m} \sumint_x \left(\nabla\psi^\dagger(x)\right)(\nabla\psi(x))$
- $U = \sumint_x \psi^\dagger(x)U(x)\psi(x)$
- $V = \frac{1}{2} \sumint_{x,x'} \psi^\dagger(x)\psi^\dagger(x')V(x,x')\psi(x')\psi(x)$

which gives an expression for the total Hamiltonian rewritten w.r.t. field-operators as

$$\begin{aligned} H &= T + U + V \\ &= \frac{\hbar^2}{2m} \sumint \Big[\left(\nabla\psi^\dagger(x)\right)(\nabla\psi(x)) + \psi^\dagger(x)U(x)\psi(x) \Big] \quad + \quad \frac{1}{2} \sumint_{x,x'} \psi^\dagger(x)\psi^\dagger(x')V(x,x')\psi(x')\psi(x) \end{aligned}$$

Proof 33.82 This is proven by straight-forward computation. For the kinetic energy operator we obtain

$$T = \sum_{i,j} c_i^\dagger T_{ij} c_j = \sum_{i,j} c_i^\dagger \langle i| - \hbar^2 \frac{\Delta}{2m} |j\rangle c_j = \sum_{i,j} \sumint_x c_i^\dagger \phi_i^*(x) \left(-\hbar^2 \frac{\Delta}{2m}\right) \phi_j(x) c_j = \frac{\hbar^2}{2m} \sumint \left(\nabla\psi^\dagger(x)\right)(\nabla\psi(x))$$

therefore we obtain for a one-body operator

$$U = \sum_{i,j} c_i^\dagger U_{ij} c_j = \sum_{i,j} c_i^\dagger \langle i|U|j\rangle c_j = \sum_{i,j} \sumint_x c_i^\dagger \phi_i^*(x) U(x) \phi_j(x) c_j = \sumint \psi^\dagger(x)U(x)\psi(x)$$

and for the two-body interaction

$$
\begin{aligned}
V = \frac{1}{2}\sum_{i,j,k,m} c_i^\dagger c_j^\dagger V_{ijkm} c_m c_k &= \frac{1}{2}\sum_{i,j,k,m} c_i^\dagger c_j^\dagger \langle i,j|V|k,m\rangle c_m c_k \\
&= \frac{1}{2}\sum_{i,j,k,m} \sum\!\!\!\!\!\!\!\int_{x,x'} c_i^\dagger c_j^\dagger \phi_i^*(x)\phi_j^*(x')V(x,x')\phi_k(x)\phi_m(x')c_m c_k \\
&= \frac{1}{2} \sum\!\!\!\!\!\!\!\int_{x,x'} \psi^\dagger(x)\psi^\dagger(x')V(x,x')\psi(x')\psi(x)
\end{aligned}
$$

Lemma 33.83 In the momentum-basis the kinetic energy operator acquires the form [Negele Eq. (1.100)].

Proof 33.83 TBD, or maybe not since it is so useless?

Definition 33.84 A two-body operator V that is diagonal in its position representation as

$$
\langle \mathbf{x}_1\mathbf{x}_2|V|\mathbf{x}_3\mathbf{x}_4\rangle = \delta(\mathbf{x}_1-\mathbf{x}_3)\delta(\mathbf{x}_2-\mathbf{x}_4)V(\mathbf{x}_1-\mathbf{x}_2) \tag{33.106}
$$

is said to be *local*[7], or *velocity independent*.

Corollary 33.85 Let V denote a two-body operator which is diagonal in the position-representation. Then, it becomes in second quantized form

$$
V = \frac{1}{2}\int \mathrm{d}^3x\mathrm{d}^3y\, V(\mathbf{x}-\mathbf{y})\,\psi^\dagger(\mathbf{x})\psi^\dagger(\mathbf{y})\psi(\mathbf{y})\psi(\mathbf{x}) \tag{33.107}
$$

Proof 33.85 TBD

[7] Cf. Marcos Marinos p. 46 ff

Identity for expectation-value of the Hamilton-operator as in Problem 1.6 of Negele.

Definition 33.86 Now we introduce the field operator in the Heisenberg-representation as

$$\psi(\mathbf{x},t) = \mathrm{e}^{\mathrm{i}Ht/\hbar}\psi(\mathbf{x},0)\mathrm{e}^{-\mathrm{i}Ht/\hbar} \tag{33.108}$$

Lemma 33.87 It follows the equation of motion

$$\mathrm{i}\partial_t\psi(\mathbf{x},t) = \left(-\frac{\hbar^2}{2m}\Delta + U(\mathbf{x})\right)\psi(\mathbf{x},t) + \int \mathrm{d}^3x'\psi^\dagger(\mathbf{x}',t)V(\mathbf{x},\mathbf{x}')\psi(\mathbf{x}',t)\psi(\mathbf{x},t) \tag{33.109}$$

Proof 33.87 TBDigitalized from Schwabl p. 24

Green-Functions

Definition 33.88 We introduce the Green-function of the field-operators as

$$G_{\sigma,\sigma'}(\mathbf{x}-\mathbf{x}',t-t') = \langle\psi^\dagger_\sigma(\mathbf{x},t)\psi_{\sigma'}(\mathbf{x}',t')\rangle \tag{33.110}$$

Remark 33.88.1 From this definition, it is actually quite non-trivial that the defined Green-function only depends on space- and time-differences. The time-difference is proven analog to Lemma TBReferenced. The space-difference can be seen as follows TBDone.

- See Schwabl 2 Eq. (2.1.6), GW01 Eq. (6) for the concrete values at equal times at zero temperature.
- Flensberg (somewhere): real-space version of the green-functions in the non-interacting electron-case.
- Imaginary time Green-function in coordinate space: Henk p. 145.

Corollary 33.89 The Green-function obeys the equation of motion

$$TBD. \tag{33.111}$$

Proof 33.89 See GW01 Eq. B(9)

Lemma 33.90 The Green-function of the field-operators has the Lehmann-representation

See GW01 Eq. (7)

Self-Energy

TBDigitalized from GW01 Eq. (81), (82).

Theorem 33.91 The Green-function can be obtained from Hedins-equations

$$TBD. \tag{33.112}$$

Proof 33.91 See GW01 Appendix.

Notice that the differential equation for the finite temperature Green-function now has the form of a heat-equation

$$TBD. \tag{33.113}$$

Example: Non-Interacting Electrons

Marcos Marinos QFT-notes p. 46f: Two-Body-Interactions in real and momentum-space

TBDigitalized from Schwabl p. 35.

Example: Free Electrons with Two-Body-Interactions

Now one can also introduce the Fourier-transform of these operators as in [Negele, page 18]. This is the so-called momentum-basis. Cf. Schwabl p. 25 onwards.

See also Schwabl p. 46 for the Coulomb-interaction in second quantization.

33.9. Generalized Matrices

As we will shortly see, we can catch two birds with one stone by introducing the following abbreviations.

Notation 33.92 We introduce a multi-index with a time-argument

$$A \equiv (\alpha, \tau) \quad \in \quad \mathscr{A} \times (0, +\hbar\beta) \equiv I_\tau$$

and consider generalized matrices with indices

$$\begin{aligned} \mathscr{G} :\ & I_\tau \times I_\tau \longrightarrow \mathbb{C} \\ & (A, A') \longmapsto \mathscr{G}(A, A') \equiv \mathscr{G}_{\alpha,\alpha'}(\tau, \tau'). \end{aligned}$$

Definition 33.93 Completely analogous to usual matrices, one defines a generalized trace and generalized matrix-multiplication[8] as

[8]This is a combination of ordinary matrix-multiplication for the α in the multi-index, and a convolution for the time-argument τ.

$$\left(\mathscr{G} \cdot \mathscr{G}'\right)(A, A') \equiv \sum\!\!\!\!\!\!\int_{B \in I_\tau} \mathscr{G}(A, B)\,\mathscr{G}'(B, A')$$

Remark 33.93.1 Note that the inversion-relation now reads

$$\left(\mathscr{G} \cdot \mathscr{G}^{-1}\right)(A, A') = \delta_{\alpha,\alpha'}\,\delta(\tau - \tau') \equiv \mathrm{Id}$$

Definition 33.94 We call a generalized matrix a *propagator in thermal equilibrium with statistics* ζ, if there exists a periodic standard-continuation[9] (denoted by the same symbol) on the temporal-axis

$$\mathbb{R} \ni \tau \mapsto \mathscr{G}_{\alpha,\alpha'}(\tau) \in \mathbb{C}$$

s.t. the following equations holds

$$\forall (\tau, \tau') \in \mathbb{R}^2 \,:\, \mathscr{G}_{\alpha,\alpha'}(\tau, \tau') = \mathscr{G}_{\alpha,\alpha'}(\tau - \tau') = \zeta \mathscr{G}_{\alpha,\alpha'}(\tau - \tau' + \hbar\beta)$$

Remark 33.94.1 We denote the set of all [P]ropagators [I]n [T]hermal [E]quilibrium with statistics ζ as PITE_ζ.

Remark 33.94.2 Note that the analytic standard-continuation of a PITE_ζ can be expanded with Matsubara-frequencies according to eq (TBD).

Theorem 33.95 Let $\mathscr{G}, \mathscr{G}' \in \mathrm{PITE}_\zeta$. Then it follows $\mathscr{G} \cdot \mathscr{G}' \in \mathrm{PITE}_\zeta$.

Proof 33.95 See Waste-Folder 1, topic 13, page 1 for notes on the substitution.

One finds by immediate calculation

$$\begin{aligned}
\left(\mathscr{G} \cdot \mathscr{G}'\right)(\tau_1, \tau_2) &= + \int_0^{+\hbar\beta} \mathrm{d}\tau\, \mathscr{G}(\tau_1 - \tau)\mathscr{G}'(\tau - \tau_2) \\
&= + \int_0^{+\hbar\beta} \mathrm{d}\tau\, \mathscr{G}(\tau_1 - \tau)\mathscr{G}'(\tau_1 - \tau_2 - (\tau_1 - \tau)) \\
&= - \int_{\tau_1}^{\tau_1 - \hbar\beta} \mathrm{d}\tau' \mathscr{G}(\tau')\mathscr{G}'(\tau_1 - \tau_2 - \tau') \\
&= + \int_0^{+\hbar\beta} \mathrm{d}\tau' \mathscr{G}(\tau')\mathscr{G}'(\tau_1 - \tau_2 - \tau')
\end{aligned}$$

[9] Construction for the anti-periodic case: TBD .

where we used the substitution $\tau' = \tau_1 - \tau$ in the second last step and we used Waste-Folder 1, Digital Waste-Folder 1 with Useful Links, topic 4 in the last step. Now it is clear, that the product depends only on the time-difference and that it obeys the same periodicity as the respective Green-function. Therefore there will be a periodic standard-continuation with the desired properties.

Corollary 33.96 It follows for the Matsubara-component of the product

$$\left(\mathscr{G} \cdot \mathscr{G}'\right)(\mathrm{i}\omega_n) = \mathscr{G}(\mathrm{i}\omega_n) \cdot \mathscr{G}'(\mathrm{i}\omega_n). \tag{33.114}$$

Proof 33.96 See Waste-Folder 1, topic 13?

Remark 33.96.1 This theorem will turn out as a central result for our later discussions with the Dyson-equation of interacting systems; It implies that even if we take only a finite number of contributions from the infinite series in the Dyson-equation, we will still end up with a physically reasonable propagator.

Theorem 33.97 Under the assumption, that the Dyson-equation holds

$$\mathscr{G}(\mathrm{i}\omega_n) = \mathscr{G}_0(\mathrm{i}\omega_n) + \left(\mathscr{G}_0 \cdot \Sigma \cdot \mathscr{G}\right)(\mathrm{i}\omega_n),$$

we obtain a corresponding equation with generalized matrix-product for the imaginary-time-version of the propagators.

Proof 33.97 Cf. Waste-Folder 1, topic 13, page 2,3 for further notes on the substitution

We perform the straight-forward calculation

$$\begin{aligned}
\mathscr{G}(\tau, \tau') &= \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} \left(\mathscr{G}_0(\mathrm{i}\omega_n) + \mathscr{G}_0(\mathrm{i}\omega_n)\Sigma(\mathrm{i}\omega_n)\mathscr{G}(\mathrm{i}\omega_n)\right) \mathrm{e}^{-\mathrm{i}\omega_n(\tau-\tau')} \\
&= \mathscr{G}_0(\tau, \tau') + \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} \int_0^{+\hbar\beta} \mathrm{d}\tau_1 \mathscr{G}_0(\tau_1)\, \mathrm{e}^{\mathrm{i}\omega_n\tau_1} \Sigma(\mathrm{i}\omega_n) \int_0^{+\hbar\beta} \mathrm{d}\tau_3 \mathscr{G}(\tau_3)\, \mathrm{e}^{\mathrm{i}\omega_n\tau_3}\, \mathrm{e}^{-\mathrm{i}\omega_n(\tau-\tau')} \\
&= \mathscr{G}_0(\tau, \tau') + \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} \int_\tau^{\tau+\hbar\beta} \mathrm{d}\tilde{\tau}_1 \mathscr{G}_0(\tau - \tilde{\tau}_1)\, \mathrm{e}^{\mathrm{i}\omega_n(\tau-\tilde{\tau}_1)} \Sigma(\mathrm{i}\omega_n) \int_0^{+\hbar\beta} \mathrm{d}\tau_3 \mathscr{G}(\tau_3)\, \mathrm{e}^{\mathrm{i}\omega_n(\tau_3-\tau+\tau')} = \ldots
\end{aligned}$$

whereas we used the substitution $\tau_1 = \tau - \tilde{\tau}_1$ in the last step. We now use the substitution $\tau_3 = \tilde{\tau}_3 - \tau'$ for the second integral and subsequently Waste-Folder 1, UL, topic 4 which yields

$$
\begin{aligned}
\ldots &= \mathscr{G}_0(\tau,\tau') + \frac{1}{\hbar\beta}\sum_{\omega_n\in\Omega_\zeta}\int_{\tau}^{\tau+\hbar\beta} \mathrm{d}\tilde{\tau}_1 \mathscr{G}_0(\tau-\tilde{\tau}_1)\, \mathrm{e}^{\mathrm{i}\omega_n(\tau-\tilde{\tau}_1)}\Sigma(\mathrm{i}\omega_n)\int_{\tau'}^{+\hbar\beta+\tau'} \mathrm{d}\tilde{\tau}_3 \mathscr{G}(\tilde{\tau}_3-\tau')\, \mathrm{e}^{\mathrm{i}\omega_n(\tilde{\tau}_3-\tau'-\tau+\tau')} \\
&= \mathscr{G}_0(\tau,\tau') + \int_0^{+\hbar\beta} \mathrm{d}\tilde{\tau}_1 \mathscr{G}_0(\tau-\tilde{\tau}_1)\,\frac{1}{\hbar\beta}\sum_{\omega_n\in\Omega_\zeta}\Sigma(\mathrm{i}\omega_n)\mathrm{e}^{-\mathrm{i}\omega_n(\tilde{\tau}_1-\tilde{\tau}_3)}\int_0^{+\hbar\beta} \mathrm{d}\tilde{\tau}_3 \mathscr{G}(\tilde{\tau}_3-\tau') \\
&\equiv \mathscr{G}_0(\tau,\tau') + \int_0^{+\hbar\beta}\int_0^{+\hbar\beta} \mathrm{d}\tau_1 \mathrm{d}\tau_2 \mathscr{G}_0(\tau,\tau_1)\,\Sigma(\tau_1,\tau_2)\mathscr{G}(\tau_2,\tau')
\end{aligned}
$$

33.10. The ("Holy") Trinity of Electronic Basis Sets in Materials Theory

Fact 33.98 There are three different levels of description for electrons in materials, which are all related to each other by a mere basis-change:

- *Continuum-basis* (cf. TBReferenced[10])
- *Bloch-basis* (cf. TBReferenced[11])
- *Wannier-basis* (cf. TBReferenced[12])

[10]Chapter about position eigenstates
[11]LDA-approximation, EPW-system, chapter about thermal QFT
[12]Cf. comment in the Green-function identities and the specifications for real materials

33.11. Discrete Identities with limited use for the EPW-System

Lemma 33.99 Notice that the latter identity can be simplified to

$$\begin{aligned}\langle H_{\mathrm{el}}\rangle = \frac{1}{\beta} & \sum_{\substack{\sigma\in\{\downarrow,\uparrow\}\\ k\in \mathrm{i}\Omega_{-1}\times\mathrm{Q}\\ n\in\{1,2,\ldots,\infty\}}} \epsilon_{n\sigma}(\mathbf{k})\mathscr{G}_{\substack{0\\ n,n\\ \sigma,\sigma}}(k,k) \\ & -\frac{1}{\beta^2 N_p}\sum_{\substack{\sigma\in\{\downarrow,\uparrow\}\\ k,k'\in \mathrm{i}\Omega_{-1}\times\mathrm{Q}\\ n,n'\in\{1,\ldots,\infty\}\\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}}} |g_{nn'\nu}(\mathbf{k},\mathbf{k}'-\mathbf{k})|^2\mathscr{G}^2_{0,n,\sigma}(k)\mathscr{G}_{0,n',\sigma}(k')\mathscr{D}_{0,\nu}(k'-k) \quad + \quad \mathcal{O}(g^4)\end{aligned} \tag{33.115}$$

Proof 33.99 TBDigitalized.

Corollary 33.100 Evaluating the Matsubara-sums simplified the individual terms as

$$\langle H_{\mathrm{el}}\rangle \approx \sum_{\substack{\mathbf{k}\in\mathrm{Q}\\ \sigma\in\{\downarrow,\uparrow\}\\ n\in\{1,2,\ldots,\infty\}}} \epsilon_{n\sigma}(\mathbf{k})\,\mathrm{F}_\beta(\xi_{n\sigma}(\mathbf{k})) \quad + \quad \delta\langle H_{\mathrm{el}}\rangle \tag{33.116}$$

with correction-term

$$\begin{aligned}&\delta\langle H_{\mathrm{el}}\rangle \\ &\approx \frac{1}{N_p}\sum_{\substack{\sigma\in\{\downarrow,\uparrow\}\\ \mathbf{k},\mathbf{q}\in\mathrm{Q}\\ n,n'\in\{1,\ldots,\infty\}\\ \nu\in\{1,\ldots,rd_{\mathbf{x}}\}}} \epsilon_{n\sigma}(\mathbf{k})|g_{nn'\nu}(\mathbf{k},\mathbf{q})|^2\cdot\Bigg[\\ &\frac{\beta}{4}\operatorname{sech}^2\left(\frac{\beta\xi_{n\sigma}(\mathbf{k})}{2}\right)\left[\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))+\mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}))}{\xi_{n\sigma}(\mathbf{k})-\xi_{n'\sigma}(\mathbf{k}+\mathbf{q})+\Omega_\nu(\mathbf{q})}+\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))+1-\mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}))}{\xi_{n\sigma}(\mathbf{k})-\xi_{n'\sigma}(\mathbf{k}+\mathbf{q})-\Omega_\nu(\mathbf{q})}\right] \\ &+\mathrm{F}_\beta(\xi_{n\sigma}(\mathbf{k}))\left[\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))+\mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}))}{\left[\xi_{n\sigma}(\mathbf{k})-\xi_{n'\sigma}(\mathbf{k}+\mathbf{q})+\Omega_\nu(\mathbf{q})\right]^2}+\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))+1-\mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}))}{\left[\xi_{n\sigma}(\mathbf{k})-\xi_{n'\sigma}(\mathbf{k}+\mathbf{q})-\Omega_\nu(\mathbf{q})\right]^2}\right] \\ &-\mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q})-\Omega_\nu(\mathbf{q}))\cdot\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))+\mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}))}{\left[\xi_{n'\sigma}(\mathbf{k}+\mathbf{q})-\xi_{n\sigma}(\mathbf{k})-\Omega_\nu(\mathbf{q})\right]^2}-\mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q})+\Omega_\nu(\mathbf{q}))\cdot\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q}))+1-\mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}))}{\left[\xi_{n'\sigma}(\mathbf{k}+\mathbf{q})-\xi_{n\sigma}(\mathbf{k})+\Omega_\nu(\mathbf{q})\right]^2}\Bigg]\end{aligned} \tag{33.117}$$

Proof 33.100 TBDigitalized, Cf. Waste-Folder 1.

Remark 33.100.1 Notice that this expression can be simplified to

$$
\begin{aligned}
&\delta\langle H_{\text{el}}\rangle \\
&\approx \frac{1}{N_p} \sum_{\substack{\sigma\in\{\downarrow,\uparrow\}\\ \mathbf{k},\mathbf{q}\in Q\\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}\\ n,n'\in\{1,\dots,\infty\}}} \epsilon_{n\sigma}(\mathbf{k})|g_{nn'\nu}(\mathbf{k},\mathbf{q})|^2 \cdot \Bigg[\\
&\quad + \frac{\beta}{4}\operatorname{sech}^2\left(\frac{\beta\xi_{n\sigma}(\mathbf{k})}{2}\right)\left[\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) + \mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}))}{\xi_{n\sigma}(\mathbf{k}) - \xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) + \Omega_\nu(\mathbf{q})} + \frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) + 1 - \mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}))}{\xi_{n\sigma}(\mathbf{k}) - \xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) - \Omega_\nu(\mathbf{q})}\right] \\
&\quad \left[\mathrm{F}_\beta(\xi_{n\sigma}(\mathbf{k})) - \mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) - \Omega_\nu(\mathbf{q}))\right] \frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) + \mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}))}{\left[\xi_{n\sigma}(\mathbf{k}) - \xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) + \Omega_\nu(\mathbf{q})\right]^2} \\
&\quad \left[\mathrm{F}_\beta(\xi_{n\sigma}(\mathbf{k})) - \mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) + \Omega_\nu(\mathbf{q}))\right] \frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) + 1 - \mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}))}{\left[\xi_{n\sigma}(\mathbf{k}) - \xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) - \Omega_\nu(\mathbf{q})\right]^2} \quad \Bigg]
\end{aligned} \tag{33.118}
$$

Remark 33.100.2 Notice however that despite it's simplicity (no integration required!) this formula is of limited practical use, since it is very difficult to converge the infinite sums without any form of smearing.

Lemma 33.101 For the temperature-derivative of the electronic correction, we obtain

$$
\begin{aligned}
&\partial_T\delta\langle H_{\text{el}}\rangle \\
&\approx \frac{1}{N_p} \sum_{\substack{\sigma\in\{\downarrow,\uparrow\}\\ \mathbf{k},\mathbf{q}\in Q\\ \nu\in\{1,\dots,rd_{\mathbf{x}}\}\\ n,n'\in\{1,\dots,\infty\}}} \epsilon_{n\sigma}(\mathbf{k})|g_{nn'\nu}(\mathbf{k},\mathbf{q})|^2 \left[\quad 1+2+3+4+5+6 \quad \right]
\end{aligned} \tag{33.119}
$$

with terms 1 to 6 defined as

$$
1 = \frac{\operatorname{sech}^2\left(\beta\xi_{n\sigma}(\mathbf{k})/2\right)}{4k_BT^2}\left(\beta\xi_{n\sigma}(\mathbf{k})\tanh\left(\beta\xi_{n\sigma}(\mathbf{k})/2\right) - 1\right)\left[\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) + \mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}))}{\xi_{n\sigma}(\mathbf{k}) - \xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) + \Omega_\nu(\mathbf{q})} + \frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) + 1 - \mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}))}{\xi_{n\sigma}(\mathbf{k}) - \xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) - \Omega_\nu(\mathbf{q})}\right] \tag{33.120}
$$

$$
2 = \frac{\beta}{4}\operatorname{sech}^2\left(\frac{\beta\xi_{n\sigma}(\mathbf{k})}{2}\right)\frac{1}{4k_BT^2}\left[\frac{\Omega_\nu(\mathbf{q})\operatorname{csch}^2(\beta\Omega_\nu(\mathbf{q})/2) + \xi_{n'\sigma}(\mathbf{k}+\mathbf{q})\operatorname{sech}^2(\beta\xi_{n'\sigma}(\mathbf{k}+\mathbf{q})/2)}{\xi_{n\sigma}(\mathbf{k}) - \xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) + \Omega_\nu(\mathbf{q})} + \frac{\Omega_\nu(\mathbf{q})\operatorname{csch}^2(\beta\Omega_\nu(\mathbf{q})/2) - \xi_{n'\sigma}(\mathbf{k}+\mathbf{q})\operatorname{sech}^2(\beta\xi_{n'\sigma}(\mathbf{k}+\mathbf{q})/2)}{\xi_{n\sigma}(\mathbf{k}) - \xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) - \Omega_\nu(\mathbf{q})}\right] \tag{33.121}
$$

$$
3 = \frac{1}{4k_BT^2}\left[\xi_{n\sigma}(\mathbf{k})\operatorname{sech}^2(\beta\xi_{n\sigma}(\mathbf{k})/2) - (\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) - \Omega_\nu(\mathbf{q}))\operatorname{sech}^2\left(\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) - \Omega_\nu(\mathbf{q}))/2\right)\right]\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) + \mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}))}{\left[\xi_{n\sigma}(\mathbf{k}) - \xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) + \Omega_\nu(\mathbf{q})\right]^2} \tag{33.122}
$$

$$
4 = \frac{1}{4k_BT^2}\left[\mathrm{F}_\beta(\xi_{n\sigma}(\mathbf{k})) - \mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) - \Omega_\nu(\mathbf{q}))\right]\frac{\Omega_\nu(\mathbf{q})\operatorname{csch}^2(\beta\Omega_\nu(\mathbf{q})/2) + \xi_{n'\sigma}(\mathbf{k}+\mathbf{q})\operatorname{sech}^2(\beta\xi_{n'\sigma}(\mathbf{k}+\mathbf{q})/2)}{\left[\xi_{n\sigma}(\mathbf{k}) - \xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) + \Omega_\nu(\mathbf{q})\right]^2} \tag{33.123}
$$

$$5 = \frac{1}{4k_B T^2}\left[\xi_{n\sigma}(\mathbf{k})\,\mathrm{sech}^2(\beta\xi_{n\sigma}(\mathbf{k})/2) - (\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) + \Omega_\nu(\mathbf{q}))\,\mathrm{sech}^2\left(\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) + \Omega_\nu(\mathbf{q}))/2\right)\right]\frac{\mathrm{B}_\beta(\Omega_\nu(\mathbf{q})) + 1 - \mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}))}{\left[\xi_{n\sigma}(\mathbf{k}) - \xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) - \Omega_\nu(\mathbf{q})\right]^2} \tag{33.124}$$

$$6 = \frac{1}{4k_B T^2}\left[\mathrm{F}_\beta(\xi_{n\sigma}(\mathbf{k})) - \mathrm{F}_\beta(\xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) + \Omega_\nu(\mathbf{q}))\right]\frac{\Omega_\nu(\mathbf{q})\,\mathrm{csch}^2(\beta\Omega_\nu(\mathbf{q})/2) - \xi_{n'\sigma}(\mathbf{k}+\mathbf{q})\,\mathrm{sech}^2(\beta\xi_{n'\sigma}(\mathbf{k}+\mathbf{q})/2)}{\xi_{n\sigma}(\mathbf{k}) - \xi_{n'\sigma}(\mathbf{k}+\mathbf{q}) - \Omega_\nu(\mathbf{q})} \tag{33.125}$$

Proof 33.101 TBDone / TBDigitalized from Waste-Folder 1.

Remark 33.101.1 We emphasize again, that this formula is of very limited use.

34. Appendix - Applications to Toy Models

34.1. The Free Electron Gas

Possible / reasonable topics:

- fermi-energy and DOS as a function of density
- Sommerfeld-approximation for the free electron-gas and comparison with experiment.
- Band-structure and reduced zone-scheme.

34.2. Superconductivity from BCS-Theory

34.2.1. BCS-Hamiltonian in First Quantization

Notation 34.1 We consider a single electron obeying the time-independent free Schroedinger-equation

$$H_0|\psi\rangle = -\hbar^2 \frac{\Delta}{2m}|\psi\rangle = E|\psi\rangle$$

subject to periodic boundary conditions in a volume $[0, L]^{d_\mathrm{x}} / \sim$.

Lemma 34.2 With the inclusion of spin, this gives us a basis for the single-particle Hilbert-space

$$\mathrm{span}_\mathbb{C} \left\{\psi_{\mathbf{k}\sigma}\right\}_{\substack{\mathbf{k}\in 2\pi\mathbb{Z}^{d_\mathrm{x}}/L \\ \sigma\in\{\downarrow,\uparrow\}}} = \mathrm{span}_\mathbb{C} \left\{\psi_{\mathbf{kn}\sigma}\right\}_{\substack{\mathbf{k}\in\mathrm{Q} \\ \mathbf{n}\in\mathbb{Z}^{d_\mathrm{x}} \\ \sigma\in\{\downarrow,\uparrow\}}} = \mathcal{H} = L^2([0, L]^{d_\mathrm{x}} / \sim) \otimes \mathbb{C}^2. \tag{34.1}$$

Proof 34.2 See [Kaul, Fischer] and [Bulla lecture notes].

Remark 34.2.1 Notice that one has to conclude from the context "which" $\mathbf{k}$ one is referring to.

Definition 34.3 We now consider an *attractive, local BCS-toy-model-interaction potential* V, which acts as a linear operator on the two-body Hilbert-space as

$$\begin{aligned} V: \mathcal{H} \otimes \mathcal{H} &\to \mathcal{H} \otimes \mathcal{H} \\ \psi \otimes \chi &\mapsto V(\psi \otimes \chi) \end{aligned}$$

with

$$V(\psi \otimes \chi)(x, x') := -|g|\delta_{\sigma \neq \sigma'}\delta(\mathbf{r} - \mathbf{r}')\psi(x)\chi(x').$$

for all $x, x' \in [0, L]^3 \times \{\downarrow, \uparrow\}$ with $x \equiv (\mathbf{r}, \sigma)$ etc.

Remark 34.3.1 Usually one refers to V simply as *BCS-interaction*.

Remark 34.3.2 Notice that the Bra-Ket-Notation gives us the integral kernel

$$\begin{aligned} -|g|\delta_{\sigma \neq \sigma'}\delta(\mathbf{r} - \mathbf{r}')\psi(x_1)\chi(x_2) &= \left\langle x_1, x_2 \middle| V \middle| \psi, \chi \right\rangle \\ &= \sum\!\!\!\!\!\!\!\int_{x_3, x_4 \in [0,L]^3 \times \{\downarrow,\uparrow\}} \left\langle x_1, x_2 \middle| V \middle| x_3, x_4 \right\rangle \left\langle x_3, x_4 \middle| \psi, \chi \right\rangle \\ &= \sum\!\!\!\!\!\!\!\int_{x_3, x_4 \in [0,L]^3 \times \{\downarrow,\uparrow\}} \left\langle x_1, x_2 \middle| V \middle| x_3, x_4 \right\rangle \psi(x_3)\chi(x_4) \end{aligned}$$

thus one can conclude that the integral kernel reads

$$\langle x_1, x_2 | V | x_3, x_4 \rangle = -|g|\delta_{\sigma_1 \neq \sigma_2}\delta(\mathbf{r}_1 - \mathbf{r}_2) \cdot \delta_{\sigma_1, \sigma_3}\delta(\mathbf{r}_1 - \mathbf{r}_3)\delta_{\sigma_2, \sigma_4}\delta(\mathbf{r}_2 - \mathbf{r}_4) \tag{34.2}$$

Remark 34.3.3 Notice that this is a special case of a "local two-body interaction" a.k.a. "velocity independent"[1].

We will now continue the many body problem defined by the single particle Hamiltonian H_0 above, combined with the many-particle interactions V.

34.2.2. BCS-Hamiltonian in Second Quantization

Lemma 34.4 One computes the matrix-elements of the BCS-interaction w.r.t. the basis in Equation (34.1) to be

$$\langle \mathbf{k}_1\sigma_1, \mathbf{k}_2\sigma_2 | V | \mathbf{k}_3\sigma_3, \mathbf{k}_4\sigma_4 \rangle = -\frac{|g|}{L^{d_\mathbf{x}}}\delta_{\sigma_3 \neq \sigma_4}\delta_{\sigma_1, \sigma_3}\delta_{\sigma_2, \sigma_4}\delta_{\mathbf{k}_1 + \mathbf{k}_2, \mathbf{k}_3 + \mathbf{k}_4} \tag{34.3}$$

Proof 34.4 See Appendix, Proofs, page 405.

[1] Cf. MarcosMarino p. 46, Negele Orland page 10.

Theorem 34.5 Using the plane basis in Equation (34.1), one can write the Hamiltonian in second quantization as

$$H_{\mathrm{BCS}} = \sum_{\substack{\mathbf{k}\in 2\pi\mathbb{Z}^{d_\mathrm{x}}/L \\ \sigma\in\{\downarrow,\uparrow\}}} \epsilon(\mathbf{k}) c^\dagger_\sigma(\mathbf{k}) c_\sigma(\mathbf{k}) - \frac{g}{L^{d_\mathrm{x}}} \sum_{\mathbf{k},\mathbf{k}',\mathbf{q}\in 2\pi\mathbb{Z}^{d_\mathrm{x}}/L} c^\dagger_\uparrow(\mathbf{k}+\mathbf{q}) c^\dagger_\downarrow(-\mathbf{k}) c_\downarrow(-\mathbf{k}'+\mathbf{q}) c_\uparrow(\mathbf{k}') \tag{34.4}$$

with fermionic creation- and annihilation-operators as usual.

Proof 34.5 See Appendix, Proofs, page 405.

Remark 34.5.1 Notice that this expression is in harmony with the literature[2].

Notation 34.6 Using the basis introduced above, and notating $V \equiv L^{d_\mathrm{x}}$, we introduce the field-operators

$$c_\sigma(\mathbf{r}) = \frac{1}{\sqrt{L^{d_\mathrm{x}}}} \sum_{\mathbf{k}\in 2\pi\mathbb{Z}^{d_\mathrm{x}}/L} \mathrm{e}^{\mathrm{i}\mathbf{k}\mathbf{r}} c_\sigma(\mathbf{k}), \qquad c^\dagger_\sigma(\mathbf{r}) = \frac{1}{\sqrt{L^{d_\mathrm{x}}}} \sum_{\mathbf{k}\in 2\pi\mathbb{Z}^{d_\mathrm{x}}/L} \mathrm{e}^{-\mathrm{i}\mathbf{k}\mathbf{r}} c^\dagger_\sigma(\mathbf{k}) \tag{34.5}$$

Lemma 34.7 Note that these field-operators obey the commutation-relations

$$\left\{c_\sigma(\mathbf{r}),\, c^\dagger_{\sigma'}(\mathbf{r}')\right\} = \delta_{\sigma,\sigma'}\delta(\mathbf{r}-\mathbf{r}'), \qquad \left\{c'_\sigma(\mathbf{r}),\, c^\dagger_{\sigma'}(\mathbf{r}')\right\} = \left\{c_\sigma(\mathbf{r}),\, c_{\sigma'}(\mathbf{r}')\right\} = 0 \tag{34.6}$$

whereas one has to conclude from the context, whether the symbol ψ denotes the single particle wavefunction or the many particle field-operator.

Proof 34.7 See Appendix, Proofs, page 406.

Corollary 34.8 The BCS-Hamiltonian can be rewritten with the field-operators as

$$\begin{aligned} H_{\mathrm{BCS}} &= \sum\!\!\!\!\!\!\int c^\dagger_\sigma(\mathbf{r}) \left(-\hbar^2\frac{\Delta}{2m} - \mu\right) c_\sigma(\mathbf{r}) \quad - \quad g\int \mathrm{d}\mathbf{r}\, c^\dagger_\uparrow(\mathbf{r}) c^\dagger_\downarrow(\mathbf{r}) c_\downarrow(\mathbf{r}) c_\uparrow(\mathbf{r}) \\ &= \sum\!\!\!\!\!\!\int c^\dagger_\sigma(\mathbf{r}) \left(-\hbar^2\frac{\Delta}{2m} - \mu\right) c_\sigma(\mathbf{r}) \quad - \quad \frac{g}{2}\int \mathrm{d}\mathbf{r}\ :\ \left[\begin{pmatrix} c^\dagger_\uparrow & c^\dagger_\downarrow \end{pmatrix} \cdot \begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix}\right]^2 (\mathbf{r}) : \end{aligned}$$

Proof 34.8 See Appendix, Proofs, page 406.

[2] Cf. Altland pages 267, 346, Flensberg page 321, Rickayzen page 209.

34.2.3. BCS-Action in the Field Integral Formalism

Fact 34.9 The action of the system reads

$$S_{\mathrm{BCS}}[\psi,\overline{\psi}] = \sum\!\!\!\!\!\!\int_{\substack{\sigma\in\{\downarrow,\uparrow\}\\ \tau\in(0,\hbar\beta)\\ \mathbf{k}\in 2\pi\mathbb{Z}^{d_\mathrm{x}}/L}} \left[\overline{\psi}_\sigma(\tau,\mathbf{k})\left(\partial_\tau + \epsilon_\sigma(\mathbf{k}) - \mu\right)\psi_\sigma(\tau,\mathbf{k}) - \frac{g}{L^{d_\mathrm{x}}} \sum\!\!\!\!\!\!\int_{\substack{\tau\in(0,\hbar\beta)\\ \mathbf{k},\mathbf{k}',\mathbf{q}\in 2\pi\mathbb{Z}^{d_\mathrm{x}}/L}} \left(\overline{\psi}_{\mathbf{k}+\mathbf{q}\uparrow}\overline{\psi}_{-\mathbf{k}\downarrow}\psi_{-\mathbf{k}'+\mathbf{q}\downarrow}\psi_{\mathbf{k}'\uparrow}\right)(\tau)\right]$$

Proof 34.9 See [Draft Summary].

Notation 34.10 Recalling the shorthand-notation[3] $x \equiv (\tau,\mathbf{r})$, we introduce the Fourier-transformed integration-variables as

$$\psi_\sigma(\tau,\mathbf{r}) = \frac{1}{\sqrt{\hbar\beta V}} \sum_{\substack{\omega_n\in\Omega_{-1}\\ \mathbf{k}\in 2\pi\mathbb{Z}^{d_\mathrm{x}}/L}} \mathrm{e}^{\mathrm{i}(\mathbf{kr}-\omega_n\tau)}\psi_\sigma(\mathrm{i}\omega_n,\mathbf{k}), \qquad \overline{\psi}_\sigma(\tau,\mathbf{r}) = \frac{1}{\sqrt{\hbar\beta V}} \sum_{\substack{\omega_n\in\Omega_{-1}\\ \mathbf{k}\in 2\pi\mathbb{Z}^{d_\mathrm{x}}/L}} \mathrm{e}^{-\mathrm{i}(\mathbf{kr}-\omega_n\tau)}\overline{\psi}_\sigma(\mathrm{i}\omega_n,\mathbf{k}). \tag{34.7}$$

and analogously (but with bosonic Matsubara-frequencies) later on for the auxiliary fields Δ and $\overline{\Delta}$.

Fact 34.11 The corresponding action of the real-space-representation in the functional integral formalism reads[4]

$$\begin{aligned}
S_{\mathrm{BCS}}[\psi,\overline{\psi}] &= \int \mathrm{d}x \left[\sum_{\sigma\in\{\downarrow,\uparrow\}} \overline{\psi}_\sigma(\mathrm{x})\left(\partial_\tau - \hbar^2\frac{\Delta}{2m} - \mu\right)\psi_\sigma(\mathrm{x}) - g\left(\overline{\psi}_\uparrow\overline{\psi}_\downarrow\psi_\downarrow\psi_\uparrow\right)(\mathrm{x})\right] \\
&= \sum\!\!\!\!\!\!\int_{\mathrm{x},\sigma} \overline{\psi}_\sigma(\mathrm{x})\left(\partial_\tau - \hbar^2\frac{\Delta}{2m} - \mu\right)\psi_\sigma(\mathrm{x}) - g\left(\overline{\psi}_\uparrow\overline{\psi}_\downarrow\psi_\downarrow\psi_\uparrow\right)(\mathrm{x})
\end{aligned}$$

Proof 34.11 By combining the expansions in TBReferenced with Corollary TBReferenced.

Corollary 34.12 For future reference we note the representation of the action using the momentum-eigenstates

$$\begin{aligned}
S_{\mathrm{BCS}}[\psi,\overline{\psi}] = &\sum_{\substack{\sigma\in\{\downarrow,\uparrow\}\\ k\in\mathrm{i}\Omega_{-1}\times 2\pi\mathbb{Z}^{d_\mathrm{x}}/L}} \overline{\psi}_\sigma(k)\left(-\mathrm{i}\omega_k + \epsilon_\sigma(\mathbf{k}) - \mu\right)\psi_\sigma(k) \\
&- \frac{g}{\hbar\beta L^{d_\mathrm{x}}} \sum_{k,k',q\in\mathrm{i}\Omega_{-1}\times 2\pi\mathbb{Z}^{d_\mathrm{x}}/L} \overline{\psi}_\uparrow(k+q)\overline{\psi}_\downarrow(-k)\psi_\downarrow(-k'+q)\psi_\uparrow(k')
\end{aligned}$$

[3] Shit... This is in conflict to earlier where this denoted space and spin-coordinate.
[4] Cf. this to Henk page 275.

Proof 34.12 See Appendix, Proofs, page 407.

Notation 34.13 We implement Graßmann-valued external sources into the partition-function as

$$Z \mapsto Z[j,\bar{j}] = \int \mathscr{D}_{-1}(\overline{\psi},\psi)\exp\left(-S_{\text{BCS}} + \sum\!\!\!\!\!\!\int \bar{j}\psi + \overline{\psi}j\right) \tag{34.8}$$

34.2.4. Hubbard-Stratonovich-Transformation

Lemma 34.14 A Hubbard-Stratonovich-transformation reads

$$\begin{aligned}\exp - S_{\text{int}}[\overline{\psi},\psi] &= \exp g\int \mathrm{d}\,\mathrm{x}\,\left(\overline{\psi}_\uparrow\overline{\psi}_\downarrow\psi_\downarrow\psi_\uparrow\right)(\mathrm{x})\\ &= \int \mathscr{D}_{+1}(\Delta,\overline{\Delta})\exp - \int \mathrm{d}\,\mathrm{x}\,\left[\frac{|\Delta|^2(\mathrm{x})}{g} - \left(\overline{\Delta}\psi_\downarrow\psi_\uparrow + \Delta\overline{\psi}_\uparrow\overline{\psi}_\downarrow\right)(\mathrm{x})\right]\end{aligned}$$

with complex integration-variables $\Delta, \overline{\Delta}$.

Proof 34.14 See Appendix, Proofs, page 408.

Fact 34.15 After decoupling the quartic contributions we obtain the following partition-function in presence of external sources

$$Z[j,\bar{j}] = \iint \mathscr{D}_{-1}(\overline{\psi},\psi)\mathscr{D}_{+1}(\Delta,\overline{\Delta})\exp\left(-\widetilde{S}_{\text{BCS}} + \sum\!\!\!\!\!\!\int \bar{j}\psi + \overline{\psi}j\right).$$

and with the new BCS-action

$$\widetilde{S}_{\text{BCS}} = \sum\!\!\!\!\!\!\int_{\mathrm{x},\sigma} \overline{\psi}_\sigma(\mathrm{x})\left(\partial_\tau - \hbar^2\frac{\Delta}{2m} - \mu\right)\psi_\sigma(\mathrm{x}) + \int \mathrm{d}\,\mathrm{x}\,\left[\frac{|\Delta|^2(\mathrm{x})}{g} - \left(\overline{\Delta}\psi_\downarrow\psi_\uparrow + \Delta\overline{\psi}_\uparrow\overline{\psi}_\downarrow\right)(\mathrm{x})\right] \tag{34.9}$$

Proof 34.15 Follows by combining the previous results.

Lemma 34.16 Switching to the Fourier-representation of the action, we obtain for the individual terms

- $\sum\!\!\!\!\!\!\int_{\mathrm{x},\sigma} \overline{\psi}_\sigma(\mathrm{x})\left(\partial_\tau - \hbar^2\frac{\Delta}{2m} - \mu\right)\psi_\sigma(\mathrm{x}) = \sum_{\substack{\sigma\in\{\downarrow,\uparrow\}\\ k\in \mathrm{i}\Omega_{-1}\times 2\pi\mathbb{Z}^{d_\mathrm{x}}/L}} \overline{\psi}_\sigma(k)\left(-\mathrm{i}\omega_k + \epsilon_\sigma(\mathbf{k}) - \mu\right)\psi_\sigma(k)$

- $\int \mathrm{d}\,\mathrm{x}\ \frac{|\Delta|^2(\mathrm{x})}{g} = \sum_{k\in \mathrm{i}\Omega_{-1}\times 2\pi\mathbb{Z}^{d_\mathrm{x}}/L} \frac{|\Delta|^2(k)}{g}$
- $\int \mathrm{d}\,\mathrm{x}\, -\left(\overline{\Delta}\psi_\downarrow\psi_\uparrow + \Delta\overline{\psi}_\uparrow\overline{\psi}_\downarrow\right)(\mathrm{x}) = \frac{1}{\sqrt{\hbar\beta L^{d_\mathrm{x}}}} \sum_{k,k'\in \mathrm{i}\Omega_{-1}\times 2\pi\mathbb{Z}^{d_\mathrm{x}}/L} -\left(\overline{\Delta}(k+k')\psi_\downarrow(k)\psi_\uparrow(k') + \Delta(k+k')\overline{\psi}_\uparrow(k)\overline{\psi}_\downarrow(k')\right)$

Proof 34.16 See the path-integral-description of BCS-theory-document, pdf-page 10.

Notation 34.17 We further introduce the *Nambu-spinors*

$$\overline{\Psi}_\mathrm{Nambu}(\mathrm{i}\omega_n,\mathbf{k}) = \left(\overline{\psi}_\uparrow(\mathrm{i}\omega_n,\mathbf{k}),\quad \psi_\downarrow(-\mathrm{i}\omega_n,-\mathbf{k})\right), \qquad \Psi_\mathrm{Nambu}(\mathrm{i}\omega_n,\mathbf{k}) = \begin{pmatrix}\psi_\uparrow(\mathrm{i}\omega_n,\mathbf{k})\\ \overline{\psi}_\downarrow(-\mathrm{i}\omega_n,-\mathbf{k})\end{pmatrix} \tag{34.10}$$

Lemma 34.18 Changing the integration-variables[5] gives the momentum-representation of the partition-function

$$Z[j,\overline{j}] = \iint \mathscr{D}_{-1}(\overline{\psi},\psi)\mathscr{D}_{+1}(\overline{\Delta},\Delta)\exp -\widetilde{S}_\mathrm{BCS} \tag{34.11}$$

with

$$\begin{aligned}\widetilde{S}_\mathrm{BCS}[\Delta,\overline{\Delta},\psi,\overline{\psi},j,\overline{j}] = &-\sum_{\substack{\mathbf{q}\in 2\pi\mathbb{Z}^{d_\mathrm{x}}/L\\ \omega_m\in\Omega_{+1}}} \frac{|\Delta(\mathrm{i}\omega_m,\mathbf{q})|^2}{g}\\ &+\sum_{\substack{\mathbf{k}',\mathbf{k}''\in 2\pi\mathbb{Z}^{d_\mathrm{x}}/L\\ \omega_{n'},\omega_{n''}\in\Omega_{-1}}} \overline{\Psi}_\mathrm{Nambu}(\mathrm{i}\omega_{n'},\mathbf{k}')\left\langle \mathrm{i}\omega_{n'},\mathbf{k}'\middle| -\frac{1}{\hbar}\mathscr{G}^{-1}_\mathrm{Gorkov}\middle|\mathrm{i}\omega_{n''},\mathbf{k}''\right\rangle \Psi_\mathrm{Nambu}(\mathrm{i}\omega_{n''},\mathbf{k}'')\\ &-\left(\overline{j}\psi + \overline{j}\psi\right)\end{aligned}$$

and Gorkov-Green-function

$$\begin{aligned}\langle \mathrm{i}\omega_n,\mathbf{k}| -\frac{1}{\hbar}\mathscr{G}^{-1}_\mathrm{Gorkov}|\mathrm{i}\omega_{n'},\mathbf{k}'\rangle = \langle \mathrm{i}\omega_n,\mathbf{k}|\mathrm{i}\omega_{n'},\mathbf{k}'\rangle &\begin{pmatrix}-\mathrm{i}\omega_n + \left(\frac{\hbar^2k^2}{2m}-\mu\right) & 0\\ 0 & +\mathrm{i}\omega_n - \left(\frac{\hbar^2k^2}{2m}-\mu\right)\end{pmatrix}\\ &+\frac{1}{\sqrt{\hbar\beta V}}\begin{pmatrix}0 & -\Delta_{n+n',\mathbf{k}+\mathbf{k}'}\\ \overline{\Delta}_{n+n',\mathbf{k}+\mathbf{k}'} & 0\end{pmatrix}\end{aligned}$$

Proof 34.18 See Appendix, Proofs, page 408.

[5]The unphysical, diverging normalization-factor is irrelevant, when we determine observables later on, as it will cancel out.

Corollary 34.19 Integrating out the fermionic fields gives the partition-function

$$Z[j,\bar{j}] = \int \mathscr{D}_{+1}(\overline{\Delta},\Delta) \exp - S_{\mathrm{BCS}}[\Delta,\overline{\Delta},j,\bar{j}] \tag{34.12}$$

with

$$\begin{aligned} S_{\mathrm{BCS}}[\Delta,\overline{\Delta},j,\bar{j}] = &- \sum_{\substack{\mathbf{q}\in \mathrm{Q} \\ \omega_m \in \Omega_{+1}}} \frac{|\Delta(\mathrm{i}\omega_m,\mathbf{q})|^2}{g} \\ &- \operatorname{Tr} \operatorname{Log} - \frac{1}{\hbar}\mathscr{G}^{-1}_{\mathrm{Gorkov}}[\Delta,\overline{\Delta}] \\ &\mp \begin{pmatrix} \bar{j}_\uparrow, & -j_\downarrow \end{pmatrix} \cdot \left(-\hbar \mathscr{G}_{\mathrm{Gorkov}}[\Delta,\overline{\Delta}]\right) \cdot \begin{pmatrix} j_\uparrow \\ -\bar{j}_\downarrow \end{pmatrix} \end{aligned}$$

Proof 34.19 We first rewrite the source-terms as linear terms in the Nambu-spinor-fields as

$$\begin{aligned} \sum\!\!\!\!\!\!\int \bar{j}\psi + \overline{\psi} j &= \sum\!\!\!\!\!\!\int_x \left(\bar{j}_\uparrow \psi_\uparrow + \bar{j}_\downarrow \psi_\downarrow + \overline{\psi}_\uparrow j_\uparrow + \overline{\psi}_\downarrow j_\downarrow\right) \\ &= \sum_k \left(\begin{pmatrix} \overline{\psi}_\uparrow, & \psi_\downarrow \end{pmatrix} \cdot \begin{pmatrix} j_\uparrow \\ -\bar{j}_\downarrow \end{pmatrix} + \begin{pmatrix} \bar{j}_\uparrow, & -j_\downarrow \end{pmatrix} \cdot \begin{pmatrix} \psi_\uparrow \\ \overline{\psi}_\downarrow \end{pmatrix} \right) \\ &= \sum_k \left(\overline{\Psi}_{\mathrm{Nambu}} \cdot \begin{pmatrix} j_\uparrow \\ -\bar{j}_\downarrow \end{pmatrix} + \begin{pmatrix} \bar{j}_\uparrow, & -j_\downarrow \end{pmatrix} \cdot \Psi_{\mathrm{Nambu}} \right) \end{aligned}$$

Now performing the integral in the fermionic variables with the usual identities and using $\operatorname{Log}\operatorname{Det}(\bullet) = \operatorname{Tr}\operatorname{Log}(\bullet)$ gives the desired result.

34.2.5. Spontaneous Symmetry Breaking

We now perform a mean-field-approximation and subsequently a saddle-point-approximation, and break the U(1)-symmetry by hand[6], in order to identify and determine the (superconducting-)order-parameter.

Corollary 34.20 The BCS-action is invariant under the unitary transformation

$$\Delta \mapsto \Delta \mathrm{e}^{\mathrm{i}\theta} \quad \text{with } \theta \in [0,2\pi). \tag{34.13}$$

Hypothesis 34.21 We make the ansatz for a mean-field-configuration as

[6] Cf. also Metsch 2020 Exercise Sheet 5 for the classical analogon.

$$\Delta(\mathrm{i}\omega_m, \mathbf{q}) \approx \sqrt{\hbar\beta V}\,\delta_{\mathbf{q},0}\delta_{\omega_m,0}\Delta$$
$$\overline{\Delta}(\mathrm{i}\omega_m, \mathbf{q}) \approx \sqrt{\hbar\beta V}\,\delta_{\mathbf{q},0}\delta_{\omega_m,0}\overline{\Delta}$$

Remark 34.21.1 Note that this ansatz assumes, that the mean-field-configuration of the order-parameter is dominated by a single contribution which is constant in the momenta.

Corollary 34.22 Using this ansatz and minimizing the action w.r.t. this configuration in the absence of external sources yields the gap-equation

$$\frac{1}{g} = \frac{1}{\hbar\beta L^{d_{\mathrm{x}}}} \sum_{k\in \mathrm{i}\Omega_{-1}\times 2\pi\mathbb{Z}^{d_{\mathrm{x}}}/L} \frac{1}{\omega_k^2 + \xi_{\mathbf{k}}^2 + |\Delta|^2} \tag{34.14}$$

with $\xi_k \equiv \frac{\hbar^2 k^2}{2m} - \mu$.

Proof 34.22 Varying the action w.r.t. Δ gives

$$\left(\frac{\delta}{\delta\Delta}S\right)[\Delta,\overline{\Delta}] = \frac{\overline{\Delta}}{g} \mp \mathrm{Tr}\left(\mathscr{G}\frac{\delta}{\delta\Delta}\mathscr{G}^{-1}\right) \tag{34.15}$$

The rest is algebra [TBDone and TBDigitalized].[7]

Notation 34.23 One can expand the action w.r.t. a reference-field Δ as

$$\begin{aligned}
S_{\mathrm{BCS}}[\Delta+\delta\Delta, \overline{\Delta}+\delta\overline{\Delta}] \equiv\; & S_{\mathrm{BCS}}[\Delta,\overline{\Delta}] \\
+\; & \sum\!\!\!\!\!\!\int \left(\frac{\delta}{\delta\Delta}S_{\mathrm{BCS}}[\Delta,\overline{\Delta}], \frac{\delta}{\delta\overline{\Delta}}S_{\mathrm{BCS}}[\Delta,\overline{\Delta}]\right)\cdot\begin{pmatrix}\delta\Delta\\ \delta\overline{\Delta}\end{pmatrix} \\
+\; & \frac{1}{2!}\sum\!\!\!\!\!\!\int\sum\!\!\!\!\!\!\int \begin{pmatrix}\delta\Delta\\ \delta\overline{\Delta}\end{pmatrix}^{\mathrm{T}}\cdot\begin{pmatrix}\frac{\delta^2}{\delta\Delta\,\delta\Delta}S_{\mathrm{BCS}}[\Delta,\overline{\Delta}] & \frac{\delta^2}{\delta\Delta\,\delta\overline{\Delta}}S_{\mathrm{BCS}}[\Delta,\overline{\Delta}]\\ \frac{\delta^2}{\delta\overline{\Delta}\,\delta\Delta}S_{\mathrm{BCS}}[\Delta,\overline{\Delta}] & \frac{\delta^2}{\delta\overline{\Delta}\,\delta\overline{\Delta}}S_{\mathrm{BCS}}[\Delta,\overline{\Delta}]\end{pmatrix}\cdot\begin{pmatrix}\delta\Delta\\ \delta\overline{\Delta}\end{pmatrix} \\
+\; & \mathcal{O}\left(\delta\Delta^3, \delta\Delta^2\delta\overline{\Delta}^1, \delta\Delta^1\delta\overline{\Delta}^2, \delta\overline{\Delta}^3\right).
\end{aligned}$$

Expanding the action w.r.t. to one of the (infinitely many! cf. Corollary 34.20) fields that fulfills the gap-equation, the contribution that is linear in $\delta\Delta$ vanishes.

Hypothesis 34.24 We perform a saddle-point-approximation and compute the partition-function as sum over mean-field-configurations with different phases of the order parameter.

[7]See Appendix, Proofs, page **??**.

$$Z[j,\bar{j}] \approx \sum_{\theta \in [0,2\pi)} Z_\theta[j,\bar{j}] \tag{34.16}$$

with

$$Z_\theta[j,\bar{j}] = \int \mathscr{D}(\delta\Delta, \delta\overline{\Delta}) \exp -S_{\mathrm{BCS}}[\Delta + \delta\Delta, \overline{\Delta} + \delta\overline{\Delta}, j, \bar{j}]. \tag{34.17}$$

Our hypothesis is now, that in the superconducting state, not all of these (infinitely many) summands contribute "matter", but only a single one. We therefore replace the partition-function, which contains all the information about the system, as

$$Z[j,\bar{j}] \mapsto Z_\theta[j,\bar{j}] \tag{34.18}$$

We are therefore left with the computation of a single functional integral again.

Remark 34.24.1 Note that this also implies that the expectation-values to determine the Green-functions have to be taken w.r.t. this new partition-function.

Lemma 34.25 In the mean-field-approximation we evaluate the partition-function as

$$Z_\theta[j,\bar{j}] \approx \exp -S_{\mathrm{BCS}}[\Delta, \overline{\Delta}, j, \bar{j}]. \tag{34.19}$$

whereas Δ solves the gap-equation.

Proof 34.25 TBD.

Lemma 34.26 One obtains the following decomposition of the action in the one-loop-approximation. Surpressing the dependence on the external sources, this yields

$$S_{\mathrm{BCS}}[\Delta + \delta\Delta, \overline{\Delta} + \delta\overline{\Delta}] \approx S_{\mathrm{BCS}}[\Delta, \overline{\Delta}] \quad + \quad \frac{1}{2!} \sum\!\!\!\!\!\!\int \sum\!\!\!\!\!\!\int \begin{pmatrix} \delta\Delta \\ \delta\overline{\Delta} \end{pmatrix}^{\mathrm{T}} \cdot \begin{pmatrix} \frac{\delta^2}{\delta\Delta\,\delta\Delta} S_{\mathrm{BCS}}[\Delta, \overline{\Delta}] & \frac{\delta^2}{\delta\Delta\,\delta\overline{\Delta}} S_{\mathrm{BCS}}[\Delta, \overline{\Delta}] \\ \frac{\delta^2}{\delta\overline{\Delta}\,\delta\Delta} S_{\mathrm{BCS}}[\Delta, \overline{\Delta}] & \frac{\delta^2}{\delta\overline{\Delta}\,\delta\overline{\Delta}} S_{\mathrm{BCS}}[\Delta, \overline{\Delta}] \end{pmatrix} \cdot \begin{pmatrix} \delta\Delta \\ \delta\overline{\Delta} \end{pmatrix}.$$

Proof 34.26 The one-loop-approximation relies on the expansion Now since Δ solves the gap-equation, the first contribution vanishes. The higher orders are being neglected within the saddle-point-approximation.

Theorem 34.27 One can also compute observables beyond the one-loop-approximation. Then one expands the action w.r.t. its interaction.

34.2.6. Landaus Free Energy Model

We finally want to illustrate where the usual Landau-model for the free-energy comes from.

Notation 34.28 We write the Gorkov-Green-function as

$$\mathscr{G}^{-1}_{\text{Gorkov}} \equiv \mathscr{G}^{-1}_{\substack{0\\ \text{Gorkov}}} + V \equiv \mathscr{G}^{-1}_{\substack{0\\ \text{Gorkov}}} \left(1 + \mathscr{G}_{\substack{0\\ \text{Gorkov}}} V\right) \tag{34.20}$$

with

$$\mathscr{G}^{-1}_{\substack{0\\ \text{Gorkov}}} \equiv \mathscr{G}^{-1}_{\text{Gorkov}}[\Delta = \overline{\Delta} = 0]. \tag{34.21}$$

Lemma 34.29 It holds

$$\operatorname{Tr} \operatorname{Log} \mathscr{G}^{-1}_{\text{Gorkov}} \equiv \text{const.} \quad + \quad \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \operatorname{Tr} \left(\mathscr{G}_{\substack{0\\ \text{Gorkov}}} V\right)^k \tag{34.22}$$

Proof 34.29 See Appendix, Proofs, page 408.

Theorem 34.30 Expanding the expansion up to fourth order in the field Δ gives the action

$$\begin{aligned} S_{\text{BCS}}[\Delta, \overline{\Delta}, j, \bar{j}] \approx & - \sum_{\substack{\mathbf{q} \in Q\\ \omega_m \in \Omega_{+1}}} \frac{|\Delta(\mathrm{i}\omega_m, \mathbf{q})|^2}{g} \\ & + TBD_1 |\Delta|^2 \\ & + TBD_2 |\Delta|^4 \\ & \mp \left(\bar{j}_\uparrow, \quad -j_\downarrow\right) \cdot \left(-\frac{1}{\hbar} \mathscr{G}_{\text{Gorkov}}[\Delta, \overline{\Delta}]\right) \cdot \begin{pmatrix} j_\uparrow \\ -\bar{j}_\downarrow \end{pmatrix} \end{aligned}$$

Proof 34.30 See Appendix, Proofs, page 408.

35. Appendix - Crystal Structure Determination with the Debye-Scherrer Method

Motivation and Overview

After this part of the notes you will understand ...

- how one determines the crystal structure of a material by evaluating so-called x-ray diffraction patterns like in Figure 35.1.

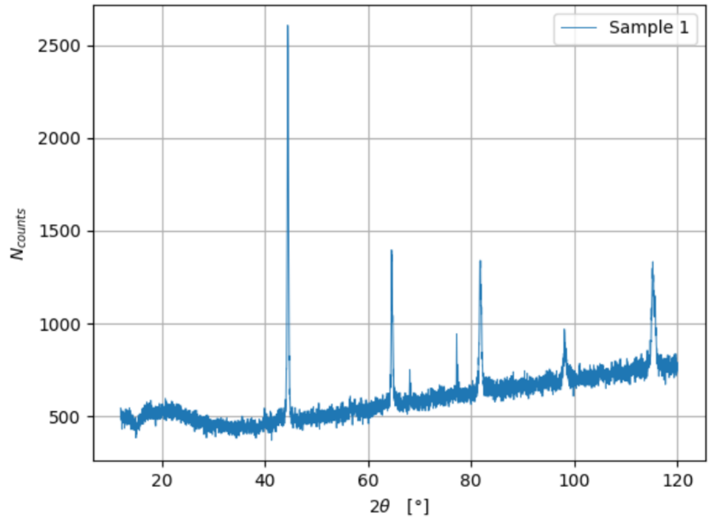


Figure 35.1.: X-Ray Diffraction Pattern of a sample that will be examined / identified in the laboratory-part of the course.

35.1. Preparation: The Miller-Indices

Notation 35.1 For the discussion of the „Miller-indices“ (to be introduced below) it is advantageous to have conventional lattice-vectors. We therefore consider a crystal with conventional lattice-vectors $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ in this subsection.

Definition 35.2 A *lattice plane* is defined by at least three grid points that do not lie on a straight line.

Lemma 35.3 Each lattice-plane has intersection-points $\vec{S}_i \equiv m_i \mathbf{a}_i$ (whereas $m_i \in \mathbb{Q} \cup \{\infty\}$) with the axes that are asssociated to the primitive basis-vectors, whereas $\vec{S}_i \equiv \infty \cdot \vec{a}_i$ is defined for planes that never intersect an axis (i.e. are parallel to it).

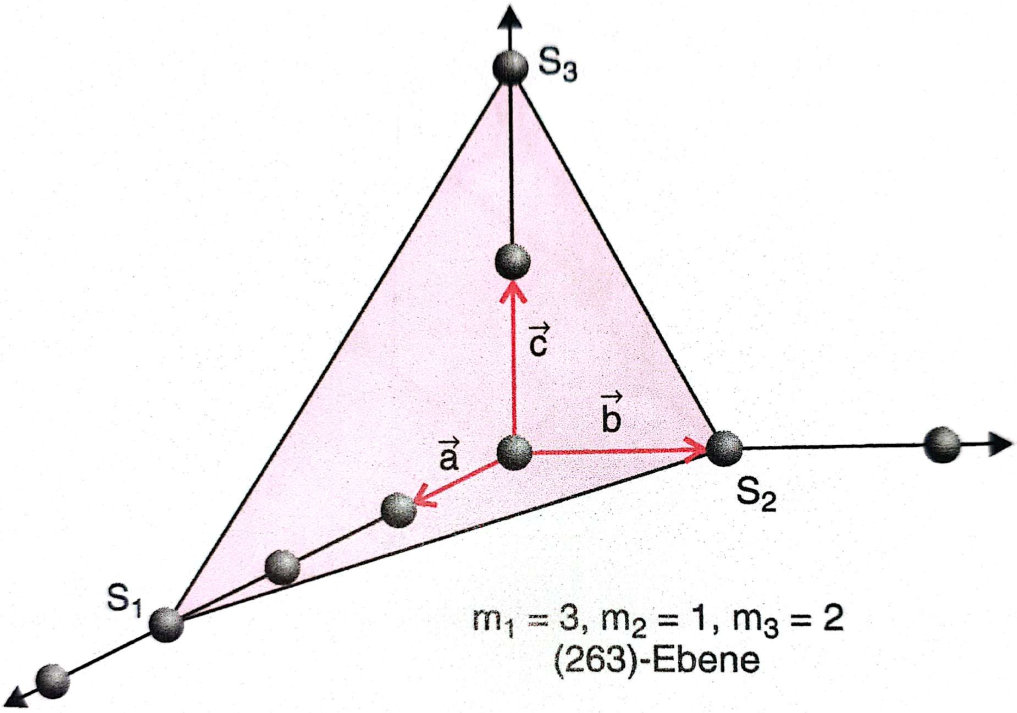


Figure 35.2.: Excerpt from Wolfang Demtröders Experimentalphysik 3.

Proof 35.3 By taking two non-identical vectors that span the plane and simultaneously represents differences between lattice-points, we can form linear combinations such that $m_i \in \mathbb{Q}$.

Definition 35.4 Let $p \in \mathbb{N}_{>0}$ denote the smallest positive natural number s.t.

$$h = \frac{p}{m_1} \qquad k = \frac{p}{m_2} \qquad l = \frac{p}{m_3} \tag{35.1}$$

are three (non-partial) integers. This triple (hkl) is called the Miller's indices associated to the lattice-plane above.

Remark 35.4.1 If one considers the lattice-plane, that has the intersection-points $\vec{S}_i \equiv nm_i\mathbf{a}_i$ with $m_i \in \mathbb{Q}$ and $n \in \mathbb{Z}$, then the number $|np|$ is the smallest integer s.t.

$$\pm h = \frac{|np|}{nm_1} \qquad \pm k = \frac{|np|}{nm_2} \qquad \pm l = \frac{|np|}{nm_3} \tag{35.2}$$

are three (non-partial) integers. We therefore see, that the tripel $\pm(hkl)$ is not in one-to-one correspondence to a single lattice-plane, but in „infinity-to-one" correspondence to infinitely many layers of lattice-planes.

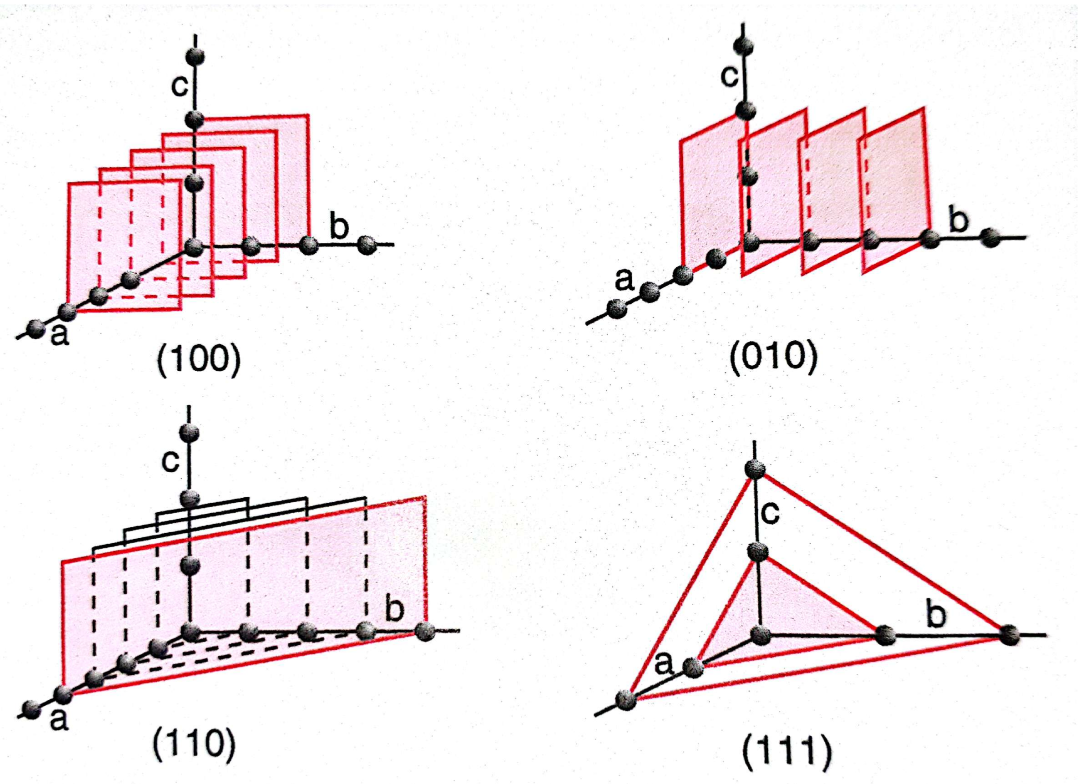


Figure 35.3.: Excerpt from Wolfang Demtröders Experimentalphysik 3.

Lemma 35.5 The reciprocal lattice-vector $\mathbf{G} = h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3$ is orthogonal to the planes (hkl).

Proof 35.5 See Appendix, p. 301.

Lemma 35.6 The absolute-value of $\mathbf{G}$ is inversely proportional to the distance d_{hkl} of two neighboring lattice-planes of the layers (hkl) as

$$|\mathbf{G}| = \frac{2\pi}{d_{hkl}}. \tag{35.3}$$

Proof 35.6 The vector $\mathbf{G}/|\mathbf{G}|$ has length one and is orthogonal to the plane. We consider now any lattice-point on the plane as „origin“. A vector, that points to a point in a neighboring plane is, for example, $\mathbf{v} \equiv \frac{\mathbf{a}_1}{h}$ whereas we assume without loss of generality that $h \neq 0$. The distance $d \equiv d_{hkl}$ between those two planes is then given as

$$d_{hkl} = \frac{\mathbf{G}}{G} \cdot \mathbf{v} = \frac{2\pi}{G}. \tag{35.4}$$

Corollary 35.7 For a three dimensional simple cubic lattice we obtain for the distance d_{hkl} between lattice-layers described by the tripel (hkl)

$$d_{hkl} = \frac{a}{\sqrt{h^2+k^2+l^2}}. \tag{35.5}$$

Proof 35.7 Follows from the orthogonality of the reciprocal lattice-vectors and the formula in the Lemma.

35.2. The Scattering-Amplitude

35.2.1. A Single Scattering-Center

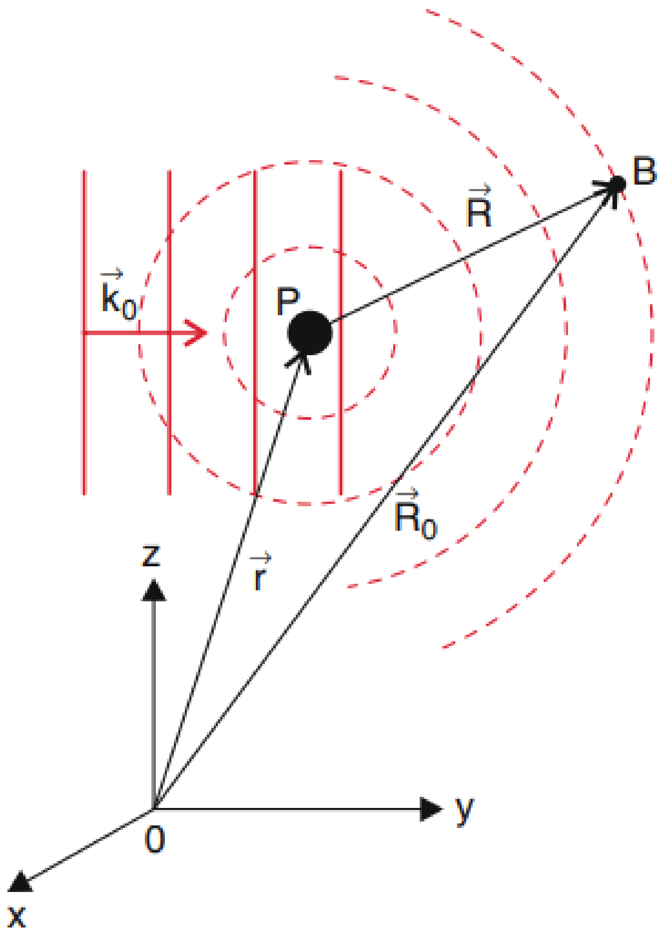


Figure 35.4.: Excerpt from Wolfang Demtröders Experimentalphysik 3.

We consider as in the figure a scattering-center (i.e. a single atom inside a crystal) at the position described by the vector $\mathbf{r}$. If a plane wave

$$\mathbf{E}(\mathbf{x}) = \mathbf{A}_e e^{i\mathbf{k}_0\cdot\mathbf{x}-\omega t} \tag{35.6}$$

falls into this scattering-center, a fraction f of this plane-wave is being scattered as a spherical wave[1] into all directions. In an observation-point $B(\mathbf{R}_0)$ with $\mathbf{R}_0 \gg \mathbf{r}$ we have for the amplitude A_B of the scattered wave

[1]This is far from obvious. ToBeElaborated.

$$
\begin{aligned}
A_B &= f\frac{\mathbf{E}(\mathbf{x})}{R}\mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{R}} \\
&= f\frac{\mathbf{A}_e}{R}\mathrm{e}^{\mathrm{i}\mathbf{k}_0\cdot\mathbf{x}-\omega t}\mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{R}} \\
&= f\frac{\mathbf{A}_e}{R}\mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{R}_0}\mathrm{e}^{\mathrm{i}\left(\mathbf{k}_0-\mathbf{k}\right)\cdot\mathbf{x}}\mathrm{e}^{-\mathrm{i}\omega t} \\
&= Cf\mathrm{e}^{\mathrm{i}\Delta\mathbf{k}\cdot\mathbf{x}}
\end{aligned}
$$

whereas we introduced the abbreviations $\Delta\mathbf{k} = \mathbf{k}_0 - \mathbf{k}$ and $C = \left(A_e/R\right)\mathrm{e}^{\mathrm{i}\left(\mathbf{k}\cdot\mathbf{R}_0-\omega t\right)}$.

35.2.2. A Crystal with a mono-atomic basis as Scattering-Centers at Lattice-Positions

In a crystal with a mono-atomic basis, we have many scattering-centers with positions $\mathrm{R}_\mathbf{n} \equiv \sum_j n_j\mathbf{a}_j$. This yields in a calculation analogous to the one above the amplitude A_B of the scattered waves

$$
\begin{aligned}
A_B &= Cf\sum_{\mathbf{R}_\mathbf{n}\in\mathrm{L_f}}\mathrm{e}^{\mathrm{i}\Delta\mathbf{k}\cdot\mathbf{R}_\mathbf{n}} \\
&= Cf\mathscr{F}_{\mathrm{Lattice}}(\Delta\mathbf{k}),
\end{aligned}
$$

whereas we introduced the so-called „lattice-factor“ $\mathscr{F}_{\mathrm{L_f}}(\Delta\mathbf{k})$. The usual way to proceed at this point is to argue that that the lattice-factor is non-zero, whenever the Laue-condition

$$
\Delta\mathbf{k}\cdot\mathbf{R}_\mathbf{n} \in 2\pi\mathbb{Z} \quad\Longleftrightarrow\quad \Delta\mathbf{k} \equiv \mathbf{G}_{hkl} \in \mathrm{RL}_\infty \tag{35.7}
$$

is fulfilled, and averages to zero (due to a so-called „random phase cancellation“) whenever this is not the case, which reads as a formula

$$
\mathscr{F}_{\mathrm{Lattice}}(\Delta\mathbf{k}) \approx \begin{cases} N_p & \text{if } \Delta\mathbf{k} \equiv \mathbf{G}_{hkl} \in \mathrm{RL}_\infty \\ 0 & \text{otherwise (due to rpc).} \end{cases} \tag{35.8}
$$

It is important to note, that this equation alone gives us nothing else but the Bragg-condition, i.e. implying that the lattice-structure does imply the existence of constructive interference. However, in the next step we will determine proper tools in order to make further conclusions about the type of the crystal-structure.

35.2.3. A Crystal with a many-atomic basis as Scattering-Centers at Crystal-Positions

In a crystal with a potentially many-atomic basis, we have many scattering-centers with positions

$$
\mathrm{R}_{\mathbf{n}\kappa} \equiv \sum_j n_j\mathbf{a}_j \quad + \quad \sum_j c_{j\kappa}\mathbf{a}_j \equiv \mathbf{R}_\mathbf{n} \quad + \quad \sum_j c_{j\kappa}\mathbf{a}_j \tag{35.9}
$$

with $c_{j\kappa} < 1$. This yields in a calculation analogous to the one above the amplitude A_B of the scattered waves

$$
\begin{aligned}
A_B &= C \sum_{\mathbf{R_n} \in \mathrm{L_f}} \mathrm{e}^{\mathrm{i}\Delta\mathbf{k}\cdot\mathbf{R_n}} \sum_{\kappa \in \{1,\ldots,r\}} f_\kappa \mathrm{e}^{\mathrm{i}\Delta\mathbf{k}\cdot(\sum_j c_{j\kappa}\mathbf{a}_j)} \\
&= C\mathscr{F}_{\text{Lattice}}(\Delta\mathbf{k})\mathscr{F}_{\text{Atomic}}(\Delta\mathbf{k}),
\end{aligned}
$$

whereas we introduced the so-called „atomic scattering-factor“ $\mathscr{F}_{\text{Atomic}}(\Delta\mathbf{k})$.

35.3. The Debye-Scherrer Setup

Copy-Pasted from [Demtröder, Experimentalphysik 3] into DeepL and subsequently Copy-Pasted (and slightly adapted) into this document:

> *In the case of polycrystalline solids (e.g. pulverized crystalline material such as common salt, diamond powder, etc.), the microcrystals and their lattice planes are are randomly oriented (...), and the Bragg Bragg method can therefore not be used. Here is a method developed by Peter J. W. Debye (1884-1966) and Paul Scherrer (1890-1969) is useful here. The general idea is shown in Fig. 11.29. A thin tube containing the powdered sample is placed placed in the axis of a cylindrical vessel and irradiated with a monochromatic parallel X-ray beam. Only those microcrystals whose orientation is suitable to fulfill the Bragg condition contribute noticeably to the scattered radiation. This means that only at certain angles $2\theta_i$ against the direction of incidence, i.e. on cones with full aperture angles $4\theta_i$, reflected intensity is to be expected. It is measured on a film strip which is attached to the inner wall of the cylindrical container with radius RZ. On the unrolled developed film you can therefore see blackened approximated circular arcs (Fig. 11.29b), from whose radii $R_i \equiv R_C \sin(2\theta_i)$ the angles $theta_i$ and thus, for a known wavelength λ the grid plane spacing d_{hkl} can be determined.*

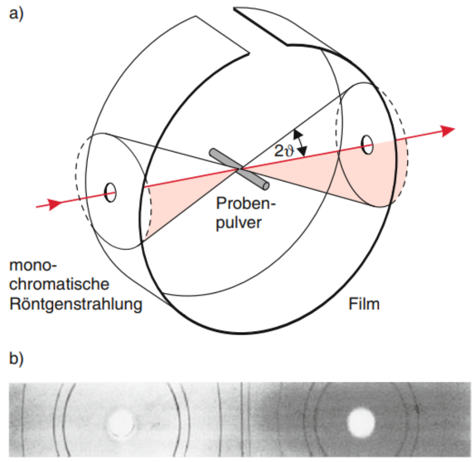


Abb. 11.29a,b. Debye-Scherrer-Verfahren: (**a**) Experimentelle Anordnung; (**b**) Aufnahme der ringförmigen Interferenzmaxima von Kupferpulver

Figure 35.5.: Excerpt from Wolfang Demtröders Experimentalphysik 3.

35.4. Cooking-Recipe to determine the Crystal-Structure from a Debye-Scherrer-Experiment

Given a diffraction pattern N_{counts} vs 2θ, we give a a cooking-recipe on how to determine the crystal-structure, whereas we restrict our attention to cubic lattices with only a single type of atom inside the crystal.

See the Lab-Sheet for the rest.

36. Work in Progress 1: The Internal Energy using the full Bethe-Salpeter-Equation in the Tree-Level-Approximation

36.1. Correcting the Internal Energy for the Phononic Contribution

We will now elaborate how we can compute the correction-terms from the Bethe-Salpeter-equation in the specific heat calculation

$$\begin{aligned}
\mathscr{G}^2_{0,\nu}(q)\hbar^2 \langle \rho_\nu(q)\rho_\nu(-q)\rangle &= \frac{\mathscr{G}^2_{0,\nu}(q)}{\hbar\beta N_p} \sum_{\substack{\sigma,\sigma' \in \{\downarrow,\uparrow\} \\ k,k' \in \mathrm{i}\Omega_{-1}\times\mathrm{Q} \\ n,n',m,m' \in \{1,\ldots,\infty\}}} g_{mn\nu}(\mathbf{k},\mathbf{q}) g_{m'n'\nu}(\mathbf{k}',-\mathbf{q}) \left\langle \overline{\psi}_{m\sigma}(k+q)\psi_{n\sigma}(k)\overline{\psi}_{m'\sigma'}(k'-q)\psi_{n'\sigma'}(k')\right\rangle \\
&= \frac{\mathscr{G}^2_{0,\nu}(q)}{\hbar\beta N_p} \sum_{\substack{\sigma,\sigma' \in \{\downarrow,\uparrow\} \\ k,k' \in \mathrm{i}\Omega_{-1}\times\mathrm{Q} \\ n,n',m,m' \in \{1,\ldots,\infty\}}} g_{mn\nu}(\mathbf{k},\mathbf{q}) g_{m'n'\nu}(\mathbf{k}',-\mathbf{q}) \left\langle \psi_{n\sigma}(k)\psi_{n'\sigma'}(k')\overline{\psi}_{m'\sigma'}(k'-q)\overline{\psi}_{m\sigma}(k+q)\right\rangle \\
&= \ldots
\end{aligned} \tag{36.1}$$

We focus on the contribution that we neglected earlier (from the Bethe-Salpeter-equation TBReferenced and Equation Exp. Value as GF) and compute

$$\begin{aligned}
&\left\langle \psi_{n\sigma}(k)\psi_{n'\sigma'}(k')\overline{\psi}_{m'\sigma'}(k'-q)\overline{\psi}_{m\sigma}(k+q)\right\rangle \\
&= \hbar^2 \mathscr{G}\left(n\sigma k,\, n'\sigma' k' \mid m\sigma k+q,\, m'\sigma' k'-q\right) \\
&= \text{"Wick-Theorem-contributions"} \\
&- \hbar^2 \sum_{\substack{\sigma_1,\sigma_2,\sigma_1',\sigma_2' \in \{\downarrow,\uparrow\} \\ n_1,n_2,n_1',n_2' \in \{1,\ldots,\infty\} \\ k_1,k_2,k_1',k_2' \in \mathrm{i}\Omega_{-1}\times\mathrm{Q}}} \mathscr{G}_{0\,\substack{n,n_1 \\ \sigma,\sigma_1 \\ k,k_1}} \mathscr{G}_{0\,\substack{n',n_2 \\ \sigma',\sigma_2 \\ k',k_2}} V_{\substack{n_1,n_2,n_1',n_2' \\ \sigma_1,\sigma_2,\sigma_1',\sigma_2' \\ k_1,k_2,k_1',k_2'}} \mathscr{G}_{0\,\substack{n_1',m \\ \sigma_1',\sigma \\ k_1',k+q}} \mathscr{G}_{0\,\substack{n_2',m' \\ \sigma_2',\sigma' \\ k_2',k'-q}}
\end{aligned} \tag{36.2}$$

inserting this into the equation above yields

$$\begin{aligned}
\ldots &= -\hbar^2 \frac{\mathscr{G}^2_{0,\nu}(q)}{\hbar\beta N_p} \sum_{\substack{\sigma,\sigma' \in \{\downarrow,\uparrow\} \\ k,k' \in \mathrm{i}\Omega_{-1}\times\mathrm{Q} \\ n,n',m,m' \in \{1,\ldots,\infty\}}} g_{mn\nu}(\mathbf{k},\mathbf{q}) g_{m'n'\nu}(\mathbf{k}',-\mathbf{q}) \sum_{\substack{\sigma_1,\sigma_2,\sigma_1',\sigma_2' \in \{\downarrow,\uparrow\} \\ n_1,n_2,n_1',n_2' \in \{1,\ldots,\infty\} \\ k_1,k_2,k_1',k_2' \in \mathrm{i}\Omega_{-1}\times\mathrm{Q}}} \mathscr{G}_{0\,\substack{n,n_1 \\ \sigma,\sigma_1 \\ k,k_1}} \mathscr{G}_{0\,\substack{n',n_2 \\ \sigma',\sigma_2 \\ k',k_2}} V^{\mathrm{as}}_{\substack{n_1,n_2,n_1',n_2' \\ \sigma_1,\sigma_2,\sigma_1',\sigma_2' \\ k_1,k_2,k_1',k_2'}} \mathscr{G}_{0\,\substack{n_1',m \\ \sigma_1',\sigma \\ k_1',k+q}} \mathscr{G}_{0\,\substack{n_2',m' \\ \sigma_2',\sigma' \\ k_2',k'-q}} \\
&= \ldots
\end{aligned} \tag{36.3}$$

now we terminate the summations over n_1, n_2, n_1', n_2', $\sigma_1, \sigma_2, \sigma_1', \sigma_2'$ and k_1, k_2, k_1', k_2' and obtain

$$\ldots = \frac{-\hbar\mathscr{G}_{0,\nu}^2(q)}{\beta N_p} \sum_{\substack{\sigma,\sigma' \in \{\downarrow,\uparrow\} \\ k,k' \in \mathrm{i}\Omega_{-1}\times\mathrm{Q} \\ n,n',m,m' \in \{1,\ldots,\infty\}}} g_{mn\nu}(\mathbf{k},\mathbf{q}) g_{m'n'\nu}(\mathbf{k}',-\mathbf{q}) \mathscr{G}^0_{\substack{n,n \\ \sigma,\sigma \\ k,k}} \mathscr{G}^0_{\substack{n',n' \\ \sigma',\sigma' \\ k',k'}} V^{\mathrm{as}}_{\substack{n,n',m,m' \\ \sigma,\sigma',\sigma,\sigma' \\ k,k',k+q,k'-q}} \mathscr{G}^0_{\substack{m,m \\ \sigma,\sigma \\ k+q,k+q}} \mathscr{G}^0_{\substack{m',m' \\ \sigma',\sigma' \\ k'-q,k'-q}} \tag{36.4}$$

In order to get the total contribution, we sum over ν and q and multiply with $1/(\hbar\beta)$ obtain

$$\begin{aligned}
&\text{"BS-Correction-Terms"} \\
&= \frac{1}{\hbar\beta} \sum_{\substack{\nu \in \{1,\ldots,rd_{\mathbf{x}}\} \\ q \in \mathrm{i}\Omega_{+1}\times\mathrm{Q}}} \frac{-\hbar\mathscr{G}_{0,\nu}^2(q)}{\beta N_p} \sum_{\substack{\sigma,\sigma' \in \{\downarrow,\uparrow\} \\ k,k' \in \mathrm{i}\Omega_{-1}\times\mathrm{Q} \\ n,n',m,m' \in \{1,\ldots,\infty\}}} g_{mn\nu}(\mathbf{k},\mathbf{q}) g_{m'n'\nu}(\mathbf{k}',-\mathbf{q}) \mathscr{G}^0_{\substack{n,n \\ \sigma,\sigma \\ k,k}} \mathscr{G}^0_{\substack{n',n' \\ \sigma',\sigma' \\ k',k'}} V^{\mathrm{as}}_{\substack{n,n',m,m' \\ \sigma,\sigma',\sigma,\sigma' \\ k,k',k+q,k'-q}} \mathscr{G}^0_{\substack{m,m \\ \sigma,\sigma \\ k+q,k+q}} \mathscr{G}^0_{\substack{m',m' \\ \sigma',\sigma' \\ k'-q,k'-q}} \\
&= \frac{-1}{\beta^2 N_p} \sum_{\substack{\sigma,\sigma' \in \{\downarrow,\uparrow\} \\ \nu \in \{1,\ldots,rd_{\mathbf{x}}\} \\ q \in \mathrm{i}\Omega_{+1}\times\mathrm{Q} \\ k,k' \in \mathrm{i}\Omega_{-1}\times\mathrm{Q} \\ n,n',m,m' \in \{1,\ldots,\infty\}}} g_{mn\nu}(\mathbf{k},\mathbf{q}) g_{m'n'\nu}(\mathbf{k}',-\mathbf{q}) \mathscr{G}_{0,\nu}^2(q) \mathscr{G}^0_{\substack{n,n \\ \sigma,\sigma \\ k,k}} \mathscr{G}^0_{\substack{n',n' \\ \sigma',\sigma' \\ k',k'}} V^{\mathrm{as}}_{\substack{n,n',m,m' \\ \sigma,\sigma',\sigma,\sigma' \\ k,k',k+q,k'-q}} \mathscr{G}^0_{\substack{m,m \\ \sigma,\sigma \\ k+q,k+q}} \mathscr{G}^0_{\substack{m',m' \\ \sigma',\sigma' \\ k'-q,k'-q}}
\end{aligned} \tag{36.5}$$

Now we use Equation (22.17) and compute the antisymmetrized part of the vertex as ...

$$\begin{aligned}
&V^{\mathrm{as}}_{\alpha_1,\alpha_2;\alpha_3,\alpha_4} \\
&= \mathrm{AS}\, \frac{1}{\hbar\beta N_p} \sum_{\nu \in \{1,\ldots,rd_{\mathbf{x}}\}} \frac{\omega_\nu(\mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})}{\omega^2_{\alpha_1-\alpha_4} + \omega^2_\nu(\mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})} \frac{g_{m_{\alpha_1} m_{\alpha_4}\nu}(\mathbf{k}_{\alpha_4}, \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4}) g^*_{m_{\alpha_3} m_{\alpha_2}\nu}(\mathbf{k}_{\alpha_2}, \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})}{\hbar^2} \delta_{k_{\alpha_3}+k_{\alpha_4}, k_{\alpha_2}+k_{\alpha_1}} \delta_{\sigma_{\alpha_1},\sigma_{\alpha_4}} \delta_{\sigma_{\alpha_2},\sigma_{\alpha_3}} \\
&= \mathrm{AS}\, \frac{1}{\hbar\beta N_p} \sum_{\nu \in \{1,\ldots,rd_{\mathbf{x}}\}} \frac{\omega_\nu(\mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})}{\omega^2_{\alpha_1-\alpha_4} + \omega^2_\nu(\mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})} \frac{g_{m_{\alpha_1} m_{\alpha_4}\nu}(\mathbf{k}_{\alpha_4}, \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4}) g_{m_{\alpha_2} m_{\alpha_3}\nu}(\mathbf{k}_{\alpha_2} + \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4}, -\mathbf{k}_{\alpha_1} + \mathbf{k}_{\alpha_4})}{\hbar^2} \delta_{k_{\alpha_3}+k_{\alpha_4}, k_{\alpha_2}+k_{\alpha_1}} \delta_{\sigma_{\alpha_1},\sigma_{\alpha_4}} \delta_{\sigma_{\alpha_2},\sigma_{\alpha_3}}
\end{aligned} \tag{36.6}$$

36.2. Correcting the Internal Energy for the Interacting Contribution

TBD.

37. Work in Progress 2: Feynman-Diagramology

37.1. Graph Theory

In quantum field theory, there are *four* famous theorems about "Feynman-diagrams", which are never proven in the generalaity in which one claims their applicability. Those are …

- The partition-function is the sum of all "diagrams" without "external vertices".
- The free energy is the sum of all "connected diagrams" without "external vertices".
- The self-energy is the sum of all "one-particle-irreducible two-point[1] diagrams".
- The n-body Green-functions are computed as sums of all "connected diagrams" with n "external vertices" (a.k.a. "Linked Cluster Theorem").

37.1.1. Fundamentals

Definition 37.1 A *statistical pre-graph* is a quintuple

$$g = (V, H, \partial, \epsilon, \tau), \tag{37.1}$$

where

- $V = \{v_1, v_2, \dots, v_{|V|}\}$ is a finite set of elements which we call *vertices*,
- $H = \{h_1, h_2, \dots, h_{|H|}\}$ is a finite set of elements which we call *oriented half-edges*,
- $(\partial, \epsilon) : H \to V \times \{\text{in}, \text{out}\}$ assigns to each half-edge its *associated vertex* and *direction*,
- $\tau : V \to \mathcal{T}$ assigns each vertex a *type* from a finite set of *vertex-types*.

Remark 37.1.1 The diagram elucidates the definitions.

$$\begin{tikzcd} H \arrow[r, "\partial"] \arrow[d, "\epsilon"'] & V \arrow[d, "\tau"] \\ \{\text{in}, \text{out}\} & \mathcal{T} \end{tikzcd}$$

Remark 37.1.2 Instead of laboriously defining the set of half-edges and their image under the map ∂, one often simply draws the half-graph as below.

[1] I.e. diagrams with two external legs.

Definition 37.2 For a vertex $v \in V$, its *valency* is defined by

$$\operatorname{val}(v) = \left|\partial^{-1}(v)\right| . \tag{37.2}$$

Fact 37.3 Vertices of valency 0 will never appear in the physical theory. Vertices of valency 1, and 2 and higher are usually denoted as ...

- A vertex v with $\operatorname{val}(v) = 1$ is attached to exactly one half-edge. These are "*source vertices*" or "*external vertices*", whereas we will clarify this distinction later on, when we introduce the underlying symbols. All remaining edges are called *internal edges*.
- A vertex v with $\operatorname{val}(v) = 2$ is (depending on the context) called a *mass* or a *potential* or an *impurity vertex* or a *one-body interaction-vertex*.
- A vertex v with $\operatorname{val}(v) = 4$ is called a *two-body interaction-vertex*.
- A vertex v with $\operatorname{val}(v) = 6$ is called a *three-body interaction-vertex*.
- $\dots$
- A vertex v with $\operatorname{val}(v) \geq 2$ can (more generally) be called a *many-body interaction-vertex* or (inclusively) *internal vertex*

Notation 37.4 The table summarizes the symbols that we will use to denote the different types of vertices and their properties.

Symbol(v)	$\tau(v)$	$\left(\epsilon \circ \partial^{-1}\right)(v)$	$\lvert\partial^{-1}(v)\rvert$
$\bullet$	external vertex	$\{\text{in}\}, \{\text{out}\}$	1
$\times$	source vertex	$\{\text{out}\}$	1
$\circ$	drain / sink vertex	$\{\text{in}\}$	1
$\boxed{m}$	mass vertex	$\{\text{in}, \text{out}\}$	2
$\boxed{p}$	potential vertex	$\{\text{in}, \text{out}\}$	2
$\boxed{\text{imp}}$	impurity vertex	$\{\text{in}, \text{out}\}$	2
$\rm{ooo}$	interaction vertex		$3, 4, 5 \dots$
$\sim$	interaction vertex		$3, 4, 5 \dots$

Examples:

Some **common examples** that one might be anticipating, if one has already seen some so-called "Feynman-diagrams", are the following:

a) Two vertices, one with outgoing one with incoming half-edge (valency 1 respectively). Cf. TBDrawn 1

b) Three vertices, one with outgoing one with incoming half-edge (valency 1 respectively) and one with two half-edges, one incoming and one outgoing (valency 2). Cf. TBDrawn 1

c) Three vertices, one with outgoing one with incoming half-edge (valency 1 respectively) and one with two half-edges, two incoming and two outgoing (valency 4). Cf. TBDrawn 1

d) Four vertices, one with outgoing one with incoming half-edge (valency 1 respectively) and one with two half-edges, two incoming and two outgoing (valency 4), and one with two half-edges, one incoming and one outgoing (valency 2). Cf. TBDrawn 1

Other, more **exotic examples**, that will however also become relevant, are the following:

a) Four vertices, two with outgoing two with incoming half-edges (valency 1 respectively). Cf. TBDrawn 2

b) Three vertices, one with outgoing one with incoming half-edges (valency 1 respectively) and one with three half-edges ...
   - three incoming, Cf. TBDrawn 2
   - two incoming and one outgoing, Cf. TBDrawn 2
   - one incoming and two outgoing, Cf. TBDrawn 2
   - three outgoing. Cf. TBDrawn 2

c) Three vertices, one with outgoing one with incoming half-edge (valency 1 respectively) and one with two half-edges, two incoming and two outgoing (valency 4). Cf. TBDrawn 2

We now extend the definition above.

Definition 37.5 A *statistical graph* is a statistical pre-graph $g = (V, H, \partial, \epsilon, \tau)$ equipped with an involution

$$\iota : H \to H,$$

that we call *pairing*, satisfying

$$\begin{aligned} \iota^2 &= \mathrm{id}_H, \\ \epsilon &\neq \epsilon \circ \iota \end{aligned} \tag{37.3}$$

Remark 37.5.1 We will notate the statistical graph as a sextuple $G = (V, H, \partial, \epsilon, \tau, \iota)$.

Remark 37.5.2 The involution tells us which half-edges are *glued* together to form a so-called "edge" (to be defined cleanly below). The condition $\iota^2 = \mathrm{id}_H$ means that if $\iota(h) = h'$, then h is said to be *glued* to h'.

Remark 37.5.3 Notice that the second conditions prohibits a half-edge from being glued to itself.

Remark 37.5.4 Notice further that it might not be always possible to find a pairing for a given statistical pre-graph. In that case, we say that the statistical pre-graph is *unpairable*.

Remark 37.5.5 Notice that the definition implies that one finds the following diagram

$$
\begin{tikzcd}
H \arrow[r, "\iota"] \arrow[dr, "\epsilon"'] & H \arrow[r, "\partial"] \arrow[d, "\epsilon"] & V \arrow[d, "\tau"] \\
 & \{\text{in}, \text{out}\} & \mathcal{T}
\end{tikzcd}
$$

Lemma 37.6 For a given statistical pre-graph $g = (V, H, \partial, \epsilon, \tau)$, in order for a pairing to exist, one needs to have

$$H_{\text{in}} \equiv |\{h \in H \mid \epsilon(h) = \text{in}\}| = |\{h \in H \mid \epsilon(h) = \text{out}\}| \equiv H_{\text{out}} \tag{37.4}$$

Proof 37.6 Suppose that a pairing $\imath$ exists. Define

$$H_{\text{in}} = \{h \in H \mid \epsilon(h) = \text{in}\}, \qquad H_{\text{out}} = \{h \in H \mid \epsilon(h) = \text{out}\}.$$

The map $\imath$ therefore restricts to a map

$$\imath|_{H_{\text{in}}} : H_{\text{in}} \to H_{\text{out}}.$$

This map is injective: if $\imath(h_1) = \imath(h_2)$, then applying $\imath$ once more gives

$$h_1 = \imath^2(h_1) = \imath^2(h_2) = h_2.$$

It is also surjective: if $h_{\text{out}} \in H_{\text{out}}$, then $\imath(h_{\text{out}}) \in H_{\text{in}}$, and

$$\imath(\imath(h_{\text{out}})) = h_{\text{out}}.$$

Hence $\imath|_{H_{\text{in}}}$ is a bijection between H_{in} and H_{out}.
Therefore

$$|H_{\text{in}}| = |H_{\text{out}}|,$$

which is exactly

$$|\{h \in H \mid \epsilon(h) = \text{in}\}| = |\{h \in H \mid \epsilon(h) = \text{out}\}|.$$

Corollary 37.7 One finds the number n_{pairs} of pairings of a statistical pre-graph $g = (V, H, \partial, \epsilon, \tau)$ to be given by the formula

$$n_{\text{pairs}} = \begin{cases} (|H|/2)! & \text{if } |H| \text{ is even,} \\ 0 & \text{if } |H| \text{ is odd.} \end{cases} \tag{37.5}$$

Proof 37.7 Follows from the size of the symmetric group.

Notation 37.8 For a given statistical pre-graph $g = (V, H, \partial, \epsilon, \tau)$, we denote the set of all statistical graphs that can be constructed from g by equipping it with a pairing ι as $C(g)$, i.e.

$$C(g) \quad = \quad \{(V, H, \partial, \epsilon, \tau, \iota) \mid \iota \text{ is a pairing on } H\} \quad \equiv \quad \{G_1(g), G_2(g), \dots G_{n_{\text{pairs}}}(g)\}. \tag{37.6}$$

Remark 37.8.1 If no such pairing exists, then we say that the statistical pre-graph is *unpairable* and we set $C(g) = \emptyset$. Oftentimes we also call $\emptyset$ the *null-graph*

Definition 37.9 For a statistical graph we define an equivalence relation $\sim$ on H by

$$h \sim h' \quad \Longleftrightarrow \quad h' = \iota(h). \tag{37.7}$$

Remark 37.9.1 Notice that this implies that half-edges that were glued together are equivalent under this relation.

Definition 37.10 For a statistical graph, we define the *set of edges* $E \equiv E(G)$ as the quotient

$$E := H/\!\sim\, \equiv \left\{ e = (h, \iota(h)) \mid h \in H \ \wedge \ \epsilon(h) = \text{in} \right\}. \tag{37.8}$$

Consequently we call an element $e = (h, \iota(h)) \in E$ an *edge*.

Remark 37.10.1 Notice that we could have alternatively also written $E \equiv H/\,\iota$.

Remark 37.10.2 Notice that this allows us to rewrite the set of all graphs as $\{G_1, G_2, \dots G_{|E|}\}$.

Remark 37.10.3 Notice that the definition implies that one finds the following diagram

$$
\begin{tikzcd}
 & E & \\
H \arrow[ur] \arrow[r, "\iota"] \arrow[dr, "\epsilon"'] & H \arrow[u] \arrow[r, "\partial"] \arrow[d, "\epsilon"] \arrow[ul, phantom, "\circlearrowright"] \arrow[dl, phantom, "\cancel{\circlearrowright}"] & V \arrow[d, "\tau"] \\
 & \{\text{in}, \text{out}\} & \mathcal{T}
\end{tikzcd}
$$

TBD: Re-creating this picture as a tikzpicture.

Definition 37.11 We now define a mapping

$$G \equiv G(g, \iota) = (V, H, \partial, \epsilon, \tau, \iota) \mapsto (V, E, \tau) \equiv \Gamma \tag{37.9}$$

and we call the pair (V, E, τ) a *statistical Feynman-diagram*.

Remark 37.11.1 Notice that if we equip the vertices with their types to a vector $\nu \equiv (v, \tau(v))$ and collect them in a set $\mathcal{V} = \{\nu \equiv (v, \tau(v)) \mid v \in V\}$ then we can also write the statistical Feynman-diagram as a pair $\Gamma \equiv (\mathcal{V}, E)$.

Remark 37.11.2 Notice that by extension we also obtain a mapping

$$C(g) \equiv \{G_1(g), G_2(g), \dots, G_{|E_{\bullet}(g)|}(g)\} \overset{\text{Feynman}}{\mapsto} \{\Gamma_i(g)\}_{i=1,2,\dots,|E_{\bullet}(g)|}. \tag{37.10}$$

Definition 37.12 We call two vertices v, w of a statistical Feynman-diagram *adjacent* if there exists an edge $e \in E$ such that $\partial(e) = (v, w)$ or $\partial(e) = (w, v)$.

Lemma 37.13 Two vertices v, w of a statistical Feynman-diagram are adjacent if and only if there exists a half-edge $h \in H$ such that $\partial(h) = v$ and $\partial(\iota(h)) = w$.

Proof 37.13 TBD.

So far we have only been considering Feynman-diagrams that originate from a single pre-graph.

Definition 37.14 We call two Feynman-diagrams $\Gamma_i \equiv (V_i, E_i, \tau_i)$ for $i \in \{1, 2\}$ *automorph*, if there is a bijection $\varphi : V_1 \to V_2$ such that

1. $\tau_1 = \tau_2 \circ \varphi$,
2. $v, w \in V_1$ are adjacent if and only if $\varphi(v), \varphi(w) \in V_2$ are adjacent.

Remark 37.14.1 For a given Feynman-diagram Γ we denote the set of all automorphisms as $\mathrm{Aut}(\Gamma)$.

Remark 37.14.2 The beautiful thing about statistical Feynman-diagrams is that many different so-called "Wick-contractions" produce automorphic graphs. Instead of computing all "Wick-contractions", one can compute only the automorphism-classes of diagrams and multiply with the number of Wick-contractions that produce the same diagram. This is the main idea behind the "diagrammatic expansion" in quantum field theory.

Lemma 37.15 The set of automorphisms $\mathrm{Aut}(\Gamma)$ forms a group under composition of maps.

Proof 37.15 TBDone.

Lemma 37.16 The automorphism-property in Definition 37.14 defines an equivalence-relation $\sim_{\mathrm{Aut}}$ on the set $\{\Gamma_i\}_{i=1,2,\ldots}$.

Proof 37.16 TBD.

Remark 37.16.1 Notice that this implies that the set of all Feynman-diagrams that are generated by a given pre-graph, i.e. the set $\{\Gamma_i\}_{i=1,2,\ldots}$ decomposes into equivalence-classes as

$$\{\Gamma_i\}_{i=1,2,\ldots} / \sim_{\mathrm{Aut}} \equiv \left\{ [\Gamma_i] \right\}_{i=1,2,\ldots} \equiv \tag{37.11}$$

Definition 37.17 For two statistical Feynman-diagrams $\Gamma_1 = (V_1, E_1, \tau_1)$ and $\Gamma_2 = (V_2, E_2, \tau_2)$ with $V_1 \cap V_2 = \emptyset$, we define the *union* of the two statistical Feynman-diagrams as

$$\Gamma_1 \cup \Gamma_2 = (V_1 \cup V_2, E_1 \cup E_2, \tau_{1\cup 2}) \tag{37.12}$$

with

$$\tau_{1\cup 2}(v) \equiv \begin{cases} \tau_1(v) & \text{if } v \in V_1, \\ \tau_2(v) & \text{if } v \in V_2. \end{cases} \tag{37.13}$$

Remark 37.17.1 A statistical Feynman-diagram $\Gamma = (V, E, \tau)$ is called *connected* if it cannot be expressed as the union of two disjoint non-empty statistical Feynman-diagrams. Otherwise, it is called *disconnected*.

Remark 37.17.2 If a statistical Feynman-diagram can be expressed as the union of connected statistical Feynman-diagrams, then those connected statistical Feynman-diagrams are called the *connected components* of the statistical Feynman-diagram.

[Maybe we will add things here later on.] [2]

37.1.2. These Wicked Diagrams...

We now define a mapping that associates a given statistical pre-graph with a set of statistical graphs, and consequently a set of statistical Feynman-diagrams.

Theorem 37.18 Let $g = (V, H, \partial, \epsilon, \tau)$ be a statistical pre-graph with $H_{\text{in}} = H_{\text{out}}$. Then considering all possible pairings, we obtain a mapping

$$g \overset{\langle\;\rangle}{\mapsto} C(g) \equiv \{G_1(g), G_2(g), \dots, G_{|E_\bullet(g)|}(g)\} \overset{\text{Feynman}}{\mapsto} \{\Gamma_i(g)\}_{i=1,2,\dots,|E_\bullet(g)|}. \tag{37.14}$$

with

$$|\langle g\rangle| = |C(g)| = |\{\Gamma_i(g)\}_{i=1,2,\dots,|E_\bullet(g)|}| = \begin{cases} |E_\bullet(g)| = |H/2|! & \text{if } |H| \text{ is even,} \\ 0 & \text{if } |H| \text{ is odd.} \end{cases} \tag{37.15}$$

Graphical Examples

TBD.

In the next sections, we will use the above definitions to formulate the diagrammatic expansion of the partition function, the free energy, and the Green-functions. We will also discuss how to compute the self-energy and the vertex functions using Feynman-diagrams. For simplicity, we will focus on systems with particle-number preserving interactions for the beginning.

[2] Cf. Book Graph Theory from Robin J. Wilson or arXiv:1301.6918v1.
Might be later relevant:

- matrix-representations of graphs
- adjacency matrices
- vector-space of a Graph
- walk in a graph
- etc.
- Cf. page 42 in the book of Salmhofer.
- page 110, 111 in Negele, Orland: Definition of a Tree-Diagram.
- page 110: reference to SSB and their implications for expectation-values.
- Definition of a pairing from Etingof.
- Cf. also Mahan page 85 for a decomposition of graphs.
- Cf. also Peskin page 305 for some nice pictures of Feynman-Diagrams.

37.2. Pure Systems

Notation 37.19 We use the notation of Notation 16.23. Thus

$$S[\psi,\overline{\psi}] = S_0[\psi,\overline{\psi}] + S_{\text{int}}[\psi,\overline{\psi}] \quad \text{with} \quad S_0[\psi,\overline{\psi}] = \psi^\dagger A\psi \tag{37.16}$$

and

$$C(i|i') := \left\langle \psi_i\overline{\psi}_{i'} \right\rangle_0 = A^{-1}_{ii'} \tag{37.17}$$

denotes the *free covariance*.

Remark 37.19.1 Notice that the free covariance is merely (in the physics notation) the Green-function. One has by Equation (16.27),

$$C(i|i') \equiv -c\,\mathscr{G}_0(i|i'). \tag{37.18}$$

Remark 37.19.2 The letter C is used here only to keep the diagrammatic combinatorics separate from the conventional Green-function prefactors. The edges of a diagram naturally carry the covariance $C = A^{-1}$; after the conventional normalization one rewrites them as free Green-functions.

Notation 37.20 We consider an index-set $\mathscr{V} = \{1,\dots,n_{\text{int}}\}$ for our interactions and we write the interaction as a finite sum of ordered monomials

$$S_{\text{int}}[\psi,\overline{\psi}] = \sum_{v\in\mathscr{V}} M_v[\psi,\overline{\psi}] \tag{37.19}$$

where every M_v is an ordered sum of monomials with degree $d_v + d'_v$ in the generators of $\mathscr{P}_\zeta$. If

$$\begin{aligned} M_v[\psi,\overline{\psi}] &\equiv \sum_{\substack{1\le j_1<\ldots<j_{d_v}\le D \\ 1\le j'_1<j'_2<\ldots<j'_{d'_v}\le D}} \overline{\psi}(j_1)\cdots\overline{\psi}(j_{d_v}) m_v(j_1,\dots,j_{d_v},j'_1,\dots j'_{d_v})\,\psi(j'_{d'_v})\cdots\psi(j'_1) \\ &= \frac{1}{d_v!\,d'_v!} \sum_{\substack{1\le j_1<\ldots<j_{d_v}\le D \\ 1\le j'_1<j'_2<\ldots<j'_{d_v}\le D}} \overline{\psi}(j_1)\cdots\overline{\psi}(j_{d_v}) m_v(j_1,\dots,j_{d_v},j'_1,\dots j'_{d_v})\,\psi(j'_{d'_v})\cdots\psi(j'_1) \end{aligned} \tag{37.20}$$

then an occurrence of M_v will be drawn as one vertex with r_v ordered half-edges.

Definition 37.21 For fixed *external indices*

$$(I|I') = (i_1,\dots,i_n\,|\,i'_n,\dots,i'_1) \tag{37.21}$$

we consider the ordered monomial

$$X_{(I|I')}[\psi,\overline{\psi}] := \psi_{i_1}\cdots\psi_{i_n}\overline{\psi}_{i'_1}\cdots\overline{\psi}_{i'_n}. \tag{37.22}$$

Fact 37.22 When it comes to the computation of ...

- partition-functions,
- free energies,
- Green-functions,

... we encounter the problem of computing the expectation-value of a polynomial expression of the form

$$P[\psi,\overline{\psi}] \equiv \left(X_{(I|I')}\mathrm{e}^{-S_{\mathrm{int}}}\right)[\psi,\overline{\psi}] = \left(\sum_{k=0}^{\infty}\frac{(-1)^k}{k!}X_{(I|I')}S_{\mathrm{int}}^k.\right)[\psi,\overline{\psi}] \tag{37.23}$$

and since with Equation (37.19) one has

$$S_{\mathrm{int}}^k[\psi,\overline{\psi}] = \left(\sum_{v\in\mathscr{V}}M_v[\psi,\overline{\psi}]\right)^k = \sum_{v_1,\ldots,v_k\in\mathscr{V}}M_{v_1}[\psi,\overline{\psi}]\cdots M_{v_k}[\psi,\overline{\psi}] \tag{37.24}$$

The goal is now to map polynomial expressions of the form

$$\left(X_{(I|I')}M_{v_1}\cdots M_{v_k}\right)[\psi,\overline{\psi}] \tag{37.25}$$

to pre-graphs (and therefore, subsequently, to Feynman-diagrams). We seek to do so, as we will encounter such polynomial expressions in the perturbative expansion of the partition function, the free energy, and the Green-functions. The following translation-key will allow us to do so. To this end, we now use Equation (37.20) and obtain the frighteningly long expression

$$\begin{aligned}
&\left(X_{(I|I')}M_{v_1}\cdots M_{v_k}\right)[\psi,\overline{\psi}]\\
&= \psi_{i_1}\cdots\psi_{i_n}\overline{\psi}_{i'_1}\cdots\overline{\psi}_{i'_n}\prod_{l=1}^{k}\frac{1}{d_{v_l}!\,d'_{v_l}!}\sum_{\substack{1\le j_{1,l}<\ldots<j_{d_{v_l},l}\le D\\ 1\le j'_{1,l}<j'_{2,l}<\ldots<j'_{d_{v_l},l}\le D}}\overline{\psi}(j_{1,l})\cdots\overline{\psi}(j_{d_{v_l},l})m_{v_l}(j_{1,l},\ldots,j_{d_{v_l},l},j'_{1,l},\ldots j'_{d_{v_l},l})\,\psi(j'_{d'_{v_l},l})\cdots\psi(j'_{1,l})\\
&= \sum_{\substack{1\le j_{1,1}<\ldots<j_{d_{v_1},l}\le D\\ 1\le j'_{1,1}<j'_{2,1}<\ldots<j'_{d_{v_1},l}\le D\\ \vdots\\ 1\le j_{1,k}<\ldots<j_{d_{v_k},k}\le D\\ 1\le j'_{1,k}<j'_{2,l}<\ldots<j'_{d_{v_k},k}\le D}}\psi_{i_1}\cdots\psi_{i_n}\overline{\psi}_{i'_1}\cdots\overline{\psi}_{i'_n}\prod_{l=1}^{k}\overline{\psi}(j_{1,l})\cdots\overline{\psi}(j_{d_{v_l},l})\frac{m_{v_l}(j_{1,l},\ldots,j_{d_{v_l},l},j'_{1,l},\ldots j'_{d_{v_l},l})}{d_{v_l}!\,d'_{v_l}!}\psi(j'_{d'_{v_l},l})\cdots\psi(j'_{1,l})\\
&= \sum_{\substack{1\le j_{1,1}<\ldots<j_{d_{v_1},l}\le D\\ 1\le j'_{1,1}<j'_{2,1}<\ldots<j'_{d_{v_1},l}\le D\\ \vdots\\ 1\le j_{1,k}<\ldots<j_{d_{v_k},k}\le D\\ 1\le j'_{1,k}<j'_{2,l}<\ldots<j'_{d_{v_k},k}\le D}}\left(\prod_{l=1}^{k}\frac{m_{v_l}(j_{1,l},\ldots,j_{d_{v_l},l},j'_{1,l},\ldots j'_{d_{v_l},l})}{d_{v_l}!\,d'_{v_l}!}\right)\psi_{i_1}\cdots\psi_{i_n}\overline{\psi}_{i'_1}\cdots\overline{\psi}_{i'_n}\prod_{l=1}^{k}\overline{\psi}(j_{1,l})\cdots\overline{\psi}(j_{d_{v_l},l})\psi(j'_{d'_{v_l},l})\cdots\psi(j'_{1,l})\\
&= \ldots
\end{aligned} \tag{37.26}$$

Remark 37.22.1 Notice that we have now for each monomial appearing in the sum of Equation (37.26) in total ...

- $N_{\text{tot}} \equiv 2n + \sum_{l=1}^{k}(d_{v_l} + d'_{v_l})$ variables
- $N_{\text{non-conj}} \equiv n + \sum_{l=1}^{k} d'_{v_l}$ are non-conjugated variables
- $N_{\text{conj}} \equiv n + \sum_{l=1}^{k} d_{v_l}$ are conjugated variables

Definition 37.23 We define *the pregraph associated to the polynomial expression in Equation* (37.26) as the statistical pre-graph with ...

- n external vertices with an outgoing half-edge
- n external vertices with an incoming half-edge
- k internal vertices of type $v_1, \ldots, v_k$, each with d_{v_l} incoming half-edges and d_{v_l} outgoing half-edges, $l = 1, \ldots, k$.

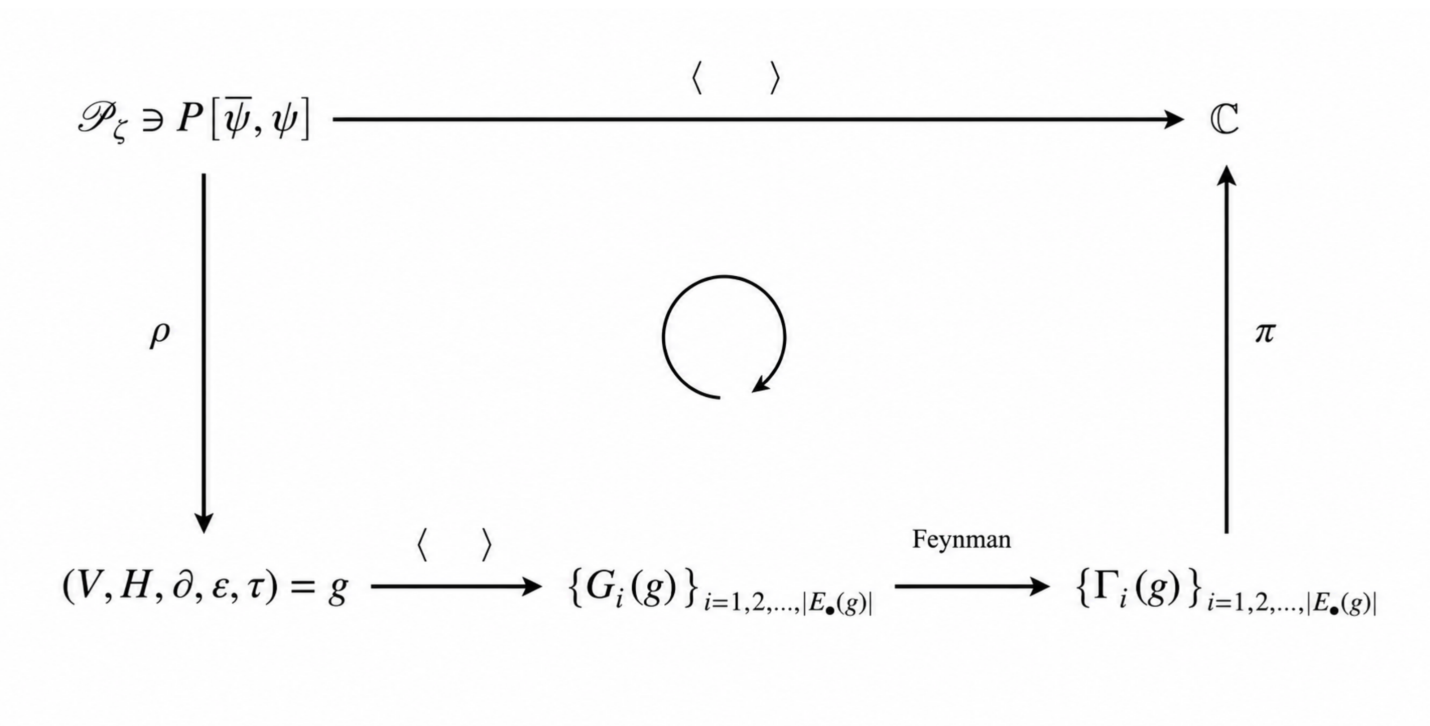


Figure 37.1.: TBD: Implementing this as a tikzpicture.

Fact 37.24 Notice that we can rewrite the polynomial expression in Equation (37.26) as

$$
\begin{aligned}
\ldots &= \sum_{\substack{1\le j_{1,1}<\ldots<j_{d_{v_1},l}\le D\\ 1\le j'_{1,1}<j'_{2,1}<\ldots<j'_{d_{v_1},l}\le D\\ \vdots\\ 1\le j_{1,k}<\ldots<j_{d_{v_k},k}\le D\\ 1\le j'_{1,k}<j'_{2,l}<\ldots<j'_{d_{v_k},k}\le D}} \left(\prod_{l=1}^{k}\frac{m_{v_l}(j_{1,l},\ldots,j_{d_{v_l},l},j'_{1,l},\ldots j'_{d_{v_l},l})}{d_{v_l}!\,d'_{v_l}!}\right)\psi_{i_1}\cdots\psi_{i_n}\overline{\psi}_{i'_1}\cdots\overline{\psi}_{i'_n}\prod_{l=1}^{k}\overline{\psi}(j_{1,l})\cdots\overline{\psi}(j_{d_{v_l},l})\psi(j'_{d'_{v_l},l})\cdots\psi(j'_{1,l})\\
&= \sum_{\substack{1\le j_{1,1}<\ldots<j_{d_{v_1},l}\le D\\ 1\le j'_{1,1}<j'_{2,1}<\ldots<j'_{d_{v_1},l}\le D\\ \vdots\\ 1\le j_{1,k}<\ldots<j_{d_{v_k},k}\le D\\ 1\le j'_{1,k}<j'_{2,l}<\ldots<j'_{d_{v_k},k}\le D}} \left(\prod_{l=1}^{k}\frac{m_{v_l}(j_{1,l},\ldots,j_{d_{v_l},l},j'_{1,l},\ldots j'_{d_{v_l},l})}{d_{v_l}!\,d'_{v_l}!}\right)\psi_{i_1}\cdots\psi_{i_n}\overline{\psi}_{i'_1}\cdots\overline{\psi}_{i'_n}\left(\prod_{l=1}^{k}\overline{\psi}(j_{1,l})\cdots\overline{\psi}(j_{d_{v_l},l})\right)\left(\prod_{l=1}^{k}\psi(j'_{d'_{v_l},l})\cdots\psi(j'_{1,l})\right)\\
&= \ldots
\end{aligned}
\tag{37.27}
$$

in order to get all the conjugated variables to the right and all the not-conjugated variables to the left, we need to apply the commutation-relations. This gives for $\psi(j'_{d'_{v_1},1})$ the prefactor $\zeta^{n+\sum_{l=1}^{k} d_{v_l}}$ this gives in total (applying the same scheme to all of the other variables) the prefactor

$$
\zeta^{d'_{v_1}\left(n+\sum_{l=1}^{k} d_{v_l}\right)+d'_{v_2}\left(n+\sum_{l=1}^{k} d_{v_l}\right)+\ldots+d'_{v_k}\left(n+\sum_{l=1}^{k} d_{v_l}\right)} = \zeta^{\sum_{\tilde{l}=1}^{k} d'_{v_{\tilde{l}}}\left(n+\sum_{l=1}^{k} d_{v_l}\right)}
\tag{37.28}
$$

such that

$$
\ldots = \zeta^{\sum_{\tilde{l}=1}^{k} d'_{v_{\tilde{l}}}\left(n+\sum_{l=1}^{k} d_{v_l}\right)} \sum_{\substack{1\le j_{1,1}<\ldots<j_{d_{v_1},l}\le D\\ 1\le j'_{1,1}<j'_{2,1}<\ldots<j'_{d_{v_1},l}\le D\\ \vdots\\ 1\le j_{1,k}<\ldots<j_{d_{v_k},k}\le D\\ 1\le j'_{1,k}<j'_{2,l}<\ldots<j'_{d_{v_k},k}\le D}} \left(\prod_{l=1}^{k}\frac{m_{v_l}(j_{1,l},\ldots,j_{d_{v_l},l},j'_{1,l},\ldots j'_{d_{v_l},l})}{d_{v_l}!\,d'_{v_l}!}\right)\left[\psi_{i_1}\cdots\psi_{i_n}\left(\prod_{l=1}^{k}\psi(j'_{d'_{v_l},l})\cdots\psi(j'_{1,l})\right)\overline{\psi}_{i'_1}\cdots\overline{\psi}_{i'_n}\left(\prod_{l=1}^{k}\overline{\psi}(j_{1,l})\cdots\overline{\psi}(j_{d_{v_l},l})\right)\right]
\tag{37.29}
$$

37.2.1. "Experimental Section": The Diagrammatic Expansions

Notation 37.25 In the following, all identities are understood as identities of formal power series in the interaction. If the series and integrals converge sufficiently strongly, the same identities hold analytically.

For fixed external indices $(I|I')$ and an ordered list $\boldsymbol{v} = (v_1, \dots, v_k)$, let

$$g(I|I'; \boldsymbol{v}) := \rho\big(X_{(I|I')} M_{v_1} \cdots M_{v_k}\big) \tag{37.30}$$

be the statistical pre-graph of Definition 37.23. We write $g(\emptyset; \boldsymbol{v})$ when there are no external vertices. As announced above, we restrict ourselves to particle-number preserving interactions, so that $d_v = d'_v$ for every vertex-type v.

Let

$$G \in \mathcal{C}(g(I|I'; \boldsymbol{v})) \tag{37.31}$$

and let $\Gamma(G)$ be its statistical Feynman-diagram. Its *labelled diagrammatic weight* $\omega(G; I|I')$ is obtained by

1. attaching the tensor m_{v_l}, including the factor $(d_{v_l}! d'_{v_l}!)^{-1}$ from Equation (37.20), to the internal vertex l;
2. attaching $C(a(h_{\text{out}})|a(h_{\text{in}}))$ to every edge $e = (h_{\text{in}}, h_{\text{out}})$, where $a(h)$ denotes the index carried by the corresponding field;
3. summing all internal indices and multiplying by $(-1)^k$ and by the Wick sign $\operatorname{sign}(\pi_\iota)^{1-\theta(\zeta)}$, where π_ι is the permutation from the outgoing to the incoming half-edges induced by the pairing ι.

Thus $\omega(G; I|I')$ is precisely the Wick contribution represented by G, before multiplication by the exponential factor $1/k!$. The empty diagram has weight one. For an automorphism-class $[\Gamma]$, we define

$$\pi([\Gamma]) := \frac{1}{k!} \sum_{\substack{\boldsymbol{v},\, G \in \mathcal{C}(g(I|I'; \boldsymbol{v})) \\ \Gamma(G) \in [\Gamma]}} \omega(G; I|I'). \tag{37.32}$$

In this last sum the external indices and the ordering of the half-edges are retained. Consequently, Equation (37.32) includes the multiplicity of all Wick contractions producing automorphic statistical Feynman-diagrams; equivalently, this is where the usual symmetry factor involving $\operatorname{Aut}(\Gamma)$ is contained.

Lemma 37.26 The Wick terms in

$$\big\langle X_{(I|I')} M_{v_1} \cdots M_{v_k} \big\rangle_0 \tag{37.33}$$

are in bijection with the statistical graphs in $\mathcal{C}(g(I|I'; \boldsymbol{v}))$. Under the Feynman mapping, their sum is therefore the sum of the weights of all statistical Feynman-diagrams generated by the associated pre-graph.

Proof 37.26 By particle-number preservation, the pre-graph has equally many incoming and outgoing half-edges. The Wick theorem, Theorem 16.21, associates every non-conjugated field with precisely one conjugated field. Hence every Wick term defines an involution ι on H satisfying $\iota^2 = \mathrm{id}_H$ and $\epsilon \neq \epsilon \circ \iota$, and therefore defines a statistical graph in $\mathcal{C}(g)$.

Conversely, a pairing ι restricts to a bijection from the outgoing to the incoming half-edges. It therefore determines one of the permutations in the Wick theorem, including its bosonic or fermionic sign. These two constructions are inverse to one another. The theorem on the pairings of a statistical pre-graph then maps these statistical graphs to the family $\{\Gamma_i(g)\}_i$ of statistical Feynman-diagrams. Finally, the definition of an edge gives exactly one covariance C for each paired pair of half-edges, so the Wick value equals the diagrammatic weight defined above.

Theorem 37.27 *Diagrammatic expansion of the partition function.*

The interacting partition function is the sum of all statistical Feynman-diagrams without external vertices:

$$\frac{Z}{Z_0} = \left\langle \mathrm{e}^{-S_{\mathrm{int}}} \right\rangle_0 = \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{v_1,\ldots,v_k \in \mathscr{V}} \sum_{G \in \mathcal{C}(g(\emptyset;v))} \omega(G;\emptyset) = \sum_{[\Gamma]\ \mathrm{vacuum}} \pi([\Gamma]). \tag{37.34}$$

Here the term $k = 0$ is the empty diagram, and "vacuum" means that the diagram has no external vertices.

Proof 37.27 From Notation 37.19 and the definition of the normalized free expectation value,

$$Z = Z_0 \left\langle \mathrm{e}^{-S_{\mathrm{int}}} \right\rangle_0 . \tag{37.35}$$

Expanding the exponential and then using Equation (37.19) gives

$$\frac{Z}{Z_0} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \sum_{v_1,\ldots,v_k \in \mathscr{V}} \left\langle M_{v_1} \cdots M_{v_k} \right\rangle_0 . \tag{37.36}$$

There is no external monomial in this expression. Lemma 37.26 turns every expectation value into the sum over all pairings of its associated pre-graph, and hence into the sum over all statistical Feynman-diagrams without external vertices. Regrouping those diagrams by the equivalence relation $\sim_{\mathrm{Aut}}$ gives the last member of Equation (37.34); no term is lost because $\pi([\Gamma])$ was defined as the sum over the entire automorphism-class.

Theorem 37.28 *Connected-diagram theorem for the free energy.*

Let $\mathfrak{D}^{\mathrm{conn}}_{\mathrm{vac}}$ denote the automorphism-classes of non-empty connected statistical Feynman-diagrams without external vertices. Then

$$\log \frac{Z}{Z_0} = \sum_{[\gamma] \in \mathfrak{D}^{\mathrm{conn}}_{\mathrm{vac}}} \pi([\gamma]). \tag{37.37}$$

Consequently the free-energy analogue $W = \log Z$ introduced in Definition 16.29 is

$$W = \log Z_0 + \sum_{[\gamma] \in \mathfrak{D}^{\mathrm{conn}}_{\mathrm{vac}}} \pi([\gamma]). \tag{37.38}$$

Thus, apart from the free contribution $\log Z_0$, the free energy is the sum of all connected vacuum diagrams. If the thermodynamic convention $F = -\beta^{-1} \log Z$ is used, the right-hand side is multiplied by $-\beta^{-1}$.

Proof 37.28 By the definitions of union and connectedness, every non-empty statistical Feynman-diagram Γ has a unique decomposition, up to the order of the factors, into connected components,

$$\Gamma = \gamma_1 \cup \cdots \cup \gamma_r. \tag{37.39}$$

The diagrammatic weight factorizes over this union. Indeed, the vertex tensors and edge covariances belonging to different components have disjoint internal indices. In the fermionic case every interaction vertex is even, so interchanging two components produces no additional sign. Therefore

$$\omega(\gamma_1 \cup \cdots \cup \gamma_r) = \prod_{a=1}^{r} \omega(\gamma_a). \tag{37.40}$$

Suppose that a connected automorphism-class $[\gamma]$ occurs m_γ times. The factor $1/k!$ in the interaction expansion, together with the number of ways of distributing the k labelled internal vertices among the connected components, leaves precisely the factor $1/m_\gamma!$ for repetitions of this component. Hence Theorem 37.27 becomes

$$\begin{aligned} \frac{Z}{Z_0} &= \sum_{(m_\gamma)} \prod_{[\gamma]\in\mathfrak{D}^{\mathrm{conn}}_{\mathrm{vac}}} \frac{\pi([\gamma])^{m_\gamma}}{m_\gamma!} \\ &= \prod_{[\gamma]\in\mathfrak{D}^{\mathrm{conn}}_{\mathrm{vac}}} \exp\big(\pi([\gamma])\big) = \exp\Bigg(\sum_{[\gamma]\in\mathfrak{D}^{\mathrm{conn}}_{\mathrm{vac}}} \pi([\gamma]) \Bigg). \end{aligned} \tag{37.41}$$

Taking the formal logarithm proves Equation (37.37).

Definition 37.29 A statistical Feynman-diagram generated by $g(I|I';\nu)$ is called *external-linked* if each of its connected components contains at least one external vertex. We denote the automorphism-classes of such diagrams by $\mathfrak{D}^{\mathrm{ext}}_{I|I'}$.

Theorem 37.30 *Cancellation of vacuum diagrams and linked-cluster theorem.*

Define the unnormalized numerator

$$\mathcal{N}(I|I') := \left\langle X_{(I|I')} \mathrm{e}^{-S_{\mathrm{int}}} \right\rangle_0 . \tag{37.42}$$

It decomposes as

$$\mathcal{N}(I|I') = \left\langle \mathrm{e}^{-S_{\mathrm{int}}} \right\rangle_0 \sum_{[\Gamma]\in\mathfrak{D}^{\mathrm{ext}}_{I|I'}} \pi([\Gamma]). \tag{37.43}$$

Therefore the denominator in the interacting expectation value cancels exactly all vacuum components, and

$$\mathcal{G}(I|I') = \left(-\frac{1}{c}\right)^n \frac{\left\langle X_{(I|I')} \mathrm{e}^{-S_{\mathrm{int}}} \right\rangle_0}{\langle \mathrm{e}^{-S_{\mathrm{int}}} \rangle_0} = \left(-\frac{1}{c}\right)^n \sum_{[\Gamma]\in\mathfrak{D}^{\mathrm{ext}}_{I|I'}} \pi([\Gamma]). \tag{37.44}$$

The connected n-body Green-function is the sum over the subfamily in which Γ itself is connected. In particular, for $n = 1$ every external-linked diagram is connected, so the ordinary one-body Green-function is the sum of all connected diagrams with its two external vertices.

Proof 37.30 Apply Lemma 37.26 to every coefficient in the interaction expansion of $\mathcal{N}(I|I')$. Every resulting diagram has a unique union decomposition

$$\Gamma = \Gamma_{\text{ext}} \cup \Gamma_{\text{vac}}, \tag{37.45}$$

where Γ_{ext} is the union of all connected components meeting an external vertex and Γ_{vac} is the union of all remaining components. By construction, Γ_{ext} is external-linked and Γ_{vac} is a possibly empty vacuum diagram.

If p of the k labelled internal vertices belong to Γ_{vac}, there are $\binom{k}{p}$ ways to choose them. Weight factorization and

$$\frac{1}{k!}\binom{k}{p} = \frac{1}{p!}\frac{1}{(k-p)!} \tag{37.46}$$

show that summing independently over the two parts gives

$$\mathcal{N}(I|I') = \left(\sum_{\Gamma_{\text{vac}}} \pi([\Gamma_{\text{vac}}])\right)\left(\sum_{[\Gamma_{\text{ext}}]\in\mathfrak{D}^{\text{ext}}_{I|I'}} \pi([\Gamma_{\text{ext}}])\right) \tag{37.47}$$

$$= \left\langle \mathrm{e}^{-S_{\text{int}}}\right\rangle_0 \sum_{[\Gamma]\in\mathfrak{D}^{\text{ext}}_{I|I'}} \pi([\Gamma]), \tag{37.48}$$

where the partition-function theorem was used in the second line. Since $Z = Z_0\langle \mathrm{e}^{-S_{\text{int}}}\rangle_0$, division by the denominator in the definition of the interacting expectation value proves Equation (37.44).

For general n, the remaining external-linked diagram may have several connected components, but each contains external vertices. The connected Green-function, equivalently the corresponding cumulant obtained from derivatives of $W[\bar{j}, j] = \log Z[\bar{j}, j]$, selects the diagrams with one connected component by the connected-diagram theorem. Finally, for $n = 1$, particle-number conservation implies that each connected component contains equally many incoming and outgoing external half-edges. The only outgoing and incoming external half-edges must consequently lie in the same component. Thus an external-linked one-body diagram is connected.

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